Morse Theory, proof by proof

The formal proof of Milnor's Morse Theory, in words

Contents

Part 0. The formal definitions **2**

Part I. Critical points and the lemma of Morse (Milnor §§1–2) **19**

Part II. Sublevel sets and attached cells (Milnor §3) **46**

Part III. Reeb, the torus, the Euler number and focal points (Milnor §§4–6) **72**

Part IV. Homology and the Morse inequalities (Milnor §5) **99**

Part V. Lefschetz, connections and curvature (Milnor §§7–9) **129**

Part VI. Geodesics and completeness (Milnor §10) **153**

Part VII. Energy, variations, Jacobi fields and the index theorem (Milnor §§11–15) **181**

Part VIII. Broken geodesics (Milnor §16) **213**

Part IX. The full path space and the fundamental theorem (Milnor §17) **237**

Part X. Non-conjugate points, topology and curvature (Milnor §§18–19) **266**

Part XI. The Appendix: the homotopy type of a monotone union **293**

Part XII. Homotopy groups, cells and fibre bundles **313**

Part XIII. Lie groups, symmetric spaces and periodicity (Milnor §§20–24) **340**

Concordance **406**

Part 0. The formal definitions

This part makes the book self-contained. It defines, in ordinary mathematical language and exactly as the formal proof defines them, the notions on which every later part is built: critical points and their indices, submanifolds of Euclidean space with their induced geometry, sublevel sets and deformation retracts, cells and cell complexes, singular homology, homotopy groups, path spaces with Milnor's metric, geodesics, Jacobi fields and the index form, and flows of vector fields. Each definition carries a trace tag. A tag whose path begins with `Mathlib/` points into Mathlib; one beginning with `Milnor/` points into the formalization of *Topology from the Differentiable Viewpoint*, called the companion project below, which is a Lake dependency of this one; every other tag points into this project. The parts that follow add the special definitions of their chapters (Milnor's function μ and the Morse charts of Theorem 3.2, focal points, broken geodesics, Clifford modules, and so on). Everything they build on is here, and for each definition the text says why it is the right formal counterpart of Milnor's notion and where it is not literally his.

Three settings recur, and it is worth separating them at once. Milnor's notions of §2 (critical points, the Hessian, its index and nullity) are defined on abstract manifolds in Mathlib's sense (§0.2), with no Hausdorff or countability hypothesis. Almost every theorem of the book from Theorem 3.1 on, the index theorem and the fundamental theorem of §17 included, is proved for a submanifold N of a Euclidean space in the sense of the companion project, with the Riemannian metric induced by the ambient inner product (§§0.1, 0.3). Every Riemannian manifold is isometric to one of these, by Nash's embedding theorem, which is quoted for orientation and used in no proof. The topology (deformation retracts, cell attachments, homology, homotopy groups, telescopes) is done for arbitrary topological spaces (§§0.4–0.7). A formal statement carries every hypothesis it uses, so Hausdorffness, compactness, finite dimension, connectedness and completeness appear exactly where they are needed.

0.1 The companion foundations

The project rests on the foundations laid in Part 0 of the companion volume. They are recalled here in outline, with tags into the companion project and into Mathlib; nothing is re-derived.

Calculus. All vector spaces are real normed spaces. A map $f : E \to F$ has derivative L at x, a continuous linear map, when $f(x+h) = f(x) + Lh + o(\|h\|)$ `Mathlib/Analysis/Calculus/FDeriv/Defs · HasFDerivAt`. The formal derivative $Df(x)$ is this L when it exists and 0 otherwise `Mathlib/Analysis/Calculus/FDeriv/Defs · fderiv`, and the derivative *within* a set uses only increments that stay in the set `Mathlib/Analysis/Calculus/FDeriv/Defs · fderivWithin`. For a function of one real variable the derivative is the vector $f'(t)$, and the derivative within $[t,\infty)$ or $(-\infty,t]$ is a one-sided derivative `Mathlib/Analysis/Calculus/Deriv/Basic · deriv` `Mathlib/Analysis/Calculus/Deriv/Basic · derivWithin`; the jumps Δ_t of §12 and the derivatives of a path at its ends are read this way. Smooth means C^∞ `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiff`. Smoothness at a point and on a set are local, order-by-order notions `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffAt` `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn`; for a function on $\mathbb{R}$, "smooth on $[a,b]$" includes one-sided derivatives of all orders at a and b. $\mathbb{R}^n$ is the space of functions $\{0,\dots,n-1\} \to \mathbb{R}$ with the Euclidean norm `Mathlib/Analysis/InnerProductSpace/PiL2 · EuclideanSpace`, so the formal text numbers

coordinates from 0. The gradient $\nabla F(x)$ of a real function on an inner product space is the vector with $\langle \nabla F(x), h\rangle = DF(x)h$ `Mathlib/Analysis/Calculus/Gradient/Basic · gradient`. Milnor's $\mathbb{R}^K$ is usually replaced by a finite-dimensional real inner product space V, which changes nothing but the notation.

Manifolds. A charted space modelled on H is a topological space M with an atlas of charts, each an open partial homeomorphism from an open subset of M onto an open subset of H `Mathlib/Topology/OpenPartialHomeomorph/Defs · OpenPartialHomeomorph`, and with a *preferred chart* at each point `Mathlib/Geometry/Manifold/ChartedSpace · ChartedSpace`. The model is embedded in a normed space E by a model with corners `Mathlib/Geometry/Manifold/IsManifold/Basic · ModelWithCorners`. In this book the model is always the identity of E, written $\mathcal{I}(\mathbb{R}, E)$ `Mathlib/Geometry/Manifold/IsManifold/Basic · modelWithCornersSelf`, which is boundaryless `Mathlib/Geometry/Manifold/IsManifold/Basic · ModelWithCorners.Boundaryless`: every manifold of the formal text is a manifold without boundary. M is a smooth manifold when any two of its charts are smoothly compatible `Mathlib/Geometry/Manifold/IsManifold/Basic · IsManifold`; the charts compatible with all of them form the maximal atlas `Mathlib/Geometry/Manifold/IsManifold/Basic · IsManifold.maximalAtlas`. Nothing else is required: unless a statement says so, a manifold need not be Hausdorff, second countable or finite-dimensional. The extended chart φ_x is the preferred chart at x, read in E `Mathlib/Geometry/Manifold/IsManifold/ExtChartAt · extChartAt`, and the local representative of $f : M \to N$ at x is $\varphi_{f(x)} \circ f \circ \varphi_x^{-1}$ `Mathlib/Geometry/Manifold/IsManifold/ExtChartAt · writtenInExtChartAt`. f is smooth at x when it is continuous at x and its local representative is smooth at $\varphi_x(x)$ `Mathlib/Geometry/Manifold/ContMDiff/Defs · ContMDiffAt`, and smooth on a set, or everywhere, in the same way `Mathlib/Geometry/Manifold/ContMDiff/Defs · ContMDiffOn` `Mathlib/Geometry/Manifold/ContMDiff/Defs · ContMDiff`. The tangent space T_xM is, by definition, the model space E `Mathlib/Geometry/Manifold/IsManifold/Basic · TangentSpace`, and the derivative $df_x : E \to F$ is the derivative of the local representative at $\varphi_x(x)$, or 0 when f is not differentiable at x `Mathlib/Geometry/Manifold/MFDeriv/Defs · mfderiv`; the companion project restates it with the type "continuous linear map $E \to F$" and nothing else `Milnor/Chapter5/Orient · mfd`. So df_x depends on the preferred charts at x and at $f(x)$, but only through composition with the invertible derivatives of chart transitions. A diffeomorphism is a smooth bijection with smooth inverse `Mathlib/Geometry/Manifold/Diffeomorph · Diffeomorph`, and a partial diffeomorphism is a smooth bijection between open subsets with smooth inverse `Mathlib/Geometry/Manifold/LocalDiffeomorph · PartialDiffeomorph`. The inverse function theorem between manifolds is the companion project's `Milnor/Chapter1/InverseFunction · isLocalDiffeomorphAt_of_isInvertible_mfderiv`.

Slice submanifolds. From Theorem 3.1 on, the manifolds of the formal text are, with few exceptions, subsets of a Euclidean space, as Milnor's are from §6 on, and the companion project describes such a subset by slice charts. Let S be a subset of a smooth manifold M (here M is always a normed space V with its standard model) and let Z, W be real normed spaces. A **slice chart** for S is a partial diffeomorphism φ from an open subset of M onto an open subset of $Z \times W$ such that, on its domain, $p \in S$ if and only if the W-component of $\varphi(p)$ vanishes; it comes with a base point of S in its domain `Milnor/Chapter2/Submanifold · SliceChart`. S is a **slice submanifold** modelled on Z, with complement W, if every point of S lies in the domain of some slice chart `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`. Such an S is a smooth manifold modelled on Z. Its charts are the restrictions to S of the maps $p \mapsto \varphi(p)_Z$, the Z-component of $\varphi(p)$ `Milnor/Chapter2/Submanifold · SliceChart.toChart`, and its preferred chart at a point is the one induced by a slice chart chosen at that point by the axiom of choice `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.chartedSpace`. The transitions are smooth `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.isManifold`, the inclusion $\iota : S \to M$ is smooth `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.contMDiff_val` and, for M finite-dimensional and without boundary, has injective derivative `Milnor/Chapter2/LevelSet · injective_mfderiv_val`, and a smooth map into M with values in S is smooth as a map into S `Milnor/Chapter2/Corestrict · IsSliceSubmanifold.contMDiff_codRestrict`. Since one subset may carry several such structures, the formal statements say which one they use, "the manifold structure given by the proof h_N that N is a slice submanifold"; this book suppresses that and writes simply N.

Throughout the book, **Milnor's "manifold $M \subseteq \mathbb{R}^K$ of dimension n"** is a set $N \subseteq V$ together with a proof h_N that it is a slice submanifold of V modelled on a space Z with complement W. Here V, Z, W are finite-dimensional wherever dimensions are counted, $\dim Z = n$, and V carries an inner product as soon as lengths or angles enter. Such an N is Hausdorff and second countable, as a subspace of V. It need

not be closed in V: an open subset of V is a slice submanifold. Compactness, closedness, connectedness and nonemptiness are separate hypotheses, stated where used. A function on N is usually the restriction $f = F|N = F \circ \iota$ of a smooth $F : V \to \mathbb{R}$. For compact nonempty $N \subseteq \mathbb{R}^{K+1}$ every smooth function on N is such a restriction `Chapter3/EulerChar · exists_contDiff_comp_val_eq`; for noncompact N the theorems assume it. Other slice submanifolds come from the companion project: the level set of a regular value `Milnor/Chapter2/LevelSet · isSliceSubmanifold_preimage`, the unit sphere $S^n = \{x \in \mathbb{R}^{n+1} : \langle x, x\rangle = 1\}$ `Milnor/Chapter5/HairyBall · Sphere.Sph`, and the tangent bundle $TN = \{(x, v) : x \in N,\ v \in T_xN\}$, a slice submanifold of $V \times V$ modelled on $Z \times Z$ `Milnor/Chapter8/TangentBundle · Problems.tangentBundleSet` `Milnor/Chapter8/TangentBundle · Problems.isSliceSubmanifold_tangentBundle`. If 0 is a regular value of a smooth g, the region $\{g \geq 0\}$ is a manifold with boundary $g^{-1}(0)$ in the companion project's sense `Milnor/Chapter2/WithBoundary · isHalfSliceSubmanifold_of_regularValue`; applied to $g = a - f$, this is Milnor's remark on p. 4 that M^a is a manifold with boundary when a is a regular value, and it is the only place where a boundary occurs.

Results imported from the companion project. Critical points and regular values in its sense, where x is critical when df_x is not surjective `Milnor/Chapter1/RegularValue · IsCriticalPoint` `Milnor/Chapter1/RegularValue · IsRegularValue`; Sard's theorem `Milnor/Chapter3/Sard · sard`; Hadamard's lemma `Milnor/Chapter6/Hadamard · exists_hadamard`; and, for a compact nonempty slice submanifold $N \subseteq \mathbb{R}^{K+1}$: the slice chart chosen at each point and its parametrization `Milnor/Chapter6/Tubular1 · Tub.σ` `Milnor/Chapter6/Tubular1 · Tub.φ`, the tangent space as a subspace of $\mathbb{R}^{K+1}$ `Milnor/Chapter6/ChartIndex · Tub.tanN`, the ε-neighbourhood theorem `Milnor/Chapter6/Tubular3 · Tub.tubular` with its radius, its nearest-point retraction and its neighbourhood N_ε `Milnor/Chapter6/Tubular4 · Tub.tubε` `Milnor/Chapter6/Tubular4 · Tub.tubr` `Milnor/Chapter6/Tubular4 · Tub.Nε`, the index of an isolated zero of a tangent vector field `Milnor/Chapter6/PoincareHopf · Tub.indexN`, the Poincaré–Hopf theorem for $K \geq 1$ `Milnor/Chapter6/PoincareHopf · Tub.poincare_hopf`, and the index of a non-degenerate zero as the sign of a determinant `Milnor/Chapter6/ChartIndex · Tub.indexN_eq_sgn_det`. Part III uses the last three to identify $\sum_\lambda (-1)^\lambda C_\lambda$ with the Euler number.

Conventions of the formal language. Four features recur, as in the companion volume.

- *Every function is total.* Where mathematics leaves a value undefined, the formal definition picks one, and this book says which. A derivative that does not exist is 0. The dimension of an infinite-dimensional space is 0 `Mathlib/LinearAlgebra/Dimension/Finrank · Module.finrank`, and so is the number of elements of an infinite set `Mathlib/Data/Set/Card · Set.ncard` and a finite sum over an infinite support `Mathlib/Algebra/BigOperators/Finprod · finsum`. The index of a quadratic form on an infinite-dimensional space is 0 (§0.2). The tangent projection, the tangent space and the curvature at a point of V off N are 0 (§0.3). $\exp_q(v)$ is q when there is no geodesic (§0.9), and the flow of a field without global integral curves is the identity (§0.10). Every theorem is stated where these values play no role, and the parts say so when a hypothesis is there only to exclude them.
- *Definitions by choice.* The preferred charts of a submanifold, the geodesic that defines exp and the integral curve that defines a flow are chosen by the axiom of choice; a theorem then shows that what matters does not depend on the choice.
- *Subsets as spaces.* A subset S of a space is also a space, whose points are points of the ambient space together with a proof that they lie in S. A sublevel set M^a is first a subset of M and then a space, and "M^a inside M^b" is the preimage of M^a under the inclusion of M^b, a subset of the space M^b (§0.4).
- *Universes.* Spaces built by pushouts live in a category of topological spaces of a fixed universe, a size bound on types, and a space compared with one of them must lie in the same universe; Betti numbers over a field need the field and the space in the same universe. The condition has no mathematical content.

0.2 Critical points, the Hessian, nullity and index, Morse functions

Milnor's definitions (pp. 4–5), for a smooth $f : M \to \mathbb{R}$: p is critical when $f_* : TM_p \to T\mathbb{R}_{f(p)}$ is zero; at a critical point the Hessian f_{**} is the bilinear form $f_{**}(v,w) = \tilde{v}_p(\tilde{w}(f))$, where $\tilde{v}$, $\tilde{w}$ are vector fields extending v, w; its matrix in local coordinates is $(\partial^2 f/\partial x^i \partial x^j(p))$, and the critical point is non-degenerate when this matrix is non-singular; the index of a bilinear form is the largest dimension of a subspace on which it is negative definite, and its nullity is the dimension of its null space. The formal proof defines everything first on a normed space, which is Milnor's situation after a choice of coordinates, and then on a manifold by reading the function in the preferred chart.

On a normed space. Let E be a real normed space and $f : E \to \mathbb{R}$.

- p is a **critical point** of f if f is differentiable at p with derivative 0 `Chapter1/Nondegenerate · IsCriticalPoint`. A point where f is not differentiable is not critical in this sense, unlike in the manifold version below; for smooth f the difference is empty.
- The **Hessian** of f at p is the second derivative $d^2 f_p = D(Df)(p)$, the continuous bilinear form $d^2 f_p(v,w) = D(x \mapsto Df(x)w)(p)\, v$ `Chapter1/Nondegenerate · hessian`. It is defined at every point, and Milnor's $\tilde{v}_p(\tilde{w}(f))$ is exactly this when $\tilde{v}$, $\tilde{w}$ are the constant extensions of v, w. It is symmetric wherever f is C^2, critical or not `Chapter1/Nondegenerate · hessian_symm`, and depends only on the germ of f at p `Chapter1/Nondegenerate · hessian_congr`. Milnor's proof of the symmetry, $\tilde{v}_p(\tilde{w}(f)) - \tilde{w}_p(\tilde{v}(f)) = [\tilde{v},\tilde{w}]_p(f) = 0$, uses that p is critical; here the symmetry is that of second derivatives, and criticality is needed only for the independence of coordinates.
- For a continuous bilinear form H on E: the **null space** of H is $\{v : H(v,w) = 0 \text{ for all } w\}$ `Chapter1/Nondegenerate · nullSpace` `Chapter1/Nondegenerate · mem_nullSpace`, and the **nullity** is its dimension `Chapter1/Nondegenerate · nullity`, by the convention above 0 if it is infinite-dimensional. H is **non-degenerate** if its null space is 0 `Chapter1/Nondegenerate · IsNondegenerate`, which for finite-dimensional E is nullity 0 `Chapter1/Nondegenerate · isNondegenerate_iff_nullity_eq_zero`. The **index** of H is Mathlib's negative index of inertia of the quadratic form $v \mapsto H(v,v)$ `Chapter1/Nondegenerate · quadForm` `Chapter1/Nondegenerate · index`.
- A **non-degenerate critical point** is a critical point at which $d^2 f_p$ is non-degenerate `Chapter1/Nondegenerate · IsNondegenerateCriticalPoint`.

Why the index is Mathlib's `sigNeg`. For a quadratic form Q on a real vector space of dimension r (so $r = 0$ for an infinite-dimensional space, by the convention of §0.1), Mathlib defines $\mathrm{sigPos}(Q)$ as the largest $k \le r$ such that Q is positive definite on some subspace of dimension k, and $\mathrm{sigNeg}(Q) = \mathrm{sigPos}(-Q)$ `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigPos` `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg`; its defining property is that it is the greatest dimension of a negative definite subspace `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg_isGreatest`. For finite-dimensional E this is literally Milnor's index: $\mathrm{index}(H)$ is the largest dimension of a subspace W with $H(v,v) < 0$ for every $0 \ne v \in W$ `Chapter1/Nondegenerate · index_isGreatest`. Taking Mathlib's notion brings Sylvester's law of inertia with it. The index is invariant under a linear change of variables `Chapter1/Nondegenerate · index_bilinearComp`, and for a diagonal form $\sum_i s_i y_i^2$ it is the number of negative s_i, which is how the λ of the Morse lemma is identified with the index (Part I). The price is that the definition is useless in infinite dimension, where the rank is 0 and so is the index. The path space of §§13–15 therefore needs a different index, defined in §0.9.

Matrices and changes of coordinates. For f of class C^2 at $p \in \mathbb{R}^n$, non-degeneracy of $d^2 f_p$ is Milnor's condition that the matrix $(\partial^2 f/\partial x^i \partial x^j(p))$ be non-singular `Chapter1/Nondegenerate · isNondegenerate_hessian_iff_det_ne_zero`. The second-order chain rule

$$d^2(f \circ \psi)_q(v,w) = d^2 f_{\psi(q)}(d\psi_q v, d\psi_q w) + df_{\psi(q)}(d^2\psi_q(v,w))$$

holds for ψ of class C^2 at q and f of class C^2 at $\psi(q)$ `Chapter1/Manifold · hessian_comp_apply`. At a critical point of f the last term vanishes, so there the Hessian transforms as a bilinear form `Chapter1/`

Nondegenerate · hessian_comp, and non-degeneracy and index are unchanged when $d\psi_q$ is a linear isomorphism Chapter1/Nondegenerate · isNondegenerate_hessian_comp_iff Chapter1/Nondegenerate · index_hessian_comp; so is the nullity Chapter1/Manifold · nullity_bilinearComp. This is Milnor's remark (p. 4) that non-degeneracy "can be checked directly" not to depend on the coordinates.

On a manifold. Let M be a charted space modelled on a real normed space E, with the boundaryless model $\mathcal{J}(\mathbb{R}, E)$. The definitions need nothing more; most theorems about them add that M is a smooth manifold. The model space is an explicit argument of every definition, because a charted space does not determine its model. Let $f : M \to \mathbb{R}$ and $p \in M$.

- The **local representative** of f at p is $f \circ \varphi_p^{-1}$, a function on E that is meaningful near $\varphi_p(p)$ Chapter1/Manifold · localRep. Its derivative at $\varphi_p(p)$ is df_p, with no hypothesis on f Chapter1/Manifold · mfderiv_eq_fderiv_localRep.
- p is a **critical point** of f if $df_p = 0$ Chapter1/Manifold · IsCriticalPointM. Since the formal df_p is 0 where f is not differentiable, such points count as critical; Milnor's f is smooth, so this never matters. The definition agrees with the companion project's, where p is critical when df_p is not surjective, because a linear functional is surjective exactly when it is nonzero Chapter1/Manifold · isCriticalPointM_iff_milnor.
- The **Hessian** f_{**} at p is the Hessian of the local representative at $\varphi_p(p)$, a continuous bilinear form on $T_pM = E$ Chapter1/Manifold · hessianM. It is symmetric wherever f is C^2 Chapter1/Manifold · hessianM_symm, and on $M = E$ with its standard chart it is $d^2 f_p$ Chapter1/Manifold · hessianM_model.
- p is a **non-degenerate critical point** if it is critical and f_{**} is non-degenerate Chapter1/Manifold · IsNondegenerateCriticalPointM. The **index** and the **nullity** of f at p are those of f_{**} Chapter1/Manifold · indexM Chapter1/Manifold · nullityM. The index is at most $\dim E$ Chapter1/Manifold · indexM_le_finrank, and on $M = E$ it is the index of $d^2 f_p$ Chapter1/Manifold · indexM_model.
- f is a **Morse function** if it is smooth and all its critical points are non-degenerate Chapter1/Manifold · IsMorseFunction: Milnor's "differentiable function with no degenerate critical points" (§§3, 6). Compact sublevel sets and distinct critical values are not part of the definition; they are hypotheses of the theorems that need them.
- The **Morse number** $C_\lambda(f)$ is the number of critical points of f of index λ Chapter3/EulerChar · critCount, counted by Mathlib's cardinality of sets, hence 0 if there are infinitely many. A Morse function on a compact smooth manifold modelled on a finite-dimensional space has finitely many critical points Chapter1/ManifoldMorse · IsMorseFunction.finite_setOf_isCriticalPointM, with no Hausdorff hypothesis, so on compact manifolds the convention plays no role.
- The **Euler number** of an abstract manifold M is $\sum_{\lambda=0}^{\dim M} (-1)^\lambda C_\lambda(f)$ for a Morse function f chosen once and for all, and 0 if M has none Chapter4/Whitney · eulerNumberM. This is Milnor's formula $\chi(M) = C_0 - C_1 + \cdots \pm C_n$ (p. 29) read as a definition; on a compact Hausdorff manifold every Morse function gives the same value Chapter4/Whitney · sum_critCount_eq_eulerNumberM, by the Poincaré–Hopf theorem on an embedded copy. For a compact nonempty submanifold $N \subseteq \mathbb{R}^{K+1}$, the integer of the Poincaré–Hopf theorem is recorded as a property of an integer χ: the indices of the zeros of every smooth tangent vector field on N with finitely many zeros add up to χ Chapter3/EulerChar · IsEulerNumber. Its identification with the homological Euler characteristic $\sum_i (-1)^i b_i$ (§0.6) is a theorem of Part IV.

Why the Hessian is defined in one chart, and why that is enough. Milnor defines f_{**} without coordinates and then computes it in coordinates. The formal definition goes the other way, fixing the preferred chart, so that the well-definedness of f_{**} *is* its independence of the chart, and that is a theorem. The coordinate systems allowed are the **smooth coordinate systems**: open partial homeomorphisms y from M to any normed space E' (for instance $\mathbb{R}^n$, as in Milnor) that are smooth on their domain, with inverse smooth on their image Chapter1/Manifold · IsSmoothCoord. Such a y is a partial diffeomorphism Chapter1/Manifold · IsSmoothCoord.toPartialDiffeomorph, and for $E' = E$ these are exactly the charts of the maximal atlas Chapter1/Manifold · isSmoothCoord_iff_mem_maximalAtlas. Let M be a smooth manifold, y a smooth coor-

dinate system whose domain contains p, and f of class C^2 at p; only C^2 is ever used.

- For f differentiable at p, p is critical iff $y(p)$ is a critical point of $f \circ y^{-1}$ `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff`; for y with values in $\mathbb{R}^n$, iff every $\partial f/\partial y^i(p)$ vanishes `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff_forall_partial`.
- At a critical point, the Hessian of $f \circ y^{-1}$ at $y(p)$ is f_{**} pulled back by $d(y^{-1})$: the matrix $(\partial^2 f/\partial y^i \partial y^j(p))$ represents f_{**} in the basis $\partial/\partial y^i|_p$ `Chapter1/Manifold · IsSmoothCoord.hessian_comp_symm`. Hence non-degeneracy, index and nullity can be computed in any coordinate system `Chapter1/Manifold · IsSmoothCoord.isNondegenerate_iff` `Chapter1/Manifold · IsSmoothCoord.index_eq` `Chapter1/Manifold · IsSmoothCoord.nullity_eq`, and non-degeneracy is the non-vanishing of $\det(\partial^2 f/\partial y^i \partial y^j(p))$ `Chapter1/Manifold · IsSmoothCoord.isNondegenerate_iff_det_ne_zero`.

Away from critical points f_{**} depends on the chart, and no statement uses it there. The same invariance holds under diffeomorphisms: if $g : M \to M'$ is a diffeomorphism, the critical points, indices, nullities, the Morse property and the Morse numbers of f and of $f \circ g^{-1}$ correspond `Chapter4/WhitneyInvariance · indexM_comp_symm` `Chapter4/WhitneyInvariance · isMorseFunction_comp_symm_iff` `Chapter4/WhitneyInvariance · critCount_comp_symm`. Milnor uses this without saying so when he replaces a manifold by an embedded copy. With Mathlib's Whitney embedding, every compact Hausdorff manifold is diffeomorphic to a compact slice submanifold of some $\mathbb{R}^{K+1}$ `Chapter4/Whitney · exists_diffeomorph_sliceSubmanifold`, and the two facts together carry the results proved for submanifolds to abstract compact manifolds (Part III).

Functions on a submanifold. For a slice submanifold $N \subseteq V$ modelled on Z and $F : V \to \mathbb{R}$, the function of the book is $f = F|N$ on the manifold N, with model Z, and its critical points, Hessian and index are those just defined. They are computed in a slice chart σ through Milnor's local parametrization $x(u) = \sigma^{-1}(u, 0)$ `Chapter1/ManifoldSubmanifold · sliceParam`. For q in the domain of σ and F differentiable at q, q is critical for $F|N$ iff the corresponding parameter is a critical point of $F \circ x$ `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_sliceParam`, and for F of class C^2 at a critical point the index is that of the Hessian of $F \circ x$ `Chapter1/ManifoldSubmanifold · index_hessian_comp_sliceParam`. Critical points can also be recognized without charts: q is critical iff $DF(q)$ vanishes on T_qN `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_forall_tangent`, iff the ambient gradient $\nabla F(q)$ is normal to N `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_gradient_mem_orthogonal`, which is how Milnor recognizes them on p. 35. The Hessian of $F|N$ is *not* the restriction of the ambient Hessian of F: in $F \circ x$ the second-order chain rule adds the term $DF(\partial^2 x/\partial u^i \partial u^j)$, which for $L_p = \|\cdot - p\|^2$ is the second fundamental form term of Milnor's formula on p. 36 `Chapter1/ManifoldSubmanifold · hessian_normSq_sub_comp`.

0.3 Submanifolds of a Euclidean space: tangent spaces, the tangent projection, the induced metric

From §6 on, and throughout §§8–19, the manifold of the formal text is a slice submanifold N of a finite-dimensional real inner product space V, modelled on Z with complement W, both finite-dimensional. Its Riemannian metric is induced from V: the inner product of two tangent vectors at q is their inner product in V. In this setting Milnor's coordinate constructions of §§8–10 have global formulas, all derived from one object, the orthogonal projection onto the tangent space.

Tangent spaces and the tangent projection.

- The **tangent space** $T_qN \subseteq V$ at $q \in N$ is the range of the derivative of the inclusion $N \to V$ at q `Chapter1/ManifoldSubmanifold · tangentSubspace`. It is the range of dx for the parametrization x of any slice chart around q `Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam`, it is the companion project's tangent space when $N \subseteq \mathbb{R}^{K+1}$ `Chapter1/ManifoldSubmanifold · tangentSubspace_eq_tanN`, and its dimension is $\dim Z$ `Chapter9/JacobiSystem · finrank_tanAt`. For a point x of V the text also uses T_xN, set to 0 off N `Chapter9/JacobiSystem · tanAt`. Milnor's tangent spaces are abstract. Identifying them with subspaces of V is what makes vector fields along curves, and the derivatives of

§8, ordinary vector-valued functions.

- The **tangent projection** $P_q : V \to V$ is the orthogonal projection onto T_qN `Chapter6/Projection · tanProj`. For the parametrization x of a slice chart, $P_{x(u)} = A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ with $A = dx_u$ `Chapter6/Projection · gramProj` `Chapter6/Projection · tanProj_eq_gramProj`, where $A^\mathsf{T}A$ is invertible because A is injective `Chapter6/Projection · gramL_bijective`. Hence $q \mapsto P_q$ **is a smooth map** from N to the operators on V `Chapter6/Projection · contMDiff_tanProj`. Milnor never needs this, because he works in coordinates; here every smoothness statement of §§8–10 is reduced to it. It is not automatic: the projection onto the range of A depends smoothly on A only while the rank does not drop. For integrands along paths it is convenient to have P_x for every $x \in V$, defined to be 0 off N `Chapter8/Energy · prj`; a vector v is tangent at $x \in N$ iff $P_x v = v$ `Chapter8/Energy · prj_eq_self_iff`.
- **Derivatives along** N**.** The slice chart chosen at q `Chapter6/Projection · sliceAt` defines the chart of N at q `Chapter6/Projection · chartAt_eq_sliceAt` and the chart coordinates $v \mapsto (d\sigma_q v)_Z$ of an ambient vector `Chapter6/Projection · tanCoord`. For a map $F : N \to E$ into a normed space, the derivative is then read on ambient vectors: $d^N F_q(v) = dF_q(\text{coordinates of } v)$ `Chapter6/Projection · dN`. Its values on vectors not tangent to N depend on the chart and are never used. On tangent vectors, if $F = G|N$ near q with $G : V \to E$ differentiable at q, then $d^N F_q(v) = DG(q)v$ `Chapter6/Projection · dN_eq_fderiv`, and all calculus along N is reduced to ambient calculus this way.

Connection, second fundamental form, curvature. A vector field on N is a map $X : N \to V$ with $X_q \in T_qN$ for every q `Chapter6/Submanifold · IsTangentField`. The Levi-Civita connection of the induced metric is Milnor's $X \vdash Y$, with $(X \vdash Y)_q = P_q(d^N Y_q(X_q))$ `Chapter6/Submanifold · leviCivita`; Part V proves Milnor's rules and Lemma 8.6 for it. The bracket is $[X,Y]_q = d^N Y_q(X_q) - d^N X_q(Y_q)$ `Chapter6/Submanifold · bracket`. The **second fundamental form** is $\mathrm{II}_q(v,w) = (d^N P_q\, v)\, w$, the derivative of the projection in the direction v, applied to w `Chapter6/Submanifold · secondFundForm`; for tangent v, w it is normal to N and symmetric. Milnor's **curvature tensor** (p. 51),

$$R(X,Y)Z = -X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X,Y] \vdash Z$$

`Chapter6/Submanifold · curvature`, is at each point $R_q(x,y) = D_y D_x - D_x D_y$ with $D_v = d^N P_q\, v$, that is, $R(x,y)z = \mathrm{II}(y,\mathrm{II}(x,z)) - \mathrm{II}(x,\mathrm{II}(y,z))$ `Chapter6/Submanifold · curvOp` `Chapter6/Submanifold · curvOp_apply` `Chapter6/Submanifold · curvature_eq`. Milnor's sign is kept, so that the unit sphere has sectional curvature $+1$ `Chapter6/SphereCurvature · sectionalCurvature_sph`. For paths, which lie in N only for $t \in [0,1]$, the curvature is used at every point of V, set to 0 off N `Chapter8/SecondVariation · curvR`.

Curves. For a curve c in N, a vector field along c is a map $X : \mathbb{R} \to V$ with $X(t) \in T_{c(t)}N$ `Chapter6/Parallel · IsFieldAlong`, and its **covariant derivative** is $DX/dt = P_{c(t)}(X'(t))$ `Chapter6/Parallel · covDeriv`, Milnor's D/dt (p. 46) for the induced metric. For paths given as functions $\mathbb{R} \to V$ the same formula is written with the total projection, $DX/dt = P_{c(t)}X'(t)$ `Chapter8/FirstVariation · covD`, and the **acceleration** of c is $D/dt\,(dc/dt)$ `Chapter8/FirstVariation · accel`. A vector field along c is **parallel** if it is differentiable with $DX/dt = 0$ `Chapter6/Parallel · IsParallel`; parallel transport is constructed in Part V `Chapter6/Parallel · parallelTransport`.

The induced metric. Lengths and distances are those of Milnor's §10, computed with the norm of V.

- The **length** of c on $[a,b]$ is $\int_a^b \|c'(t)\|\, dt$ `Chapter7/Path · pathLength`.
- c is **piecewise smooth** on $[a,b]$ if $a \le b$, c is continuous on $[a,b]$, and there is a finite set B of break points such that c is smooth, with one-sided derivatives at the ends, on every closed subinterval $[s,t] \subseteq [a,b]$ whose interior contains no point of B `Chapter7/Path · IsPiecewiseSmooth`. For Milnor's subdivision $a = t_0 < \cdots < t_k = b$ (p. 67) take $B = \{t_i\}$; formulating it with a set rather than an ordered list makes restriction and concatenation easy. Lower bounds for lengths are proved for the larger class of **admissible** paths, continuous on $[a,b]$, differentiable off a countable set and with integrable speed `Chapter7/Path · IsAdmissible`; every piecewise smooth path is admissible `Chapter7/Path · IsPiecewiseSmooth.isAdmissible`.

- The **distance** $\rho(p,q)$ is the infimum of the lengths of the piecewise smooth paths in N from p to q `Chapter7/Distance · pathLengths Chapter7/Distance · riemDist`. It lies in $[0,\infty]$ and is ∞ when there is no such path, for instance between two components of N, or when $q \notin N$. Milnor assumes M connected (p. 55); the formal ρ needs no such assumption and is finite on a connected N `Chapter7/Metric · riemDist_ne_top`. It is symmetric `Chapter7/Distance · riemDist_comm`, satisfies the triangle inequality `Chapter7/Distance · riemDist_triangle` and separates points `Chapter7/Distance · eq_of_riemDist_eq_zero`, so it makes N an extended metric space whose topology is, by construction, that of N as a subspace of V `Chapter7/Metric · riemEMetricSpace`; for connected N it is a metric space `Chapter7/Metric · riemMetricSpace Chapter7/Metric · riemMetricSpace_topology`. These are not declared as *the* metric of N, which as a subspace already carries the distance of V; statements name them explicitly. The metric balls about a point form a basis of its neighbourhoods `Chapter7/Distance · hasBasis_nhdsWithin_riemDist`, and ρ is continuous `Chapter7/Metric · continuous_riemDist`.
- N is **complete** if every ρ-Cauchy sequence of points of N converges for ρ to a point of N `Chapter7/Metric · IsRiemComplete`; for connected N this is completeness of the metric space (N,ρ) in Mathlib's sense `Chapter7/Metric · isRiemComplete_iff_completeSpace`. A set is **bounded** if it has finite ρ-diameter `Chapter7/Metric · IsRiemBounded`. A closed submanifold of V is complete `Chapter7/Metric · isRiemComplete_of_isClosed`, and this is where the complete examples of the book, spheres and compact groups of matrices, come from.

0.4 Sublevel sets, strong deformation retracts, homotopy equivalences

Sublevel sets. For any function $F : T \to \mathbb{R}$ on any set T and any $c \in \mathbb{R}$, $T^c = \{x : F(x) \le c\}$ `Chapter2/Crossing · sublevel`, increasing in c `Chapter2/Crossing · sublevel_mono`. This is Milnor's M^a (p. 12). For $f = F|N$ on a submanifold, N^a is the sublevel set of $F \circ \iota$, a subset of N, and as a space it carries the subspace topology. When a statement concerns $M^a \subseteq M^b$, M^a is viewed inside the space M^b as the preimage of M^a under the inclusion $M^b \to M$.

Homotopies. A homotopy from f_0 to $f_1 : X \to Y$ is a continuous $H : [0,1] \times X \to Y$ with $H(0,\cdot) = f_0$ and $H(1,\cdot) = f_1$, time coming first `Mathlib/Topology/Homotopy/Basic · ContinuousMap.Homotopy`, and f_0, f_1 are homotopic if one exists `Mathlib/Topology/Homotopy/Basic · ContinuousMap.Homotopic`. Here $[0,1]$ is Mathlib's unit interval, written I `Mathlib/Topology/UnitInterval · unitInterval`. A homotopy *relative to* $S \subseteq X$ is one with $H(t,x) = f_0(x)$ for all t and all $x \in S$ `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopicRel`, a special case of homotopies all of whose stages have a given property `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyWith`.

Strong deformation retracts. $A \subseteq X$ is a strong deformation retract of X if there are a continuous $r : X \to X$ with values in A and a homotopy from the identity of X to r relative to A `Chapter2/Crossing · IsStrongDeformationRetract`. In Milnor's notation (p. 13) the homotopy is $(t,x) \mapsto r_t(x)$ with $r_0 = \mathrm{id}$, $r_1 = r$, $r_1(X) \subseteq A$ and every r_t the identity on A; r is then a retraction onto A. Milnor says "deformation retract", but his deformations fix A at all times, and the formal definition records this because the homology of pairs in §5 needs it: to compare $H_*(M^b, A)$ with $H_*(M^a, A)$ for $A \subseteq M^a$, the deformation must be a homotopy of pairs, keeping A in place (Part IV). The inclusion of a strong deformation retract is a homotopy equivalence `Chapter2/Crossing · IsStrongDeformationRetract.exists_homotopyEquiv`; that statement is made for subsets $A \subseteq B$ of a space, with A a strong deformation retract of the space B, which is the form in which sublevel sets occur.

Homotopy equivalences. Mathlib's homotopy equivalence between X and Y is a structure: a pair of continuous maps $f : X \to Y$, $g : Y \to X$ with $g \circ f$ and $f \circ g$ homotopic to the identities `Mathlib/Topology/Homotopy/Equiv · ContinuousMap.HomotopyEquiv`. "X has the homotopy type of Y" is the existence of one. The project also uses the corresponding property of a given map: f is a homotopy equivalence if it has a homotopy inverse `Appendix/Telescope · IsHomotopyEquiv`; the two notions correspond `Appendix/Telescope · IsHomotopyEquiv.of_homotopyEquiv Appendix/Telescope · IsHomotopyEquiv.exists_homotopyEquiv`. Milnor's Theorem 3.1 states that M^a is diffeomorphic to M^b and a deformation retract of it, so that the inclusion

is a homotopy equivalence; the formal theorem has the three conclusions separately: a diffeomorphism of the whole manifold carrying M^a onto M^b, M^a a strong deformation retract of M^b, and the inclusion $M^a \to M^b$ a homotopy equivalence `Chapter2/Regular · theorem_3_1` `Chapter2/Submanifold · theorem_3_1_submanifold`.

Telescopes and homotopy direct limits (Milnor, Appendix, p. 149). For an increasing sequence $X_0 \subseteq X_1 \subseteq \cdots$ of subsets of X, the telescope is $X_\Sigma = \bigcup_n X_n \times [n, n+1] \subseteq X \times \mathbb{R}$, with the subspace topology `Appendix/Telescope · telescope`, and X is the **homotopy direct limit** of the X_n if the projection $X_\Sigma \to X$ `Appendix/Telescope · telescopeProj` is a homotopy equivalence `Appendix/Telescope · IsHomotopyDirectLimit`. If $X^{a_n} = \{E \le a_n\}$ for a continuous $E : X \to \mathbb{R}$ and $a_0 \le a_1 \le \cdots \to \infty$, then X is the homotopy direct limit of the X^{a_n}, with no paracompactness hypothesis `Appendix/Examples · isHomotopyDirectLimit_sublevel`. This is how the path space of §17 is exhausted by its sublevel sets (Parts IX and XI), and Milnor's Theorem A is stated in these terms `Appendix/TheoremA · theoremA`.

0.5 Cell attachments and cell complexes

The cell. Mathlib's disk $\mathbb{D}^n$ is the closed unit ball of $\mathbb{R}^n$ and $\partial\mathbb{D}^n$ its unit sphere, both as objects of the category of topological spaces, with the inclusion $\partial\mathbb{D}^n \to \mathbb{D}^n$ `Mathlib/Topology/Category/TopCat/Sphere · TopCat.disk` `Mathlib/Topology/Category/TopCat/Sphere · TopCat.diskBoundary` `Mathlib/Topology/Category/TopCat/Sphere · TopCat.diskBoundaryInclusion`. These are the basic cells of Mathlib's CW complexes. For $n = 0$ the sphere is empty and the disk is a point, which is Milnor's convention for 0-cells.

Attaching a cell. For a space X and a continuous $\varphi : \partial\mathbb{D}^n \to X$, the space $X \cup_\varphi e^n$ is the pushout, in the category of topological spaces, of φ and the inclusion $\partial\mathbb{D}^n \to \mathbb{D}^n$ `Chapter2/Attaching · AttachCell`: the quotient of $X \sqcup \mathbb{D}^n$ by $u \sim \varphi(u)$ for $u \in \partial\mathbb{D}^n$, with the quotient topology. It comes with the inclusion of X and the characteristic map of the cell `Chapter2/Attaching · cellInl` `Chapter2/Attaching · cellInr`. This is Milnor's definition (pp. 2–3). Working with the categorical pushout gives the universal property for free: a map out of $X \cup_\varphi e^n$ is a pair of maps on X and on $\mathbb{D}^n$ that agree along φ `Chapter2/Attaching · cellDesc`. A homotopy out of it is given in the same way `Chapter2/Attaching · cellHomotopy`, which needs one fact more than Milnor states: the product of the quotient map $X \sqcup \mathbb{D}^n \to X \cup_\varphi e^n$ with $[0,1]$ is again a quotient map, because $[0,1]$ is locally compact (J. H. C. Whitehead's lemma, here the exponential law). Lemmas 3.6 and 3.7, that homotopic attaching maps give homotopy equivalent spaces and that a homotopy equivalence extends over an attached cell, are proved for this construction `Chapter2/Attaching · lemma_3_6` `Chapter2/Attaching · lemma_3_7`.

Cells inside a space. What the proof of Theorem 3.2 produces is not a pushout but a subspace $M^{c-\varepsilon} \cup e^\lambda$ of M: a map χ from the closed unit ball of $\mathbb{R}^\lambda$ to M, continuous and injective on the ball, with $\chi(z) \in M^{c-\varepsilon}$ exactly when $\|z\| = 1$, such that $M^{c-\varepsilon} \cup \chi(\text{ball})$ is a strong deformation retract of $M^{c+\varepsilon}$ `Chapter2/CellAttachSubmanifold · theorem_3_2_submanifold`. The two forms are identified by Lemma D of the exposition. Let a Hausdorff space W be the union of a closed subset A and the image of a compact space D under a continuous injective map χ, let $i : S \to D$ be such that $\chi(d) \in A$ only for d in the image of i, and let $\varphi : S \to A$ satisfy $\varphi(s) = \chi(i(s))$ for every $s \in S$. Then W is the pushout of $A \xleftarrow{\varphi} S \xrightarrow{i} D$ `Chapter2/CellAdjunction · isPushout_of_cell`. This is Milnor's sentence "so that e^λ is attached to $M^{c-\varepsilon}$ as required" (p. 16). With it, Theorem 3.2 gives a homotopy equivalence from $N^{c+\varepsilon}$ to the pushout along an injective attaching map `Chapter2/CellAttachSubmanifold · theorem_3_2_attach_submanifold`. The homology computations of Part IV use the subspace form, the homotopy computations of Part XII the pushout form.

Finite cell complexes. K is an **iterated adjunction space** with cells of dimensions $(n_1, \dots, n_k)$ if it is obtained from the empty space by attaching cells of these dimensions one at a time, in this order, each along an arbitrary continuous map of its boundary sphere into the space built so far `Chapter2/CWType · IsCellAttachment`. A space X **has the homotopy type of a finite cell complex with cells** $(n_1, \dots, n_k)$ if it is homotopy equivalent to such a K, in the same universe `Chapter2/CWType · IsFiniteCellComplexType`; the property is invariant under homotopy equivalence `Chapter2/`

CWType · IsFiniteCellComplexType.of_homotopyEquiv. Milnor's Theorem 3.5 concludes that M has the homotopy type of a CW complex with one λ-cell for each critical point of index λ. The formal Theorem 3.5 concludes this property for the list of the indices of the critical points in order of critical value, hence with exactly C_λ cells of dimension λ Chapter2/CWType · theorem_3_5_count'. The difference is twofold. The attaching maps are arbitrary continuous maps, not cellular ones: Milnor makes them cellular with the cellular approximation theorem, which Mathlib does not have, and then applies Lemma 3.6; the formal induction uses only Lemma 3.7. And the cells are attached in order of critical value, not of dimension. The number of cells in each dimension is what Milnor's theorem is used for (the Morse inequalities, the Euler characteristic, the homotopy groups), and that is exactly what the formal statement records.

Relative cell complexes. K is obtained from L by attaching cells $(n_1, \dots, n_k)$ along $j : L \to K$ if it is reached from L by attaching these cells one at a time, j being the composite of the inclusions $K' \to K' \cup_\varphi e^n$ Appendix/MonotoneUnion · IsRelCellAttachment. A continuous $\iota : A \to X$ is **up to homotopy the inclusion of A with cells attached** if there are such a $j : A \to K$ and a homotopy equivalence $e : X \to K$ with $e \circ \iota$ homotopic to j Chapter10/RelativeCells · IsRelCellComplexType. This is the form in which one step $M^{a_n} \subseteq M^{a_{n+1}}$ of the relative Theorem 3.5, used for Theorem 17.3, is recorded, and it is transported through homotopy equivalences by an iterated, relative Lemma 3.7 Chapter10/RelativeCells · IsRelCellAttachment.transport.

Countable cell complexes. A **cell tower** with cells $c_0, c_1, \dots$, each a finite list of dimensions, is a sequence of spaces and maps $K_0 \to K_1 \to \cdots$ in which K_0 is an iterated adjunction space with cells c_0 and K_{n+1} is obtained from K_n by attaching the cells c_{n+1}, the map being the inclusion Appendix/MonotoneUnion · CellTower; each stage is a finite cell complex with the cells of the stages up to it Appendix/MonotoneUnion · CellTower.isCellAttachment. The **mapping telescope** of a sequence of spaces $F_0 \to F_1 \to \cdots$ with maps c_n is the quotient of $\bigsqcup_n F_n \times [0,1]$ by $(k,1)_n \sim (c_n(k),0)_{n+1}$ Appendix/MappingTelescope · MapTelRel Appendix/MappingTelescope · MappingTelescope, the point $(k,t)_n$ lying at level $n+t$ Appendix/MappingTelescope · MappingTelescope.level. A space **has the homotopy type of a countable cell complex with cells** $c_0, c_1, \dots$ if it is homotopy equivalent to the mapping telescope of a cell tower with these cells Appendix/MonotoneUnion · IsCountableCellComplexType. Milnor's Theorem 17.3 speaks of a CW complex $K = \bigcup K_i$ with the weak topology. Two facts make the mapping telescope the right formal model. The stages of a formal tower are pushouts connected by maps, not subspaces of one space, so there is no union to take. And the union with the weak topology is homotopy equivalent to the mapping telescope by Example 2 of the Appendix, which rests on Whitehead's theorem and is not formalized, whereas the telescope is exactly what the Appendix handles directly. The topological half of Theorem 17.3 is then: if X is exhausted by the sublevel sets X^{a_n} of a continuous function, $a_n \to \infty$, and homotopy equivalences $X^{a_n} \to K_n$ to the stages of a tower extend one another, X has the homotopy type of the countable complex Appendix/MonotoneUnion · isCountableCellComplexType_of_sublevel.

0.6 Singular chains, homology, relative homology, Betti numbers

Mathlib has the singular chain complex and the singular homology of a space, with coefficients in an object R of a category with coproducts Mathlib/AlgebraicTopology/SingularHomology/Basic · AlgebraicTopology.singularChainComplexFunctor Mathlib/AlgebraicTopology/SingularHomology/Basic · AlgebraicTopology.singularHomologyFunctor: its n-chains are a coproduct of copies of R indexed by the singular n-simplices, the continuous maps from the standard simplex. It has no relative homology, no long exact sequence of a pair and no excision. The project supplies them.

Pairs. A pair of spaces is an embedding $A \to X$ in the category of topological spaces, the morphisms of pairs being the commutative squares Mathlib/Topology/Category/TopPair · TopPair. For subsets $A \subseteq B$ of a space E, the pair (B, A) is the inclusion of the subspace A in the subspace B Homology/Relative · subsetPair; this is the form of Milnor's (M^b, M^a).

Relative chains and relative homology. Let the coefficients R lie in an abelian category $\mathcal{C}$ with coproducts. The singular chain functor Homology/Relative · singularChains sends the inclusion of a pair to

a monomorphism $C(A) \to C(X)$, and the relative chain complex $C(X, A)$ is defined as its cokernel. So $0 \to C(A) \to C(X) \to C(X, A) \to 0$ is a short exact sequence of chain complexes, natural in the pair `Homology/Relative · relShortComplex` `Homology/Relative · relShortComplex_shortExact` `Homology/Relative · relChains`. **Relative homology** $H_n(X, A; R)$ is the n-th homology of $C(X, A)$, a functor on pairs `Homology/Relative · relHomology`, and the connecting morphism $H_{n+1}(X, A) \to H_n(A)$ is Mathlib's connecting morphism of that short exact sequence `Homology/Relative · relδ`. The long exact sequence of the pair is exact at every place `Homology/Relative · relHomology_exact₁` `Homology/Relative · relHomology_exact₂` `Homology/Relative · relHomology_exact₃`, and $H_n(X) \cong H_n(X, \emptyset)$ naturally `Homology/Relative · relHomologyInclIso`. Defining $C(X, A)$ as a cokernel in an abelian category, rather than as a quotient of free modules, is what makes the exact sequence and its naturality available at once from Mathlib's homological algebra.

Concrete chains. For subdivision and excision the textbook description is used: an n-chain is a finite formal combination $\sum r_i \sigma_i$ of continuous maps $\sigma_i : \Delta^n \to X$ from the standard simplex `Homology/Simplex · Simplex` `Homology/Simplex · Chains`, with boundary $\partial\sigma = \sum_i (-1)^i \sigma \circ \delta^i$ `Homology/Simplex · sbd`. These form a chain complex `Homology/Simplex · singChains` naturally isomorphic to Mathlib's `Homology/Simplex · singChainsIso`, so that results proved with one description transfer to the other.

The axioms. With these definitions, singular homology is an instance of Mathlib's Eilenberg–Steenrod homology pretheory `Homology/Pretheory · singularHomologyPretheory`; homotopic maps of pairs induce the same maps in relative homology `Homology/RelativeHomotopy · relHomology_map_eq_of_homotopy`; and excision holds `Homology/Excision · excision`. A triple $Z \subseteq Y \subseteq X$ is a pair of composable embeddings `Homology/Triple · TopTriple`, with its own connecting morphism and exact sequence `Homology/Triple · TopTriple.δ`. For subsets $A \subseteq B \subseteq X$ with B a strong deformation retract of X, $H_*(B, A) \cong H_*(X, A)$ `Homology/RelativeRetract · relHomologyIsoOfIsStrongDeformationRetract`.

Field coefficients and Betti numbers. Milnor's §5 works with homology over a field (p. 28). The formal text takes the coefficients in a field $\Bbbk$, with homology groups as $\Bbbk$-vector spaces: $H_i(B, A; \Bbbk)$ for subsets $A \subseteq B$ `Homology/TripleFiltration · subsetHomology`, and the absolute $H_i(X; \Bbbk)$, Mathlib's singular homology with coefficients $\Bbbk$ `Homology/MorseHomologyBasic · spaceHomology`. The **Betti number** $b_i(X) = \dim_{\Bbbk} H_i(X; \Bbbk)$ `Homology/MorseHomologyBasic · bettiNumber` is Milnor's $R_i(X)$. By the convention on dimensions it is 0 when $H_i(X; \Bbbk)$ is infinite-dimensional; for the spaces of the book finiteness is proved, not assumed (Part IV). Betti numbers are invariant under homeomorphisms `Homology/MorseHomologyBasic · bettiNumber_eq_of_homeomorph`. The field and the space must lie in the same universe. The Euler characteristic $\sum_i (-1)^i b_i$ and the Morse inequalities are in Part IV `Homology/MorseHomology · bettiNumber_le_critCount` `Homology/MorseHomology · sum_bettiNumber_eq_sum_critCount`.

0.7 Homotopy groups and induced maps

Milnor uses homotopy groups in §19 and throughout his Part IV (§§22–24). Mathlib defines them; the project adds the maps they induce and the exact sequence of a fibration.

- **Generalized loops.** For an index type N, I^N is the cube of functions $N \to [0, 1]$, and its boundary is the set of points with some coordinate equal to 0 or 1 `Mathlib/Topology/Homotopy/HomotopyGroup · Cube.boundary`. A generalized loop at $x \in X$ is a continuous $\gamma : I^N \to X$ equal to x on the boundary `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop`, and two generalized loops are equivalent if they are homotopic relative to the boundary `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.Homotopic`.
- **Homotopy groups.** $\pi_N(X, x)$ is the set of equivalence classes `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup`, and $\pi_n(X, x)$ is the case $N = \{0, \dots, n-1\}$ `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.Pi`. For nonempty N it is a group `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.group`, abelian when N has at least two elements `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.commGroup`, and π_1 is the fundamental group `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.pi1EquivFundamentalGroup`. A path is a continuous map from $[0, 1]$ with prescribed ends `Mathlib/Topology/Path · Path`, and the loop space ΩX at x is the space of paths from x to x `Mathlib/Topology/Homotopy/HomotopyGroup · LoopSpace`, with the compact-open topology.

- **Induced maps.** A continuous $f : X \to Y$ carries a generalized loop γ at x to the generalized loop $f \circ \gamma$ at $f(x)$ `Homotopy/Basic · genLoopMap`, and homotopies relative to the boundary to homotopies relative to the boundary, so it induces $f_* : \pi_N(X, x) \to \pi_N(Y, f(x))$ `Homotopy/Basic · homotopyGroupMap`, functorially `Homotopy/Basic · homotopyGroupMap_comp` `Homotopy/Basic · homotopyGroupMap_id`. It is defined as a map of sets, and most results about induced maps are stated as bijections or surjections of the underlying sets; for nonempty N it is a group homomorphism `Homotopy/ExactSequenceIso · homotopyGroupMap_mul` `Homotopy/ExactSequenceIso · homotopyGroupHom`, so a bijective f_* is an isomorphism of groups. For finite N, a path ω from y_0 to y_1 gives a bijection $\pi_N(Y, y_0) \to \pi_N(Y, y_1)$ `Homotopy/ChangeOfBase · changeBaseEquiv`, and a homotopy equivalence e gives bijections $\pi_N(X, x) \to \pi_N(Y, e(x))$ `Homotopy/ChangeOfBase · homotopyGroupEquivOfHomotopyEquiv`.
- **Compressions.** An embedding $j : Y \to X$ compresses maps from P if every map $P \to X$ is homotopic, relative to the preimage of $j(Y)$, to a map into $j(Y)$ `Homotopy/Basic · Compresses`. Compressing maps from the cube gives surjectivity on π_N, and compressing maps from $I \times I^N$ gives injectivity `Homotopy/Basic · surjective_homotopyGroupMap_of_compresses` `Homotopy/Basic · injective_homotopyGroupMap_of_compresses`. This is how attaching a cell of dimension λ is shown to induce bijections on π_i for $i \leq \lambda - 2$ `Homotopy/CellAttach · bijective_homotopyGroupMap_cellInl` (Part XII).
- **Loop spaces.** $\pi_{i+1}(\Omega X) \cong \pi_{i+2}(X)$ as groups `Homotopy/LoopSpace · homotopyGroupPiLoopMulEquiv`, and in a path-connected space the space of paths from p to q is homotopy equivalent to the loop space at p `Homotopy/LoopSpace · pathHomotopyEquivLoop`. These pass between Milnor's path spaces $\Omega(M; p, q)$ and his loop spaces.
- **Fibrations.** In §§23–24 Milnor quotes the exact homotopy sequence of a fibre bundle. A map $p : E \to B$ is **locally trivial** with fibre F if every point of B has an open neighbourhood U with a homeomorphism $p^{-1}(U) \cong U \times F$ whose first component is p `Homotopy/Fibration · IsLocallyTrivial`; this is a property of an arbitrary map, so that it applies directly to maps between concrete spaces such as a unitary group and a sphere, where Mathlib's fibre bundles are a structure on a total space built for the purpose. p is a **Serre fibration** if it has the homotopy lifting property for cubes relative to the open box $I^n \times \{0\} \cup \partial I^n \times I$ `Homotopy/Fibration · openBox` `Homotopy/Fibration · IsSerreFibration`; locally trivial maps, and in particular Mathlib's fibre bundles, are Serre fibrations `Homotopy/HomotopyLifting · IsLocallyTrivial.isSerreFibration` `Homotopy/HomotopyLifting · isSerreFibration_proj`. For a continuous Serre fibration p and $e_0 \in E$, with fibre $p^{-1}(p(e_0))$ `Homotopy/ExactSequence · fiber`, the boundary map $\partial : \pi_{n+1}(B) \to \pi_n(\text{fibre})$ `Homotopy/ExactSequence · boundaryMap` makes the homotopy sequence exact at the fibre, the total space and the base `Homotopy/ExactSequence · exact_fiber` `Homotopy/ExactSequence · exact_total` `Homotopy/ExactSequence · exact_base`, as pointed sets (Part XII).

0.8 Paths, energy, the path spaces Ω and Ω^*, Milnor's metric

Paths as functions on $\mathbb{R}$. A path in the formal text is a function $c : \mathbb{R} \to V$ whose values matter only on $[0, 1]$, or on whatever interval is named. This is a deliberate choice. Derivatives at interior points are ordinary derivatives of functions on $\mathbb{R}$; one-sided derivatives at the ends and at break points are derivatives within half-lines; and paths can be added, rescaled, reparametrized and concatenated without subtypes. The cost is that two paths that agree on $[0, 1]$ are different functions. The book then speaks of paths that "agree on $[0, 1]$", and a few notions normalize a function by extending its restriction to $[0, 1]$ as a constant outside (§0.9).

- **The path space** $\Omega(N; p, q)$ (Milnor, p. 67) is the set of paths c that are piecewise smooth on $[0, 1]$ in the sense of §0.3, with $c(t) \in N$ for $t \in [0, 1]$, $c(0) = p$ and $c(1) = q$ `Chapter8/Energy · pathSpace`.
- The **energy** of c from a to b is $E_a^b(c) = \int_a^b \|c'(t)\|^2 \, dt$, and Milnor's E is E_0^1 `Chapter8/Energy · energy`; at the finitely many break points c' takes a junk value, which does not affect the integral. Schwarz's inequality $L^2 \leq (b-a)E$ holds for piecewise smooth c `Chapter8/Energy · pathLength_sq_le_energy`, with equality exactly when the speed $\|c'(t)\|$ is the same at every point of (a, b) where c is differentiable `Chapter8/Energy · pathLength_sq_eq_energy_iff`.
- The **jump** of a velocity at t is $\Delta_t c' = c'(t^+) - c'(t^-)$, computed with one-sided derivatives `Chapter8/`

`Energy · jump`; this is Milnor's Δ_t (p. 71).

- **Milnor's "tangent space"** $T\Omega_c$ (p. 67) is the vector space of fields X, piecewise smooth on $[0,1]$, tangent along c ($P_{c(t)}X(t) = X(t)$ for $t \in [0,1]$), with $X(0) = X(1) = 0$ `Chapter8/Energy · pathTangent`. As in Milnor's §§11–15, Ω carries no manifold structure: $T\Omega_c$ is a definition, related to Ω only through variations.
- A **variation** with fixed end points (Milnor, p. 68) has its parameters in an open set $U \ni 0$ of a normed space P; $P = \mathbb{R}$ gives Milnor's $(-\varepsilon, \varepsilon)$, and $P = \mathbb{R}^2$ the two-parameter variations of §13. It is a map α from P to paths with $\alpha(u)(t) \in N$ for $u \in U$ and $t \in [0,1]$, with end points independent of u, and with $(u,t) \mapsto \alpha(u)(t)$ smooth on $U \times [s,t]$ for every closed strip whose interior contains no point of a finite set B of break points `Chapter8/Energy · IsVariation`. It is a variation of the path $\alpha(0)$. Its **variation field** is $W_t = \partial\alpha/\partial u\,(0,t)$ `Chapter8/Energy · varField`, which lies in $T\Omega$, as Milnor says "clearly" (p. 68) `Chapter8/Energy · IsVariation.varField_mem`. Conversely, a piecewise smooth family of tangent fields depending linearly on a parameter $u \in P$ is the derivative at $u = 0$ of a variation built with the exponential map (Lemma 11.A of the exposition) `Chapter8/FirstVariation · exists_variation`; in particular every finite family of elements of $T\Omega_c$ is the family of variation fields of one variation. Derivatives "of E on Ω" are always derivatives of $u \mapsto E(\alpha(u))$, functions of finitely many real variables.

Milnor's metric. In §16 (p. 88) Milnor topologizes Ω by

$$d(\omega,\omega') = \max_t \rho(\omega(t),\omega'(t)) + \Big[\int_0^1 \Big(\frac{ds}{dt} - \frac{ds'}{dt}\Big)^2 dt\Big]^{1/2},$$

s and s' being the arc lengths.

- $\Omega(N;p,q)$ as a space is the type of its elements `Chapter10/PathMetric · PathSp`. For two paths γ, η the formal d is built from the speed $ds/dt = \|\gamma'(t)\|$ `Chapter10/PathMetric · PathSp.speed`, the uniform part $\sup_{t\in[0,1]} \rho(\gamma(t),\eta(t)) \in [0,\infty]$ `Chapter10/PathMetric · PathSp.supDist`, and the L^2 part $\big(\int_0^1 (\|\gamma'(t)\| - \|\eta'(t)\|)^2\,dt\big)^{1/2}$ `Chapter10/PathMetric · PathSp.l2Dist`. Their sum is an extended **pseudometric** on Ω `Chapter10/PathMetric · PathSp.instPseudoEMetricSpace`. It is only a pseudometric because paths are functions on $\mathbb{R}$: two paths that agree on $[0,1]$ are at distance 0. So the formal Ω is not Hausdorff. This matters in the proof of Theorem 17.1, where Example 1 of the Appendix (for paracompact Hausdorff spaces) is applied in Ω^* and pulled back to Ω (Part IX). The triangle inequality for the L^2 part is Minkowski's inequality, proved in the project for piecewise continuous integrands.
- The point of the L^2 term is that it makes the energy continuous: $\big|\sqrt{E_a^b(\gamma)} - \sqrt{E_a^b(\eta)}\big|$ is at most the L^2 part of $d(\gamma,\eta)$ for $0 \le a \le b \le 1$ `Chapter10/PathMetric · PathSp.abs_sqrt_energy_sub_le`. So E is continuous on Ω `Chapter10/PathMetric · PathSp.continuous_energy` `Chapter10/FundamentalTheorem · PathSp.energyC`, $\Omega^c = E^{-1}[0,c]$ is closed `Chapter10/PathMetric · PathSp.isClosed_sublevel`, and $\operatorname{Int}\Omega^c = E^{-1}[0,c)$ is open `Chapter10/PathMetric · PathSp.isOpen_sublevel_lt`.

The full path space. Milnor's $\Omega^* = \Omega^*(M;p,q)$ (p. 93) is the space of all continuous paths from p to q with the compact-open topology. Formally it is the set of continuous maps $\gamma : [0,1] \to V$ with values in N, $\gamma(0) = p$ and $\gamma(1) = q$ `Chapter10/FullPathSpace · pathCSet` `Chapter10/FullPathSpace · PathC`, as a subspace of the space of continuous maps $[0,1] \to V$ with the compact-open topology `Mathlib/Topology/CompactOpen · ContinuousMap.compactOpen`, which, $[0,1]$ being compact, is the topology of uniform convergence. Here paths are genuine maps on $[0,1]$, so Ω^* is Hausdorff. Restricting a piecewise smooth path to $[0,1]$ `Chapter10/FullPathSpace · PathSp.toC` gives the inclusion $i : \Omega \to \Omega^*$ `Chapter10/FullPathSpace · PathSp.toPathC`, which does not increase distances `Chapter10/FullPathSpace · PathSp.edist_toPathC_le` and is therefore continuous `Chapter10/FullPathSpace · PathSp.continuous_toPathC`. Theorem 17.1 says that it is a homotopy equivalence `Chapter10/FullPathTheorem · pathIncl` `Chapter10/FullPathTheorem · theorem_17_1`.

0.9 Geodesics, the exponential map, Jacobi fields, conjugate points, the index form E_{**}

Throughout this section $N \subseteq V$ is a slice submanifold of a finite-dimensional inner product space, with the induced metric, as in §0.3.

Geodesics.

- $\gamma : \mathbb{R} \to V$ is a **geodesic on an open set** $s \subseteq \mathbb{R}$, in practice an open interval, if γ is smooth on s, $\gamma(s) \subseteq N$, and the acceleration $\gamma''(t)$ is normal to $T_{\gamma(t)}N$ for every $t \in s$ `Chapter7/Geodesic · IsGeodesicOn`. Since $D/dt = P \circ d/dt$, this is Milnor's $D/dt\,(d\gamma/dt) = 0$ (p. 55) for the induced metric `Chapter7/Geodesic · isGeodesicOn_univ_iff_covDeriv`: the motion of a particle constrained to N and subject to no other force. Equivalently $\gamma'' = \mathrm{II}(\gamma', \gamma')$ `Chapter7/Geodesic · IsGeodesicOn.deriv_deriv_eq Chapter7/Geodesic · isGeodesicOn_of_deriv_deriv_eq`, that is, (γ, γ') is an integral curve of the geodesic spray $(x, v) \mapsto (v, \mathrm{II}_x(v, v))$ on TN `Chapter7/Geodesic · geodesicSpray Chapter7/Geodesic · isGeodesicOn_iff_hasDerivAt_geodesicSpray`. A geodesic on an open interval has constant speed `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq`, and two geodesics on an open interval with the same position and velocity at one time agree on it `Chapter7/Geodesic · IsGeodesicOn.eqOn`. Milnor allows any interval, with one-sided derivatives at closed ends; the formal definition uses open sets, which avoids them.
- A path of Ω is an **unbroken geodesic** if it is smooth on the closed interval $[0, 1]$, with one-sided derivatives at the ends, and a geodesic on $(0, 1)$ `Chapter8/Critical · IsUnbrokenGeodesic`. A path is a **critical path** of E if, for every one-parameter variation of it, the derivative of $u \mapsto E(\alpha(u))$ at 0 exists and is 0 `Chapter8/Critical · IsCriticalPath` (Milnor, p. 69). Corollary 12.3 says that on Ω the two notions coincide `Chapter8/Critical · corollary_12_3`.
- A **minimal geodesic** on $[a, b]$ is the restriction of a geodesic on an open set containing $[a, b]$ whose length is at most that of every piecewise smooth path in N with the same ends `Chapter7/HopfRinow · IsMinimalGeodesic`. A path of Ω whose length is at most that of every path of Ω is a **minimal path** `Chapter8/Hessian · IsMinimalPath`.

The exponential map.

- γ is a **geodesic from** q **with initial velocity** v if for some $\delta > 0$ it is a geodesic on $(-\delta, 1 + \delta)$ with $\gamma(0) = q$ and $\gamma'(0) = v$ `Chapter7/Exp · IsGeodesicFrom`. The **domain of the exponential map** is the set of pairs $(q, v) \in V \times V$ for which such a geodesic exists `Chapter7/Exp · expDom`. For such a pair, $\exp_q(v) = \gamma(1)$ for a geodesic γ chosen by the axiom of choice; otherwise $\exp_q(v) = q$ `Chapter7/Exp · exp`. The value does not depend on the choice, by uniqueness of geodesics `Chapter7/Exp · IsGeodesicFrom.exp_eq`, and $\gamma(t) = \exp_q(tv)$ `Chapter7/Exp · IsGeodesicFrom.exp_smul`. Milnor (p. 57) asks for a geodesic on the closed interval $[0, 1]$; the formal domain asks for one on an open interval containing $[0, 1]$, which avoids one-sided derivatives in the definition. The two domains are the same. A geodesic on $[0, 1]$, that is, a curve smooth on $[0, 1]$ with values in N and a geodesic on $(0, 1)$, coincides on $[0, 1]$ with the radial curve $t \mapsto \exp_{\gamma(0)}(t\gamma'(0))$, $\gamma'(0)$ the one-sided velocity, and that curve is a geodesic on an open interval containing $[0, 1]$ `Chapter9/GeodesicVariation · exists_initial_ray`; a radial curve that is a geodesic on such an interval puts its initial data in the domain of exp `Chapter11/ExpSmooth · mem_expDom_of_isGeodesicOn`. So $\exp_q(v)$ is defined in the formal sense exactly when it is defined in Milnor's.
- exp is defined and smooth on an open neighbourhood of the zero section of TN, in the sense that near each point it is the restriction of a smooth map on an open set of $V \times V$ `Chapter7/Exp · exists_isOpen_exp`, and $d(\exp_q)_0$ is the inclusion $T_qN \to V$ `Chapter7/Exp · hasFDerivAt_exp_zero`. In fact exp is smooth on its whole domain `Chapter11/ExpSmooth · exists_contDiffOn_exp`: for $q \in N$ the domain of $\exp_q$ in T_qN `Chapter11/ExpSmooth · expDomAt` is open `Chapter11/ExpSmooth · isOpen_expDomAt` and star-shaped about 0 `Chapter11/ExpSmooth · smul_mem_expDomAt`, and $\exp_q$ is smooth on it `Chapter11/`

ExpSmooth · contDiffOn_expAt. As a map into the manifold N, $\exp_q : T_qN \to N$ takes the junk value q where $\exp_q(v) \notin N$ Chapter7/NormalNbhd · expToN.

- N is **geodesically complete** if (q, v) lies in the domain of exp for every $q \in N$ and every $v \in T_qN$ Chapter7/GeodesicFlow · IsGeodesicallyComplete (Milnor, p. 58), equivalently if every geodesic on an open interval is the restriction of a geodesic on all of $\mathbb{R}$ Chapter7/GeodesicFlow · isGeodesicallyComplete_iff. For connected nonempty N the Hopf–Rinow theorem makes this equivalent to completeness of (N, ρ) and to compactness of the closures of bounded sets Chapter7/HopfRinow · hopf_rinow.

Jacobi fields and conjugate points.

- A **smooth curve in N on $[0,1]$** is a $\gamma : \mathbb{R} \to V$, smooth on $[0,1]$, with values in N there Chapter9/JacobiSystem · IsSmoothCurveIn: Milnor's $\gamma : [0,1] \to M$.
- The **Jacobi operator** along γ is $J \mapsto D^2J/dt^2 + R(V,J)V$, with $V = d\gamma/dt$ and D/dt, R as in §0.3 Chapter8/Jacobi · jacobiExpr. J is a **Jacobi field** along γ (Milnor, p. 77) if it is smooth on $[0,1]$, tangent to N along γ there, and satisfies the Jacobi equation on $(0,1)$, hence by continuity on $[0,1]$ Chapter8/Jacobi · IsJacobiField. For a smooth curve γ, the Jacobi fields normalized to be constant outside $[0,1]$ form a vector space Chapter9/JacobiSpace · jacobiSpace of dimension $2n$, $n = \dim N$ Chapter9/JacobiSpace · finrank_jacobiSpace, a field being determined by $J(s)$ and $DJ/dt(s)$ for any $s \in [0,1]$ Chapter9/JacobiSpace · jacEquiv. The normalization makes "a field along $\gamma|[0,1]$" a single function; without it, every Jacobi field would have infinitely many representatives, differing off $[0,1]$.
- Two parameter values $a \neq b$ in $[0,1]$ are **conjugate along** γ if some Jacobi field vanishes at a and at b without vanishing identically on $[0,1]$ Chapter9/JacobiSpace · IsConjugate. Milnor speaks of the points $\gamma(a)$ and $\gamma(b)$, and adds in a footnote that for a curve with self-intersections one should speak of parameter values; the formal definition takes the footnote's form. The **multiplicity** is the dimension of the space of Jacobi fields vanishing at a and b Chapter9/JacobiSpace · conjSpace Chapter9/JacobiSpace · conjMultiplicity; for $a \neq b$ in $[0,1]$ it is positive exactly when a and b are conjugate Chapter9/JacobiSpace · isConjugate_iff. The points q conjugate to p along some geodesic from p to q form the conjugate locus of p Chapter11/NonConjugate · conjLocus.

The Hessian of the energy. Milnor defines $E_{**} : T\Omega_\gamma \times T\Omega_\gamma \to \mathbb{R}$ at a geodesic γ (p. 74) as $\partial^2 E(\bar\alpha(u_1,u_2))/\partial u_1 \partial u_2$ at $(0,0)$, for a two-parameter variation with variation fields W_1 and W_2, and Theorem 13.1 computes it. The formal definition takes the computed formula as the definition and proves the rest.

- The **second variation** at a path γ is the right side of Milnor's Theorem 13.1,

$$-\sum_t \langle W_2(t), \Delta_t \tfrac{DW_1}{dt}\rangle - \int_0^1 \langle W_2, \tfrac{D^2W_1}{dt^2} + R(V,W_1)V\rangle\, dt,$$

  where $\Delta_t\, DW_1/dt = P_{\gamma(t)}(\Delta_t W_1')$ and the sum runs over the finitely many $t \in (0,1)$ at which the term is not zero Chapter8/SecondVariation · secondVariation. Its integral form is $\int_0^1 \left(\langle DW_1/dt, DW_2/dt\rangle - \langle R(V,W_1)V, W_2\rangle\right) dt$ Chapter8/Hessian · hessForm, and the two agree for every $\gamma \in \Omega$ and all $W_1, W_2 \in T\Omega_\gamma$ Chapter8/Hessian · secondVariation_eq_hessForm.
- E_{**} is the bilinear form on $T\Omega_\gamma$ given by twice the second variation Chapter8/Hessian · energyHessian, defined for every $\gamma \in \Omega$. Bilinearity comes from the integral form, and so does symmetry Chapter8/Hessian · energyHessian_isSymm, together with the symmetry of $\langle R(V,W_1)V, W_2\rangle$ in W_1, W_2. That this is Milnor's E_{**} is Theorem 13.1: at an unbroken geodesic, for every two-parameter variation with fields W_1 and W_2, the iterated derivative $\partial/\partial u_1(\partial E/\partial u_2\,(u_1,0))$ at $u_1 = 0$ exists and equals $E_{**}(W_1,W_2)$ Chapter8/SecondVariation · second_variation Chapter8/Hessian · hasDerivAt_energyHessian, and every pair (W_1,W_2) arises in this way Chapter8/FirstVariation · exists_variation_two. Milnor derives the symmetry from $\partial^2/\partial u_1\partial u_2 = \partial^2/\partial u_2\partial u_1$, which would need $E \circ \bar\alpha$ to be C^2 near $(0,0)$; the integral form avoids this.

- **Null space and nullity.** Elements of $T\Omega_\gamma$ are functions on $\mathbb{R}$ whose values off $[0,1]$ play no role, so the null space of the bilinear form E_{**} itself contains every field that vanishes on $[0,1]$ and is infinite-dimensional. The **null space** of E_{**} in Milnor's sense is therefore its image under restriction to $[0,1]$, $X \mapsto X \circ \pi$ with $\pi : \mathbb{R} \to [0,1]$ the nearest-point projection `Chapter9/JacobiSpace · restrictIcc` `Chapter9/JacobiSpace · hessNullSpace`, and the **nullity** is the dimension of this space `Chapter9/JacobiSpace · hessNullity`. Theorem 14.1 identifies the null space with the space of Jacobi fields vanishing at 0 and 1 `Chapter9/JacobiSpace · hessNullSpace_eq`, and the nullity with the multiplicity of 0 and 1 as conjugate parameter values `Chapter9/JacobiSpace · theorem_14_1_nullity`.

The index of E_{**}. Milnor defines it (p. 83) as the maximum dimension of a subspace of $T\Omega_\gamma$ on which E_{**} is negative definite; that this maximum is finite is part of Theorem 15.1. Mathlib's index of §0.2 is 0 on the infinite-dimensional space $T\Omega_\gamma$, so a different definition is needed.

- For a bilinear form B on any real vector space, B is **negative definite** on a subspace S if $B(x,x) < 0$ for every $0 \neq x \in S$ `Chapter9/FormIndex · IsNegDefOn`, and the **index** of B is the supremum, in $\{0,1,2,\dots\} \cup \{\infty\}$, of the dimensions of the finite-dimensional subspaces on which B is negative definite `Chapter9/FormIndex · formIndex`. This is Milnor's index with the value ∞ allowed: if B is negative definite on an infinite-dimensional subspace, that subspace has finite-dimensional subspaces of every dimension, and the supremum is ∞. On a finite-dimensional space it is Mathlib's index `Chapter9/FormIndex · formIndex_eq_sigNeg`. Milnor's $\lambda(\tau)$, the index of E_{**} on the fields along $\gamma|[0,\tau]$, is the same notion for the restricted form `Chapter9/IndexReduction · indexUpTo`.
- The **index of a geodesic** $\gamma \in \Omega(N;p,q)$, as a natural number, is the index of E_{**} at γ, converted to a natural number with $\infty \mapsto 0$, and 0 if $\gamma \notin \Omega$ `Chapter10/FundamentalTheorem · geodesicIndex`. For an unbroken geodesic the index is finite, so this is the index itself `Chapter10/FundamentalTheorem · geodesicIndex_eq`: the index theorem, Theorem 15.1, states that the parameter values in $(0,1)$ conjugate to 0 are finitely many and that the index of E_{**} equals the sum of their multiplicities `Chapter9/IndexTheorem · theorem_15_1`.

0.10 Flows of vector fields

Milnor's Lemma 2.4 (p. 10) produces a one-parameter group of diffeomorphisms from a compactly supported vector field on a manifold, quoting the smooth dependence of solutions of differential equations on their initial conditions. The formal flows live on a normed space E, in the applications $\mathbb{R}^K$ or V; a flow of a submanifold is the restriction of an ambient flow that preserves it.

- A vector field on E is a map $X : E \to E$, and an integral curve of X is a $\gamma : \mathbb{R} \to E$ with $\gamma'(t) = X(\gamma(t))$ for every t `Mathlib/Analysis/ODE/Basic · IsIntegralCurve`. The standing hypothesis for global flows is that X is **bounded and globally Lipschitz** `Flow/Global · IsBddLipschitz`, which a C^1 field with compact support is `Flow/Global · IsBddLipschitz.of_contDiff`.
- The **flow** $\varphi_t(x)$ is the value at time t of an integral curve defined on all of $\mathbb{R}$ and passing through x at time 0, chosen by the axiom of choice, and x if there is none `Flow/Global · flow`. For bounded Lipschitz X there is exactly one such curve `Flow/Global · IsBddLipschitz.flow_spec` `Flow/Global · IsBddLipschitz.eq_flow`; $\varphi_{s+t} = \varphi_s \circ \varphi_t$ `Flow/Global · IsBddLipschitz.flow_add`; $(t,x) \mapsto \varphi_t(x)$ is continuous `Flow/Global · IsBddLipschitz.continuous_flow`; each φ_t is a homeomorphism with inverse φ_{-t} `Flow/Global · IsBddLipschitz.flowHomeomorph`; and zeros of X are fixed `Flow/Global · IsBddLipschitz.flow_of_eq_zero`. A closed set through each point of which there is a local integral curve staying in the set is invariant under the flow `Flow/Global · IsBddLipschitz.flow_mem`; this is how a field tangent to a submanifold N gives a flow of N (Part II, `Chapter2/Submanifold · flow_mem_of_tangent`). Since a bounded field has solutions on time intervals of any length, Milnor's iteration of $\varphi_{\varepsilon_0/2}$, and the check that it is well defined, are not needed.
- **Smoothness.** For X of class C^n, $1 \le n \le \infty$, with compact support on a complete space, each φ_t is of class C^n `Flow/Smooth · contDiff_flow` and so is $(t,x) \mapsto \varphi_t(x)$ `Flow/Smooth · contDiff_flow_uncurry`; each φ_t is therefore a C^n diffeomorphism `Flow/Smooth · flowDiffeomorph`. Milnor's "well known" smooth de-

pendence is proved, through the inverse function theorem in the Banach space $C([0,1],E)$ applied to $\gamma \mapsto \gamma - \int_0^{\cdot} X(\gamma)$ `Flow/Volterra · voltHomeomorph` (Part I).

- A **one-parameter group of diffeomorphisms** (Milnor, pp. 9–10) is a family (φ_t) such that $(t,x) \mapsto \varphi_t(x)$ is smooth, each φ_t is a diffeomorphism, and $\varphi_{s+t} = \varphi_s \circ \varphi_t$ `Flow/Smooth · IsOneParameterGroup`; then φ_0 is the identity `Flow/Smooth · IsOneParameterGroup.zero`. It is **generated** by X if $\frac{d}{dh}\varphi_h(x)\big|_{h=0} = X(x)$ for every x `Flow/Smooth · Generates`. Milnor's Lemma 2.4, for $M = E$ a complete real normed space (finite-dimensional in every application): a smooth vector field with compact support generates exactly one one-parameter group of diffeomorphisms, its flow `Flow/Smooth · lemma_2_4`.
- **Local flows and linear equations.** For a smooth field on an open subset of a finite-dimensional space, solutions exist near any point for a uniform time and depend smoothly on the initial point `Chapter7/ODE · exists_localFlow`, and two solutions that agree at one time agree on every open interval on which both are defined `Chapter7/ODE · ode_eqOn`; the second-order form of this is Milnor's Theorem 10.1 `Chapter7/ODE · theorem_10_1`. A linear equation $x' = A(t)x$ with continuous A has solutions on all of $\mathbb{R}$ `Chapter9/JacobiODE · exists_linearODE`, and its propagator sends (s,ξ) to the solution with value ξ at time s `Chapter9/JacobiODE · linSol`. Jacobi fields (§0.9) are components of the solutions of such a system `Chapter9/JacobiSystem · jacSol`.
- **Gradient flows and slabs.** Milnor's field in the proof of Theorem 3.1 is $X = \rho \cdot \operatorname{grad} f$ `Chapter2/Regular · gradField`. What the proof uses of its flow is recorded abstractly: a continuous flow φ on a space T is a **slab flow** for a continuous $F : T \to \mathbb{R}$ and an interval $[a,b]$ if $t \mapsto F(\varphi_t(x))$ has derivative 1 at every time at which its value lies in $[a,b]$ `Chapter2/Crossing · IsSlabFlow`, and the topological half of Theorem 3.1 is proved for any slab flow (Part II). Finally, the geodesic flow, the position and velocity at time t of $s \mapsto \exp_y(sw)$, is read off from exp `Chapter7/GeodesicFlow · geoFlow`.

Part I. Critical points and the lemma of Morse (Milnor §§1–2)

Milnor's second section fixes the vocabulary of the whole book: critical points, the Hessian f_{**}, its index and nullity, Hadamard's lemma (Lemma 2.1), the lemma of Morse (Lemma 2.2), the isolation of non-degenerate critical points (Corollary 2.3), five examples of degenerate critical points, and the one-parameter group generated by a compactly supported vector field (Lemma 2.4). The formal proof establishes all of this in three settings: in a real normed space E, which is Milnor's situation once coordinates have been chosen; on an abstract smooth manifold without boundary modelled on E, with no Hausdorff or countability hypothesis; and for functions $F|N$ on a slice submanifold N of a Euclidean space, the setting of most later parts. The torus of §1 is not told here: its Lean file uses Theorem 3.1, and it appears in Part III.

Two proofs take a different road from Milnor's. The lemma of Morse is not proved by his induction, which completes one square at a time. A single application of the inverse function theorem, in the space of linear maps of E, produces a smooth family of linear maps that turns the quadratic form with variable coefficients into a constant one, and Sylvester's law of inertia, from Mathlib, finishes. The smooth dependence of solutions of an ordinary differential equation on the initial point, which Milnor quotes as well known and which Mathlib does not have, is proved by the inverse function theorem in the Banach space $C([0,1],E)$. The global flow itself comes from the Picard–Lindelöf theorem on time intervals of any length, which a bounded field allows, and this replaces Milnor's iteration of $\varphi_{\varepsilon_0/2}$. Smaller departures are flagged where they occur. Hadamard's lemma to second order comes from Taylor's formula, Corollary 2.3 is proved without the lemma of Morse, and the Hessian is defined in a chart, its independence of the chart being a theorem.

Map.

- `Chapter1/Nondegenerate`: critical points, the Hessian, null space, nullity and index in a normed space; linear changes of variables; the second-order chain rule at a critical point; the matrix criterion; Corollary 2.3.
- `Chapter1/Hadamard`: parametric integrals on an open set; Lemma 2.1, coordinate free, in coordinates, and to second order.
- `Chapter1/SquareRoot`: the inverse function theorem with an inverse smooth on its whole domain; the smooth square root of a family of bilinear forms.
- `Chapter1/MorseLemma`: the normal form and its index (Sylvester); diagonalization with the negative squares first; Lemma 2.2, coordinate free and in Milnor's form; the index is forced by the normal form.
- `Chapter1/Examples`: Milnor's examples (a), (c), (d), (e).
- `Chapter1/Manifold`: the same notions on a manifold, through the local representative; smooth coordinate systems and the independence of the chart.
- `Chapter1/ManifoldMorse`: Corollary 2.3 on a manifold, the critical set, finiteness in compact sets; Lemma 2.2 on a manifold and its converse.

- `Chapter1/ManifoldSubmanifold`: functions $F|N$ on a slice submanifold $N \subseteq V$; critical points through the gradient; Milnor's formulas of pp. 35–36 for L_p.
- `Flow/Global`: global flows of bounded Lipschitz fields, the group law, continuity, invariant sets.
- `Flow/Nemytskii`: the operator $\gamma \mapsto Y \circ \gamma$ on $C([0,1], E)$ and its smoothness.
- `Flow/Volterra`: the operator $\gamma \mapsto \gamma - \int_0^{\cdot} X(\gamma)$, a homeomorphism with smooth inverse.
- `Flow/Smooth`: smooth dependence on the initial point, joint smoothness in (t, x), and Lemma 2.4.

I.1 Conventions

The three settings share one set of conventions; Part 0 recalls them (§0.1, §0.2, §0.10), and they are repeated here only as far as the proofs below need them.

Normed spaces. E is a real normed space, finite-dimensional wherever a dimension is counted; the formal statements say so each time. In statements written in coordinates, Milnor's $\mathbb{R}^n$ is the space of functions $\{0, \dots, n-1\} \to \mathbb{R}$ with the maximum norm; only its linear and smooth structure matters. The formal coordinates are numbered from 0, so Milnor's negative squares $(y^1)^2, \dots, (y^\lambda)^2$ are those with index $i < \lambda$. The derivative df_x is Mathlib's, the continuous linear map L with $f(x+h) = f(x) + Lh + o(\|h\|)$ `Mathlib/Analysis/Calculus/FDeriv/Defs · HasFDerivAt`, and by convention it is 0 where f is not differentiable `Mathlib/Analysis/Calculus/FDeriv/Defs · fderiv`. The second derivative $d^2 f_x$ is the derivative of $x \mapsto df_x$, a continuous bilinear form, written $d^2 f_x(v, w)$. Being C^n at x is Mathlib's notion `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiff`. For a finite order it propagates to a neighbourhood: a map that is C^2 at x is C^2 at every point near x `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffAt.eventually`. For the order ∞ it does not, which is why the inverse function theorem of §I.7 asks for smoothness on an open set.

Manifolds. A manifold M is a charted space modelled on E itself, through the identity model, which has no boundary `Mathlib/Geometry/Manifold/ChartedSpace · ChartedSpace`. Where the atlas is used it is a smooth manifold `Mathlib/Geometry/Manifold/IsManifold/Basic · IsManifold`. No Hausdorff or second countability hypothesis is ever assumed in this Part. Each point p has a preferred chart `Mathlib/Geometry/Manifold/ChartedSpace · chartAt` and the extended chart κ_p built from it `Mathlib/Geometry/Manifold/IsManifold/ExtChartAt · extChartAt`. The tangent space T_pM is, by definition, E `Mathlib/Geometry/Manifold/IsManifold/Basic · TangentSpace`. The derivative df_p of $f : M \to \mathbb{R}$ is the derivative at $\kappa_p(p)$ of f read in the charts, and 0 where f is not differentiable `Mathlib/Geometry/Manifold/MFDeriv/Defs · mfderiv`. Because the formal statements must name the model space, the project's definitions on M take E as an explicit argument.

Submanifolds. The submanifolds of a Euclidean space are the slice submanifolds of the companion project (§0.1). A set $N \subseteq V$ is one, modelled on Z with complement W, if every point of N lies in the domain of a *slice chart*: a diffeomorphism σ of an open subset of V onto an open subset of $Z \times W$ which carries the points of N in its domain exactly onto $Z \times \{0\}$ `Milnor/Chapter2/Submanifold · SliceChart` `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`. Such an N is a smooth manifold modelled on Z, whose charts are the maps $q \mapsto \sigma(q)_Z$ `Milnor/Chapter2/Submanifold · SliceChart.toChart` `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.chartedSpace` `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.isManifold`, and the inclusion $N \to V$ is smooth `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.contMDiff_val`.

I.2 Critical points, the Hessian, the index and the nullity (Milnor pp. 4–5)

Milnor defines these notions on a manifold and computes them in coordinates. The formal text starts in coordinates, on a normed space E, and transports the definitions to manifolds in §I.13. The definitions are collected in §0.2.

Definitions (V1.3). Let $f : E \to \mathbb{R}$ and $p \in E$.

- p is a *critical point* of f if f is differentiable at p with derivative 0 `Chapter1/Nondegenerate · IsCriticalPoint`. Then $df_p = 0$ `Chapter1/Nondegenerate · IsCriticalPoint.fderiv_eq`, and for f differentiable at p the converse holds `Chapter1/Nondegenerate · isCriticalPoint_iff`. A critical point stays critical under a differentiable change of variables, by the chain rule `Chapter1/Nondegenerate · IsCriticalPoint.comp`.
- The *Hessian* of f at p is the second derivative d^2f_p, a continuous bilinear form on E `Chapter1/Nondegenerate · hessian`. Its value $d^2f_p(v,w)$ is the derivative at p, in the direction v, of $x \mapsto df_x(w)$. That is Milnor's $f_{**}(v,w) = \tilde{v}_p(\tilde{w}(f))$ for the constant extensions $\tilde{v} = v$, $\tilde{w} = w$. The formal Hessian is defined at every point and is used only at critical points.
- For a continuous bilinear form H on E: its *quadratic form* is $Q_H(v) = H(v,v)$ `Chapter1/Nondegenerate · quadForm`; its *null space* is $\{v : H(v,w) = 0 \text{ for all } w\}$, the kernel of $v \mapsto H(v,\cdot)$ `Chapter1/Nondegenerate · nullSpace Chapter1/Nondegenerate · mem_nullSpace`; its *nullity* is the dimension of the null space `Chapter1/Nondegenerate · nullity`; H is *non-degenerate* if the null space is 0 `Chapter1/Nondegenerate · IsNondegenerate Chapter1/Nondegenerate · isNondegenerate_iff`.
- The *index* of H is Mathlib's negative index of inertia of Q_H `Chapter1/Nondegenerate · index Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg`. Mathlib defines it as the positive index of $-Q_H$. The positive index of a quadratic form Q is the largest $r \le \dim E$ for which some r-dimensional subspace carries Q positive definite `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigPos`. On an infinite-dimensional space the formal dimension is 0, so the index is 0 there; this is a junk value, and every statement about indices assumes finite dimension.
- A *non-degenerate critical point* is a critical point at which d^2f_p is non-degenerate `Chapter1/Nondegenerate · IsNondegenerateCriticalPoint`.

Nothing in these definitions asks H to be symmetric. A Hessian is symmetric wherever f is C^2, by the symmetry of second derivatives `Chapter1/Nondegenerate · hessian_symm Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`. Milnor proves symmetry differently, through $\tilde{v}_p(\tilde{w}(f)) - \tilde{w}_p(\tilde{v}(f)) = [\tilde{v},\tilde{w}]_p(f) = 0$, which uses that p is critical. In the formal text criticality is not needed for symmetry. It is needed for the independence of the coordinates, and §I.4 shows exactly where it enters. Two more facts are used throughout. The Hessian depends only on the germ of f at p `Chapter1/Nondegenerate · hessian_congr`. And for a continuous bilinear form B, the function $x \mapsto B(x,x)$ has Hessian $(v,w) \mapsto B(v,w) + B(w,v)$ at every point `Chapter1/Nondegenerate · hessian_bilinear`. The proof: by the product rule for bilinear maps `Mathlib/Analysis/Calculus/FDeriv/Bilinear · ContinuousLinearMap.hasFDerivAt_of_bilinear`, the derivative of $x \mapsto B(x,x)$ at x is $v \mapsto B(x,v) + B(v,x)$. As a function of x this is a continuous linear map, so its derivative is the map itself.

Theorem (the index in Milnor's words). Let E be finite-dimensional and H a continuous bilinear form on E. The index of H is the largest k such that some k-dimensional subspace W has $H(v,v) < 0$ for every $v \in W \setminus \{0\}$. `Chapter1/Nondegenerate · index_isGreatest`

The proof as formalized. Mathlib's defining property of the negative index `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg_isGreatest` makes it the largest dimension of a subspace W on which the restriction of $-Q_H$ is positive definite. For a subspace W this restriction is positive definite exactly when $H(v,v) < 0$ for all non-zero $v \in W$ `Chapter1/Nondegenerate · negDef_iff_posDef_restrict`. So the two sets of dimensions coincide. Two consequences are used later: some negative definite subspace has dimension equal to the index `Chapter1/Nondegenerate · exists_negDef_finrank_eq_index`, and every negative definite subspace has dimension at most the index `Chapter1/Nondegenerate · finrank_le_index_of_negDef`. The index never exceeds $\dim E$ `Chapter1/Manifold · index_le_finrank Mathlib/LinearAlgebra/QuadraticForm/Signature · sigPos_le_finrank`.

Milnor's remark that a critical point is non-degenerate exactly when f_{**} has nullity 0 is, for finite-dimensional E, the fact that a subspace is 0 exactly when its dimension is 0 `Chapter1/Nondegenerate · isNondegenerate_iff_nullity_eq_zero`.

I.3 Linear changes of variables, and the matrix criterion

Three invariance facts are used over and over, two for a linear change of variables and one for multiplication by a scalar. Let $L : E' \to E$ be a linear isomorphism and H a continuous bilinear form on E. Its pull-back is $H(L\cdot, L\cdot)$.

- *Index.* L is an isometry from the quadratic form of the pull-back to Q_H, because $Q_H(Lv) = H(Lv, Lv)$ `Chapter1/Nondegenerate · quadForm_bilinearComp_equivalent`. Mathlib's negative index is invariant under such equivalences `Mathlib/LinearAlgebra/QuadraticForm/Signature · QuadraticMap.Equivalent.sigNeg_eq`, since the isometry carries negative definite subspaces to negative definite subspaces of the same dimension. So the pull-back has the same index as H `Chapter1/Nondegenerate · index_bilinearComp`.
- *Null space and nullity.* The null space of the pull-back is the preimage under L of the null space of H: if $H(Lv, Lw) = 0$ for all w, then $H(Lv, u) = 0$ for all u, taking $w = L^{-1}u$ `Chapter1/Manifold · nullSpace_bilinearComp`. A linear isomorphism preserves the dimension of subspaces, so the two forms have the same nullity `Chapter1/Manifold · nullity_bilinearComp` and are non-degenerate together `Chapter1/Nondegenerate · isNondegenerate_bilinearComp_iff`.
- *Scalars.* For $c > 0$, multiplication by $\sqrt{c}$ is an isometry from Q_{cH} to Q_H, since $H(\sqrt{c}\,v, \sqrt{c}\,v) = c\,H(v, v)$. Hence cH and H have the same index `Chapter1/Nondegenerate · index_smul`. For $c \neq 0$, $cH(v, w) = 0$ exactly when $H(v, w) = 0$, so cH and H are non-degenerate together `Chapter1/Nondegenerate · isNondegenerate_smul_iff`. These two facts absorb the factors 2 and $\frac{1}{2}$ that separate the Hessian from the forms of the lemma of Morse.

Theorem (Milnor's matrix criterion, p. 4). A symmetric continuous bilinear form H on $\mathbb{R}^n$ is non-degenerate if and only if the matrix $\big(H(e_i, e_j)\big)$ has non-zero determinant. Consequently, if f is C^2 at p, then $d^2 f_p$ is non-degenerate if and only if $\det\big(\partial^2 f/\partial x^i \partial x^j(p)\big) \neq 0$. `Chapter1/Nondegenerate · isNondegenerate_iff_det_ne_zero` `Chapter1/Nondegenerate · isNondegenerate_hessian_iff_det_ne_zero`

The proof as formalized. View H as a bilinear form in Mathlib's sense. Its matrix in the standard basis is $\big(H(e_i, e_j)\big)$, by the definition of that matrix. Mathlib proves that a bilinear form on a finite-dimensional space is non-degenerate on both sides exactly when the determinant of its matrix is non-zero `Mathlib/LinearAlgebra/Matrix/BilinearForm · LinearMap.BilinForm.nondegenerate_iff_det_ne_zero`. The project's non-degeneracy is literally non-degeneracy on the left: $H(v, \cdot) = 0$ forces $v = 0$. Symmetry turns it into non-degeneracy on the right. For a Hessian the symmetry is that of §I.2. Milnor *defines* a non-degenerate critical point through this matrix; here it is a theorem.

I.4 The second-order chain rule, and the independence of the coordinates

Milnor says that one can check directly that non-degeneracy does not depend on the coordinates (p. 4), and later he computes f_{**} in whatever coordinates are at hand. The check is the chain rule for second derivatives, and its first-order term is where criticality enters.

Theorem (the second-order chain rule). Let $\psi : E_1 \to E_2$ be C^2 at q and $f : E_2 \to \mathbb{R}$ be C^2 at $\psi(q)$. Then for all $v, w \in E_1$

$$d^2(f \circ \psi)_q(v, w) = d^2 f_{\psi(q)}\big(d\psi_q v,\ d\psi_q w\big) + df_{\psi(q)}\big(d^2\psi_q(v, w)\big).$$

If $df_{\psi(q)} = 0$, the last term vanishes:

$$d^2(f \circ \psi)_q(v, w) = d^2 f_{\psi(q)}\big(d\psi_q v,\ d\psi_q w\big).$$

`Chapter1/Manifold · hessian_comp_apply` `Chapter1/Nondegenerate · hessian_comp`

The proof as formalized. The project proves the two statements separately, the second in `Chapter1/Nondegenerate` and the first, which only §I.17 needs, in `Chapter1/Manifold`. The proofs are the same.

1. A finite order of smoothness propagates to a neighbourhood, so ψ is C^2 at every point near q, and, ψ being continuous at q, f is C^2 at $\psi(x)$ for every x near q `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffAt.eventually`.
2. At each such x the first-order chain rule gives $d(f\circ\psi)_x = df_{\psi(x)}\circ d\psi_x$ `Mathlib/Analysis/Calculus/FDeriv/Comp · fderiv_comp`. The Hessian of $f\circ\psi$ at q depends only on the germ of $x\mapsto d(f\circ\psi)_x$, so it is the derivative at q of $x\mapsto df_{\psi(x)}\circ d\psi_x$.
3. This map composes two operator-valued maps. The first, $x\mapsto df_{\psi(x)}$, has derivative $v\mapsto d^2f_{\psi(q)}(d\psi_q v)$ at q, by the chain rule, since f is C^2 at $\psi(q)$. The second, $x\mapsto d\psi_x$, has derivative $d^2\psi_q$. The product rule for the composition of operators `Mathlib/Analysis/Calculus/FDeriv/CompCLM · HasFDerivAt.clm_comp` gives the derivative
$$v\mapsto d^2f_{\psi(q)}(d\psi_q v)\circ d\psi_q + df_{\psi(q)}\circ d^2\psi_q(v),$$
and evaluating at w gives the formula.
4. When $df_{\psi(q)}=0$ the second term is zero.

Corollary (non-degeneracy and index do not depend on the coordinates). If moreover $d\psi_q$ is a linear isomorphism L and $df_{\psi(q)}=0$, then $d^2(f\circ\psi)_q$ is the pull-back of $d^2f_{\psi(q)}$ by L. Hence the two are non-degenerate together and have the same index. `Chapter1/Nondegenerate · isNondegenerate_hessian_comp_iff` `Chapter1/Nondegenerate · index_hessian_comp`

This is §I.3 applied to the displayed identity.

Where the formal proof departs from Milnor. Milnor's Hessian is intrinsic: $f_{**}(v,w)=\tilde v_p(\tilde w(f))$, for vector fields $\tilde v$, $\tilde w$ extending v and w. He shows that it is symmetric and independent of the extensions by the bracket identity. The formal Hessian is defined in coordinates, and the two facts become theorems of a different kind. Symmetry is the symmetry of second derivatives (§I.2). Independence of the extensions is replaced by independence of the coordinate system, which is the identity above. The definition through extensions is not formalized. The only use of criticality is to kill the term $df_{\psi(q)}(d^2\psi_q(v,w))$, and only C^2 at the point is needed. The first formula, with that term kept, is not idle. In §6 Milnor computes the Hessian of the squared distance from a point, restricted to a submanifold, and there the extra term is the second fundamental form (§I.17).

I.5 Lemma 2.1: Hadamard's lemma

Lemma 2.1 (Milnor, p. 5). Let f be C^∞ on a convex neighbourhood V of 0 in $\mathbb{R}^n$, with $f(0)=0$. Then $f(x)=\sum_i x_i\,g_i(x)$ on V for some C^∞ functions g_i on V with $g_i(0)=\partial f/\partial x_i(0)$.

Milnor writes $f(x)=\int_0^1 \frac{d}{dt}f(tx)\,dt$ and takes $g_i(x)=\int_0^1 \frac{\partial f}{\partial x_i}(tx)\,dt$. That the g_i are smooth is not argued. The companion project proved Hadamard's lemma for a smooth map defined on the whole space, $f(x)=L(x)\,x$ with $L(x)=\int_0^1 df_{tx}\,dt$ smooth and $L(0)=df_0$ `Milnor/Chapter6/Hadamard · exists_hadamard`, on top of a theorem on parametric integrals with a globally C^n integrand `Milnor/Chapter6/Hadamard · contDiff_intervalIntegral` `Milnor/Chapter6/Hadamard · contDiff_intervalIntegral_infty`. Milnor's lemma is local. The integrand $(x,t)\mapsto\partial f/\partial x_i(tx)$ exists only on the open set $\{(x,t) : tx\in V\}$, so the project proves local versions of both theorems by the same method.

A tube around a segment. Let $h : E\times\mathbb{R}\to G$, with G a normed group, be continuous on an open set U containing $\{x_0\}\times[0,1]$. Then there are an M, an open $s\ni x_0$ and an open $w\supseteq[0,1]$ with $s\times w\subseteq U$ and $\|h\|\le M$ on $s\times w$. `Chapter1/Hadamard · exists_tube_bound`

Proof. The segment $\{x_0\}\times[0,1]$ is compact and lies in U, so $\|h\|$ has a bound M_0 on it. The set of points of U where $\|h\| < M_0+1$ is open, since h is continuous on the open set U, and it contains the segment. The generalized tube lemma `Mathlib/Topology/Compactness/Compact · generalized_tube_lemma` fits a product $s \times w$ of open sets between the product of compact sets $\{x_0\} \times [0,1]$ and that open set. Take $M = M_0 + 1$.

Theorem (parametric integrals on an open set). Let F be a complete normed space, $U \subseteq E \times \mathbb{R}$ open, and $h : E \times \mathbb{R} \to F$ of class C^n on U, with n finite. Let $V \subseteq E$ be open with $V \times [0,1] \subseteq U$. Then $x \mapsto \int_0^1 h(x,t)\,dt$ is C^n on V, and the same holds with C^∞ in place of C^n. `Chapter1/Hadamard · contDiffOn_intervalIntegral` `Chapter1/Hadamard · contDiffOn_intervalIntegral_infty`

The proof as formalized is an induction on n, stated for all complete targets at once `Chapter1/Hadamard · contDiffOn_intervalIntegral_aux`.

1. $n = 0$. Fix $x_0 \in V$ and a tube $s \times w$ around $\{x_0\}\times[0,1]$ on which $\|h\| \le M$. For $x \in s$ the integrand $t \mapsto h(x,t)$ is continuous on $[0,1]$ and bounded by M. For each t, $x \mapsto h(x,t)$ is continuous at x_0, because h is continuous at the point (x_0,t) of the open set U. Dominated convergence with the constant bound M `Mathlib/MeasureTheory/Integral/DominatedConvergence · intervalIntegral.continuousAt_of_dominated_interval` makes the integral continuous at x_0.
2. *From n to $n+1$.* The partial derivative in x, $D(x,t) = dh_{(x,t)} \circ \iota$ with $\iota(v) = (v,0)$, is C^n on U, being built from the derivative of a C^{n+1} map on an open set. At $x_0 \in V$ a tube bounds $\|D\|$ by some M, and Mathlib's differentiation under the integral sign `Mathlib/Analysis/Calculus/ParametricIntervalIntegral · intervalIntegral.hasFDerivAt_integral_of_dominated_of_fderiv_le` shows that $x \mapsto \int_0^1 h(x,t)\,dt$ is differentiable at x_0 with derivative $\int_0^1 D(x_0,t)\,dt$. So the integral is differentiable on V, and its derivative is $x \mapsto \int_0^1 D(x,t)\,dt$. That map is C^n on V by the induction hypothesis applied to D, whose target is now the space of linear maps $E \to F$. A map on an open set is C^{n+1} exactly when it is differentiable with a C^n derivative `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_fderiv_of_isOpen`.
3. *Universes.* The induction replaces the target G by the space of linear maps $E \to G$ at each step, so it must range over all targets of one size. It is stated for targets in the universe $\max(u,v)$, where u and v are the universes of E and F; that universe contains the space of linear maps $E \to G$ whenever it contains G. The theorem for an arbitrary F lifts F into that universe by a continuous linear equivalence, applies the induction, and comes back, commuting the equivalence with the integral `Mathlib/MeasureTheory/Integral/IntervalIntegral/Basic · ContinuousLinearMap.intervalIntegral_comp_comm`. This is the companion's device, and the size condition has no mathematical content.
4. C^∞ is C^n for every finite n.

The domination is on a tube $s \times w$, not on a compact neighbourhood of x_0, so E need not be locally compact; it is any real normed space.

Corollary (the integrals of Hadamard's lemma). Let $V \subseteq E$ be open and star-shaped about 0, g smooth on V with values in a complete space, and $a : \mathbb{R} \to \mathbb{R}$ smooth. Then $x \mapsto \int_0^1 a(t)\,g(tx)\,dt$ is smooth on V. `Chapter1/Hadamard · contDiffOn_integral_smul_comp`

Proof. Apply the theorem to $h(x,t) = a(t)\,g(tx)$ on $U = \{(x,t) : tx \in V\}$, open as the preimage of V under the continuous map $(x,t) \mapsto tx$. Star-shapedness gives $V \times [0,1] \subseteq U$, and h is smooth on U as a product and composite of smooth maps.

Theorem (Lemma 2.1, coordinate free; V1.7). Let $V \subseteq E$ be open and star-shaped about 0, and $f : E \to F$ smooth on V with $f(0) = 0$, F complete. Then $f(x) = L(x)\,x$ for $x \in V$, where $L(x) = \int_0^1 df_{tx}\,dt$ is smooth on V and $L(0) = df_0$. `Chapter1/Hadamard · exists_hadamard_on`

The proof as formalized.

1. df is smooth on the open set V, so L is smooth on V by the corollary with $a \equiv 1$.

2. Let $x \in V$. The segment $\{tx : 0 \le t \le 1\}$ lies in V, so f is C^1 at each of its points. Taylor's formula with integral remainder at order 0 `Mathlib/Analysis/Calculus/TaylorIntegral · map_add_eq_sum_add_integral_iteratedFDeriv` is then the fundamental theorem of calculus along the segment: $f(x) = f(0) + \int_0^1 df_{tx}(x)\,dt$.
3. The integrand $t \mapsto df_{tx}$ is continuous on $[0,1]$, hence integrable, and evaluation at x is a continuous linear map, which commutes with the integral `Mathlib/MeasureTheory/Integral/IntervalIntegral/Basic · ContinuousLinearMap.intervalIntegral_comp_comm`. With $f(0) = 0$ this gives $f(x) = L(x)\,x$.
4. $L(0) = \int_0^1 df_0\,dt = df_0$.

Theorem (Lemma 2.1 as Milnor states it). Let $V \subseteq \mathbb{R}^n$ be open and convex with $0 \in V$, and f smooth on V with $f(0) = 0$. Then $f(x) = \sum_i x_i\, g_i(x)$ on V for functions g_i smooth on V with $g_i(0) = \partial f/\partial x_i(0)$, the derivative df_0 applied to the i-th basis vector. `Chapter1/Hadamard · exists_hadamard_coord`

Proof. A convex set containing 0 is star-shaped about 0. Put $g_i(x) = L(x)\,e_i$, smooth since evaluation at e_i is linear. Writing $x = \sum_i x_i e_i$, $f(x) = L(x)\,x = \sum_i x_i\,L(x)\,e_i = \sum_i x_i\,g_i(x)$, and $g_i(0) = df_0(e_i)$.

Hidden hypothesis, and a generalization. The neighbourhood V must be open, or at least the statement must be read on an open set; a convex neighbourhood of 0 contains an open convex one, its interior, so nothing is lost. Only star-shapedness about 0 is used, the source may be any normed space, and the target any complete normed space.

Theorem (Hadamard's lemma to second order). Let V, f be as in the coordinate-free form, and assume also $df_0 = 0$. Then $f(x) = B(x)(x,x)$ for $x \in V$, where

$$B(x) = \int_0^1 (1-t)\,d^2 f_{tx}\,dt$$

is smooth on V, each $B(x)$ with $x \in V$ is a symmetric bilinear map, and $B(0) = \frac{1}{2}\,d^2 f_0$. `Chapter1/Hadamard · exists_hadamard_two_on`

The proof as formalized.

1. $d^2 f$ is smooth on V, so B is smooth on V by the corollary with $a(t) = 1 - t$.
2. Evaluating $B(x)$ at u and then at w commutes with the integral twice: $B(x)(u,w) = \int_0^1 (1-t)\,d^2 f_{tx}(u,w)\,dt$.
3. f is C^2 along the segment from 0 to x, and Taylor's formula at order 1 `Mathlib/Analysis/Calculus/TaylorIntegral · map_add_eq_sum_add_integral_iteratedFDeriv` gives $f(x) = f(0) + df_0(x) + \int_0^1 (1-t)\,d^2 f_{tx}(x,x)\,dt$. The first two terms vanish, so $f(x) = B(x)(x,x)$.
4. For $t \in [0,1]$, f is smooth at $tx \in V$, so $d^2 f_{tx}$ is symmetric `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`. The integrands of $B(x)(u,w)$ and $B(x)(w,u)$ therefore agree.
5. $B(0) = (\int_0^1 (1-t)\,dt)\,d^2 f_0 = \frac{1}{2}\,d^2 f_0$.

Where the formal proof departs from Milnor. On p. 6 Milnor applies Lemma 2.1 twice, obtaining $f = \sum_{i,j} x_i x_j h_{ij}$. He symmetrizes, $H_{ij} = \frac{1}{2}(h_{ij} + h_{ji})$, and asserts without proof that $(H_{ij}(0)) = (\frac{1}{2}\,\partial^2 f/\partial x^i \partial x^j(0))$. That follows by differentiating $f = \sum x_i x_j H_{ij}$ twice at 0, but he does not say so. Taylor's formula produces a coefficient that is symmetric from the start and gives its value at 0 in one line. The two coefficients agree: writing out Milnor's two integrals and changing variables gives $h_{ij}(x) = \int_0^1 (1-t)\,\partial_i \partial_j f(tx)\,dt$.

I.6 The index through Sylvester's law of inertia

In the first paragraph of his proof of Lemma 2.2 (p. 6) Milnor shows that a normal form $f(p) - \sum_{i\le\lambda}(z^i)^2 + \sum_{i>\lambda}(z^i)^2$ forces λ to be the index. In the coordinates z the matrix of f_{**} is $\operatorname{diag}(-2,\dots,-2,2,\dots,2)$. A negative definite subspace of dimension greater than λ would meet the positive definite span of the last $n-\lambda$ coordinate vectors. This is the uniqueness half of Sylvester's law of inertia, and Mathlib has it.

Definitions. For $k \le n$, the *Morse signs* are $s_i = -1$ for $i < k$ and $s_i = +1$ otherwise `Chapter1/MorseLemma · morseSign`. The *diagonal form* with weights s is $D_s(y,z) = \sum_i s_i\, y_i z_i$ on $\mathbb{R}^n$ `Chapter1/MorseLemma · diagForm` `Chapter1/MorseLemma · diagForm_apply`. It is symmetric `Chapter1/MorseLemma · diagForm_symm`, and with the Morse signs it is Milnor's normal form on the diagonal: $D(y,y) = -\sum_{i<k} y_i^2 + \sum_{i\ge k} y_i^2$ `Chapter1/MorseLemma · diagForm_morseSign_apply_self`. Its quadratic form is Mathlib's weighted sum of squares $\sum_i s_i y_i^2$ `Chapter1/MorseLemma · quadForm_diagForm` `Mathlib/LinearAlgebra/QuadraticForm/Basic · QuadraticMap.weightedSumSquares`.

Theorem (the index of the normal form). For $k \le n$, the diagonal form with k minus signs on $\mathbb{R}^n$ has index k. `Chapter1/MorseLemma · index_diagForm_morseSign`

The proof as formalized. The index is the negative index of the weighted sum of squares, which Mathlib computes as the number of negative weights `Mathlib/LinearAlgebra/QuadraticForm/Signature · QuadraticForm.sigNeg_weightedSumSquares`. Here that is the number of $i < k$, namely k. Mathlib's proof is Milnor's argument, made for the positive index and transferred to the negative one by changing the signs of the weights. The coordinate span of the positive weights is positive definite, which bounds the positive index from below. For the bound from above, a positive definite subspace and a negative semidefinite subspace meet only in 0, so their dimensions add up to at most n `Mathlib/LinearAlgebra/QuadraticForm/Signature · QuadraticForm.sigPos_add_finrank_le_of_nonpos`.

Theorem (diagonalization, negative squares first). Let E be finite-dimensional, $n = \dim E$, and H a symmetric non-degenerate continuous bilinear form on E of index λ. There is a linear isomorphism $e : E \to \mathbb{R}^n$ with

$$H(v,v) = -\sum_{i<\lambda}(ev)_i^2 + \sum_{i\ge\lambda}(ev)_i^2 \qquad \text{for all } v.$$

`Chapter1/MorseLemma · exists_linearEquiv_diagForm`

The proof as formalized.

1. The symmetric bilinear form associated with Q_H is H itself, because H is symmetric `Mathlib/LinearAlgebra/QuadraticForm/Basic · QuadraticMap.associated_left_inverse`. So the non-degeneracy of H is that of the associated form.
2. The existence half of Sylvester's law, in Mathlib `Mathlib/LinearAlgebra/QuadraticForm/Real · QuadraticForm.equivalent_one_neg_one_weighted_sum_squared`, gives weights $w_i \in \{-1,+1\}$ and an isometric linear isomorphism e_0 from (E, Q_H) to $(\mathbb{R}^n, \sum_i w_i y_i^2)$. Non-degeneracy is what excludes the weight 0.
3. The uniqueness half `Mathlib/LinearAlgebra/QuadraticForm/Signature · QuadraticForm.sigNeg_of_equiv_weightedSumSquares` identifies the number of negative weights: it is λ.
4. Let $k = \#\{j : w_j < 0\}$. A permutation σ of the indices brings the negative weights to the front: $w_{\sigma(i)} < 0$ exactly when $i < k$ `Chapter1/MorseLemma · exists_perm_neg_first`. It is built by matching the indices $i < k$ with the indices of negative weight, and the remaining indices with the remaining ones, both pairs of sets having equal cardinality.
5. Then $w_{\sigma(i)}$ is the i-th Morse sign for every i. Put $(ev)_i = (e_0 v)_{\sigma(i)}$. Reindexing the sum by σ gives $H(v,v) = \sum_i w_i (e_0 v)_i^2 = \sum_i w_{\sigma(i)} (ev)_i^2$, the normal form.

Milnor cites Birkhoff–MacLane for the diagonalization of a quadratic form, and his induction imitates it with smooth coefficients. In the formal proof the linear algebra is used once, at the end, on the constant

form $d^2 f_p$.

I.7 The inverse function theorem with a smooth inverse

Mathlib's inverse function theorem `Mathlib/Analysis/Calculus/InverseFunctionTheorem/ContDiff · ContDiffAt.toOpenPartialHomeomorph` produces a local inverse that is known to be C^n only at the image of the base point. A coordinate system must be smooth on its whole domain, so the project proves a strengthened form, which this Part uses twice.

Theorem. Let E be complete, $f : E \to F$ of class C^n ($n \neq 0$) on an open set $U \ni a$, and suppose f has an invertible derivative at a. Then f restricts to an open partial homeomorphism Φ with a in its domain, its domain inside U, Φ of class C^n on its domain and Φ^{-1} of class C^n on its image. `Chapter1/SquareRoot · exists_openPartialHomeomorph_of_contDiffOn`

The proof as formalized.

1. Let Φ_0 be the open partial homeomorphism that Mathlib's theorem provides `Mathlib/Topology/OpenPartialHomeomorph/Defs · OpenPartialHomeomorph`: its forward map is f, its domain is a neighbourhood of a, and its inverse is a local inverse of f.
2. The derivative of f is invertible at every point near a. This is a lemma of the companion project `Milnor/Chapter1/InverseFunction · eventually_isInvertible_fderiv`: the invertible operators form an open set, and df is continuous at a.
3. Let V be the intersection of U, the domain of Φ_0, and the interior of the set where df is invertible. It is open and contains a. Restrict Φ_0 to V `Mathlib/Topology/OpenPartialHomeomorph/IsImage · OpenPartialHomeomorph.restrOpen`, and call the result Φ.
4. Φ is C^n on its domain, which lies in U. For b in the image of Φ, the point $x = \Phi^{-1}(b)$ lies in V, so f is C^n at x with an invertible derivative there. The inverse of a partial homeomorphism is C^n at the image of any such point `Mathlib/Analysis/Calculus/ContDiff/Operations · OpenPartialHomeomorph.contDiffAt_symm`.

This is the Euclidean core of the companion's inverse function theorem on manifolds `Milnor/Chapter1/InverseFunction · isLocalDiffeomorphAt_of_isInvertible_mfderiv`: step 2 makes the inverse smooth at every point of its domain, and not only at $f(a)$.

I.8 A smooth square root of a family of bilinear forms

The heart of Milnor's proof of Lemma 2.2 is a diagonalization with smooth coefficients, done one square at a time. The formal proof does it at once.

Theorem (the square root). Let E be finite-dimensional, and B_0 a symmetric non-degenerate continuous bilinear form on E. Let $x \mapsto B(x)$ be a family of continuous bilinear forms, smooth on an open set $U \ni 0$, with $B(0) = B_0$. Then there are an open W with $0 \in W \subseteq U$ and a family $x \mapsto C(x)$ of linear maps $E \to E$, smooth on W, with $C(0) = \mathrm{id}$ and

$$B(x)(v, v) = B_0(C(x)v,\ C(x)v) \qquad \text{for } x \in W \text{ and all } v.$$

`Chapter1/SquareRoot · exists_sqrt_family`

The idea. Consider the map from linear maps of E to bilinear forms on E

$$\Psi(T) = B_0(T\cdot, T\cdot) + B_0(T\cdot, \cdot) - B_0(\cdot, T\cdot)$$

`Chapter1/SquareRoot · sqrtAux` `Chapter1/SquareRoot · sqrtAux_apply`. Then $\Psi(\mathrm{id}) = B_0$ `Chapter1/SquareRoot · sqrtAux_id`. Because B_0 is symmetric, the last two terms cancel on the diagonal, so $\Psi(T)(v, v) = B_0(Tv, Tv)$ `Chapter1/SquareRoot · sqrtAux_apply_self`. If Ψ has a smooth local inverse near B_0, then $C(x) = \Psi^{-1}(B(x))$ works.

The derivative. Regard B_0 as a linear map $E \to E^*$ and write T^{T} for precomposition with T on E^* `Chapter1/SquareRoot · precomp`. Then $\Psi(T) = T^{\mathsf{T}} B_0 T + B_0 T - T^{\mathsf{T}} B_0$ is a polynomial in T, hence smooth `Chapter1/SquareRoot · contDiff_sqrtAux`. Differentiating at $T = \mathrm{id}$ in the direction K, the three terms contribute $B_0(K\cdot,\cdot) + B_0(\cdot, K\cdot)$, then $B_0(K\cdot,\cdot)$, then $-B_0(\cdot, K\cdot)$. The derivative is therefore

$$K \mapsto 2\, B_0(K\cdot,\cdot)$$

`Chapter1/SquareRoot · sqrtAuxDeriv` `Chapter1/SquareRoot · hasFDerivAt_sqrtAux`, a computation that does not use symmetry. It is invertible. Since E^* has the dimension of E `Chapter1/SquareRoot · finrank_dual_eq`, the injective map $v \mapsto B_0(v,\cdot)$ is a linear isomorphism $E \to E^*$ `Chapter1/SquareRoot · dualEquiv`, and the inverse of the derivative is $S \mapsto \frac{1}{2}\, B_0^{-1} \circ S$, checked by composing both ways `Chapter1/SquareRoot · sqrtAuxDerivEquiv`.

Why the skew term. Without $B_0(T\cdot,\cdot) - B_0(\cdot, T\cdot)$ the derivative at the identity would be $K \mapsto B_0(K\cdot,\cdot) + B_0(\cdot, K\cdot)$, which kills every K that is skew for B_0. The inverse function theorem would then not apply. The added term is skew-symmetric, so it is invisible on the diagonal and costs nothing.

The proof as formalized.

1. The space of linear maps of E is finite-dimensional, hence complete. Apply §I.7 to Ψ on the whole space, at id, with the derivative just computed. This gives Φ, equal to Ψ on its domain, with id in its domain and Φ^{-1} smooth on its image. The image contains $\Phi(\mathrm{id}) = B_0$.
2. Let W be the set of $x \in U$ with $B(x)$ in the image of Φ. It is open because B is continuous on the open set U, and it contains 0 because $B(0) = B_0$.
3. Put $C(x) = \Phi^{-1}(B(x))$. It is smooth on W as a composite, and $C(0) = \Phi^{-1}(\Phi(\mathrm{id})) = \mathrm{id}$.
4. For $x \in W$, $\Psi(C(x)) = \Phi(\Phi^{-1}(B(x))) = B(x)$, so $B(x)(v,v) = \Psi(C(x))(v,v) = B_0(C(x)v, C(x)v)$.

The forms $B(x)$ need not be symmetric; the statement concerns only the diagonal. When $B(x)$ is symmetric, $C(x)$ is a square root of $B(x)$ relative to B_0, whence the name.

I.9 Lemma 2.2: the lemma of Morse

Lemma 2.2 (the lemma of Morse, Milnor p. 6; V1.8). Let E be finite-dimensional, $n = \dim E$, and $f : E \to \mathbb{R}$ smooth on an open set $U \ni p$. Let p be a critical point of f at which $d^2 f_p$ is non-degenerate, and let λ be its index. Then there is an open partial homeomorphism $y : E \to \mathbb{R}^n$, smooth on its domain with an inverse smooth on its image, whose domain contains p and lies in U, such that $y(p) = 0$ and

$$f(x) = f(p) - \sum_{i<\lambda} y_i(x)^2 + \sum_{i\geq\lambda} y_i(x)^2$$

throughout its domain. `Chapter1/MorseLemma · morse_lemma`

The formal proof first proves a coordinate-free form and then diagonalizes.

Theorem (the Morse chart, coordinate free). Under the same hypotheses there is an open partial homeomorphism $\varphi : E \to E$, smooth with smooth inverse, whose domain contains p and lies in U, with $\varphi(p) = 0$, $d\varphi_p = \mathrm{id}$, and

$$f(x) = f(p) + \tfrac{1}{2}\, d^2 f_p(\varphi(x), \varphi(x))$$

throughout its domain. `Chapter1/MorseLemma · exists_morse_chart`

The idea. Write $f(p+x) - f(p) = B(x)(x,x)$ by Hadamard's lemma to second order, with $B(0) = \frac{1}{2} d^2 f_p =: B_0$. Take the square root C of the family B relative to B_0 (§I.8). Then $f(p+x) - f(p) = B_0(C(x)x, C(x)x)$, and $x \mapsto C(x)x$ has derivative $C(0) = \mathrm{id}$ at 0, so it is a coordinate system.

The proof as formalized.

1. *A ball.* Choose $r > 0$ with the ball of radius r about p inside U, and let V be the ball of radius r about 0. It is open and convex, hence star-shaped about 0.
2. *Translate.* Put $g(x) = f(p + x) - f(p)$. It is smooth on V, $g(0) = 0$, and $dg_0 = df_p = 0$ `Chapter1/MorseLemma · fderiv_translate`. Its Hessian at 0 is $d^2 f_p$ `Chapter1/MorseLemma · hessian_translate`.
3. *Hadamard.* By §I.5, $g(x) = B(x)(x, x)$ on V with B smooth on V and $B(0) = \frac{1}{2}\, d^2 f_p = B_0$ `Chapter1/Hadamard · exists_hadamard_two_on`.
4. *B_0 is admissible.* f is C^2 at p, so $d^2 f_p$ is symmetric `Chapter1/Nondegenerate · hessian_symm`, and so is B_0. And B_0 is non-degenerate because $d^2 f_p$ is and $\frac{1}{2} \neq 0$ `Chapter1/Nondegenerate · isNondegenerate_smul_iff`.
5. *Square root.* §I.8 gives an open W with $0 \in W \subseteq V$ and C smooth on W, with $C(0) = \mathrm{id}$ and $B(x)(v, v) = B_0(C(x)v, C(x)v)$ `Chapter1/SquareRoot · exists_sqrt_family`.
6. *The chart before inversion.* Put $\psi(x) = C(x-p)(x-p)$, smooth on the open set $p+W$. Its derivative at p comes from the product rule. The term in which C is differentiated is applied to $x - p$, which is 0 at $x = p$. What remains is $C(0) = \mathrm{id}$.
7. *Inversion.* §I.7, applied to ψ on $(p + W) \cap U$ at p with derivative id, gives Φ, smooth with smooth inverse, equal to ψ on its domain, which contains p and lies in U `Chapter1/SquareRoot · exists_openPartialHomeomorph_of_contDiffOn`.
8. *The identity.* $\Phi(p) = C(0)\, 0 = 0$ and $d\Phi_p = \mathrm{id}$. For x in the domain, $x - p \in W \subseteq V$, so

$$f(x) = f(p) + g(x - p) = f(p) + B(x - p)(x - p, x - p) = f(p) + B_0(\Phi(x), \Phi(x)),$$

   and $B_0 = \frac{1}{2}\, d^2 f_p$.

The proof of Lemma 2.2 as formalized.

1. Take φ from the coordinate-free form.
2. Diagonalize $d^2 f_p$, which is symmetric and non-degenerate of index λ (§I.6): there is $e : E \to \mathbb{R}^n$ with $d^2 f_p(v, v) = D(ev, ev)$, where D is the diagonal form with λ minus signs `Chapter1/MorseLemma · exists_linearEquiv_diagForm`.
3. Absorb the factor $\frac{1}{2}$: with $L(z) = e(z/\sqrt{2})$, $\frac{1}{2}\, d^2 f_p(z, z) = D(Lz, Lz)$.
4. Put $y = L \circ \varphi$, that is, φ followed by the linear homeomorphism L. It has the domain of φ, it is smooth with smooth inverse $\varphi^{-1} \circ L^{-1}$, and $y(p) = L(0) = 0$. On its domain $f = f(p) + D(y, y)$, which is the normal form `Chapter1/MorseLemma · diagForm_morseSign_apply_self`.

Where the formal proof departs from Milnor. Milnor proves the existence of the coordinates by induction on r. At stage r the function is $\pm(u_1)^2 \pm \cdots \pm (u_{r-1})^2 + \sum_{i,j \geq r} u_i u_j H_{ij}(u)$ with (H_{ij}) symmetric. A linear change of the last coordinates makes $H_{rr}(0) \neq 0$. One then completes the square with $g = \sqrt{|H_{rr}|}$, and the inverse function theorem shows that the new functions are coordinates. The linear change is asserted without proof. It needs the block $\big(H_{ij}(0)\big)_{i,j \geq r}$ to be non-singular, hence non-zero, which the induction must carry along, and when all its diagonal entries vanish a substitution such as $u_r = w_r + w_l$ is needed first. Here no such step occurs. The square root of §I.8 performs the whole diagonalization with smooth coefficients in one application of the inverse function theorem, and the only linear algebra, Sylvester's law, is applied once to the constant form $d^2 f_p$. The statement is also slightly more general than Milnor's: f need only be smooth on a neighbourhood of p, and the coordinate-free form holds in any finite-dimensional normed space.

I.10 The index is forced by the normal form

This is the first paragraph of Milnor's proof of Lemma 2.2, which the formal text proves after the existence.

Theorem (V1.9). Let $f : E \to \mathbb{R}$ be C^2 at a critical point p. Let $z : E \to \mathbb{R}^n$ be an open partial homeomorphism, smooth with smooth inverse, $n = \dim E$, with p in its domain and $z(p) = 0$. Suppose

that for some $k \le n$

$$f(x) = f(p) - \sum_{i<k} z_i(x)^2 + \sum_{i\ge k} z_i(x)^2$$

throughout the domain of z. Then k is the index of $d^2 f_p$. Chapter1/MorseLemma · index_eq_of_morse_chart

The proof as formalized. Let D be the diagonal form with k minus signs.

1. On the image of z, an open set containing 0, $f \circ z^{-1}(w) = f(p) + D(w, w)$.
2. So the Hessian of $f \circ z^{-1}$ at 0 is $2D$. It depends only on the germ Chapter1/Nondegenerate · hessian_congr, the constant does not contribute, and the Hessian of $w \mapsto D(w, w)$ is $D(v, w) + D(w, v) = 2D(v, w)$ Chapter1/Nondegenerate · hessian_bilinear Chapter1/MorseLemma · diagForm_symm.
3. The derivative of z^{-1} at 0 is a linear isomorphism, with inverse the derivative of z at p. This follows from the chain rule applied to $z^{-1} \circ z = \mathrm{id}$ near p and to $z \circ z^{-1} = \mathrm{id}$ near 0.
4. By the independence of the coordinates (§I.4), applied to $\psi = z^{-1}$ at 0, with $z^{-1}(0) = p$ and $df_p = 0$, the Hessian of $f \circ z^{-1}$ at 0 has the index of $d^2 f_p$ Chapter1/Nondegenerate · index_hessian_comp.
5. The index of $2D$ is that of D Chapter1/Nondegenerate · index_smul, which is k Chapter1/MorseLemma · index_diagForm_morseSign.

Milnor computes the matrix of f_{**} in the coordinates z (p. 6). That presupposes the independence of the coordinates, which he does not prove; the formal proof invokes it in step 4. The formal statement does not assume E finite-dimensional: the derivative of the chart provides the linear isomorphism with $\mathbb{R}^n$ that step 4 needs.

I.11 Corollary 2.3: non-degenerate critical points are isolated

Milnor states Corollary 2.3 right after the lemma of Morse, from which it follows (in Morse coordinates $df = \sum \pm 2y^i\, dy^i$ vanishes only at $y = 0$), but he gives no argument. The formal proof does not use the lemma of Morse.

Theorem. Let E be finite-dimensional, f of class C^2 at p, and $d^2 f_p$ non-degenerate. Then no point $x \ne p$ sufficiently close to p is a critical point of f. Chapter1/Nondegenerate · eventually_not_isCriticalPoint

That p itself is critical is not needed.

Corollary 2.3 (Milnor, p. 8; V1.10). If E is finite-dimensional, f is C^2 at p, and p is a non-degenerate critical point, then p has a neighbourhood in which a point is critical exactly when it is p. Chapter1/Nondegenerate · IsNondegenerateCriticalPoint.exists_nhds_isCriticalPoint_eq

The idea. The map $x \mapsto df_x$, from E to its dual, has the injective derivative $d^2 f_p$ at p. A map with an injective derivative at a point does not take its value at that point again nearby. This is the injectivity half of the inverse function theorem.

The proof as formalized.

1. Since f is C^2 at p, the map $x \mapsto df_x$ is differentiable at p, with derivative $d^2 f_p$, viewed as the linear map $v \mapsto d^2 f_p(v, \cdot)$ from E to E^*.
2. Non-degeneracy says that this linear map has kernel 0. In finite dimension an injective linear map is bounded below: $\|v\| \le K \|d^2 f_p(v, \cdot)\|$ for some K Mathlib/Analysis/Normed/Module/FiniteDimension · LinearMap.exists_antilipschitzWith.
3. For each value c, a map whose derivative at p is bounded below avoids c at all points $x \ne p$ close enough to p Mathlib/Analysis/Calculus/FDeriv/Equiv · HasFDerivAt.eventually_ne. For c different from the value at p this is continuity. For $c = df_p$ it is the estimate $df_x - df_p = d^2 f_p(x - p, \cdot) + o(\|x - p\|)$, whose first term has norm at least $\|x - p\|/K$. Applied with $c = 0$: $df_x \ne 0$ for $x \ne p$ near p.
4. A critical point x has $df_x = 0$ Chapter1/Nondegenerate · IsCriticalPoint.fderiv_eq.

For the corollary, step 3 provides a punctured neighbourhood of p free of critical points, and p is critical by hypothesis.

The companion project proves the analogous statement for vector fields, that a zero z of v with dv_z bijective is isolated `Milnor/Chapter6/Nondegenerate · Sphere.exists_radius_of_nondegenerate`. In coordinates on $\mathbb{R}^n$, Corollary 2.3 is its case $v = \operatorname{grad} f$. Working with $x \mapsto df_x$ in E^* avoids choosing an inner product. *Deviation (stronger):* only C^2 at p is assumed.

I.12 The examples of pp. 8–9

After Corollary 2.3 Milnor lists five examples of degenerate critical points, with pictures (V1.12). In (a) and (c) the formal proofs compute the Hessian. In (d) and (e) they go through one general principle, the contrapositive of the theorem of §I.11, and a device for producing accumulating critical points.

Contrapositive of Corollary 2.3. If E is finite-dimensional, f is C^2 at p, and critical points $x \neq p$ accumulate at p, then $d^2 f_p$ is degenerate. `Chapter1/Examples · not_isNondegenerate_of_frequently`

Lines. If a property holds at every point $p + tv$ of a line through p, with $v \neq 0$, then it holds at points $x \neq p$ arbitrarily close to p, because $t \mapsto p + tv$ maps punctured neighbourhoods of 0 into punctured neighbourhoods of p `Chapter1/Examples · frequently_of_line`.

- **(a)** $f(x) = x^3$ on $\mathbb{R}$. Its derivative is $3x^2$ `Chapter1/Examples · Examples.hasFDerivAt_cube`. A point is critical exactly when it is 0 `Chapter1/Examples · Examples.isCriticalPoint_cube_iff`: by uniqueness of the derivative, a critical x has $3x^2 = 0$. The Hessian at 0 is 0, the derivative of $x \mapsto 3x^2$ at 0 `Chapter1/Examples · Examples.hessian_cube`. So 0 is a degenerate critical point, the vector 1 lying in the null space `Chapter1/Examples · Examples.cube_degenerate`.
- **(b)** $f(x) = e^{-1/x^2} \sin^2(1/x)$ is not formalized. Its smoothness at 0 needs the flatness of e^{-1/x^2} against the growth of the derivatives of $\sin^2(1/x)$. Degeneracy and non-isolation would then follow as in (d), since the points $1/(k\pi)$ are minima, hence critical.
- **(c)** The monkey saddle $f(x, y) = x^3 - 3xy^2$ `Chapter1/Examples · Examples.monkey`. Its derivative is $(3x^2 - 3y^2)\,dx - 6xy\,dy$ `Chapter1/Examples · Examples.hasFDerivAt_monkey`. The origin is its only critical point `Chapter1/Examples · Examples.isCriticalPoint_monkey_iff`. At a critical point, evaluating the derivative on $(1, 0)$ and $(0, 1)$ gives $x^2 = y^2$ and $xy = 0$. Hence $x^4 = x^2 y^2 = 0$, so $x = 0$, and then $y^2 = 0$. Both coefficients of the derivative have derivative 0 at the origin, so the Hessian there is 0 `Chapter1/Examples · Examples.hessian_monkey`, and the origin is degenerate `Chapter1/Examples · Examples.monkey_degenerate`.
- **(d)** $f(x, y) = x^2$ `Chapter1/Examples · Examples.sqFst`. Its derivative is $2x\,dx$ `Chapter1/Examples · Examples.hasFDerivAt_sqFst`. Its critical points are exactly the points with $x = 0$ `Chapter1/Examples · Examples.isCriticalPoint_sqFst_iff`. That set is the y axis. Milnor writes that the critical points form the x axis, and the formalization found the slip. Every critical point is degenerate `Chapter1/Examples · Examples.not_isNondegenerate_sqFst`: the critical points fill the vertical line through it, so they accumulate there.
- **(e)** $f(x, y) = x^2 y^2$ `Chapter1/Examples · Examples.sqMul`. Its derivative is $2xy^2\,dx + 2x^2 y\,dy$ `Chapter1/Examples · Examples.hasFDerivAt_sqMul`. Its critical points are the points of the two axes `Chapter1/Examples · Examples.isCriticalPoint_sqMul_iff`: if $x \neq 0$ and $y \neq 0$, the coefficient $2xy^2$ is non-zero. Every critical point is degenerate `Chapter1/Examples · Examples.not_isNondegenerate_sqMul`, by the vertical line on the axis $x = 0$ and the horizontal line on the axis $y = 0$.

In (a) and (c) the formal statements are stronger than the pictures: the origin is the only critical point. Milnor's remarks that the critical set of (d) is a submanifold and that of (e) is not are not formalized.

I.13 Critical points, the Hessian and the index on a manifold

Let M be a manifold modelled on E as in §I.1, with no Hausdorff or countability hypothesis, and $f : M \to \mathbb{R}$. The definitions transport those of §I.2 through the preferred chart.

Definitions (V1.4).

- The *local representative* of f at p is $f \circ \kappa_p^{-1}$, a function on E that is meaningful on the image of the chart, an open neighbourhood of $\kappa_p(p)$ `Chapter1/Manifold · localRep`. For the identity model it is f composed with the inverse of the preferred chart `Chapter1/Manifold · localRep_eq`.
- p is a *critical point* of f if $df_p = 0$ `Chapter1/Manifold · IsCriticalPointM`. Since Mathlib's df_p is 0 where f is not differentiable, such points count as critical. This junk case never matters, because Milnor's f is smooth.
- The *Hessian* f_{**} at p is the Hessian of the local representative at $\kappa_p(p)$, a continuous bilinear form on $T_pM = E$ `Chapter1/Manifold · hessianM`.
- *Non-degenerate critical point*, *index* and *nullity* are those of f_{**} `Chapter1/Manifold · IsNondegenerateCriticalPointM` `Chapter1/Manifold · indexM` `Chapter1/Manifold · nullityM`.
- A *Morse function* is a smooth f all of whose critical points are non-degenerate `Chapter1/Manifold · IsMorseFunction` `Chapter1/Manifold · IsMorseFunction.isNondegenerateCriticalPointM`. These are the differentiable functions with no degenerate critical points of Milnor's Theorem 3.5 (p. 20) and of §6.

As in §I.2, this is Milnor's $\tilde{v}_p(\tilde{w}(f))$ for extensions with constant coefficients in the preferred chart. It is defined at every point, and at a critical point it does not depend on the chart (§I.14).

The local representative computes everything. On the chart's domain f equals its local representative read through the chart: $f(x) = (f \circ \kappa_p^{-1})(\kappa_p(x))$ `Chapter1/Manifold · localRep_chartAt`, so f agrees with $(f \circ \kappa_p^{-1}) \circ \kappa_p$ near p `Chapter1/Manifold · eventuallyEq_localRep_comp`. Mathlib's representation of f in charts, with the chart of $\mathbb{R}$ at $f(p)$, which is the identity, is the local representative `Chapter1/Manifold · writtenInExtChartAt_eq_localRep` `Mathlib/Geometry/Manifold/IsManifold/ExtChartAt · writtenInExtChartAt`. Hence:

- f is differentiable at p exactly when its local representative is differentiable at $\kappa_p(p)$, and C^n at p exactly when the representative is C^n there `Chapter1/Manifold · mdifferentiableAt_iff_differentiableAt_localRep` `Chapter1/Manifold · contMDiffAt_iff_contDiffAt_localRep`. Mathlib's characterizations `Mathlib/Geometry/Manifold/MFDeriv/Defs · mdifferentiableAt_iff` `Mathlib/Geometry/Manifold/ContMDiff/Defs · contMDiffAt_iff` also ask for continuity of f at p. That follows from the representative and the chart. The identity model has all of E as its image, so differentiability within the image is plain differentiability.
- df_p is the derivative of the local representative at $\kappa_p(p)$, with no hypothesis at all `Chapter1/Manifold · mfderiv_eq_fderiv_localRep`. Where f is not differentiable both sides are 0 `Mathlib/Geometry/Manifold/MFDeriv/Basic · mfderiv_zero_of_not_mdifferentiableAt`.
- So p is critical exactly when the local representative has derivative 0 at $\kappa_p(p)$ `Chapter1/Manifold · isCriticalPointM_iff_fderiv`. For f differentiable at p, this says that $\kappa_p(p)$ is a critical point of the local representative `Chapter1/Manifold · isCriticalPointM_iff`, or that f has derivative 0 at p `Chapter1/Manifold · isCriticalPointM_iff_hasMFDerivAt`.

Agreement with the companion volume. A linear functional is onto exactly when it is non-zero: if $\ell(v) \neq 0$, then $t = \ell\big((t/\ell(v))\, v\big)$ `Chapter1/Manifold · surjective_iff_ne_zero`. So p is critical in Milnor's sense here ($f_* = 0$) exactly when it is critical in the sense of *Topology from the Differentiable Viewpoint* (df_p not onto) `Chapter1/Manifold · isCriticalPointM_iff_milnor` `Milnor/Chapter1/RegularValue · Milnor.IsCriticalPoint`.

Elementary facts. f_{**} is symmetric where f is C^2 `Chapter1/Manifold · hessianM_symm`. The index is at most $\dim E$ `Chapter1/Manifold · indexM_le_finrank`. Non-degeneracy is nullity 0 `Chapter1/Manifold · isNondegenerate_hessianM_iff_nullityM_eq_zero`. On the model space itself the preferred chart is the identity, so the local representative is f `Chapter1/Manifold · localRep_model` and the manifold notions are those of §I.2 `Chapter1/Manifold · hessianM_model` `Chapter1/Manifold · isCriticalPointM_model_iff` `Chapter1/Manifold · indexM_model`.

Not restated. Milnor remarks on p. 4 that for a regular value a, $M^a = f^{-1}(-\infty, a]$ is a smooth manifold with boundary $f^{-1}(a)$. This is the companion's Lemma 3 `Milnor/Chapter2/WithBoundary · isHalfSliceSubmanifold_of_regularValue` applied to $a - f$, and is not restated in this project.

I.14 Smooth coordinate systems: the independence of the chart

Since f_{**} is defined in one chart, its well-definedness *is* its independence of the chart. Milnor gives no argument. The formal proof is the second-order chain rule of §I.4, applied to the transition map between two coordinate systems.

Definition. A *smooth coordinate system* on M with values in a normed space E' is an open partial homeomorphism y from M to E' that is smooth on its domain and has an inverse smooth on its image `Chapter1/Manifold · IsSmoothCoord`. It is the same thing as a partial diffeomorphism onto an open subset of E' `Chapter1/Manifold · IsSmoothCoord.toPartialDiffeomorph` `Chapter1/Manifold · isSmoothCoord_toOpenPartialHomeomorph`. With $E' = \mathbb{R}^n$ it is Milnor's local coordinate system $(y^1, \dots, y^n)$. With $E' = E$ the smooth coordinate systems are exactly the charts of the maximal smooth atlas `Chapter1/Manifold · isSmoothCoord_iff_mem_maximalAtlas` `Mathlib/Geometry/Manifold/ContMDiff/Atlas · IsManifold.mem_maximalAtlas_iff_contMDiffOn`. In particular the preferred chart at p is one `Chapter1/Manifold · isSmoothCoord_chartAt`. A smooth coordinate system followed by a diffeomorphism between open subsets of normed spaces is again one `Chapter1/Manifold · IsSmoothCoord.trans`.

Transition maps. Let y, z be smooth coordinate systems with values in E', E'', both defined at p.

- The domain $y(\operatorname{dom} y \cap \operatorname{dom} z)$ of the transition map $z \circ y^{-1}$ is open `Chapter1/Manifold · IsSmoothCoord.isOpen_trans_source`. The transition map is smooth there `Chapter1/Manifold · IsSmoothCoord.contDiffOn_trans` `Chapter1/Manifold · IsSmoothCoord.contDiffAt_trans`.
- Its derivative at $y(p)$ is a linear isomorphism, whose inverse is the derivative of $y \circ z^{-1}$ at $z(p)$ `Chapter1/Manifold · IsSmoothCoord.exists_fderiv_trans_eq`. This is a lemma of calculus `Chapter1/Manifold · exists_fderiv_eq_of_localInverse`: if ψ and χ are inverse to each other near a and $\psi(a)$ and differentiable there, the chain rule applied to $\chi \circ \psi = \mathrm{id}$ and $\psi \circ \chi = \mathrm{id}$ shows that $d\psi_a$ and $d\chi_{\psi(a)}$ are inverse isomorphisms.
- Near $y(p)$, $f \circ y^{-1} = (f \circ z^{-1}) \circ (z \circ y^{-1})$ `Chapter1/Manifold · IsSmoothCoord.comp_symm_eventuallyEq`.

Theorem (two coordinate systems). In this situation, with $f \circ z^{-1}$ differentiable at $z(p)$: $y(p)$ is a critical point of $f \circ y^{-1}$ exactly when $z(p)$ is a critical point of $f \circ z^{-1}$ `Chapter1/Manifold · IsSmoothCoord.isCriticalPoint_trans_iff`. If moreover $f \circ z^{-1}$ is C^2 at $z(p)$ with derivative 0 there, then the Hessian of $f \circ y^{-1}$ at $y(p)$ is the Hessian of $f \circ z^{-1}$ at $z(p)$ pulled back by the derivative of the transition map `Chapter1/Manifold · IsSmoothCoord.hessian_trans` `Chapter1/Manifold · IsSmoothCoord.exists_hessian_trans_eq_bilinearComp`. Hence they are non-degenerate together and have the same index and the same nullity `Chapter1/Manifold · IsSmoothCoord.isNondegenerate_trans_iff` `Chapter1/Manifold · IsSmoothCoord.index_trans_eq` `Chapter1/Manifold · IsSmoothCoord.nullity_trans_eq`.

The proof as formalized. For criticality, the derivative of $f \circ y^{-1}$ at $y(p)$ is that of $f \circ z^{-1}$ composed with an isomorphism, and it vanishes exactly when that of $f \circ z^{-1}$ does. For the Hessian, the germ identity reduces it to the Hessian of a composite with the smooth transition map, and §I.4 applies at the critical point $z(p)$ `Chapter1/Nondegenerate · hessian_comp`. Non-degeneracy, index and nullity then follow from §I.3.

Theorem (any coordinate system computes f_{**}; V1.5). Let y be a smooth coordinate system with values in E', defined at p.

- The derivative of y at p is a linear isomorphism $T_pM \to E'$, and that of y^{-1} at $y(p)$ is a linear isomorphism $E' \to T_pM$ `Chapter1/Manifold · IsSmoothCoord.exists_mfderiv_eq` `Chapter1/Manifold · IsSmoothCoord.exists_mfderiv_symm_eq`. Read in the preferred chart, they are the derivatives of the transition maps `Chapter1/Manifold · IsSmoothCoord.mfderiv_eq` `Chapter1/Manifold · IsSmoothCoord.mfderiv_symm_eq`.
- If f is differentiable at p, then p is critical exactly when $y(p)$ is a critical point of $f \circ y^{-1}$ `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff`. For $E' = \mathbb{R}^n$ this is Milnor's criterion $\partial f/\partial y^1(p) = \cdots = \partial f/\partial y^n(p) = 0$ (p. 4), because a linear functional on $\mathbb{R}^n$ that vanishes on the basis vectors is 0 `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff_forall_partial`.

- If f is C^2 at the critical point p, the Hessian of $f \circ y^{-1}$ at $y(p)$ is f_{**} pulled back by the derivative of y^{-1} `Chapter1/Manifold · IsSmoothCoord.hessian_comp_symm`. In Milnor's words (p. 5), the matrix $(\partial^2 f/\partial y^i \partial y^j(p))$ represents f_{**} in the basis $\partial/\partial y^i|_p$. So non-degeneracy, index and nullity can be computed in y `Chapter1/Manifold · IsSmoothCoord.isNondegenerate_iff` `Chapter1/Manifold · IsSmoothCoord.index_eq` `Chapter1/Manifold · IsSmoothCoord.nullity_eq` `Chapter1/Manifold · IsSmoothCoord.isNondegenerateCriticalPointM_iff`. For $E' = \mathbb{R}^n$, p is non-degenerate exactly when $\det(\partial^2 f/\partial y^i \partial y^j(p)) \neq 0$, which is Milnor's definition `Chapter1/Manifold · IsSmoothCoord.isNondegenerate_iff_det_ne_zero`.

The proof as formalized. Take z to be the preferred chart at p and apply the previous theorem. The Hessian f_{**} is by definition the Hessian of f read in that chart `Chapter1/Manifold · hessianM_eq_chartAt`. If f is C^2 at p, so is that representative `Chapter1/Manifold · contDiffAt_comp_chartAt_symm`, and at a critical point its derivative is 0 `Chapter1/Manifold · fderiv_comp_chartAt_symm_eq_zero`. Representatives in any coordinate system inherit the smoothness of f at p `Chapter1/Manifold · IsSmoothCoord.contDiffAt_comp_symm` `Chapter1/Manifold · IsSmoothCoord.differentiableAt_comp_symm`. The matrix criterion is §I.3 applied in the coordinates y.

The same statements are recorded for the charts of the maximal atlas `Chapter1/Manifold · isCriticalPointM_iff_of_mem_maximalAtlas` `Chapter1/Manifold · hessian_comp_symm_of_mem_maximalAtlas` `Chapter1/Manifold · isNondegenerate_iff_of_mem_maximalAtlas` `Chapter1/Manifold · indexM_eq_of_mem_maximalAtlas` `Chapter1/Manifold · nullityM_eq_of_mem_maximalAtlas`.

Deviation (stronger). f need only be C^2 at p, and the coordinates may take values in any normed space. The only use of criticality is the one of §I.4.

I.15 Corollary 2.3 on a manifold, and the finiteness of the critical set

Theorem (criticality near p can be read in the chart at p). If f is C^1 at p, then for every x near p: x is critical exactly when $\kappa_p(x)$ is a critical point of the local representative at p. `Chapter1/ManifoldMorse · eventually_isCriticalPointM_iff`

Proof. A finite order of smoothness propagates, so f is C^1 at every x near p `Mathlib/Geometry/Manifold/ContMDiff/Defs · contMDiffAt_iff_contMDiffAt_nhds`. For such x in the domain of the preferred chart at p, that chart is a smooth coordinate system defined at x (§I.14), and criticality at x can be computed in it `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff`.

Theorem. If E is finite-dimensional, f is C^2 at p and f_{**} is non-degenerate at p, then no $x \neq p$ near p is critical. `Chapter1/ManifoldMorse · eventually_not_isCriticalPointM`

Proof. Put $a = \kappa_p(p)$. The local representative is C^2 at a with non-degenerate Hessian f_{**}, so the Euclidean theorem of §I.11 gives a punctured neighbourhood of a free of its critical points `Chapter1/Nondegenerate · eventually_not_isCriticalPoint`. The chart maps a punctured neighbourhood of p into a punctured neighbourhood of a: it is continuous at p, and injective on its domain, so $x \neq p$ gives $\kappa_p(x) \neq a$. The previous theorem translates back.

Corollary 2.3 on a manifold. If E is finite-dimensional, f is C^2 at p, and p is a non-degenerate critical point, then p has a neighbourhood in which it is the only critical point. `Chapter1/ManifoldMorse · IsNondegenerateCriticalPointM.exists_nhds_isCriticalPointM_eq`

Milnor never states the global consequence, but he uses it repeatedly. On p. 23 the critical values have no cluster point because each M^a is compact. In §5 a compact manifold carries finitely many critical points, arranged in levels. In §6 (p. 36) one sums over the critical points.

Theorem (V1.11). Let $f : M \to \mathbb{R}$.

- If f is C^1, its critical set is closed. `Chapter1/ManifoldMorse · isClosed_setOf_isCriticalPointM`

- If E is finite-dimensional, f is C^2 at each point of a set $S \subseteq M$, and every critical point in S is non-degenerate, then the critical points in S form a discrete set. `Chapter1/ManifoldMorse · isDiscrete_inter_setOf_isCriticalPointM`
- If E is finite-dimensional, f is C^2 and all its critical points are non-degenerate, then every compact set contains only finitely many critical points. `Chapter1/ManifoldMorse · finite_inter_setOf_isCriticalPointM` In particular a Morse function has finitely many critical points in each compact set, and finitely many in all if M is compact. `Chapter1/ManifoldMorse · IsMorseFunction.finite_inter_setOf_isCriticalPointM` `Chapter1/ManifoldMorse · IsMorseFunction.finite_setOf_isCriticalPointM`

The proof as formalized.

1. *Closed.* Let x_0 be non-critical, g the local representative at x_0 and $a = \kappa_{x_0}(x_0)$. Since f is C^1, dg is continuous at a, and $dg_a \neq 0$. So $dg \neq 0$ near a, and, pulled back by the chart, at $\kappa_{x_0}(x)$ for x near x_0. By the first theorem of this section such x are not critical. So the complement of the critical set is open.
2. *Discrete.* By Mathlib's criterion `Mathlib/Topology/DiscreteSubset · isDiscrete_iff_forall_mem_exists_isOpen` it suffices that each critical $x \in S$ have an open set meeting the critical points of S only in x. The interior of the neighbourhood given by Corollary 2.3 is such a set.
3. *Finite.* The critical points in a compact K form a closed subset of K, hence a compact set, and a discrete one by step 2. A compact discrete set is finite `Mathlib/Topology/Compactness/Compact · IsCompact.finite`.

Hidden hypotheses: none. Hausdorffness is not needed. A closed subset of a compact set is compact, and a compact discrete set is finite, in any topological space. In §5 (p. 29) Milnor asks the critical points to be isolated and non-degenerate; the first condition follows from the second.

I.16 The lemma of Morse on a manifold, and its converse

Theorem (the Morse chart on a manifold). Let E be finite-dimensional, f smooth on an open set $U \ni p$ of M, and p a non-degenerate critical point. There is a chart φ of the maximal atlas, with p in its domain and its domain inside U, such that $\varphi(p) = 0$ and $f = f(p) + \frac{1}{2}\, f_{**}(\varphi, \varphi)$ throughout its domain. `Chapter1/ManifoldMorse · exists_morse_chartM`

Lemma 2.2 on a manifold (V1.8). Under the same hypotheses, with λ the index of f at p and $n = \dim E$, there is a smooth coordinate system $y : M \to \mathbb{R}^n$, with p in its domain and its domain inside U, such that $y(p) = 0$ and $f = f(p) - \sum_{i<\lambda} y_i^2 + \sum_{i\geq\lambda} y_i^2$ throughout its domain. `Chapter1/ManifoldMorse · morse_lemmaM`

The proof as formalized. Both are obtained by applying the Euclidean theorems of §I.9 to the local representative in the preferred chart at p, and composing with that chart.

1. On the open set of points of the chart's image whose preimage lies in U, which contains $\kappa_p(p)$, the representative $f \circ \kappa_p^{-1}$ is smooth. The point $\kappa_p(p)$ is critical for it, and its Hessian there is f_{**} `Chapter1/ManifoldMorse · morse_lemmaM_aux`.
2. The Euclidean Morse chart, or Morse coordinates, for $f \circ \kappa_p^{-1}$ at $\kappa_p(p)$ give φ_0, respectively y_0 `Chapter1/MorseLemma · exists_morse_chart` `Chapter1/MorseLemma · morse_lemma`.
3. The composite of the preferred chart with φ_0, respectively y_0, is a smooth coordinate system (§I.14). With values in E it lies in the maximal atlas `Chapter1/Manifold · isSmoothCoord_iff_mem_maximalAtlas`. Its domain lies in U because that of φ_0 lies in the preimage of U. The value at p and the formula for f follow by cancelling the chart with its inverse on the chart's domain.

The converse (Milnor p. 6, first paragraph of the proof; V1.9). Let $y : M \to \mathbb{R}^n$ be a smooth coordinate system with p in its domain and $y(p) = 0$, and suppose that for some $k \leq n$ the function f has the normal form $f(p) - \sum_{i<k} y_i^2 + \sum_{i\geq k} y_i^2$ throughout the domain of y. Then p is a non-degenerate critical

point of f of index k. No hypothesis is made on f. Chapter1/ManifoldMorse · isNondegenerateCriticalPointM_of_morse_chart Chapter1/ManifoldMorse · indexM_eq_of_morse_chart

The proof as formalized. Let D be the diagonal form with k minus signs and $Q(w) = f(p) + D(w,w)$ the normal form as a function on $\mathbb{R}^n$ Chapter1/ManifoldMorse · morseNormalForm Chapter1/ManifoldMorse · morseNormalForm_eq.

1. Q is smooth, as a constant plus a continuous bilinear form on the diagonal Chapter1/ManifoldMorse · contDiff_morseNormalForm. On the domain of y, $f = Q \circ y$, so f agrees near p with a smooth function and is smooth at p. Smoothness is a conclusion, not a hypothesis.
2. Near $0 = y(p)$, $f \circ y^{-1} = Q$.
3. 0 is a critical point of Q: the derivative of $w \mapsto D(w,w)$ at 0 is $D(0,\cdot) + D(\cdot,0) = 0$ Chapter1/ManifoldMorse · isCriticalPoint_morseNormalForm. By §I.14, p is critical.
4. The Hessian of Q at 0 is $2D$ Chapter1/ManifoldMorse · hessian_morseNormalForm.
5. D is non-degenerate: $D(v, e_i) = \pm v_i$, so a vector in the null space has every coordinate 0 Chapter1/ManifoldMorse · isNondegenerate_diagForm_morseSign. By §I.14 and §I.3, f_{**} is non-degenerate.
6. The index, computed in y (§I.14), is that of $2D$, which is k Chapter1/Nondegenerate · index_smul Chapter1/MorseLemma · index_diagForm_morseSign.

This is Milnor's argument with the independence of the coordinates made explicit. *Deviation (stronger):* smoothness, criticality and non-degeneracy are conclusions, not hypotheses.

I.17 Functions on a submanifold of Euclidean space

Milnor's §6 studies functions $f = F|N$ on a manifold N in Euclidean space, chiefly $L_p(x) = \|x - p\|^2$, by computing in coordinates $(u^1, \dots, u^k)$ on a region of N. The inclusion becomes a parametrization $u \mapsto x(u)$, and he states two formulas. On p. 35, $\partial f/\partial u^i = 2\,(\partial x/\partial u^i) \cdot (x - p)$, so L_p has a critical point at q exactly when $q - p$ is normal to N at q. On p. 36, at a critical point,

$$\frac{\partial^2 f}{\partial u^i\,\partial u^j} = 2\Big(\frac{\partial x}{\partial u^i}\cdot\frac{\partial x}{\partial u^j} + \frac{\partial^2 x}{\partial u^i\,\partial u^j}\cdot(x-p)\Big),$$

and the index and nullity of L_p at q are those of this matrix. The file `Chapter1/ManifoldSubmanifold` connects such computations with the manifold notions of §§I.13–1.14, for the slice submanifolds of the companion project. Most later parts work in this setting.

Setting (V1.6). V is a real normed space, an inner product space from the paragraph on gradients onwards (Milnor's $\mathbb{R}^n$ of §6). $N \subseteq V$ is a slice submanifold modelled on Z with complement W (§I.1), a manifold modelled on Z through the charts $q \mapsto \sigma(q)_Z$ induced by its slice charts σ. For $F : V \to \mathbb{R}$ the function on N is $F|N = F \circ \iota$, with $\iota : N \to V$ the inclusion. N need not be closed.

The parametrization of a slice chart. For a slice chart σ put $x(z) = \sigma^{-1}(z, 0)$ Chapter1/ManifoldSubmanifold · sliceParam. This is Milnor's $u \mapsto x(u)$, and it is meaningful on the open set of z with $(z, 0)$ in the image of σ Chapter1/ManifoldSubmanifold · isOpen_toChart_target. There it takes values in N Chapter1/ManifoldSubmanifold · sliceParam_mem Milnor/Chapter2/Submanifold · SliceChart.symm_mem and is the inverse of the induced chart of N Chapter1/ManifoldSubmanifold · val_toChart_symm Milnor/Chapter2/Submanifold · SliceChart.toChart_symm_apply, so $x(\sigma(q)_Z) = q$ Chapter1/ManifoldSubmanifold · sliceParam_toChart. It is smooth, as σ^{-1} composed with $z \mapsto (z, 0)$ Chapter1/ManifoldSubmanifold · contDiffOn_sliceParam Chapter1/ManifoldSubmanifold · contDiffAt_sliceParam. Near any point of its domain, $F|N$ read in the induced chart is $F \circ x$ Chapter1/ManifoldSubmanifold · comp_val_comp_toChart_symm_eventuallyEq.

Slice charts are coordinate systems of N. The induced chart of every slice chart belongs to the atlas of N Milnor/Chapter2/Submanifold · IsSliceSubmanifold.mem_atlas_iff, hence to the maximal atlas Chapter1/ManifoldSubmanifold · toChart_mem_maximalAtlas Mathlib/Geometry/Manifold/IsManifold/Basic · IsManifold.subset_maximalAtlas. It is therefore a smooth coordinate system in the sense of §I.14 Chapter1/

ManifoldSubmanifold · isSmoothCoord_toChart. An ambient function that is C^n at q, or differentiable at q, or smooth, restricts to one on N Chapter1/ManifoldSubmanifold · contMDiffAt_comp_val Chapter1/ManifoldSubmanifold · mdifferentiableAt_comp_val Chapter1/ManifoldSubmanifold · contMDiff_comp_val.

Theorem (computing in a parametrization). Let $q \in N$ lie in the domain of the slice chart σ, with x its parametrization and $u = \sigma(q)_Z$.

- If F is differentiable at q, then q is a critical point of $F|N$ exactly when u is a critical point of $F \circ x$. Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_sliceParam
- If F is C^2 at q and q is a critical point of $F|N$, the Hessian of $F \circ x$ at u is f_{**} pulled back by the derivative of the inverse chart Chapter1/ManifoldSubmanifold · hessian_comp_sliceParam. Hence non-degeneracy, index and nullity of $F|N$ at q are those of $F \circ x$ at u Chapter1/ManifoldSubmanifold · isNondegenerate_hessian_comp_sliceParam_iff Chapter1/ManifoldSubmanifold · index_hessian_comp_sliceParam Chapter1/ManifoldSubmanifold · nullity_hessian_comp_sliceParam Chapter1/ManifoldSubmanifold · isNondegenerateCriticalPointM_comp_val_iff_sliceParam.

Proof. Apply §I.14 to the coordinate system induced by σ Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff Chapter1/Manifold · IsSmoothCoord.hessian_comp_symm Chapter1/Manifold · IsSmoothCoord.index_eq. The representative $(F|N) \circ (\text{chart})^{-1}$ agrees with $F \circ x$ near u, and Hessians and criticality depend only on germs.

Milnor works in one fixed coordinate system; that any parametrization gives the same answer is the chart independence of §I.14.

The tangent space. $T_qN \subseteq V$ is the range of the derivative of the inclusion at q Chapter1/ManifoldSubmanifold · tangentSubspace Chapter1/ManifoldSubmanifold · mem_tangentSubspace_iff (§0.3), the derivative being read as a linear map $Z \to V$ through the companion's restatement of the manifold derivative Milnor/Chapter5/Orient · mfd. It is spanned by the partial derivatives of any parametrization: T_qN is the range of dx_u Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam. *Proof:* near q the inclusion is x composed with the induced chart. The chain rule Mathlib/Geometry/Manifold/MFDeriv/Basic · mfderiv_comp gives $d\iota_q = dx_u \circ d(\text{chart})_{q'}$, and the second factor is a linear isomorphism (§I.14). Composing with a surjection does not change the range.

Theorem (critical points through the ambient derivative). Let F be differentiable at $q \in N$.

- q is a critical point of $F|N$ exactly when dF_q vanishes on T_qN. Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_forall_tangent
- If V is a complete inner product space, this holds exactly when the gradient $\nabla F(q)$ is orthogonal to T_qN, the condition Milnor uses on p. 35. Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_gradient_mem_orthogonal
- If V is finite-dimensional, this holds exactly when the orthogonal projection of $\nabla F(q)$ on T_qN is 0. This is the fact Milnor uses on p. 36: the zeros of the gradient field of f on N are the critical points of f. Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_starProjection_gradient_eq_zero

Proof. The chain rule gives $d(F|N)_q = dF_q \circ d\iota_q$, which is 0 exactly when dF_q vanishes on the range of $d\iota_q$. The gradient is the vector with $\langle \nabla F(q), v \rangle = dF_q(v)$ Mathlib/Analysis/Calculus/Gradient/Basic · gradient, so dF_q vanishes on T_qN exactly when $\nabla F(q)$ lies in the orthogonal complement Mathlib/Analysis/InnerProductSpace/Orthogonal · Submodule.orthogonal. A vector lies in $T_qN^\perp$ exactly when its orthogonal projection on T_qN is 0 Mathlib/Analysis/InnerProductSpace/Projection/Basic · Submodule.starProjection_apply_eq_zero_iff.

The function L_p. On an inner product space, $L_p(y) = \|y - p\|^2$ has derivative $v \mapsto 2\langle y - p, v \rangle$ at y Chapter1/ManifoldSubmanifold · hasFDerivAt_normSq_sub Chapter1/ManifoldSubmanifold · fderiv_normSq_sub_apply Mathlib/Analysis/InnerProductSpace/Calculus · HasFDerivAt.norm_sq. Its Hessian is $2\langle \cdot, \cdot \rangle$ at every point Chapter1/ManifoldSubmanifold · hessian_normSq_sub Chapter1/ManifoldSubmanifold · hessian_normSq_sub_apply, because the derivative is an affine function of y. And L_p is smooth Chapter1/ManifoldSubmanifold · contDiff_normSq_sub.

Theorem (Milnor, p. 35). A point $q \in N$ is a critical point of $L_p|N$ exactly when $q - p$ is orthogonal to

T_qN. Chapter1/ManifoldSubmanifold · isCriticalPointM_normSq_sub_iff

Proof. By the theorem above, q is critical exactly when $2\langle q-p, v\rangle = 0$ for every $v \in T_qN$, that is, when $q - p \in T_qN^{\perp}$.

Theorem (Milnor, p. 36). Let $x : Z \to V$ be C^2 at u. Then for all $v, w \in Z$

$$d^2(L_p \circ x)_u(v,w) = 2\Big(\langle dx_u v,\ dx_u w\rangle + \langle d^2x_u(v,w),\ x(u)-p\rangle\Big).$$

Chapter1/ManifoldSubmanifold · hessian_normSq_sub_comp

Proof. This is the second-order chain rule of §I.4 *with* its first-order term Chapter1/Manifold · hessian_comp_apply. The term $d^2L_p(dx\,v, dx\,w)$ is $2\langle dx\,v, dx\,w\rangle$, and the term $dL_p(d^2x(v,w))$ is $2\langle x(u)-p, d^2x(v,w)\rangle$. Milnor states the formula at a critical point; it holds at every u.

Corollary. At a critical point q of $L_p|N$ in the domain of a slice chart with parametrization x and $u = \sigma(q)_Z$, the index, the nullity and the non-degeneracy of $L_p|N$ at q are those of the bilinear form $(v,w) \mapsto 2(\langle dx_u v, dx_u w\rangle + \langle d^2x_u(v,w), x(u)-p\rangle)$. Chapter1/ManifoldSubmanifold · indexM_normSq_sub_eq Chapter1/ManifoldSubmanifold · nullityM_normSq_sub_eq Chapter1/ManifoldSubmanifold · isNondegenerate_hessianM_normSq_sub_iff

The nullity statement is the computation behind Milnor's Lemma 6.5 (p. 36). Part III uses these statements for focal points.

What Milnor leaves implicit. The Hessian of $F \circ x$ is not the restriction of the ambient Hessian of F. The chain rule adds $dF_q(d^2x(v,w))$, which does not vanish at a critical point of $F|N$, since only the tangential part of dF_q does. For L_p this added term is the second-fundamental-form term of Milnor's formula, and it is why §I.4 keeps the first-order term.

In the notation of the companion's tubular neighbourhoods. The companion's Poincaré–Hopf theorem Milnor/Chapter6/PoincareHopf · Tub.poincare_hopf works with $N \subseteq \mathbb{R}^{K+1}$, its chosen slice chart at q Milnor/Chapter6/Tubular1 · Tub.σ, the parametrization Milnor/Chapter6/Tubular1 · Tub.φ, the point over q Milnor/Chapter6/Tubular3 · Tub.z₀ and the tangent space Milnor/Chapter6/ChartIndex · Tub.tanN. These are, by definition, the objects above for that slice chart Chapter1/ManifoldSubmanifold · sliceParam_tub Chapter1/ManifoldSubmanifold · tangentSubspace_eq_tanN. So the index and the non-degeneracy of $F|N$ at a critical point can be computed in the companion's parametrization Chapter1/ManifoldSubmanifold · indexM_comp_val_eq_tub Chapter1/ManifoldSubmanifold · isNondegenerateCriticalPointM_comp_val_iff_tub Milnor/Chapter6/Tubular1 · Tub.mem_source. This is how §6, Application 2 (the alternating sum of the numbers of critical points equals the Euler characteristic) is connected to Poincaré–Hopf in Part III.

I.18 Global flows of bounded Lipschitz vector fields

Milnor's proof of Lemma 2.4 (pp. 10–11) rests on two facts he quotes, citing Graves: locally the equation $d\varphi_t(q)/dt = X_{\varphi_t(q)}$, $\varphi_0(q) = q$, has a unique solution, and that solution depends smoothly on the initial condition. He covers the compact set K outside which X vanishes by finitely many neighbourhoods on which solutions exist for $|t| < \varepsilon$, and lets ε_0 be the least of these ε. He sets $\varphi_t(q) = q$ for $q \notin K$. Larger times come from iterating $\varphi_{\varepsilon_0/2}$, and he leaves the reader to verify that the result is well defined, smooth, and a group. The formal proof works on a Banach space E; for the lemma itself only $E = \mathbb{R}^n$ matters (§I.22). This section proves everything except smoothness, for every *bounded, globally Lipschitz* field, without any iteration. Flows are recalled in §0.10.

Definitions. A vector field $X : E \to E$ is *bounded and Lipschitz* if $\|X\| \le M$ everywhere and X is K-Lipschitz for some M, K Flow/Global · IsBddLipschitz. An *integral curve* of X is a curve $\gamma : \mathbb{R} \to E$ with $\gamma'(t) = X(\gamma(t))$ for every t, in Mathlib's sense with a field that does not depend on time Mathlib/Analysis/ODE/Basic · IsIntegralCurve.

A C^1 field that vanishes outside a compact set is bounded and Lipschitz `Flow/Global · IsBddLipschitz.of_contDiff`. X and dX are continuous with compact support, hence bounded, and a bound C on $\|dX\|$ is a Lipschitz constant by the mean value inequality `Mathlib/Analysis/Calculus/MeanValue · lipschitzWith_of_nnnorm_fderiv_le`. A scalar multiple of a bounded Lipschitz field is one `Flow/Global · IsBddLipschitz.smul`.

Uniqueness. Two solutions of $\gamma' = X(\gamma)$ on an open interval that agree at one time agree on the whole interval `Flow/Global · eqOn_Ioo_of_hasDerivAt`. This is Mathlib's uniqueness theorem, proved by Grönwall's inequality `Mathlib/Analysis/ODE/ExistUnique · ODE_solution_unique_of_mem_Ioo`. The same holds for integral curves defined on all of $\mathbb{R}$ `Flow/Global · IsBddLipschitz.isIntegralCurve_eq` `Mathlib/Analysis/ODE/ExistUnique · ODE_solution_unique_univ`. On a manifold this step would need the Hausdorff property; a normed space has it.

Theorem (local existence on any interval). Let E be complete and X bounded and Lipschitz. For every $x \in E$ and every a there is a solution α of $\alpha' = X(\alpha)$ on $(-a, a)$ with $\alpha(0) = x$. `Flow/Global · IsBddLipschitz.exists_local`

Proof. For $a \le 0$ the interval is empty. For $a > 0$, apply the Picard–Lindelöf theorem `Mathlib/Analysis/ODE/PicardLindelof · IsPicardLindelof` to the time-independent field X `Mathlib/Analysis/ODE/PicardLindelof · IsPicardLindelof.of_time_independent` on the time interval $[-a, a]$, with initial point x and the closed ball of radius Ma about x. Its condition, that the speed bound times the time from 0 to either end of the interval does not exceed the radius, reads $M \cdot a \le Ma$: a solution moving at speed at most M cannot leave the ball before time a. So no shrinking of the time interval is needed. Mathlib provides a solution with a one-sided derivative within $[-a, a]$ `Mathlib/Analysis/ODE/ExistUnique · IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt₀`, which is a genuine derivative in the interior.

Theorem (global existence). Under the same hypotheses, through every point there is an integral curve defined on all of $\mathbb{R}$. `Flow/Global · IsBddLipschitz.exists_isIntegralCurve`

Proof. Choose a solution α_a on $(-a, a)$ for each a. If $|t| < a$ and $|t| < b$, then $\alpha_a(t) = \alpha_b(t)$, by uniqueness on $(-\min(a, b), \min(a, b))$, where both start at x. Put $\gamma(t) = \alpha_{|t|+1}(t)$. Near any t, γ agrees with $\alpha_{|t|+1}$, because $|s| < |t| + 1$ for s near t. So γ has derivative $X(\gamma(t))$ at t.

Definition (the flow). $\varphi_t(x)$ is the value at time t of an integral curve of X through x at time 0, chosen when one exists; otherwise $\varphi_t(x) = x$, a junk value `Flow/Global · flow`. For a bounded Lipschitz field the curve exists and is unique, so the choice is immaterial.

Theorem (the flow, without smoothness). Let E be complete and X bounded and Lipschitz.

- $\varphi_0 = \mathrm{id}$, and $t \mapsto \varphi_t(x)$ is an integral curve `Flow/Global · IsBddLipschitz.flow_zero` `Flow/Global · IsBddLipschitz.isIntegralCurve_flow` `Flow/Global · IsBddLipschitz.hasDerivAt_flow`.
- Every integral curve is a flow line: $\gamma(t) = \varphi_{t-t_0}(\gamma(t_0))$ `Flow/Global · IsBddLipschitz.eq_flow`. Both sides are integral curves that agree at t_0.
- The group law: $\varphi_{s+t} = \varphi_s \circ \varphi_t$ `Flow/Global · IsBddLipschitz.flow_add` `Flow/Global · IsBddLipschitz.flow_comp`. Apply the previous item to the flow line through x, read from time t. In particular φ_{-t} inverts φ_t `Flow/Global · IsBddLipschitz.flow_neg_flow` `Flow/Global · IsBddLipschitz.flow_flow_neg`, and $\varphi_{nt} = (\varphi_t)^n$ `Flow/Global · IsBddLipschitz.flow_nat_mul`.
- Rescaling: the flow of cX at time t is the flow of X at time ct `Flow/Global · IsBddLipschitz.flow_smul`, because $s \mapsto \varphi_{cs}(x)$ is an integral curve of cX through x.
- Zeros are fixed: if $X(x) = 0$, the constant curve is an integral curve, so $\varphi_t(x) = x$ `Flow/Global · IsBddLipschitz.flow_of_eq_zero`. This gives Milnor's $\varphi_t(q) = q$ for q outside the support `Flow/Global · IsBddLipschitz.flow_of_notMem_tsupport`.
- Lipschitz dependence: $d(\varphi_t x, \varphi_t y) \le d(x, y)\, e^{K|t|}$ `Flow/Global · IsBddLipschitz.dist_flow_le` `Flow/Global · IsBddLipschitz.lipschitzWith_flow`. This is Grönwall's inequality `Flow/Global · dist_le_of_hasDerivAt` `Mathlib/Analysis/ODE/Gronwall · dist_le_of_trajectories_ODE`. For negative t it is applied to the curves $s \mapsto \varphi_{-s}(x)$, integral curves of $-X$, which is also K-Lipschitz.

- In time, flow lines are M-Lipschitz, by the mean value inequality `Flow/Global · IsBddLipschitz.dist_flow_flow_le`.
- Joint continuity: $(t, x) \mapsto \varphi_t(x)$ is continuous `Flow/Global · IsBddLipschitz.continuous_flow`. Within distance 1 of (t_0, x_0), for the maximum distance on $\mathbb{R} \times E$, the two previous items give $d(\varphi_t x, \varphi_{t_0} x_0) \le (e^{K(|t_0|+1)} + M)\, d((t, x), (t_0, x_0))$. A locally Lipschitz map is continuous `Mathlib/Topology/Algebra/MetricSpace/Lipschitz · continuousAt_of_locally_lipschitz`. Each φ_t is a homeomorphism with inverse φ_{-t} `Flow/Global · IsBddLipschitz.flowHomeomorph`.

Theorem (invariant closed sets). Let $S \subseteq E$ be closed, and suppose that through every point $y \in S$ there is a solution of $\gamma' = X(\gamma)$ on some $(-\varepsilon, \varepsilon)$ that starts at y and stays in S. Then $\varphi_t(S) \subseteq S$ for every t. `Flow/Global · IsBddLipschitz.flow_mem`

Proof. By uniqueness on $(-\varepsilon, \varepsilon)$, the flow line through y is the given curve for small times, so it stays in S for small $|s|$. Then, for $x \in S$, the set $T = \{t : \varphi_t(x) \in S\}$ is closed, since S is closed and $t \mapsto \varphi_t(x)$ is continuous. It is open: if $t \in T$, then $\varphi_u(x) = \varphi_{u-t}(\varphi_t(x)) \in S$ for u near t, by the group law. It contains 0. A non-empty clopen subset of $\mathbb{R}$ is $\mathbb{R}$ `Flow/Global · IsBddLipschitz.flow_mem_of_eventually` `Mathlib/Topology/Connected/Clopen · IsClopen.eq_univ`.

This is not in Milnor. Part II uses it to keep the flow of a field tangent to a submanifold $N \subseteq \mathbb{R}^K$ inside N.

Where the formal proof departs from Milnor. There is no iteration of $\varphi_{\varepsilon_0/2}$. Milnor's definition $\varphi_t = (\varphi_{\varepsilon_0/2})^k \circ \varphi_r$ depends on a decomposition $t = k(\varepsilon_0/2) + r$ that is not unique ($\varepsilon_0/4 = 0 \cdot \varepsilon_0/2 + \varepsilon_0/4 = 1 \cdot \varepsilon_0/2 - \varepsilon_0/4$), and his remark that well-definedness is not difficult to verify (p. 11) passes over that. A field bounded by M has solutions on intervals of every length, so the flow is defined directly. Compactness of the support is used only to make X bounded and globally Lipschitz. Milnor's remark on p. 11 that the hypothesis cannot be omitted (d/dt on $(0, 1)$) shows that some such global condition is needed; the remark itself is not formalized.

I.19 The Nemytskii operator

Smoothness of the flow is where Milnor's proof of Lemma 2.4 hides real work, and Mathlib proves only Lipschitz dependence on the initial point. The formal proof writes the flow line through x on $[0, 1]$ as the solution of the integral equation $\gamma(s) = x + \int_0^s X(\gamma(r))\, dr$ in the Banach space $C([0, 1], E)$ of continuous curves on Mathlib's unit interval `Mathlib/Topology/UnitInterval · unitInterval`, with the supremum norm. It then applies the inverse function theorem there. The only non-linear ingredient of the equation is the *Nemytskii* (or superposition) operator $\gamma \mapsto Y \circ \gamma$.

Definitions. For a continuous $Y : E \to F$, $N_Y(\gamma) = Y \circ \gamma$ `Flow/Nemytskii · nem` `Flow/Nemytskii · nem_apply`. It is continuous `Flow/Nemytskii · continuous_nem` `Mathlib/Topology/CompactOpen · ContinuousMap.continuous_postcomp`, K-Lipschitz if Y is `Flow/Nemytskii · lipschitzWith_nem`, and bounded by any bound of Y `Flow/Nemytskii · norm_nem_le`. For a continuous curve A of operators $E \to F$, write $\mathcal{M}_A$ for pointwise application, $\mathcal{M}_A \eta = (s \mapsto A(s)\, \eta(s))$. The map $(A, \eta) \mapsto \mathcal{M}_A \eta$ is a continuous bilinear map of norm at most 1 `Flow/Nemytskii · pmul` `Flow/Nemytskii · norm_pmul _le` `Flow/Nemytskii · pmul` `Flow/Nemytskii · pmul_apply` `Flow/Nemytskii · norm_pmul_le`. It is therefore smooth `Flow/Nemytskii · contDiff_pmul`. For Y of class C^1, dY is a continuous map from E to the operators `Flow/Nemytskii · dmap` `Flow/Nemytskii · dmap_apply`.

Theorem (the derivative of N_Y). If Y is C^1, then N_Y is differentiable at every γ, with derivative $\mathcal{M}_{N_{dY}(\gamma)} : \eta \mapsto (s \mapsto dY_{\gamma(s)}\, \eta(s))$. `Flow/Nemytskii · hasFDerivAt_nem`

The proof as formalized. By the definition of the derivative `Mathlib/Analysis/Calculus/FDeriv/Basic · hasFDerivAt_iff_isLittleO_nhds_zero`, it must be shown that for every $\varepsilon > 0$ there is $\delta > 0$

such that $\|\eta\| < \delta$ implies

$$\sup_s \left\| Y(\gamma(s) + \eta(s)) - Y(\gamma(s)) - dY_{\gamma(s)}\, \eta(s) \right\| \le \varepsilon \, \|\eta\|.$$

1. dY is continuous and $\gamma([0,1])$ is compact, so by the Heine–Cantor theorem in its form near a compact set `Mathlib/Topology/UniformSpace/HeineCantor · IsCompact.uniformContinuousAt_of_continuousAt` there is $\delta > 0$ with $\|dY_z - dY_{\gamma(s)}\| \le \varepsilon$ whenever $\|z - \gamma(s)\| < \delta$, for every s at once.
2. Let $\|\eta\| < \delta$ and fix s. The point $\gamma(s) + \eta(s)$ lies in the ball of radius δ about $\gamma(s)$. On that convex ball the derivative of $z \mapsto Y(z) - dY_{\gamma(s)} z$ has norm at most ε. The mean value inequality `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasFDerivWithin_le'` bounds the expression above by $\varepsilon\|\eta(s)\| \le \varepsilon\|\eta\|$.

The estimate is uniform in s, which is what the supremum norm demands. The operator norm of the derivative is at most $\sup \|dY\|$ `Flow/Nemytskii · norm_pmul_nem_dmap_le`.

Theorem (smoothness of N_Y). If Y is C^n, $n \in \mathbb{N} \cup \{\infty\}$, then N_Y is C^n. `Flow/Nemytskii · contDiff_nem_nat` `Flow/Nemytskii · contDiff_nem`

The proof as formalized. By induction on finite n, for all targets F of the size of E at once, since dY takes values in the operators $E \to F$. For $n = 0$ this is continuity. For $n+1$, the derivative of N_Y is $\gamma \mapsto \mathcal{M}_{N_{dY}(\gamma)}$. Here N_{dY} is C^n by the induction hypothesis, dY being C^n, and $A \mapsto \mathcal{M}_A$ is continuous linear. A map is C^{n+1} when it has a derivative that is C^n `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiff_succ_iff_hasFDerivAt`. The case $n = \infty$ follows from all finite cases.

Neither theorem is in Mathlib. Milnor's reference proves smooth dependence by another route, the variational equation.

I.20 The Volterra operator, and the inverse function theorem in $C([0,1], E)$

Definitions. The primitive $P\gamma(s) = \int_0^s \gamma(r)\, dr$ `Flow/Volterra · primFun` `Flow/Volterra · primFun_apply` uses γ extended to $\mathbb{R}$ by composing with the projection of $\mathbb{R}$ onto $[0,1]$, and is continuous in s `Mathlib/MeasureTheory/Integral/DominatedConvergence · intervalIntegral.continuous_primitive`. Since $s \le 1$, $\|P\gamma\| \le \|\gamma\|$ `Flow/Volterra · norm_primFun_le`. P is linear `Flow/Volterra · prim`□, hence a continuous linear operator of norm at most 1 `Flow/Volterra · prim` `Flow/Volterra · prim_apply` `Flow/Volterra · norm_prim_le` `Flow/Volterra · norm_prim_apply_le`, and smooth `Flow/Volterra · contDiff_prim`. The *Volterra operator* of a continuous $Y : E \to E$ is

$$\Psi_Y(\gamma) = \gamma - P(Y \circ \gamma)$$

`Flow/Volterra · volt` `Flow/Volterra · volt_apply`. A curve γ solves $\gamma' = Y(\gamma)$ on $[0,1]$ with $\gamma(0) = x$ exactly when $\Psi_Y(\gamma)$ is the constant curve x; one direction is checked in §I.21.

Theorem (Ψ is a homeomorphism). Let E be complete and Y Lipschitz with constant $c < 1$. Then Ψ_Y is a homeomorphism of $C([0,1], E)$, and its inverse is $(1-c)^{-1}$-Lipschitz. `Flow/Volterra · voltHomeomorph` `Flow/Volterra · coe_voltHomeomorph`

The proof as formalized.

1. $\gamma \mapsto P(Y \circ \gamma)$ is c-Lipschitz, since P has norm at most 1 and N_Y is c-Lipschitz `Flow/Volterra · dist_prim_nem_le`.
2. *The key estimate.* $\|\gamma - \delta\| \le (1-c)^{-1} \|\Psi\gamma - \Psi\delta\|$ `Flow/Volterra · dist_le_dist_volt`. Indeed $\gamma - \delta = (\Psi\gamma - \Psi\delta) + (P(Y\gamma) - P(Y\delta))$, so $\|\gamma - \delta\| \le \|\Psi\gamma - \Psi\delta\| + c\,\|\gamma - \delta\|$. Hence Ψ is injective `Flow/Volterra · volt_injective`.

3. *Surjectivity.* For fixed ψ, the map $\gamma \mapsto \psi + P(Y \circ \gamma)$ is a contraction with constant c Flow/Volterra · contractingWith_volt. $C([0,1],E)$ is complete, so the contraction principle gives a fixed point Mathlib/Topology/MetricSpace/Contracting · ContractingWith.fixedPoint Mathlib/Topology/MetricSpace/Contracting · ContractingWith.fixedPoint_isFixedPt, and this fixed point defines $\Psi^{-1}(\psi)$ Flow/Volterra · voltInv. Then $\Psi(\Psi^{-1}\psi) = \psi$ Flow/Volterra · volt_voltInv, and $\Psi^{-1}(\Psi\gamma) = \gamma$ by injectivity Flow/Volterra · voltInv_volt.
4. Ψ is continuous. By the key estimate, its inverse is $(1-c)^{-1}$-Lipschitz.

Theorem (Ψ^{-1} is as smooth as X). Let E be complete and $X : E \to E$ of class C^n, $1 \le n \le \infty$, and Lipschitz with constant $c < 1$. Then Ψ_X^{-1} is C^n. Flow/Volterra · contDiff_voltHomeomorph_symm

The proof as formalized.

1. Ψ_X is C^n, since N_X is (§I.19) and P is continuous linear Flow/Volterra · contDiff_volt.
2. Its derivative at γ is $1 - P\,\mathcal{M}_{N_{dX}(\gamma)}$ Flow/Volterra · hasFDerivAt_volt.
3. A c-Lipschitz map has $\|dX\| \le c$ everywhere Mathlib/Analysis/Calculus/FDeriv/Basic · norm_fderiv_le_of_lipschitz. So the operator $P\,\mathcal{M}_{N_{dX}(\gamma)}$ has norm at most $1 \cdot c < 1$ Flow/Volterra · norm_prim_comp_pmul_lt, and the Neumann series Mathlib/Analysis/SpecificLimits/Normed · Units.oneSub makes the derivative an invertible operator Flow/Volterra · voltDeriv Flow/Volterra · coe_voltDeriv Mathlib/Topology/Algebra/Module/Equiv · ContinuousLinearEquiv.ofUnit.
4. A homeomorphism that is C^n with an invertible derivative at every point has a C^n inverse Mathlib/Analysis/Calculus/ContDiff/Operations · Homeomorph.contDiff_symm. This is the easy half of the inverse function theorem, available because Ψ is already known to be a homeomorphism.

Why $c < 1$. With $c < 1$ the operator $P\,\mathcal{M}_{N_{dX}(\gamma)}$ is a contraction, so no estimate of its iterates (of the classical form $c^n/n!$) is needed. It costs nothing, because the flow at time t is the time-one map of $(t/N)X$ iterated N times (§I.21).

I.21 Smooth dependence on the initial point, and the joint smoothness

Let E be complete; Mathlib infers completeness from finite dimension, and the applications have $E = \mathbb{R}^K$.

The flow line solves the integral equation. For X bounded and Lipschitz, hence continuous, the flow line through x, restricted to $[0,1]$, is a point of $C([0,1],E)$ Flow/Smooth · IsBddLipschitz.flowCurve Flow/Smooth · IsBddLipschitz.flowCurve_apply, and Ψ_X maps it to the constant curve x Flow/Smooth · IsBddLipschitz.volt_flowCurve. On $[0,s] \subseteq [0,1]$ the projection does nothing, and the fundamental theorem of calculus Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus · intervalIntegral.integral_eq_sub_of_hasDerivAt gives $\int_0^s X(\varphi_r x)\,dr = \varphi_s(x) - \varphi_0(x)$, with $\varphi_0(x) = x$. Constant curves form a continuous linear map $x \mapsto \underline{x}$ from E to $C([0,1],E)$ Flow/Smooth · constCLM Flow/Smooth · constCLM_apply.

Step 1 (the time-one map of a slow field). If X is C^n with $n \ge 1$, bounded and Lipschitz with constant $c < 1$, then φ_1 is C^n. Flow/Smooth · contDiff_flow_one_of_lipschitz

Proof. Ψ_X maps the flow line through x to $\underline{x}$, so the flow line is $\Psi_X^{-1}(\underline{x})$ and $\varphi_1(x) = \Psi_X^{-1}(\underline{x})(1)$. This composes the continuous linear map $x \mapsto \underline{x}$, the C^n map Ψ_X^{-1} (§I.20), and evaluation at 1, which is continuous linear Mathlib/Topology/ContinuousMap/Algebra · ContinuousMap.evalCLM.

Step 2 (all times; V1.14). If X is C^n, $1 \le n \le \infty$, and vanishes outside a compact set, then each φ_t is C^n. Flow/Smooth · contDiff_flow

Proof.

1. X is bounded and Lipschitz with some constant K (§I.18).
2. Choose an integer $N > |t|K$ and put $\tau = t/N$. The field τX is C^n, bounded and Lipschitz with constant $|\tau|K < 1$.

3. By Step 1, its time-one map is C^n. By rescaling (§I.18) that map is φ_τ.
4. $\varphi_t = (\varphi_\tau)^N$ `Flow/Global · IsBddLipschitz.flow_nat_mul`, and an iterate of a C^n map is C^n `Flow/Smooth · contDiff_iterate`.

Step 3 (joint smoothness). If X is C^n, $1 \le n \le \infty$, and vanishes outside a compact set, then $(t, x) \mapsto \varphi_t(x)$ is C^n on $\mathbb{R} \times E$. `Flow/Smooth · contDiff_flow_uncurry`

The idea (the Z-trick). Trade the time variable for a space variable. On $E \times \mathbb{R}$ consider the field $Z(x, \tau) = (\rho(\tau)X(x), 0)$, where $\rho(\tau) = \tau$ for $|\tau| \le R$ and ρ has compact support. Along a flow line of Z the coordinate τ is constant, and the E-coordinate follows the flow of X at speed $\rho(\tau)$. So the time-one map of Z is $(x, \tau) \mapsto (\varphi_{\rho(\tau)}(x), \tau)$, which for $|\tau| \le R$ is $(\varphi_\tau(x), \tau)$. Step 2, at the single time 1, now gives smoothness in (x, τ) jointly.

The proof as formalized, at a point (t_0, x_0).

1. Let $R = |t_0| + 1$. A smooth bump function χ `Mathlib/Analysis/Calculus/BumpFunction/Basic · ContDiffBump` equals 1 on $[-R, R]$ `Mathlib/Analysis/Calculus/BumpFunction/Basic · ContDiffBump.one_of_mem_closedBall` and vanishes outside $[-R - 1, R + 1]$. Then $\rho(\tau) = \chi(\tau)\,\tau$ is smooth with compact support, and $\rho(\tau) = \tau$ for $|\tau| \le R$.
2. $Z(x, \tau) = (\rho(\tau)X(x), 0)$ is C^n, and it vanishes outside the product of the supports of X and ρ, a compact set. So it is bounded and Lipschitz.
3. For each (x, τ), the curve $s \mapsto (\varphi_{\rho(\tau)s}(x), \tau)$ is an integral curve of Z through (x, τ). Its derivative is $(\rho(\tau)X(\varphi_{\rho(\tau)s}(x)), 0)$. Since integral curves are flow lines (§I.18), the time-one map of Z is $(x, \tau) \mapsto (\varphi_{\rho(\tau)}(x), \tau)$.
4. On the open set $|t| < R$, which contains (t_0, x_0), $\rho(t) = t$, so $\varphi_t(x)$ is the first component of $\varphi_1^Z(x, t)$.
5. φ_1^Z is C^n by Step 2 applied to Z on $E \times \mathbb{R}$. So $(t, x) \mapsto \varphi_t(x)$ agrees near (t_0, x_0) with a C^n map, and is C^n there.

Diffeomorphisms. Each φ_t is a C^n diffeomorphism of E with inverse φ_{-t} `Flow/Smooth · flowDiffeomorph` `Flow/Smooth · coe_flowDiffeomorph` `Flow/Smooth · coe_flowDiffeomorph_symm` `Mathlib/Geometry/Manifold/Diffeomorph · Diffeomorph`, by Step 2 and the group law.

What Milnor leaves implicit. The smooth dependence on the initial condition is a theorem, not a remark. Smoothness of $(t, q) \mapsto \varphi_t(q)$ for large t needs the joint smoothness of the local flow. Milnor's iteration $\varphi_t = (\varphi_{\varepsilon_0/2})^k \circ \varphi_r$, with k and r depending on t, gives it only after one checks that the decomposition of t can be chosen locally constant in t. Step 3 avoids the issue, and Step 1 is needed only for a fixed time. The argument gives C^n dependence for a C^n field, $1 \le n \le \infty$.

I.22 Lemma 2.4

Definitions (Milnor pp. 9–10, on $M = E$; V1.13). A *one-parameter group of diffeomorphisms* of E is a map $\varphi : \mathbb{R} \to (E \to E)$ such that $(t, x) \mapsto \varphi_t(x)$ is C^∞, each φ_t is a C^∞ diffeomorphism of E, and $\varphi_{s+t} = \varphi_s \circ \varphi_t$ `Flow/Smooth · IsOneParameterGroup`. A vector field X *generates* φ if for every x the curve $h \mapsto \varphi_h(x)$ has derivative $X(x)$ at $h = 0$ `Flow/Smooth · Generates`. On E a tangent vector is a vector, so Milnor's $X_q(f) = \lim_{h \to 0} (f(\varphi_h(q)) - f(q))/h$ becomes the derivative of $h \mapsto \varphi_h(q)$.

Two facts Milnor leaves implicit:

- $\varphi_0 = \mathrm{id}$ `Flow/Smooth · IsOneParameterGroup.zero`. The group law gives $\varphi_0(\varphi_0(x)) = \varphi_0(x)$, and φ_0 is injective, being a diffeomorphism. With the group law, injectivity of φ_0 alone would suffice.
- If X generates φ, every orbit $t \mapsto \varphi_t(x)$ is an integral curve of X `Flow/Smooth · IsOneParameterGroup.isIntegralCurve`. Write $\varphi_s(x) = \varphi_{s-t}(\varphi_t(x))$ and differentiate at $s = t$ with the generating property at the point $\varphi_t(x)$. This is Milnor's computation on p. 10, which uses $\varphi_0 = \mathrm{id}$.

Lemma 2.4 (Milnor, p. 10; V1.15). Let E be a complete real normed space and $X : E \to E$ a C^∞ vector field that vanishes outside a compact set. Then X generates exactly one one-parameter group of diffeomorphisms of E, namely its flow. `Flow/Smooth · lemma_2_4`

The proof as formalized.

1. *Existence.* The flow φ of X is jointly C^∞ (§I.21, Step 3) `Flow/Smooth · contDiff_flow_uncurry`. Each φ_t is a C^∞ diffeomorphism `Flow/Smooth · flowDiffeomorph`. The group law holds `Flow/Global · IsBddLipschitz.flow_comp`. X generates φ, because $h \mapsto \varphi_h(x)$ has derivative $X(\varphi_0(x)) = X(x)$ at $h = 0$ `Flow/Global · IsBddLipschitz.hasDerivAt_flow`.
2. *Uniqueness.* Let ψ be a one-parameter group generated by X. Its orbits are integral curves of X, and $\psi_0 = \mathrm{id}$. Since every integral curve is a flow line (§I.18) `Flow/Global · IsBddLipschitz.eq_flow`, $\psi_t(x) = \varphi_{t-0}(\psi_0(x)) = \varphi_t(x)$.

Deviation (narrower), and what is implicit. Milnor's lemma is about a manifold M; the formal one is about $M = E$. If E is infinite-dimensional the hypothesis forces $X = 0$: the support of a continuous field is open, and a non-empty open set in an infinite-dimensional normed space has non-compact closure. So the content is the case $E = \mathbb{R}^n$. For submanifolds $N \subseteq \mathbb{R}^K$, Part II uses instead the ambient flow of a field tangent to N, kept inside N by the invariance theorem of §I.18. Milnor's statement asserts uniqueness, and his proof leaves it implicit. It holds among all one-parameter groups generated by X, because uniqueness of integral curves is global in time.

I.23 What the formalization taught us

- **The Hessian is the second derivative.** Its symmetry holds at every point where f is C^2, and criticality is used exactly once, to kill the first-order term $df(d^2\psi)$ of the second-order chain rule, which is what makes the Hessian independent of the coordinates `Chapter1/Nondegenerate · hessian_comp`. Milnor's definition through extending vector fields is replaced by a definition in one chart plus this theorem `Chapter1/Manifold · IsSmoothCoord.hessian_comp_symm`.
- **The index is Mathlib's negative index of inertia.** Milnor's definition, the maximal dimension of a subspace on which the form is negative definite, is a theorem `Chapter1/Nondegenerate · index_isGreatest`, and his argument for the index of the normal form is the uniqueness half of Sylvester's law `Chapter1/MorseLemma · index_diagForm_morseSign`.
- **Hadamard's lemma needs differentiation under the integral sign on an open set,** which Milnor does not argue. It is proved by domination on a tube around the segment, with no local compactness `Chapter1/Hadamard · contDiffOn_intervalIntegral`. The second-order form comes from Taylor's formula, which yields the symmetric coefficient and its value $\frac{1}{2}d^2f_0$ that Milnor asserts `Chapter1/Hadamard · exists_hadamard_two_on`.
- **The lemma of Morse in one step.** The inverse function theorem in the space of linear maps of E, applied to $T \mapsto B_0(T\cdot, T\cdot) + B_0(T\cdot, \cdot) - B_0(\cdot, T\cdot)$, diagonalizes a smooth family of quadratic forms at once `Chapter1/SquareRoot · exists_sqrt_family`. The gap in Milnor's induction, the unjustified linear change making $H_{rr}(0) \neq 0$, does not arise `Chapter1/MorseLemma · exists_morse_chart`.
- **That the normal form forces the index uses the independence of the coordinates,** which Milnor takes for granted `Chapter1/MorseLemma · index_eq_of_morse_chart`. On a manifold the converse needs no hypothesis on f at all `Chapter1/ManifoldMorse · isNondegenerateCriticalPointM_of_morse_chart`.
- **Corollary 2.3 needs only C^2 at the point and no lemma of Morse** `Chapter1/Nondegenerate · eventually_not_isCriticalPoint`. Its global consequence, finitely many critical points in every compact set, which Milnor uses silently, needs no Hausdorff hypothesis `Chapter1/ManifoldMorse · finite_inter_setOf_isCriticalPointM`.
- **One possible misprint:** in example (d) the critical set of x^2, which the text calls the x axis, is the y axis `Chapter1/Examples · Examples.isCriticalPoint_sqFst_iff`.
- **On a submanifold the Hessian of $F \circ x$ is not the ambient Hessian restricted.** The first-order term of the chain rule survives, and for L_p it is the second fundamental form `Chapter1/`

`ManifoldSubmanifold · hessian_normSq_sub_comp`.

- **Smooth dependence on initial conditions is a theorem that Mathlib lacked.** It is proved by the inverse function theorem in $C([0,1],E)$ `Flow/Volterra · contDiff_voltHomeomorph_symm`, with joint smoothness from the field $(\rho(\tau)X(x),0)$ `Flow/Smooth · contDiff_flow_uncurry`. Milnor's iteration, whose well-definedness he leaves unchecked, is replaced by Picard–Lindelöf on intervals of any length `Flow/Global · IsBddLipschitz.exists_isIntegralCurve`. The uniqueness in Lemma 2.4 is global in time `Flow/Smooth · lemma_2_4`.

Part II. Sublevel sets and attached cells (Milnor §3)

Milnor's §3 watches the sublevel sets $M^a = f^{-1}(-\infty, a]$ of a smooth function grow. Across an interval without critical values nothing happens: M^a and M^b are diffeomorphic, and the larger set deforms onto the smaller one (Theorem 3.1). Across a non-degenerate critical point of index λ a λ-cell is attached (Theorem 3.2, with Remark 3.3 for several critical points on one level and Remark 3.4 for the critical level itself). With two lemmas on attaching cells (Lemmas 3.6 and 3.7), an induction over the critical values then shows that a manifold carrying a function with non-degenerate critical points has the homotopy type of a complex with one λ-cell for each critical point of index λ (Theorem 3.5). This part retells the formal proofs of all of these, and of the counterexample behind the Remark on p. 13.

The formal proofs work in three settings. Theorems 3.1 and 3.2 and Remark 3.4 are proved on a finite-dimensional real inner product space E, which is Milnor's $\mathbb{R}^n$ with a Riemannian metric. Theorems 3.1, 3.2 and 3.5 and Remark 3.3 are proved for $f = F|N$, the restriction of a smooth function F on a finite-dimensional real normed space V to a slice submanifold $N \subseteq V$ in the sense of the companion project (§0.1). These are Milnor's manifolds in $\mathbb{R}^K$, with the extra hypothesis that f extends to a smooth function on V. Theorem 3.1 is moreover transferred to abstract compact Hausdorff manifolds through Whitney's embedding; Part III tells that transfer. The formal route departs from Milnor's in four places, each explained where it occurs.

1. Everything in the proof of Theorem 3.1 after the computation $\frac{d}{dt} f(\varphi_t(q)) = 1$ is proved once, for an abstract *slab flow* on a topological space. Milnor's "clearly" becomes a lemma about real functions of one variable which have unit speed while their values lie in a slab.
2. On a submanifold there is no Riemannian metric and no intrinsic flow. Milnor's field ρ grad f is replaced by an ambient field built from slice charts, tangent to N, whose flow preserves N.
3. All computations of Theorem 3.2 in Morse coordinates are done once, on a product $X \times Y$ of inner product spaces, and then transported: to E, to a coordinate patch of an arbitrary Hausdorff space, and to submanifolds. Milnor's function μ is constructed, his three-case deformation r_t is a single formula, and the identification of the subspace $M^{c-\epsilon} \cup e^\lambda$ with $M^{c-\epsilon}$ with a cell attached is a lemma (Lemma D).
4. Mathlib has no cellular approximation theorem. In Theorem 3.5 "CW-complex" is therefore replaced by an iterated adjunction space with exactly the right number of cells in each dimension, and only compact N is treated.

Results carry the labels V2.1–V2.15 of the verified edition (`Exposition/Verification.md`), so that one can move between the two texts.

Map.

- `Chapter2/Crossing`: sublevel sets, strong deformation retracts, the one-crossing lemma, slab flows.
- `Chapter2/Regular`: the cut-off ρ, the field $\rho\nabla f$, Theorem 3.1 on E.
- `Chapter2/Submanifold`: tangent fields from slice charts, invariance of N, Theorem 3.1 on N.

- `Chapter2/Remark`: the punctured plane of the Remark on p. 13.
- `Chapter2/CellModel`: μ, F and Assertions 1, 2 and 4 in Morse coordinates.
- `Chapter2/CellAttach`: Theorem 3.2 on E.
- `Chapter2/CellAdjunction`: Lemma D, and Theorem 3.2 with an attaching space.
- `Chapter2/CriticalLevel`: Remark 3.4.
- `Chapter2/CellPatch`: Theorem 3.2 in a coordinate patch of a Hausdorff space.
- `Chapter2/CellAttachSubmanifold`: Theorem 3.2 on N.
- `Chapter2/SeveralCritical`: Remark 3.3 on N.
- `Chapter2/Attaching`: Lemmas 3.6 and 3.7.
- `Chapter2/CWType`: iterated adjunction spaces and Theorem 3.5.

II.1 Sublevel sets and strong deformation retracts

For a real function F on a set T and a real number c, the **sublevel set** is $T^c = \{x \in T : F(x) \le c\}$, Milnor's M^c `Chapter2/Crossing · sublevel` (§0.4). It increases with c `Chapter2/Crossing · sublevel_mono`.

A subset A of a topological space X is a **strong deformation retract** of X if there are a continuous map $r : X \to X$ with values in A and a homotopy $H : [0,1] \times X \to X$ from the identity of X to r such that $H(t,a) = a$ for every t and every $a \in A$ `Chapter2/Crossing · IsStrongDeformationRetract`. The homotopy is Mathlib's homotopy relative to A `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel`. In Milnor's notation $r_t = H(t,\cdot)$: r_0 is the identity, $r_1 = r$ maps X into A, and every r_t fixes A, so that r is a retraction onto A. Milnor uses "deformation retract" without a definition. Every deformation he constructs fixes the smaller set at all times, so every formal statement asserts this strong form, which implies the weaker ones.

The sets that occur are subsets of one ambient space, as in $M^a \subseteq M^b \subseteq M$. For $A \subseteq B$, "A is a strong deformation retract of B" then means the property above for the space B, with the subspace topology, and the subset of B formed by the points of A. All statements below are of this kind.

Lemma (V2.1). If $A \subseteq B$ and A is a strong deformation retract of B, the inclusion $A \to B$ is a homotopy equivalence `Chapter2/Crossing · IsStrongDeformationRetract.exists_homotopyEquiv`.

Proof. This is the last clause of Milnor's statement of Theorem 3.1 (p. 12). Let $i : A \to B$ be the inclusion and $j : B \to A$ the map $x \mapsto r(x)$, continuous because r is continuous with values in A. Then $j \circ i$ is the identity of A: the homotopy fixes A at all times, in particular $r(a) = a$ at time 1. And $i \circ j = r$, which is homotopic to the identity of B by H run backwards. So i and j are mutually inverse up to homotopy, which is a homotopy equivalence in Mathlib's sense `Mathlib/Topology/Homotopy/Equiv · ContinuousMap.HomotopyEquiv`. The homotopy inverse is Milnor's r_1. □

Two closure properties are used throughout. Milnor uses both without stating them; the exposition calls them Lemma A.

Lemma A(i). If $A \subseteq B \subseteq C$, A is a strong deformation retract of B, and B is one of C, then A is one of C `Chapter2/CellAttach · IsStrongDeformationRetract.trans`.

Proof. Let G deform C onto B, with end map s, and let H deform B onto A, with end map r. A homotopy relative to B is also a homotopy relative to the smaller set A `Chapter2/CellAttach · homotopyRelMono`. First follow G from the identity of C to s; then follow $K(t,x) = H(t, s(x))$, which runs from s to $r \circ s$. Both halves fix A: G fixes $B \supseteq A$, and for $x \in A$ one has $s(x) = x$ and so $K(t,x) = H(t,x) = x$. The concatenation is a homotopy relative to A from the identity of C to $r \circ s$, which takes values in A. □

Lemma A(ii). If $A \subseteq B \subseteq C$ and both A and B are strong deformation retracts of C, then A is a strong deformation retract of B `Chapter2/CriticalLevel · IsStrongDeformationRetract.of_retract`.

Proof. Of the second hypothesis only the retraction $\kappa : C \to B$ is used, which fixes B pointwise. Let H deform C onto A, with end map r, and put $K(t,b) = \kappa(H(t,b))$ for $b \in B$. At $t = 0$ this is $\kappa(b) = b$.

For $a \in A$ it is $\kappa(a) = a$ at every time. At $t = 1$ it is $\kappa(r(b))$, and since $r(b) \in A \subseteq B$ is fixed by κ, the end map takes values in A. So K deforms B onto A relative to A. □

This is what Milnor's "we see easily" in Remark 3.4 needs (§II.12).

II.2 The one-crossing lemma

Milnor's proof of Theorem 3.1 (p. 13) computes that along a trajectory $t \mapsto \varphi_t(q)$ of his field $X = \rho \operatorname{grad} f$ the derivative of $f(\varphi_t(q))$ is $+1$ whenever $\varphi_t(q)$ lies in $f^{-1}[a, b]$. He concludes that $t \mapsto f(\varphi_t(q))$ is linear with slope one while its value lies between a and b, and reads off the rest. Outside the slab nothing is known about the derivative, and Milnor does not even require $\rho \geq 0$. That a trajectory which is in the slab stays in it, moving linearly, until it leaves through the top, and that it never crosses a level of the slab downwards, needs a continuity argument. The formal proof isolates it as a statement about one real function.

Definition. A function $g : \mathbb{R} \to \mathbb{R}$ **has unit speed in the slab** $[a, b]$ if it is continuous and has derivative 1 at every t with $a \leq g(t) \leq b$ `Chapter2/Crossing · UnitSpeedInSlab`. Nothing is assumed about g while its value lies outside $[a, b]$.

The lemmas below are proved by *continuous induction*: to show that a set which is closed in $[s, t]$ contains all of $[s, t]$, it suffices that it contains s and contains a right neighbourhood of each of its points in $[s, t)$ `Mathlib/Topology/Order/IntermediateValue · IsClosed.Icc_subset_of_forall_mem_nhdsWithin`.

1. *A positive derivative pushes upward.* If $g'(t) > 0$, then $g(s) > g(t)$ for all s slightly larger than t `Chapter2/Crossing · eventually_lt_of_hasDerivAt_pos`. The difference quotients $(g(s) - g(t))/(s - t)$ tend to $g'(t) > 0$, so they are positive for s near t, and $s - t > 0$.
2. *Time reversal.* If g has unit speed in $[a, b]$, then $t \mapsto -g(-t)$ has unit speed in $[-b, -a]$ `Chapter2/Crossing · UnitSpeedInSlab.reverse`. Indeed $-g(-t) \in [-b, -a]$ exactly when $g(-t) \in [a, b]$, and then the chain rule gives the derivative $-(1 \cdot (-1)) = 1$. So every statement about the future has a mirror statement about the past.
3. *No downward crossing.* If $c \in [a, b]$, $s \leq t$ and $c \leq g(s)$, then $c \leq g(t)$ `Chapter2/Crossing · UnitSpeedInSlab.le_of_le`. The set $\{r : c \leq g(r)\}$ is closed and contains s. At one of its points r, either $c < g(r)$, and then $c < g$ near r by continuity, or $g(r) = c$; in the second case $g(r)$ lies in the slab, so $g'(r) = 1 > 0$ and $g > c$ just to the right of r by step 1. Continuous induction gives all of $[s, t]$. In mirror form: if $g(t) \leq c$, then $g(s) \leq c$ for every $s \leq t$ `Chapter2/Crossing · UnitSpeedInSlab.le_of_le'`, which is step 3 for $-g(-\cdot)$ and the level $-c$.
4. *Linear while in the slab.* If $a \leq g(s)$, $s \leq t$ and $g(s) + (t - s) \leq b$, then $g(t) = g(s) + (t - s)$ `Chapter2/Crossing · UnitSpeedInSlab.eq_add_of_le`. The hypotheses force $a \leq b$. The set of r with $g(r) = g(s) + (r - s)$ is closed and contains s. Let $r \in [s, t)$ belong to it. Then $g(r) = g(s) + (r - s) < g(s) + (t - s) \leq b$, so by continuity there is $\delta > 0$ with $g < b$ on $(r - \delta, r + \delta)$. Let $r' \in (r, r + \delta)$. On $[r, r']$ the function stays in the slab: below b by the choice of δ, and at or above a by step 3, because it was at or above the level a at the earlier time s. So $g' = 1$ on $[r, r')$, and the function $x \mapsto g(x) - x$, continuous on $[r, r']$ with right derivative 0 on $[r, r')$, is constant there `Mathlib/Analysis/Calculus/MeanValue · constant_of_has_deriv_right_zero`. Hence $g(r') = g(r) + (r' - r) = g(s) + (r' - s)$, and the set contains $(r, r + \delta)$.

Putting the two directions of time together gives the statement Milnor uses.

Lemma C (one crossing). If g has unit speed in $[a, b]$ and $g(s) \in [a, b]$, then $g(s + \tau) = g(s) + \tau$ for every τ with $a - g(s) \leq \tau \leq b - g(s)$ `Chapter2/Crossing · UnitSpeedInSlab.eq_add`.

Proof. For $\tau \geq 0$ this is step 4 from s to $s + \tau$. For $\tau \leq 0$, apply step 4 to the reversed function $-g(-\cdot)$, which has unit speed in $[-b, -a]$, from the time $-s$ to the time $-(s + \tau)$: the hypothesis $-g(s) \geq -b$ is $g(s) \leq b$, and $-g(s) + (-\tau) \leq -a$ is $\tau \geq a - g(s)$. □

The exposition states Lemma C as a trichotomy: below a before some time α, equal to $a + (t - \alpha)$ on

$[\alpha, \alpha + b - a]$, above b afterwards. The formal proof never needs the trichotomy. It uses steps 3 and 4 and Lemma C directly, as the next section shows.

II.3 Slab flows: the topological half of Theorem 3.1

Definition. Let T be a topological space, $F : T \to \mathbb{R}$, and $(t, x) \mapsto \varphi_t(x)$ a map $\mathbb{R} \times T \to T$. This is a **slab flow** for F and the slab $[a, b]$ if $(t, x) \mapsto \varphi_t(x)$ is continuous, φ_0 is the identity, $\varphi_{s+t} = \varphi_s \circ \varphi_t$, F is continuous, and for all x and t with $F(\varphi_t(x)) \in [a, b]$ the function $s \mapsto F(\varphi_s(x))$ has derivative 1 at t `Chapter2/Crossing · IsSlabFlow`. For Milnor, $T = M$, $F = f$, and φ is the flow of ρ grad f (§II.4). The same abstraction serves for a submanifold that is invariant under an ambient flow (§II.5). Along every trajectory, $t \mapsto F(\varphi_t(x))$ has unit speed in the slab `Chapter2/Crossing · IsSlabFlow.unitSpeed`, and the lemmas of §II.2 apply to it.

Milnor's deformation as one formula. Milnor's $r_t(q)$ is q when $f(q) \le a$ and $\varphi_{t(a-f(q))}(q)$ when $a \le f(q) \le b$. The formal proof writes it as a single formula `Chapter2/Crossing · deformation`,

$$r_t(x) = \varphi_{\,t \cdot \min(0,\ a - F(x))}(x).$$

It is visibly continuous in (t, x), as a composite of continuous maps `Chapter2/Crossing · IsSlabFlow.continuous_deformation`, so no pasting lemma is needed. It agrees with Milnor's two cases: if $F(x) \le a$, the time is 0 and $r_t(x) = \varphi_0(x) = x$ `Chapter2/Crossing · IsSlabFlow.deformation_of_le`; if $F(x) \ge a$, it is $\varphi_{t(a-F(x))}(x)$ `Chapter2/Crossing · deformation_of_ge`. At $t = 0$ it is the identity `Chapter2/Crossing · IsSlabFlow.deformation_zero`.

Four consequences of §II.2 along trajectories carry the whole argument.

- *Linear in the slab.* If $F(x) \in [a, b]$ and $a - F(x) \le t \le b - F(x)$, then $F(\varphi_t(x)) = F(x) + t$ `Chapter2/Crossing · IsSlabFlow.apply_eq_add`. This is Lemma C at the time 0, since $\varphi_0(x) = x$.
- *No downward crossing.* If $c \in [a, b]$, $t \ge 0$ and $F(\varphi_t(x)) \le c$, then $F(x) \le c$ `Chapter2/Crossing · IsSlabFlow.le_of_apply_le`. This is the mirror form of step 3.
- *Rising at most at unit speed.* If $a \le c \le c' \le b$ and $F(x) \le c$, then $F(\varphi_{c'-c}(x)) \le c'$ `Chapter2/Crossing · IsSlabFlow.apply_le_of_le`. If $F(\varphi_{c'-c}(x)) \le c$ there is nothing to prove. Otherwise the continuous function $u \mapsto F(\varphi_u(x))$ is at most c at $u = 0$ and larger than c at $u = c' - c$, so by the intermediate value theorem it equals c at some $u \in [0, c' - c]$ `Mathlib/Topology/Order/IntermediateValue · intermediate_value_Icc`. From the time u on, step 4 applies: the value c is at least a, and $c + (c' - c - u) \le c' \le b$. So at the time $c' - c$ the value is $c + (c' - c - u) = c' - u \le c'$.
- *Backwards.* If $a \le c \le c' \le b$ and $F(y) \le c'$, then $F(\varphi_{c-c'}(y)) \le c$ `Chapter2/Crossing · IsSlabFlow.le_of_apply_le'`. Put $x = \varphi_{c-c'}(y)$, so that $\varphi_{c'-c}(x) = y$ by the group law. If $F(y) \le c$, no downward crossing, for the time $c' - c \ge 0$, gives $F(x) \le c$. If $F(y) > c$, then $F(y) \in (c, c'] \subseteq [a, b]$, and going back from y for the time $F(y) - c$ reaches the level c: $F(\varphi_{c-F(y)}(y)) = c$ by linearity in the slab. Since $\varphi_{c'-F(y)}(x) = \varphi_{c-F(y)}(y)$ and $c' - F(y) \ge 0$, the trajectory of x is at the level c at a later time, and no downward crossing gives $F(x) \le c$.

Theorem 3.1, first half, for a slab flow. If $a \le c \le c' \le b$, then $\varphi_{c'-c}(T^c) = T^{c'}$ `Chapter2/Crossing · IsSlabFlow.image_sublevel`.

Proof. The inclusion $\subseteq$ is "rising at most at unit speed". For $\supseteq$, a point y with $F(y) \le c'$ is the image under $\varphi_{c'-c}$ of $\varphi_{c-c'}(y)$, which lies in T^c by the backwards statement. □

With $c = a$ and $c' = b$ this is Milnor's "clearly" (p. 13): φ_{b-a} carries M^a onto M^b. Both inclusions are needed. A trajectory starting below a must not be above b at the time $b - a$, and a trajectory ending below b must not have started above a. Both rest on the facts that a trajectory crosses the slab at most once, upwards, and at unit speed.

Theorem 3.1, second half, for a slab flow. T^a is a strong deformation retract of T^b `Chapter2/Crossing · IsSlabFlow.isStrongDeformationRetract`, and if $a \le b$ the inclusion $T^a \to T^b$ is a homotopy equivalence `Chapter2/Crossing · IsSlabFlow.exists_homotopyEquiv`.

Proof.

1. *Where r_t goes.* For x in the slab and $0 \le t \le 1$, $F(r_t(x)) = F(x) + t\,(a - F(x))$ `Chapter2/Crossing · IsSlabFlow.apply_deformation`. Indeed $r_t(x) = \varphi_{t(a-F(x))}(x)$, and the time $t(a - F(x))$ lies in $[a - F(x), 0]$, inside the interval where linearity in the slab applies. The value lies between a and $F(x)$.
2. Hence r_t maps T^b into T^b for $0 \le t \le 1$ `Chapter2/Crossing · IsSlabFlow.deformation_mem`, points below a not moving at all, and r_1 maps T^b into T^a `Chapter2/Crossing · IsSlabFlow.deformation_one_mem`, because $F(r_1(x)) = a$ for x in the slab.
3. Milnor's retraction r_1 is a continuous self-map of T^b with values in T^a `Chapter2/Crossing · IsSlabFlow.retraction Chapter2/Crossing · IsSlabFlow.retraction_mem`. The map $(t, x) \mapsto r_t(x)$, $0 \le t \le 1$, is a homotopy from the identity of T^b to r_1, and it fixes T^a at all times because $r_t(x) = x$ when $F(x) \le a$ `Chapter2/Crossing · IsSlabFlow.homotopyRel`.
4. The homotopy equivalence follows by the lemma of §II.1, with homotopy inverse r_1. □

Milnor's hypothesis $a < b$ is not needed: $a \le b$ suffices, and for $a = b$ the statements are trivial.

II.4 Theorem 3.1 in a Euclidean space

Let E be a finite-dimensional real inner product space. A point p is a **critical point** of $f : E \to \mathbb{R}$ if f is differentiable at p with derivative 0 `Chapter1/Nondegenerate · IsCriticalPoint` (§0.2). The **gradient** $\nabla f(x)$ is the vector with $\langle \nabla f(x), h \rangle = df_x(h)$ for all h `Mathlib/Analysis/Calculus/Gradient/Basic · gradient`; it is Milnor's grad f for the Riemannian metric given by the inner product. Three facts about it are used.

- If f is C^{m+1}, then ∇f is C^m `Chapter2/Regular · contDiff_gradient`: it is $x \mapsto df_x$ followed by the inverse of the Riesz isomorphism $E \to E^*$, a continuous linear map.
- $df_x(\nabla f(x)) = \|\nabla f(x)\|^2$ `Chapter2/Regular · fderiv_apply_gradient`, from $\langle \nabla f(x), h \rangle = df_x(h)$ `Mathlib/Analysis/Calculus/Gradient/Basic · inner_gradient_left` with $h = \nabla f(x)$.
- At a point where f is differentiable, p is critical exactly when $\nabla f(p) = 0$ `Chapter2/Regular · isCriticalPoint_iff_gradient_eq_zero`, since the Riesz map is injective. This is Milnor's remark that the gradient vanishes precisely at the critical points (p. 12).

The cut-off. Milnor takes a smooth $\rho : M \to \mathbb{R}$ equal to $1/\langle \operatorname{grad} f, \operatorname{grad} f \rangle$ on $f^{-1}[a, b]$ and vanishing outside a compact neighbourhood of it, and does not show that one exists. The formal proof constructs it in a more general form, which §II.5 and §II.13 reuse. It needs a product rule for functions that are smooth only on part of the space.

A product lemma. If χ is smooth on a normed space and g is smooth on an open set U containing the closed support of χ, then $\chi \cdot g$ is smooth everywhere `Chapter2/Regular · contDiff_smul_of_tsupport`. At a point of U it is a product of two functions smooth near that point. A point outside U lies outside the closed support of χ, so χ vanishes on a neighbourhood of it, and there the product is identically 0.

Lemma B. Let g be smooth on a finite-dimensional real normed space, with no zero on the compact set K, and let $O \supseteq K$ be open. There is a smooth ρ with compact support contained in O, with $\rho \ge 0$ wherever $g \ge 0$, and with $\rho\, g = 1$ on K `Chapter2/Regular · exists_cutoff_mul`.

Proof.

1. The set $U = O \cap \{g \ne 0\}$ is open and contains K. As the space is locally compact, there is a compact L with $K \subseteq \operatorname{int} L \subseteq L \subseteq U$ `Mathlib/Topology/Compactness/LocallyCompact · exists_compact_between`.

2. A smooth Urysohn function: there is a smooth χ with values in $[0,1]$, support exactly int L, and $\chi(x) = 1$ exactly for $x \in K$ `Mathlib/Geometry/Manifold/PartitionOfUnity · exists_contMDiff_support_eq_eq_one_iff`. Its closed support lies in L.
3. Put $\rho = \chi \cdot g^{-1}$. It is smooth by the product lemma, as g^{-1} is smooth on U, which contains L. Its closed support lies in that of χ, hence in the compact set $L \subseteq O$. It is ≥ 0 where $g \geq 0$, since $\chi \geq 0$. On K, $\rho\, g = 1 \cdot g^{-1} g = 1$. □

For Milnor's ρ apply Lemma B with $g = \|\nabla f\|^2$, which is smooth and has no zero on $K = f^{-1}[a,b]$ when K contains no critical point, and with $O = E$. This gives a smooth $\rho \geq 0$ with compact support and $\rho\,\|\nabla f\|^2 = 1$ on K `Chapter2/Regular · exists_cutoff`. The sign $\rho \geq 0$ is never used.

The field and its flow. Milnor's vector field is $X = \rho\,\nabla f$ `Chapter2/Regular · gradField`. It is smooth `Chapter2/Regular · contDiff_gradField` with compact support `Chapter2/Regular · hasCompactSupport_gradField`. A C^1 field with compact support is bounded and globally Lipschitz `Chapter2/Regular · isBddLipschitz_gradField Flow/Global · IsBddLipschitz.of_contDiff`, which is the hypothesis under which Part I constructs the global flow of Lemma 2.4 `Flow/Global · flow` (§0.10). Each $t \mapsto \varphi_t(x)$ is an integral curve of X `Flow/Global · IsBddLipschitz.hasDerivAt_flow`, and outside the support of ρ the flow is stationary `Chapter2/Regular · flow_gradField_of_notMem Flow/Global · IsBddLipschitz.flow_of_notMem_tsupport`. Along X, $df(X) = \rho\,\|\nabla f\|^2$ `Chapter2/Regular · fderiv_apply_gradField`.

Lemma (the derivative along a trajectory). If $\rho\,\|\nabla f\|^2 = 1$ wherever $a \leq f \leq b$, then the flow of X is a slab flow for f and $[a,b]$ `Chapter2/Regular · isSlabFlow_gradField`.

Proof. Continuity, $\varphi_0 = \mathrm{id}$ and the group law are proved in Part I `Flow/Global · IsBddLipschitz.continuous_flow Flow/Global · IsBddLipschitz.flow_zero Flow/Global · IsBddLipschitz.flow_add`. By the chain rule the derivative of $t \mapsto f(\varphi_t(x))$ is $df(X)$ evaluated at $\varphi_t(x)$, that is $\rho\,\|\nabla f\|^2 = 1$ when $f(\varphi_t(x)) \in [a,b]$. This is Milnor's computation $\langle d\varphi_t(q)/dt, \operatorname{grad} f\rangle = \langle X, \operatorname{grad} f\rangle = +1$. □

In particular $f(\varphi_t(x)) = f(x) + t$ for x in the slab and $a - f(x) \leq t \leq b - f(x)$ `Chapter2/Regular · apply_flow_gradField`, and $\varphi_{c'-c}(M^c) = M^{c'}$ for $a \leq c \leq c' \leq b$ `Chapter2/Regular · flow_gradField_image_sublevel`.

Theorem 3.1 (V2.2). Let $f : E \to \mathbb{R}$ be C^∞ and $a \leq b$, and suppose that $f^{-1}[a,b]$ is compact and contains no critical point of f. Then the following hold `Chapter2/Regular · theorem_3_1`:

1. there is a C^∞ diffeomorphism Φ of E with $\Phi(M^a) = M^b$;
2. M^a is a strong deformation retract of M^b;
3. the inclusion $M^a \to M^b$ is a homotopy equivalence;
4. M^a and M^b are homeomorphic.

Proof. By the characterization of critical points, ∇f has no zero on $K = f^{-1}[a,b]$. Lemma B gives ρ, and the flow of $X = \rho\nabla f$ is a slab flow for f. Its time-$(b-a)$ map is a C^∞ diffeomorphism of E, with inverse the time-$(a-b)$ map `Flow/Smooth · flowDiffeomorph`; that the flow is smooth is Part I's version of Lemma 2.4. The equality $\Phi(M^a) = M^b$ is the first half for slab flows with $c = a$, $c' = b$; items 2 and 3 are the second half. For item 4, Φ restricts to a homeomorphism of M^a onto its image, which is M^b. □

Read backwards, the inverse φ_{a-b} pushes M^b down onto M^a `Chapter2/Regular · exists_diffeomorph_image_sublevel_eq`, which is the picture Milnor draws. *Restated:* Milnor's "M^a is diffeomorphic to M^b" is formalized as what his proof gives, a diffeomorphism of the whole space carrying M^a onto M^b. Here M^a and M^b are manifolds with boundary, since a and b are regular values; recording an ambient diffeomorphism avoids giving them that structure.

II.5 Theorem 3.1 on a submanifold of Euclidean space

Lemma 2.4 is formalized for compactly supported fields on a vector space (Part I), not on manifolds. So on a submanifold the formal proof runs Milnor's argument with an *ambient* flow which preserves the submanifold.

Setting. V is a finite-dimensional real normed space, and $N \subseteq V$ is a slice submanifold modelled on Z with normal model W `Milnor/Chapter2/Submanifold · IsSliceSubmanifold` (§0.1): every point of N lies in the domain of a *slice chart*, a diffeomorphism σ from an open subset of V onto an open subset of $Z \times W$ such that a point x of its domain lies in N exactly when the W-component of $\sigma(x)$ is 0 `Milnor/Chapter2/Submanifold · SliceChart`. N carries the manifold structure of the companion project `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.chartedSpace`, and need not be closed in V. $F : V \to \mathbb{R}$ is smooth and $f = F|N$. Critical points of f are critical points of the manifold derivative `Chapter1/Manifold · IsCriticalPointM`, which are the points q at which dF_q vanishes on the tangent space $T_qN \subseteq V$ `Chapter1/ManifoldSubmanifold · tangentSubspace` `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_forall_tangent` (§0.3). A slice chart σ gives a parametrization of N, $x(z) = \sigma^{-1}(z, 0)$ `Chapter1/ManifoldSubmanifold · sliceParam`, and T_qN is the range of its derivative `Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam`.

Derivatives of slice charts. σ and σ^{-1} are smooth on their domains `Chapter2/Submanifold · contDiffOn_sliceChart` `Chapter2/Submanifold · contDiffOn_sliceChart_invFun`. By the chain rule applied to $\sigma \circ \sigma^{-1} = \mathrm{id}$ near $\sigma(x)$, $d\sigma_x \circ d(\sigma^{-1})_{\sigma(x)}$ is the identity `Chapter2/Submanifold · fderiv_sliceChart_fderiv_invFun`. The derivative of the parametrization at z is $w \mapsto d(\sigma^{-1})_{(z,0)}(w, 0)$ `Chapter2/Submanifold · fderiv_sliceParam_apply`.

Coordinate fields. For $u \in Z$, the coordinate field of σ in the direction u is

$$Y_u(x) = d(\sigma^{-1})_{\sigma(x)}(u, 0),$$

defined on the domain of σ `Chapter2/Submanifold · chartField`. It is smooth there `Chapter2/Submanifold · contDiffOn_chartField` and linear in u; in particular $Y_{-u} = -Y_u$ `Chapter2/Submanifold · chartField_neg`. At a point $q \in N$ of the domain, $\sigma(q)$ has W-component 0, so $Y_u(q)$ is the derivative of the parametrization applied to u `Chapter2/Submanifold · chartField_eq_fderiv_sliceParam`. Hence the coordinate fields are tangent to N along N `Chapter2/Submanifold · chartField_mem_tangentSubspace`. And if q is not a critical point of f, some coordinate field has $dF_q(Y_u(q)) > 0$ `Chapter2/Submanifold · exists_chartField_pos`: dF_q does not vanish on T_qN, so it is nonzero on the image $dx(w)$ of some w, and one takes $u = w$ or $u = -w$ according to the sign.

The chart domain. Let O be the union of the domains of all slice charts of N `Chapter2/Submanifold · chartDomain`. It is open `Chapter2/Submanifold · isOpen_chartDomain`, contains N `Chapter2/Submanifold · subset_chartDomain`, and N is closed in it: $O \setminus N$ is open `Chapter2/Submanifold · isOpen_chartDomain_diff`. Indeed, if x lies in the domain of σ but not in N, then $\sigma(x)$ has nonzero W-component; by continuity the same holds on a neighbourhood of x inside the domain, and every point there is outside N.

Step 1: a tangent field along which F increases. If $K \subseteq N$ is compact and contains no critical point of f, there is a smooth field Y on V, tangent to N at every point of N, with $dF(Y) > 0$ on K `Chapter2/Submanifold · exists_tangent_field`.

Proof. For $q \in K$ choose a slice chart σ_q at q and a direction u_q with $dF_q(Y_{u_q}(q)) > 0$. The set of points of the domain of σ_q where $dF(Y_{u_q}) > 0$ is open and contains q, so it contains the closed support of a smooth bump $\chi_q \geq 0$ with $\chi_q(q) = 1$ `Mathlib/Analysis/Calculus/BumpFunction/FiniteDimension · exists_contDiff_tsupport_subset`. The open sets $\{\chi_q > 0\}$ cover the compact K; choose finitely many points $q_1, \dots, q_m$ whose sets cover it, and put

$$Y = \sum_{i=1}^{m} \chi_{q_i}\, Y_{u_{q_i}}.$$

Each term is smooth on all of V by the product lemma of §II.4, the coordinate field being smooth on the chart domain, an open set containing the closed support of χ_{q_i}. At a point of N each term is 0 or a multiple of a tangent vector, because $\chi_{q_i} \neq 0$ forces the point into the chart domain; since T_qN is a

linear subspace, Y is tangent. At a point of K every term of $dF(Y)$ is ≥ 0, because where $\chi_{q_i} \neq 0$ the point lies in the closed support, on which $dF(Y_{u_{q_i}}) > 0$; and at least one term is > 0, because the sets $\{\chi_{q_i} > 0\}$ cover K. □

Step 2: normalization. The function $dF(Y)$ is smooth and positive on K. Lemma B of §II.4, with O the chart domain, gives a smooth ρ with compact support inside O and $\rho\, dF(Y) = 1$ on K. The field $X = \rho Y$ is smooth, has compact support inside O, is tangent to N along N, and satisfies $dF(X) = 1$ on K. It plays the role of Milnor's ρ grad f. For the metric induced by an inner product on V, grad f is the orthogonal projection of ∇F to T_qN, but the proof only uses $df(X) = 1$ on the slab, so no metric is needed.

Step 3: local integral curves inside N. Let X be a smooth field on V which is tangent to N along N, and let $y \in N$. There are $\varepsilon > 0$ and a curve γ with $\gamma(0) = y$, $\gamma'(t) = X(\gamma(t))$ and $\gamma(t) \in N$ for $|t| < \varepsilon$ `Chapter2/Submanifold · exists_integralCurve_mem`.

Proof. Take a slice chart σ at y, with parametrization x. Read X in the chart: let $\tilde{X}(z)$ be the Z-component of $d\sigma(X(x(z)))$, taken at the point $x(z)$. For z in the domain of the parametrization, $dx_z(\tilde{X}(z)) = X(x(z))$. Indeed $X(x(z))$ is tangent, so it equals $dx_z(w) = d(\sigma^{-1})(w, 0)$ for some w; applying $d\sigma$ gives $(w, 0)$, so $\tilde{X}(z) = w$. The field $\tilde{X}$ is C^1 at the Z-component z_0 of $\sigma(y)$, as a composite of smooth maps. Mathlib's Picard–Lindelöf theorem `Mathlib/Analysis/ODE/ExistUnique · ContDiffAt.exists_forall_mem_closedBall_exists_eq_forall_mem_Ioo_hasDerivAt₀` gives a solution α of $\alpha' = \tilde{X}(\alpha)$ with $\alpha(0) = z_0$ on an interval $(-\varepsilon, \varepsilon)$. Shrinking ε, α stays in the open domain of the parametrization, by continuity at 0. Then $\gamma = x \circ \alpha$ lies in N `Chapter1/ManifoldSubmanifold · sliceParam_mem`, starts at y, and $\gamma' = dx(\alpha') = dx(\tilde{X}(\alpha)) = X(\gamma)$. □

Step 4: invariance of N. Let X be smooth with compact support, tangent to N along N, with closed support inside the chart domain O. Then $\varphi_t(x) \in N$ for every $x \in N$ and every t `Chapter2/Submanifold · flow_mem_of_tangent`.

Proof. Part I proves an invariance criterion: a closed set through each point of which some local integral curve stays in the set is invariant under the flow `Flow/Global · IsBddLipschitz.flow_mem`. (Uniqueness makes the flow line agree with the local curve, so the set of times at which the flow line lies in the set is open; it is closed by continuity; it contains 0.) But N need not be closed, so the criterion is applied to $S = N \cup (V \setminus O)$, which is closed because its complement $O \setminus N$ is open. Every point of S has a local integral curve inside S: a point of N by Step 3, a point outside O by the constant curve, since X vanishes there. So $\varphi_t(x) \in S$. If $\varphi_t(x)$ were outside O, it would be a zero of X and hence a fixed point of the flow `Flow/Global · IsBddLipschitz.flow_of_eq_zero`, and then $x = \varphi_{-t}(\varphi_t(x)) = \varphi_t(x)$ `Flow/Global · IsBddLipschitz.flow_neg_flow` would lie outside $O \supseteq N$: a trajectory cannot reach a zero of X in finite time. So $\varphi_t(x) \in N$. □

Theorem 3.1 on a submanifold (V2.3). Let $N \subseteq V$ be a slice submanifold modelled on a finite-dimensional Z, $F : V \to \mathbb{R}$ smooth and $f = F|N$. Let $a \leq b$, and suppose that $f^{-1}[a, b] \subseteq N$ is compact and contains no critical point of f. Then there is a C^∞ diffeomorphism Φ of the manifold N with $\Phi(N^a) = N^b$; N^a is a strong deformation retract of N^b; and the inclusion $N^a \to N^b$ is a homotopy equivalence `Chapter2/Submanifold · theorem_3_1_submanifold`.

Proof.

1. $K = f^{-1}[a, b]$, viewed in V, is compact, as the image of a compact set under the inclusion, and lies in N.
2. Steps 1 and 2 give the field $X = \rho Y$: smooth, compactly supported inside the chart domain, tangent to N, with $dF(X) = 1$ on K. It is bounded and Lipschitz, so it has a global flow, and by Step 4 the flow preserves N.
3. The restricted flow $\varphi_t|N$ is a slab flow for f. Continuity and the group law are inherited from V. When $F(\varphi_t(q)) \in [a, b]$, the point $\varphi_t(q)$ lies in K, and the chain rule gives the derivative $dF(X) = 1$.

4. Each $\varphi_t|N$ is smooth as a map of manifolds $N \to N$: the ambient φ_t is smooth `Flow/Smooth · contDiff_flow`, the inclusion of N is smooth, the composite takes values in N, and a smooth map with values in a slice submanifold is smooth into it `Milnor/Chapter2/Corestrict · IsSliceSubmanifold.contMDiff_codRestrict`. So $\Phi = \varphi_{b-a}|N$ is a diffeomorphism of N, with inverse $\varphi_{a-b}|N$.
5. The two halves for slab flows (§II.3) give $\Phi(N^a) = N^b$, the strong deformation retraction and the homotopy equivalence. □

When N is compact, $f^{-1}[a,b]$ is a closed subset of a compact space, and the compactness hypothesis can be dropped `Chapter2/Submanifold · theorem_3_1_submanifold_of_isCompact`. *Narrower than the book:* f must be the restriction of a smooth function on V. On a compact Hausdorff manifold M without boundary this costs nothing: Whitney's embedding makes M diffeomorphic to a compact slice submanifold of some $\mathbb{R}^{K+1}$, every smooth function on which is such a restriction, and Theorem 3.1 follows on M for any $a \le b$ with $f^{-1}[a,b]$ free of critical points `Chapter4/Whitney · theorem_3_1_of_compactSpace` (V2.4). Part III tells that transfer. For non-compact abstract manifolds the theorem is not formalized: it would need a closed embedding into Euclidean space, which Mathlib does not have.

II.6 The Remark on p. 13: compactness cannot be omitted

Milnor's Remark after Theorem 3.1 says that the compactness of $f^{-1}[a,b]$ cannot be dropped. His Diagram 3 (p. 14) shows a surface from which a point between the levels a and b has been removed, and for it M^a is "clearly" not a deformation retract of M^b. The formal version replaces the pictured surface by a punctured plane, and proves more: the two sublevel sets are not even homotopy equivalent.

The example (V2.5). $M = \mathbb{C} \setminus \{i\}$ `Chapter2/Remark · RemarkP13.M`, $f(z) = |z|^2$ `Chapter2/Remark · RemarkP13.F`, $a = \frac{1}{4}$ and $b = 4$. The removed point has $f(i) = 1$.

1. *No critical point in the slab.* If $\frac{1}{4} \le |z|^2 \le 4$, then z is not a critical point of $|z|^2$ on $\mathbb{C}$ `Chapter2/Remark · RemarkP13.remark_p13_not_isCriticalPoint`. The derivative of $|z|^2$ at z is $h \mapsto 2\langle z, h\rangle$ `Mathlib/Analysis/InnerProductSpace/Calculus · hasStrictFDerivAt_norm_sq`; if it vanished, its value $2|z|^2$ at $h = z$ would be 0, so $z = 0$, which lies at level $0 < \frac{1}{4}$. Since M is open in $\mathbb{C}$, the critical points of f on the surface M are those of $|z|^2$, and the formal statement is made in $\mathbb{C}$.
2. *The slab is not compact* `Chapter2/Remark · RemarkP13.remark_p13_not_isCompact`. If $f^{-1}[\frac{1}{4}, 4] \subseteq M$ were compact, its image in $\mathbb{C}$ would be compact, hence closed, and it does not contain i; so some disc of radius ε about i would miss it. But for $\delta = \min(\varepsilon/2, \frac{1}{2})$ the point $(1+\delta)i$ lies in M, at distance $\delta < \varepsilon$ from i, and $|(1+\delta)i|^2 = (1+\delta)^2 \in [1, \frac{9}{4}] \subseteq [\frac{1}{4}, 4]$.
3. *$M^{1/4}$ is contractible.* As $|i| = 1 > \frac{1}{2}$, $M^{1/4}$ is the closed disc of radius $\frac{1}{2}$ `Chapter2/Remark · RemarkP13.sublevelHomeomorph`, which is convex and nonempty, hence contractible `Mathlib/Analysis/Convex/Contractible · Convex.contractibleSpace`.
4. *A loop of winding number one is not null-homotopic* `Chapter2/Remark · RemarkP13.not_nullhomotopic`. Let γ be a loop in a space X and $j : X \to \mathbb{C}\setminus\{0\}$ a continuous map with $j(\gamma(s)) = e^{2\pi i s}$. Suppose H is a homotopy from j to a constant map `Mathlib/Topology/Homotopy/Contractible · ContinuousMap.Nullhomotopic`, and put $G(t,s) = H(t, \gamma(s))$, a map $[0,1]^2 \to \mathbb{C} \setminus \{0\}$ with $G(0,s) = e^{2\pi i s}$.
   - The exponential map $\mathbb{C} \to \mathbb{C} \setminus \{0\}$ is a covering map `Mathlib/Analysis/Complex/CoveringMap · Complex.isCoveringMap_exp`, so G lifts: there is a continuous $L : [0,1]^2 \to \mathbb{C}$ with $e^L = G$ and $L(0,s) = 2\pi i s$ `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.liftHomotopy`.
   - Two logarithms of the same number differ by an integer multiple of $2\pi i$ `Mathlib/Analysis/SpecialFunctions/Complex/Log · Complex.exp_eq_exp_iff_exists_int`. Since $G(t,1) = G(t,0)$, the winding number $k(t) = \operatorname{Im}\big(L(t,1) - L(t,0)\big)/2\pi$ of the loop $G(t,\cdot)$ is an integer for every t, and k is continuous.
   - A continuous integer-valued function on a connected space is constant `Chapter2/Remark · RemarkP13.eq_of_forall_eq_intCast`: if it took two integer values $m < n$, it would take the value $m + \frac{1}{2}$ in between, by the intermediate value theorem.

- $k(0) = 1$, because $L(0,1) - L(0,0) = 2\pi i$. And $k(1) = 0$: the loop $G(1,\cdot)$ is constant, so $s \mapsto \operatorname{Im}\big(L(1,s) - L(1,0)\big)/2\pi$ is continuous, integer-valued and 0 at $s = 0$, hence 0 at $s = 1$. This contradicts the constancy of k.

5. M^4 *is not contractible* `Chapter2/Remark · RemarkP13.not_contractibleSpace`. Apply step 4 to $j(z) = z - i$, a map $M^4 \to \mathbb{C} \setminus \{0\}$ `Chapter2/Remark · RemarkP13.j`, and to the loop $\gamma(s) = i + e^{2\pi i s}$ around the removed point `Chapter2/Remark · RemarkP13.γ`, which lies in M^4 because $|\gamma(s)| \le |i| + 1 = 2$ and is a closed loop, $\gamma(0) = \gamma(1)$ `Chapter2/Remark · RemarkP13.γ_zero_eq_one`. If M^4 were contractible, its identity map would be null-homotopic, and so would j; but $j(\gamma(s)) = e^{2\pi i s}$.
6. *Conclusion.* $M^{1/4}$ and M^4 are not homotopy equivalent `Chapter2/Remark · RemarkP13.remark_p13`, because a space homotopy equivalent to a contractible space is contractible. Hence, by the lemma of §II.1, $M^{1/4}$ is not a strong deformation retract of M^4 `Chapter2/Remark · RemarkP13.remark_p13_not_isStrongDeformationRetract`, and the two are not homeomorphic either `Chapter2/Remark · RemarkP13.remark_p13_not_homeomorph`: the first half of Theorem 3.1 fails as well.
7. *With the point put back*, on all of $\mathbb{C}$, $f^{-1}[\frac{1}{4}, 4]$ is a closed bounded annulus, hence compact, still without critical points, and Theorem 3.1 of §II.4 applies: the disc of radius $\frac{1}{2}$ is a strong deformation retract of the disc of radius 2 `Chapter2/Remark · RemarkP13.remark_p13_plane`.

So Milnor's "clearly" hides the fact that a punctured disc is not contractible, which the formal proof derives from the homotopy lifting property of the exponential map.

II.7 Milnor's function μ

Theorem 3.2 modifies f near the critical point with the help of a function μ of one variable. Milnor requires (p. 16) that μ be C^∞ with

$$\mu(0) > \epsilon, \qquad \mu(r) = 0 \ \text{ for } r \ge 2\epsilon, \qquad -1 < \mu'(r) \le 0 \ \text{ for all } r,$$

and he does not show that such a function exists. These four conditions are the formal definition `Chapter2/CellModel · IsMilnorMu`. Everything in the proof of Theorem 3.2 uses μ only through them, and through three consequences.

- μ is decreasing, since $\mu' \le 0$ `Chapter2/CellModel · IsMilnorMu.antitone`.
- $\mu \ge 0$, since it decreases to $\mu(2\epsilon) = 0$ `Chapter2/CellModel · IsMilnorMu.nonneg`. Milnor uses this in Assertion 1 without stating it.
- $s \mapsto s + \mu(s)$ is increasing, since its derivative $1 + \mu'$ is positive `Chapter2/CellModel · IsMilnorMu.monotone_add`. This is Milnor's $\partial F/\partial \xi < 0$ along $\eta = 0$.

Why a construction is needed. μ must drop by more than ϵ on $[0, 2\epsilon]$ with a slope of absolute value less than 1. So its average slope there is below $-\frac{1}{2}$, and the slope has to be controlled everywhere. The first candidate, $A\,(1 - \psi(r/2\epsilon))$ with a smooth step ψ and $A > \epsilon$, fails: Mathlib's smooth step has slope 2 at its midpoint, which gives the slope $-A/\epsilon < -1$ at $r = \epsilon$. The formal construction integrates a plateau instead.

The construction. Let ψ be Mathlib's smooth transition function: smooth, with values in $[0,1]$, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`. Put `Chapter2/CellModel · muBump` `Chapter2/CellModel · milnorMu`

$$\beta(s) = \psi\Big(4 - \frac{2s}{\epsilon}\Big), \qquad \mu(r) = \frac{3}{4}\int_r^{2\epsilon} \beta(s)\, ds.$$

For $\epsilon > 0$, β is smooth `Chapter2/CellModel · contDiff_muBump` with values in $[0,1]$ `Chapter2/CellModel · muBump_nonneg` `Chapter2/CellModel · muBump_le_one`; it equals 1 for $s \le \frac{3}{2}\epsilon$, where $4 - 2s/\epsilon \ge 1$ `Chapter2/CellModel · muBump_of_le`, and 0 for $s \ge 2\epsilon$, where $4 - 2s/\epsilon \le 0$ `Chapter2/CellModel · muBump_of_ge`.

Lemma. For every $\epsilon > 0$ this μ satisfies Milnor's four conditions `Chapter2/CellModel · isMilnorMu_milnorMu`.

Proof.

1. By the fundamental theorem of calculus, with the lower limit of integration varying, $\mu'(r) = -\frac{3}{4}\,\beta(r)$ `Chapter2/CellModel · hasDerivAt_milnorMu` `Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus · intervalIntegral.integral_hasDerivAt_left`. Hence $-1 < -\frac{3}{4} \le \mu' \le 0$.
2. μ' is smooth, so μ is C^∞ `Chapter2/CellModel · contDiff_milnorMu`.
3. For $r \ge 2\epsilon$ the integrand vanishes between 2ϵ and r, so $\mu(r) = 0$ `Chapter2/CellModel · milnorMu_of_ge`.
4. Splitting the integral at $\frac{3}{2}\epsilon$, $\mu(0) = \frac{3}{4}\big(\int_0^{3\epsilon/2} 1\,ds + \int_{3\epsilon/2}^{2\epsilon} \beta\big) \ge \frac{3}{4}\cdot\frac{3\epsilon}{2} = \frac{9}{8}\,\epsilon > \epsilon$ `Chapter2/CellModel · lt_milnorMu_zero`. □

The exposition normalizes a different plateau. Any function with Milnor's four properties serves, and the formal proof fixes this one.

II.8 Theorem 3.2 in Morse coordinates: F, Assertions 1 and 2, the cell and the handle

Milnor's proof of Theorem 3.2 (pp. 15–19) works in coordinates $u^1, \dots, u^n$ around p in which $f = c - \xi + \eta$, with $\xi = (u^1)^2 + \cdots + (u^\lambda)^2$ and $\eta = (u^{\lambda+1})^2 + \cdots + (u^n)^2$. All his formulas make sense on the whole coordinate space. The formal proof therefore states and proves every computation once, on a product of two inner product spaces, and transports the results afterwards (§II.10 and §II.13).

Setting. X and Y are real inner product spaces, Milnor's $\mathbb{R}^\lambda$ and $\mathbb{R}^{n-\lambda}$. A point of $X \times Y$ is written $v = (x, y)$, and Milnor's two sums of squares become `Chapter2/CellModel · CellModel.xi` `Chapter2/CellModel · CellModel.eta`

$$\xi(v) = \|x\|^2, \qquad \eta(v) = \|y\|^2.$$

In these coordinates f becomes $f(v) = c - \xi + \eta$ `Chapter2/CellModel · CellModel.morseQuad`, and Milnor's modified function (p. 16) is `Chapter2/CellModel · CellModel.morseF`

$$F(v) = c - \xi + \eta - \mu(\xi + 2\eta).$$

With $\mu = 0$ it is f `Chapter2/CellModel · CellModel.morseQuad_eq_morseF`. Three sets enter:

- the **ellipsoid** $\{\xi + 2\eta \le 2\epsilon\}$ `Chapter2/CellModel · CellModel.ellipsoid`, which is closed `Chapter2/CellModel · CellModel.isClosed_ellipsoid` and lies in the ball $\{\xi + \eta \le 2\epsilon\}$ `Chapter2/CellModel · CellModel.ellipsoid_subset_ball`;
- Milnor's **cell** $e^\lambda = \{\xi \le \epsilon,\ y = 0\}$ `Chapter2/CellModel · CellModel.cellModel`, which lies in the ellipsoid `Chapter2/CellModel · CellModel.cellModel_subset_ellipsoid`;
- Milnor's **handle** H, the closure of $F^{-1}(-\infty, c-\epsilon] \setminus f^{-1}(-\infty, c-\epsilon]$ `Chapter2/CellModel · CellModel.handleModel`.

Let μ satisfy Milnor's conditions for ϵ. Then $F \le f$, because $\mu \ge 0$ `Chapter2/CellModel · CellModel.morseF_le_morseQuad`, and $F = f$ outside the ellipsoid, where $\xi + 2\eta > 2\epsilon$ and therefore $\mu(\xi + 2\eta) = 0$ `Chapter2/CellModel · CellModel.morseF_eq_of_notMem_ellipsoid`.

Assertion 1, in coordinates. $F(v) \le c + \epsilon$ if and only if $f(v) \le c + \epsilon$ `Chapter2/CellModel · CellModel.morseF_le_iff`.

Proof. Outside the ellipsoid $F = f$. Inside it both inequalities hold: $\xi + 2\eta \le 2\epsilon$ gives $\frac{1}{2}\xi + \eta \le \epsilon$, so $F \le f = c - \xi + \eta \le c + \frac{1}{2}\xi + \eta \le c + \epsilon$. □

Assertion 2, in coordinates. The derivatives of ξ and η are $d\xi(h,k) = 2\langle x, h\rangle$ and $d\eta(h,k) = 2\langle y, k\rangle$ `Chapter2/CellModel · CellModel.hasFDerivAt_xi` `Chapter2/CellModel · CellModel.hasFDerivAt_eta`. Writing μ' for $\mu'(\xi + 2\eta)$, the chain rule gives `Chapter2/CellModel · CellModel.hasFDerivAt_morseF`

$$dF = -(1 + \mu')\,d\xi + (1 - 2\mu')\,d\eta.$$

This is Milnor's $dF = \frac{\partial F}{\partial \xi}\, d\xi + \frac{\partial F}{\partial \eta}\, d\eta$, with $\partial F/\partial \xi = -1 - \mu' < 0$ and $\partial F/\partial \eta = 1 - 2\mu' \geq 1$. If μ is differentiable with $-1 < \mu' \leq 0$ and $dF = 0$ at v, then $v = 0$ `Chapter2/CellModel · CellModel.eq_zero_of_hasFDerivAt_morseF`: evaluating dF on $(x, 0)$ gives $-2(1 + \mu')\|x\|^2 = 0$ with $1 + \mu' > 0$, so $x = 0$; evaluating it on $(0, y)$ gives $2(1 - 2\mu')\|y\|^2 = 0$ with $1 - 2\mu' \geq 1$, so $y = 0$. Milnor's conclusion, that dF vanishes only where $d\xi$ and $d\eta$ vanish simultaneously, uses that $d\xi$ and $d\eta$ involve disjoint sets of coordinates; the evaluations on $(x, 0)$ and $(0, y)$ make this explicit. The origin is a critical point `Chapter2/CellModel · CellModel.hasFDerivAt_morseF_zero`. So for μ satisfying Milnor's conditions the only critical point of F is the origin `Chapter2/CellModel · CellModel.hasFDerivAt_morseF_zero_iff`, and the same computation with $\mu = 0$ gives the same for f `Chapter2/CellModel · CellModel.hasFDerivAt_morseQuad_zero_iff`.

The bound $\partial F/\partial \eta \geq 1$. As reproduced in the scan, Milnor's display reads > 1, with the "∂F" of the numerator dropped, so the scan may also have lost the bar of a $\geq$. Since $\mu' = 0$ on $[2\epsilon, \infty)$, the correct bound is ≥ 1, and that is all the proof uses.

The cell lies below the level $c - \epsilon$ of F. On e^λ one has $\eta = 0$, so $F = c - (\xi + \mu(\xi))$, and since $s \mapsto s + \mu(s)$ is increasing and $\xi \geq 0$, $F \leq c - \mu(0) < c - \epsilon$ `Chapter2/CellModel · CellModel.morseF_lt_of_mem_cellModel`. This is Milnor's inequality $F(q) \leq F(p) < c - \epsilon$ (p. 18), there justified by $\partial F/\partial \xi < 0$, made into a statement about the whole cell. In particular $F(p) = c - \mu(0) < c - \epsilon$.

The cell meets $M^{c-\epsilon}$ in its boundary sphere. On the cell $f = c - \xi$, which is at most $c - \epsilon$ exactly when $\xi \geq \epsilon$; with $\xi \leq \epsilon$ this means $\xi = \epsilon$. So $e^\lambda \cap M^{c-\epsilon} = \{\xi = \epsilon,\ y = 0\}$ `Chapter2/CellModel · CellModel.cellModel_inter_sublevel`, the boundary sphere of the cell, as Milnor notes on p. 16.

$F^{-1}(-\infty, c - \epsilon] = M^{c-\epsilon} \cup H$. If $A \subseteq B$ and B is closed, then $B = A \cup \overline{B \setminus A}$ `Chapter2/CellModel · eq_union_closure_diff`: a point of B not in A lies in $B \setminus A$, and the closure of $B \setminus A$ lies in the closed set B. Here $A = M^{c-\epsilon}$ is contained in $B = F^{-1}(-\infty, c - \epsilon]$ because $F \leq f$, and B is closed because F is continuous `Chapter2/CellModel · CellModel.sublevel_morseF_eq_union`. Milnor writes this equality (p. 17) as notation; it needs exactly these two facts.

$e^\lambda \subseteq H$ (p. 18). The points of the cell with $\xi < \epsilon$ lie in $F^{-1}(-\infty, c - \epsilon] \setminus M^{c-\epsilon}$: there $F < c - \epsilon$ while $f = c - \xi > c - \epsilon$ `Chapter2/CellModel · CellModel.mem_diff_of_mem_cellModel`. Every point $(x, 0)$ of the cell is the limit, as $s \to 1$ from below, of the points $(sx, 0)$, which lie in the cell with $\xi = s^2\|x\|^2 < \epsilon$ for $0 < s < 1$ `Chapter2/CellModel · CellModel.tendsto_smul_cellModel`. So the whole cell lies in the closure H `Chapter2/CellModel · CellModel.cellModel_subset_handleModel`. The points with $\xi = \epsilon$ are in $M^{c-\epsilon}$ and belong to H only as such limits; Milnor's text does not distinguish them. (The scan's "$f(q) \geq c - e$" on p. 18 reads $c - \epsilon$.)

II.9 Assertion 4 in Morse coordinates: r_t as one formula

Milnor's Assertion 4 (pp. 18–19, Diagram 7) deforms $M^{c-\epsilon} \cup H$ onto $M^{c-\epsilon} \cup e^\lambda$ by moving points "vertically", shrinking y and keeping x. He gives three cases: in Case 1, $\xi \leq \epsilon$, $r_t(x, y) = (x, ty)$; in Case 2, $\epsilon \leq \xi \leq \eta + \epsilon$, $r_t(x, y) = (x, s_t\, y)$ with $s_t = t + (1 - t)((\xi - \epsilon)/\eta)^{1/2}$; in Case 3, $\eta + \epsilon \leq \xi$, r_t is the identity. His r_1 is the identity and r_0 the retraction. The formal proof lets the parameter run the other way, as in Theorem 3.1 and in Mathlib's homotopies, and writes the three cases as one formula `Chapter2/CellModel · CellModel.retractScale` `Chapter2/CellModel · CellModel.retractModel`:

$$\sigma_t(v) = 1 - t + t\sqrt{\min\Big(1,\ \frac{\xi - \epsilon}{\eta}\Big)}, \qquad r_t(x, y) = (x,\ \sigma_t(v)\, y).$$

Two conventions of the formal language, under which every function is total, make the single formula work: division by 0 gives 0, and the square root of a negative number is 0.

The three cases.

- *Case 1,* $\xi \le \epsilon$: $r_t(x,y) = (x, (1-t)\,y)$ `Chapter2/CellModel · CellModel.retractModel_of_xi_le`. If $\eta = 0$ then $y = 0$ and nothing moves. Otherwise $(\xi-\epsilon)/\eta \le 0$, so the minimum is ≤ 0, its square root is 0, and $\sigma_t = 1 - t$.
- *Case 2,* $\xi \le \eta + \epsilon$ with $\eta > 0$: then $(\xi-\epsilon)/\eta \le 1$ and $\sigma_t = 1 - t + t\,((\xi-\epsilon)/\eta)^{1/2}$ `Chapter2/CellModel · CellModel.retractModel_of_case2`, which is Milnor's s_t with t replaced by $1-t$. The hypothesis $\epsilon \le \xi$ is not needed: for $\xi < \epsilon$ the square root is 0 and this is Case 1.
- *Case 3,* $\eta + \epsilon \le \xi$: r_t is the identity `Chapter2/CellModel · CellModel.retractModel_of_le_xi`. If $\eta > 0$, then $(\xi-\epsilon)/\eta \ge 1$ and the minimum is 1; if $\eta = 0$ nothing moves.

The single formula agrees with Milnor's three on their overlaps automatically, so the compatibility checks he makes (on $\xi = \epsilon$ Cases 1 and 2 agree, on $\xi = \eta + \epsilon$ Cases 2 and 3 agree) are built in. *The point* $\xi = \epsilon$, $\eta = 0$, where Milnor's Case 2 reads $0/0$ and which he does not discuss, is harmless: there $y = 0$, and a point with $y = 0$ is fixed by every r_t `Chapter2/CellModel · CellModel.retractModel_of_snd_eq_zero`.

Elementary properties. For $0 \le t \le 1$, $0 \le \sigma_t \le 1$ `Chapter2/CellModel · CellModel.retractScale_nonneg` `Chapter2/CellModel · CellModel.retractScale_le_one`. $\sigma_0 = 1$ `Chapter2/CellModel · CellModel.retractScale_zero`, so r_0 is the identity `Chapter2/CellModel · CellModel.retractModel_zero`. r_t keeps ξ `Chapter2/CellModel · CellModel.xi_retractModel` and multiplies η by σ_t^2 `Chapter2/CellModel · CellModel.eta_retractModel`, so it does not increase η `Chapter2/CellModel · CellModel.eta_retractModel_le`. It fixes $M^{c-\epsilon}$, which in coordinates is $\{f \le c-\epsilon\} = \{\eta + \epsilon \le \xi\}$, Case 3 `Chapter2/CellModel · CellModel.retractModel_of_morseQuad_le`, and it fixes the cell, where $y = 0$ `Chapter2/CellModel · CellModel.retractModel_of_mem_cellModel`.

Continuity. Milnor leaves it to the reader to check that the functions $s_t u^i$ stay continuous as $\xi \to \epsilon$ and $\eta \to 0$ (p. 19). Formally, $(t,v) \mapsto r_t(v)$ is continuous on all of $[0,1] \times X \times Y$ `Chapter2/CellModel · CellModel.continuous_retractModel`, not only on $F^{-1}(-\infty, c-\epsilon]$. At a point (t_0, v_0) with $\eta(v_0) \neq 0$, the quotient $(\xi-\epsilon)/\eta$ is continuous nearby, and so are the minimum, the square root, the product with t, and the product with y. At a point with $\eta(v_0) = 0$, that is $v_0 = (x_0, 0)$, the factor σ_t itself is discontinuous. But $r_{t_0}(v_0) = v_0$, and, because $0 \le \sigma_t \le 1$, the distance from $r_t(v)$ to v_0 in the maximum norm of $X \times Y$ is at most that from v to v_0: the x-components are unchanged, and $\|\sigma_t\, y - 0\| \le \|y\|$. So $r_t(v) \to v_0$ as $(t,v) \to (t_0, v_0)$.

What r_t does.

- r_t *maps every sublevel set of* F *into itself* (p. 19): $F(r_t(v)) \le F(v)$ for $0 \le t \le 1$ `Chapter2/CellModel · CellModel.morseF_retractModel_le`. The deformation keeps ξ and replaces η by some $\eta' \le \eta$. As μ is decreasing, $\mu(\xi+2\eta') \ge \mu(\xi+2\eta)$, and therefore $c-\xi+\eta'-\mu(\xi+2\eta') \le c-\xi+\eta-\mu(\xi+2\eta)$. This is Milnor's "$F$ is increasing in η", in integrated form.
- r_1 *lands in* $M^{c-\epsilon} \cup e^\lambda$ `Chapter2/CellModel · CellModel.retractModel_one_mem`, for every point of $X \times Y$. Case 1 goes to $(x, 0)$ with $\xi \le \epsilon$, a point of the cell. In Case 2 with $\eta > 0$ the new value of η is $\frac{\xi-\epsilon}{\eta} \cdot \eta = \xi - \epsilon$, so $f = c - \xi + (\xi - \epsilon) = c - \epsilon$: the point lands on the hypersurface $f^{-1}(c-\epsilon)$, as Milnor says. Case 2 with $\eta = 0$ is the point $\xi = \epsilon$, $y = 0$, already at level $c - \epsilon$. Case 3 is already in $M^{c-\epsilon}$.
- r_t *keeps the ellipsoid in itself* `Chapter2/CellModel · CellModel.retractModel_mem_ellipsoid`, since it keeps ξ and does not increase η. This holds for the deformation built with any level δ in place of ϵ.
- *On* $F^{-1}(-\infty, c-\delta]$, *the deformation of level* δ *moves only points of the ellipsoid* `Chapter2/CellModel · CellModel.retractModel_of_notMem_ellipsoid`. Outside the ellipsoid $F = f$, so $f \le c - \delta$, which is Case 3 for the level δ. This is what will make "the identity outside the coordinate neighbourhood" continuous (§II.10).

Assertion 4, in coordinates. $M^{c-\epsilon} \cup e^\lambda$ is a strong deformation retract of $F^{-1}(-\infty, c-\epsilon] = M^{c-\epsilon} \cup H$, in $X \times Y$ `Chapter2/CellModel · CellModel.isStrongDeformationRetract_model`. Each r_t, $0 \le t \le 1$, maps $F^{-1}(-\infty, c-\epsilon]$ into itself `Chapter2/CellModel · CellModel.retractModel_mem_sublevel`; r_1 is the retraction `Chapter2/CellModel · CellModel.retractOne`; and $(t,v) \mapsto r_t(v)$ is a homotopy from the identity to r_1 which fixes $M^{c-\epsilon} \cup e^\lambda$ at all times `Chapter2/CellModel · CellModel.retractHomotopy`.

II.10 Theorem 3.2 in a Euclidean space

Let E be a finite-dimensional real inner product space; the inner product is needed only for Theorem 3.1, which is applied to F.

Morse charts in split form. The lemma of Morse (Lemma 2.2, proved in Part I) gives, for a non-degenerate critical point p of a smooth f on E, coordinates $y = (y^0, \dots, y^{n-1})$ near p, smooth with smooth inverse, with $y(p) = 0$ and

$$f = f(p) - \sum_{i<\lambda} (y^i)^2 + \sum_{i\geq\lambda} (y^i)^2$$

throughout their domain, where λ is the index of the Hessian `Chapter1/MorseLemma · morse_lemma` `Chapter1/Nondegenerate · hessian` `Chapter1/Nondegenerate · index` (§0.2). Milnor's splitting $u = (x, y)$ is the linear isomorphism of $\mathbb{R}^n$ with the product of the Euclidean spaces of the first λ and of the last $n - \lambda$ coordinates `Chapter2/CellAttach · splitCoord`, under which $\xi = \sum_{i<\lambda} u_i^2$ and $\eta = \sum_{i\geq\lambda} u_i^2$ `Chapter2/CellAttach · norm_sq_splitCoord_fst` `Chapter2/CellAttach · norm_sq_splitCoord_snd`. A **split Morse chart** at p is an open partial homeomorphism φ from E to $X \times Y$ with p in its domain, $\varphi(p) = 0$, φ smooth on its domain, φ^{-1} smooth on its image, and $f = f(p) - \xi \circ \varphi + \eta \circ \varphi$ on its domain `Chapter2/CellAttach · IsSplitMorseChart`. The Morse lemma followed by the splitting provides one with $X = \mathbb{R}^\lambda$, $Y = \mathbb{R}^{n-\lambda}$ `Chapter2/CellAttach · exists_isSplitMorseChart`. Critical points depend only on the germ of a function `Chapter2/CellAttach · isCriticalPoint_congr`, and they correspond under a Morse chart: $g \circ \varphi$ is critical at q if and only if g is critical at $\varphi(q)$ `Chapter2/CellAttach · IsSplitMorseChart.isCriticalPoint_comp_iff`, by the chain rule applied with φ in one direction and with φ^{-1} in the other.

Everything below except the final assembly is stated for an arbitrary split Morse chart, so that it can be reused for charts of a manifold. The standing hypothesis, Milnor's condition (2) on p. 15, is that the image of φ contains the ball $\{\xi + \eta \leq 2\epsilon\}$. Then the ellipsoid lies in the image `Chapter2/CellAttach · ellipsoid_subset_target`, its preimage, the ellipsoid in E `Chapter2/CellAttach · ellipsoidE`, lies in the domain of φ `Chapter2/CellAttach · ellipsoidE_subset_source`, and a point q of the domain lies in it exactly when $\varphi(q)$ lies in the ellipsoid `Chapter2/CellAttach · mem_ellipsoidE_iff`.

Milnor's F on E. The correction $\mu(\xi + 2\eta)$ is transported by the chart and extended by 0 outside its domain `Chapter2/CellAttach · milnorBump`, and $F = f$ minus this correction `Chapter2/CellAttach · milnorF`. This is Milnor's definition: $F = f - \mu(\xi + 2\eta)$ within the coordinate neighbourhood U, and $F = f$ outside it. The cell and the handle in E are $e^\lambda = \varphi^{-1}(\text{cell})$ `Chapter2/CellAttach · cellE` and $H = \overline{F^{-1}(-\infty, c - \epsilon] \setminus M^{c-\epsilon}}$ `Chapter2/CellAttach · handleE`.

F is smooth. Milnor says this is easily verified. The verification needs the ellipsoid in E to be closed. The ellipsoid in $X \times Y$ is closed and bounded in a finite-dimensional space, hence compact `Chapter2/CellAttach · isCompact_ellipsoid`; its image under the continuous map φ^{-1} is compact `Chapter2/CellAttach · isCompact_ellipsoidE`, hence closed in the Hausdorff space E `Chapter2/CellAttach · isClosed_ellipsoidE`. Outside it the correction vanishes `Chapter2/CellAttach · milnorBump_eq_zero`. Near a point of the chart domain the correction equals the smooth function $\mu(\xi \circ \varphi + 2\,\eta \circ \varphi)$. Near a point outside the domain, which lies in the open complement of the closed ellipsoid, it is identically 0. So the correction is smooth `Chapter2/CellAttach · contDiff_milnorBump`, and so is F `Chapter2/CellAttach · contDiff_milnorF`.

Moreover $F = (\text{the model } F) \circ \varphi$ on the chart domain `Chapter2/CellAttach · milnorF_eq`, $F = f$ outside the ellipsoid `Chapter2/CellAttach · milnorF_eq_of_notMem`, $F \leq f$ `Chapter2/CellAttach · milnorF_le`, and $F(p) = c - \mu(0)$ `Chapter2/CellAttach · milnorF_apply_self`.

Assertion 1. $F^{-1}(-\infty, c + \epsilon] = M^{c+\epsilon}$ `Chapter2/CellAttach · sublevel_milnorF_add`: on the chart domain this is Assertion 1 in coordinates, and outside it $F = f$.

Assertion 2. F and f have the same critical points `Chapter2/CellAttach · isCriticalPoint_milnorF_iff`. At a point q of the chart domain, F and f agree near q with the model functions composed with φ; by the

correspondence of critical points under φ, q is critical for either of them exactly when $\varphi(q) = 0$, by the computation in coordinates. At a point outside the domain, which lies outside the closed ellipsoid, $F = f$ on a neighbourhood.

The handle and the cell. $F^{-1}(-\infty, c-\epsilon] = M^{c-\epsilon} \cup H$ `Chapter2/CellAttach · sublevel_milnorF_eq_union`, by the closure argument of §II.8. $e^\lambda \subseteq H$ `Chapter2/CellAttach · cellE_subset_handleE`: the curve $s \mapsto \varphi^{-1}(sx, 0)$ converges, by the continuity of φ^{-1} on its domain, and lies in $F^{-1}(-\infty, c-\epsilon] \setminus M^{c-\epsilon}$ for $s < 1$. $e^\lambda \subseteq F^{-1}(-\infty, c-\epsilon]$ `Chapter2/CellAttach · cellE_subset_sublevel_milnorF`. And $e^\lambda \cap M^{c-\epsilon}$ is the image of the sphere $\{\xi = \epsilon,\ y = 0\}$ `Chapter2/CellAttach · cellE_inter_sublevel`.

Assertion 3. If $f^{-1}[c-\epsilon, c+\epsilon]$ is compact and contains no critical point other than p, then $F^{-1}(-\infty, c-\epsilon]$ is a strong deformation retract of $M^{c+\epsilon}$ `Chapter2/CellAttach · isStrongDeformationRetract_milnorF`.

Proof. First, $F^{-1}[c-\epsilon, c+\epsilon] \subseteq f^{-1}[c-\epsilon, c+\epsilon]$: $F \geq c-\epsilon$ gives $f \geq F \geq c-\epsilon$, and $F \leq c+\epsilon$ gives $f \leq c+\epsilon$ by Assertion 1. Being closed in a compact set, $F^{-1}[c-\epsilon, c+\epsilon]$ is compact; Milnor needs this closedness and does not mention it. A critical point of F in it would be a critical point of f (Assertion 2, used in the direction from F to f), hence p; but $F(p) = c - \mu(0) < c-\epsilon$, so p is not in it. Theorem 3.1 (§II.4), applied to F with $a = c-\epsilon$ and $b = c+\epsilon$, makes $F^{-1}(-\infty, c-\epsilon]$ a strong deformation retract of $F^{-1}(-\infty, c+\epsilon]$, which is $M^{c+\epsilon}$ by Assertion 1. □

Assertion 4 on E: the identity outside the chart. Milnor lets r_t be the identity outside U. The formal deformation is `Chapter2/CellAttach · handleRetract` `Chapter2/CellAttach · handleRetract_of_mem` `Chapter2/CellAttach · handleRetract_of_notMem`

$$r_t(q) = \varphi^{-1}(r_t(\varphi(q))) \text{ for } q \text{ in the chart domain}, \qquad r_t(q) = q \text{ otherwise.}$$

It is built with a level δ ($\delta = \epsilon$ here, $\delta = 0$ for Remark 3.4), while the chart, μ and the ellipsoid keep their ϵ. Milnor does not check that this is well defined and continuous. The formal proof does, from one observation: *on $F^{-1}(-\infty, c-\delta]$, the deformation never leaves the chart*, that is, $r_t(\varphi(q))$ lies in the image of φ `Chapter2/CellAttach · retractModel_mem_target`. If $\varphi(q)$ is in the ellipsoid, r_t keeps it in the ellipsoid, which lies in the image of φ; this is a second use of condition (2), which Milnor does not mention. If $\varphi(q)$ is not in the ellipsoid, r_t fixes it (§II.9). From this:

- r_t does not increase F on $F^{-1}(-\infty, c-\delta]$ `Chapter2/CellAttach · milnorF_handleRetract_le`, so it maps this set into itself for $0 \leq t \leq 1$ `Chapter2/CellAttach · handleRetract_mem`;
- r_0 is the identity `Chapter2/CellAttach · handleRetract_zero`, and r_t fixes $M^{c-\delta}$ `Chapter2/CellAttach · handleRetract_of_mem_sublevel` and the cell `Chapter2/CellAttach · handleRetract_of_mem_cellE`;
- on $F^{-1}(-\infty, c-\delta]$, r_t is the identity outside the ellipsoid `Chapter2/CellAttach · handleRetract_of_notMem_ellipsoidE`;
- r_1 maps $F^{-1}(-\infty, c-\delta]$ into $M^{c-\delta}$ together with the cell of radius δ `Chapter2/CellAttach · handleRetract_one_mem`: on the chart domain this is the statement in coordinates, and outside it, hence outside the ellipsoid, $f = F \leq c-\delta$;
- *continuity:* $(t, q) \mapsto r_t(q)$ is continuous on $[0,1] \times F^{-1}(-\infty, c-\delta]$ `Chapter2/CellAttach · continuous_handleRetract`. Near a point (t_0, q_0) with q_0 in the open chart domain, $r_t = \varphi^{-1} \circ r_t \circ \varphi$ there, which is continuous because φ is continuous at q_0, the deformation in coordinates is continuous, and φ^{-1} is continuous at $r_{t_0}(\varphi(q_0))$, a point of the image of φ by the observation. Near a point with q_0 outside the domain, q_0 lies outside the closed ellipsoid; the points of $F^{-1}(-\infty, c-\delta]$ outside the ellipsoid form an open neighbourhood of it, and there r_t is the identity. This step uses the closedness of the ellipsoid, hence that E is Hausdorff.

Assertion 4. $M^{c-\epsilon} \cup e^\lambda$ is a strong deformation retract of $F^{-1}(-\infty, c-\epsilon] = M^{c-\epsilon} \cup H$ `Chapter2/CellAttach · isStrongDeformationRetract_handle`, with the retraction r_1 `Chapter2/CellAttach · handleRetractOne` and the homotopy $(t, q) \mapsto r_t(q)$, which fixes every set that each r_t fixes `Chapter2/CellAttach · handleHomotopy`.

The conclusion, given a chart. If the image of the chart contains $\{\xi + \eta \leq 2\epsilon\}$ and $f^{-1}[c-\epsilon, c+\epsilon]$ is compact with no critical point other than p, then $M^{c-\epsilon} \cup e^\lambda$ is a strong deformation retract of $M^{c+\epsilon}$

Chapter2/CellAttach · isStrongDeformationRetract_union_cellE. Indeed $M^{c-\epsilon} \cup e^\lambda \subseteq F^{-1}(-\infty, c-\epsilon] \subseteq M^{c+\epsilon}$, because $F \le f$, the cell lies below $c-\epsilon$ for F, and Assertion 1; and Lemma A(i) (§II.1) composes Assertion 3 with Assertion 4. This is the composition behind Milnor's closing sentence of the proof (p. 19), that Assertion 4 together with Assertion 3 completes it.

"For all sufficiently small ϵ." Milnor chooses ϵ so small that (1) $f^{-1}[c-\epsilon, c+\epsilon]$ is compact and contains no critical point other than p, and (2) the image of the chart contains the ball $\sum (u^i)^2 \le 2\epsilon$. Both conditions persist for smaller ϵ, and this is how "for all sufficiently small ϵ" is formalized: there is $\epsilon_1 > 0$ such that the conclusion holds for every $\epsilon \in (0, \epsilon_1]$. Condition (2): the image of φ is open and contains 0, so it contains a ball of some radius r, and for $\epsilon \le r^2/4$ the set $\{\xi + \eta \le 2\epsilon\}$ lies in that ball, because then $\|x\|^2$ and $\|y\|^2$ are both less than r^2 Chapter2/CellAttach · exists_forall_ball_subset_target. Condition (1): $f^{-1}[c-\epsilon, c+\epsilon]$ is closed in the compact $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ of the hypothesis. Together, with $\epsilon_1 = \min(\epsilon_0, r^2/4)$ Chapter2/CellAttach · exists_morseChart_forall_small.

The cell as an embedded disk. For a linear isometry ι from X onto a Euclidean space, the map

$$\chi(z) = \varphi^{-1}(\sqrt{\epsilon}\, \iota^{-1}(z),\ 0)$$

sends the closed unit ball onto e^λ, is continuous and injective on the ball, and satisfies $f(\chi(z)) = c - \epsilon|z|^2$ Chapter2/CellAttach · cellE_parametrization. The point $(\sqrt{\epsilon}\,\iota^{-1}z, 0)$ has $\xi = \epsilon|z|^2 \le \epsilon$, so it lies in the cell and in the image of φ, on which φ^{-1} is continuous and injective; conversely a point $(x, 0)$ of the cell is $\chi(\iota(x/\sqrt{\epsilon}))$. In the final statement, the space of the first λ coordinates is identified with $\mathbb{R}^\lambda$ by renumbering Chapter2/CellAttach · negIdxEquiv.

Theorem 3.2 (V2.6). Let $f : E \to \mathbb{R}$ be C^∞, p a non-degenerate critical point of index λ, $c = f(p)$, and suppose that for some $\epsilon_0 > 0$ the set $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ is compact and contains no critical point other than p. Then there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$ there is a map χ from the closed unit ball D^λ of $\mathbb{R}^\lambda$ to E, continuous and injective on D^λ, with $\chi(z) \in M^{c-\epsilon}$ exactly when $|z| = 1$ and with $\chi(D^\lambda) \subseteq M^{c+\epsilon}$, such that $M^{c-\epsilon} \cup \chi(D^\lambda)$ is a strong deformation retract of $M^{c+\epsilon}$ Chapter2/CellAttach · theorem_3_2. In particular the inclusion of $M^{c-\epsilon} \cup \chi(D^\lambda)$ in $M^{c+\epsilon}$ is a homotopy equivalence Chapter2/CellAttach · theorem_3_2_homotopyEquiv.

Proof. Choose the chart and ϵ_1 as above. For $\epsilon \in (0, \epsilon_1]$ take for μ the function of §II.7 for this ϵ, and χ as above. For $|z| \le 1$, $\chi(z) \in M^{c-\epsilon}$ means $c - \epsilon|z|^2 \le c - \epsilon$, that is $|z|^2 \ge 1$, that is $|z| = 1$; and $c - \epsilon|z|^2 \le c + \epsilon$ always. The strong deformation retraction is the conclusion given a chart. The homotopy equivalence follows by the lemma of §II.1. □

Stronger than the book. Milnor states only that $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon}$ with a λ-cell attached. The formal statement records what the proof produces, a strong deformation retraction onto $M^{c-\epsilon}$ together with an embedded cell; §II.11 derives Milnor's statement from it. As reproduced in the scan, the statement of Theorem 3.2 ends with "a *k*-cell" and renders the index as "A"; the proof requires a λ-cell, and both are read so.

II.11 Lemma D: the subspace is the attached cell

Milnor's theorem speaks of $M^{c-\epsilon}$ *with a λ-cell attached*: the space obtained from the disjoint union of $M^{c-\epsilon}$ and the closed disk D^λ by identifying each $u \in S^{\lambda-1}$ with $\varphi(u)$, for an attaching map φ (§0.5). His proof produces the subspace $M^{c-\epsilon} \cup e^\lambda$ of M, and the passage from one to the other is a single remark, that $e^\lambda \cap M^{c-\epsilon}$ is the boundary of the cell, "so that" the cell is attached as required (p. 16). The exposition states the missing lemma as Lemma D.

In the formal proof the attaching space is Mathlib's pushout, in the category of topological spaces, of $\varphi : S^{\lambda-1} \to M^{c-\epsilon}$ and the inclusion $S^{\lambda-1} \to D^\lambda$ Mathlib/Topology/Category/TopCat/Sphere · TopCat.diskBoundaryInclusion, where D^λ and $S^{\lambda-1}$ are the closed unit ball and the unit sphere of $\mathbb{R}^\lambda$ Mathlib/Topology/Category/TopCat/Sphere · TopCat.disk Mathlib/Topology/Category/TopCat/Sphere · TopCat.diskBoundary, the

basic cell of Mathlib's CW complexes. A commutative square is a pushout when it has the universal property of one `Mathlib/CategoryTheory/Limits/Shapes/Pullback/IsPullback/Defs · CategoryTheory.IsPushout`.

Lemma D (V2.7). Let W be a Hausdorff space, $j : A \to W$ a closed embedding, D a compact space, $\chi : D \to W$ continuous and injective, and $i : S \to D$, $\varphi : S \to A$ continuous with $j \circ \varphi = \chi \circ i$. Suppose that $W = j(A) \cup \chi(D)$ and that $\chi(d) \in j(A)$ only for $d \in i(S)$. Then the square formed by φ, i, j and χ is a pushout: W is A with D attached along φ `Chapter2/CellAdjunction · isPushout_of_cell`.

Proof. The exposition argues with a closed quotient map; the formal proof checks the universal property directly.

1. χ is a continuous injection of a compact space into a Hausdorff space, hence a closed embedding, a homeomorphism onto its closed image $\chi(D)$. Likewise j is a homeomorphism onto its closed image $j(A)$.
2. Let $a : A \to P$ and $d : D \to P$ be continuous with $a \circ \varphi = d \circ i$. Define $g : W \to P$ by $g(w) = a(j^{-1}(w))$ if $w \in j(A)$, and $g(w) = d(\chi^{-1}(w))$ otherwise; such w lie in $\chi(D)$ by the covering hypothesis.
3. $g \circ j = a$ by definition. Also $g \circ \chi = d$: if $\chi(x) \notin j(A)$ this is the definition; if $\chi(x) \in j(A)$, then $x = i(s)$ for some s, so $\chi(x) = j(\varphi(s))$ and $g(\chi(x)) = a(\varphi(s)) = d(i(s)) = d(x)$.
4. g is continuous by the pasting lemma for the closed cover $W = j(A) \cup \chi(D)$ `Mathlib/Topology/ContinuousOn · ContinuousOn.union_of_isClosed`: on $j(A)$ it is $a \circ j^{-1}$, and on $\chi(D)$ it is $d \circ \chi^{-1}$ by step 3, both continuous.
5. Uniqueness: a map which agrees with a on $j(A)$ and with d on $\chi(D)$ agrees with g everywhere, since these sets cover W. □

Lemma D for a cell in a normed space. Let A be a closed subset of a normed space E, and $\chi : \mathbb{R}^n \to E$ continuous and injective on D^n with $\chi(z) \in A$ exactly when $|z| = 1$, for $z \in D^n$. Then $A \cup \chi(D^n)$ is the pushout of $\varphi = \chi|_{S^{n-1}} : S^{n-1} \to A$ and $S^{n-1} \to D^n$, and φ is injective `Chapter2/CellAdjunction · exists_isPushout_union_cell`. This is Lemma D with $W = A \cup \chi(D^n)$, which is Hausdorff as a subspace of E; j the inclusion of A, a closed embedding because A is closed; and χ and i the evident maps. W is covered by A and $\chi(D^n)$, and $\chi(z) \in A$ forces $|z| = 1$, that is $z \in S^{n-1}$. The same proof works in any Hausdorff space `Chapter2/CellPatch · exists_isPushout_union_cell_of_t2`. *Hidden hypotheses:* the ambient space is Hausdorff and A is closed, so that the cell is closed and χ is a homeomorphism onto it.

Theorem 3.2 with an attaching space (V2.6). Under the hypotheses of Theorem 3.2, there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$ there is an injective continuous $\varphi : S^{\lambda-1} \to M^{c-\epsilon}$ for which $M^{c+\epsilon}$ is homotopy equivalent to the pushout of $D^\lambda \leftarrow S^{\lambda-1} \to M^{c-\epsilon}$, that is, to $M^{c-\epsilon} \cup_\varphi e^\lambda$ `Chapter2/CellAdjunction · theorem_3_2_attach`.

Proof. Theorem 3.2 gives χ and the homotopy equivalence of $M^{c-\epsilon} \cup \chi(D^\lambda)$ with $M^{c+\epsilon}$. The set $M^{c-\epsilon}$ is closed, as f is continuous. Lemma D identifies $M^{c-\epsilon} \cup \chi(D^\lambda)$ with the pushout up to the canonical isomorphism of two pushouts of the same diagram `Mathlib/CategoryTheory/Limits/Shapes/Pullback/IsPullback/Defs · CategoryTheory.IsPushout.isoPushout`, which in the category of topological spaces is a homeomorphism. Composing gives the homotopy equivalence. □

The same deduction, for any continuous function on a Hausdorff space and levels $a \le b$, is `Chapter2/CellPatch · exists_homotopyEquiv_pushout_of_cell`: if $M^a \cup \chi(D^n)$ is a strong deformation retract of M^b, for a cell $\chi(D^n) \subseteq M^b$ meeting M^a exactly in $\chi(S^{n-1})$, then M^b is homotopy equivalent to $M^a \cup_\varphi e^n$ with φ injective.

II.12 Remark 3.4: the critical level

Milnor's Remark 3.4 (p. 20; printed "Remark 2.4" in the scan, and cited as 3.4 on p. 24) says that a simple modification of the proof of Theorem 3.2 shows M^c to be a deformation retract of $M^{c+\epsilon}$, because M^c is one of $F^{-1}(-\infty, c]$, which is one of $M^{c+\epsilon}$ (Diagram 8); and that, combining this with Theorem 3.2, one

sees easily that $M^{c-\epsilon} \cup e^\lambda$ is a deformation retract of M^c. Neither half is written out. The formal proofs work on E, with the function F, the chart and the cell of §II.10.

1. $F^{-1}(-\infty, c]$ *is a strong deformation retract of* $M^{c+\epsilon}$ `Chapter2/CriticalLevel · isStrongDeformationRetract_milnorF_level`. This is Theorem 3.1 for F on the slab $F^{-1}[c, c+\epsilon]$, argued as Assertion 3: the slab lies in $f^{-1}[c-\epsilon, c+\epsilon]$ (since $F \geq c$ gives $f \geq c$, and Assertion 1), so it is compact; a critical point of F in it would be p, which lies at the level $c - \mu(0) < c$.
2. *The simple modification:* M^c *is a strong deformation retract of* $F^{-1}(-\infty, c]$ `Chapter2/CriticalLevel · isStrongDeformationRetract_level`. It is the deformation of §II.10 with the level $\delta = 0$, while the chart, μ and the ellipsoid keep their ϵ. In coordinates it is $(x, y) \mapsto (x, s\, y)$ with $s = 1 - t + t\,(\xi/\eta)^{1/2}$ where $\xi \leq \eta$ and $\eta > 0$, and the identity where $\eta \leq \xi$, that is on M^c. All the lemmas of §II.10 hold for $\delta = 0$. In particular r_1 maps $F^{-1}(-\infty, c]$ into M^c together with the cell of radius 0; that cell, $\{\xi \leq 0,\ y = 0\}$, is the single point p, at which $f = c$. So r_1 lands in M^c, and M^c is fixed throughout. The same deformation in coordinates is `Chapter2/CellModel · CellModel.isStrongDeformationRetract_model_level`, with r_1 landing in $\{f \leq c\}$ `Chapter2/CellModel · CellModel.morseQuad_retractModel_zero_one_le`.
3. M^c *is a strong deformation retract of* $M^{c+\epsilon}$ `Chapter2/CriticalLevel · isStrongDeformationRetract_sublevel_crit`: Lemma A(i) for $M^c \subseteq F^{-1}(-\infty, c] \subseteq M^{c+\epsilon}$, composing steps 1 and 2.
4. $M^{c-\epsilon} \cup e^\lambda$ *is a strong deformation retract of* M^c `Chapter2/CriticalLevel · isStrongDeformationRetract_union_cellE_crit`. Apply Lemma A(ii) (§II.1) with $A = M^{c-\epsilon} \cup e^\lambda$, $B = M^c$ and $C = M^{c+\epsilon}$: A is a strong deformation retract of C by Theorem 3.2 (§II.10), B is one of C by step 3, and $A \subseteq B$ because $f = c - \xi \leq c$ on the cell. Milnor's "easily" is exactly Lemma A(ii), which uses the retraction of $M^{c+\epsilon}$ onto M^c.

Remark 3.4 (V2.10). Under the hypotheses of Theorem 3.2 on E, there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$: M^c is a strong deformation retract of $M^{c+\epsilon}$; and there is a map χ from D^λ to E, continuous and injective on D^λ, with $\chi(z) \in M^{c-\epsilon}$ exactly when $|z| = 1$, such that $M^{c-\epsilon} \cup \chi(D^\lambda)$ is a strong deformation retract of M^c `Chapter2/CriticalLevel · remark_3_4`. The proof assembles steps 3 and 4 with the chart, the choice of ϵ_1 and the parametrization of §II.10. The whole cell lies in M^c; the proof uses this, but the formal statement does not record it. *Narrower than the book:* only on E. On submanifolds the Remark is not formalized, and with it the Remark of p. 24 at a critical value (§II.16).

II.13 Theorem 3.2 on a submanifold, through one chart

The proof of Theorem 3.2 has two kinds of steps. Assertions 2 and 3 use the smooth structure: critical points of F, and Theorem 3.1 applied to F. Everything else (the definition of F, Assertion 1, the cell, the deformation of Assertion 4 and Lemma D) takes place in the coordinate neighbourhood of p and uses only topology. The formal proof first redoes the topological part in a coordinate patch of an arbitrary Hausdorff space, then supplies Assertions 2 and 3 on a submanifold.

Morse patches. Let M be a topological space, $f : M \to \mathbb{R}$ and $p \in M$. A **Morse patch** at p is an open partial homeomorphism φ from M to $X \times Y$ with p in its domain, $\varphi(p) = 0$, and $f = f(p) - \xi \circ \varphi + \eta \circ \varphi$ on its domain `Chapter2/CellPatch · IsMorsePatch`. No smoothness is required. The constructions of §II.10 carry over word for word: the correction `Chapter2/CellPatch · patchBump`, Milnor's F `Chapter2/CellPatch · patchF`, the ellipsoid `Chapter2/CellPatch · patchEllipsoid` and the cell `Chapter2/CellPatch · patchCell` in M, and the deformation, $\varphi^{-1} \circ r_t \circ \varphi$ on the patch and the identity elsewhere `Chapter2/CellPatch · patchRetract`. So do the proofs, with E replaced by M: Assertion 1 `Chapter2/CellPatch · sublevel_patchF_add`; the cell lies below $c - \epsilon$ for F `Chapter2/CellPatch · patchCell_subset_sublevel_patchF`; on $F^{-1}(-\infty, c - \delta]$ the deformation never leaves the patch `Chapter2/CellPatch · retractModel_mem_target_of_patchF_le`; Assertion 4 `Chapter2/CellPatch · isStrongDeformationRetract_patch_handle`; the choice of ϵ `Chapter2/CellPatch · exists_forall_ball_subset_patch_target`; and the parametrization of the cell `Chapter2/CellPatch · patchCell_parametrization`.

Hausdorffness is used twice. The ellipsoid in M is compact, as the image of a compact set under φ^{-1} `Chapter2/CellPatch · isCompact_patchEllipsoid`, and closed when M is Hausdorff `Chapter2/CellPatch · isClosed_`

`patchEllipsoid`; that makes the deformation continuous `Chapter2/CellPatch · continuous_patchRetract`, and, on a manifold, F smooth. And Lemma D needs a Hausdorff space. A manifold in Milnor's sense, a subset of some $\mathbb{R}^K$, is Hausdorff; an abstract manifold need not be.

Theorem 3.2 in a patch, given Assertion 3 (V2.8 (i)). Let M be Hausdorff, φ a Morse patch at p whose image contains $\{\xi + \eta \le 2\epsilon\}$, $\epsilon > 0$, μ a function with Milnor's four properties, and ι a linear isometry from X onto $\mathbb{R}^k$. Suppose that $F^{-1}(-\infty, c-\epsilon]$ is a strong deformation retract of $M^{c+\epsilon}$. Then $\chi(z) = \varphi^{-1}(\sqrt{\epsilon}\,\iota^{-1}z, 0)$ is continuous and injective on D^k, meets $M^{c-\epsilon}$ exactly in $\chi(S^{k-1})$, lies in $M^{c+\epsilon}$, and $M^{c-\epsilon} \cup \chi(D^k)$ is a strong deformation retract of $M^{c+\epsilon}$ `Chapter2/CellPatch · patch_theorem_3_2`. The proof is that of §II.10: Lemma A(i) composes the hypothesis with Assertion 4.

On a submanifold. Now $N \subseteq V$ is a slice submanifold, $F : V \to \mathbb{R}$ is smooth and $f = F|N$, so f is smooth on N `Chapter1/ManifoldSubmanifold · contMDiff_comp_val`. Three things remain to be supplied: a Morse patch of N, the smoothness of Milnor's F on N, and Assertion 3.

The Morse chart of a manifold. Milnor applies Lemma 2.2 in an unspecified coordinate system. The formal proof applies the lemma of Morse on the manifold N (Part I) `Chapter1/ManifoldMorse · morse_lemmaM` inside the domain of one slice chart σ of N, and then splits the coordinates. The result is a Morse patch ψ of f at p, with values in $\mathbb{R}^\lambda \times \mathbb{R}^{n-\lambda}$, where λ is the index of f at p `Chapter1/Manifold · indexM` and p is a non-degenerate critical point in the sense of §0.2 `Chapter1/Manifold · IsNondegenerateCriticalPointM`. The patch is a smooth coordinate system of N `Chapter1/Manifold · IsSmoothCoord` whose domain lies inside the domain of σ `Chapter2/CellAttachSubmanifold · exists_isMorsePatch_submanifold`. So the whole construction takes place where N is the graph described by σ.

F is smooth on N `Chapter2/CellAttachSubmanifold · contMDiff_patchBump`. On the domain of ψ the correction is $\mu(\xi + 2\eta) \circ \psi$, smooth because ψ is a smooth coordinate system; on the open complement of the closed ellipsoid it is 0.

Assertion 2 on N `Chapter2/CellAttachSubmanifold · isCriticalPointM_patchF_iff`. In a smooth coordinate system, the critical points of a function on N are those of the function read in the coordinates `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff`. Read in ψ, Milnor's F and f are the model functions of §II.8 near $\psi(q)$, and the only critical point of either is the origin. This transports Milnor's computation of dF in the coordinates $u^1, \dots, u^n$ (p. 17). Outside the domain of ψ, and hence outside the closed ellipsoid, $F = f$ near q, and their manifold derivatives agree.

Extending F to V. Milnor's Theorem 3.1 is intrinsic. The formal Theorem 3.1 on a submanifold is for restrictions of smooth functions on V (§II.5), so Assertion 3 needs Milnor's F on N to be such a restriction, a step Milnor does not need.

Lemma. Let $b : N \to \mathbb{R}$ be smooth and zero outside a compact set $K \subseteq N$ contained in the domain of a slice chart σ. Then b is the restriction of a smooth function B on V `Chapter2/CellAttachSubmanifold · exists_contDiff_comp_val_eq_of_chart`.

Proof. Let O be the set of points x of the domain of σ such that $(\sigma(x)_Z, 0)$ lies in the image of σ, where $\sigma(x)_Z$ is the Z-component; O is open by continuity, and it contains K, because for $q \in N$ in the domain of σ the point $\sigma(q)$ is itself $(\sigma(q)_Z, 0)$ `Milnor/Chapter2/Submanifold · SliceChart.mk_zero_mem_target`. On O the function $g(x) = b(\sigma_N^{-1}(\sigma(x)_Z))$ is defined, where σ_N is the chart of N induced by σ `Milnor/Chapter2/Submanifold · SliceChart.toChart`. It is smooth: b is smooth, σ_N^{-1} is smooth because σ_N belongs to the maximal atlas of N `Chapter1/ManifoldSubmanifold · toChart_mem_maximalAtlas`, and $x \mapsto \sigma(x)_Z$ is smooth. Lemma B of §II.4 with $g = 1$ gives a smooth cut-off χ with closed support in O and $\chi = 1$ on K. Then $B = \chi\, g$ is smooth on V by the product lemma. On N: for q in the domain of σ, $\sigma_N^{-1}(\sigma(q)_Z) = q$, so $B(q) = \chi(q)\, b(q)$, which is $b(q)$ if $q \in K$ and $0 = b(q)$ if $q \notin K$; for q outside the domain, $q \notin K$ and $\chi(q) = 0 = b(q)$. □

Theorem 3.2 on a submanifold (V2.8 (ii)). Let V be a finite-dimensional real normed space, $N \subseteq V$ a slice submanifold modelled on a finite-dimensional Z, $F : V \to \mathbb{R}$ smooth, $f = F|N$, $p \in N$ a non-degenerate critical point of f of index λ, and $c = f(p)$. Suppose that for some $\epsilon_0 > 0$ the set

$f^{-1}[c-\epsilon_0, c+\epsilon_0] \subseteq N$ is compact and contains no critical point of f other than p. Then there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$ there is a map $\chi : D^\lambda \to N$, continuous and injective on D^λ, meeting $N^{c-\epsilon}$ exactly in $\chi(S^{\lambda-1})$ and contained in $N^{c+\epsilon}$, such that $N^{c-\epsilon} \cup \chi(D^\lambda)$ is a strong deformation retract of $N^{c+\epsilon}$ `Chapter2/CellAttachSubmanifold · theorem_3_2_submanifold`.

Proof.

1. Take the Morse patch ψ inside a slice chart σ, and let $\epsilon_1 = \min(\epsilon_0, \epsilon_2)$, where ϵ_2 is given by the ball condition for ψ. Fix $\epsilon \in (0, \epsilon_1]$ and let μ be the function of §II.7 for ϵ.
2. The correction of ψ vanishes outside the ellipsoid, which is compact and lies in the domain of ψ, hence of σ. By the lemma it is the restriction of a smooth B on V. So $G = F - B$ is smooth on V, and $G|N$ is Milnor's F on N.
3. By Assertion 2 on N, $G|N$ and f have the same critical points.
4. Assertion 3: $(G|N)^{-1}[c-\epsilon, c+\epsilon]$ lies in $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ ($G \geq c-\epsilon$ gives $f \geq c-\epsilon$, and $G \leq c+\epsilon$ gives $f \leq c+\epsilon$ by Assertion 1), so it is compact; a critical point of $G|N$ in it would be p, which lies at the level $c - \mu(0) < c - \epsilon$. Theorem 3.1 on N (§II.5), applied to G, makes $(G|N)^{-1}(-\infty, c-\epsilon]$ a strong deformation retract of $(G|N)^{-1}(-\infty, c+\epsilon]$, which is $N^{c+\epsilon}$ by Assertion 1.
5. Theorem 3.2 in a patch, for the Hausdorff space N, with ι the renumbering isometry, which exists because the index is at most the dimension `Chapter1/Manifold · indexM_le_finrank`, gives the conclusion. □

With Lemma D in a Hausdorff space (§II.11), $N^{c+\epsilon}$ is homotopy equivalent to $N^{c-\epsilon} \cup_\varphi e^\lambda$ for an injective $\varphi : S^{\lambda-1} \to N^{c-\epsilon}$ `Chapter2/CellAttachSubmanifold · theorem_3_2_attach_submanifold`. *Narrower than the book:* f is the restriction of a smooth function on V, and abstract manifolds are not treated.

II.14 Remark 3.3: several critical points on one level

Milnor's Remark 3.3 (p. 19) says that if $f^{-1}(c)$ contains k non-degenerate critical points $p_1, \dots, p_k$, of indices $\lambda_1, \dots, \lambda_k$, a similar proof shows that $M^{c+\epsilon}$ has the homotopy type of $M^{c-\epsilon} \cup e^{\lambda_1} \cup \cdots \cup e^{\lambda_k}$. The hypotheses of Theorem 3.2 are understood: $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ is compact and the p_i are all its critical points. The exposition proves it by redoing Assertions 1–4 chart by chart for $F = f - \sum_i \mu(\xi_i + 2\eta_i)$. The formal proof redoes nothing: it subtracts the bumps one at a time and applies the single-patch statements of §II.13 at each step.

Extending a deformation by the identity. If $A \subseteq B$ is a strong deformation retract of B, B and C are closed, and $B \cap C \subseteq A$, then $A \cup C$ is a strong deformation retract of $B \cup C$ `Chapter2/SeveralCritical · IsStrongDeformationRetract.union_of_isClosed`. The deformation H of B, extended by the identity on C, is well defined because the two agree on $B \cap C \subseteq A$, where H fixes every point; it is continuous by the pasting lemma for the closed cover of $[0,1] \times (B \cup C)$ by $[0,1] \times B$ and $[0,1] \times C$; its end map takes values in $A \cup C$; and it fixes $A \cup C$. A set which is the whole space is trivially a strong deformation retract `Chapter2/SeveralCritical · isStrongDeformationRetract_of_eq_univ`.

The functions F_s. Given smooth functions B_i on V, the extensions of the corrections of the patches, put $F_s = F - \sum_{i \in s} B_i$ for every finite set s of indices `Chapter2/SeveralCritical · multiF`; it is smooth `Chapter2/SeveralCritical · contDiff_multiF`.

Lemma (Assertions 1, 2 and 4 for several patches). Let ψ_i be Morse patches of f at points p_i with $f(p_i) = c$, smooth coordinate systems of N with pairwise disjoint domains, each image containing $\{\xi + \eta \leq 2\epsilon\}$; let μ have Milnor's properties; and let B_i be smooth on V with $B_i|N$ the correction of ψ_i. Then for every finite set s: (1) $(F_s|N)^{-1}(-\infty, c+\epsilon] = N^{c+\epsilon}$; (2) $F_s|N$ and f have the same critical points; (3) $F_s(p_i) = c - \mu(0)$ for $i \in s$; (4) the cells e_i, $i \in s$, lie in $(F_s|N)^{-1}(-\infty, c-\epsilon]$; (5) $N^{c-\epsilon} \cup \bigcup_{i \in s} e_i$ is a strong deformation retract of $(F_s|N)^{-1}(-\infty, c-\epsilon]$ `Chapter2/SeveralCritical · multi_induction`.

Proof, by induction on s. For $s = \emptyset$, $F_\emptyset = F$, and (5) holds because $N^{c-\epsilon}$ is the whole space $F^{-1}(-\infty, c-\epsilon]$. Suppose the claims hold for s, let $i \notin s$, and write $g = F_s|N$.

- Every B_j, $j \in s$, vanishes on the domain of ψ_i, by disjointness; so $g = f$ there. Hence ψ_i is a Morse patch of g at p_i with $g(p_i) = c$, and $F_{s\cup\{i\}}|N$ is Milnor's modification of g in the patch ψ_i.
- (1) is Assertion 1 for g and ψ_i, followed by (1) for s. (2) is Assertion 2 on N for g and ψ_i, followed by (2) for s. (3): at p_i the new value is $g(p_i) - \mu(0) = c - \mu(0)$; a point p_j with $j \in s$ lies outside the domain of ψ_i, where the new correction vanishes, so (3) for s applies. (4): the new cell lies below $c - \epsilon$ for the modification of g; the old cells lie below $c - \epsilon$ for g, hence for the smaller function $F_{s\cup\{i\}}$.
- (5) in two steps. First, extend the deformation of claim (5) for s by the identity over the new cell e_i. This is allowed: $g^{-1}(-\infty, c-\epsilon]$ is closed, e_i is compact, hence closed, and a point of e_i with $g \le c - \epsilon$ lies in $N^{c-\epsilon}$, because $g = f$ on the domain of ψ_i. So $N^{c-\epsilon} \cup \bigcup_{j\in s} e_j \cup e_i$ is a strong deformation retract of $g^{-1}(-\infty, c-\epsilon] \cup e_i$. Second, Assertion 4 in the patch ψ_i for the function g: $g^{-1}(-\infty, c-\epsilon] \cup e_i$ is a strong deformation retract of $(F_{s\cup\{i\}}|N)^{-1}(-\infty, c-\epsilon]$. Lemma A(i) composes the two. □

Remark 3.3 (V2.9). Let $N \subseteq V$, F and $f = F|N$ be as in §II.13, and S a finite set of non-degenerate critical points of f, all at the level c. Suppose that for some $\epsilon_0 > 0$ the set $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ is compact and all its critical points lie in S. Then there is $\epsilon_1 > 0$ such that for every $\epsilon \in (0, \epsilon_1]$ there are maps $\chi_p : D^{\lambda_p} \to N$, $p \in S$, with λ_p the index of p, each continuous and injective on the ball, with $\chi_p(z) \in N^{c-\epsilon}$ exactly when $|z| = 1$ and $\chi_p(D^{\lambda_p}) \subseteq N^{c+\epsilon}$, the cells pairwise disjoint, such that $N^{c-\epsilon} \cup \bigcup_p \chi_p(D^{\lambda_p})$ is a strong deformation retract of $N^{c+\epsilon}$ `Chapter2/SeveralCritical · remark_3_3_submanifold`.

Proof.

1. *Disjoint Morse charts.* N is Hausdorff, so the finite set S has pairwise disjoint open neighbourhoods U_p `Mathlib/Topology/Separation/Hausdorff · Set.Finite.t2_separation`. Morse patches can be taken inside given open neighbourhoods `Chapter2/SeveralCritical · exists_isMorsePatch_submanifold_subset`, so patches ψ_p with domains inside the U_p have pairwise disjoint domains. Milnor's "similar proof" needs these disjoint charts, hence Hausdorffness.
2. *One ϵ for all patches:* ϵ_1 is the minimum of ϵ_0 and the radii given by the ball condition for the finitely many patches.
3. Extend each correction to a smooth B_p on V (the lemma of §II.13), apply the multi-patch lemma with $s = S$, and let $G = F_S$, smooth on V with $G \le F$ on N.
4. *Assertion 3:* $(G|N)^{-1}[c-\epsilon, c+\epsilon]$ lies in $f^{-1}[c-\epsilon_0, c+\epsilon_0]$, so it is compact; a critical point of $G|N$ in it is a critical point of f by claim (2), hence a point of S, but there $G = c - \mu(0) < c - \epsilon$ by claim (3). Theorem 3.1 on N applied to G, with claim (1), makes $(G|N)^{-1}(-\infty, c-\epsilon]$ a strong deformation retract of $N^{c+\epsilon}$.
5. The cells are parametrized as before `Chapter2/SeveralCritical · patchCell_props`; they are pairwise disjoint because each lies in the domain of its own patch. Lemma A(i) composes step 4 with claim (5). □

Narrower than the book: only on slice submanifolds. That the union is $N^{c-\epsilon}$ with the cells attached one at a time is part of the proof of Theorem 3.5 (§II.16).

II.15 Lemmas 3.6 and 3.7: attaching along homotopic maps, and to equivalent spaces

The attaching space. For a space X and a continuous $\varphi : S^{n-1} \to X$, the space $X \cup_\varphi e^n$ is the pushout of $X \leftarrow S^{n-1} \to D^n$ `Chapter2/Attaching · AttachCell` (§0.5), with the inclusion of X `Chapter2/Attaching · cellInl`

and the characteristic map of the cell Chapter2/Attaching · cellInr, which agree on the sphere: the point $u \in S^{n-1}$ of the cell is the point $\varphi(u)$ of X Chapter2/Attaching · cellInl_apply_eq.

Maps and homotopies out of $X \cup_\varphi e^n$. Milnor uses two facts silently. (Q1) A map given on X and on the cell, compatibly on S^{n-1}, is a continuous map out of $X \cup_\varphi e^n$ Chapter2/Attaching · cellDesc; this is the universal property of the pushout, and two maps agreeing on X and on the cell are equal Chapter2/Attaching · attachCell_hom_ext. (Q2) A *homotopy* $[0,1]\times(X\cup_\varphi e^n) \to Z$ given on $[0,1]\times X$ and on $[0,1]\times D^n$, compatibly, is continuous too Chapter2/Attaching · cellHomotopy. This is not the universal property: it needs the product of $[0,1]$ with a quotient map to be a quotient map, which holds because $[0,1]$ is locally compact (a lemma of J. H. C. Whitehead). The formal proof uses the exponential law instead: because $[0,1]$ is locally compact, a continuous map $[0,1] \times P \to Z$ is the same as a continuous map from P to the space $C([0,1], Z)$ Mathlib/Topology/CompactOpen · ContinuousMap.curry. The two pieces of the homotopy become maps $X \to C([0,1], Z)$ and $D^n \to C([0,1], Z)$, compatible on S^{n-1}; they glue by (Q1), and uncurrying gives the homotopy. Its values at times 0 and 1 are checked on X and on the cell.

Milnor's formulas on the cell. Write a point of D^n as $z = tu$, with $t = |z|$ and $u = z/|z|$. All Milnor's maps and homotopies have the same shape on the cell: for a radius $a \in (0,1]$ and a map Ψ with values in X such that $\Psi(a,u) = \varphi(u)$,

$$tu \;\mapsto\; \text{the point } (t/a)\,u \text{ of the cell, for } t \le a; \qquad tu \;\mapsto\; \Psi(t,u) \in X, \text{ for } t \ge a,$$

with a and Ψ possibly depending on a parameter Chapter2/Attaching · radial. The formal definition uses the first formula for $t \le a$ and the second for $t > a$; at $t = a$ the two agree, because $\Psi(a,u) = \varphi(u)$ is the boundary point u Chapter2/Attaching · radial_of_le Chapter2/Attaching · radial_of_ge. On the boundary sphere, when $a \le 1$, the formula is $\Psi(1,u)$ Chapter2/Attaching · radial_inclusion. It is jointly continuous in the parameter and the point when a and Ψ are Chapter2/Attaching · continuous_radial: each piece is continuous on its closed set, $\{t \le a\}$ or $\{t \ge a\}$, where $u = z/|z|$ is continuous because $t \ge a > 0$, and they agree on $t = a$, so the pasting lemma applies Chapter2/Attaching · continuous_dite_le. Two instances with the same a and Ψ agree Chapter2/Attaching · radial_congr, and with $a = 1$ the formula is the identity of the cell Chapter2/Attaching · radial_of_eq_one. The case $n = 0$, where S^{-1} is empty, needs no separate treatment. Homotopy parameters given by formulas are clamped to $[0,1]$ Chapter2/Attaching · clampI Mathlib/Order/Interval/Set/ProjIcc · Set.projIcc; in every use the formula already lies in $[0,1]$.

Lemma 3.6 (V2.13). If $\varphi_0, \varphi_1 : S^{n-1} \to X$ are homotopic, there is a homotopy equivalence $k : X \cup_{\varphi_0} e^n \to X \cup_{\varphi_1} e^n$ which is the identity on X Chapter2/Attaching · lemma_3_6.

Proof. Let Φ be a homotopy from φ_0 to φ_1, and $\varphi_s = \Phi(s,\cdot)$.

1. *Milnor's k* (p. 21) is the identity on X and, on the cell, $k(tu) = 2tu$ for $t \le \frac12$ and $k(tu) = \varphi_{2-2t}(u)$ for $t \ge \frac12$ Chapter2/Attaching · lemma36Map: the formula above with $a = \frac12$ and $\Psi(t,u) = \varphi_{2-2t}(u)$ Chapter2/Attaching · kOuter Chapter2/Attaching · kCell. At $t = \frac12$, $\Psi = \varphi_1(u)$ Chapter2/Attaching · kOuter_half, which is identified with u in the target. At $t = 1$, $\Psi = \varphi_0(u)$ Chapter2/Attaching · kOuter_one, as the identification $u \sim \varphi_0(u)$ in the source requires. So (Q1) applies, and k is the identity on X Chapter2/Attaching · lemma36Map_inl.
2. *Milnor's ℓ*, "defined by similar formulas", is formally k for the reversed homotopy, from φ_1 to φ_0. Then ℓk is the identity on X, and on the cell Chapter2/Attaching · lemma36_zero
$$\ell k(tu) = 4tu\ (t \le \tfrac14), \qquad \varphi_{4t-1}(u)\ (\tfrac14 \le t \le \tfrac12), \qquad \varphi_{2-2t}(u)\ (t \ge \tfrac12).$$
Along each radius, the outer three quarters run out along the homotopy to φ_1 and back.
3. *ℓk is homotopic to the identity*, which Milnor calls not difficult to verify. The homotopy H_τ, $\tau \in [0,1]$, is the identity on X and, on the cell, the radial formula with $a = (1+3\tau)/4$ Chapter2/Attaching · aH and Chapter2/Attaching · hOuter
$$\Psi(t,u) = \varphi_{\min(4(t-a),\ 2-2t)}(u).$$

The two terms of the minimum cross at $t = (1+\tau)/2$, where both equal $1-\tau$: so along each radius the subscript rises from 0 to $1-\tau$ and returns to 0, and the excursion shrinks as τ grows. The checks: $\Psi(a,u) = \varphi_0(u)$ `Chapter2/Attaching · hOuter_aH`, because $\min(0, 2-2a) = 0$ for $a \le 1$; on the boundary sphere $\Psi(1,u) = \varphi_0(u)$ `Chapter2/Attaching · hOuter_one`, which is the identity of X at $\varphi_0(u)$ under $u \sim \varphi_0(u)$; and Ψ is continuous `Chapter2/Attaching · continuous_hOuter`. So (Q2) applies. At $\tau = 0$, $a = \frac{1}{4}$ and the formula is ℓk (step 2); at $\tau = 1$, $a = 1$ and it is the identity. So H is a homotopy from ℓk to the identity `Chapter2/Attaching · lemma36Homotopy`. The exposition writes H_τ in three pieces, with a break at $(1+\tau)/2$; the minimum is the same function.

4. $k\ell$ is homotopic to the identity by step 3 for the reversed homotopy, whose reverse is Φ. So k is a homotopy equivalence with homotopy inverse ℓ. □

Lemma 3.7 (V2.14). Let $\varphi : S^{n-1} \to X$ be continuous and $f : X \to Y$ a homotopy equivalence. The map $F : X \cup_\varphi e^n \to Y \cup_{f\varphi} e^n$ which is f on X and the identity on the cell is a homotopy equivalence `Chapter2/Attaching · lemma_3_7`.

Proof (Milnor's, after Hilton).

1. F is well defined: the point $u \sim \varphi(u)$ goes to $u \sim f\varphi(u)$ `Chapter2/Attaching · attachMap` `Chapter2/Attaching · attachMap_inl` `Chapter2/Attaching · attachMap_inr`. Let g be a homotopy inverse of f, h_t a homotopy from gf to the identity of X, and $G : Y \cup_{f\varphi} e^n \to X \cup_{gf\varphi} e^n$ the analogous map, g on Y and the identity on the cell.
2. $h_t \circ \varphi$ is a homotopy from $gf\varphi$ to φ `Chapter2/Attaching · homotopyAttach`, so Lemma 3.6 provides $k : X \cup_{gf\varphi} e^n \to X \cup_\varphi e^n$. Lemma 3.7 needs the formula for k, not only the statement of Lemma 3.6: Milnor's next step uses the specific definitions of k, G and F (p. 22).
3. *kGF is homotopic to the identity* `Chapter2/Attaching · lemma37Homotopy`. Milnor's homotopy q_τ is h_τ on X and, on the cell, the radial formula with $a = (1+\tau)/2$ `Chapter2/Attaching · aQ` and $\Psi(t,u) = h_{2-2t+\tau}(\varphi(u))$ `Chapter2/Attaching · qOuter`, that is, $tu \mapsto \frac{2}{1+\tau}\,tu$ for $t \le \frac{1+\tau}{2}$ and $tu \mapsto h_{2-2t+\tau}(\varphi(u))$ for $t \ge \frac{1+\tau}{2}$. The checks: $\Psi(a,u) = h_1(\varphi(u)) = \varphi(u)$ `Chapter2/Attaching · qOuter_aQ`; on the sphere $\Psi(1,u) = h_\tau(\varphi(u))$ `Chapter2/Attaching · qOuter_one`, which is q_τ on X at $\varphi(u)$; the subscript $2-2t+\tau$ lies in $[\tau, 1]$. At $\tau = 0$: on X, $h_0 = gf$, which is kGF on X; on the cell, $a = \frac{1}{2}$ and $\Psi(t,u) = h_{2-2t}(\varphi(u))$, which is k's formula for the homotopy $h_t\varphi$, and $kGF = k$ on the cell. At $\tau = 1$: $a = 1$, the identity on the cell, and h_1 is the identity on X.
4. *Milnor's Assertion* (p. 22): if $LF \simeq \mathrm{id}$ and $FR \simeq \mathrm{id}$, then $RF \simeq \mathrm{id}$ `Chapter2/Attaching · homotopic_of_leftInv_rightInv`. Indeed $L \simeq L(FR) = (LF)R \simeq R$, so $RF \simeq LF \simeq \mathrm{id}$, homotopy being compatible with composition on either side.
5. *G has a left homotopy inverse*, by "a similar proof": step 3 for $(Y, f\varphi, g, f)$ in place of (X, φ, f, g), with a homotopy from fg to the identity of Y, gives maps G' and k' with $k'G'G \simeq \mathrm{id}$.
6. Milnor's three steps. k has the left homotopy inverse ℓ (Lemma 3.6) and the right homotopy inverse GF (step 3), so $(GF)k \simeq \mathrm{id}$ by step 4. Then G has the left homotopy inverse $k'G'$ and the right homotopy inverse Fk, so $(Fk)G \simeq \mathrm{id}$. Finally $(kG)F \simeq \mathrm{id}$ by step 3 and $F(kG) = (Fk)G \simeq \mathrm{id}$: F is a homotopy equivalence with homotopy inverse kG. □

II.16 Theorem 3.5: one cell for each critical point

Iterated adjunction spaces (V2.11). A space K is an *iterated adjunction space with cells* $(\lambda_1, \dots, \lambda_k)$ if it is obtained from the empty space by attaching a λ_1-cell, then a λ_2-cell, and so on, each along an arbitrary continuous map of its boundary sphere into the space built so far `Chapter2/CWType · IsCellAttachment`. A space *X has the homotopy type of a finite cell complex with cells* $(\lambda_1, \dots, \lambda_k)$ if it is homotopy equivalent to such a K `Chapter2/CWType · IsFiniteCellComplexType` (§0.5). The empty space has this property with no cells `Chapter2/CWType · IsFiniteCellComplexType.of_isEmpty`, the property is invariant under homotopy equivalence `Chapter2/CWType · IsFiniteCellComplexType.of_homotopyEquiv`, and it survives attaching a cell: if $X \simeq K$ by a homotopy equivalence h, then $X \cup_\varphi e^n \simeq K \cup_{h\varphi} e^n$ by Lemma 3.7 `Chapter2/CWType · IsFiniteCellComplexType.attach`. This is Milnor's appeal to Lemmas 3.6 and 3.7 on p. 23; only Lemma 3.7 is needed, because the attaching

maps are not made cellular.

Why this is not literally a CW complex. Milnor's statement is about a CW complex in J. H. C. Whitehead's sense. The formal statement differs in two ways. First, the attaching maps are arbitrary continuous maps, not cellular ones. Milnor makes them cellular with the cellular approximation theorem: the attaching map of a λ-cell is homotopic to a map into the $(\lambda-1)$-skeleton, and Lemma 3.6 then replaces it. Mathlib has no cellular approximation theorem. Second, the cells are attached in order of increasing critical value, not of dimension. An iterated adjunction space whose cells come in non-decreasing dimension, each attached along a map into the union of the cells of smaller dimension, is a finite CW complex. Without that condition it is still homotopy equivalent to a CW complex with the same number of cells in each dimension, by cellular approximation and Lemmas 3.6 and 3.7, but it need not itself carry a CW structure with these cells: a 1-cell attached after a 2-cell may have its ends inside the 2-cell. So Theorem 3.5 is verified with "CW-complex" replaced by "iterated adjunction space with exactly C_λ cells of dimension λ"; the passage to a CW complex is not formalized.

A minimum is a critical point. If $f(p) \le f(x)$ for every x, then p is a critical point `Chapter2/CWType · isCriticalPointM_of_forall_le`. The function read in the preferred chart at p `Chapter1/Manifold · localRep` takes the value $f(p)$ at the image of p `Chapter1/Manifold · localRep_self` and is nowhere smaller, so its derivative vanishes there `Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.fderiv_eq_zero`, which is criticality `Chapter1/Manifold · isCriticalPointM_iff_fderiv`. Milnor's remark that M^a is vacuous below the first critical value (p. 23) rests on this.

Setting. $N \subseteq V$ is a *compact* slice submanifold modelled on a finite-dimensional Z, $F : V \to \mathbb{R}$ is smooth, $f = F|N$, and every critical point of f is non-degenerate. Then f is a Morse function (§0.2) `Chapter1/Manifold · IsMorseFunction`, and as N is compact it has finitely many critical points `Chapter1/ManifoldMorse · IsMorseFunction.finite_setOf_isCriticalPointM` (Part I).

Passing one critical level. Let p be a non-degenerate critical point, $c = f(p)$, and $a' < c < a$ such that every other critical point has its value below a' or above a. If $N^{a'}$ has the homotopy type of a cell complex with cells $(\lambda_1, \dots, \lambda_j)$, then N^a has that of one with cells $(\lambda_1, \dots, \lambda_j, \lambda)$, λ the index of p `Chapter2/CWType · IsFiniteCellComplexType.sublevel_step`.

Proof. Put $\epsilon_0 = \min(c-a', a-c)$. Then $f^{-1}[c-\epsilon_0, c+\epsilon_0]$ is compact, being closed in the compact N, and contains no critical point other than p. Theorem 3.2 with an attaching space on N (§II.13) gives ϵ_1; take $\epsilon = \min(\epsilon_1, \epsilon_0/2)$, so that $a' < c-\epsilon$ and $c+\epsilon < a$, and an attaching map φ with $N^{c+\epsilon} \simeq N^{c-\epsilon} \cup_\varphi e^\lambda$. No critical value lies in $[a', c-\epsilon]$ or in $[c+\epsilon, a]$, so Theorem 3.1 for compact N (§II.5) makes the inclusions $N^{a'} \to N^{c-\epsilon}$ and $N^{c+\epsilon} \to N^a$ homotopy equivalences. Hence $N^{c-\epsilon}$ has the homotopy type of the complex with cells $(\lambda_1, \dots, \lambda_j)$, $N^{c-\epsilon} \cup_\varphi e^\lambda$ that of the complex with one more cell (attaching), and $N^a \simeq N^{c+\epsilon} \simeq N^{c-\epsilon} \cup_\varphi e^\lambda$. □

Regular sublevel sets, distinct critical values. Suppose moreover that distinct critical points have distinct values, and let a be a regular value. Then the critical points with value below a can be listed without repetition as $p_1, \dots, p_j$ with $f(p_1) < \cdots < f(p_j)$, so that N^a has the homotopy type of a cell complex with cells $(\lambda(p_1), \dots, \lambda(p_j))$ `Chapter2/CWType · isFiniteCellComplexType_sublevel`.

Proof, by strong induction on the number of critical points below a. If there is none, N^a is empty: otherwise the minimum of f on the compact N, attained at a critical point, would be at most a, and not equal to a as a is regular, so below a. The empty space has the homotopy type of the complex without cells. Otherwise, let p be the critical point with the largest value below a, and choose a regular value $a' < f(p)$ above every smaller critical value: the midpoint between $f(p)$ and the largest critical value below it, or $f(p) - 1$ if there is none. Every critical point other than p then lies below a' or above a (values are distinct, and none lies in $(f(p), a]$ by the maximality of p and the regularity of a). There is one critical point fewer below a'; the induction hypothesis applies to $N^{a'}$, and passing one critical level adds the cell of p at the end of the list. □

Theorem 3.5 (V2.12). Let $N \subseteq V$ be a compact slice submanifold, $F : V \to \mathbb{R}$ smooth, and suppose that all critical points of $f = F|N$ are non-degenerate.

- If distinct critical points have distinct values, the critical points can be listed without repetition as $p_1, \dots, p_k$ with $f(p_1) < \cdots < f(p_k)$, so that N has the homotopy type of the iterated adjunction space with cells $(\lambda(p_1), \dots, \lambda(p_k))$ `Chapter2/CWType · theorem_3_5`; in particular N has the homotopy type of one in which, for every λ, the number of cells of dimension λ is the number C_λ of critical points of index λ `Chapter2/CWType · theorem_3_5_count`.
- In general, the critical points can be listed without repetition with $f(p_1) \le \dots \le f(p_k)$, with the same conclusion `Chapter2/CWType · theorem_3_5'`, and in particular with exactly C_λ cells of dimension λ `Chapter2/CWType · theorem_3_5_count'`.

Proof of the first case. f is bounded on the compact N, say by M_0; the value $a = M_0 + 1$ is regular and $N^a = N$, so the previous result applies, and N is homeomorphic to N^a. For the count, the list of critical points has no repetition and contains each critical point exactly once, so the number of entries λ in the list of indices is the number of critical points of index λ. □

The general case replaces Theorem 3.2 by Remark 3.3. Two further steps are needed.

1. *Attaching disjoint cells one at a time* `Chapter2/CWType · isFiniteCellComplexType_union_cells`. Let A be a closed subset of a Hausdorff space, and χ_i, for i in a list, maps from closed unit balls, continuous and injective, with pairwise disjoint images, each meeting A exactly in the image of the unit sphere. If A has the homotopy type of a complex with cells $(\lambda_1, \dots, \lambda_j)$, then $A \cup \bigcup_i \chi_i(D)$ has that of one with the dimensions of the χ_i appended. The proof is by induction on the list. Lemma D (§II.11) makes $A \cup \chi_{i_1}(D)$ the space A with a cell attached, so it has the homotopy type of the complex with one more cell; it is closed, the image of a compact ball being closed; and, because the cells are disjoint, each later χ_i still meets $A \cup \chi_{i_1}(D)$ exactly in the image of its sphere. The induction hypothesis applies to it.
2. *Passing a critical level with several critical points* `Chapter2/CWType · IsFiniteCellComplexType.sublevel_step_finset`. If S is the set of critical points with value c and every other critical point lies below a' or above a, then, for some enumeration of S, N^a has the homotopy type of the complex with the indices of S appended. The proof is that of the single-point step, with Remark 3.3 (§II.14) in place of Theorem 3.2 and step 1 in place of attaching one cell.

The induction over regular values then runs as before, now removing all the critical points of the highest critical value below a at once `Chapter2/CWType · isFiniteCellComplexType_sublevel'`, and a above the maximum of f gives the theorem. □

The Remark on p. 24 (V2.15). Milnor remarks at the end of §3 that each M^a has the homotopy type of a finite complex with one cell of dimension λ for each critical point of index λ in M^a, even when a is a critical value. For a not a critical value this is the induction itself `Chapter2/CWType · isFiniteCellComplexType_sublevel'`, with the critical points below a listed by non-decreasing value. At a critical value it would need Remark 3.4 on submanifolds, which is not formalized.

What Milnor's proof uses that the formal one supplies. The emptiness of M^a below the first critical value needs the minimum to be a critical point. The induction runs over regular values with exactly one critical value between consecutive ones, which needs the finiteness of the set of critical points. The ϵ of Theorem 3.2 must also be smaller than the distances to the neighbouring regular values. *Narrower than the book:* N is compact and f extends to V. The non-compact case, with infinitely many critical points, needs Whitehead's theorem and the fact that a manifold has the homotopy type of a CW complex, neither of which Mathlib has; it is not formalized, and neither is the passage from an iterated adjunction space to a CW complex.

II.17 What the formalization taught us

- **One crossing.** Milnor's "clearly" in Theorem 3.1 rests on the fact that a trajectory crosses the slab $f^{-1}[a, b]$ only once, upwards and at unit speed. It is a continuity argument about one real function (§II.2), and once isolated it yields both inclusions of $\varphi_{b-a}(M^a) = M^b$ and the properties of r_t for any slab flow (§II.3).
- **Two auxiliary functions exist.** The cut-off ρ of Theorem 3.1 comes from a smooth Urysohn function on a compact neighbourhood (§II.4). Milnor's μ of Theorem 3.2 needs a slope bound everywhere, and the obvious candidate violates it; an integrated plateau satisfies it (§II.7).
- **Hausdorffness, used three times.** The ellipsoid $\xi + 2\eta \leq 2\epsilon$ must be closed for F to be smooth and for "the identity outside U" to be continuous (§II.10, §II.13); Lemma D needs the cell to be closed (§II.11); Remark 3.3 needs disjoint Morse charts (§II.14).
- **The deformation of Assertion 4 is one formula**, continuous on the whole coordinate space; it never leaves the chart on $F^{-1}(-\infty, c - \epsilon]$, and the point $\xi = \epsilon$, $\eta = 0$, where Milnor's Case 2 reads $0/0$, is fixed (§II.9).
- **Unstated lemmas.** The composition of strong deformation retractions and its converse with a retraction (Lemma A, §II.1); the identification of $M^{c-\epsilon} \cup e^\lambda$ with an attaching space (Lemma D, §II.11); the fact that $[0, 1]$ times a quotient map is a quotient map, needed for homotopies on attaching spaces (§II.15); and the fact that a global minimum is a critical point (§II.16).
- **Lemma 3.7 uses the proof of Lemma 3.6**, namely the formula for k, not only its statement (§II.15).
- **Small corrections.** The bound in Assertion 2 is $\partial F/\partial \eta \geq 1$, not > 1 (§II.8); the scan's "k-cell" in Theorem 3.2 is a λ-cell (§II.10).
- **On submanifolds** the intrinsic flow is replaced by an ambient flow tangent to N, whose invariance needs a closed set larger than N (§II.5), and Assertion 3 needs Milnor's F to be extended to the ambient space (§II.13).
- **What is not done.** Theorem 3.1 on non-compact abstract manifolds (§II.5); Theorems 3.2 and 3.5 and Remarks 3.3 and 3.4 on abstract manifolds; Remark 3.4 on submanifolds, and with it the Remark of p. 24 at a critical value; Theorem 3.5 for non-compact manifolds; the CW structure, which needs the cellular approximation theorem (§II.16); and the remark on Smale's handles (p. 17), that $M^{c-\epsilon} \cup H$ is diffeomorphic to $M^{c+\epsilon}$, which is Theorem 3.1 for F with its diffeomorphism kept, a step the formal proof of Assertion 3 discards.

Part III. Reeb, the torus, the Euler number and focal points (Milnor §§4–6)

This part retells the formal proofs of Milnor's §4 (Reeb's theorem), of the opening example of §1 (the torus standing on a plane, whose Lean files live with §4 because they use the results of §5), of the half of §5 that needs no homology (the alternating count $\sum_\lambda (-1)^\lambda C_\lambda$ of the critical points of a Morse function is a topological invariant, read through Application 2 of §6), and of the whole of §6: Sard's theorem on manifolds, the normal bundle and its endpoint map, focal points and their multiplicities, principal curvatures, Lemmas 6.3–6.5, Theorem 6.6, Corollaries 6.7 and 6.8, and Lemma 6.9. The homological half of §5, the Morse inequalities in singular homology and the identification of the Euler number with the homological Euler characteristic, is Part IV.

The setting is that of the companion project, the formalization of Milnor's *Topology from the Differentiable Viewpoint* (the companion book below): a manifold is a slice submanifold of a finite-dimensional real normed space, usually of $\mathbb{R}^n$, with its induced charts (§0.1, §0.3), and critical points, Hessians, indices, nullities and Morse functions are the intrinsic notions of §0.2, read in charts. Most proofs are Milnor's, with every step he leaves to the reader carried out. There are five departures. (1) In Reeb's theorem the gluing of the two cells, which Milnor calls easy, is Alexander's trick followed by the pasting lemma, and Theorem 3.1 is re-run to see that the flow also carries the boundary sphere of one cell onto that of the other. (2) Formula (3) of §5 is proved by running Application 2 of §6 backwards: the index sum of the gradient field of a Morse function is $\sum_\lambda (-1)^\lambda C_\lambda$, and by the Poincaré–Hopf theorem of the companion project every tangent field has the same index sum; no homology is used. (3) In §6 no normal frame is chosen: the tangent space of the normal bundle is the kernel of the derivative of a local defining function, and the principal curvatures are the eigenvalues of a self-adjoint shape operator on the tangent space, which removes Milnor's normalization $g_{ij} = \delta_{ij}$. (4) Corollary 6.8 is proved by a linear perturbation $f + \langle a, \cdot \rangle$, composed with a diffeomorphism of $\mathbb{R}^n$ onto its unit ball, instead of Milnor's construction with an embedding whose first coordinate is f; this removes his hypothesis that f be bounded. (5) The results proved for submanifolds are carried to abstract compact Hausdorff manifolds through Mathlib's Whitney embedding, after proving that the image of an embedding is a submanifold diffeomorphic to the manifold, and that every Morse notion is invariant under diffeomorphisms.

Results carry the labels V3.1–V3.25 and V4.1–V4.25 of the verified edition (`Exposition/verification/ch03.md`, `ch04.md`), so that one can move between the two texts; V3.26–V3.40 belong to Part IV.

Map.

- `Chapter3/ReebMorse`: extrema are critical, of index 0 and n; Morse charts at extrema; small sublevel sets are closed balls.
- `Chapter3/ReebGluing`: Alexander's trick; two balls glued along their boundary spheres form a sphere.
- `Chapter3/Reeb`: Theorem 3.1 with the boundary levels; the two critical points are the extrema;

Theorem 4.1.

- `Chapter3/TorusSurface`, `Chapter3/Torus`: the torus as a regular level set; its height function by Lagrange's rule; four critical points of indices $0, 1, 1, 2$; its Euler number.
- `Chapter3/EulerCharAlgebra`: $\operatorname{sgn}\det A = (-1)^{\text{index}}$ for the operator of a non-degenerate form.
- `Chapter3/EulerCharGradient`: the tangential gradient, and its index $(-1)^\lambda$ at a critical point of index λ.
- `Chapter3/EulerChar`, `Chapter3/EulerCharSphere`: $\sum_\lambda (-1)^\lambda C_\lambda$ is the Poincaré–Hopf integer; the sphere.
- `Chapter4/Sard`: Theorem 6.1. `Chapter4/NormalBundle`: the normal bundle, the endpoint map, the fundamental forms. `Chapter4/Focal`: focal points, Corollary 6.2, Assertion 6.4, Lemma 6.5, Theorem 6.6. `Chapter4/Curvature`: the shape operator, Lemma 6.3, Lemma 6.9.
- `Chapter4/Existence`, `Chapter4/Compact`: Corollary 6.7. `Chapter4/Approximation`, `Chapter4/Uniform`: Corollary 6.8.
- `Chapter4/WhitneySlice`, `Chapter4/WhitneyInvariance`, `Chapter4/Whitney`: the bridge to abstract compact manifolds.

III.1 Extrema: Fermat's rule and the indices 0 and n

Reeb's proof opens with a sentence Milnor does not justify: the two critical points must be the minimum and the maximum, and near each of them the lemma of Morse produces a small closed cell. The second half presupposes that the index is 0 at a non-degenerate minimum and n at a non-degenerate maximum; for any other index the set $\{f \le f(p) + \varepsilon\}$ is not a ball near p. Both facts are proved for a manifold M modelled on a real normed space E, finite-dimensional where indices occur, with $n = \dim E$.

Extrema are critical points. Let φ_p be the extended chart at p and $f_p = f \circ \varphi_p^{-1}$ the local representative of f (§0.2) `Chapter1/Manifold · localRep`. If p is a local minimum of f, then $\varphi_p(p)$ is a local minimum of f_p: the inverse chart is continuous at $\varphi_p(p)$ and sends it back to p, so the inequality $f(p) \le f$, valid near p, pulls back to a neighbourhood of $\varphi_p(p)$, where $f_p(\varphi_p(p)) = f(p)$ `Chapter3/ReebMorse · isLocalMin_localRep` `Chapter3/ReebMorse · isLocalMax_localRep`. Mathlib's form of Fermat's rule makes the derivative of f_p vanish there, and a point is critical exactly when this derivative vanishes `Chapter1/Manifold · isCriticalPointM_iff_fderiv`. So local minima and maxima are critical points `Chapter3/ReebMorse · isCriticalPointM_of_isLocalMin` `Chapter3/ReebMorse · isCriticalPointM_of_isLocalMax`. No differentiability is needed: where f_p is not differentiable its formal derivative is 0, and Fermat's rule holds trivially.

The normal form along an axis. Write Milnor's normal form on $\mathbb{R}^n$ as

$$\nu_{c,k}(w) = c - \sum_{i<k} w_i^2 + \sum_{i\ge k} w_i^2$$

`Chapter1/ManifoldMorse · morseNormalForm` (coordinates are numbered from 0, as in the formal text). On the i-th axis, $\nu_{c,k}(s e_i)$ is $c - s^2$ if $i < k$ and $c + s^2$ otherwise `Chapter3/ReebMorse · morseNormalForm_single`. Let y be a chart around p with $y(p) = 0$ and $f = \nu_{f(p),k} \circ y$ on its domain. Then $y^{-1}(s e_i) \to p$ as $s \to 0$, the inverse chart being continuous at $0 = y(p)$; and for all small s the point $s e_i$ lies in the open target of y, so that $f(y^{-1}(s e_i)) = \nu_{f(p),k}(s e_i)$ is $f(p) - s^2$ or $f(p) + s^2$ according to the sign carried by the axis `Chapter3/ReebMorse · tendsto_and_eventually_morse_axis`.

Proposition (V3.1, the index at an extremum). Let M be a smooth manifold without boundary modelled on a finite-dimensional real space E, $n = \dim E$, let f be smooth on an open set $U \ni p$, and let p be a non-degenerate critical point of f. If p is a local minimum, the index of f at p is 0; if it is a local maximum, the index is n `Chapter3/ReebMorse · indexM_eq_zero_of_isLocalMin` `Chapter3/ReebMorse · indexM_eq_finrank_of_isLocalMax`.

Proof as formalized. The lemma of Morse on manifolds (Part I) gives smooth coordinates y around p with values in $\mathbb{R}^n$, domain inside U, $y(p) = 0$ and $f = \nu_{f(p),\lambda} \circ y$, λ the index of f at p `Chapter1/ManifoldMorse · morse_lemmaM`. Suppose that p is a local minimum and $\lambda \ne 0$. As $\lambda \le n$ `Chapter1/`

`Manifold · indexM_le_finrank`, $n \geq 1$ and the axis $i = 0$ carries a minus sign: along it $f(y^{-1}(se_0)) = f(p) - s^2$ for small s. Minimality, pulled back along the curve $s \mapsto y^{-1}(se_0)$, which tends to p, gives $f(p) \leq f(y^{-1}(se_0))$ for small s. A small $s \neq 0$ exists (the punctured neighbourhoods of 0 form a proper filter), and for it $f(p) \leq f(p) - s^2 < f(p)$, a contradiction. At a local maximum with $\lambda < n$ the last axis $i = n - 1$ carries a plus sign, and $f(p) + s^2 \leq f(p)$ is the contradiction.

Morse charts at the extrema, in Euclidean coordinates (V3.2 (i)). With $\lambda = 0$ the normal form is $f(p) + \sum_i y_i^2$; with $\lambda = n$ it is $f(p) - \sum_i y_i^2$. The lemma of Morse produces coordinates in $\{0, \dots, n-1\} \to \mathbb{R}$, whose Mathlib norm is the maximum norm, so $\sum_i y_i^2$ is not the square of that norm and its sublevel sets are not balls for it. The coordinates are therefore followed by the identification with Euclidean $\mathbb{R}^n$; this is still a smooth coordinate system `Chapter3/ReebMorse · IsSmoothCoord.transHomeomorph_euclidean`, and in it $\sum_i y_i^2 = \|y\|^2$ `Chapter3/ReebMorse · norm_sq_toLp`. The result: at a non-degenerate local minimum there are smooth Euclidean coordinates y with $y(p) = 0$, domain inside U, and $f = f(p) + \|y\|^2$ on that domain; at a non-degenerate local maximum, $f = f(p) - \|y\|^2$ `Chapter3/ReebMorse · exists_morse_chart_of_isLocalMin` `Chapter3/ReebMorse · exists_morse_chart_of_isLocalMax`. These are the two cases of Lemma 2.2 that Milnor's appeal to it requires.

III.2 Small sublevel sets are closed balls

Proposition (V3.2 (ii)). Let X be a compact topological space, F a real normed space, $f : X \to \mathbb{R}$ continuous with a strict global minimum at p (that is, $f(p) < f(x)$ for every $x \neq p$), and y a homeomorphism from an open neighbourhood of p onto an open subset of F with $y(p) = 0$ and $f = f(p) + \|y\|^2$ on its domain. There is $\delta > 0$ such that for every $0 < \varepsilon < \delta$ the map

$$h_\varepsilon(v) = y^{-1}(\sqrt{\varepsilon}\, v)$$

is continuous and injective on the closed unit ball D of F, maps D onto the sublevel set $\{f \leq f(p) + \varepsilon\}$, and maps the unit sphere S onto the level set $\{f = f(p) + \varepsilon\}$ `Chapter3/ReebMorse · exists_image_closedBall_eq_sublevel`.

The lemma of Morse speaks about f near p only. Milnor's proviso that ε be small enough hides that the *whole* sublevel set has to lie inside the Morse chart, and this is where the compactness of X and the uniqueness of the minimum enter.

Proof as formalized.

1. The target of y is open and contains 0, so it contains a ball $B(0, r)$, $r > 0$. Let U be the set of points of the domain of y with $\|y\| < r$; it is open and contains p.
2. There is $\eta > 0$ with $f \geq f(p) + \eta$ off U. If $U = X$, take $\eta = 1$. Otherwise $X \setminus U$ is closed in the compact space X, hence compact and nonempty, and the continuous f attains its minimum on it at some x_1 `Mathlib/Topology/Order/Compact · IsCompact.exists_isMinOn`. As $p \in U$, $x_1 \neq p$, and $\eta = f(x_1) - f(p) > 0$ by the strictness of the minimum.
3. Put $\delta = \min(r^2, \eta)$ and fix $0 < \varepsilon < \delta$, so that $\sqrt{\varepsilon} < r$ and $\varepsilon < \eta$.
4. For $\|v\| \leq 1$, $\|\sqrt{\varepsilon}\, v\| \leq \sqrt{\varepsilon} < r$, so $\sqrt{\varepsilon}\, v$ lies in the target of y and $f(h_\varepsilon(v)) = f(p) + \|\sqrt{\varepsilon}\, v\|^2 = f(p) + \varepsilon\|v\|^2$.
5. Conversely let $f(x) \leq f(p) + \varepsilon$. Then $x \in U$, since off U the value is at least $f(p) + \eta > f(p) + \varepsilon$; in particular x lies in the domain of y. The vector $v = y(x)/\sqrt{\varepsilon}$ satisfies $\varepsilon\|v\|^2 = \|y(x)\|^2 = f(x) - f(p) \leq \varepsilon$, so $v \in D$, and $h_\varepsilon(v) = y^{-1}(y(x)) = x$.
6. Continuity and injectivity on D come from those of y^{-1} on its target, composed with the scaling by $\sqrt{\varepsilon} \neq 0$. By 4 the image of D lies in the sublevel set and by 5 it covers it. On the sphere $\|v\| = 1$, step 4 gives the value $f(p) + \varepsilon$; and if $f(x) = f(p) + \varepsilon$, the vector v of step 5 has $\varepsilon\|v\|^2 = \varepsilon$, hence $\|v\| = 1$.

Applied to $-f$, at a strict maximum q with $f = f(q) - \|y\|^2$, the proposition gives the same description of $\{f \geq f(q) - \varepsilon\}$ and of the level $\{f = f(q) - \varepsilon\}$. Milnor only says that the small sublevel sets are

closed n-cells. The gluing of §III.3 needs more: a parametrization of the cell by the ball that carries the boundary sphere onto the level set, and this is what the proposition provides.

III.3 Two balls make a sphere: Alexander's trick

Milnor ends the proof of Theorem 4.1 by declaring a homeomorphism with S^n easy to construct, and constructs none. The file `Chapter3/ReebGluing` proves the purely topological statement behind that sentence.

Alexander's cone. Let E be a real normed space with unit sphere S, and $g : S \to S$ any map. Its cone is

$$G(x) = \|x\|\, g\!\left(\frac{x}{\|x\|}\right) \quad (x \neq 0), \qquad G(0) = 0$$

`Chapter3/ReebGluing · alexanderCone`. Four identities are proved by direct computation. The cone preserves norms, $\|G(x)\| = \|x\|$, because g takes values in the unit sphere `Chapter3/ReebGluing · norm_alexanderCone`. It extends g, because $x/\|x\| = x$ on S `Chapter3/ReebGluing · alexanderCone_of_mem_sphere`. The cone of a composite is the composite of the cones: for $x \neq 0$ the point $G_{g'}(x)$ is nonzero, its norm being $\|x\|$, and its direction is $g'(x/\|x\|)$, so applying G_g to it gives $\|x\|\, g(g'(x/\|x\|))$ `Chapter3/ReebGluing · alexanderCone_alexanderCone`. And the cone of the identity is the identity `Chapter3/ReebGluing · alexanderCone_id`. If g is continuous, so is G `Chapter3/ReebGluing · continuous_alexanderCone`: on the open set $\{x \neq 0\}$ it is the composite $x \mapsto \|x\|\, g(\|x\|^{-1}x)$ of continuous maps, and at 0 it is continuous *only* because $\|G(x)\| = \|x\| \to 0$.

Alexander's trick (V3.5 (i)). If g is a homeomorphism of S, its cone is a homeomorphism of E, with inverse the cone of g^{-1} `Chapter3/ReebGluing · alexanderConeHomeomorph`. The two inverse identities are the composition rule and the identity rule above. The homeomorphism preserves norms, so it maps the closed unit ball onto itself and the unit sphere onto itself by g. It is in general not differentiable at 0: this is the place in the proof of Reeb's theorem where smoothness is lost, in accordance with Milnor's Remark 2.

Coordinates on the sphere. Write a point of $\mathbb{R}^{n+1} = \mathbb{R}^n \times \mathbb{R}$ by its projection to the first n coordinates and its last coordinate, its height `Chapter3/ReebGluing · hemiProj` `Chapter3/ReebGluing · hemiHeight`, and conversely form the point (v, s) `Chapter3/ReebGluing · hemiLift`. Then $\|z\|^2 = \|\mathrm{proj}\, z\|^2 + \mathrm{height}(z)^2$ `Chapter3/ReebGluing · norm_sq_eq_hemiProj_hemiHeight`. On S^n the projection has norm at most 1 and the height satisfies $\mathrm{height}^2 = 1 - \|\mathrm{proj}\|^2$ `Chapter3/ReebGluing · norm_hemiProj_le` `Chapter3/ReebGluing · hemiHeight_sq`; (v, s) lies on S^n when $s^2 = 1 - \|v\|^2$ `Chapter3/ReebGluing · hemiLift_mem_sphere`; and two points of S^n with the same projection and heights of the same sign are equal, since the heights have the same square `Chapter3/ReebGluing · eq_of_hemiProj_eq_of_sameSign`.

Proposition (V3.5 (ii), two balls glued along their boundary spheres). Let X be a Hausdorff space, $n \geq 0$, D and S the closed unit ball and the unit sphere of Euclidean $\mathbb{R}^n$, and $h_1, h_2 : \mathbb{R}^n \to X$ maps that are continuous and injective on D, such that $h_1(D) \cup h_2(D) = X$, $h_1(D) \cap h_2(D) = h_1(S)$ and $h_2(S) = h_1(S)$. Then X is homeomorphic to the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ `Chapter3/ReebGluing · nonempty_homeomorph_sphere_of_union_closedBall`.

Proof as formalized.

1. *A point of the second ball lying in the first comes from the boundary.* If $x \in D$ and $h_2(x) \in h_1(D)$, then $h_2(x) \in h_1(D) \cap h_2(D) = h_1(S) = h_2(S)$, so $h_2(x) = h_2(w)$ with $w \in S$, and injectivity of h_2 on D gives $x = w \in S$.
2. *The boundary identification.* Since $h_1(S) = h_2(S)$, each $v \in S$ has some $g(v) \in S$ with $h_2(g(v)) = h_1(v)$, and each $w \in S$ some $g'(w) \in S$ with $h_1(g'(w)) = h_2(w)$. Injectivity gives $g \circ g' = \mathrm{id}$ and $g' \circ g = \mathrm{id}$.
3. *It is a homeomorphism of S.* The restrictions of h_1 and h_2 to the compact sphere are continuous injections into a Hausdorff space, hence closed embeddings `Mathlib/Topology/Separation/`

Hausdorff · Continuous.isClosedEmbedding. As $h_2 \circ g = h_1$ is continuous and $h_2|_S$ is an embedding, g is continuous; likewise g'. So g is a homeomorphism of S.

4. *Alexander's trick.* Let G be the cone of g, a homeomorphism of $\mathbb{R}^n$. It preserves norms, so it maps D into D, and on S it is g, so $h_2(G(x)) = h_1(x)$ for $x \in S$.
5. *The map $H : S^n \to X$.* Put $H(z) = h_1(\operatorname{proj} z)$ if height$(z) \geq 0$ and $H(z) = h_2(G(\operatorname{proj} z))$ otherwise. Both formulas are continuous on all of S^n, because the projection of a point of S^n lies in D, where h_1 and h_2 are continuous. Where the height is 0, the projection has norm 1, so lies in S, and the two formulas agree by step 4. The pasting lemma for a condition $u \leq w$ between continuous functions then makes H continuous Mathlib/Topology/Order/OrderClosed · Continuous.if_le.
6. *H is injective.* If both points are in the closed upper hemisphere, injectivity of h_1 on D gives equal projections, and the heights have the same sign, so the points are equal. If z is upper and z' lower with $H(z) = H(z')$, then $h_2(G(\operatorname{proj} z')) \in h_1(D)$, so $G(\operatorname{proj} z') \in S$ by step 1; G preserves norms, so $\|\operatorname{proj} z'\| = 1$ and the height of z' is 0, contradicting that it is negative. If both are lower, injectivity of h_2 on D and of G gives equal projections, and again the heights have the same sign.
7. *H is surjective.* Every point of X is $h_1(v)$ or $h_2(w)$ with $v, w \in D$. The point $h_1(v)$ is $H(v, \sqrt{1 - \|v\|^2})$. If $w \in S$, then $h_2(w) \in h_2(S) = h_1(S)$ is of the first kind. If $\|w\| < 1$, put $v = G^{-1}(w)$; then $\|v\| = \|w\| < 1$, the point $(v, -\sqrt{1 - \|v\|^2})$ has negative height, and H sends it to $h_2(G(v)) = h_2(w)$.
8. A continuous bijection from the compact space S^n to the Hausdorff space X is a homeomorphism Mathlib/Topology/Homeomorph/Lemmas · Continuous.homeoOfEquivCompactToT2, and its inverse is the required homeomorphism $X \cong S^n$.

The argument works for $n = 0$ as well, where D is a point and S is empty.

III.4 Reeb's theorem

Two ingredients remain: the extrema of Step 1, and a version of Theorem 3.1 that also matches the boundary levels.

Slab flows carry levels to levels. Part II proves Theorem 3.1 for a *slab flow*: a continuous flow φ on a space T and a continuous $F : T \to \mathbb{R}$ such that $t \mapsto F(\varphi_t(x))$ has derivative 1 at every time at which its value lies in $[a, b]$ Chapter2/Crossing · IsSlabFlow. For such a flow, $F(\varphi_t(x)) = F(x) + t$ whenever $F(x) \in [a, b]$ and $a - F(x) \leq t \leq b - F(x)$ Chapter2/Crossing · IsSlabFlow.apply_eq_add, and $\varphi_{c'-c}$ carries the sublevel set T^c onto $T^{c'}$ for $a \leq c \leq c' \leq b$ Chapter2/Crossing · IsSlabFlow.image_sublevel. The new statement is that φ_{b-a} carries the level $F^{-1}(a)$ onto $F^{-1}(b)$ Chapter3/Reeb · IsSlabFlow.image_level. If $F(x) = a$, the time $t = b - a$ lies in $[a - F(x), b - F(x)] = [0, b - a]$, so $F(\varphi_{b-a}(x)) = a + (b - a) = b$. Conversely, if $F(y) = b$, the time $a - b$ lies in $[a - b, 0]$, so $F(\varphi_{a-b}(y)) = a$, and $\varphi_{b-a}(\varphi_{a-b}(y)) = \varphi_0(y) = y$ by the group law.

Theorem 3.1 with the boundary levels (V3.4). Let N be a slice submanifold of a finite-dimensional real normed space V, modelled on a finite-dimensional space Z, let $F : V \to \mathbb{R}$ be smooth and $f = F|_N$, and let $a \leq b$ with $f^{-1}[a, b]$ compact and free of critical points of f. There is a smooth diffeomorphism Φ of N with $\Phi(N^a) = N^b$ and $\Phi(f^{-1}(a)) = f^{-1}(b)$ Chapter3/Reeb · exists_diffeomorph_image_sublevel_level.

The *statement* of Theorem 3.1 says nothing about the level $f^{-1}(a)$; its *proof* does, since the flow moves the levels of the slab at unit speed. The formal statement of Theorem 3.1 on submanifolds records only the sublevel sets Chapter2/Submanifold · theorem_3_1_submanifold, so its construction is repeated here, with one more conclusion. *The proof as formalized*, in the steps of Part II: the set $K = f^{-1}[a, b]$, compact in V, carries no critical point, so there is a smooth field Y on V, tangent to N along N, with $dF(Y) > 0$ on K Chapter2/Submanifold · exists_tangent_field. A smooth, compactly supported ρ with support in the union of the slice-chart domains of N Chapter2/Submanifold · chartDomain and $\rho\, dF(Y) = 1$ on K exists Chapter2/Regular · exists_cutoff_mul. The field $X = \rho Y$ is smooth, compactly supported, tangent to N, and satisfies $dF(X) = 1$ on K. Its global flow Flow/Global · flow preserves N Chapter2/Submanifold · flow_mem_of_tangent, and restricted

to N it is a slab flow for f on $[a,b]$. Each time-t map is smooth on N, as the restriction of a smooth map of V `Flow/Smooth · contDiff_flow` with values in N `Milnor/Chapter2/Corestrict · contMDiff_codRestrict`. So $\Phi = \varphi_{b-a}$, with inverse $\varphi_{-(b-a)}$, is a smooth diffeomorphism of N; $\Phi(N^a) = N^b$ by the slab-flow lemma on sublevel sets, and $\Phi(f^{-1}(a)) = f^{-1}(b)$ by the one on levels.

Step 1: the two critical points are the extrema (V3.3). Let M be a compact smooth manifold without boundary modelled on a finite-dimensional E with $n = \dim E \geq 1$, and f smooth with exactly two critical points, both non-degenerate. Then there are p, q with $f(p) < f(q)$, the critical points are exactly p and q, $f(p) < f(x)$ for every $x \neq p$, $f(x) < f(q)$ for every $x \neq q$, the index is 0 at p and n at q `Chapter3/Reeb · exists_min_max_of_ncard_eq_two`.

Proof as formalized. Exactly two means formally that the set of critical points has cardinality 2; in particular M is nonempty. By compactness f attains its minimum at some p and its maximum at some q. Every point where the minimum (maximum) value is attained is a local minimum (maximum), hence critical, and p, q have indices 0 and n (§III.1, with $U = M$). If $f(p) = f(q)$, then p also attains the maximum value, so its index is also n, and $n = 0$, contrary to $n \geq 1$. Hence $f(p) < f(q)$ and $p \neq q$. The set $\{p, q\}$ consists of critical points, has two elements, and the critical set is finite with two elements, so the two sets are equal. Finally, if $x \neq p$ had $f(x) = f(p)$, it would be a minimum, hence critical, hence equal to q, whose value is larger; so $f(p) < f(x)$, and symmetrically $f(x) < f(q)$ for $x \neq q$.

Two hypotheses that Milnor does not state appear here. That the minimum and the maximum are *different* points needs $n \geq 1$ (in dimension 0 every point is critical and f may be constant) and the non-degeneracy, which forbids a point to be at the same time a minimum of index 0 and a maximum of index n. And that *no other point* takes an extreme value is what §III.2 needs.

Theorem 4.1 (Reeb; V3.6). Let N be a compact slice submanifold of a finite-dimensional real normed space V, modelled on a space Z of dimension n, and $f = F|_N$ with $F : V \to \mathbb{R}$ smooth. If f has exactly two critical points, both non-degenerate, then N is homeomorphic to the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ `Chapter3/Reeb · reeb`.

Proof as formalized, for $n \geq 1$ `Chapter3/Reeb · reeb_of_finrank_pos`. The function f is smooth on the compact manifold N `Chapter1/ManifoldSubmanifold · contMDiff_comp_val`.

1. Step 1 gives the strict minimum p and the strict maximum q, of indices 0 and n.
2. §III.1 gives Euclidean Morse coordinates y_p at p with $f = f(p) + \|y_p\|^2$ and y_q at q with $f = f(q) - \|y_q\|^2$. §III.2 applied to f at p gives δ_1, and applied to $-f$ at q (a strict minimum of $-f$, where $-f = -f(q) + \|y_q\|^2$) gives δ_2.
3. Put $\varepsilon = \frac{1}{4}\min(\delta_1, \delta_2, f(q) - f(p))$. Then $0 < \varepsilon < \delta_1, \delta_2$ and $f(p) + \varepsilon \leq f(q) - \varepsilon$. The map $h_0(v) = y_p^{-1}(\sqrt{\varepsilon}\, v)$ carries D onto $\{f \leq f(p) + \varepsilon\}$ and S onto $\{f = f(p) + \varepsilon\}$; the map $h_2(v) = y_q^{-1}(\sqrt{\varepsilon}\, v)$ carries D onto $\{f \geq f(q) - \varepsilon\}$ and S onto $\{f = f(q) - \varepsilon\}$.
4. The slab $f^{-1}[f(p)+\varepsilon, f(q)-\varepsilon]$ is closed in the compact N, hence compact, and contains no critical point, since the only critical values $f(p)$ and $f(q)$ lie outside it. Theorem 3.1 with the boundary levels gives a diffeomorphism Φ with $\Phi\{f \leq f(p)+\varepsilon\} = \{f \leq f(q)-\varepsilon\}$ and $\Phi\{f = f(p)+\varepsilon\} = \{f = f(q) - \varepsilon\}$.
5. Apply the gluing proposition of §III.3 to $h_1 = \Phi \circ h_0$ and h_2: both are continuous and injective on D. The images $h_1(D) = \{f \leq f(q) - \varepsilon\}$ and $h_2(D) = \{f \geq f(q) - \varepsilon\}$ cover N, they meet in $\{f = f(q) - \varepsilon\} = \Phi\{f = f(p) + \varepsilon\} = h_1(S)$, and $h_2(S) = \{f = f(q) - \varepsilon\} = h_1(S)$.

Milnor's normalization $f(p) = 0$, $f(q) = 1$ is not needed, and the sphere is S^n with $n = \dim N$.

Dimension 0 `Chapter3/Reeb · reeb_of_finrank_eq_zero`. Milnor's proof does not apply, since f may be constant, but the conclusion holds for every function $f : N \to \mathbb{R}$. On a 0-dimensional manifold every point is critical, the derivative being a linear map on the zero space; so N has exactly two points $p \neq q$. The gluing proposition with $n = 0$, the constant maps $h_1 \equiv p$, $h_2 \equiv q$, the one-point ball $D = \{0\}$ and the empty sphere S, gives $N \cong S^0$.

Intrinsic functions Chapter3/Reeb · reeb_of_contMDiff. For a compact $N \subseteq \mathbb{R}^{K+1}$ of dimension $d+1$ the theorem holds for every smooth $f : N \to \mathbb{R}$ with exactly two critical points, both non-degenerate: such an f is the restriction of a smooth function on $\mathbb{R}^{K+1}$ (§III.9) Chapter3/EulerChar · exists_contDiff_comp_val_eq, and N is nonempty because it carries critical points. For an abstract compact manifold the theorem is §III.21.

Not formalized. Milnor's Remark 1, which allows the two critical points to be degenerate, needs M. Brown's theorem on monotone unions of open cells, which is not in Mathlib; Remark 2, a manifold homeomorphic but not diffeomorphic to S^7, needs characteristic classes.

III.5 The torus as a regular level set

Milnor's book opens with a torus touching a plane, drawn upright, and its height function f. He gives no equations and does not say why the torus is a smooth compact surface. The formal text fixes the surface. For $0 < r < R$ let

$$T = \{x \in \mathbb{R}^3 : (\sqrt{x_1^2 + x_2^2} - R)^2 + x_3^2 = r^2\},$$

the torus of revolution about the x_3-axis, swept out by the circle of radius r whose centre runs along the circle of radius R in the plane $x_3 = 0$ Chapter3/TorusSurface · torus. (The formal coordinates are numbered $0, 1, 2$; here they are x_1, x_2, x_3.) The function inside the definition is not differentiable on the x_3-axis. Instead of working off the axis, the formal proof uses a quartic polynomial with the same zero set,

$$g(x) = Q(x)^2 - 4R^2(x_1^2 + x_2^2), \qquad Q(x) = \|x\|^2 + R^2 - r^2$$

Chapter3/TorusSurface · quad Chapter3/TorusSurface · poly.

Derivatives. Explicit formulas are proved for $dQ_x = 2\langle x, \cdot\rangle$ Chapter3/TorusSurface · hasFDerivAt_quad, for

$$dg_x(w) = 4(Q - 2R^2)(x_1 w_1 + x_2 w_2) + 4Q\, x_3 w_3$$

Chapter3/TorusSurface · dpoly Chapter3/TorusSurface · hasFDerivAt_poly Chapter3/TorusSurface · dpoly_apply, and for the second derivative $d^2 g_x$ as the derivative of $x \mapsto dg_x$, computed coefficient by coefficient Chapter3/TorusSurface · d2poly Chapter3/TorusSurface · hasFDerivAt_coef Chapter3/TorusSurface · hasFDerivAt_dpoly, so that the Hessian of g is this bilinear form Chapter3/TorusSurface · hessian_poly. The polynomial is smooth Chapter3/TorusSurface · contDiff_poly.

The zero set. With $\rho = \sqrt{x_1^2 + x_2^2}$ one has the factorization

$$g = ((\rho - R)^2 + x_3^2 - r^2)((\rho + R)^2 + x_3^2 - r^2),$$

a polynomial identity once $\rho^2 = x_1^2 + x_2^2$ is used Chapter3/TorusSurface · poly_eq_mul. The second factor is at least $R^2 - r^2 > 0$, so $x \in T$ if and only if $g(x) = 0$ Chapter3/TorusSurface · mem_torus_iff. Also $Q > 0$ everywhere, since $R^2 - r^2 > 0$ Chapter3/TorusSurface · quad_pos.

Lemma (0 is a regular value). If $g(x) = 0$, then $dg_x \neq 0$ Chapter3/TorusSurface · dpoly_ne_zero.

Proof as formalized. Suppose $dg_x = 0$. Its values on the three basis vectors give $(Q - 2R^2)x_1 = 0$, $(Q - 2R^2)x_2 = 0$ and $Qx_3 = 0$; as $Q > 0$, $x_3 = 0$. Then $g(x) = 0$ reads $Q^2 = 4R^2(x_1^2 + x_2^2)$ with $Q = x_1^2 + x_2^2 + R^2 - r^2$. If $Q = 2R^2$, then $4R^4 = 4R^2(x_1^2 + x_2^2)$ forces $x_1^2 + x_2^2 = R^2$, and then $Q = 2R^2 - r^2 = 2R^2$ forces $r = 0$, which is excluded. Otherwise $x_1 = x_2 = 0$, so $Q^2 = 0$, contradicting $Q > 0$.

Proposition (V3.22). T is a 2-dimensional slice submanifold of $\mathbb{R}^3$ (slice charts onto $\mathbb{R}^2 \times \mathbb{R}^1$), compact and nonempty Chapter3/TorusSurface · isSliceSubmanifold_torus Chapter3/TorusSurface · isCompact_torus Chapter3/TorusSurface · nonempty_torus.

Proof as formalized. The level set theorem of the companion project wants a map into its model space, so g is read with values in $\mathbb{R}^1$ `Chapter3/TorusSurface · poly1`; its preimage of 0 is T `Chapter3/TorusSurface · preimage_poly1`. At each point of T its derivative is dg_x followed by the isomorphism $\mathbb{R} \cong \mathbb{R}^1$, and a nonzero linear functional is onto, so 0 is a regular value `Chapter3/TorusSurface · isRegularValue_poly1`. Milnor's Lemma 1 of the companion book, that the preimage of a regular value is a submanifold of dimension $3-1$, gives the slice charts `Milnor/Chapter2/LevelSet · isSliceSubmanifold_preimage`. T is closed, as the preimage of a point `Chapter3/TorusSurface · isClosed_torus`, and bounded: on T, $\rho \le R + r$ and $x_3^2 \le r^2$, so $\|x\|^2 = \rho^2 + x_3^2 \le (R+2r)^2$ `Chapter3/TorusSurface · norm_le_of_mem_torus`. Closed and bounded in $\mathbb{R}^3$, it is compact. The point $(R+r, 0, 0)$ lies on it.

Points on the axis. The point $(a,0,0)$ of the x_1-axis `Chapter3/TorusSurface · onAxis` lies on T exactly when $(|a| - R)^2 = r^2$, in particular for $|a| = R \pm r$ `Chapter3/TorusSurface · onAxis_mem_iff` `Chapter3/TorusSurface · onAxis_mem`. There

$$dg = 4a(a^2 - R^2 - r^2)\,dx_1, \qquad d^2g(v,w) = 4(a^2 - R^2 - r^2)\,v_2w_2 + 4(a^2 + R^2 - r^2)\,v_3w_3 \quad (v_1 = 0)$$

`Chapter3/TorusSurface · dpoly_onAxis` `Chapter3/TorusSurface · d2poly_onAxis`.

Proposition (the tangent plane). For $q \in T$ the tangent plane $T_qT \subseteq \mathbb{R}^3$, the range of the derivative of the inclusion (§0.3) `Chapter1/ManifoldSubmanifold · tangentSubspace`, has dimension 2 and equals $\ker dg_q$ `Chapter3/TorusSurface · finrank_tangentSubspace_torus` `Chapter3/TorusSurface · mem_tangentSubspace_torus_iff`.

Proof as formalized. The derivative of the inclusion of a slice submanifold is injective `Milnor/Chapter2/LevelSet · injective_mfderiv_val`, so its range has dimension 2. Since g vanishes on T, the restriction $g|_T$ is constant and every point is a critical point of it; a point is critical for the restriction of an ambient function exactly when the ambient derivative vanishes on the tangent space `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_forall_tangent`. So $T_qT \subseteq \ker dg_q$. The kernel has dimension $3 - 1 = 2$ by rank–nullity, dg_q being a nonzero functional. A subspace contained in another of the same finite dimension is equal to it. This is Lemma 2 of the companion book, re-proved for the set T rather than for the preimage $g^{-1}(0)$.

III.6 The four critical points by Lagrange's rule

Milnor reads off the picture that the critical points of the height function are the four points p, q, r, s marked on it, and that f has the normal forms $x^2 + y^2$, const $+ x^2 - y^2$ (twice) and const $- x^2 - y^2$ there; he verifies none of this. The height is formalized as

$$f(x) = x_1 + R + r,$$

the height above the plane $V = \{x_1 = -(R+r)\}$ `Chapter3/Torus · height`. The axis of revolution is parallel to V, so the torus stands upright, as in Milnor's figure. Lying flat, its height function would have whole circles of degenerate critical points. The four points are

$$p = (-(R+r), 0, 0), \quad q = (-(R-r), 0, 0), \quad r = (R-r, 0, 0), \quad s = (R+r, 0, 0)$$

`Chapter3/Torus · pointP` `Chapter3/Torus · pointQ` `Chapter3/Torus · pointR` `Chapter3/Torus · pointS`, at heights $0, 2r, 2R, 2R + 2r$ `Chapter3/Torus · height_pointP` `Chapter3/Torus · height_pointQ` `Chapter3/Torus · height_pointR` `Chapter3/Torus · height_pointS`. (The letter r is both the tube radius, as in the formal text, and Milnor's name for the third point; the context tells which.) For these four values of a one has $a \neq 0$ and $a^2 - R^2 - r^2 = \pm 2Rr \neq 0$ `Chapter3/Torus · coef_ne_zero`. On T, $|x_1| \le \rho \le R + r$, so $0 \le f \le 2R + 2r$ `Chapter3/Torus · height_mem_Icc`. Hence $T^a = \emptyset$ for $a < 0$ and $T^a = T$ for $a \ge 2R + 2r$ `Chapter3/Torus · sublevel_height_eq_empty` `Chapter3/Torus · sublevel_height_eq_univ`: these are Milnor's cases (1) and (5). The scan prints the hypothesis of (1) as $a < 0 < f(p)$; since V touches T at p, $f(p) = 0$ and it must read $a < 0 = f(p)$. Case (5) holds for $a = f(s)$ as well.

Proposition (V3.23, the critical points). A point of T is a critical point of $f|_T$ if and only if it is one of p, q, r, s `Chapter3/Torus · isCriticalPointM_height_iff` `Chapter3/Torus · isCriticalPointM_height_iff'`.

Proof as formalized. By §III.5, x is critical iff dx_1 vanishes on $T_xT = \ker dg_x$ `Chapter3/Torus · isCriticalPointM_height_iff_forall`. Suppose it does. Two explicit vectors lie in $\ker dg_x$: the rotation vector $(-x_2, x_1, 0)$, on which dg_x gives $4(Q-2R^2)(-x_1x_2+x_2x_1) = 0$, and the vector $(Qx_3, 0, -(Q-2R^2)x_1)$, on which it gives $4(Q-2R^2)x_1Qx_3 - 4Qx_3(Q-2R^2)x_1 = 0$. Their first coordinates must vanish: $x_2 = 0$, and $Qx_3 = 0$, so $x_3 = 0$. The equation of T then reads $(|x_1| - R)^2 = r^2$, so $|x_1| = R \pm r$, and x is one of the four points. Conversely, at $(a,0,0)$ with $a = \pm(R \pm r)$, $dg = \kappa\, dx_1$ with $\kappa = 4a(a^2 - R^2 - r^2) \neq 0$, so $\ker dg = \{w_1 = 0\}$, on which dx_1 vanishes.

Lagrange's rule, as a lemma about Hessians. Two general facts about a slice submanifold N of a normed space are proved first. A critical point of an ambient function G at a point of N is a critical point of $G|_N$ `Chapter3/Torus · isCriticalPointM_comp_val_of_fderiv_eq_zero`. And if $dG_q = 0$ and x is the parametrization of N given by a slice chart at q ($x(u) = \sigma^{-1}(u,0)$; §0.3) `Chapter1/ManifoldSubmanifold · sliceParam`, then

$$\mathrm{Hess}(G \circ x)(v,w) = d^2G_q(dx\, v, dx\, w)\,:$$

the second-order chain rule has a further term $dG_q(d^2x(v,w))$ `Chapter1/Manifold · hessian_comp_apply`, and it vanishes because $dG_q = 0$ `Chapter1/Nondegenerate · hessian_comp` `Chapter3/Torus · hessian_comp_sliceParam_of_fderiv_eq_zero`. For $G = f - \mu g$ with $N = g^{-1}(0)$ and $df_q = \mu\, dg_q$ this computes the Hessian of $f|_N$: it is Lagrange's rule.

Diagonal forms on a plane. On a 2-dimensional space, a symmetric form B with $B(u,u) < 0$ and $B(w,w) > 0$ for some u, w has index 1 `Chapter3/Torus · index_eq_one_of_finrank_eq_two`. Recall that the index is the largest dimension of a subspace on which the form is negative definite (§0.2) `Chapter1/Nondegenerate · index_isGreatest`. A negative definite subspace of dimension 2 would be the whole plane and contain w; and the line through u is negative definite, since $B(cu,cu) = c^2B(u,u) < 0$ for $c \neq 0$. Next, if $B(v,w) = \alpha\,\ell_1(v)\ell_1(w) + \beta\,\ell_2(v)\ell_2(w)$ with $\alpha, \beta \neq 0$, functionals ℓ_1, ℓ_2 that vanish together only at 0, and vectors v_1, v_2 with $\ell_i(v_j) = \delta_{ij}$, then B is non-degenerate and its index is the number of negative coefficients among α, β `Chapter3/Torus · isNondegenerate_and_index_of_diag`. If $B(v,\cdot) = 0$, then $0 = B(v,v_1) = \alpha\ell_1(v)$ and $0 = B(v,v_2) = \beta\ell_2(v)$, so $v = 0$. For the index, $B(v,v) = \alpha\ell_1(v)^2 + \beta\ell_2(v)^2$. With both coefficients negative the form is negative definite and its index is the dimension 2 `Chapter3/EulerCharAlgebra · index_eq_finrank_of_negDef`; with one negative it is 1 by the previous lemma, tested on v_1 and v_2; with both positive it is positive semi-definite and its index is 0 `Chapter3/EulerCharAlgebra · index_eq_zero_of_nonneg`.

Proposition (the Hessian on the axis). Let $(a,0,0) \in T$ with $a \neq 0$ and $a^2 \neq R^2 + r^2$. Then it is a non-degenerate critical point of $f|_T$, of index $[a > 0] + [a(a^2 - R^2 - r^2) > 0]$, where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise `Chapter3/Torus · onAxis_spec`.

Proof as formalized.

1. Put $\kappa = 4a(a^2 - R^2 - r^2) \neq 0$ and $G = f - \kappa^{-1}g$. It agrees with f on T, where $g = 0$. Its derivative is $dx_1 - \kappa^{-1}dg$, which vanishes at $P = (a,0,0)$ because $dg_P = \kappa\, dx_1$. As f is affine, $d^2G_P = -\kappa^{-1}d^2g_P$. So P is a critical point of $G|_T$.
2. Take a slice chart σ at P and let L be the derivative of its parametrization at the point over P. Its range is T_PT `Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam`, which is $\ker dg_P = \{w_1 = 0\}$. It is injective, its range having dimension $2 = \dim \mathbb{R}^2$. Choose v_1, v_2 with $Lv_1 = (0,1,0)$ and $Lv_2 = (0,0,1)$.
3. By Lagrange's rule and the formula for d^2g on the axis, for all v, w,

$$\mathrm{Hess}(G \circ x)(v,w) = -\frac{1}{a}\,(Lv)_2(Lw)_2 \;-\; \frac{a^2 + R^2 - r^2}{a(a^2 - R^2 - r^2)}\,(Lv)_3(Lw)_3.$$

This is a diagonal form with $\ell_1 = (L\,\cdot)_2$ and $\ell_2 = (L\,\cdot)_3$, which vanish together only at 0 because $(Lv)_1 = 0$ and L is injective, and which are dual to v_1, v_2. The coefficients are nonzero, since $a^2 + R^2 - r^2 > 0$. So the form is non-degenerate, and its index counts the negative coefficients: $-1/a < 0$ iff $a > 0$, and the second is negative iff $a(a^2 - R^2 - r^2) > 0$.

4. Non-degeneracy and index of the Hessian of $G \circ x$ at the point over P are those of the Hessian of $G|_T$ at P (§0.2: the Hessian at a critical point may be read in any parametrization) `Chapter1/ManifoldSubmanifold · isNondegenerate_hessian_comp_sliceParam_iff` `Chapter1/ManifoldSubmanifold · index_hessian_comp_sliceParam`, and $G|_T = f|_T$.

At the four points the two diagonal entries are $(1/(R+r), 1/r)$ at p, $(1/(R-r), -1/r)$ at q, $(-1/(R-r), 1/r)$ at r and $(-1/(R+r), -1/r)$ at s: the principal curvatures of the torus there, with signs. Hence p, q, r, s are non-degenerate critical points of indices $0, 1, 1, 2$ `Chapter3/Torus · pointP_spec` `Chapter3/Torus · pointQ_spec` `Chapter3/Torus · pointR_spec` `Chapter3/Torus · pointS_spec`, read off from the signs of a and of $a^2 - R^2 - r^2 = \pm 2Rr$. So $f|_T$ is a Morse function `Chapter3/Torus · isMorseFunction_height`, with critical values $0, 2r, 2R, 2R + 2r$ `Chapter3/Torus · criticalValues_height` and Morse numbers $C_0 = 1$, $C_1 = 2$, $C_2 = 1$, $C_\lambda = 0$ for $\lambda \geq 3$ `Chapter3/Torus · critCount_height`. The count $C_1 = 2$ uses $q \neq r$, whose first coordinates $-(R-r)$ and $R-r$ differ. Milnor's normal forms are the lemma of Morse at these indices; they are not restated.

Theorem 3.1 on the torus (V3.24). If $[a, b]$ contains none of the four critical values, a diffeomorphism of T carries T^a onto T^b, T^a is a strong deformation retract of T^b, and the inclusion is a homotopy equivalence `Chapter3/Torus · theorem_3_1_torus`. This is Theorem 3.1 for a compact submanifold (Part II) `Chapter2/Submanifold · theorem_3_1_submanifold_of_isCompact`, the absence of critical points in $f^{-1}[a, b]$ following from the list of critical points. So the sets T^a with a in one of $(0, 2r)$, $(2r, 2R)$, $(2R, 2R+2r)$ are diffeomorphic to one another. That they are a disk, a cylinder and a punctured torus, Milnor's (2)–(4), and the attachments of cells of dimensions $0, 1, 1, 2$ are not proved; Milnor's phrase that the homotopy type changes when a crosses a critical value also needs $[a, b]$ to contain no *other* critical value.

III.7 The sign of the determinant: $\operatorname{sgn}\det A = (-1)^{\text{index}}$

Milnor's §5 reads $\chi(M) = C_0 - C_1 + \cdots \pm C_n$ off the homology of the slabs between critical levels; that route is Part IV. In Application 2 of §6 he goes the other way: for a function f the vector field grad f has index $(-1)^\lambda$ at a critical point of index λ, and the index sum of a vector field is a topological invariant. The files `Chapter3/EulerChar*` run this argument with the Poincaré–Hopf theorem of the companion project, and so obtain the invariance of $\sum_\lambda (-1)^\lambda C_\lambda$ without homology. The first ingredient, which Milnor uses without comment, is linear algebra.

Proposition (V3.18). Let H be a non-degenerate symmetric bilinear form on a finite-dimensional real inner product space V, and A the operator with $H(v, w) = \langle Av, w\rangle$. Then $\operatorname{sgn}\det A = (-1)^{\text{index}\,H}$ `Chapter3/EulerCharAlgebra · sgn_det_eq_neg_one_pow_index` `Chapter3/EulerCharAlgebra · sgn_det_eq_neg_one_pow_index_of_symm`.

Proof as formalized. An operator representing a symmetric form is self-adjoint `Chapter3/EulerCharAlgebra · isSymmetric_of_symm`. By the spectral theorem in Mathlib, A has an orthonormal basis of eigenvectors $b_1, \dots, b_n$ with eigenvalues $\mu_1, \dots, \mu_n$ `Mathlib/Analysis/InnerProductSpace/Spectrum · LinearMap.IsSymmetric.eigenvectorBasis`.

1. In the coordinates $c_i = \langle b_i, v\rangle$, $\langle Av, v\rangle = \sum_i \mu_i c_i^2$: expand $\langle Av, v\rangle = \sum_i \langle Av, b_i\rangle\langle b_i, v\rangle$ in the orthonormal basis and use $\langle Av, b_i\rangle = \langle v, Ab_i\rangle = \mu_i\langle b_i, v\rangle$ `Chapter3/EulerCharAlgebra · inner_apply_self_eq_sum_eigenvalues`. So the coordinate isomorphism $v \mapsto (c_i)$ makes the quadratic form $v \mapsto H(v, v)$ equivalent to the weighted sum of squares $\sum_i \mu_i c_i^2$ `Chapter3/EulerCharAlgebra · quadForm_equivalent_weightedSumSquares`.
2. The index of H is Mathlib's negative index of inertia of that quadratic form (§0.2) `Chapter1/Nondegenerate · index`. By the uniqueness half of Sylvester's law of inertia `Mathlib/LinearAlgebra/`

QuadraticForm/Signature · QuadraticForm.sigNeg_of_equiv_weightedSumSquares it is the number of negative weights: index $H = \#\{i : \mu_i < 0\}$ Chapter3/EulerCharAlgebra · index_eq_ncard_eigenvalues_neg.

3. No eigenvalue vanishes: if $\mu_i = 0$, then $Ab_i = 0$, so $H(b_i, w) = \langle Ab_i, w\rangle = 0$ for all w, and non-degeneracy gives $b_i = 0$, which is impossible for a unit vector Chapter3/EulerCharAlgebra · eigenvalues_ne_zero.
4. $\det A = \prod_i \mu_i$ Mathlib/Analysis/InnerProductSpace/Spectrum · det_eq_prod_eigenvalues. The sign of a product is the product of the signs Chapter3/EulerCharAlgebra · sgn_prod, and a product of signs of nonzero numbers is (-1) raised to the number of negative factors Chapter3/EulerCharAlgebra · prod_sgn_eq_neg_one_pow.

Two remarks. First, the right-hand side $(-1)^{\operatorname{index} H}$ is defined from H alone; so the sign of $\det A$ is the same for the operators representing H with respect to any two inner products. This is what makes the argument work on a submanifold, where the operator lives on a tangent space with the induced inner product, while Milnor's Hessian is read in a chart, where that inner product is the first fundamental form. Second, H is non-degenerate exactly when A is bijective Chapter3/EulerCharAlgebra · isNondegenerate_iff_bijective: the kernel of A lies in the null space of H, and an injective endomorphism of a finite-dimensional space is onto; conversely, if $H(v, \cdot) = 0$, then $\|Av\|^2 = H(v, Av) = 0$. Two facts about definite forms are recorded for later use: a negative definite form has index the dimension, and a positive semi-definite form has index 0 Chapter3/EulerCharAlgebra · index_eq_finrank_of_negDef Chapter3/EulerCharAlgebra · index_eq_zero_of_nonneg.

III.8 The tangential gradient and its index

From here to §III.10, $N \subseteq \mathbb{R}^{K+1}$ is a compact nonempty slice submanifold modelled on $\mathbb{R}^{d+1}$, and $F : \mathbb{R}^{K+1} \to \mathbb{R}$ is smooth with $f = F|_N$. Milnor's sentence that the index of grad f at a critical point of index λ is $(-1)^\lambda$ hides four points: which metric defines the gradient; that the Hessian of $F|_N$ is not the restriction of the Hessian of F; that the derivative of a vector field at a zero is an endomorphism of the tangent space; and the linear algebra of §III.7.

The metric and the field. The metric is the one induced from $\mathbb{R}^{K+1}$. The **tangential gradient** is $v(q) = P_q \nabla F(q)$, with P_q the orthogonal projection onto T_qN Chapter3/EulerCharGradient · tangentialGradient. It is tangent Chapter3/EulerCharGradient · tangentialGradient_mem, and it is the gradient of f for the induced metric: $\langle v(q), t\rangle = dF_q(t)$ for $t \in T_qN$, because P_q is self-adjoint and the identity on T_qN Chapter3/EulerCharGradient · inner_tangentialGradient. Its zeros are exactly the critical points of f Chapter3/EulerCharGradient · tangentialGradient_eq_zero_iff: q is critical iff dF_q vanishes on T_qN, iff $\nabla F(q) \perp T_qN$, iff $P_q \nabla F(q) = 0$ Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_gradient_mem_orthogonal Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_starProjection_gradient_eq_zero.

The device: $F \circ r$. Let r be the nearest-point retraction of the tubular neighbourhood N_ε of N, which the companion project constructs (its ε-neighbourhood theorem) Milnor/Chapter6/Tubular3 · tubular Milnor/Chapter6/Tubular4 · tubr Milnor/Chapter6/Tubular4 · Nε, read as a map of $\mathbb{R}^{K+1}$ Chapter3/EulerCharGradient · tubRet. It is the identity on N and smooth on the open set $N_\varepsilon \supseteq N$ Chapter3/EulerCharGradient · tubRet_of_mem Chapter3/EulerCharGradient · contDiffAt_tubRet. Its derivative dr_q at $q \in N$ is the orthogonal projection P_q Chapter3/EulerCharGradient · dret_eq_starProjection: dr_q takes values in T_qN Milnor/Chapter6/ChartIndex · dret_mem_tanN, it is idempotent Milnor/Chapter6/ChartIndex · dret_dret, and its kernel is $(T_qN)^\perp$ Milnor/Chapter6/ChartIndex · ker_dret. So $w - dr_q w$ lies in the kernel, hence is orthogonal to T_qN, and that characterizes the orthogonal projection. Now $F \circ r$ agrees with F on N Chapter3/EulerCharGradient · comp_tubRet_comp_val, its derivative at $q \in N$ is $dF_q \circ P_q$ Chapter3/EulerCharGradient · fderiv_comp_tubRet, and therefore:

- $v(q) = \nabla(F \circ r)(q)$ for $q \in N$ Chapter3/EulerCharGradient · tangentialGradient_eq_gradient: for every w, $\langle \nabla(F \circ r)(q), w\rangle = dF_q(P_q w) = \langle \nabla F(q), P_q w\rangle = \langle P_q \nabla F(q), w\rangle$;
- at a critical point q of f, $d(F \circ r)_q = dF_q \circ P_q = 0$ Chapter3/EulerCharGradient · fderiv_comp_tubRet_eq_zero: q is a critical point of the *ambient* function $F \circ r$, while $\nabla F(q)$ itself is normal to N and need

not vanish.

The first item makes the field smooth: v is the restriction to N of $\nabla(F\circ r)$, which is smooth on N_ε, the gradient of a smooth function being smooth `Chapter3/EulerCharGradient · contDiffAt_gradient`. This is the smoothness required by the Poincaré–Hopf theorem `Chapter3/EulerCharGradient · contMDiff_tangentialGradient` `Chapter3/EulerCharGradient · isTangent_tangentialGradient` `Chapter3/EulerCharGradient · tangentialGradient_mem_range`. The second makes the Hessian computable.

The derivative of the field. The companion project reads the derivative of a tangent field v at q as the derivative dv_q of $x \mapsto v(r(x))$ `Milnor/Chapter6/ChartIndex · dvr`. For the tangential gradient, $v \circ r = \nabla(F \circ r) \circ r$, so $dv_q = d(\nabla(F \circ r))_q \circ dr_q$ `Chapter3/EulerCharGradient · dvr_tangentialGradient`. The derivative of a gradient is the Hessian: $\langle d(\nabla h)_x t, s\rangle = d^2h_x(t,s)$, since ∇h is the inverse Riesz map applied to dh `Chapter3/EulerCharGradient · inner_fderiv_gradient`. Let hess_q be the second derivative of $F \circ r$ at q restricted to $T_qN \times T_qN$ `Chapter3/EulerCharGradient · hessT`; it is symmetric `Chapter3/EulerCharGradient · hessT_symm`. As dr_q is the identity on T_qN,

$$\mathrm{hess}_q(t,s) = \langle dv_q\, t, s\rangle \qquad (t, s \in T_qN)$$

`Chapter3/EulerCharGradient · hessT_apply_eq_inner_dvr`. At a zero of v, Milnor's Lemma 5 of the companion book makes dv_q map T_qN into itself `Milnor/Chapter6/ChartIndex · dvr_mem_tanN`, so it restricts to an endomorphism dv_q^T of T_qN `Milnor/Chapter6/ChartIndex · dvT`, and dv_q^T is the self-adjoint operator of hess_q `Chapter3/EulerCharGradient · inner_dvT_tangentialGradient`.

The Hessian of f. Let ϕ be the parametrization of N by the slice chart that the companion project chooses at q, and z_0 the point over q `Milnor/Chapter6/Tubular1 · φ` `Milnor/Chapter6/Tubular3 · z₀`. Its derivative is an isomorphism $\mathbb{R}^{d+1} \cong T_qN$ `Chapter3/EulerCharGradient · tanN_eq_range` `Chapter3/EulerCharGradient · tanEquiv`. At a critical point of f, the Hessian of $F \circ \phi$ at z_0 is hess_q pulled back by this isomorphism `Chapter3/EulerCharGradient · hessian_comp_φ_eq`. Indeed $F \circ \phi = (F \circ r) \circ \phi$ near z_0, because ϕ takes values in N, where r is the identity; so the two Hessians coincide `Chapter1/Nondegenerate · hessian_congr`. And q is a critical point of $F \circ r$, so the second-order term of the chain rule vanishes `Chapter1/Nondegenerate · hessian_comp`. Milnor's Hessian f_{**} at q may be computed in this parametrization `Chapter1/ManifoldSubmanifold · indexM_comp_val_eq_tub` `Chapter1/ManifoldSubmanifold · isNondegenerate_hessian_comp_sliceParam_iff`, and index and non-degeneracy are invariant under a linear change of variables `Chapter1/Nondegenerate · index_bilinearComp` `Chapter1/Nondegenerate · isNondegenerate_bilinearComp_iff`. Hence the index of hess_q is the index of f at q, and hess_q is non-degenerate iff q is `Chapter3/EulerCharGradient · index_hessT` `Chapter3/EulerCharGradient · isNondegenerate_hessT_iff`.

Theorem (V3.19, the index of the gradient field). Let $K \geq 1$ and let q be a non-degenerate critical point of $f = F|_N$ of index λ. Then dv_q^T is invertible with $\operatorname{sgn}\det dv_q^T = (-1)^\lambda$, and the tangential gradient has index $(-1)^\lambda$ at q `Chapter3/EulerCharGradient · bijective_dvT_tangentialGradient` `Chapter3/EulerCharGradient · indexN_tangentialGradient`.

Proof as formalized. $\mathrm{hess}_q(t,s) = \langle dv_q^T t, s\rangle$ with hess_q symmetric and non-degenerate, so dv_q^T is bijective and $\operatorname{sgn}\det dv_q^T = (-1)^{\mathrm{index}\,\mathrm{hess}_q} = (-1)^\lambda$ (§III.7). The index of a tangent field at a zero is defined in the companion project through the extension $x \mapsto v(r(x)) + (x - r(x))$ to the tubular neighbourhood `Milnor/Chapter6/PoincareHopf · indexN`. At a non-degenerate zero it equals the sign of the determinant of dv_q^T on the tangent space `Milnor/Chapter6/ChartIndex · indexN_eq_sgn_det` (Milnor's Lemma 4 of the companion book, together with the fact that the extension has derivative dv_q^T on T_qN and the identity on the normal space `Milnor/Chapter6/ChartIndex · det_fderiv_w₀`). This is where $K \geq 1$ enters: the companion project computes indices of non-degenerate zeros only in ambient dimension at least 2.

III.9 $\sum_\lambda (-1)^\lambda C_\lambda$ is the Poincaré–Hopf integer

Morse numbers (V3.16). For a function f on a manifold modelled on E, $C_\lambda(f)$ is the number of critical points of index λ, set to 0 if there are infinitely many `Chapter3/EulerChar · critCount`. If f has finitely many

critical points, grouping them by index gives

$$\sum_{p\ \text{critical}} (-1)^{\operatorname{index}_p f} = \sum_{\lambda=0}^{\dim E} (-1)^\lambda C_\lambda(f),$$

the index never exceeding $\dim E$ `Chapter3/EulerChar · finsum_neg_one_pow_indexM_eq_sum_critCount`. A Morse function on a compact manifold has finitely many critical points (Corollary 2.3 with compactness; Part I) `Chapter1/ManifoldMorse · IsMorseFunction.finite_setOf_isCriticalPointM`, so Milnor's word *isolated* in the hypothesis on p. 29 is redundant.

Euler numbers (V3.17). An integer χ is an **Euler number** of N if for every smooth tangent field v on N whose zeros are finitely many distinct points $z_1, \dots, z_j$, the indices add up to χ `Chapter3/EulerChar · IsEulerNumber`. For $K \geq 1$ one exists: this is the Poincaré–Hopf theorem of the companion project, where χ is the degree of the Gauss map of the boundary of a tubular neighbourhood `Milnor/Chapter6/PoincareHopf · poincare_hopf` `Chapter3/EulerChar · exists_isEulerNumber`. It plays the part of Milnor's appeal to the topological invariance of the index sum, which he quotes from Steenrod.

Theorem (V3.20, formula (3) without homology). Let $K \geq 1$ and χ be an Euler number of N. For every Morse function f on N,

$$\sum_{q\ \text{critical}} (-1)^{\operatorname{index}_q f} = \sum_{\lambda=0}^{d+1} (-1)^\lambda C_\lambda(f) = \chi$$

`Chapter3/EulerChar · IsEulerNumber.finsum_eq` `Chapter3/EulerChar · IsEulerNumber.sum_critCount_eq`.

Proof as formalized. First for $f = F|_N$ with F smooth on $\mathbb{R}^{K+1}$ `Chapter3/EulerChar · IsEulerNumber.finsum_eq_of_ambient`. Then f is a Morse function, and its critical set is finite; enumerate it injectively as $z_1, \dots, z_j$. The tangential gradient of F is a smooth tangent field whose zeros are exactly the critical points (§III.8), that is, exactly the z_i. By the definition of an Euler number its index sum is χ; by the theorem of §III.8 its index at z_i is $(-1)^{\operatorname{index}_{z_i} f}$. Rewriting the sum over the critical set as the sum over the enumeration gives the first equality. A general smooth f on N is the restriction of a smooth function on $\mathbb{R}^{K+1}$ `Chapter3/EulerChar · exists_contDiff_comp_val_eq`: the function $f \circ r$ is smooth on the tubular neighbourhood N_ε, the retraction being smooth there as a map into the manifold N `Milnor/Chapter2/Corestrict · contMDiff_codRestrict`, and a cut-off provides a smooth function on all of $\mathbb{R}^{K+1}$ that agrees with $f \circ r$ near a closed tube around N `Milnor/Chapter6/Extend · exists_extension`; on N, r is the identity. The second equality is the regrouping by index, with $\dim N = d + 1$.

Consequently there is an integer which is simultaneously the index sum of every tangent field and the alternating count of every Morse function `Chapter3/EulerChar · exists_eulerNumber_eq_sum_critCount`; the alternating count is the same for all Morse functions on N `Chapter3/EulerChar · sum_critCount_eq_of_isMorseFunction`; and once a Morse function exists (Corollary 6.7, §III.16) every Euler number is determined by it `Chapter3/EulerChar · IsEulerNumber.eq_sum_critCount`, so the Euler number is unique. Below it is written $\chi(N)$. Its identification with $\sum_i (-1)^i \dim H_i(N; \Bbbk)$ over any field $\Bbbk$ is in Part IV `Homology/MorseHomologyManifold · IsEulerNumber.eq_sum_bettiNumber`.

Three readings of Milnor are fixed here. His function in Application 2, said to be any differentiable function, must be a Morse function, since otherwise neither C_λ nor the index $(-1)^\lambda$ makes sense; the critical points of the vector field are its zeros; and the last term $\pm C_n$ of the alternating sum takes $n = \dim M$. The vector-field argument does not order the critical points, so distinct critical values, which Milnor's slab argument uses, are not needed. Dimension 0 is excluded here, because the companion project computes indices of non-degenerate zeros only in positive dimension; there every point is critical of index 0 and the alternating count is the number of points (§III.21).

III.10 Two checks: the sphere and the torus

The sphere (V3.21). Let $S^n \subseteq \mathbb{R}^{n+1}$, $n = d + 1 \geq 1$, be the unit sphere, as the slice submanifold of the companion project `Milnor/Chapter6/EulerSphere · sphSub`. The companion project computes its index sum to be $1 + (-1)^n$ `Milnor/Chapter6/EulerSphere · indexSum_sph`, so $1 + (-1)^n$ is an Euler number of S^n `Chapter3/EulerCharSphere · isEulerNumber_sph`, and every Morse function on S^n has $\sum_\lambda (-1)^\lambda C_\lambda = 1+(-1)^n$ `Chapter3/EulerCharSphere · sum_critCount_sph`.

The height function $h(x) = \langle e, x\rangle$, e a unit vector `Chapter3/EulerCharSphere · heightFun`, checks this directly. Its gradient is e `Chapter3/EulerCharSphere · gradient_heightFun`, and its tangential gradient is the field $e - \langle e, x\rangle x$ that the companion book uses for the sphere: $e - (e - \langle e, q\rangle q) = \langle e, q\rangle q$ is orthogonal to $T_q S^n = q^\perp$ `Chapter3/EulerCharSphere · tangentialGradient_heightFun`. Its zeros, hence the critical points of h, are $\pm e$ `Chapter3/EulerCharSphere · isCriticalPointM_heightFun_iff`. The derivative of that field on the tangent space is $t \mapsto -(\langle e, q\rangle t + \langle e, t\rangle q)$ `Milnor/Chapter6/EulerSphere · dvr_gradSph_apply`. Taking the inner product with $s \perp q$ gives, by §III.8, $\mathrm{hess}_q(t, s) = -\langle e, q\rangle\langle t, s\rangle$ `Chapter3/EulerCharSphere · hessT_heightFun`, non-degenerate wherever $\langle e, q\rangle \neq 0$ `Chapter3/EulerCharSphere · isNondegenerate_hessT_heightFun`. So h is a Morse function `Chapter3/EulerCharSphere · isMorseFunction_heightFun`. At e the form is $-\langle\cdot,\cdot\rangle$, negative definite on the n-dimensional tangent space, so the index is n; at $-e$ it is $\langle\cdot,\cdot\rangle$, and the index is 0 `Chapter3/EulerCharSphere · indexM_heightFun_pos Chapter3/EulerCharSphere · indexM_heightFun_neg`. Hence $C_0 = C_n = 1$ and all other $C_\lambda = 0$, the points $e \neq -e$ being distinct `Chapter3/EulerCharSphere · critCount_heightFun`, so $\sum_\lambda (-1)^\lambda C_\lambda = 1 + (-1)^n$ directly `Chapter3/EulerCharSphere · sum_critCount_heightFun`. Conversely, every Euler number of S^n equals this count (§III.9), which recomputes the companion project's value by Morse theory `Chapter3/EulerCharSphere · IsEulerNumber.eq_sph`. The height function also satisfies the hypotheses of Reeb's theorem, and its indices 0 and n illustrate §III.1.

The torus (V3.25). By §III.6 and §III.9, $C_0 - C_1 + C_2 = 1 - 2 + 1 = 0$ is an Euler number of T, every Euler number of T is 0, and every Morse function on T has $C_0 - C_1 + C_2 = 0$ `Chapter3/Torus · isEulerNumber_torus Chapter3/Torus · IsEulerNumber.eq_torus Chapter3/Torus · sum_critCount_torus`. An independent check avoids Morse theory: the rotation field $(-x_2, x_1, 0)$ `Chapter3/Torus · Torus.rotField` is smooth and tangent to T, since dg vanishes on it, and it has no zeros on T, since $x_1 = x_2 = 0$ would give $R^2 + x_3^2 = r^2 < R^2$. Its index sum is the empty sum, so every Euler number of T is 0 `Chapter3/Torus · IsEulerNumber.eq_torus_of_rotation`.

III.11 Sard's theorem on a manifold (Theorem 6.1)

Milnor quotes Theorem 6.1 for C^1 maps between manifolds of the same dimension with countable bases, refers to de Rham, and uses it once, for a smooth map into $\mathbb{R}^n$ (Corollary 6.2). The formal theorem treats smooth maps into a vector space, of any dimensions. Write $\mathrm{crit}(f)$ for the set of points where the manifold derivative of f is not surjective (§0.1) `Milnor/Chapter2/Brown · critPoints`.

Theorem 6.1 (V4.3). Let M be a second countable smooth manifold without boundary, modelled on a finite-dimensional real space E; F a finite-dimensional real normed space with a Haar measure μ; $U \subseteq M$ open; and $f : M \to F$ smooth on U. Then $\mu(f(U \cap \mathrm{crit}(f))) = 0$. In particular, for f smooth on M the set of critical values is μ-null `Chapter4/Sard · measure_image_inter_critPoints_eq_zero Chapter4/Sard · measure_image_critPoints_eq_zero`.

(E and F are required to lie in the same universe of the type theory, as in the companion project; this has no mathematical content.)

Proof as formalized.

1. *Countably many charts.* By second countability, countably many chart domains, those of the charts at points x_0 of a countable set, cover U `Mathlib/Topology/Bases · TopologicalSpace.countable_cover_nhdsWithin`.

2. *Sard in a chart.* For a chart φ at x_0, the set W of points of its target whose image under φ^{-1} lies in U is open, and $f \circ \varphi^{-1}$ is smooth on it. The Euclidean Sard theorem of the companion project `Milnor/Chapter3/Sard · sard` makes the image of its critical set in W null.
3. *Critical points go to critical points.* If $x \in U$ lies in the domain of φ and df_x is not onto, then $\varphi(x)$ is a critical point of $f \circ \varphi^{-1}$ in W `Chapter4/Sard · mem_critSet_of_mem_critPoints_of_mdifferentiableAt`: by the chain rule, $d(f \circ \varphi^{-1})$ at $\varphi(x)$ is $df_x \circ d\varphi^{-1}$, and if this composite were onto, df_x would be onto.
4. So $f(U \cap \mathrm{crit}(f))$ lies in a countable union of null sets, which is null.

The proof is that of A. B. Brown's theorem in the companion project, which records only the density of regular values. Only $\mathbb{R}^n$ is used as a target in §6, and there all Haar measures, Lebesgue measure among them, have the same null sets.

III.12 The normal bundle, the endpoint map and Milnor's coordinates

From here to §III.18, $M \subseteq \mathbb{R}^n$ is a slice submanifold modelled on $\mathbb{R}^k$ with complement $\mathbb{R}^w$. It need not be closed or compact, and $k = n$ is allowed. If M is nonempty, then $k + w = n$, because the derivative of a slice chart at a point of M is an isomorphism $\mathbb{R}^n \cong \mathbb{R}^k \times \mathbb{R}^w$ `Chapter4/Focal · slice_dim_add_eq_of_nonempty`; for empty M nothing constrains k and w, so the formal statements carry $k + w = n$ as a hypothesis where it is used.

The normal bundle and the endpoint map (V4.1). The normal bundle is the set

$$N = \{(q, v) \in M \times \mathbb{R}^n : v \perp T_q M\}$$

`Milnor/Chapter8/NormalBundle · normalBundleSet` `Chapter4/NormalBundle · mem_normalBundleSet_iff`. When $k + w = n$ it is a slice submanifold of $\mathbb{R}^n \times \mathbb{R}^n$ of dimension n `Milnor/Chapter8/NormalBundle · isSliceSubmanifold_normalBundle`; Milnor calls this not difficult, and it is Problem 11 of the companion book, formalized there. Its proof is the source of everything below. Near $p \in M$ let σ be the slice chart chosen at p, $c(a) = (\sigma a)_1$ the first slice coordinate, x the parametrization $u \mapsto \sigma^{-1}(u, 0)$, and $B(a) = (dx_{c(a)})^\dagger$ the adjoint of its derivative at $c(a)$ `Milnor/Chapter8/NormalBundle · NormalAux.B`. On the chart domain, N is the zero set of

$$\Theta(a, v) = \big((\sigma a)_2,\ B(a)\, v\big)$$

`Milnor/Chapter8/NormalBundle · NormalAux.g` `Milnor/Chapter8/NormalBundle · NormalAux.mem_normalBundleSet_iff`: the first component cuts out M, and as $T_a M = \operatorname{range} dx_{c(a)}$, $v \perp T_a M$ iff $B(a)v = 0$. The derivative $d\Theta_{(a,v)}(u, h) = \big((d\sigma\, u)_2,\ B(a)h + (dB_a u)v\big)$ is onto `Milnor/Chapter8/NormalBundle · NormalAux.fderiv_g_apply` `Milnor/Chapter8/NormalBundle · NormalAux.surjective_fderiv_g`, and a local form of Lemma 1 gives the slice charts `Milnor/Chapter8/LocalSlice · IsSliceSubmanifold.of_locally_preimage`. Milnor's endpoint map is $E(q, v) = q + v$ `Milnor/Chapter8/NormalBundle · normalSum`. The formal text also uses the generalized endpoint map $E_\Gamma(q, v) = \Gamma(q) + v$ for $\Gamma : \mathbb{R}^n \to \mathbb{R}^n$ `Chapter4/NormalBundle · normalEnd`, with $E_{\mathrm{id}} = E$ `Chapter4/NormalBundle · normalEnd_id`; $\Gamma = -\nabla F$ serves Corollary 6.8 (§III.17). E_Γ is smooth over the open set where Γ is smooth `Chapter4/NormalBundle · contMDiffOn_normalEnd`.

Milnor's coordinates and the fundamental forms (V4.5). For $q \in M$, Milnor's coordinates $u \mapsto x(u)$ are the inverse of the slice chart chosen at q, with $x(u_q) = q$ `Chapter4/NormalBundle · xParam` `Chapter4/NormalBundle · uParam` `Chapter4/NormalBundle · xParam_uParam`. Write dx and d^2x for its first and second derivatives at u_q, Milnor's vectors $\partial x/\partial u^i$ and $\partial^2 x/\partial u^i \partial u^j$ `Chapter4/NormalBundle · dParam` `Chapter4/NormalBundle · d2Param`. The map x is smooth near u_q `Chapter4/NormalBundle · contDiffAt_xParam`, and dx is injective `Chapter4/NormalBundle · injective_dParam`. Since $\sigma \circ x = (u \mapsto (u, 0))$ near u_q, $d\sigma_q \circ dx = (\xi \mapsto (\xi, 0))$ `Chapter4/NormalBundle · fderiv_σφ_comp_dParam`, and $d\sigma_q$ is injective `Chapter4/NormalBundle · injective_fderiv_σφ`. The **first fundamental form** is $g(\xi, \eta) = \langle dx\, \xi, dx\, \eta \rangle$ on $\mathbb{R}^k$ `Chapter4/NormalBundle · firstFF`; the **second fundamental form in the direction** v is $\ell_v(\xi, \eta) = \langle v, d^2x(\xi, \eta) \rangle$ `Chapter4/NormalBundle · secondFF`, linear in v `Chapter4/NormalBundle · secondFF_smul`.

For v normal at q this is Milnor's $v \cdot \ell_{ij}$: the tangential part of d^2x is orthogonal to v. For a linear map T of $\mathbb{R}^n$, put

$$B_{T,v}(\xi,\eta) = \langle T\,dx\,\xi, dx\,\eta\rangle - \ell_v(\xi,\eta)$$

`Chapter4/NormalBundle · normalForm`; for $T = \mathrm{id}$ it is $g - \ell_v$, Milnor's matrix $(g_{ij} - v\cdot\ell_{ij})$ `Chapter4/NormalBundle · normalForm_id`.

Proposition (V4.8, the tangent space of N). Assume $k+w=n$ and $(q,v)\in N$. A pair $(u,h)\in\mathbb{R}^n\times\mathbb{R}^n$ is tangent to N at (q,v) if and only if $u = dx\,\xi$ for some $\xi\in\mathbb{R}^k$ and

$$\langle h, dx\,\eta\rangle + \langle v, d^2x(\xi,\eta)\rangle = 0 \quad \text{for every } \eta\in\mathbb{R}^k$$

`Chapter4/NormalBundle · mem_range_mfd_val_normalBundle_iff`.

Proof as formalized. By Lemma 2 for a local defining function, the tangent space of N at (q,v) is the kernel of $d\Theta$ there `Milnor/Chapter8/LocalSlice · IsSliceSubmanifold.range_mfderiv_val_eq_ker`, the dimensions matching since $n+(w+k)=2n$ `Chapter4/NormalBundle · finrank_normalBundle_add`. So (u,h) is tangent iff $(d\sigma\,u)_2 = 0$ and $B(q)h + (dB_q u)v = 0$. The first condition says $d\sigma\,u = (\xi,0)$ for $\xi = (d\sigma\,u)_1$, that is, $u = dx\,\xi$, by $d\sigma\circ dx = (\cdot,0)$ and the injectivity of $d\sigma$. For the second, $\langle B(q)h,\eta\rangle = \langle h, dx\,\eta\rangle$ by the definition of the adjoint `Chapter4/NormalBundle · B_self_inner`. The derivative of $a\mapsto B(a)$ at q is $u\mapsto (d^2x((d\sigma\,u)_1,\cdot))^\dagger$ `Chapter4/NormalBundle · hasFDerivAt_B`, so $\langle (dB_q u)v,\eta\rangle = \langle v, d^2x(\xi,\eta)\rangle$. A vector of $\mathbb{R}^k$ vanishes iff its inner products with all η do. The second condition is the derivative of Milnor's identity $w_\alpha\cdot\partial x/\partial u^j = 0$, here without the normal frame w_α.

Proposition (V4.8, the kernel of dE_Γ). Assume $k+w=n$, $(q,v)\in N$, and Γ differentiable at q. The derivative of E_Γ at (q,v) is $(u,h)\mapsto d\Gamma_q(u)+h$ on the tangent space `Chapter4/NormalBundle · hasMFDerivAt_normalEnd`, and the dimension of its kernel is the nullity of $B_{d\Gamma_q,v}$ `Chapter4/NormalBundle · finrank_ker_mfd_normalEnd`.

Proof as formalized. E_Γ is the map $(y,z)\mapsto\Gamma(y)+z$ of $\mathbb{R}^n\times\mathbb{R}^n$ composed with the inclusion of N, so by the chain rule its derivative is $S\circ L$, with $S(u,h) = d\Gamma_q(u)+h$ and L the injective derivative of the inclusion `Milnor/Chapter2/LevelSet · injective_mfderiv_val`. The kernel of $S\circ L$ is carried by L isomorphically onto range $L\cap\ker S$. By the previous proposition this is the image of the null space of $B_{d\Gamma_q,v}$ under $\iota(\xi) = (dx\,\xi, -d\Gamma_q\,dx\,\xi)$. Indeed a tangent pair with $d\Gamma_q(u)+h=0$ has $u = dx\,\xi$ and $h = -d\Gamma_q\,dx\,\xi$, and the tangency condition then reads $-\langle d\Gamma_q\,dx\,\xi, dx\,\eta\rangle + \langle v, d^2x(\xi,\eta)\rangle = 0$ for all η, that is, ξ is in the null space; and conversely. As ι is injective, its first component dx being injective, the dimensions agree.

For $\Gamma = \mathrm{id}$ this is Milnor's computation of pp. 34–35. He chooses orthonormal normal fields $w_\alpha(u)$, coordinates $(u,t)\mapsto(x(u),\sum_\alpha t^\alpha w_\alpha(u))$ on N, and a block triangular $n\times n$ matrix whose upper left block is $(g_{ij} - \sum_\alpha t^\alpha w_\alpha\cdot\ell_{ij})$. Two things in that argument need correction. The fields w_α exist only on a coordinate patch (Gram–Schmidt applied to a completion of the $\partial x/\partial u^i$), and Milnor does not say why they exist. And his sentence that the rank of the matrix equals the rank of the upper left block does not hold as written: the ranks differ by $n-k$; it is the *nullities* that agree, and that is what is used.

III.13 Focal points and Corollary 6.2

Assume $k+w=n$ (V4.2). The **nullity of** E at $x\in N$ is the dimension of the kernel of the derivative of E at x, the tangent space of N being identified with $\mathbb{R}^n$ by the chart of N at x `Chapter4/Focal · endpointNullity`. Another chart changes the derivative by an isomorphism of $\mathbb{R}^n$, which preserves the dimension of the kernel. Following Milnor, $e\in\mathbb{R}^n$ is a **focal point of** (M,q) **with multiplicity** μ if $(q, e-q)\in N$ and the nullity of E at $(q,e-q)$ is $\mu>0$ `Chapter4/Focal · IsFocalPointMult`; a focal point of (M,q) if this holds for some μ `Chapter4/Focal · IsFocalPoint`; and a focal point of M if it is a focal point of (M,q) for some $q\in M$ `Chapter4/Focal · focalSet`. Given q and e, the normal vector $e-q$ is determined.

Focal points are critical values of E. The nullity of E at x is positive iff x is a critical point of E `Chapter4/Focal · endpointNullity_pos_iff`: N and $\mathbb{R}^n$ have the same dimension, so for the derivative not injective, not surjective and singular are the same. Hence e is a focal point of (M, q) iff $(q, e - q) \in N$ is a critical point of E `Chapter4/Focal · isFocalPoint_iff`, and the set of focal points of M is exactly the set of critical values of E `Chapter4/Focal · focalSet_eq_image_critPoints`. For the inclusion of the critical values, a critical point $x = (q, v)$ has foot point q and $E(x) - q = v$.

Corollary 6.2 (V4.4). If $k + w = n$, the set of focal points of M is null for every Haar measure on $\mathbb{R}^n$ `Chapter4/Focal · measure_focalSet_eq_zero`.

Proof as formalized. It is the set of critical values of the smooth map $E : N \to \mathbb{R}^n$ `Milnor/Chapter8/NormalBundle · contMDiff_normalSum`, and N, a subset of $\mathbb{R}^{2n}$, is second countable. Theorem 6.1 applies.

III.14 Assertion 6.4, the shape operator and Lemma 6.3

Assertion 6.4 (V4.9). Assume $k + w = n$. The nullity of E at $(q, v) \in N$ is the nullity of $g - \ell_v$ `Chapter4/Focal · endpointNullity_eq_nullity`. Hence, for $q \in M$, $v \perp T_qM$, $t \in \mathbb{R}$ and $\mu \in \mathbb{N}$: $q + tv$ is a focal point of (M, q) with multiplicity μ iff $g - t\ell_v$ has nullity $\mu > 0$ `Chapter4/Focal · isFocalPointMult_add_smul_iff`.

Proof as formalized. The first statement is §III.12 with $\Gamma = \mathrm{id}$, whose derivative is the identity. For the second, $(q, tv) \in N$, the normal vectors forming a subspace, and $\ell_{tv} = t\ell_v$. Milnor takes v a unit vector; that is not needed.

In particular q **is never a focal point of** (M, q) `Chapter4/Focal · not_isFocalPoint_self`: for $e = q$ the normal vector is 0 and the form is g, positive definite because $g(\xi, \xi) = \|dx\,\xi\|^2$ and dx is injective, so its nullity is 0. Milnor leaves this case, $t = 0$, implicit; his passage from the singularity of $(g_{ij} - t\,v \cdot \ell_{ij})$ to the eigenvalue $1/t$ divides by t.

The shape operator (V4.6). Milnor defines the principal curvatures in coordinates with $g_{ij} = \delta_{ij}$ at q, without saying why such coordinates exist or why the eigenvalues do not depend on them. The formal text works on the tangent space instead. Let $T_qM = \mathrm{range}(dx) \subseteq \mathbb{R}^n$ `Chapter4/Curvature · tanSpace`, an inner product space with the inner product of $\mathbb{R}^n$, of dimension k `Chapter4/Curvature · finrank_tanSpace`. The map dx is a linear isomorphism $\mathbb{R}^k \cong T_qM$ `Chapter4/Curvature · dParamEquiv`, and an isometry from $(\mathbb{R}^k, g)$ by the definition of g `Chapter4/Curvature · inner_dParamEquiv`. Transport ℓ_v to T_qM by dx^{-1} `Chapter4/Curvature · secondFFTan` and let the **shape operator** S_v be the endomorphism of T_qM that represents it, $\langle S_v a, b\rangle = \ell_v(dx^{-1}a, dx^{-1}b)$ `Chapter4/Curvature · shapeOp` `Mathlib/Analysis/InnerProductSpace/Dual · InnerProductSpace.continuousLinearMapOfBilin` `Chapter4/Curvature · inner_shapeOp_dParamEquiv`. It is self-adjoint `Chapter4/Curvature · isSymmetric_shapeOp`, because d^2x is symmetric, by the symmetry of second derivatives of a C^2 map `Chapter4/Curvature · d2Param_symm`. The **principal curvatures** of M at q in the direction v are its k eigenvalues, with multiplicity, in decreasing order `Chapter4/Curvature · principalCurvatures`, and the multiplicity of K, the number of indices i with $\kappa_i = K$, is the dimension of the eigenspace of K `Chapter4/Curvature · finrank_eigenspace_shapeOp` `Mathlib/Analysis/InnerProductSpace/Spectrum · LinearMap.IsSymmetric.card_filter_eigenvalues_eq`. In coordinates with $g = I$ at q these are Milnor's K_i; in general they are the eigenvalues of $g^{-1}(v \cdot \ell)$. That S_v does not depend on the parametrization is not proved, and nothing uses it. The scan lists the curvatures as $K_1, \dots, K_n$; there are k of them.

Null spaces are eigenspaces. For every t, ξ lies in the null space of $g - t\ell_v$ iff $(1 - tS_v)(dx\,\xi) = 0$ `Chapter4/Curvature · mem_nullSpace_firstFF_sub_iff`: for every η, $\langle dx\,\xi - tS_v\,dx\,\xi,\ dx\,\eta\rangle = (g - t\ell_v)(\xi, \eta)$, and dx is onto T_qM. So for $t \neq 0$, dx maps the null space of $g - t\ell_v$ onto the eigenspace of S_v for the eigenvalue $1/t$ `Chapter4/Curvature · map_nullSpace_firstFF_sub_eq_eigenspace`, and the nullity is the dimension of that eigenspace `Chapter4/Curvature · nullity_firstFF_sub_smul_secondFF`; for $t = 0$ the nullity is 0 `Chapter4/Curvature · nullity_firstFF_sub_zero_smul_secondFF`. This replaces Milnor's normalized coordinates. It also settles a point of language: the multiplicity of an eigenvalue of a symmetric matrix as a root of the characteristic polynomial equals the dimension of its eigenspace, and the latter is what the argument produces.

Multiplicity as a number. Let $\mathrm{mult}_q(e)$ be the nullity of E at $(q, e-q)$ if this point lies in N, and 0 otherwise `Chapter4/Curvature · focalMult`. Then e is a focal point of (M, q) with multiplicity μ iff $\mathrm{mult}_q(e) = \mu > 0$, and a focal point iff $\mathrm{mult}_q(e) > 0$ `Chapter4/Curvature · isFocalPointMult_iff_focalMult Chapter4/Curvature · isFocalPoint_iff_focalMult_pos`.

Lemma 6.3 (V4.7). Assume $k + w = n$, $q \in M$ and $v \perp T_qM$. Then $\mathrm{mult}_q(q + tv)$ is 0 for $t = 0$ and the dimension of the eigenspace of S_v for $1/t$ otherwise `Chapter4/Curvature · focalMult_add_smul`. Hence $q + tv$ is a focal point of (M, q) iff $t = K^{-1}$ for a nonzero principal curvature K in the direction v `Chapter4/Curvature · isFocalPoint_add_smul_iff`, the multiplicity of $q + K^{-1}v$ is the multiplicity of K `Chapter4/Curvature · focalMult_add_inv_smul`, and for any finitely many values of t the multiplicities of the points $q+tv$ add up to at most k `Chapter4/Curvature · sum_focalMult_add_smul_le`.

Proof as formalized. $(q, tv) \in N$, and its nullity is that of $g - \ell_{tv} = g - t\ell_v$; the two cases are the computation just made. A nonzero eigenspace for $1/t$ is an eigenvalue $K = 1/t \neq 0$. For the bound, the multiplicity at $q + tv$ is the number of indices i with $t \neq 0$ and $\kappa_i = t^{-1}$ `Chapter4/Curvature · focalMult_add_smul_eq_card`. For distinct values of t these index sets are disjoint, $t \mapsto t^{-1}$ being injective, so their sizes add up to at most the number k of indices. Milnor takes v to be a unit vector; for any normal v the principal curvatures in the direction sv are s times those in the direction v, and nothing changes.

III.15 The squared distance: Lemma 6.5 and Theorem 6.6

For $p \in \mathbb{R}^n$ let $L_p(q) = \|q - p\|^2$ on M. It is smooth, as the restriction of a smooth function `Chapter1/ManifoldSubmanifold · contDiff_normSq_sub`. On $\mathbb{R}^n$ its derivative at y is $u \mapsto 2\langle y - p, u\rangle$ `Chapter1/ManifoldSubmanifold · hasFDerivAt_normSq_sub` and its Hessian is $2\langle\cdot,\cdot\rangle$ `Chapter1/ManifoldSubmanifold · hessian_normSq_sub`.

Critical points (p. 35; V4.10). q is a critical point of L_p iff $q - p \perp T_qM$ `Chapter1/ManifoldSubmanifold · isCriticalPointM_normSq_sub_iff`, that is, iff $(q, p - q) \in N$ `Chapter4/Focal · isCriticalPointM_normSq_sub_iff_mem_normalBundleSet`: the derivative $2\langle q - p, \cdot\rangle$ must vanish on T_qM.

The Hessian (p. 36). In any parametrization x,

$$\mathrm{Hess}(L_p \circ x)(\xi, \eta) = 2\big(\langle dx\,\xi, dx\,\eta\rangle + \langle d^2x(\xi, \eta), x - p\rangle\big)$$

at *every* point `Chapter1/ManifoldSubmanifold · hessian_normSq_sub_comp`, by the second-order chain rule, whose extra term $d(L_p)(d^2x)$ does not vanish here. In Milnor's coordinates at q this is

$$\mathrm{Hess}(L_p \circ x)(u_q) = 2\,(g - \ell_{p-q})$$

`Chapter4/Focal · hessian_normSq_sub_xParam`. Milnor writes $p = q + tv$ with v a unit normal, which presupposes that q is critical and $p \neq q$. Using ℓ_{p-q} directly, which is linear in $p - q$, removes both conditions. The index and the nullity of L_p at a critical point are those of this form `Chapter1/ManifoldSubmanifold · indexM_normSq_sub_eq Chapter1/ManifoldSubmanifold · nullityM_normSq_sub_eq`, the Hessian at a critical point being readable in any slice parametrization (§0.2).

Lemma 6.5 (V4.11). Assume $k+w = n$, $p \in \mathbb{R}^n$, $q \in M$. If $(q, p-q) \in N$, the nullity of L_p at q equals the nullity of E at $(q, p - q)$ `Chapter4/Focal · nullityM_normSq_sub_eq_endpointNullity`. Hence p is a focal point of (M, q) with multiplicity μ iff q is a critical point of L_p with nullity $\mu > 0$ `Chapter4/Focal · isFocalPointMult_iff`, and q is a degenerate critical point of L_p iff p is a focal point of (M, q) `Chapter4/Focal · isDegenerate_normSq_sub_iff_isFocalPoint`.

Proof as formalized. The nullity of L_p at q is that of $2(g - \ell_{p-q})$, and a nonzero factor does not change a null space `Chapter4/Focal · nullity_smul_of_ne_zero`; by Assertion 6.4 the nullity of $g - \ell_{p-q}$ is that of E at $(q, p - q)$. The last statement uses that non-degeneracy is nullity 0 `Chapter1/Manifold · isNondegenerate_`

`hessianM_iff_nullityM_eq_zero`. If q is not critical, $p-q$ is not normal at q and p is not a focal point of (M,q), by the definition; Milnor leaves this case implicit.

Theorem 6.6 (V4.12). For every slice submanifold $M \subseteq \mathbb{R}^n$ and every Haar measure μ on $\mathbb{R}^n$ (Lebesgue measure in particular), L_p is a Morse function on M for μ-almost every p `Chapter4/Focal · ae_isMorseFunction_normSq_sub` `Chapter4/Focal · ae_volume_isMorseFunction_normSq_sub`.

Proof as formalized. If M is empty, every L_p is a Morse function vacuously. Otherwise $k+w=n$. If L_p is not a Morse function, then, L_p being smooth, some critical point q is degenerate, and by Lemma 6.5 p is a focal point of (M,q); so p is a focal point of M `Chapter4/Focal · mem_focalSet_of_not_isMorseFunction`. The exceptional set lies in the focal set, which is null by Corollary 6.2. Neither closedness nor compactness of M is used, and $k=n$ is allowed.

Density. A nonempty open subset of $\mathbb{R}^n$ has positive Lebesgue measure, so it contains a point of any set of full measure `Mathlib/MeasureTheory/Measure/Restrict · Measure.exists_mem_of_measure_ne_zero_of_ae`. Hence every nonempty open set contains a p with L_p a Morse function, and in particular such a p exists `Chapter4/Focal · exists_mem_isMorseFunction_normSq_sub` `Chapter4/Focal · exists_isMorseFunction_normSq_sub`. Milnor uses the density in the proof of Corollary 6.8.

III.16 Corollary 6.7: Morse functions with compact sublevel sets

Milnor deduces Corollary 6.7 from Theorem 6.6 and Whitney's theorem that an n-manifold embeds as a closed subset of $\mathbb{R}^{2n+1}$. What the argument uses about the embedding is that its image is *closed*. The formal statement is for a closed slice submanifold $M \subseteq \mathbb{R}^n$; abstract compact manifolds are §III.21.

Compact sublevel sets. The set $\{y \in \mathbb{R}^n : \|y-p\|^2 \le a\}$ is closed and bounded, lying in the closed ball of radius $\max(1,a)$ about p (if $\|y-p\| > 1$, then $\|y-p\| \le \|y-p\|^2 \le a$), hence compact `Chapter4/Existence · isCompact_setOf_normSq_sub_le`. If M is closed, its inclusion is a closed embedding, and the preimage of a compact set under a closed embedding is compact. So every sublevel set $\{q \in M : L_p(q) \le a\}$ is compact `Chapter4/Existence · isCompact_sublevel_normSq_sub`.

Corollary 6.7 (V4.13). If M is closed in $\mathbb{R}^n$, then for almost every p, L_p is a Morse function on M all of whose sublevel sets are compact `Chapter4/Existence · ae_isMorseFunction_normSq_sub_and_isCompact_sublevel`; so M carries a Morse function f with every $M^a = f^{-1}(-\infty,a]$ compact `Chapter4/Existence · exists_isMorseFunction_isCompact_sublevel`, and each M^a then contains only finitely many critical points `Chapter4/Existence · exists_isMorseFunction_finite_critPoints_sublevel`.

Proof as formalized. Theorem 6.6 and the compactness just proved. For the finiteness: non-degenerate critical points are isolated (Corollary 2.3) `Chapter1/ManifoldMorse · IsNondegenerateCriticalPointM.exists_nhds_isCriticalPointM_eq`, the critical set of a C^1 function is closed `Chapter1/ManifoldMorse · isClosed_setOf_isCriticalPointM`, so the critical points in the compact set M^a form a compact discrete set, hence a finite one `Chapter1/ManifoldMorse · finite_inter_setOf_isCriticalPointM` `Chapter1/ManifoldMorse · IsMorseFunction.finite_inter_setOf_isCriticalPointM`. This is the form in which §§3 and 5 use the corollary. For compact M: almost every L_p is a Morse function with finitely many critical points `Chapter4/Compact · ae_isMorseFunction_normSq_sub_finite_critPoints`, and M carries one `Chapter4/Compact · exists_isMorseFunction_finite_critPoints`.

Two hypotheses Milnor does not state are visible here. The closedness of the embedding is essential: for the open interval $(-1,1) \subseteq \mathbb{R}$ every L_p has a non-compact sublevel set (all of the interval, once a is large). And a manifold in Milnor's sense must be Hausdorff with a countable basis: on the long line every continuous real function is eventually constant, so some sublevel set is not compact. For non-compact abstract manifolds the corollary is not formalized, because Mathlib has Whitney's theorem only for compact manifolds.

III.17 Corollary 6.8: approximation, without boundedness

Milnor's Corollary 6.8: a bounded smooth f on M is a uniform limit of functions without degenerate critical points, and the approximating g can be chosen so that on a given compact set its derivatives up to a given order approximate those of f. His proof embeds M as a bounded set by an embedding h with first coordinate f, and takes $g = (L_p \circ h - c^2)/2c$ with p near $(-c, 0, \dots, 0)$ and c large. The formal proof is different: it perturbs f by a small linear function, $g = f + \langle a, \cdot \rangle$, and repeats the proof of Theorem 6.6 with E replaced by the generalized endpoint map E_Γ, $\Gamma = -\nabla F$. For $f = \frac{1}{2}\|x\|^2$ this is Theorem 6.6 itself, since $\frac{1}{2}\|x\|^2 + \langle a, x\rangle = \frac{1}{2}L_{-a}(x) - \frac{1}{2}\|a\|^2$.

Gradients. The Riesz isomorphism $u \mapsto \langle u, \cdot\rangle$ of $\mathbb{R}^n$ onto its dual `Chapter4/Approximation · rieszEquiv` defines the negative gradient $-\nabla F$ `Chapter4/Approximation · negGrad`, with $\langle -\nabla F(y), u\rangle = -dF_y(u)$ `Chapter4/Approximation · inner_negGrad`. It is smooth where F is `Chapter4/Approximation · contDiffOn_negGrad`, and its derivative is minus the Hessian, $\langle d(-\nabla F)_y u, w\rangle = -d^2F_y(u, w)$ `Chapter4/Approximation · hasFDerivAt_negGrad` `Chapter4/Approximation · inner_fderiv_negGrad`. Adding a linear function changes the derivative, $d(F + \langle a, \cdot\rangle)_y(u) = \langle a - (-\nabla F(y)), u\rangle$, but not the Hessian `Chapter4/Approximation · fderiv_add_inner` `Chapter4/Approximation · hessian_add_inner`. In Milnor's coordinates the Hessian of $G \circ x$ is $d^2G_q(dx\,\xi, dx\,\eta) + dG_q(d^2x(\xi, \eta))$ `Chapter4/Approximation · hessian_comp_xParam`.

Local extensions. A smooth $f : M \to \mathbb{R}$ extends near each $p \in M$ to a smooth F on an open set $U \subseteq \mathbb{R}^n$: in the slice chart σ at p, put $F(y) = f(\sigma^{-1}((\sigma y)_1, 0))$, which is constant along the slices $\{(\sigma y)_1 = \text{const}\}$ and agrees with f on $M \cap U$ `Chapter4/Approximation · exists_local_extension_of_contMDiff`.

The local argument. Assume $k + w = n$. Let F be smooth on an open $U \subseteq \mathbb{R}^n$ and agree with the smooth f on $M \cap U$, and let $q \in M \cap U$ be a degenerate critical point of $f + \langle a, \cdot\rangle$. Then $(q, a + \nabla F(q))$ is a critical point of $E_{-\nabla F}$, lying over U, with value a `Chapter4/Approximation · mem_image_critPoints_of_not_isNondegenerate`.

Proof as formalized.

1. Put $g = f + \langle a, \cdot\rangle$ on M and $G = F + \langle a, \cdot\rangle$ on U; $g = G$ on M near q, so q is a critical point of $G|_M$, and dG_q vanishes on T_qM. As $dG_q = \langle a - (-\nabla F(q)), \cdot\rangle$, the vector $v = a - (-\nabla F(q)) = a + \nabla F(q)$ is normal at q, so $(q, v) \in N$; and $E_{-\nabla F}(q, v) = -\nabla F(q) + v = a$.
2. In Milnor's coordinates at q, $\mathrm{Hess}(G \circ x)(\xi, \eta) = d^2F_q(dx\,\xi, dx\,\eta) + \langle v, d^2x(\xi, \eta)\rangle = -(\langle d(-\nabla F)_q\, dx\,\xi, dx\,\eta\rangle - \ell_v(\xi, \eta))$. So the Hessian is $-B_{d(-\nabla F)_q, v}$, and the nullity of g at q, read in the slice chart `Chapter1/Manifold · IsSmoothCoord.nullity_eq`, is the nullity of $B_{d(-\nabla F)_q, v}$.
3. By §III.12 this is the dimension of the kernel of $dE_{-\nabla F}$ at (q, v). It is positive because q is degenerate, and an endomorphism of $\mathbb{R}^n$ with nonzero kernel is not onto `Chapter4/Approximation · finrank_ker_pos_iff_not_surjective`.

Theorem (V4.20). For every slice submanifold $M \subseteq \mathbb{R}^n$, every smooth $f : M \to \mathbb{R}$ and every Haar measure on $\mathbb{R}^n$, $f + \langle a, \cdot\rangle$ is a Morse function for almost every a; such a exist in every nonempty open set `Chapter4/Approximation · ae_isMorseFunction_add_inner` `Chapter4/Approximation · exists_mem_isMorseFunction_add_inner`.

Proof as formalized. If M is empty there is nothing to prove; otherwise $k + w = n$. Choose a local extension (U_p, F_p) at each $p \in M$. As M is second countable, countably many of the open sets $M \cap U_p$ cover M `Mathlib/Topology/Bases · TopologicalSpace.countable_cover_nhds`. For each of them, $E_{-\nabla F_p}$ is smooth on the open part of N over U_p, and Theorem 6.1 on that open set makes the image of its critical points there null. If $f + \langle a, \cdot\rangle$ is not a Morse function, it has a degenerate critical point q, which lies in some U_p of the countable family, and the local argument puts a in the corresponding null set. A countable union of null sets is null. No boundedness, compactness or closedness of anything is used.

Corollary 6.8 for compact M (V4.21). Let $M \subseteq \mathbb{R}^n$ be compact, f smooth, $\varepsilon > 0$, $r \in \mathbb{N}$, and y_j finitely many smooth coordinate systems of M with compact sets K_j in their ranges. There is a Morse function

$g = f + \langle a, \cdot \rangle$ with $|g - f| < \varepsilon$ on M and $\|D^i((g-f) \circ y_j^{-1})(u)\| < \varepsilon$ for all j, all $i \le r$ and all $u \in K_j$ `Chapter4/Approximation · exists_isMorseFunction_approx`.

Proof as formalized. Bound $\|x\| \le R$ on M, and the iterated derivatives of order $i \le r$ of the inclusion read in y_j by $C_{j,i}$ on K_j; these exist because those derivatives are continuous on the open target of y_j `Chapter4/Approximation · exists_bound_iteratedFDeriv_val_comp_symm`. Let $D \ge 0$ dominate all these bounds and $\delta = \varepsilon/(D+1)$, so that $\delta D < \varepsilon$. Choose a with $\|a\| < \delta$ and $f + \langle a, \cdot \rangle$ a Morse function. Then $|g - f| = |\langle a, x \rangle| \le \|a\|\|x\| \le \delta D < \varepsilon$. And $(g-f) \circ y_j^{-1}$ is the linear map $\langle a, \cdot \rangle$ composed with the inclusion read in y_j, so its i-th derivative is $\langle a, \cdot \rangle$ applied to the i-th derivative of the inclusion, of norm at most $\|a\| C_{j,i} < \varepsilon$.

Derivatives on a manifold exist only in coordinates, so Milnor's approximation of derivatives on a compact K has to be read in fixed coordinate systems covering K, finitely many by compactness: that is how it is stated.

Why boundedness is unnecessary. On a non-compact M the perturbation $\langle a, x \rangle$ is unbounded, so the theorem above gives uniform approximation only for compact M. Milnor's boundedness hypothesis serves exactly to keep his embedding $h = (f, \dots)$ bounded, so that the error terms $\sum h_i^2/2c$ are uniformly small. The formal proof makes the *perturbation* bounded instead. Let

$$\beta(y) = \frac{y}{\sqrt{1 + \|y\|^2}},$$

Mathlib's diffeomorphism of $\mathbb{R}^n$ onto its open unit ball `Mathlib/Analysis/Normed/Module/Ball/Homeomorph · OpenPartialHomeomorph.univUnitBall`, here as a partial diffeomorphism `Chapter4/Uniform · ballDiffeo`; $\|\beta(y)\| < 1$ `Chapter4/Uniform · norm_ballDiffeo_lt_one`. Then $g = f + \langle a, \beta(\cdot) \rangle$ satisfies $|g - f| \le \|a\|$, whatever f is.

1. *$\beta(M)$ is a submanifold.* For a slice chart σ of M, $\sigma \circ \beta^{-1}$, on the part of the unit ball over its domain, is a slice chart of $\beta(M)$: $z \in \beta(M)$ iff $\beta^{-1}(z) \in M$ iff $(\sigma\beta^{-1}z)_2 = 0$ `Chapter4/Uniform · ballSliceChart` `Chapter4/Uniform · isSliceSubmanifold_ballImage`. The maps $\beta : M \to \beta(M)$ and $\beta^{-1} : \beta(M) \to M$ are mutually inverse `Chapter4/Uniform · toBall` `Chapter4/Uniform · fromBall`, and β^{-1} is smooth `Chapter4/Uniform · contMDiff_fromBall`.
2. *The Morse property comes back along β.* If g' is a Morse function on $\beta(M)$ and $g = g' \circ \beta$ is smooth, then g is a Morse function on M `Chapter4/Uniform · isMorseFunction_of_ballImage`. Read g in the chart σ of M at x and g' in the chart $\sigma \circ \beta^{-1}$ of $\beta(M)$ at $\beta(x)$. The two charts send x and $\beta(x)$ to the same point `Chapter4/Uniform · ballSliceChart_toChart_toBall`, and since $(\sigma\beta^{-1})^{-1}(u) = \beta(\sigma^{-1}(u))$ the two local representatives agree near it `Chapter4/Uniform · fromBall_ballSliceChart_toChart_symm`. Critical points and non-degenerate critical points may be read in any smooth coordinate system `Chapter1/Manifold · IsSmoothCoord.isCriticalPointM_iff` `Chapter1/Manifold · IsSmoothCoord.isNondegenerateCriticalPointM_iff`, and functions agreeing near a point have the same critical and non-degenerate critical points there `Chapter4/Uniform · isCriticalPoint_iff_of_eventuallyEq` `Chapter4/Uniform · isNondegenerateCriticalPoint_iff_of_eventuallyEq`. So a critical point x of g gives a critical point $\beta(x)$ of g', which is non-degenerate, and hence x is non-degenerate.
3. *Genericity.* Apply the theorem above to the submanifold $\beta(M)$ and the smooth function $f \circ \beta^{-1}$: for almost every a, $f \circ \beta^{-1} + \langle a, \cdot \rangle$ is a Morse function on $\beta(M)$, and composed with β it is $f + \langle a, \beta(\cdot) \rangle$, which is smooth, $y \mapsto \langle a, \beta(y) \rangle$ being smooth `Chapter4/Uniform · contDiff_inner_ballDiffeo`; by step 2 it is a Morse function on M `Chapter4/Uniform · ae_isMorseFunction_add_inner_ballDiffeo`.

Corollary 6.8 (V4.21). Let $M \subseteq \mathbb{R}^n$ be any slice submanifold and $f : M \to \mathbb{R}$ any smooth function, bounded or not. For every $\varepsilon > 0$, $r \in \mathbb{N}$ and finitely many smooth coordinate systems y_j with compact sets K_j in their ranges, there is a Morse function g with $|g - f| < \varepsilon$ on all of M and $\|D^i((g-f) \circ y_j^{-1})(u)\| < \varepsilon$ for $i \le r$ and $u \in K_j$; one can take $g = f + \langle a, \beta(\cdot) \rangle$ with a small `Chapter4/Uniform · exists_isMorseFunction_approx_uniform`.

Proof as formalized. As for compact M, with $\beta \circ \iota \circ y_j^{-1}$ in place of $\iota \circ y_j^{-1}$ (ι the inclusion), smooth on the target of y_j, and with $D \geq 1$ dominating the derivative bounds on the K_j. Then $|g - f| = |\langle a, \beta(x)\rangle| \leq \|a\| < \delta \leq \delta D < \varepsilon$, and the derivative estimate is as before. Milnor's own route, L_p on an embedding with first coordinate f, is not formalized, although its two ingredients, that the image of an embedding is a submanifold and that the Morse notions transport along it, are proved (§III.19, §III.20).

III.18 Lemma 6.9: the index of L_p

Lemma (the index of $g - \ell_v$). For $q \in M$ and $v \in \mathbb{R}^n$, the index of $g - \ell_v$ is the number of principal curvatures in the direction v, with multiplicity, that are greater than 1 `Chapter4/Curvature · index_firstFF_sub_secondFF`.

Proof as formalized. Let $e_1, \dots, e_k$ be an orthonormal eigenbasis of S_v in T_qM, with eigenvalues $\kappa_1, \dots, \kappa_k$, and let $\Phi : \mathbb{R}^k \to \mathbb{R}^k$ send ξ to the coordinates c_i of $dx\,\xi$ in this basis, a linear isomorphism. Then $g(\xi,\xi) = \|dx\,\xi\|^2 = \sum_i c_i^2$, the basis being orthonormal, and $\ell_v(\xi,\xi) = \langle S_v\,dx\,\xi, dx\,\xi\rangle = \sum_i \kappa_i c_i^2$, S_v acting diagonally `Mathlib/Analysis/InnerProductSpace/Spectrum · LinearMap.IsSymmetric.eigenvectorBasis_apply_self_apply`. So the quadratic form of $g - \ell_v$ is equivalent, through Φ, to $\sum_i (1-\kappa_i)c_i^2$, and by Sylvester's law of inertia `Mathlib/LinearAlgebra/QuadraticForm/Signature · QuadraticForm.sigNeg_of_equiv_weightedSumSquares` its index is $\#\{i : 1 - \kappa_i < 0\} = \#\{i : \kappa_i > 1\}$.

At a critical point q of L_p, the index of L_p is the index of $2(g - \ell_{p-q})$ (§III.15), a positive multiple `Chapter1/Nondegenerate · index_smul`, hence the number of principal curvatures of M at q in the direction $p - q$ that are greater than 1 `Chapter4/Curvature · indexM_normSq_sub_eq_card`. If $p - q = tv$ with $\|v\| = 1$, the curvatures in the direction $p - q$ are t times those in the direction v, so the condition $\kappa > 1$ in the direction $p - q$ is Milnor's condition $K > 1/t$ in the direction v.

Lemma 6.9 (V4.23). Assume $k + w = n$, and let q be a non-degenerate critical point of L_p. Then the index of L_p at q is

$$\sum_{t \in [0,1]} \operatorname{mult}_q\,(q + t(p - q)),$$

a sum with finitely many nonzero terms: the number of focal points of (M, q) on the segment from q to p, each counted with its multiplicity `Chapter4/Curvature · indexM_normSq_sub_eq_finsum_focalMult`.

Proof as formalized. Put $v = p - q$, normal at q, and let $\kappa_1, \dots, \kappa_k$ be the principal curvatures in the direction v.

1. *No κ_i equals* 1. Otherwise the multiplicity of $q + 1 \cdot v = p$ would count the index i and be positive. But it is the nullity of E at $(q, p-q)$, which by Lemma 6.5 is the nullity of L_p at q, which is 0 by non-degeneracy. This is where non-degeneracy enters, and it is why Milnor's count of the eigenvalues that are at least $1/t$ must be a count of those that are greater than $1/t$: equality would make p a focal point of (M, q).
2. By the previous lemma the index is $\#\{i : \kappa_i > 1\}$. Group these indices by the value $t = \kappa_i^{-1}$: the index is $\sum_t \#\{i : t \neq 0,\ \kappa_i = t^{-1}\}$, summed over the finite set of values κ_i^{-1} with $\kappa_i > 1$, and each summand is $\operatorname{mult}_q(q + tv)$ by Lemma 6.3 `Chapter4/Curvature · focalMult_add_smul_eq_card`.
3. *The support.* If $t \in [0,1]$ and $\operatorname{mult}_q(q + tv) > 0$, then $t \neq 0$ and $\kappa_i = t^{-1} \geq 1$ for some i, hence $\kappa_i > 1$ by step 1, and t is one of the values of step 2. Conversely those values lie in $(0,1)$. So the sum over $t \in [0,1]$ has its support in the finite set of step 2, and the two sums agree.

The case $p = q$ is included: the index is 0, the segment is $\{q\}$, and q is not a focal point of (M, q). Milnor's proof sets it aside implicitly by writing $p = q + tv$ with $t > 0$.

III.19 The Whitney bridge: the image of an embedding is a submanifold

Milnor's manifolds are abstract; in the proof of Corollary 6.7 he quotes Whitney's embedding theorem and from then on identifies a manifold with its image in Euclidean space. Two facts are taken for granted there: that the image of a differentiable embedding is a manifold in Euclidean space, and that the embedding identifies the two, so that critical points, indices and sublevel sets correspond. This section proves the first, §III.20 the second.

Let M be a smooth manifold without boundary modelled on a finite-dimensional real space E, F a finite-dimensional real normed space (Milnor's $\mathbb{R}^K$), and $e : M \to F$ smooth, a topological embedding (a homeomorphism onto its image), with injective derivative everywhere. Let Z, W be finite-dimensional with $\dim Z = \dim E$ and $\dim Z + \dim W = \dim F$.

Linear algebra. If $A : E \to F$ is injective and $T : Z \cong E$, there is $L : W \to F$ such that $(z, w) \mapsto A(Tz) + Lw$ is an isomorphism $Z \times W \cong F$ `Chapter4/WhitneySlice · exists_continuousLinearEquiv_coprod`. Choose a complement C of range A. It has the dimension of W, since range A has that of Z; take L to be an isomorphism $W \cong C$ followed by the inclusion. The map $(z, w) \mapsto A(Tz) + Lw$ is injective: if $A(Tz) = -Lw$, this vector lies in range $A \cap C = 0$, so $z = 0$ and $w = 0$. Equal dimensions make it an isomorphism.

The local immersion theorem, as a slice chart (V4.14). For every $x \in M$ there is a slice chart σ of $e(M) \subseteq F$ whose domain contains $e(x)$, such that for each p in its domain $T(\sigma p)_1$ lies in the target of the chart φ of M at x, and $e(\varphi^{-1}(T(\sigma p)_1)) = p$ whenever $p \in e(M)$ `Chapter4/WhitneySlice · exists_sliceChart_range`.

Proof as formalized.

1. Let L complete de_x as above, and $\Psi(z, w) = e(\varphi^{-1}(Tz)) + Lw$, smooth on the open set $S_0 = \{(z, w) : Tz \in \text{target of } \varphi\}$. At $x_0 = (T^{-1}\varphi(x), 0)$ its derivative is $(z, w) \mapsto de_x(Tz) + Lw$, an isomorphism. Here the derivative of $e \circ \varphi^{-1}$ at $\varphi(x)$ is de_x, the derivative of a map read in charts `Milnor/Chapter1/InverseFunction · hasFDerivAt_writtenInExtChartAt`.
2. By the inverse function theorem on manifolds of the companion project `Milnor/Chapter1/InverseFunction · isLocalDiffeomorphAt_of_isInvertible_mfderiv`, Ψ agrees near x_0 with a partial diffeomorphism Φ; restrict it to S_0.
3. Let U be the set of y in the domain of φ with $(T^{-1}\varphi(y), 0)$ in the domain of Φ; it is open and contains x. **Here the embedding hypothesis enters:** e is a homeomorphism onto its image, so $U = e^{-1}(O)$ for some open $O \subseteq F$.
4. The slice chart is Φ^{-1} restricted to O, with base point $e(x)$. If $p = e(y)$ is in its domain, then $y \in U$ and $\Phi(T^{-1}\varphi(y), 0) = \Psi(T^{-1}\varphi(y), 0) = e(y) + L0 = p$, so $\Phi^{-1}(p)$ has second component 0. Conversely, if $\Phi^{-1}(p) = (z, 0)$, then $p = \Psi(z, 0) = e(\varphi^{-1}(Tz))$ lies in $e(M)$. The same computation gives the last assertion.

So $e(M)$ is a slice submanifold of F modelled on Z with complement W `Chapter4/WhitneySlice · isSliceSubmanifold_range`, and $e : M \to e(M)$ **is a diffeomorphism** `Chapter4/WhitneySlice · diffeomorphRange` `Chapter4/WhitneySlice · exists_diffeomorph_range`. The forward map is smooth because e is smooth with values in $e(M)$ `Milnor/Chapter2/Corestrict · contMDiff_codRestrict`. The inverse is smooth because near $e(x)$ it is $p \mapsto \varphi^{-1}(T(\sigma p)_1)$, a composite of the inclusion of $e(M)$, the chart σ, a linear map and φ^{-1} `Chapter4/WhitneySlice · contMDiff_ofInjective_symm`.

Three remarks. An injective immersion need not have a submanifold as image (the figure eight); the hypothesis that e is a homeomorphism onto its image is used once, in step 3, and for compact M it is automatic. The inverse of e is not smooth merely because e is a smooth bijection onto $e(M)$ ($x \mapsto x^3$ on $\mathbb{R}$); it is read off the slice chart. And the model of $e(M)$ may be any Z of the right dimension, which is how an abstract manifold modelled on E is read on $\mathbb{R}^{d+1}$, the model of §III.9.

III.20 Invariance of the Morse notions under diffeomorphisms

Milnor defines critical points, non-degeneracy and the index intrinsically, and in §6 treats a function on a manifold and the corresponding function on its image as the same; the invariance this needs is never stated. Let $g : M \to M'$ be a smooth diffeomorphism between smooth manifolds without boundary modelled on real normed spaces E, E', and $f : M \to \mathbb{R}$. The function f on M corresponds to $f \circ g^{-1}$ on M', and all the statements below are V4.15.

Critical points `Chapter4/WhitneyInvariance · isCriticalPointM_comp_symm_iff`. q is a critical point of $f \circ g^{-1}$ iff $g^{-1}(q)$ is a critical point of f, for *every* f. If f is differentiable at $g^{-1}(q)$, the chain rule gives $d(f \circ g^{-1})_q = df_{g^{-1}(q)} \circ d(g^{-1})_q$, and $d(g^{-1})_q$ is onto, so one side vanishes iff the other does. If f is not differentiable there, neither is $f \circ g^{-1}$ at q, since composing back with g would make f differentiable; and both derivatives are 0 by convention. Hence g maps the critical set of f onto that of $f \circ g^{-1}$ `Chapter4/WhitneyInvariance · setOf_isCriticalPointM_comp_symm`, and the two critical sets have the same number of elements `Chapter4/WhitneyInvariance · ncard_setOf_isCriticalPointM_comp_symm`.

The Hessian. The Hessian of $f \circ g^{-1}$ at q is read in the chart of M' at q (§0.2) `Chapter1/Manifold · hessianM_eq_chartAt`. That is the same as reading f in the coordinate system $\mathrm{chart}_q \circ g$ on M around $g^{-1}(q)$ `Chapter4/WhitneyInvariance · diffeoCoord` `Chapter4/WhitneyInvariance · hessianM_comp_symm_eq`, which is a smooth coordinate system `Chapter4/WhitneyInvariance · isSmoothCoord_diffeoCoord`. By the chart independence of Part I `Chapter1/Manifold · IsSmoothCoord.hessian_comp_symm`, at a critical point where f is C^2 that Hessian is the Hessian of f pulled back by the derivative of the inverse coordinates. That derivative is $d(g^{-1})_q$: the derivative of the inverse chart at the centre of the chart is the identity `Chapter4/WhitneyInvariance · mfderiv_chartAt_symm_self_apply`. So

$$(f \circ g^{-1})_{**}(v, w) = f_{**}(d(g^{-1})_q v,\ d(g^{-1})_q w)$$

`Chapter4/WhitneyInvariance · hessianM_comp_symm_apply`. Non-degeneracy, non-degenerate critical points, index and nullity correspond, through the same coordinate system `Chapter4/WhitneyInvariance · isNondegenerate_hessianM_comp_symm_iff` `Chapter4/WhitneyInvariance · isNondegenerateCriticalPointM_comp_symm_iff` `Chapter4/WhitneyInvariance · indexM_comp_symm` `Chapter4/WhitneyInvariance · nullityM_comp_symm` `Chapter1/Manifold · IsSmoothCoord.index_eq`.

Morse functions and Morse numbers. A Morse function composed with g^{-1} is a Morse function `Chapter4/WhitneyInvariance · IsMorseFunction.comp_diffeomorph_symm`, and applying this to g^{-1} gives the converse `Chapter4/WhitneyInvariance · isMorseFunction_comp_symm_iff`. For f of class C^2, g maps the critical points of f of index λ bijectively onto those of $f \circ g^{-1}$, so $C_\lambda(f \circ g^{-1}) = C_\lambda(f)$ `Chapter4/WhitneyInvariance · critCount_comp_symm`. Only C^2 at the point is used for the Hessian statements, criticality needs no regularity at all, and the model spaces may differ.

III.21 Part I on an abstract compact manifold

In this section M is a compact Hausdorff smooth manifold without boundary modelled on a finite-dimensional real space E: one of Milnor's manifolds, read as Hausdorff, the countable basis being automatic by compactness.

Embeddings (V4.16). Mathlib's Whitney theorem for compact manifolds, `exists_embedding_euclidean_of_compact`, gives a smooth closed embedding $e : M \to \mathbb{R}^n$ with injective derivative, for some n; by §III.19 its image is a slice submanifold modelled on E `Chapter4/Whitney · exists_isSliceSubmanifold_whitney`, the dimension count $\dim E \le n$ coming from the injectivity of a derivative `Chapter4/Whitney · finrank_le_of_injective_clm`. Following e by $v \mapsto (v, 0)$ into a larger Euclidean space gives an embedding into $\mathbb{R}^{K+1}$ with $K \ge 1$, and with $\dim M + m = K + 1$ for some m `Chapter4/Whitney · exists_embedding_euclidean`. The two extra coordinates are there for the Poincaré–Hopf theorem of the companion project, which needs $K \ge 1$; Milnor's $\mathbb{R}^{2n+1}$ plays no role. Together with §III.19: M **is diffeomorphic to a compact, closed slice submanifold** $N \subseteq \mathbb{R}^{K+1}$, $K \ge 1$, **modelled on** $\mathbb{R}^{\dim M}$ `Chapter4/Whitney · exists_diffeomorph_sliceSubmanifold`. Below, $\Phi : M \to N$ is such a diffeomorphism.

Theorem 6.6 and Corollary 6.7 (V4.17). M carries a Morse function, with finitely many critical points `Chapter4/Whitney · exists_isMorseFunction_of_compactSpace`: take a Morse function f_0 on N (§III.16) and pull it back to $f_0 \circ \Phi$, a Morse function by §III.20; its critical set is finite by compactness. So the function is $L_p \circ \Phi$, for a p taken from a set of full measure.

Corollary 6.8 (V4.22). For any embedding $e : M \to \mathbb{R}^n$ as in §III.19 and any smooth f, $f + \langle a, e(\cdot)\rangle$ is a Morse function for almost every a `Chapter4/Whitney · ae_isMorseFunction_add_inner_comp`; this statement needs neither compactness nor the Hausdorff property, nor a closed embedding. If M is empty it is vacuous. Otherwise $e(M)$ is a slice submanifold diffeomorphic to M by $\Phi = e$ (§III.19), $f \circ \Phi^{-1}$ is smooth on it, the theorem of §III.17 applies to it, and §III.20 carries the Morse property back, since $f + \langle a, e(\cdot)\rangle = (f \circ \Phi^{-1} + \langle a, \cdot\rangle) \circ \Phi$. With Mathlib's embedding, the estimates of §III.17 then give Corollary 6.8 on M: every smooth f is a uniform limit of Morse functions $f + \langle a, e(\cdot)\rangle$, with derivatives up to any order close on compact sets of finitely many coordinate systems, which may now take values in any real normed space `Chapter4/Whitney · exists_isMorseFunction_approx_of_compactSpace`. Milnor's boundedness hypothesis is automatic for compact M.

The Euler number (V4.18). In dimension 0 every point is critical for every function `Chapter4/Whitney · isCriticalPointM_of_finrank_eq_zero`, of index 0, so $\sum_\lambda (-1)^\lambda C_\lambda(f) = C_0(f) = \#M$ for every f `Chapter4/Whitney · sum_critCount_of_finrank_eq_zero`. Define

$$\chi(M) = \sum_{\lambda=0}^{\dim M} (-1)^\lambda C_\lambda(f_M)$$

for one Morse function f_M chosen once and for all, and $\chi(M) = 0$ if M has none `Chapter4/Whitney · eulerNumberM`. This is Milnor's formula $\chi(M) = C_0 - C_1 + \cdots \pm C_n$ taken as a definition; the next result shows the choice does not matter.

Theorem (V4.19, formula (3) on a compact manifold). For any two Morse functions f, g on M, $\sum_\lambda (-1)^\lambda C_\lambda(f) = \sum_\lambda (-1)^\lambda C_\lambda(g)$ `Chapter4/Whitney · sum_critCount_eq_of_isMorseFunction_of_compactSpace`. Hence every Morse function f on M has $\sum_\lambda (-1)^\lambda C_\lambda(f) = \chi(M)$ `Chapter4/Whitney · sum_critCount_eq_eulerNumberM` `Chapter4/Whitney · exists_eulerNumber_of_compactSpace`.

Proof as formalized. In dimension 0 both sides are $\#M$. If M is empty, every C_λ is 0. Otherwise $\dim M = d + 1$; take the embedded copy N, nonempty, with $K \geq 1$. The functions $f \circ \Phi^{-1}$ and $g \circ \Phi^{-1}$ are Morse functions on N with the same Morse numbers as f and g (§III.20), and on N the alternating count does not depend on the Morse function (§III.9).

Application 2, on an embedded copy. Let $\dim M = d + 1 \geq 1$, and let $N \subseteq \mathbb{R}^{K+1}$, $K \geq 1$, be any compact nonempty slice submanifold modelled on $\mathbb{R}^{d+1}$ and diffeomorphic to M, for instance the image of any embedding. Then every Euler number of N equals $\chi(M)$ `Chapter4/Whitney · IsEulerNumber.eq_eulerNumberM`. Indeed, for a Morse function f on M, the Euler number is $\sum_\lambda (-1)^\lambda C_\lambda(f \circ \Phi^{-1})$ by §III.9, which is $\sum_\lambda (-1)^\lambda C_\lambda(f) = \chi(M)$ by §III.20. So the Poincaré–Hopf integer does not depend on the embedding. And M of positive dimension is diffeomorphic to a compact N on which every smooth tangent field with finitely many zeros has index sum $\chi(M)$ `Chapter4/Whitney · exists_isEulerNumber_eulerNumberM`: the Euler number of the copy exists by Poincaré–Hopf, and it equals $\chi(M)$. Vector fields and their indices are defined in the companion project only for submanifolds of Euclidean space, which is why Application 2 is stated on a copy. Finally χ is invariant under diffeomorphisms of compact Hausdorff manifolds `Chapter4/Whitney · eulerNumberM_eq_of_diffeomorph`: compute both sides with f and $f \circ g^{-1}$, whose Morse numbers agree; the dimensions agree when M is nonempty, a derivative of g being an isomorphism, and both counts vanish when it is empty. That $\chi(M) = \sum_i (-1)^i \dim H_i(M; \Bbbk)$ for every field $\Bbbk$ is in Part IV `Homology/MorseHomologyManifold · eulerNumberM_eq_sum_bettiNumber`.

Theorem 4.1 (Reeb) on a compact manifold (V4.24). If a smooth f on M, $\dim M = n$, has exactly two critical points, both non-degenerate, then M is homeomorphic to S^n `Chapter4/Whitney · reeb_of_compactSpace`.

Proof as formalized. On the copy $N \subseteq \mathbb{R}^{K+1}$, modelled on $\mathbb{R}^n$, the function $f \circ \Phi^{-1}$ is smooth, has exactly two critical points (§III.20, the counts of critical points agree), and they are non-degenerate (§III.20). For $n = 0$, §III.4 in dimension 0 applies; for $n \geq 1$, Reeb's theorem for intrinsic functions on a compact submanifold of $\mathbb{R}^{K+1}$ (§III.4). The homeomorphism is Φ followed by $N \cong S^n$.

Theorem 3.1 on a compact manifold (V2.4, V4.25). If f is smooth on M, $a \leq b$, and $f^{-1}[a, b]$ contains no critical point, then a diffeomorphism Ψ of M has $\Psi(M^a) = M^b$, M^a is a strong deformation retract of M^b, and the inclusion $M^a \to M^b$ is a homotopy equivalence `Chapter4/Whitney · theorem_3_1_of_compactSpace`.

Proof as formalized.

1. The homotopy equivalence follows from the strong deformation retraction (Part II) `Chapter2/Crossing · IsStrongDeformationRetract.exists_homotopyEquiv`, so two statements remain.
2. *The empty slab.* If f takes no value in $[a, b]$, then $M^a = M^b$: a point with $f \leq b$ has $f < a$. The identity diffeomorphism works, and M^a is trivially a strong deformation retract of itself `Chapter2/Crossing · IsStrongDeformationRetract`.
3. Otherwise some p_0 has $f(p_0) \in [a, b]$. It is not critical, so $\dim M \geq 1$, since in dimension 0 every point is critical. Take the copy N, nonempty, and extend $f \circ \Phi^{-1}$ to a smooth F on $\mathbb{R}^{K+1}$ (§III.9). The critical points of $F|_N$ correspond to those of f (§III.20), so $F|_N$ has none in $f^{-1}[a, b]$. Theorem 3.1 on the compact submanifold N (Part II) gives a diffeomorphism Φ_N with $\Phi_N(N^a) = N^b$ and the strong deformation retraction of N^b onto N^a.
4. *Transport.* Φ maps M^c onto N^c for every c, since $F \circ \Phi = f$. So $\Psi = \Phi^{-1} \circ \Phi_N \circ \Phi$ maps M^a onto M^b. And Φ restricts to a homeomorphism $M^b \cong N^b$ under which M^a corresponds to N^a. A strong deformation retract is carried back by a homeomorphism h, with retraction $h^{-1} \circ r \circ h$ and homotopy $h^{-1} \circ H \circ (\mathrm{id} \times h)$ `Chapter4/Whitney · IsStrongDeformationRetract.preimage_homeomorph`.

Theorems 3.2 and 3.5 are not transferred. Nor is anything for non-compact abstract manifolds: Mathlib has no closed embedding of a non-compact manifold into Euclidean space.

III.22 What the formalization taught us

- **Reeb's eight lines.** The indices 0 and n at the extrema are used without comment (§III.1); minimum and maximum are distinct only for $n \geq 1$ and by non-degeneracy (§III.4); a small ε must put the whole sublevel set in the Morse chart, which needs compactness and a unique minimum (§III.2); the level $f^{-1}(\varepsilon)$ goes onto $f^{-1}(1 - \varepsilon)$ by the proof of Theorem 3.1, not by its statement (§III.4); the easy gluing is Alexander's trick, the pasting lemma and compact-to-Hausdorff, and the cone is where smoothness is lost (§III.3).
- **The torus** must stand upright with $r < R$; its four critical points, indices and normal forms are asserted by Milnor and verified by Lagrange's rule (§III.6).
- **Application 2** needs a metric, $\operatorname{sgn}\det = (-1)^{\text{index}}$, and the replacement of F by $F \circ r$, because the Hessian of $F|_N$ is not the restriction of that of F (§III.7, §III.8); its function must be a Morse function, and *isolated* in §5 is redundant (§III.9).
- **§6, possible misprints and gaps:** $K_1, \ldots, K_k$ for $K_1, \ldots, K_n$; nullity for rank; normal frames only local, and unnecessary (§III.12); $t = 0$ and $p = q$ set aside (§III.14, §III.18); $> 1/t$ for $\geq 1/t$, equality being excluded exactly by non-degeneracy (§III.18). The shape operator removes the normalized coordinates (§III.14).
- **Corollaries 6.7 and 6.8** need Hausdorff, second countable manifolds and a closed embedding, and fixed charts for derivatives; the boundedness of f in 6.8 is unnecessary (§III.16, §III.17).
- **Abstract manifolds** need two facts Milnor never states: the image of an embedding (not merely of an injective immersion) is a submanifold diffeomorphic to the manifold, and the Morse notions are invariant under diffeomorphisms (§III.19, §III.20).

Not formalized: Remarks 1 and 2 on Reeb's theorem; the sublevel sets (2)–(4) of the torus and its cell attachments; Milnor's own constructions in Corollary 6.8; Theorem 6.1 for C^1 maps between manifolds;

Corollary 6.7 for non-compact abstract manifolds and Application 1; and the independence of the shape operator from the parametrization, which nothing uses.

Part IV. Homology and the Morse inequalities (Milnor §5)

Milnor's §5 turns the cell decompositions of §3 into inequalities between numbers. Let M be a compact manifold, f a function on M whose critical points are non-degenerate, C_λ the number of critical points of index λ, and $R_\lambda(M) = \dim_F H_\lambda(M;F)$ the Betti numbers over a field F. Theorem 5.2 gives the weak inequalities $R_\lambda(M) \le C_\lambda$ and the equality $\sum_\lambda (-1)^\lambda R_\lambda(M) = \sum_\lambda (-1)^\lambda C_\lambda$; Lemma 5.3 leads to the strong inequalities (4_λ); Corollary 5.4 turns some of them into equalities. The proof has two halves of very unequal weight in the formal text. The algebraic half (subadditive functions, Lemmas 5.1 and 5.3, and the bookkeeping of pp. 28–31) takes a page of the book and one file here. The topological half is a single line on p. 29, which identifies the homology of a slab $(M^{a_i}, M^{a_{i-1}})$ with that of a cell modulo its boundary "by excision"; it rests on relative singular homology, its exact sequences, homotopy invariance, excision and the homology of $(D^\lambda, S^{\lambda-1})$, all quoted as standard.

Mathlib, the library underneath, has the singular chain complex of a space, absolute singular homology with its homotopy invariance, the category of pairs of spaces with the data of an Eilenberg–Steenrod theory, and the long exact homology sequence of a short exact sequence of complexes. It has no relative singular homology, no exact sequence of a pair or of a triple, no excision, and no homology of a sphere. For most of its length this part is therefore an account of the singular homology that the project built: relative homology as a cokernel complex and the sequences of pairs and triples (§§IV.5–IV.6), homotopy invariance for pairs (§IV.7), a concrete model of singular chains (§IV.8), barycentric subdivision and the small-simplex theorem (§§IV.9–IV.12), excision and Mayer–Vietoris (§§IV.13–IV.14), and cells, disks and spheres (§§IV.15–IV.17). The Morse inequalities in singular homology then take the last four sections.

The formal route departs from Milnor's in five places, each explained where it occurs.

1. The algebra is proved once, for an abstract *homology of a filtration* whose only axioms are finite-dimensionality and the exact sequences of triples (§§IV.1–IV.4). Singular homology over a field is an instance (§IV.18).
2. Milnor chooses levels a_i with exactly one critical point between consecutive ones, which presupposes distinct critical values. Here a slab contains one critical *value* and any number of critical points on it, whose cells are attached at once (Remark 3.3; Part II, §II.14), and the homology of the slab comes from a cell-attachment theorem for several disjoint cells (§IV.17).
3. "By excision" is not literally excision. The cell-attachment theorem first removes the centre of the cell and pushes the rest radially onto the bottom set, and then excises (§IV.15).
4. The top of a pair is moved along strong deformation retractions that fix the bottom; the bottom is moved in homology, through the exact sequence of a triple (§IV.19).
5. Everything is proved first for a compact submanifold of a Euclidean space and a function that extends smoothly to the ambient space, and then carried to abstract compact manifolds by a Whitney embedding (§IV.21). There it also identifies the Euler number of Part III with $\sum_i (-1)^i \dim H_i(M;\Bbbk)$, for every field $\Bbbk$.

Results carry the labels V3.7–V3.15 and V3.26–V3.40 of the verified edition

(`Exposition/verification/ch03.md`). Throughout, $\Bbbk$ denotes a field (Milnor's F; the letter F is kept for ambient functions). The general constructions allow coefficients in an object R of an abelian category with coproducts; from excision on, R is a commutative ring, viewed as a module over itself; the Morse inequalities use a field. One formality of type theory recurs: from excision on, the spaces whose homology is taken live in the universe of the coefficients, as the comparison of chain models in §IV.8 requires; §IV.19 explains how the final statements accommodate it. Singular chains, homology and Betti numbers were introduced in §0.6; this part says how they are built.

Map.

- `Chapter3/Inequalities`: alternating sums, truncated exact sequences, subadditive functions, Lemma 5.1, the homology of a filtration, Lemma 5.3, Theorem 5.2, (4_λ), Corollary 5.4 (§§IV.1–IV.4).
- `Homology/Relative`, `Homology/Triple`: relative singular homology as a cokernel complex; the sequences of a pair and of a triple; $H_*(X,\emptyset)$, $H_*(X,X)$; the five lemma for pairs (§§IV.5–IV.6).
- `Homology/RelativeHomotopy`, `Homology/RelativeRetract`, `Homology/Pretheory`: homotopy invariance for pairs through the cylinder; strong deformation retracts; the Eilenberg–Steenrod pretheory (§IV.7).
- `Homology/ModuleComplex`, `Homology/Simplex`: chain maps and homotopies degree by degree, subcomplexes and quotients; singular chains as finite combinations of maps from the simplex, and their isomorphism with Mathlib's complex (§IV.8).
- `Homology/LinearChains`, `Homology/AffineSimplex`: linear chains, the cone, barycentric subdivision; affine simplices, barycentres, the diameter bound (§§IV.9–IV.10).
- `Homology/Subdivision`, `Homology/SmallChains`: subdivision of singular chains and the homotopy T; the small-simplex theorem (§§IV.11–IV.12).
- `Homology/Excision`, `Homology/MayerVietoris`: excision; the Mayer–Vietoris sequence (§§IV.13–IV.14).
- `Homology/CellExcision`, `Homology/Spheres`, `Homology/CellAttach`: excision for an attached cell; disks and spheres; the cell-attachment theorem (§§IV.15–IV.17).
- `Homology/TripleFiltration`: singular homology as a homology of a filtration (§IV.18).
- `Homology/MorseHomologyBasic`, `Homology/MorseHomologyCells`, `Homology/MorseHomology`, `Homology/MorseHomologyManifold`: the slab, the Morse filtration, Theorem 5.2, (4_λ) and Corollary 5.4 in singular homology, compact manifolds, $\chi=\sum_i(-1)^i b_i$, examples (§§IV.19–IV.21).

IV.1 Alternating sums and finite exact sequences

The file `Chapter3/Inequalities` contains no space at all: only integers, finite-dimensional vector spaces over a field, and linear maps. Everything Milnor does on pp. 28–31 once the homology of the slabs is known is proved there.

Definition. For a sequence of integers $d=(d_0,d_1,\dots)$ and $n\in\mathbb{N}$, the *truncated alternating sum* is

$$\operatorname{alt}_n(d)=\sum_{i=0}^{n}(-1)^{n-i}d_i=d_n-d_{n-1}+d_{n-2}-\cdots\pm d_0$$

`Chapter3/Inequalities · altSum`. With $d_i=R_i(X,Y)$ and $n=\lambda$ it is Milnor's $S_\lambda(X,Y)$ of Lemma 5.3; with $d_i=C_i$ it is the right side of (4_λ).

Its algebra is elementary, and each fact is a lemma. $\operatorname{alt}_0(d)=d_0$ `Chapter3/Inequalities · altSum_zero`. Since $(-1)^{n+1-i}=-(-1)^{n-i}$ for $i\le n$, a new term changes the sign of all earlier ones: $\operatorname{alt}_{n+1}(d)=d_{n+1}-\operatorname{alt}_n(d)$ `Chapter3/Inequalities · altSum_succ`, hence

$$\operatorname{alt}_{n+1}(d)+\operatorname{alt}_n(d)=d_{n+1}$$

`Chapter3/Inequalities · altSum_succ_add_altSum`, the identity behind Milnor's remark that adding $(4_{\lambda+1})$ and (4_λ) gives the weak inequality in degree $\lambda+1$. The sum depends only on $d_0,\dots,d_n$ `Chapter3/Inequalities · altSum_congr`, is additive and subtractive in d `Chapter3/Inequalities · altSum_add Chapter3/`

`Inequalities · altSum_sub`, and commutes with finite sums, $\mathrm{alt}_n\left(\sum_k d^{(k)}\right) = \sum_k \mathrm{alt}_n(d^{(k)})$, by an exchange of two finite sums `Chapter3/Inequalities · altSum_sum`. The last is Milnor's rewriting of $\sum_i S_\lambda(M^{a_i}, M^{a_{i-1}})$ as $C_\lambda - C_{\lambda-1} + \cdots \pm C_0$ (p. 30). The Euler characteristic truncated at N is a signed alternating sum,

$$\sum_{i \le N} (-1)^i d_i = (-1)^N \,\mathrm{alt}_N(d)$$

`Chapter3/Inequalities · sum_neg_one_pow_mul_eq`: writing $N = i + k$, the term $(-1)^N(-1)^{N-i}d_i$ is $(-1)^{i+2k}d_i = (-1)^i d_i$.

Lemma (telescoping). If $d_0 = r_0$ and $d_{i+1} = r_{i+1} + r_i$ for all $i < n$, then $\mathrm{alt}_n(d) = r_n$ `Chapter3/Inequalities · altSum_eq_of_telescope`.

Proof. Induction on n: $\mathrm{alt}_{n+1}(d) = d_{n+1} - \mathrm{alt}_n(d) = (r_{n+1} + r_n) - r_n$. □

Lemma (Milnor's computation, pp. 29–30; V3.12). Let $V_0, V_1, \ldots$ be finite-dimensional vector spaces over a field and $d_k : V_{k+1} \to V_k$ linear maps such that

$$V_{n+1} \xrightarrow{d_n} V_n \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_0} V_0 \to 0$$

is exact at $V_0, \ldots, V_n$: d_0 is onto, and $\ker d_k = \operatorname{im} d_{k+1}$ for $k < n$. Then

$$\operatorname{rank} d_n = \mathrm{alt}_n(\dim V_\bullet) = \dim V_n - \dim V_{n-1} + \cdots \pm \dim V_0$$

`Chapter3/Inequalities · finrank_range_eq_altSum`.

Proof. The telescoping lemma with $r_k = \operatorname{rank} d_k$. First, $r_0 = \dim V_0$ because d_0 is onto. Second, rank–nullity for d_k `Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · finrank_range_add_finrank_ker` together with $\ker d_k = \operatorname{im} d_{k+1}$ gives $\dim V_{k+1} = r_k + r_{k+1}$. □

In Milnor's notation $h = d_n$, $A = V_n$, $B = V_{n-1}$, …, $D = V_0$. His first step, $\operatorname{rank} h + \operatorname{rank} i = \dim A$, is the second equation for $k = n - 1$, and his unwinding along the sequence is the telescoping lemma. If moreover $d_n = 0$, so that the sequence is a finite exact sequence $0 \to V_n \to \ldots \to V_0 \to 0$, the alternating sum of the dimensions vanishes `Chapter3/Inequalities · altSum_finrank_eq_zero_of_exact`. This is the argument behind the additivity of the Euler characteristic, which Milnor asserts without proof on p. 28. (Mathlib has the vanishing for sequences indexed by a finite interval; the triples need the form for a sequence indexed by $\mathbb{N}$ and cut off at n.)

Definition (long exact sequences; V3.9). A *long exact sequence* over a field $\Bbbk$ `Chapter3/Inequalities · LongExactSeq` consists of three families A_i, B_i, C_i ($i \in \mathbb{N}$) of $\Bbbk$-vector spaces and linear maps $f_i : A_i \to B_i$, $g_i : B_i \to C_i$, $\delta_i : C_{i+1} \to A_i$, such that

$$\cdots \to A_{i+1} \xrightarrow{f} B_{i+1} \xrightarrow{g} C_{i+1} \xrightarrow{\delta_i} A_i \xrightarrow{f} B_i \xrightarrow{g} C_i \to \cdots \to A_0 \to B_0 \to C_0 \to 0$$

is exact at every node: $\ker g_i = \operatorname{im} f_i$, $\ker \delta_i = \operatorname{im} g_{i+1}$, $\ker f_i = \operatorname{im} \delta_i$, and g_0 is onto. For a triple $X \supset Y \supset Z$ the intended values are $A_i = H_i(Y, Z)$, $B_i = H_i(X, Z)$, $C_i = H_i(X, Y)$, with δ the boundary of the triple. Milnor's sequence ends with "$\to D \to 0$" (p. 29); for a triple this is the surjectivity of $H_0(X, Z) \to H_0(X, Y)$, true because there is no homology in degree -1. Here it is an axiom of the structure, and §IV.6 proves it for singular homology.

Three lemmas about a long exact sequence carry Milnor's algebra.

- *The middle is finite-dimensional when the ends are.* If A_i and C_i are finite-dimensional, so is B_i `Chapter3/Inequalities · LongExactSeq.finiteDimensional_middle`, since $A_i \to B_i \to \operatorname{im} g_i \to 0$ is exact and $\operatorname{im} g_i$ is a subspace of C_i. Milnor defines R_λ only for pairs whose λ-th homology is finite-dimensional (p. 28). This lemma is the closedness of that class of pairs under $(X, Y), (Y, Z) \mapsto (X, Z)$, which his statement that R_λ is subadditive presupposes.

- *R_λ is subadditive* (p. 28, left to the reader; V3.10). If all the spaces are finite-dimensional, then $\dim B_i \le \dim A_i + \dim C_i$ `Chapter3/Inequalities · LongExactSeq.finrank_le`: by rank–nullity for g_i and $\ker g_i = \operatorname{im} f_i$, $\dim B_i = \operatorname{rank} g_i + \operatorname{rank} f_i \le \dim C_i + \dim A_i$.
- *The key identity of Lemma 5.3* (p. 30). If all the spaces are finite-dimensional, then for every λ

$$\operatorname{rank} \delta_\lambda = \operatorname{alt}_\lambda(\dim A_\bullet) - \operatorname{alt}_\lambda(\dim B_\bullet) + \operatorname{alt}_\lambda(\dim C_\bullet)$$

`Chapter3/Inequalities · LongExactSeq.finrank_range_δ`. In Milnor's notation, $\operatorname{rank} \partial = S_\lambda(Y,Z) - S_\lambda(X,Z) + S_\lambda(X,Y)$.

Proof of the key identity. Rank–nullity at the three kinds of nodes gives

$$\dim A_i = \operatorname{rank} f_i + \operatorname{rank} \delta_i, \quad \dim B_i = \operatorname{rank} g_i + \operatorname{rank} f_i, \quad \dim C_{i+1} = \operatorname{rank} \delta_i + \operatorname{rank} g_{i+1}, \quad \dim C_0 = \operatorname{rank} g_0.$$

So $e_i = \dim A_i - \dim B_i + \dim C_i$ satisfies $e_0 = \operatorname{rank} \delta_0$ and $e_{i+1} = \operatorname{rank} \delta_{i+1} + \operatorname{rank} \delta_i$. The telescoping lemma gives $\operatorname{alt}_\lambda(e) = \operatorname{rank} \delta_\lambda$, and linearity splits $\operatorname{alt}_\lambda(e)$ into the three sums. □

Milnor unwinds the sequence $H_\lambda(Y,Z) \to H_\lambda(X,Z) \to H_\lambda(X,Y) \to H_{\lambda-1}(Y,Z) \to \ldots \to 0$ term by term; the formal proof groups the terms by degree. Two corollaries follow at once. A rank is non-negative, so

$$\operatorname{alt}_\lambda(\dim B_\bullet) \le \operatorname{alt}_\lambda(\dim A_\bullet) + \operatorname{alt}_\lambda(\dim C_\bullet)$$

`Chapter3/Inequalities · LongExactSeq.altSum_le`, which is Lemma 5.3 for one triple. And if $C_{N+1} = 0$, then δ_N has rank 0 and the truncated sums are additive, $\operatorname{alt}_N(\dim B_\bullet) = \operatorname{alt}_N(\dim A_\bullet) + \operatorname{alt}_N(\dim C_\bullet)$ `Chapter3/Inequalities · LongExactSeq.altSum_eq_of_finrank_eq_zero`. The additivity of the Euler characteristic on a triple therefore needs only $H_{N+1}(X,Y) = 0$, not the vanishing of all homology above N.

IV.2 Subadditive functions and Lemma 5.1

The formalization replaces spaces by indices. A preorder ι indexes the spaces, $a \le b$ standing for $X_a \subseteq X_b$; for Milnor, $\iota = \mathbb{R}$ with $X_a = M^a$, or $\iota = \mathbb{N}$ with $X_k = M^{a_k}$. A function on pairs is a function S of two indices with integer values, of which only the values at $a \le b$ are ever used, and $S(a,b)$ stands for Milnor's $S(X_b, X_a)$: Milnor writes the bigger space first, the formal text lists the indices in increasing order.

Definition (p. 28; V3.7). S is *subadditive* if $S(a,c) \le S(a,b) + S(b,c)$ whenever $a \le b \le c$ `Chapter3/Inequalities · IsSubadditive`, and *additive* if equality always holds `Chapter3/Inequalities · IsAdditive`. An additive function is subadditive `Chapter3/Inequalities · IsAdditive.isSubadditive`. In Milnor's notation, $S(X,Z) \le S(X,Y) + S(Y,Z)$ for $X \supset Y \supset Z$.

Milnor does not say on which pairs S is defined. Here S is defined on all pairs, and the question which pairs have finite-dimensional homology is moved into the homology structure of §IV.3, where it is settled once.

Lemma 5.1 (V3.8). Let S be subadditive and $X_0 \le X_1 \le \ldots \le X_n$ with $n \ge 1$. Then

$$S(X_0, X_n) \le \sum_{k=0}^{n-1} S(X_k, X_{k+1}),$$

with equality if S is additive `Chapter3/Inequalities · IsSubadditive.le_sum` `Chapter3/Inequalities · IsAdditive.eq_sum`.

The proof as formalized is Milnor's induction, with the hypothesis it actually uses isolated: subadditivity on the triples $X_0 \le X_m \le X_{m+1}$, $0 < m < n$, suffices `Chapter3/Inequalities · le_sum_of_chain`, and in the

equality case additivity on those triples `Chapter3/Inequalities · eq_sum_of_chain`. For $n = 1$ both sides are $S(X_0, X_1)$. From n to $n+1$,

$$S(X_0, X_{n+1}) \le S(X_0, X_n) + S(X_n, X_{n+1}) \le \sum_{k<n} S(X_k, X_{k+1}) + S(X_n, X_{n+1}).$$

That these are triples, $X_0 \le X_m$, follows from the chain by induction `Chapter3/Inequalities · chain_le`. □

The hypothesis $n \ge 1$ is not decoration. For the empty chain the statement would read $S(X_0, X_0) \le 0$, which subadditivity does not give: it gives only $S(X, X) \le 2S(X, X)$, that is, $S(X, X) \ge 0$. Milnor's induction starts at $n = 1$. With $X_0 = \emptyset$, Lemma 5.1 is Milnor's inequality (1).

IV.3 The homology of a filtration: Lemma 5.3, Theorem 5.2 and the inequalities (4_λ)

Definition (V3.9). A *homology of a filtration* over a field $\Bbbk$, indexed by a preorder ι `Chapter3/Inequalities · FiltrationHomology`, is a family of $\Bbbk$-vector spaces $H(i, a, b)$, $i \in \mathbb{N}$, $a, b \in \iota$, to be read as $H_i(X_b, X_a; \Bbbk)$, such that

- $H(i, a, b)$ is finite-dimensional whenever $a \le b$, and
- for every $a \le b \le c$ there is a long exact sequence (§IV.1) with $A_i = H(i, a, b)$, $B_i = H(i, a, c)$ and $C_i = H(i, b, c)$: the sequence of the triple $X_c \supseteq X_b \supseteq X_a$.

Nothing else is assumed: no functoriality, no compatibility between the sequences of different triples, no vanishing of $H(i, a, a)$, no excision. Values with $a \not\le b$ are never used. The structure records exactly what §5 uses, and the topology enters only through two hypotheses kept apart: the sequences of triples, an axiom here, and the homology of the slabs, a hypothesis of the final theorems.

For a chain $X_0 \le X_1 \le \dots \le X_n$ in ι (a sequence X indexed by $\mathbb{N}$ with $X_k \le X_{k+1}$ for $k < n$), the *Morse numbers* are

$$C_i = \sum_{k<n} \dim H(i, X_k, X_{k+1})$$

`Chapter3/Inequalities · morseNumber`, the total dimension of the homology of the slabs in degree i. When the slabs have the homology of attached cells, C_i is Milnor's number of critical points of index i (below).

Fix a homology of a filtration. Three functions on pairs are subadditive or additive (V3.10, V3.12).

- $R_\lambda(a, b) = \dim H(\lambda, a, b)$ is subadditive `Chapter3/Inequalities · FiltrationHomology.isSubadditive_finrank`, by the second lemma of §IV.1 for the sequence of $a \le b \le c$, all three of whose families are finite-dimensional by the first axiom.
- **Lemma 5.3.** $S_\lambda(a, b) = \operatorname{alt}_\lambda(\dim H(\bullet, a, b)) = R_\lambda - R_{\lambda-1} + \dots \pm R_0$ is subadditive `Chapter3/Inequalities · FiltrationHomology.isSubadditive_altSum`, by the corollary of the key identity.
- *The Euler characteristic is additive* (p. 28). Truncate it at N: $\chi_N(a, b) = \sum_{i \le N} (-1)^i \dim H(i, a, b)$. If $H(N+1, a, b) = 0$ for all $a \le b$, then χ_N is additive `Chapter3/Inequalities · FiltrationHomology.isAdditive_eulerChar`: by $\chi_N = (-1)^N \operatorname{alt}_N$ the claim is the additivity of alt_N on a triple whose third family vanishes in degree $N+1$. If the homology vanishes in all degrees above N, χ_N is Milnor's χ.

Lemma 5.1 then gives the statements of §5 for an arbitrary chain $X_0 \le \dots \le X_n$ with $n \ge 1$, writing $b_i = \dim H(i, X_0, X_n)$ for its Betti numbers and C_i for its Morse numbers.

- **Theorem 5.2 (2), for any filtration (V3.11):** $b_\lambda \le C_\lambda$ `Chapter3/Inequalities · FiltrationHomology.finrank_le_morseNumber`, Lemma 5.1 for R_λ.
- **The inequalities (4_λ), for any filtration (V3.13):** $\operatorname{alt}_\lambda(b) \le \operatorname{alt}_\lambda(C)$ `Chapter3/Inequalities · FiltrationHomology.altSum_le_morseNumber`, Lemma 5.1 for S_λ, after which $\operatorname{alt}_\lambda$ is moved inside the sum over the slabs.

- **Theorem 5.2 (3), for any filtration (V3.11):** if no slab has homology in degree $N+1$, then $\sum_{i\le N}(-1)^i b_i = \sum_{i\le N}(-1)^i C_i$ `Chapter3/Inequalities · FiltrationHomology.sum_neg_one_pow_finrank_eq`. The proof is the equality case of Milnor's induction for χ_N on the triples $X_0 \le X_m \le X_{m+1}$. On each of them additivity needs only $H(N+1, X_m, X_{m+1}) = 0$, a slab, which is why the hypothesis concerns slabs and not the pairs (X_m, X_0). An exchange of two finite sums then rewrites $\sum_k \chi_N(X_k, X_{k+1})$ as $\sum_{i\le N}(-1)^i C_i$.

Finiteness along a chain. Let $\iota = \mathbb{N}$, and suppose only that the sequences of triples exist and that every slab $H(i,k,k+1)$ and every $H(i,k,k)$ is finite-dimensional. Then every $H(i,a,b)$ with $a \le b$ is finite-dimensional `Chapter3/Inequalities · finiteDimensional_of_steps`: by induction on $b \ge a$, starting from $H(i,a,a)$ and passing from b to $b+1$ through the sequence of $a \le b \le b+1$, whose middle is finite-dimensional when its ends are (§IV.1). This gives a constructor `Chapter3/Inequalities · FiltrationHomology.ofSteps` that produces a homology of a filtration from the sequences of triples and the finiteness of the slabs alone. In singular homology $H_*(X_k, X_k) = 0$, so the finiteness of $H_*(M;\Bbbk)$ is, as in Milnor, a consequence of the computation, not an assumption.

Critical points. The topological input of p. 29 enters as a hypothesis about a finite set P (the critical points) carrying two functions, an index $\mathrm{ind}(p)$ and a step $\mathrm{step}(p) < n$ (the k such that p lies in the slab $a_k < f(p) < a_{k+1}$):

$$\dim H(i, X_k, X_{k+1}) = \#\{p \in P : \mathrm{step}(p) = k,\ \mathrm{ind}(p) = i\} \qquad (k < n,\ i \in \mathbb{N}). \tag{cells}$$

In words, step k attaches one i-cell for each critical point of index i in the slab. Under (cells) the Morse numbers count critical points, $C_i = \#\{p \in P : \mathrm{ind}(p) = i\}$ `Chapter3/Inequalities · morseNumber_eq_card`: the sizes of the fibres of step over $k < n$, within the points of index i, add up to the size of that set, because step maps it into $\{0, \dots, n-1\}$. In Milnor's case of one critical point per slab, of index $\mathrm{idx}(k)$, (cells) reads $\dim H(i, X_k, X_{k+1}) = [\mathrm{idx}(k) = i]$, and $C_i = \#\{k < n : \mathrm{idx}(k) = i\}$ `Chapter3/Inequalities · morseNumber_eq_card_of_single`.

Substituting, for every homology of a filtration and every chain satisfying (cells):

- **Theorem 5.2 (2):** $b_\lambda \le \#\{p \in P : \mathrm{ind}(p) = \lambda\}$ `Chapter3/Inequalities · FiltrationHomology.weak_morse_inequality`;
- (4_λ), with C_i the number of points of index i `Chapter3/Inequalities · FiltrationHomology.morse_inequality`;
- if no point of P has index l, then $b_l = 0$ `Chapter3/Inequalities · FiltrationHomology.finrank_eq_zero_of_forall_ne`, by the weak inequality;
- **Theorem 5.2 (3):** if every index is at most N, then $\sum_{i\le N}(-1)^i b_i = \sum_{p\in P}(-1)^{\mathrm{ind}(p)}$ `Chapter3/Inequalities · FiltrationHomology.euler_characteristic_eq`. No slab has homology in degree $N+1$, its dimension being by (cells) a number of points of index $N+1$; so Theorem 5.2 (3) above applies, and its right side, regrouped by the value of ind, is $\sum_{p\in P}(-1)^{\mathrm{ind}(p)}$. By the previous item, the left side is the whole Euler characteristic of (X_n, X_0).

Where the formal statements depart from Milnor. Milnor chooses $a_1 < \dots < a_k$ so that M^{a_i} contains exactly i critical points (p. 29), which requires distinct critical points to have distinct critical values; that is not assumed, and (cells) allows any number of critical points per slab, the form in which the computation of a slab holds in general (§IV.19). Milnor's chain starts at $M^{a_0} = \emptyset$ although only $a_1, \dots, a_k$ were introduced; here the bottom X_0 is arbitrary, and b_i is the homology of the pair (X_n, X_0). The last term $\pm C_n$ of his formula for $\chi(M)$ silently takes $n = \dim M$; here the bound N on the indices is a hypothesis.

IV.4 Corollary 5.4 and the lacunary principle

The consequences on pp. 30–31 concern only two integer sequences b and c that satisfy (4_l) for every l, $\mathrm{alt}_l(b) \le \mathrm{alt}_l(c)$, and they are proved in that generality (V3.14, V3.15). For a filtration, $b_i = \dim H(i, X_0, X_n) \ge 0$ and $c_i = C_i$.

- *The strong inequalities imply the weak ones* (p. 30): $b_l \le c_l$ for every l `Chapter3/Inequalities · le_of_altSum_le`. For $l = 0$, (4_0) is $b_0 \le c_0$. For $l+1$, add (4_{l+1}) and (4_l) and use $\mathrm{alt}_{l+1} + \mathrm{alt}_l =$ (term of index $l+1$).
- *An equality* (p. 30): if $c_{l+1} = 0$ and $b_{l+1} \ge 0$, then (4_l) is an equality, $\mathrm{alt}_l(b) = \mathrm{alt}_l(c)$ `Chapter3/Inequalities · altSum_eq_of_altSum_le`. Indeed (4_{l+1}) reads $b_{l+1} - \mathrm{alt}_l(b) \le 0 - \mathrm{alt}_l(c)$, so $\mathrm{alt}_l(c) \le \mathrm{alt}_l(b) - b_{l+1} \le \mathrm{alt}_l(b)$, and (4_l) is the reverse inequality. Milnor first deduces $R_{\lambda+1} = 0$; only $R_{\lambda+1} \ge 0$ is needed.
- *Formula (3) from the strong inequalities* (p. 30): if $b \ge 0$ and $c_{N+1} = 0$, then $\sum_{i \le N}(-1)^i b_i = \sum_{i \le N}(-1)^i c_i$ `Chapter3/Inequalities · sum_neg_one_pow_eq_of_altSum_le`, by the previous item and $\chi_N = (-1)^N \mathrm{alt}_N$. Milnor compares (4_λ) and $(4_{\lambda-1})$ for all $\lambda > n$; only $\lambda = n+1$ matters.

Corollary 5.4 (p. 31), for $\lambda = l+1 \ge 1$. If $b \ge 0$ and $c_{l+2} = c_l = 0$, then $b_{l+1} = c_{l+1}$ and $b_{l+2} = b_l = 0$ `Chapter3/Inequalities · eq_of_altSum_le_succ`.

Proof. $b_{l+2} = b_l = 0$, because $0 \le b \le c$ there. (4_{l+1}) is an equality by the second item, since $c_{l+2} = 0$. (4_l) is an equality too: for $l = 0$ both sides are $b_0 = c_0 = 0$; for $l = m+1$, $\mathrm{alt}_l =$ (term of index l) $- \mathrm{alt}_m$, the terms of index l vanish on both sides, and (4_m) is an equality by the second item, since $c_{m+1} = c_l = 0$. Subtracting, $b_{l+1} = \mathrm{alt}_{l+1}(b) + \mathrm{alt}_l(b) = \mathrm{alt}_{l+1}(c) + \mathrm{alt}_l(c) = c_{l+1}$. □

Corollary 5.4 for $\lambda = 0$, where $C_{\lambda-1}$ is meaningless and is read as 0: if $b \ge 0$ and $c_1 = 0$, then $b_0 = c_0$ and $b_1 = 0$ `Chapter3/Inequalities · eq_of_altSum_le_zero`.

The lacunary principle (the remark on $\mathbb{C}P^n$, p. 31). If $b \ge 0$ and no two consecutive c_i are nonzero ($c_i = 0$ or $c_{i+1} = 0$ for every i), then $b_i = c_i$ for every i `Chapter3/Inequalities · eq_of_altSum_le_of_lacunary`. If $c_i = 0$, then $0 \le b_i \le c_i = 0$. Otherwise $c_{i+1} = 0$, and for $i \ge 1$ also $c_{i-1} = 0$, since c_{i-1} or c_i vanishes; Corollary 5.4 applies at i. Sequences vanishing in all odd degrees, such as the Morse numbers of $\mathbb{C}P^n$, satisfy the hypothesis `Chapter3/Inequalities · lacunary_of_odd`.

For a homology of a filtration and a chain, $b_i = \dim H(i, X_0, X_n)$ is non-negative and the inequalities (4_l) hold for every l (§IV.3), so these statements apply verbatim: Corollary 5.4 `Chapter3/Inequalities · FiltrationHomology.finrank_eq_morseNumber_of_succ` `Chapter3/Inequalities · FiltrationHomology.finrank_eq_morseNumber_of_zero` and the lacunary principle `Chapter3/Inequalities · FiltrationHomology.finrank_eq_morseNumber_of_lacunary`. Milnor remarks that Corollary 5.4 computes the homology of $\mathbb{C}P^n$ without Theorem 3.5. What it gives, here as in the book, is the Betti numbers over every field, not the integral groups. $\mathbb{C}P^n$ is not constructed in the project; the spheres, for which the lacunary principle gives the whole homology, appear in §IV.21.

IV.5 Relative singular homology and the exact sequence of a pair

What Mathlib provides. For an object R of a preadditive category with coproducts, Mathlib's singular chain complex with coefficients R is a functor from topological spaces to chain complexes indexed by $\mathbb{N}$ `Mathlib/AlgebraicTopology/SingularHomology/Basic · singularChainComplexFunctor`, and singular homology is its homology `Mathlib/AlgebraicTopology/SingularHomology/Basic · singularHomologyFunctor`. A *pair* is an embedding $A \hookrightarrow X$ of topological spaces, viewed as an arrow; a map of pairs is a commutative square `Mathlib/Topology/Category/TopPair · TopPair`. In an abelian category, a short exact sequence of chain complexes has a connecting morphism `Mathlib/Algebra/Homology/HomologySequence · ShortExact.δ` and a long exact homology sequence, exact at each of its three kinds of places `Mathlib/Algebra/Homology/HomologySequence · ShortExact.homology_exact₁` `Mathlib/Algebra/Homology/HomologySequence · ShortExact.homology_exact₂` `Mathlib/Algebra/Homology/HomologySequence · ShortExact.homology_exact₃`. In this section the coefficients are an object R of an abelian category with coproducts; the case in view is the category of modules over a commutative ring, with R the ring itself.

The cokernel sequence. In any abelian category, sending an arrow $f : A \to B$ to the short complex $A \xrightarrow{f} B \to \mathrm{coker}\, f$ is a functor from arrows to short complexes `Homology/Relative · cokernelSequence`. When

f is a monomorphism the short complex is short exact `Homology/Relative · cokernelSequence_shortExact`: it is exact at B, being a map followed by its cokernel; f is mono by hypothesis; and the projection onto a cokernel is epi.

Definition (relative chains and relative homology; V3.26). Write $C_\bullet(X)$ for the singular chain complex with coefficients R `Homology/Relative · singularChains`. For a pair (X, A) the inclusion $A \hookrightarrow X$ is injective, hence a monomorphism of spaces, and $C_\bullet(A) \to C_\bullet(X)$ is a monomorphism of complexes, because Mathlib's singular chain functor preserves monomorphisms: a singular simplex of A is determined by its image in X. The *relative short complex* is the composite functor from pairs to arrows of spaces, to arrows of complexes, to short complexes `Homology/Relative · relShortComplex`:

$$0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0,$$

short exact for every pair `Homology/Relative · relShortComplex_shortExact`. The *relative chain complex* $C_\bullet(X, A)$ is its third term `Homology/Relative · relChains`, by definition the cokernel of $C_\bullet(A) \to C_\bullet(X)$ `Homology/Relative · relChains_obj`, and the *relative homology* $H_n(X, A; R)$ is its n-th homology `Homology/Relative · relHomology` `Homology/Relative · relHomology_obj`. Both are functors on pairs; a map of pairs acts on relative homology through the homology of the induced map of relative chains `Homology/Relative · relHomology_map`.

Three facts make the cokernel description usable. The projection $\pi : C_\bullet(X) \to C_\bullet(X, A)$ `Homology/Relative · relπ` is an epimorphism that kills $C_\bullet(A)$ `Homology/Relative · map_relπ`. The map of relative chains induced by $\varphi : (X, A) \to (Y, B)$ is characterized by $\varphi_\sharp \circ \pi_{(X,A)} = \pi_{(Y,B)} \circ (\varphi_X)_\sharp$, where $\varphi_X : X \to Y$ is the map of the big spaces `Homology/Relative · relπ_relChains_map`. And a map out of $C_\bullet(X, A)$ is determined by its composite with π `Homology/Relative · relChains_hom_ext`. Most identities between maps of relative chains below are proved by this last fact: compose with π and compute in $C_\bullet(X)$.

Theorem (the exact sequence of a pair; V3.26). For every pair (X, A) the sequence

$$\cdots \to H_{n+1}(X, A) \xrightarrow{\partial} H_n(A) \to H_n(X) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots \to H_0(X) \to H_0(X, A) \to 0$$

is exact at every place, and ∂ is natural in maps of pairs.

The proof as formalized. The boundary $\partial : H_{n+1}(X, A) \to H_n(A)$ is Mathlib's connecting morphism of the short exact sequence of the pair `Homology/Relative · relδ`. Consecutive composites vanish `Homology/Relative · relδ_comp` `Homology/Relative · comp_relδ` `Homology/Relative · homologyMap_map_comp_relπ`, and exactness at $H_n(A)$, at $H_n(X)$ and at $H_{n+1}(X, A)$ is Mathlib's exactness of the homology sequence, specialized `Homology/Relative · relHomology_exact₁` `Homology/Relative · relHomology_exact₂` `Homology/Relative · relHomology_exact₃`; a six-term form around degree n is recorded for diagram lemmas `Homology/Relative · relHomologySequence_exact`. The end of the sequence, the surjectivity of $H_0(X) \to H_0(X, A)$ `Homology/Relative · epi_homologyMap_relπ_zero`, is not part of Mathlib's sequence, which stops at the last connecting morphism. With the homological grading no differential leaves degree 0, so H_0 is a quotient of the 0-chains, and an epimorphism of 0-chains induces an epimorphism of H_0 `Mathlib/Algebra/Homology/ShortComplex/HomologicalComplex · epi_homologyMap_of_epi_of_not_rel`. Naturality, $\partial \circ \varphi_* = (\varphi|_A)_* \circ \partial$, is Mathlib's naturality of connecting morphisms for morphisms of short exact sequences `Mathlib/Algebra/Homology/HomologySequenceLemmas · HomologySequence.δ_naturality`, applied to the morphism of relative short complexes induced by φ `Homology/Relative · relδ_naturality`. It is packaged as a natural transformation of functors on pairs `Homology/Relative · relδNatTrans` `Homology/Relative · relδNatTrans_app_succ`; the projections $H_n(X) \to H_n(X, A)$ form one as well `Homology/Relative · homologyToRelHomology`. □

Three consequences of exactness are recorded. If $H_n(A) \to H_n(X)$ is onto and $H_{n-1}(A) \to H_{n-1}(X)$ is injective, then $H_n(X, A) = 0$ `Homology/Relative · isZero_relHomology_of_epi_of_mono`; in particular, if $A \hookrightarrow X$ induces isomorphisms in all degrees (for instance if it is a homotopy equivalence), then $H_*(X, A) = 0$ `Homology/Relative · isZero_relHomology_of_isIso_homologyMap`. If $H_{n+1}(X) = H_n(X) = 0$, then $\partial : H_{n+1}(X, A) \to H_n(A)$ is an isomorphism `Homology/Relative · isIso_relδ`.

The pair $(X, \emptyset)$. The chains of an empty space vanish in every degree Homology/Relative · isZero_singularChains_X_of_isEmpty: a generator of $C_n(X)$ is a singular simplex $\Delta^n \to X$, and there is none, Δ^n being nonempty. So the whole complex is zero Homology/Relative · isZero_singularChains_of_isEmpty. For $A = \emptyset$ the projection π is then the cokernel of a zero map, an isomorphism Homology/Relative · isIso_relπ_of_isEmpty, and so is its effect on homology Homology/Relative · isIso_homologyMap_relπ_of_isEmpty. This is the natural isomorphism $H_n(X) \cong H_n(X, \emptyset)$ Homology/Relative · relHomologyInclIso, natural because the projections are.

The pair (X, X). If $A \hookrightarrow X$ is an isomorphism, then so is $C_\bullet(A) \to C_\bullet(X)$; in particular it is an epimorphism, and its cokernel is zero Homology/Relative · isZero_relChains_of_isIso. So $H_n(X, A) = 0$ Homology/Relative · isZero_relHomology_of_isIso; in particular $H_n(X, X) = 0$ Homology/Relative · isZero_relHomology_diag.

The five lemma for pairs. If a map of pairs $\varphi : (X, A) \to (Y, B)$ induces quasi-isomorphisms $C_\bullet(A) \to C_\bullet(B)$ and $C_\bullet(X) \to C_\bullet(Y)$, it induces a quasi-isomorphism of relative chains Homology/Relative · quasiIso_relChains_map, by Mathlib's five lemma for morphisms of short exact sequences of complexes Mathlib/Algebra/Homology/HomologySequenceLemmas · HomologySequence.quasiIso_τ₃, and therefore isomorphisms in relative homology Homology/Relative · isIso_relHomology_map_of_quasiIso. Stated with homology: if φ induces isomorphisms $H_*(A) \cong H_*(B)$ and $H_*(X) \cong H_*(Y)$ in all degrees, then $H_n(X, A) \cong H_n(Y, B)$ for every n Homology/Relative · isIso_relHomology_map. A version in one degree asks for an epimorphism on $H_n(A)$, isomorphisms on $H_n(X)$ and $H_{n-1}(A)$, and a monomorphism on $H_{n-1}(X)$ Homology/Relative · isIso_relHomology_map_of_degree.

Pairs of subsets. Milnor's pairs are pairs of subsets, (M^b, M^a). For subsets $A \subseteq B$ of a space E, the inclusion of subspaces Homology/Relative · subsetInclusion is an embedding, and (B, A) is a pair Homology/Relative · subsetPair. The inclusions $(B, A) \to (B', A')$ for $A \subseteq A'$, $B \subseteq B'$ are maps of pairs Homology/Relative · subsetPairHom, and so are the restrictions of a continuous map f with $f(A) \subseteq A'$, $f(B) \subseteq B'$ Homology/Relative · subsetMap Homology/Relative · subsetPairMap. Equal subsets give isomorphic pairs Homology/Relative · subsetPairIsoOfEq, which is needed because two pairs whose subsets are equal only by a proof are not literally the same pair.

IV.6 The exact sequence of a triple

Milnor uses the sequence of a triple $X \supset Y \supset Z$ (p. 30). It comes from a short exact sequence of relative chain complexes, the third isomorphism theorem $\big(C(X)/C(Z)\big)\big/\big(C(Y)/C(Z)\big) \cong C(X)/C(Y)$.

Lemma (the third isomorphism theorem). In an abelian category, let $f : A \to B$, let $g : B \to C$ be a monomorphism, and $h = g \circ f$. Then $0 \to \operatorname{coker} f \to \operatorname{coker} h \to \operatorname{coker} g \to 0$, with the maps induced by g and by the identity of C, is short exact Homology/Triple · cokernel_comp_shortExact' Homology/Triple · cokernel_comp_shortExact.

Proof. Mathlib has the exact sequence of a composition Mathlib/CategoryTheory/Abelian/DiagramLemmas/KernelCokernelComp · kernelCokernelCompSequence_exact,

$$0 \to \ker f \to \ker h \to \ker g \to \operatorname{coker} f \to \operatorname{coker} h \to \operatorname{coker} g \to 0.$$

Exactness at $\operatorname{coker} h$ is part of it. The map $\operatorname{coker} f \to \operatorname{coker} h$ is mono by exactness at $\operatorname{coker} f$, the map into $\operatorname{coker} f$ coming from $\ker g = 0$. The map $\operatorname{coker} h \to \operatorname{coker} g$ is epi because its composite with the epimorphism $C \to \operatorname{coker} h$ is the epimorphism $C \to \operatorname{coker} g$. □

Definition (triples; V3.27). A *triple of spaces* $Z \subseteq Y \subseteq X$ Homology/Triple · TopTriple consists of embeddings $i : Z \to Y$ and $j : Y \to X$ and a map $k : Z \to X$ with $k = j \circ i$, which is then an embedding too Homology/Triple · TopTriple.isEmbedding_k. The redundant field k allows the pair (X, Z) to be literally the pair one wants. The three pairs Homology/Triple · TopTriple.pairYZ Homology/Triple · TopTriple.pairXZ Homology/Triple · TopTriple.pairXY are joined by the inclusions $(Y, Z) \to (X, Z)$ Homology/Triple · TopTriple.

map₁ and $(X, Z) \to (X, Y)$ Homology/Triple · TopTriple.map₂. For subsets $A \subseteq B \subseteq D$ of a space there is a triple Homology/Triple · TopTriple.ofSubsets whose three pairs are exactly the pairs of subsets (B, A), (D, A), (D, B) of §IV.5 Homology/Triple · TopTriple.ofSubsets_pairYZ Homology/Triple · TopTriple.ofSubsets_pairXZ Homology/Triple · TopTriple.ofSubsets_pairXY.

Theorem (the exact sequence of a triple; V3.27). For every triple the sequence

$$\cdots \to H_{n+1}(X, Y) \xrightarrow{\partial} H_n(Y, Z) \to H_n(X, Z) \to H_n(X, Y) \xrightarrow{\partial} H_{n-1}(Y, Z) \to \cdots \to H_0(X, Z) \to H_0(X, Y) \to 0$$

is exact at every place, and its ∂ is the boundary of the pair (X, Y) followed by $H_n(Y) \to H_n(Y, Z)$.

The proof as formalized.

1. The two inclusions of pairs give a short complex $C_\bullet(Y, Z) \to C_\bullet(X, Z) \to C_\bullet(X, Y)$ Homology/Triple · TopTriple.shortComplex whose composite vanishes Homology/Triple · TopTriple.relChains_map_map₁_comp_map₂: composed with the projection from $C_\bullet(Y)$, it is $C_\bullet(Y) \to C_\bullet(X) \to C_\bullet(X, Y)$, which is zero (§IV.5).
2. The two maps are the cokernel maps of the lemma: $C_\bullet(Y, Z) \to C_\bullet(X, Z)$ is induced on cokernels by $C_\bullet(Y) \to C_\bullet(X)$ Homology/Triple · TopTriple.relChains_map_map₁_eq, and $C_\bullet(X, Z) \to C_\bullet(X, Y)$ by the identity of $C_\bullet(X)$ Homology/Triple · TopTriple.relChains_map_map₂_eq; both identities are checked after the projections.
3. The lemma, with $f = C_\bullet(Z) \to C_\bullet(Y)$, the monomorphism $g = C_\bullet(Y) \to C_\bullet(X)$, and $g \circ f = C_\bullet(Z) \to C_\bullet(X)$ Homology/Triple · TopTriple.singularChains_map_fac, makes the short complex short exact Homology/Triple · TopTriple.shortComplex_shortExact.
4. The boundary of the triple is the connecting morphism Homology/Triple · TopTriple.δ. The composites vanish Homology/Triple · TopTriple.δ_comp Homology/Triple · TopTriple.comp_δ Homology/Triple · TopTriple.map₁_comp_map₂, and exactness at the three kinds of places is Mathlib's Homology/Triple · TopTriple.exact₁ Homology/Triple · TopTriple.exact₂ Homology/Triple · TopTriple.exact₃. The end, the surjectivity of $H_0(X, Z) \to H_0(X, Y)$, is the degree-0 argument of §IV.5 for the epimorphism $C_\bullet(X, Z) \to C_\bullet(X, Y)$ Homology/Triple · TopTriple.epi_map₂_zero.
5. The projections $C_\bullet(Y) \to C_\bullet(Y, Z)$ and $C_\bullet(X) \to C_\bullet(X, Z)$, with the identity of $C_\bullet(X, Y)$, form a morphism from the short exact sequence of the pair (X, Y) to that of the triple Homology/Triple · TopTriple.fromPair. Naturality of connecting morphisms, with the identity on the third terms, gives $\partial_{\text{triple}} = \pi_* \circ \partial_{(X,Y)}$ Homology/Triple · TopTriple.δ_eq. □

For $Z = \emptyset$ the sequence of the triple is that of the pair (X, Y), by $H_*(X, \emptyset) = H_*(X)$.

IV.7 Homotopy invariance for pairs, strong deformation retracts, and the Eilenberg–Steenrod pretheory

Mathlib proves that homotopic maps induce chain-homotopic maps of singular chain complexes Mathlib/AlgebraicTopology/SingularHomology/HomotopyInvariance · singularChainComplexFunctorObjMap. Everything in this section is deduced from that one theorem.

Homotopy equivalences. Maps $f : X \to Y$ and $g : Y \to X$ with homotopies $g \circ f \simeq \mathrm{id}$ and $f \circ g \simeq \mathrm{id}$ induce a homotopy equivalence of singular chain complexes Homology/RelativeHomotopy · singularChainsHomotopyEquivOfHomotopy: by functoriality $(g \circ f)_\sharp = g_\sharp \circ f_\sharp$, Mathlib's theorem turns $g \circ f \simeq \mathrm{id}$ into a chain homotopy $(g \circ f)_\sharp \simeq \mathrm{id}_\sharp$, and $\mathrm{id}_\sharp = \mathrm{id}$. A homotopy equivalence of spaces (Mathlib's notion Mathlib/Topology/Homotopy/Equiv · HomotopyEquiv) therefore gives one of chain complexes Homology/RelativeHomotopy · singularChainsHomotopyEquiv; a map that is a homotopy equivalence induces a quasi-isomorphism Homology/RelativeHomotopy · quasiIso_map_of_homotopyEquiv and isomorphisms $H_n(X) \cong H_n(Y)$ Homology/RelativeHomotopy · isIso_homologyMap_of_homotopyEquiv; and a map of pairs both of whose components are homotopy equivalences induces isomorphisms in relative homology Homology/RelativeHomotopy · isIso_relHomology_map_of_homotopyEquiv, by the five lemma of §IV.5. A contractible space has the homology of

a point, $H_n = 0$ for $n \neq 0$ `Homology/RelativeHomotopy · isZero_homology_of_contractibleSpace`: it is homotopy equivalent to a point `Mathlib/Topology/Homotopy/Contractible · ContractibleSpace.hequiv`, and a point, being totally disconnected, has no homology in positive degrees `Mathlib/AlgebraicTopology/SingularHomology/Basic · isZero_singularHomologyFunctor_of_totallyDisconnectedSpace`. Hence, for a pair (X, A) with X contractible, $\partial : H_{n+1}(X, A) \to H_n(A)$ is an isomorphism for $n \neq 0$ `Homology/RelativeHomotopy · isIso_relδ_of_contractibleSpace`, as for (D^n, S^{n-1}).

Theorem (homotopy invariance for pairs; V3.26). Maps of pairs $f, g : (X, A) \to (Y, B)$ that are homotopic through maps of pairs induce the same maps $H_n(X, A) \to H_n(Y, B)$ in every degree `Homology/RelativeHomotopy · relHomology_map_eq_of_homotopy` `Homology/RelativeHomotopy · relHomology_map_eq_of_homotopic`. A homotopy of maps of pairs is Mathlib's: a homotopy of the maps of the big spaces and one of the maps of the small spaces, compatible with the inclusions `Mathlib/Topology/Category/TopPair · TopPair.Homotopy`.

The idea. The textbook proof restricts the chain homotopy of X to A and passes to quotients. Mathlib's chain homotopy is built through several layers (singular simplicial sets, simplicial homotopies, the alternating face map complex), and its naturality in the space, which the restriction needs, is not available. The formal proof avoids it, using only the absolute theorem and the five lemma, applied to the cylinder of the pair.

The proof as formalized.

1. *The cylinder contracts onto its base.* For a space Z, $((z, s), t) \mapsto (z, ts)$ is a homotopy from $(z, s) \mapsto (z, 0)$ to the identity of $Z \times I$ `Homology/RelativeHomotopy · cylinderContraction`. So the projection $Z \times I \to Z$ is a homotopy equivalence, with inverse $z \mapsto (z, 0)$, and it induces isomorphisms in homology `Homology/RelativeHomotopy · isIso_homologyMap_fst`.
2. *The cylinder of a pair* is $(X \times I, A \times I)$ `Homology/RelativeHomotopy · cylinder`, a pair because a product of embeddings is an embedding. It has two ends $\iota_0, \iota_1 : (X, A) \to (X \times I, A \times I)$, $x \mapsto (x, 0)$ and $x \mapsto (x, 1)$ `Homology/RelativeHomotopy · cylinderι₀` `Homology/RelativeHomotopy · cylinderι₁`, and a projection π `Homology/RelativeHomotopy · cylinderπ`, with $\pi \circ \iota_0 = \mathrm{id} = \pi \circ \iota_1$ `Homology/RelativeHomotopy · cylinderι₀_π` `Homology/RelativeHomotopy · cylinderι₁_π`. Both components of π are projections of cylinders, so by step 1 and the five lemma π induces isomorphisms in relative homology `Homology/RelativeHomotopy · isIso_relHomology_map_cylinderπ`.
3. *The two ends agree in homology.* From $\pi_* \circ (\iota_0)_* = \mathrm{id}$ and the invertibility of π_*, $(\iota_0)_* = \pi_*^{-1}$; likewise $(\iota_1)_* = \pi_*^{-1}$ `Homology/RelativeHomotopy · relHomology_map_cylinderι₀_eq_ι₁`.
4. *A homotopy is a map out of the cylinder.* A homotopy of pairs from f to g is a map of pairs $F : (X \times I, A \times I) \to (Y, B)$ `Homology/RelativeHomotopy · cylinderHom` with $F \circ \iota_0 = f$ and $F \circ \iota_1 = g$ `Homology/RelativeHomotopy · cylinderι₀_cylinderHom` `Homology/RelativeHomotopy · cylinderι₁_cylinderHom`; that F is a map of pairs is the compatibility built into a homotopy of pairs.
5. Hence $f_* = F_* \circ (\iota_0)_* = F_* \circ (\iota_1)_* = g_*$. □

Strong deformation retracts. Recall from §0.4 (and Part II, §II.1) that $A \subseteq X$ is a strong deformation retract if there are a continuous $r : X \to X$ with values in A and a homotopy from the identity to r fixing A pointwise at all times `Chapter2/Crossing · IsStrongDeformationRetract`; the inclusion is then a homotopy equivalence, with homotopy inverse r `Chapter2/Crossing · IsStrongDeformationRetract.exists_homotopyEquiv`. This is the form in which Theorem 3.1 and Remark 3.3 are proved (Part II, §§II.5, 2.14). For subsets $A \subseteq B \subseteq X$ of a space, with B a strong deformation retract of X (inside the subspace X), the following hold (V3.26):

- $H_n(B) \to H_n(X)$ is an isomorphism `Homology/RelativeRetract · isIso_homologyMap_of_isStrongDeformationRetract`;
- $(B, A) \to (X, A)$ induces isomorphisms $H_n(B, A) \cong H_n(X, A)$ `Homology/RelativeRetract · isIso_relHomology_map_of_isStrongDeformationRetract` `Homology/RelativeRetract · relHomologyIsoOfIsStrongDeformationRetract`, by the five lemma with the identity of A;
- $H_n(X, B) = 0$ `Homology/RelativeRetract · isZero_relHomology_of_isStrongDeformationRetract`, by §IV.5.

The bottom of a pair can be moved as well: if $A \subseteq B \subseteq X$ and A is a strong deformation re-

tract of B, then $(X, A) \to (X, B)$ induces isomorphisms `Homology/CellExcision · isIso_relHomology_map_of_isStrongDeformationRetract_snd`, by the five lemma with the identity of X on top. Nothing requires the deformation of B onto A to extend to X. (Over a field the same fact is proved a second time in §IV.19, from the sequence of a triple.) Inclusions of pairs of subsets compose `Homology/CellExcision · subsetPairHom_comp`, which is used to compare routes around squares of inclusions.

The Eilenberg–Steenrod pretheory. Mathlib formulates the data of an Eilenberg–Steenrod homology theory as a *homology pretheory* `Mathlib/AlgebraicTopology/EilenbergSteenrod · HomologyPretheory`: relative homology functors on pairs, absolute homology functors on spaces, natural isomorphisms between absolute homology and the relative homology of $(X, \emptyset)$, and natural boundary maps from the relative homology of (X, A) to the absolute homology of A, zero unless the degrees are adjacent. Homotopy invariance, the first axiom, is a separate class `Mathlib/AlgebraicTopology/EilenbergSteenrod · IsHomotopyInvariant`. Mathlib has no instance. The project provides singular homology as one `Homology/Pretheory · singularHomologyPretheory`: the relative homology of §IV.5 `Homology/Pretheory · singularHomologyPretheory_H□`, Mathlib's absolute homology, the isomorphism $H_n(X) \cong H_n(X, \emptyset)$, and as boundary in degrees $(n+1, n)$ the connecting morphism of the pair `Homology/Pretheory · singularHomologyPretheory_δ_app`, zero in all other degrees. It is homotopy invariant `Homology/Pretheory · isHomotopyInvariant_singularHomologyPretheory`, by the theorem above. Exactness and excision are not part of Mathlib's pretheory; here they are the theorems of §§IV.5–IV.6 and §IV.13.

IV.8 Singular chains as combinations of maps from the simplex

Mathlib's $C_\bullet(X; R)$ is the alternating face map complex of the simplicial object $n \mapsto \coprod_\sigma R$, a categorical coproduct indexed by the n-simplices of the singular simplicial set. This is convenient for functoriality, and inconvenient for barycentric subdivision, which needs a chain to be a finite formal combination $\sum r_i \sigma_i$ of continuous maps $\sigma_i : \Delta^n \to X$ and operates on the maps σ_i themselves. The project sets up the textbook description and proves it isomorphic to Mathlib's, naturally in the space.

Complexes of modules, degree by degree. A chain map between complexes indexed by $\mathbb{N}$ is given by components commuting with the differentials `Homology/ModuleComplex · natHomMk`, and a homotopy $f \simeq g$ by components $h_n : C_n \to D_{n+1}$ with $f_0 = dh_0 + g_0$ and $f_{n+1} = h_n d + dh_{n+1} + g_{n+1}$ `Homology/ModuleComplex · natHomotopyMk`. For a complex K of modules and submodules $L_n \subseteq K_n$ with $d(L_{n+1}) \subseteq L_n$ there are the subcomplex L `Homology/ModuleComplex · subcomplex`, with its injective inclusion `Homology/ModuleComplex · subcomplexι`, and the quotient complex K/L, degreewise K_n/L_n `Homology/ModuleComplex · quotient`, with its surjective projection `Homology/ModuleComplex · quotientπ`, which kills every chain map with values in L `Homology/ModuleComplex · quotientπ_comp_eq_zero`. If $f : K' \to K$ is injective in each degree with image L, then $0 \to K' \to K \to K/L \to 0$ is short exact `Homology/ModuleComplex · shortExact_quotient` and K/L is a cokernel of f `Homology/ModuleComplex · isColimitQuotient`; this is the bridge between concrete quotients and the cokernels of §IV.5. Chain maps preserving the submodules restrict to the subcomplexes `Homology/ModuleComplex · subcomplexMap` and descend to the quotients `Homology/ModuleComplex · quotientMap`, functorially `Homology/ModuleComplex · quotientMap_id` `Homology/ModuleComplex · quotientMap_comp`; so do homotopies whose components preserve the submodules `Homology/ModuleComplex · quotientHomotopy`.

Free chains of a simplicial set. Let $S \mapsto (S \to_0 R)$ be the free module on a set `Homology/Simplex · freeFunctor`. The free chain complex of a simplicial set K is the alternating face map complex of K followed by this functor `Homology/Simplex · freeChainsFunctor` `Homology/Simplex · freeChains`. Its n-chains are the finite combinations of n-simplices, and its differential is $\partial x = \sum_i (-1)^i d_i x$ on simplices `Homology/Simplex · bd` `Homology/Simplex · freeChains_d` `Homology/Simplex · bd_single`. That $\partial\partial = 0$ comes for free from Mathlib's alternating face map complex `Homology/Simplex · bd_bd`, and simplicial maps commute with ∂ `Homology/Simplex · lmapDomain_bd`.

The standard simplex. $\Delta^n = \{t \in \mathbb{R}^{n+1} : t_i \geq 0,\ \sum_i t_i = 1\}$ `Homology/Simplex · Simplex` is Mathlib's standard simplex `Mathlib/Analysis/Convex/StdSimplex · stdSimplex`, a subset of the space of functions

$\{0,\dots,n\} \to \mathbb{R}$, whose norm is the maximum norm; distances and diameters below are taken for that norm. A map of vertices f defines the affine map $\Delta^m \to \Delta^n$ with $e_i \mapsto e_{f(i)}$ `Homology/Simplex · simplexMap`, functorially `Homology/Simplex · simplexMap_id` `Homology/Simplex · simplexMap_comp`. The face $\delta^i : \Delta^n \to \Delta^{n+1}$ is the affine map of the increasing injection that skips i `Homology/Simplex · faceMap`.

Singular chains. The singular simplicial set of X is redefined so that its n-simplices are literally the continuous maps $\Delta^n \to X$, simplicial operators acting by precomposition with the affine maps `Homology/Simplex · sing`; its faces are $\sigma \mapsto \sigma \circ \delta^i$ `Homology/Simplex · sing_δ`, and it is a functor of X `Homology/Simplex · singFunctor`. The n-chains are $C_n(X) = C(\Delta^n, X) \to_0 R$ `Homology/Simplex · Chains`, the boundary is $\partial\sigma = \sum_i (-1)^i \sigma \circ \delta^i$ `Homology/Simplex · sbd` `Homology/Simplex · sbd_single`, with $\partial\partial = 0$ `Homology/Simplex · sbd_sbd`, and the complex `Homology/Simplex · singChains` is defined directly, so that its modules of chains are the free modules themselves and elementwise reasoning needs no unfolding. A continuous map f acts by $\sigma \mapsto f \circ \sigma$ `Homology/Simplex · push` `Homology/Simplex · push_single`, functorially `Homology/Simplex · push_id` `Homology/Simplex · push_comp` and commuting with ∂ `Homology/Simplex · push_sbd`; this gives a chain map $f_\sharp$ `Homology/Simplex · pushHom` and the singular chain functor `Homology/Simplex · singChainsFunctor`. It is the free chain complex of the singular simplicial set, by the identity in each degree `Homology/Simplex · freeChainsIso`.

Theorem (comparison with Mathlib). For a commutative ring R, Mathlib's singular chain functor with coefficients R is naturally isomorphic to the concrete one `Homology/Simplex · singChainsIso`, and on the summand indexed by a singular simplex the isomorphism sends the generator to that simplex, viewed as a map $\Delta^n \to X$ `Homology/Simplex · ι_singChainsIso_hom_app_f`.

The proof as formalized. Three natural isomorphisms are composed. First, the coproduct of copies of R indexed by a set S is the free module $S \to_0 R$ `Homology/Simplex · sigmaIsoFinsupp`, by Mathlib's description of that coproduct `Mathlib/Algebra/Category/ModuleCat/Colimits · finsuppCoconeIsColimit`, naturally in S `Homology/Simplex · sigmaConstIsoFree` `Homology/Simplex · ι_sigmaConstIsoFree_hom_app`; whiskering, Mathlib's chain complex functor on simplicial sets is isomorphic to the free one `Homology/Simplex · chainComplexFunctorIso`. Second, Mathlib's singular simplicial set is isomorphic to the concrete one through Mathlib's own identification of its simplices with continuous maps, naturality holding by definition `Homology/Simplex · toSSetIsoSing`. Third, the free chains of the concrete singular simplicial set are the concrete singular chains, as above. □

The comparison is stated for spaces in the universe of the ring. This is why excision, and everything that uses it, is stated with the coefficients and the spaces in one universe.

IV.9 Linear chains, the cone and barycentric subdivision

Hatcher's proof of excision (*Algebraic Topology*, Prop. 2.21) subdivides *linear* chains first. A linear simplex $[w_0,\dots,w_k]$ in a convex set is determined by its ordered tuple of vertices, and the operations of the subdivision are combinatorial operations on tuples together with one geometric input, the barycentre. The formalization separates the two. This section is the combinatorics; §IV.10 is the geometry.

Linear chains. For a set V of vertices, the simplicial set of tuples has as k-simplices the tuples $w = (w_0,\dots,w_k)$ of elements of V, simplicial operators acting by precomposition `Homology/LinearChains · tuples`; the i-th face deletes w_i `Homology/LinearChains · tuples_δ`. The linear k-chains are its free chains, finite combinations of $(k+1)$-tuples `Homology/LinearChains · LinChain`, with boundary

$$\partial[w_0,\dots,w_{k+1}] = \sum_i (-1)^i [w_0,\dots,\widehat{w_i},\dots,w_{k+1}]$$

`Homology/LinearChains · lbd` `Homology/LinearChains · lbd_single`; $\partial\partial = 0$ is inherited from §IV.8 `Homology/LinearChains · lbd_lbd`. A map of vertex sets acts on linear chains, commuting with ∂ `Homology/LinearChains · lmap` `Homology/LinearChains · lmap_lbd`.

The cone. For $b \in V$, $b \cdot [w_0,\dots,w_k] = [b, w_0,\dots,w_k]$ `Homology/LinearChains · cone` `Homology/LinearChains · cone_single`. Let ε be the augmentation of 0-chains, the sum of the coefficients `Homology/LinearChains · aug`, which vanishes on boundaries `Homology/LinearChains · aug_lbd`. Then:

- in degree 0, $\partial(b \cdot \alpha) = \alpha - \varepsilon(\alpha)[b]$ `Homology/LinearChains · lbd_cone_zero`, since $\partial[b, w_0] = [w_0] - [b]$;
- in positive degrees, $\partial(b \cdot \alpha) = \alpha - b \cdot \partial\alpha$ `Homology/LinearChains · lbd_cone_succ`. In $\partial[b, w_0, \dots, w_{k+1}]$, deleting b gives $[w]$ with the sign $+$; deleting w_j gives the cone over the j-th face of w `Homology/LinearChains · cons_comp_succAbove_succ` with the sign $(-1)^{j+1}$, and these terms add up to $-b \cdot \partial[w]$.

Hatcher combines the two formulas through the augmented complex; the formalization keeps them apart and never needs degree -1. The cone is natural for vertex maps `Homology/LinearChains · lmap_cone`.

Definition (subdivision and the subdivision homotopy). Let β assign a vertex $\beta(w)$, a "barycentre", to every tuple. The subdivision operator S and the homotopy T are defined by recursion on the degree `Homology/LinearChains · sd` `Homology/LinearChains · sdH`:

$$S = \mathrm{id} \text{ and } T = 0 \text{ in degree } 0, \qquad S[w] = \beta(w) \cdot S(\partial[w]), \quad T[w] = \beta(w) \cdot ([w] - T(\partial[w])) \text{ in degree } k+1$$

`Homology/LinearChains · sd_succ_single` `Homology/LinearChains · sdH_succ_single`. Hatcher's T in degree 0 is $[w_0] \mapsto [w_0, w_0]$; with $T = 0$ the identities below still hold, because $S = \mathrm{id}$ in degree 0.

Theorem. S is a chain map, $\partial S = S\partial$ `Homology/LinearChains · lbd_sd`, and $\partial T + T\partial = 1 - S$ in every positive degree `Homology/LinearChains · lbd_sdH`; in degree 0 both sides vanish `Homology/LinearChains · lbd_sdH_zero`. Both operators are natural for vertex maps f compatible with the barycentres, $f(\beta(w)) = \beta'(f \circ w)$ `Homology/LinearChains · lmap_sd` `Homology/LinearChains · lmap_sdH`.

The proof as formalized. Everything is checked on generators $[w]$, by induction on the degree.

1. $\partial S = S\partial$. In degree 1: $\partial S[w] = \partial(\beta \cdot \partial[w]) = \partial[w] - \varepsilon(\partial[w])[\beta] = \partial[w] = S\partial[w]$, because ε kills boundaries and $S = \mathrm{id}$ on 0-chains. In degree $k+2$: $\partial(\beta \cdot S\partial[w]) = S\partial[w] - \beta \cdot \partial S\partial[w] = S\partial[w] - \beta \cdot S\partial\partial[w] = S\partial[w]$, by the induction hypothesis.
2. *Reduction to a key identity.* $\partial T + T\partial = 1 - S$ in degree $k+1$ follows from the identity $\partial([w] - T\partial[w]) = S\partial[w]$ for the generators of that degree `Homology/LinearChains · lbd_sdH_of_key`: granted it,

   $$\partial T[w] = ([w] - T\partial[w]) - \beta \cdot \partial([w] - T\partial[w]) = [w] - T\partial[w] - \beta \cdot S\partial[w] = [w] - T\partial[w] - S[w].$$

3. *The key identity*, by induction. For 1-simplices, $T = 0$ on 0-chains and $\partial[w] = S\partial[w]$. In higher degrees the identity $\partial T + T\partial = 1 - S$ one degree down, applied to $\partial[w]$ with $T\partial\partial[w] = 0$, reads $\partial T\partial[w] = \partial[w] - S\partial[w]$; hence $\partial([w] - T\partial[w]) = S\partial[w]$. □

IV.10 Affine simplices and the diameter bound

Let s be a convex set in a real topological vector space E. A tuple w of points of s spans the *affine simplex* $[w] : \Delta^k \to s$, $t \mapsto \sum_i t_i w_i$ `Homology/AffineSimplex · aff`, which lands in s by convexity `Homology/AffineSimplex · sum_smul_mem` and sends the vertex e_i to w_i `Homology/AffineSimplex · aff_vertex`. Δ^n itself is convex `Homology/AffineSimplex · convex_simplex`, with vertices $e_0, \dots, e_n$ `Homology/AffineSimplex · verts`. The calculus of affine simplices:

- they compose, $[u] \circ [v] = [[u] \circ v]$ `Homology/AffineSimplex · aff_comp_aff`, an exchange of two finite sums;
- the identity of Δ^n is $[e_0, \dots, e_n]$ `Homology/AffineSimplex · aff_verts`, and the affine map of a vertex map f is $[e_{f(0)}, \dots, e_{f(m)}]$ `Homology/AffineSimplex · simplexMap_eq_aff`, so precomposition reorders or repeats vertices, $[w] \circ \Delta(f) = [w \circ f]$ `Homology/AffineSimplex · aff_comp_simplexMap`, where $\Delta(f)$ denotes the affine map of f;
- the barycentre $b(w) = (w_0 + \cdots + w_k)/(k+1)$ is the image of the barycentre of Δ^k `Homology/AffineSimplex · bary` `Homology/AffineSimplex · coe_bary` `Mathlib/Analysis/Convex/StdSimplex · stdSimplex.barycenter`, and affine simplices preserve barycentres `Homology/AffineSimplex · aff_bary`.

Realization. $w \mapsto [w]$ is a morphism of simplicial sets from the tuples of s to the singular simplicial set of s `Homology/AffineSimplex · realize`, naturality being the reordering rule; on chains it is a chain

map from linear to singular chains Homology/AffineSimplex · lreal Homology/AffineSimplex · lreal_single Homology/AffineSimplex · sbd_lreal. Pushing a realized linear chain of Δ^n forward along $[u]$ realizes the chain whose vertices are mapped by $[u]$ Homology/AffineSimplex · push_aff_lreal. Barycentric subdivision of linear chains in s is §IV.9 with $\beta = b$ Homology/AffineSimplex · bsd Homology/AffineSimplex · bsdH; since affine simplices preserve barycentres, it is natural with respect to them Homology/AffineSimplex · lmap_aff_bsd Homology/AffineSimplex · lmap_aff_bsdH.

Supports. A chain is supported in a set P of generators if all its terms lie in P. A linear map between free modules sends chains supported in P to chains supported in Q as soon as it does so on the generators in P Homology/AffineSimplex · map_mem_supported Homology/AffineSimplex · mapDomain_mem_supported. Statements of the form "every simplex of the chain c has property Q" are formalized as "c is supported in Q".

The diameter bound (Hatcher, step (1) of the proof of Prop. 2.21). Let now E be normed, and write $\mathrm{vset}(w) = \{w_0, \dots, w_k\}$ Homology/AffineSimplex · vset. The image of $[w]$ lies in the convex hull of $\mathrm{vset}(w)$ Homology/AffineSimplex · range_aff_subset and has the same diameter Homology/AffineSimplex · diam_range_aff_le Homology/AffineSimplex · diam_range_aff, because a convex hull has the diameter of the set Mathlib/Analysis/Normed/Module/Convex · convexHull_diam. The barycentre lies in the hull Homology/AffineSimplex · bary_mem_convexHull.

Lemma (the barycentre is central). For points $w_0, \dots, w_n$ and x in their convex hull,

$$\operatorname{dist}(x, b(w)) \le \frac{n}{n+1} \operatorname{diam} \mathrm{vset}(w)$$

Homology/AffineSimplex · dist_bary_le.

Proof. Some vertex w_j is at least as far from $b(w)$ as x is Mathlib/Analysis/Normed/Module/Convex · convexHull_exists_dist_ge. And $w_j - b(w) = \frac{1}{n+1}\sum_i (w_j - w_i)$, a sum in which the term $i = j$ vanishes and each of the other n terms has norm at most the diameter. □

Theorem (the diameter bound). Let $\mathrm{Shrunk}(w, r)$ be the set of tuples u with $\mathrm{vset}(u)$ in the convex hull of $\mathrm{vset}(w)$ and $\operatorname{diam}\mathrm{vset}(u) \le r$ Homology/AffineSimplex · Shrunk, monotone in the hull and in r Homology/AffineSimplex · Shrunk.mono. For a linear n-simplex $[w]$ and every m, every simplex of $S^m[w]$ lies in $\mathrm{Shrunk}\left(w, (\frac{n}{n+1})^m \operatorname{diam}\mathrm{vset}(w)\right)$ Homology/AffineSimplex · bsd_pow_mem_supported. For the images: every simplex $[u]$ of $S^m[w]$ has $\operatorname{diam}[u] \le (\frac{n}{n+1})^m \operatorname{diam}[w]$ Homology/AffineSimplex · diam_range_aff_le_of_mem_support.

The proof as formalized. One subdivision first Homology/AffineSimplex · bsd_mem_supported, by induction on n; write $D = \operatorname{diam}\mathrm{vset}(w)$.

1. For $n = 0$, $S[w] = [w]$, a point, of diameter 0.
2. For $n = k+1$, $S[w] = b \cdot S(\partial[w])$ with $b = b(w)$. The simplices u of $S(\partial[w])$ come from the faces of w, which are k-simplices; by induction each lies in the hull of a face, which lies in the hull of w Homology/AffineSimplex · convexHull_vset_comp_subset, and has diameter at most $\frac{k}{k+1}$ times that of the face, which is at most D Homology/AffineSimplex · diam_vset_comp_le.
3. The coned simplex $[b, u]$ has vertex set $\{b\} \cup \mathrm{vset}(u)$ Homology/AffineSimplex · vset_cons, contained in the hull of w. Two vertices of u are at distance at most $\frac{k}{k+1}D \le \frac{k+1}{k+2}D$; a vertex of u is at distance at most $\frac{k+1}{k+2}D$ from b, by the lemma, since it lies in the hull of w. So $\operatorname{diam}\left(\{b\} \cup \mathrm{vset}(u)\right) \le \frac{n}{n+1}D$.

Iterating, by induction on m: every simplex u of $S^m[w]$ lies in the hull of w with diameter at most $(\frac{n}{n+1})^m D$; one more subdivision of u multiplies the bound by $\frac{n}{n+1}$, and the hull of u lies in that of w. □

Two small facts complete the geometry. Every point of $[u]$ is within $\operatorname{diam}\mathrm{vset}(u)$ of the vertex u_0 Homology/AffineSimplex · dist_aff_le, and the vertices of Δ^n are at distance at most 1 from each other in the maximum norm Homology/AffineSimplex · diam_vset_verts_le.

IV.11 Subdivision of singular chains and the homotopy T

Natural operators from models. A singular simplex $\sigma : \Delta^n \to X$ is the image of the identity simplex $\iota_n = \mathrm{id}_{\Delta^n}$ `Homology/Subdivision · idSimplex` under $\sigma_\sharp$ `Homology/Subdivision · push_idSimplex`. So an operator on singular chains that is natural in X is determined by its value on ι_n, its *model*. Conversely every chain $m \in C_k(\Delta^n)$ defines a natural operator $\sigma \mapsto \sigma_\sharp m$ `Homology/Subdivision · modelOp` `Homology/Subdivision · modelOp_single` `Homology/Subdivision · push_modelOp`. The model ι_n gives the identity `Homology/Subdivision · modelOp_idSimplex`; operators add and subtract with their models `Homology/Subdivision · modelOp_add_model` `Homology/Subdivision · modelOp_sub_model` `Homology/Subdivision · modelOp_zero_model`; composites of such operators are such operators `Homology/Subdivision · modelOp_modelOp`; and so are the boundary after an operator, whose model is ∂m `Homology/Subdivision · sbd_modelOp`, and an operator after the boundary, whose model is the operator applied to $\partial\iota_{n+1}$ `Homology/Subdivision · modelOp_sbd`. Identities between natural operators thus reduce to identities between chains of standard simplices. Every simplex of $\sigma_\sharp m$ has the form $\sigma\circ\tau$, so these operators preserve chains made of simplices from any family closed under precomposition `Homology/Subdivision · IsPrecompClosed` `Homology/Subdivision · modelOp_mem_supported`, for instance the simplices lying in a given subset.

The models. The model of subdivision is the barycentric subdivision of Δ^n, the linear chain $S[e_0,\dots,e_n]$ realized as a singular chain on Δ^n `Homology/Subdivision · sdModel`; that of the homotopy is $T[e_0,\dots,e_n]$, realized `Homology/Subdivision · sdHModel`. Pushed forward along an affine simplex $[u]$ they give the subdivision and the homotopy of $[u]$ `Homology/Subdivision · push_aff_sdModel` `Homology/Subdivision · push_aff_sdHModel`, by the naturality of linear subdivision for affine maps and $[u](e_i) = u_i$ `Homology/Subdivision · aff_comp_verts`. The identity simplex is the realization of $[e_0,\dots,e_n]$ `Homology/Subdivision · lreal_verts`, and its boundary is the alternating sum of its faces, each an affine simplex `Homology/Subdivision · sbd_idSimplex`. From the linear identities of §IV.9 follow the model identities

$$\partial(S\iota_{n+1}) = S(\partial\iota_{n+1}), \qquad \partial(T\iota_{n+1}) + T(\partial\iota_{n+1}) = \iota_{n+1} - S\iota_{n+1}$$

`Homology/Subdivision · sbd_sdModel` `Homology/Subdivision · sbd_sdHModel`, where S and T on the right act on the faces through their models; and in degree 0, $S\iota_0 = \iota_0$ and $T\iota_0 = 0$ `Homology/Subdivision · sdModel_zero` `Homology/Subdivision · sdHModel_zero`.

Theorem (subdivision of singular chains). The operators $S\sigma = \sigma_\sharp(S\iota_n)$ `Homology/Subdivision · ssd` and $T\sigma = \sigma_\sharp(T\iota_n)$ `Homology/Subdivision · ssdH` satisfy:

- $\partial S = S\partial$ `Homology/Subdivision · sbd_ssd` and $\partial T + T\partial = 1 - S$ `Homology/Subdivision · sbd_ssdH`, with $S = 1$ and $T = 0$ in degree 0 `Homology/Subdivision · ssd_zero` `Homology/Subdivision · ssdH_zero`;
- both are natural `Homology/Subdivision · push_ssd` `Homology/Subdivision · push_ssdH` `Homology/Subdivision · push_ssd_pow`, and both preserve chains made of simplices from a family closed under precomposition `Homology/Subdivision · ssd_mem_supported` `Homology/Subdivision · ssdH_mem_supported` `Homology/Subdivision · ssd_pow_mem_supported`;
- on realized linear chains S is linear subdivision `Homology/Subdivision · ssd_lreal` `Homology/Subdivision · ssd_pow_lreal`, so that $S^m\sigma$ is $\sigma_\sharp$ of the realized m-th barycentric subdivision of Δ^n `Homology/Subdivision · ssd_pow_single`.

Proof. By the rules for models, $\partial(Sc)$ is the operator with model $\partial(S\iota_{n+1})$ applied to c, and $S(\partial c)$ the operator with model $S(\partial\iota_{n+1})$; the first model identity concludes. The same argument with the second identity, whose right side is the model of $1 - S$, gives $\partial T + T\partial = 1 - S$. The rest is the naturality of $\sigma_\sharp$. □

As morphisms of complexes: subdivision is a chain map `Homology/Subdivision · subdivisionHom`, natural in the space `Homology/Subdivision · subdivision`, and T is a chain homotopy from the identity to S `Homology/Subdivision · subdivisionHomotopy`, assembled degree by degree as in §IV.8.

IV.12 The small-simplex theorem

Let $U = (U_i)$ be a family of subsets of a space X whose interiors cover X. A singular simplex is *U-small* if its image lies in some U_i `Homology/SmallChains · smallSet`. The simplices lying in a set A `Homology/SmallChains · inSet` form a family closed under precomposition `Homology/SmallChains · isPrecompClosed_inSet`, and so do the small ones `Homology/SmallChains · isPrecompClosed_smallSet` `Homology/SmallChains · inSet_subset_smallSet`. The boundary preserves chains made of simplices from such a family `Homology/SmallChains · sbd_mem_supported`, and $\partial S^m = S^m \partial$ `Homology/SmallChains · sbd_ssd_pow`.

Theorem (Hatcher, Prop. 2.21; V3.28). The inclusion $\iota : C_\bullet^U(X) \to C_\bullet(X)$ of the complex of U-small chains is a chain homotopy equivalence `Homology/SmallChains · smallChainsHomotopyEquiv`, with a homotopy inverse ρ satisfying $\rho\iota = 1$; in particular it induces isomorphisms in homology `Homology/SmallChains · quasiIso_smallChainsι`. Moreover ρ and the homotopy preserve the chains of every family of simplices closed under precomposition, so that for such a family P the induced map $C_\bullet^U(X)/C_\bullet^P \to C_\bullet(X)/C_\bullet^P$ is again a chain homotopy equivalence `Homology/SmallChains · smallQuotHomotopyEquiv` `Homology/SmallChains · quasiIso_smallQuotMap`. Here $C_\bullet^P$ denotes the chains made of simplices in P.

The proof as formalized.

1. *Fineness.* Let $r(n,m) = (\frac{n}{n+1})^m$ `Homology/SmallChains · subdivRadius`, increasing in n `Homology/SmallChains · subdivRadius_mono_left`. A simplex $\sigma : \Delta^n \to X$ is *m-fine* if it maps every closed ball of radius $r(n,m)$ of Δ^n into some U_i `Homology/SmallChains · IsFine`. Small simplices are 0-fine `Homology/SmallChains · isFine_zero_of_mem_smallSet`.
2. *Every simplex is fine for some m* `Homology/SmallChains · exists_isFine`. The sets $\sigma^{-1}(\operatorname{int} U_i)$ form an open cover of the compact metric space Δ^n, which has a Lebesgue number $\delta > 0$ `Mathlib/Topology/MetricSpace/Pseudo/Lemmas · lebesgue_number_lemma_of_metric`. Since $\frac{n}{n+1} < 1$ there is m with $r(n,m) < \delta$ `Mathlib/Algebra/Order/Archimedean/Basic · exists_pow_lt_of_lt_one`, and a closed ball of radius $r(n,m)$ lies in the open ball of radius δ about the same centre.
3. *Fineness passes to faces.* The affine map of an injective vertex map does not increase distances `Homology/SmallChains · dist_simplexMap_le`: each coordinate of its value is a coordinate of the argument or 0. So δ^i maps the closed ball of radius $r(n,m)$ about x into the closed ball of radius $r(n,m) \le r(n+1,m)$ about $\delta^i x$; hence $\sigma \circ \delta^i$ is m-fine if σ is `Homology/SmallChains · IsFine.comp_faceMap`.
4. *A fine simplex becomes small* `Homology/SmallChains · IsFine.ssd_pow_mem`. If σ is m-fine, then $S^m\sigma = \sigma_\sharp$ (realized m-th subdivision of Δ^n) by §IV.11. Each simplex $[u]$ of that subdivision has $\operatorname{diam} \operatorname{vset}(u) \le r(n,m) \operatorname{diam}\{e_0, \dots, e_n\} \le r(n,m)$ by §IV.10, and every point of $[u]$ is within that distance of u_0. So $[u]$ lies in the closed ball of radius $r(n,m)$ about u_0, which σ maps into some U_i.
5. *The order of fineness.* Let $m(\sigma)$ be the least m for which σ is m-fine `Homology/SmallChains · fineOrder` `Homology/SmallChains · isFine_fineOrder` `Homology/SmallChains · fineOrder_le`. By step 3, $m(\sigma \circ \delta^i) \le m(\sigma)$ `Homology/SmallChains · fineOrder_comp_faceMap_le`; by step 1, $m(\sigma) = 0$ for small σ `Homology/SmallChains · fineOrder_eq_zero`; and $S^m\sigma$ is small for every $m \ge m(\sigma)$ `Homology/SmallChains · ssd_pow_mem_of_le`, by step 4 and because S preserves small chains.
6. *Hatcher's D and ρ.* Let $D_m = \sum_{i<m} TS^i$ `Homology/SmallChains · partialH` `Homology/SmallChains · partialH_apply`. Summing $\partial T + T\partial = 1 - S$ at $S^i c$, $i < m$, and using $\partial S^i = S^i \partial$ telescopes to $\partial D_m + D_m \partial = 1 - S^m$ `Homology/SmallChains · partialH_telescope`. D_m vanishes on 0-chains `Homology/SmallChains · partialH_zero_degree`, and $TS^i\sigma$ is small for $i \ge m(\sigma)$ `Homology/SmallChains · ssdH_ssd_pow_mem`. Define $D\sigma = D_{m(\sigma)}\sigma$ `Homology/SmallChains · smallH` `Homology/SmallChains · smallH_single` and $\rho = 1 - \partial D - D\partial$ `Homology/SmallChains · smallRetr` `Homology/SmallChains · smallRetr_zero_apply` `Homology/SmallChains · smallRetr_succ_apply`; $D = 0$ in degree 0 `Homology/SmallChains · smallH_zero_degree`. Then ρ is a chain map `Homology/SmallChains · sbd_smallRetr`, because $\partial\partial = 0$.
7. *ρ lands in the small chains* `Homology/SmallChains · smallRetr_mem_supported`. In degree 0, $\rho = 1$, and every 0-simplex is small: Δ^0 is a point, whose image lies in some $\operatorname{int} U_i$. In degree $n+1$, with $m = m(\sigma)$

and $\sigma_j = \sigma \circ \delta^j$, the telescope gives Hatcher's formula

$$\rho\sigma = S^m\sigma + (D_m(\partial\sigma) - D(\partial\sigma)) = S^m\sigma + \sum_j (-1)^j \sum_{m(\sigma_j)\le i<m} TS^i\sigma_j,$$

the inner sums making sense because $m(\sigma_j) \le m$. Every term is small by step 5.

8. *The homotopy equivalence.* $D = 0$ on small chains, because $m(\sigma) = 0$ there `Homology/SmallChains · smallH_eq_zero`, so $\rho = 1$ on them `Homology/SmallChains · smallRetr_eq_self`; and D and ρ preserve the chains of every family closed under precomposition `Homology/SmallChains · smallH_mem_supported Homology/SmallChains · smallRetr_mem_supported_of`. As complexes: the chains of such a family form a subcomplex `Homology/SmallChains · supp Homology/SmallChains · suppChains` with a quotient `Homology/SmallChains · quotChains`; $C_\bullet^U(X)$ is one of them `Homology/SmallChains · smallChains`, with inclusion ι `Homology/SmallChains · smallChainsι`; ρ is a chain map into it `Homology/SmallChains · smallRetraction` with $\rho\iota = 1$ `Homology/SmallChains · smallChainsι_smallRetraction`; and D is a homotopy $1 \simeq \iota\rho$, $1 - \iota\rho = \partial D + D\partial$ `Homology/SmallChains · smallRetractionHomotopy`. For the relative statement `Homology/SmallChains · suppSmall Homology/SmallChains · smallQuotChains Homology/SmallChains · smallQuotMap Homology/SmallChains · smallQuotRetraction`, the homotopy descends to the quotients because D preserves the chains of the family (§IV.8). □

Where the formal proof departs from Hatcher. Hatcher takes $m(\sigma)$ to be the least m with $S^m\sigma \in C_\bullet^U(X)$ and uses $m(\sigma_j) \le m(\sigma)$ for the faces. For that definition the inequality is not obvious: $S^m\sigma$ is a formal sum in which terms may cancel, so its being small says nothing directly about the subdivisions of the faces. The geometric $m(\sigma)$ of step 5 makes the inequality immediate (step 3).

IV.13 Excision

Theorem (excision; Hatcher, Thm. 2.20; V3.28). Let R be a commutative ring and E a space in the universe of R; all the sets below are subsets of E, and interiors and closures are taken in the subspace X unless said otherwise.

1. If $A, B \subseteq X$ have interiors covering X, then the inclusion $(B, A\cap B) \to (X, A)$ induces isomorphisms $H_n(B, A\cap B) \cong H_n(X, A)$ for all n `Homology/Excision · excision_of_union`; more generally so does $(B, C) \to (X, A)$ for any C equal to $A \cap B$ `Homology/Excision · excision_of_eq_inter`.
2. If $Z \subseteq A \subseteq X$ and the closure of Z lies in the interior of A, then $(X\setminus Z, A\setminus Z) \to (X, A)$ induces isomorphisms in all degrees `Homology/Excision · excision`; the same holds with closure and interior taken in E `Homology/Excision · excision_of_closure_subset_interior`.

The proof as formalized has two halves: excision for the concrete chains of §IV.8, and a bridge from the relative chains of §IV.5 to concrete quotients.

1. *Simplices along an embedding.* If $j : Y \to X$ is an embedding, a singular simplex of X with image in $j(Y)$ lifts to Y `Homology/Excision · exists_comp_eq_of_range_subset`: choose preimages pointwise; the lift is continuous because j is an embedding. $j_\sharp$ is injective on chains `Homology/Excision · push_injective`, its image is the set of chains made of simplices in $j(Y)$ `Homology/Excision · range_push`, and $j_\sharp c$ is made of simplices in a set S exactly when c is made of simplices in $j^{-1}(S)$ `Homology/Excision · push_mem_supported_iff`.
2. *The excision map at the chain level.* For a continuous $j : Y \to X$ and $A \subseteq X$, $j_\sharp$ induces $C_\bullet(Y)/C_\bullet(j^{-1}A) \to C_\bullet(X)/C_\bullet(A)$ `Homology/Excision · excisionMap Homology/Excision · push_mem_inSet_preimage`, where $C_\bullet(A)$ now means the chains made of simplices in A; for the inclusion of B it is $C_\bullet(B)/C_\bullet(A\cap B) \to C_\bullet(X)/C_\bullet(A)$. Let $U = \{A, B\}$, indexed by the two booleans `Homology/Excision · coverOf Homology/Excision · iUnion_interior_coverOf`; a simplex is U-small exactly when it lies in A or in B `Homology/Excision · mem_smallSet_coverOf`. When j is an embedding with image B, $j_\sharp$ lands in the U-small chains `Homology/Excision · push_mem_smallSet Homology/Excision · pushSmall Homology/`

Excision · pushSmall_ι, and the excision map factors as

$$C_\bullet(B)/C_\bullet(A\cap B)\to C_\bullet^U(X)/C_\bullet(A)\to C_\bullet(X)/C_\bullet(A)$$

Homology/Excision · excisionMapSmall Homology/Excision · excisionMapSmall_comp.

3. *The first factor is an isomorphism* Homology/Excision · bijective_excisionMapSmall_f Homology/Excision · isIso_excisionMapSmall. It is injective: a chain of B whose image is made of simplices in A is made of simplices in $A\cap B$, by step 1. It is surjective: a U-small chain x splits as the part x_A of its terms that lie in A plus the rest, whose terms are small and not in A, hence in B. By step 1 the rest is $j_\sharp c$ for a chain c of B, and $x-j_\sharp c=x_A$ lies in $C_\bullet(A)$, so x and $j_\sharp c$ have the same class.
4. *Excision at the chain level* Homology/Excision · quasiIso_excisionMap. The second factor is a chain homotopy equivalence by the relative small-simplex theorem of §IV.12, with P the simplices in A; so the excision map is a quasi-isomorphism.
5. *Relative chains are concrete quotients.* For a pair (X,A) with $S\subseteq X$ the image of A, let $C_\bullet(X)\to C_\bullet(X)/C_\bullet(S)$ be the concrete projection Homology/Excision · quotπ, which kills the image of $C_\bullet(A)$ Homology/Excision · pushHom_comp_quotπ. The concrete sequence $0\to C_\bullet(A)\to C_\bullet(X)\to C_\bullet(X)/C_\bullet(S)\to 0$ is short exact Homology/Excision · concreteRelShortComplex_shortExact, by §IV.8 and step 1. Transported through the comparison isomorphism of §IV.8 Homology/Excision · mathlibRelShortComplexIso, so is the sequence that starts with Mathlib's $C_\bullet(A)\to C_\bullet(X)$ Homology/Excision · mathlibRelShortComplex_shortExact, and the concrete quotient is a cokernel of Mathlib's map Homology/Excision · isColimitMathlibRelShortComplex. Two cokernels of one map are canonically isomorphic: $C_\bullet(X,A)\cong C_\bullet(X)/C_\bullet(S)$ Homology/Excision · relChainsIsoQuot Homology/Excision · relπ_relChainsIsoQuot_hom, compatibly with maps of pairs Homology/Excision · relChainsIsoQuot_naturality.
6. *Excision* Homology/Excision · excision_of_eq_inter. Let $j:B\to X$ be the inclusion, with image B; the image of $C=A\cap B$ in B is $j^{-1}(A)$. By step 5 the map of relative chains induced by $(B,C)\to(X,A)$ is isomorphic, as an arrow, to the excision map of step 2, a quasi-isomorphism by step 4; so it is a quasi-isomorphism and induces isomorphisms in relative homology. Form 2 Homology/Excision · excision is form 1 with $B=X\setminus Z$: then $A\cap B=A\setminus Z$, and the interior of $X\setminus Z$ is the complement of the closure of Z, which together with the interior of A, containing that closure, covers X. For closures and interiors in E Homology/Excision · excision_of_closure_subset_interior: the closure in X of $Z\cap X$ lies in the closure of Z in E, and the interior of A in E, intersected with X, lies in the interior of A in X. □

IV.14 The Mayer–Vietoris sequence

Theorem (V3.29). With coefficients in a commutative ring, if $A,B\subseteq X$ have interiors covering X, the sequence

$$\cdots\to H_{n+1}(X)\xrightarrow{\partial}H_n(A\cap B)\xrightarrow{(i_*,-i'_*)}H_n(A)\oplus H_n(B)\xrightarrow{j_*+j'_*}H_n(X)\xrightarrow{\partial}H_{n-1}(A\cap B)\to\cdots$$

is exact at every place Homology/MayerVietoris · mv_exact₁ Homology/MayerVietoris · mv_exact₂ Homology/MayerVietoris · mv_exact₃, where i,i',j,j' are the inclusions Homology/MayerVietoris · mvMap₁ Homology/MayerVietoris · mvMap₂ and ∂ is defined in the proof Homology/MayerVietoris · mvδ.

The proof as formalized.

1. Inside the concrete chains of X, the chains made of simplices in a subset S form a subcomplex $C_\bullet(S)$ Homology/MayerVietoris · chainsIn, with inclusions for $S\subseteq S'$ Homology/MayerVietoris · chainsInIncl and into the $\{A,B\}$-small chains for $S=A,B$ Homology/MayerVietoris · chainsInToSmall.
2. The sequence $0\to C_\bullet(A\cap B)\to C_\bullet(A)\oplus C_\bullet(B)\to C_\bullet^U(X)\to 0$, with maps $x\mapsto(x,-x)$ and $(y,z)\mapsto y+z$ Homology/MayerVietoris · mvShortComplex, is short exact Homology/MayerVietoris · mvShortComplex_shortExact, degree by degree. The first map is injective. If $y+z=0$ with y made of simplices in A and z of simplices in B, then $y=-z$ is made of simplices lying

in both, hence in $A \cap B$ Homology/MayerVietoris · inSet_inter, and (y, z) is the image of y; an element of the biproduct is determined by its two components Homology/MayerVietoris · biprod_X_eq_add. And a small chain is a chain in A plus a chain in B, by splitting off its terms in A as in §IV.13.

3. The homology of the concrete subcomplexes is identified with Mathlib's homology of the subspaces. The singular chains of a subspace S are the chains of X made of simplices in S Homology/MayerVietoris · subspaceChainsHom Homology/MayerVietoris · bijective_subspaceChainsHom_f Homology/MayerVietoris · subspaceChainsIso; composed with the comparison of §IV.8 this gives Homology/MayerVietoris · mathlibChainsIso, compatibly with inclusions Homology/MayerVietoris · mathlibChainsIso_incl Homology/MayerVietoris · mathlibChainsIso_val. Homology commutes with biproducts, being an additive functor, and $C_\bullet^U(X)$ has the homology of X by the small-simplex theorem Homology/MayerVietoris · isIso_homologyMap_smallChains₁. These give three isomorphisms Homology/MayerVietoris · mvIso₁ Homology/MayerVietoris · mvIso₂ Homology/MayerVietoris · mvIso₃ under which the maps of the homology sequence become $(i_*, -i'_*)$ and $j_* + j'_*$ Homology/MayerVietoris · mvIso₁_hom_comp_mvMap₁ Homology/MayerVietoris · mvIso₂_hom_comp_mvMap₂, and ∂ is the connecting morphism transported by them.
4. Exactness is transported along isomorphisms of short complexes Homology/MayerVietoris · exact_mk_of_iso. □

Mayer–Vietoris is not used in §5; it belongs to the infrastructure, and the naturality of its ∂ is not proved.

IV.15 Excision for an attached cell

Milnor's "by excision" on p. 29 identifies $H_*(M^{a_{i-1}} \cup e^\lambda, M^{a_{i-1}})$ with $H_*(e^\lambda, \dot{e}^\lambda)$. As it stands this is not an excision. One would excise $M^{a_{i-1}} \setminus \dot{e}^\lambda$, but the closure of that set contains points of $\dot{e}^\lambda$, and these are not interior points of $M^{a_{i-1}}$ in $M^{a_{i-1}} \cup e^\lambda$, being limits of points of the open cell. The formal proof first removes the centre of the cell and deforms what is left onto the bottom set; then excision applies.

Setting. E is a Hausdorff space and $Y \subseteq X \subseteq E$, with Y closed in X. V is a real normed space whose closed balls are compact (for instance a finite-dimensional one), with closed unit ball D and unit sphere S. $\chi : V \to E$ is continuous and injective on D, with $\chi(z) \in Y \iff \|z\| = 1$ for $z \in D$, and $X = Y \cup \chi(D)$. So $e = \chi(D)$ is a closed cell meeting Y exactly in its boundary sphere $\chi(S)$. The pair $(\chi(D), \chi(S))$ is the *cell pair* of χ Homology/CellExcision · cellPair, and $\chi(S) \subseteq Y$ Homology/CellExcision · image_sphere_subset.

Lemma (the inverse of a parametrization). If f is continuous and injective on a compact set s, with values in a Hausdorff space, its inverse is continuous on $f(s)$ Homology/CellExcision · continuousOn_invFunOn_image: the preimage under the inverse of a closed set t is $f(s \cap t)$, which is compact and hence closed.

The radial deformation. For $z \neq 0$ and $0 \le t \le 1$ let $\rho_t(z) = ((1-t) + t/\|z\|)\, z$ Homology/CellExcision · radialPush. Then $\rho_0 = \mathrm{id}$ Homology/CellExcision · radialPush_zero; ρ_t fixes the unit sphere Homology/CellExcision · radialPush_of_norm_eq_one; $\|\rho_t(z)\| = (1-t)\|z\| + t$ Homology/CellExcision · norm_radialPush, so $\rho_t(z) \neq 0$ Homology/CellExcision · radialPush_ne_zero, ρ_t maps the punctured ball into the ball Homology/CellExcision · radialPush_mem_closedBall, and $\|\rho_1(z)\| = 1$ Homology/CellExcision · norm_radialPush_one; and $(t, z) \mapsto \rho_t(z)$ is continuous away from $z = 0$ Homology/CellExcision · continuousOn_radialPush.

Lemma (the space minus the centre retracts onto Y; V3.31). In the setting above, with $p = \chi(0)$, Y is a strong deformation retract of $X \setminus \{p\}$ Homology/CellExcision · isStrongDeformationRetract_sdiff_center.

Proof. Let g be the inverse of χ on $\chi(D)$, continuous by the lemma, D being compact. Define

$$k(t,x) = x \ \text{ for } x \in Y, \qquad k(t,x) = \chi(\rho_t(g(x))) \ \text{ for } x \in \chi(D) \setminus Y.$$

For $x \neq p$ in the cell, $g(x) \neq 0$, so the second formula makes sense. At $t = 0$ it gives $\chi(g(x)) = x$; at $t = 1$, $\rho_1(g(x))$ lies on the sphere, so $k(1,x) \in Y$; and $k(t,x) \neq p$, because $\rho_t(g(x)) \neq 0$ and χ is injective on D. The second formula also gives x on $Y \cap \chi(D)$: there $\chi(g(x)) = x \in Y$ forces $\|g(x)\| = 1$,

which ρ_t fixes. Y is closed in X by hypothesis, and $\chi(D)$ is compact, hence closed in the Hausdorff space E; these two closed sets cover $X \setminus \{p\}$, and on each of them k is continuous, so k is continuous by the pasting lemma. It fixes Y at all times. □

The same tools show that a cell contracts onto any of its points, by the straight-line homotopy $(t, x) \mapsto \chi\big((1-t)g(x) + tz_0\big)$ in the convex ball `Homology/CellExcision · isStrongDeformationRetract_singleton_image`. Excision is used in a form that accepts the excised pair up to equality of sets `Homology/CellExcision · excision_of_eq`.

Theorem (excision for an attached cell; V3.31). In the setting above, with coefficients in a commutative ring and E in its universe, the inclusion $(\chi(D), \chi(S)) \to (X, Y)$ induces isomorphisms $H_n(\chi(D), \chi(S)) \cong H_n(X, Y)$ in all degrees `Homology/CellExcision · isIso_relHomology_map_cellPair`.

The proof as formalized. Let $p = \chi(0)$ and $e = \chi(D)$; $p \notin Y$ since $\|0\| \neq 1$.

1. $(X, Y) \to (X, X \setminus \{p\})$ induces isomorphisms: Y is a strong deformation retract of $X \setminus \{p\}$ by the lemma, and the bottom of a pair moves along such retracts (§IV.7).
2. $(e, e \setminus \{p\}) \to (X, X \setminus \{p\})$ induces isomorphisms, by excision of $Z = X \setminus e$. Indeed $X \setminus Z = e$ and $(X \setminus \{p\}) \setminus Z = e \setminus \{p\}$. Since $X = Y \cup e$, $Z \subseteq Y$, so the closure of Z lies in the closed set Y, hence in $X \setminus \{p\}$, which is open in X (points are closed in a Hausdorff space) and is therefore its own interior.
3. $(e, \chi(S)) \to (e, e \setminus \{p\})$ induces isomorphisms: step 1 for the cell alone, with e in place of X and $\chi(S)$ in place of Y. The sphere $\chi(S)$ is closed, as a compact image, and $\chi(z) \in \chi(S) \iff \|z\| = 1$ by injectivity.
4. The square of inclusions, $(e, \chi(S)) \to (X, Y) \to (X, X \setminus \{p\})$ against $(e, \chi(S)) \to (e, e \setminus \{p\}) \to (X, X \setminus \{p\})$, commutes `Homology/CellExcision · subsetPairHom_comp`. The second route induces isomorphisms by steps 3 and 2, and the second map of the first route by step 1; so the first map of the first route does. □

The domain of χ plays no role in the homology: only subsets of E appear.

IV.16 The homology of disks and spheres

Theorem (V3.30). Let R be a commutative ring (in the lowest universe, where Euclidean space lives), and D^n, S^{n-1} the closed unit ball and the unit sphere of Euclidean $\mathbb{R}^n$, with $S^{-1} = \emptyset$. Then $H_n(D^n, S^{n-1}) \cong R$ `Homology/Spheres · relHomologyDiskPairIso` and $H_i(D^n, S^{n-1}) = 0$ for $i \neq n$ `Homology/Spheres · isZero_relHomology_diskPair`; over a field, $\dim H_i(D^n, S^{n-1}) = [i = n]$ `Homology/Spheres · finrank_relHomology_diskPair`. For $n \geq 1$, $H_0(S^n) \cong R$ `Homology/Spheres · sphereHomologyZeroIso` and $H_n(S^n) \cong R$ `Homology/Spheres · sphereHomologyTopIso`; $H_0(S^0) \cong R \oplus R$ `Homology/Spheres · sphereHomologyZeroIsoBiprod`; $H_i(S^n) = 0$ for $i \neq 0, n$ `Homology/Spheres · isZero_sphereHomology`; and over a field $\dim H_i(S^n) = [i = 0] + [i = n]$ `Homology/Spheres · finrank_sphereHomology`.

Everything is proved for embedded cells and spheres in a Hausdorff space E in the universe of R. If $\chi : \mathbb{R}^k \to E$ is continuous and injective on D^k, the cell pair $(\chi(D^k), \chi(S^{k-1}))$ has the homology of (D^k, S^{k-1}); the theorem is the case $\chi = \mathrm{id}$ `Homology/Spheres · diskPair` `Homology/Spheres · diskPairIsoCellPair` `Homology/Spheres · sphereHomology`.

Hemispheres. For $s = \pm 1$ the map $h_s : z \mapsto \big(z, s\sqrt{1 - \|z\|^2}\big)$ from $\mathbb{R}^n$ to $\mathbb{R}^{n+1}$ `Homology/Spheres · hemisphere` is continuous `Homology/Spheres · continuous_hemisphere` and injective `Homology/Spheres · hemisphere_injective`; since $\|(z, t)\|^2 = \|z\|^2 + t^2$ `Homology/Spheres · norm_sq_hemisphere`, it maps D^n into S^n `Homology/Spheres · hemisphere_mem_sphere`. S^n is the union of the lower hemisphere $h_{-1}(D^n)$ and the upper hemisphere $h_1(D^n)$ `Homology/Spheres · sphere_eq_union_hemisphere`: for $x \in S^n$ with first coordinates z, $\|z\|^2 = 1 - x_{n+1}^2 \leq 1$ and $\sqrt{1 - \|z\|^2} = |x_{n+1}|$, so $x = h_{\pm 1}(z)$ according to the sign of x_{n+1}. A point of the upper hemisphere lies in the lower one exactly on the equator `Homology/Spheres · hemisphere_one_mem_iff`: equal first coordinates and $\sqrt{1 - \|z\|^2} = -\sqrt{1 - \|z\|^2}$ force $\|z\| = 1$.

Splitting off a retract. If the inclusion $A \hookrightarrow X$ of a pair has a retraction r, the sequence of the pair splits, $H_n(X) \cong H_n(A) \oplus H_n(X, A)$ Homology/Spheres · homologyIsoBiprodOfRetraction. Indeed $r_* \circ i_* = 1$ makes i_* injective in every degree, so $\partial = 0$, being followed by the injective i_* with $i_* \circ \partial = 0$; hence $H_n(X) \to H_n(X, A)$ is onto (for $n = 0$ by §IV.5, for $n = m + 1$ by exactness at $H_{m+1}(X, A)$); and an exact sequence $H_n(A) \to H_n(X) \to H_n(X, A) \to 0$ whose first map has a retraction splits Mathlib/Algebra/Homology/ShortComplex/Exact · Splitting.ofExactOfRetraction. For a point $q \in B$ and the constant retraction Homology/Spheres · retractionSingleton Homology/Spheres · map_retractionSingleton this is reduced homology, $H_n(B) \cong H_n(\{q\}) \oplus H_n(B, \{q\})$ Homology/Spheres · homologyIsoBiprodSingleton. A point has $H_n = 0$ for $n \neq 0$ Homology/Spheres · isZero_homology_singleton, by §IV.7, and $H_0 \cong R$ Homology/Spheres · homologyZeroSingletonIso, by Mathlib's augmentation, an isomorphism for path-connected spaces Mathlib/AlgebraicTopology/SingularHomology/HomologyZero · singularHomology₀ε. Over a field, dimensions are read off from such isomorphisms Homology/Spheres · finrank_eq_one_of_iso Homology/Spheres · finrank_eq_zero_of_isZero Homology/Spheres · finrank_biprod Homology/Spheres · finite_and_finrank_eq_ite.

Theorem (cells; V3.30). If $\chi : \mathbb{R}^k \to E$ is continuous and injective on D^k, then $H_k(\chi(D^k), \chi(S^{k-1})) \cong R$ Homology/Spheres · cellPairIso and $H_i(\chi(D^k), \chi(S^{k-1})) = 0$ for $i \neq k$ Homology/Spheres · isZero_relHomology_cellPair; over a field the dimension is $[i = k]$ Homology/Spheres · finrank_relHomology_cellPair.

The proof as formalized, by induction on k.

1. *The base case.* For $k = 0$, D^0 is a point and S^{-1}, a sphere of radius 1 in the zero space, is empty Homology/Spheres · image_closedBall_fin_zero Homology/Spheres · image_sphere_fin_zero; so the pair is $(\{\chi(0)\}, \emptyset)$, whose homology is that of a point Homology/Spheres · relHomologyCellPairZeroIso.
2. *A cell has no homology relative to one of its points:* $H_*(\chi(D), \{\chi(z_0)\}) = 0$ Homology/Spheres · isZero_relHomology_cell_singleton, the point being a strong deformation retract of the cell (§IV.15).
3. *The boundary of a triple.* For χ on $\mathbb{R}^{k+1}$ let $q = \chi(h_{-1}(0))$, the image of the south pole $-e_{k+1}$ Homology/Spheres · hemisphere_neg_one_zero_mem_sphere Homology/Spheres · singleton_subset_image_sphere. In the triple $\{q\} \subseteq \chi(S^k) \subseteq \chi(D^{k+1})$ Homology/Spheres · cellTriple, the middle terms $H_*(\chi(D^{k+1}), \{q\})$ vanish by step 2. So the boundary $H_{i+1}(\chi(D^{k+1}), \chi(S^k)) \to H_i(\chi(S^k), \{q\})$ is an isomorphism Homology/Spheres · isIso_cellTriple_δ (a connecting morphism is invertible when the middle complex has no homology in the two degrees involved), and $H_0(\chi(D^{k+1}), \chi(S^k))$, a quotient of $H_0(\chi(D^{k+1}), \{q\}) = 0$ by §IV.6, vanishes.
4. *The sphere lemma* Homology/Spheres · sphereRelIso. Let χ be continuous and injective on $S^k \subseteq \mathbb{R}^{k+1}$, and q as in step 3, with the pair $(\chi(S^k), \{q\})$ Homology/Spheres · spherePointPair. Then $H_i(\chi(S^k), \{q\}) \cong H_i(\chi(D_+), \chi(S^{k-1}))$, where $\chi(D_+) = \chi(h_1(D^k))$ is the upper hemisphere, the cell of the map $\chi \circ h_1$ on $\mathbb{R}^k$. First, $(\chi(S^k), \{q\}) \to (\chi(S^k), \chi(D_-))$ induces isomorphisms, because the lower hemisphere $\chi(D_-) = \chi(h_{-1}(D^k))$ Homology/Spheres · lowerHemisphere_subset Homology/Spheres · singleton_subset_lowerHemisphere contracts onto q (§§IV.7, IV.15). Second, $\chi(S^k) = \chi(D_-) \cup \chi(D_+)$ Homology/Spheres · image_sphere_eq_union, $\chi(D_-)$ is closed, being compact Homology/Spheres · isClosed_lowerHemisphere, and $\chi(h_1(z))$ lies in $\chi(D_-)$ exactly when $\|z\| = 1$ Homology/Spheres · upperHemisphere_mem_iff, by the injectivity of χ on S^k and the hemisphere lemma. So the upper hemisphere is a k-cell attached to the lower one along the equator, and excision for an attached cell (§IV.15) makes $(\chi(D_+), \chi(S^{k-1})) \to (\chi(S^k), \chi(D_-))$ an isomorphism in homology. The maps $\chi \circ h_{\pm 1}$ are continuous and injective on D^k Homology/Spheres · continuousOn_comp_hemisphere Homology/Spheres · injOn_comp_hemisphere.
5. *Induction.* $H_{k+1}(\chi(D^{k+1}), \chi(S^k)) \cong H_k(\chi(S^k), \{q\}) \cong H_k(\chi(D_+), \chi(S^{k-1})) \cong R$, the last by induction for the k-cell $\chi \circ h_1$. For $i+1 \neq k+1$ the same chain gives $H_{i+1}(\chi(D^{k+1}), \chi(S^k)) \cong H_i$ of the k-cell pair, which vanishes by induction since $i \neq k$; and $H_0 = 0$ by step 3. □

Embedded spheres. For χ continuous and injective on S^n: $H_n(\chi(S^n), \{q\}) \cong R$ Homology/Spheres · spherePointPairIso and $H_i(\chi(S^n), \{q\}) = 0$ for $i \neq n$ Homology/Spheres · isZero_relHomology_spherePointPair, by the sphere lemma and the theorem on cells. Reduced homology then gives $H_0(\chi(S^n)) \cong R$ for $n \neq 0$ Homology/Spheres · homologyImageSphereZeroIso, $H_n(\chi(S^n)) \cong R$ for $n \neq 0$ Homology/Spheres · homologyImageSphereTopIso, $H_i(\chi(S^n)) = 0$ for $i \neq 0, n$ Homology/Spheres · isZero_homology_

imageSphere, $H_0(\chi(S^0)) \cong R \oplus R$ Homology/Spheres · homologyImageSphereZeroIsoBiprod, and over a field $\dim H_i(\chi(S^n)) = [i = 0] + [i = n]$ Homology/Spheres · finrank_homology_imageSphere.

Mathlib's disk, in any universe. Mathlib's CW complexes use the pair $(\mathbb{D}^n, \partial\mathbb{D}^n)$, the closed ball and the sphere of $\mathbb{R}^n$ lifted to an arbitrary universe Mathlib/Topology/Category/TopCat/Sphere · disk Mathlib/Topology/Category/TopCat/Sphere · diskBoundary. The project makes it a pair Homology/Spheres · topDiskPair Homology/Spheres · isEmbedding_diskBoundaryInclusion, and shows that χ induces a map of pairs $(\mathbb{D}^k, \partial\mathbb{D}^k) \to (\chi(D^k), \chi(S^{k-1}))$ Homology/Spheres · diskToCell that is an isomorphism in relative homology Homology/Spheres · isIso_relHomology_map_diskToCell: both components are continuous bijections from compact spaces onto Hausdorff spaces, hence homeomorphisms, and in particular homotopy equivalences Homology/Spheres · homotopyEquivOfBijective. With χ the lift of $\mathbb{R}^n$ into the universe, this gives $H_n(\mathbb{D}^n, \partial\mathbb{D}^n) \cong R$ and $H_i = 0$ for $i \neq n$, in every universe Homology/Spheres · relHomologyTopDiskPairIso Homology/Spheres · isZero_relHomology_topDiskPair Homology/Spheres · finrank_relHomology_topDiskPair.

IV.17 The cell-attachment theorem

On p. 29 Milnor needs the homology of $M^{c-\varepsilon}$ with cells attached, relative to $M^{c-\varepsilon}$; with Remark 3.3 several cells are attached on one level at once.

Theorem (one cell; V3.31). Let E be a Hausdorff space, $Y \subseteq X \subseteq E$ with Y closed, and $\chi : \mathbb{R}^k \to E$ continuous and injective on D^k, with $\chi(z) \in Y \iff \|z\| = 1$ for $z \in D^k$ and $X = Y \cup \chi(D^k)$. Let the coefficients be a commutative ring R in the universe of E. Then the characteristic map $\chi : (\mathbb{D}^k, \partial\mathbb{D}^k) \to (X, Y)$ Homology/CellAttach · cellAttachMap, which passes through the cell pair Homology/CellAttach · cellPairHom, induces isomorphisms in all degrees Homology/CellAttach · relHomology_cellAttach_iso Homology/CellAttach · relHomologyCellAttachIso. Hence $H_k(X, Y) \cong R$ Homology/CellAttach · relHomologyCellAttachTopIso, $H_i(X, Y) = 0$ for $i \neq k$ Homology/CellAttach · isZero_relHomology_cellAttach, and over a field $H_i(X, Y)$ is finite-dimensional, of dimension $[i = k]$ Homology/CellAttach · finite_and_finrank_relHomology_cellAttach Homology/CellAttach · finrank_relHomology_cellAttach.

Proof. The map factors as $(\mathbb{D}^k, \partial\mathbb{D}^k) \to (\chi(D^k), \chi(S^{k-1})) \to (X, Y)$. The first factor is an isomorphism in homology by §IV.16, the second by excision for an attached cell (§IV.15). □

Theorem (several disjoint cells, the situation of Remark 3.3; V3.31). Let moreover $\chi_j : \mathbb{R}^{k_j} \to E$, for j in a finite set s, be maps of the same kind with pairwise disjoint cells $e_j = \chi_j(D^{k_j})$, and $X = Y \cup \bigcup_{j \in s} e_j$. Over a field, $H_i(X, Y)$ is finite-dimensional, of dimension $\#\{j \in s : k_j = i\}$ Homology/CellAttach · finite_and_finrank_relHomology_cellsUnion Homology/CellAttach · finrank_relHomology_cellsAttach Homology/CellAttach · finiteDimensional_relHomology_cellsAttach Homology/CellAttach · finrank_relHomology_cellsAttach_univ. With any commutative ring of coefficients, $H_i(X, Y) \cong R^{c_i}$, c_i the number of cells of dimension i Homology/CellAttach · nonempty_relHomology_cellsUnion_iso Homology/CellAttach · relHomology_cellsAttach_iso; in particular $H_i(X, Y) = 0$ if no cell has dimension i Homology/CellAttach · isZero_relHomology_cellsUnion Homology/CellAttach · isZero_relHomology_cellsAttach.

The idea. Attach the cells one at a time. The exact sequence of the triple $Y \subseteq X_s \subseteq X_{s\cup\{a\}}$ alone gives only an inequality of dimensions. Disjointness makes the sequence split, and then dimensions add.

The proof as formalized, by induction on s, writing $X_s = Y \cup \bigcup_{j \in s} e_j$ Homology/CellAttach · cellsUnion.

1. *Bookkeeping.* $X_{s\cup\{a\}} = X_s \cup e_a$ Homology/CellAttach · cellsUnion_insert Homology/CellAttach · cellsUnion_subset_insert. X_s is closed, the cells being compact Homology/CellAttach · isClosed_cellsUnion. A cell disjoint from the others meets X_s only in Y Homology/CellAttach · inter_cellsUnion_subset, so for $a \notin s$, $\chi_a(z) \in X_s \iff \|z\| = 1$ Homology/CellAttach · mem_cellsUnion_iff: e_a is a cell attached to the closed set X_s.
2. *The base case.* $X_\emptyset = Y$, and $H_*(Y, Y) = 0$ Homology/CellAttach · isZero_relHomology_cellsUnion_empty.

3. *The splitting* Homology/CellAttach · relHomologyCellsUnionInsertIso. For $a \notin s$ and $X = X_{s\cup\{a\}}$, $H_i(X,Y) \cong H_i(X_s,Y) \oplus H_i(X,X_s)$. The inclusion $\varepsilon : (X_s,Y) \to (X, Y\cup e_a)$ is the excision of $Z = e_a \setminus Y$: $X \setminus Z = X_s$, and $(Y\cup e_a)\setminus Z = Y$ because $e_a \cap X_s \subseteq Y$. The closure of Z lies in the closed set e_a, which lies in $X \setminus \bigcup_{j\in s} e_j$, the cells being disjoint; that set is open in X and contained in $Y\cup e_a$, so e_a lies in the interior of $Y\cup e_a$. So ε induces isomorphisms. Now ε is the inclusion $(X_s,Y)\to(X,Y)$ followed by $\pi : (X,Y)\to(X,Y\cup e_a)$, so $(\varepsilon_*)^{-1}\circ\pi_*$ is a retraction of $H_*(X_s,Y)\to H_*(X,Y)$. A triple whose first map has a retraction in every degree splits Homology/CellAttach · TopTriple.isoBiprodOfRetraction, by the argument of §IV.16 for the triple $Y\subseteq X_s\subseteq X$.
4. *Induction.* $H_i(X,X_s)$ is the homology of one cell attached to the closed set X_s (step 1 and the one-cell theorem): of dimension $[k_a = i]$ over a field, and R in degree k_a, 0 elsewhere, in general. Dimensions add over the direct sum, and $\#\{j\in s\cup\{a\} : k_j = i\} = \#\{j\in s : k_j = i\} + [k_a = i]$; with general coefficients, $R^c\oplus R \cong R^{c+1}$. □

The disjointness of the cells is used exactly in step 3: it makes e_a lie in the interior of $Y\cup e_a$ inside X, which is what makes the connecting maps of the triple vanish.

The form in which §IV.19 uses this is a property of a field $\mathbb{k}$ Homology/MorseHomologyBasic · CellAttachHomology: for every Hausdorff space in the universe of $\mathbb{k}$, every closed Y and every finite family of pairwise disjoint cells χ_j, $j<m$, attached to Y as above, $H_i(Y\cup\bigcup_j e_j, Y;\mathbb{k})$ is finite-dimensional, of dimension $\#\{j : k_j = i\}$. It holds for every field Homology/MorseHomologyCells · cellAttachHomology, by the theorem above.

IV.18 Singular homology as a homology of a filtration

With the sequences of triples in hand, singular homology over a field is an instance of the abstract structure of §IV.3, and all of §§IV.3–IV.4 applies to it (V3.32).

The sequence of a triple as a long exact sequence. For every triple, with coefficients in a field $\mathbb{k}$, the spaces $A_i = H_i(Y,Z)$, $B_i = H_i(X,Z)$, $C_i = H_i(X,Y)$ and the maps of §IV.6 form a long exact sequence in the sense of §IV.1 Homology/TripleFiltration · TopTriple.longExactSeq: exactness in the category of $\mathbb{k}$-modules is exactness of the underlying linear maps, and an epimorphism of modules is a surjection. For subsets $A\subseteq B$ of a space, write $H_i(B,A;\mathbb{k})$ for the homology of the pair of subsets Homology/TripleFiltration · subsetHomology. It depends only on the subsets Homology/TripleFiltration · finrank_subsetHomology_congr Homology/TripleFiltration · finiteDimensional_subsetHomology_congr, $H_i(A,A) = 0$ Homology/TripleFiltration · subsingleton_subsetHomology_self, and a triple of subsets $A\subseteq B\subseteq D$ has its sequence Homology/TripleFiltration · subsetTripleLongExactSeq, also when the pair (D,B) is replaced by an equal one Homology/TripleFiltration · subsetTripleLongExactSeq'.

The filtration. Let $X_0\subseteq X_1\subseteq\cdots$ be subsets of a space. The abstract structure needs a group for every pair of indices, and the choice is $H(i,a,b) = H_i(X_{\max(a,b)}, X_a;\mathbb{k})$ Homology/TripleFiltration · filtrationGroup. This is a pair of subsets for all a,b, and equals $H_i(X_b,X_a)$ when $a\le b$ Homology/TripleFiltration · finrank_filtrationGroup. Because $\max(a,b) = b$ holds only by a proof, the sequences of triples are transported along equalities of subsets. If every step (X_{k+1},X_k) has finite-dimensional homology, the result is a homology of a filtration Homology/TripleFiltration · filtrationHomology, built by the constructor of §IV.3, with $H_*(X_k,X_k) = 0$ on the diagonal.

The final theorems should assume finiteness only for the steps $k<n$ that are used. So the construction is applied to the truncated filtration $k\mapsto X_{\min(k,n)}$ Homology/TripleFiltration · monotone_truncate Homology/TripleFiltration · truncatedFiltrationHomology, constant from n on. Its steps are the steps $k<n$ of X followed by copies of (X_n,X_n), which have zero homology Homology/TripleFiltration · finiteDimensional_truncate_step; and it has the same total groups $H_i(X_n,X_0)$ Homology/TripleFiltration · finrank_truncate_total and the same Morse numbers $C_i = \sum_{k<n}\dim H_i(X_{k+1},X_k;\mathbb{k})$ Homology/TripleFiltration · singularMorseNumber Homology/TripleFiltration · morseNumber_truncate.

Theorem (the Morse inequalities for a filtration of a space; V3.32). Let $X_0\subseteq X_1\subseteq\cdots$ be subsets of

a space, $n \geq 1$, and suppose that the steps (X_{k+1}, X_k), $k < n$, have finite-dimensional homology over $\Bbbk$. Then $H_i(X_n, X_0; \Bbbk)$ is finite-dimensional `Homology/TripleFiltration · finiteDimensional_subsetHomology_of_steps`, and with $b_i = \dim H_i(X_n, X_0; \Bbbk)$:

- $b_l \leq C_l$ `Homology/TripleFiltration · singular_weak_morse_inequality`;
- $\mathrm{alt}_l(b) \leq \mathrm{alt}_l(C)$ `Homology/TripleFiltration · singular_morse_inequality`;
- $\sum_{i \leq N} (-1)^i b_i = \sum_{i \leq N} (-1)^i C_i$ if no step has homology in degree $N+1$ `Homology/TripleFiltration · singular_euler_characteristic`.

Under the cell hypothesis of §IV.3 for a finite set P with an index and a step, $C_i = \#\{p \in P : \mathrm{ind}(p) = i\}$ `Homology/TripleFiltration · singularMorseNumber_eq_card`, and the three statements hold with these numbers `Homology/TripleFiltration · singular_weak_morse_inequality_of_cells` `Homology/TripleFiltration · singular_morse_inequality_of_cells`, the alternating sum being $\sum_{p \in P} (-1)^{\mathrm{ind}(p)}$ when every index is at most N `Homology/TripleFiltration · singular_euler_characteristic_of_cells`.

Proof. §§IV.3–IV.4 for the truncated filtration and the chain $0 \leq 1 \leq \ldots \leq n$, rewritten with the equalities of the totals and of the Morse numbers. For the version with cells, (cells) for the steps of X gives (cells) for the abstract groups, and the count of §IV.3 applies; the Euler characteristic is regrouped by index as there. □

This holds for any filtration of any space by subsets, and the finiteness of $H_*(X_n, X_0)$ is a consequence, not an assumption.

IV.19 The homology of a slab

Milnor computes the homology of $(M^{a_i}, M^{a_{i-1}})$ in one line (p. 29): it equals that of $(M^{a_{i-1}} \cup e^{\lambda_i}, M^{a_{i-1}})$ by Theorems 3.1 and 3.2, which equals $H_*(e^{\lambda_i}, \dot{e}^{\lambda_i})$ by excision. The formal proof makes each equality a lemma.

What the first equality hides. It is a statement about pairs, not spaces. Theorem 3.2 is stated up to homotopy type, and the cell is attached to $M^{c-\varepsilon}$, c the critical value, not to $M^{a_{i-1}}$. The top of the pair moves along deformations that fix $M^{c-\varepsilon}$: the deformation of M^{a_i} onto $M^{c+\varepsilon}$ (Theorem 3.1) and that of $M^{c+\varepsilon}$ onto $M^{c-\varepsilon} \cup e$ (the proof of Theorem 3.2, in the strong form of Part II, §§II.13–2.14). The bottom cannot be moved along a deformation of pairs, since the deformation of $M^{c-\varepsilon}$ onto $M^{a_{i-1}}$ does not extend to $M^{c+\varepsilon}$. It moves in homology all the same, because $H_*(M^{c-\varepsilon}, M^{a_{i-1}}) = 0$.

Lemma (moving the bottom over a field; V3.33). In a long exact sequence (§IV.1) all of whose terms A_i vanish, every $g_i : B_i \to C_i$ is bijective `Homology/MorseHomologyBasic · LongExactSeq.bijective_g_of_subsingleton` `Homology/MorseHomologyBasic · LongExactSeq.equivOfSubsingleton`: it is injective because $\ker g_i = \mathrm{im}\, f_i = 0$, and surjective for $i = 0$ by the end of the sequence and for $i+1$ because δ_i lands in $A_i = 0$. Hence, for subsets $A \subseteq B \subseteq D$ with A a strong deformation retract of B, $H_i(D, A; \Bbbk) \cong H_i(D, B; \Bbbk)$ `Homology/MorseHomologyBasic · subsetHomologyEquivOfIsSDRBottom`: the sequence of the triple $A \subseteq B \subseteq D$ has $H_*(B, A) = 0$ `Homology/MorseHomologyBasic · subsingleton_subsetHomology_of_isSDR`. The top moves along strong deformation retracts by §IV.7 `Homology/MorseHomologyBasic · subsetHomologyEquivOfIsSDR`.

Lemma (the slab, abstractly; V3.33). Let $A \subseteq B \subseteq X' \subseteq X$ be subsets of a Hausdorff space with B closed, and let $e_j = \chi_j(D^{k_j})$, $j < m$, be pairwise disjoint cells attached to B as in §IV.17, with $B \cup \bigcup_j e_j \subseteq X'$. If A is a strong deformation retract of B, $B \cup \bigcup_j e_j$ one of X', and X' one of X, then $H_i(X, A; \Bbbk)$ is finite-dimensional, of dimension $\#\{j : k_j = i\}$ `Homology/MorseHomologyBasic · finrank_subsetHomology_of_cells`.

Proof.

$$H_i(X, A) \cong H_i(X, B) \cong H_i(X', B) \cong H_i(B \cup \textstyle\bigcup_j e_j,\ B),$$

by moving the bottom, then the top twice; the last group is given by the cell-attachment theorem (§IV.17). □

For the slab, $A = M^{a_{i-1}}$, $B = M^{c-\varepsilon}$, $X' = M^{c+\varepsilon}$ and $X = M^{a_i}$.

The submanifold setting. Part II proves Theorems 3.1 and 3.2 and Remark 3.3 for a slice submanifold N of a finite-dimensional real normed space V, modelled on a finite-dimensional Z (§0.1) `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`, and a function $f = F|_N$ restricted from a smooth $F : V \to \mathbb{R}$; here N is moreover compact. The sublevel sets are $N^a = \{q \in N : f(q) \le a\}$ `Chapter2/Crossing · sublevel`. Homology is taken over a field $\Bbbk$ of some universe, while N lives in V, possibly in another. The statements are therefore made for a space T in the universe of $\Bbbk$ together with a homeomorphism $e : T \cong N$, homology being read in T through e; for N itself one takes $e = \mathrm{id}$ (§IV.20), and §IV.21 uses the freedom for abstract manifolds. Strong deformation retracts are carried by homeomorphisms `Chapter4/Whitney · IsStrongDeformationRetract.preimage_homeomorph Homology/MorseHomologyBasic · IsStrongDeformationRetract.preimage_subtype`, and the number of elements satisfying a predicate does not depend on the enumeration `Homology/MorseHomologyBasic · card_filter_equiv`.

Theorem (one critical level; V3.34). Let S be a finite set of non-degenerate critical points of f `Chapter1/Manifold · IsNondegenerateCriticalPointM`, all with $f = c$, and $a' < c < a$ such that every critical point outside S has value $< a'$ or $> a$. Then $H_i(N^a, N^{a'}; \Bbbk)$, read in T, is finite-dimensional, of dimension the number of points of S of index i `Homology/MorseHomology · sublevel_step_critical`.

The proof as formalized.

1. Let $\varepsilon_0 = \min(c-a', a-c) > 0$. The set $f^{-1}[c-\varepsilon_0, c+\varepsilon_0]$ is compact, being closed in the compact N, and its critical points lie in S, the others having values outside $[a', a]$.
2. Remark 3.3 at c (Part II, §II.14) `Chapter2/SeveralCritical · remark_3_3_submanifold` gives $\varepsilon_1 > 0$ such that for every $0 < \varepsilon \le \varepsilon_1$ there are cells χ_p, $p \in S$, of dimension the index of p: continuous and injective on the closed ball, meeting $N^{c-\varepsilon}$ exactly in the image of the sphere, contained in $N^{c+\varepsilon}$, pairwise disjoint, and with $N^{c-\varepsilon} \cup \bigcup_p e_p$ a strong deformation retract of $N^{c+\varepsilon}$. Take $\varepsilon = \min(\varepsilon_1, \varepsilon_0/2)$, so that $a' < c - \varepsilon$ and $c + \varepsilon < a$.
3. Theorem 3.1 for a compact submanifold (Part II, §II.5) `Chapter2/Submanifold · theorem_3_1_submanifold_of_isCompact`, on $[a', c-\varepsilon]$ and on $[c+\varepsilon, a]$, which contain no critical values: $N^{a'}$ is a strong deformation retract of $N^{c-\varepsilon}$, and $N^{c+\varepsilon}$ one of N^a.
4. Index the cells by $\{0, \dots, \#S - 1\}$ through an enumeration of S, and carry everything to T by e^{-1}: the maps $e^{-1} \circ \chi_p$ are continuous and injective on the ball, their cells are pairwise disjoint and attached to the closed set $e^{-1}(N^{c-\varepsilon})$, and the strong deformation retracts are carried along.
5. The slab lemma with $A = e^{-1}(N^{a'})$, $B = e^{-1}(N^{c-\varepsilon})$, $X' = e^{-1}(N^{c+\varepsilon})$, $X = e^{-1}(N^a)$ gives the dimension $\#\{j : k_j = i\}$, which is the number of points of S of index i, counted through the enumeration. □

Theorem (the homology of a slab, p. 29; V3.34). Let every critical point of f be non-degenerate, P the finite set of critical points, and $a' \le a$ two values taken by no critical point, such that all critical values in (a', a) are equal. Then $H_i(N^a, N^{a'}; \Bbbk)$ is finite-dimensional, of dimension the number of critical points of index i in $f^{-1}(a', a)$ `Homology/MorseHomology · sublevel_step`.

Proof. If some critical point q_0 has $a' < f(q_0) < a$, let $c = f(q_0)$ and let S be the set of critical points on the level c. A critical point outside S cannot have its value in (a', a), where all critical values equal c, nor at a' or a, which are no critical values; so the previous theorem applies, and its count is the stated one, the critical points in $f^{-1}(a', a)$ being exactly those of S. Otherwise no critical point has its value in $[a', a]$; Theorem 3.1 makes $N^{a'}$ a strong deformation retract of N^a, the group vanishes (§IV.7), and the count is 0. □

IV.20 The Morse filtration and the inequalities on a compact submanifold

Lemma (separating values: Milnor's choice of the a_i). For a finite set s of reals all larger than L there are $n > 0$ and a monotone sequence $a_0 \le a_1 \le \cdots$ with $L < a_0$, no a_k in s, every element of s in (a_0, a_n), and at most one element of s in each interval (a_k, a_{k+1}), $k < n$ `Homology/MorseHomologyBasic · exists_separating_seq`.

Proof. Induction on s, adding each time a new minimum m. For s empty, take $n = 1$ and the constant sequence $L + 1$. Given, by induction, a sequence for s with lower bound m, so that $m < a_0$, prepend $(L+m)/2$: the new first interval $((L+m)/2, a_0)$ contains m and no element of s, all of which lie above a_0. □

If $a_0 < c < a_n$ and c is none of the a_k, then c lies in some (a_k, a_{k+1}) with $k < n$ `Homology/MorseHomologyBasic · exists_mem_Ioo_of_lt`, $k + 1$ being the least index with $c < a_{k+1}$; for a monotone sequence these intervals are disjoint `Homology/MorseHomologyBasic · eq_of_mem_Ioo_of_monotone`. A global maximum is a critical point `Homology/MorseHomologyBasic · isCriticalPointM_of_forall_ge`, like a global minimum `Chapter2/CWType · isCriticalPointM_of_forall_le`: the local representative has a local extremum there, so its derivative vanishes.

Theorem (the Morse filtration, p. 29; V3.34). Let N be a compact slice submanifold of V and $F : V \to \mathbb{R}$ smooth with every critical point of $f = F|_N$ non-degenerate. There are $n > 0$ and values $a_0 \le a_1 \le \cdots$, taken by no critical point, with $N^{a_0} = \emptyset$, $N^{a_n} = N$, at most one critical value in each (a_k, a_{k+1}), and $H_i(N^{a_{k+1}}, N^{a_k}; \Bbbk)$ finite-dimensional, of dimension the number of critical points of index i in $f^{-1}(a_k, a_{k+1})$ `Homology/MorseHomology · exists_morse_filtration` `Homology/MorseHomology · exists_morse_filtration'`.

Proof. A Morse function on a compact manifold has finitely many critical points `Chapter1/ManifoldMorse · IsMorseFunction.finite_setOf_isCriticalPointM`; let P be their set and s the set of critical values, all larger than $L = -\sum_{c \in s} |c| - 1$. Take the separating sequence of the lemma. $N^{a_0} = \emptyset$, because the minimum of f is a critical value and so lies above a_0; $N^{a_n} = N$, because the maximum is a critical value below a_n. Each step is a slab of §IV.19, its ends being taken by no critical point and its interior containing at most one critical value. □

Milnor starts his chain at $M^{a_0} = \emptyset$ without naming a_0, and ends it at $M^{a_k} = M$; both rest on the minimum and the maximum of f being critical values. The filtration is repackaged for §IV.18 `Homology/MorseHomology · exists_cells_filtration`: $X_k = N^{a_k}$ read in T, with $X_0 = \emptyset$ and $X_n = T$, steps with finite-dimensional homology, and the cell hypothesis of §IV.3 with step(p) the unique $k < n$ such that $a_k < f(p) < a_{k+1}$; it exists because $a_0 < f(p) < a_n$, and is unique because the intervals are disjoint. The numbers C_λ of §0.2, the numbers of critical points of index λ `Chapter3/EulerChar · critCount`, are the counts in P `Homology/MorseHomology · critCount_eq_card`.

Betti numbers (V3.33). $H_i(X; \Bbbk)$ is Mathlib's singular homology `Homology/MorseHomologyBasic · spaceHomology`, and the Betti number $b_i(X) = \dim_{\Bbbk} H_i(X; \Bbbk)$ `Homology/MorseHomologyBasic · bettiNumber` (§0.6) is Milnor's $R_i(X)$; by convention it is 0 when $H_i(X; \Bbbk)$ is infinite-dimensional, which does not happen for the spaces below. Homeomorphic spaces have isomorphic homology and the same Betti numbers `Homology/MorseHomologyBasic · spaceHomologyEquivOfHomeomorph` `Homology/MorseHomologyBasic · bettiNumber_eq_of_homeomorph`, and the pair of subsets $(E, \emptyset)$ has the homology of E `Homology/MorseHomologyBasic · subsetHomologyUnivEquiv` `Homology/MorseHomologyBasic · finrank_subsetHomology_eq_bettiNumber` `Homology/MorseHomologyBasic · finiteDimensional_spaceHomology_of_subsetHomology`.

Theorem 5.2, (4_λ) and Corollary 5.4 on a compact submanifold (V3.35–V3.37). In this setting, for every field $\Bbbk$, with $b_i = \dim H_i(N; \Bbbk)$, C_λ the number of critical points of index λ, and $n = \dim N$:

- $H_i(N; \Bbbk)$ is finite-dimensional `Homology/MorseHomology · finiteDimensional_spaceHomology`;

- (2) $b_\lambda \le C_\lambda$ Homology/MorseHomology · bettiNumber_le_critCount Homology/MorseHomology · bettiNumber_le_critCount';
- (4_λ): $b_\lambda - b_{\lambda-1} + \cdots \pm b_0 \le C_\lambda - C_{\lambda-1} + \cdots \pm C_0$ Homology/MorseHomology · altSum_bettiNumber_le Homology/MorseHomology · altSum_bettiNumber_le';
- (3) $\sum_{i\le n}(-1)^i b_i = \sum_{\lambda\le n}(-1)^\lambda C_\lambda$ Homology/MorseHomology · sum_bettiNumber_eq_sum_critCount Homology/MorseHomology · sum_bettiNumber_eq_sum_critCount';
- $b_i = 0$, indeed $H_i(N;\Bbbk) = 0$, for $i > n$ Homology/MorseHomology · bettiNumber_eq_zero_of_finrank_lt Homology/MorseHomology · subsingleton_spaceHomology_of_finrank_lt, so that the left side of (3) is the whole alternating sum;
- **Corollary 5.4:** if $\lambda \ge 1$ and $C_{\lambda+1} = C_{\lambda-1} = 0$, then $b_\lambda = C_\lambda$ and $b_{\lambda+1} = b_{\lambda-1} = 0$ Homology/MorseHomology · bettiNumber_eq_critCount_of_succ; if $C_1 = 0$, then $b_0 = C_0$ and $b_1 = 0$ Homology/MorseHomology · bettiNumber_eq_critCount_of_zero; and if no two consecutive C_i are nonzero, $b_i = C_i$ for every i Homology/MorseHomology · bettiNumber_eq_critCount_of_lacunary.

All of these hold for every space homeomorphic to N (the forms with e) and for N itself (the primed forms).

Proof. The filtration above in §IV.18: $X_0 = \emptyset$ and $X_n = T$, so $H_i(X_n, X_0) = H_i(T)$. For (3), no step has homology in degree $\dim N + 1$, because no index exceeds $\dim N$ Chapter1/Manifold · indexM_le_finrank. Vanishing above the dimension is (2) with $C_i = 0$, and Corollary 5.4 and the lacunary principle are §IV.4 for b and C, which satisfy (4_l) for every l. □

Milnor's hypothesis that the critical points be isolated (p. 29) is redundant: non-degenerate critical points are isolated, so on a compact manifold there are finitely many (Part I, §§I.11, 1.15).

IV.21 Compact manifolds, $\chi = \sum_i (-1)^i b_i$, and three examples

Compact submanifolds of $\mathbb{R}^n$ (V3.39). For a compact slice submanifold $N \subseteq \mathbb{R}^n$ of dimension k, some $L_p = \|\cdot - p\|^2$ restricts to a Morse function on N (Theorem 6.6; Part III, §III.15) Chapter4/Focal · exists_isMorseFunction_normSq_sub, and L_p is smooth on $\mathbb{R}^n$ Homology/MorseHomologyManifold · contDiff_normSq_sub. Hence $H_i(N;\Bbbk)$ is finite-dimensional Homology/MorseHomologyManifold · finiteDimensional_spaceHomology_of_isCompact and $b_i(N) = 0$ for $i > k$ Homology/MorseHomologyManifold · bettiNumber_eq_zero_of_isCompact, for N and for every space homeomorphic to it.

Theorem (χ is the homological Euler characteristic; V3.38).

1. Let $N \subseteq \mathbb{R}^{K+1}$, $K \ge 1$, be a compact nonempty slice submanifold of dimension $d+1$, and χ_N an Euler number of N in the sense of Part III, §III.9 Chapter3/EulerChar · IsEulerNumber: the index sum of every smooth tangent field with finitely many zeros. Then $\chi_N = \sum_{i=0}^{d+1}(-1)^i \dim H_i(N;\Bbbk)$ for every field $\Bbbk$ Homology/MorseHomologyManifold · IsEulerNumber.eq_sum_bettiNumber.
2. For a compact Hausdorff smooth manifold M without boundary, modelled on a finite-dimensional E, the Euler number $\chi(M)$ of Part III, §III.21 Chapter4/Whitney · eulerNumberM, which is the alternating count $\sum_\lambda (-1)^\lambda C_\lambda$ of every Morse function Chapter4/Whitney · sum_critCount_eq_eulerNumberM, equals $\sum_{i\le \dim M}(-1)^i \dim H_i(M;\Bbbk)$ for every field $\Bbbk$ Homology/MorseHomologyManifold · eulerNumberM_eq_sum_bettiNumber.

Proof. (1) Take L_p Morse on N. The gradient field and Poincaré–Hopf give $\sum_\lambda (-1)^\lambda C_\lambda(L_p) = \chi_N$ Chapter3/EulerChar · IsEulerNumber.sum_critCount_eq, and formula (3) of §IV.20 for L_p gives $\sum_i (-1)^i b_i(N) = \sum_\lambda (-1)^\lambda C_\lambda(L_p)$. (2) By a Whitney embedding, M is diffeomorphic to a compact slice submanifold $N \subseteq \mathbb{R}^{K+1}$ modelled on $\mathbb{R}^{\dim M}$ Chapter4/Whitney · exists_diffeomorph_sliceSubmanifold, on which some L_p is Morse; through the diffeomorphism $\Phi : M \to N$, $L_p \circ \Phi$ is a Morse function on M Chapter4/WhitneyInvariance · IsMorseFunction.comp_diffeomorph_symm with the same numbers C_λ Chapter4/

WhitneyInvariance · critCount_comp_symm Homology/MorseHomologyManifold · exists_whitney_normSq. Then

$$\chi(M) = \sum_\lambda (-1)^\lambda C_\lambda(L_p \circ \Phi) = \sum_\lambda (-1)^\lambda C_\lambda(L_p|_N) = \sum_i (-1)^i b_i(M),$$

the last by formula (3) for N, read in M through the homeomorphism Φ. □

The left side of (2) does not involve the field, so the alternating sum of the Betti numbers is the same for every field. Milnor computes $\chi(M)$ over a field F on p. 29 and uses it in §6 without reference to F. For the individual b_i no such statement holds, and none is claimed.

Theorem (the Morse inequalities on a compact manifold; V3.39). Let M be a compact Hausdorff smooth manifold without boundary, modelled on a finite-dimensional real normed space E, of dimension n, and $\Bbbk$ a field in the universe of M. Then $H_i(M;\Bbbk)$ is finite-dimensional Homology/MorseHomologyManifold · finiteDimensional_spaceHomology_of_compactSpace and vanishes for $i > n$ Homology/MorseHomologyManifold · bettiNumber_eq_zero_of_compactSpace. For every Morse function g on M (§0.2) Chapter1/Manifold · IsMorseFunction, with $b_i = \dim H_i(M;\Bbbk)$:

- (3) $\sum_{i\le n}(-1)^i b_i = \sum_{\lambda\le n}(-1)^\lambda C_\lambda(g)$ Homology/MorseHomologyManifold · sum_bettiNumber_eq_sum_critCount_of_compactSpace;
- (4_λ) Homology/MorseHomologyManifold · altSum_bettiNumber_le_of_compactSpace;
- (2) $b_\lambda \le C_\lambda(g)$ Homology/MorseHomologyManifold · bettiNumber_le_critCount_of_compactSpace;
- the lacunary principle Homology/MorseHomologyManifold · bettiNumber_eq_critCount_of_lacunary_of_compactSpace.

The proof as formalized. Finiteness and vanishing come from the Whitney embedding and the first paragraph of this section, carried back by the homeomorphism. Formula (3) is part 2 of the previous theorem: both sides equal $\chi(M)$. For (4_λ) there are two cases. In dimension 0 the inequalities are equalities, $b_i = C_i$ for every i: for $i = 0$ by (3), whose sums then have a single term; for $i > 0$ both sides vanish, there being no homology above the dimension and no index above 0. In dimension $d + 1 \ge 1$, g is realized as an ambient function on an embedded copy Homology/MorseHomologyManifold · exists_ambient_of_isMorseFunction: $M \cong N \subseteq \mathbb{R}^{K+1}$, $g \circ \Phi^{-1}$ is Morse on N, and it is the restriction of a smooth G on $\mathbb{R}^{K+1}$ Chapter3/EulerChar · exists_contDiff_comp_val_eq (built from the nearest-point retraction and a cut-off; for N empty, $G = 0$), whose restriction has non-degenerate critical points and the numbers $C_\lambda(g)$. Then (4_λ) of §IV.20 applies. (2) and the lacunary principle follow from (4_λ) by §IV.4. □

The detour through ambient functions is needed because Part II proves Theorems 3.1 and 3.2 for restrictions $F|_N$. Milnor's appeal to Whitney (p. 36) is silent about the invariance of homology under homeomorphisms, which carries the results back to M.

Examples (V3.40).

- *Two critical points* (Reeb's situation, p. 25, with the lacunary principle). If $\dim M = n \ge 2$ and a smooth f on M has exactly two critical points, both non-degenerate, then $b_0 = b_n = 1$ and $b_i = 0$ otherwise, over every field Homology/MorseHomologyManifold · bettiNumber_of_ncard_eq_two. The two critical points are the minimum, of index 0, and the maximum, of index n (Part III, §III.4) Chapter3/Reeb · exists_min_max_of_ncard_eq_two, so $C_\lambda = [\lambda = 0] + [\lambda = n]$, and for $n \ge 2$ no two consecutive C_λ are nonzero.
- *The torus* of Milnor's first diagram (Part III, §§III.5–III.6). Its height function has $(C_0, C_1, C_2) = (1, 2, 1)$ Chapter3/Torus · Torus.critCount_height, so $b_0 \le 1$, $b_1 \le 2$, $b_2 \le 1$, $b_i = 0$ for $i \ge 3$, and $b_0 - b_1 + b_2 = 0$ Homology/MorseHomologyManifold · torus_bettiNumber, in agreement with its Euler number 0 Chapter3/Torus · isEulerNumber_torus. The Betti numbers $1, 2, 1$ themselves do not follow from these inequalities and are not derived.
- *The spheres.* $\sum_i (-1)^i b_i(S^n) = 1 + (-1)^n$ for $n \ge 1$ Homology/MorseHomologyManifold · sphere_sum_bettiNumber, from the Euler number of the sphere (Part III, §III.10) Chapter3/EulerCharSphere · isEulerNumber_sph. For $n \ge 2$ the height function, with $C_0 = C_n = 1$ and all other $C_\lambda = 0$ Chapter3/EulerCharSphere · critCount_heightFun, is lacunary, so $b_0 = b_n = 1$ and

$b_i = 0$ otherwise `Homology/MorseHomologyManifold · sphere_bettiNumber`, in agreement with the direct computation of §IV.16. For the circle the principle does not apply ($C_0 = C_1 = 1$), and Morse theory gives only $b_0 = b_1 \leq 1$ `Homology/MorseHomologyManifold · circle_bettiNumber`; §IV.16 gives $b_0 = b_1 = 1$.

IV.22 What the formalization taught us

- **The homology had to be built.** Relative singular homology, the sequences of pairs and triples, excision, Mayer–Vietoris and the homology of spheres were absent from Mathlib, and Milnor's single line on p. 29 rests on all of them `Homology/Relative · relHomology` `Homology/Excision · excision` `Homology/Spheres · cellPairIso`.
- **The algebra needs only exact sequences of triples**: no functoriality, no compatibility between triples, no excision `Chapter3/Inequalities · FiltrationHomology`. The finiteness of $H_*(M)$ follows from that of the slabs `Chapter3/Inequalities · finiteDimensional_of_steps`.
- **Small hypotheses of the algebra**: Lemma 5.1 needs $n \geq 1$ `Chapter3/Inequalities · le_sum_of_chain`; the additivity of χ on a triple needs the vanishing of one group `Chapter3/Inequalities · LongExactSeq.altSum_eq_of_finrank_eq_zero`; Corollary 5.4 needs only $R_{\lambda+1} \geq 0$ `Chapter3/Inequalities · altSum_eq_of_altSum_le`.
- **Homotopy invariance for pairs comes from the absolute theorem**, through the cylinder and the five lemma, without the naturality of Mathlib's chain homotopy `Homology/RelativeHomotopy · relHomology_map_eq_of_homotopy`.
- **Hatcher's order of fineness was replaced** by a geometric one, for which the inequality on faces is immediate `Homology/SmallChains · fineOrder_comp_faceMap_le`.
- **"By excision" is excision after a deformation**: remove the centre of the cell, push the rest radially onto the bottom set, which must be closed, and then excise `Homology/CellExcision · isIso_relHomology_map_cellPair`.
- **Several cells on one level need disjointness**, which splits the sequence of a triple; exactness alone gives only an inequality of dimensions `Homology/CellAttach · relHomologyCellsUnionInsertIso`.
- **The first equality of p. 29 is about pairs**: the top moves along strong deformation retracts that fix the bottom, the bottom only in homology `Homology/MorseHomologyBasic · subsetHomologyEquivOfIsSDRBottom` `Homology/MorseHomologyBasic · finrank_subsetHomology_of_cells`.
- **Milnor's levels need distinct critical values**; one critical value per slab, with any number of critical points on it, removes the hypothesis `Homology/MorseHomology · sublevel_step`, and the two ends of the filtration rest on the extrema being critical `Homology/MorseHomology · exists_morse_filtration`.
- $\chi(M)$ **does not depend on the field**, a consequence of formula (3) `Homology/MorseHomologyManifold · eulerNumberM_eq_sum_bettiNumber`; the passage from an abstract manifold to an embedded copy uses the invariance of homology under homeomorphisms `Homology/MorseHomologyBasic · bettiNumber_eq_of_homeomorph`.

Part V. Lefschetz, connections and curvature (Milnor §§7–9)

Milnor's §7 applies the Morse theory of the distance function $L_p = \|\cdot - p\|^2$ to complex submanifolds of $\mathbb{C}^n$: a closed complex submanifold of complex dimension k has the homotopy type of a CW-complex of dimension at most k (Theorem 7.2), a non-singular affine variety of real dimension $2k$ has no homology above dimension k (Theorem 7.1), and Lefschetz's theorems on hyperplane sections follow (Corollary 7.3, Theorem 7.4). The only new mathematics in the section is algebraic: at a critical point of L_p on a complex submanifold of complex dimension k, the Hessian has index at most k. The formal proof establishes that algebraic core in a form stronger than Milnor's: at every point of a holomorphic chart, critical or not, degenerate or not. It does so by the identity of Andreotti and Frankel, not by counting focal points. Milnor's Assertions 1 and 2 are proved beside it, read as statements about quadratic forms and about Hessians in a holomorphic chart. **The topological results of §7, Theorems 7.1, 7.2 and 7.4 and Corollary 7.3, are not formalized.** Complex submanifolds are not formalized as manifolds, Theorem 3.5 is available only for compact manifolds and produces adjunction spaces rather than CW-complexes, and cellular homology, relative homology, Lefschetz duality, relative homotopy groups and projective varieties are all missing; §V.6 lists what each statement would need.

Milnor's §§8–9 are a rapid course on connections on an abstract Riemannian manifold. Mathlib has covariant derivatives on vector bundles, their torsion and the notion of a metric connection. It has no curvature tensor, no Christoffel symbols, no covariant derivative along a curve, no parallel transport and no Levi-Civita connection. The formal proof therefore works on three levels. *At a point* (§§V.7–V.8), any trilinear map on an inner product space that satisfies (1)–(3) of Lemma 9.3 also satisfies (4), by Milnor's octahedron, and the theory of sectional curvature is developed for it: independence of the basis of the plane, polarization, determination of R, constant curvature. *On an abstract manifold* (§V.9), Milnor's notions are matched with Mathlib's: his rules for a connection, symmetric connections as torsion-free ones, compatible connections as metric ones, the uniqueness half of Lemma 8.6, and the sign of his curvature tensor. *On a submanifold N of a Euclidean space with the induced metric* (§§V.10–V.21), the whole substance of §§8–9 is proved. There the Levi-Civita connection is $X \vdash Y = P(dY(X))$, with P_q the orthogonal projection onto T_qN, and $q \mapsto P_q$ is smooth by the explicit formula $A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$. On that basis the formalization proves the second fundamental form and its symmetry, Lemma 8.6 with both halves, Lemmas 8.1–8.3, parallel transport as a linear isometry, Lemmas 8.7 and 9.2, the formula $R(X,Y)Z = \mathrm{II}(Y,\mathrm{II}(X,Z)) - \mathrm{II}(X,\mathrm{II}(Y,Z))$, the Gauss equation, Lemmas 9.1 and 9.3, and the constant curvature 1 of the round sphere. Every Riemannian manifold is isometric to such an N by Nash's embedding theorem, which the Lean files quote but neither prove nor use. The embedded results are therefore statements about embedded manifolds with the induced metric. The bridge between the second and the third level, the induced connection as a Mathlib covariant derivative on the tangent bundle of N, is not built.

The embedded route also changes the proofs. Milnor computes in local coordinates throughout, and several of his proofs are a computation in coordinates left to the reader. The formal proofs use no coordinates. Everything is reduced to the calculus of one smooth operator field $q \mapsto P_q$ and of its derivative

D_v, whose values $D_v w$ are the second fundamental form. Several steps that the book treats as evident turn out to need an argument, and the sections below point them out. The symmetry of the induced connection is the symmetry of the second fundamental form, a genuine second-order property of the embedding. Milnor's remark that the linear equations of a parallel field have solutions for all t needs a different equation, whose coefficient is skew-adjoint. The tensor property of R in its third argument is a different computation from the one in the first.

Map. Paths are relative to `Morse/`.

- `Chapter5/IndexBound`: Assertions 1 and 2, the Hessian of L_p in holomorphic coordinates, the Andreotti–Frankel identity and the index bound.
- `Chapter6/CurvatureAlgebra`: curvature tensors at a point, the octahedron, sectional curvature, polarization, constant curvature.
- `Chapter6/Connection`: Milnor's §8 in the language of Mathlib's covariant derivatives; the Koszul formula; the sign of R.
- `Chapter6/Projection`: derivatives along N, the tangent projection P_q and its smoothness, the derivative D_v of P, the chain rules for curves and surfaces.
- `Chapter6/Submanifold`: the connection $P(dY(X))$ and Milnor's rules, the second fundamental form, the bracket, Lemma 8.6, the curvature of N, Lemmas 9.1 and 9.3.
- `Chapter6/LinearODE`: uniqueness for linear equations and global existence for skew-adjoint coefficients.
- `Chapter6/Parallel`: covariant derivatives along curves, Lemmas 8.1–8.3, parallel transport.
- `Chapter6/Surface`: parametrized surfaces, Lemmas 8.7 and 9.2.
- `Chapter6/SphereCurvature`: the second fundamental form and the curvature of S^n.

V.1 Quadratic forms on a complex vector space

The setting of §7, as formalized, is linear algebra. V is a complex normed space: Milnor's $\mathbb{C}^k$, the domain of a holomorphic chart, finite-dimensional wherever dimensions are counted. V is also regarded as a real vector space, by restriction of scalars. F is a complex inner product space (Milnor's $\mathbb{C}^n$), and $w : V \to F$ is a map that is complex C^2 at a point q, hence near it: Milnor's inclusion $M \to \mathbb{C}^n$, read in holomorphic coordinates. "Bilinear form" means a continuous real-bilinear form $H : V \times V \to \mathbb{R}$.

The notions of §0.2 are used unchanged. The Hessian of $f : E \to \mathbb{R}$ at q is the second Fréchet derivative, $d^2 f_q(v, w) = d(x \mapsto df_x(w))_q(v)$, defined at every point `Chapter1/Nondegenerate · hessian`. The null space of a bilinear form H is $\{v : H(v, w) = 0 \text{ for all } w\}$ and its nullity is the dimension of the null space `Chapter1/Nondegenerate · nullSpace` `Chapter1/Nondegenerate · nullity`. The index is the negative index of inertia of $v \mapsto H(v, v)$ `Chapter1/Nondegenerate · index` `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg`. On a finite-dimensional space it is the largest dimension of a subspace on which H is negative definite `Chapter1/Nondegenerate · index_isGreatest`, and such a subspace of dimension index H exists `Chapter1/Nondegenerate · exists_negDef_finrank_eq_index`.

Two facts about the nullity are proved in the file itself. *The nullity does not change under a linear change of variables*: if L is a linear isomorphism, the form $H(L\cdot, L\cdot)$ has the same nullity as H `Chapter5/IndexBound · nullity_bilinearComp_equiv`. Indeed v is in the null space of $H(L\cdot, L\cdot)$ iff $H(Lv, Lw) = 0$ for all w, iff Lv is in the null space of H, because Lw runs through all vectors as w does. So the null space of the new form is L^{-1} of the old one, and L^{-1} preserves dimension. *The nullity of $-H$ is that of H*, the two null spaces being equal `Chapter5/IndexBound · nullity_neg`.

A *complex structure* on a real normed space E is a continuous real-linear map J with $J(Jv) = -v$. It is an isomorphism, with inverse $-J$ `Chapter5/IndexBound · complexStructureEquiv`. On a complex vector space the complex structure is multiplication by i, regarded as a real-linear map `Chapter5/IndexBound · realMulI`, with $i(iv) = -v$ `Chapter5/IndexBound · realMulI_realMulI` and injective `Chapter5/IndexBound · realMulI_injective`. A bilinear form G is *J-invariant* if $G(Jv, Jw) = G(v, w)$, and L is *J-anti-invariant* if $L(Jv, Jw) =$

$-L(v,w)$.

The two prototypes come from Milnor's p. 39. For a continuous complex-bilinear $B : V \times V \to \mathbb{C}$, the real part $\operatorname{Re} B$ is a real bilinear form `Chapter5/IndexBound · reBilin`, with $(\operatorname{Re} B)(v,w) = \operatorname{Re}(B(v,w))$ `Chapter5/IndexBound · reBilin_apply`. If $Q(z) = \sum b_{hj} z^h z^j$ with $b_{hj} = b_{jh}$ and $B(z,z') = \sum b_{hj} z^h z'^j$, then Milnor's real form $Q'(x,y)$, obtained by substituting $z = x + iy$ and taking real parts, is $\operatorname{Re} B(z,z)$. Milnor's identity $Q(iz) = -Q(z)$ becomes, polarized, $\operatorname{Re} B(iv,iw) = -\operatorname{Re} B(v,w)$, because $B(iv,iw) = i^2 B(v,w)$ `Chapter5/IndexBound · reBilin_I_smul`. So $\operatorname{Re} B$ is J-anti-invariant for $J = i\cdot$. When V is a complex inner product space it carries the real inner product $\operatorname{Re}\langle v,w\rangle$ `Chapter5/IndexBound · reInner` `Chapter5/IndexBound · reInner_apply`. This is the Euclidean inner product of $\mathbb{R}^{2k}$ when $V = \mathbb{C}^k$: it satisfies $\operatorname{Re}\langle v,v\rangle = \|v\|^2$ `Chapter5/IndexBound · reInner_self`, it is symmetric `Chapter5/IndexBound · reInner_comm`, and it is homogeneous for real scalars `Chapter5/IndexBound · re_inner_real_smul_left`. Mathlib's Hermitian product is conjugate-linear in the first slot, Milnor's $\sum_\alpha a_\alpha \bar{b}_\alpha$ in the second. Their real parts agree, and only real parts enter the proofs. Multiplication by i is orthogonal for the real inner product, since $\langle iv, iw\rangle = \bar{\imath}\, i \langle v,w\rangle = \langle v,w\rangle$ `Chapter5/IndexBound · reInner_I_smul`. So $\operatorname{Re}\langle\cdot,\cdot\rangle$ is J-invariant.

V.2 Assertion 1

Milnor's Assertion 1 (p. 39) says that if e is an eigenvalue of Q' of multiplicity μ, then so is $-e$. His proof observes that an orthogonal change of variables carries Q' into $-Q'$, and says that the assertion "clearly follows". The formal proof splits the statement in two: a version relative to an arbitrary invariant form, which is what §7 actually needs, and the matrix version Milnor states.

Assertion 1, relative form (V5.4). Let J be a complex structure on a real normed space E, G a J-invariant and L a J-anti-invariant bilinear form. Then for every real e the forms $L - eG$ and $L + eG$ have the same nullity `Chapter5/IndexBound · nullity_sub_smul_eq_nullity_add_smul'`.

The proof as formalized. Pull $L - eG$ back by J:

$$(L - eG)(Jv, Jw) = -L(v,w) - eG(v,w) = -(L + eG)(v,w).$$

So the change of variables J carries $L - eG$ into $-(L + eG)$. The nullity is unchanged by a linear change of variables `Chapter5/IndexBound · nullity_bilinearComp_equiv` and by a change of sign `Chapter5/IndexBound · nullity_neg`.

When G is positive definite, the nullity of $L - eG$ is the multiplicity of e as an eigenvalue of L *relative to* G: the dimension of the space of v with $L(v,\cdot) = eG(v,\cdot)$.

Assertion 1 (V5.3). Let V be a complex inner product space, B a continuous complex-bilinear form on V and e real. Then $\operatorname{Re} B - e \operatorname{Re}\langle\cdot,\cdot\rangle$ and $\operatorname{Re} B + e \operatorname{Re}\langle\cdot,\cdot\rangle$ have the same nullity `Chapter5/IndexBound · nullity_reBilin_sub_smul_reInner_eq`. This is the relative form with $J = i\cdot$, $G = \operatorname{Re}\langle\cdot,\cdot\rangle$, invariant by §V.1, and $L = \operatorname{Re} B$, anti-invariant by §V.1.

Milnor speaks of the eigenvalues of Q', that is, of its matrix. The formal statement takes a real-linear operator A on V with $\operatorname{Re}\langle Au, w\rangle = \operatorname{Re} B(u,w)$ for all u, w: the symmetric matrix of Q' in orthonormal real coordinates. The e-eigenspace of such an A is the null space of $\operatorname{Re} B - e \operatorname{Re}\langle\cdot,\cdot\rangle$ `Chapter5/IndexBound · eigenspace_eq_nullSpace_sub_smul_reInner`. If $Au = eu$, then $\operatorname{Re} B(u,w) - e \operatorname{Re}\langle u,w\rangle = \operatorname{Re}\langle Au - eu, w\rangle = 0$ for every w. Conversely, if u is in the null space, test against $w = Au - eu$: this gives $\|Au - eu\|^2 = 0$.

Assertion 1, matrix form (V5.3). With A as above, the e-eigenspace and the $(-e)$-eigenspace of A have the same dimension `Chapter5/IndexBound · finrank_eigenspace_neg_eq`. Both are null spaces, by the previous lemma. Since $\operatorname{Re} B - (-e) \operatorname{Re}\langle\cdot,\cdot\rangle = \operatorname{Re} B + e \operatorname{Re}\langle\cdot,\cdot\rangle$, Assertion 1 gives the equality. The multiplicity is read as the dimension of the eigenspace, which for a symmetric matrix is the usual multiplicity.

What "clearly follows" contains. The orthogonal map J with $Q'(Ju) = -Q'(u)$ anticommutes with A, so it maps the e-eigenspace injectively into the $(-e)$-eigenspace and back. The formal proof performs the

same exchange on null spaces, without mentioning commutation. *Why the relative form matters.* Assertion 1 concerns eigenvalues for the coordinate inner product of $\mathbb{R}^{2k}$. Focal points are governed by the eigenvalues of the second fundamental form $(v \cdot \ell_{ij})$ relative to the first fundamental form (g_{ij}) (Assertion 6.4 and Lemma 6.3, Part III). The two agree only if the real coordinates are orthonormal at q, which Milnor assumes in §6 (p. 34) but not on p. 40. What Assertion 2 needs is the relative form, with $G = (g_{ij})$, which is J-invariant by the Cauchy–Riemann equations.

V.3 The Hessian of L_p in holomorphic coordinates

On p. 40 Milnor takes holomorphic coordinates centred at q and a unit normal v. The Hermitian product $\sum_\alpha w_\alpha(z)\bar{v}_\alpha$ is a constant plus $Q(z)$ plus higher terms, the linear terms vanishing because v is orthogonal to M. Its real part, the real inner product $w \cdot v$, is a constant plus $Q'(x, y)$ plus higher terms, and Milnor says that the quadratic terms "clearly" determine the second fundamental form in the direction v. The formal proof replaces the remark by an explicit formula for the whole Hessian of $L_p \circ w$ in the chart.

Theorem (V5.6). Let V be a complex normed space, F a complex inner product space, $w : V \to F$ complex C^2 at q, and $p \in F$. For all real directions $\xi, \xi' \in V$,

$$d^2\big(\|w(\cdot) - p\|^2\big)_q(\xi, \xi') = 2\Big(\operatorname{Re}\langle dw_q\xi,\, dw_q\xi'\rangle + \operatorname{Re}\,\langle w(q) - p,\, d^2w_q(\xi, \xi')\rangle\Big),$$

where dw_q and d^2w_q are the complex first and second derivatives of w `Chapter5/IndexBound · hessian_normSq_sub_holomorphic`.

The proof as formalized.

1. Write $\|w(z) - p\|^2 = \operatorname{Re}\langle w(z) - p, w(z) - p\rangle$ `Chapter5/IndexBound · reInner_self`.
2. A map that is C^2 at a point, for a finite order, is C^2 on a neighbourhood. So w is complex differentiable at every z near q, and its complex derivative dw_z, restricted to real scalars, is its real derivative.
3. Near q, the real derivative of $z \mapsto \operatorname{Re}\langle w(z)-p, w(z)-p\rangle$ is $\xi \mapsto 2\operatorname{Re}\langle w(z)-p, dw_z\xi\rangle$. The rule for a bilinear form evaluated on two differentiable maps gives $\operatorname{Re}\langle dw_z\xi, w(z)-p\rangle + \operatorname{Re}\langle w(z)-p, dw_z\xi\rangle$, and the two terms are equal by symmetry of the real inner product `Chapter5/IndexBound · reInner_comm`.
4. The Hessian at q is the derivative at q of $z \mapsto df_z$. Near q this map coincides with $z \mapsto 2\operatorname{Re}\langle w(z) - p, \cdot\rangle \circ dw_z$, a product of two operator-valued maps, and the Hessian depends only on the germ. The product rule differentiates it. The factor $z \mapsto \operatorname{Re}\langle w(z) - p, \cdot\rangle$ has derivative $\xi \mapsto \operatorname{Re}\langle dw_q\xi, \cdot\rangle$ at q. The factor $z \mapsto dw_z$ has derivative $\xi \mapsto d^2w_q(\xi, \cdot)$ at q. Here the second complex derivative enters: C^2 at q makes $z \mapsto dw_z$ of class C^1 at q, hence differentiable there. Evaluating the derivative in the direction ξ at ξ' gives the formula.

The two terms behave oppositely under $J = i\cdot$. The first is unchanged by $(\xi, \xi') \mapsto (i\xi, i\xi')$, because $dw_q(i\xi) = i\,dw_q\xi$ and multiplication by i is orthogonal `Chapter5/IndexBound · reInner_I_smul`. It is the first fundamental form of w, and its J-invariance is the Cauchy–Riemann equations in disguise. The second changes sign, because d^2w_q is complex bilinear and $\langle w(q) - p, \cdot\rangle$ is complex-linear in its second slot: it is the real part of a complex bilinear form. For $p = w(q) + tv$ with v a unit normal, the second term on the diagonal is $-2t\,Q'(\xi)$, Q being the quadratic part of Milnor's $\sum_\alpha w_\alpha\bar{v}_\alpha$. This is the precise content of "the quadratic terms determine the second fundamental form".

The formula holds at every point, and it never uses the vanishing of the linear terms of Milnor's expansion. The fact behind that vanishing is nevertheless proved. **Real orthogonality to a complex subspace is Hermitian orthogonality (V5.5):** if T is a complex subspace and $\operatorname{Re}\langle a, v\rangle = 0$ for every $a \in T$, then $\langle a, v\rangle = 0$ for every $a \in T$ `Chapter5/IndexBound · inner_eq_zero_of_re_inner_eq_zero`. The real part vanishes by hypothesis. For the imaginary part, apply the hypothesis to $ia \in T$: $\langle ia, v\rangle = \bar{i}\langle a, v\rangle = -i\langle a, v\rangle$,

whose real part is $\mathrm{Im}\langle a, v\rangle$. Milnor's normal vector is orthogonal to TM_q in the real sense, while the linear coefficients are Hermitian products; since TM_q is a complex subspace, the lemma with $T = TM_q$ closes the gap. Nothing below uses it, since the formal route avoids power series.

V.4 Assertion 2

Milnor's Assertion 2 (p. 40) says that the focal points of (M, q) along a normal line occur in pairs symmetric about q, with equal multiplicities. The formal proof gives its algebraic content and its form in a holomorphic chart.

Assertion 2, algebraic content (V5.4). Let J be a complex structure, G a J-invariant and L a J-anti-invariant bilinear form. For every real t, $G - tL$ and $G + tL$ have the same nullity `Chapter5/IndexBound · nullity_sub_smul_eq_nullity_add_smul`.

Proof. $(G - tL)(Jv, Jw) = G(v, w) + tL(v, w)$, so J carries $G - tL$ into $G + tL$, and the nullity is invariant `Chapter5/IndexBound · nullity_bilinearComp_equiv`. With $G = (g_{ij})$ and $L = (v \cdot \ell_{ij})$ these two nullities are the multiplicities of $q + tv$ and $q - tv$ as focal points, by Assertion 6.4, which is formalized for real slice submanifolds `Chapter4/Focal · isFocalPointMult_add_smul_iff` (Part III).

Assertion 2 in a holomorphic chart (V5.7). Let $w : V \to F$ be complex C^2 at q, $n \in F$ and t real. The Hessians at q of $\|w(\cdot) - (w(q) + tn)\|^2$ and of $\|w(\cdot) - (w(q) - tn)\|^2$ have the same nullity `Chapter5/IndexBound · nullity_hessian_normSq_sub_add_eq_sub`.

The proof as formalized. Call the two Hessians H_+ and H_-, and let $J = i\cdot$, an isomorphism `Chapter5/IndexBound · complexStructureEquiv`. By §V.3,

$$H_+(i\xi, i\xi') = 2\Big(\mathrm{Re}\langle dw_q(i\xi), dw_q(i\xi')\rangle + \mathrm{Re}\,\big\langle w(q) - (w(q) + tn),\ d^2w_q(i\xi, i\xi')\big\rangle \Big).$$

The first term is $\mathrm{Re}\langle dw_q\xi, dw_q\xi'\rangle$, by complex linearity of dw_q and orthogonality of i. In the second, $w(q) - (w(q) + tn) = -tn$ and $d^2w_q(i\xi, i\xi') = -d^2w_q(\xi, \xi')$, so it is $\mathrm{Re}\langle tn, d^2w_q(\xi, \xi')\rangle$. That is exactly the second term of $H_-(\xi, \xi')$, since $w(q) - (w(q) - tn) = tn$. So $H_+(J\cdot, J\cdot) = H_-$, and the nullity is invariant under the change of variables J `Chapter5/IndexBound · nullity_bilinearComp_equiv`.

The formal statement assumes nothing of n, which need be neither normal nor a unit vector, nor of q, which need not be critical. For a unit normal n at $w(q)$, the two nullities are the multiplicities of $w(q) \pm tn$ as focal points, by Assertion 6.4 and the formula of p. 36 for the Hessian of L_p. That translation is not formalized, because a holomorphic chart of a complex submanifold is not identified with a chart of the real slice submanifold (§V.6). Milnor's route through the eigenvalues of Q' is not used.

V.5 The Andreotti–Frankel identity and the index bound

Milnor ends the argument on p. 41 by counting. There are at most $2k$ focal points on the normal line, symmetric about q, so at most k lie between q and p, and by Lemma 6.9 the index of L_p at q is at most k. The formal proof uses instead the argument of Andreotti and Frankel, to whom Milnor credits the proof of §7 (p. 39). It needs no focal points.

The Andreotti–Frankel identity (V5.8). Let G, L, H be bilinear forms on a real normed space E and $J : E \to E$ any map such that, for all v,

$$H(v, v) = G(v, v) - L(v, v), \qquad G(Jv, Jv) = G(v, v), \qquad L(Jv, Jv) = -L(v, v).$$

Then $H(Jv, Jv) = 2G(v, v) - H(v, v)$ `Chapter5/IndexBound · apply_J_apply_J_eq`.

Proof. $H(Jv, Jv) = G(Jv, Jv) - L(Jv, Jv) = G(v, v) + L(v, v) = 2G(v, v) - H(v, v)$. Neither linearity of J nor anything off the diagonal is used.

On a complex vector space, with $J = i\cdot$ and $L = \operatorname{Re} B$ for a complex bilinear B, the condition on L is automatic `Chapter5/IndexBound · apply_I_smul_apply_I_smul_eq`. On a complex inner product space with $G(v,v) = \|v\|^2$ it reads $Q(iv) = 2\|v\|^2 - Q(v)$ for $Q(v) = \|v\|^2 - \operatorname{Re} B(v,v)$ `Chapter5/IndexBound · apply_I_smul_apply_I_smul_eq_two_mul_norm_sq_sub`.

The index bound (V5.9). Let E be a finite-dimensional real normed space, H a bilinear form and J an injective linear map of E with $H(v,v) + H(Jv,Jv) \geq 0$ for all v. Then $2 \cdot \operatorname{index} H \leq \dim E$ `Chapter5/IndexBound · two_mul_index_le_finrank`.

The proof as formalized.

1. Choose a subspace W of dimension $\operatorname{index} H$ on which H is negative definite `Chapter1/Nondegenerate · exists_negDef_finrank_eq_index`.
2. $J(W)$ has the same dimension as W, because J is injective.
3. $W \cap J(W) = 0$. Let $v \in W \cap J(W)$, say $v = Jw$ with $w \in W$, and suppose $v \neq 0$. Then $w \neq 0$, since $J0 = 0$. So $H(w,w) < 0$, and $H(Jw,Jw) = H(v,v) < 0$ because $v \in W$. This contradicts $H(w,w) + H(Jw,Jw) \geq 0$.
4. By the dimension formula $\dim(W + JW) + \dim(W \cap JW) = \dim W + \dim JW$ `Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · Submodule.finrank_sup_add_finrank_inf_eq`, $\dim(W + JW) = 2\dim W$, which is at most $\dim E$.

On a finite-dimensional complex vector space V: if $H(v,v) + H(iv,iv) \geq 0$ for all v, then $\operatorname{index} H \leq \dim_{\mathbb{C}} V$ `Chapter5/IndexBound · index_le_finrank_complex`. Apply the theorem to $J = i\cdot$ and use $\dim_{\mathbb{R}} V = 2\dim_{\mathbb{C}} V$ `Mathlib/LinearAlgebra/Complex/FiniteDimensional · finrank_real_of_complex`.

The index bound for Milnor's shape (V5.10). Let V be a complex vector space of finite dimension k, G a bilinear form with $G(v,v) \geq 0$ and $G(iv,iv) = G(v,v)$, and B a continuous complex-bilinear form. If $H(v,v) = G(v,v) - \operatorname{Re} B(v,v)$ for all v, then $\operatorname{index} H \leq k$ `Chapter5/IndexBound · index_le_finrank_of_eq_sub_re`.

Proof. The identity, with $L = \operatorname{Re} B$, whose anti-invariance on the diagonal is §V.1, gives $H(v,v) + H(iv,iv) = 2G(v,v) \geq 0$; then apply the bound on a complex space. Two cases are recorded: every form whose quadratic form is $\|v\|^2 - \operatorname{Re} B(v,v)$ has index at most k `Chapter5/IndexBound · index_le_finrank_of_hermitian`; and so does the Hessian at q of any $f : V \to \mathbb{R}$ whose Hessian has, on the diagonal, the shape $G - \operatorname{Re} B$ `Chapter5/IndexBound · index_hessian_le_finrank_complex`. Neither criticality, nor non-degeneracy, nor any smoothness of f is assumed: only the shape of the Hessian matters.

Milnor's index bound in holomorphic coordinates (V5.11). Let V be a complex vector space of finite dimension k, F a complex inner product space, $w : V \to F$ complex C^2 at q, and $p \in F$. Then the Hessian of $z \mapsto \|w(z) - p\|^2$ at q has index at most k `Chapter5/IndexBound · index_hessian_normSq_sub_holomorphic_le`.

The proof as formalized. Let H be the Hessian. By §V.3 at (ξ,ξ) and at $(i\xi,i\xi)$,

$$H(\xi,\xi) = 2(\|dw_q\xi\|^2 + \operatorname{Re}\langle w(q) - p, d^2w_q(\xi,\xi)\rangle), \qquad H(i\xi,i\xi) = 2(\|dw_q\xi\|^2 - \operatorname{Re}\langle w(q) - p, d^2w_q(\xi,\xi)\rangle).$$

The first terms agree because $dw_q(i\xi) = i\,dw_q\xi$ and i is orthogonal; the second terms are opposite because $d^2w_q(i\xi,i\xi) = i^2d^2w_q(\xi,\xi)$. So $H(\xi,\xi) + H(i\xi,i\xi) = 4\|dw_q\xi\|^2 \geq 0$, and the bound on a complex space applies `Chapter5/IndexBound · index_le_finrank_complex`.

This is the formal counterpart of Milnor's "the index of L_p at q is at most k", and it is stronger: q need not be a critical point, the Hessian need not be non-degenerate, and dw_q need not be injective. When w parametrizes a complex submanifold M near a critical point $w(q)$ of $L_p|M$, this Hessian is the Hessian of $L_p|M$ in the real coordinates (x,y). Its index is then the index of L_p at $w(q)$, by the chart independence of the index in a parametrization `Chapter1/ManifoldSubmanifold · indexM_normSq_sub_eq` (Part I). Making that identification formal needs M as a real slice submanifold with w^{-1} among its charts, which is not done (§V.6). Milnor's count also needs facts he does not state: that neither q nor p is focal (p is not, by Lemma 6.5, since q is non-degenerate), and the case $p = q$. The formal route needs none of them.

No negative semidefinite Hessian (V5.12). If in V5.10 the form G is moreover positive definite and $V \neq 0$, then $H(v,v) > 0$ for some v `Chapter5/IndexBound · exists_pos_apply_of_eq_sub_re`. Take $v \neq 0$. If $H(v,v) \leq 0$, then $H(iv,iv) = 2G(v,v) - H(v,v) > 0$. With $G(\xi,\xi) = 2\|dw_q\xi\|^2$ (§V.3), positive definite where dw_q is injective, this says that the Hessian of L_p in a holomorphic chart of positive dimension is never negative semidefinite. The consequence drawn in the exposition, that L_p has no local maximum on a complex submanifold of positive dimension, is not formalized.

V.6 What is not formalized in §7

None of the four topological statements of §7 is formalized. Each needs something that neither this project nor Mathlib provides.

- **Theorem 7.2** (p. 39), the homotopy type of a closed complex submanifold, is missing three things. (i) *Complex submanifolds as manifolds.* Holomorphic charts would have to be charts of a real slice submanifold of $\mathbb{R}^{2n}$. This is the bridge that would carry V5.7 and V5.11 to focal points and to the index of $L_p : M \to \mathbb{R}$, which the project computes in a real parametrization `Chapter1/ManifoldSubmanifold · indexM_normSq_sub_eq` and relates to focal points by Lemma 6.9 `Chapter4/Curvature · indexM_normSq_sub_eq_finsum_focalMult`. (ii) *Theorem 3.5 for non-compact manifolds.* The formal Theorem 3.5 assumes the submanifold compact `Chapter2/CWType · theorem_3_5` (Part II). A closed complex submanifold of $\mathbb{C}^n$ of positive dimension is never compact, and the non-compact case needs the Appendix and Whitehead's theorem, which Mathlib has only in an abstract model-category form. (iii) *CW-complexes.* Even for compact manifolds the formal Theorem 3.5 produces an iterated adjunction space, not a CW-complex, for lack of cellular approximation. The remaining ingredient is available: for almost every p, L_p is a Morse function with compact sublevel sets on a closed slice submanifold `Chapter4/Existence · ae_isMorseFunction_normSq_sub_and_isCompact_sublevel` (Theorem 6.6, Part III).
- **Theorem 7.1** (p. 39) needs Theorem 7.2 and the vanishing of the homology of a CW-complex above its dimension. Cellular homology is not available.
- **Corollary 7.3** (p. 41) needs relative singular homology, the exact sequence of a pair and Lefschetz duality.
- **Theorem 7.4** (pp. 41–42) needs relative homotopy groups and a deformation of a neighbourhood of $V \cap P$ into $V \cap P$, which Milnor obtains from the triangulability of algebraic varieties (quoted). It also needs Theorems 3.1–3.2 for $1/L_p$ on V in $\mathbb{C}P_n$. Algebraic varieties and complex projective space are not formalized here.

The statement of Theorem 7.2 also says more than its proof gives: the proof yields a CW-complex of dimension *at most* k ($\mathbb{C}$ is contractible). Assertion 2 as a statement about focal points, and Milnor's count on p. 41, are replaced by V5.7–V5.11.

V.7 Curvature tensors at a point and Milnor's octahedron

By Lemma 9.1, the curvature of a connection at a point p is a trilinear map $TM_p^3 \to TM_p$. The fourth identity of Lemma 9.3 and the whole theory of sectional curvature are therefore linear algebra on the inner product space TM_p. The formal text develops them once, for any real inner product space E and any map $R : E \times E \times E \to E$ that is linear in each argument, $R(X,Y)Z$ standing for Milnor's notation. For submanifolds, Lemma 9.1 itself is proved in §V.16, which then applies everything here.

Definition. R is a *curvature tensor* if it satisfies (1)–(3) of Lemma 9.3 (p. 53): (1) $R(X,Y)Z + R(Y,X)Z = 0$; (2) $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$; (3) $\langle R(X,Y)Z, W\rangle + \langle R(X,Y)W, Z\rangle = 0$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor`.

At this level the three identities are hypotheses. Milnor proves (2) and (3) on a manifold by extending vectors to fields with vanishing brackets, which is differential geometry; for sub-

manifolds the formal proofs of (2) and (3) are in §V.16. The definition has immediate consequences: $R(Y,X) = -R(X,Y)$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.apply_swap`, and $R(X,X) = 0$, because (1) gives $2R(X,X)Z = 0$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.apply_self`. Also $\langle R(X,Y)W,Z\rangle = -\langle R(X,Y)Z,W\rangle$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_swap`, $\langle R(X,Y)Z,Z\rangle = 0$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_apply_self`, and (2) paired with W, $\langle R(X,Y)Z,W\rangle + \langle R(Y,Z)X,W\rangle + \langle R(Z,X)Y,W\rangle = 0$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_bianchi`.

Milnor calls (3) "clearly equivalent" to $\langle R(X,Y)Z,Z\rangle = 0$ (p. 53). The formal statement: for a linear map T, the form $\langle TZ,W\rangle$ is skew-symmetric iff $\langle TZ,Z\rangle = 0$ for every Z `Chapter6/CurvatureAlgebra · inner_add_inner_eq_zero_iff`. Taking $W = Z$ gives $2\langle TZ,Z\rangle = 0$. Conversely, expand $\langle T(Z+W),Z+W\rangle = 0$ by linearity of T; the terms $\langle TZ,Z\rangle$ and $\langle TW,W\rangle$ vanish and what remains is $\langle TZ,W\rangle + \langle TW,Z\rangle = 0$. The equivalence uses linearity of R in Z.

Milnor's octahedron (V6.21). Let α be any set and $T : \alpha^4 \to \mathbb{R}$ any function that is skew-symmetric in its first two arguments, skew-symmetric in its last two, and satisfies the cyclic identity $T(X,Y,Z,W) + T(Y,Z,X,W) + T(Z,X,Y,W) = 0$. Then $T(X,Y,Z,W) = T(Z,W,X,Y)$ `Chapter6/CurvatureAlgebra · octahedron`.

The proof as formalized is Milnor's (p. 54), with the signs he leaves to the reader written out. Write $[XYZW]$ for $T(X,Y,Z,W)$ and name the six vertices of the octahedron

$$T = [XYZW], \quad B = [ZWXY], \quad L = [XWYZ], \quad K = [YWZX], \quad Q = [YZXW], \quad F = [ZXYW].$$

Each shaded face is an instance of the cyclic identity, brought into shape by the two skew-symmetries.

- *Face W:* the cyclic identity for X,Y,Z with last argument W is $T + Q + F = 0$ as it stands.
- *Face Z:* the cyclic identity for X,Y,W with last argument Z reads $[XYWZ] + [YWXZ] + [WXYZ] = 0$. Skew-symmetry in the last pair gives $[XYWZ] = -T$ and $[YWXZ] = -K$; skew-symmetry in the first pair gives $[WXYZ] = -L$. So $T + K + L = 0$.
- *Face X:* the cyclic identity for Y,Z,W with last argument X reads $[YZWX] + [ZWYX] + [WYZX] = 0$, where $[YZWX] = -Q$ and $[ZWYX] = -B$ by the last pair and $[WYZX] = -K$ by the first. So $Q + B + K = 0$.
- *Face Y:* the cyclic identity for Z,W,X with last argument Y reads $[ZWXY] + [WXZY] + [XZWY] = 0$. Here $[ZWXY] = B$; $[WXZY] = -[XWZY] = [XWYZ] = L$, using both skew-symmetries; and likewise $[XZWY] = -[ZXWY] = [ZXYW] = F$. So $B + L + F = 0$.

The two faces at the top vertex T are added and the two at the bottom vertex B subtracted:

$$(T+Q+F) + (T+K+L) - (Q+B+K) - (B+L+F) = 2T - 2B = 0.$$

No linearity of T is used, nor any structure on α; the formal proof is the list of the fourteen instances above, four of the cyclic identity and ten of the skew-symmetries, followed by linear arithmetic.

Lemma 9.3 (4) (V6.21). Every curvature tensor satisfies $\langle R(X,Y)Z,W\rangle = \langle R(Z,W)X,Y\rangle$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_comm_pairs`. Apply the octahedron to $T(X,Y,Z,W) = \langle R(X,Y)Z,W\rangle$. Skew-symmetry in the first pair is (1) paired with W. Skew-symmetry in the last pair is (3). The cyclic identity is (2) paired with W.

V.8 Sectional curvature, polarization and constant curvature

Milnor defines the sectional curvature only on p. 100, in a remark after Lemma 19.1. For orthonormal $A, B \in TM_p$, $\langle R(A,B)A,B\rangle$ is the sectional curvature "determined by A and B". He does not say that it depends only on the plane of A and B, nor that these numbers determine R. Both are proved here, as Lemmas 7.1 and 7.2 of the exposition's Chapter 6.

Definition. For vectors X, Y,

$$K(X,Y) = \frac{\langle R(X,Y)X, Y\rangle}{\|X\|^2\|Y\|^2 - \langle X,Y\rangle^2}$$

`Chapter6/CurvatureAlgebra · sectionalCurvature`. For an orthonormal pair this is Milnor's $\langle R(A,B)A, B\rangle$. For a dependent pair the denominator vanishes, and the formal convention for division makes $K = 0$.

Lemma (V6.22). If R is a curvature tensor and $ad - bc \neq 0$, then $K(aX + bY, cX + dY) = K(X,Y)$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.sectionalCurvature_smul_add`.

The proof as formalized.

1. The Gram determinant of $(aX + bY, cX + dY)$ is $(ad - bc)^2$ times that of (X, Y), by expanding the inner products `Chapter6/CurvatureAlgebra · gram_smul_add`.
2. $R(aX + bY, cX + dY) = (ad - bc)R(X,Y)$, by bilinearity, $R(X,X) = R(Y,Y) = 0$ and $R(Y,X) = -R(X,Y)$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.apply_smul_add`.
3. Hence the numerator is $(ad-bc)\langle R(X,Y)(aX+bY), cX+dY\rangle$. Of the four terms of the expansion, $\langle R(X,Y)X, X\rangle$ and $\langle R(X,Y)Y, Y\rangle$ vanish, and $\langle R(X,Y)Y, X\rangle = -\langle R(X,Y)X, Y\rangle$, by (3). What remains is $(ad - bc)^2\langle R(X,Y)X, Y\rangle$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_apply_smul_add` `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_apply_self_eq`.
4. Numerator and denominator are both multiplied by $(ad - bc)^2 \neq 0$.

The Gram determinant of an independent pair is positive `Chapter6/CurvatureAlgebra · gram_pos`. $Y \neq 0$, and $u = \langle Y,Y\rangle X - \langle X,Y\rangle Y$ is not 0 by independence; then $\langle u, u\rangle = \langle Y,Y\rangle\,(\|X\|^2\|Y\|^2 - \langle X,Y\rangle^2)$ is positive, and so is $\langle Y,Y\rangle$. For a dependent pair the numerator vanishes as well, because $R(X,Y) = 0$ when one vector is a multiple of the other `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_apply_self_eq_zero_of_not_linearIndependent`.

Polarization. The *second mixed difference* of a function k of two vectors is

$$\Delta_{A,C}\Delta_{B,D}k = k(A{+}C, B{+}D){-}k(A, B{+}D){-}k(C, B{+}D){-}k(A{+}C, B){+}k(A, B){+}k(C, B){-}k(A{+}C, D){+}k(A, D){+}k$$

`Chapter6/CurvatureAlgebra · mixedDiff`. For $k(A,B) = \langle R(A,B)A, B\rangle$, with R a curvature tensor,

$$\Delta_{A,C}\Delta_{B,D}k = 2(\langle R(A,B)C, D\rangle + \langle R(A,D)C, B\rangle)$$

`Chapter6/CurvatureAlgebra · IsCurvatureTensor.mixedDiff_eq`. The function k is quadratic in its first argument, which fills the first and third slots, and in its second, which fills the second and fourth. Of the sixteen terms of $k(A + C, B + D)$, the mixed difference keeps the four in which A and C both occur among the first and third slots and B and D both among the second and fourth: $\langle R(A,B)C, D\rangle + \langle R(A,D)C, B\rangle + \langle R(C,B)A, D\rangle + \langle R(C,D)A, B\rangle$. Pair symmetry (Lemma 9.3 (4)) identifies the last two with the first two.

Polarization (V6.23). For a curvature tensor,

$$6\langle R(X,Y)Z, W\rangle = 2\,\Delta_{X,Z}\Delta_{Y,W}k + \Delta_{Y,X}\Delta_{Z,W}k$$

`Chapter6/CurvatureAlgebra · IsCurvatureTensor.polarization`.

Proof. Write $T = \langle R(X,Y)Z, W\rangle$, $Q = \langle R(Y,Z)X, W\rangle$, $F = \langle R(Z,X)Y, W\rangle$. By the previous lemma the right side is $4(T + \langle R(X,W)Z, Y\rangle) + 2(Q + \langle R(Y,W)X, Z\rangle)$. Pair symmetry and (1) give $\langle R(X,W)Z, Y\rangle = \langle R(Z,Y)X, W\rangle = -Q$ and $\langle R(Y,W)X, Z\rangle = \langle R(X,Z)Y, W\rangle = -F$. So the right side is $4T - 2Q - 2F$, which is $6T$ by (2) in the form $T + Q + F = 0$.

The sectional curvatures determine R (V6.23). Two curvature tensors with the same $\langle R(X,Y)X, Y\rangle$ for all X, Y are equal `Chapter6/CurvatureAlgebra · IsCurvatureTensor.ext`. By polarization the two have the same $\langle R(X,Y)Z, W\rangle$ for every W, hence the same $R(X,Y)Z$ `Mathlib/Analysis/InnerProductSpace/Basic · ext_`

`inner_right`. The same holds if they have the same sectional curvature on all independent pairs `Chapter6/CurvatureAlgebra · IsCurvatureTensor.ext_of_sectionalCurvature`: on an independent pair the Gram determinant is positive, so the numerators agree, and on a dependent pair both numerators are 0. Finally, it holds for Milnor's own definition: if $\langle R(A,B)A,B\rangle = \langle R'(A,B)A,B\rangle$ for all orthonormal A, B, then $R = R'$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.ext_of_orthonormal`. This rests on Gram–Schmidt for the sectional numerator `Chapter6/CurvatureAlgebra · inner_apply_self_eq_zero_or_exists_orthonormal`. If $X = 0$, or Y is a multiple of X, then $\langle R(X,Y)X,Y\rangle = 0$ for every curvature tensor. Otherwise put $A = X/\|X\|$, $Y' = Y - \langle A,Y\rangle A$ and $B = Y'/\|Y'\|$. These are orthonormal, $X = \|X\|A$ and $Y = \langle A,Y\rangle A + \|Y'\|B$, so $\langle R(X,Y)X,Y\rangle = (\|X\|\,\|Y'\|)^2\langle R(A,B)A,B\rangle$ for every curvature tensor at once.

Constant curvature (V6.24). Put $R_1(X,Y)Z = \langle X,Z\rangle Y - \langle Y,Z\rangle X$ `Chapter6/CurvatureAlgebra · curvatureOne` `Chapter6/CurvatureAlgebra · curvatureOne_apply`. It is a curvature tensor `Chapter6/CurvatureAlgebra · isCurvatureTensor_curvatureOne`. (1) is visible. In (2) the six terms cancel in pairs by symmetry of the inner product. (3) is $\langle X,Z\rangle\langle Y,W\rangle - \langle Y,Z\rangle\langle X,W\rangle + \langle X,W\rangle\langle Y,Z\rangle - \langle Y,W\rangle\langle X,Z\rangle = 0$. Its sectional numerator is $\langle R_1(X,Y)X,Y\rangle = \|X\|^2\|Y\|^2 - \langle X,Y\rangle^2$ `Chapter6/CurvatureAlgebra · inner_curvatureOne_self`, so κR_1, again a curvature tensor `Chapter6/CurvatureAlgebra · IsCurvatureTensor.smul`, has sectional curvature κ on every independent pair `Chapter6/CurvatureAlgebra · sectionalCurvature_smul_curvatureOne`. Conversely, let R be a curvature tensor. If $\langle R(A,B)A,B\rangle = \kappa$ for every orthonormal pair, then $R = \kappa R_1$ `Chapter6/CurvatureAlgebra · IsCurvatureTensor.eq_smul_curvatureOne_of_orthonormal`: compare with κR_1, whose value on an orthonormal pair is $\kappa(1\cdot 1 - 0^2) = \kappa$. If $K(X,Y) = \kappa$ for every independent pair, then

$$R(X,Y)Z = \kappa\big(\langle X,Z\rangle Y - \langle Y,Z\rangle X\big)$$

`Chapter6/CurvatureAlgebra · IsCurvatureTensor.apply_eq_of_sectionalCurvature_eq`.

V.9 Milnor's §8 in Mathlib's language, and the sign of R

On an abstract manifold the formal text matches Milnor's definitions with Mathlib's covariant derivatives. Let M be a smooth manifold in Mathlib's sense, modelled on a real normed space E. A Mathlib *covariant derivative* on the tangent bundle is a function ∇ that assigns to every vector field Y and every point x a continuous linear map $\nabla Y(x) : T_xM \to T_xM$ `Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Basic · CovariantDerivative`. It must be additive in Y and satisfy the Leibniz rule at every point where the sections involved are differentiable `Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Basic · IsCovariantDerivativeOn`. Milnor's $X_p \vdash Y$ (p. 43) is the value of $\nabla Y(p)$ at X_p; Mathlib's argument order is the reverse of the usual one.

Milnor's rules (p. 44; V6.3). Since $\nabla Y(x)$ is a linear map on T_xM, three things are built into the definition: dependence on X_p only, linearity in X_p, and rule (3), $(fX) \vdash Y = f\,(X \vdash Y)$ `Chapter6/Connection · cov_smul_left`. Rule (2), additivity in Y, holds at every point where both summands are differentiable `Chapter6/Connection · cov_add_right`. Rule (4), $X \vdash (fY) = (Xf)\,Y + f\,(X \vdash Y)$, holds at every point where f and Y are differentiable `Chapter6/Connection · cov_smul_right`. Here Xf is the derivative of f at x, read as a linear form on T_xM, applied to X_x `Mathlib/Geometry/Manifold/MFDeriv/NormedSpace · mvfderiv`. Rule (1), smoothness of $X \vdash Y$ for smooth X and Y, is a separate class `Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Basic · ContMDiffCovariantDerivative`. The hidden hypothesis is differentiability at the point. A Mathlib covariant derivative is defined on all sections and is unconstrained on sections that are not differentiable at x. Christoffel symbols, the covariant derivative along a curve and parallel transport do not exist in Mathlib, so Lemmas 8.1–8.3 are not stated at this level.

Symmetric connections (Definition 8.5). ∇ is *symmetric* if $X \vdash Y - Y \vdash X = [X,Y]$ at every point where X and Y are differentiable `Chapter6/Connection · IsSymmetric`. Here $[X,Y]$ is Mathlib's Lie bracket of vector fields on a manifold `Mathlib/Geometry/Manifold/VectorField/LieBracket · VectorField.mlieBracket`, which has Milnor's sign, $[X,Y]f = X(Yf) - Y(Xf)$. Mathlib's torsion is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$ `Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Torsion · CovariantDerivative.torsion`. For a

finite-dimensional model space, Milnor's symmetric connections are exactly the torsion-free ones `Chapter6/Connection · isSymmetric_iff_torsion_eq_zero`, by Mathlib's characterization of vanishing torsion `Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Torsion · CovariantDerivative.torsion_eq_zero_iff`.

Compatible connections (Corollary 8.4). Milnor defines compatibility by "parallel translation preserves inner products" (p. 47). That cannot be stated without parallel transport. Its consequence, Corollary 8.4 (p. 48), is taken as the definition instead. With a Riemannian metric on the tangent bundle `Mathlib/Topology/VectorBundle/Riemannian · RiemannianBundle`, ∇ is *compatible* if

$$X_x\langle Y, Y'\rangle = \langle X_x \vdash Y, Y'_x\rangle + \langle Y_x, X_x \vdash Y'\rangle$$

at every point x, for all vector fields differentiable there `Chapter6/Connection · IsCompatible`. For a finite-dimensional model space and a C^1 metric `Mathlib/Geometry/Manifold/VectorBundle/Riemannian · IsContMDiffRiemannianBundle` this is Mathlib's notion of a metric connection, `IsMetricCompatible` `Chapter6/Connection · isCompatible_iff_isMetricCompatible`.

The Koszul formula. Put

$$\kappa(X,Y,Z) = X\langle Y,Z\rangle + Y\langle X,Z\rangle - Z\langle X,Y\rangle + \langle [X,Y],Z\rangle - \langle [X,Z],Y\rangle - \langle [Y,Z],X\rangle$$

at x `Chapter6/Connection · koszul`. It involves only the metric and the bracket. For a symmetric compatible connection and X, Y, Z differentiable at x, $2\langle X \vdash Y, Z\rangle = \kappa(X,Y,Z)$ `Chapter6/Connection · two_mul_inner_cov_eq_koszul`. *Proof.* Rewrite the three derivative terms by compatibility and the three brackets by symmetry. Twelve inner products result. Ten cancel in pairs, and two copies of $\langle X \vdash Y, Z\rangle$ remain. For coordinate fields, whose brackets vanish, this is Milnor's first Christoffel identity (p. 49), without his linear system. That system has a misprint: of the three unknowns as printed, the second equals the first by the symmetry he quotes; they should be $\langle \partial_i \vdash \partial_j, \partial_k\rangle$, $\langle \partial_i \vdash \partial_k, \partial_j\rangle$ and $\langle \partial_j \vdash \partial_k, \partial_i\rangle$.

Lemma 8.6, uniqueness, on an abstract manifold (V6.14). Two symmetric connections compatible with the metric agree on every vector field Y differentiable at x: $\nabla Y(x) = \nabla' Y(x)$ `Chapter6/Connection · cov_eq_of_isSymmetric_of_isCompatible`.

The proof as formalized. Fix $X_0 \in T_xM$ and put $Z_0 = \nabla Y(x)X_0 - \nabla' Y(x)X_0$. Mathlib extends any vector at x to a section that is constant in the trivialization of the tangent bundle at x `Mathlib/Topology/FiberBundle/Basic · FiberBundle.extend`. That section is differentiable at x `Mathlib/Geometry/Manifold/VectorBundle/MDifferentiable · mdifferentiableAt_extend` and takes the given value there `Mathlib/Topology/FiberBundle/Basic · FiberBundle.extend_apply_self`. Extend X_0 and Z_0 to fields X and Z in this way. The Koszul formula for both connections gives $2\langle \nabla Y(x)X_0, Z_0\rangle = \kappa(X,Y,Z)(x) = 2\langle \nabla' Y(x)X_0, Z_0\rangle$, the middle term not involving the connection. So $\langle Z_0, Z_0\rangle = 0$. With Mathlib's predicates: two torsion-free metric connections agree on fields differentiable at the point `Chapter6/Connection · cov_eq_of_torsion_eq_zero_of_isMetricCompatible`.

Uniqueness holds only on sections differentiable at the point: as functions on all sections, two Levi-Civita connections need not agree, and Milnor never uses them elsewhere. *Existence* is not proved at this level. Mathlib has no Levi-Civita connection, and Milnor's "one can verify" (p. 49) hides two things: a converse of Lemma 8.3, which he does not state, and the patching of the local connections defined by the Christoffel symbols. In the embedded setting both halves are proved (§V.15).

The curvature tensor and its sign (V6.16). Milnor's curvature of vector fields (p. 51) is

$$R(X,Y)Z = -X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X,Y] \vdash Z$$

`Chapter6/Connection · milnorCurvature`. Nomizu's, which most modern texts use, is $\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$ `Chapter6/Connection · nomizuCurvature`. Milnor's footnote on p. 51, that Nomizu gives R the opposite sign, is the identity $R_{\text{Milnor}} = -R_{\text{Nomizu}}$, proved by rearranging the terms `Chapter6/Connection · milnorCurvature_eq_neg`. Lemma 9.3 (1) holds for vector fields with no hypothesis, $R(Y,X)Z = -R(X,Y)Z$ `Chapter6/Connection · milnorCurvature_swap`, because $[Y,X] = -[X,Y]$ `Mathlib/`

Geometry/Manifold/VectorField/LieBracket · VectorField.mlieBracket_swap. The tensor property (Lemma 9.1) and (2)–(4) are not proved at this level.

Mathlib defines no curvature, so there is no Mathlib sign to reconcile. Three statements record how the convention enters. Identities (1)–(3) are invariant under $R \mapsto -R$ Chapter6/CurvatureAlgebra · IsCurvatureTensor.neg. The sectional curvature is the same number in both conventions: Milnor writes $\langle R(X,Y)X, Y\rangle$, the standard convention writes $\langle R^{\text{std}}(X,Y)Y, X\rangle$, and by (3) the two are equal when $R^{\text{std}} = -R$ Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_neg_apply_swap. Constant curvature κ reads $R(X,Y)Z = \kappa(\langle X,Z\rangle Y - \langle Y,Z\rangle X)$ with Milnor's sign and $\kappa(\langle Y,Z\rangle X - \langle X,Z\rangle Y)$ with the standard one Chapter6/CurvatureAlgebra · neg_smul_curvatureOne_apply. The Gauss equation of §V.16 confirms that Milnor's convention gives the unit sphere curvature $+1$ (§V.21).

V.10 Submanifolds of a Euclidean space: derivatives along N

From here on, V is a finite-dimensional real inner product space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ a slice submanifold in the sense of the companion project (§0.1), modelled on Z with complementary space W Milnor/Chapter2/Submanifold · IsSliceSubmanifold Milnor/Chapter2/Submanifold · SliceChart. N carries the induced metric: tangent vectors at q are vectors of V, with the inner product of V. Z is finite-dimensional throughout, and W wherever the symmetry of the second fundamental form is used; when N is nonempty both hold automatically, since $Z \times W$ is isomorphic to V. No closedness or compactness of N is assumed anywhere. A map $F : N \to E$ into a normed space is smooth, or differentiable at q, as a map of manifolds.

Tangent spaces and parametrizations (§0.3). $T_qN \subseteq V$ is the range of the derivative at q of the inclusion $\iota : N \to V$ Chapter1/ManifoldSubmanifold · tangentSubspace Milnor/Chapter5/Orient · mfd. A slice chart σ has the parametrization $x(u) = \sigma^{-1}(u, 0)$, Milnor's $u \mapsto x(u)$ Chapter1/ManifoldSubmanifold · sliceParam, smooth on its domain Chapter1/ManifoldSubmanifold · contDiffOn_sliceParam. For any slice chart σ around q, T_qN is the range of dx_u at $u = (\sigma q)_1$ Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam.

Chart coordinates of an ambient vector. For each q the formal text uses the slice chart chosen at q Chapter6/Projection · sliceAt Chapter6/Projection · mem_sliceAt_source, whose induced chart is the preferred chart of the manifold N at q Chapter6/Projection · chartAt_eq_sliceAt. With σ this chart, the chart coordinates of $v \in V$ are $\tau_q(v) = (d\sigma_q v)_1 \in Z$ Chapter6/Projection · tanCoord. They invert the derivative of the inclusion: $\tau_q(d\iota_q z) = z$ for $z \in Z$ Chapter6/Projection · tanCoord_mfderiv_val, because the preferred chart of N at q is $x \mapsto (\sigma x)_1$ restricted to N, whose derivative at q is $\tau_q \circ d\iota_q$ by the chain rule, while the derivative of a chart at its own base point, read in that chart, is the identity. Hence $d\iota_q(\tau_q v) = v$ for $v \in T_qN$ Chapter6/Projection · mfderiv_val_tanCoord.

Definition (V6.1). For $F : N \to E$ the linear map $d^N F_q = dF_q \circ \tau_q : V \to E$ is *the derivative of F at q on ambient vectors* Chapter6/Projection · dN Chapter6/Projection · dN_apply. On $v \in T_qN$ it is the manifold derivative of F applied to the chart coordinates of v; we write it $dF_q(v)$. Off T_qN its values depend on the chart, and no statement uses them. It depends only on the germ of F Chapter6/Projection · dN_congr and vanishes on constants Chapter6/Projection · dN_const.

The bridge to ambient calculus is the chain rule. If $G : V \to E$ is differentiable at q, then $d(G|_N)_q(v) = dG_q(v)$ for $v \in T_qN$ Chapter6/Projection · dN_comp_val, by the manifold chain rule and $d\iota_q(\tau_q v) = v$. Hence if $F = G$ on N near q, then $dF_q(v) = dG_q(v)$ for tangent v Chapter6/Projection · dN_eq_fderiv. In particular, an ambient map that vanishes on N near q has zero derivative at q in tangent directions Chapter6/Projection · fderiv_eq_zero_of_eventually. Every F differentiable at q has such a representative: with σ the slice chart chosen at q, put $G(y) = F(\text{the point of } N \text{ with chart coordinate } (\sigma y)_1)$ Chapter6/Projection · exists_differentiableAt_rep. It is differentiable at q, as F read in the chart composed with $y \mapsto (\sigma y)_1$, and it equals F on N near q, because the chart is inverted by its inverse. If F is smooth, the same formula is smooth on the open set of y in the domain of σ with $(\sigma y)_1$ in the image of the chart Chapter6/Projection · exists_contDiffOn_rep.

All the calculus of d^N is reduced to ambient calculus in this way. Sums and differences behave as expected Chapter6/Projection · dN_add Chapter6/Projection · dN_neg Chapter6/Projection · dN_sub. Continuous linear maps commute with d^N, on all of V: $d(L \circ F) = L \circ dF$ Chapter6/Projection · dN_clm_comp. The key rule is the Leibniz rule for a continuous bilinear map β:

$$d\big(\beta(F_1, F_2)\big)_q(v) = \beta(dF_{1,q}(v), F_2(q)) + \beta(F_1(q), dF_{2,q}(v)) \qquad (v \in T_qN)$$

Chapter6/Projection · dN_bilinear. Choose representatives G_1, G_2; then $\beta(G_1, G_2)$ represents $\beta(F_1, F_2)$, the ambient product rule differentiates it, and the chain rule translates back. It specializes to the rules for a function times a vector Chapter6/Projection · dN_smul, for inner products Chapter6/Projection · dN_inner, for an operator-valued map applied to a vector-valued one Chapter6/Projection · dN_clm_apply, and for a composition of operator-valued maps Chapter6/Projection · dN_clm_comp'. The derivative of the inclusion is the identity on T_qN Chapter6/Projection · dN_val.

V.11 The tangent projection $P_q = A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$

Let P_q be the orthogonal projection of V onto T_qN Chapter6/Projection · tanProj Mathlib/Analysis/InnerProductSpace/Projection/Basic · Submodule.starProjection. It takes values in T_qN Chapter6/Projection · tanProj_apply_mem, fixes exactly the tangent vectors Chapter6/Projection · tanProj_eq_self Chapter6/Projection · tanProj_eq_self_iff, is idempotent Chapter6/Projection · tanProj_tanProj Chapter6/Projection · tanProj_comp_tanProj and self-adjoint Chapter6/Projection · inner_tanProj_left Chapter6/Projection · inner_tanProj_of_mem. It kills exactly the normal space Chapter6/Projection · tanProj_eq_zero_iff, $v - P_qv$ is normal Chapter6/Projection · sub_tanProj_mem_orthogonal Chapter6/Projection · tanProj_sub_tanProj, and a vector both tangent and normal is 0 Chapter6/Projection · eq_zero_of_mem_of_mem_orthogonal.

In the embedded setting every smoothness question of §8 reduces to one fact: $q \mapsto P_q$ is smooth. Milnor checks the smoothness of what he builds chart by chart (condition (1), p. 44) and never needs this statement. It is not automatic, because the projection onto the range of an operator A is a smooth function of A only where the rank of A does not drop.

The Gram formula. Let Y be a finite-dimensional normed space and $A : Y \to V$ linear. Its transpose is $A^\mathsf{T} : V \to Y^*$, $v \mapsto \langle v, A\cdot\rangle$ Chapter6/Projection · transposeL Chapter6/Projection · transposeL_apply, and its Gram operator is $A^\mathsf{T}A : Y \to Y^*$, $y \mapsto \langle Ay, A\cdot\rangle$ Chapter6/Projection · gramL Chapter6/Projection · gramL_apply. If A is injective, so is $A^\mathsf{T}A$ Chapter6/Projection · gramL_injective: $A^\mathsf{T}A(y - y') = 0$ gives $\|A(y - y')\|^2 = 0$. As $\dim Y^* = \dim Y$ Chapter6/Projection · finrank_clmDual_eq, it is then bijective Chapter6/Projection · gramL_bijective, an isomorphism Chapter6/Projection · gramEquiv Chapter6/Projection · coe_gramEquiv. The formal inverse of a linear map is total, 0 where there is no inverse, but at an injective A it is the true inverse Chapter6/Projection · gramL_inverse_apply. Put $\Pi(A) = A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ Chapter6/Projection · gramProj.

Lemma. If A is injective, $\Pi(A)$ is the orthogonal projection onto the range of A Chapter6/Projection · gramProj_eq_starProjection Chapter6/Projection · gramProj_eq_starProjection'. *Proof.* $y = (A^\mathsf{T}A)^{-1}A^\mathsf{T}v$ satisfies $\langle Ay, Ay'\rangle = \langle v, Ay'\rangle$ for every y'. So $v - Ay$ is orthogonal to the range of A, and Ay lies in it: Ay is the projection of v Mathlib/Analysis/InnerProductSpace/Projection/Basic · Submodule.eq_starProjection_of_mem_of_inner_eq_zero.

Lemma. Π is smooth near every injective A Chapter6/Projection · contDiffAt_gramProj. $A \mapsto A^\mathsf{T}$ is linear, $A \mapsto A^\mathsf{T}A$ is a product of smooth maps, inversion is smooth near the isomorphism $A^\mathsf{T}A$ Mathlib/Analysis/Calculus/ContDiff/Operations · contDiffAt_map_inverse, and Π is a composite of these.

In a slice chart. The parametrization of a slice chart has injective derivative at every point of its domain Chapter6/Projection · injective_fderiv_sliceParam. Near u, $\sigma(x(u')) = (u', 0)$, so by the chain rule $d\sigma \circ dx_u$ is the inclusion $Z \to Z \times W$, which is injective. Since T_qN is the range of dx_u, $P_q = \Pi(dx_u)$ at $u = (\sigma q)_1$ Chapter6/Projection · tanProj_eq_gramProj.

Theorem (V6.2). The map $q \mapsto P_q$ from N to the operators on V is smooth Chapter6/Projection · contMDiff_tanProj.

The proof as formalized. A slice chart σ gives the ambient map $y \mapsto \Pi(dx_{(\sigma y)_1})$ `Chapter6/Projection · projRep`. It equals P at the points of N in the domain of σ `Chapter6/Projection · projRep_eq`. It is smooth at those points `Chapter6/Projection · contDiffAt_projRep`, as a composite of three smooth maps: $y \mapsto (\sigma y)_1$, smooth because σ is smooth on its open domain; $u \mapsto dx_u$, the derivative of a map smooth on an open set; and Π, smooth at the injective dx_u. Near q, with σ the chart chosen at q, P coincides with this ambient map composed with the smooth inclusion. In particular P is differentiable at every point `Chapter6/Projection · mdifferentiableAt_tanProj`, and its derivative along N in a tangent direction is the ambient derivative of the extension `Chapter6/Projection · dN_tanProj_eq_fderiv`.

For compact N the project also has the nearest-point retraction r of the companion project, with $dr_q = P_q$ `Chapter3/EulerCharGradient · dret_eq_starProjection` (Part III). The construction here needs no compactness and applies to every slice submanifold.

Every tangent vector extends to a smooth vector field. For $u \in V$ the field $q \mapsto P_q u$ `Chapter6/Submanifold · projField` is tangent `Chapter6/Submanifold · isTangentField_projField`, smooth `Chapter6/Submanifold · contMDiff_projField`, and equal to u at every q with $u \in T_qN$ `Chapter6/Submanifold · projField_apply_of_mem`. These global fields replace Milnor's coordinate fields cut off by bump functions in all the uniqueness arguments below.

V.12 The derivative of the projection

For $v \in V$ write $D_v = d^N P_{\cdot}\, v$, the derivative of P at q in the direction v, an operator on V. Differentiating the identities $P^* = P$ and $P^2 = P$ along N gives its structure.

Lemma. D_v is self-adjoint, for every $v \in V$ `Chapter6/Projection · inner_dN_tanProj_comm`. *Proof.* Fix x, y. The linear functional $\ell(T) = \langle Tx, y\rangle - \langle x, Ty\rangle$ on operators vanishes on every P_q. Since $d(\ell \circ P) = \ell \circ dP$ `Chapter6/Projection · dN_clm_comp` and the derivative of the zero function is 0 `Chapter6/Projection · dN_const`, $\ell(D_v) = 0$.

Lemma. For $v \in T_qN$, $D_v P_q + P_q D_v = D_v$ `Chapter6/Projection · dN_tanProj_comp_add` `Chapter6/Projection · dN_tanProj_apply_add`, by the Leibniz rule for compositions `Chapter6/Projection · dN_clm_comp'` applied to $P \circ P = P$.

Multiplying by P_q on the left gives $P_q D_v P_q = 0$ `Chapter6/Projection · tanProj_dN_tanProj`. So D_v maps tangent vectors to normal vectors `Chapter6/Projection · tanProj_dN_tanProj_of_mem` `Chapter6/Projection · dN_tanProj_mem_orthogonal_of_mem`. It also maps normal vectors to tangent vectors `Chapter6/Projection · dN_tanProj_mem_of_mem_orthogonal`: for ν normal, $P_q \nu = 0$ and the identity becomes $P_q D_v \nu = D_v \nu$.

V.13 The Levi-Civita connection of the induced metric

A *vector field* on N is a map $X : N \to V$ with $X_q \in T_qN$ for every q `Chapter6/Submanifold · IsTangentField`; it is smooth if it is smooth as a map of manifolds $N \to V$.

Definition. For $v \in T_qN$ and $Y : N \to V$, $v \vdash Y = P_q(dY_q(v))$ `Chapter6/Submanifold · covAt`, and $(X \vdash Y)_q = P_q(dY_q(X_q))$ `Chapter6/Submanifold · leviCivita`. Both are tangent `Chapter6/Submanifold · covAt_mem` `Chapter6/Submanifold · leviCivita_mem` `Chapter6/Submanifold · isTangentField_leviCivita`.

Milnor's rules (V6.4). (1) $X \vdash Y$ is smooth when X is a smooth vector field and Y a smooth map `Chapter6/Submanifold · contMDiff_leviCivita`. First, $q \mapsto dY_q(X_q)$ is smooth `Chapter6/Submanifold · contMDiff_dN_apply`: near q it equals $x \mapsto dG_x(X_x)$ for a smooth representative G of Y on an open set U, and $x \mapsto dG_x$ is smooth on U. Then apply the smooth P. (2) $X \vdash Y$ is additive in X with no hypothesis `Chapter6/Submanifold · leviCivita_add_left`, and additive in Y at points where both summands are differentiable `Chapter6/Submanifold · leviCivita_add_right`. (3) $(fX) \vdash Y = f\,(X \vdash Y)$ for every function f `Chapter6/Submanifold · leviCivita_smul_left`, since dY_q is linear. (4) $X \vdash (fY) = (Xf)\,Y + f\,(X \vdash Y)$

at q, with $Xf = df_q(X_q)$, when f and Y are differentiable at q and X_q, Y_q are tangent `Chapter6/Submanifold · leviCivita_smul_right`. Indeed $d(fY)(X) = df(X)\,Y + f\,dY(X)$ `Chapter6/Projection · dN_smul`, and P_q fixes Y_q. The hypotheses of (2) and (4) are Milnor's tacit ones: his fields are smooth.

Corollary 8.4 for the induced metric (V6.11). Let Y, Y' be differentiable at q with Y_q, Y'_q tangent, and $v \in T_qN$. Then

$$v\langle Y, Y'\rangle = \langle v \vdash Y, Y'_q\rangle + \langle Y_q, v \vdash Y'\rangle$$

`Chapter6/Submanifold · dN_inner_eq_covAt`. *Proof.* The Leibniz rule for inner products `Chapter6/Projection · dN_inner` gives $\langle dY_q(v), Y'_q\rangle + \langle Y_q, dY'_q(v)\rangle$. Since Y'_q is tangent and P_q is self-adjoint, $\langle dY_q(v), Y'_q\rangle = \langle P_q dY_q(v), Y'_q\rangle$, and likewise for the other term. For the induced connection, compatibility is a one-line computation.

V.14 The second fundamental form

Definition. $\mathrm{II}_q(v, w) = D_v w$ `Chapter6/Submanifold · secondFundForm`. It is bilinear `Chapter6/Submanifold · secondFundForm_add_left` `Chapter6/Submanifold · secondFundForm_smul_left` `Chapter6/Submanifold · secondFundForm_add_right` `Chapter6/Submanifold · secondFundForm_smul_right` `Chapter6/Submanifold · secondFundForm_sub_right`.

Milnor meets the second fundamental form only in §6, as the normal component ℓ_{ij} of $\partial^2 x/\partial u^i \partial u^j$ (p. 34), and never mentions it in Part II. In the embedded setting it is the tool for symmetry and curvature.

Lemma (V6.12). For a vector field W differentiable at q and $v \in T_qN$, $\mathrm{II}_q(v, W_q) = dW_q(v) - P_q\,dW_q(v)$, the normal part of $dW_q(v)$ `Chapter6/Submanifold · secondFundForm_eq_sub_tanProj`. *Proof.* $W = PW$ as maps on N, since W is tangent. The Leibniz rule `Chapter6/Projection · dN_clm_apply` gives $dW_q(v) = D_v W_q + P_q dW_q(v)$.

So the normal part of $dW_q(v)$ depends only on v and W_q `Chapter6/Submanifold · sub_tanProj_dN_eq_of_eq`: this is the tensoriality of the second fundamental form. For tangent v, w, $\mathrm{II}_q(v, w)$ is normal `Chapter6/Submanifold · secondFundForm_mem_orthogonal` `Chapter6/Submanifold · tanProj_secondFundForm`, and $\mathrm{II}_q(v, \cdot)$ is self-adjoint on all of V `Chapter6/Submanifold · inner_secondFundForm_comm`. One more identity is used in the curvature computation: for a vector field W differentiable at q and $u, v \in T_qN$,

$$P_q\,\mathrm{II}_q(u, dW_q(v)) = \mathrm{II}_q(u, \mathrm{II}_q(v, W_q))$$

`Chapter6/Submanifold · tanProj_secondFundForm_dN`. Split $dW_q(v) = P_q dW_q(v) + \mathrm{II}_q(v, W_q)$. The first part contributes $P_q D_u P_q(\cdots) = 0$. The second is normal, so D_u maps it to a tangent vector, which P_q fixes.

Theorem (symmetry of II, V6.12). For $v, w \in T_qN$, $\mathrm{II}_q(v, w) = \mathrm{II}_q(w, v)$ `Chapter6/Submanifold · secondFundForm_comm`.

The idea. T_xN is the kernel of the derivative of the normal coordinate of a slice chart. Differentiating that statement along N expresses the normal coordinate of $\mathrm{II}(v, w)$ through a second derivative, which is symmetric.

The proof as formalized.

1. Let σ be the slice chart chosen at q and $g(y) = (\sigma y)_2 \in W$ its normal coordinate. Then $dg_y = \mathrm{pr}_2 \circ d\sigma_y$ on the domain of σ, and g is smooth near q, because σ is smooth on its open domain `Milnor/Chapter8/TangentBundle · Problems.sliceChart_contDiffOn`.
2. For $x \in N$ in the domain of σ, $T_xN = \ker dg_x$, a lemma of the companion project `Milnor/Chapter8/TangentBundle · Problems.sliceChart_mem_tangentSpace_iff`. Hence $dg_x \circ P_x = 0$ for such x.
3. Replace P near q by its smooth ambient extension $\Pi(dx_{(\sigma y)_1})$ of §V.11 `Chapter6/Projection · projRep_eq` `Chapter6/Projection · contDiffAt_projRep`. The ambient map $y \mapsto dg_y \circ \Pi(\ldots)$ is differentiable at q and

vanishes on N near q, so its derivative in a tangent direction a is 0 `Chapter6/Projection · fderiv_eq_zero_of_eventually`. By the product rule, applied to a tangent b with $P_q b = b$: $d^2 g_q(a,b) + dg_q(D_a b) = 0$, where the derivative of the extension in the direction a is D_a `Chapter6/Projection · dN_tanProj_eq_fderiv`. So $dg_q(\mathrm{II}_q(a,b)) = -d^2 g_q(a,b)$ for all tangent a, b.

4. $d^2 g_q$ is symmetric, g being smooth `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`. Hence $dg_q(\mathrm{II}_q(v,w) - \mathrm{II}_q(w,v)) = 0$, so the difference lies in $\ker dg_q = T_q N$.
5. The difference is also normal, as a difference of normal vectors. So it is 0 `Chapter6/Projection · eq_zero_of_mem_of_mem_orthogonal`.

V.15 The bracket, symmetry, and Lemma 8.6

Definition. For maps $X, Y : N \to V$, $[X,Y]_q = dY_q(X_q) - dX_q(Y_q)$, the bracket computed in V `Chapter6/Submanifold · bracket`.

Milnor defines $[X,Y]$ by its action on functions. That for fields on N it is computed by the ambient formula, and that this ambient vector is tangent to N, are both statements that need proof.

Lemma. If X, Y are vector fields differentiable at q, then $[X,Y]_q \in T_q N$ `Chapter6/Submanifold · bracket_mem`.
Proof. By §V.14 the normal part of $dY_q(X_q)$ is $\mathrm{II}_q(X_q, Y_q)$ and that of $dX_q(Y_q)$ is $\mathrm{II}_q(Y_q, X_q)$. They are equal by the symmetry of II. So the bracket equals its own projection $P_q dY_q(X_q) - P_q dX_q(Y_q)$.

Definition 8.5: the induced connection is symmetric (V6.13). For vector fields differentiable at q, $X \vdash Y - Y \vdash X = [X,Y]$ `Chapter6/Submanifold · leviCivita_sub_leviCivita`: the left side is $P_q[X,Y]_q$, and the bracket is tangent.

Lemma (the bracket is Milnor's Poisson bracket). For a smooth function f and smooth vector fields X, Y, $X(Yf) - Y(Xf) = [X,Y]f$ at every q, where $Xf = df(X)$ `Chapter6/Submanifold · dN_bracket`.

Proof. Take a smooth representative G of f on an open $U \ni q$. Near q, $Af = (x \mapsto dG_x(A_x))$ for any smooth vector field A. So, by the Leibniz rule for an operator applied to a vector `Chapter6/Projection · dN_clm_apply` and the chain rule `Chapter6/Projection · dN_comp_val`, $B(Af)$ at q is $d^2 G_q(B_q, A_q) + dG_q(dA_q(B_q))$ for B_q tangent. Therefore $X(Yf) - Y(Xf) = d^2 G_q(X_q, Y_q) - d^2 G_q(Y_q, X_q) + dG_q([X,Y]_q)$. The second derivative is symmetric `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`, and $dG_q = df_q$ on the tangent vector $[X,Y]_q$.

Lemma 8.6 in the embedded setting (V6.14). Call an operation D on maps $N \to V$ *symmetric and compatible* if, for all smooth vector fields X, Y, Y', $D(X,Y)$ is tangent, $D(X,Y) - D(Y,X) = [X,Y]$, and $X\langle Y, Y'\rangle = \langle D(X,Y), Y'\rangle + \langle Y, D(X,Y')\rangle$ `Chapter6/Submanifold · IsSymmCompat`. Then:

- *Existence.* $P(dY(X))$ is symmetric and compatible `Chapter6/Submanifold · isSymmCompat_leviCivita`, by §V.13 and the symmetry just proved.
- *The Koszul formula.* Every symmetric compatible D satisfies, on smooth vector fields,

$$2\langle D(X,Y), Z\rangle = X\langle Y,Z\rangle + Y\langle X,Z\rangle - Z\langle X,Y\rangle + \langle [X,Y],Z\rangle - \langle [X,Z],Y\rangle - \langle [Y,Z],X\rangle$$

  `Chapter6/Submanifold · IsSymmCompat.two_mul_inner`, by the same rewriting as in §V.9.
- *Uniqueness.* Every symmetric compatible D agrees with $P(dY(X))$ on smooth vector fields `Chapter6/Submanifold · IsSymmCompat.eq_leviCivita`. Put $u = D(X,Y)_q - (X \vdash Y)_q$, a tangent vector, and let Z be the smooth field $x \mapsto P_x u$, equal to u at q (§V.11). The Koszul formula for D and for $P(dY(X))$ has the same right side, so $2\langle D(X,Y)_q, u\rangle = 2\langle (X \vdash Y)_q, u\rangle$, that is, $\langle u, u\rangle = 0$.

The uniqueness assumes no linearity or tensoriality of D, only the three properties on smooth fields. Milnor proves uniqueness with the Christoffel identities (p. 49) and asserts the converse, that the connection they define is symmetric and compatible, with "one can verify". In the embedded setting the converse is the computation of §V.13 together with the symmetry of II. The abstract uniqueness of §V.9 and this

embedded statement are parallel results, not one theorem, since the induced connection is not packaged as a Mathlib covariant derivative.

V.16 The curvature of N: Lemma 9.1, the Gauss equation, Lemma 9.3

Definition. With the induced connection and the ambient bracket, Milnor's curvature of maps $X, Y, Z : N \to V$ is

$$R(X,Y)Z = -X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X,Y] \vdash Z$$

`Chapter6/Submanifold · curvature`. At a point, put $R_q(x,y) = D_y D_x - D_x D_y$ `Chapter6/Submanifold · curvOp`, so that $R_q(x,y)z = \mathrm{II}_q(y, \mathrm{II}_q(x,z)) - \mathrm{II}_q(x, \mathrm{II}_q(y,z))$ `Chapter6/Submanifold · curvOp_apply`.

Theorem (Lemma 9.1 made explicit, V6.17). For smooth vector fields X, Y, Z and every q,

$$R(X,Y)Z\,(q) = \mathrm{II}_q(Y_q, \mathrm{II}_q(X_q, Z_q)) - \mathrm{II}_q(X_q, \mathrm{II}_q(Y_q, Z_q))$$

`Chapter6/Submanifold · curvature_eq`.

The proof as formalized.

1. Take a smooth representative G of Z on an open $U \ni q$ `Chapter6/Projection · exists_contDiffOn_rep`. For $x \in N \cap U$ and $v \in T_xN$, $dZ_x(v) = dG_x(v)$.
2. For smooth vector fields A and B, at q,

   $$B \vdash (A \vdash Z) = \mathrm{II}_q(B_q, \mathrm{II}_q(A_q, Z_q)) + P_q\, d^2G_q(B_q, A_q) + P_q\, dG_q(dA_q(B_q)).$$

   By definition $A \vdash Z = x \mapsto P_x(dZ_x(A_x))$, and $x \mapsto dZ_x(A_x)$ is smooth `Chapter6/Submanifold · contMDiff_dN_apply`. The Leibniz rule for P applied to it gives $d(A \vdash Z)_q(B_q) = D_{B_q}(dZ_q(A_q)) + P_q\, d(x \mapsto dZ_x(A_x))_q(B_q)$. Near q the map $x \mapsto dZ_x(A_x)$ equals $x \mapsto dG_x(A_x)$, whose derivative in the direction B_q is $d^2G_q(B_q, A_q) + dG_q(dA_q(B_q))$, by the Leibniz rule for the operator field $x \mapsto dG_x$ applied to A. Apply P_q and use $P_q^2 = P_q$ and $P_q\,\mathrm{II}_q(B_q, dZ_q(A_q)) = \mathrm{II}_q(B_q, \mathrm{II}_q(A_q, Z_q))$ (§V.14).
3. The bracket is tangent (§V.15), so $[X,Y] \vdash Z = P_q dG_q(dY_q(X_q)) - P_q dG_q(dX_q(Y_q))$.
4. In $-X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X,Y] \vdash Z$, step 2 for $(A,B) = (Y,X)$ and (X,Y) and step 3 leave

   $$\mathrm{II}(Y, \mathrm{II}(X,Z)) - \mathrm{II}(X, \mathrm{II}(Y,Z)) + P_q(d^2G_q(Y_q, X_q) - d^2G_q(X_q, Y_q)),$$

   the first-derivative terms of G cancelling against the bracket. The last term vanishes by symmetry of d^2G_q `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`.

The Gauss equation (V6.18). For all $x, y, z, w \in V$,

$$\langle R_q(x,y)z, w\rangle = \langle \mathrm{II}_q(x,z), \mathrm{II}_q(y,w)\rangle - \langle \mathrm{II}_q(x,w), \mathrm{II}_q(y,z)\rangle$$

`Chapter6/Submanifold · inner_curvOp`, by self-adjointness of $\mathrm{II}_q(y,\cdot)$ and $\mathrm{II}_q(x,\cdot)$. Hence, for smooth vector fields X, Y, Z and any $W : N \to V$, the same identity holds for $\langle R(X,Y)Z, W\rangle$ at every point `Chapter6/Submanifold · gauss_equation`. With $Z = X$, $W = Y$ and the symmetry of II,

$$\langle R(X,Y)X, Y\rangle = \langle \mathrm{II}(X,X), \mathrm{II}(Y,Y)\rangle - \|\mathrm{II}(X,Y)\|^2,$$

Gauss's curvature for a surface in $\mathbb{R}^3$. With Milnor's sign of R, his sectional curvature $\langle R(A,B)A, B\rangle$ is therefore Gauss's curvature, positive on the sphere (§V.21). The Gauss equation is not in Milnor's text.

Lemma 9.1 (V6.17). R_q maps tangent vectors to tangent vectors `Chapter6/Submanifold · curvOp_mem`, since $\mathrm{II}_q(x,z)$ is normal and D_y maps normal vectors to tangent ones. It is linear in x and in y `Chapter6/`

Submanifold · curvOp_add_left Chapter6/Submanifold · curvOp_smul_left Chapter6/Submanifold · curvOp_add_right Chapter6/Submanifold · curvOp_smul_right. So it defines a trilinear map $T_qN \times T_qN \times T_qN \to T_qN$, the curvature tensor of N at q Chapter6/Submanifold · curvTensor Chapter6/Submanifold · coe_curvTensor_apply. By the theorem, $R(X,Y)Z$ at q is this tensor applied to X_q, Y_q, Z_q Chapter6/Submanifold · curvature_eq_curvTensor, and so depends only on those values Chapter6/Submanifold · curvature_congr. Milnor proves tensoriality in X and says the identities for Y and Z are "easily obtained by similar computations" (p. 51). For Z the computation is a different one: the terms with two derivatives of f cancel only by $[X,Y]f = X(Yf) - Y(Xf)$. In the formal proof this shows up as the symmetry of d^2G in step 4.

Lemma 9.3 (1)–(3) (V6.20). The curvature tensor of N at each point is a curvature tensor in the sense of §V.7 Chapter6/Submanifold · isCurvatureTensor_curvTensor. (1) is visible from the formula. (2) is the symmetry of II: the six terms of $R(x,y)z + R(y,z)x + R(z,x)y$ cancel in pairs once $\mathrm{II}(y,x)$, $\mathrm{II}(z,y)$, $\mathrm{II}(x,z)$ are replaced by $\mathrm{II}(x,y)$, $\mathrm{II}(y,z)$, $\mathrm{II}(z,x)$. (3) is the Gauss equation, whose right side is visibly skew in (z,w). For vector fields the identities read: (1) with no hypothesis at all Chapter6/Submanifold · curvature_add_swap, since $[Y,X] = -[X,Y]$; (2) Chapter6/Submanifold · curvature_bianchi and (3) Chapter6/Submanifold · inner_curvature_add_inner for smooth vector fields. Milnor proves (2) and (3) for extensions of the given vectors with vanishing brackets, which must be constructed: coordinate fields cut off by a bump function. No extension is needed here.

Lemma 9.3 (4) (V6.21). For smooth vector fields, $\langle R(X,Y)Z, W\rangle = \langle R(Z,W)X, Y\rangle$ Chapter6/Submanifold · inner_curvature_comm_pairs, by the octahedron of §V.7 applied to the tensor at q. The theory of §V.8 (sectional curvature, polarization, constant curvature) applies to the curvature tensor of N at every point.

V.17 Covariant derivatives along curves: Lemmas 8.1 and 8.3

A curve in N is a map $c : \mathbb{R} \to N$ such that $t \mapsto c(t) \in V$ is smooth. It is then smooth as a map into the manifold N Chapter6/Projection · contMDiff_curve. Its velocity $c'(t)$ Chapter6/Parallel · velocity is tangent to N Chapter6/Projection · deriv_mem_tangentSubspace: it is the image under $d\iota$ of the manifold derivative of c. The chain rule along c reads $(F \circ c)'(t) = dF_{c(t)}(c'(t))$ for F differentiable at $c(t)$ Chapter6/Projection · hasDerivAt_comp_curve, proved with an ambient representative of F. Smooth maps compose with c to smooth functions of t Chapter6/Projection · contDiff_comp_curve. In particular $t \mapsto P_{c(t)}$ is smooth Chapter6/Projection · contDiff_tanProj_comp, with derivative $D_{c'(t)}$ Chapter6/Projection · hasDerivAt_tanProj_comp.

Definition (V6.5). A *vector field along* c (p. 45) is a map $X : \mathbb{R} \to V$ with $X(t) \in T_{c(t)}N$ for every t Chapter6/Parallel · IsFieldAlong. Its *covariant derivative* is $DX/dt\,(t) = P_{c(t)}(X'(t))$ Chapter6/Parallel · covDeriv, a tangent vector Chapter6/Parallel · covDeriv_mem. Smoothness of a field along c is smoothness of $t \mapsto X(t) \in V$.

Milnor characterizes D/dt by three axioms (p. 46), and $P(X')$ satisfies them. (a) $D(X+Y)/dt = DX/dt + DY/dt$ at t when X, Y are differentiable at t Chapter6/Parallel · covDeriv_add. (b) $D(fX)/dt = f'X + f\,DX/dt$ at t when f, X are differentiable at t and $X(t)$ is tangent Chapter6/Parallel · covDeriv_smul: the product rule, and P fixes $X(t)$. (c) For c smooth and a map $Y : N \to V$ differentiable at $c(t)$, $D(Y \circ c)/dt\,(t) = c'(t) \vdash Y$ Chapter6/Parallel · covDeriv_comp, by the chain rule along c.

Lemma 8.3 for the induced metric (V6.10). For any curve c in N, smooth or not, and fields X, Y along c differentiable at t,

$$\frac{d}{dt}\langle X, Y\rangle = \left\langle \frac{DX}{dt}, Y \right\rangle + \left\langle X, \frac{DY}{dt} \right\rangle$$

at t Chapter6/Parallel · hasDerivAt_inner_covDeriv. *Proof.* The product rule in V gives $\langle X', Y\rangle + \langle X, Y'\rangle$. Since $Y(t)$ is tangent and P is self-adjoint, $\langle X', Y\rangle = \langle PX', Y\rangle$; likewise for the other term. Milnor's proof (p. 47) expands the two fields in a parallel orthonormal frame, which needs Lemma 8.2 for n initial

vectors and the smoothness of the coefficients. Here Lemma 8.3 is a direct computation and comes *before* parallel transport, which uses it.

Lemma 8.1 (V6.6). Let c be a smooth curve. Call an operation D on maps $\mathbb{R} \to V$ *admissible* if it satisfies (a) for smooth fields along c, (b) for smooth functions f and smooth fields along c, and (c) for smooth vector fields Y on N `Chapter6/Parallel · IsCovDerivAlong`. Then $X \mapsto P(X')$ is admissible `Chapter6/Parallel · isCovDerivAlong_covDeriv`, and every admissible D agrees with it on every smooth field along c `Chapter6/Parallel · IsCovDerivAlong.eq_covDeriv`.

The proof as formalized. Milnor's uniqueness argument works in one coordinate neighbourhood. It uses that D/dt is local in t, which follows from (b), and that the coordinate fields ∂_j extend to global vector fields; neither is mentioned when he deduces uniqueness from the axioms. The formal proof uses global fields instead of coordinate fields. Let (e_i) be an orthonormal basis of V `Mathlib/Analysis/InnerProductSpace/PiL2 · stdOrthonormalBasis` and E_i the smooth vector field $q \mapsto P_q e_i$ (§V.11).

1. A smooth field X along c is tangent, so $X = PX = \sum_i \langle e_i, X\rangle\,(E_i \circ c)$: expand $X(s)$ in the basis and apply $P_{c(s)}$. Each summand is a smooth field along c.
2. Axiom (a) gives $D(0) = 0$, from $0 = 0 + 0$, and then, by induction over finite sums, D of the sum is the sum of the D's.
3. Axiom (b) gives $D(\langle e_i, X\rangle (E_i \circ c)) = \langle e_i, X\rangle'\,(E_i \circ c) + \langle e_i, X\rangle\, D(E_i \circ c)$, and axiom (c) gives $D(E_i \circ c)(t) = c'(t) \vdash E_i$.

So every admissible D satisfies

$$DX(t) = \sum_i \Big(\frac{d}{dt}\langle e_i, X\rangle\, P_{c(t)} e_i + \langle e_i, X(t)\rangle\; c'(t) \vdash E_i \Big)$$

`Chapter6/Parallel · IsCovDerivAlong.eq_sum`. The right side does not involve D, so any two admissible operations agree.

V.18 Linear equations with skew-adjoint coefficients

In the proof of Lemma 8.2, Milnor writes the equations of a parallel field in coordinates, observes that they are linear, and concludes, citing Graves, that the solutions "can be defined for all relevant values of t" (p. 47). The local existence theorem gives solutions only on intervals of length about $1/\sup\|A\|$, and a global statement needs an a priori bound and a continuation argument. Mathlib has the Picard–Lindelöf theorem on intervals limited by a bound for the field on a ball `Mathlib/Analysis/ODE/PicardLindelof · IsPicardLindelof` and Grönwall-type uniqueness `Mathlib/Analysis/ODE/ExistUnique · ODE_solution_unique_of_mem_Ioo`. It has no global existence theorem for linear equations. The project supplies what parallel transport needs, for a continuous family $A : \mathbb{R} \to \mathrm{End}(E)$.

Uniqueness (V6.7 (i)). On a real normed space, two solutions of $x' = A(t)x$ on an open interval that agree at one time agree on the whole interval `Chapter6/LinearODE · linearODE_eqOn`; the same holds on $\mathbb{R}$ `Chapter6/LinearODE · linearODE_eq`. A continuous family is bounded on a compact interval `Chapter6/LinearODE · exists_nnnorm_le_Icc`, so the field $(t, x) \mapsto A(t)x$ is uniformly Lipschitz in x there, and Mathlib's uniqueness theorem applies. On $\mathbb{R}$, restrict to an interval containing both times.

The radial truncation. For $R > 0$ put $\rho_R(x) = \frac{R}{\max(R, \|x\|)}\, x$ `Chapter6/LinearODE · radTrunc`, the identity on the closed ball of radius R `Chapter6/LinearODE · radTrunc_of_norm_le`, with values in that ball `Chapter6/LinearODE · norm_radTrunc_le`. It is 2-Lipschitz `Chapter6/LinearODE · lipschitzWith_radTrunc`. With $m_x = \max(R, \|x\|)$ and $m_y = \max(R, \|y\|)$,

$$\rho_R(x) - \rho_R(y) = \frac{R}{m_x}(x - y) + \frac{R(m_y - m_x)}{m_x m_y}\, y.$$

The first term has norm at most $\|x-y\|$, since $R \le m_x$. The second has norm at most $\frac{R}{m_x}|m_y - m_x|$, since $\|y\| \le m_y$, and so at most $|m_y - m_x| \le \big|\|y\| - \|x\|\big| \le \|x-y\|$.

Global existence for skew-adjoint coefficients (V6.7 (ii)). Let E be a complete real inner product space and A continuous with $\langle A(t)x, x\rangle = 0$ for all t, x. For every t_0 and x_0 there is a solution of $x' = A(t)x$ on all of $\mathbb{R}$ with $x(t_0) = x_0$ `Chapter6/LinearODE · exists_skewLinearODE`.

The proof as formalized.

1. *On a bounded interval* `Chapter6/LinearODE · exists_skewLinearODE_Ioo`. Fix $a > 0$, a bound C for $\|A\|$ on $[t_0 - a, t_0 + a]$, and $R = \|x_0\| + 1$. The truncated field $f(t,x) = A(t)\rho_R(x)$ is $2C$-Lipschitz in x, continuous in t, and bounded by $L = CR$ everywhere. So the Picard–Lindelöf conditions hold on the whole interval in one step, with a ball of radius La around x_0. Mathlib's theorem gives α on $[t_0 - a, t_0 + a]$ with $\alpha(t_0) = x_0$ and $\alpha' = f(t, \alpha)$ within the interval `Mathlib/Analysis/ODE/ExistUnique · IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt₀`.
2. *The norm is constant.* $\frac{d}{dt}\|\alpha\|^2 = 2\langle \alpha, f(t,\alpha)\rangle = 2\frac{R}{\max(R,\|\alpha\|)}\langle \alpha, A(t)\alpha\rangle = 0$ by skewness. A function continuous on the interval with zero right derivative is constant `Mathlib/Analysis/Calculus/MeanValue · constant_of_has_deriv_right_zero`. So $\|\alpha(t)\| = \|x_0\| < R$ throughout.
3. *The truncation is never active.* Hence $f(t, \alpha(t)) = A(t)\alpha(t)$, and at interior points the derivative within the interval is a genuine derivative.
4. *Gluing.* Choose such a solution β_a for every $a > 0$. Two of them agree wherever both are defined, by uniqueness on the smaller interval. Put $\alpha(t) = \beta_{|t-t_0|+1}(t)$. Near each t, α coincides with the single solution $\beta_{|t-t_0|+1}$, so it solves the equation at t, and $\alpha(t_0) = x_0$.

Later, for the Jacobi equation, the project proves global existence for every continuous linear equation, with the a priori bound supplied by Grönwall's inequality `Chapter9/JacobiODE · exists_linearODE` (Part VII). Parallel transport does not use it: the skew-adjoint case gets its bound for free.

V.19 Parallel transport: Lemma 8.2

Definition. A *parallel field* along c (p. 46) is a differentiable field along c whose covariant derivative vanishes at every t `Chapter6/Parallel · IsParallel`.

Fix a smooth curve c and write $P(t) = P_{c(t)}$. Its derivative is $P'(t) = D_{c'(t)}$ `Chapter6/Parallel · dProjAlong` `Chapter6/Parallel · hasDerivAt_tanProj_along` `Chapter6/Parallel · deriv_tanProj_along`, a smooth `Chapter6/Parallel · contDiff_dProjAlong`, hence continuous `Chapter6/Parallel · continuous_dProjAlong` family of operators.

A parallel field satisfies $X' = P'X$. If X is parallel, then $X = PX$, so $X' = P'X + PX' = P'X$, because $PX' = DX/dt = 0$ `Chapter6/Parallel · IsParallel.hasDerivAt`. **Uniqueness in Lemma 8.2:** two parallel fields along c with the same value at one time are equal `Chapter6/Parallel · IsParallel.eq`, by uniqueness for the linear equation $X' = P'(t)X$ `Chapter6/LinearODE · linearODE_eq`.

Why existence needs another equation. The natural equation $X' = P'X$ is linear with continuous coefficients, but §V.18 gives global solutions only for skew-adjoint coefficients, and P' is self-adjoint (§V.12). Its solutions have no a priori bound: $\frac{d}{dt}\|X\|^2 = 2\langle P'X, X\rangle$, which vanishes for tangent X but not in general, and one does not know in advance that a solution stays tangent. The formal proof therefore solves instead

$$X' = \Omega(t)X, \qquad \Omega = P'P - PP'$$

`Chapter6/Parallel · skewAlong`. Its coefficient is skew-adjoint `Chapter6/Parallel · inner_skewAlong_self`: $\langle P'Px, x\rangle = \langle Px, P'x\rangle = \langle PP'x, x\rangle$, by self-adjointness of P' and of P. On tangent vectors the two equations agree, since $PP'X = PP'PX = 0$ there, so $\Omega X = P'X$.

Lemma 8.2 (V6.8). For a smooth curve c, a time t_0 and $v \in T_{c(t_0)}N$, there is exactly one parallel field X along c with $X(t_0) = v$ `Chapter6/Parallel · exists_unique_isParallel`. It is defined for all t, and it is smooth `Chapter6/Parallel · IsParallel.contDiff`.

The proof of existence as formalized `Chapter6/Parallel · exists_isParallel`.

1. Ω is continuous and skew-adjoint, so §V.18 gives $\alpha : \mathbb{R} \to V$ with $\alpha' = \Omega\alpha$ everywhere and $\alpha(t_0) = v$ `Chapter6/LinearODE · exists_skewLinearODE`.
2. *The normal part $U = \alpha - P\alpha$ satisfies $U' = -P'U$.* By the product rule $U' = \Omega\alpha - P'\alpha - P\Omega\alpha$. Since $c'(t)$ is tangent, $PP'P = 0$ (§V.12), so $P\Omega\alpha = PP'P\alpha - P^2P'\alpha = -PP'\alpha$. Hence $U' = P'P\alpha - PP'\alpha - P'\alpha + PP'\alpha = -P'(\alpha - P\alpha)$.
3. $U(t_0) = v - Pv = 0$, since v is tangent. The zero function solves the same linear equation, so $U \equiv 0$ by uniqueness `Chapter6/LinearODE · linearODE_eq`. Thus $\alpha = P\alpha$: the solution stays tangent.
4. α is parallel: $D\alpha/dt = P\alpha' = PP'P\alpha - PP'\alpha$, and both terms vanish because $\alpha = P\alpha$ and $PP'P = 0$.

Uniqueness was proved above, among differentiable fields, a stronger statement than Milnor's among smooth ones. Smoothness follows by bootstrapping $X' = P'(t)X$ with P' smooth: if X is C^n, then X' is C^n, so X is C^{n+1}. Milnor states the lemma with initial time 0; any initial time is allowed here. On a manifold his coordinate equations would also have to be patched from chart to chart along the curve. In the embedded setting there is one equation in V.

Compatibility in Milnor's sense (V6.9). For parallel fields X, Y along any curve, $\langle X, Y\rangle$ is constant `Chapter6/Parallel · IsParallel.inner_eq`. Its derivative is 0 by Lemma 8.3, and a function with zero derivative is constant. This is Milnor's definition of a connection compatible with the metric (p. 47), proved here for the induced connection. Parallel fields form a vector space: sums `Chapter6/Parallel · IsParallel.add` and scalar multiples `Chapter6/Parallel · IsParallel.const_smul` of parallel fields are parallel.

Parallel translation (V6.9). Let X_v be the parallel field along c through v at time t_0, chosen once and for all `Chapter6/Parallel · parallelField`; formally it is 0 when there is none, which happens only if v is not tangent or c is not smooth. For tangent v it is parallel with $X_v(t_0) = v$ `Chapter6/Parallel · parallelField_spec`, and every parallel field is the chosen one through any of its values `Chapter6/Parallel · IsParallel.eq_parallelField`. Translation from $c(t_0)$ to $c(t_1)$ sends v to $X_v(t_1)$ `Chapter6/Parallel · parallelTransportL` `Chapter6/Parallel · parallelTransportL_apply`.

- It is *linear*: $X_v + X_w$ is parallel with value $v + w$ at t_0, so it is X_{v+w} by uniqueness; likewise for scalars.
- It *preserves norms*: $\|X_v(t_1)\|^2 = \|X_v(t_0)\|^2 = \|v\|^2$ by compatibility.
- It is *onto* `Chapter6/Parallel · surjective_parallelTransportL`: given w at t_1, take $v = X_w(t_0)$, where X_w is the parallel field through w at time t_1. By uniqueness $X_v = X_w$, whose value at t_1 is w.

So parallel translation is a linear isometry $T_{c(t_0)}N \to T_{c(t_1)}N$ `Chapter6/Parallel · parallelTransport` `Chapter6/Parallel · parallelTransport_apply` `Mathlib/Analysis/Normed/Operator/LinearIsometry · LinearIsometryEquiv.ofSurjective`. It preserves inner products `Chapter6/Parallel · inner_parallelTransport`, is the identity when $t_1 = t_0$ `Chapter6/Parallel · parallelTransport_self`, and is transitive `Chapter6/Parallel · parallelTransport_trans`: the parallel field through $X_v(t_1)$ at time t_1 is X_v itself, so translating to t_1 and then to t_2 gives $X_v(t_2)$. Milnor uses linearity, surjectivity and transitivity without comment.

V.20 Parametrized surfaces: Lemmas 8.7 and 9.2

For a map F on $\mathbb{R}^2$, $\partial F/\partial x = dF(1,0)$ and $\partial F/\partial y = dF(0,1)$ `Chapter6/Surface · partialX` `Chapter6/Surface · partialY`. They are the derivatives along the horizontal and vertical lines `Chapter6/Surface · hasDerivAt_partialX` `Chapter6/Surface · hasDerivAt_partialY`, smooth for smooth F `Chapter6/Surface · contDiff_partialX` `Chapter6/Surface · contDiff_partialY`. The mixed partials are second derivatives `Chapter6/Surface · partialY_partialX_eq` `Chapter6/Surface · partialX_partialY_eq`, equal by Schwarz `Chapter6/Surface · partialY_partialX`. Each partial derivative obeys the product rule for an operator-valued map applied to a vector-valued one `Chapter6/Surface · partialX_clm_apply` `Chapter6/Surface · partialY_clm_apply`.

A *parametrized surface* is a map $s : \mathbb{R}^2 \to N$ with $p \mapsto s(p) \in V$ smooth. It is smooth into N `Chapter6/`

Projection · contMDiff_map, its partial derivatives are tangent Chapter6/Projection · fderiv_mem_tangentSubspace, the chain rule $d(F \circ s)_p = dF_{s(p)} \circ ds_p$ holds Chapter6/Projection · hasFDerivAt_comp_map, and smooth maps compose to smooth maps Chapter6/Projection · contDiff_comp_map. The velocity fields are $\partial s/\partial x$ and $\partial s/\partial y$ (p. 49) Chapter6/Surface · velX Chapter6/Surface · velY, tangent Chapter6/Surface · velX_mem Chapter6/Surface · velY_mem. Milnor's $DW/\partial x$ at (x, y_0) is the covariant derivative of $x \mapsto W(x, y_0)$ along the curve $x \mapsto s(x, y_0)$, and the formal definition is literally that Chapter6/Surface · covDerivX Chapter6/Surface · covDerivY. At a point where W is differentiable it equals $P_{s(p)}(\partial W/\partial x)$ Chapter6/Surface · covDerivX_eq Chapter6/Surface · covDerivY_eq. Milnor defines $DW/\partial x$ one curve at a time, but Lemma 9.2 applies $D/\partial y$ to it, which needs joint smoothness in (x, y). Here it is visible from the formula.

Lemma 8.7 (V6.15). $\frac{D}{\partial x}\frac{\partial s}{\partial y} = \frac{D}{\partial y}\frac{\partial s}{\partial x}$ at every point Chapter6/Surface · covDerivX_velY. Both sides are $P(\partial^2 s/\partial x \partial y)$, by the formula above and Schwarz. Milnor states the lemma for a symmetric connection and proves it by a computation in coordinates. The induced connection is symmetric (§V.15), but the formal proof does not need it, only the symmetry of second derivatives in V.

Lemma 9.2 (V6.19). For a smooth map $W : \mathbb{R}^2 \to V$ with $W(p) \in T_{s(p)}N$ for all p,

$$\frac{D}{\partial y}\frac{D}{\partial x}W - \frac{D}{\partial x}\frac{D}{\partial y}W = R\Big(\frac{\partial s}{\partial x}, \frac{\partial s}{\partial y}\Big)W$$

at every point, with $R_q(x, y) = D_y D_x - D_x D_y$ the curvature of §V.16 Chapter6/Surface · covDerivY_covDerivX_sub.

The proof as formalized. Write $\tilde{P}(p) = P_{s(p)}$, and $D_x = D_{\partial s/\partial x}$, $D_y = D_{\partial s/\partial y}$ at $s(p)$.

1. $\tilde{P}$ is smooth, and its partial derivatives at p are D_x and D_y, by the chain rule for s.
2. $DW/\partial x = \tilde{P}\,\partial W/\partial x$ and $DW/\partial y = \tilde{P}\,\partial W/\partial y$ everywhere. These are smooth, so the outer covariant derivatives are again given by the formula.
3. Since $W = \tilde{P}W$, the product rule gives $\partial W/\partial x = D_x W + \tilde{P}\,\partial W/\partial x$ at p, and likewise for y.
4. *Key identity:* for tangent a, b at $s(p)$ and any w with $w = D_a W(p) + P_{s(p)}w$, $P_{s(p)}(D_b w) = D_b D_a W(p)$. Indeed $P D_b P = 0$ kills the second part, and $D_a W(p)$ is normal, $W(p)$ being tangent, so $D_b D_a W(p)$ is tangent and fixed by $P_{s(p)}$.
5. By the product rule, $\frac{D}{\partial y}(\tilde{P}\,\partial W/\partial x) = P(D_y\,\partial W/\partial x) + P\,\partial^2 W/\partial y \partial x$. By step 4 with $w = \partial W/\partial x$, the first term is $D_y D_x W$. Symmetrically, $\frac{D}{\partial x}\frac{D}{\partial y}W = D_x D_y W + P\,\partial^2 W/\partial x \partial y$. The second-derivative terms cancel by Schwarz, leaving $D_y D_x W - D_x D_y W = R(\partial s/\partial x, \partial s/\partial y)W$.

Lemma 9.2 needs no symmetry of the connection, and the formal proof makes no use of the symmetry of II. Milnor again leaves a computation in local coordinates to the reader; in the embedded setting the computation is steps 3–5.

V.21 The round sphere

The unit sphere $S^n = \{x \in \mathbb{R}^{n+1} : \langle x, x\rangle - 1 = 0\}$ Milnor/Chapter5/HairyBall · Sphere.Sph is a slice submanifold of $\mathbb{R}^{n+1}$ with complementary space $\mathbb{R}^1$ Milnor/Chapter6/EulerSphere · Sphere.sphSub. Its points satisfy $\langle q, q\rangle = 1$ Chapter6/SphereCurvature · inner_self_sph Milnor/Chapter5/HairyBall · Sphere.inner_self_eq_one_of_mem, and its tangent space at q is $q^\perp$ Chapter6/SphereCurvature · mem_tangentSubspace_sph_iff Milnor/Chapter6/EulerSphere · Sphere.mem_range_mfd_val_iff, a result of the companion project. Milnor uses the constant curvature of the sphere in §§19–20 without computing it. The formal computation is coordinate-free.

The projection. $P_q v = v - \langle q, v\rangle q$ Chapter6/SphereCurvature · tanProj_sph. The right side is tangent, as $\langle q, v - \langle q, v\rangle q\rangle = 0$, and v minus it, $\langle q, v\rangle q$, is orthogonal to every tangent vector; this characterizes the orthogonal projection Mathlib/Analysis/InnerProductSpace/Projection/Basic · Submodule.eq_starProjection_of_mem_of_inner_eq_zero. In particular $P_q q = 0$ Chapter6/SphereCurvature · tanProj_sph_self.

The second fundamental form (V6.25). For tangent v, w at q, $\mathrm{II}_q(v,w) = -\langle v, w\rangle\, q$ `Chapter6/SphereCurvature · secondFundForm_sph`.

Proof. Use the smooth field $x \mapsto P_x w$, equal to w at q (§V.11). By §V.14, $\mathrm{II}_q(v,w)$ is the normal part of its derivative in the direction v. On the sphere this field is $G(x) = w - \langle w, x\rangle x$, an ambient map whose derivative at q is $v \mapsto -\langle w, q\rangle v - \langle w, v\rangle q = -\langle w, v\rangle q$, since $w \perp q$. The normal part of $-\langle w, v\rangle q$ is itself, because $P_q q = 0$.

The curvature. For tangent x, y, z, $R(x,y)z = \langle x, z\rangle y - \langle y, z\rangle x$, with Milnor's sign `Chapter6/SphereCurvature · curvOp_sph`. Both sides are tangent `Chapter6/Submanifold · curvOp_mem`. For tangent w, the Gauss equation (§V.16) and $\langle q, q\rangle = 1$ give

$$\langle R(x,y)z, w\rangle = \langle x, z\rangle\langle y, w\rangle - \langle x, w\rangle\langle y, z\rangle = \big\langle \langle x, z\rangle y - \langle y, z\rangle x,\ w\big\rangle.$$

So the difference of the two sides is tangent and orthogonal to every tangent vector, hence 0. On each tangent space the curvature tensor of S^n is therefore the tensor R_1 of §V.8 `Chapter6/SphereCurvature · curvTensor_sph`. S^n **has constant sectional curvature** 1, both as $K(x,y) = 1$ for independent tangent vectors `Chapter6/SphereCurvature · sectionalCurvature_sph` and in Milnor's form, $\langle R(A,B)A, B\rangle = 1$ for orthonormal A, B (p. 100) `Chapter6/SphereCurvature · inner_curvTensor_sph_orthonormal`. With Milnor's sign the sphere comes out positively curved, as it must.

V.22 What the formalization taught us

- **The index bound needs no focal points.** Andreotti and Frankel's identity $H(iv, iv) = 2G(v,v) - H(v,v)$ bounds the index by k at every point of a holomorphic chart, critical or not, degenerate or not `Chapter5/IndexBound · index_hessian_normSq_sub_holomorphic_le`. Milnor's count also needs p and q non-focal and the case $p = q$, which he does not mention.
- **Assertion 1 must be relative.** Focal points are eigenvalues of the second fundamental form relative to the first, not relative to the coordinate inner product. What Assertion 2 needs is the version for a J-invariant form `Chapter5/IndexBound · nullity_sub_smul_eq_nullity_add_smul`, and the vanishing of Milnor's linear terms needs real orthogonality to imply Hermitian orthogonality `Chapter5/IndexBound · inner_eq_zero_of_re_inner_eq_zero`.
- **The octahedron needs no linearity**, and its signs, left to the reader, are fourteen instances of three identities `Chapter6/CurvatureAlgebra · octahedron`. That the sectional curvature depends only on the plane and determines R is not stated in the book; both are proved `Chapter6/CurvatureAlgebra · IsCurvatureTensor.ext_of_orthonormal`.
- **One smooth operator field replaces the Christoffel symbols.** All smoothness in §8 reduces to that of $q \mapsto P_q$, which holds because the projection onto the range of an injective operator is $A(A^{\mathsf T}A)^{-1}A^{\mathsf T}$ `Chapter6/Projection · contMDiff_tanProj`.
- **Symmetry of the connection is symmetry of** II, a second-order property of the embedding that Part II never mentions `Chapter6/Submanifold · secondFundForm_comm`. That the ambient bracket is tangent and acts on functions as Milnor's bracket also needs proof `Chapter6/Submanifold · bracket_mem` `Chapter6/Submanifold · dN_bracket`.
- **Global existence of parallel fields needs another equation.** Milnor's appeal to linearity hides a global existence theorem, and $X' = P'X$ has no a priori bound. The skew equation $X' = (P'P - PP')X$ preserves norms, and a second application of uniqueness keeps its solutions tangent `Chapter6/Parallel · exists_isParallel`.
- **The order of the lemmas changes.** Lemma 8.3 is a direct computation and comes before parallel transport `Chapter6/Parallel · hasDerivAt_inner_covDeriv`. Lemma 8.1 needs global fields instead of coordinate fields `Chapter6/Parallel · IsCovDerivAlong.eq_sum`.
- **Milnor's "similar computations" hide different ones.** Tensoriality of R in Z uses the symmetry of second derivatives `Chapter6/Submanifold · curvature_eq`. Lemma 9.3 (2) and (3) need no extensions with vanishing brackets. Lemmas 8.7 and 9.2 need $DW/\partial x$ jointly smooth, which the formula $P(\partial W/\partial x)$ provides `Chapter6/Surface · covDerivX_eq`.

- **Signs.** Milnor's R is the negative of Nomizu's `Chapter6/Connection · milnorCurvature_eq_neg`. The sectional curvature is the same number in both conventions, and with Milnor's sign the Gauss equation makes the unit sphere curvature $+1$ `Chapter6/SphereCurvature · inner_curvTensor_sph_orthonormal`.

Part VI. Geodesics and completeness (Milnor §10)

Milnor's §10 (pp. 55–66) turns a Riemannian manifold into a metric space whose shortest paths are geodesics: geodesics solve a differential equation (Theorem 10.1, Lemma 10.2), are collected into the exponential map, a diffeomorphism near the origin of each tangent space uniformly in the base point (Lemma 10.3), are locally shortest (Theorem 10.4, from Lemmas 10.5 and 10.6), and shortest paths are geodesics (Corollary 10.7); the infimum of lengths is a distance ρ (Corollary 10.8), and geodesic and metric completeness agree (Theorem 10.9, Corollary 10.10 and its converse). This part retells the formal proof of every numbered statement from Theorem 10.1 to Corollary 10.10, of the converse of Corollary 10.10, of the four-way Hopf–Rinow equivalence, and of the completeness of closed submanifolds, among them the round sphere.

The setting is that of Part V. V is a finite-dimensional real inner product space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ a slice submanifold modelled on a finite-dimensional Z, with finite-dimensional complement W (§0.1, §0.3), with the induced metric. N need not be closed, compact or connected; connectedness is assumed exactly where it is used. Every Riemannian manifold is isometric to such an N by Nash's embedding theorem, which is neither in Mathlib nor proved here, so what follows are theorems about embedded manifolds. Two of Milnor's silent hypotheses disappear: N is Hausdorff, which uniqueness of geodesics and the closedness of normal balls need, and its topology is metric from the start.

The formal route differs from Milnor's in these places, each explained where it occurs. A geodesic is a curve whose acceleration in V is normal to N, and no Christoffel symbols are defined. Theorem 10.1 is proved, not quoted, with a cut-off and the flow library of Part I. Existence of geodesics is solved in a slice chart, uniqueness in $V \times V$ and on whole intervals. exp is defined through geodesics on open intervals containing $[0, 1]$. Lemma 10.3 uses the inverse function theorem. The Gauss lemma is proved for every direction, and Lemma 10.6 follows from it by Cauchy–Schwarz. Theorem 10.4 is proved for a class of paths closed under restriction, with a sharp equality case that Corollary 10.7 needs. ρ takes values in $[0, \infty]$ and its continuity is proved; the continuity of exp on its whole domain is proved by continuing geodesics; Theorem 10.9 needs completeness at one point only.

Map.

- `Chapter7/ODE`: Theorem 10.1, existence with smooth dependence and uniqueness on intervals.
- `Chapter7/Geodesic`: geodesics, the geodesic equation, the spray, constant speed, homogeneity, uniqueness.
- `Chapter7/GeodesicLocal`: the pseudo-inverse, the geodesic equation in a chart, Lemma 10.2.
- `Chapter7/Exp`: the exponential map, smoothness near the zero section, $d(\exp_q)_0 = \mathrm{id}$.
- `Chapter7/NormalNbhd`: normal radii, exp read in a chart, Milnor's Jacobian, Lemma 10.3, $\exp_q$ as a partial diffeomorphism.
- `Chapter7/GaussLemma`: the Gauss lemma and the pointwise form of Lemma 10.6.
- `Chapter7/Path`: piecewise smooth and admissible paths, their lengths.

- `Chapter7/Shortest`: Lemma 10.6, paths leaving a normal ball, Theorem 10.4 with its sharp equality case.
- `Chapter7/Minimal`: Corollary 10.7.
- `Chapter7/Distance`: the distance ρ, its axioms, normal balls as metric balls, Corollary 10.8.
- `Chapter7/Metric`: continuity and finiteness of ρ, the metric space (N, ρ), metric completeness.
- `Chapter7/GeodesicFlow`: the geodesic flow, continuation of geodesics, continuity of exp, geodesic completeness, the converse of Corollary 10.10.
- `Chapter7/HopfRinow`: minimal geodesics, Theorem 10.9, Corollary 10.10, the Hopf–Rinow equivalence, closed submanifolds and the sphere.

The labels V7.1–V7.30 are those of the verified edition.

VI.1 The setting: charts, tangent spaces, the projection and the second fundamental form

Recalled from §0.3 and Part V (§V.10–§V.14, §V.17), for use throughout.

Charts and parametrizations. A slice chart of N is a partial diffeomorphism σ from an open subset of V onto an open subset of $Z \times W$ which, on its domain, carries N exactly onto $Z \times \{0\}$ `Milnor/Chapter2/Submanifold · SliceChart`; N is a slice submanifold if every point of N lies in the domain of one `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`. For each $q \in N$ one such chart is chosen once and for all; call it σ_q `Chapter6/Projection · sliceAt Chapter6/Projection · mem_sliceAt_source`. A slice chart has a parametrization $x(u) = \sigma^{-1}(u, 0)$ `Chapter1/ManifoldSubmanifold · sliceParam`, meaningful on the open set $T_\sigma = \{u \in Z : (u, 0) \text{ is in the image of } \sigma\}$ `Chapter1/ManifoldSubmanifold · isOpen_toChart_target`. It is smooth there `Chapter1/ManifoldSubmanifold · contDiffOn_sliceParam`, takes values in N `Chapter1/ManifoldSubmanifold · sliceParam_mem`, satisfies $x((\sigma q)_1) = q$ for $q \in N$ in the domain of σ `Chapter1/ManifoldSubmanifold · sliceParam_toChart`, and has injective derivative, as one sees by differentiating $\sigma(x(u)) = (u, 0)$ `Chapter6/Projection · injective_fderiv_sliceParam`. Here and below $(a, b)_1 = a$ and $(a, b)_2 = b$.

Tangent spaces. $T_qN \subseteq V$ is the range of the derivative at q of the inclusion $N \to V$ `Chapter1/ManifoldSubmanifold · tangentSubspace`. For any slice chart σ around q it is the range of dx at $(\sigma q)_1$ `Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam` and the kernel of $v \mapsto (d\sigma_q v)_2$ `Milnor/Chapter8/TangentBundle · sliceChart_mem_tangentSpace_iff`. Formally T_qN is indexed by a point of N (a point with a proof of membership), and equal points have equal tangent spaces `Chapter7/Geodesic · tangentSubspace_congr`.

The tangent projection. P_q is the orthogonal projection of V onto T_qN `Chapter6/Projection · tanProj`. In a chart it is a formula. For an injective linear map A into V, the operator $A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ `Chapter6/Projection · gramProj` is the orthogonal projection onto the range of A `Chapter6/Projection · gramProj_eq_starProjection` and depends smoothly on A `Chapter6/Projection · contDiffAt_gramProj`. Taking $A = dx$ at $(\sigma y)_1$ defines an operator $Q_\sigma(y)$ for every y in the domain of σ `Chapter6/Projection · projRep`. It equals P_y for $y \in N$ `Chapter6/Projection · projRep_eq` and is smooth near every point of N in the domain `Chapter6/Projection · contDiffAt_projRep`. So Q_σ extends $q \mapsto P_q$ smoothly to an open set of V (used in §VI.4).

The second fundamental form. For $v \in T_qN$ let D_v be the derivative of $q \mapsto P_q$ along N in the direction v `Chapter6/Projection · dN`; it is the ambient derivative of any smooth extension, $D_v = dQ_\sigma(v)$ at q `Chapter6/Projection · dN_tanProj_eq_fderiv`. The second fundamental form is $\mathrm{II}_q(v, w) = D_v w$ `Chapter6/Submanifold · secondFundForm`, and for tangent v, w it is normal to N `Chapter6/Submanifold · secondFundForm_mem_orthogonal`. The covariant derivative of a field X along a curve c in N is $\frac{D}{dt}X = P_{c(t)}X'(t)$ `Chapter6/Parallel · covDeriv`, and the velocity of c is c' `Chapter6/Parallel · velocity` (Part V).

The tangent bundle. $TN = \{(q, v) \in V \times V : q \in N,\ v \in T_qN\}$ `Milnor/Chapter8/TangentBundle · tangentBundleSet` is a slice submanifold of $V \times V$ modelled on $Z \times Z$ `Milnor/Chapter8/`

`TangentBundle · isSliceSubmanifold_tangentBundle` (Problem 10 of the companion volume). It is Milnor's tangent manifold (p. 58).

Three small lemmas.

- *Velocities are tangent.* If $c : \mathbb{R} \to V$ lies in N near t and is differentiable at t, then $c'(t) \in T_{c(t)}N$ `Chapter7/Geodesic · hasDerivAt_mem_tangentSubspace`. *Proof.* Let $\sigma = \sigma_{c(t)}$. For s near t, $c(s)$ lies in N and in the domain of σ, so the W-component of $\sigma(c(s))$ vanishes identically near t. Its derivative at t, which is $(d\sigma\, c'(t))_2$ by the chain rule, is therefore 0, and the kernel description of $T_{c(t)}N$ applies.
- *The projection along a curve.* Under the same hypotheses, with $\sigma = \sigma_{c(t)}$, the operator-valued curve $s \mapsto Q_\sigma(c(s))$ has derivative $D_{c'(t)}$ at t `Chapter7/Geodesic · hasDerivAt_projRep_comp`: this is the chain rule for the smooth Q_σ, together with $dQ_\sigma(v) = D_v$ for tangent v.
- *Curves in a subspace.* A curve that stays near t in a linear subspace K of V and is differentiable at t has its derivative in K `Chapter7/GaussLemma · hasDerivAt_mem_of_eventually_mem`: the projection onto K fixes the curve near t, hence fixes its derivative.

VI.2 Geodesics, the geodesic equation, constant speed and homogeneity

Definition (V7.1; §0.9). Let $s \subseteq \mathbb{R}$ be open, in practice an open interval. A curve $\gamma : \mathbb{R} \to V$ is a *geodesic on s* if γ is C^∞ on s, $\gamma(t) \in N$ for every $t \in s$, and the acceleration $\gamma''(t)$ is orthogonal to $T_{\gamma(t)}N$ for every $t \in s$ `Chapter7/Geodesic · IsGeodesicOn`.

Curves are maps on all of $\mathbb{R}$, but only their values on s matter: a curve that agrees on the open set s with a geodesic is one `Chapter7/Geodesic · IsGeodesicOn.congr`. A geodesic on s is a geodesic on every open subset of s `Chapter7/Geodesic · IsGeodesicOn.mono`, constant curves at points of N are geodesics `Chapter7/Geodesic · isGeodesicOn_const`, and being a geodesic is a local property: if every $t \in s$ has an open neighbourhood $s' \subseteq s$ on which γ is a geodesic, then γ is a geodesic on s `Chapter7/Minimal · isGeodesicOn_of_locally`, because smoothness on an open set is local `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_of_locally_contDiffOn` and the other two conditions are pointwise. Milnor allows any interval; a geodesic on a closed interval is represented by one on an open interval containing it (§VI.6).

The velocity of a geodesic is tangent, $\gamma'(t) \in T_{\gamma(t)}N$ `Chapter7/Geodesic · IsGeodesicOn.deriv_mem` (the first small lemma of §VI.1), and the normality of γ'' says exactly that $P_{\gamma(t)}\gamma''(t) = 0$ `Chapter7/Geodesic · IsGeodesicOn.tanProj_deriv_deriv` `Chapter6/Projection · tanProj_eq_zero_iff`. As $\frac{D}{dt} = P \circ \frac{d}{dt}$, this is Milnor's definition: a smooth curve $c : \mathbb{R} \to N$ is a geodesic on $\mathbb{R}$ if and only if $\frac{D}{dt}\frac{dc}{dt} = 0$ at every time `Chapter7/Geodesic · isGeodesicOn_univ_iff_covDeriv`.

Proposition (the geodesic equation, V7.3). Let γ be C^∞ on the open set s with $\gamma(s) \subseteq N$. Then γ is a geodesic on s if and only if

$$\gamma''(t) = \mathrm{II}_{\gamma(t)}(\gamma'(t), \gamma'(t)) \qquad (t \in s)$$

`Chapter7/Geodesic · IsGeodesicOn.deriv_deriv_eq` `Chapter7/Geodesic · isGeodesicOn_of_deriv_deriv_eq`.

Proof. Let γ be a geodesic, $t \in s$, $q = \gamma(t)$, $\sigma = \sigma_q$ and $Q = Q_\sigma$. For τ near t, $\gamma(\tau)$ is in N and in the domain of σ, so $Q(\gamma(\tau)) = P_{\gamma(\tau)}$, and $\gamma'(\tau)$ is tangent; hence

$$Q(\gamma(\tau))\,\gamma'(\tau) = \gamma'(\tau) \qquad \text{for } \tau \text{ near } t$$

`Chapter6/Projection · projRep_eq` `Chapter6/Projection · tanProj_eq_self`. Differentiate at t with the product rule. The derivative of $\tau \mapsto Q(\gamma(\tau))$ is $D_{\gamma'(t)}$ (§VI.1), so the left side has derivative $D_{\gamma'(t)}\gamma'(t) + P_q\gamma''(t) = \mathrm{II}_q(\gamma'(t), \gamma'(t)) + P_q\gamma''(t)$, and the right side has derivative $\gamma''(t)$. By uniqueness of derivatives these agree, and $P_q\gamma''(t) = 0$. Conversely, if $\gamma'' = \mathrm{II}(\gamma', \gamma')$, then $\gamma'(t)$ is tangent (the curve lies in N near t), and II of two tangent vectors is normal, so $\gamma''(t)$ is normal. □

The geodesic spray. The vector field

$$(x,v)\longmapsto (v,\mathrm{II}_x(v,v))\quad (x\in N),\qquad (x,v)\longmapsto (v,0)\quad (x\notin N)$$

on $V\times V$ `Chapter7/Geodesic · geodesicSpray` is the geodesic spray; its values off N are a placeholder and never used. The lift $t\mapsto(\gamma(t),\gamma'(t))$ of a geodesic lies in TN `Chapter7/Geodesic · IsGeodesicOn.mem_tangentBundleSet`, and a smooth curve γ in N on an open set s is a geodesic if and only if its lift is an integral curve of the spray on s `Chapter7/Geodesic · isGeodesicOn_iff_hasDerivAt_geodesicSpray`. This is Milnor's first-order system (p. 56), and the proof is the geodesic equation read in both directions. As the spray involves the derivative of P, which exists only along N, the equations actually solved are a smooth extension of it to an open set of $V\times V$ (§VI.4) and its expression in a chart (§VI.5).

Proposition (constant speed, V7.2). If γ is a geodesic on s, then $\frac{d}{dt}\|\gamma'(t)\|^2=0$ for $t\in s$ `Chapter7/Geodesic · IsGeodesicOn.hasDerivAt_norm_sq`. If s is preconnected (an open interval), the speed $\|\gamma'\|$ is constant on s `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq`.

Proof. $\frac{d}{dt}\|\gamma'\|^2=2\langle\gamma'',\gamma'\rangle$, and γ'' is normal while γ' is tangent. A function with zero derivative on an open preconnected subset of $\mathbb{R}$ is constant there `Mathlib/Analysis/Calculus/MeanValue · IsOpen.is_const_of_deriv_eq_zero`; apply it to $\|\gamma'\|^2$ and take square roots. □

This is Milnor's computation (p. 55). The length of a geodesic on $[a,b]\subseteq s$ is then its speed times $b-a$ (§VI.10).

Proposition (homogeneity, V7.2). If γ is a geodesic on s, then for all real a,b the curve $t\mapsto\gamma(at+b)$ is a geodesic on $\{t : at+b\in s\}$ `Chapter7/Geodesic · IsGeodesicOn.comp_mul_add`.

Proof. The preimage of an open set under a continuous map is open, and smoothness and membership in N are inherited. By the chain rule the velocity at t is $a\gamma'(at+b)$, and near any t of the preimage this holds identically, so the acceleration is $a^2\gamma''(at+b)$, a multiple of a normal vector. □

Milnor states the case $b=0$ (p. 57); shifts ($a=1$) and reversals ($a=-1$) are used below.

VI.3 Theorem 10.1: Milnor's facts about ordinary differential equations

Milnor's Theorem 10.1 (p. 56) gives, for $u''=F(u,u')$ with F smooth near $(u_1,v_1)\in\mathbb{R}^{2n}$, unique solutions for a uniform time, depending smoothly on the initial data; he passes to the system $u'=v$, $v'=F(u,v)$ and quotes Graves. The formal proof has four steps: a connectedness principle, local uniqueness by Grönwall's inequality, local existence with smooth dependence by a cut-off and the global flow of Part I (§I.18–§I.21), and the reduction to first order.

Lemma (a connectedness principle). Let $s\subseteq\mathbb{R}$ be open and preconnected, F a Hausdorff space, and $f,g:\mathbb{R}\to F$ continuous on s. Suppose $f(t_0)=g(t_0)$ for some $t_0\in s$, and that whenever f and g agree at a point of s they agree near it. Then $f=g$ on s `Chapter7/ODE · eqOn_of_eventuallyEq`.

Proof. The set u of times near which f and g agree is open `Mathlib/Topology/Neighborhoods · isOpen_setOfPred_eventually_nhds` and contains t_0. If $t\in s$ lies in the closure of u, then f and g agree at points arbitrarily close to t; both are continuous at t and limits in a Hausdorff space are unique, so $f(t)=g(t)$, and by hypothesis $t\in u$. An open set that meets s and contains every point of s in its closure contains the preconnected set s `Mathlib/Topology/Connected/Basic · IsPreconnected.subset_of_closure_inter_subset`. □

Lemma (local uniqueness). Let E be a real normed space and $G:E\to E$ of class C^1 at $f(t_0)$. If f and g solve $x'=G(x)$ near t_0 and $f(t_0)=g(t_0)$, then $f=g$ near t_0 `Chapter7/ODE · ode_eventuallyEq`.

Proof. A C^1 map is Lipschitz, with some constant K, on a neighbourhood S of the point `Mathlib/Analysis/Calculus/ContDiff/RCLike · ContDiffAt.exists_lipschitzOnWith`. Both solutions are continuous at t_0 with the common value $f(t_0)\in S$, so for some $\varepsilon>0$ both solve the equation and stay in S on $(t_0-\varepsilon,t_0+\varepsilon)$.

Grönwall's inequality, in Mathlib's form for a field that is K-Lipschitz on the sets where the solutions live, gives $f = g$ on that interval `Mathlib/Analysis/ODE/ExistUnique · ODE_solution_unique_of_mem_Ioo`. □

Lemma (uniqueness on an interval). Let G be C^1 on an open set $U \subseteq E$ and $s \subseteq \mathbb{R}$ open and preconnected. Two solutions of $x' = G(x)$ on s with values in U that agree at one time of s agree on s `Chapter7/ODE · ode_eqOn`.

Proof. Solutions are continuous; where they agree, local uniqueness applies (at a point of U, where G is C^1); the connectedness principle spreads the agreement over s. □

Proposition (local existence with smooth dependence). Let E be finite-dimensional, $U \subseteq E$ open, G smooth on U and $x_1 \in U$. There are a smooth map $\varphi : \mathbb{R} \times E \to E$ and numbers $r, \tau > 0$ such that for every x with $\|x - x_1\| < r$: $\varphi(0, x) = x$, and for $|t| < \tau$, $\varphi(t, x) \in U$ and $\frac{\partial}{\partial t}\varphi(t, x) = G(\varphi(t, x))$ `Chapter7/ODE · exists_localFlow`.

Proof.

1. *Cut-off.* Choose $\rho > 0$ with the ball $B(x_1, \rho) \subseteq U$ and put $R = \rho/3$. Mathlib's smooth bump function χ with inner radius R and outer radius $2R$ about x_1 `Mathlib/Analysis/Calculus/BumpFunction/Basic · ContDiffBump` equals 1 on the closed ball $\bar{B}(x_1, R)$ and has support in $\bar{B}(x_1, 2R) \subseteq U$.
2. *A global field.* $X = \chi G$ is smooth on all of E: at points of U it is a product of smooth maps, and a point outside U lies outside the closed support of χ, near which X vanishes identically. X has compact support, so it is bounded, say by $M \geq 0$, and globally Lipschitz `Flow/Global · IsBddLipschitz.of_contDiff`.
3. *Its flow.* By the flow library (Part I, §I.18–§I.21), X has a global flow $\varphi = \varphi^X$ `Flow/Global · flow` with $\varphi(0, x) = x$ `Flow/Global · IsBddLipschitz.flow_zero` whose flow lines are integral curves `Flow/Global · IsBddLipschitz.hasDerivAt_flow`, and $(t, x) \mapsto \varphi(t, x)$ is smooth `Flow/Smooth · contDiff_flow_uncurry`.
4. *The flow lines stay where $\chi = 1$.* The flow lines are M-Lipschitz in time, $\|\varphi(t, x) - x\| \leq M|t|$ `Flow/Global · IsBddLipschitz.dist_flow_flow_le`. Put $r = R/2$ and $\tau = R/(2(M + 1))$. For $\|x - x_1\| < R/2$ and $|t| < \tau$ one gets $M|t| < R/2$, so $\|\varphi(t, x) - x_1\| < R$. There $\chi = 1$, so $X = G$ at $\varphi(t, x)$, which lies in U, and the flow line solves $x' = G(x)$. □

The uniform time τ, which Milnor's product $W \times (-\varepsilon, \varepsilon)$ requires, falls out of the bound on the speed.

Theorem 10.1 (V7.4). Let E be a finite-dimensional real normed space (Milnor's $\mathbb{R}^n$), $U \subseteq E \times E$ open, $F : E \times E \to E$ smooth on U and $z_1 \in U$.

1. (Existence and smooth dependence.) There are a neighbourhood W of z_1, an $\varepsilon > 0$ and a map $(z, t) \mapsto u_z(t)$, smooth on $W \times (-\varepsilon, \varepsilon)$, such that for every $z = (u_0, v_0) \in W$: $u_z(0) = u_0$, $u_z'(0) = v_0$, and for $|t| < \varepsilon$, $(u_z(t), u_z'(t)) \in U$ and $u_z''(t) = F(u_z(t), u_z'(t))$ `Chapter7/ODE · theorem_10_1`.
2. (Uniqueness.) Here E may be any real normed space and F need only be C^1 on U. If (u_1, v_1) and (u_2, v_2) satisfy $u_i' = v_i$, $v_i' = F(u_i, v_i)$ and $(u_i, v_i) \in U$ on an open preconnected $s \subseteq \mathbb{R}$, and $u_1(t_0) = u_2(t_0)$, $v_1(t_0) = v_2(t_0)$ for some $t_0 \in s$, then $u_1 = u_2$ on s `Chapter7/ODE · theorem_10_1_unique`.

Proof. Both parts pass to $G(u, v) = (v, F(u, v))$ on U, smooth (respectively C^1) where F is. For (1), take φ, r, τ from the proposition at z_1, put $W = B(z_1, r)$, $\varepsilon = \tau$ and let $u_z(t)$ be the first component of $\varphi(t, z)$, a smooth function of (z, t) everywhere. For $z \in W$ and $|t| < \tau$, the first component of an integral curve of G has as derivative the second component `Chapter7/ODE · hasDerivAt_fst_comp`, so $u_z' = (\varphi(\cdot, z))_2$ near t; hence $(u_z(t), u_z'(t)) = \varphi(t, z) \in U$, $u_z'(0) = v_0$, and the derivative of u_z' is the second component of $G(\varphi(t, z))$ `Chapter7/ODE · hasDerivAt_snd_comp`, which is $F(u_z(t), u_z'(t))$. For (2), the pairs (u_i, v_i) are solutions of $x' = G(x)$ in U that agree at t_0, so they agree on s by uniqueness on an interval; take first components. □

Part (2) is stronger than Milnor's uniqueness for small times: it holds on every common open interval,

for C^1 equations on any normed space.

VI.4 Uniqueness of geodesics

Theorem (uniqueness of geodesics, V7.5). (1) Let γ_1, γ_2 be geodesics on open sets s_1, s_2 containing t_0, with $\gamma_1(t_0) = \gamma_2(t_0)$ and $\gamma_1'(t_0) = \gamma_2'(t_0)$. Then $\gamma_1 = \gamma_2$ near t_0 `Chapter7/Geodesic · IsGeodesicOn.eventuallyEq`. (2) If γ_1, γ_2 are geodesics on the same open preconnected set s (an open interval) with the same position and velocity at one time of s, then $\gamma_1 = \gamma_2$ on s `Chapter7/Geodesic · IsGeodesicOn.eqOn`.

Proof of (1). II is defined only on N; the proof replaces it by an expression defined on an open set of $V \times V$.

1. Let $q = \gamma_1(t_0)$, $\sigma = \sigma_q$ and $Q = Q_\sigma$, the smooth extension of P near q (§VI.1). Define
$$G(x, v) = (v,\ dQ_x(v)\, v) \qquad (x, v \in V),$$
where $dQ_x(v)$ is the derivative of the operator-valued map Q at x in the direction v, an operator, applied to v. Since Q is smooth near q, so is dQ, and G is C^1 (indeed smooth) at $(q, \gamma_1'(t_0))$.
2. For each i, near t_0 the point $\gamma_i(t)$ lies in the domain of σ, and there the geodesic equation and $D_v = dQ(v)$ for tangent v `Chapter6/Projection · dN_tanProj_eq_fderiv` give
$$\gamma_i''(t) = \mathrm{II}_{\gamma_i(t)}(\gamma_i'(t), \gamma_i'(t)) = dQ_{\gamma_i(t)}(\gamma_i'(t))\, \gamma_i'(t).$$
So both lifts $t \mapsto (\gamma_i(t), \gamma_i'(t))$ solve $x' = G(x)$ near t_0. The same chart σ serves both curves, because the identity $D_v = dQ_\sigma(v)$ holds at every point of N in the domain of σ, not only at q.
3. The lifts agree at t_0, so they agree near t_0 by local uniqueness; take first components. □

Proof of (2). Apply the connectedness principle to the lifts $t \mapsto (\gamma_i(t), \gamma_i'(t))$, which are continuous on s because γ_i and γ_i' are smooth there. If the lifts agree at $t \in s$, part (1) gives $\gamma_1 = \gamma_2$ near t; then also $\gamma_1' = \gamma_2'$ near t, so the lifts agree near t. □

What Milnor leaves implicit. "The unique geodesic", and the uniqueness of $\exp_q(v)$ when defined (p. 58), need uniqueness on whole intervals, not only for small times in one chart; on an abstract manifold also the Hausdorff property. The verified edition calls this Lemma A.

VI.5 The geodesic equation in a chart, and Lemma 10.2

The extension Q of §VI.4 makes the geodesic equation an ordinary differential equation on an open set of $V \times V$, but nothing would keep its solutions in TN: the invariance criterion of the flow library needs a closed set, and TN need not be closed in $V \times V$. So existence is solved in a chart, where solutions lie in N by construction.

The pseudo-inverse. For a linear map $A : Y \to V$, Y finite-dimensional, put $A^+ = (A^\mathsf{T}A)^{-1}A^\mathsf{T}$, where A^T is the transpose and $A^\mathsf{T}A$ the Gram operator `Chapter7/GeodesicLocal · pinvL` `Chapter6/Projection · transposeL` `Chapter6/Projection · gramL`. If A is injective then $A^+A = \mathrm{id}$ `Chapter7/GeodesicLocal · pinvL_apply_self` (the Gram operator is then injective `Chapter6/Projection · gramL_injective`, hence invertible), and by definition AA^+ is the operator $A(A^\mathsf{T}A)^{-1}A^\mathsf{T}$ of §VI.1 `Chapter7/GeodesicLocal · gramProj_eq_comp_pinvL`, the orthogonal projection onto the range of A. Near an injective A, $A \mapsto A^+$ is smooth `Chapter7/GeodesicLocal · contDiffAt_pinvL`: the transpose and the Gram operator depend smoothly on A, and inversion is smooth at an invertible operator `Mathlib/Analysis/Calculus/ContDiff/Operations · contDiffAt_map_inverse`.

Chart facts. Let σ be a slice chart with parametrization x on $T = T_\sigma$. Then $\sigma(x(u)) = (u, 0)$ `Chapter7/GeodesicLocal · sliceChart_sliceParam`; differentiating, $d\sigma \circ dx = (\,\cdot\,, 0)$ `Chapter7/GeodesicLocal · fderiv_sliceChart_comp_fderiv_sliceParam`, that is $(d\sigma\, dx_u w)_1 = w$ `Chapter7/GeodesicLocal · fst_fderiv_sliceChart_fderiv_sliceParam`. The tangent space at $x(u)$ is the range of dx_u `Chapter7/GeodesicLocal · tangentSubspace_`

sliceParam. And the chart coordinates of a tangent vector $v \in T_pN$ are $(d\sigma_p v)_1$, in the sense that $dx_{(\sigma p)_1}((d\sigma_p v)_1) = v$ Chapter7/GeodesicLocal · fderiv_sliceParam_fst_fderiv: write $v = dx\, z$ and use the previous identity.

The chart equation. Define the Christoffel term and the chart spray

$$\Gamma(u)(w,w) = (dx_u)^+(d^2x_u(w,w)), \qquad (u,w) \longmapsto (w,\ -\Gamma(u)(w,w))$$

Chapter7/GeodesicLocal · chartAcc Chapter7/GeodesicLocal · chartSpray. The term is quadratic in w: $\Gamma(u)(aw,aw) = a^2\,\Gamma(u)(w,w)$ Chapter7/GeodesicLocal · chartAcc_smul and $\Gamma(u)(0,0) = 0$ Chapter7/GeodesicLocal · chartAcc_zero. The spray is smooth on $T \times Z$ Chapter7/GeodesicLocal · contDiffOn_chartSpray: dx and d^2x are smooth on T, and $u \mapsto (dx_u)^+$ is smooth because each dx_u is injective.

$(dx_u)^+ d^2x_u(w,w)$ is the coordinate vector of the tangential part of $d^2x_u(w,w)$, with components $\sum_{i,j}\Gamma^k_{ij}w^iw^j$ for the Christoffel symbols of p. 55; as no Christoffel symbols are defined formally, this is a remark, not a theorem.

Proposition (solutions of the chart equation are geodesics, V7.3). Let $\zeta = (u,w)$ be C^∞ on an open set s, with $u(t) \in T$ and $\zeta'(t) = (w(t), -\Gamma(u(t))(w(t),w(t)))$ for $t \in s$. Then $t \mapsto x(u(t))$ is a geodesic on s Chapter7/GeodesicLocal · isGeodesicOn_chart, with velocity $dx_{u(t)}w(t)$ Chapter7/GeodesicLocal · hasDerivAt_sliceParam_chart.

Proof. The curve lies in N and is smooth as a composite. Its velocity is $dx_u u' = dx_u w$ by the chain rule, and this holds near every t, so its acceleration is the derivative of $t \mapsto dx_{u(t)}w(t)$:

$$d^2x_u(w,w) + dx_u w' = y - dx_u(dx_u)^+ y = y - P\,y, \qquad y = d^2x_u(w,w),$$

where $dx_u(dx_u)^+$ is the orthogonal projection onto the range of dx_u Chapter6/Projection · gramProj_eq_starProjection, which is the tangent space at $x(u)$. So the acceleration is y minus its tangential part, a normal vector. □

Only this direction is stated in the chart; it is what existence needs, and uniqueness comes from §VI.4.

A local flow of the chart spray. For $u_0 \in T$, a *local spray* at u_0 Chapter7/GeodesicLocal · LocalSpray consists of a map $\varphi : \mathbb{R} \times (Z \times Z) \to Z \times Z$, smooth in all variables, and numbers $r, \tau > 0$, such that for every z in the ball $B((u_0,0),r)$ (for the maximum norm on $Z \times Z$): $\varphi(0,z) = z$, and for $|t| < \tau$, the first component of $\varphi(t,z)$ lies in T and $t \mapsto \varphi(t,z)$ solves the chart equation. One exists Chapter7/GeodesicLocal · nonempty_localSpray, by the local existence proposition of §VI.3 applied to the chart spray on the open set $T \times Z$ at $(u_0,0)$.

Milnor's rescaling. Let L be a local spray at u_0 and $c = \tau/2$ Chapter7/GeodesicLocal · LocalSpray.c. Milnor obtains geodesics on $(-2,2)$ by rescaling, $\gamma_v(t) = \gamma_{v/c}(ct)$ (pp. 56–57). Formally:

$$\mathrm{geo}(u,w)(t) = x\Big(\big(\varphi(ct,(u,w/c))\big)_1\Big)$$

Chapter7/GeodesicLocal · LocalSpray.geo, with chart position and velocity $\big((\varphi(ct,(u,w/c)))_1,\ c\,(\varphi(ct,(u,w/c)))_2\big)$ Chapter7/GeodesicLocal · LocalSpray.chartCurve. The admissible chart data are $\mathcal{G} = B(u_0,r) \times B(0,cr)$ Chapter7/GeodesicLocal · LocalSpray.good, an open set containing $(u_0,0)$ Chapter7/GeodesicLocal · LocalSpray.isOpen_good Chapter7/GeodesicLocal · LocalSpray.mem_good. For $(u,w) \in \mathcal{G}$ the rescaled datum $(u,w/c)$ lies in $B((u_0,0),r)$ Chapter7/GeodesicLocal · LocalSpray.mem_ball_of_good, and for $|t| < 2$, $|ct| < \tau$ Chapter7/GeodesicLocal · LocalSpray.mul_mem_Ioo. Then:

- the rescaled chart curve solves the chart equation on $(-2,2)$ Chapter7/GeodesicLocal · LocalSpray.hasDerivAt_chartCurve. With $\xi = \varphi(ct,(u,w/c))$, the chain rule gives the derivative $(c\,\xi_2,\ -c^2\,\Gamma(\xi_1)(\xi_2,\xi_2))$, because ξ solves the chart equation; by quadratic homogeneity this is the chart spray at the rescaled point $(\xi_1, c\,\xi_2)$;
- it starts at (u,w) Chapter7/GeodesicLocal · LocalSpray.chartCurve_zero;

- hence $\mathrm{geo}(u,w)$ is a geodesic on $(-2,2)$ `Chapter7/GeodesicLocal · LocalSpray.isGeodesicOn_geo` with $\mathrm{geo}(u,w)(0) = x(u)$ `Chapter7/GeodesicLocal · LocalSpray.geo_zero` and initial velocity $dx_u w$ `Chapter7/GeodesicLocal · LocalSpray.hasDerivAt_geo_zero`;
- $(u,w,t) \mapsto \mathrm{geo}(u,w)(t)$ is smooth on $\mathcal{G} \times (-2,2)$ `Chapter7/GeodesicLocal · LocalSpray.contDiffOn_geo`, a composite of φ, linear maps and x, the first component staying in T.

Lemma 10.2 (V7.6). For every $q_0 \in N$ there are an open set $U \ni q_0$ of V, an $\varepsilon > 0$ and a map $\Theta : V \times V \times \mathbb{R} \to V$, smooth on $U \times B(0,\varepsilon) \times (-2,2)$, such that for every $p \in N \cap U$ and every $v \in T_pN$ with $\|v\| < \varepsilon$, the curve $\gamma_v = \Theta(p,v,\cdot)$ is a geodesic on $(-2,2)$ with $\gamma_v(0) = p$ and $\gamma_v'(0) = v$, and every geodesic on $(-2,2)$ with this position and velocity at 0 equals γ_v on $(-2,2)$ `Chapter7/GeodesicLocal · lemma_10_2`.

Proof.

1. Let $\sigma = \sigma_{q_0}$, $u_0 = (\sigma q_0)_1 \in T$ `Chapter1/ManifoldSubmanifold · toChart_apply_mem_target`, and L a local spray at u_0, with r, τ, c.
2. *From lengths to coordinates.* Put $C = \|d\sigma_{q_0}\| + 1$ and
$$U = \{p \in \operatorname{dom}\sigma : (\sigma p)_1 \in B(u_0,r),\ \|d\sigma_p\| < C\},$$
open because $p \mapsto ((\sigma p)_1, d\sigma_p)$ is continuous on the open domain of σ, and containing q_0. Put $\varepsilon = cr/C$. For $p \in U$ and $\|v\| < \varepsilon$,
$$\|(d\sigma_p v)_1\| \le \|d\sigma_p v\| \le \|d\sigma_p\|\,\|v\| < C\varepsilon = cr,$$
so the chart data $((\sigma p)_1, (d\sigma_p v)_1)$ lie in $\mathcal{G}$.
3. Define $\Theta(p,v,t) = \mathrm{geo}((\sigma p)_1, (d\sigma_p v)_1)(t)$. It is smooth on $U \times B(0,\varepsilon) \times (-2,2)$, as the smooth map of the last list composed with the smooth map $(p,v,t) \mapsto (((\sigma p)_1, (d\sigma_p v)_1), t)$, which lands in $\mathcal{G} \times (-2,2)$ by step 2.
4. For $p \in N \cap U$ and tangent v with $\|v\| < \varepsilon$: $\Theta(p,v,\cdot)$ is a geodesic on $(-2,2)$; its value at 0 is $x((\sigma p)_1) = p$; its velocity at 0 is $dx((d\sigma_p v)_1) = v$ by the chart facts. Uniqueness is the theorem of §VI.4 on the interval $(-2,2)$. □

What Milnor leaves implicit. To pass from the coordinate neighbourhood W of Theorem 10.1 to "$p \in U$, $\|v\| < \varepsilon_1$" one must bound the coordinates of v by $\|v\|$, uniformly near p_0: step 2. The smooth dependence of $\gamma_v(t)$ on (p,v,t) is not in Milnor's statement but is used on p. 58; the smoothness of exp near the zero section rests on it.

VI.6 The exponential map

Milnor (p. 57) puts $\exp_q(v) = \gamma(1)$ for the geodesic $\gamma : [0,1] \to M$ with $\gamma(0) = q$ and $\frac{d\gamma}{dt}(0) = v$, if there is one, and observes $\gamma(t) = \exp_q(tv)$. The formal definition replaces $[0,1]$, with its one-sided derivatives at the ends, by an open interval around it.

Definition (V7.7; §0.9). For $q, v \in V$, a curve γ is a *geodesic from q with initial velocity v* if for some $\delta > 0$ it is a geodesic on $(-\delta, 1+\delta)$ with $\gamma(0) = q$ and $\gamma'(0) = v$ `Chapter7/Exp · IsGeodesicFrom`. The *domain of* exp is the set of pairs (q,v) for which such a geodesic exists `Chapter7/Exp · expDom`, and
$$\exp_q(v) = \gamma(1)$$
for one of them, chosen; off the domain $\exp_q(v) = q$, a placeholder `Chapter7/Exp · exp`.

If γ is a geodesic from q with velocity v, then (q,v) is in the domain `Chapter7/Exp · IsGeodesicFrom.mem_expDom`, $q \in N$ `Chapter7/Exp · IsGeodesicFrom.mem` and $v \in T_qN$; and $\exp_q(v) \in N$ whenever (q,v) is in the domain

Chapter7/HopfRinow · exp_mem_of_mem_expDom. A geodesic on any open preconnected set s containing 0 and 1 is a geodesic from $\gamma(0)$ with initial velocity $\gamma'(0)$ Chapter7/Exp · isGeodesicFrom_of_isGeodesicOn: s contains balls around 0 and 1 and, being an interval, everything between, so it contains $(-\delta, 1+\delta)$ for small δ.

Proposition (exp **is well defined, V7.7).** Two geodesics from q with initial velocity v agree on an open interval containing $[0,1]$ Chapter7/Exp · IsGeodesicFrom.eqOn. Consequently $\exp_q(v) = \gamma(1)$ for *every* geodesic γ from q with initial velocity v Chapter7/Exp · IsGeodesicFrom.exp_eq Chapter7/Exp · exp_eq_of_mem_expDom.

Proof. Restrict both to $(-\delta, 1+\delta)$ with δ the smaller of their two margins and apply uniqueness of geodesics (§VI.4) at $t = 0$. The chosen geodesic in the definition of exp is one of them. □

This is Milnor's remark that $\exp_q(v)$, if defined, is uniquely defined (p. 58).

For $q \in N$ the constant curve is a geodesic from q with velocity 0 Chapter7/Exp · isGeodesicFrom_const, so $(q, 0)$ is in the domain Chapter7/Exp · zero_mem_expDom and $\exp_q(0) = q$ Chapter7/Exp · exp_zero.

Proposition ($\gamma(t) = \exp_q(tv)$, V7.7). Let γ be a geodesic from q with initial velocity v, on $(-\delta, 1+\delta)$. For each t in this interval, $s \mapsto \gamma(ts)$ is a geodesic from q with initial velocity tv, and $\exp_q(tv) = \gamma(t)$ Chapter7/Exp · IsGeodesicFrom.exp_smul. In particular $t \mapsto \exp_q(tv)$ is a geodesic on $(-\delta, 1+\delta)$, equal there to γ Chapter7/Exp · IsGeodesicFrom.isGeodesicOn_exp_smul.

Proof. By homogeneity (§VI.2), $\eta(s) = \gamma(ts)$ is a geodesic on $\{s : ts \in (-\delta, 1+\delta)\}$. This set is convex, hence preconnected, and contains 0 and 1 (the latter because t is in the interval). So η is a geodesic from $\eta(0) = q$ with initial velocity $\eta'(0) = t\gamma'(0) = tv$ (chain rule), and $\exp_q(tv) = \eta(1) = \gamma(t)$ by well-definedness. The second statement: a curve equal on an open set to a geodesic is a geodesic there. □

The same argument works for any geodesic, not only near $[0,1]$: if γ is a geodesic on an open interval $s \ni 0$, then

$$\exp_{\gamma(0)}\left(t\,\gamma'(0)\right) = \gamma(t) \qquad (t \in s)$$

Chapter7/GeodesicFlow · IsGeodesicOn.exp_smul_eq Chapter7/GeodesicFlow · IsGeodesicOn.eqOn_exp_smul, the preimage of s under $\sigma \mapsto t\sigma$ being again an interval containing 0 and 1 Chapter7/GeodesicFlow · isPreconnected_preimage_mul_add. So the radial curves $t \mapsto \exp_y(tw)$ are all the geodesics there are; §VI.16 builds the geodesic flow on this observation.

The two definitions of the domain. A geodesic on an open interval containing $[0,1]$ restricts to one on $[0,1]$ in Milnor's sense, with the same value at 1. Conversely a geodesic on $[0,1]$ extends beyond both ends, so the two domains coincide; this is proved later, with the tools of Chapters 9 and 11 Chapter9/GeodesicVariation · exists_initial_ray Chapter11/ExpSmooth · mem_expDom_of_isGeodesicOn, and nothing in this part needs it.

VI.7 Smoothness near the zero section, and $d(\exp_q)_0 = \mathrm{id}$

Milnor (p. 58) says that, by Lemma 10.2, $(q, v) \mapsto \exp_q(v)$ is defined and differentiable on a neighbourhood of $(p, 0)$ in TM. For a subset of $V \times V$, "smooth" means what it means for Milnor: near each point, the restriction of a smooth map on an open set.

Proposition (Lemma 10.2 in terms of exp**, V7.8).** For every $q_0 \in N$ there are an open $U \ni q_0$, an $\varepsilon > 0$ and a map $F : V \times V \to V$, smooth on $U \times B(0,\varepsilon)$, such that for every $p \in N \cap U$ and $v \in T_pN$ with $\|v\| < \varepsilon$, (p, v) is in the domain of exp and $\exp_p(v) = F(p, v)$ Chapter7/Exp · exists_exp_eq.

Proof. Take U, ε, Θ from Lemma 10.2 and $F(p,v) = \Theta(p, v, 1)$, smooth because $1 \in (-2, 2)$. The geodesic $\Theta(p, v, \cdot)$ on $(-2, 2)$ is a geodesic from p with initial velocity v (take $\delta = 1$), so $\exp_p(v) = \Theta(p, v, 1)$. □

Theorem (exp **near the zero section, V7.8).** There is an open set $O \subseteq V \times V$ containing $(q, 0)$ for every $q \in N$ such that every point of $TN \cap O$ is in the domain of exp, and near each point of $TN \cap O$, exp is the restriction to TN of a smooth map on an open set of $V \times V$ `Chapter7/Exp · exists_isOpen_exp`. Moreover $(q, v) \mapsto \exp_q(v)$ is smooth on the open subset $TN \cap O$ of the manifold TN `Chapter7/Exp · contMDiffOn_exp`.

Proof. Choose, for each $q_0 \in N$, the data $U(q_0)$, $\varepsilon(q_0)$, F_{q_0} of the last proposition, and let O be the union of the open sets $U(q_0) \times B(0, \varepsilon(q_0))$. It contains every $(q, 0)$. A point z of $TN \cap O$ lies in one of these sets, so it is in the domain, and on that open set $\exp = F_{q_0}$ along TN. For the manifold statement: on the open neighbourhood of z in TN cut out by that set, exp agrees with F_{q_0} composed with the inclusion $TN \to V \times V$, which is smooth `Milnor/Chapter2/Submanifold · IsSliceSubmanifold.contMDiff_val`. □

This is stronger than Milnor's statement in one respect: one open set serves the whole zero section. The smoothness of exp on its whole domain, not only near the zero section, is proved in Chapter 11 `Chapter11/ExpSmooth · exists_contDiffOn_exp` `Chapter11/ExpSmooth · contMDiffOn_exp_expDom`; §VI.16 proves the continuity on the whole domain that Corollary 10.10 needs.

Theorem ($d(\exp_q)_0 = \mathrm{id}$**, V7.8).** For $q \in N$, the map $\exp_q$, regarded as a map on the vector space T_qN, has at 0 the derivative the inclusion $T_qN \to V$ `Chapter7/Exp · hasFDerivAt_exp_zero`.

Proof.

1. Take U, ε, F from the proposition at $q_0 = q$ and let $F_q = F(q, \cdot)$, smooth near 0. On the ball of radius ε in T_qN, $\exp_q = F_q$.
2. *Radial directions.* Let $v \in T_qN$ with $\|v\| < \varepsilon$, and γ a geodesic from q with velocity v. For $|t| < 1$, $\|tv\| < \varepsilon$, so $F_q(tv) = \exp_q(tv) = \gamma(t)$ for t near 0 (§VI.6). The curve $t \mapsto F_q(tv)$ has derivative $dF_q(0)\, v$ at 0 by the chain rule, and γ has derivative v. So $dF_q(0)\, v = v$.
3. *All directions.* For $v \neq 0$ in T_qN, apply step 2 to av with $a = \varepsilon/(2\|v\|)$ and divide by a, using linearity. So $dF_q(0)$ restricted to T_qN is the inclusion.
4. The derivative at 0 of F_q composed with the inclusion of T_qN is therefore the inclusion, and $\exp_q$ agrees with this composite near 0. □

In Milnor's computation (p. 58), the Jacobian of $(q, v) \mapsto (q, \exp_q v)$ at $(p, 0)$ has $F_*(\partial/\partial t^j) = \partial/\partial u_2^j$, which is this statement.

VI.8 Normal radii and Lemma 10.3

Definition (normal coordinates, V7.9). Let $q \in N$ and $\varepsilon > 0$, and write $B_\varepsilon(q) = \{v \in T_qN : \|v\| < \varepsilon\}$. *Normal coordinates of radius ε at q* consist of open sets $O, U \subseteq V$ and maps F, smooth on O, and G, smooth on U, such that `Chapter7/NormalNbhd · NormalCoord`:

- for $v \in B_\varepsilon(q)$: (q, v) is in the domain of exp, $v \in O$, $F(v) = \exp_q(v)$, $\exp_q(v) \in U$, and $G(\exp_q v) = v$;
- for $y \in N \cap U$: $G(y) \in T_qN$, $\|G(y)\| < \varepsilon$, and $\exp_q(G(y)) = y$.

So $\exp_q$ is a bijection of $B_\varepsilon(q)$ onto the open subset $U_q = N \cap U$ of N, and it and its inverse are restrictions of smooth maps on open sets of V: Milnor's "maps diffeomorphically onto an open set U_q" (Lemma 10.3 (3), p. 59), made precise in his own sense of smoothness for subsets of Euclidean space. ε is a *normal radius* at q if such coordinates exist `Chapter7/NormalNbhd · IsNormalRadius`. (The name is the exposition's.)

The proof of Lemma 10.3 reads exp in the chart of §VI.5 and inverts Milnor's map $(q, v) \mapsto (q, \exp_q v)$ there.

exp **in the chart.** Let σ be a slice chart, x its parametrization, L a local spray at u_0 with constants r, τ, c. Points z of $B((u_0, 0), r)$ have first component in T `Chapter7/NormalNbhd · LocalSpray.fst_mem_target`, and the

ball is stable under $(u, w) \mapsto (u, aw)$ for $|a| \le 1$ `Chapter7/NormalNbhd · LocalSpray.mem_ball_smul`.

- *Homogeneity of the chart flow.* For (u, w) in the ball, $|a| \le 1$ and $|s| < \tau$,

$$\varphi(s, (u, aw)) = \Big((\varphi(as, (u, w)))_1,\ a\,(\varphi(as, (u, w)))_2 \Big)$$

  `Chapter7/NormalNbhd · LocalSpray.φ_smul`. Both sides solve the chart equation in $T \times Z$ on $(-\tau, \tau)$ (the right side by the chain rule and quadratic homogeneity) and agree at $s = 0$; apply uniqueness on an interval (§VI.3).
- *The chart expression of* exp. $E(u, w) = (\varphi(c, (u, w/c)))_1$ `Chapter7/NormalNbhd · LocalSpray.chartExp` is smooth on $Z \times Z$ `Chapter7/NormalNbhd · LocalSpray.contDiff_chartExp`, lies in T for $(u, w) \in \mathcal{G}$ `Chapter7/NormalNbhd · LocalSpray.chartExp_mem`, and

$$\exp(x(u),\ dx_u w) = x(E(u, w)) \qquad ((u, w) \in \mathcal{G})$$

  `Chapter7/NormalNbhd · LocalSpray.exp_eq_chartExp`, because $\mathrm{geo}(u, w)$ is a geodesic from $x(u)$ with initial velocity $dx_u w$ `Chapter7/NormalNbhd · LocalSpray.isGeodesicFrom_geo` and its value at 1 is $x(E(u, w))$.
- $E(u, 0) = u$ for $u \in B(u_0, r)$ `Chapter7/NormalNbhd · LocalSpray.chartExp_zero`: $(u, 0)$ is a zero of the chart spray, so the constant curve solves the chart equation, and by uniqueness the flow line through $(u, 0)$ is constant.
- For $(u_0, b) \in \mathcal{G}$, the curve $t \mapsto E(u_0, tb)$ has velocity b at $t = 0$ `Chapter7/NormalNbhd · LocalSpray.hasDerivAt_chartExp_line`. For $|t| < 1$ the homogeneity of the flow gives $E(u_0, tb) = (\varphi(ct, (u_0, b/c)))_1$; at $t = 0$ its derivative is c times the first component of the chart spray at $(u_0, b/c)$, that is $c \cdot b/c = b$.

Milnor's Jacobian. Let $\Phi(z) = (z_1, E(z))$ `Chapter7/NormalNbhd · LocalSpray.chartF`, the map $(q, v) \mapsto (q, \exp_q v)$ in the chart. Its derivative at $(u_0, 0)$ is

$$(a, b) \longmapsto (a,\ a + b),$$

an isomorphism with inverse $(a, b) \mapsto (a, b - a)$ `Chapter7/NormalNbhd · LocalSpray.jacobian` `Chapter7/NormalNbhd · LocalSpray.hasFDerivAt_chartF`. *Proof.* In the direction $(a, 0)$: $E(u, 0) = u$ near u_0, so $dE(a, 0) = a$. In the direction $(0, b)$ with $(u_0, b) \in \mathcal{G}$: the chain rule along $t \mapsto (u_0, tb)$ and the velocity just computed give $dE(0, b) = b$; for arbitrary b scale it into $\mathcal{G}$ and use linearity. Add. □

These are the images of $\partial/\partial u^i$ and $\partial/\partial t^j$ in Milnor's matrix $\begin{pmatrix} I & I \\ 0 & I \end{pmatrix}$ (p. 58).

Lemma 10.3 (V7.10). For each $p \in N$ there are an open set $O \ni p$ of V and an $\varepsilon > 0$ such that, with $W = N \cap O$:

1. for all $q_1, q_2 \in W$ there is exactly one $v \in T_{q_1} N$ with $\|v\| < \varepsilon$ and $\exp_{q_1}(v) = q_2$;
2. there are an open set $O_2 \subseteq V \times V$ and a map G, smooth on O_2, such that for all $q_1, q_2 \in W$: $(q_1, q_2) \in O_2$ and $G(q_1, q_2)$ is the vector of (1);
3. ε is a normal radius at every $q \in W$

`Chapter7/NormalNbhd · lemma_10_3`. By (1), each U_q of (3) contains W.

Proof. The proof is Milnor's, in the slice chart $\sigma = \sigma_p$, with $u_0 = (\sigma p)_1$, a local spray L at u_0 and the parametrization x.

1. *The inverse function theorem.* Φ is smooth, and its derivative at $(u_0, 0)$ is invertible. The companion volume's inverse function theorem for manifolds `Milnor/Chapter1/InverseFunction · ContMDiff.isLocalDiffeomorphAt_of_isInvertible_mfderiv` gives a partial diffeomorphism $\hat{\Phi}$ between open subsets of $Z \times Z$, whose domain contains $(u_0, 0)$ and on which it agrees with Φ; its inverse $\hat{\Phi}^{-1}$ is smooth on the image, and $\hat{\Phi}^{-1}(\Phi(z)) = z$ on the domain. Milnor says "implicit function theorem"; it is this one.

2. *The shape of V'.* Milnor takes $V' = \{(q,v) : q \in U',\ \|v\| < \varepsilon\}$; that such sets form a neighbourhood base needs a uniform comparison of $\|v\|$ with the coordinates of v, made explicit here. Let $C_2 = \|dx_{u_0}\| + 1$, and let A be the set of z in the domain of $\hat\Phi$ and in $\mathcal{G}$ with $z_1 \in T$ and $\|dx_{z_1}\| < C_2$; it is open and contains $(u_0, 0)$, so it contains a ball $B((u_0,0),\rho)$. Let $C_1 = \|d\sigma_p\| + 1$, $\varepsilon = \rho/C_1$, and
$$S = \{z \in B((u_0,0),\rho) : \|dx_{z_1} z_2\| < \varepsilon\},$$
the chart picture of $\{(q,v) : \|v\| < \varepsilon\}$. S is open, contains $(u_0,0)$ and lies in A. Its image $\Phi(S) = \hat\Phi(S)$ is open (a partial homeomorphism maps open subsets of its domain to open sets) and contains $\Phi(u_0,0) = (u_0,u_0)$, hence a ball $B((u_0,u_0),\rho_2)$. Put $\rho' = \min(\rho,\rho_2)$ and
$$O = \{y \in \operatorname{dom}\sigma : \|(\sigma y)_1 - u_0\| < \rho',\ \|d\sigma_y\| < C_1\},$$
open (by continuity of $y \mapsto ((\sigma y)_1, d\sigma_y)$) and containing p.
3. *Four facts.* (K1) For $y \in O$ and $\|v\| < \varepsilon$, the chart data $((\sigma y)_1, (d\sigma_y v)_1)$ lie in $B((u_0,0),\rho)$, since $\|(d\sigma_y v)_1\| \le \|d\sigma_y\|\,\|v\| < C_1\varepsilon = \rho$. (K2) Points of that ball are in the domain of $\hat\Phi$, in $\mathcal{G}$, and have first component in T. (K3) For $q \in N \cap O$ and $v \in T_qN$ with $\|v\| < \varepsilon$, the chart data of (q,v) lie in S (the condition $\|dx\,(d\sigma_q v)_1\| = \|v\| < \varepsilon$ holds by the chart facts of §VI.5), and $\exp_q(v) = x(E((\sigma q)_1, (d\sigma_q v)_1))$ by the chart expression of exp. (K4) For $q_1, q_2 \in N \cap O$, the point $((\sigma q_1)_1, (\sigma q_2)_1)$ lies within ρ_2 of (u_0,u_0), hence in $\Phi(S)$: there is w with $((\sigma q_1)_1, w) \in S$ and $\Phi((\sigma q_1)_1, w) = ((\sigma q_1)_1, (\sigma q_2)_1)$ (the first component of Φ is the identity), and then $\exp_{q_1}(dx\,w) = q_2$.
4. *Item (1).* Existence: $v = dx_{(\sigma q_1)_1} w$ with w from (K4) is tangent at q_1 (it is in the range of dx), has $\|v\| < \varepsilon$ (definition of S), and $\exp_{q_1}(v) = q_2$. Uniqueness: if v' is tangent, $\|v'\| < \varepsilon$ and $\exp_{q_1}(v') = q_2$, then by (K3) the chart data $((\sigma q_1)_1, (d\sigma_{q_1} v')_1)$ lie in S and E of them is $(\sigma q_2)_1$; so Φ takes the same value at these chart data and at $((\sigma q_1)_1, w)$. Applying $\hat\Phi^{-1}$, $(d\sigma_{q_1} v')_1 = w$, and $v' = dx\,w = v$ by the chart facts.
5. *Item (2).* Let O_2 be the set of pairs (y_1,y_2) of points of the domain of σ with $((\sigma y_1)_1, (\sigma y_2)_1)$ in $T \times Z$ and in the image of $\hat\Phi$; it is open. Put
$$G(y_1,y_2) = dx_{(\sigma y_1)_1}\Big((\hat\Phi^{-1}((\sigma y_1)_1, (\sigma y_2)_1))_2\Big),$$
smooth on O_2 as a composite of smooth maps. For $q_1, q_2 \in N \cap O$, step 3 shows $(q_1,q_2) \in O_2$ and $\hat\Phi^{-1}((\sigma q_1)_1, (\sigma q_2)_1) = ((\sigma q_1)_1, w)$, so $G(q_1,q_2)$ is the vector of item (1).
6. *Item (3).* Fix $q \in N \cap O$ and write $u_q = (\sigma q)_1$. The normal coordinates are: $O_q = \{v : (u_q, (d\sigma_q v)_1) \in B((u_0,0),\rho)\}$, open; $F_q(v) = x(E(u_q, (d\sigma_q v)_1))$, smooth on O_q by (K2); $U_q = \{y \in \operatorname{dom}\sigma : (u_q, (\sigma y)_1) \in \Phi(S)\}$, open; and $G_q(y) = dx_{u_q}((\hat\Phi^{-1}(u_q, (\sigma y)_1))_2)$, smooth on U_q. The ten conditions follow from steps 3–4: vectors of $B_\varepsilon(q)$ are in the domain of exp (the geodesics geo) and in O_q (K1); $F_q = \exp_q$ there (K3); $\exp_q$ maps them into U_q, because $(u_q, (\sigma(\exp_q v))_1) = \Phi(u_q, (d\sigma_q v)_1)$ with the argument in S; $G_q \circ \exp_q = \mathrm{id}$ on $B_\varepsilon(q)$ because $\hat\Phi^{-1} \circ \Phi = \mathrm{id}$; and for $y \in N \cap U_q$, writing $(u_q, (\sigma y)_1) = \Phi(u_q, w)$ with $(u_q, w) \in S$, $G_q(y) = dx\,w$ is tangent at q, has norm $< \varepsilon$, and $\exp_q(G_q(y)) = x(E(u_q,w)) = x((\sigma y)_1) = y$. □

Geodesics or vectors. Milnor's item (1) speaks of "a unique geodesic of length $< \varepsilon$", the formal item (1) of initial vectors. They agree: a geodesic from q_1 with initial velocity v, $\|v\| < \varepsilon$, ending at q_2 has $\exp_{q_1}(v) = q_2$, so v is the vector of (1) `Chapter7/NormalNbhd · lemma_10_3_geodesic`; the geodesic is $t \mapsto \exp_{q_1}(tv)$ on $[0,1]$ (§VI.6), of length $\|v\|$ (§VI.10).

Corollary. Every point of N has a normal radius `Chapter7/NormalNbhd · exists_isNormalRadius` (item (3) at $q = p$).

$\exp_q$ **as a partial diffeomorphism (V7.9, V7.10).** The same statement in Mathlib's language: let $\exp_q$ be regarded as a map from the vector space T_qN to the manifold N (with a placeholder value where $\exp_q(v) \notin N$) `Chapter7/NormalNbhd · expToN Chapter7/NormalNbhd · coe_expToN`. If ε is a normal radius at q, then $\exp_q$ restricts to a partial diffeomorphism from the open ball of radius ε in T_qN onto the open subset $\{y \in N : y \in U\}$ of N `Chapter7/NormalNbhd · NormalCoord.expDiffeo`. Its inverse is $y \mapsto$ the orthogonal projection of $G(y)$ onto T_qN, which is $G(y)$ itself because $G(y)$ is tangent `Chapter7/NormalNbhd · NormalCoord.coe_orthogonalProjectionOnto_G`; $\exp_q$ takes values in N on the ball `Chapter7/NormalNbhd · NormalCoord.exp_mem`. Smoothness into the manifold N uses that a map into N which agrees near a point with a smooth map into V is smooth into N there `Chapter7/NormalNbhd · contMDiffAt_of_val_eventuallyEq` (read it in the slice chart at the image point), applied to F; the inverse is smooth because G composed with the inclusion of N is. Hence $\exp_q$ is a local diffeomorphism at 0: for every $q \in N$ there are $\varepsilon > 0$ and a partial diffeomorphism with domain the ε-ball of T_qN, whose image contains q, equal to $\exp_q$ on that ball `Chapter7/NormalNbhd · exists_expDiffeo`. This is what Milnor obtains on p. 58 from $d(\exp_q)_0 = \mathrm{id}$ and the inverse function theorem; formally it comes out of Lemma 10.3.

VI.9 The Gauss lemma (Lemma 10.5)

Milnor's Lemma 10.5 (p. 60) says that in U_q the geodesics through q are the orthogonal trajectories of the hypersurfaces $\{\exp_q(v) : \|v\| = \text{const}\}$. His proof takes a curve $v(t)$ of unit vectors and the surface $f(r,t) = \exp_q(rv(t))$, and shows that $\langle \partial f/\partial r, \partial f/\partial t\rangle$ does not depend on r. The formal statement is stronger and needs no curve $v(t)$.

Fix normal coordinates (O, U, F, G) of radius ε at q and write $D_w = dF_w$, $D_w^2 = d^2F_w$ for $w \in O$. For $w \in B_\varepsilon(q)$, F is smooth near w `Chapter7/GaussLemma · NormalCoord.contDiffAt_F Chapter7/GaussLemma · NormalCoord.hasFDerivAt_F`, and along a line through w in a direction of T_qN, $F = \exp_q$ near w `Chapter7/GaussLemma · NormalCoord.eventually_F_eq` (the points stay in the ball for small parameter).

Lemma (radial speed). For $u \in B_\varepsilon(q)$ and $0 \le r \le 1$, $\|D_{ru}(u)\| = \|u\|$ `Chapter7/GaussLemma · NormalCoord.norm_fderiv_F_apply`.

Proof. $t \mapsto \exp_q(tu)$ is a geodesic near $[0,1]$, with velocity u at 0 (§VI.6). Near $t = r$ it equals $t \mapsto F(tu)$, whose velocity at r is $D_{ru}(u)$. By constant speed (§VI.2) the speed at r is the speed at 0. □

Lemma (G inverts F infinitesimally). For $w \in B_\varepsilon(q)$ and $v \in T_qN$, $dG_{F(w)}(D_w v) = v$; in particular D_w is injective on T_qN `Chapter7/GaussLemma · NormalCoord.fderiv_G_fderiv_F Chapter7/GaussLemma · NormalCoord.injective_fderiv_F`.

Proof. $G(F(w+sv)) = G(\exp_q(w+sv)) = w + sv$ for s near 0; differentiate at $s = 0$ with the chain rule. □

Lemma 10.5, the Gauss lemma (V7.12). Let ε be a normal radius at q. For $w, x \in T_qN$ with $\|w\| < \varepsilon$,

$$\big\langle d(\exp_q)_w(w),\ d(\exp_q)_w(x)\big\rangle = \langle w, x\rangle,$$

where $\exp_q$ is regarded as a map on the vector space T_qN `Chapter7/GaussLemma · gauss_lemma`. In particular, if $x \perp w$, the radial velocity $d(\exp_q)_w(w)$ is orthogonal to $d(\exp_q)_w(x)$ `Chapter7/GaussLemma · gauss_lemma_orthogonal`.

The vectors $x \perp w$ form the tangent space at w of the sphere $\{\|v\| = \|w\|\}$, so the special case is Milnor's statement; the general one holds for every $x \in T_qN$. The statement for $\exp_q$ follows from the one for F `Chapter7/GaussLemma · NormalCoord.inner_fderiv_F`, since near w in T_qN, $\exp_q = F$ and the derivative of $\exp_q$ is D_w restricted to T_qN.

Proof for F. The proof is Milnor's, with $f(r,s)=\exp_q(r(w+sx))$ and everything computed in V. Put

$$g(r)=\big\langle D_{rw}(w),\ D_{rw}(rx)\big\rangle \qquad (0\le r\le 1),$$

which is $\langle \partial f/\partial r,\partial f/\partial s\rangle$ at $s=0$. For $r\in[0,1]$ the point rw lies in $B_\varepsilon(q)$, so F is smooth near it and g is continuous on $[0,1]$. Clearly $g(0)=0$ and $g(1)=\langle D_w w, D_w x\rangle$. For $0<r<1$, the product rule gives

$$g'(r)=\big\langle D^2_{rw}(w,w),\ D_{rw}(rx)\big\rangle+\big\langle D_{rw}(w),\ D^2_{rw}(w,rx)+D_{rw}(x)\big\rangle,$$

because $r\mapsto D_{rw}(w)$ has derivative $D^2_{rw}(w,w)$ and $r\mapsto D_{rw}(rx)$ has derivative $D^2_{rw}(w,rx)+D_{rw}(x)$.

1. *The first term vanishes.* Near $t=r$, $\exp_q(tw)=F(tw)$, so the acceleration of the geodesic $t\mapsto\exp_q(tw)$ at r is $D^2_{rw}(w,w)$, and it is normal to N at $\exp_q(rw)$. On the other hand $D_{rw}(rx)$ is the velocity at $s=0$ of $s\mapsto F(rw+s\,rx)=\exp_q(rw+s\,rx)$, a curve in N through $\exp_q(rw)$, so it is tangent there (§VI.1). Milnor's version: $\frac{D}{\partial r}\frac{\partial f}{\partial r}=0$ because the radial curves are geodesics.
2. *The second term is $\langle w,x\rangle$.* Consider $h(s)=\|D_{r(w+sx)}(w+sx)\|^2$. For s near 0, $w+sx\in B_\varepsilon(q)$, and the radial speed lemma gives $h(s)=\|w+sx\|^2$. Differentiate at $s=0$: the left side gives $2\langle D_{rw}w,\ D^2_{rw}(rx,w)+D_{rw}x\rangle$, the right side $2\langle w,x\rangle$. Second derivatives of a smooth map are symmetric `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`, so $D^2_{rw}(rx,w)=D^2_{rw}(w,rx)$, and the second term equals $\langle w,x\rangle$. Milnor's version: $\langle\frac{\partial f}{\partial r},\frac{D}{\partial r}\frac{\partial f}{\partial s}\rangle=\frac12\frac{\partial}{\partial s}\|\frac{\partial f}{\partial r}\|^2$, with his symmetry Lemma 8.7 replaced by the symmetry of second derivatives in V.
3. So $g'=\langle w,x\rangle$ on $(0,1)$. By the mean value theorem `Mathlib/Analysis/Calculus/Deriv/MeanValue · exists_hasDerivAt_eq_slope`, $g(1)-g(0)=\langle w,x\rangle$, that is $\langle D_w w,D_w x\rangle=\langle w,x\rangle$. □

What Milnor leaves implicit. That the sets $\{\exp_q(v):\|v\|=r\}$ are hypersurfaces with tangent vectors $\frac{d}{dt}\exp_q(r_0v(t))$ uses that $\exp_q$ is a diffeomorphism on the ball; the form proved here sidesteps the hypersurfaces entirely.

VI.10 Paths and their lengths

Theorem 10.4 compares the geodesic with every *other piecewise smooth path*. Milnor defines these only on p. 67 (§11) and uses them from p. 59 on; the formal text defines them, and a wider class, here.

Definitions (V7.11; §0.3). Let $c:\mathbb{R}\to V$ and $a\le b$.

- The *length* of c on $[a,b]$ is $L(c|[a,b])=\int_a^b\|c'(t)\|\,dt$ `Chapter7/Path · pathLength`, where $c'(t)$ is taken to be 0 where c is not differentiable, the convention of the formal language (§0.1); for the paths below this happens only on a null set.
- c is *piecewise smooth* on $[a,b]$ if it is continuous on $[a,b]$ and there is a finite set B of break points such that c is C^∞ (with one-sided derivatives at the ends) on every $[s,t]\subseteq[a,b]$, $s<t$, whose interior contains no point of B `Chapter7/Path · IsPiecewiseSmooth`. For Milnor's subdivision $a=t_0<\cdots<t_k=b$ take $B=\{t_i\}$.
- c is *admissible* on $[a,b]$ if it is continuous on $[a,b]$, differentiable at all but countably many points of (a,b), and $\|c'\|$ is integrable on $[a,b]$ `Chapter7/Path · IsAdmissible`.

A path in N is one with $c([a,b])\subseteq N$. The lower bounds for lengths below are proved for admissible paths, a wider class than the piecewise smooth paths, stable under restriction to subintervals `Chapter7/Path · IsAdmissible.mono`, in which the fundamental theorem of calculus holds.

Proposition. Piecewise smooth paths are admissible `Chapter7/Path · IsPiecewiseSmooth.isAdmissible`.

Proof. Off the finite set B, a point $t\in(a,b)$ has a closed interval around it free of break points, on which c is smooth; so c is differentiable at t. For integrability, induct on the number of break points in the open

interval: with none, c is smooth on the closed piece, its one-sided derivative is continuous there and agrees with c' inside, so $\|c'\|$ is integrable `Chapter7/Path · IsPiecewiseSmooth.intervalIntegrable_of_contDiffOn`; with some, split at one of them, which leaves fewer on each side, and integrability is additive. □

Properties of length (V7.11). For an admissible c on $[a,b]$ (with V complete):

- *Fundamental theorem of calculus*: $\int_a^b c' = c(b) - c(a)$ `Chapter7/Path · IsAdmissible.integral_deriv`, from Mathlib's version with countably many exceptional points `Mathlib/MeasureTheory/Integral/DivergenceTheorem · integral_eq_of_hasDerivAt_off_countable_of_le`. Hence $\|c(b) - c(a)\| \le L(c|[a,b])$ `Chapter7/Path · IsAdmissible.norm_sub_le_pathLength`.
- $L \ge 0$ `Chapter7/Path · IsAdmissible.pathLength_nonneg`; $L(c|[a,m]) + L(c|[m,b]) = L(c|[a,b])$ for $m \in [a,b]$ `Chapter7/Path · IsAdmissible.pathLength_add`; lengths of subintervals are smaller `Chapter7/Path · IsAdmissible.pathLength_le`.
- A path of length 0 is constant `Chapter7/Path · IsAdmissible.eq_of_pathLength_eq_zero` (by the chord bound on each $[a,t]$).
- $t \mapsto L(c|[a,t])$ is continuous `Chapter7/Path · IsAdmissible.continuousOn_pathLength` and nondecreasing `Chapter7/Path · IsAdmissible.monotoneOn_pathLength` on $[a,b]$.
- If $\|c'(t)\| = k$ wherever c is differentiable in (a,b), then $L(c|[a,b]) = k(b-a)$ `Chapter7/Path · pathLength_eq_of_norm_deriv`: the integrand is k off a countable, hence null, set `Chapter7/Path · ae_uIoc_of_forall_Ioo_diff`.

Operations (V7.11). A path smooth on an open set containing $[a,b]$ is piecewise smooth with no break `Chapter7/Path · IsPiecewiseSmooth.of_contDiffOn`; restriction to a subinterval `Chapter7/Path · IsPiecewiseSmooth.mono`, concatenation of c_1 on $[a,m]$ and c_2 on $[m,b]$ with $c_1(m) = c_2(m)$ (break set $B_1 \cup B_2 \cup \{m\}$) `Chapter7/Path · IsPiecewiseSmooth.concat`, reversal $t \mapsto c(a+b-t)$ `Chapter7/Path · IsPiecewiseSmooth.reverse` and translation of the parameter `Chapter7/Path · IsPiecewiseSmooth.translate` preserve piecewise smoothness. Under concatenation lengths add `Chapter7/Path · pathLength_concat` (the concatenation has the derivative of c_1 before m and of c_2 after, and $\{m\}$ is null); under reversal and translation they are unchanged `Chapter7/Path · pathLength_reverse` `Chapter7/Path · pathLength_translate` (substitution in the integral).

The length of a geodesic (V7.2). If γ is a geodesic on an open preconnected $s \supseteq [a,b]$ and $t_0 \in s$, then $L(\gamma|[a,b]) = \|\gamma'(t_0)\|\,(b-a)$ `Chapter7/Shortest · IsGeodesicOn.pathLength_eq`: constant speed and the last bullet.

VI.11 Lemma 10.6, and paths that leave a normal ball

Fix normal coordinates (O,U,F,G) of radius ε at q. Then $q \in U$ `Chapter7/Shortest · NormalCoord.mem_U_self`, $G(q) = 0$ `Chapter7/Shortest · NormalCoord.G_self`, and $G(y) \neq 0$ for $y \in N \cap U$, $y \neq q$ `Chapter7/Shortest · NormalCoord.G_ne_zero` (since $y = \exp_q(Gy)$ and $\exp_q(0) = q$). In Milnor's notation a point $y \neq q$ of U_q is $\exp_q(rv)$ with $r = \|G(y)\| \in (0,\varepsilon)$ and $v = G(y)/r$ a unit vector.

Lemma 10.6, pointwise. Let c lie in N near t, be differentiable at t, with $c(t) \in U$ and $c(t) \neq q$. Then the radius $\rho(s) = \|G(c(s))\|$ is differentiable at t with $|\rho'(t)| \le \|c'(t)\|$; and if $\rho'(t) = \|c'(t)\|$, the direction $s \mapsto G(c(s))/\|G(c(s))\|$ has derivative 0 at t `Chapter7/GaussLemma · NormalCoord.hasDerivAt_radius`.

Proof. Put $w = G(c(t))$, nonzero, and $w' = dG_{c(t)}(c'(t))$, the derivative of $s \mapsto G(c(s))$ at t; it lies in T_qN, because $G(c(s)) \in T_qN$ near t (the third small lemma of §VI.1). Near t, $F(G(c(s))) = c(s)$, so $D_w(w') = c'(t)$. The radius has derivative $\rho'(t) = \langle w, w'\rangle/\|w\|$ `Chapter7/GaussLemma · hasDerivAt_norm_of_ne`. By the Gauss lemma,

$$\langle D_w w,\ c'(t)\rangle = \langle D_w w,\ D_w w'\rangle = \langle w, w'\rangle, \qquad \|D_w w\|^2 = \langle w, w\rangle,$$

the second by the Gauss lemma with $x = w$. The Cauchy–Schwarz inequality `Mathlib/Analysis/InnerProductSpace/Basic · abs_real_inner_le_norm` gives $|\langle w, w'\rangle| \le \|w\|\,\|c'(t)\|$, that is $|\rho'(t)| \le \|c'(t)\|$. If

$\rho'(t) = \|c'(t)\|$, then $\langle D_w w, c'(t)\rangle = \|D_w w\|\,\|c'(t)\|$, and equality in Cauchy–Schwarz `Mathlib/Analysis/InnerProductSpace/Basic · inner_eq_norm_mul_iff_real` makes $c'(t) = l\,D_w w$ with $l = \|c'(t)\|/\|w\|$. Then $D_w(w' - lw) = 0$, and injectivity of D_w on T_qN (§VI.9) gives $w' = lw$. The derivative of $G(c)/\|G(c)\|$ at t is

$$\frac{w'}{\|w\|} - \frac{\langle w, w'\rangle}{\|w\|^3}\,w = \frac{l\,w}{\|w\|} - \frac{l\,\|w\|^2}{\|w\|^3}\,w = 0. \qquad \square$$

Milnor (p. 61) splits $\frac{d\omega}{dt}$ into orthogonal radial and spherical parts. The formal proof never splits: the stronger Gauss lemma and Cauchy–Schwarz do the work. Milnor's step from $\partial f/\partial t = 0$ to $dv/dt = 0$ needs $d(\exp_q)$ injective and $r > 0$; both appear above.

Lemma 10.6 (V7.13). Let c be admissible on $[a, b]$ with $c(t) \in N \cap U$ and $c(t) \neq q$ for $t \in [a, b]$. Then

$$L(c|[a, b]) \;\geq\; \|G(c(b))\| - \|G(c(a))\|$$

`Chapter7/Shortest · NormalCoord.radius_sub_le_pathLength`; and if equality holds, the direction $G(c(t))/\|G(c(t))\|$ is constant on $[a, b]$ `Chapter7/Shortest · NormalCoord.direction_eq_of_pathLength_eq`.

Proof of the inequality. The radius ρ is continuous on $[a, b]$, and off the countable exceptional set of c it is differentiable with $|\rho'| \leq \|c'\|$ by the pointwise lemma. So ρ' is integrable (dominated by $\|c'\|$), the fundamental theorem of calculus off a countable set gives $\int_a^b \rho' = \rho(b) - \rho(a)$, and $\rho' \leq |\rho'| \leq \|c'\|$ almost everywhere gives $\int_a^b \rho' \leq L(c|[a, b])$. $\square$

Proof of the equality clause. With equality, $\int_a^b (\|c'\| - \rho') = 0$ with a nonnegative integrand, so $\|c'\| = \rho'$ almost everywhere `Mathlib/MeasureTheory/Integral/IntervalIntegral/Basic · integral_eq_zero_iff_of_le_of_nonneg_ae`. At those points (off the countable set) the pointwise lemma gives that the direction $u = G(c)/\|G(c)\|$ has derivative 0. So $u' = 0$ almost everywhere; u is continuous and differentiable off a countable set, and the fundamental theorem of calculus on $[t, b]$ gives $u(b) - u(t) = \int_t^b u' = 0$. $\square$

The formal Lemma 10.6 is one-sided ($r(b) - r(a)$ rather than $|r(b) - r(a)|$), the case Theorem 10.4 uses; the monotone parametrization of the equality case is obtained in §VI.12.

Normal balls. For $0 < s$, the *open normal ball* of radius s is cut out of N by the open set

$$\text{ball}_s = U \cap G^{-1}(B(0, s))$$

`Chapter7/Shortest · NormalCoord.ballSet` `Chapter7/Shortest · NormalCoord.isOpen_ballSet` `Chapter7/Shortest · NormalCoord.mem_ballSet`, and the *closed normal ball* is $\overline{\text{ball}}_s = F(\bar{B}(0, s) \cap T_qN)$ `Chapter7/Shortest · NormalCoord.closedBallSet`. For $s < \varepsilon$ the closed normal ball is compact, as the continuous image of a compact set `Chapter7/Shortest · NormalCoord.isCompact_closedBallSet`, hence closed in V; and it consists of the points $y \in N \cap U$ with $\|G(y)\| \leq s$ `Chapter7/Shortest · NormalCoord.mem_closedBallSet`.

Lemma (paths leaving a normal ball, V7.14). Let c be admissible on $[a, b]$ with $c([a, b]) \subseteq N$, $c(a) = q$, $0 < s < \varepsilon$, and $c(b) \notin \text{ball}_s$. There are $a \leq t_1 < t_2 \leq b$ such that $c(t_1) = q$; $c(t) \in U \setminus \{q\}$ for $t_1 < t \leq t_2$; $\|G(c(t))\| < s$ for $t_1 \leq t < t_2$; $\|G(c(t_2))\| = s$; and $L(c|[t_1, t_2]) \geq s$ `Chapter7/Shortest · NormalCoord.exists_crossing`.

Here t_2 is the first exit from the open normal ball and t_1 the last visit of q before it. After t_2 the path may leave U_q altogether, and that is exactly the case Milnor's proof does not discuss.

Proof.

1. *The first exit.* The set S_2 of $t \in [a, b]$ with $c(t) \notin \text{ball}_s$ is closed (the preimage of a closed set under a map continuous on a closed interval) and contains b; let t_2 be its infimum, which belongs to it. Since $c(a) = q \in \text{ball}_s$, $a < t_2$, and $c(t) \in \text{ball}_s$ for $a \leq t < t_2$.

2. *The exit point is on the sphere.* For $a \le t < t_2$, $c(t)$ is in N and in ball_s, hence in the closed normal ball. That ball is closed in V, and $c(t_2)$ is the limit of $c(t)$ as $t \uparrow t_2$, so $c(t_2)$ lies in it: $c(t_2) \in U$ and $\|G(c(t_2))\| \le s$. As $t_2 \in S_2$, $c(t_2) \notin \mathrm{ball}_s$, so $\|G(c(t_2))\| = s$; in particular $c(t_2) \neq q$. This is where compactness of closed normal balls, and Hausdorffness, enter.
3. *The last visit of q.* The set S_1 of $t \in [a, t_2]$ with $c(t) = q$ is closed and contains a; let t_1 be its supremum, which belongs to it, and $t_1 < t_2$ since $c(t_2) \neq q$. For $t_1 < t \le t_2$, $c(t) \neq q$, and $c(t) \in U$ (from step 1, or step 2 at $t = t_2$).
4. *The length.* Let $\eta > 0$. The radius $\|G(c(t))\|$ is continuous at t_1 from the right, with value $\|G(q)\| = 0$, so there is $t' \in (t_1, t_2)$ with $\|G(c(t'))\| < \eta$. On $[t', t_2]$ the path avoids q and stays in $N \cap U$, so Lemma 10.6 gives $L(c|[t', t_2]) \ge s - \|G(c(t'))\| > s - \eta$, and $L(c|[t_1, t_2]) \ge L(c|[t', t_2])$. As η was arbitrary, $L(c|[t_1, t_2]) \ge s$. □

Milnor (p. 61) argues that for every $\delta > 0$ the path contains a segment joining the shells of radii δ and r and lying between them, of length $\ge r - \delta$, and lets $\delta \to 0$. The formal proof replaces the shells of radius δ by the last visit of q and takes the limit inside the argument. The verified edition calls this lemma Lemma C.

VI.12 Theorem 10.4, with the sharp equality case

Lemma (the radial geodesic). If ε is a normal radius at q and $v \in B_\varepsilon(q)$, then $t \mapsto \exp_q(tv)$ is a geodesic on some $(-\delta, 1+\delta)$ with velocity v at 0 `Chapter7/Shortest · NormalCoord.radial`, and its length on $[0, 1]$ is $\|v\|$ `Chapter7/Shortest · NormalCoord.pathLength_exp_smul` (constant speed $\|v\|$, §VI.10).

Theorem 10.4, inequality (V7.15). Let ε be a normal radius at q, $v \in T_qN$ with $\|v\| < \varepsilon$, and c an admissible path in N, on any $[a, b]$, from q to $\exp_q(v)$. Then $L(c|[a, b]) \ge \|v\|$ `Chapter7/Shortest · NormalCoord.norm_le_pathLength`, whether or not c stays in U_q.

Proof. For $v = 0$ there is nothing to prove. Otherwise $s = \|v\|$ satisfies $0 < s < \varepsilon$, and $c(b) = \exp_q(v)$ has $\|G(c(b))\| = \|v\| = s$, so $c(b) \notin \mathrm{ball}_s$. The lemma of §VI.11 gives $t_1 < t_2$ with $L(c|[a, b]) \ge L(c|[t_1, t_2]) \ge s$. □

Theorem 10.4, equality, sharp form (V7.16). In the same situation, suppose $v \neq 0$ and $L(c|[a, b]) = \|v\|$. Then

$$c(t) = \exp_q(\rho(t)\, v), \qquad \rho(t) = \frac{L(c|[a,t])}{\|v\|} \qquad (a \le t \le b)$$

`Chapter7/Shortest · NormalCoord.eq_exp_of_pathLength_eq`, where ρ is continuous and nondecreasing with $\rho(a) = 0$ and $\rho(b) = 1$ (§VI.10). So $c = \gamma \circ \rho$ is a monotone reparametrization of the geodesic $\gamma(t) = \exp_q(tv)$, $0 \le t \le 1$.

Proof. Let $s = \|v\|$ and take $t_1 < t_2$ from the crossing lemma.

1. *Splitting the length.* $L(c|[a, t_1]) + L(c|[t_1, t_2]) + L(c|[t_2, b]) = s$, the middle term is $\ge s$ and the others are ≥ 0. So $L(c|[a, t_1]) = 0 = L(c|[t_2, b])$ and $L(c|[t_1, t_2]) = s$. Paths of length 0 are constant: $c = q$ on $[a, t_1]$ and $c = c(b) = \exp_q(v)$ on $[t_2, b]$, so $G(c(t_2)) = v$.
2. *The middle.* Let $t_1 < t \le t_2$. The point $c(t)$ is $\exp_q(G(c(t)))$ with $G(c(t)) \in B_\varepsilon(q)$, so the inequality on $[t_1, t]$ gives $L(c|[t_1, t]) \ge \|G(c(t))\|$, and Lemma 10.6 on $[t, t_2]$ (where c avoids q and stays in U) gives $L(c|[t, t_2]) \ge s - \|G(c(t))\|$. The two lengths add up to s, so both inequalities are equalities: $L(c|[t_1, t]) = \|G(c(t))\|$, and Lemma 10.6 holds with equality on $[t, t_2]$. Its equality clause gives $G(c(t))/\|G(c(t))\| = v/s$, hence $G(c(t)) = (\|G(c(t))\|/s)\, v$ and $c(t) = \exp_q\big((L(c|[t_1, t])/s)\, v\big)$.
3. *All t.* For $t \le t_1$: $L(c|[a, t]) = 0$ and $c(t) = q = \exp_q(0)$. For $t_1 < t \le t_2$: $L(c|[a, t]) = L(c|[t_1, t])$ because the length on $[a, t_1]$ is 0, and step 2 applies. For $t > t_2$: $L(c|[a, t]) = s$, so $\rho(t) = 1$, and $c(t) = \exp_q(v)$. □

Milnor's equality clause (p. 59) is that the point sets coincide; he says a strict inequality is easily obtained when they differ (p. 61). The sharp form implies it: $c([a,b]) = \gamma([0,1])$ `Chapter7/Shortest · NormalCoord.image_eq_of_pathLength_eq`, because ρ maps $[a,b]$ into $[0,1]$ by monotonicity and onto it by the intermediate value theorem. Equal images are, however, too weak for Corollary 10.7: a path that runs along γ at variable speed, or backwards and forwards, has the same image without being a geodesic.

Theorem 10.4 (V7.15). For each $p \in N$ there are an open set $O \ni p$ of V and an $\varepsilon > 0$ such that for any $q_1, q_2 \in N \cap O$ there is $v \in T_{q_1}N$ with $\|v\| < \varepsilon$ and $\exp_{q_1}(v) = q_2$, the geodesic $\gamma(t) = \exp_{q_1}(tv)$, $0 \le t \le 1$, has length $\|v\|$, and every piecewise smooth path c in N, on any $[a,b]$, from q_1 to q_2 satisfies $L(c|[a,b]) \ge \|v\|$, with equality only if $c([a,b]) = \gamma([0,1])$ `Chapter7/Shortest · theorem_10_4`.

Proof. Take O and ε from Lemma 10.3, v from its item (1), and normal coordinates at q_1 from its item (3). The length of γ is $\|v\|$ by the radial lemma; piecewise smooth paths are admissible, so the inequality is the one above. For equality: if $v = 0$, then $q_1 = q_2$, the path has length 0 and is constant at $q_1 = \gamma(t)$; otherwise apply the sharp form and its corollary on images. □

The formal statement is stronger than Milnor's in allowing any parameter interval and, in the two lemmas behind it, admissible paths. Milnor's proof needs the crossing lemma (V7.14), which his text does not supply.

VI.13 Corollary 10.7: shortest paths are geodesics

Lemma (the segment lemma). Let c be piecewise smooth on $[0,\ell]$ with $c([0,\ell]) \subseteq N$, parametrized by arc length ($\|c'(t)\| = 1$ wherever c is differentiable in $(0,\ell)$), and *minimal*: $L(c|[0,\ell]) \le L(\tilde{c}|[a,b])$ for every piecewise smooth path $\tilde{c}$ in N, on any $[a,b]$, from $c(0)$ to $c(\ell)$. Let O, ε satisfy items (1) and (3) of Lemma 10.3, and let $0 \le \alpha < \beta \le \ell$ with $c([\alpha,\beta]) \subseteq O$. Then c agrees on $[\alpha,\beta]$ with a geodesic defined on an open set containing $[\alpha,\beta]$, namely $t \mapsto \exp_{c(\alpha)}\left(\frac{t-\alpha}{\beta-\alpha}v\right)$ for the vector v of item (1) joining $c(\alpha)$ to $c(\beta)$ `Chapter7/Minimal · minimal_segment`.

Proof.

1. *Arc length.* For $0 \le s \le t \le \ell$, $L(c|[s,t]) = t - s$, by the constant-speed formula of §VI.10.
2. *The geodesic segment.* Let $q_1 = c(\alpha)$, $q_2 = c(\beta)$, v the vector of item (1) and normal coordinates of radius ε at q_1 (item (3)). The radial geodesic $t \mapsto \exp_{q_1}(tv)$ lives on some $(-\delta, 1+\delta)$, so by homogeneity $\tilde{\gamma}(t) = \exp_{q_1}(k(t-\alpha)v)$, $k = (\beta-\alpha)^{-1}$, is a geodesic on the preimage of $(-\delta, 1+\delta)$ under $t \mapsto k(t-\alpha)$, an open interval containing $[\alpha,\beta]$. It runs from $\tilde{\gamma}(\alpha) = q_1$ to $\tilde{\gamma}(\beta) = \exp_{q_1}(v) = q_2$, lies in N on $[\alpha,\beta]$, and has speed $k\|v\|$, hence length $\|v\|$ on $[\alpha,\beta]$.
3. *The competitor.* Let c_2 be c on $[0,\alpha]$, $\tilde{\gamma}$ on $[\alpha,\beta]$ and c on $[\beta,\ell]$. It is piecewise smooth (two concatenations), lies in N, joins $c(0)$ to $c(\ell)$, and has length $L(c|[0,\alpha]) + \|v\| + L(c|[\beta,\ell])$.
4. *Equality.* Minimality of c against c_2 and additivity of length give $L(c|[\alpha,\beta]) \le \|v\|$; Theorem 10.4's inequality, applied to $c|[\alpha,\beta]$, gives the reverse. So $L(c|[\alpha,\beta]) = \|v\| = \beta - \alpha > 0$, and $v \neq 0$.
5. *Identification.* The sharp equality case gives, for $t \in [\alpha,\beta]$, $c(t) = \exp_{q_1}\left(\frac{L(c|[\alpha,t])}{\|v\|}v\right) = \exp_{q_1}\left(\frac{t-\alpha}{\beta-\alpha}v\right) = \tilde{\gamma}(t)$, by arc length. □

The arc-length hypothesis is what turns the monotone reparametrization of the sharp equality case into an affine one.

Corollary 10.7 (V7.17). Let c be a piecewise smooth path in N on $[0,\ell]$, parametrized by arc length, whose length is at most that of every piecewise smooth path in N, on any parameter interval, from $c(0)$ to $c(\ell)$. Then c is a geodesic on $(0,\ell)$ `Chapter7/Minimal · corollary_10_7`; in particular it is smooth there, also at its break points. Moreover c agrees on $[0,\ell]$ with a geodesic defined on an open interval $(-\delta, \ell+\delta)$ `Chapter7/Minimal · corollary_10_7_extend`.

Proof. Let $t_0 \in (0,\ell)$. Take O, ε from Lemma 10.3 at $c(t_0)$. By continuity of c at t_0 there is $\eta > 0$, also at most $t_0/2$ and $(\ell - t_0)/2$, with $c([t_0-\eta, t_0+\eta]) \subseteq O$. The segment lemma on $[t_0-\eta, t_0+\eta]$ gives a geodesic equal to c there; so c is a geodesic on $(t_0-\eta, t_0+\eta)$. Being a geodesic is local (§VI.2), so c is a geodesic on $(0,\ell)$.

For the extension, the case $\ell = 0$ is a constant geodesic. Otherwise the segment lemma at the endpoints, on $[0,\eta_0]$ and on $[\ell-\eta_1,\ell]$ (with η_0, η_1 chosen as above from Lemma 10.3 at $c(0)$ and $c(\ell)$, using continuity within $[0,\ell]$), gives geodesics γ_0, γ_1 on open sets containing these intervals, equal to c there. Choose $\delta > 0$ so that $(-\delta, 0]$ lies in the domain of γ_0 and $[\ell, \ell+\delta)$ in that of γ_1, and let g be γ_0 before 0, c on $[0,\ell]$ and γ_1 after ℓ. Near any $t \in (-\delta, \ell+\delta)$, g coincides with γ_0 (if $t < \eta_0$), with c (if $\eta_0 \le t \le \ell - \eta_1$, inside $(0,\ell)$) or with γ_1 (if $t > \ell - \eta_1$), each a geodesic there, so g is a geodesic on $(-\delta, \ell+\delta)$ by locality. □

What Milnor leaves implicit (p. 61). Applying Theorem 10.4 to short segments needs three more things: the sharp equality case, since equal images do not make a segment a geodesic; the minimality of each segment, from the competitor of step 3; and segments with the points of $(0,\ell)$, break points included, in their interiors.

VI.14 The distance ρ

Milnor (p. 62) says that the infimum of lengths "clearly" makes M a metric space, compatible with its topology by Theorem 10.4. Finiteness needs connectedness, positivity needs the crossing lemma, and the topology needs small metric balls to be normal balls; Theorem 10.9 also uses the continuity of ρ, which the book never states.

Definition (V7.18; §0.3). For $p, q \in V$, let $\Lambda(p,q)$ be the set of lengths $L(c|[a,b])$, as elements of $[0,\infty]$, of the piecewise smooth paths c in N, on any $[a,b]$, with $c(a) = p$ and $c(b) = q$ `Chapter7/Distance · pathLengths`, and

$$\rho(p,q) = \inf \Lambda(p,q) \in [0,\infty]$$

`Chapter7/Distance · riemDist`, which is ∞ when no path joins p to q, for instance when $q \notin N$ `Chapter7/Distance · riemDist_eq_top_of_notMem` or q lies in another component of N.

The axioms of an extended metric (V7.18).

- $\rho(c(a), c(b)) \le L(c|[a,b])$ for every piecewise smooth path in N `Chapter7/Distance · riemDist_le_pathLength`.
- $\rho(p,p) = 0$ for $p \in N$ `Chapter7/Distance · riemDist_self`: the constant path on $[0,0]$ has length 0.
- *Symmetry* `Chapter7/Distance · riemDist_comm`: reversal $t \mapsto c(a+b-t)$ turns a path from p to q into one from q to p of the same length, so $\Lambda(p,q) \subseteq \Lambda(q,p)$ `Chapter7/Distance · pathLengths_comm`, and by symmetry the two sets are equal.
- *The triangle inequality* `Chapter7/Distance · riemDist_triangle`. Given c_1 from p to q on $[a_1,b_1]$ and c_2 from q to r on $[a_2,b_2]$, translate the parameter of c_2 by $d = a_2 - b_1$ so that it starts at time b_1, and concatenate. The result is a piecewise smooth path in N from p to r of length $L(c_1)+L(c_2)$ (§VI.10). So $\rho(p,r) \le L(c_1) + L(c_2)$ for all such pairs, and taking infima gives $\rho(p,r) \le \rho(p,q) + \rho(q,r)$; if either set of lengths is empty the right side is ∞.

Normal balls are metric balls (V7.18, V7.19). Let ε be a normal radius at p, with normal coordinates (O, U, F, G).

- *A lower bound.* If $0 < s < \varepsilon$ and $y \notin \text{ball}_s$, then $\rho(p,y) \ge s$ `Chapter7/Distance · NormalCoord.ofReal_le_riemDist`: every piecewise smooth path from p to y is admissible, and the crossing lemma (§VI.11) gives a segment of length $\ge s$.
- *The radial distance.* For $v \in B_\varepsilon(p)$, $\rho(p, \exp_p v) = \|v\|$ `Chapter7/Distance · NormalCoord.riemDist_exp`. The radial geodesic on $[0,1]$ is a piecewise smooth path in N of length $\|v\|$, which gives $\le$ `Chapter7/`

Distance · NormalCoord.riemDist_exp_le; for $v \neq 0$, $\exp_p v \notin \mathrm{ball}_{\|v\|}$ because $G(\exp_p v) = v$, and the lower bound gives $\geq$.

- *Small metric balls.* For $0 < r \leq \varepsilon$, $\rho(p,y) < r$ if and only if $y \in N \cap \mathrm{ball}_r$ Chapter7/Distance · NormalCoord.riemDist_lt_iff. If $\rho(p,y) < r$, then $y \in N$ (else $\rho = \infty$); and if y were outside ball_r, choosing s with $\rho(p,y) < s < r$ would put y outside ball_s and give $\rho(p,y) \geq s$. Conversely, $y \in N \cap \mathrm{ball}_r$ is $\exp_p(Gy)$ with $\|Gy\| < r$, at distance at most $\|Gy\|$.

Positivity (V7.18). For $p, q \in N$, $\rho(p,q) = 0$ implies $p = q$ Chapter7/Distance · eq_of_riemDist_eq_zero. *Proof.* Suppose $q \neq p$, and take a normal radius ε at p. There is s with $0 < s < \varepsilon$ and $q \notin \mathrm{ball}_s$: if $q \in U$, then $G(q) \neq 0$ and $s = \min(\|G(q)\|, \varepsilon/2)$ works; otherwise $s = \varepsilon/2$. Then $\rho(p,q) \geq s > 0$. □

The topology (V7.19). For every $p \in N$, the metric balls $\{y : \rho(p,y) < r\}$, $r > 0$, form a basis of the neighbourhoods of p in N, for the topology of N as a subspace of V Chapter7/Distance · hasBasis_nhdsWithin_riemDist.

Proof. Take a normal radius ε at p. A neighbourhood of p in N contains $N \cap O'$ for an open $O' \ni p$. F is continuous at 0 with $F(0) = p$, so $F(B(0, r_0)) \subseteq O'$ for some $r_0 > 0$; with $r = \min(r_0, \varepsilon)$, the metric ball of radius r is $N \cap \mathrm{ball}_r$, whose points are $F(Gy)$ with $\|Gy\| < r$, so it lies in O'. Conversely a metric ball of radius r contains $N \cap \mathrm{ball}_{\min(r,\varepsilon)}$, the trace on N of an open set containing p. □

Milnor's "follows easily from 10.4" is these two facts: small metric balls are normal balls, and small normal balls lie in any given neighbourhood of p. Consequently, for $p \in N$ and every $r > 0$, all points of N near p are at distance $< r$ from p Chapter7/Metric · eventually_riemDist_lt.

ρ dominates the distance of V (V7.21). $\|q - p\| \leq \rho(p,q)$ for all $p, q \in V$ Chapter7/Metric · ofReal_norm_sub_le_riemDist: a path is at least as long as its chord (§VI.10).

Continuity of ρ (V7.21). The function $(x,y) \mapsto \rho(x,y) \in [0,\infty]$ is continuous on $N \times N$ Chapter7/Metric · continuousOn_riemDist, in particular $y \mapsto \rho(y,q)$ is continuous on N for every q Chapter7/Metric · continuousOn_riemDist_left (constant ∞ if $q \notin N$), and ρ is continuous on the product space $N \times N$ Chapter7/Metric · continuous_riemDist.

Proof. Fix $(x,y) \in N \times N$. By the neighbourhood-basis property, for every $r > 0$ the pairs $(z_1, z_2) \in N \times N$ near (x,y) have $\rho(x,z_1) < r/2$ and $\rho(y,z_2) < r/2$. For such pairs the triangle inequality gives

$$\rho(z_1,z_2) \leq \rho(x,y) + r \qquad \text{and} \qquad \rho(x,y) \leq \rho(z_1,z_2) + r.$$

If $\rho(x,y) < b$, choose r with $\rho(x,y) + r < b$: the first inequality gives $\rho(z_1,z_2) < b$ near (x,y). If $a < \rho(x,y)$, choose r with $a + r < \rho(x,y)$: the second inequality excludes $\rho(z_1,z_2) \leq a$ near (x,y). This is continuity in the order topology of $[0,\infty]$. □

Finiteness (V7.21). If N is preconnected, $\rho(p,q) < \infty$ for all $p, q \in N$ Chapter7/Metric · riemDist_ne_top. *Proof.* Let T be the set of $y \in N$ at finite distance from p. Points near y are at distance < 1 from y; so if $y \in T$ they are in T, and if $y \notin T$ they are not in T (else $\rho(p,y) \leq \rho(p,y') + \rho(y',y) < \infty$). So T is open and closed in N, it contains p, and N is preconnected. □

This is what Milnor's standing assumption that M is connected (p. 55) is for. "Connected" is preconnected in the formal statements, which also admits $N = \emptyset$.

The metric spaces (N,ρ) (V7.21). ρ restricted to N satisfies the axioms of an extended metric Chapter7/Metric · riemEMetricSpace₀, and the topology it defines is the topology of N as a subspace of V Chapter7/Metric · topology_eq_riemEMetricSpace₀, by the neighbourhood-basis property. Replacing the topology of the metric structure by the (equal) subspace topology Mathlib/Topology/EMetricSpace/Defs · EMetricSpace.replaceTopology gives an extended metric space whose topology is the topology of N by definition Chapter7/Metric · riemEMetricSpace. For preconnected N, where ρ is finite, it becomes a metric space with distance ρ Chapter7/Metric · riemMetricSpace Chapter7/Metric · riemMetricSpace_dist Chapter7/Metric · riemMetricSpace_edist, again with the topology of N by definition Chapter7/Metric · riemMetricSpace_topology. They are not instances, since N, a subset of V, already carries the metric of V.

VI.15 Corollary 10.8

Milnor's Corollary 10.8 (p. 62): for compact K there is $\delta > 0$ such that two points of K at distance $< \delta$ are joined by a unique geodesic of length $< \delta$, which is minimal and depends differentiably on its endpoints. His proof covers K by sets W of Lemma 10.3 and takes δ so small that two points of K at distance $< \delta$ lie in a common W.

Definition. For $\delta > 0$ and $y, z \in N$, say y and z are *δ-joined* `Chapter7/Distance · IsShortJoin` if there is $v \in T_yN$ with $\|v\| < \delta$ and $\exp_y(v) = z$, such that: $\rho(y,z) = \|v\|$ (the geodesic $t \mapsto \exp_y(tv)$, $0 \le t \le 1$, of length $\|v\|$, is minimal); every $w \in T_yN$ with $\|w\| < \delta$ and $\exp_y(w) = z$ equals v; and there are an open $O_2 \ni (y,z)$ of $V \times V$ and a map G, smooth on O_2, with $G(y,z) = v$ and, for all $y', z' \in N$ with $(y',z') \in O_2$ and $\rho(y',z') < \delta$, $G(y',z') \in T_{y'}N$, $\|G(y',z')\| < \delta$ and $\exp_{y'}(G(y',z')) = z'$.

Uniqueness is stated for initial vectors, which covers geodesics as after Lemma 10.3; "depends differentiably" becomes the smooth G.

Lemma (Lemma 10.3 read with the distance). Around each $p \in N$ there are $s > 0$ and an open $\Omega \ni p$ of V such that points of $N \cap \Omega$ are at distance $< s$ from p, and for every $0 < \delta \le s$, every $y \in N \cap \Omega$ and every $z \in N$ with $\rho(y,z) < \delta$, y and z are δ-joined `Chapter7/Distance · exists_isShortJoin`.

Proof.

1. Take O, ε and items (1)–(3) from Lemma 10.3 at p, and $r > 0$ with the metric ball of radius r about p inside O (§VI.14). Let $s = \min(r,\varepsilon)/2$ and $\Omega = \mathrm{ball}_s$ for normal coordinates of radius ε at p; so $N \cap \Omega$ is the metric ball of radius s, and $2s \le r$, $s \le \varepsilon$.
2. *Both points lie in O.* $\rho(p,y) < s$, and $\rho(p,z) \le \rho(p,y) + \rho(y,z) < s + \delta \le 2s \le r$.
3. *The vector.* Item (1) gives the unique $v \in T_yN$ with $\|v\| < \varepsilon$ and $\exp_y(v) = z$. As ε is a normal radius at y (item (3)), $\rho(y,z) = \|v\|$ (§VI.14); so $\|v\| < \delta$.
4. *Uniqueness* among vectors of norm $< \delta \le \varepsilon$ is item (1).
5. *Smooth dependence.* Item (2) gives O_2' and G; take $O_2 = O_2' \cap (O \times O)$. $G(y,z) = v$ by item (1). For $(y',z') \in O_2$ with $y', z' \in N$ and $\rho(y',z') < \delta$: $G(y',z')$ is the vector of item (1) for (y',z'), and $\|G(y',z')\| = \rho(y',z') < \delta$ by step 3 at y'. □

Corollary 10.8 (V7.20). Let $K \subseteq N$ be compact. There is $\delta > 0$ such that any $y, z \in K$ with $\rho(y,z) < \delta$ are δ-joined `Chapter7/Distance · corollary_10_8`.

Proof. If K is empty there is nothing to prove. Otherwise apply the lemma at each $x \in K$, obtaining s_x and Ω_x; finitely many $\Omega_{x_1}, \dots, \Omega_{x_k}$ cover K `Mathlib/Topology/Compactness/Compact · IsCompact.elim_finite_subcover`. Let $\delta = \min_i s_{x_i} > 0$. A point $y \in K$ lies in some Ω_{x_i}, and $\delta \le s_{x_i}$, so the lemma applies to y and any z with $\rho(y,z) < \delta$. □

Milnor's choice of δ is this Lebesgue-number argument; uniqueness "of length $< \delta$" needs $\delta \le \varepsilon$ for each set of the cover, and minimality is Theorem 10.4 through $\rho(y,z) = \|v\|$. Only y needs to lie in K.

VI.16 The geodesic flow, and the continuity of exp on its domain

Corollary 10.10 needs $\exp_p$ continuous on a large disk of T_pM, about which Lemma 10.2 says nothing. The formal proof obtains it by continuing geodesics (the verified edition's Lemma B, in a continuous form).

The geodesic flow. By §VI.6 every geodesic is a radial curve $t \mapsto \exp_y(tw)$. The geodesic flow is therefore read off from exp:

$$\Psi_t(y,w) = \Big(\exp_y(tw),\ \tfrac{d}{dt} \exp_y(tw) \Big)$$

`Chapter7/GeodesicFlow · geoFlow`, the position and velocity at time t of $s \mapsto \exp_y(sw)$; where the geodesic with initial data (y, w) exists, this is its position and velocity at time t.

Proposition (the local flow, uniformly). For $x_0 \in N$ and $R > 0$ there are an open set $U \ni x_0$ of V, a time $\tau > 0$ and a map Φ, continuous on $U \times B(0,R) \times (-\tau, \tau)$, such that for every $y \in N \cap U$ and $w \in T_yN$ with $\|w\| < R$: $t \mapsto \exp_y(tw)$ is a geodesic on $(-\tau, \tau)$ with velocity w at 0, and $\Psi_t(y,w) = \Phi(y,w,t)$ for $|t| < \tau$ `Chapter7/GeodesicFlow · exists_localFlow_exp`.

Proof. Take U, ε, Θ from Lemma 10.2 at x_0 and rescale by homogeneity. Let $a = \varepsilon/(2R)$, so $a\|w\| < \varepsilon$ for $\|w\| < R$, and $\tau = 2a$. The map $\iota(y,w,t) = (y, aw, t/a)$ sends $U \times B(0,R) \times (-2a, 2a)$ into $U \times B(0,\varepsilon) \times (-2,2)$, where Θ is smooth. Put

$$\Phi(y,w,t) = \Big(\Theta(\iota(y,w,t)),\ a^{-1}\,\partial_t\Theta(\iota(y,w,t))\Big),$$

continuous since Θ and its derivative are. For $y \in N \cap U$ and tangent w with $\|w\| < R$, the curve $\eta(t) = \Theta(y, aw, t/a)$ is a geodesic on $(-2a, 2a)$ by homogeneity, with $\eta(0) = y$ and $\eta'(t) = a^{-1}\partial_t\Theta(\iota(y,w,t))$ by the chain rule, so $\eta'(0) = a^{-1}(aw) = w$. Every geodesic is radial, so $\exp_y(tw) = \eta(t)$ on $(-2a, 2a)$, and its position and velocity at t are $\Phi(y,w,t)$. □

In particular, for $y \in N$ and $w \in T_yN$, $t \mapsto \exp_y(tw)$ is a geodesic on some $(-\tau, \tau)$ with velocity w at 0 `Chapter7/GeodesicFlow · exists_isGeodesicOn_exp_smul` `Chapter7/GeodesicFlow · hasDerivAt_exp_smul_zero` (take $R = \|w\| + 1$), and Ψ_0 is the identity on TN `Chapter7/GeodesicFlow · geoFlow_zero`.

Proposition (continuing a geodesic). Suppose $t \mapsto \exp_y(tw)$ is a geodesic on an open interval $I \ni 0$, let $s \in I$ and $(y', w') = \Psi_s(y,w)$. If $t \mapsto \exp_{y'}(tw')$ is a geodesic on an open interval $J \ni 0$, then $t \mapsto \exp_y(tw)$ is a geodesic on $I \cup (s + J)$, and $\Psi_{s+t}(y,w) = \Psi_t(y', w')$ for $t \in J$ `Chapter7/GeodesicFlow · isGeodesicOn_exp_smul_union`.

Proof. Let $\gamma_1(t) = \exp_y(tw)$ and $\gamma_2(t) = \exp_{y'}((t-s)w')$, a geodesic on $s + J$ by homogeneity. At time s both have position y' and velocity w'. On the open interval $I \cap (s + J)$ (an intersection of intervals), uniqueness of geodesics gives $\gamma_1 = \gamma_2$. The curve g equal to γ_1 on I and to γ_2 elsewhere is then a geodesic on $I \cup (s + J)$ by locality, with $g(0) = y$ and $g'(0) = w$. The union is an interval (two intervals sharing the point s) `Mathlib/Topology/Connected/Basic · IsPreconnected.union`, so $g(t) = \exp_y(tw)$ on all of it because every geodesic is radial. Near $s+t$, $t \in J$, the curve $\exp_y(\cdot\, w)$ coincides with γ_2, whose position and velocity at $s + t$ are those of $\exp_{y'}(\cdot\, w')$ at t. □

This gluing replaces the maximal geodesics that Milnor's talk of extending geodesics (pp. 58, 62) presupposes; maximal geodesics are not constructed.

Theorem (the domain of $\exp$ is open in TN and $\exp$ is continuous on it, V7.23). Let (p, v) be in the domain of exp. Then every $z \in TN$ near (p, v) is in the domain, and exp, as a function on TN, is continuous at (p, v) `Chapter7/GeodesicFlow · eventually_mem_expDom_and_continuousWithinAt_exp`. In particular, if $\exp_p$ is defined on all of T_pN, it is continuous there `Chapter7/GeodesicFlow · continuousOn_exp`.

Proof.

1. *The given geodesic.* Let γ be a geodesic from p with initial velocity v on $(-\delta_0, 1+\delta_0)$. Then $v \in T_pN$, $\exp_p(tv) = \gamma(t)$ there, so $\Psi_s(p,v) = (\gamma(s), \gamma'(s))$, and $\|\gamma'(s)\| = \|v\|$. Put $R = \|v\| + 1$.
2. *A uniform time.* For each $x \in N$ the local flow proposition gives U_x, τ_x, Φ_x for this R. The arc $\gamma([0,1])$ is compact, so finitely many $U_{x_1}, \dots, U_{x_k}$ cover it. Let $\tau_0 = \frac{1}{2}\min_i \tau_{x_i}$, so $\tau_0 < \tau_{x_i}$ for each i.
3. *The induction statement.* For $s \in [0,1]$ let $P(s)$ say: for all $z = (y,w) \in TN$ near (p,v) (eventually in the neighbourhood filter of (p,v) within TN), the curve $t \mapsto \exp_y(tw)$ is a geodesic on some $(-\delta, s+\delta)$; and $z \mapsto \Psi_s(z)$ is continuous within TN at (p,v).

4. $P(0)$. For $z = (y, w) \in TN$ near (p, v), $y \in U_p$ and $\|w\| < R$, so $t \mapsto \exp_y(tw)$ is a geodesic on $(-\tau_p, \tau_p)$. And Ψ_0 is the identity on TN, hence continuous.
5. *The step:* $P(s) \Rightarrow P(\min(s+\tau_0, 1))$ *for* $s \in [0, 1)$. Let $s' = \min(s+\tau_0, 1)$ and $h = s' - s \in (0, \tau_0]$. Choose i with $\gamma(s) \in U_{x_i}$. By continuity of Ψ_s, for $z = (y, w)$ near (p, v) the point $(y', w') = \Psi_s z$ has $y' \in U_{x_i}$; also $y' \in N$, w' is tangent at y', and $\|w'\| = \|w\| < R$ by constant speed. So $t \mapsto \exp_{y'}(tw')$ is a geodesic on $(-\tau_{x_i}, \tau_{x_i})$ with $\Psi_t(y', w') = \Phi_{x_i}(y', w', t)$. Continuing (previous proposition), $t \mapsto \exp_y(tw)$ is a geodesic on $(-\delta, s+\delta) \cup (s - \tau_{x_i}, s + \tau_{x_i})$, which contains $(-\delta', s' + \delta')$ for $\delta' = \min(\delta, \tau_{x_i} - \tau_0)$, and

$$\Psi_{s'}(z) = \Psi_h(\Psi_s z) = \Phi_{x_i}(\Psi_s z, h).$$

   The right side is continuous within TN at (p, v): Φ_{x_i} is continuous at $(\gamma(s), \gamma'(s), h)$, a point of its domain because $\|\gamma'(s)\| = \|v\| < R$ and $|h| < \tau_{x_i}$, and $z \mapsto (\Psi_s z, h)$ is continuous at (p, v) by $P(s)$.
6. *Iteration.* By induction on k, $P(\min(k\tau_0, 1))$ for every natural k; for $k \geq 1/\tau_0$ this is $P(1)$.
7. *Conclusion.* For $z = (y, w) \in TN$ near (p, v), $t \mapsto \exp_y(tw)$ is a geodesic on $(-\delta, 1+\delta)$ starting at y with velocity w, so z is in the domain. And $\exp(z) = (\Psi_1 z)_1$ is continuous within TN at (p, v). For the last statement compose with $w \mapsto (p, w)$, which maps T_pN into TN. □

This is the global dependence of geodesics on their initial data. The smoothness of exp on its whole domain is proved in Chapter 11 (§VI.7); continuity is all that Corollary 10.10 uses.

VI.17 Completeness

Definitions (V7.22).

- (N, ρ) is *complete* if every sequence (x_n) of points of N that is Cauchy for ρ (for every $\varepsilon > 0$, $\rho(x_m, x_n) < \varepsilon$ for all large m, n) converges for ρ to some $y \in N$, that is $\rho(x_n, y) \to 0$ `Chapter7/Metric · IsRiemComplete`. For preconnected N this is completeness of the metric space (N, ρ) in Mathlib's sense `Chapter7/Metric · isRiemComplete_iff_completeSpace` (a translation between the two formulations of Cauchy sequences and convergence).
- A set $X \subseteq V$ is *bounded* if $\rho(x, y) \leq d$ for all $x, y \in X$ and some real d `Chapter7/Metric · IsRiemBounded`. A nonempty bounded set lies in N, since $\rho(x, x) = \infty$ for $x \notin N$.
- N is *geodesically complete* if $\exp_q(v)$ is defined for every $q \in N$ and every $v \in T_qN$ `Chapter7/GeodesicFlow · IsGeodesicallyComplete` (Milnor, p. 58).

Proposition. If $\exp_q$ is defined on all of T_qN, then for every $v \in T_qN$ the curve $t \mapsto \exp_q(tv)$ is a geodesic on $\mathbb{R}$ `Chapter7/GeodesicFlow · isGeodesicOn_univ_exp_smul`.

Proof. For $T > 0$ and $w \in T_qN$: $\exp_q(Tw)$ is defined, so there is a geodesic from q with velocity Tw on some $(-\delta, 1+\delta)$, equal there to $\sigma \mapsto \exp_q(\sigma T w)$ (§VI.6); rescaling time by $1/T$ (homogeneity), $t \mapsto \exp_q(tw)$ is a geodesic on $(-T\delta, T + T\delta)$. For $t \geq 0$ take $T = t + 1$ and $w = v$. For $t < 0$ take $T = -t + 1$ and $w = -v$, and reverse time: $\exp_q(-s(-v)) = \exp_q(sv)$. So $t \mapsto \exp_q(tv)$ is a geodesic near every t, hence on $\mathbb{R}$ by locality. □

Proposition (Milnor's two definitions agree, V7.22). N is geodesically complete if and only if every geodesic on an open preconnected nonempty $s \subseteq \mathbb{R}$ is the restriction to s of a geodesic on $\mathbb{R}$ `Chapter7/GeodesicFlow · isGeodesicallyComplete_iff`.

Proof. ($\Rightarrow$) Let γ be a geodesic on s and $t_0 \in s$, $q = \gamma(t_0)$, $v = \gamma'(t_0)$. By the last proposition and a shift of time, $t \mapsto \exp_q((t - t_0)v)$ is a geodesic on $\mathbb{R}$ with position q and velocity v at t_0; by uniqueness of geodesics on s it equals γ there. ($\Leftarrow$) For $q \in N$ and $v \in T_qN$, $t \mapsto \exp_q(tv)$ is a geodesic on some

$(-\tau, \tau)$; it extends to a geodesic $\bar{\gamma}$ on $\mathbb{R}$, which on $(-1, 2)$ is a geodesic from q with velocity v (its value and derivative at 0 are those of $\exp_q(tv)$). □

Milnor calls the two "clearly equivalent" (p. 58). The equivalence glues geodesics, which needs uniqueness in the large.

VI.18 Theorem 10.9 (Hopf–Rinow)

Definition (V7.24). A curve γ is a *minimal geodesic on* $[a, b]$, $a \le b$, if it is a geodesic on some open set containing $[a, b]$ and $L(\gamma|[a, b]) \le L(c|[a', b'])$ for every piecewise smooth path c in N, on any $[a', b']$, with $c(a') = \gamma(a)$ and $c(b') = \gamma(b)$ `Chapter7/HopfRinow · IsMinimalGeodesic` (Milnor, pp. 61–62).

Lemma. If $v \in T_pN$, $t \mapsto \exp_p(tv)$ is a geodesic on $\mathbb{R}$ and $\rho(p, \exp_p v) = \|v\|$, then $t \mapsto \exp_p(tv)$ is a minimal geodesic on $[0, 1]$, of length $\|v\|$ `Chapter7/HopfRinow · isMinimalGeodesic_exp_smul`. *Proof.* Its speed is $\|v\|$, so its length on $[0, 1]$ is $\|v\|$; a competitor c has $\|v\| = \rho(p, \exp_p v) \le L(c)$. □

Milnor's proof (pp. 62–64, Diagram 10) has two steps that the formal proof isolates as lemmas.

Step 1: the nearest point of a small sphere (V7.25). Let ε be a normal radius at $p \in N$, $0 < \delta < \varepsilon$, and q with $\delta < \rho(p, q) < \infty$. There is $w \in T_pN$ with $\|w\| = \delta$ and

$$\rho(p, q) = \delta + \rho(\exp_p w, q)$$

`Chapter7/HopfRinow · NormalCoord.exists_riemDist_eq_add`: the point $p_0 = \exp_p w$ of the sphere S of radius δ about p is a point of S nearest to q.

Proof.

1. $q \in N$, since $\rho(p, q) < \infty$; and $q \notin \mathrm{ball}_\delta$, since otherwise $\rho(p, q) < \delta$ (§VI.14). This is where $\delta < \rho(p, q)$ is needed: otherwise q lies inside the shell and paths from p to q need not meet it.
2. $S = \exp_p(\{w \in T_pN : \|w\| = \delta\}) = F(\dots)$ is compact, as the continuous image of a compact set, and lies in N.
3. *Every path crosses S.* Let c be a piecewise smooth path in N from p to q on $[a, b]$. The crossing lemma with $s = \delta$ gives $t_1 < t_2$ with $\|G(c(t_2))\| = \delta$, so $y = c(t_2) = \exp_p(G(c(t_2))) \in S$, and $L(c|[t_1, t_2]) \ge \delta$. Then $L(c|[a, b]) \ge L(c|[t_1, t_2]) + L(c|[t_2, b]) \ge \delta + \rho(y, q)$.
4. *S is nonempty*: $\rho(p, q) < \infty$, so some path from p to q exists (the infimum of the empty set is ∞), and step 3 produces a point of S.
5. *The nearest point.* $y \mapsto \rho(y, q)$ is continuous on $N \supseteq S$ (§VI.14), and S is compact and nonempty, so it has a minimum at some $y_0 = \exp_p(w_0)$, $\|w_0\| = \delta$ `Mathlib/Topology/Order/Compact · IsCompact.exists_isMinOn`.
6. *Equality.* $\rho(p, q) \le \rho(p, y_0) + \rho(y_0, q) = \delta + \rho(y_0, q)$, as $\rho(p, \exp_p w_0) = \|w_0\| = \delta$. Conversely, for every path length $L \in \Lambda(p, q)$, step 3 gives $y \in S$ with $\delta + \rho(y_0, q) \le \delta + \rho(y, q) \le L$; take the infimum. □

Step 2: a broken geodesic of minimal length is unbroken (V7.25). Let γ be a geodesic on $\mathbb{R}$ with $\|\gamma'(0)\| = 1$, let $0 < t_0 < t_1$, and let σ be a geodesic on an open preconnected set containing $[t_0, t_1]$ with $\|\sigma'(t_0)\| = 1$ and $\sigma(t_0) = \gamma(t_0)$. If $\rho(\gamma(0), \sigma(t_1)) \ge t_1$, then $\sigma(t_1) = \gamma(t_1)$ `Chapter7/HopfRinow · broken_geodesic_eq`.

Proof.

1. *The broken path.* Let $c = \gamma$ on $t \le t_0$ and $c = \sigma$ after. It is piecewise smooth on $[0, t_1]$ (concatenation) and lies in N.
2. *Unit speed.* Both pieces have constant speed 1. At a point $t \ne t_0$ where c is differentiable, c agrees near t with γ or with σ. At $t = t_0$, if c is differentiable there, its derivative is also its left derivative,

which is that of γ, of norm 1 (left derivatives on $(-\infty, t_0]$ are unique `Mathlib/Analysis/Calculus/TangentCone/Real · uniqueDiffWithinAt_Iic`). So $L(c|[0,t_1]) = t_1$.

3. *Minimality.* For any piecewise smooth competitor $\tilde{c}$ from $c(0) = \gamma(0)$ to $c(t_1) = \sigma(t_1)$: $t_1 \le \rho(\gamma(0), \sigma(t_1)) \le L(\tilde{c})$.
4. *Corollary 10.7* makes c a geodesic on $(0, t_1)$. It agrees with γ near $t_0/2$, in position and velocity, so by uniqueness of geodesics $c = \gamma$ on $(0, t_1)$, and by continuity on $[0, t_1]$ `Mathlib/Topology/Separation/Hausdorff · Set.EqOn.of_subset_closure`. In particular $\sigma(t_1) = c(t_1) = \gamma(t_1)$. □

Milnor (pp. 63–64) concludes by Corollary 10.7 that the broken geodesic is unbroken "and hence coincides with γ"; this needs unit speed on both pieces and uniqueness in the large.

Theorem 10.9, at one point (V7.26). Let $p \in N$ be such that $\exp_p$ is defined on all of T_pN. Every $q \in N$ with $\rho(p,q) < \infty$ is $\exp_p(v)$ for some $v \in T_pN$ with $\|v\| = \rho(p,q)$, and $t \mapsto \exp_p(tv)$ is a minimal geodesic on $[0,1]$ from p to q `Chapter7/HopfRinow · theorem_10_9`.

Proof. Let $r = \rho(p,q)$. Once a $v \in T_pN$ with $\exp_p(v) = q$ and $\|v\| = r$ is found, the lemma before Step 1 finishes the proof, $t \mapsto \exp_p(tv)$ being a geodesic on $\mathbb{R}$ (§VI.17).

1. $r = 0$. Then $p = q$ by positivity, and $v = 0$.
2. *The first sphere.* Let $r > 0$, ε a normal radius at p and $\delta = \min(\varepsilon, r)/2$, so that $\delta < \varepsilon$ and $\delta < r$. Step 1 gives w_0 with $\|w_0\| = \delta$ and $r = \delta + \rho(\exp_p w_0, q)$. Let $u = w_0/\delta$, a unit vector, and $\gamma(t) = \exp_p(tu)$, a geodesic on $\mathbb{R}$ of unit speed with $\gamma(0) = p$ and $\gamma(\delta) = \exp_p(w_0)$.
3. *The set of good times.* Let $A = \{t : \rho(\gamma(t), q) = r - t\}$. It is closed, because $t \mapsto \rho(\gamma(t), q)$ is continuous (§VI.14) and $t \mapsto r - t$, read in $[0, \infty]$, is continuous. By Step 1, $\delta \in A$.
4. *The inductive step.* Let $t_0 \in A$ with $\delta \le t_0 < r$, and $y > t_0$; we find a point of A in $(t_0, y]$. Let $x_0 = \gamma(t_0)$, ε' a normal radius at x_0, and $\delta' = \min(\varepsilon', r - t_0, y - t_0)/2$. Since $\rho(x_0, q) = r - t_0 > \delta'$, Step 1 at x_0 gives w with $\|w\| = \delta'$ and $\rho(x_0, q) = \delta' + \rho(p_0', q)$ for $p_0' = \exp_{x_0}(w)$; so $\rho(p_0', q) = r - (t_0 + \delta')$, and the triangle inequality $r \le \rho(p, p_0') + \rho(p_0', q)$ gives $\rho(p, p_0') \ge t_0 + \delta'$. Let $\sigma(t) = \exp_{x_0}(\frac{t - t_0}{\delta'} w)$, the radial geodesic of the normal ball at x_0 rescaled to unit speed: $\sigma(t_0) = x_0 = \gamma(t_0)$, $\sigma(t_0 + \delta') = p_0'$, and σ is a geodesic on an open interval containing $[t_0, t_0 + \delta']$. Step 2, with $t_1 = t_0 + \delta'$, gives $\gamma(t_0 + \delta') = p_0'$, hence $t_0 + \delta' \in A$; and $t_0 < t_0 + \delta' \le y$.
5. *Continuous induction.* A closed set containing δ that, with each of its points $t_0 \in [\delta, r)$, meets every interval $(t_0, y]$, contains $[\delta, r]$ `Mathlib/Topology/Order/IntermediateValue · IsClosed.Icc_subset_of_forall_exists_gt`. So $r \in A$: $\rho(\gamma(r), q) = 0$, and $\gamma(r) = q$ by positivity.
6. Take $v = ru$: $\exp_p(v) = \gamma(r) = q$ and $\|v\| = r$. □

What Milnor leaves implicit. Besides $\delta < \rho(p,q)$ in Step 1, the inductive step needs $\delta' < r - t_0$, else $t_0 + \delta'$ overshoots r; the supremum "by continuity" is the continuous induction of step 5, resting on the continuity of ρ. Completeness is used only at p: the proof uses the single geodesic $t \mapsto \exp_p(tu)$.

Theorem 10.9, Milnor's statement (V7.26). If N is preconnected and geodesically complete, any two points $p, q \in N$ are joined by a minimal geodesic γ on $[0,1]$, $\gamma(0) = p$, $\gamma(1) = q$, with $\rho(p,q) = L(\gamma|[0,1])$ `Chapter7/HopfRinow · theorem_10_9'`. *Proof.* $\rho(p,q) < \infty$ by connectedness (§VI.14); apply the theorem at p; the length of $t \mapsto \exp_p(tv)$ on $[0,1]$ is $\|v\| = \rho(p,q)$. □

VI.19 Corollary 10.10, its converse, and the Hopf–Rinow theorem

Corollary 10.10, first part, at one point (V7.27). Let N be preconnected and $p \in N$ such that $\exp_p$ is defined on all of T_pN. Every bounded set $X \subseteq V$ has compact closure, and the closure lies in N `Chapter7/HopfRinow · isCompact_closure_of_isRiemBounded`.

Proof. The empty set is trivial. Let $x_0 \in X$ and d a bound for the diameter. $X \subseteq N$, as noted in §VI.17. With $R = \rho(p, x_0) + \max(d, 0)$, finite by connectedness, every $x \in X$ has $\rho(p,x) \le \rho(p,x_0) + \rho(x_0, x) \le$

R. The set

$$K = \exp_p\left(\bar{B}(0,R) \cap T_pN\right)$$

is compact, as the image of a compact set under $\exp_p$, which is continuous on T_pN (§VI.16); and $K \subseteq N$. By Theorem 10.9 at p, each $x \in X$ is $\exp_p(v)$ with $\|v\| = \rho(p,x) \le R$, so $X \subseteq K$. K is closed in V, so the closure of X lies in K, is compact, and lies in N. □

Milnor's proof (p. 64) is this one; it needs $\exp_p$ continuous on the whole disk, which Lemma 10.2 does not give, and M connected, so that distances are finite and Theorem 10.9 applies to all of X.

Corollary 10.10, second part (V7.27). If every bounded set has compact closure contained in N, then (N,ρ) is complete `Chapter7/HopfRinow · isRiemComplete_of_forall_isCompact_closure`.

Proof. Let (x_n) be ρ-Cauchy in N and n_0 such that $\rho(x_m,x_n) < 1$ for $m,n \ge n_0$. The tail $X = \{x_{n_0+k}\}$ is bounded, so its closure is compact and lies in N, and a subsequence of the tail converges in V to some y in the closure `Mathlib/Topology/Sequences · IsCompact.tendsto_subseq`, hence $y \in N$. Convergence in N implies convergence for ρ `Chapter7/Metric · tendsto_riemDist_of_tendsto`, by the neighbourhood-basis property. Finally a Cauchy sequence with a convergent subsequence converges: given $\eta > 0$, choose n_1 with $\rho(x_m,x_n) < \eta/2$ for $m,n \ge n_1$ and a term $x_{\varphi(k)+n_0}$ of the subsequence with index $\ge n_1$ and $\rho(x_{\varphi(k)+n_0},y) < \eta/2$; then $\rho(x_n,y) < \eta$ for $n \ge n_1$. □

Corollary 10.10 (V7.27). If N is preconnected and geodesically complete, every bounded set has compact closure contained in N, and (N,ρ) is complete `Chapter7/HopfRinow · corollary_10_10`. *Proof.* If N is empty, a bounded set is empty (a point x would have $\rho(x,x) = \infty$), and both conclusions hold. Otherwise use the two parts at any $p \in N$. □

The converse of Corollary 10.10 (V7.28). If (N,ρ) is complete, then N is geodesically complete `Chapter7/GeodesicFlow · isGeodesicallyComplete_of_isRiemComplete`. No connectedness is assumed.

Milnor (p. 64) calls this not difficult, using Lemma 10.3, and leaves the details to Hopf and Rinow. The formal proof is the exposition's, and uses only the uniform local existence of Lemma 10.2.

Proof. Let $q \in N$, $v \in T_qN$ and $c(t) = \exp_q(tv)$. It suffices that c is a geodesic near $[0,1]$: then it is a geodesic from q with initial velocity v.

1. *The set of good times.* Let A be the set of $T \ge 0$ such that c is a geodesic on some $(-\delta, T+\delta)$. It is downward closed in $[0,\infty)$ and contains 0 (local existence). Suppose $1 \notin A$; then $A \subseteq [0,1)$. Let $T_0 = \sup A$; $T_0 > 0$, since A contains a small positive time.
2. *Below T_0.* For $0 \le t < T_0$ there is $T > t$ in A, so c is a geodesic near $[0,T]$, of speed $\|v\|$, lying in N. For $0 \le t \le t' < T_0$ the restriction of c to $[t,t']$ is a piecewise smooth path of length $\|v\|(t'-t)$, so $\rho(c(t),c(t')) \le \|v\|\,|t'-t|$.
3. *A Cauchy sequence.* Let $t_n = T_0 - T_0/(n+2)$, which increases to T_0. By step 2, $(c(t_n))$ is ρ-Cauchy, and by completeness $\rho(c(t_n),x) \to 0$ for some $x \in N$.
4. *Uniform time near the limit.* The local flow proposition at x with $R = \|v\|+1$ gives $U \ni x$ and $\tau > 0$: every geodesic $t \mapsto \exp_y(tw)$ with $y \in N \cap U$, $w \in T_yN$, $\|w\| < R$ exists on $(-\tau,\tau)$. Metric balls about x are neighbourhoods in N, so $c(t_n) \in U$ for large n; also $t_n > T_0 - \tau/2$ for large n. Fix such an n.
5. *Continuation.* c is a geodesic on some $(-\delta, T+\delta)$ with $T > t_n$, and $\Psi_{t_n}(q,v)$ has position $c(t_n) \in N \cap U$ and velocity of norm $\|v\| < R$. So the geodesic starting there exists on $(-\tau,\tau)$, and continuing (§VI.16) makes c a geodesic on $(-\delta, T+\delta) \cup (t_n-\tau, t_n+\tau)$. Since $t_n + \tau > T_0 + \tau/2$, this union contains $(-\delta', T_0 + \tau/4 + \delta')$ for $\delta' = \min(\delta, \tau/4)$. So $T_0 + \tau/4 \in A$, contradicting $T_0 = \sup A$. □

Bounded closed sets are compact. If every bounded set has compact closure contained in N, then every subset of N that is closed in N and bounded is compact `Chapter7/HopfRinow · isCompact_of_isClosed_of_isRiemBounded`: its closure in V is compact and lies in N, so it equals the closure in N, which is the set itself.

The Hopf–Rinow theorem (V7.29). Let N be preconnected and nonempty. The following are equivalent `Chapter7/HopfRinow · hopf_rinow`:

1. (N, ρ) is complete;
2. N is geodesically complete;
3. $\exp_p$ is defined on all of T_pN for some $p \in N$;
4. every bounded set has compact closure, contained in N.

Proof. (1) $\Rightarrow$ (2) is the converse of Corollary 10.10; (2) $\Rightarrow$ (3) because N is nonempty; (3) $\Rightarrow$ (4) is the first part of Corollary 10.10 at p; (4) $\Rightarrow$ (1) is its second part. □

Condition (3) is not in the book. After Corollary 10.10 and its converse Milnor speaks simply of complete Riemannian manifolds (p. 64).

Consequences of completeness (V7.29). If N is preconnected and (N, ρ) is complete, then N is geodesically complete; every geodesic on an open interval is the restriction of a geodesic on $\mathbb{R}$; any two points p, q are joined by a minimal geodesic on $[0, 1]$ whose length is $\rho(p, q)$; and every bounded subset of N that is closed in N is compact `Chapter7/HopfRinow · hopf_rinow_of_isRiemComplete`. *Proof.* The converse of Corollary 10.10, the two definitions of §VI.17, Theorem 10.9, and Corollary 10.10 with the last remark. □

VI.20 Closed submanifolds and the sphere

In the embedded setting completeness has a simple sufficient condition, the source of the complete manifolds used later.

Theorem (closed submanifolds are complete, V7.30). If N is closed in V, then (N, ρ) is complete `Chapter7/Metric · isRiemComplete_of_isClosed`.

Proof. A ρ-Cauchy sequence is Cauchy in V, because $\|x - y\| \le \rho(x, y)$ (§VI.14). V is complete, so the sequence converges in V `Mathlib/Topology/UniformSpace/Cauchy · cauchySeq_tendsto_of_complete`, to a limit that lies in N because N is closed. Convergence in N implies convergence for ρ. □

Corollary (V7.30). A preconnected N that is closed in V, in particular a preconnected compact one, has all the properties listed under "consequences of completeness": it is geodesically complete, its geodesics extend to $\mathbb{R}$, any two of its points are joined by a minimal geodesic whose length is their distance, and its closed bounded subsets are compact `Chapter7/HopfRinow · hopf_rinow_of_isClosed` `Chapter7/HopfRinow · hopf_rinow_of_isCompact`. This covers, for instance, compact matrix groups embedded in a space of matrices.

The round sphere (V7.30). Let $n \ge 1$ and $S^n = \{x \in \mathbb{R}^{n+1} : \langle x, x\rangle - 1 = 0\}$ `Milnor/Chapter5/HairyBall · sphG` `Milnor/Chapter5/HairyBall · Sph`, with its structure of a slice submanifold of $\mathbb{R}^{n+1}$ modelled on $\mathbb{R}^n$ with complement $\mathbb{R}^1$, from the companion volume `Milnor/Chapter6/EulerSphere · sphSub`. Then S^n is geodesically complete, and any two of its points are joined by a minimal geodesic on $[0, 1]$ whose length is their distance `Chapter7/HopfRinow · sphere_hopf_rinow`.

Proof. S^n is the metric sphere of radius 1 `Milnor/Chapter5/HairyBall · preimage_eq_sphere`, compact `Milnor/Chapter6/EulerSphere · isCompact_sph` and connected, because the dimension $n + 1$ of the ambient space exceeds 1 `Mathlib/Analysis/Normed/Module/Connected · isConnected_sphere`. Apply the corollary. □

Milnor (p. 65) shows that the geodesics of S^n are the great circles and discusses which arcs of them are minimal; that identification is not formalized.

VI.21 What the formalization taught us

The hidden hypotheses of §10 are of four kinds, each recorded where it is used: uniqueness of geodesics in the large (§VI.4); uniform comparisons between tangent vectors and their coordinates, in Lemmas 10.2 and 10.3 (§VI.5, §VI.8); the crossing lemma, needed wherever a competing path may leave a normal ball,

with the compactness of closed normal balls behind it (§VI.11); and two continuity statements the book uses without proof, of ρ and of $\exp_p$ on large disks (§VI.14, §VI.16). Three statements needed sharpening: the equality case of Theorem 10.4 (§VI.12), the radii $\delta < \rho(p, q)$ and $\delta' < r - t_0$ in Theorem 10.9 (§VI.18), and δ below every ε of the cover in Corollary 10.8 (§VI.15). And two results need less than the book uses: Theorem 10.9 needs completeness at one point only, and the converse of Corollary 10.10 only Lemma 10.2, not Lemma 10.3.

Not formalized: maximal geodesics (gluing replaces them); the remark after Lemma 10.3 that W can be chosen geodesically convex (p. 59); the remarks on minimal geodesics of p. 62; the examples of pp. 65–66 beyond the completeness of the sphere; Lemma 10.6 in its two-sided form; and §10 on an abstract Riemannian manifold, which would need a Levi-Civita connection and an exponential map that Mathlib does not have, or Nash's embedding theorem.

Part VII. Energy, variations, Jacobi fields and the index theorem (Milnor §§11–15)

Milnor's §§11–15 set up a calculus of variations on the space of paths and use it to compute the index of a geodesic. §11 introduces the path space Ω, its "tangent spaces" and its variations; §12 defines the energy, proves the first variation formula and identifies the critical paths of the energy with the geodesics; §13 defines the Hessian E_{**} at a geodesic through two-parameter variations and proves the second variation formula; §14 introduces Jacobi fields and shows that they are the null vectors of E_{**} and the variation fields of families of geodesics; §15 proves the Morse index theorem: the index of E_{**} is the number of points conjugate to the initial point in the open segment, counted with multiplicity. The formal proof does all of this for a submanifold N of a Euclidean space with the induced metric, the setting of Parts V and VI. Every Riemannian manifold is of this kind by Nash's embedding theorem, which is quoted and not proved. Every numbered statement of §§11–15 is proved, with three exceptions: Examples 1 and 2 of §14, Lemma 12.1, of which only the inequality $E(\gamma) \le E(\omega)$ is proved, and the part of Remark 14.6 about negative parameters and cluster points. Remark 14.5 is proved in a restated form (§VII.17), and the vanishing of the index in Lemma 13.6 in its equivalent form, positive semi-definiteness (§VII.10).

The formal route departs from Milnor's in more places than in any earlier part, and each departure has a reason. **(1)** As in the book, Ω carries no topology and no manifold structure. $T\Omega_\omega$ is *defined* as a vector space of fields along ω and is tied to Ω by variations in both directions: the variation field of every variation lies in it, and every finite family of its elements is the family of variation fields of a single variation (Lemma 11.A). A derivative "on Ω" is always the derivative of a function of finitely many real variables. **(2)** E_{**} is *defined* as twice the right side of Milnor's second variation formula. Theorem 13.1 and Lemma 11.A then show that Milnor's own definition, through two-parameter variations, is well defined and agrees with it. Symmetry and bilinearity come from an integral form, not from the symmetry of the mixed partials of $E \circ \bar{\alpha}$, a function the book never shows to be of class C^2. **(3)** The Jacobi equation is solved without a parallel frame: the formal proof writes it as a first-order linear system for the pair $(J, DJ/dt)$ in $V \times V$, with a correction term that keeps tangent data tangent, and obtains global solutions from Grönwall's inequality. **(4)** Lemma 14.4 is proved from initial data rather than from end points. This avoids minimal geodesics, the restriction to a short initial segment, and a step that Milnor's proof leaves unargued. **(5)** The two properties of a "sufficiently fine" subdivision, and Assertion (2), both come from a single Poincaré-type estimate instead of from minimal geodesics. **(6)** $\lambda(\tau)$ is defined as the index of E_{**} restricted to the fields that vanish outside $[0, \tau]$, so no Hessian on $[0, \tau]$ and no rescaling is needed. **(7)** Theorem 15.1 needs the conjugate points to be finite in number before the jumps of λ can be added up, and the formal proof establishes this first.

Map.

- `Chapter8/Piecewise`: induction over the pieces of a subdivision; integrability and integration by parts over the pieces; partial derivatives on closed strips and their symmetry up to the edges; differentiation under the integral sign.
- `Chapter8/Energy`: the tangent projection along paths; jumps; Ω, $T\Omega_\omega$, variations and variation

fields; the energy and Schwarz's inequality.
- `Chapter8/FirstVariation`: covariant derivative and acceleration; the first variation formula at every value of the parameter; Theorem 12.2; Lemma 11.A.
- `Chapter8/Critical`: critical paths, unbroken geodesics, Corollary 12.3.
- `Chapter8/SecondVariation`: Milnor's curvature tensor; the curvature identity (13.3); Theorem 13.1; Remark 13.5.
- `Chapter8/Hessian`: the integral form of the Hessian; E_{**} as a symmetric bilinear form (Corollary 13.4); the inequality of Lemma 12.1; Lemma 13.6.
- `Chapter8/Jacobi`: Jacobi fields; Theorem 14.1.
- `Chapter9/FormIndex`: the index of a bilinear form; Lemma 15.B of the exposition; Sylvester's law; persistence of definiteness.
- `Chapter9/JacobiODE`: global solutions of linear equations; the propagator.
- `Chapter9/JacobiSystem`: the Jacobi equation as a linear system; Jacobi fields from initial data; uniqueness.
- `Chapter9/JacobiSpace`: the $2n$-dimensional space of Jacobi fields; conjugate points and multiplicity; the nullity of E_{**}; Remark 14.2.
- `Chapter9/IndexForm`: the second variation formula for broken Jacobi fields; the Poincaré estimate; no conjugate points closer than h_0.
- `Chapter9/BrokenJacobi`: fields along $\gamma|[0,\tau]$; fine subdivisions; broken Jacobi fields; Lemma 15.3.
- `Chapter9/IndexReduction`: the splitting; Lemma 15.4; $\lambda(\tau)$ and Assertions (1) and (2).
- `Chapter9/IndexContinuity`: the forms H_τ and their continuity.
- `Chapter9/IndexAssertions`: the compound matrix; Assertions (3) and (4).
- `Chapter9/IndexTheorem`: from the Assertions to Theorem 15.1; Corollary 15.2; Remark 14.6.
- `Chapter9/JacobiVariation`: Lemma 14.3; Remark 14.5.
- `Chapter9/GeodesicVariation`: the local smooth flow, the extension of geodesic segments, smooth dependence on initial data over a compact interval; Lemma 14.4.

VII.1 The setting: tangent projections, jumps and covariant derivatives

Throughout, V is a finite-dimensional real inner product space (Milnor's $\mathbb{R}^K$) and $N \subseteq V$ is a slice submanifold modelled on a finite-dimensional space Z with complement W (§§0.1, 0.3) `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`. We write $n = \dim Z$. N carries the induced metric. It need not be closed, compact or connected, and no completeness is assumed anywhere in this part. For $x \in N$ the tangent space $T_xN \subseteq V$ is the range of the derivative of the inclusion `Chapter1/ManifoldSubmanifold · tangentSubspace`. That derivative is injective, so $\dim T_xN = n$ `Chapter9/JacobiSystem · finrank_tanAt`.

The tangent projection. The orthogonal projection of V onto T_xN is P_x (Part V, §V.11) `Chapter6/Projection · tanProj`. The formal proofs use a version defined at every point of V: P_x for $x \in N$, and $P_x = 0$ for $x \notin N$ `Chapter8/Energy · prj`. Paths lie in N only for $t \in [0,1]$, and integrands are easier to handle when $x \mapsto P_x$ is defined everywhere. The basic facts are that $P_x v = v$ if and only if $v \in T_xN$ `Chapter8/Energy · prj_eq_self_iff`, that $P_x^2 = P_x$ `Chapter8/Energy · prj_prj`, and that P_x is self-adjoint `Chapter8/Energy · inner_prj_left`. Hence $\langle v, P_x w\rangle = \langle v, w\rangle$ for tangent v `Chapter8/Energy · inner_prj_right_of_eq`. Below, this identity is what lets an ordinary derivative replace a covariant one inside an inner product with a tangent field.

Two lemmas about P are used throughout the part.

(a) Derivatives of maps into N are tangent, including derivatives within a set. If g maps a set s into N and has derivative g' within s at a point $p \in s$ at which s has unique derivatives, then $g'(w) \in T_{g(p)}N$ for every w `Chapter8/Energy · hasFDerivWithinAt_mem_tangentSubspace`. The proof takes the slice chart σ chosen at $g(p)$ `Chapter6/Projection · sliceAt`. Near p in s, $g(x)$ lies in the domain of σ and in N, so the W-component of $\sigma(g(x))$ vanishes. Its derivative within s at p is therefore both the W-component of $d\sigma \circ g'$ and 0, and the two agree because derivatives within s are unique. Vanishing of the W-component of $d\sigma(g'w)$

characterizes $T_{g(p)}N$. The one-variable form Chapter8/Energy · hasDerivWithinAt_mem_tangentSubspace makes one-sided velocities at break points tangent, and also the one-sided t-derivatives along the edges of the strips used below.

(b) Along a smooth map into N, P is smooth. If g is C^∞ within s at p and maps s into N, then so is $x \mapsto P_{g(x)}$ Chapter8/Energy · contDiffWithinAt_prj_comp Chapter8/Energy · ContDiffOn.prj_comp. Near $g(p)$, P coincides on N with a smooth map defined on an open subset of V, the local ambient extension built from a slice chart and written $\mathrm{projRep}_\sigma$ Chapter6/Projection · projRep Chapter6/Projection · projRep_eq Chapter6/Projection · contDiffAt_projRep, and the composite with g is smooth. In particular $t \mapsto P_{\omega(t)}$ is piecewise smooth along a piecewise smooth path in N Chapter8/Energy · IsPiecewiseSmooth.prj_comp.

The second fundamental form and the curvature. $\mathrm{II}_x(v,w) = (dP)_x(v)\,w$ is the derivative of P along N in the direction v, applied to w (Part V, §V.14) Chapter6/Submanifold · secondFundForm Chapter6/Projection · dN. For tangent v it is the derivative of the extension $\mathrm{projRep}_\sigma$ Chapter6/Projection · dN_tanProj_eq_fderiv. It is symmetric on tangent vectors Chapter6/Submanifold · secondFundForm_comm, takes two tangent vectors to a normal one Chapter6/Projection · dN_tanProj_mem_orthogonal_of_mem and a normal vector to a tangent one Chapter6/Projection · dN_tanProj_mem_of_mem_orthogonal, and satisfies $P\,\mathrm{II}(v, Pw) = 0$ for tangent v Chapter6/Projection · tanProj_dN_tanProj. The curvature tensor is

$$R(a,b)c = \mathrm{II}\big(b, \mathrm{II}(a,c)\big) - \mathrm{II}\big(a, \mathrm{II}(b,c)\big)$$

at $x \in N$ (Part V, §V.16) Chapter6/Submanifold · curvOp Chapter6/Submanifold · curvOp_apply, extended by 0 off N Chapter8/SecondVariation · curvR Chapter8/SecondVariation · curvR_of_mem. This is Milnor's sign convention, $R(X,Y)Z = -\nabla_X\nabla_Y Z + \nabla_Y\nabla_X Z + \nabla_{[X,Y]}Z$ (p. 51; Part V, §V.9), which is opposite to that of many modern texts. With it the unit sphere has curvature $+1$, that is $\langle R(a,b)a, b\rangle = \|a\|^2\|b\|^2 - \langle a,b\rangle^2$, and the Gauss equation reads $\langle R(x,y)z, w\rangle = \langle \mathrm{II}(x,z), \mathrm{II}(y,w)\rangle - \langle \mathrm{II}(x,w), \mathrm{II}(y,z)\rangle$ Chapter6/Submanifold · inner_curvOp. For tangent arguments $R(a,b)c$ is tangent Chapter6/Submanifold · curvOp_mem.

Paths and piecewise smoothness. A path, and likewise a vector field along a path, is a map $\mathbb{R} \to V$ of which only the values on $[0,1]$ matter (§0.8). As everywhere in Mathlib, $c'(t)$ denotes the derivative of c at t where it exists and 0 where it does not. A map c is *piecewise smooth* on $[a,b]$ (Part VI, §VI.10) if $a \le b$, c is continuous on $[a,b]$, and there is a finite set B of break points such that c is C^∞, with one-sided derivatives at the ends, on every closed interval $[s,t] \subseteq [a,b]$ whose interior contains no point of B Chapter7/Path · IsPiecewiseSmooth. Such an interval is called a *piece*. The class is closed under the operations this part needs. A smooth function of two piecewise smooth maps is piecewise smooth, with the union of the two break sets Chapter8/Energy · IsPiecewiseSmooth.map₂; hence so are sums, products with scalar functions, and applications of operator-valued maps Chapter8/Energy · IsPiecewiseSmooth.add Chapter8/Energy · IsPiecewiseSmooth.smul Chapter8/Energy · IsPiecewiseSmooth.clm_apply. On a piece, the derivative computed within the piece is smooth and equals the derivative at interior points Chapter8/Energy · IsPiecewiseSmooth.derivWithin_piece.

Jumps. The jump of the derivative of f at t is $\Delta_t f = f'(t^+) - f'(t^-)$, the difference of the right and the left derivative Chapter8/Energy · jump. It vanishes where f is differentiable Chapter8/Energy · jump_eq_zero and depends only on f near t Chapter8/Energy · jump_congr. If f is smooth on a piece $[s,t]$, the right derivative at s and the left derivative at t are the derivatives within the piece Chapter8/Energy · derivWithin_Ici_eq_of_piece Chapter8/Energy · derivWithin_Iic_eq_of_piece, and at a point of the piece where f is differentiable the derivative within the piece is the derivative Chapter8/Energy · derivWithin_eq_deriv_of_piece.

Covariant derivative and acceleration. Along a path c, the covariant derivative of a field X is $\frac{DX}{dt}(t) = P_{c(t)}X'(t)$ Chapter8/FirstVariation · covD. For a curve with values in N this is the covariant derivative of the Levi-Civita connection of the induced metric constructed in Part V (§§V.13, V.17) Chapter8/FirstVariation · covD_eq_covDeriv Chapter6/Parallel · covDeriv. The acceleration is $A = \frac{D}{dt}c'$ Chapter8/FirstVariation · accel. Both are meaningful only where the derivatives exist, which for piecewise smooth data means everywhere except at finitely many points.

VII.2 Calculus on closed strips

Milnor's proofs of the variation formulas differentiate under the integral sign, integrate by parts "on each $[t_{i-1}, t_i]$" and add up the results (pp. 71–72, 74–76). The file `Chapter8/Piecewise` proves what these steps rest on.

Induction over the pieces. Let B be a finite set and let $\mathcal{P}(a,b)$ be a property of intervals $[a,b] \subseteq [lo,hi]$. Suppose $\mathcal{P}$ holds for every interval whose interior contains no point of B, and that $\mathcal{P}(a,x)$ and $\mathcal{P}(x,b)$ imply $\mathcal{P}(a,b)$ whenever $a < x < b$ and $x \in B$. Then $\mathcal{P}(a,b)$ holds for all $lo \le a < b \le hi$ `Chapter8/Piecewise · piecewise_induction`. The proof is an induction on the number of points of B in (a,b). If there is such a point x, then (a,x) and (x,b) each contain fewer. Some companion facts locate pieces: a point of $(a,b]$ has a piece ending at it, a point of $[a,b)$ has a piece starting at it, a point of (a,b) outside B lies in the interior of a piece, and every point of $[a,b]$ lies in some piece `Chapter8/Piecewise · exists_piece_left` `Chapter8/Piecewise · exists_piece_right` `Chapter8/Piecewise · exists_piece_around` `Chapter8/Piecewise · exists_piece_mem`. Each piece is found by taking the largest point of $B \cup \{a\}$ below the given point and the smallest point of $B \cup \{b\}$ above it.

Three gluing results follow from the induction.

1. A function continuous on every piece of $[lo,hi]$ is continuous on $[lo,hi]$, because a function continuous on two closed intervals is continuous on their union `Chapter8/Piecewise · continuousOn_of_piecewise`.
2. If on every piece $[s,t]$ the function g agrees on (s,t) with a function continuous on $[s,t]$, then g is integrable on every $[a,b] \subseteq [lo,hi]$ `Chapter8/Piecewise · intervalIntegrable_of_piecewise`.
3. *The telescoping.* Suppose that on every piece $[s,t]$ one has $\int_s^t g = F^-(t) - F^+(s)$, where F^- and F^+ are functions of the end point, the left and the right boundary values. Then

$$\int_a^b g = F^-(b) - F^+(a) - \sum_{x \in B \cap (a,b)} \bigl(F^+(x) - F^-(x)\bigr)$$

   `Chapter8/Piecewise · integral_eq_of_piecewise`. The proof is the induction again, using additivity of the integral over adjacent intervals and the splitting of the sum at the gluing point `Chapter8/Piecewise · sum_filter_Ioo_split`. In the variation formulas $F^{\pm}(x) = \langle Y(x), f'(x^{\pm})\rangle$ for a continuous field Y, and the sum becomes $\sum_x \langle Y(x), \Delta_x f\rangle$. This is how the boundary terms of the integrations by parts "telescope into the jump terms".

Partial derivatives on a closed strip. Let $S = U \times [a,b]$ with $U \subseteq \mathbb{R}$ open and $a < b$. For F defined on S, $\partial F/\partial u$ and $\partial F/\partial t$ are the derivative of F within S, applied to $(1,0)$ and to $(0,1)$ `Chapter8/Piecewise · pdU` `Chapter8/Piecewise · pdT`. Since S has unique derivatives `Chapter8/Piecewise · uniqueDiffOn_strip`, both are smooth on S when F is `Chapter8/Piecewise · contDiffOn_pdU` `Chapter8/Piecewise · contDiffOn_pdT`. They are honest partial derivatives. $\frac{\partial F}{\partial u}(u,t)$ is the derivative of $u \mapsto F(u,t)$ at u, because the u-direction is open `Chapter8/Piecewise · hasDerivAt_pdU` `Chapter8/Piecewise · deriv_eq_pdU`. $\frac{\partial F}{\partial t}$ is the derivative within $[a,b]$ of $t \mapsto F(u,t)$ `Chapter8/Piecewise · hasDerivWithinAt_pdT`: the two-sided derivative inside `Chapter8/Piecewise · hasDerivAt_pdT`, the right derivative on the edge $t = a$ and the left derivative on the edge $t = b$ `Chapter8/Piecewise · derivWithin_Ici_eq_pdT` `Chapter8/Piecewise · derivWithin_Iic_eq_pdT`.

Symmetry of the mixed partials on the closed strip, edges included. For F smooth on S, $\partial_u \partial_t F = \partial_t \partial_u F$ at every point of S `Chapter8/Piecewise · pdU_pdT`. Both sides are values of the second derivative within S, namely at the pairs $((1,0),(0,1))$ and $((0,1),(1,0))$, because the derivative of $q \mapsto D_S F(q)v$ is $D_S(D_S F)(\cdot)\, v$. Mathlib's symmetry theorem for second derivatives within a set applies at every point of the closure of the interior of a set with unique derivatives `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffWithinAt.isSymmSndFDerivWithinAt`. The interior of S is $U \times (a,b)$, whose closure contains S. Milnor never comments on this point. It matters because the t-derivatives on the edges are one-sided, and the jump terms are made of exactly these derivatives.

Differentiation under the integral sign on a closed strip. Let F and F' be continuous on S, with $F'(u,t)$ the derivative in u of F at every (u,t), $t \in (a,b)$. Then $u \mapsto \int_a^b F(u,t)\,dt$ has derivative $\int_a^b F'(u_0,t)\,dt$ at every $u_0 \in U$ `Chapter8/Piecewise · hasDerivAt_integral_strip`. To prove it, choose δ with the δ-ball around u_0 inside U. Then F' is bounded by some M on the compact set $\bar{B}(u_0,\delta/2)\times[a,b]$. This constant is the domination required by Mathlib's theorem on differentiation of parametric integrals `Mathlib/Analysis/Calculus/ParametricIntervalIntegral · hasDerivAt_integral_of_dominated_loc_of_deriv_le`. That theorem needs the hypothesis on derivatives only for almost every t, so the end point $t = b$, a null set, is excluded. The piecewise version `Chapter8/Piecewise · hasDerivAt_integral_of_piecewise` assumes that on every piece $[s,t]$ the integrand $g(u,\tau)$ agrees, for $\tau \in (s,t)$, with a function of class C^1 on the closed strip $U\times[s,t]$. It concludes that $u \mapsto \int_a^b g(u,\tau)\,d\tau$ has derivative $\int_a^b \partial_u g(u_0,\tau)\,d\tau$. On each piece this is the strip theorem applied to the C^1 representative and its partial derivative in u, and replacing g by the representative inside the piece does not change the integrals. The pieces are then glued by the induction and additivity, with integrability of g and of $\partial_u g$ coming from the second gluing result. Milnor uses all of this without comment. The compactness behind the domination is the reason the strips must be closed in t.

VII.3 The path space, its "tangent spaces", and variations

Definition (Milnor p. 67; V8.1; §0.8). $\Omega(N;p,q)$ is the set of maps $c:\mathbb{R}\to V$ that are piecewise smooth on $[0,1]$, with $c(t)\in N$ for $t\in[0,1]$, $c(0)=p$ and $c(1)=q$ `Chapter8/Energy · pathSpace`. For a map ω, $T\Omega_\omega$ is the set of fields $X:\mathbb{R}\to V$ that are piecewise smooth on $[0,1]$, tangent along ω ($P_{\omega(t)}X(t)=X(t)$ for $t\in[0,1]$), and vanish at 0 and 1 `Chapter8/Energy · pathTangent Chapter8/Energy · mem_pathTangent`. It is a real vector space, by the closure properties of §VII.1 and the linearity of each P_x, and it depends only on ω on $[0,1]$ `Chapter8/FirstVariation · mem_pathTangent_congr`. The break points of a field need not be those of ω.

How $T\Omega_\omega$ is formalized. Milnor thinks of Ω as a kind of infinite-dimensional manifold (p. 67) but gives it a topology only in §16. The formal proof follows him. No topology or manifold structure is put on Ω, and $T\Omega_\omega$ is not a tangent space of anything: it is the vector space just defined. It is connected to Ω through variations, in two directions. The variation field of every variation of ω lies in $T\Omega_\omega$ (below), and every finite family of elements of $T\Omega_\omega$ is the family of variation fields of one variation (Lemma 11.A, §VII.5). So each statement Milnor makes about derivatives on Ω becomes a statement about the derivatives of functions $u\mapsto E(\bar{\alpha}(u))$ of one or two real variables. The second half of the connection is what turns these statements into statements about all of $T\Omega_\omega$. Elements of $T\Omega_\omega$ are functions on $\mathbb{R}$, and only their values on $[0,1]$ enter the first and second variations `Chapter8/FirstVariation · firstVariation_congr_field Chapter8/Hessian · secondVariation_congr`. This convention has one visible consequence, met in §VII.14: the literal null space of the formal E_{**} contains every field vanishing on $[0,1]$, and the nullity has to be read on $[0,1]$.

Definition (Milnor p. 68; V8.2). Let $\mathcal{P}$ be a real normed space, $U\subseteq\mathcal{P}$ an open set containing 0, and B a finite set. A map $\alpha(u,t)$, $u\in\mathcal{P}$, $t\in\mathbb{R}$, is a *variation with break points B* if $\alpha(u,t)\in N$ for $u\in U$ and $t\in[0,1]$; the end points $\alpha(u,0)$ and $\alpha(u,1)$ do not depend on $u\in U$; and $(u,t)\mapsto\alpha(u,t)$ is C^∞ on every closed strip $U\times[s,t]$ with $0\le s<t\le 1$ and no point of B in (s,t) `Chapter8/Energy · IsVariation`. It is a variation *of* the path $\alpha(0,\cdot)$. For $\mathcal{P}=\mathbb{R}$ and $U=(-\varepsilon,\varepsilon)$ this is Milnor's definition; §13 uses $\mathcal{P}=\mathbb{R}^2$. The formal definition is more general than the book's, since it allows any open parameter set in any normed space.

Each path $\alpha(u,\cdot)$, $u\in U$, is piecewise smooth, as the restriction of the strip smoothness to the segment $\{u\}\times[s,t]$ `Chapter8/Energy · IsVariation.isPiecewiseSmooth`, and lies in $\Omega(N;\alpha(0,0),\alpha(0,1))$ `Chapter8/Energy · IsVariation.mem_pathSpace`. A variation can be reparametrized: for a smooth φ from another normed space into $\mathcal{P}$ with $\varphi(0)\in U$, the map $u\mapsto\alpha(\varphi(u),\cdot)$ is a variation on $\varphi^{-1}(U)$ with the same break points `Chapter8/Energy · IsVariation.comp`, its strip smoothness coming from composition with $(u,t)\mapsto(\varphi(u),t)$.

The slices $u_2 \mapsto \alpha(u_1, u_2, \cdot)$ of a two-parameter variation, the restriction $u_1 \mapsto \alpha(u_1, 0, \cdot)$, and Milnor's $(u_1, u_2) \mapsto \bar{\alpha}(u_1 + u_2)$ of Remark 13.5 are all instances of this.

Definition. The *variation field* of a one-parameter variation is $W_t = \frac{\partial \alpha}{\partial u}(0, t)$, the derivative of $u \mapsto \alpha(u, t)$ at 0 `Chapter8/Energy · varField`. For a one-parameter variation, $u \mapsto \alpha(u, t)$ is differentiable at every $u \in U$ when $t \in [0, 1]$, since t lies in some piece and the derivative there is ∂_u on the strip `Chapter8/Energy · IsVariation.hasDerivAt_param`. Moreover $\frac{\partial \alpha}{\partial u}(u, t) \in T_{\alpha(u,t)}N$, as the velocity of a curve in N `Chapter8/Energy · IsVariation.prj_varDeriv`.

Lemma ("clearly $W \in T\Omega_\omega$", Milnor p. 68; V8.2). The variation field of a variation of ω lies in $T\Omega_\omega$ `Chapter8/Energy · IsVariation.varField_mem`.

The proof as formalized. On a piece $[s, t]$, W coincides on the whole closed piece with $\tau \mapsto \partial_u \alpha(0, \tau)$ computed within the strip `Chapter8/Piecewise · deriv_eq_pdU`, and that function is smooth (§VII.2). Hence W is smooth on every piece. Continuity at a break point t_i comes for free, which is the content hidden behind Milnor's "clearly". $W(t_i)$ is the derivative of the single curve $u \mapsto \alpha(u, t_i)$, and the two adjacent strips both compute that same derivative. So W is a single function, smooth on every closed piece and therefore continuous `Chapter8/Piecewise · continuousOn_of_piecewise`. Tangency is the previous fact at $u = 0$ `Chapter8/Energy · IsVariation.prj_varDeriv`. At $t = 0$ and $t = 1$ the curve $u \mapsto \alpha(u, t)$ is constant near 0, so its derivative vanishes.

VII.4 Energy and Schwarz's inequality

Definition (Milnor p. 70; V8.5; §0.8). For a map c into a normed space and $a \le b$, the energy is $E_a^b(c) = \int_a^b \|c'(t)\|^2 \, dt$ `Chapter8/Energy · energy`, and $E = E_0^1$. The length is $L_a^b(c) = \int_a^b \|c'(t)\| \, dt$ `Chapter7/Path · pathLength` (Part VI, §VI.10). For piecewise smooth c the function $\|c'\|^2$ is integrable, because on each piece it agrees inside with the squared norm of the derivative within the piece, which is continuous `Chapter8/Energy · IsPiecewiseSmooth.intervalIntegrable_norm_deriv_sq`. Piecewise smooth paths are admissible in the sense of Part VI (§VI.10), so that their speed is integrable `Chapter7/Path · IsPiecewiseSmooth.isAdmissible` `Chapter7/Path · IsAdmissible`.

Lemma 12.A of the exposition (Schwarz's inequality and its equality case; V8.5). Let c be piecewise smooth on $[a, b]$. Then $(L_a^b)^2 \le (b - a) E_a^b$ `Chapter8/Energy · pathLength_sq_le_energy`, with equality if and only if there is a number k such that $\|c'(t)\| = k$ at every $t \in (a, b)$ at which c is differentiable `Chapter8/Energy · pathLength_sq_eq_energy_iff`.

The proof as formalized. It runs through the variance identity

$$\int_a^b \big(\|c'\| - k\big)^2 = E_a^b - 2kL_a^b + k^2(b - a),$$

obtained by expanding, all terms being integrable `Chapter8/Energy · integral_norm_deriv_sub_sq`. For $a < b$ put $k = L/(b - a)$, the mean speed. Then $(b - a)(E - 2kL + k^2(b - a)) = (b - a)E - L^2$, and the left side is $(b - a)$ times the integral of a square, so it is nonnegative. For the equality case, equality makes that integral zero, so $(\|c'\| - k)^2 = 0$ almost everywhere. Turning this into a statement at a given $t \in (a, b)$ where c is differentiable takes a continuity argument. Take a piece $[s, t]$ ending at t `Chapter8/Piecewise · exists_piece_left`. On it, the function $(\|c'_{[s,t]}\| - k)^2$ built from the derivative within the piece is continuous on $[s, t]$ and almost everywhere 0 on (s, t), hence 0 on all of $[s, t]$, which lies in the closure of its interior. At t the derivative within the piece is $c'(t)$, because c is differentiable at t `Chapter8/Energy · derivWithin_eq_deriv_of_piece`. Conversely, a constant speed k gives $L = k(b - a)$ `Chapter7/Path · pathLength_eq_of_norm_deriv` and $E = k^2(b - a)$ `Chapter8/Energy · energy_eq_of_norm_deriv`, the speed being k off a countable set.

A hidden hypothesis. Milnor's equality case reads "$\|d\omega/dt\|$ constant", that is, parameter proportional to arc length. For a speed that is only piecewise continuous, "constant" can only mean constant away from

the break points, which is the formal statement. *Deviation (stronger).* Any $a \le b$ (Milnor: $0 \le a < b \le 1$) and any normed target space. The inequality of Lemma 12.1 is proved in §VII.10, where it is used.

VII.5 Lemma 11.A: a variation for every family of fields

Milnor observes on p. 68 that every $W \in T\Omega_\omega$ is the variation field of some variation, namely of $\bar\alpha(u)(t) = \exp_{\omega(t)}(uW_t)$. On p. 74 he takes two-parameter variations with given fields without comment. The exposition states both as Lemma 11.A, for any number of fields. The formal proof proves it with Milnor's formula, after supplying the one missing ingredient: a radius ε that works uniformly along ω.

Recall from Part VI (§VI.6; see also §0.9) that $\exp_q(v) = \gamma(1)$ for the geodesic γ defined on an open interval containing $[0,1]$ with $\gamma(0) = q$ and $\gamma'(0) = v$, and $\exp_q(v) = q$ if there is no such geodesic `Chapter7/Exp · exp`. The pairs (q, v) for which the geodesic exists form the *domain* of exp `Chapter7/Exp · expDom`. Two facts from Part VI (§VI.7) are used. First, exp is defined and smooth on an open neighbourhood of the zero section: there is an open set $O \subseteq V \times V$ containing every $(q, 0)$, $q \in N$, such that every point of $TN \cap O$ is in the domain of exp, and near every point of $TN \cap O$, exp agrees on TN with a smooth map defined on an open subset of $V \times V$ `Chapter7/Exp · exists_isOpen_exp`. Here TN is the set of pairs (q, v) with $q \in N$ and $v \in T_qN$ `Milnor/Chapter8/TangentBundle · tangentBundleSet`. Second, $d(\exp_q)_0$ is the inclusion of T_qN in V `Chapter7/Exp · hasFDerivAt_exp_zero`.

Lemma 11.A of the exposition (V8.3). Let $\omega \in \Omega(N; p, q)$, let $\mathcal{P}$ be a real normed space, and let $\Lambda(t) : \mathcal{P} \to V$ be continuous linear maps, depending piecewise smoothly on $t \in [0,1]$, with values in $T_{\omega(t)}N$ and with $\Lambda(0) = \Lambda(1) = 0$. There are $\varepsilon > 0$ and a finite set B such that

$$\alpha(u,t) = \exp_{\omega(t)}\big(\Lambda(t)u\big), \qquad \|u\| < \varepsilon,$$

is a variation with break points B, with $\alpha(0,t) = \omega(t)$ and $\frac{\partial\alpha}{\partial u}(0,t) = \Lambda(t)$ for $t \in [0,1]$ `Chapter8/FirstVariation · exists_variation`.

The proof as formalized.

1. *A uniform radius.* The set $K = \{(\omega(t), 0) : t \in [0,1]\}$ is compact, as the continuous image of $[0,1]$, and it is contained in O. Some δ-thickening of K therefore lies in O. Since the distance on $V \times V$ is the maximum of the two distances, $(\omega(t), v) \in O$ whenever $t \in [0,1]$ and $\|v\| < \delta$.
2. *A bound for Λ.* $\|\Lambda(t)\| \le C$ on $[0,1]$ for some $C \ge 0$, by continuity on a compact interval. Put $\varepsilon = \delta/(C+1)$. Then $\|\Lambda(t)u\| \le C\varepsilon < \delta$ for $\|u\| < \varepsilon$.
3. *Values in N.* Hence $(\omega(t), \Lambda(t)u) \in TN \cap O$ for $t \in [0,1]$ and $\|u\| < \varepsilon$. The pair is in the domain of exp, and $\alpha(u,t) \in N$ because it is the value at time 1 of a geodesic in N `Chapter8/FirstVariation · exp_mem_of_mem_expDom` `Chapter7/Exp · exp_eq_of_mem_expDom`.
4. *Fixed end points.* Since $\Lambda(0) = \Lambda(1) = 0$, $\alpha(u,0) = \exp_{\omega(0)}(0)$ for every u, and likewise at $t = 1$.
5. *Smoothness on the strips.* Take for B the union of the break points of ω and of Λ. On a strip $\{\|u\| < \varepsilon\} \times [s,t]$ free of B, the map $(u,\tau) \mapsto (\omega(\tau), \Lambda(\tau)u)$ is smooth. Near each point of the strip, exp agrees on TN with a smooth map F `Chapter7/Exp · exists_isOpen_exp`, so α agrees near that point, within the strip, with the composite of F with a smooth map, and is smooth there.
6. *The derivative at $u = 0$.* $\alpha(0,t) = \exp_{\omega(t)}(0) = \omega(t)$ `Chapter7/Exp · exp_zero`. Regard $\Lambda(t)$ as a linear map into $T_{\omega(t)}N$ and compose it with $d(\exp_{\omega(t)})_0$, which is the inclusion. The chain rule gives $\Lambda(t)$ as the derivative in u at 0.

What Milnor leaves implicit. $\exp_{\omega(t)}(uW_t)$ makes sense only near the zero section (Lemma 10.2). A uniform ε therefore has to be found, here by the compactness of $\omega([0,1])$ together with a bound for $\|\Lambda\|$. Completeness of N is never used.

Two special cases are what the rest of the part uses. For one field $W \in T\Omega_\omega$, take $\Lambda(t)u = uW(t)$, which is piecewise smooth as the image of W under a fixed continuous linear map `Chapter8/FirstVariation · IsPiecewiseSmooth.smulRight`. The variation $\exp_{\omega(t)}(uW(t))$, $|u| < \varepsilon$, then has variation field W on $[0,1]$ `Chapter8/FirstVariation · exists_variation_one`. For two fields W_1, W_2, take $\Lambda(t)(u_1,u_2) = u_1W_1(t) + u_2W_2(t)$, which gives a two-parameter variation with $\frac{\partial\alpha}{\partial u_i}(0,0,t) = W_i(t)$, the partial derivatives being derivatives along the coordinate axes by the chain rule `Chapter8/FirstVariation · exists_variation_two`.

VII.6 The first variation formula

Definition (Milnor p. 71; V8.7). The *first variation* of a path ω in the direction of a field X is

$$\mathrm{FV}(\omega, X) = -\sum_{0<t<1} \langle X(t), \Delta_t\omega' \rangle - \int_0^1 \langle X(t), A(t)\rangle\, dt,$$

with A the acceleration `Chapter8/FirstVariation · firstVariation`. The sum runs over all of $(0,1)$ and is Mathlib's sum of the finitely many nonzero terms (by convention 0 if there are infinitely many). If ω is differentiable off a finite set B, it equals the finite sum over $B \cap (0,1)$ `Chapter8/FirstVariation · finsum_jump_eq_sum` `Chapter8/FirstVariation · finsum_mem_Ioo_eq_sum`. $\mathrm{FV}(\omega, X)$ depends only on X on $[0,1]$ `Chapter8/FirstVariation · firstVariation_congr_field`. Milnor writes the sum over the break points t_i. The formal sum runs over the whole open interval, where its terms vanish away from the break points.

The proof works on one strip at a time, and a group of small lemmas translates between a variation and its strips. Inside a piece, with $c = \alpha(u_0,\cdot)$ and $Y = \frac{\partial\alpha}{\partial u}(u_0,\cdot)$, one has $c' = \partial_t\alpha$, $Y = \partial_u\alpha$ on the closed piece, $Y' = \partial_t\partial_u\alpha$ and $c'' = \partial_t\partial_t\alpha$, all partials computed within the strip `Chapter8/FirstVariation · piece_deriv` `Chapter8/FirstVariation · piece_varDeriv` `Chapter8/FirstVariation · piece_deriv_varDeriv` `Chapter8/FirstVariation · piece_deriv_deriv`. The one-sided velocities of c at the ends of a piece are $\partial_t\alpha$ on the edges `Chapter8/FirstVariation · piece_derivWithin_Ici` `Chapter8/FirstVariation · piece_derivWithin_Iic`. Away from the break points, both c and Y are differentiable `Chapter8/FirstVariation · IsVariation.differentiableAt` `Chapter8/FirstVariation · IsVariation.differentiableAt_varDeriv`.

Formula (12.2′) of the exposition: the first variation at every u (V8.9). Let α be a one-parameter variation with parameter set $U \subseteq \mathbb{R}$. For every $u_0 \in U$, the function $u \mapsto E(\alpha(u,\cdot))$ is differentiable at u_0, and

$$\frac{1}{2}\frac{d}{du}E(\alpha(u,\cdot))\Big|_{u=u_0} = -\sum_t \Big\langle \frac{\partial\alpha}{\partial u}, \Delta_t\frac{\partial\alpha}{\partial t}\Big\rangle - \int_0^1 \Big\langle \frac{\partial\alpha}{\partial u}, \frac{D}{\partial t}\frac{\partial\alpha}{\partial t}\Big\rangle dt,$$

everything evaluated at (u_0,t). In other words, the derivative is $2\,\mathrm{FV}(\alpha(u_0,\cdot), \frac{\partial\alpha}{\partial u}(u_0,\cdot))$ `Chapter8/FirstVariation · IsVariation.hasDerivAt_energy`.

The proof as formalized. Write $c = \alpha(u_0,\cdot)$ and $Y = \frac{\partial\alpha}{\partial u}(u_0,\cdot)$.

1. *Differentiation under the integral sign.* On each piece, the integrand $\|\partial_t\alpha(u,\tau)\|^2$ agrees inside with the same expression computed within the strip, which is smooth on the closed strip. So by the piecewise theorem of §VII.2 `Chapter8/Piecewise · hasDerivAt_integral_of_piecewise`, $\frac{d}{du}E(\alpha(u))(u_0) = \int_0^1 \partial_u(\|\partial_t\alpha\|^2)(u_0,\tau)\,d\tau$.
2. *The integrand.* Inside a piece, $\partial_u\|\partial_t\alpha\|^2 = 2\langle\partial_u\partial_t\alpha, \partial_t\alpha\rangle = 2\langle\partial_t\partial_u\alpha, \partial_t\alpha\rangle = 2\langle Y', c'\rangle$ at u_0, by the symmetry of the mixed partials on the strip `Chapter8/Piecewise · pdU_pdT`. Milnor's proof here uses Lemma 8.3 and "Lemma 8.5", that is, $\frac{D}{\partial u}\frac{\partial\alpha}{\partial t} = \frac{D}{\partial t}\frac{\partial\alpha}{\partial u}$. The reference should be to Lemma 8.7, since Lemma 8.5 is the definition of a symmetric connection. In V no projection is needed at this stage.
3. *Integration by parts on each piece.* On a piece $[s,t]$, the function $\tau \mapsto \langle Y(\tau), c'(\tau)\rangle$, with the strip's partials in place of Y and c', is continuous on the closed piece and has derivative $\langle Y', c'\rangle + \langle Y, c''\rangle$

inside. Hence $\int_s^t(\langle Y',c'\rangle+\langle Y,c''\rangle)=\langle Y(t),c'(t^-)\rangle-\langle Y(s),c'(s^+)\rangle$. The telescoping of §VII.2 `Chapter8/Piecewise · integral_eq_of_piecewise` gives

$$\int_0^1(\langle Y',c'\rangle+\langle Y,c''\rangle)=\langle Y(1),c'(1^-)\rangle-\langle Y(0),c'(0^+)\rangle-\sum_{x\in B\cap(0,1)}\langle Y(x),\Delta_x c'\rangle.$$

4. *The ends.* $Y(0)=Y(1)=0$, because $u\mapsto\alpha(u,0)$ and $u\mapsto\alpha(u,1)$ are constant near u_0.
5. *The covariant derivative appears.* Y is tangent `Chapter8/Energy · IsVariation.prj_varDeriv`, so $\langle Y,c''\rangle=\langle Y,Pc''\rangle=\langle Y,A\rangle$ on $[0,1]$ `Chapter8/Energy · inner_prj_right_of_eq`.
6. *Assembly.* Off the finite set $B\cup\{1\}$ the integrand of step 1 is $2\langle Y',c'\rangle$. Its integral is $2(-\sum_x\langle Y(x),\Delta_x c'\rangle-\int_0^1\langle Y,A\rangle)$, which is $2\,\mathrm{FV}(c,Y)$.

Theorem 12.2 (the first variation formula, Milnor p. 71; V8.9). Let α be a variation with $\alpha(0,t)=\omega(t)$ for $t\in[0,1]$ and variation field W. Then $u\mapsto E(\alpha(u,\cdot))$ has derivative $2\,\mathrm{FV}(\omega,W)$ at $u=0$, that is,

$$\frac{1}{2}\frac{dE(\bar\alpha(u))}{du}(0)=-\sum_t\langle W_t,\Delta_t V\rangle-\int_0^1\langle W_t,A_t\rangle\,dt$$

`Chapter8/FirstVariation · first_variation`.

Proof. This is (12.2′) at $u_0=0$ `Chapter8/FirstVariation · IsVariation.hasDerivAt_energy`. The first variation of $\alpha(0,\cdot)$ equals that of ω, because jumps and accelerations at points of $(0,1)$ depend only on the path near those points, and $\alpha(0,\cdot)=\omega$ on $[0,1]$. *Remarks.* In the scan, the leading minus sign of the displayed statement (p. 71) is lost; the end of the proof on p. 72 has it. Milnor states the formula at $u=0$ only, but his proof gives it at every u, and §13 needs it there: the proof of the second variation formula differentiates (12.2′) in a second parameter (§VII.8).

VII.7 Critical paths: Corollary 12.3

Definitions (Milnor pp. 69, 73; V8.4, V8.10). A map c is a *critical path* of the energy if, for every variation α (any parameter set, any break points) with $\alpha(0,t)=c(t)$ on $[0,1]$, the function $u\mapsto E(\alpha(u,\cdot))$ has derivative 0 at $u=0$ `Chapter8/Critical · IsCriticalPath`. Milnor defines critical paths for every function on Ω; the formal definition is for E only, which is what Corollary 12.3 needs. A map γ is an *(unbroken) geodesic* if it is C^∞ on $[0,1]$, with one-sided derivatives at the ends, and is a geodesic on $(0,1)$ `Chapter8/Critical · IsUnbrokenGeodesic`. Here (Part VI, §VI.2) a geodesic on an open set s is a curve that is smooth on s, takes values in N there, and has acceleration normal to N: $\gamma''(t)\perp T_{\gamma(t)}N$ `Chapter7/Geodesic · IsGeodesicOn`. This is Milnor's $\frac{D}{dt}\frac{d\gamma}{dt}=0$, since $D/dt=P\circ d/dt$.

Criticality through the first variation (V8.10). For $c\in\Omega(N;p,q)$, c is critical if and only if $\mathrm{FV}(c,X)=0$ for every $X\in T\Omega_c$ `Chapter8/Critical · isCriticalPath_iff`. Suppose c is critical and $X\in T\Omega_c$. Lemma 11.A gives the variation $\exp_{c(t)}(uX(t))$ `Chapter8/FirstVariation · exists_variation_one`. Criticality says $u\mapsto E$ has derivative 0 at 0, Theorem 12.2 says the derivative is $2\,\mathrm{FV}(c,W)$ `Chapter8/FirstVariation · first_variation` with $W=X$ on $[0,1]$ `Chapter8/FirstVariation · firstVariation_congr_field`, and derivatives are unique. Conversely, if $\mathrm{FV}(c,\cdot)$ vanishes on $T\Omega_c$, Theorem 12.2 gives derivative 0 for any variation, whose variation field lies in $T\Omega_c$ `Chapter8/Energy · IsVariation.varField_mem`. The first direction needs Lemma 11.A, which is hidden behind Milnor's "clearly".

A geodesic is critical (Milnor p. 73, "clearly"). For an unbroken geodesic, $\mathrm{FV}(\gamma,X)=0$ for every field X `Chapter8/Critical · IsUnbrokenGeodesic.firstVariation_eq_zero`. The jumps vanish because γ is differentiable on $(0,1)$, and the integrand vanishes on $(0,1)$ because $P\gamma''=0$ there `Chapter7/Geodesic · IsGeodesicOn.tanProj_deriv_deriv`.

Corollary 12.3 (Milnor p. 73; V8.11). A path $c\in\Omega(N;p,q)$ is a critical path of E if and only if it is an unbroken geodesic `Chapter8/Critical · corollary_12_3`.

The idea is Milnor's. Test fields concentrated on one piece make the acceleration vanish there; test fields concentrated at a break point make the velocity jump vanish; and a path that is a geodesic on both sides of a point and C^1 across it is a geodesic across it. The formal proof does the three steps separately. Let c be critical, with break points B.

Step 1: the acceleration vanishes off B `Chapter8/Critical · accel_eq_zero_of_critical`. Let $x \in (0,1) \setminus B$, lying inside a piece (s, s'). Put $r = \min(x-s, s'-x)/2$ and let φ be a smooth bump function centred at x, equal to 1 on $[x-r/2, x+r/2]$ and supported in $(x-r, x+r)$ `Chapter8/Critical · bumpAt` `Mathlib/Analysis/Calculus/BumpFunction/Basic · ContDiffBump`. The acceleration $A = P_c c''$ is smooth on (s, s'). The test field $X = \varphi A$ is smooth on all of $\mathbb{R}$, because φ vanishes outside a compact subset of (s, s'), where A is smooth `Chapter8/Critical · contDiff_smul_of_tsupport_subset`. It is tangent, since $PA = A$, and it vanishes at 0 and 1. So $X \in T\Omega_c$ and $\mathrm{FV}(c, X) = 0$. The jump terms vanish: inside (s, s') the path is differentiable, and outside it $\varphi = 0$. What remains is $\int_0^1 \varphi \|A\|^2 = 0$. The integrand is continuous and nonnegative, hence zero on $(0,1)$, and at x, where $\varphi(x) = 1$, this gives $A(x) = 0$. Milnor uses $W = f(t)A(t)$ with f positive off the break points. That field must be checked to be continuous at the t_i, where the one-sided accelerations differ. The formal proof avoids the issue by treating one point at a time.

Step 2: no velocity jumps `Chapter8/Critical · jump_eq_zero_of_critical`. At $x \notin B$ the path is differentiable. Let $x \in B$, with pieces $[s, x]$ and $[x, s']$ on either side, and $\Delta = \Delta_x c'$. The one-sided velocities of a path in N are tangent by lemma (a) of §VII.1, hence so is Δ `Chapter8/Critical · prj_jump`. Take the test field $X(t) = \varphi(t) P_{c(t)} \Delta$, with φ a bump at x of radius $r = \min(x-s, s'-x)/2$. It is piecewise smooth, being a smooth function times the piecewise smooth $t \mapsto P_{c(t)}\Delta$, tangent, and zero at the ends. In $\mathrm{FV}(c, X) = 0$ the integral term vanishes by Step 1, the acceleration being zero off $B \cup \{1\}$ `Chapter8/Critical · accel_eq_zero_of_critical`. The sum reduces to the term at x: at any other t, either $\varphi(t) = 0$, or t lies in $(x-r, x)$ or in $(x, x+r)$, where c is differentiable. The term at x is $\langle P\Delta, \Delta\rangle = \|\Delta\|^2$. Hence $\Delta = 0$. Milnor asks for a variation whose field takes the value $\Delta_{t_i} V$ at t_i; this needs an explicit field, and here the projection P plays the role that parallel transport would play on an abstract manifold.

Step 3: across a former break point, by uniqueness of geodesics `Chapter8/Critical · isGeodesicOn_of_glue`. Let γ be smooth on $[x-r, x]$ and on $[x, x+r]$, a geodesic on $(x-r, x)$ and on $(x, x+r)$, with $\gamma(x) \in N$ and γ differentiable at x. Then γ is a geodesic on a neighbourhood of x.

1. *The lift.* Let $Q = \mathrm{projRep}_\sigma$ for the slice chart σ at $\gamma(x)$, and $G(y, v) = (v, dQ_y(v)v)$ on $V \times V$ `Chapter8/Critical · liftField`. By the geodesic equation $\gamma'' = \mathrm{II}(\gamma', \gamma')$ of Part VI (§VI.2) `Chapter7/Geodesic · IsGeodesicOn.deriv_deriv_eq`, rewritten with $\mathrm{II}(v, \cdot) = dQ(v)$, the lift $f = (\gamma, \gamma')$ satisfies $f' = G(f)$ on both sides of x `Chapter8/Critical · IsGeodesicOn.hasDerivAt_lift`. G is C^1 near $f(x)$ `Chapter8/Critical · contDiffAt_liftField`.
2. *The lift solves the system at x too.* γ' is continuous at x: from each side the derivative within the piece is continuous and equals $\gamma'(x)$ at x, since γ is differentiable there. So f is continuous at x and $f' = G \circ f$ on a punctured neighbourhood of x, with $G \circ f$ continuous at x. A derivative extension theorem, applied from the right and from the left `Mathlib/Analysis/Calculus/FDeriv/Extend · hasDerivWithinAt_Ici_of_tendsto_deriv`, shows that f is differentiable at x with $f'(x) = G(f(x))$.
3. *Comparison.* Let σ be the geodesic through $\gamma(x)$ at time x with velocity $\gamma'(x)$ `Chapter8/Critical · exists_isGeodesicOn_through`. It comes from Lemma 10.2 (Part VI, §VI.5) `Chapter7/GeodesicLocal · lemma_10_2`, which gives geodesics $\Gamma(q, w, \cdot)$ on $(-2, 2)$ for $\|w\| < \varepsilon$, by rescaling: with $k = 2(\|v\| + 1)/\varepsilon$ and $w = v/k$, the curve $\sigma(t) = \Gamma(q, w, k(t-x))$ has $\sigma(x) = q$ and $\sigma'(x) = v$, and is a geodesic on $(x - 1/k, x + 1/k)$. The lifts of γ and σ solve $f' = G(f)$ near x with the same value at x, so by local uniqueness for an autonomous C^1 system (Part VI, §VI.3) `Chapter7/ODE · ode_eventuallyEq` they agree near x. Hence $\gamma = \sigma$ near x, and γ is a geodesic there.

Assembly. Near each $x \in (0,1)$, c is a geodesic. If $x \notin B$, this is Step 1 on the interior of the piece, because a smooth path in N with zero acceleration is a geodesic `Chapter8/Critical · isGeodesicOn_of_accel`. If $x \in B$, Step 2 makes the left and right derivatives agree `Chapter8/Critical · jump_eq_zero_of_critical`, so they

combine into a derivative at x, and Step 3 applies `Chapter8/Critical · isGeodesicOn_of_glue`. Being a geodesic is a local property `Chapter7/Minimal · isGeodesicOn_of_locally`, so c is a geodesic on $(0,1)$. It is smooth on $[0,1]$: near 0 and 1 by the first and last pieces, and inside because it is a geodesic.

What Milnor leaves implicit. The book settles the last step by an appeal to uniqueness for differential equations (p. 73). At a former break point the path is known to be a geodesic only on each side. Before uniqueness can be applied, the lift has to be shown to solve the geodesic system at the break point itself, which is the derivative-extension step 2.

VII.8 The curvature identity and the second variation formula

Milnor proves the second variation formula by differentiating (12.2′) in a second parameter. He rearranges the jump terms (p. 75) and, through Lemmas 8.7 and 9.2, reaches the identity (13.3), $\frac{D}{\partial u_1}\frac{D}{\partial t}V = \frac{D^2W_1}{dt^2} + R(V,W_1)V$. The formal proof follows the same plan. It needs (13.3) for a surface that is defined only on the strips, and it needs the jump terms to be differentiable in the parameter.

The curvature identity (13.3) (V8.12 (i)). Let $F(u,t)$ be smooth on an open set $O \subseteq \mathbb{R}^2$ with values in N. At every point of O,

$$\frac{D}{\partial u}\frac{D}{\partial t}\frac{\partial F}{\partial t} = \frac{D}{\partial t}\frac{D}{\partial t}\frac{\partial F}{\partial u} + R\Big(\frac{\partial F}{\partial t}, \frac{\partial F}{\partial u}\Big)\frac{\partial F}{\partial t},$$

where $\frac{D}{\partial u}X = P(\partial X/\partial u)$ and $\frac{D}{\partial t}X = P(\partial X/\partial t)$ `Chapter8/SecondVariation · curvature_identity`. The partial derivatives are the ordinary ones of Part V (§V.20) `Chapter6/Surface · partialX` `Chapter6/Surface · partialY`.

The proof as formalized is a direct computation in V, with the second fundamental form doing the work of Lemmas 8.7 and 9.2. Fix p_0, shrink O so that F lands in the domain of the slice chart σ at $F(p_0)$, and let $Q = \mathrm{projRep}_\sigma \circ F$, which equals P_F there. Write $a = \partial F/\partial t$, $b = \partial F/\partial u$, and $D_v w = \mathrm{II}_{F(p_0)}(v,w)$.

1. a and b are tangent everywhere, as derivatives of a map into N, so $Qa = a$ and $Qb = b$. By the chain rule, $\partial_u Q(p_0) = D_b$ and $\partial_t Q(p_0) = D_a$ `Chapter6/Projection · dN_tanProj_eq_fderiv`.
2. *Normal parts.* Differentiating $a = Qa$ in t by the product rule for partial derivatives `Chapter6/Surface · partialY_clm_apply` gives $\partial_t a = D_a a + P\partial_t a$, and likewise $\partial_t b = D_a b + P\partial_t b$.
3. *Symmetry of mixed partials* at an interior point `Chapter8/SecondVariation · partialX_partialY_of_contDiffOn` `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`, used once for a and once for F near p_0: $\partial_u\partial_t a = \partial_t\partial_t b$. This is Lemma 8.7 in V.
4. *The left side.* $P\partial_u(Q\partial_t a) = PD_b(\partial_t a) + P\partial_u\partial_t a$. By step 2, $PD_b(\partial_t a) = PD_b(P\partial_t a) + PD_b(D_a a)$. The first term is 0 because $PD_bP = 0$ `Chapter6/Projection · tanProj_dN_tanProj`. In the second, $D_a a$ is normal, so $D_b(D_a a)$ is tangent and P fixes it. The left side is therefore $D_bD_a a + P\partial_u\partial_t a$.
5. *The first term on the right* is, in the same way, $D_aD_a b + P\partial_t\partial_t b$.
6. By steps 3–5, the difference of the two is $D_bD_a a - D_aD_a b$. Since $D_a b = D_b a$ (symmetry of II), this is $\mathrm{II}(b,\mathrm{II}(a,a)) - \mathrm{II}(a,\mathrm{II}(b,a)) = R(a,b)a$ by the definition of R (§VII.1).

Milnor applies Lemma 9.2, which Part V proves for surfaces defined on all of $\mathbb{R}^2$, to a variation defined only on strips. Proving the identity at an interior point of an arbitrary open set removes the difficulty.

Definition (V8.13). For a path γ with velocity $V = \gamma'$ and fields X_1, X_2,

$$\Phi(X_1,X_2) = -\sum_{0<t<1}\Big\langle X_2(t), \Delta_t\frac{DX_1}{dt}\Big\rangle - \int_0^1 \Big\langle X_2, \frac{D^2X_1}{dt^2} + R(V,X_1)V\Big\rangle\,dt,$$

where $\Delta_t\frac{DX_1}{dt} = P_{\gamma(t)}(\Delta_t X_1')$ `Chapter8/SecondVariation · secondVariation`. Since $DX_1/dt = PX_1'$ and P is continuous along γ, the jump of DX_1/dt is the projection of the jump of X_1'.

Jumps under differentiation (V8.12 (ii)). For a one-parameter variation α, a parameter $u_0 \in U$ and $x \in (0,1)$, the jump $\Delta_x\frac{\partial\alpha}{\partial t}(u,\cdot)$ is differentiable in u at u_0, with derivative $\Delta_x(\frac{\partial}{\partial t}\frac{\partial\alpha}{\partial u})(u_0,\cdot)$ `Chapter8/`

SecondVariation · IsVariation.hasDerivAt_jump. Take the pieces $[s, x]$ and $[x, s']$ on either side of x. The jump is the difference between $\partial_t \alpha(u, x)$ computed within the right strip and within the left strip. Each is smooth in u, and its u-derivative is $\partial_u \partial_t$ on that strip. By the symmetry of the mixed partials *on the edge of the strip* Chapter8/Piecewise · pdU_pdT, this equals $\partial_t \partial_u$, which is the one-sided derivative at x of $t \mapsto \partial_u \alpha(u_0, t)$. This is what Milnor calls "rearranging the second term" (p. 75).

For a two-parameter variation, $\frac{\partial \alpha}{\partial u_2}$ on the slice $u_2 = 0$ agrees, on each strip, with a function of (u_1, τ) that is smooth on the closed strip, namely the derivative of α within the strip applied to the direction u_2 Chapter8/SecondVariation · IsVariation.exists_param₂.

Theorem 13.1 (the second variation formula, Milnor p. 74; V8.13). Let γ be an unbroken geodesic and $\alpha(u_1, u_2, t)$ a two-parameter variation, with parameter set $U \subseteq \mathbb{R}^2$, such that $\alpha(0, 0, t) = \gamma(t)$ on $[0, 1]$. Let $W_1 = \frac{\partial \alpha}{\partial u_1}(0, 0, \cdot)$ and $W_2 = \frac{\partial \alpha}{\partial u_2}(0, 0, \cdot)$. Then $u_1 \mapsto \frac{\partial}{\partial u_2} E(\alpha(u_1, u_2, \cdot))\big|_{u_2 = 0}$ is differentiable at $u_1 = 0$, with derivative $2\Phi(W_1, W_2)$ Chapter8/SecondVariation · second_variation:

$$\frac{1}{2} \frac{\partial^2 E}{\partial u_1 \, \partial u_2}(0, 0) = -\sum_t \left\langle W_2(t), \Delta_t \frac{DW_1}{dt} \right\rangle - \int_0^1 \left\langle W_2, \frac{D^2 W_1}{dt^2} + R(V, W_1)V \right\rangle dt.$$

The factor and the sign. Milnor's right side is *half* the mixed second derivative, so the Hessian is $E_{**} = 2\Phi$, and this factor reappears in every formula below. The curvature term carries Milnor's sign. With the opposite convention it would read $-R(V, W_1)V$, and the same is true of the Jacobi equation. The mixed second derivative is formalized as the iterated derivative $\partial/\partial u_1$ of $\partial E/\partial u_2$ along $u_2 = 0$, which is exactly what Milnor's proof computes. Its existence is part of the conclusion, and $E \circ \bar{\alpha}$ is not assumed to be C^2. The scan drops the "$-\sum_t$" in front of the first term of the displayed statement on p. 75; (13.2) on the same page has it.

The proof as formalized. Let $U_1 = \{u_1 : (u_1, 0) \in U\}$, $\sigma(u_1) = \alpha(u_1, 0, \cdot)$, and $Y(u_1)(t) = \frac{\partial \alpha}{\partial u_2}(u_1, 0, t)$. By reparametrization Chapter8/Energy · IsVariation.comp, σ is a variation of γ with variation field W_1, and W_2, the field of the slice $u_2 \mapsto \alpha(0, u_2)$, is tangent along γ Chapter8/Energy · IsVariation.varField_mem.

1. *(12.2′) at every u_1.* For $u_1 \in U_1$, the slice $u_2 \mapsto \alpha(u_1, u_2, \cdot)$ is a variation, and (12.2′) at $u_2 = 0$ gives $\frac{\partial E}{\partial u_2}(u_1, 0) = 2\,\mathrm{FV}(\sigma(u_1), Y(u_1))$ Chapter8/FirstVariation · IsVariation.hasDerivAt_energy. It therefore suffices to differentiate $u_1 \mapsto \mathrm{FV}(\sigma(u_1), Y(u_1))$ at 0. This is where Milnor's proof uses (12.2′) away from $u = 0$.
2. *The jump terms.* $\mathrm{FV}(\sigma(u_1), Y(u_1))$ contains $-\sum_{x \in B} \langle Y(u_1)(x), \Delta_x \sigma(u_1)' \rangle$. The map $u_1 \mapsto Y(u_1)(x)$ is differentiable, by the smooth representative on the strip to the right of x Chapter8/SecondVariation · IsVariation.exists_param₂. By the product rule, the derivative of each term at 0 is $\langle \partial_{u_1} Y(x), \Delta_x \gamma' \rangle + \langle W_2(x), \partial_{u_1} \Delta_x \sigma(u_1)' \rangle$. The first part vanishes because the geodesic γ is differentiable at x. The second part is $\langle W_2(x), \Delta_x W_1' \rangle$ by the lemma on jumps under differentiation Chapter8/SecondVariation · IsVariation.hasDerivAt_jump, and this equals $\langle W_2(x), P \Delta_x W_1' \rangle$ because W_2 is tangent.
3. *The integral term.* On each piece, the integrand $\langle Y(u_1)(\tau), P_{\sigma(u_1)(\tau)} \partial_t \partial_t \sigma(u_1, \tau) \rangle$ agrees inside with a function that is smooth on the closed strip: the representative of $\partial\alpha/\partial u_2$ from the lemma above, paired with $P \circ \sigma$ applied to the second t-derivative computed within the strip. Differentiation under the integral sign Chapter8/Piecewise · hasDerivAt_integral_of_piecewise gives the derivative $\int_0^1 \partial_{u_1}(\ldots)(0, \tau)\, d\tau$.
4. *The integrand at $u_1 = 0$.* Fix $t \notin B$ in a piece (s, s') and work on the open set $U_1 \times (s, s')$, where all partials are ordinary. By the product rule, the derivative is $\langle \partial_{u_1} Y, \mathrm{Acc}(0, t) \rangle + \langle W_2(t), \partial_{u_1} \mathrm{Acc}(0, t) \rangle$, with $\mathrm{Acc} = P(\partial_t \partial_t \sigma)$. Here $\mathrm{Acc}(0, t)$ is the acceleration of γ, which is 0. The second term equals $\langle W_2, P\, \partial_{u_1} \mathrm{Acc} \rangle$ because W_2 is tangent. By the curvature identity at $(0, t)$ Chapter8/SecondVariation · curvature_identity it equals $\langle W_2, P \partial_t (P \partial_t \partial_{u_1} \sigma) + R(\partial_t \sigma, \partial_{u_1} \sigma) \partial_t \sigma \rangle$. On

the line $u_1 = 0$ one has $\partial_{u_1}\sigma = W_1$, $\partial_t\sigma = \gamma'$, and $P\partial_t\partial_{u_1}\sigma = DW_1/dt$ near t, so $P\partial_t$ of the latter is D^2W_1/dt^2. The integrand is therefore $\langle W_2, \frac{D^2W_1}{dt^2} + R(V, W_1)V\rangle$.

5. *Assembly.* The sum in Φ reduces to the finite sum over B, since W_1 is differentiable off B, and the integrands of steps 3 and 4 agree off the finite set $B \cup \{1\}$.

What Milnor leaves implicit: (12.2′) at every $(u_1, 0)$; the differentiation of the jump and integral terms in u_1, which needs smoothness on closed strips and the symmetry of mixed partials on their edges; the curvature identity on strips; and the fact that $\frac{D}{\partial t}\frac{D}{\partial u_1}V$ at $u_1 = 0$ is $\frac{D^2W_1}{dt^2}$, because $D/\partial t$ along $u_1 = 0$ involves only values on that line.

Well-definedness (Milnor p. 74). Two two-parameter variations of γ whose variation fields agree on $[0, 1]$ have the same $\frac{\partial^2 E}{\partial u_1 \partial u_2}(0, 0)$ `Chapter8/SecondVariation · second_variation_eq`. The reason is that Φ depends only on the fields on $[0, 1]$: jumps and covariant derivatives at interior points are local.

Remark 13.5 (Milnor p. 76; V8.17). Let α be a one-parameter variation of the unbroken geodesic γ with field W. Then $u \mapsto \frac{d}{du}E(\alpha(u))$ is differentiable at 0, with derivative $2\Phi(W, W)$ `Chapter8/SecondVariation · remark_13_5`. Milnor's proof works as stated. $\beta(u_1, u_2) = \alpha(u_1 + u_2)$ is a two-parameter variation `Chapter8/Energy · IsVariation.comp` with both fields equal to W, so Theorem 13.1 applies to it `Chapter8/SecondVariation · second_variation`, and $\frac{\partial}{\partial u_2}E(\beta(u_1, u_2))\big|_{u_2=0} = \frac{dE(\alpha)}{du}(u_1)$, the derivative of a translate.

VII.9 The Hessian E_{**}: integral form, bilinearity, symmetry

Definition (Milnor p. 74; V8.14; §0.9). For $\gamma \in \Omega(N; p, q)$, the Hessian is the bilinear form on $T\Omega_\gamma$

$$E_{**}(W_1, W_2) = 2\Phi(W_1, W_2)$$

`Chapter8/Hessian · energyHessian` `Chapter8/Hessian · energyHessian_apply`, with bilinearity proved below. It is defined for every $\gamma \in \Omega$. Only at a geodesic is it the second derivative of E, and there it agrees with Milnor's definition, in three senses. (i) The variation fields of every two-parameter variation of γ lie in $T\Omega_\gamma$ `Chapter8/Hessian · IsVariation.partial₁_mem` `Chapter8/Hessian · IsVariation.partial₂_mem`, and Milnor's $\frac{\partial^2 E}{\partial u_1 \partial u_2}(0, 0)$ exists and equals $E_{**}(W_1, W_2)$ `Chapter8/Hessian · hasDerivAt_energyHessian`. (ii) Every pair $W_1, W_2 \in T\Omega_\gamma$ arises in this way, by Lemma 11.A for two fields `Chapter8/FirstVariation · exists_variation_two`. (iii) Remark 13.5, in terms of E_{**}: $\frac{d^2}{du^2}E(\bar\alpha(u))(0) = E_{**}(W, W)$ `Chapter8/Hessian · hasDerivAt_deriv_energy`. E_{**} depends only on the fields on $[0, 1]$ `Chapter8/Hessian · secondVariation_congr`.

Proposition 13.A of the exposition (the Hessian as an integral; V8.15). For $\gamma \in \Omega(N; p, q)$ and $X_1, X_2 \in T\Omega_\gamma$,

$$\frac{1}{2}E_{**}(X_1, X_2) = \int_0^1 \left(\left\langle \frac{DX_1}{dt}, \frac{DX_2}{dt} \right\rangle - \langle R(V, X_1)V, X_2\rangle \right) dt$$

`Chapter8/Hessian · hessForm` `Chapter8/Hessian · secondVariation_eq_hessForm` `Chapter8/Hessian · energyHessian_eq_integral`. γ need not be a geodesic. The formula is not in Milnor's book.

The proof as formalized. Take a common subdivision B for γ, X_1 and X_2, the union of their break sets `Chapter8/Hessian · exists_pieceData`. On each piece $[s, t]$ the three maps are smooth, γ lies in N, and the fields are tangent `Chapter8/Hessian · PieceData`. On a piece, $DX = PX'_{[s,t]}$, built from the derivative within the piece `Chapter8/Hessian · PieceData.D`, is smooth on the closed piece `Chapter8/Hessian · PieceData.contDiffOn_D` and equals the covariant derivative inside `Chapter8/Hessian · PieceData.D_eq`; the same holds for D^2X `Chapter8/Hessian · PieceData.covD_covD_eq`.

1. *Integration by parts on a piece* (Lemma 8.3):

$$\int_s^t \Big(\Big\langle \frac{DX_2}{dt}, \frac{DX_1}{dt} \Big\rangle + \Big\langle X_2, \frac{D^2X_1}{dt^2} \Big\rangle \Big) = \langle X_2(t), PX_1'(t^-) \rangle - \langle X_2(s), PX_1'(s^+) \rangle$$

   `Chapter8/Hessian · PieceData.integral_eq`. The derivative of $\langle X_2, DX_1\rangle$ is $\langle X_2', DX_1\rangle + \langle X_2, (DX_1)'\rangle$. Since P is self-adjoint and $P^2 = P$, $\langle X_2', P(...)\rangle = \langle DX_2, DX_1\rangle$. Since X_2 is tangent, $\langle X_2, (DX_1)'\rangle = \langle X_2, P(DX_1)'\rangle = \langle X_2, D^2X_1\rangle$.
2. *Integrability.* The curvature term is continuous on the closed piece `Chapter8/Hessian · PieceData.continuousOn_curv`. Near $\gamma(\tau_0)$, $R(a,b)c$ is an expression in the derivative of $\mathrm{projRep}_\sigma$ at $\gamma(\tau)$, valid for tangent a and b `Chapter8/Hessian · continuousWithinAt_curvR`. All integrands are thus integrable by the gluing of §VII.2 `Chapter8/Piecewise · intervalIntegrable_of_piecewise`.
3. *The telescoping* `Chapter8/Piecewise · integral_eq_of_piecewise`, with X_2 vanishing at the ends: $\int_0^1 (\langle DX_2, DX_1\rangle + \langle X_2, D^2X_1\rangle) = -\sum_x \langle X_2(x), P\Delta_x X_1'\rangle$.
4. *Substitution.* In $\Phi = -\sum - \int\langle X_2, D^2X_1\rangle - \int\langle X_2, R(V,X_1)V\rangle$ the jump terms cancel, leaving $\int\langle DX_2, DX_1\rangle - \int\langle R(V,X_1)V, X_2\rangle$.

Corollary 13.4 (Milnor p. 76; V8.16). For $\gamma \in \Omega(N;p,q)$, E_{**} is a symmetric bilinear form on $T\Omega_\gamma$ `Chapter8/Hessian · energyHessian_isSymm`.

The proof as formalized. Symmetry. For tangent v, w_1, w_2 at a point of N, $\langle R(v,w_1)v, w_2\rangle = \langle R(v,w_2)v, w_1\rangle$ `Chapter8/Hessian · inner_curvR_comm`. This is Lemma 9.3 (4) with $X = Z = V$, obtained here from the Gauss equation: the left side is $\langle \mathrm{II}(v,v), \mathrm{II}(w_1,w_2)\rangle - \langle \mathrm{II}(v,w_2), \mathrm{II}(w_1,v)\rangle$, which is symmetric in w_1, w_2 by the symmetry of II. Hence the integrand of the integral form is symmetric wherever γ is differentiable, its velocity then being tangent, and the integral form is symmetric `Chapter8/Hessian · hessForm_comm`. *Linearity in the second argument.* Where the fields are differentiable, which is almost everywhere `Chapter8/Hessian · IsPiecewiseSmooth.ae_differentiableAt`, the covariant derivative is additive and homogeneous `Chapter8/Hessian · covD_add` `Chapter8/Hessian · covD_smul`. With integrability `Chapter8/Hessian · intervalIntegrable_hessIntegrand` this gives linearity in X_2 `Chapter8/Hessian · hessForm_add_right` `Chapter8/Hessian · hessForm_smul_right`. *Linearity in the first argument* follows by symmetry, and this is how E_{**} is constructed as a bilinear form: Proposition 13.A, then symmetry, then linearity in the second argument.

What Milnor leaves implicit. Milnor derives symmetry from $\frac{\partial^2 E}{\partial u_1 \partial u_2} = \frac{\partial^2 E}{\partial u_2 \partial u_1}$, which needs $E \circ \bar\alpha$ to be C^2 near $(0,0)$, a point he does not address. Bilinearity in W_1 is not visible in his formula without knowing that $W \mapsto \Delta_t \frac{DW}{dt}$ and $W \mapsto \frac{D^2W}{dt^2}$ are linear on piecewise smooth fields, which involves one-sided differentiability at the break points. The integral form gives both properties at once.

VII.10 The inequality of Lemma 12.1, and Lemma 13.6

Definition (V8.6). A path γ is *minimal* from p to q if $\gamma \in \Omega(N;p,q)$ and $L(\gamma) \le L(\omega)$ for every $\omega \in \Omega(N;p,q)$ `Chapter8/Hessian · IsMinimalPath`. Up to affine reparametrization, an equivalence that is not stated formally, this is Milnor's notion.

The inequality of Lemma 12.1 (Milnor p. 70; V8.6). If γ is minimal and an unbroken geodesic, then $E(\gamma) \le E(\omega)$ for every $\omega \in \Omega(N;p,q)$ `Chapter8/Hessian · energy_le_of_isMinimalPath`.

Proof. This is Milnor's chain $E(\gamma) = L(\gamma)^2 \le L(\omega)^2 \le E(\omega)$. A geodesic has constant speed on $(0,1)$ `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq`, so Lemma 12.A holds with equality for γ (with $b-a = 1$) `Chapter8/Energy · pathLength_sq_eq_energy_iff` and with inequality for ω `Chapter8/Energy · pathLength_sq_le_energy`. Minimality gives $0 \le L(\gamma) \le L(\omega)$. Completeness serves in the book only to guarantee that a minimal geodesic exists (Part VI, §§VI.18–VI.19); it plays no role in the inequality. The value d^2 of the minimum and the converse (a path of energy d^2 is a minimal geodesic) are not formalized.

Lemma 13.6 (Milnor p. 76; V8.18). If γ is a minimal unbroken geodesic, then $E_{**}(W,W) \geq 0$ for every $W \in T\Omega_\gamma$ `Chapter8/Hessian · lemma_13_6`.

The proof as formalized. Suppose $E_{**}(W,W) < 0$. Lemma 11.A gives the variation $\alpha(u,t) = \exp_{\gamma(t)}(uW(t))$, $|u| < \varepsilon$ `Chapter8/FirstVariation · exists_variation_one`. Put $f(u) = E(\alpha(u))$.

1. f has a local minimum at 0. Every $\alpha(u)$ lies in $\Omega(N;p,q)$ `Chapter8/Energy · IsVariation.mem_pathSpace`, so $f(u) \geq E(\gamma) = f(0)$ by the inequality above `Chapter8/Hessian · energy_le_of_isMinimalPath`, the energy depending only on the values on $[0,1]$ `Chapter8/Hessian · energy_congr`.
2. f is differentiable at 0 by (12.2′) `Chapter8/FirstVariation · IsVariation.hasDerivAt_energy`, so $f'(0) = 0$. Its second derivative at 0 exists and equals $E_{**}(W,W)$ by Remark 13.5 `Chapter8/Hessian · hasDerivAt_deriv_energy`, since the variation field equals W on $[0,1]$ `Chapter8/Hessian · secondVariation_congr`. So it is negative.
3. By the second derivative test `Mathlib/Analysis/Calculus/DerivativeTest · isLocalMax_of_deriv_deriv_neg`, 0 is also a local maximum of f. Then f is constant near 0 and $f''(0) = 0$, a contradiction.

What Milnor leaves implicit. That a minimum forces a nonnegative second derivative needs an argument when only the second derivative *at* 0 is known to exist: Taylor's formula with the second derivative at intermediate points would need more regularity than has been proved. Milnor concludes that the index of E_{**} is zero. The index is defined formally in §VII.18, but its vanishing at a minimal geodesic is not stated as a separate theorem. What is proved is the positive semi-definiteness, which is equivalent to it.

VII.11 Theorem 14.1: the null vectors of E_{**} are the Jacobi fields

Definitions (Milnor p. 77; V9.2; §0.9). Along a map γ with $V = \gamma'$, the *Jacobi operator* of a field J is $\frac{D^2J}{dt^2} + R(V,J)V$ `Chapter8/Jacobi · jacobiExpr`. J is a *Jacobi field along* γ if it is C^∞ on $[0,1]$, with one-sided derivatives at the ends, tangent along γ on $[0,1]$, and satisfies the Jacobi equation $\frac{D^2J}{dt^2} + R(V,J)V = 0$ on $(0,1)$ `Chapter8/Jacobi · IsJacobiField`. Milnor defines Jacobi fields along geodesics. The formal definition makes sense along any curve, and the theory of §§VII.12–VII.13 is developed in that generality.

The easy half. A Jacobi field vanishing at 0 and 1 lies in $T\Omega_\gamma$ `Chapter8/Jacobi · IsJacobiField.mem_pathTangent`. For a Jacobi field J, $\Phi(J,Y) = 0$ for *every* field Y `Chapter8/Jacobi · IsJacobiField.secondVariation_eq_zero`: J is smooth on $(0,1)$, so it has no jumps, and the integrand vanishes by the Jacobi equation.

Theorem 14.1 (Milnor p. 78; V9.5). Let $\gamma \in \Omega(N;p,q)$ be an unbroken geodesic and $W_1 \in T\Omega_\gamma$. Then $E_{**}(W_1,W_2) = 0$ for every $W_2 \in T\Omega_\gamma$ if and only if W_1 is a Jacobi field along γ `Chapter8/Jacobi · theorem_14_1`. In particular a null vector is C^∞ on all of $[0,1]$.

The idea is Milnor's. Test fields concentrated inside a piece show that W_1 satisfies the Jacobi equation there, and test fields concentrated at a break point show that DW_1/dt does not jump. The pieces then "fit together". The formal proof replaces this last step, which in the book rests on the existence and uniqueness of Jacobi fields along all of γ, by a local regularity argument. Let $X \in T\Omega_\gamma$ be a null vector, smooth on the pieces of a finite set B.

Step 1: the Jacobi equation off B `Chapter8/Jacobi · jacobiExpr_eq_zero_of_null`. As in Step 1 of Corollary 12.3, with the test field $\varphi\,\mathcal{J}(X)$ `Chapter8/Critical · bumpAt` `Chapter8/Critical · contDiff_smul_of_tsupport_subset`, where $\mathcal{J}(X)$ is the Jacobi operator of X and φ a bump function inside the piece around x. On the interior of the piece $\mathcal{J}(X)$ is smooth `Chapter8/Jacobi · contDiffOn_jacobiExpr`, the curvature term being smooth along smooth tangent data `Chapter8/Jacobi · contDiffAt_curvR`. It is tangent `Chapter8/Jacobi · prj_jacobiExpr`, since D^2X/dt^2 is a projection and $R(V,X)V$ is tangent for tangent V and X `Chapter6/Submanifold · curvOp_mem` `Chapter8/Jacobi · IsUnbrokenGeodesic.prj_deriv`. The test field lies in $T\Omega_{\gamma'}$, the jump terms vanish, and $0 = \Phi(X, \varphi\mathcal{J}(X)) = -\int_0^1 \varphi\|\mathcal{J}(X)\|^2$ forces $\mathcal{J}(X)(x) = 0$.

Step 2: no jumps `Chapter8/Jacobi · jump_eq_zero_of_null`. At a break point x, the test field $\varphi(t)P_{\gamma(t)}\Delta$ with $\Delta =$

$P_{\gamma(x)}\Delta_x X'$ and φ a bump at x gives $\|\Delta\|^2 = 0$, exactly as in Step 2 of Corollary 12.3; the integral term vanishes by Step 1 `Chapter8/Jacobi · jacobiExpr_eq_zero_of_null`. This kills the tangential part of the jump. The normal part vanishes for another reason: $X = PX$ near x, so $X'(x^{\pm}) = P'(x)X(x) + PX'(x^{\pm})$, and the two one-sided derivatives have the same normal part, hence $\Delta_x X' = P\Delta_x X'$ `Chapter8/Jacobi · jump_eq_prj_jump`. In Milnor's language both normal parts are $\mathrm{II}(V, W_1)$. So X is C^1 at x. In $\mathbb{R}^K$, the absence of jumps of DW_1/dt gives $W_1 \in C^1$ only because of this observation.

Step 3: smoothness across the former break points `Chapter8/Jacobi · smooth_near_of_null`.

1. *The Jacobi equation as a second-order equation in V* `Chapter8/Jacobi · deriv_deriv_eq_of_jacobiExpr`. For a smooth tangent field X, write $P(t) = P_{\gamma(t)}$. Then $X = PX$ gives $X' = P'X + PX'$ and $X'' = P''X + 2P'X' + PX''$, so $(1-P)(P''X + 2P'X') = X'' - PX''$. The Jacobi equation $P(P'X' + PX'') + R(V,X)V = 0$ gives PX''. Adding,
$$X'' = -P(P'X') - R(V,X)V + (1-P)(P''X + 2P'X').$$
2. *A smooth right side.* Near a break point x, define $F(t, J, J')$ by this right side, with $R(V,J)V$ written through the derivative of $\mathrm{projRep}_\sigma$ applied to $P(t)J$. Then F is smooth in $(t, J, J') \in O \times V \times V$ for an interval O around x, and it agrees with the formula above for tangent J.
3. *The equation holds at x too.* By Step 2, X is differentiable at x and X' is continuous there, since the one-sided derivatives agree and are continuous from each side. At $t \neq x$ in O, Step 1 and the second-order form give $X'' = F(t, X, X')$. A derivative extension lemma `Chapter8/Jacobi · hasDerivAt_of_hasDerivAt_punctured` says that if f and g are continuous at x and $f' = g$ on a punctured neighbourhood, then $f'(x) = g(x)$. It shows that X' is differentiable at x with derivative $F(x, X(x), X'(x))$.
4. *Bootstrap.* A solution of $J'' = F(t, J, J')$ with F smooth is smooth `Chapter8/Jacobi · contDiffOn_of_ode`, by induction: if J and J' are C^k, then $J'' = F(t, J, J')$ is C^k, so J and J' are C^{k+1}. So X is smooth on O. Its Jacobi operator is then continuous on O `Chapter8/Jacobi · contDiffOn_jacobiExpr` and vanishes off x, hence at x.

Assembly. X is smooth on $[0,1]$: at 0 and 1 by the first and last pieces, inside by Step 3. The Jacobi equation holds on $(0,1)$ by Steps 1 and 3 `Chapter8/Jacobi · smooth_near_of_null`. The converse is the easy half above `Chapter8/Jacobi · IsJacobiField.secondVariation_eq_zero`.

What Milnor leaves implicit. The test fields must be continuous at the break points, which the bump functions ensure. The field with prescribed values at the break points has to be constructed. And "fit together", justified in the book because a Jacobi field is determined by its value and derivative at one point, needs the global existence and uniqueness theory of §VII.12. That theory is formalized but not used here. The local argument above needs no existence theorem at all.

VII.12 Linear equations on the whole line

Milnor writes the Jacobi equation in a parallel orthonormal frame as a linear second-order system and concludes that it has $2n$ independent solutions, all defined along the whole of γ (p. 77). The local existence theorem gives solutions only on intervals of length about $1/\sup\|A\|$, so a global theorem for linear equations is needed. Part V (§V.18) proved one only for skew-adjoint coefficients, those of parallel transport. The file `Chapter9/JacobiODE` proves it in general.

Lemma 14.A of the exposition (V9.1). Let E be a complete real normed space and $t \mapsto A(t)$ a continuous map from $\mathbb{R}$ to the continuous linear maps of E.

(i) *Global existence.* For every t_0 and x_0 there is a solution of $x' = A(t)x$ on all of $\mathbb{R}$ with $x(t_0) = x_0$ `Chapter9/JacobiODE · exists_linearODE`.

The proof as formalized.

1. *On a bounded interval* $(t_0 - a, t_0 + a)$ Chapter9/JacobiODE · exists_linearODE_Ioo. Let $\|A(t)\| \le C$ on $[t_0-a, t_0+a]$ Chapter6/LinearODE · exists_nnnorm_le_Icc, and put $R = \|x_0\|e^{Ca}+1$. Truncate the field radially: $f(t,x) = A(t)\rho_R(x)$, with $\rho_R(x) = \frac{R}{\max(R,\|x\|)}x$ the radial retraction onto the ball of radius R Chapter6/LinearODE · radTrunc. ρ_R is 2-Lipschitz and bounded by R Chapter6/LinearODE · lipschitzWith_radTrunc Chapter6/LinearODE · norm_radTrunc_le, so f is bounded by CR and Lipschitz in x with constant $2C$. The Picard–Lindelöf theorem Mathlib/Analysis/ODE/PicardLindelof · IsPicardLindelof Mathlib/Analysis/ODE/ExistUnique · IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt₀ then gives a solution on the whole closed interval in one step.
2. *The truncation is never active.* ρ_R does not increase norms Chapter9/JacobiODE · norm_radTrunc_le_norm, so $\|f(t,\alpha(t))\| \le C\|\alpha(t)\|$. Grönwall's inequality forward in time Chapter9/JacobiODE · norm_le_of_norm_deriv_le Mathlib/Analysis/ODE/Gronwall · norm_le_gronwallBound_of_norm_deriv_right_le gives $\|\alpha(t)\| \le \|x_0\|e^{C(t-t_0)} < R$ for $t \ge t_0$. Applied to $s \mapsto \alpha(t_0 - s)$, it gives the same backward in time. So $\rho_R(\alpha(t)) = \alpha(t)$ Chapter6/LinearODE · radTrunc_of_norm_le, and α solves the true equation on the open interval.
3. *On* $\mathbb{R}$. Choose such a solution β_a for every $a > 0$. Two of them agree where both are defined, by uniqueness for linear equations on open intervals (Part V, §V.18) Chapter6/LinearODE · linearODE_eqOn. So $\alpha(t) = \beta_{|t-t_0|+1}(t)$ solves the equation at every t. (The exposition uses Picard iteration on the whole interval instead.)

(ii) *The propagator.* $\mathrm{sol}_s(\xi)$ is a chosen global solution with value ξ at s Chapter9/JacobiODE · linSol Chapter9/JacobiODE · linSol_self Chapter9/JacobiODE · hasDerivAt_linSol. By uniqueness on $\mathbb{R}$ Chapter6/LinearODE · linearODE_eq, every global solution is the propagated solution through any of its values Chapter9/JacobiODE · eq_linSol. Hence sol_s is linear in ξ Chapter9/JacobiODE · linSol_add Chapter9/JacobiODE · linSol_smul: a sum of solutions is a solution, with the summed value at s. For finite-dimensional E, the propagator $\xi \mapsto \mathrm{sol}_s(\xi)(t)$ is a continuous linear map Chapter9/JacobiODE · linFlow that depends continuously on t Chapter9/JacobiODE · continuous_linFlow, since in finite dimension an operator-valued map is continuous when its values on each vector are.

(iii) *Uniqueness on closed intervals, end points included.* A function continuous on $[a,b]$ that solves the equation on (a,b) agrees on $[a,b]$ with the propagated solution through its value at any $s \in [a,b]$ Chapter9/JacobiODE · eqOn_Icc_linSol. It agrees on (a,b) with the solution through its value at the midpoint Chapter6/LinearODE · linearODE_eqOn, hence on $[a,b]$ by continuity Chapter9/JacobiODE · continuous_linSol, and that solution is the one through $f(s)$ Chapter9/JacobiODE · eq_linSol.

(iv) *Regularity.* If A is smooth on $[a,b]$, with one-sided derivatives at the ends, and $f' = Af$ at every point of $[a,b]$, then f is smooth on $[a,b]$ Chapter9/JacobiSystem · contDiffOn_Icc_of_hasDerivAt. This is a bootstrap with derivatives within $[a,b]$.

VII.13 The Jacobi equation as a linear system

The setting along a curve (V9.2). A *smooth curve in* N *on* $[0,1]$ is a map γ that is C^∞ on $[0,1]$, with one-sided derivatives at the ends, and takes values in N there Chapter9/JacobiSystem · IsSmoothCurveIn. Every unbroken geodesic of Ω is one Chapter9/JacobiSystem · IsUnbrokenGeodesic.isSmoothCurveIn, and every such curve lies in $\Omega(N;\gamma(0),\gamma(1))$ Chapter9/IndexForm · IsSmoothCurveIn.mem_pathSpace. Along it, the velocity $V(t)$ is the derivative within $[0,1]$ Chapter9/JacobiSystem · velAlong, $P'(t)$ is the derivative within $[0,1]$ of $t \mapsto P_{\gamma(t)}$ Chapter9/JacobiSystem · dpAlong, and $\frac{DX}{dt}(t) = P(t)X'(t)$ with the derivative taken within $[0,1]$ Chapter9/JacobiSystem · covAlong. All three are therefore one-sided at the ends, as Milnor's initial conditions at $t = 0$ require. T_xN is also regarded as a subspace of V, the zero subspace off N Chapter9/JacobiSystem · tanAt. Differentiating $P^2 = P$ within $[0,1]$ gives $P' = P'P + PP'$ Chapter9/JacobiSystem · IsSmoothCurveIn.dp_eq, hence $PP'P = 0$ Chapter9/JacobiSystem · IsSmoothCurveIn.prj_dp_prj. The velocity is tangent Chapter9/JacobiSystem · IsSmoothCurveIn.velAlong_mem.

The coefficients. $K(t)v = R(V(t),P(t)v)V(t)$ Chapter9/JacobiSystem · curvAlong Chapter9/

JacobiSystem · curvAlong_apply, built from the linear map $b \mapsto R(a,b)a$ Chapter9/JacobiSystem · curvMid. The projection $P(t)$ inside makes K defined on all of V. Put $S = P'P - PP'$ Chapter9/JacobiSystem · jacS. The *Jacobi system* is the first-order linear system

$$J' = SJ + Y, \qquad Y' = SY - KJ$$

for $(J, Y) \in V \times V$ Chapter9/JacobiSystem · jacSys Chapter9/JacobiSystem · jacSys_apply. Its coefficient is extended to $\mathbb{R}$ as a constant outside $[0,1]$, by evaluating it at the nearest point of $[0,1]$ Chapter9/JacobiSystem · jacA. The algebra used below is that $Sv = P'v$ and $PSv = 0$ for tangent v Chapter9/JacobiSystem · IsSmoothCurveIn.jacS_of_tangent Chapter9/JacobiSystem · IsSmoothCurveIn.prj_jacS_of_tangent, that $(1-P)Sv = P'Pv$ for every v Chapter9/JacobiSystem · IsSmoothCurveIn.jacS_sub_prj, and that K takes tangent values Chapter9/JacobiSystem · IsSmoothCurveIn.prj_curvAlong. The coefficients are smooth on $[0,1]$ Chapter9/JacobiSystem · IsSmoothCurveIn.contDiffOn_curvAlong Chapter9/JacobiSystem · IsSmoothCurveIn.contDiffOn_jacSys: near $\gamma(t_0)$, K is an expression in the derivative of $\mathrm{projRep}_\sigma$ along γ. The extended coefficient is therefore continuous on $\mathbb{R}$ Chapter9/JacobiSystem · IsSmoothCurveIn.continuous_jacA, and §VII.12 applies. The solution with value ζ at s is the propagated one Chapter9/JacobiSystem · jacSol. It is linear in ζ Chapter9/JacobiSystem · jacSol_add Chapter9/JacobiSystem · jacSol_smul Chapter9/JacobiSystem · jacSol_zero and smooth on $[0,1]$ by (iv) Chapter9/JacobiSystem · contDiffOn_jacSol.

Why the term S. Without a frame, a solution of a linear system in $V \times V$ has no reason to stay tangent to N. The term S is chosen to make the tangent bundle along γ invariant.

Tangent data stay tangent Chapter9/JacobiSystem · jacSol_mem_tanAt. Let $s \in [0,1]$ and $\zeta = (x, y)$ with $x, y \in T_{\gamma(s)}N$. Then both components of the solution are tangent on $[0,1]$.

Proof. Let $a = (1-P)J$ and $b = (1-P)Y$ be the normal parts. On $(0,1)$,

$$a' = (1-P)J' - P'J = (1-P)(SJ+Y) - P'J = P'PJ + (1-P)Y - P'J = -P'a + b,$$

and $b' = (1-P)(SY - KJ) - P'Y = P'PY - P'Y = -P'b$, because $(1-P)KJ = 0$. So (a,b) solves a homogeneous linear system with continuous coefficients, and it vanishes at s. Uniqueness on $[0,1]$, in the form (iii) of §VII.12 Chapter9/JacobiODE · eqOn_Icc_linSol, makes it vanish on $[0,1]$.

Solutions with tangent data are Jacobi fields Chapter9/JacobiSystem · isJacobiField_jacSol. For such ζ, the first component J is a Jacobi field and the second is DJ/dt on $[0,1]$. Indeed $DJ/dt = PJ' = P(SJ + Y) = Y$, because $PSJ = 0$ and Y is tangent, both components being tangent by the previous result Chapter9/JacobiSystem · jacSol_mem_tanAt. Then $D^2J/dt^2 = PY' = P(SY - KJ) = -KJ = -R(V,J)V$.

Every Jacobi field is a solution Chapter9/JacobiSystem · IsJacobiField.eqOn_jacSol. Let J be a Jacobi field along γ and $s \in [0,1]$. On $[0,1]$, $(J, DJ/dt)$ agrees with the solution whose value at s is $(J(s), \frac{DJ}{dt}(s))$.

Proof. The pair is continuous on $[0,1]$, and on $(0,1)$ it solves the system. First, $J = PJ$ gives $J' = P'J + PJ'$, and $SJ = P'J - PP'PJ = P'J$ for tangent J, so $J' = SJ + DJ/dt$. Second, $(PJ')' = P'J' + PJ''$, and the Jacobi equation says $PP'J' + PJ'' = -KJ$. Together with $P' = P'P + PP'$ this gives $(DJ/dt)' = S(DJ/dt) - KJ$. Uniqueness on closed intervals, end points included Chapter9/JacobiODE · eqOn_Icc_linSol, concludes the proof. This is why s may be 0 or 1.

Uniqueness (Milnor p. 77). A Jacobi field with $J(s) = 0$ and $\frac{DJ}{dt}(s) = 0$ at some $s \in [0,1]$ vanishes on $[0,1]$ Chapter9/JacobiSystem · IsJacobiField.eqOn_zero: it is the solution with data 0, which is 0 Chapter9/JacobiSystem · jacSol_zero.

Deviation (restated). Milnor's coordinates need orthonormal parallel fields along γ (Lemma 8.2), and Part V (§V.19) provides parallel transport only for curves defined on all of $\mathbb{R}$. Here there is no frame: the unknown is $(J, DJ/dt)$ in $V \times V$, and the invariance of the tangent bundle under the system replaces the frame. *Deviation (stronger).* Everything holds along any smooth curve in N, not only along geodesics, with initial values at any $s \in [0,1]$.

VII.14 The $2n$-dimensional space of Jacobi fields; conjugate points; nullity equals multiplicity

Definition (V9.3). For a smooth curve γ in N on $[0,1]$, $\mathcal{J}(\gamma)$ is the set of Jacobi fields along γ that are constant off $[0,1]$, that is, $J(t) = J(\pi(t))$ with π the nearest-point map of $\mathbb{R}$ onto $[0,1]$ `Chapter9/JacobiSpace · jacobiSpace Chapter9/JacobiSpace · mem_jacobiSpace`. The normalization makes a field along $\gamma : [0,1] \to N$ a single element. Without it, functions that differ only off $[0,1]$ would be distinct elements, and the space would be infinite-dimensional. Being a Jacobi field depends only on the values on $[0,1]$ `Chapter9/JacobiSpace · IsJacobiField.congr`, and the Jacobi fields are exactly the first components of the solutions of the Jacobi system with tangent data at 0 `Chapter9/JacobiSpace · isJacobiField_iff`. Since those solutions depend linearly on the data, $\mathcal{J}(\gamma)$ is a vector space. Every Jacobi field agrees on $[0,1]$ with its normalization, which lies in $\mathcal{J}(\gamma)$ `Chapter9/JacobiSpace · IsJacobiField.comp_projIcc_mem`, and an element of $\mathcal{J}(\gamma)$ that vanishes on $[0,1]$ is 0 `Chapter9/JacobiSpace · jacobiSpace_eq_zero`.

Corollary 14.B of the exposition (Milnor p. 77; V9.3). For every $s \in [0,1]$, the map $J \mapsto (J(s), \frac{DJ}{dt}(s))$ is a linear isomorphism $\mathcal{J}(\gamma) \to T_{\gamma(s)}N \times T_{\gamma(s)}N$ `Chapter9/JacobiSpace · jacEval Chapter9/JacobiSpace · jacEval_bijective Chapter9/JacobiSpace · jacEquiv`. Hence $\dim \mathcal{J}(\gamma) = 2n$ `Chapter9/JacobiSpace · finrank_jacobiSpace`, and Jacobi fields with any prescribed tangent data at s exist `Chapter9/JacobiSpace · exists_jacobi`.

Proof. The map is linear because D/dt is additive and homogeneous on smooth fields `Chapter9/JacobiSpace · covAlong_add Chapter9/JacobiSpace · covAlong_smul`. It is injective by uniqueness `Chapter9/JacobiSystem · IsJacobiField.eqOn_zero` and the normalization. It is surjective because the solution with data (x, y) at s is a Jacobi field `Chapter9/JacobiSystem · isJacobiField_jacSol`; its normalization has the same value and the same covariant derivative at s, since the derivative is computed within $[0,1]$. The dimension count uses $\dim T_xN = n$ `Chapter9/JacobiSystem · finrank_tanAt`. *Hidden hypothesis.* That every pair of initial values occurs needs the global existence theorem of §VII.12. *Deviation (stronger).* The isomorphism holds at every $s \in [0,1]$, along any smooth curve.

Definition (Milnor p. 77; V9.4; §0.9). Parameters a and b are *conjugate along* γ if $a, b \in [0,1]$, $a \neq b$, and some Jacobi field vanishes at a and at b without vanishing identically on $[0,1]$ `Chapter9/JacobiSpace · IsConjugate`. The space of fields of $\mathcal{J}(\gamma)$ vanishing at a and b `Chapter9/JacobiSpace · conjSpace Chapter9/JacobiSpace · mem_conjSpace` has finite dimension, being a subspace of $\mathcal{J}(\gamma)$, and that dimension is the *multiplicity* `Chapter9/JacobiSpace · conjMultiplicity`. For $a \neq b$ in $[0,1]$, a and b are conjugate if and only if the multiplicity is positive `Chapter9/JacobiSpace · isConjugate_iff`. Below, $\nu(t)$ is the multiplicity of 0 and t, which is 0 when t is not conjugate to 0. Milnor's footnote on p. 77 advises speaking of conjugate parameter values when γ has self-intersections, and the formal notion follows that advice: in §15 the points $\gamma(t)$ conjugate to $\gamma(0)$ are read as the parameters t conjugate to 0.

Theorem 14.1, the null space and the nullity (Milnor pp. 77–79; V9.6). The fields of $T\Omega_\gamma$ are functions on $\mathbb{R}$ whose values off $[0,1]$ play no role. So the literal kernel of the formal E_{**} contains every field vanishing on $[0,1]$, and it is infinite-dimensional. The *null space* is therefore read on $[0,1]$: it is the image of that kernel under $X \mapsto X \circ \pi$, restriction to $[0,1]$ followed by constant extension `Chapter9/JacobiSpace · restrictIcc Chapter9/JacobiSpace · hessNullSpace`. The *nullity* is its dimension `Chapter9/JacobiSpace · hessNullity`. If $\gamma \in \Omega(N; p, q)$ is an unbroken geodesic, the null space is the space of fields of $\mathcal{J}(\gamma)$ vanishing at 0 and 1 `Chapter9/JacobiSpace · hessNullSpace_eq`. This follows from Theorem 14.1 in both directions `Chapter8/Jacobi · theorem_14_1` together with the normalization. Hence the nullity is the multiplicity of 0 and 1, in particular finite `Chapter9/JacobiSpace · theorem_14_1_nullity`, and E_{**} is degenerate if and only if 0 and 1 are conjugate `Chapter9/JacobiSpace · hessNullity_pos_iff`. Milnor's remark that the nullity is always finite (p. 79) needs the identification with a subspace of the $2n$-dimensional $\mathcal{J}(\gamma)$.

VII.15 Remark 14.2

Remark 14.2 (Milnor p. 79; V9.7). Let $\gamma \in \Omega(N; p, q)$ be an unbroken geodesic and $n \geq 1$. Then the nullity satisfies $\nu < n$ `Chapter9/JacobiSpace · remark_14_2`.

The proof as formalized. Let $\mathcal{J}_0 = \{J \in \mathcal{J}(\gamma) : J(0) = 0\}$. The map $J \mapsto \frac{DJ}{dt}(0)$ is an isomorphism of $\mathcal{J}_0$ onto $T_{\gamma(0)}N$, injective by uniqueness and onto by existence `Chapter9/JacobiSpace · exists_jacobi`, so $\dim \mathcal{J}_0 = n$ `Chapter9/JacobiSpace · finrank_jacobiSpace_zero`. The null space is contained in $\mathcal{J}_0$. The inclusion is strict as soon as some Jacobi field vanishes at 0 but not at 1, and two cases produce one.

1. If $V(1) \neq 0$, take $J(t) = tV(t)$ `Chapter9/JacobiSpace · isJacobiField_smul_vel`. Indeed $\frac{D}{dt}(tV) = V + tP\gamma'' = V$, then $\frac{D}{dt}V = P\gamma'' = 0$, and $R(V, tV)V = tR(V, V)V = 0$ by antisymmetry.
2. If $V(1) = 0$, then γ is constant on $[0, 1]$ `Chapter9/JacobiSpace · eq_const_of_velAlong_one_eq_zero`. Its speed is constant on $(0, 1)$ `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq`, and by continuity of V on $[0, 1]$ at 1 it equals $\|V(1)\| = 0$, so $\gamma' = 0$ on $(0, 1)$. Along a constant curve, $t \mapsto ty$ is a Jacobi field for every tangent vector y `Chapter9/JacobiSpace · isJacobiField_const_curve`, and $n \geq 1$ provides some $y \neq 0$.

Hidden hypotheses. Milnor's witness $J_t = tV_t$ has $J_1 = V_1$, which is non-zero only if γ is non-constant. For the constant geodesic ($p = q$) the Jacobi fields are $t \mapsto x + ty$ and another witness is needed. And $\nu < n$ needs $n \geq 1$, since for $n = 0$, $\nu = 0 = n$. *Deviation (stronger).* The constant geodesic is included.

VII.16 Lemma 14.3, and Lemma 14.4 from initial data

Lemma 14.3 (Milnor p. 80; V9.8). Let $\varepsilon > 0$ and let $\alpha(u, t)$ be C^∞ on $(-\varepsilon, \varepsilon) \times [0, 1]$, with one-sided derivatives along the edges $t = 0, 1$ and values in N there, such that each $t \mapsto \alpha(u, t)$ is a geodesic on $(0, 1)$. Then $W(t) = \frac{\partial \alpha}{\partial u}(0, t)$ is a Jacobi field along $\alpha(0, \cdot)$ `Chapter9/JacobiVariation · lemma_14_3`. Being a Jacobi field along γ depends only on γ on $[0, 1]$ `Chapter9/JacobiVariation · IsJacobiField.congr_curve`.

The proof as formalized is Milnor's. W is smooth on $[0, 1]$, as $\partial_u \alpha(0, \cdot)$ on the closed strip `Chapter8/Piecewise · contDiffOn_pdU`, and tangent, as the derivative of a curve in N `Chapter7/Geodesic · hasDerivAt_mem_tangentSubspace`. For the Jacobi equation at $t \in (0, 1)$, apply the curvature identity `Chapter8/SecondVariation · curvature_identity` at $(0, t)$ on the open strip $(-\varepsilon, \varepsilon) \times (0, 1)$. The left side is $P\partial_u(P\partial_t\partial_t\alpha)$, and $P\partial_t\partial_t\alpha$ vanishes identically on the open strip because each $\alpha(u, \cdot)$ is a geodesic, so the left side is 0. On the right side, $\partial_t\alpha(0, t) = \gamma'(t)$, and $P\partial_t\partial_u\alpha(0, \cdot) = \frac{DW}{dt}$ near t, so the right side is $\frac{D^2W}{dt^2} + R(V, W)V$. Milnor applies Lemmas 9.2 and 8.7 to a map defined on a strip and smooth up to its edges; the curvature identity on the open strip is what makes that legitimate.

Lemma 14.4 (Milnor p. 81; V9.9). Let γ be a smooth curve in N on $[0, 1]$ that is a geodesic on $(0, 1)$, and J a Jacobi field along γ. There are $\varepsilon > 0$ and a map α, C^∞ on $(-\varepsilon, \varepsilon) \times [0, 1]$ with values in N, such that each $\alpha(u, \cdot)$ is a geodesic on $(0, 1)$, $\alpha(0, t) = \gamma(t)$, and $\frac{\partial \alpha}{\partial u}(0, t) = J(t)$ for $t \in [0, 1]$ `Chapter9/GeodesicVariation · lemma_14_4`. This is the converse of Lemma 14.3.

Milnor's proof, and the step it leaves unargued. Milnor takes a neighbourhood U of $\gamma(0)$ in which any two points are joined by a unique minimal geodesic depending differentiably on its end points, and δ with $\gamma([0, \delta]) \subseteq U$. Moving the end points along curves with prescribed velocities and joining them by minimal geodesics gives variations through geodesics whose fields take arbitrary values at 0 and δ. So $J \mapsto (J(0), J(\delta))$ is onto, hence an isomorphism, both spaces having dimension $2n$, and every Jacobi field along $\gamma|[0, \delta]$ arises. Compactness of $[0, 1]$ then extends the geodesics. The construction returns $\bar{\alpha}(0) = \gamma|[0, \delta]$ only if $\gamma|[0, \delta]$ is *the* minimal geodesic from $\gamma(0)$ to $\gamma(\delta)$, and $\gamma([0, \delta]) \subseteq U$ does not say this. For Milnor's U it is true, by an argument the book omits and the exposition supplies. Two further steps are left implicit. The compactness step is the smooth dependence of geodesics on their initial data over a compact interval, together with the openness of the domain of exp, and neither is among the

statements of §10. And identifying the field of the extended variation with J needs the uniqueness of Jacobi fields.

The formal proof builds the variation from initial data, as $\alpha(u,t) = \exp_{c(u)}(t\,w(u))$. It needs three tools.

(i) The local smooth flow, uniformly `Chapter9/GeodesicVariation · exists_smooth_ray`. For $x_1 \in N$ and $R > 0$ there are an open set $U \ni x_1$, a time $h > 0$ and a map Φ, smooth on $U \times \{\|w\| < R\} \times (-h,h)$, such that for $y \in N \cap U$ and $w \in T_yN$ with $\|w\| < R$, $t \mapsto \exp_y(tw)$ is a geodesic on $(-h,h)$ and equals $\Phi(y,w,t)$ there. *Proof.* Lemma 10.2 (Part VI, §VI.5) `Chapter7/GeodesicLocal · lemma_10_2` gives Γ smooth on $U \times \{\|v\| < \varepsilon\} \times (-2,2)$, with $\Gamma(y,v,\cdot)$ the geodesic through y with velocity v. Put $c = \varepsilon/R$, $\Phi(y,w,t) = \Gamma(y,cw,t/c)$ and $h = 2c$. Since $\Gamma(y,cw,\cdot)$ is a geodesic with initial velocity cw, $\exp_y(s\,cw) = \Gamma(y,cw,s)$ `Chapter7/GeodesicFlow · IsGeodesicOn.exp_smul_eq`, which is the claim with $s = t/c$. The rescaled curve is again a geodesic.

(ii) A geodesic segment extends past its end points `Chapter9/GeodesicVariation · exists_midpoint_ray` `Chapter9/GeodesicVariation · exists_initial_ray`. A smooth curve γ in N on $[0,1]$ that is a geodesic on $(0,1)$ agrees on $[0,1]$ with $t \mapsto \exp_{\gamma(0)}(tV(0))$, which is a geodesic on an open interval containing $[0,1]$. Part VI leaves this extension unformalized. *Proof.* Start at the midpoint. With $x = \gamma(1/2)$ and $v = \gamma'(1/2)$, the curve $\gamma(\cdot + 1/2)$ equals the ray $t \mapsto \exp_x(tv)$ on $(-1/2,1/2)$ `Chapter7/GeodesicFlow · IsGeodesicOn.eqOn_exp_smul`. To continue past an end $e \in \{0,1\}$, apply (i) at $\gamma(e)$ with $R = \|V(e)\| + 1$, obtaining U and h. By continuity of (γ, V) on $[0,1]$ at e, choose $t \in (0,1)$ within $h/2$ of e with $\gamma(t) \in U$ and $\|\gamma'(t)\| < R$. The geodesic flow of (x,v) at time $t-1/2$ is $(\gamma(t),\gamma'(t))$ `Chapter7/GeodesicFlow · geoFlow` `Chapter9/GeodesicVariation · geoFlow_eq_exp`, and the ray it starts is a geodesic on $(-h,h)$. The gluing lemma of Part VI (§VI.16) `Chapter7/GeodesicFlow · isGeodesicOn_exp_smul_union`, used first at the right end and then at the left, makes the ray from the midpoint a geodesic on $(-1/2-\eta, 1/2+\eta)$, and it equals γ on $[0,1]$ by continuity at the ends. Recentred at 0, $g(t) = \exp_x((t-1/2)v)$ is a geodesic on $(-\eta, 1+\eta)$ that agrees with γ on $[0,1]$. Hence $g(0) = \gamma(0)$, and $g'(0) = V(0)$, the one-sided velocity, so that $g(t) = \exp_{\gamma(0)}(tV(0))$.

(iii) Smooth dependence on initial data over a compact interval. Let c, w be smooth on an open set $U_0 \ni 0$, with $c(u) \in N$ and $w(u) \in T_{c(u)}N$. Write $\mathrm{rays}(u,t) = \exp_{c(u)}(t\,w(u))$ `Chapter9/GeodesicVariation · rays`. Say the rays are *good up to* b if there are $\varepsilon, \eta > 0$ with $(-\varepsilon,\varepsilon) \subseteq U_0$ such that rays is smooth on $(-\varepsilon,\varepsilon) \times (-\eta, b+\eta)$ and every ray is a geodesic on $(-\eta, b+\eta)$ `Chapter9/GeodesicVariation · RaysGood` `Chapter9/GeodesicVariation · RaysGood.mono`. **Claim:** if the ray of $(c(0),w(0))$ is a geodesic on $(-\delta, B+\delta)$ with $B \geq 0$, the rays are good up to B `Chapter9/GeodesicVariation · raysGood_of_isGeodesicOn`.

1. *Start* `Chapter9/GeodesicVariation · raysGood_zero`. Apply (i) `Chapter9/GeodesicVariation · exists_smooth_ray` at $c(0)$ with $R = \|w(0)\| + 1$. For small u, $c(u) \in U_1$ and $\|w(u)\| < R$, so $\mathrm{rays}(u,t) = \Phi(c(u),w(u),t)$ for $|t| < h$. This is smooth, and each ray is a geodesic on $(-h/2, h/2)$.
2. *One step* `Chapter9/GeodesicVariation · RaysGood.step`. Suppose the rays are good up to $b \geq 0$, and the position and velocity of the base ray at time b lie in the domain $U_1 \times \{\|v\| < R\}$ of a local flow Φ of time h. The position $F(u) = \mathrm{rays}(u,b)$ and the velocity $G(u) = \partial_t \mathrm{rays}(u,b)$ `Chapter9/GeodesicVariation · deriv_rays_eq` are smooth in u, and for small u they stay in that domain. Each ray continues, by the gluing lemma `Chapter7/GeodesicFlow · isGeodesicOn_exp_smul_union`, as the ray of $(F(u),G(u))$, and there it equals $\Phi(F(u),G(u),t-b)$, which is smooth in (u,t). Smoothness is a local property, so rays is smooth on the union, and with $\eta' = \min(\eta, h/4)$ the rays are good up to $b + h/2$.
3. *Supremum.* Let b_0 be the supremum of the $b \in [0,B]$ up to which the rays are good; 0 is such a b by step 1. Take the local flow at the base ray's point at time b_0, and $r > 0$ such that its position and velocity at times within r of b_0 lie in the domain of that flow. Pick an admissible $s > b_0 - \min(r, h/4)$. Step 2 from s `Chapter9/GeodesicVariation · RaysGood.step` makes the rays good up to $s + h/2 > b_0$, hence up to $\min(s + h/2, B)$. That number is admissible and at most $b_0 < s + h/2$, so it equals B.

The supremum argument replaces Milnor's appeal to the compactness of $[0,1]$.

The proof of Lemma 14.4 as formalized.

1. *Initial points.* $c(u) = \exp_{\gamma(0)}(uJ(0))$ is smooth for $|u| < h$, by (i) at $\gamma(0)$ with $R = \|J(0)\| + 1$. Moreover $c(0) = \gamma(0)$ and $c'(0) = J(0)$ `Chapter7/GeodesicFlow · hasDerivAt_exp_smul_zero`.
2. *Initial velocities.* $w(u) = P_{c(u)}(V(0) + u\frac{DJ}{dt}(0))$ is smooth and tangent, and $w(0) = V(0)$.
3. *The variation.* By (ii) `Chapter9/GeodesicVariation · exists_initial_ray`, the ray of $(c(0), w(0))$ is γ on $[0,1]$ and a geodesic on $(-\eta, 1+\eta)$. By (iii) with $B = 1$ `Chapter9/GeodesicVariation · raysGood_of_isGeodesicOn`, $\alpha(u,t) = \exp_{c(u)}(t\,w(u))$ is smooth on $(-\varepsilon,\varepsilon) \times (-\eta', 1+\eta')$, which contains $(-\varepsilon,\varepsilon) \times [0,1]$, takes values in N, consists of geodesics, and satisfies $\alpha(0,\cdot) = \gamma$ on $[0,1]$.
4. *Its field is a Jacobi field.* $W = \frac{\partial\alpha}{\partial u}(0,\cdot)$ is a Jacobi field along γ, by Lemma 14.3 `Chapter9/JacobiVariation · lemma_14_3` `Chapter9/JacobiVariation · IsJacobiField.congr_curve`.
5. *Same value at* 0. $\alpha(u,0) = c(u)$ `Chapter7/Exp · exp_zero`, so $W(0) = c'(0) = J(0)$.
6. *Same covariant derivative at* 0. $\frac{DW}{dt}(0) = P\,\partial_t\partial_u\alpha(0,0) = P\,\partial_u\partial_t\alpha(0,0)$ by the symmetry of mixed partials at an interior point `Chapter8/SecondVariation · partialX_partialY_of_contDiffOn`. Since $\partial_t\alpha(u,0) = w(u)$, the velocity at 0 of the ray `Chapter7/GeodesicFlow · hasDerivAt_exp_smul_zero`, this equals $P\,w'(0)$. Now $w'(0) = (P_c)'(0)V(0) + P_{c(0)}\frac{DJ}{dt}(0)$. The projection of the first term vanishes: differentiating $P_c \circ P_c = P_c$ gives $(P_c)' = (P_c)'P_c + P_c(P_c)'$, so $P(P_c)'P = 0$, and $V(0)$ is tangent. Hence $\frac{DW}{dt}(0) = \frac{DJ}{dt}(0)$.
7. *Uniqueness.* W and J have the same initial data at 0, so on $[0,1]$ both are the first component of the same solution of the Jacobi system `Chapter9/JacobiSystem · IsJacobiField.eqOn_jacSol`. Hence $W = J$ there.

Why this avoids the gap. No end-point data are used, so no minimal geodesic enters, and the question whether $\gamma|[0,\delta]$ is minimal never arises. The formal statement has no minimality hypothesis, and neither the restriction to $[0,\delta]$ nor the dimension count is needed. The only global ingredient is the smooth dependence (iii), which is exactly the compactness step that Milnor's proof also needs.

VII.17 Remark 14.5, and two non-conjugate parameters

Two non-conjugate parameters (Milnor p. 81, a parenthetical remark; V9.10). If $a \neq b$ in $[0,1]$ are not conjugate along γ, then $J \mapsto (J(a), J(b))$ is a linear isomorphism $\mathcal{J}(\gamma) \to T_{\gamma(a)}N \times T_{\gamma(b)}N$ `Chapter9/IndexForm · jacEvalEnds` `Chapter9/GeodesicVariation · jacEvalEnds_bijective_of_not_isConjugate`. Injectivity is the definition of non-conjugacy together with the normalization. Surjectivity follows from the dimension count $2n = n + n$ `Chapter9/JacobiSpace · finrank_jacobiSpace`. Milnor states that the values determine the field; the formal statement adds that they can be prescribed.

Remark 14.5 (Milnor pp. 81–82; V9.11). (i) Let ε be a normal radius at q, meaning that $\exp_q$ maps the ε-ball of T_qN diffeomorphically onto an open subset of N, with smooth extensions F of $\exp_q$ and G of its inverse `Chapter7/NormalNbhd · NormalCoord` `Chapter7/NormalNbhd · IsNormalRadius`. Let $v \in T_qN$ with $\|v\| < \varepsilon$. Then $\sigma(t) = \exp_q(tv)$ is a geodesic on an open interval containing $[0,1]$, hence a smooth curve in N on $[0,1]$ `Chapter9/JacobiVariation · IsNormalRadius.isGeodesicOn_ray` `Chapter9/JacobiVariation · IsNormalRadius.isSmoothCurveIn_ray`. A Jacobi field along σ that vanishes at 0 and 1 vanishes on $[0,1]$ `Chapter9/JacobiVariation · remark_14_5_of_isNormalRadius`, and $J \mapsto (J(0), J(1))$ is an isomorphism `Chapter9/JacobiVariation · remark_14_5_bijective`. (ii) Every $p \in N$ has a neighbourhood O in V, with some $\varepsilon > 0$, such that any $q_1, q_2 \in N \cap O$ are joined by $t \mapsto \exp_{q_1}(tv)$ with $\|v\| < \varepsilon$, where ε is a normal radius at q_1, and Jacobi fields along this segment vanishing at both ends vanish `Chapter9/JacobiVariation · remark_14_5`. (ii) follows from (i) and Lemma 10.3 (Part VI, §VI.8) `Chapter7/NormalNbhd · lemma_10_3`.

The proof of (i) as formalized. Let $w = \frac{DJ}{dt}(0) \in T_qN$. Since $J(0) = 0$, it suffices to show $w = 0$, by uniqueness `Chapter9/JacobiSystem · IsJacobiField.eqOn_zero`.

1. *A variation through geodesics.* For $|u| < \varepsilon_1 = (\varepsilon - \|v\|)/(\|w\| + 1)$ one has $\|t(v + uw)\| < \varepsilon$ for

$t \in [0,1]$. So $\alpha(u,t) = \exp_q(t(v+uw)) = F(t(v+uw))$ is smooth, and each $\alpha(u,\cdot)$ is a geodesic on an interval containing $[0,1]$.

2. *Its field.* By Lemma 14.3 `Chapter9/JacobiVariation · lemma_14_3`, $J'(t) = \frac{\partial \alpha}{\partial u}(0,t) = t\, dF_{tv}(w)$ is a Jacobi field along σ. It has $J'(0) = 0$ and $\frac{DJ'}{dt}(0) = P\, dF_0(w) = w$, because dF_0 is the identity on T_qN `Chapter9/JacobiVariation · fderiv_extension_exp_zero`, as $d(\exp_q)_0$ is.
3. *It is J.* By uniqueness $J = J'$ `Chapter9/JacobiSystem · IsJacobiField.eqOn_jacSol`, and $J(1) = 0$ gives $dF_v(w) = 0$.
4. *The local inverse.* $G \circ F$ is the identity on the ε-ball of T_qN. Differentiating along $u \mapsto v + uw$ gives $dG(dF_v(w)) = w$, so $w = 0$.

Relation with Milnor's argument. Milnor reads Remark 14.5 off his proof of Lemma 14.4: surjectivity comes from the end-point construction, and injectivity from the dimension count. The formal proof goes the other way, with injectivity from the local inverse of $\exp_q$ and surjectivity from the count. Milnor's geodesic segments inside U become the segment $t \mapsto \exp_{q_1}(tv)$, $\|v\| < \varepsilon$, of Lemma 10.3, which is what his argument actually uses. The index theorem itself does not use Remark 14.5. It needs only the form of §VII.19, no conjugate parameters closer than h_0, which is proved by an estimate.

VII.18 The index of a bilinear form

Milnor defines the index of E_{**} as the largest dimension of a subspace of $T\Omega_\gamma$ on which E_{**} is negative definite (p. 83). $T\Omega_\gamma$ is infinite-dimensional, so this maximum need not exist a priori, and its finiteness is part of Theorem 15.1. Mathlib's signature of a quadratic form is meaningful only in finite dimension. The file `Chapter9/FormIndex` therefore defines the index for any bilinear form and proves the linear algebra that §15 uses without comment.

Definitions (V9.13; §0.9). Let B be a bilinear form on a real vector space X. B is *negative definite* on a subspace S if $B(x,x) < 0$ for all non-zero $x \in S$, and *positive definite* if $B(x,x) > 0$ for all such x `Chapter9/FormIndex · IsNegDefOn` `Chapter9/FormIndex · IsPosDefOn`. The *index* of B is the supremum, in $\{0,1,2,\ldots\} \cup \{\infty\}$, of the dimensions of the finite-dimensional subspaces on which B is negative definite `Chapter9/FormIndex · formIndex`. Each such subspace bounds the index from below `Chapter9/FormIndex · le_formIndex`, and a bound valid for all of them bounds it from above `Chapter9/FormIndex · formIndex_le`. A finite index is attained by some subspace `Chapter9/FormIndex · exists_isNegDefOn_of_formIndex_eq`, since otherwise every negative definite subspace would have dimension at most $k-1$. In finite dimension the index is Mathlib's negative index of inertia of $x \mapsto B(x,x)$ `Chapter9/FormIndex · formIndex_eq_sigNeg` `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg` `Mathlib/LinearAlgebra/QuadraticForm/Signature · sigNeg_isGreatest`. The *nullity* is the dimension of the null space $\{x : B(x,y) = 0 \text{ for all } y\}$, read on $[0,1]$ for E_{**} (§VII.14).

Form-preserving maps. If L satisfies $B'(Lx,Ly) = B(x,y)$, then the index of B is at most that of B' `Chapter9/FormIndex · formIndex_le_of_map`. The point is that L is automatically injective on every negative definite subspace: $Lx = 0$ would give $B(x,x) = B'(0,0) = 0$. So the image of such a subspace is negative definite of the same dimension `Chapter9/FormIndex · finrank_map_of_injOn`. Consequently restriction to a subspace does not increase the index `Chapter9/FormIndex · formIndex_restrict_le`, a linear isomorphism carrying one form to the other preserves it `Chapter9/FormIndex · formIndex_eq_of_equiv`, and so does a positive factor `Chapter9/FormIndex · formIndex_smul`.

Lemma 15.B of the exposition (V9.14 (4)). Let $X = A \oplus C$ with $B(a,c) = B(c,a) = 0$ for $a \in A$, $c \in C$, and B positive definite on C. Then B and its restriction to A have the same index `Chapter9/FormIndex · formIndex_eq_restrict_of_isCompl`, and the null space of B is that of its restriction to A `Chapter9/FormIndex · ker_eq_map_ker_restrict`.

Proof. Let π be the projection onto A along C. Then $B(x,x) = B(\pi x, \pi x) + B(x-\pi x, x-\pi x) \geq B(\pi x, \pi x)$. On a negative definite subspace S this makes π injective and $\pi(S)$ negative definite for $B|_A$, which gives one inequality; restriction gives the other. For the null space, if $x = a + c$ lies in it, then

$B(c,c) = B(x,c) - B(a,c) = 0$, so $c = 0$; and a null vector of $B|_A$ is null for B because $A \perp C$. This is what lies behind Milnor's remark that the proof of Lemma 15.4 is "straightforward".

Sylvester's law of inertia, in the form of p. 86 (V9.14 (2)). A symmetric form on an m-dimensional space is positive definite on a subspace of dimension m — index — nullity `Chapter9/FormIndex · exists_isPosDefOn_add`. The proof combines Mathlib's positive definite subspace of dimension equal to the positive index `Mathlib/LinearAlgebra/QuadraticForm/Signature · exists_finrank_eq_sigPos_and_posDef`, the identity $\mathrm{sig}^+ + \mathrm{sig}^- + \dim \mathrm{radical} = m$ `Mathlib/LinearAlgebra/QuadraticForm/Signature · QuadraticForm.sigPos_add_sigNeg_add_radical`, and the fact that the radical of the quadratic form of a symmetric bilinear form is its null space `Chapter9/FormIndex · radical_toQuadraticMap`. **Lemma 15.A (1).** In finite dimension, a negative definite and a positive definite subspace meet only in 0, so their dimensions add up to at most m `Chapter9/FormIndex · finrank_add_finrank_le`.

Lemma 15.A (3): definiteness persists (V9.14 (3)). Let $\tau \mapsto H_\tau$ be a family of bilinear forms indexed by a topological space, S a finite-dimensional subspace, and suppose $\tau \mapsto H_\tau(x,y)$ is continuous at τ_0 for $x, y \in S$. If H_{τ_0} is negative (positive) definite on S, so is H_τ for all τ near τ_0 `Chapter9/FormIndex · eventually_isNegDefOn` `Chapter9/FormIndex · eventually_isPosDefOn`. *Proof.* In a basis b_i of S, $(\tau, c) \mapsto H_\tau(\sum c_i b_i, \sum c_i b_i) = \sum c_i c_j H_\tau(b_i, b_j)$ is continuous at every (τ_0, c). It is negative on the compact unit sphere of coefficients at τ_0, hence negative on the whole sphere for τ near τ_0, and by homogeneity on all of $S \setminus 0$. The versions within a set of parameters `Chapter9/IndexContinuity · eventually_isNegDefOn_within` `Chapter9/IndexContinuity · eventually_isPosDefOn_within` follow by restricting to the subspace. This is the perturbation step that Milnor uses without proof on p. 86.

Milnor's compound matrix as a lemma (p. 87; V9.14 (5)). Let B be symmetric on a space $\mathcal{P}$, $U : \mathbb{R}^k \to \mathcal{P}$ linear with $B(Ua, Ua) < 0$ for $a \neq 0$, and $J, X : \mathbb{R}^\nu \to \mathcal{P}$ linear with

$$B(Jb, Ua) = 0, \qquad B(Jb, Jb') = 0, \qquad B(Jb, Xb') = 2\langle b, b' \rangle.$$

Then the index of B is at least $k + \nu$ `Chapter9/IndexAssertions · le_formIndex_of_compound`. Here $\langle b, b' \rangle = \sum_h b_h b'_h$ is the standard dot product on $\mathbb{R}^\nu$.

Proof. For $c \neq 0$ let $\Phi_c(a,b) = Ua + c^{-1} Jb - cXb$. Expanding with the three hypotheses,

$$B(\Phi_c(a,b), \Phi_c(a',b')) = B(Ua,Ua') - c(B(Ua,Xb') + B(Xb,Ua')) - 4\langle b,b'\rangle + c^2 B(Xb,Xb') =: G_c((a,b),(a',b')),$$

the two cross terms $c^{-1}(-c)B(Jb,Xb')$ and $(-c)c^{-1}B(Xb,Jb')$ each contributing $-2\langle b,b'\rangle$. The form G_0 is negative definite on $\mathbb{R}^k \times \mathbb{R}^\nu$: if $a \neq 0$, then $B(Ua,Ua) < 0$ and $-4\langle b,b\rangle \leq 0$, and if $a = 0$ and $b \neq 0$, then $-4\langle b,b\rangle < 0$. G_c depends continuously on c, so by Lemma 15.A (3) it stays negative definite for small c `Chapter9/FormIndex · eventually_isNegDefOn`; choose such a $c \neq 0$. Then Φ_c carries G_c to B, and the index of B is at least that of G_c `Chapter9/FormIndex · formIndex_le_of_map`, namely $k + \nu$ `Chapter9/FormIndex · le_formIndex`. The linear independence of the $k + \nu$ fields, which Milnor's argument needs and does not mention, comes for free from the injectivity of form-preserving maps on negative definite subspaces.

VII.19 Broken Jacobi fields in the second variation; positivity by a Poincaré estimate

The second variation formula for piecewise Jacobi fields (Milnor pp. 84, 87; V9.15). Let $\gamma \in \Omega(N; p, q)$, $X, Y \in T\Omega_\gamma$ and B a finite set such that near every $t \in (0,1) \setminus B$ the field X agrees with some Jacobi field along γ. Then

$$E_{**}(X,Y) = -2 \sum_{b \in B \cap (0,1)} \langle Y(b), P\Delta_b X' \rangle$$

`Chapter9/IndexForm · energyHessian_eq_jumps`. γ need not be a geodesic. *Proof.* In Φ the integrand vanishes off $B \cup \{1\}$. The Jacobi operator depends only on X near t `Chapter9/IndexForm · jacobiExpr_congr`, and the local

Jacobi field satisfies the equation. The sum reduces to B, since X is differentiable wherever it agrees with a smooth field. The zero field counts as a Jacobi field `Chapter9/IndexForm · isJacobiField_zero`, which is how fields that vanish on part of the interval are handled.

The mesh h_0 (V9.16). Along a smooth curve γ in N on $[0,1]$, let $C \geq 0$ bound $\|P'(t)\|^2 + \|K(t)\|$ on $[0,1]$; such a bound exists because the function is continuous on a compact interval `Chapter9/IndexForm · exists_jacBound Chapter9/IndexForm · jacBound Chapter9/IndexForm · jacBound_nonneg Chapter9/IndexForm · le_jacBound`. Put $h_0 = 1/(C+1) > 0$ `Chapter9/IndexForm · fineMesh Chapter9/IndexForm · fineMesh_pos`. It depends only on γ, through bounds for P' and for the curvature along γ.

The integrand bound `Chapter9/IndexForm · integrand_ge`. For a field X tangent along γ on $[0,1]$ and differentiable at $t \in (0,1)$,

$$\Big\|\frac{DX}{dt}\Big\|^2 - \langle R(V,X)V, X\rangle \;\geq\; \|X'\|^2 - C\|X\|^2.$$

Since $X = PX$, $X' = P'X + PX'$, so the normal part of X' is $P'X$. Pythagoras gives $\|X'\|^2 = \|PX'\|^2 + \|P'X\|^2 \leq \|\frac{DX}{dt}\|^2 + \|P'\|^2\|X\|^2$, and $\langle R(V,X)V,X\rangle = \langle KX, X\rangle \leq \|K\|\,\|X\|^2$.

Poincaré's inequality on short pieces `Chapter9/IndexForm · integral_norm_sq_le`. Let X be piecewise smooth on $[0,1]$, $h \geq 0$, and suppose every $t \in [0,1]$ has a zero $s \in [0,t]$ of X with $t - s \leq h$. Then $\int_0^1 \|X\|^2 \leq h\int_0^1 \|X'\|^2$. Indeed $\|X(t)\| = \|X(t) - X(s)\| \leq L_s^t(X)$ `Chapter7/Path · IsAdmissible.norm_sub_le_pathLength`, and Lemma 12.A gives $L_s^t(X)^2 \leq (t-s)E_s^t(X) \leq hE_0^1(X)$. Integrating over $[0,1]$ gives the claim.

The estimate (V9.16 (i)). For $X \in T\Omega_\gamma$ with a zero within h_0 before every point of $[0,1]$,

$$\int_0^1 \|X\|^2\,dt \;\leq\; \int_0^1 \Big(\Big\|\frac{DX}{dt}\Big\|^2 - \langle R(V,X)V,X\rangle\Big)dt = \tfrac{1}{2}E_{**}(X,X)$$

`Chapter9/IndexForm · hessForm_ge`. By the integrand bound, the integral is at least $\int\|X'\|^2 - C\int\|X\|^2$. By Poincaré, with $h_0(C+1) = 1$, $\int\|X'\|^2 \geq (C+1)\int\|X\|^2$. Hence the integral is at least $\int\|X\|^2$. So if moreover $E_{**}(X,X) \leq 0$, then $X = 0$ on $[0,1]$ `Chapter9/IndexForm · eq0n_zero_of_hessForm`: the continuous function $\|X\|^2$ has integral 0.

No two parameters closer than h_0 are conjugate (V9.16 (ii)). If $J \in \mathcal{J}(\gamma)$ vanishes at a and b with $0 \leq a < b \leq 1$ and $b - a \leq h_0$, then $J = 0$ `Chapter9/IndexForm · eq_zero_of_short`.

Proof. Let X equal J on $[a,b]$ and 0 elsewhere. It is piecewise smooth with break points $\{a,b\}$ `Chapter9/IndexForm · isPiecewiseSmooth_indicator` and lies in $T\Omega_\gamma$ `Chapter9/IndexForm · indicator_mem_pathTangent`. Off $\{a,b\}$ it agrees locally with a Jacobi field, J or 0, so the formula above `Chapter9/IndexForm · energyHessian_eq_jumps` gives $E_{**}(X,X) = -2\sum\langle X(\cdot), \ldots\rangle = 0$, since X vanishes at a and b. Every point has a zero of X within h_0 before it: a for points of $[a,b]$, the point itself elsewhere. So $X = 0$ on $[0,1]$ by the estimate `Chapter8/Hessian · secondVariation_eq_hessForm Chapter9/IndexForm · eq0n_zero_of_hessForm`. Then J vanishes near the midpoint of $[a,b]$, so J and DJ/dt vanish there, and $J = 0$ by uniqueness `Chapter9/JacobiSystem · IsJacobiField.eq0n_zero`.

Consequence: two-point boundary values on short segments `Chapter9/IndexForm · jacEvalEnds_bijective Chapter9/IndexForm · exists_jacobi_ends`. For $b - a \leq h_0$, $J \mapsto (J(a), J(b))$ is an isomorphism $\mathcal{J}(\gamma) \to T_{\gamma(a)}N \times T_{\gamma(b)}N$. It is injective by the previous result `Chapter9/IndexForm · eq_zero_of_short` and surjective by the dimension count `Chapter9/JacobiSpace · finrank_jacobiSpace`. This is the form of Remark 14.5 that the index theorem uses.

Relation with Milnor's proof. Milnor chooses the subdivision "sufficiently fine" (p. 83) that each segment lies in one of his neighbourhoods U and is minimal. He then uses Remark 14.5 on each segment and the minimality of the segments (p. 84). Behind this lie a Lebesgue number for a finite cover and a length bound for minimality (Theorem 10.4, Part VI, §VI.12). The formal proof replaces all of it by the explicit

mesh h_0, which comes from bounds on P' and on the curvature along γ. Neither exp nor minimality is used, and γ may be any smooth curve.

VII.20 Fine subdivisions; Lemma 15.3 and Lemma 15.4

Milnor proves Lemmas 15.3 and 15.4 for γ on $[0,1]$ and applies them on p. 86 to the restrictions $\gamma_\tau = \gamma|[0,\tau]$. The formal proof works along $\gamma|[0,\tau]$ from the start, for every $\tau \in (0,1]$. It represents a field along γ_τ by a field along γ that vanishes beyond τ.

Definition (V9.17). The *fields along* $\gamma|[0,\tau]$ are the elements of $T\Omega_\gamma$ that vanish outside $[0,\tau]$ `Chapter9/BrokenJacobi · fieldsUpTo`. They increase with τ `Chapter9/BrokenJacobi · fieldsUpTo_mono`, and for $0 \le \tau \le 1$ they vanish at τ `Chapter9/BrokenJacobi · fieldsUpTo_apply_self`, by continuity from the right, or from $X(1) = 0$ if $\tau = 1$. These are Milnor's fields along γ_τ vanishing at 0 and τ, extended by zero, which is the extension he uses in Assertion (1). By Proposition 13.A, E_{**} on them is $2\int_0^\tau(\dots)$, the integral that represents $(E_0^\tau)_{**}$. The Hessian on $[0,\tau]$ itself is never formalized and is not needed.

Fine subdivisions (V9.16 (iii)). A finite set $B \subseteq (0,\tau)$ is *fine for* $[0,\tau]$ if every interval $[s,s'] \subseteq [0,\tau]$ whose interior contains no point of B has length at most h_0 `Chapter9/BrokenJacobi · IsFine`, that is, consecutive points of $\{0\} \cup B \cup \{\tau\}$ are at most h_0 apart. Every $t \in [0,\tau]$ lies between two consecutive points of this set `Chapter9/BrokenJacobi · exists_node_piece`. Fine subdivisions exist, with the last point of $\{0\} \cup B$ within $h_0/2$ of τ: take the grid points $kh_0/2$ in $(0,\tau)$ `Chapter9/BrokenJacobi · exists_isFine`.

Broken Jacobi fields (V9.17). A map X is a *broken Jacobi field on* $[0,\tau]$ for B if on every piece it agrees with a field of $\mathcal{J}(\gamma)$ `Chapter9/BrokenJacobi · IsBrokenJacobi`. A piece is thus the restriction of a Jacobi field along all of γ. By §§VII.12–VII.13 this is equivalent to being a Jacobi field along $\gamma|[t_{i-1},t_i]$, an equivalence not stated formally.

- *Uniqueness* (for which Milnor refers to 14.5): a broken Jacobi field vanishing at 0, at τ and at all points of B vanishes on $[0,\tau]$ `Chapter9/BrokenJacobi · IsBrokenJacobi.eqOn_zero`. On each piece between consecutive division points `Chapter9/BrokenJacobi · exists_node_piece` it is a Jacobi field vanishing at both ends of a segment of length at most h_0, hence zero `Chapter9/IndexForm · eq_zero_of_short`.
- *Existence* with arbitrary tangent values at 0, τ and the points of B `Chapter9/BrokenJacobi · exists_isBrokenJacobi`. The proof is an induction that adds the largest point m of B. $B \setminus \{m\}$ is fine for $[0,m]$, so the induction hypothesis gives a broken field X_0 on $[0,m]$. On $[m,\tau]$, of length at most h_0, take the Jacobi field with the prescribed values at m and τ `Chapter9/IndexForm · exists_jacobi_ends`. Glue them: X_0 for $t \le m$, the new field for $t > m$, both having the value $x(m)$ at m.

The broken Jacobi fields along $\gamma|[0,\tau]$, Milnor's $T\Omega_\gamma(t_0,\dots,t_k)$, form the space `Chapter9/BrokenJacobi · brokenSpace`, and the fields along $\gamma|[0,\tau]$ vanishing at the points of B, Milnor's T', form `Chapter9/BrokenJacobi · nodeZero`. Both lie among the fields along $\gamma|[0,\tau]$ `Chapter9/BrokenJacobi · brokenSpace_le_fieldsUpTo Chapter9/BrokenJacobi · nodeZero_le_fieldsUpTo`. A broken Jacobi field vanishing at 0, at τ and outside $[0,\tau]$ lies in $T\Omega_\gamma$, with break points $B \cup \{\tau\}$ `Chapter9/BrokenJacobi · IsBrokenJacobi.mem_pathTangent`. Near every point of $(0,1)$ other than τ and the points of B, it agrees with a Jacobi field, namely 0 beyond τ `Chapter9/BrokenJacobi · IsBrokenJacobi.eventuallyEq`. Hence, by the jump formula of §VII.19 `Chapter9/IndexForm · energyHessian_eq_jumps`, for a broken X and any Y along $\gamma|[0,\tau]$,

$$E_{**}(X,Y) = -2\sum_{b\in B}\langle Y(b), P\Delta_b X'\rangle$$

`Chapter9/BrokenJacobi · energyHessian_broken_eq`. The term at τ drops out because $Y(\tau) = 0$ `Chapter9/BrokenJacobi · fieldsUpTo_apply_self`.

Lemma 15.3 (Milnor p. 84; V9.18). Let γ be a smooth curve in N on $[0,1]$, $0 < \tau \le 1$, and B fine for $[0,\tau]$. (i) The fields along $\gamma|[0,\tau]$ are the direct sum of the broken Jacobi fields and T' `Chapter9/IndexReduction · isCompl_broken_nodeZero`. (ii) $E_{**}(W_1,W_2) = 0$ for W_1 broken and

$W_2 \in T'$ Chapter9/BrokenJacobi · energyHessian_broken_nodeZero. (iii) E_{**} is positive definite on T' Chapter9/BrokenJacobi · energyHessian_nodeZero_pos.

The proof as formalized. (i) Every X along $\gamma|[0,\tau]$ is $W_1 + (X - W_1)$, with W_1 the broken Jacobi field taking the values $X(b)$ at the points of B Chapter9/IndexReduction · exists_brokenSpace_eq. That field is built by the existence result with value 0 at 0 and τ Chapter9/BrokenJacobi · exists_isBrokenJacobi, then cut off outside $[0,\tau]$ by an indicator, which keeps it in $T\Omega_\gamma$ Chapter9/BrokenJacobi · IsBrokenJacobi.mem_pathTangent. The intersection of the two summands is 0 by uniqueness Chapter9/BrokenJacobi · IsBrokenJacobi.eqOn_zero. (ii) is the jump formula, with $W_2(b) = 0$. (iii) departs from Milnor's proof. A field of T' has a zero within h_0 before every point: the point of $\{0\} \cup B$ that starts its piece, or the point itself beyond τ. So the estimate of §VII.19 Chapter9/IndexForm · eqOn_zero_of_hessForm gives $E_{**}(W,W) > 0$ unless $W = 0$ on $[0,1]$, and such a W vanishes everywhere, since it vanishes outside $[0,\tau]$. Milnor argues instead that $E_{**}(W,W) \geq 0$ because each segment is minimal, and that $E_{**}(W,W) = 0$ puts W in the null space, so that W is a Jacobi field (Theorem 14.1), hence 0 by Remark 14.5. The formal proof uses no minimal geodesics, no variations and no Theorem 14.1.

Lemma 15.4 (Milnor p. 85; V9.19). In the same situation, the index of E_{**} on the fields along $\gamma|[0,\tau]$ equals its index on the broken Jacobi fields Chapter9/IndexReduction · lemma_15_4. The null space of E_{**} on the fields along $\gamma|[0,\tau]$ is that of its restriction to the broken Jacobi fields Chapter9/IndexReduction · lemma_15_4_ker.

Proof. This is Lemma 15.B Chapter9/FormIndex · formIndex_eq_restrict_of_isCompl Chapter9/FormIndex · ker_eq_map_ker_restrict applied to Lemma 15.3, with A the broken fields and $C = T'$, both viewed inside the fields along $\gamma|[0,\tau]$. Perpendicularity in the other order follows from the symmetry of E_{**} Chapter8/Hessian · energyHessian_isSymm. A linear isomorphism then identifies the two descriptions of the broken fields Chapter9/FormIndex · formIndex_eq_of_equiv.

The finite-dimensional space Σ (Milnor p. 86). $\Sigma = \bigoplus_{b\in B} T_{\gamma(b)}N$ Chapter9/IndexReduction · nodeSpace. The map $W \mapsto (W(b))_{b\in B}$ Chapter9/IndexReduction · nodeEval is an isomorphism from the broken Jacobi fields onto Σ Chapter9/IndexReduction · nodeEval_bijective Chapter9/IndexReduction · nodeEquiv, by uniqueness and existence Chapter9/BrokenJacobi · IsBrokenJacobi.eqOn_zero Chapter9/IndexReduction · exists_brokenSpace_eq. So the broken Jacobi fields form a finite-dimensional space Chapter9/IndexReduction · finiteDimensional_brokenSpace.

The nullity on the broken Jacobi fields is $\nu(\tau)$ (V9.19) Chapter9/IndexReduction · finrank_ker_broken. Map $J \mapsto \mathbf{1}_{[0,\tau]}J$, from the Jacobi fields vanishing at 0 and τ into $T\Omega_\gamma$. It is injective, because a field of $\mathcal{J}(\gamma)$ that vanishes on an interval vanishes Chapter9/IndexReduction · jacobiSpace_eq_zero_of_eqOn. Its range is the null space of E_{**} on the broken fields, by two inclusions.

1. $\mathbf{1}_{[0,\tau]}J$ is broken, being one Jacobi field on every piece, and it has no jumps at the points of B, where it is differentiable. So it is null by the jump formula Chapter9/BrokenJacobi · energyHessian_broken_eq.
2. Let X be broken and null. Test it against the broken field Y with $Y(b) = P\Delta_b X'$ Chapter9/IndexReduction · exists_brokenSpace_eq: the jump formula gives $-2\sum_b \|P\Delta_b X'\|^2 = 0$, so X has no tangential jumps. A broken Jacobi field without tangential jumps is a single Jacobi field on $[0,\tau]$ Chapter9/IndexReduction · eq_jacobi_of_jumps_zero. The proof is an induction on the largest point m of B. If $X = J_1$ on $[0,m]$ and $X = J_2$ on $[m,\tau]$, then $J_2 - J_1$ vanishes at m and has covariant derivative $P(J_2'(m) - J_1'(m)) = P\Delta_m X' = 0$ there, so $J_1 = J_2$ by uniqueness Chapter9/JacobiSystem · IsJacobiField.eqOn_zero. That Jacobi field vanishes at 0 and at τ, so $X = \mathbf{1}_{[0,\tau]}J$.

Milnor obtains this nullity on p. 86 from Theorem 14.1 applied to γ_τ. The formal proof gets it directly, along any smooth curve.

VII.21 $\lambda(\tau)$: monotonicity, finiteness, small τ

Definition (Milnor p. 85; V9.20). For a smooth curve γ in N on $[0,1]$ and $\tau \in \mathbb{R}$, $\lambda(\tau)$ is the index of E_{**} restricted to the fields along $\gamma|[0,\tau]$ Chapter9/IndexReduction · indexUpTo. In particular $\lambda(1)$ is the index

of E_{**} on all of $T\Omega_\gamma$ Chapter9/IndexReduction · indexUpTo_one. One inequality holds because restriction does not increase the index Chapter9/FormIndex · formIndex_restrict_le. For the other, $X \mapsto \mathbf{1}_{[0,1]}X$ maps $T\Omega_\gamma$ into the fields along $\gamma|[0,1]$ and preserves E_{**}, which depends only on the values on $[0,1]$ Chapter8/Hessian · secondVariation_congr Chapter9/FormIndex · formIndex_le_of_map. Milnor's $\lambda(\tau)$ is the index of the Hessian $(E_0^\tau)_{**}$ of γ_τ. Defining it by restriction makes Assertion (1) immediate, and no rescaling of γ_τ to $[0,1]$ (the exposition's Lemma 13.B) is needed.

Assertion (1) (Milnor p. 85). $\lambda(\tau) \le \lambda(\tau')$ for $\tau \le \tau'$ Chapter9/IndexReduction · indexUpTo_mono: the inclusion of fields preserves E_{**}, so the index can only grow Chapter9/FormIndex · formIndex_le_of_map. Milnor's argument extends fields by zero and needs this extension to preserve the Hessian. That is the integral form, and here it is built into the definition.

Finiteness (Lemma 15.4). For $0 < \tau \le 1$ and B fine for $[0,\tau]$, $\lambda(\tau)$ equals the index on the broken Jacobi fields Chapter9/IndexReduction · lemma_15_4, so it is at most their finite dimension Chapter9/IndexReduction · indexUpTo_le_finrank. Since fine subdivisions exist Chapter9/BrokenJacobi · exists_isFine, $\lambda(\tau) < \infty$ Chapter9/IndexReduction · indexUpTo_lt_top.

Assertion (2) (Milnor p. 85). $\lambda(\tau) = 0$ for $0 < \tau \le h_0$ Chapter9/IndexReduction · indexUpTo_eq_zero. A field along $\gamma|[0,\tau]$ with $\tau \le h_0$ has a zero within h_0 before every point: 0 for points up to τ, the point itself beyond. By the estimate of §VII.19 Chapter9/IndexForm · eqOn_zero_of_hessForm, $E_{**}(X,X) < 0$ forces $X = 0$, so every negative definite subspace is 0. Milnor's argument is that γ_τ is a minimal geodesic for small τ, which needs Theorem 10.4, so that $\lambda(\tau) = 0$ by Lemma 13.6. Here E_{**} is even positive definite on these fields, and minimality is not used.

VII.22 The forms H_τ and their continuity

On p. 86 Milnor chooses the subdivision with $t_i < \tau < t_{i+1}$. He observes that $\lambda(\tau)$ is the index of a quadratic form H_τ on the broken Jacobi fields along γ_τ, a space isomorphic to $\Sigma = TM_{\gamma(t_1)} \oplus \cdots \oplus TM_{\gamma(t_i)}$, which does not depend on τ, and that "evidently" H_τ varies continuously with τ. The formal proof makes H_τ precise and proves the continuity.

Definitions (V9.21). Fix a finite set B. A parameter τ is *valid* if $0 < \tau \le 1$ and B is fine for $[0,\tau]$ Chapter9/IndexContinuity · validFor. For valid τ and $x \in \Sigma$, W_x^τ is the broken Jacobi field along $\gamma|[0,\tau]$ with values x at the points of B Chapter9/IndexContinuity · brokenOf Chapter9/IndexContinuity · brokenOf_mem Chapter9/IndexContinuity · brokenOf_apply_node, and

$$H_\tau(x,y) = E_{**}(W_x^\tau, W_y^\tau)$$

Chapter9/IndexContinuity · Hform Chapter9/IndexContinuity · Hform_apply, with $H_\tau = 0$ for τ not valid. H_τ is symmetric Chapter9/IndexContinuity · Hform_isSymm. For valid τ, its index is $\lambda(\tau)$ Chapter9/IndexContinuity · formIndex_Hform, by Lemma 15.4 Chapter9/IndexReduction · lemma_15_4 and the isomorphism with Σ Chapter9/FormIndex · formIndex_eq_of_equiv, and its nullity is $\nu(\tau)$ Chapter9/IndexContinuity · finrank_ker_Hform, the null space being transported by the same isomorphism Chapter9/IndexReduction · finrank_ker_broken. By the jump formula Chapter9/BrokenJacobi · energyHessian_broken_eq, $H_\tau(x,y) = -2\sum_b \langle y_b, P\Delta_b(W_x^\tau)'\rangle$ Chapter9/IndexContinuity · Hform_eq_jumps.

Validity near τ Chapter9/IndexContinuity · isFine_of_isFine Chapter9/IndexContinuity · mem_validFor_of_near. Suppose B is fine for $[0,\tau]$ and the last point m of $\{0\}\cup B$ is within $h_0/2$ of τ. Then every $\tau' \in (m, \tau + h_0/2)$ with $\tau' \le 1$ is valid. The pieces before m are unchanged, and the last piece $[m,\tau']$ is shorter than h_0. Milnor's argument silently needs the subdivision to remain admissible for τ' near τ, and this is that fact.

Lemma 15.C of the exposition (Milnor p. 86, "evidently"; V9.21). For $x, y \in \Sigma$, $\tau \mapsto H_\tau(x,y)$ is continuous on the set of valid τ Chapter9/IndexContinuity · continuousWithinAt_Hform.

The proof as formalized.

1. *Reduction to the jumps.* In $H_\tau(x,y) = -2\sum_b \langle y_b, P\Delta_b(W_x^\tau)'\rangle$ the vectors y_b are fixed. It suffices that $\tau \mapsto P\Delta_b(W_x^\tau)'$ be continuous on the valid τ for each $b \in B$ `Chapter9/IndexContinuity · continuousWithinAt_jump`.
2. *Agreement up to the last division point* `Chapter9/IndexContinuity · brokenOf_eqOn`. For valid τ and τ', W_x^τ and $W_x^{\tau'}$ agree on $[0,m]$, where $m = \max B$. Their difference is a broken Jacobi field on $[0,m]$ for $B \setminus \{m\}$, which is fine for $[0,m]$, and it vanishes at 0, at m and at the other points, so it is 0 `Chapter9/BrokenJacobi · IsBrokenJacobi.eqOn_zero`. Hence for $b < m$ the jump does not depend on τ. At $b = m$ the left derivative is fixed, and only the right derivative depends on τ.
3. *The right derivative at m solves a linear system.* On $[m,\tau]$, W_x^τ is the Jacobi field Y^τ with $Y^\tau(m) = x_m$ and $Y^\tau(\tau) = 0$. Let $z(\tau) = \frac{DY^\tau}{dt}(m)$, the projected right derivative. Y^τ is the first component of the solution with data $(x_m, z(\tau))$ at m `Chapter9/JacobiSystem · IsJacobiField.eqOn_jacSol`, which by linearity is the solution with data $(x_m, 0)$ plus the solution with data $(0, z(\tau))$ `Chapter9/JacobiSystem · jacSol_add`. Evaluating at τ gives $L(\tau)z(\tau) = c(\tau)$. Here $L(\tau)z$ is the first component at time τ of the solution with data $(0,z)$ at m, and $c(\tau)$ is minus the first component at time τ of the solution with data $(x_m,0)$. Both L and c are continuous in τ, by the continuity of the propagator `Chapter9/JacobiODE · continuous_linFlow` `Chapter9/JacobiODE · continuous_linSol`.
4. *$L(\tau_0)$ is injective.* If $L(\tau_0)w = 0$, the Jacobi field with $J(m) = 0$ and $\frac{DJ}{dt}(m) = w$ `Chapter9/JacobiSystem · isJacobiField_jacSol` vanishes at m and at τ_0, which are at most h_0 apart by fineness. So $J = 0$ `Chapter9/IndexForm · eq_zero_of_short`, and $w = 0$.
5. *Continuity of the solution of an injective linear system* `Chapter9/IndexContinuity · continuousWithinAt_of_injective`. Suppose $L(\tau)z(\tau) = c(\tau)$, with L and c continuous at τ_0 and $L(\tau_0)$ injective on a finite-dimensional space. Then $\|v\| \le K\|L(\tau_0)v\|$ for some K `Mathlib/Analysis/Normed/Module/FiniteDimension · LinearMap.exists_antilipschitzWith`. For τ near τ_0, $\|L(\tau) - L(\tau_0)\| < 1/(2K)$, hence $\|v\| \le 2K\|L(\tau)v\|$. Then $\|z(\tau) - z(\tau_0)\| \le 2K\|c(\tau) - L(\tau)z(\tau_0)\|$, which tends to $2K\|c(\tau_0) - L(\tau_0)z(\tau_0)\| = 0$. This replaces Cramer's rule in the exposition's proof.

What "evidently" hides: the non-conjugacy of t_i and τ, without which Y^τ is not even defined; the linear dependence of Jacobi fields on their initial data; and the continuity of the solution of an injective linear system.

VII.23 Assertions (3) and (4): the jumps at conjugate points

Assertion (3) (Milnor p. 85; V9.22). For $0 < \tau \le 1$, $\lambda(\tau') = \lambda(\tau)$ for all $\tau' < \tau$ close enough to τ `Chapter9/IndexAssertions · assertion_3`.

Proof (Milnor's). Take B fine for $[0,\tau]$ with last point $m < \tau$ within $h_0/2$ of τ `Chapter9/BrokenJacobi · exists_isFine` `Chapter9/IndexAssertions · max_lt_of_isFine`. Then $\lambda(\tau) = k$ is finite `Chapter9/IndexReduction · indexUpTo_lt_top` and attained by a subspace $S \subseteq \Sigma$ on which H_τ is negative definite `Chapter9/IndexContinuity · formIndex_Hform` `Chapter9/FormIndex · exists_isNegDefOn_of_formIndex_eq`. By continuity `Chapter9/IndexContinuity · continuousWithinAt_Hform` and Lemma 15.A (3) `Chapter9/IndexContinuity · eventually_isNegDefOn_within`, $H_{\tau'}$ is negative definite on S for valid τ' near τ, and every $\tau' \in (m,\tau)$ is valid `Chapter9/IndexContinuity · mem_validFor_of_near`. So $\lambda(\tau') \ge k$, and Assertion (1) gives $\lambda(\tau') \le \lambda(\tau)$ `Chapter9/IndexReduction · indexUpTo_mono`. *Hidden hypothesis:* that $\lambda(\tau)$ is finite and attained (§§VII.18, VII.21).

Assertion (4), the upper bound (Milnor p. 86). For $0 < \tau < 1$, $\lambda(\tau') \le \lambda(\tau) + \nu(\tau)$ for all $\tau' > \tau$ close enough to τ `Chapter9/IndexAssertions · assertion_4_le`.

Proof. By Sylvester's law `Chapter9/FormIndex · exists_isPosDefOn_add`, H_τ is positive definite on a subspace P of dimension $\dim\Sigma - \lambda(\tau) - \nu(\tau)$, its index being $\lambda(\tau)$ `Chapter9/FormIndex · formIndex_eq_sigNeg` and its nullity $\nu(\tau)$ `Chapter9/IndexContinuity · finrank_ker_Hform`. For valid τ' near τ, which includes all $\tau' \in (\tau, \min(\tau + h_0/2, 1))$ `Chapter9/IndexContinuity · mem_validFor_of_near`, P stays positive definite for $H_{\tau'}$ `Chapter9/IndexContinuity · eventually_isPosDefOn_within`. A negative definite subspace for $H_{\tau'}$ then has dimension at most $\dim\Sigma - \dim P = \lambda(\tau) + \nu(\tau)$ by Lemma 15.A (1) `Chapter9/FormIndex · finrank_add_finrank_le`.

This is Milnor's count with $\dim \Sigma = ni$. *Hidden hypotheses:* Sylvester's law, and the fact that the nullity of H_τ is $\nu(\tau)$, which needs Lemma 15.4 and Theorem 14.1 for γ_τ in the book and §VII.20 here.

The Hessian against a cut-off Jacobi field (Milnor p. 87; V9.23 (ii)). Let $0 < \tau < 1$, let $J \in \mathcal{J}(\gamma)$ vanish at 0 and τ, and let $\hat{J} = \mathbf{1}_{[0,\tau]} J \in T\Omega_\gamma$. For every $Z \in T\Omega_\gamma$,

$$E_{**}(\hat{J}, Z) = 2\Big\langle Z(\tau), \frac{DJ}{dt}(\tau)\Big\rangle$$

`Chapter9/IndexAssertions · energyHessian_cut`. Off τ, $\hat{J}$ is locally J or 0. At τ its left derivative is $J'(\tau)$, and its right derivative is 0, since $\hat{J}$ vanishes from τ on. So $P\Delta_\tau \hat{J}' = -\frac{DJ}{dt}(\tau)$, and the jump formula `Chapter9/IndexForm · energyHessian_eq_jumps` gives $E_{**}(\hat{J}, Z) = -2\langle Z(\tau), -\frac{DJ}{dt}(\tau)\rangle$.

The fields X_k (Milnor p. 87; V9.23 (iii)) `Chapter9/IndexAssertions · exists_dual`. Linearly independent tangent vectors $d_1, \dots, d_\nu$ at a point of N have tangent vectors $x_1, \dots, x_\nu$ at that point with $\langle d_h, x_k\rangle = \delta_{hk}$. The map $u \mapsto (\langle d_h, u\rangle)_h$ from the span of the d_h to $\mathbb{R}^\nu$ is injective, since a vector orthogonal to all d_h in their span is orthogonal to itself. It is therefore onto, and the preimages of the standard basis are the x_k.

Assertion (4), the lower bound, for every $\tau' > \tau$ (Milnor pp. 86–87; V9.23 (iv)). For $0 < \tau < \tau' \le 1$, $\lambda(\tau) + \nu(\tau) \le \lambda(\tau')$ `Chapter9/IndexAssertions · assertion_4_ge`.

The proof as formalized.

1. *The negative directions up to τ.* $\lambda(\tau) = k$ is attained `Chapter9/FormIndex · exists_isNegDefOn_of_formIndex_eq` by a subspace S of fields along $\gamma|[0,\tau]$ on which E_{**} is negative definite. A basis $W_1, \dots, W_k$ of S gives the linear map $U : \mathbb{R}^k \to \{\text{fields along } \gamma|[0,\tau']\}$.
2. *The new directions.* Let $J_1, \dots, J_\nu$ be a basis of the Jacobi fields vanishing at 0 and τ, and $d_h = \frac{DJ_h}{dt}(\tau)$. The d_h are linearly independent, because $J \mapsto \frac{DJ}{dt}(\tau)$ `Chapter9/JacobiSpace · jacEval` is injective on these fields by uniqueness `Chapter9/JacobiSystem · IsJacobiField.eqOn_zero`, as $J(\tau) = 0$. Take the dual vectors x_k at $\gamma(\tau)$ `Chapter9/IndexAssertions · exists_dual` and a bump function φ at τ `Chapter8/Critical · bumpAt` of radius $r = \min(\tau, \tau' - \tau)$, equal to 1 at τ. Then $X_k = \varphi\, P x_k$ is smooth, tangent, zero at 0, at 1 and outside $[0,\tau']$, and satisfies $X_k(\tau) = x_k$. The cut-off fields $\hat{J}_h = \mathbf{1}_{[0,\tau]} J_h$ `Chapter9/IndexForm · indicator_mem_pathTangent` are fields along $\gamma|[0,\tau] \subseteq$ fields along $\gamma|[0,\tau']$, and $\hat{J}_h(\tau) = 0$.
3. *The compound matrix.* Apply the lemma of §VII.18 `Chapter9/IndexAssertions · le_formIndex_of_compound` to E_{**} on the fields along $\gamma|[0,\tau']$, with U as above, $Jb = \sum_h b_h \hat{J}_h$ and $Xb = \sum_k b_k X_k$. By the cut-off formula `Chapter9/IndexAssertions · energyHessian_cut`, $E_{**}(Jb, Z) = 2\langle Z(\tau), \sum_h b_h d_h\rangle$. Hence $E_{**}(Jb, Ua) = 0$ because fields along $\gamma|[0,\tau]$ vanish at τ `Chapter9/BrokenJacobi · fieldsUpTo_apply_self`; $E_{**}(Jb, Jb') = 0$ because $\hat{J}_h(\tau) = 0$; and $E_{**}(Jb, Xb') = 2\langle \sum_k b'_k x_k, \sum_h b_h d_h\rangle = 2\langle b, b'\rangle$. So the index on the fields along $\gamma|[0,\tau']$ is at least $k + \nu$.

What Milnor leaves implicit: the linear independence of the $DJ_h/dt(\tau)$ (uniqueness), the construction of the X_k, and the linear independence of the $\lambda + \nu$ fields W_i, $c^{-1}J_h - cX_h$, which the compound-matrix lemma supplies. *Deviation (stronger).* The lower bound holds for every $\tau' \in (\tau, 1]$, not only for τ' near τ.

Assertion (4) (Milnor p. 86; V9.23). For $0 < \tau < 1$, $\lambda(\tau') = \lambda(\tau) + \nu(\tau)$ for all $\tau' > \tau$ close enough to τ `Chapter9/IndexAssertions · assertion_4`, by the upper bound and the lower bound `Chapter9/IndexAssertions · assertion_4_le` `Chapter9/IndexAssertions · assertion_4_ge`. Milnor's ν is the nullity of $(E_0^\tau)_{**}$. Here it is the multiplicity $\nu(\tau)$ of 0 and τ, which by §§VII.20 and VII.22 is the nullity of E_{**} along $\gamma|[0,\tau]$. So λ jumps by $\nu(\tau)$ just after a conjugate parameter τ and is continuous from the left.

VII.24 Theorem 15.1, Corollary 15.2 and Remark 14.6

Milnor writes that the index theorem "clearly follows" from Assertions (2), (3) and (4) (p. 87). The formal proof isolates the deduction as a statement about integer-valued functions.

The deduction from the Assertions (V9.24 (i)). Let $\ell, \nu : \mathbb{R} \to \{0, 1, 2, \ldots\}$ and $h_0 > 0$. Suppose ℓ is non-decreasing on $(0, 1]$, vanishes on $(0, h_0]$, satisfies $\ell(\tau') = \ell(\tau)$ for $\tau' < \tau$ near τ when $0 < \tau \le 1$, and satisfies $\ell(\tau') = \ell(\tau) + \nu(\tau)$ for $\tau' > \tau$ near τ when $0 < \tau < 1$. Then $\nu(t) \neq 0$ for only finitely many $t \in (0, 1)$, and $\ell(1) = \sum_{0<t<1} \nu(t)$ `Chapter9/IndexTheorem · eq_finsum_of_jumps`.

Proof.

1. *The jumps before τ add up to at most $\ell(\tau)$.* For a finite $F \subseteq (0, 1)$ with all points below $\tau \le 1$, $\sum_F \nu \le \ell(\tau)$. By induction, adding the largest point a of F: there is $\tau' \in (a, \tau)$ with $\ell(\tau') = \ell(a) + \nu(a)$, so $\ell(\tau) \ge \ell(a) + \nu(a) \ge \sum_{F \setminus a} \nu + \nu(a)$.
2. *Finiteness.* If infinitely many $t \in (0, 1)$ had $\nu(t) \neq 0$, then $\ell(1) + 1$ of them would give $\sum \nu \ge \ell(1) + 1$, contradicting step 1.
3. *No jumps near 0.* For $t < \min(h_0, 1)$, take $\tau' \in (t, h_0)$ with $\ell(\tau') = \ell(t) + \nu(t)$. Both values of ℓ are 0, so $\nu(t) = 0$.
4. *A locally constant function.* With C the finite set of jumps, $g(\tau) = \ell(\tau) - \sum_{t \in C,\, t<\tau} \nu(t)$ is locally constant on $(0, 1]$. From the left this follows from the left continuity, the sum being unchanged just below τ. From the right, ℓ gains $\nu(\tau)$ and so does the sum; if $\tau \notin C$, then $\nu(\tau) = 0$. A locally constant integer-valued function on the connected set $(0, 1]$ is constant. Since $g(\min(h_0, 1)/2) = 0$, we get $g(1) = 0$, that is, $\ell(1) = \sum_C \nu$.

What "clearly" hides. The conjugate parameters must first be shown to be finite in number, from Assertions (1) and (4) and the finiteness of $\lambda(1)$. Without that, the jumps could not be added up. Near 0, Assertion (4) also excludes conjugate parameters, which Assertion (2) alone does not.

Theorem 15.1 along any smooth curve (V9.24 (ii)). For every smooth curve γ in N on $[0, 1]$, $\nu(t) \neq 0$ for only finitely many $t \in (0, 1)$, and $\lambda(1) = \sum_{0<t<1} \nu(t)$ `Chapter9/IndexTheorem · indexUpTo_one_eq`. Apply the deduction `Chapter9/IndexTheorem · eq_finsum_of_jumps` to $\ell = \lambda$, which takes natural-number values on $(0, 1]$ by finiteness `Chapter9/IndexTheorem · indexUpTo_toNat`, and to h_0 `Chapter9/IndexForm · fineMesh_pos`, using Assertions (1)–(4) `Chapter9/IndexReduction · indexUpTo_mono` `Chapter9/IndexReduction · indexUpTo_eq_zero` `Chapter9/IndexAssertions · assertion_3` `Chapter9/IndexAssertions · assertion_4`.

Theorem 15.1 (the Morse index theorem, Milnor p. 83; V9.24). Let $\gamma \in \Omega(N; p, q)$ be an unbroken geodesic. The parameters $t \in (0, 1)$ conjugate to 0 along γ form a finite set, and the index of E_{**} on $T\Omega_\gamma$ is

$$\lambda = \sum_{0<t<1} \nu(t),$$

the number of these parameters, each counted with its multiplicity. In particular λ is finite `Chapter9/IndexTheorem · theorem_15_1`.

Proof. Apply the previous theorem `Chapter9/IndexTheorem · indexUpTo_one_eq` to the smooth curve γ `Chapter9/JacobiSystem · IsUnbrokenGeodesic.isSmoothCurveIn`. The conjugate parameters in $(0, 1)$ are those with $\nu(t) \neq 0$ `Chapter9/JacobiSpace · isConjugate_iff`, and the index of E_{**} is $\lambda(1)$ `Chapter9/IndexReduction · indexUpTo_one`. Milnor's statement concerns the Hessian of the energy at a critical point, and that is where the geodesic hypothesis enters. The proof uses only the integral form of E_{**} along a smooth curve. Along a curve that is not a geodesic, the same equality holds for the form E_{**} of §VII.9, which is then not a Hessian.

Corollary 15.2 (Milnor p. 83; V9.25). For an unbroken geodesic $\gamma \in \Omega(N; p, q)$, the set of parameters conjugate to 0 along γ is finite `Chapter9/IndexTheorem · corollary_15_2`. It consists of the finitely many conjugate parameters in $(0, 1)$ `Chapter9/IndexTheorem · theorem_15_1`, together with possibly 1.

Remark 14.6 (Milnor p. 82; V9.12). For every smooth curve γ in N on $[0,1]$, no $t \in (0,\delta)$ is conjugate to 0, where $\delta = \min(h_0, 1)$ `Chapter9/IndexTheorem · remark_14_6`. A Jacobi field vanishing at 0 and at $t \leq h_0$ vanishes `Chapter9/IndexForm · eq_zero_of_short`, while conjugacy means a non-zero one exists `Chapter9/JacobiSpace · isConjugate_iff`. Milnor's neighbourhood U is not used. Two parts of the remark are not formalized: negative parameters, and the absence of cluster points of conjugate points along an entire geodesic, since the geodesics of Ω are given on $[0,1]$.

VII.25 What the formalization taught us

- $T\Omega_\omega$ **is a definition.** It is connected to Ω through variations in both directions, and the direction "every field is a variation field" (Lemma 11.A) needs a radius uniform along ω, obtained from compactness. It does not need completeness. Every derivative on Ω is a derivative of a function of finitely many real variables.
- **The variation formulas rest on calculus on closed strips.** Differentiation under the integral sign needs the compactness of closed strips. The jump terms need the symmetry of mixed partials on the edges of the strips, where the t-derivatives are one-sided. (12.2′) is needed at every u, since §13 differentiates it. Milnor's reference to "Lemma 8.5" should be to Lemma 8.7. The scan loses a minus sign in the displayed statements of 12.2 and 13.1.
- **Test fields need care.** In Corollary 12.3 and Theorem 14.1 the test fields exist only by Lemma 11.A. They must be continuous at the break points and must be constructed explicitly. The step "by uniqueness" needs a derivative-extension argument at a former break point.
- **Symmetry of** E_{**} follows in the book from mixed partials of $E \circ \bar{\alpha}$, a function never shown to be C^2. The integral form gives symmetry and bilinearity directly.
- **Jacobi fields need a global theory.** Solutions defined along all of γ need global existence for linear equations. Remark 14.2 fails for a constant geodesic and needs $n \geq 1$.
- **Lemma 14.4** needs $\gamma|[0,\delta]$ to be *the* minimal geodesic between its end points, which Milnor does not argue, and a compactness step that is really smooth dependence on initial data. Building the variation from initial data avoids the first issue and makes the second explicit.
- **One estimate replaces minimality three times.** The two properties of a fine subdivision and Assertion (2) all follow from the Poincaré estimate of §VII.19, with an explicit mesh depending only on bounds for P' and the curvature. "Evidently continuous" needs non-conjugacy on the last segment and a perturbation lemma. Lemma 15.4's "straightforward" proof is an abstract splitting lemma.
- **Theorem 15.1** needs the conjugate parameters to be finite in number before the jumps of λ can be added up. Assembled correctly, the proof works along any smooth curve, with λ defined by restriction to the fields vanishing beyond τ.

Part VIII. Broken geodesics (Milnor §16)

Milnor's §16 makes the path space finite dimensional. Below a fixed energy c, a path from p to q is straightened segment by segment into a *broken geodesic* with prescribed break times; the broken geodesics form a manifold B of dimension $n(k-1)$, parametrized by their break points; the energy restricted to B is a smooth function E' with compact sublevel sets, whose critical points are exactly the geodesics from p to q, with the index and nullity of E_{**}; and the straightening is a deformation retraction of Ω^a onto B^a. Hence, when p and q are not conjugate along the geodesics in Ω^a, Ω^a has the homotopy type of a finite cell complex with one cell of dimension λ for each geodesic of index λ (Theorem 16.3). This Part retells the formal proof of all of it: Lemma 16.1, every assertion of Theorem 16.2, Theorem 16.3 for B^a and for Ω^a, and the Remark on p. 92. The labels V10.1–V10.14 are those of the verified edition.

The setting is that of Parts VI and VII. Milnor's M is replaced by a slice submanifold N, modelled on Z with complement W (§0.3), of a finite-dimensional real inner product space V, with the induced metric and its distance ρ, which takes values in $[0,\infty]$ `Chapter7/Distance · riemDist`; every Riemannian manifold is of this kind by Nash's embedding theorem, which is quoted, not proved. Further, $p, q \in V$, and $\Omega = \Omega(N; p, q)$ is the set of maps $\mathbb{R} \to V$ that are piecewise smooth on $[0,1]$, lie in N there and run from p to q `Chapter8/Energy · pathSpace` (§0.8), with energy $E(\omega) = E_0^1(\omega) = \int_0^1 \|\omega'\|^2$ `Chapter8/Energy · energy`. Milnor's completeness of M enters only through the compactness of one ball, and neither connectedness nor Milnor's assumption $\Omega^c \neq \emptyset$ is used.

The formal route departs from Milnor's in five places. (1) The manifold B is represented by its set U of break points, an open subset of N^m and hence a slice submanifold; no structure is transported to B. (2) Paths are maps on $\mathbb{R}$, so Milnor's metric is a pseudometric, and his homeomorphism between B and U becomes a homotopy equivalence, a homeomorphism up to paths at distance 0. (3) Lemma 15.4 does not apply as proved in §15 and is proved anew for the subdivision of §16. (4) Theorem 16.3 is proved through a smooth extension of E' and a version of Theorem 3.5 for compact sublevel sets, assuming that a is not the energy of a geodesic. (5) The continuity of Milnor's deformation r_u, left to the reader in the book, occupies two files. The formal development proves Lemma 16.1, Theorem 16.2 on B and Theorem 16.3 for B^a before Ω carries any topology; the metric enters only for the deformation retraction and for the passage from B^a to Ω^a. This Part follows that order.

Map.

- `Chapter10/Product`: finite products and open parts of slice submanifolds, their tangent spaces, and functions smooth in Milnor's sense.
- `Chapter10/Broken`: subdivisions, the ball S, Corollary 10.8 with normal radii, Milnor's data, the short join vector, E', the set U, and the broken geodesic with given break points.
- `Chapter10/BrokenPaths`: Lemma 16.1.
- `Chapter10/BrokenEnergy`: U is open in N^m, E' is smooth, B^a is compact.
- `Chapter10/BrokenVariation`: variations through broken geodesics, and the critical points of E'.
- `Chapter10/BrokenSplitting`: broken Jacobi fields for the subdivision of §16, and Lemma 15.4 for it.
- `Chapter10/BrokenIndex`: the Hessian of E' is E_{**} on broken Jacobi fields; index and nullity.

- `Chapter10/SublevelCW`: smooth extension from a compact set, and Theorem 3.5 for a compact sublevel set.
- `Chapter10/Finite`: Theorem 16.3 for B^a, and the Remark on p. 92.
- `Chapter10/PathMetric`: Milnor's metric, and the exposition's Lemmas 16.A and 16.B.
- `Chapter10/Retraction`: Milnor's deformation r_u and its algebraic properties.
- `Chapter10/RetractionCont`, `Chapter10/RetractionConv`: the continuity of r_u in both variables.
- `Chapter10/Deformation`: Ω^a deformation retracts onto B^a.
- `Chapter10/Omega`: the two descriptions of B^a, and Theorem 16.3 for Ω^a.

VIII.1 Subdivisions, the ball S and the choice of δ

Subdivisions. A subdivision with m interior points is a strictly increasing $t : \{0, \dots, m+1\} \to \mathbb{R}$ with $t_0 = 0$ and $t_{m+1} = 1$ `Chapter10/Broken · Subdivision`; Milnor's k is $m+1$, the number of segments. The i-th segment is $[t_i, t_{i+1}]$, $0 \le i \le m$, of length $\ell_i = t_{i+1} - t_i > 0$ `Chapter10/Broken · Subdivision.len` `Chapter10/Broken · Subdivision.len_pos`, and the lengths add up to 1 by telescoping `Chapter10/Broken · Subdivision.sum_len`. All t_i lie in $[0,1]$ `Chapter10/Broken · Subdivision.t_nonneg` `Chapter10/Broken · Subdivision.t_le_one` and are ordered as their indices `Chapter10/Broken · Subdivision.t_lt_iff`. The interior break points $t_1, \dots, t_m$ form a finite set `Chapter10/Broken · Subdivision.breaks` `Chapter10/Broken · Subdivision.mem_breaks` contained in $(0,1)$ `Chapter10/Broken · Subdivision.breaks_subset`. Every number τ is assigned a segment: the first i with $\tau < t_{i+1}$, and the last segment if there is none `Chapter10/Broken · Subdivision.segOf`. So $t_i \le \tau < t_{i+1}$ puts τ in segment i `Chapter10/Broken · Subdivision.segOf_eq`, $\tau \ge 1$ in the last `Chapter10/Broken · Subdivision.segOf_of_one_le`, and every $\tau \in [0,1]$ lies in the closed interval of its segment `Chapter10/Broken · Subdivision.mem_Icc_segOf`. The convention is asymmetric: an interior break point t_{i+1} belongs to segment $i+1$, not i, which is why several proofs below check that two formulas agree at a division point. A closed interval $[a,b] \subseteq [0,1]$, $a < b$, with no break point in its interior lies in one segment `Chapter10/Broken · Subdivision.exists_seg_of_piece`: take the segment i of a; if $b > t_{i+1}$, then t_{i+1} is an interior break point in (a,b), unless $a = t_{i+1}$, which is impossible because then a would belong to segment $i+1$. Subdivisions of every mesh exist: the uniform one, $t_i = i/(m+1)$, has all lengths $1/(m+1)$ `Chapter10/Broken · Subdivision.uniform` `Chapter10/Broken · Subdivision.uniform_len`.

Vertices. For $y = (y_1, \dots, y_m) \in V^m$ the vertices of a broken path with break points y are $x_0 = p$, $x_j = y_j$ for $1 \le j \le m$, and $x_{m+1} = q$ `Chapter10/Broken · vert` `Chapter10/Broken · vert_zero` `Chapter10/Broken · vert_inner` `Chapter10/Broken · vert_last`. (The formal indices of y start at 0; this Part numbers them from 1, so that y_j sits at t_j.)

The ball S. Milnor's S is $\{x \in N : \rho(p,x) \le \sqrt{c}\}$ `Chapter10/Broken · closedRiemBall`, a subset of N `Chapter10/Broken · closedRiemBall_subset`.

Step 1 (Milnor p. 89). *If $\omega \in \Omega$ has $E(\omega) \le c$, then $\omega(t) \in S$ for every $t \in [0,1]$* `Chapter10/Broken · mem_closedRiemBall_of_energy_le`. The restriction of ω to $[0,t]$ is a piecewise smooth path in N from p to $\omega(t)$, so $\rho(p,\omega(t)) \le L_0^t(\omega)$ `Chapter7/Distance · riemDist_le_pathLength` `Chapter7/Path · pathLength`. Schwarz's inequality gives $(L_0^t)^2 \le t\,E_0^t$ `Chapter8/Energy · pathLength_sq_le_energy`, and $t\,E_0^t \le E_0^t \le E_0^1 \le c$, the middle inequality because the integral of the nonnegative $\|\omega'\|^2$ grows with the interval. Hence $L_0^t \le \sqrt{c}$.

S is compact. Milnor uses that M is complete; the formal proof offers two embedded forms of that hypothesis. If $p \notin N$, then $\rho(p,\cdot) = \infty$ `Chapter7/Distance · riemDist_eq_top_of_notMem` and S is empty, so assume $p \in N$.

- *N closed in V* `Chapter10/Broken · isCompact_closedRiemBall_of_isClosed`. The function $\rho(p,\cdot)$ is continuous on N `Chapter7/Metric · continuousOn_riemDist_left`, so S is closed in the closed set N, hence in V. Since $\|x-p\| \le \rho(p,x)$ `Chapter7/Metric · ofReal_norm_sub_le_riemDist`, S lies in the closed ball of radius $\sqrt{c}$ about p, a compact set.
- *N preconnected and (N,ρ) complete* `Chapter10/Broken · isCompact_closedRiemBall_of_isRiemComplete` `Chapter7/Metric · IsRiemComplete`. The Hopf–Rinow theorem of Part VI says that closed ρ-bounded subsets

of N are compact `Chapter7/HopfRinow · hopf_rinow_of_isRiemComplete`. Here S, as a subset of N, is closed by continuity and ρ-bounded by $2\sqrt{c}$ through the triangle inequality at p `Chapter7/Distance · riemDist_triangle` `Chapter7/Metric · IsRiemBounded`.

Corollary 10.8 with normal radii. Part VI packages the conclusion of Milnor's Corollary 10.8, for $y, z \in N$ and a bound δ, as one predicate `Chapter7/Distance · IsShortJoin`: there is $v \in T_yN$ with $\|v\| < \delta$, $\exp_y v = z$ `Chapter7/Exp · exp` and $\rho(y,z) = \|v\|$; v is the only tangent vector at y of norm $< \delta$ with $\exp_y v = z$; and v depends smoothly on the end points: there are an open $O_2 \ni (y,z)$ in $V \times V$ and a map G, smooth on O_2, with $G(y,z) = v$, such that for $y', z' \in N$ with $(y',z') \in O_2$ and $\rho(y',z') < \delta$, $G(y',z')$ is tangent at y', of norm $< \delta$, and $\exp_{y'} G(y',z') = z'$. Part VI proves it locally: around each $p \in N$ there are $s > 0$ and an open $\Omega \ni p$ such that the predicate holds for $y \in N \cap \Omega$, $\rho(y,z) < \delta \le s$ `Chapter7/Distance · exists_isShortJoin`.

The §16 file adds a requirement that Milnor never mentions, at no cost: δ must also be at most a normal radius at every point of S, an ε such that $\exp_x$ maps the open ε-ball of T_xN diffeomorphically onto an open set `Chapter7/NormalNbhd · IsNormalRadius`. In the local statement, Ω is shrunk into the open set O of Lemma 10.3, on which a single ε is a normal radius at every point `Chapter7/NormalNbhd · lemma_10_3`, and s is replaced by $\min(s, \varepsilon)$ `Chapter10/Broken · exists_isShortJoin'`. Then **Corollary 10.8 with normal radii** `Chapter10/Broken · corollary_10_8'`: *for compact $K \subseteq N$ there is $\delta > 0$ such that any $y, z \in K$ with $\rho(y,z) < \delta$ satisfy the predicate at scale δ, and every point of K has a normal radius $\ge \delta$.* For empty K, $\delta = 1$ works vacuously. Otherwise the open sets Ω_x, $x \in K$, cover K; a finite subcover is chosen `Mathlib/Topology/Compactness/Compact · IsCompact.elim_finite_subcover` and δ is the least of the corresponding s_x. A point $y \in K$ lies in some Ω_x of the subcover, so the joins from y hold at the scale $\delta \le s_x$, and the normal radius at y is $\varepsilon_x \ge s_x \ge \delta$. The normal radius is used only for Jacobi fields along the segments (VIII.6, VIII.7).

Milnor's data. All results of this Part are stated for a structure that collects Milnor's choices on p. 89 `Chapter10/Broken · BrokenSetup`, which this book calls *Milnor's data*: a subdivision; end points $p, q \in N$; an energy bound $c > 0$ such that S is compact; a number $\delta > 0$, Milnor's ε, such that any two points of S at distance $< \delta$ satisfy the conclusion of Corollary 10.8 at scale δ, and such that every point of S has a normal radius $\ge \delta$; and the fineness condition $c\,\ell_i < \delta^2$ for every segment, Milnor's $t_i - t_{i-1} < \varepsilon^2/c$. Trivially $p \in S$ `Chapter10/Broken · BrokenSetup.p_mem_ball`.

Sufficiently fine subdivisions (V10.3). *If $p, q \in N$, $c > 0$ and S is compact, there is $\eta > 0$ such that every subdivision with all $\ell_i < \eta$ is part of Milnor's data* `Chapter10/Broken · exists_brokenSetup`. Take δ from Corollary 10.8 with normal radii for $K = S$ and $\eta = \delta^2/c$; then $c\,\ell_i < c \cdot \delta^2/c = \delta^2$. Some uniform subdivision qualifies, by choosing m with $1/(m+1) < \eta$ `Chapter10/Broken · exists_brokenSetup_uniform`. This makes exact the quantifier of Lemma 16.1 over all sufficiently fine subdivisions, and by the compactness of S above, Milnor's data exist for every $c > 0$ when N is closed in V, or connected and complete.

The short join vector. For $x, x' \in V$, $v(x,x')$ is a tangent vector at x of norm $< \delta$ with $\exp_x v = x'$, chosen when one exists, and 0 otherwise `Chapter10/Broken · BrokenSetup.joinVec`. Tangency is expressed with the total tangent projection P_x, the orthogonal projection onto T_xN for $x \in N$ and 0 off N `Chapter8/Energy · prj`, for which $P_x v = v$ if and only if $v \in T_xN$ `Chapter8/Energy · prj_eq_self_iff`. *For $x, x' \in S$ with $\rho(x,x') < \delta$, $v(x,x')$ is tangent, of norm $< \delta$, $\exp_x v(x,x') = x'$, $\rho(x,x') = \|v(x,x')\|$, and it is the only tangent vector of norm $< \delta$ that $\exp_x$ sends to x'* `Chapter10/Broken · BrokenSetup.joinVec_spec`. The data supply such a vector with all these properties, so the choice is made, and by uniqueness the chosen vector is that one. One more fact is needed because exp is a total function with a default value: $(x, v(x,x'))$ lies in the domain of exp, the pairs from which a geodesic runs on a neighbourhood of $[0,1]$ `Chapter7/Exp · expDom` `Chapter10/Broken · BrokenSetup.mem_expDom_joinVec`. Otherwise $\exp_x v = x$ by the default, so $x' = x$, $\rho(x,x') = 0$, $v = 0$; but $(x, 0)$ always lies in the domain `Chapter7/Exp · zero_mem_expDom`. Only the domain fact guarantees that $s \mapsto \exp_x(s\,v)$ is a geodesic.

VIII.2 Lemma 16.1: a broken geodesic is its break points

Definitions. The energy of the broken geodesic with break points y is given by formula $(*)$ of the exposition,

$$E'(y) = \sum_{i=0}^{m} \frac{\rho(x_i, x_{i+1})^2}{\ell_i},$$

the distances being read as real numbers `Chapter10/Broken · BrokenSetup.energy'`. Milnor's open set of break points is `Chapter10/Broken · BrokenSetup.nodes`

$$U = \{y \in N^m : \rho(x_i, x_{i+1}) < \infty \text{ for all } i,\ E'(y) < c\}.$$

The finiteness clause is a formal necessity: ρ is infinite between points that no path joins, and the real part of ∞ is 0 in the formal conventions, so without the clause a tuple with an infinite gap would look like one of small energy. A path γ is a *broken geodesic* for the subdivision when it is a geodesic `Chapter7/Geodesic · IsGeodesicOn` on every open segment (t_i, t_{i+1}) `Chapter10/Broken · BrokenSetup.IsBroken`, and Milnor's $B = \operatorname{Int}\Omega(t_0, \dots, t_k)^c$ is the set of broken geodesics $\gamma \in \Omega$ with $E(\gamma) < c$ `Chapter10/BrokenPaths · BrokenSetup.brokenPaths`. Milnor asks for a geodesic on the closed segment; the open one suffices, because *a piecewise smooth path that is a geodesic on (a, b) is smooth on $[a, b]$* `Chapter10/BrokenPaths · contDiffOn_Icc_of_isGeodesicOn`: at an interior point the geodesic is smooth, and at a (or b) the path's own smooth piece to the right of a (left of b) `Chapter8/Piecewise · exists_piece_right Chapter8/Piecewise · exists_piece_left` gives smoothness on a neighbourhood within $[a, b]$; local smoothness at every point of $[a, b]$ is smoothness on $[a, b]$. The break points of a path are $(\gamma(t_1), \dots, \gamma(t_m))$ `Chapter10/Broken · BrokenSetup.nodeVal`, and for $\gamma \in \Omega$ their vertices are the values of γ at all division points, $x_k = \gamma(t_k)$ `Chapter10/BrokenPaths · BrokenSetup.vert_nodeVal`.

From break points to a path. Two estimates make every $y \in U$ the break points of a broken geodesic.

- *Consecutive vertices are at distance $< \delta$* `Chapter10/Broken · BrokenSetup.riemDist_vert_lt`. The terms of E' are nonnegative `Chapter10/Broken · BrokenSetup.energy'_nonneg_terms`, so each satisfies $\rho_i^2/\ell_i \le E'(y) < c$, and $\rho_i^2 < c\,\ell_i < \delta^2$ by fineness.
- *All vertices lie in S* `Chapter10/Broken · BrokenSetup.vert_mem_ball`, the exposition's Step 4, which Milnor leaves implicit. The triangle inequality along the chain of vertices, by induction on k, gives $\rho(p, x_k) \le \sum_i \rho_i$ `Chapter10/Broken · riemDist_le_sum_filter Chapter10/Broken · riemDist_le_sum`, and Cauchy–Schwarz in Sedrakyan's form `Mathlib/Algebra/Order/BigOperators/Ring/Finset · sq_sum_div_le_sum_sq_div` gives

  $$\Big(\sum_i \rho_i\Big)^2 \le \Big(\sum_i \ell_i\Big)\Big(\sum_i \frac{\rho_i^2}{\ell_i}\Big) = E'(y) < c.$$

  The vertices lie in N `Chapter10/Broken · BrokenSetup.vert_mem`.

So the join vector specification applies to every pair of consecutive vertices `Chapter10/Broken · BrokenSetup.joinVec_vert`. Write $v_i = v(x_i, x_{i+1})$. The i-th segment of the broken geodesic is

$$\tau \mapsto \exp_{x_i}\Big(\frac{\tau - t_i}{\ell_i}\, v_i\Big)$$

`Chapter10/Broken · BrokenSetup.segPath`, and the broken geodesic ω_y takes at τ the value of the segment of τ `Chapter10/Broken · BrokenSetup.path`. Its properties are checked one at a time.

1. The i-th segment starts at x_i, since $\exp_x 0 = x$ `Chapter7/Exp · exp_zero Chapter10/Broken · BrokenSetup.segPath_start`, and ends at x_{i+1} `Chapter10/Broken · BrokenSetup.segPath_end`.
2. *Each segment is a geodesic on an open interval $(t_i - \eta, t_{i+1} + \eta)$, with velocity v_i/ℓ_i at t_i* `Chapter10/Broken · BrokenSetup.segPath_isGeodesicOn`. The ray $s \mapsto \exp_{x_i}(s v_i)$ is a geodesic on some $(-\delta', 1 + \delta')$ `Chapter7/Exp · IsGeodesicFrom.isGeodesicOn_exp_smul`, which needs the domain fact above; an affine

change of parameter $s = (\tau - t_i)/\ell_i$ preserves geodesics `Chapter7/Geodesic · IsGeodesicOn.comp_mul_add`, and $\eta = \delta'\ell_i$. The velocity follows from the chain rule and the velocity v of $s \mapsto \exp_x(sv)$ at $s = 0$ `Chapter7/GeodesicFlow · hasDerivAt_exp_smul_zero`.

3. *ω_y agrees with segment i on the closed interval $[t_i, t_{i+1}]$* `Chapter10/Broken · BrokenSetup.path_eq_segPath`. Only the right end needs an argument: at an interior t_{i+1} the definition uses segment $i+1$, which starts at x_{i+1}, where segment i ends; at $t_{m+1} = 1$ the definition uses the last segment.
4. ω_y is smooth on every closed piece free of break points, which lies in one segment `Chapter10/Broken · BrokenSetup.contDiffOn_path`; it takes values in N on $[0,1]$ `Chapter10/Broken · BrokenSetup.path_mem`; it is continuous on $[0,1]$ because it is continuous on each piece `Chapter8/Piecewise · continuousOn_of_piecewise`; and $\omega_y(0) = x_0 = p$, $\omega_y(1) = x_{m+1} = q$. So $\omega_y \in \Omega$, with the interior break points as break set `Chapter10/Broken · BrokenSetup.path_mem_pathSpace` `Chapter7/Path · IsPiecewiseSmooth`. It is a broken geodesic `Chapter10/Broken · BrokenSetup.isBroken_path`, and its break points are y, since $\omega_y(t_j)$ is the start $x_j = y_j$ of segment j `Chapter10/Broken · BrokenSetup.nodeVal_path`.
5. *Its speed on (t_i, t_{i+1}) is $\|v_i\|/\ell_i$* `Chapter10/Broken · BrokenSetup.norm_deriv_path`, because a geodesic has constant speed `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq` and its velocity at t_i is v_i/ℓ_i.
6. **Formula** $(*)$ (Milnor p. 90): $E(\omega_y) = E'(y)$ `Chapter10/Broken · BrokenSetup.energy_path`. The integral over $[0,1]$ splits over the segments `Chapter10/Broken · sum_integral_fin` (additivity over a subdivision indexed by $\{0, \dots, m+1\}$, by induction from the additivity over two adjacent intervals `Mathlib/MeasureTheory/Integral/IntervalIntegral/Basic · intervalIntegral.integral_add_adjacent_intervals`). On a segment the speed is the constant $\|v_i\|/\ell_i$, so the energy there is $(\|v_i\|/\ell_i)^2\ell_i$ `Chapter8/Energy · energy_eq_of_norm_deriv`, which is $\|v_i\|^2/\ell_i = \rho_i^2/\ell_i$.

From a path to its break points: Step 3 (Milnor p. 89). *Let $\gamma \in B$ and let $[t_i, t_{i+1}]$ be a segment. There is $w \in T_{\gamma(t_i)}N$ with $\|w\| < \delta$ such that $\gamma(\tau) = \exp_{\gamma(t_i)}(\frac{\tau - t_i}{\ell_i} w)$ for $\tau \in [t_i, t_{i+1}]$, $E_{t_i}^{t_{i+1}}(\gamma) = \|w\|^2/\ell_i$, and $\rho(\gamma(t_i), \gamma(t_{i+1})) \le \|w\|$* `Chapter10/BrokenPaths · BrokenSetup.exists_segVec`.

Write $a = t_i$, $b = t_{i+1}$, $h = \ell_i$. The path is smooth on $[a,b]$ (above), so $\tilde\gamma(s) = \gamma(hs + a)$ is a smooth curve in N on $[0,1]$ `Chapter9/JacobiSystem · IsSmoothCurveIn`, and a geodesic on $(0,1)$ by the affine change of parameter. Part VII proves that a geodesic segment is the radial geodesic of its initial velocity `Chapter9/GeodesicVariation · exists_initial_ray`: $\tilde\gamma(s) = \exp_{\gamma(a)}(sw)$ on $[0,1]$, where w is the one-sided velocity of $\tilde\gamma$ at 0 `Chapter9/JacobiSystem · velAlong`, a tangent vector `Chapter9/JacobiSystem · IsSmoothCurveIn.velAlong_mem`. The ray has the constant speed $\|w\|$ of its velocity at 0, while $\tilde\gamma'(s) = h\,\gamma'(hs+a)$; so $\|\gamma'\| = \|w\|/h$ on (a,b). Hence $E_a^b(\gamma) = (\|w\|/h)^2 h = \|w\|^2/h$ and $L_a^b(\gamma) = (\|w\|/h)\,h = \|w\|$ `Chapter7/Path · pathLength_eq_of_norm_deriv`, and $\rho(\gamma(a), \gamma(b)) \le L_a^b(\gamma) = \|w\|$. Finally $\|w\|^2 = h\,E_a^b(\gamma) \le h\,E(\gamma) < h\,c < \delta^2$, which is Milnor's estimate $(L_{t_{i-1}}^{t_i})^2 = (t_i - t_{i-1})E_{t_{i-1}}^{t_i} \le (t_i - t_{i-1})\,c < \varepsilon^2$.

The break points of $\gamma \in B$ lie in U `Chapter10/BrokenPaths · BrokenSetup.nodeVal_mem_nodes`. They lie in N because γ does on $[0,1]$; the distances $\rho(\gamma(t_i), \gamma(t_{i+1})) \le \|w_i\|$ are finite; and, splitting E over the segments,

$$E'(\gamma(t_1), \dots, \gamma(t_m)) \le \sum_i \frac{\|w_i\|^2}{\ell_i} = \sum_i E_{t_i}^{t_{i+1}}(\gamma) = E(\gamma) < c.$$

$\gamma \in B$ is the broken geodesic of its break points (on $[0,1]$) `Chapter10/BrokenPaths · BrokenSetup.path_nodeVal`. For τ in segment i, $\omega_y(\tau) = \exp_{\gamma(t_i)}(\frac{\tau - t_i}{\ell_i}\, v(\gamma(t_i), \gamma(t_{i+1})))$ with y the break points of γ. Both ends lie in S by Step 1, and $\rho(\gamma(t_i), \gamma(t_{i+1})) \le \|w\| < \delta$; the vector w of Step 3 is tangent, of norm $< \delta$, and $\exp_{\gamma(t_i)} w = \gamma(t_{i+1})$. By the uniqueness clause of the join vector, $v(\gamma(t_i), \gamma(t_{i+1})) = w$, and Step 3 gives $\omega_y(\tau) = \gamma(\tau)$. This is Milnor's remark that a segment of energy this small is determined by its end points.

Lemma 16.1 (V10.4). *Given Milnor's data: (a) for every $y \in U$, the broken geodesic ω_y lies in B, its break points are y, and $E(\omega_y) = E'(y)$; (b) for every $\gamma \in B$, the break points of γ lie in U and $\omega_{(\gamma(t_1), \dots, \gamma(t_m))} = \gamma$ on $[0,1]$.* `Chapter10/BrokenPaths · BrokenSetup.lemma_16_1`

So $y \mapsto \omega_y$ and $\gamma \mapsto (\gamma(t_1), \dots, \gamma(t_m))$ are inverse bijections between U and B, paths being compared on $[0,1]$. Milnor's assumption $\Omega^c \neq \emptyset$ is not used, so the formal lemma is slightly stronger; the image, which Milnor calls an open subset of $M \times \cdots \times M$, is identified as U (open by VIII.3); and the claim that the correspondence is a homeomorphism needs the metric of Ω (VIII.10) and is proved in VIII.14.

VIII.3 The manifold of break points and the smoothness of E'

Milnor gives B the differentiable structure of an open subset of $M \times \cdots \times M$ (p. 89). For subsets of Euclidean space nothing needs to be said; for slice submanifolds the products and their open parts must be built, and so must the notion of a function that is smooth because it is locally the restriction of an ambient smooth function.

Products and open parts. Given slice charts σ_j of N, j in a finite index set, their product $y \mapsto \big((\sigma_j(y_j))_1, (\sigma_j(y_j))_2\big)_j$ is a slice chart of $N^\iota \subseteq V^\iota$ `Chapter10/Product · SliceChart.pi`. Its domain is the product of the domains `Chapter10/Product · SliceChart.pi_source`; its target $\{(z,w) : (z_j, w_j) \in$ target of σ_j for all $j\}$ is open as a finite intersection of preimages of open sets; it is smooth in both directions coordinate by coordinate; and $y \in N^\iota$ exactly when every second component vanishes, which is the slice condition. Its parametrization is the product of the parametrizations `Chapter10/Product · sliceParam_pi` `Chapter1/ManifoldSubmanifold · sliceParam`. Hence *N^ι is a slice submanifold of V^ι modelled on Z^ι* `Chapter10/Product · isSliceSubmanifold_pi` `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`. A slice chart of S restricted to an open set O, with any point of $S \cap O$ in its domain as base point, is a slice chart of $S \cap O$ `Chapter10/Product · SliceChart.inter` `Chapter10/Product · sliceParam_inter`: on O, membership in $S \cap O$ is membership in S. So *an open part of a slice submanifold is a slice submanifold* `Chapter10/Product · IsSliceSubmanifold.inter_isOpen`, and in particular $N^m \cap O$ is one for every open $O \subseteq V^m$ `Chapter10/Product · isSliceSubmanifold_pi_inter`. The chart used below at a point y is the product of the charts chosen at the y_j, restricted to O `Chapter10/Product · piInterChart` `Chapter10/Product · mem_piInterChart_source`.

The tangent space. *The tangent space of $N^m \cap O$ at y is $T_{y_1}N \times \cdots \times T_{y_m}N$* `Chapter10/Product · tangentSubspace_pi_inter`. The tangent space of a slice submanifold, as a subspace of the ambient space `Chapter1/ManifoldSubmanifold · tangentSubspace`, is the range of the derivative of the parametrization of any slice chart `Chapter1/ManifoldSubmanifold · tangentSubspace_eq_range_fderiv_sliceParam`. For the product chart the parametrization is $z \mapsto (x_j(z_j))_j$, whose derivative is $z \mapsto (dx_j(z_j))_j$, component by component, and the range of that map is the product of the ranges.

Functions smooth in Milnor's sense. *A function f on a slice submanifold S that agrees, on the points of S in an open set $O \ni y$, with a function F smooth on O is smooth at y as a function on the manifold S* `Chapter10/Product · contMDiffAt_of_locally_eq`: F composed with the inclusion is smooth at y `Chapter1/ManifoldSubmanifold · contMDiffAt_comp_val`, and f agrees with it on the neighbourhood $\{x \in S : x \in O\}$ of y.

U is open in N^m (V10.4 (i)). The vertices depend affinely on the break points `Chapter10/BrokenEnergy · contDiff_vert`. The gaps $g(y) = (\rho(x_i, x_{i+1}))_{0 \le i \le m}$, a point of $[0,\infty]^{m+1}$ `Chapter10/BrokenEnergy · BrokenSetup.gaps`, depend continuously on $y \in N^m$ `Chapter10/BrokenEnergy · BrokenSetup.continuousOn_gaps`, because ρ is continuous on $N \times N$ with values in $[0,\infty]$ `Chapter7/Metric · continuousOn_riemDist`. *There is an open $O \subseteq V^m$ with $U = N^m \cap O$* `Chapter10/BrokenEnergy · BrokenSetup.exists_isOpen_nodes`. The set

$$T = \Big\{ r \in [0,\infty]^{m+1} : r_i < \infty \text{ for all } i,\ \sum_i \frac{r_i^2}{\ell_i} < c \Big\}$$

is open: the condition that all r_i are finite defines an open set, the sum is continuous on it (the real part is continuous away from ∞), and T is the part of that open set where a continuous function is $< c$. Continuity of g on N^m produces an open O with $g^{-1}(T) \cap N^m = O \cap N^m$, and $g^{-1}(T) \cap N^m$ is U by

definition. So U is a slice submanifold of V^m of dimension m dim Z, with tangent space $\prod_j T_{y_j}N$: the manifold whose differentiable structure Milnor takes over from the product.

E' is smooth (Theorem 16.2, first assertion; V10.6 (i)). *Every $y_0 \in U$ has an open neighbourhood O in V^m with a function F, smooth on O, such that $E' = F$ on $U \cap O$* `Chapter10/BrokenEnergy · BrokenSetup.exists_smooth_energy'`. For each i, the consecutive vertices $x_i(y_0), x_{i+1}(y_0)$ lie in S at distance $< \delta$ (VIII.2), so the data provide an open $O_{2,i} \ni (x_i(y_0), x_{i+1}(y_0))$ and a map G_i smooth on it, as in Corollary 10.8. With the affine maps $\varphi_i(y) = (x_i(y), x_{i+1}(y))$, put

$$O = \bigcap_i \varphi_i^{-1}(O_{2,i}), \qquad F(y) = \sum_i \frac{\|G_i(\varphi_i(y))\|^2}{\ell_i},$$

smooth on O because the squared norm is smooth. For $y \in U \cap O$, $G_i(\varphi_i(y))$ is a tangent vector at x_i of norm $< \delta$ with $\exp_{x_i} G_i(\varphi_i(y)) = x_{i+1}$, and $\rho(x_i, x_{i+1}) < \delta$; by the uniqueness clause of the join vector it is $v(x_i, x_{i+1})$, whose norm is $\rho(x_i, x_{i+1})$. So $E'(y) = F(y)$. Milnor says that E' depends smoothly on the break points because of the explicit formula. The formula alone does not show it, since ρ is not smooth on the diagonal: what is smooth is $\rho^2 = \|v\|^2$, with v smooth by Corollary 10.8.

Two consequences: *E' is continuous on U* `Chapter10/BrokenEnergy · BrokenSetup.continuousOn_energy'`, since near each point it agrees on U with a continuous function; and *for every open $O \subseteq V^m$ with $N^m \cap O \subseteq U$, E' is a smooth function on the manifold $N^m \cap O$* `Chapter10/BrokenEnergy · BrokenSetup.contMDiff_energy'`, by the criterion for functions smooth in Milnor's sense at each point. The manifold U itself is one such $N^m \cap O$.

VIII.4 B^a is compact, and a possible misprint on p. 91

Theorem 16.2, compactness (V10.6 (ii)). *For $a < c$ the set $\{y \in U : E'(y) \le a\}$ is compact.* `Chapter10/BrokenEnergy · BrokenSetup.isCompact_sublevel`

This is compactness for the set of break points; the set $B^a \subseteq \Omega$ is identified with it in VIII.14.

Proof as formalized. Let $K = S^m$, compact as a product of compact sets `Mathlib/Topology/Compactness/Compact · isCompact_univ_pi`. The set lies in K, because the vertices of a point of U lie in S (VIII.2). Two cases.

1. $\rho(p, q) = \infty$. Then the set is empty: for $y \in U$, $\rho(p, x_m) \le \sqrt{c}$ because $x_m \in S$, and $\rho(x_m, q) < \infty$ by the definition of U, so $\rho(p, q) < \infty$ by the triangle inequality, a contradiction.
2. $\rho(p, q) < \infty$. Then every gap is finite on all of K: $\rho(x_i, x_{i+1}) \le \rho(x_i, p) + \rho(p, x_{i+1})$, where $x_i \in S$ for $i \le m$, and x_{i+1} lies in S or is q. So E' is continuous on K, as a composite of the continuous gaps and the real part on finite values; and the set equals $K \cap E'^{-1}(-\infty, a]$, because a point of K with $E' \le a < c$ has finite gaps and hence lies in U. A closed subset of the compact K is compact.

What Milnor leaves implicit, and a misprint. Milnor identifies B^a with a set of tuples closed in the compact $S \times \cdots \times S$ (p. 91). The set is closed only because the distances there are finite; with $[0, \infty]$-valued distances a tuple with an infinite gap would otherwise pass the test. Only the last gap can be infinite, and only if $\rho(p, q) = \infty$, which never happens for connected M; the formal proof treats that case separately. As reproduced in the scan, the condition defining the tuples carries the exponent c instead of 2 and has no visible summation sign. The exponent looks like a misprint in the printed text; the missing Σ is probably a defect of the scan, which drops symbols from displays (Findings, §§16–19, marked *(scan?)*). The correct condition is the energy $(*)$, $\sum_i \rho(p_{i-1}, p_i)^2/(t_i - t_{i-1}) \le a$, and that is the formal one: $E'(p_1, \dots, p_{k-1}) \le a$ `Chapter10/Broken · BrokenSetup.energy'`.

VIII.5 The critical points of E'

Milnor (p. 91) observes that every geodesic is a broken geodesic, and that by the first variation formula the critical points of E' are clearly the unbroken geodesics. The formal proof needs four things the sentence takes for granted: every tangent vector of the manifold of break points is the velocity of a curve; a smooth curve of break points yields a variation through broken geodesics, smooth on each strip; the first variation at a broken geodesic reduces to its velocity jumps; and criticality must force every jump to vanish, which the proof obtains by testing the derivative against the vector of all the jumps.

Curves. *Every tangent vector w of a slice submanifold S at y_0 is the velocity at 0 of a smooth curve in S* `Chapter10/BrokenVariation · exists_curve_of_mem_tangentSubspace`. With x the parametrization of the slice chart chosen at y_0 and z_0 the chart value of y_0, write $w = dx_{z_0}(z)$ (the tangent space is the range of dx_{z_0}). The curve $u \mapsto x(z_0 + uz)$ is defined on the open set of u for which $z_0 + uz$ lies in the chart's target `Chapter1/ManifoldSubmanifold · isOpen_toChart_target`, is smooth there `Chapter1/ManifoldSubmanifold · contDiffOn_sliceParam`, takes values in S `Chapter1/ManifoldSubmanifold · sliceParam_mem`, passes through y_0 at $u = 0$ `Chapter1/ManifoldSubmanifold · sliceParam_toChart`, and has velocity $dx_{z_0}(z) = w$ there by the chain rule `Chapter1/ManifoldSubmanifold · contDiffAt_sliceParam`.

Variations through broken geodesics. *Let $u \mapsto y(u)$ be smooth on an open set $U_0 \ni 0$ with values in U. Then there is $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq U_0$ such that for every segment i the map $(u, s) \mapsto \omega_{y(u)}(t_i + s\,\ell_i)$ is smooth on $(-\varepsilon, \varepsilon) \times [0, 1]$ and a geodesic in s for each u, and $u \mapsto \omega_{y(u)}$ is a variation with break points $t_1, \dots, t_m$* `Chapter10/BrokenVariation · BrokenSetup.exists_variation_path` `Chapter8/Energy · IsVariation`.

Proof as formalized. Fix a segment i, and let $c_0(u)$, $c_1(u)$ be the i-th and $(i+1)$-th vertices of $y(u)$, smooth in u.

1. Corollary 10.8 at $(c_0(0), c_1(0))$ gives an open O_2 and a map G smooth on it. On $U_1 = U_0 \cap (c_0, c_1)^{-1}(O_2)$, open and containing 0, the join vector $v_i(u) = v(c_0(u), c_1(u))$ equals $G(c_0(u), c_1(u))$ by uniqueness, so $u \mapsto v_i(u)$ is smooth on U_1, with tangent values.
2. The base ray $s \mapsto \exp_{c_0(0)}(s\, v_i(0))$ is a geodesic on some $(-\delta', 1 + \delta')$. Part VII's theorem on the smooth dependence of geodesics on their initial data over a compact interval `Chapter9/GeodesicVariation · raysGood_of_isGeodesicOn` `Chapter9/GeodesicVariation · RaysGood` provides $\varepsilon_i, \eta_i > 0$ such that $(u, s) \mapsto \exp_{c_0(u)}(s\, v_i(u))$ is smooth on $(-\varepsilon_i, \varepsilon_i) \times (-\eta_i, 1 + \eta_i)$ and a geodesic in s for each u.
3. Let ε be the least of the ε_i. On the i-th strip, $\omega_{y(u)}(t_i + s\ell_i)$ is that ray at time s (VIII.2, item 3), so the strip map is smooth on $(-\varepsilon, \varepsilon) \times [0, 1]$ and a geodesic in s.
4. The axioms of a variation: the paths take values in N; their ends are p and q for every u, since each $\omega_{y(u)}$ lies in Ω; and on every strip $(-\varepsilon, \varepsilon) \times [a, b]$ with no break point inside (a, b) the map is smooth, because $[a, b]$ lies in one segment and the map is the strip map composed with $(u, \tau) \mapsto (u, (\tau - t_i)/\ell_i)$.

Step 2 is the content that Milnor's sentence hides: a geodesic segment depends smoothly on its end points over the whole segment, not only near its initial point, and this is a compactness argument on $[0, 1]$.

The variation field at the break points is the velocity of the curve of break points `Chapter10/BrokenVariation · BrokenSetup.varField_path_node`: if $y'(0)$ is the velocity of y at 0, the variation field `Chapter8/Energy · varField` at t_j is $y'(0)_j$, because $\omega_{y(u)}(t_j) = y(u)_j$ for all u near 0 (VIII.2, item 4).

The first variation at a broken geodesic. The first variation of Part VII at a path ω in the direction of a field X is `Chapter8/FirstVariation · firstVariation`

$$\mathrm{FV}_\omega(X) = -\sum_{t \in (0,1)} \langle X(t), \Delta_t \omega' \rangle - \int_0^1 \langle X(t), A(t) \rangle \, dt,$$

with Δ_t the jump of the velocity `Chapter8/Energy · jump` and A the acceleration `Chapter8/FirstVariation · accel`. *For $y \in U$ and every X, $\mathrm{FV}_{\omega_y}(X) = -\sum_j \langle X(t_j), \Delta_{t_j}\omega'_y\rangle$* `Chapter10/BrokenVariation · BrokenSetup.firstVariation_path`. A point $\tau \in (0,1)$ that is not an interior break point lies in an open segment (t_i, t_{i+1}) (if $\tau = t_i$ with $i > 0$, or $\tau = t_{i+1}$ with $i < m$, it would be a break point); there ω_y is a geodesic, so its acceleration, the tangential part of ω''_y, vanishes `Chapter7/Geodesic · IsGeodesicOn.tanProj_deriv_deriv`. The integrand of the integral term therefore vanishes off the finite, hence null, set of break points and 1, and the integral is 0. The sum over the jumps reduces to the finite sum over the break points, ω_y being differentiable elsewhere in $(0,1)$ `Chapter8/FirstVariation · finsum_jump_eq_sum`; all break points lie in $(0,1)$, and the sum is reindexed by j since t is injective. *The jumps are tangent at the break points* `Chapter10/BrokenVariation · BrokenSetup.prj_jump_path`, as differences of one-sided velocities of a path in N `Chapter8/Critical · prj_jump`.

The derivative of E'. *Let $O \subseteq V^m$ be open with $N^m \cap O \subseteq U$, and F smooth on V^m with $F = E'$ on $N^m \cap O$. For $y_0 \in N^m \cap O$ and every $w \in T_{y_0}(N^m \cap O) = \prod_j T_{y_{0,j}}N$,*

$$dF_{y_0}(w) = -2\sum_j \langle w_j, \Delta_{t_j}\omega'_{y_0}\rangle.$$

`Chapter10/BrokenVariation · BrokenSetup.fderiv_eq_jumps`. Take a curve $y(u)$ in $N^m \cap O$ with velocity w at 0, and the variation $\alpha(u) = \omega_{y(u)}$. The first variation formula, Theorem 12.2 of Part VII `Chapter8/FirstVariation · first_variation`, says that $u \mapsto E(\alpha(u))$ has derivative $2\,\mathrm{FV}_{\omega_{y_0}}(W)$ at 0, with W the variation field. By the chain rule $u \mapsto F(y(u))$ has derivative $dF_{y_0}(w)$ at 0. The two functions agree for $|u| < \varepsilon$, because $F(y(u)) = E'(y(u)) = E(\omega_{y(u)})$ by formula $(*)$. Derivatives are unique, so $dF_{y_0}(w) = 2\,\mathrm{FV}_{\omega_{y_0}}(W)$, and the first variation at a broken geodesic, with $W(t_j) = w_j$, gives the formula.

Theorem 16.2, critical points (V10.9). *Let O and F be as above and $y_0 \in N^m \cap O$. Then y_0 is a critical point* `Chapter1/Manifold · IsCriticalPointM` *of $E' = F|(N^m \cap O)$ if and only if the broken geodesic ω_{y_0} is an unbroken geodesic* `Chapter8/Critical · IsUnbrokenGeodesic`*, that is, by Corollary 12.3, a critical point of E on Ω.* `Chapter10/BrokenVariation · BrokenSetup.isCriticalPointM_iff`

Proof as formalized. A point of a submanifold is critical for the restriction of an ambient function exactly when the ambient derivative vanishes on the tangent space `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_forall_tangent`.

- ($\Rightarrow$) Test with the jumps themselves: $\Delta = (\Delta_{t_j}\omega'_{y_0})_j$ is tangent to $N^m \cap O$ at y_0, since each jump is tangent at $y_{0,j}$ and the tangent space is the product. Then $0 = dF_{y_0}(\Delta) = -2\sum_j \|\Delta_j\|^2$, so every jump vanishes. By Corollary 12.3 `Chapter8/Critical · corollary_12_3`, an unbroken geodesic is the same as a critical path, and a path is critical exactly when its first variation vanishes on every $X \in T\Omega$ `Chapter8/Critical · isCriticalPath_iff Chapter8/Energy · pathTangent`; with all jumps zero, the first variation at the broken geodesic is 0 for every X.
- ($\Leftarrow$) An unbroken geodesic is differentiable at every $t_j \in (0,1)$, so its jumps vanish `Chapter8/Energy · jump_eq_zero` and dF_{y_0} vanishes on the tangent space.

The formal statement reads E' through an ambient smooth extension F, as Part I reads functions on submanifolds, and holds for every such F and O; such an F exists near any compact sublevel set (VIII.8).

Every geodesic of energy $< c$ is a broken geodesic (Milnor p. 91; V10.9 (iv)). An unbroken geodesic is a broken geodesic for every subdivision `Chapter10/Finite · BrokenSetup.isBroken_of_isUnbrokenGeodesic`, so a geodesic $\gamma \in \Omega$ with $E(\gamma) < c$ lies in B, and Lemma 16.1 applies: *its break points y lie in U, $\omega_y = \gamma$ on $[0,1]$, ω_y is an unbroken geodesic, and $E'(y) = E(\gamma)$* `Chapter10/Finite · BrokenSetup.eqOn_path_nodeVal_of_isUnbrokenGeodesic`. Smoothness on $[0,1]$ and the geodesic property pass along agreement on $[0,1]$, and the energy only sees $[0,1]$ `Chapter8/Hessian · energy_congr`. So the critical points of E' correspond to the

geodesics from p to q of energy $< c$, that is, of length $< \sqrt{c}$, an unbroken geodesic having constant speed and so $E = L^2$.

VIII.6 Broken Jacobi fields for the subdivision of §16, and Lemma 15.4 for it

Milnor finishes Theorem 16.2 in two sentences (p. 92): the tangent space of B at a geodesic γ is identified with the space $T\Omega_\gamma(t_0, \dots, t_k)$ of broken Jacobi fields of §15, and the equality of the indices and nullities is then an immediate consequence of Lemma 15.4. But Lemma 15.4, as proved in §15 and in Part VII `Chapter9/IndexReduction · lemma_15_4`, concerns a subdivision fine in the sense of §15 `Chapter9/BrokenJacobi · IsFine`: every piece has length at most $h_0 = 1/(K+1)$, with K a bound along γ for $\|P'\|^2 + \|R(V, P\cdot)V\|$, which makes E_{**} positive definite on the fields vanishing at the division points. The subdivision of §16 was chosen for a different purpose and need not be that fine. This is a hidden hypothesis. The formal proof isolates the two facts that Lemma 15.4 uses: (1) broken Jacobi fields with arbitrary values at the break points exist and are unique; (2) E_{**} is positive definite on the fields vanishing at the break points. For §16, (1) holds because each segment is a radial geodesic shorter than a normal radius, and (2) because a broken geodesic minimizes the energy among the paths through its break points, with (1) excluding null vectors.

The objects are those of Part VII (§0.9): Jacobi fields along γ, smooth and tangent on $[0,1]$ and solving the Jacobi equation `Chapter8/Jacobi · IsJacobiField Chapter8/Jacobi · jacobiExpr`; conjugate parameters $a \neq b$, at which a nonzero Jacobi field vanishes `Chapter9/JacobiSpace · IsConjugate`; broken Jacobi fields, which agree on every piece with a Jacobi field `Chapter9/BrokenJacobi · IsBrokenJacobi`; the space $T\Omega_\gamma(t_0, \dots, t_k)$ of fields of $T\Omega_\gamma$ that vanish outside $[0,\tau]$ and are broken Jacobi fields there `Chapter9/BrokenJacobi · brokenSpace`, here with $\tau = 1$ and the interior break points as division points; Milnor's T', the fields vanishing at the division points and outside $[0,\tau]$ `Chapter9/BrokenJacobi · nodeZero`; the fields vanishing outside $[0,\tau]$ `Chapter9/BrokenJacobi · fieldsUpTo`; and the index form E_{**} `Chapter8/Hessian · energyHessian`.

Throughout this section $y \in U$ and $\gamma = \omega_y$ is an unbroken geodesic, hence a smooth curve in N `Chapter10/BrokenSplitting · BrokenSetup.isSmoothCurveIn_path Chapter9/JacobiSystem · IsUnbrokenGeodesic.isSmoothCurveIn`.

Jacobi fields under an affine change of parameter. Milnor's Lemma 14.3 is stated for variations parametrized by $[0,1]$, so every segment must be rescaled, and Jacobi fields must follow. The derivative of $s \mapsto f(hs+a)$ is $hf'(hs+a)$ `Chapter10/BrokenSplitting · deriv_comp_affine`, and the curvature term satisfies $R(h\,u_1, u_2)(h\,u_3) = h^2 R(u_1,u_2)u_3$ `Chapter10/BrokenSplitting · curvR_smul_smul Chapter8/SecondVariation · curvR`. Hence *if J is a Jacobi field along γ and $0 \le a$, $0 < h$, $a + h \le 1$, then $s \mapsto J(hs+a)$ is a Jacobi field along $s \mapsto \gamma(hs+a)$* `Chapter10/BrokenSplitting · IsJacobiField.comp_affine`: the second covariant derivative and the curvature term both scale by h^2. On $[0,1]$ the rescaled i-th segment of γ is the radial geodesic of the join vector, $\gamma(\ell_i s + t_i) = \exp_{x_i}(s\,v_i)$ `Chapter10/BrokenSplitting · BrokenSetup.path_affine_eq`.

No conjugate points at the ends of a segment. *For every i, t_i and t_{i+1} are not conjugate along γ* `Chapter10/BrokenSplitting · BrokenSetup.not_isConjugate_seg`. Suppose J is a Jacobi field along γ with $J(t_i) = J(t_{i+1}) = 0$ and $J(t) \neq 0$ somewhere. Its rescaling $\tilde J(s) = J(\ell_i s + t_i)$ is a Jacobi field along the rescaled segment, hence along the ray $s \mapsto \exp_{x_i}(sv_i)$, since being a Jacobi field depends only on the curve on $[0,1]$ `Chapter9/JacobiVariation · IsJacobiField.congr_curve`. Milnor's data give a normal radius $\varepsilon \ge \delta > \|v_i\|$ at x_i. Remark 14.5 for a normal radius `Chapter9/JacobiVariation · remark_14_5_of_isNormalRadius` says that a Jacobi field along $s \mapsto \exp_q(sv)$, $\|v\| < \varepsilon$, vanishing at both ends vanishes on $[0,1]$. So $J = 0$ on $[t_i, t_{i+1}]$. A Jacobi field that vanishes on a non-degenerate interval vanishes identically `Chapter9/IndexReduction · jacobiSpace_eq_zero_of_eq0n` (uniqueness for the linear Jacobi equation), applied to J read on $[0,1]$ `Chapter9/JacobiSpace · IsJacobiField.comp_projIcc_mem`; this contradicts $J(t) \neq 0$. Here, and in the companion lemma of VIII.7, the normal radius in Milnor's data is used.

Uniqueness. *A broken Jacobi field X (division points $t_1, \dots, t_m$, $\tau = 1$) that vanishes at every t_j is zero*

Chapter10/BrokenSplitting · BrokenSetup.brokenSpace_eq_zero. For $t \in [0,1]$ in segment i, no break point lies in (t_i, t_{i+1}) (index arithmetic), so X agrees on $[t_i, t_{i+1}]$ with a Jacobi field J. X vanishes at every division point: at $t_0 = 0$ and $t_{m+1} = 1$ because it lies in $T\Omega_\gamma$, at the t_j by hypothesis. So $J(t_i) = J(t_{i+1}) = 0$, $J = 0$ on $[0,1]$ by the previous step, and $X(t) = J(t) = 0$. Outside $[0,1]$, X vanishes by definition.

Existence. *For tangent vectors $\xi_j \in T_{y_j}N$ there is a broken Jacobi field X with $X(t_j) = \xi_j$ for all j* Chapter10/BrokenSplitting · BrokenSetup.exists_brokenSpace_node.

1. Prescribe values at all division points, 0 at t_0 and t_{m+1} and ξ_j at t_j; they are tangent at the corresponding points of γ Chapter9/JacobiSystem · mem_tanAt Chapter9/JacobiSystem · tanAt.
2. On each segment the ends are not conjugate, so $J \mapsto (J(t_i), J(t_{i+1}))$ is a bijection from the Jacobi fields along γ Chapter9/JacobiSpace · jacobiSpace onto $T_{\gamma(t_i)}N \times T_{\gamma(t_{i+1})}N$ Chapter9/GeodesicVariation · jacEvalEnds_bijective_of_not_isConjugate. Choose J_i with the prescribed values at t_i and t_{i+1}.
3. Glue: $X(t) = J_{i(t)}(t)$ on $[0,1]$, $i(t)$ the segment of t, and $X = 0$ outside. On each closed segment $X = J_i$; at an interior t_{i+1} the formula uses J_{i+1}, but $J_{i+1}(t_{i+1}) = J_i(t_{i+1})$, both being the prescribed value. So X is a broken Jacobi field with the prescribed values.
4. X lies in $T\Omega_\gamma$: it is smooth on every piece free of break points and therefore continuous on $[0,1]$ Chapter8/Piecewise · continuousOn_of_piecewise, it is tangent because Jacobi fields are, and it vanishes at 0 and 1.

The values at the break points are tangent Chapter10/BrokenSplitting · BrokenSetup.brokenSpace_node_mem, so $X \mapsto (X(t_1), \dots, X(t_m))$ is a linear map into $\prod_j T_{y_j}N$ Chapter10/BrokenSplitting · BrokenSetup.brokenNodeMap, injective by uniqueness and surjective by existence Chapter10/BrokenSplitting · BrokenSetup.brokenNodeMap_bijective. This gives the linear isomorphism

$$T\Omega_\gamma(t_0, \dots, t_k) \cong T_{y_1}N \times \cdots \times T_{y_m}N$$

Chapter10/BrokenSplitting · BrokenSetup.brokenEquiv Chapter10/BrokenSplitting · BrokenSetup.brokenEquiv_apply, the first half of Milnor's identification of TB_γ with the broken Jacobi fields. The second half, that the variation fields of curves in B are these fields, is VIII.7.

E_{**} on broken Jacobi fields. *For a broken Jacobi field X and any $Y \in T\Omega_{\gamma'}$*

$$E_{**}(X, Y) = -2 \sum_j \left\langle Y(t_j),\ P\,\Delta_{t_j} X' \right\rangle$$

Chapter10/BrokenSplitting · BrokenSetup.energyHessian_brokenSpace_eq, with P the tangent projection at $\gamma(t_j)$. This is Part VII's second variation formula for fields that agree near every point of $(0,1)$ off a finite set with a Jacobi field Chapter9/IndexForm · energyHessian_eq_jumps, the local Jacobi fields being supplied by the broken Jacobi structure Chapter9/BrokenJacobi · IsBrokenJacobi.eventuallyEq; the sum is then restricted to the break points, all in $(0,1)$, and reindexed. In particular *broken Jacobi fields are perpendicular to the fields vanishing at the break points* Chapter10/BrokenSplitting · BrokenSetup.energyHessian_brokenSpace_nodeZero, Lemma 15.3 for §16.

Positivity, first the semi-definite form. *The broken geodesic minimizes the energy among paths through its break points*: for every $\omega \in \Omega$,

$$E'(\omega(t_1), \dots, \omega(t_m)) \le E(\omega)$$

Chapter10/BrokenSplitting · BrokenSetup.energy'_nodeVal_le. Split E over the segments; on each, $\rho(\omega(t_i), \omega(t_{i+1}))^2 \le (L_{t_i}^{t_{i+1}})^2 \le \ell_i E_{t_i}^{t_{i+1}}(\omega)$ by Schwarz's inequality. *Consequently $E_{**}(Y, Y) \ge 0$ for every $Y \in T\Omega_\gamma$ vanishing at the t_j* Chapter10/BrokenSplitting · BrokenSetup.energyHessian_nodeZero_nonneg.

1. Lemma 11.A for one field Chapter8/FirstVariation · exists_variation_one gives the variation $\alpha(u)(t) = \exp_{\gamma(t)}(u\,Y(t))$, $|u| < \varepsilon$, with variation field Y on $[0,1]$.

2. It keeps the break points fixed: $\alpha(u)(t_j) = \exp_{y_j}(0) = y_j$. So $f(u) = E(\alpha(u)) \geq E'(y) = E(\gamma) = f(0)$: f has a local minimum at 0.
3. Remark 13.5 `Chapter8/Hessian · hasDerivAt_deriv_energy` says that $f''(0) = E_{**}(W, W)$ for the variation field W, and $W = Y$ on $[0, 1]$, which is all E_{**} sees `Chapter8/Hessian · secondVariation_congr` `Chapter8/Hessian · energyHessian_apply`.
4. *A local minimum has nonnegative second derivative* `Chapter10/BrokenSplitting · nonneg_of_isLocalMin_of_hasDerivAt`, the step of Milnor's Lemma 13.6. If $f''(0) < 0$, then, since $f'(0) = 0$ at a minimum, the second derivative test `Mathlib/Analysis/Calculus/DerivativeTest · isLocalMax_of_deriv_deriv_neg` makes 0 a local maximum as well; f is then constant near 0, and $f''(0) = 0$, a contradiction.

Milnor's own argument in §15 is the minimality of each short segment; here it is the minimality of the broken geodesic among the paths with the same break points, which needs no fineness.

Positivity, then the definite form. *If moreover Y vanishes outside $[0, 1]$ and $Y \neq 0$, then $E_{**}(Y, Y) > 0$* `Chapter10/BrokenSplitting · BrokenSetup.energyHessian_nodeZero_pos`. Suppose $E_{**}(Y, Y) = 0$. Then Y is a null vector of E_{**}. Let X be arbitrary, and X_b the broken Jacobi field with $X_b(t_j) = X(t_j)$ (existence); then $X - X_b$ vanishes at the t_j.

- $E_{**}(Y, X_b) = 0$ by perpendicularity and the symmetry of E_{**} `Chapter8/Hessian · energyHessian_isSymm`.
- $E_{**}(Y, Z) = 0$ for $Z = X - X_b$, by Cauchy–Schwarz for a semi-definite form. With $b = E_{**}(Y, Z)$ and $a = E_{**}(Z, Z) \geq 0$, every $Y + rZ$ vanishes at the t_j, so $0 \leq E_{**}(Y + rZ, Y + rZ) = 2rb + r^2 a$ for all r. If $b \neq 0$, the choice $r = -b/(a + 1)$ gives $2rb + r^2 a = -b^2(a + 2)/(a + 1)^2 < 0$.

So $E_{**}(Y, X) = 0$ for all X, and by Theorem 14.1 `Chapter8/Jacobi · theorem_14_1` the null vector Y is a Jacobi field. It vanishes at $t_0 = 0$ and at t_1, which is 1 if $m = 0$ and a break point otherwise; since t_0 and t_1 are not conjugate, $Y = 0$ on $[0, 1]$, and outside by hypothesis. That contradicts $Y \neq 0$.

The splitting. *Inside the fields vanishing outside $[0, 1]$, the broken Jacobi fields and the fields vanishing at the break points are complementary subspaces* `Chapter10/BrokenSplitting · BrokenSetup.isCompl_brokenSpace_nodeZero`, Milnor's $T\Omega_\gamma = T\Omega_\gamma(t_0, \dots, t_k) \oplus T'$. They meet in 0 by uniqueness, and $X = X_b + (X - X_b)$, with X_b the broken Jacobi field matching X at the t_j.

Lemma 15.4 for the subdivision of §16, the index. *The index of E_{**} at γ equals the index of its restriction to the broken Jacobi fields* `Chapter10/BrokenSplitting · BrokenSetup.formIndex_brokenSpace`. The index of a bilinear form is the supremum, in $\{0, 1, \dots, \infty\}$, of the dimensions of the finite-dimensional subspaces on which it is negative definite `Chapter9/FormIndex · formIndex`. The index of E_{**} equals its index on the fields vanishing outside $[0, 1]$ `Chapter9/IndexReduction · indexUpTo_one` `Chapter9/IndexReduction · indexUpTo`. On that space Part VII's abstract Lemma 15.B applies `Chapter9/FormIndex · formIndex_eq_restrict_of_isCompl`: if the space is $A \oplus C$, with A and C perpendicular in both orders and the form positive definite on C, then the form and its restriction to A have the same index. Here A is the broken Jacobi fields and C the fields vanishing at the break points, perpendicularity is the previous step, and positivity on C is the definite form above, applicable because the elements of C vanish outside $[0, 1]$. The index is finally transported along the identification of A with the broken Jacobi fields themselves `Chapter9/FormIndex · formIndex_eq_of_equiv`.

Lemma 15.4 for the subdivision of §16, the nullity. *The null space of E_{**} restricted to the broken Jacobi fields has dimension the nullity of E_{**}* `Chapter10/BrokenSplitting · BrokenSetup.finrank_ker_brokenSpace` `Chapter9/JacobiSpace · hessNullity`. By Theorem 14.1 in its nullity form `Chapter9/JacobiSpace · theorem_14_1_nullity` `Chapter9/JacobiSpace · conjMultiplicity`, the nullity is the dimension of the space of Jacobi fields vanishing at 0 and 1 `Chapter9/JacobiSpace · conjSpace`. The map φ that cuts such a J off to $[0, 1]$ lands in $T\Omega_\gamma$ `Chapter9/IndexForm · indicator_mem_pathTangent`, is linear, and is injective, a Jacobi field vanishing on $[0, 1]$ being zero. Its range is exactly the null space of the restricted form:

- φJ is a broken Jacobi field (it is J on every piece), and $E_{**}(\varphi J, Y) = -2\sum_j \langle Y(t_j), P\Delta_{t_j}(\varphi J)' \rangle = 0$, since $\varphi J = J$ near each interior t_j `Chapter8/Energy · jump_congr` and J is differentiable there.

- Conversely let X be a broken Jacobi field in the null space. Test it against the broken Jacobi field Y with $Y(t_j) = P\Delta_{t_j}X'$, which is tangent because P is idempotent `Chapter8/Energy · prj_prj`: $0 = E_{**}(X,Y) = -2\sum_j \|P\Delta_{t_j}X'\|^2$. So every projected jump vanishes, and a broken Jacobi field without projected jumps agrees on $[0,1]$ with a single Jacobi field J `Chapter9/IndexReduction · eq_jacobi_of_jumps_zero`, which vanishes at 0 and 1 because X does. So $X = \varphi J$.

An injective linear map preserves dimension, which proves the claim. The file is Lemma 15.4 proved again, not applied: the fineness of §15 is replaced by the normal radius in Milnor's data and by the minimality of broken geodesics.

VIII.7 Theorem 16.2: the index and the nullity of E'

Throughout, $O \subseteq V^m$ is open with $N^m \cap O \subseteq U$, F is smooth on V^m with $F = E'$ on $N^m \cap O$, and $y \in N^m \cap O$ is such that $\gamma = \omega_y$ is an unbroken geodesic, that is (VIII.5), a critical point of E'. Let σ be the product chart at y restricted to O, x its parametrization, and z_0 the chart value of y, so that $x(z_0) = y$.

Four lemmas.

- *The second derivative along a line is the Hessian*: for G of class C^2 at z_0, the second derivative of $u \mapsto G(z_0 + uz)$ at 0 is $H_G(z_0)(z,z)$ `Chapter10/BrokenIndex · hasDerivAt_deriv_line`, where the Hessian is the derivative of the derivative `Chapter1/Nondegenerate · hessian`. Near 0 the first derivative is $u \mapsto dG_{z_0+uz}(z)$ (a function C^2 at a point is C^2, hence differentiable, nearby), and dG has derivative $H_G(z_0)$ at z_0; evaluate at z.
- *Polarization*: two symmetric biadditive forms with the same values on the diagonal are equal `Chapter10/BrokenIndex · eq_of_quadratic_eq`, by expanding both at $(x+y, x+y)$.
- The derivative of the parametrization of a slice chart is injective `Chapter10/BrokenIndex · injective_fderiv_sliceParam'`: near the chart value, the chart composed with the parametrization is $z \mapsto (z,0)$, so by the chain rule the derivative of the chart composed with dx is the inclusion of Z into $Z \times W$.
- *Jacobi fields along a segment are determined by their end values* `Chapter10/BrokenIndex · BrokenSetup.eqOn_of_isJacobiField_seg`: two Jacobi fields along the ray $s \mapsto \exp_{x_i}(sv_i)$ that agree at $s = 0$ and $s = 1$ agree on $[0,1]$. The ray is a smooth curve because $\|v_i\|$ is below a normal radius `Chapter9/JacobiVariation · IsNormalRadius.isSmoothCurveIn_ray`, and along it $J \mapsto (J(0), J(1))$ is a bijection, Remark 14.5 in its existence form `Chapter9/JacobiVariation · remark_14_5_bijective`; injectivity is what is used.

The isomorphism Λ. The derivative dx_{z_0} is a linear isomorphism of Z^m onto $\prod_j T_{y_j}N$ `Chapter10/BrokenIndex · chartTangentEquiv` `Chapter10/BrokenIndex · chartTangentEquiv_apply`: its range is the tangent space, which is the product (VIII.3), and it is injective. Composing with the inverse of the isomorphism of VIII.6 gives

$$\Lambda : Z^m \to T\Omega_\gamma(t_0, \dots, t_k), \qquad \Lambda z = \text{the broken Jacobi field with values } dx_{z_0}(z)_j \text{ at the } t_j$$

`Chapter10/BrokenIndex · BrokenSetup.chartBrokenEquiv` `Chapter10/BrokenIndex · BrokenSetup.chartBrokenEquiv_node`.

The second derivative of E' along a line is E_{**} on Λz. *With H the Hessian of $F \circ x$ at z_0, $H(z,z) = E_{**}(\Lambda z, \Lambda z)$ for every $z \in Z^m$* `Chapter10/BrokenIndex · BrokenSetup.hessian_apply_self`.

1. The line $\ell(u) = z_0 + uz$ stays in the chart's target for u in an open set $U_0 \ni 0$. The curve of break points $y(u) = x(\ell(u))$ is smooth, lies in $N^m \cap O \subseteq U$, has $y(0) = y$ and velocity $dx_{z_0}(z)$ at 0.
2. By VIII.5, $\alpha(u) = \omega_{y(u)}$, $|u| < \varepsilon$, is a variation of γ through broken geodesics, smooth on the strips.
3. Remark 13.5 `Chapter8/Hessian · hasDerivAt_deriv_energy`: the second derivative of $u \mapsto E(\alpha(u))$ at 0 is $E_{**}(W,W)$, with W the variation field.
4. The function $G = F \circ x$ is C^2 at z_0, so by the first lemma the second derivative of $u \mapsto G(z_0 + uz)$ at 0 is $H(z,z)$.

5. For $|u| < \varepsilon$, $G(z_0 + uz) = F(y(u)) = E'(y(u)) = E(\alpha(u))$ by formula $(*)$. The two functions agree near 0, so do their derivatives, and the second derivatives at 0 coincide: $H(z, z) = E_{**}(W, W)$.
6. It remains to show $W = \Lambda z$ on $[0, 1]$ `Chapter8/Hessian · secondVariation_congr`. Fix t in segment i.
   - At every division point the two fields agree: at t_0 and t_{m+1} both vanish, W because it is the variation field of a variation with fixed ends `Chapter8/Energy · IsVariation.varField_mem`; at t_j, $W(t_j) = y'(0)_j = dx_{z_0}(z)_j = \Lambda z(t_j)$ by VIII.5 and the definition of Λ.
   - On the i-th strip, Lemma 14.3 of Part VII `Chapter9/JacobiVariation · lemma_14_3`, applied to the strip variation $(u, s) \mapsto \omega_{y(u)}(t_i + s\ell_i)$ (smooth, with values in N, a geodesic in s), shows that $s \mapsto W(t_i + s\ell_i)$ is a Jacobi field along the rescaled segment, which is the ray $s \mapsto \exp_{x_i}(sv_i)$.
   - On segment i, Λz agrees with a Jacobi field J along γ, and $s \mapsto J(\ell_i s + t_i)$ is a Jacobi field along the same ray (affine change of parameter).
   - The two rescaled fields agree at $s = 0$ and $s = 1$, the division-point values; by the fourth lemma they agree on $[0, 1]$, and substituting $s = (t - t_i)/\ell_i$ gives $W(t) = \Lambda z(t)$.

This is Milnor's statement that the variation field of a variation through broken geodesics is clearly a broken Jacobi field, made precise: it needs Lemma 14.3 on each segment after rescaling, and uniqueness to identify the result with Λz.

Hence $H(z, z') = E_{**}(\Lambda z, \Lambda z')$ for all z, z' `Chapter10/BrokenIndex · BrokenSetup.hessian_eq_energyHessian`, by polarization: both sides are symmetric, the Hessian by the symmetry of second derivatives `Chapter1/Nondegenerate · hessian_symm` and E_{**} as before.

Theorem 16.2, index and nullity (V10.11). *Let O, F and y be as above, with ω_y an unbroken geodesic. Then the index of $E' = F|(N^m \cap O)$ at y* `Chapter1/Manifold · indexM` *equals the index of E_{**} at ω_y, and the nullity of E' at y* `Chapter1/Manifold · nullityM` *equals the nullity of E_{**} at ω_y.* `Chapter10/BrokenIndex · BrokenSetup.theorem_16_2_index`

Proof as formalized. The point y is critical (VIII.5).

- *Index.* The index of a critical point of a function on a manifold, defined in the preferred chart, equals the index of the Hessian of $F \circ x$ in the slice chart `Chapter1/ManifoldSubmanifold · index_hessian_comp_sliceParam` (chart independence at a critical point, Part I). On the finite-dimensional Z^m, the index of the continuous form H is its index in the sense of the supremum above `Chapter9/FormIndex · formIndex_eq_sigNeg`. Λ carries H to E_{**} restricted to the broken Jacobi fields, so the two indices agree `Chapter9/FormIndex · formIndex_eq_of_equiv`, and by Lemma 15.4 for §16 the latter is the index of E_{**}.
- *Nullity.* The nullity of E' at y is the nullity of H `Chapter1/ManifoldSubmanifold · nullity_hessian_comp_sliceParam`, the dimension of its null space `Chapter1/Nondegenerate · nullSpace`. Since Λ is onto, $H(z, \cdot) = 0$ if and only if $E_{**}(\Lambda z, \cdot) = 0$ on the broken Jacobi fields, so the null space of H is the preimage under Λ of the null space of the restricted form; an isomorphism preserves dimension; and that dimension is the nullity of E_{**} by Lemma 15.4 for §16.

What Milnor calls an immediate consequence of Lemma 15.4 thus takes Lemma 15.4 proved again (VIII.6), the identification of variation fields with broken Jacobi fields, and polarization.

VIII.8 Theorem 16.3 in the finite-dimensional model

Milnor's proof of Theorem 16.3 is one sentence (p. 90): it follows from 16.2 and 3.5. But Theorem 3.5, as proved in Part II `Chapter2/CWType · theorem_3_5'`, concerns a *compact* submanifold, a function that is the restriction of a smooth ambient function, and *all* critical points non-degenerate; here U is not compact, E' lives only on U, and only the critical points in B^a are assumed non-degenerate. The exposition reparametrizes E' on $\{E' < a'\}$, for a level $a' > a$ below the next critical value, by an increasing diffeomorphism onto $\mathbb{R}$. The formal proof instead extends E' smoothly from a compact sublevel set to

all of V^m, and proves Theorem 3.5 for a compact sublevel set of a function on a manifold that need not be compact.

Smooth extension from a compact set. *Let $K \subseteq S'$ be subsets of a finite-dimensional normed space E, with K compact, and suppose that near every point of K the function f agrees on S' with a function smooth on an open set. Then some smooth $F : E \to \mathbb{R}$ agrees with f on K* `Chapter10/SublevelCW · exists_contDiff_eq_of_locally`. Choose, for each $x \in K$, an open $O_x \ni x$ and G_x smooth on O_x with $f = G_x$ on $S' \cap O_x$; finitely many O_x cover K; a smooth partition of unity (φ_i) subordinate to this finite cover on the closed set K exists `Mathlib/Geometry/Manifold/PartitionOfUnity · SmoothPartitionOfUnity.exists_isSubordinate`. Put $F = \sum_i \varphi_i G_i$, each term smooth on all of E because the support of φ_i lies in the open set where G_i is smooth `Chapter2/Regular · contDiff_smul_of_tsupport`. For $y \in K$, $\sum_i \varphi_i(y) = 1$, and $\varphi_i(y) \neq 0$ forces $y \in O_i$, where $G_i(y) = f(y)$; hence $F(y) = f(y)$.

Passing one critical level, relative to the sublevel set below. *Let $N \subseteq V$ be a slice submanifold, F smooth on V, $f = F|N$, with $N^{a_0} = f^{-1}(-\infty, a_0]$ compact. Let S be a finite set of non-degenerate critical points of f, all at level c, and $a' < c < a \leq a_0$ such that every other critical point lies below a' or above a. If $N^{a'}$ has the homotopy type of a finite cell complex with cell dimensions cells, then N^a has that of one with cell dimensions cells followed by the indices of the points of S, in some order* `Chapter10/SublevelCW · IsFiniteCellComplexType.sublevel_step_of_isCompact`. Here *having the homotopy type of a finite cell complex* means being homotopy equivalent to a space obtained from the empty space by attaching cells of the listed dimensions one at a time along arbitrary continuous maps `Chapter2/CWType · IsFiniteCellComplexType` (§0.5), and the sublevel set is Milnor's M^a `Chapter2/Crossing · sublevel` (§0.4).

1. Every slab $f^{-1}[b, b']$ with $b' \leq a_0$ is compact, as a closed subset of N^{a_0}.
2. Let $\varepsilon_0 = \min(c - a', a - c)$. The slab $f^{-1}[c - \varepsilon_0, c + \varepsilon_0]$ is compact and all its critical points lie in S. Remark 3.3 for a submanifold `Chapter2/SeveralCritical · remark_3_3_submanifold` gives $\varepsilon_1 > 0$ and, for $\varepsilon = \min(\varepsilon_1, \varepsilon_0/2)$, pairwise disjoint cells e_p, $p \in S$, of dimension the index of p, each meeting $N^{c-\varepsilon}$ exactly in its boundary sphere and lying in $N^{c+\varepsilon}$, such that $N^{c-\varepsilon} \cup \bigcup_p e_p$ is a strong deformation retract `Chapter2/Crossing · IsStrongDeformationRetract` of $N^{c+\varepsilon}$.
3. The slabs $[a', c - \varepsilon]$ and $[c + \varepsilon, a]$ are compact and free of critical points, so Theorem 3.1 for a submanifold `Chapter2/Submanifold · theorem_3_1_submanifold` gives homotopy equivalences $N^{a'} \simeq N^{c-\varepsilon}$ and $N^{c+\varepsilon} \simeq N^a$.
4. Attaching the disjoint cells one at a time to the closed set $N^{c-\varepsilon}$ of a Hausdorff space `Chapter2/CWType · isFiniteCellComplexType_union_cells` appends their dimensions to the cell list; the inclusion of a strong deformation retract is a homotopy equivalence `Chapter2/Crossing · IsStrongDeformationRetract.exists_homotopyEquiv`; and the property is invariant under homotopy equivalence `Chapter2/CWType · IsFiniteCellComplexType.of_homotopyEquiv`.

The step is relative: up to homotopy, N^a is $N^{a'}$ with the cells of the level c attached, whatever happens above a.

Theorem 3.5 for a compact sublevel set. *Let N, F, f be as above, with N^{a_0} compact, every critical point of f in N^{a_0} non-degenerate, and a_0 not a critical value. Then the critical points of f below a_0 form a list $p_1, \dots, p_j$ without repetitions, with $f(p_1) \leq \dots \leq f(p_j)$, and N^{a_0} has the homotopy type of the space obtained from the empty space by attaching cells of dimensions* $\mathrm{index}(p_1), \dots, \mathrm{index}(p_j)$*, one at a time* `Chapter10/SublevelCW · isFiniteCellComplexType_sublevel_of_isCompact`.

Proof as formalized. The critical points in N^{a_0} form a compact set (the critical set is closed `Chapter1/ManifoldMorse · isClosed_setOf_isCriticalPointM` and N^{a_0} is compact) which is discrete, non-degenerate critical points being isolated `Chapter1/ManifoldMorse · isDiscrete_inter_setOf_isCriticalPointM`; so it is finite. The statement is then proved for every level $a \leq a_0$ that is not a critical value, by strong induction on the number of critical points below a.

- If there is one, let c be the largest critical value below a and S the critical points at level c. A level $a' < c$ above all smaller critical values is chosen (the midpoint between c and the next critical value

below it, or $c-1$ if there is none); it is not a critical value, every critical point outside S lies below a' or above a, and there are fewer critical points below a' than below a. The induction hypothesis at a' and the step above at the level c give the claim at a; the list is the one for a' followed by the points of S, which keeps it sorted.
- If there is none, N^a is empty: otherwise the compact N^a has a minimum point q_0 of f, which is a global minimum (outside N^a, $f > a \geq f(q_0)$), hence a critical point `Chapter2/CWType · isCriticalPointM_of_forall_le`, and below a because a is not a critical value. The empty space has the empty cell list `Chapter2/CWType · IsFiniteCellComplexType.of_isEmpty`.

This is the induction of Part II's Theorem 3.5 for regular sublevel sets `Chapter2/CWType · isFiniteCellComplexType_sublevel'`, without compactness of N and with non-degeneracy required only in N^{a_0}.

The pieces for E'. *For every b, $\{y \in U : E'(y) < b\} = N^m \cap O$ for an open O* `Chapter10/Finite · BrokenSetup.exists_isOpen_sublevel_lt`, *from $U = N^m \cap O_0$ and the continuity of E' on U. For $b < c$ there is a smooth F on V^m with $F = E'$ on B^b* `Chapter10/Finite · BrokenSetup.exists_contDiff_eq_energy'`, by the extension lemma with $K = B^b$ (compact, VIII.4), $S' = U$, and the local smooth representatives of VIII.3. And the index of a broken geodesic is recorded as a natural number `Chapter10/Finite · BrokenSetup.geodIndex`: the index of E_{**} at ω_y, which is finite at a geodesic by Theorem 16.2, and 0 for $y \notin U$.

Theorem 16.3 in the model (V10.12). *Given Milnor's data and $a < c$, suppose that $E'(y) \neq a$ for every $y \in U$ with ω_y a geodesic, and that E_{**} has nullity 0 at every geodesic ω_y, $y \in U$, with $E'(y) \leq a$. Then there is a list of break points $y^{(1)}, \dots, y^{(j)}$ without repetitions whose members are exactly the $y \in U$ with ω_y an unbroken geodesic and $E'(y) \leq a$; each carries as recorded index the index of E_{**} at ω_y; the list is sorted by energy; and $B^a = \{y \in U : E'(y) \leq a\}$ has the homotopy type of a finite cell complex with one cell of dimension* $\mathrm{index}(\omega_{y^{(i)}})$ *for each i.* `Chapter10/Finite · BrokenSetup.theorem_16_3`

Proof as formalized.

1. Let $c_1 = (a+c)/2$. Take F smooth on V^m with $F = E'$ on B^{c_1}, and the open O with $\{y \in U : E' < c_1\} = N^m \cap O$. The manifold $S'' = N^m \cap O$ is a slice submanifold, contained in U, on which $F = E'$.
2. The sublevel set of $F|S''$ at a is B^a, since $a < c_1$; it is compact by VIII.4.
3. The critical points of $F|S''$ are exactly the y with ω_y an unbroken geodesic (VIII.5). At those with $F(y) \leq a$ the nullity is that of E_{**} (VIII.7), which is 0, so they are non-degenerate `Chapter1/Nondegenerate · isNondegenerate_iff_nullity_eq_zero`. The level a is not a critical value, by hypothesis.
4. Theorem 3.5 for a compact sublevel set applies with $a_0 = a$, and the index of each critical point is the index of E_{**} at the geodesic (VIII.7).
5. The cell structure is transported along the evident homeomorphism between $B^a \subseteq V^m$ and the sublevel set of $F|S''$ at a in S''.

Counting, and Milnor's hypothesis. The same statement in counted form `Chapter10/Finite · BrokenSetup.theorem_16_3_count`: the set of break points of geodesics with $E' \leq a$ is finite, and the number of cells of each dimension λ is the number of these geodesics of index λ. With Milnor's hypothesis `Chapter10/Finite · BrokenSetup.theorem_16_3_nonconj`: if 0 and 1 are not conjugate along any geodesic ω_y with $E'(y) \leq a$, the nullity there is 0, since by Theorem 14.1 the nullity is the multiplicity of conjugacy `Chapter9/JacobiSpace · theorem_14_1_nullity` and a positive multiplicity means conjugate `Chapter9/JacobiSpace · isConjugate_iff`.

The hypothesis on a. Milnor's non-conjugacy hypothesis does not exclude a geodesic of energy exactly a; then a is a critical value of E' and Theorem 3.5 does not describe B^a directly. The formal statements assume that a is not the energy of a geodesic, which is narrower than the book; the critical level is not formalized (V10.12).

VIII.9 Milnor's Remark on p. 92: a minimum of E'

For every $\omega \in \Omega$ with $E(\omega) < c$, the break points of ω lie in U and $E'(\omega(t_1), \dots, \omega(t_m)) \le E(\omega)$ `Chapter10/Finite · BrokenSetup.nodeVal_mem_nodes_of_energy_lt` (V10.14): the inequality is the minimality of VIII.6, and the distances are finite because each is at most the length of a piece of ω.

The Remark (V10.14). *If some $\gamma_0 \in \Omega$ has $E(\gamma_0) < c$, there is $y \in U$ such that ω_y is an unbroken geodesic and $E(\omega_y) \le E(\gamma)$ for every $\gamma \in \Omega$.* `Chapter10/Finite · BrokenSetup.exists_energy_minimizing`

Proof as formalized. Let $a = E(\gamma_0)$. The set B^a is compact (VIII.4), non-empty because it contains the break points of γ_0, and E' is continuous on it (VIII.3), so E' attains its minimum on B^a at some y `Mathlib/Topology/Order/Compact · IsCompact.exists_isMinOn`. This minimum is global over Ω. For $\gamma \in \Omega$ with $E(\gamma) < c$: if E' of its break points is $\le a$, then $E'(y) \le E'(\gamma(t_1), \dots) \le E(\gamma)$; otherwise $E'(y) \le a < E'(\gamma(t_1), \dots) \le E(\gamma)$. If $E(\gamma) \ge c$, then $E'(y) \le a < c \le E(\gamma)$. Since $E(\omega_y) = E'(y)$, ω_y minimizes E on Ω. To see that it is a geodesic, use Corollary 12.3: it suffices that ω_y is a critical path `Chapter8/Critical · IsCriticalPath`, that is, that along every variation α of ω_y the derivative of $u \mapsto E(\alpha(u))$ at 0, which exists `Chapter8/FirstVariation · IsVariation.hasDerivAt_energy`, vanishes. The paths $\alpha(u)$ lie in Ω, so $u \mapsto E(\alpha(u))$ has a local minimum at 0, and its derivative there is 0.

Milnor calls the minimum the required minimal geodesic, a second proof of the existence part of Hopf–Rinow. The formal statement proves minimality of the energy on Ω; that ω_y is also shortest needs a reparametrization by arc length and is not stated. Nothing beyond Milnor's data and one path of energy $< c$ is assumed.

VIII.10 Milnor's metric on the path space

So far Ω has been a set. Milnor topologizes it (p. 88) by the distance

$$d(\omega, \omega') = \max_{0 \le t \le 1} \rho(\omega(t), \omega'(t)) + \Big[\int_0^1 \Big(\frac{ds}{dt} - \frac{ds'}{dt} \Big)^2 dt \Big]^{1/2},$$

s, s' the arc lengths, and adds the second term so that the energy becomes continuous.

Two integral inequalities. *Cauchy–Schwarz for interval integrals*: $(\int_a^b fg)^2 \le \int_a^b f^2 \int_a^b g^2$ `Chapter10/PathMetric · integral_mul_sq_le`. For every real x, $0 \le \int_a^b (f - xg)^2$, a quadratic in x whose discriminant must therefore be ≤ 0 `Mathlib/Algebra/QuadraticDiscriminant · discrim_le_zero`. *Minkowski's inequality*: $\big(\int (f+g)^2\big)^{1/2} \le \big(\int f^2\big)^{1/2} + \big(\int g^2\big)^{1/2}$ `Chapter10/PathMetric · sqrt_integral_add_sq_le`, by expanding the square and bounding the cross term with Cauchy–Schwarz.

Velocities of piecewise smooth paths. On each open piece the velocity of a piecewise smooth path agrees with a function continuous on the closed piece, its one-sided derivative within the piece `Chapter10/PathMetric · IsPiecewiseSmooth.exists_deriv_piece` `Chapter8/Energy · IsPiecewiseSmooth.derivWithin_piece`. Hence a continuous function of the velocities of two, or three, piecewise smooth paths is integrable on every subinterval `Chapter10/PathMetric · intervalIntegrable_comp_deriv` `Chapter10/PathMetric · intervalIntegrable_comp_deriv₃`: take the union of the break sets and integrate piece by piece `Chapter8/Piecewise · intervalIntegrable_of_piecewise`. And *the speed of a piecewise smooth path on $[a, b]$ is bounded* `Chapter10/PathMetric · IsPiecewiseSmooth.exists_bound_deriv`: on each piece the one-sided derivative is continuous on a compact interval, the bounds of adjacent pieces combine by taking the maximum `Chapter8/Piecewise · piecewise_induction`, and at a point where the path is not differentiable the formal derivative is 0.

The space. Ω becomes a type of its own, its points the maps $\mathbb{R} \to V$ in Ω `Chapter10/PathMetric · PathSp` `Chapter10/PathMetric · PathSp.mem`, with speed $\dot{s}(t) = \|\omega'(t)\|$ `Chapter10/PathMetric · PathSp.speed`, so that $E_a^b(\omega) = \int_a^b \dot{s}^2$ `Chapter10/PathMetric · PathSp.energy_eq_integral_speed`. The uniform part of d

is $\sup_{t\in[0,1]}\rho(\omega(t),\omega'(t))$, taken in $[0,\infty]$ Chapter10/PathMetric · PathSp.supDist, and the L^2 part is $\big(\int_0^1(\dot s-\dot s')^2\big)^{1/2}$ Chapter10/PathMetric · PathSp.l2Dist. The L^2 part is nonnegative, symmetric and zero on the diagonal Chapter10/PathMetric · PathSp.l2Dist_nonneg Chapter10/PathMetric · PathSp.l2Dist_comm Chapter10/PathMetric · PathSp.l2Dist_self; continuous functions of two speeds are integrable on subintervals of $[0,1]$ Chapter10/PathMetric · PathSp.intervalIntegrable_speed; and the triangle inequality is Minkowski's inequality applied to $(\dot s-\dot s')+(\dot s'-\dot s'')$, the integrability of the cross term coming from the three-path lemma Chapter10/PathMetric · PathSp.l2Dist_triangle. The uniform part vanishes on the diagonal, the paths lying in N on $[0,1]$ Chapter10/PathMetric · PathSp.supDist_self, is symmetric Chapter10/PathMetric · PathSp.supDist_comm, satisfies the triangle inequality pointwise and hence for the supremum Chapter10/PathMetric · PathSp.supDist_triangle, and dominates each $\rho(\omega(t),\omega'(t))$ Chapter10/PathMetric · PathSp.riemDist_le_supDist.

Milnor's metric (V10.1). *$d = $ uniform part $+$ L^2 part is an extended pseudometric on Ω* Chapter10/PathMetric · PathSp.instPseudoEMetricSpace Chapter10/PathMetric · PathSp.edist_def Mathlib/Topology/EMetricSpace/Defs · PseudoEMetricSpace, with both parts bounded by d Chapter10/PathMetric · PathSp.ofReal_l2Dist_le_edist Chapter10/PathMetric · PathSp.supDist_le_edist. From now on Ω and its subsets carry this topology (§0.8). It is a *pseudo*metric: maps that agree on $[0,1]$ are at distance 0 Chapter10/Omega · PathSp.edist_eq_zero_of_eqOn, because both parts see only $[0,1]$ Chapter10/Deformation · PathSp.supDist_congr Chapter10/Deformation · PathSp.l2Dist_congr. On paths $[0,1]\to N$ it is Milnor's metric. Formally its values lie in $[0,\infty]$, because ρ does.

Lemma 16.A of the exposition: the energy is continuous (V10.2 (i)). *For $\omega,\omega'\in\Omega$ and $0\le a\le b\le 1$,*

$$\big|\sqrt{E_a^b(\omega)}-\sqrt{E_a^b(\omega')}\big|\le\Big(\int_0^1(\dot s-\dot s')^2\Big)^{1/2}.$$

Chapter10/PathMetric · PathSp.abs_sqrt_energy_sub_le. Write $\dot s=(\dot s-\dot s')+\dot s'$; Minkowski on $[a,b]$ gives $\sqrt{E_a^b(\omega)}\le\|\dot s-\dot s'\|_{L^2[a,b]}+\sqrt{E_a^b(\omega')}$, the first term is at most the L^2 part of d because the integral of a nonnegative function grows with the interval, and the roles of ω, ω' can be exchanged. Consequences: $\sqrt{E_a^b}$ is 1-Lipschitz for d Chapter10/PathMetric · PathSp.lipschitzWith_sqrt_energy; $E_a^b\ge 0$ Chapter10/PathMetric · PathSp.energy_nonneg_of_le, so E_a^b, the square of a continuous function, is continuous Chapter10/PathMetric · PathSp.continuous_energy; Ω^c is closed Chapter10/PathMetric · PathSp.isClosed_sublevel and $\operatorname{Int}\Omega^c=E^{-1}[0,c)$ is open Chapter10/PathMetric · PathSp.isOpen_sublevel_lt. This is what Milnor's L^2 term is for: with the uniform term alone, a small wiggle of high frequency added to a path converges uniformly while its energy blows up.

Lemma 16.B of the exposition: no energy in short intervals (V10.2 (ii)). *If $\omega_n\to\omega$ in Ω, then for every $\eta>0$ there is $\delta>0$ such that $E_a^b(\omega_n)<\eta$ for all n and all $0\le a\le b\le 1$ with $b-a<\delta$.* Chapter10/PathMetric · PathSp.exists_energy_lt_of_tendsto

Proof as formalized. Choose n_0 with L^2 distance from ω_n to ω less than $\sqrt{\eta}/2$ for $n\ge n_0$. The speeds of ω and of $\omega_0,\dots,\omega_{n_0-1}$ are bounded; let K be the sum of the bounds, so each of these paths has $E_a^b\le K^2(b-a)$ Chapter10/PathMetric · PathSp.energy_le_of_bound. Take $\delta=\eta/(4(K^2+1))$, so that $K^2(b-a)<\eta/4$. For $n<n_0$, $E_a^b(\omega_n)<\eta/4$. For $n\ge n_0$, $\sqrt{E_a^b(\omega)}<\sqrt{\eta}/2$, and Lemma 16.A gives $\sqrt{E_a^b(\omega_n)}<\sqrt{\eta}/2+\sqrt{\eta}/2$. Milnor does not state this lemma; the continuity of r_u needs it.

VIII.11 Milnor's deformation r_u

Two lemmas on the energy: on $[a,b]$ it depends only on the path on $[a,b]$, the derivatives agreeing on the open interval Chapter10/Retraction · energy_congr_Icc; and *Schwarz's inequality with the distance*, $\rho(\omega(a),\omega(b))^2\le(b-a)E_a^b(\omega)$ Chapter10/Retraction · riemDist_sq_le_energy, from $\rho\le L$ and $L^2\le(b-a)E$.

The short geodesic over an interval. For x,x' and $a<b$, $\sigma(\tau)=\exp_x\big(\frac{\tau-a}{b-a}\,v(x,x')\big)$ Chapter10/Retraction · BrokenSetup.sgeoOver runs from x at $\tau=a$ Chapter10/Retraction · BrokenSetup.sgeoOver_left to x'

at $\tau = b$ when $x, x' \in S$ and $\rho(x,x') < \delta$ Chapter10/Retraction · BrokenSetup.sgeoOver_right. It is then a geodesic on a neighbourhood of $[a,b]$ with speed $\|v(x,x')\|/(b-a)$ Chapter10/Retraction · BrokenSetup.sgeoOver_isGeodesicOn, proved like the segments of VIII.2, and its energy over $[a,b]$ is $\rho(x,x')^2/(b-a)$ Chapter10/Retraction · BrokenSetup.energy_sgeoOver.

Definition. For a segment i, a number u and a path ω, the path $r_u^{(i)}(\omega)$ is Chapter10/Retraction · BrokenSetup.retrSeg

$$r_u^{(i)}(\omega)(\tau) = \begin{cases} \omega_y(\tau), & \tau \le t_i, \\ \exp_{\omega(t_i)}\left(\frac{\tau - t_i}{u - t_i}\, v(\omega(t_i), \omega(u))\right), & t_i < \tau \le u, \\ \omega(\tau), & \tau > u, \end{cases}$$

with $y = (\omega(t_1), \dots, \omega(t_m))$ the break points of ω; and Milnor's $r_u(\omega)$ is $r_u^{(i)}(\omega)$ for the segment i of u Chapter10/Retraction · BrokenSetup.retr. On $[0, t_i]$ the formal definition uses the broken geodesic of *all* break points of ω, which on $[0, t_i]$ is Milnor's $r(\omega)$.

Standing hypotheses: $\omega \in \Omega$, $E(\omega) < c$, and $u \in [t_i, t_{i+1}]$, which lies in $[0,1]$ Chapter10/Retraction · BrokenSetup.u_mem. Then:

- *Two points of ω in one segment are at distance $< \delta$* Chapter10/Retraction · BrokenSetup.riemDist_lt_of_seg: $\rho^2 \le (b-a)E_a^b(\omega) \le \ell_i\, c < \delta^2$. So the short geodesic in the middle formula exists: Milnor's $\rho(\omega(t_{i-1}), \omega(u))^2 \le (u - t_{i-1})E < \varepsilon^2$.
- The break points of ω lie in U Chapter10/Retraction · BrokenSetup.nodeVal_mem (VIII.9), and the broken geodesic of the break points passes through the vertices Chapter10/Retraction · BrokenSetup.path_t, which are the points $\omega(t_k)$ Chapter10/Retraction · BrokenSetup.path_nodeVal_t; the energy of its k-th segment is $\rho(x_k, x_{k+1})^2/\ell_k$ Chapter10/Retraction · BrokenSetup.energy_path_seg.
- *The three pieces.* $r_u^{(i)}(\omega)$ is the broken geodesic of the break points on $(-\infty, t_i]$ Chapter10/Retraction · BrokenSetup.retrSeg_eqOn_left; the short geodesic from $\omega(t_i)$ to $\omega(u)$ over $[t_i, u]$ on the *closed* interval $[t_i, u]$ Chapter10/Retraction · BrokenSetup.retrSeg_eqOn_mid, because at $\tau = t_i$ the first formula gives $\omega(t_i)$, the start of the short geodesic; and ω on $[u, \infty)$ Chapter10/Retraction · BrokenSetup.retrSeg_eqOn_right, because at $\tau = u$ the middle formula gives $\omega(u)$ (and, when $u = t_i$, the first formula gives $\omega(t_i)$).
- $r_u^{(i)}(\omega) \in \Omega$ Chapter10/Retraction · BrokenSetup.retrSeg_mem_pathSpace: it is the concatenation Chapter7/Path · IsPiecewiseSmooth.concat of three piecewise smooth pieces that match at t_i and at u; the middle piece is a smooth geodesic Chapter7/Path · IsPiecewiseSmooth.of_contDiffOn, or a single point when $u = t_i$; all values lie in N; it starts at $\omega_y(0) = p$ and ends at $\omega(1) = q$ (at the short geodesic's end $\omega(1)$ when $u = 1$).
- *The energy does not increase* (Milnor p. 91). On $[t_i, u]$, $E(r_u^{(i)}(\omega)) = \rho(\omega(t_i), \omega(u))^2/(u - t_i) \le E_{t_i}^u(\omega)$ by Schwarz, both sides being 0 when $u = t_i$ Chapter10/Retraction · BrokenSetup.energy_retrSeg_mid_le. Overall Chapter10/Retraction · BrokenSetup.energy_retrSeg_le, split both energies over the segments: for $k < i$ the path is the broken geodesic there, with energy $\rho(\omega(t_k), \omega(t_{k+1}))^2/\ell_k \le E_{t_k}^{t_{k+1}}(\omega)$; for $k = i$ split once more at u, the middle part decreasing and the rest equal; for $k > i$ the path is ω.

The formulas of neighbouring segments agree. On a segment $[t_k, t_{k+1}]$ the broken geodesic of the break points of ω is the short geodesic from $\omega(t_k)$ to $\omega(t_{k+1})$ over $[t_k, t_{k+1}]$ Chapter10/Retraction · BrokenSetup.path_nodeVal_eq_sgeoOver. *Hence for every $u \in [t_i, t_{i+1}]$, $r_u^{(i)}(\omega) = r_u(\omega)$* Chapter10/Retraction · BrokenSetup.retrSeg_eq_retr. For $u < t_{i+1}$ the segment of u is i, and for $u = t_{i+1} = 1$ it is the last one, which is i. For an interior $u = t_{i+1}$ the segment of u is $i+1$, and the formula of segment $i+1$ with $u = t_{i+1}$ is the broken geodesic on $(-\infty, t_{i+1}]$ and ω beyond; the formula of segment i is the broken geodesic on $(-\infty, t_i]$, the short geodesic over $[t_i, t_{i+1}]$, and ω beyond, and the short geodesic is the broken geodesic on that segment. This is the agreement at $u = t_i$ that the exposition notes Milnor leaves implicit; it is what makes r_u well defined as a function of u, and what lets the continuity, proved segment by segment, glue (VIII.13).

r_u **for** $u \in [0,1]$ (V10.7). $r_u(\omega) \in \Omega$ `Chapter10/Retraction · BrokenSetup.retr_mem_pathSpace` and $E(r_u(\omega)) \le E(\omega)$ `Chapter10/Retraction · BrokenSetup.energy_retr_le`, from the segment of u. r_0 *is the identity on* $[0,1]$ `Chapter10/Retraction · BrokenSetup.retr_zero`: the segment of 0 is the first, $t_0 = 0$, and $r_0(\omega)$ is $\omega_y(0) = \omega(0)$ at $\tau = 0$ and $\omega(\tau)$ for $\tau > 0$. r_1 *is Milnor's retraction* r: $r_1(\omega)$ is the broken geodesic of the break points of ω on $[0,1]$ `Chapter10/Retraction · BrokenSetup.retr_one`, since the segment of 1 is the last, its middle formula is the short geodesic over $[t_m, 1]$, and that is the broken geodesic there.

r_u **fixes** B. *For* $\gamma \in B$ *and* $u \in [0,1]$, $r_u(\gamma) = \gamma$ *on* $[0,1]$ `Chapter10/Retraction · BrokenSetup.retr_of_mem_brokenPaths`. For $\tau \le t_i$ this is Lemma 16.1 (b). For $t_i < \tau \le u$, take w from Step 3 of VIII.2 on segment i and $s = (u - t_i)/\ell_i \in (0,1]$. Then $s\,w$ is tangent, $\|s\,w\| \le \|w\| < \delta$, and $\exp_{\gamma(t_i)}(s\,w) = \gamma(u)$; by uniqueness $v(\gamma(t_i), \gamma(u)) = s\,w$, and

$$\exp_{\gamma(t_i)}\Big(\frac{\tau - t_i}{u - t_i}\, s\, w\Big) = \exp_{\gamma(t_i)}\Big(\frac{\tau - t_i}{\ell_i}\, w\Big) = \gamma(\tau).$$

For $\tau > u$ there is nothing to prove. Milnor's tacit reason is that a sub-arc of a short geodesic is the short geodesic between its own ends, which is the uniqueness in Corollary 10.8.

VIII.12 The continuity of r_u in both variables

Milnor says that $r_u(\omega)$ is continuous in both variables, and that "it is easily verified" (p. 91). The formal proof fixes a segment i and, along an arbitrary filter, numbers $u_j \to u$ in $[t_i, t_{i+1}]$ and paths $\omega_j \to \omega$ in Ω, all of energy $< c$, and shows that $d(r^{(i)}_{u_j}(\omega_j), r^{(i)}_u(\omega)) \to 0$, the uniform part and the L^2 part separately.

Ingredients.

- *The domain of* exp *is star-shaped along* $[0,1]$ `Chapter10/RetractionCont · mem_expDom_smul`: if $\exp_x(v)$ is defined, so is $\exp_x(\theta v)$ for $\theta \in [0,1]$ `Chapter7/Exp · IsGeodesicFrom.exp_smul`.
- *Convergence for* ρ *implies convergence in* V `Chapter10/RetractionCont · tendsto_of_tendsto_riemDist`, since $\|x - y\| \le \rho(x,y)$; the converse holds within N `Chapter7/Metric · tendsto_riemDist_of_tendsto`.
- *The short geodesics depend continuously on their end points.* On the set of *joinable* pairs, pairs of points of S at distance $< \delta$ `Chapter10/RetractionCont · BrokenSetup.joinable`, the join vector is continuous `Chapter10/RetractionCont · BrokenSetup.continuousWithinAt_joinVec`, because near a joinable pair it agrees with the smooth map G of Corollary 10.8 (uniqueness again). And $(x, x', \theta) \mapsto \exp_x(\theta\, v(x,x'))$ is continuous on joinable pairs times $[0,1]$ `Chapter10/RetractionCont · BrokenSetup.continuousWithinAt_shortGeo`, because exp is continuous on its domain, which is open in TN `Chapter7/GeodesicFlow · eventually_mem_expDom_and_continuousWithinAt_exp`, and $(x, \theta v)$ lies in the domain with θv tangent.
- *Paths of bounded energy are equicontinuous*: $\rho(\omega(s), \omega(\tau)) \le \sqrt{|\tau - s|\,E(\omega)}$ `Chapter10/RetractionCont · PathSp.riemDist_le_sqrt`, from $\rho \le L$ and $L^2 \le |\tau - s| E^\tau_s \le |\tau - s| E$. Hence $\omega_j(\sigma_j) \to \omega(\sigma)$ *for* ρ *when* $\omega_j \to \omega$ *and* $\sigma_j \to \sigma$ *in* $[0,1]$ `Chapter10/RetractionCont · PathSp.tendsto_riemDist_apply`: $\rho(\omega_j(\sigma_j), \omega(\sigma)) \le d(\omega_j, \omega) + \sqrt{|\sigma - \sigma_j| E(\omega)}$.
- *Lemma 16.B along a filter* `Chapter10/RetractionCont · PathSp.eventually_energy_lt`: if $\omega_j \to \omega$, there is $\delta > 0$ such that eventually $E^b_a(\omega_j) < \eta$ whenever $b - a < \delta$; the proof of VIII.10 with a bound for the speed of ω alone.
- L^2 *bookkeeping.* Speeds agree inside an interval where the paths agree `Chapter10/RetractionCont · PathSp.speed_congr`; the L^2 distance of the speeds over a subinterval is at most the full one `Chapter10/RetractionCont · PathSp.integral_speed_sub_sq_le`; and $\int_a^b (\dot s - \dot s')^2 \le 2E^b_a(\omega) + 2E^b_a(\omega')$ `Chapter10/RetractionCont · PathSp.integral_speed_sub_sq_le_energy`, from $(x-y)^2 \le 2x^2 + 2y^2$.
- *The speeds of broken geodesics depend continuously on the break points.* For $y, y' \in U$, $\int_0^1 (\dot s_{y'} - \dot s_y)^2 = \sum_k (\rho'_k - \rho_k)^2/\ell_k$ `Chapter10/RetractionCont · BrokenSetup.integral_speed_path_sub_sq`, the speeds being

the constants ρ_k/ℓ_k on the segments; so if $y_j \to y$ in U the integral tends to 0 Chapter10/RetractionCont · BrokenSetup.tendsto_integral_speed_path, the gaps being continuous and finite at y.

- *Uniform convergence from pointwise convergence* Chapter10/RetractionConv · PathSp.eventually_supDist_le_of_tendsto_apply: if $\alpha_j(s) \to \alpha(s)$ for every $s \in [0,1]$ and all energies are $\le C$, then eventually $\sup_s \rho(\alpha_j(s), \alpha(s)) \le e$. With $h = (e/3)^2/(C+1)$ and a finite h-net of $[0,1]$ Mathlib/Topology/MetricSpace/Pseudo/Basic · finite_cover_balls_of_compact, eventually $\rho(\alpha_j(g), \alpha(g)) \le e/3$ on the net, equicontinuity gives $\rho(\beta(s), \beta(\tau)) \le e/3$ for $|s-\tau| < h$ and every β of energy $\le C$, and three thirds add up. Paths of energy $< c$ are uniformly $\frac{1}{2}$-Hölder for ρ; that makes the uniform part easy.
- *Short geodesics with converging data converge* Chapter10/RetractionConv · BrokenSetup.tendsto_sgeoOver: if $x_j \to x$, $x'_j \to x'$ (joinable pairs), $b_j \to b > a$ and $\tau_j \to \tau$, $\tau_j \in [a, b_j]$, the short geodesic from x_j to x'_j over $[a, b_j]$ at τ_j tends to the limit one at τ, since $(\tau_j - a)/(b_j - a) \to (\tau - a)/(b - a) \in [0,1]$. Points of one segment of a path of energy $< c$ are joinable Chapter10/RetractionConv · BrokenSetup.mem_joinable.
- The speed of $r_u^{(i)}(\omega)$ on (t_i, u) is $\|v(\omega(t_i), \omega(u))\|/(u - t_i)$ Chapter10/RetractionConv · BrokenSetup.speed_retrSeg_mid, and its energy on $[s,t] \subseteq [t_i, u]$ is that speed squared times $t - s$ Chapter10/RetractionConv · BrokenSetup.energy_retrSeg_mid. If $\omega_j \to \omega$, the L^2 distance tends to 0 Chapter10/RetractionConv · BrokenSetup.tendsto_l2Dist, $\omega_j(\sigma_j) \to \omega(\sigma)$ in V Chapter10/RetractionConv · BrokenSetup.tendsto_apply, and the break points converge Chapter10/RetractionConv · BrokenSetup.tendsto_nodeVal.

Pointwise convergence. *For every $s \in [0,1]$, $r_{u_j}^{(i)}(\omega_j)(s) \to r_u^{(i)}(\omega)(s)$ in V* Chapter10/RetractionConv · BrokenSetup.tendsto_retrSeg_apply. Four cases.

1. $s \le t_i$: on the segment k of s all the paths are short geodesics over the fixed interval $[t_k, t_{k+1}]$ between the converging points $\omega_j(t_k) \to \omega(t_k)$ and $\omega_j(t_{k+1}) \to \omega(t_{k+1})$.
2. $t_i < s < u$: eventually $s < u_j$, and all are middle formulas, short geodesics with ends $\omega_j(t_i) \to \omega(t_i)$ and $\omega_j(u_j) \to \omega(u)$ over $[t_i, u_j]$, $u_j \to u > t_i$.
3. $s = u > t_i$: the limit is $\omega(u)$. The value $r_{u_j}^{(i)}(\omega_j)(s)$ is the short geodesic at $\min(s, u_j)$ when $s \le u_j$, which tends to the end $\omega(u)$ of the limit geodesic, and $\omega_j(s)$ otherwise, which tends to $\omega(s) = \omega(u)$; both branches converge to the same limit.
4. $s > u$: eventually $s > u_j$, and the values are $\omega_j(s) \to \omega(s)$.

Uniform convergence. *Eventually $\sup_s \rho(r_{u_j}^{(i)}(\omega_j)(s), r_u^{(i)}(\omega)(s)) \le e$* Chapter10/RetractionConv · BrokenSetup.eventually_supDist_retrSeg_le: all these paths lie in Ω with energy $\le c$ (VIII.11), so pointwise convergence is uniform.

Convergence of the speeds in L^2. *Eventually $\int_0^1 (\dot s_j - \dot s)^2 \le e$, with $\dot s_j$, $\dot s$ the speeds of $r_{u_j}^{(i)}(\omega_j)$ and $r_u^{(i)}(\omega)$* Chapter10/RetractionConv · BrokenSetup.eventually_integral_retrSeg_le. This is the delicate part, and the one where Milnor's two terms of d really interact. The integral is split at t_i.

- *On $[0, t_i]$* both paths are broken geodesics of the break points of ω_j and of ω, and the break points converge, so the integral is eventually $\le e/4$.
- *Beyond* $\max(u_j, u)$ the paths are ω_j and ω, so the integral there is at most the square of the L^2 distance of ω_j and ω, eventually $\le e/4$.
- *On $[t_i, 1]$ when $u = t_i$.* The short geodesic over $[t_i, u_j]$ may be arbitrarily fast as $u_j \to t_i$, so its speed is not controlled; its energy is. Split at u_j. On $[t_i, u_j]$, the integral is at most $2E(r_{u_j}^{(i)}(\omega_j)) + 2E(r_u^{(i)}(\omega))$ over that interval, which is at most $2E_{t_i}^{u_j}(\omega_j) + 2E_{t_i}^{u_j}(\omega)$, because the middle energy decreases and $r_u^{(i)}(\omega) = \omega$ past $u = t_i$; since $u_j - t_i \to 0$, Lemma 16.B along the filter, for (ω_j) and for the constant family ω, makes each energy $< e/16$, so the sum is $< e/4$. The tail beyond u_j contributes $\le e/4$. This is the L^2 speed control: the contribution of a fast short geodesic is bounded by its energy $\rho(\omega(t_i), \omega(u))^2/(u - t_i) \le E_{t_i}^u(\omega)$, which is uniformly small for paths near a given

one.

- *On* $[t_i, 1]$ *when* $t_i < u$. Split at $\min(u_j, u)$ and $\max(u_j, u)$. On $[t_i, \min]$ both paths are short geodesics of constant speeds $\sigma_j = \|v(\omega_j(t_i), \omega_j(u_j))\|/(u_j - t_i)$ and σ, and $\sigma_j \to \sigma$ by the continuity of the join vector and $u - t_i > 0$; so the integral is at most $(\sigma_j - \sigma)^2 \le e/8$ eventually, the interval having length ≤ 1. On $[\min, \max]$, an interval of length $|u_j - u| \to 0$, the integral is at most $2E + 2E$: the energy of $r_u^{(i)}(\omega)$ there is $< e/32$ by Lemma 16.B for that fixed path; the energy of $r_{u_j}^{(i)}(\omega_j)$ is $< e/32$ by Lemma 16.B for (ω_j) if $u_j \le u$ (the path is ω_j there), and is $\sigma_j^2(u_j - u) \le e/32$ eventually if $u < u_j$ (the path is the short geodesic there). So the middle part is $\le e/8$, and the tail beyond max is $\le e/4$.

In both cases $[t_i, 1]$ contributes $\le e/2 \le 3e/4$, and with $[0, t_i]$ the total is $\le e$.

Continuity on a segment. $d(r_{u_j}^{(i)}(\omega_j), r_u^{(i)}(\omega)) \to 0$ `Chapter10/RetractionConv · BrokenSetup.tendsto_retrSeg`: given ε, choose $0 < r < \varepsilon$; eventually the uniform part is $\le r/2$ and the integral is $\le (r/2)^2$, so the L^2 part is $\le r/2$.

VIII.13 Ω^a deformation retracts onto B^a

Paths are maps on $\mathbb{R}$, and $r_u(\omega)$ agrees with Milnor's only on $[0, 1]$. To obtain a self-map of Ω for which $r_0 = \mathrm{id}$ holds exactly and r_u fixes B exactly, $r_u(\omega)$ is completed by ω off $[0, 1]$. Membership in Ω and both parts of d depend only on $[0, 1]$ `Chapter10/Deformation · PathSp.pathSpace_congr` `Chapter10/Deformation · PathSp.supDist_congr` `Chapter10/Deformation · PathSp.l2Dist_congr`, and a map at distance 0 from a continuous map is continuous `Chapter10/Deformation · PathSp.continuous_of_edist_eq_zero`.

The deformation as a self-map of Ω. For $u \in [0, 1]$ and $E(\omega) < c$, $\bar{r}_u(\omega)$ is $r_u(\omega)$ on $[0, 1]$ and ω off $[0, 1]$; otherwise it is ω `Chapter10/Deformation · BrokenSetup.deformFun` `Chapter10/Deformation · BrokenSetup.deformFun_mem` `Chapter10/Deformation · BrokenSetup.deform` `Chapter10/Deformation · BrokenSetup.coe_deform` `Chapter10/Deformation · BrokenSetup.deform_of_not`. On $[0, 1]$ it is $r_u(\omega)$ `Chapter10/Deformation · BrokenSetup.eqOn_deform`, so the distance between two of its values is the distance between the corresponding values of r `Chapter10/Deformation · BrokenSetup.edist_deform`. The following hold exactly, not only on $[0, 1]$: $\bar{r}_0 = \mathrm{id}$ `Chapter10/Deformation · BrokenSetup.deform_zero`; $\bar{r}_u$ fixes every point of B for $u \in [0, 1]$ `Chapter10/Deformation · BrokenSetup.deform_of_mem_brokenPaths`; $E(\bar{r}_u(\omega)) \le E(\omega)$ for all u and ω `Chapter10/Deformation · BrokenSetup.energy_deform_le` `Chapter8/Hessian · energy_congr`; and $\bar{r}_1$ *retracts* $\operatorname{Int}\Omega^c$ *into* B `Chapter10/Deformation · BrokenSetup.deform_one_mem_brokenPaths`: on $[0, 1]$, $\bar{r}_1(\omega)$ is the broken geodesic of the break points of ω, which lies in B by Lemma 16.1, and both the broken geodesic property and the energy pass along agreement on $[0, 1]$.

Lemma 16.C of the exposition: continuity in both variables (V10.7). $(u, \omega) \mapsto \bar{r}_u(\omega)$ *is continuous on* $[0, 1] \times \operatorname{Int}\Omega^c$ *for Milnor's metric.* `Chapter10/Deformation · BrokenSetup.continuousOn_deform`

Proof as formalized. At (u_0, ω_0), it suffices to show, for each segment k, that eventually (in the neighbourhood filter within the domain) $u \in [t_k, t_{k+1}]$ implies $d(\bar{r}_u(\omega), \bar{r}_{u_0}(\omega_0)) \le \varepsilon$. Then, since every $u \in [0, 1]$ lies in its own segment and there are finitely many segments, the claim follows.

- If $u_0 \in [t_k, t_{k+1}]$, restrict the filter to $u \in [t_k, t_{k+1}]$ and apply the continuity on a segment (VIII.12) to the two projections. The distance between the values is the distance between $r_u^{(k)}(\omega)$ and $r_{u_0}^{(k)}(\omega_0)$, because $r^{(k)}$ is r on the closed segment, at both points (VIII.11). This is where the agreement at the division points is used: when $u_0 = t_k$, the nearby u lie in segment $k - 1$ or segment k, and both formulas give r_{u_0} at u_0.
- If $u_0 \notin [t_k, t_{k+1}]$, then eventually $u \notin [t_k, t_{k+1}]$, the complement of a closed interval being open, and the implication is vacuous.

Theorem 16.2, deformation retraction (V10.8). *For $a < c$: $(u, \omega) \mapsto \bar{r}_u(\omega)$ is continuous on $[0,1] \times \Omega^a$; it does not increase the energy, so each $\bar{r}_u$ maps Ω^a into itself; $\bar{r}_0 = \mathrm{id}$; $\bar{r}_1$ maps Ω^a into B, hence into $B^a = \Omega^a \cap B$; and $\bar{r}_u$ fixes every point of B for $u \in [0,1]$.* `Chapter10/Deformation · BrokenSetup.theorem_16_2_deformationRetract`

So B^a is a strong deformation retract of Ω^a (§0.4). The continuity on $[0,1] \times \operatorname{Int}\Omega^c$ and the same identities make B a strong deformation retract of $\operatorname{Int}\Omega^c$, Milnor's footnote. In homotopy terms `Chapter10/Deformation · BrokenSetup.homotopyEquivBroken` `Mathlib/Topology/Homotopy/Equiv · ContinuousMap.HomotopyEquiv`: Milnor's retraction $r = \bar{r}_1$, as a continuous map $\Omega^a \to B^a$ `Chapter10/Deformation · BrokenSetup.retrMap`, and the inclusion $B^a \to \Omega^a$ `Chapter10/Deformation · BrokenSetup.inclMap` are inverse homotopy equivalences, B^a and Ω^a being subspaces of Ω. The homotopy from the inclusion composed with r to the identity of Ω^a is $(t, \omega) \mapsto \bar{r}_{1-t}(\omega)$, continuous with values in Ω^a `Chapter10/Deformation · BrokenSetup.continuous_deform_sublevel`; and r composed with the inclusion is the identity of B^a exactly, because $\bar{r}_1$ fixes B.

VIII.14 The two descriptions of B^a, and Theorem 16.3 for Ω^a

Milnor says (p. 89) that $\omega \mapsto (\omega(t_1), \dots, \omega(t_{k-1}))$ evidently defines a homeomorphism of B onto an open subset of the product. Both directions need an argument for the metric d.

- *$y \mapsto \omega_y(s)$ is continuous on U, for each $s \in [0,1]$* `Chapter10/Omega · BrokenSetup.tendsto_path_apply`: on the segment k of s, $\omega_{y'}(s)$ is the short geodesic from $x_k(y')$ to $x_{k+1}(y')$ over $[t_k, t_{k+1}]$, the vertices depend continuously on y', and short geodesics with converging data converge.
- *$y \mapsto \omega_y$ is continuous from U to Ω* `Chapter10/Omega · BrokenSetup.pathP` `Chapter10/Omega · BrokenSetup.continuous_pathP`: the uniform part by uniform convergence from pointwise convergence, all energies being $E' < c$; the L^2 part by the continuity of the speeds of broken geodesics in their break points; $r/2 + r/2$.
- *$\omega \mapsto (\omega(t_1), \dots, \omega(t_m))$ is continuous from Ω to V^m* `Chapter10/Omega · BrokenSetup.continuous_nodeVal`, since $\|x - y\| \le \rho(x,y) \le d$.

These restrict to continuous maps between B^a as a subspace of Ω and B^a as a subset of N^m: the break points `Chapter10/Omega · BrokenSetup.nodeMap`, which land in $\{y \in U : E'(y) \le a\}$ by VIII.9, and the broken geodesics `Chapter10/Omega · BrokenSetup.pathMap`, which land in B^a by formula $(*)$ and Lemma 16.1.

The correspondence is a homeomorphism up to paths at distance 0 (V10.5). *For every a, the two descriptions of B^a are homotopy equivalent* `Chapter10/Omega · BrokenSetup.homotopyEquivNodes`. Taking break points after the broken geodesic is the identity, exactly (Lemma 16.1 (a)). Taking the broken geodesic after the break points is homotopic to the identity of $B^a \subseteq \Omega$ through the homotopy $H(t, \gamma) = \omega_{(\gamma(t_j))_j}$ for $t < 1$ and $H(1, \gamma) = \gamma$. As a map it jumps at $t = 1$, but it is continuous for the pseudometric: every value $H(t, \gamma)$ is at distance 0 from γ, because $\omega_{(\gamma(t_j))_j} = \gamma$ on $[0,1]$ (Lemma 16.1 (b)), and a map at distance 0 from the continuous projection $(t, \gamma) \mapsto \gamma$ is continuous. Since paths are maps on $\mathbb{R}$, Milnor's homeomorphism becomes this homotopy equivalence, which is a homeomorphism once paths that agree on $[0,1]$ are identified.

Theorem 16.3 (V10.13). *Given Milnor's data with $a < c$, suppose that no geodesic $\gamma \in \Omega$ has $E(\gamma) = a$, and that E_{**} has nullity 0 at every geodesic $\gamma \in \Omega$ with $E(\gamma) \le a$. Then the geodesics in Ω^a are finitely many, with break points forming a list without repetitions, each recorded with its index; and $\Omega^a = \{\omega \in \Omega : E(\omega) \le a\}$, with Milnor's metric, has the homotopy type of a finite cell complex with one cell of dimension λ for each geodesic in Ω^a of index λ.* `Chapter10/Omega · BrokenSetup.theorem_16_3_Omega`

Proof as formalized. The hypotheses transfer to the model through formula $(*)$, $E(\omega_y) = E'(y)$. Theorem 16.3 in the model (VIII.8) gives the cell structure of $\{y \in U : E'(y) \le a\}$; it is carried to $B^a \subseteq \Omega$ along the homotopy equivalence of the two descriptions, and to Ω^a along the homotopy equivalence $\Omega^a \simeq B^a$ of Theorem 16.2. In counted form `Chapter10/Omega · BrokenSetup.theorem_16_3_Omega_count`: the number of cells of dimension λ is the number of these geodesics of index λ. By VIII.5 every geodesic in Ω^a agrees on $[0,1]$ with some ω_y, y in the list, so one cell for each geodesic means one cell for each class of geodesics

that agree on $[0,1]$, the only sensible count when paths are maps on $\mathbb{R}$. Milnor's data exist for every $c > a$ when N is closed in V, or connected and complete (VIII.1); his finite CW-complex is an iterated adjunction space with arbitrary attaching maps (§0.5); non-conjugacy may replace nullity 0 by Theorem 14.1; and a must not be the energy of a geodesic, as in VIII.8.

What the formalization taught us

- **Completeness is only the compactness of one ball** `Chapter10/Broken · isCompact_closedRiemBall_of_isClosed` `Chapter10/Broken · isCompact_closedRiemBall_of_isRiemComplete`, Milnor's $\Omega^c \neq \emptyset$ is not needed `Chapter10/BrokenPaths · BrokenSetup.lemma_16_1`, and every subdivision of mesh $< \delta^2/c$ is fine enough `Chapter10/Broken · exists_brokenSetup`, with δ also below a normal radius on S at no cost `Chapter10/Broken · corollary_10_8'`.
- **The image of Lemma 16.1 is the open set** U, which needs Cauchy–Schwarz to put the vertices in S `Chapter10/Broken · BrokenSetup.vert_mem_ball` and finite distances in its definition `Chapter10/Broken · BrokenSetup.nodes`; and E' is smooth because $\rho^2 = \|v\|^2$, ρ itself not being smooth on the diagonal `Chapter10/BrokenEnergy · BrokenSetup.exists_smooth_energy'`.
- **The description of B^a on p. 91 has a misprint**, the exponent c for 2 (the missing Σ is probably the scan), and its closedness needs finite distances `Chapter10/BrokenEnergy · BrokenSetup.isCompact_sublevel`.
- **The critical points of E'** need a variation through broken geodesics, smooth on each strip, resting on the smooth dependence of a whole geodesic segment on its end points `Chapter10/BrokenVariation · BrokenSetup.exists_variation_path`, and a test against the vector of jumps `Chapter10/BrokenVariation · BrokenSetup.isCriticalPointM_iff`.
- **Lemma 15.4 does not apply as proved in §15.** It is proved anew from the normal radius and the minimality of broken geodesics `Chapter10/BrokenSplitting · BrokenSetup.formIndex_brokenSpace` `Chapter10/BrokenSplitting · BrokenSetup.finrank_ker_brokenSpace`; variation fields are identified with broken Jacobi fields by Lemma 14.3 on rescaled segments and uniqueness, and the Hessians are compared as quadratic forms, then polarized `Chapter10/BrokenIndex · BrokenSetup.hessian_apply_self` `Chapter10/BrokenIndex · BrokenSetup.theorem_16_2_index`.
- **Theorem 16.3** needs Theorem 3.5 for a compact sublevel set of a non-compact manifold and a smooth extension of E', and is proved when a is not the energy of a geodesic `Chapter10/SublevelCW · isFiniteCellComplexType_sublevel_of_isCompact` `Chapter10/Finite · BrokenSetup.theorem_16_3`.
- **Milnor's metric is a pseudometric on maps of** $\mathbb{R}$, whose triangle inequality is Minkowski's and whose L^2 term makes $\sqrt{E}$ 1-Lipschitz `Chapter10/PathMetric · PathSp.abs_sqrt_energy_sub_le`; his homeomorphism becomes a homotopy equivalence through a homotopy that jumps but stays at distance 0 `Chapter10/Omega · BrokenSetup.homotopyEquivNodes`.
- **The continuity of r_u** needs the formulas of neighbouring segments to agree at $u = t_i$ `Chapter10/Retraction · BrokenSetup.retrSeg_eq_retr`, equicontinuity for the uniform part, and, near $u = t_i$, the control of a fast short geodesic by its energy through Lemma 16.B `Chapter10/RetractionConv · BrokenSetup.eventually_integral_retrSeg_le`.
- **The Remark on p. 92** is proved for the energy; the shortest-length statement is not formalized `Chapter10/Finite · BrokenSetup.exists_energy_minimizing`.

Part IX. The full path space and the fundamental theorem (Milnor §17)

Milnor's §17 draws the consequences of the finite-dimensional approximation of §16. Theorem 17.1 compares the space Ω of piecewise smooth paths with the space Ω^* of all continuous paths. Theorem 17.3, the fundamental theorem of Morse theory, assembles the sublevel sets Ω^a into one countable cell complex with a λ-cell for each geodesic of index λ. The example of the round sphere gives Corollary 17.4 and the homology of ΩS^n for $n > 2$, and Corollary 17.5 draws a consequence for manifolds with the homotopy type of a sphere. This part retells the formal proofs of everything in §17 that is formalized. The setting is the embedded one of Parts VII and VIII: $N \subseteq V$ is a slice submanifold of a finite-dimensional real inner product space (§0.1, §0.3), with the induced metric and its distance ρ; $\Omega = \Omega(N; p, q)$ carries Milnor's metric d, and Ω^* the topology of uniform convergence (§0.8). Every Riemannian manifold is such an N by Nash's embedding theorem, which is quoted, not proved.

The formal route departs from Milnor's in four places.

- **Theorem 17.1 and the book's one genuine error.** Milnor proves 17.1 with a map $h : \Omega^* \to \Omega$ that replaces a continuous path by the broken geodesic through its values at the multiples of a number $A(\omega)$, and asserts without proof that h is continuous (p. 94). For the metric d of Ω, the topology in which the theorem is stated, this is false. §IX.12 retells the formal proof that it is false, for Milnor's own construction on $M = \mathbb{R}$ `Chapter10/MilnorH · milnor_h_not_continuous`. The theorem is true, and §§IX.13–9.18 retell its formal proof by another route: dyadic subdivisions kept fixed on open sets $Y_k \subseteq \Omega^*$, convex *good balls* each carrying one smooth minimal-geodesic map, and Theorem A of the Appendix applied through telescopes `Chapter10/FullPathTheorem · theorem_17_1`.
- **Theorem 17.3 by Milnor's second route.** Milnor's main proof of 17.3 uses Whitehead's theorem and Corollary 17.2, which rests on 17.1. The formal proof takes the alternative through the Appendix, to which Milnor refers and which the Appendix carries out (pp. 149–150). It builds a tower of relative cell attachments that commutes only up to homotopy, analyses each step with its own manifold of broken geodesics, and needs neither Whitehead's theorem nor 17.2 (§§IX.2–9.6).
- **Homology without cellular chains.** The homology of $\Omega(S^n; p, q)$ is computed from the filtration of Ω by the sets Ω^a, the compactness of singular chains, and the lacunary principle (§§IX.8–9.10). The cell complexes of 17.3 have arbitrary attaching maps, not cellular ones, so cellular homology is not available.
- **The sphere in full, for paths rather than loops.** The geodesics of S^n from p to $q \neq \pm p$, their conjugate points with multiplicities, and their indices are all derived (§IX.7). Corollary 17.4 is proved for the path space $\Omega(S^n; p, q)$, which is what Milnor's argument gives; the passage to the loop space needs a lemma moving an end point, which is not formalized.

Not formalized: Corollary 17.2, Theorem 17.3 for Ω^*, the loop-space form of 17.4, the homology of ΩS^2, integral homology, Corollary 17.5 itself, and Serre's remark. What is formalized about 17.5 is that its printed hypotheses are not enough: completeness, which Milnor omits there, cannot be dropped (§IX.11).

Map. Paths are relative to `Morse/`.

- `Chapter10/RelativeCells`: relative cell complexes, Lemma 3.7 relative, the relative Theorem 3.5, towers.
- `Chapter10/FundamentalTheorem`: one step of the energy filtration, and Theorem 17.3.
- `Chapter10/SphereGeodesics`: great circles, their conjugate points, the geodesics γ_k of S^n and their indices.
- `Chapter10/SublevelHomology`: the homology of an exhaustion, and of a slab below a compact sublevel set.
- `Chapter10/PathHomology`: the homology of a slab of Ω; one geodesic per window; Betti numbers.
- `Chapter10/SpherePaths`: Corollary 17.4 and the homology of $\Omega(S^n; p, q)$.
- `Chapter10/Incomplete`: $\mathbb{R}^{n+1} \setminus \{0\}$, and the completeness hypothesis of 17.5.
- `Chapter10/MilnorH`: Milnor's map h, and the proof that it is not continuous.
- `Chapter10/FullPathSpace`: Ω^*, the inclusion i, dyadic subdivisions, control of velocities, Example 1 pulled back.
- `Chapter10/GoodBalls`: minimal join vectors, good balls, convexity.
- `Chapter10/GoodJoin`: smooth gluing, and the good balls as a system of joins.
- `Chapter10/FullPathDeform`: join systems, the sets Y_k, the deformation inside Ω^*.
- `Chapter10/FullPathOmega`: the deformation inside Ω, continuous for d; $X_k \simeq Y_k$.
- `Chapter10/FullPathTheorem`: Theorem 17.1.

IX.1 The setting

The submanifold. $N \subseteq V$ is a slice submanifold modelled on Z, with complementary space W `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`; V, Z and W are finite-dimensional. The distance $\rho(x, y)$ is the infimum of the lengths of piecewise smooth paths in N from x to y, and ∞ if there is none `Chapter7/Distance · riemDist`. It dominates the norm of V, $\|x - y\| \le \rho(x, y)$ `Chapter7/Metric · ofReal_norm_sub_le_riemDist`, and its balls form a basis of the neighbourhoods of a point in N `Chapter7/Distance · hasBasis_nhdsWithin_riemDist`. Completeness, where it is needed, is assumed as closedness of N in V, which makes the closed ρ-balls about p compact `Chapter10/Broken · isCompact_closedRiemBall_of_isClosed`.

Piecewise smooth paths. A path of $\Omega(N; p, q)$ is a function $c : \mathbb{R} \to V$ that is piecewise smooth on $[0, 1]$ `Chapter7/Path · IsPiecewiseSmooth`, takes values in N there, and has $c(0) = p$, $c(1) = q$ `Chapter8/Energy · pathSpace`. Its values off $[0, 1]$ play no role. Its energy is $E(c) = \int_0^1 \|c'\|^2 \, dt$ `Chapter8/Energy · energy`. The type of these paths carries Milnor's metric

$$d(\alpha, \beta) = \max_{t \in [0,1]} \rho(\alpha(t), \beta(t)) + \Big(\int_0^1 \big(\|\alpha'(t)\| - \|\beta'(t)\| \big)^2 \, dt \Big)^{1/2}$$

`Chapter10/PathMetric · PathSp` `Chapter10/PathMetric · supDist` `Chapter10/PathMetric · l2Dist` `Chapter10/PathMetric · instPseudoEMetricSpace`, where the second term compares the speeds. It is an extended *pseudo*metric: two functions that agree on $[0, 1]$ are at distance 0 `Chapter10/Omega · edist_eq_zero_of_eqOn`, so Ω is not Hausdorff. This matters twice in this part, in the gluing of §IX.18 and in a homotopy of §IX.17, and two small facts handle it. A function that agrees on $[0, 1]$ with a path of Ω is itself in Ω `Chapter10/Deformation · pathSpace_congr`. A map into Ω at distance 0 from a continuous map is continuous `Chapter10/Deformation · continuous_of_edist_eq_zero`.

The energy. By Lemma 16.A of the exposition, $\sqrt{E_a^b}$ is 1-Lipschitz for d whenever $0 \le a \le b \le 1$ `Chapter10/PathMetric · lipschitzWith_sqrt_energy`, so the energy is continuous on Ω `Chapter10/PathMetric · continuous_energy` (Part VIII). As a bundled continuous map it is written $E : \Omega \to \mathbb{R}$ `Chapter10/FundamentalTheorem · energyC`, and its sublevel sets are $\Omega^a = \{E \le a\}$ `Chapter2/Crossing · sublevel` `Chapter2/Crossing · sublevel_mono` (§0.4).

Geodesics. A geodesic of Ω is an unbroken geodesic: smooth on $[0, 1]$, with zero tangential acceleration on $(0, 1)$ `Chapter8/Critical · IsUnbrokenGeodesic`. At a geodesic γ the Hessian E_{**} has an index `Chapter8/`

Hessian · energyHessian Chapter9/FormIndex · formIndex and a nullity Chapter9/JacobiSpace · hessNullity (§0.9, Part VII). The point q lies in the *conjugate locus* of p if some geodesic from p to q has 0 and 1 conjugate along it Chapter11/NonConjugate · conjLocus. Outside the conjugate locus every geodesic from p to q has nullity 0, by Theorem 14.1 Chapter11/NonConjugate · hessNullity_eq_zero_of_notMem_conjLocus. This is Milnor's hypothesis in 17.3.

The space of continuous paths. $\Omega^* = \Omega^*(N; p, q)$ is the set of continuous maps $\omega : [0,1] \to V$ with values in N, $\omega(0) = p$, $\omega(1) = q$, as a subspace of the space of continuous maps $[0,1] \to V$ with the uniform distance d^* Chapter10/FullPathSpace · pathCSet Chapter10/FullPathSpace · PathC. That is the compact-open topology. Milnor measures d^* with ρ and the formalization with the norm of V; on N the two distances define the same topology, and the inequality $\|x - y\| \le \rho(x, y)$ is what is used. The inclusion $i : \Omega \to \Omega^*$ restricts a path to $[0,1]$ Chapter10/FullPathSpace · toPathC. It satisfies $d^*(i\gamma, i\gamma') \le d(\gamma, \gamma')$, because $\|\gamma(t) - \gamma'(t)\| \le \rho(\gamma(t), \gamma'(t))$ is bounded by the first term of d Chapter10/FullPathSpace · edist_toPathC_le. So i is Lipschitz, hence continuous Chapter10/FullPathSpace · continuous_toPathC; as a continuous map it is written i throughout Chapter10/FullPathTheorem · pathIncl.

Homotopy equivalences. A continuous f is a homotopy equivalence if it has a homotopy inverse Appendix/Telescope · IsHomotopyEquiv. Mathlib's bundled homotopy equivalences carry the inverse and both homotopies as data Mathlib/Topology/Homotopy/Equiv · ContinuousMap.HomotopyEquiv. The inclusion of a strong deformation retract (§0.4) is a homotopy equivalence Chapter2/Crossing · IsStrongDeformationRetract.exists_homotopyEquiv.

IX.2 Relative cell complexes

Milnor proves 17.3 by referring to the proof of Theorem 3.5 (p. 95). Theorem 3.5 is about one sublevel set at a time: M^a has the homotopy type of a cell complex. The tower of 17.3 needs a statement about *pairs* $\Omega^{a_n} \subseteq \Omega^{a_{n+1}}$, with control of the inclusion. The formalization introduces the notion that carries this control and proves the three lemmas it needs.

Adjunction spaces. For a space X and a continuous $\varphi : S^{n-1} \to X$, the space $X \cup_\varphi e^n$ is the pushout in the category of topological spaces of φ and the inclusion $S^{n-1} \to D^n$ Chapter2/Attaching · AttachCell, with its inclusion $X \to X \cup_\varphi e^n$ Chapter2/Attaching · cellInl (§0.5, Part II). An *iterated adjunction space* with cells $(\lambda_1, \dots, \lambda_k)$ is obtained from the empty space by attaching a λ_1-cell, then a λ_2-cell, and so on, each along an arbitrary continuous map Chapter2/CWType · IsCellAttachment; a space has the homotopy type of a finite cell complex with these cells if it is homotopy equivalent to one Chapter2/CWType · IsFiniteCellComplexType. The relative version starts from a space L instead of the empty space: K is obtained from L by attaching cells of the listed dimensions one at a time, and $j : L \to K$ is the composite of the inclusions Appendix/MonotoneUnion · IsRelCellAttachment.

Definition. A continuous map $\iota : A \to X$ is, *up to homotopy, the inclusion of A with cells $(\lambda_1, \dots, \lambda_k)$ attached*, if there are a relative cell attachment $j : A \to K$ with these cells and a homotopy equivalence $e : X \to K$ with $e \circ \iota \simeq j$ Chapter10/RelativeCells · IsRelCellComplexType. For subsets $A \subseteq B$ the map ι is usually the inclusion of subspaces Chapter10/RelativeCells · inclC.

Four bookkeeping facts follow at once. A homotopy equivalence $e : A \to X$ is such a map with no cells: take $K = A$, $j = \mathrm{id}$ and e^{-1} Chapter10/RelativeCells · IsRelCellComplexType.of_homotopyEquiv. The property depends only on the homotopy class of ι Chapter10/RelativeCells · IsRelCellComplexType.congr, and in particular only on ι as a function Chapter10/RelativeCells · IsRelCellComplexType.of_eq. If ι has it and $e' : X \to Y$ is a homotopy equivalence, then $e' \circ \iota$ has it, with the homotopy equivalence $e \circ e'^{-1} : Y \to K$, since $ee'^{-1}e'\iota \simeq e\iota \simeq j$ Chapter10/RelativeCells · IsRelCellComplexType.comp_homotopyEquiv. For subsets $A \subseteq B \subseteq C$, if $A \to B$ has the property and the inclusion $B \to C$ is a homotopy equivalence, then $A \to C$ has it with the same cells Chapter10/RelativeCells · IsRelCellComplexType.mono. This last form is how Theorems 3.1 and 3.2 enter: they enlarge a sublevel set by an inclusion that is a homotopy equivalence.

Lemma 3.7, relative and iterated. Let $j : L \to K$ attach cells $(\lambda_1, \dots, \lambda_k)$ to L, and let $e : L \to L'$ be a

homotopy equivalence. Then the same cells can be attached to L', giving $j' : L' \to K'$, and there is a homotopy equivalence $E : K \to K'$ with $E \circ j = j' \circ e$ exactly `Chapter10/RelativeCells · IsRelCellAttachment.transport`.

The proof as formalized is an induction on the attachment. With no cells, $K' = L'$, $j' = \mathrm{id}$ and $E = e$. For the inductive step, suppose $K = K_1 \cup_\varphi e^n$ and that $E_1 : K_1 \to K_1'$ has been built with $E_1 j_1 = j_1' e$. Lemma 3.7 (Part II) extends the homotopy equivalence E_1 over the new cell: it gives a homotopy equivalence $F : K_1 \cup_\varphi e^n \to K_1' \cup_{E_1 \varphi} e^n$ that is E_1 on K_1 `Chapter2/Attaching · lemma_3_7`. Attach the cell to K_1' along $E_1 \circ \varphi$. Then $F(jx) = F(j_1 x) = E_1(j_1 x) = j_1'(ex)$, which is the required identity. Milnor uses Lemma 3.7 one cell at a time in the proof of Theorem 3.5; the lemma packages that use, relative to a base space.

Invariance under homotopy-commutative squares. Let $\iota' : A' \to X'$ be, up to homotopy, the inclusion of A' with cells attached, and let $\alpha : A \to A'$ and $\beta : X \to X'$ be homotopy equivalences with $\beta \circ \iota \simeq \iota' \circ \alpha$. Then ι is, up to homotopy, the inclusion of A with the same cells attached `Chapter10/RelativeCells · IsRelCellComplexType.of_square`.

The proof as formalized. Let $j : A' \to K$ and $e : X' \to K$ be the data for ι'. Transport the attachment along the homotopy inverse $\alpha^{-1} : A' \to A$ by the previous lemma: this gives $j'' : A \to K''$ with the same cells and a homotopy equivalence $E : K \to K''$ with $Ej = j''\alpha^{-1}$. The homotopy equivalence for ι is $E \circ e \circ \beta : X \to K''$, and

$$Ee\beta\iota \simeq Ee\iota'\alpha \simeq Ej\alpha = j''\alpha^{-1}\alpha \simeq j''.$$

Only homotopy commutativity of the square is used. That is the point of the notion: the homotopy equivalences between Ω^a and models of it need not commute with the inclusions on the nose.

Attaching a cell. If $\iota : A \to X$ has the property with cells c and $\varphi : S^{n-1} \to X$ is continuous, then $A \to X \cup_\varphi e^n$ has it with cells c followed by n `Chapter10/RelativeCells · IsRelCellComplexType.attach`. Lemma 3.7 extends $e : X \to K$ to a homotopy equivalence $F : X \cup_\varphi e^n \to K \cup_{e\varphi} e^n$ that agrees with e on X, so $F \circ (\text{inclusion}) \circ \iota = (\text{inclusion}) \circ e \circ \iota \simeq (\text{inclusion}) \circ j$, and the new relative attachment is j followed by the new cell.

Lemma D, relative. Let $A_0 \subseteq A$ be subsets of a Hausdorff space with A closed. Let $\chi_i : \mathbb{R}^{k_i} \to M$, for i in a list l, be continuous and injective on the closed unit ball D^{k_i}, with pairwise disjoint images $\chi_i(D^{k_i})$, each meeting A exactly in the image of the unit sphere: $\chi_i(z) \in A$ iff $\|z\| = 1$, for $z \in D^{k_i}$. If $A_0 \to A$ is, up to homotopy, the inclusion with cells c, then $A_0 \to A \cup \bigcup_{i \in l} \chi_i(D^{k_i})$ is, up to homotopy, the inclusion with cells c followed by k_i, $i \in l$ `Chapter10/RelativeCells · isRelCellComplexType_union_cells`.

The proof as formalized is an induction on the list. For the empty list, $A \cup \emptyset = A$ and nothing is attached. For a list $i :: l$, the case of one cell of Lemma D (Part II) identifies $A \cup \chi_i(D^{k_i})$ with the adjunction space $A \cup_\varphi e^{k_i}$, $\varphi = \chi_i | S^{k_i - 1}$: they form a pushout square in which $A \to A \cup \chi_i(D^{k_i})$ is the inclusion `Chapter2/CellPatch · exists_isPushout_union_cell_of_t2`. So the lemma on attaching a cell applies, followed by the homeomorphism from the pushout to the union, which composed with the inclusion of A is the inclusion of A. The new set $A' = A \cup \chi_i(D^{k_i})$ is closed, since a compact image is closed in a Hausdorff space. The remaining cells still meet A' exactly in their boundary spheres, because they are disjoint from $\chi_i(D^{k_i})$. The induction hypothesis for l and A' finishes, after rewriting $A' \cup \bigcup_{j \in l} = A \cup \bigcup_{j \in i :: l}$.

IX.3 The relative Theorem 3.5

Milnor's Theorem 3.5 and Remark 3.3 describe one sublevel set. The tower of 17.3 needs the inclusion of one sublevel set into another, and on the manifold of broken geodesics, which is not compact: only its sublevel sets below the energy bound are. The formalization states the relative Theorem 3.5 under exactly that hypothesis, for $f = F|N$ with $F : V \to \mathbb{R}$ smooth, $N \subseteq V$ a slice submanifold, and a level a_0 with N^{a_0} compact.

One critical level, relative to a lower level. Let S be a finite set of non-degenerate critical points of f, all with value c. Let $b \le a' < c < a \le a_0$, assume that every critical point outside S has value $< a'$ or $> a$, and that N^{a_0} is compact. If $N^b \to N^{a'}$ is, up to homotopy, the inclusion with cells c, then there is a list l of the points of S, without repetition, such that $N^b \to N^a$ is, up to homotopy, the inclusion with cells c followed by the indices of the points of l `Chapter10/RelativeCells · IsRelCellComplexType.sublevel_step_of_isCompact`.

The proof as formalized.

1. Every slab $f^{-1}[\beta, \beta']$ with $\beta' \le a_0$ is compact, as a closed subset of N^{a_0}. Put $\varepsilon_0 = \min(c - a', a - c) > 0$. The slab $[c - \varepsilon_0, c + \varepsilon_0]$ is compact, and its only critical points are those of S.
2. Remark 3.3 for a submanifold (Part II) gives $\varepsilon_1 > 0$ such that, for $0 < \varepsilon \le \varepsilon_1$, there are cells $\chi_p(D^{\lambda_p})$, $p \in S$, λ_p the index of p. Each is continuous and injective on the ball, meets $N^{c-\varepsilon}$ exactly in the image of the sphere, and lies in $N^{c+\varepsilon}$; they are pairwise disjoint; and $N^{c-\varepsilon} \cup \bigcup_p \chi_p(D^{\lambda_p})$ is a strong deformation retract of $N^{c+\varepsilon}$ `Chapter2/SeveralCritical · remark_3_3_submanifold`. Take $\varepsilon = \min(\varepsilon_1, \varepsilon_0/2)$, so that $a' < c - \varepsilon$ and $c + \varepsilon < a$.
3. Theorem 3.1 on the compact slabs $[a', c - \varepsilon]$ and $[c + \varepsilon, a]$, which contain no critical points, makes the inclusions $N^{a'} \to N^{c-\varepsilon}$ and $N^{c+\varepsilon} \to N^a$ homotopy equivalences `Chapter2/Submanifold · theorem_3_1_submanifold`.
4. By the first inclusion and the enlargement fact of §IX.2, $N^b \to N^{c-\varepsilon}$ has cells c. List the points of S without repetition as l. Lemma D, relative, attaches the cells χ_p one at a time to the closed set $N^{c-\varepsilon}$ (closed since f is continuous), so $N^b \to N^{c-\varepsilon} \cup \bigcup_p \chi_p(D^{\lambda_p})$ has cells c followed by the λ_p, $p \in l$.
5. The strong deformation retraction of step 2 makes the inclusion of that union into $N^{c+\varepsilon}$ a homotopy equivalence, and step 3 makes $N^{c+\varepsilon} \to N^a$ one. Two more uses of the enlargement fact give the claim.

The relative Theorem 3.5. Let N^{a_0} be compact, every critical point of f in N^{a_0} non-degenerate, and $b \le a_0$ with neither b nor a_0 a critical value. Then the critical points of f in $f^{-1}(b, a_0)$ can be listed without repetition as $p_1, \dots, p_j$ with $f(p_1) \le \dots \le f(p_j)$, so that the inclusion $N^b \to N^{a_0}$ is, up to homotopy, the inclusion of N^b with cells of dimensions $\mathrm{index}(p_1), \dots, \mathrm{index}(p_j)$ attached in this order `Chapter10/RelativeCells · isRelCellComplexType_sublevel_of_isCompact`.

The proof as formalized.

1. *Finitely many critical points.* The critical points of f in N^{a_0} form a compact set, as a closed subset of a compact one `Chapter1/ManifoldMorse · isClosed_setOf_isCriticalPointM`, and a discrete one, since non-degenerate critical points are isolated `Chapter1/ManifoldMorse · isDiscrete_inter_setOf_isCriticalPointM`. So they form a finite set C.
2. *Induction.* The statement is proved for every level $a \le a_0$ that is not a critical value, by strong induction on the number m of points of C with value in (b, a).
3. *No critical point in* (b, a). Theorem 3.1 on the compact slab $[b, a]$ makes the inclusion $N^b \to N^a$ a homotopy equivalence, which is a relative cell complex with no cells.
4. *Otherwise* let p be a point of C with the largest value $c \in (b, a)$, and S the points of C at level c. Choose a level $a' \in [b, c)$ that is not a critical value and lies above every other critical value in (b, c). If there are such values, let q_0 have the largest of them and put $a' = (f(q_0) + c)/2$; a critical point at that level would lie strictly between $f(q_0)$ and c. If there are none, put $a' = b$.
5. Every critical point outside S then has value $< a'$ or $> a$, and the points with value in (b, a') are those with value in (b, a) minus S, strictly fewer. The induction hypothesis gives the list for (b, a') and the relative cell structure of $N^b \to N^{a'}$. The one-level step adds the points of S, all at the level c, which is above everything already listed. Concatenating keeps the list sorted and without repetition.

The level a_0 plays two roles: its sublevel set is compact, and all the hypotheses concern critical points in

it. Nothing is assumed about critical points above a_0, of which there may be infinitely many.

IX.4 Towers and countable cell complexes

Milnor's conclusion of 17.3 is a countable CW complex $K = \bigcup K_i$ with homotopy equivalences $\Omega^{a_i} \to K_i$ whose restrictions to the smaller sublevel sets are the previous maps, a strictly commutative ladder (p. 95). The formalization replaces both the complex and the ladder.

The countable cell complex is given by its finite stages, as in the Appendix (Part XI). A *cell tower* with cells $\kappa_0, \kappa_1, \ldots$ is a sequence $K_0 \to K_1 \to \cdots$ in which K_0 is an iterated adjunction space with cells κ_0 and each map $K_n \to K_{n+1}$ is a relative cell attachment with cells κ_{n+1} `Appendix/MonotoneUnion · CellTower`. A tower may be given by its stages and the maps between them `Appendix/MonotoneUnion · CellTower.ofSequence` `Appendix/MonotoneUnion · CellTower.tmap_ofSequence`. A space X *has the homotopy type of the countable cell complex* with these cells if it is homotopy equivalent to the mapping telescope of such a tower `Appendix/MonotoneUnion · IsCountableCellComplexType`. The telescope stands for the union $\bigcup K_n$ with the weak topology. The two are homotopy equivalent by Example 2 of the Appendix, which rests on Whitehead's theorem and is not formalized; that is why the telescope, and not the union, appears in the formal statement. The attaching maps are arbitrary continuous maps, since cellular approximation is not in Mathlib.

The comparison. A space X is the *homotopy direct limit* of subsets $X_0 \subseteq X_1 \subseteq \cdots$ if the projection to X of Milnor's telescope $X_\Sigma = \bigcup_n X_n \times [n, n+1] \subseteq X \times \mathbb{R}$ is a homotopy equivalence `Appendix/Telescope · IsHomotopyDirectLimit` (Part XI). The Appendix proves that if X is the homotopy direct limit of the X_n and $g_n : X_n \to K_n$ are homotopy equivalences to the stages of a cell tower with $c_n \circ g_n \simeq g_{n+1}$ on X_n, where $c_n : K_n \to K_{n+1}$ is the map of the tower, then X has the homotopy type of the countable cell complex `Appendix/MonotoneUnion · isCountableCellComplexType_of_tower` `Appendix/MonotoneUnion · nonempty_homotopyEquiv_mappingTelescope`. Only homotopy-commutative squares are needed. This is the formal form of Milnor's alternative proof on p. 150, and it is what makes a strictly commuting ladder unnecessary.

The inductive construction. A *stage* n of the construction is a space K with a homotopy equivalence $g : X_n \to K$ `Chapter10/RelativeCells · TowerStage`. A *step* from a stage, with cells c, is a space K', a relative cell attachment $j : K \to K'$ with cells c, and a homotopy equivalence $g' : X_{n+1} \to K'$ with $g' \circ (X_n \subseteq X_{n+1}) \simeq j \circ g$ `Chapter10/RelativeCells · TowerStep` `Appendix/Telescope · up`.

The inductive step (Milnor p. 95). If the inclusion $X_n \to X_{n+1}$ is, up to homotopy, the inclusion with cells c attached, then every stage (K, g) at n has a step with cells c `Chapter10/RelativeCells · nonempty_towerStep`.

The proof as formalized. Let $j_A : X_n \to K_A$ and $e : X_{n+1} \to K_A$ be the data of the relative cell structure, with $e \circ (X_n \subseteq X_{n+1}) \simeq j_A$. Transport the attachment j_A along the homotopy equivalence $g : X_n \to K$ (Lemma 3.7, relative): this gives $j' : K \to K'$ with the same cells and a homotopy equivalence $E : K_A \to K'$ with $Ej_A = j'g$. Put $g' = E \circ e$. Then $g' \circ (X_n \subseteq X_{n+1}) = E \circ e \circ (X_n \subseteq X_{n+1}) \simeq Ej_A = j'g$.

The stages are then defined by recursion, choosing a step at each stage `Chapter10/RelativeCells · towerStages` `Chapter10/RelativeCells · towerStepAt`.

The countable cell complex of a filtered space. Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$. Suppose X_0 has the homotopy type of a finite cell complex with cells κ_0, and each inclusion $X_n \to X_{n+1}$ is, up to homotopy, the inclusion with cells κ_{n+1} attached. Then X has the homotopy type of the countable cell complex with cells $\kappa_0, \kappa_1, \ldots$ `Chapter10/RelativeCells · isCountableCellComplexType_of_rel`.

The proof as formalized. Stage 0 is the finite cell complex K_0 with a homotopy equivalence $X_0 \to K_0$. The recursion produces stages K_n, relative cell attachments $K_n \to K_{n+1}$ and homotopy equivalences $g_n : X_n \to K_n$ with homotopy-commutative squares. These stages and maps form a cell tower, and the comparison theorem applies, the map of the tower being the attachment $K_n \to K_{n+1}$ by construction.

Sublevel sets. If $E : X \to \mathbb{R}$ is continuous and $a_0 \le a_1 \le \cdots$ tend to ∞, then X is the homotopy direct limit of the sublevel sets X^{a_n}, for every space X, with no paracompactness `Appendix/Examples · isHomotopyDirectLimit_sublevel`. The proof in the Appendix uses the locally finite sum $f = \sum_j \min(1, \max(0, E - a_j + 1))$, which is continuous and is $\ge n+1$ off X^{a_n}, so Lemma 2.1 of the exposition applies `Appendix/Telescope · isHomotopyDirectLimit_of_level`. Consequently, for such levels, a finite cell complex at the bottom and relative cell structures on the inclusions $X^{a_n} \to X^{a_{n+1}}$ give X the homotopy type of the countable cell complex `Chapter10/RelativeCells · isCountableCellComplexType_of_relSublevel`.

What Milnor leaves implicit. His ladder asks for g_{n+1} to extend g_n. Lemma 3.7 extends a homotopy equivalence over the cells of $X_n \cup e^{\lambda_1} \cup \cdots$, but X_{n+1} is only homotopy equivalent to this space, by the retractions of Theorems 3.1 and 3.2, which move X_n. So the extension holds only up to homotopy. Strict compatibility can be arranged with nested subdivisions (Lemma 17.E of the exposition), and that is neither formalized nor needed: the comparison theorem only uses homotopy-commutative squares, which is exactly what the inductive step produces.

IX.5 One step of the energy filtration

The input of the tower is the relative cell structure of each inclusion $\Omega^a \to \Omega^b$. It comes from the manifold of broken geodesics of Part VIII.

Milnor's data (Part VIII). For end points $p, q \in N$, an energy bound $c > 0$ such that the closed ρ-ball of radius $\sqrt{c}$ about p is compact, and a sufficiently fine subdivision $0 = t_0 < \cdots < t_{m+1} = 1$, the data of §16 are recorded in one structure `Chapter10/Broken · BrokenSetup`. Such data exist, with a uniform subdivision, as soon as the ball is compact `Chapter10/Broken · exists_brokenSetup_uniform`, which holds for every c when N is closed in V `Chapter10/Broken · isCompact_closedRiemBall_of_isClosed`. The break points $y = (x_1, \dots, x_m) \in N^m$ of broken geodesics of energy $< c$ form Milnor's open set U `Chapter10/Broken · BrokenSetup.nodes`. On it the energy of the broken geodesic with break points y is $E'(y) = \sum_i \rho(x_i, x_{i+1})^2/(t_{i+1} - t_i)$ `Chapter10/Broken · BrokenSetup.energy'`. The broken geodesic itself is a path of Ω `Chapter10/Broken · BrokenSetup.path` `Chapter10/Broken · BrokenSetup.path_mem_pathSpace` with energy $E'(y)$ `Chapter10/Broken · BrokenSetup.energy_path` and break points y `Chapter10/Broken · BrokenSetup.nodeVal_path`. Its index, as a natural number, is the index of E_{**} at it `Chapter10/Finite · BrokenSetup.geodIndex`.

Four results of Part VIII are used. (a) E' agrees on the compact set $B^{c_1} = \{y \in U : E'(y) \le c_1\}$, $c_1 < c$, with a smooth function F on all of V^m `Chapter10/Finite · BrokenSetup.exists_contDiff_eq_energy'`, and $\{y \in U : E'(y) < c_1\} = N^m \cap O$ for an open O `Chapter10/Finite · BrokenSetup.exists_isOpen_sublevel_lt`, which is a slice submanifold of V^m `Chapter10/Product · isSliceSubmanifold_pi_inter`. (b) The sublevel sets B^b, $b < c$, are compact `Chapter10/BrokenEnergy · BrokenSetup.isCompact_sublevel`. (c) The critical points of F on $S = N^m \cap O$ are exactly the break points of unbroken geodesics `Chapter10/BrokenVariation · BrokenSetup.isCriticalPointM_iff`, and at such a point the index and nullity of the Hessian of $F|S$ are those of E_{**} at the geodesic `Chapter10/BrokenIndex · BrokenSetup.theorem_16_2_index` (Theorem 16.2). (d) Every geodesic of energy $< c$ is, on $[0,1]$, the broken geodesic of its break points, and those lie in U `Chapter10/Finite · BrokenSetup.eqOn_path_nodeVal_of_isUnbrokenGeodesic`. Finally, Ω^a and B^a have the same homotopy type for $a < c$: Milnor's retraction r_1 `Chapter10/Deformation · BrokenSetup.homotopyEquivBroken` followed by the identification of broken geodesics with their break points `Chapter10/Omega · BrokenSetup.homotopyEquivNodes`.

The relative Theorem 16.3, in the space of break points. Let $a \le b < c$ be levels that are not energies of geodesics, and let every geodesic of energy $\le b$ have nullity 0. Then the break points of the geodesics with energy in (a, b) form a list $\mathcal{Y}$ without repetition, ordered by non-decreasing energy, and the inclusion $B^a \to B^b$ is, up to homotopy, the inclusion of B^a with one cell of dimension $\mathrm{index}(\gamma)$ attached for each of them, in this order `Chapter10/FundamentalTheorem · BrokenSetup.isRelCellComplexType_nodes`.

The proof as formalized.

1. Put $c_1 = (b + c)/2$, so $b < c_1 < c$. Take the smooth extension F of E' from B^{c_1} and the slice

submanifold $S = N^m \cap O = \{y \in U : E'(y) < c_1\}$ of (a). On S, $F = E'$.

2. For $t \le b$, the sublevel set of $F|S$ at t is B^t, since a point of U with $E' \le t \le b$ lies in S. In particular $(F|S)^b = B^b$ is compact by (b).
3. By (c), the critical points of $F|S$ are the break points of geodesics, and their nullities are those of E_{**}, which are 0 below b; so every critical point of $F|S$ in $(F|S)^b$ is non-degenerate `Chapter1/Nondegenerate · isNondegenerate_iff_nullity_eq_zero`. Neither a nor b is a critical value, because neither is the energy of a geodesic.
4. The relative Theorem 3.5 (§IX.3), with $a_0 = b$, gives the sorted list of critical points in (a, b) and the relative cell structure of $(F|S)^a \to (F|S)^b$, with cells the indices of the Hessian of $F|S$. By (c) these are the indices index(γ).
5. The identifications $B^t \cong (F|S)^t$, $t = a, b$, are homeomorphisms given by the same formula, $y \mapsto y$, so they commute with the inclusions. Invariance under squares (§IX.2) transports the relative cell structure to $B^a \to B^b$.

One formula for all levels. The homotopy equivalence $\Omega^a \to B^a$ is the composite of Milnor's retraction and the identification of a broken geodesic with its break points `Chapter10/FundamentalTheorem · BrokenSetup.omegaEquivNodes`. As a map it is $\gamma \mapsto ((r_1\gamma)(t_1), \dots, (r_1\gamma)(t_m))$ for every $a < c$ `Chapter10/FundamentalTheorem · BrokenSetup.omegaEquivNodes_apply`. Because one formula serves all levels, the square formed with the inclusions $\Omega^a \to \Omega^b$ and $B^a \to B^b$ commutes on the nose.

The relative Theorem 16.3 for Ω (the step of Milnor's proof of 17.3). Under the same hypotheses, the inclusion $\Omega^a \to \Omega^b$ of sublevel sets of the energy on Ω, with Milnor's metric, is, up to homotopy, the inclusion of Ω^a with one cell of dimension index(γ) attached for each geodesic with energy in (a, b), in order of non-decreasing energy `Chapter10/FundamentalTheorem · BrokenSetup.isRelCellComplexType_Omega`. The proof is invariance under squares, applied to the previous statement and the two homotopy equivalences $\Omega^a \to B^a$ and $\Omega^b \to B^b$.

Where this departs from Milnor. Milnor applies 16.2 with 3.3 and 3.7 to each slab and requires one critical value in it. Here an interval may contain any finite number of critical values, whose cells are attached in order of energy by the induction of §IX.3. Each step uses its own manifold of broken geodesics, with an energy bound $c > b$; no relation between the subdivisions of different steps is needed.

Finitely many geodesics below each level. If every geodesic of energy $\le b < c$ has nullity 0, then the break points $y \in U$ of geodesics with $E'(y) \le b$ form a finite set `Chapter10/FundamentalTheorem · BrokenSetup.finite_geodesics`. With S and F as above, they are the critical points of $F|S$ in the compact set $(F|S)^b$, which form a closed, discrete set by non-degeneracy, hence a finite one. A broken geodesic is determined by its values on $[0, 1]$, since those determine its break points `Chapter10/FundamentalTheorem · BrokenSetup.eq_of_eq0n_path`. For N closed in V, $p, q \in N$ and every geodesic from p to q of nullity 0, it follows that the energies $\le b$ of geodesics from p to q form a finite set, for every b: take data with $c = \max(b, 0) + 1$ and map the finite set of break points by E', using (d) `Chapter10/FundamentalTheorem · finite_geodesicEnergies`. This is the discreteness of the critical values that Milnor draws from 16.3 at the start of the proof of 17.3.

The index of a geodesic. For $\gamma \in \Omega$ the index of E_{**} at γ is a priori in $\mathbb{N} \cup \{\infty\}$; the natural number ind$(\gamma)$ is defined as its truncation `Chapter10/FundamentalTheorem · geodesicIndex`. At a geodesic the index is finite by the index theorem (Theorem 15.1, Part VII) `Chapter9/IndexTheorem · theorem_15_1`, so the natural number is the index `Chapter10/FundamentalTheorem · geodesicIndex_eq`, and it agrees with the index of the broken geodesic with the same break points `Chapter10/FundamentalTheorem · BrokenSetup.geodIndex_eq_geodesicIndex`.

IX.6 Theorem 17.3

Theorem 17.3 with given levels. Let N be closed in V, $p, q \in N$, and every geodesic from p to q of nullity 0. Let $a_0 \le a_1 \le \cdots$ be levels tending to ∞, with $a_0 < 0$, none of them the energy of a geodesic. Then there are finite lists $G_0 = \emptyset, G_1, G_2, \dots$ of paths such that `Chapter10/FundamentalTheorem · theorem_17_3_of_levels`:

- every entry of G_{n+1} is a geodesic from p to q with energy in (a_n, a_{n+1});
- every geodesic from p to q with energy in (a_n, a_{n+1}) agrees on $[0,1]$ with an entry of G_{n+1};
- no two entries of one list agree on $[0,1]$, and each list is ordered by non-decreasing energy;
- $\Omega(N; p, q)$, with Milnor's metric, has the homotopy type of the countable cell complex with one cell of dimension index(γ) for each entry γ of G_0, then G_1, and so on.

The proof as formalized.

1. For each n choose data of §16 with the energy bound $c_n = \max(a_{n+1}, 0) + 1 > a_{n+1}$. They exist because N is closed, so the ρ-balls about p are compact.
2. With the data B_n, the relative Theorem 16.3 for Ω applies to $a_n \le a_{n+1} < c_n$: the levels are not energies of geodesics, and the nullities are 0. It gives a sorted list $\mathcal{Y}_n$ of break points and the relative cell structure of $\Omega^{a_n} \to \Omega^{a_{n+1}}$.
3. Put $G_0 = \emptyset$ and let G_{n+1} be the list of broken geodesics with break points in $\mathcal{Y}_n$. Their energies are the E'-values, in (a_n, a_{n+1}). A geodesic with energy in $(a_n, a_{n+1}) \subseteq (-\infty, c_n)$ is, on $[0,1]$, the broken geodesic of its break points by (d) of §IX.5, and these break points are in $\mathcal{Y}_n$. Two entries that agree on $[0,1]$ have the same break points, so they are equal entries, and $\mathcal{Y}_n$ has no repetition.
4. *The tower.* E is continuous on Ω and the a_n tend to ∞. Ω^{a_0} is empty, since $a_0 < 0 \le E$, so it has the homotopy type of the complex with no cells `Chapter2/CWType · IsFiniteCellComplexType.of_isEmpty`. The steps are those of step 2, with the indices read as natural numbers. The countable cell complex of sublevel sets (§IX.4) concludes.

Theorem 17.3 (Milnor p. 95). Let N be closed in V, $p, q \in N$, and every geodesic from p to q of nullity 0. Then there are finite lists $G_0, G_1, \ldots$ of geodesics from p to q such that every geodesic from p to q agrees on $[0,1]$ with an entry of some list; entries of different lists never agree on $[0,1]$, so the list is unique; no two entries of one list agree on $[0,1]$; the energies increase strictly from one list to the next and weakly within a list; and $\Omega(N; p, q)$, with Milnor's metric, has the homotopy type of the countable cell complex with one cell of dimension index(γ) for each entry γ `Chapter10/FundamentalTheorem · theorem_17_3`. That is, one λ-cell for each geodesic from p to q of index λ, attached in order of increasing energy.

The proof as formalized.

1. *The levels.* For each n the interval $(n, n+1)$ is infinite and the energies $\le n+1$ of geodesics form a finite set (§IX.5), so there is $x_n \in (n, n+1)$ that is not such an energy. Put $a_0 = -1$ and $a_{n+1} = x_n$. Then $n - 1 \le a_n < n$; the levels are monotone, tend to ∞, have $a_0 < 0$, and none is the energy of a geodesic (for a_0 because energies are ≥ 0).
2. Theorem 17.3 with these levels gives the lists. Every entry lies in a list G_{k+1}, since G_0 is empty.
3. *Every geodesic appears.* For a geodesic γ, choose K with $E(\gamma) + 1 < K$. Then $a_0 < E(\gamma) < a_K$, and as $E(\gamma)$ is none of the a_k there is k with $a_k < E(\gamma) < a_{k+1}$ `Chapter10/FundamentalTheorem · exists_mem_Ioo_succ_of_lt` (the least j with $E(\gamma) < a_j$ is positive; take $k = j - 1$). So γ agrees on $[0,1]$ with an entry of G_{k+1}.
4. *Exactly one list.* Paths agreeing on $[0,1]$ have the same energy `Chapter8/Hessian · energy_congr`. Entries of G_{k+1} and $G_{k'+1}$ have energies in (a_k, a_{k+1}) and $(a_{k'}, a_{k'+1})$, which are disjoint for $k \ne k'$ by monotonicity. The same observation gives the strict increase of energies between lists.

With Milnor's hypothesis. If q is not conjugate to p along any geodesic, every geodesic from p to q has nullity 0 (Theorem 14.1, §IX.1), and the theorem applies `Chapter10/FundamentalTheorem · theorem_17_3_nonconj`. By Corollary 18.2 this holds for almost every q (Part X).

What the proof needs that Milnor does not say, and what it does not need.

- **The levels must tend to infinity**, or the sets Ω^{a_i} do not exhaust Ω. Milnor's sequence $a_0 < a_1 < \cdots$ is not required to.
- **Exactly one critical value in each interval**, as Milnor requires (p. 95), is impossible when there are only finitely many geodesics, and at most one is what a cell-by-cell argument needs. The formal proof needs neither: the relative Theorem 3.5 attaches the cells of any finite number of critical values in one interval, in order of energy, so the levels can be taken in $(n, n+1)$ without looking at

the critical values. The condition of at most one critical value returns in the homology computation of §IX.9, where the homology of a slab is read off from its cells only when they all sit at one critical level.

- **The tower commutes only up to homotopy** (§IX.4), and each step uses its own manifold of broken geodesics. Milnor's strict ladder would need nested subdivisions.
- **No Whitehead theorem, no Corollary 17.2.** Milnor's main route proves that $\Omega \to K$ induces isomorphisms of homotopy groups and invokes Whitehead's theorem and Corollary 17.2 (Ω has the homotopy type of a CW complex), which rests on 17.1 and on Milnor's 1959 paper. Neither is used: the Appendix's comparison of telescopes needs only that Ω is the homotopy direct limit of the Ω^{a_n}, which holds because E is continuous for d.
- **Completeness** enters only through the compactness of the ρ-balls about p, which follows from closedness of N in V. Connectedness is not needed.
- **Geodesics are counted by their values on** $[0,1]$**.** Paths are functions on $\mathbb{R}$, so the cells correspond to classes of geodesics that agree on $[0,1]$.
- **Theorem 17.3 for** Ω^* would follow by composing with Theorem 17.1 (§IX.18); it is not stated.

IX.7 The geodesics of the round sphere

Milnor's example on pp. 95–96 lists the geodesics from p to $q \neq p, p'$ on S^n (p' the antipode of p) as $\gamma_0, \gamma_1, \ldots$, says that the interior of γ_k meets $\{p, p'\}$ exactly k times, each time at a point conjugate to p with multiplicity $n-1$, and concludes $\lambda(\gamma_k) = k(n-1)$. The formal proof derives each of these statements, plus the two that the argument needs and does not state: the list is complete, and p and q are not conjugate along any geodesic.

The sphere. $S^n \subseteq \mathbb{R}^{n+1}$ is the unit sphere `Milnor/Chapter5/HairyBall · Sph` with its slice structure `Milnor/Chapter6/EulerSphere · sphSub` and the induced metric. Its tangent projection is $P_x v = v - \langle x, v\rangle x$ `Chapter6/SphereCurvature · tanProj_sph`, its second fundamental form $\mathrm{II}_x(v,w) = -\langle v, w\rangle x$ `Chapter6/SphereCurvature · secondFundForm_sph`, and its curvature tensor, with Milnor's sign, $R(a,b)c = \langle a, c\rangle b - \langle b, c\rangle a$ on tangent vectors `Chapter6/SphereCurvature · curvOp_sph` (Part V). In what follows $n \geq 1$, which is the hypothesis wherever a unit vector orthogonal to a given one must exist.

Great circles. For $p \in S^n$, a unit vector $u \perp p$ and $L \in \mathbb{R}$, the *great circle* is $c(t) = \cos(Lt)\, p + \sin(Lt)\, u$ `Chapter10/SphereGeodesics · greatCircle`. Its velocity is $c'(t) = L(\cos(Lt)\, u - \sin(Lt)\, p)$ `Chapter10/SphereGeodesics · hasDerivAt_greatCircle`, of norm $|L|$ and orthogonal to $c(t)$, and $c'' = -L^2 c$ `Chapter10/SphereGeodesics · deriv_deriv_greatCircle`. Since $c'' = -\|c'\|^2 c = \mathrm{II}(c', c')$, the acceleration is normal and c is a geodesic on all of $\mathbb{R}$ `Chapter10/SphereGeodesics · isGeodesicOn_greatCircle` `Chapter7/Geodesic · isGeodesicOn_of_deriv_deriv_eq`. Its energy on $[0,1]$ is L^2 `Chapter10/SphereGeodesics · energy_greatCircle`.

Every geodesic from p is a great circle. If $\gamma \in \Omega(S^n; p, q)$ is a geodesic, there are a unit vector $u \perp p$ and $L \geq 0$ with $\gamma = c$ on $[0,1]$ `Chapter10/SphereGeodesics · exists_eqOn_greatCircle`.

The proof as formalized. Milnor's assertion that the geodesics are great circles rests on uniqueness of geodesics, but γ is only given on $[0,1]$, where its velocity at 0 is a one-sided derivative. So the proof first extends the segment past its ends: γ agrees on $[0,1]$ with the radial geodesic $t \mapsto \exp_p(tv)$, $v = \gamma'(0)$, which is a geodesic on some $(-\eta, 1+\eta)$ `Chapter9/GeodesicVariation · exists_initial_ray`. Its derivative at 0 is v, by uniqueness of derivatives within $[0,1]$, and v is tangent at p, so $\langle p, v\rangle = 0$. Let $u = v/\|v\|$, or any unit vector orthogonal to p if $v = 0$ `Chapter10/SphereGeodesics · exists_unit_orthogonal`, and $L = \|v\|$. The extension and the great circle are geodesics on $(-\eta, 1+\eta)$ with the same position and velocity at 0, so they agree there `Chapter7/Geodesic · IsGeodesicOn.eqOn`.

Two Jacobi fields. Let c be a great circle and e a vector orthogonal to p and u. Then $J(t) = \sin(Lt)\, e$ is a Jacobi field along c `Chapter10/SphereGeodesics · isJacobiField_sin_smul`. Indeed $e \perp c(t)$ for all t, so J is tangent and the covariant derivative is the ordinary one: $DJ/dt = L\cos(Lt)\, e$ and $D^2J/dt^2 = -L^2 J$. As $J \perp c'$ and $\|c'\|^2 = L^2$, the curvature term is $R(c', J)c' = \|c'\|^2 J - \langle J, c'\rangle c' = L^2 J$, and the

Jacobi equation $D^2J/dt^2 + R(c', J)c' = 0$ holds `Chapter8/Jacobi · IsJacobiField`. Milnor obtains such fields from rotations and Lemma 14.3; the formal proof checks the equation directly. The field $t\,c'(t)$ is a Jacobi field along any geodesic `Chapter9/JacobiSpace · isJacobiField_smul_vel`, in particular along c `Chapter10/SphereGeodesics · isJacobiField_smul_deriv_greatCircle`.

Conjugate points along a great circle. Let $L > 0$ and $0 < t \le 1$. The multiplicity of 0 and t as conjugate parameters along c (the dimension of the space of Jacobi fields vanishing at both `Chapter9/JacobiSpace · conjSpace` `Chapter9/JacobiSpace · conjMultiplicity`) is $n-1$ if Lt is a positive multiple of π, and 0 otherwise `Chapter10/SphereGeodesics · conjMultiplicity_greatCircle`.

The proof as formalized.

1. Let D send a Jacobi field J of the conjugate space to its initial covariant derivative $DJ/dt(0)$. It is linear and injective: a Jacobi field with $J(0) = 0$ and $DJ/dt(0) = 0$ vanishes `Chapter9/JacobiSpace · jacobiSpace_eq_zero` `Chapter9/JacobiSystem · IsJacobiField.eqOn_zero`. So the multiplicity is the dimension of the range of D.
2. Let J be in the conjugate space and $y = DJ/dt(0)$. Then $y \perp p$, being tangent at p. Write $a = \langle u, y\rangle$ and $e = y - au$, so $e \perp p, u$. The field $K = L^{-1}\sin(Ls)\,e + (a/L)\,s\,c'(s)$ is a Jacobi field (the two fields above, made fields on $[0,1]$ `Chapter9/JacobiSpace · IsJacobiField.comp_projIcc_mem`) with $K(0) = 0$ and $DK/dt(0) = e + au = y$. A Jacobi field is determined by its value and covariant derivative at 0 `Chapter9/JacobiSpace · jacEval_bijective`, so $J = K$.
3. From $J(t) = 0$: $L^{-1}\sin(Lt)\,e + at\,w(t) = 0$, with $w(t) = c'(t)/L$ a unit vector orthogonal to e. Pairing with $w(t)$ gives $at = 0$, so $a = 0$ as $t > 0$. Then $\sin(Lt)\,e = 0$, so if $\sin(Lt) \neq 0$ then $e = 0$ and $y = 0$.
4. Conversely, if $\sin(Lt) = 0$, every $y \perp p, u$ is $DJ/dt(0)$ for $J = L^{-1}\sin(Ls)\,y$, which vanishes at 0 and at t.
5. So the range of D is $\{p, u\}^\perp$, of dimension $n-1$ `Chapter10/SphereGeodesics · finrank_orthogonal_span_pair`, if $\sin(Lt) = 0$, and 0 otherwise. As $Lt > 0$, $\sin(Lt) = 0$ means $Lt = k\pi$ with $k \in \mathbb{N}$.

On p. 80 Milnor obtains $n-1$ independent Jacobi fields from rotations, which shows multiplicity at least $n-1$, and only for the antipode at the end of a semicircle. The statement used on p. 96 needs it at p and p' after any number of half-turns, needs equality, and needs that no other point of the circle is conjugate to p. Determining the whole n-dimensional space of Jacobi fields vanishing at 0, as above, gives all three.

Milnor's geodesics γ_k. For $p, q \in S^n$ with $q \neq \pm p$, let $\theta = \arccos\langle p, q\rangle \in (0, \pi)$ `Chapter10/SphereGeodesics · sphAngle` `Chapter10/SphereGeodesics · sphAngle_pos` `Chapter10/SphereGeodesics · sphAngle_lt_pi`, and $u = (q - \cos\theta\, p)/\sin\theta$ `Chapter10/SphereGeodesics · sphDir`, a unit vector orthogonal to p with $q = \cos\theta\, p + \sin\theta\, u$ `Chapter10/SphereGeodesics · norm_sphDir` `Chapter10/SphereGeodesics · inner_sphDir` `Chapter10/SphereGeodesics · eq_sphDir`. Put

$$L_{2j} = \theta + 2\pi j, \qquad L_{2j+1} = 2\pi(j+1) - \theta$$

`Chapter10/SphereGeodesics · sphLen`, and let γ_k be the great circle along u with $L = L_k$ for even k, and along $-u$ with $L = L_k$ for odd k `Chapter10/SphereGeodesics · sphGeod`. Then $\gamma_k(1) = q$ by periodicity of cos and sin `Chapter10/SphereGeodesics · sphGeod_one`, so γ_k is a geodesic in $\Omega(S^n; p, q)$ `Chapter10/SphereGeodesics · sphGeod_mem_pathSpace` `Chapter10/SphereGeodesics · isUnbrokenGeodesic_sphGeod` with energy L_k^2 `Chapter10/SphereGeodesics · energy_sphGeod`. From $0 < \theta < \pi$,

$$k\pi < L_k < (k+1)\pi$$

`Chapter10/SphereGeodesics · mul_pi_lt_sphLen` `Chapter10/SphereGeodesics · sphLen_lt`, so the lengths are positive and strictly increasing `Chapter10/SphereGeodesics · sphLen_pos` `Chapter10/SphereGeodesics · sphLen_strictMono`. Milnor's description of k as the number of passages of γ_k through p or p' in its interior amounts to this inequality: $\gamma_k(t) = \pm p$ exactly when $L_k t \in \pi\mathbb{Z}$, that is at $t = j\pi/L_k$, $1 \le j \le k$, in $(0, 1)$.

The γ_k are all the geodesics from p to q. Every geodesic $\gamma \in \Omega(S^n; p, q)$ agrees on $[0,1]$ with some γ_k `Chapter10/SphereGeodesics · exists_eqOn_sphGeod`.

The proof as formalized. γ is a great circle $\cos(Lt)\,p + \sin(Lt)\,w$ with $L \geq 0$. From $\gamma(1) = q$, pairing with p gives $\cos L = \cos\theta$, and then $\sin L \cdot w = \sin\theta \cdot u$. The equality of cosines gives an integer m with $L = \theta + 2\pi m$ or $L = 2\pi m - \theta$ Mathlib/Analysis/SpecialFunctions/Trigonometric/Complex · Real.cos_eq_cos_iff. In the first case $\sin L = \sin\theta > 0$, so $w = u$; and $m \geq 0$, since $m \leq -1$ would make $L < 0$. So $\gamma = \gamma_{2m}$. In the second case $\sin L = -\sin\theta$, so $w = -u$; and $m \geq 1$, since $L \geq 0$ and $\theta > 0$. So $\gamma = \gamma_{2(m-1)+1}$. This uses $q \neq \pm p$ twice: for $\sin\theta > 0$, and through γ not being constant.

The index of γ_k is $k(n-1)$. By the index theorem, the index of E_{**} at γ_k is $\sum_{0<t<1}$ of the multiplicity of 0 and t Chapter9/IndexTheorem · theorem_15_1. By the conjugate points along a great circle Chapter10/SphereGeodesics · conjMultiplicity_sphGeod, the summand is $n-1$ when $L_k t \in \pi\mathbb{N}$ and 0 otherwise. For $k\pi < L < (k+1)\pi$ such a sum is $k(n-1)$: the support is $\{j\pi/L : 1 \leq j \leq k\}$, which has k elements Chapter10/SphereGeodesics · finsum_mem_Ioo_ite_eq. Hence the index is $k(n-1)$ Chapter10/SphereGeodesics · formIndex_sphGeod. Milnor uses the index theorem here without saying so.

The index only depends on $[0,1]$. For any N, two geodesics of Ω that agree on $[0,1]$ have the same index Chapter10/SphereGeodesics · formIndex_energyHessian_congr: by the index theorem both indices count conjugate points, and the Jacobi fields along two curves agreeing on $[0,1]$ are the same Chapter9/JacobiVariation · IsJacobiField.congr_curve. So every geodesic from p to q agreeing with γ_k on $[0,1]$ has index $k(n-1)$ Chapter10/SphereGeodesics · formIndex_eq_of_eqOn_sphGeod. This is needed only because paths are functions on $\mathbb{R}$.

p and q are not conjugate along any geodesic Chapter10/SphereGeodesics · notMem_conjLocus_sph. Suppose q were in the conjugate locus of p, along a geodesic γ. Then $\gamma = \gamma_k$ on $[0,1]$ for some k, and conjugacy only depends on the curve on $[0,1]$ Chapter11/NonConjugate · IsConjugate.congr_curve_iff. So 0 and 1 would be conjugate along γ_k, with positive multiplicity Chapter9/JacobiSpace · isConjugate_iff. But at $t = 1$ the multiplicity is 0 unless $L_k \in \pi\mathbb{N}$, and $k\pi < L_k < (k+1)\pi$ excludes that. Milnor rephrases non-conjugacy as $q \neq p, p'$; the direction proved here is the one that Theorem 17.3 needs.

IX.8 The homology of an exhaustion, and of a slab

Milnor reads the homology of ΩS^n off the cells of Corollary 17.4 (p. 96). In the formalization the cell complexes have non-cellular attaching maps, so the homology is computed from the energy filtration, with two general tools.

Homology is singular homology with coefficients in a field $\Bbbk$ (§0.6, Part IV). For subsets $A \subseteq B$ of a space, $H_i(B, A; \Bbbk)$ is the relative homology of the pair Homology/TripleFiltration · subsetHomology, $H_i(X; \Bbbk)$ the homology of a space Homology/MorseHomologyBasic · spaceHomology, and $b_i(X) = \dim_{\Bbbk} H_i(X; \Bbbk)$ its Betti number Homology/MorseHomologyBasic · bettiNumber. A triple $A \subseteq B \subseteq D$ has the long exact sequence $\cdots \to H_i(B, A) \to H_i(D, A) \to H_i(D, B) \to H_{i-1}(B, A) \to \cdots$ Homology/TripleFiltration · subsetTripleLongExactSeq Chapter3/Inequalities · LongExactSeq. Two facts about such sequences are used: the map $H_i(B, A) \to H_i(D, A)$ is bijective when $H_i(D, B)$ and $H_{i+1}(D, B)$ vanish Chapter10/SublevelHomology · LongExactSeq.bijective_f_of_subsingleton, and $H_i(D, A)$ vanishes when $H_i(B, A)$ and $H_i(D, B)$ do Chapter10/SublevelHomology · LongExactSeq.subsingleton_B.

Exactness passes to an exhausted complex. Let $\varphi_k : Q_k \to Q$ be chain maps of complexes of modules, injective in every degree, such that every chain of Q in degree j comes from some Q_k. If every Q_k is exact in degree j, so is Q Chapter10/SublevelHomology · exactAt_of_injective_of_exists. A cycle x of Q is $\varphi_k(y)$; y is a cycle, because φ_k commutes with the differential and is injective; y bounds in Q_k, so x bounds in Q.

Homology commutes with exhaustions (Hatcher, Proposition 3.33). Let $A_0 \subseteq A_1 \subseteq \cdots$ be subsets of B such that every compact subset of B lies in some A_k. If $H_j(A_k, A_{k_0}; \Bbbk) = 0$ for all $k \geq k_0$, then $H_j(B, A_{k_0}; \Bbbk) = 0$ Chapter10/SublevelHomology · subsingleton_subsetHomology_of_exhaustion Chapter10/SublevelHomology · subsingleton_subsetHomology_univ_of_exhaustion.

The proof as formalized. The relative homology $H_j(B, A)$ vanishes iff the quotient complex of the singular chains of B by the chains made of simplices in A is exact in degree j `Chapter10/SublevelHomology · subsingleton_subsetHomology_iff_exactAt Homology/Excision · relChainsIsoQuot`. Apply the previous lemma to the maps $C(A_k)/C(A_{k_0}) \to C(B)/C(A_{k_0})$, $k \geq k_0$ `Homology/Excision · excisionMap`. They are injective: a chain of A_k whose image in B is made of simplices in A_{k_0} is itself made of such simplices `Homology/Excision · push_mem_supported_iff`. Every chain of B comes from some A_k: it is a finite combination of singular simplices, the union of their images is compact, so it lies in some A_k by hypothesis, and then in $A_{\max(k,k_0)}$ `Homology/Excision · range_push`. Each quotient for A_k is exact in degree j by hypothesis.

From the steps to all pairs. If $H_j(A_{k+1}, A_k; \Bbbk) = 0$ for all $k \geq k_0$, then $H_j(A_k, A_{k_0}; \Bbbk) = 0$ for all $k \geq k_0$ `Chapter10/SublevelHomology · subsingleton_subsetHomology_of_steps`. This is an induction on k, starting from $H_j(A_{k_0}, A_{k_0}) = 0$ `Homology/TripleFiltration · subsingleton_subsetHomology_self`, along the exact sequences of the triples $A_{k_0} \subseteq A_k \subseteq A_{k+1}$.

The homology of an exhausted space. If every compact subset of X lies in some A_k and the groups H_i and H_{i+1} of the pairs (A_k, A_{k_0}), $k \geq k_0$, vanish, then the inclusion induces $H_i(A_{k_0}, \emptyset; \Bbbk) \cong H_i(X; \Bbbk)$ `Chapter10/SublevelHomology · subsetHomologyEquivOfExhaustion Chapter10/SublevelHomology · spaceHomologyEquivOfExhaustion Chapter10/SublevelHomology · nonempty_linearEquiv_spaceHomology_of_exhaustion Chapter10/SublevelHomology · spaceHomologyEquivOfExhaustion'`. The proof is the exact sequence of the triple $\emptyset \subseteq A_{k_0} \subseteq X$, whose third terms $H_i(X, A_{k_0})$ and $H_{i+1}(X, A_{k_0})$ vanish by the previous results, with $H_i(X, \emptyset) \cong H_i(X)$ `Homology/MorseHomologyBasic · subsetHomologyUnivEquiv`. The consequences for Betti numbers and finite dimension follow `Chapter10/SublevelHomology · bettiNumber_eq_of_exhaustion Chapter10/SublevelHomology · finiteDimensional_spaceHomology_of_exhaustion`, as do the versions with hypotheses on the steps (A_{k+1}, A_k) only `Chapter10/SublevelHomology · spaceHomologyEquivOfSteps Chapter10/SublevelHomology · bettiNumber_eq_of_steps`. The vanishing is needed in degree $i + 1$ for injectivity and in degree i for surjectivity.

Sublevel sets exhaust the compact sets. If $f : X \to \mathbb{R}$ is continuous and the levels a_k are unbounded above, every compact $K \subseteq X$ lies in some X^{a_k}, since f is bounded on K `Chapter10/SublevelHomology · exists_subset_sublevel_of_isCompact`. For the path space this is where the topology enters: with Milnor's metric E is continuous, so compact subsets of Ω have bounded energy. For uniform convergence alone, in which E is only lower semicontinuous, this fails: a uniformly convergent sequence of paths whose energies tend to ∞, together with its limit, is compact.

One critical level inside a compact sublevel set (Milnor p. 29, for a manifold that is not compact). Let $f = F|N$ as in §IX.3, N^{a_0} compact, S a finite set of non-degenerate critical points all at the level c, and $a' < c < a \leq a_0$ with no other critical value in $[a', a]$. Then $H_i(N^a, N^{a'}; \Bbbk)$, read in any space T homeomorphic to N, is finite-dimensional, of dimension the number of points of S of index i `Chapter10/SublevelHomology · Homology.sublevel_step_critical_of_isCompact`. The proof is that of §IX.3, with homology in place of homotopy type: Remark 3.3 at the level c, Theorem 3.1 on the compact slabs $[a', c-\varepsilon]$ and $[c+\varepsilon, a]$, and the homology of finitely many disjoint cells attached to a closed set `Homology/MorseHomologyBasic · finrank_subsetHomology_of_cells Homology/MorseHomologyCells · cellAttachHomology` (Part IV).

The homology of a slab inside a compact sublevel set. Let the critical points of f in N^{a_0} be non-degenerate, let P be a finite set whose points of value $\leq a$ are exactly the critical points of value $\leq a$, let $a' \leq a \leq a_0$ be regular values, and suppose **all critical values in (a', a) are equal**. Then $H_i(N^a, N^{a'}; \Bbbk)$ is finite-dimensional, of dimension the number of critical points of index i in $f^{-1}(a', a)$ `Chapter10/SublevelHomology · Homology.sublevel_step_of_isCompact`. If some critical value lies in (a', a), all those critical points sit at one level c and the previous result applies. If none does, Theorem 3.1 on the compact slab $[a', a]$ makes $N^{a'}$ a strong deformation retract of N^a, so the relative homology vanishes `Homology/MorseHomologyBasic · subsingleton_subsetHomology_of_isSDR`, and so does the count.

IX.9 The homology of the path space

The homology of a slab of Ω (Milnor p. 29, in the path space). Let $a \le b < c$ be levels that are not energies of geodesics, every geodesic of energy $\le b$ of nullity 0, and **all geodesics with energy in (a,b) of the same energy**. Then $H_j(\Omega^b, \Omega^a; \Bbbk)$ is finite-dimensional, of dimension the number of break points $y \in U$ of geodesics with energy in (a,b) and index j `Chapter10/PathHomology · BrokenSetup.finrank_subsetHomology_Omega`.

The proof as formalized. Take c_1, F and S as in §IX.5, and P the finite set of critical points of $F|S$ in the compact set $(F|S)^b$. The slab homology of the previous section, with $a_0 = b$ and $T = S$, gives $H_j((F|S)^b, (F|S)^a)$; its hypothesis that all critical values in (a,b) are equal is the hypothesis on the energies. Then compare pairs. The homotopy equivalences $\alpha : \Omega^a \to (F|S)^a$ and $\beta : \Omega^b \to (F|S)^b$, built from the one formula of §IX.5, form a map of pairs, and a map of pairs whose two components are homotopy equivalences induces isomorphisms of relative homology, by the five lemma `Homology/RelativeHomotopy · isIso_relHomology_map_of_homotopyEquiv`. Finally the critical points of index j are counted as break points, by Theorem 16.2.

This is where Milnor's requirement of one critical value per interval is genuinely used, in the weak form of at most one. With all the critical points of a slab on one level, their cells are attached simultaneously to the same sublevel set, and the homology of the slab is spanned by them, one generator per cell. With two critical levels in one slab, the boundary of a cell at the upper level could meet a cell at the lower level, and the homology of the slab would no longer be read off from the list of cells.

One geodesic in each window. Let $g_0, g_1, \ldots$ be geodesics from p to q with strictly increasing energies $E_k = E(g_k)$. The *window levels* are $a_0 = -1$ and $a_{k+1} = (E_k + E_{k+1})/2$ `Chapter10/PathHomology · windowLevels`. Then $E_k < a_{k+1} < E_{k+1}$ `Chapter10/PathHomology · lt_windowLevels_succ Chapter10/PathHomology · windowLevels_succ_lt`, so E_j lies in the window (a_k, a_{k+1}) iff $j = k$ `Chapter10/PathHomology · mem_window_iff`; the levels are monotone `Chapter10/PathHomology · windowLevels_monotone` and tend to ∞ when the energies do `Chapter10/PathHomology · tendsto_windowLevels`. The cells of such a space are recorded as no cell at stage 0 and one cell of dimension d_k at stage $k+1$ `Chapter10/PathHomology · oneCellEach`.

One cell for each geodesic, one geodesic per window. Let N be closed in V, $p, q \in N$, every geodesic from p to q of nullity 0, and suppose the geodesics from p to q are, up to their values off $[0,1]$, a sequence $g_0, g_1, \ldots$ with strictly increasing energies tending to ∞, every geodesic agreeing with g_k on $[0,1]$ having index d_k. Then $\Omega(N; p, q)$ has the homotopy type of the countable cell complex with one cell of dimension d_k at stage $k+1$ `Chapter10/PathHomology · omega_cells_of_unique`.

The proof as formalized. No geodesic energy is a window level: a geodesic agrees with some g_j on $[0,1]$, so has energy E_j, which is $\ge 0 > a_0$ and differs from every a_{k+1}. Theorem 17.3 with the window levels gives lists G_{k+1}. Every entry of G_{k+1} has energy in the k-th window, so it agrees with g_k on $[0,1]$. Some entry agrees with g_k, since g_k has energy in the window. No two entries agree on $[0,1]$, while all agree with g_k, so the list has exactly one entry `Chapter10/PathHomology · List.eq_singleton_of_pairwise_not`, of index d_k by §IX.5.

The homology of each window. Under the same hypotheses, except that the energies need not tend to ∞, $H_j(\Omega^{a_{k+1}}, \Omega^{a_k}; \Bbbk)$ is $\Bbbk$ for $j = d_k$ and 0 otherwise `Chapter10/PathHomology · finrank_window`. Take data of §16 with $c = \max(a_{k+1}, 0) + 1$ and apply the slab homology: all geodesics in the window agree with g_k on $[0,1]$, so they have one energy. The break points in the window with index j are those of g_k if $j = d_k$, and there are none otherwise, because a broken geodesic is determined by its values on $[0,1]$.

Lacunary indices (Milnor p. 96, for ΩS^n, $n \ge 3$). If moreover the energies tend to ∞ and $d_k = kr$ with $r \ge 2$, then for every field $\Bbbk$ and every i, $b_i(\Omega(N; p, q)) = 1$ if r divides i, and 0 otherwise `Chapter10/PathHomology · bettiNumber_omega_of_unique`.

The proof as formalized.

1. Let $X_k = \Omega^{a_k}$ for the window levels. Then $X_0 = \emptyset$, as $a_0 = -1 < 0 \le E$, and every compact subset of Ω lies in some X_k, because E is continuous and the levels tend to ∞.
2. Fix i and put $K = i + 2$. For $k \ge K$ the steps (X_{k+1}, X_k) have homology only in degree $kr \ge Kr \ge 2i + 4 > i + 1$, so they vanish in degrees i and $i + 1$. Passing to the union, $b_i(\Omega) = \dim H_i(X_K, \emptyset) = \dim H_i(X_K, X_0)$ `Chapter10/SublevelHomology · bettiNumber_eq_of_steps` `Homology/TripleFiltration · finrank_subsetHomology_congr`. This uses step 1 only through compactness: no cell structure on Ω is needed.
3. The Morse numbers of the finite filtration $X_0 \subseteq \cdots \subseteq X_K$ are $C_j = \sum_{k<K} \dim H_j(X_{k+1}, X_k)$ `Homology/TripleFiltration · singularMorseNumber`. By the window homology, $C_j = 1$ if $r \mid j$ and $j/r < K$, and $C_j = 0$ otherwise.
4. The strong Morse inequalities for singular homology hold for the filtration `Homology/TripleFiltration · singular_morse_inequality` (Part IV). No two consecutive C_j are nonzero: $r \mid j$ and $r \mid j + 1$ would give $r \mid 1$. So the lacunary principle (the remark on $\mathbb{C}P^n$, p. 31) gives $\dim H_j(X_K, X_0) = C_j$ for every j `Chapter3/Inequalities · eq_of_altSum_le_of_lacunary`.
5. For $j = i$: if $r \mid i$ then $i/r \le i < K$, so $C_i = 1$; otherwise $C_i = 0$.

Milnor computes integral homology from the cellular chain complex, whose differentials vanish because no two cells have consecutive dimensions. Here the coefficients form a field, and the same gap is used through the lacunary principle on finite stages. The integral statement is not formalized.

IX.10 Corollary 17.4 and the homology of $\Omega(S^n; p, q)$

Let $p, q \in S^n$ with $q \ne \pm p$. The energies $E(\gamma_k) = L_k^2$ increase strictly, as the L_k are positive and increase `Chapter10/SpherePaths · strictMono_energy_sphGeod`, and tend to ∞, as $L_k > k\pi$ `Chapter10/SpherePaths · tendsto_energy_sphGeod`.

Corollary 17.4 (Milnor p. 96), for the path space. For $n \ge 2$ and $p, q \in S^n$ with $q \ne \pm p$, $\Omega(S^n; p, q)$, with Milnor's metric, has the homotopy type of the countable cell complex with one cell in each of the dimensions $0, n - 1, 2(n - 1), \ldots$, attached in this order `Chapter10/SpherePaths · corollary_17_4`.

The proof as formalized applies the theorem on one geodesic per window (§IX.9) with $N = S^n$, which is closed in $\mathbb{R}^{n+1}$ as it is compact `Milnor/Chapter6/EulerSphere · isCompact_sph`. Every geodesic from p to q has nullity 0, because q is not in the conjugate locus of p (§IX.7). The sequence is $g_k = \gamma_k$: these are all the geodesics, with strictly increasing energies tending to ∞, and every geodesic agreeing with γ_k on $[0, 1]$ has index $d_k = k(n - 1)$.

The homology of $\Omega(S^n; p, q)$ (Milnor p. 96). For $n \ge 3$, $q \ne \pm p$ and every field $\Bbbk$, $\dim_{\Bbbk} H_i(\Omega(S^n; p, q); \Bbbk)$ is 1 if $n - 1$ divides i, and 0 otherwise `Chapter10/SpherePaths · bettiNumber_sphere_paths`. This is the lacunary case of §IX.9 with $r = n - 1 \ge 2$.

What Milnor leaves implicit. Milnor states 17.4 for the loop space $\Omega(S^n)$, but his proof applies 17.3, which needs non-conjugate end points, while p is conjugate to itself along every great circle. Passing to loops needs the independence of $\Omega^*(M; p, q)$ from q up to homotopy (Lemma 17.D of the exposition) together with Theorem 17.1; that lemma is not formalized, so the formal statements are for $q \ne \pm p$, which is what the argument gives. The case $n = 1$, where every cell is a point, is not stated. For $n = 2$ there is a cell in every dimension and the lacunary principle does not apply, which is why the homology needs $n \ge 3$, Milnor's $n > 2$.

IX.11 Completeness cannot be dropped: $\mathbb{R}^{n+1} \setminus \{0\}$

Corollary 17.5 (p. 96) asserts that two non-conjugate points of a manifold with the homotopy type of S^n, $n > 2$, are joined by infinitely many geodesics. Its proof rests on Theorem 17.3, which assumes

completeness; the statement does not. The formalization proves that the hypothesis cannot be dropped, with $N = \mathbb{R}^{n+1} \setminus \{0\}$ and the flat metric.

Open sets are submanifolds of codimension 0. For a zero-dimensional space W, the map $x \mapsto (x, 0)$ is a global diffeomorphism $V \to V \times W$ `Chapter10/Incomplete · zeroGraph`. Restricted to an open set O it is a slice chart for O whose slice condition holds automatically, as its whole domain lies in O `Chapter10/Incomplete · openSliceChart`. So O is a slice submanifold of V modelled on V `Chapter10/Incomplete · isSliceSubmanifold_of_isOpen`, and every tangent space of O is all of V, computed in that chart `Chapter10/Incomplete · tangentSubspace_of_isOpen`.

Two facts of calculus. A function with vanishing second derivative on an open interval is affine there `Chapter10/Incomplete · eq_add_smul_of_hasDerivAt`, and a function continuous on $[0, 1]$ with vanishing second derivative on $(0, 1)$ is $f(t) = f(0)+t(f(1)-f(0))$ on $[0, 1]$ `Chapter10/Incomplete · eqOn_affine_of_hasDerivAt`.

Codimension 0 *means flat.* Suppose every tangent space of N is V. Then:

- the tangent projection is the identity `Chapter10/Incomplete · tanProj_eq_id_of_top` `Chapter10/Incomplete · prj_eq_id_of_top`, so the second fundamental form $\mathrm{II} = dP$ vanishes `Chapter10/Incomplete · secondFundForm_eq_zero_of_top`, and the curvature $R(a, b)c = \mathrm{II}(b, \mathrm{II}(a, c)) - \mathrm{II}(a, \mathrm{II}(b, c))$ vanishes `Chapter6/Submanifold · curvOp_apply` `Chapter10/Incomplete · curvR_eq_zero_of_top`;
- a geodesic has acceleration normal to V, hence zero `Chapter10/Incomplete · IsGeodesicOn.deriv_deriv_eq_zero_of_top`, so it is a straight line `Chapter10/Incomplete · IsGeodesicOn.eq_add_smul_of_top`, and a geodesic from p to q is the segment $t \mapsto p + t(q - p)$ on $[0, 1]$ `Chapter10/Incomplete · IsUnbrokenGeodesic.eqOn_segment_of_top`;
- the Jacobi equation becomes $J'' = 0$, since $D/dt = d/dt$ and $R = 0$ `Chapter10/Incomplete · IsJacobiField.deriv_deriv_eq_zero_of_top`, so Jacobi fields are affine `Chapter10/Incomplete · IsJacobiField.eqOn_affine_of_top`. An affine field vanishing at two parameters vanishes identically, so no two parameters are conjugate along any curve `Chapter10/Incomplete · not_isConjugate_of_top` and every conjugate locus is empty `Chapter10/Incomplete · notMem_conjLocus_of_top`;
- by the index theorem every geodesic has index 0 `Chapter10/Incomplete · formIndex_energyHessian_eq_zero_of_top`, and by Theorem 14.1 nullity 0 `Chapter10/Incomplete · hessNullity_eq_zero_of_top`;
- if the segment from p to q lies in N, it is a geodesic in $\Omega(N; p, q)$ `Chapter10/Incomplete · segment_mem_pathSpace` `Chapter10/Incomplete · isUnbrokenGeodesic_segment`, and every geodesic in $\Omega(N; p, q)$ agrees with it on $[0, 1]$ `Chapter10/Incomplete · unique_geodesic_of_top`.

The punctured space. $N = \mathbb{R}^{n+1} \setminus \{0\}$ is open, so all of this applies `Chapter10/Incomplete · puncturedSub` `Chapter10/Incomplete · tangentSubspace_puncturedSub` `Chapter10/Incomplete · curvR_puncturedSub` `Chapter10/Incomplete · notMem_conjLocus_puncturedSub` `Chapter10/Incomplete · formIndex_energyHessian_puncturedSub` `Chapter10/Incomplete · eqOn_segment_puncturedSub` `Chapter10/Incomplete · unique_geodesic_puncturedSub`. It is not complete in either sense. The geodesic from q with initial velocity $-q$ is straight, $t \mapsto (1 - t)q$, and reaches the missing origin at $t = 1$, so $\exp_q(-q)$ is undefined `Chapter10/Incomplete · neg_notMem_expDom_puncturedSub` `Chapter7/Exp · expDom` and N is not geodesically complete `Chapter10/Incomplete · not_isGeodesicallyComplete_puncturedSub` `Chapter7/GeodesicFlow · IsGeodesicallyComplete`; nor is it closed in $\mathbb{R}^{n+1}$ `Chapter10/Incomplete · not_isClosed_punctured`. For $n \geq 1$ it is path-connected `Chapter10/Incomplete · isPathConnected_punctured` `Mathlib/Analysis/Normed/Module/Connected · isPathConnected_compl_singleton_of_one_lt_rank`, and it has the homotopy type of S^n: the radial projection $x \mapsto x/\|x\|$ `Chapter10/Incomplete · radialProj` and the inclusion `Chapter10/Incomplete · sphIncl` are inverse homotopy equivalences, through the straight-line homotopy $((1 - t)/\|x\| + t)\, x$, whose coefficient is positive so the homotopy avoids the origin `Chapter10/Incomplete · radialCoeff_pos` `Chapter10/Incomplete · puncturedHomotopyEquiv`.

The counterexample, packaged. For $n \geq 1$: N has the homotopy type of S^n and is path-connected; it is neither closed in $\mathbb{R}^{n+1}$ nor geodesically complete; no point of N is conjugate to another along any geodesic; and there are $p \neq q$ in N, namely $p = e$ and $q = 2e$ for the first standard basis vector e, joined by a geodesic with which every geodesic from p to q agrees on $[0, 1]$ `Chapter10/Incomplete · remark_17_5_completeness`. For $n \geq 3$, N satisfies the hypotheses of Corollary 17.5 as printed, and not its conclusion.

For every $n \geq 1$ this also contradicts the conclusion of Theorem 17.3 without completeness, which would make $\Omega(N; p, q)$ a single 0-cell up to homotopy, whereas it has the homotopy type of the loop space of S^n, which is not contractible. That comparison is not formalized; what is proved is that N satisfies every hypothesis of the formal Theorem 17.3 except closedness in the ambient space, which is where completeness enters it.

IX.12 Milnor's proof of Theorem 17.1, and the step that does not hold as we read it

Theorem 17.1 (p. 93). The inclusion $i : \Omega \to \Omega^*$ is a homotopy equivalence.

Milnor's construction (pp. 93–94), paraphrased. Choose a continuous $f : M \to [0, \infty)$ whose sublevel sets $f^{-1}[0, r]$ are compact. For each r let $\varepsilon_1(r) \leq \infty$ be the largest number such that any two points of $f^{-1}[0, r]$ at distance less than $\varepsilon_1(r)$ are joined by a unique minimal geodesic, depending differentiably on its end points; Corollary 10.8 makes ε_1 positive. Choose a continuous ε_2 with $0 < \varepsilon_2 < \varepsilon_1$. For $\omega \in \Omega^*$ put $\varepsilon(\omega) = \varepsilon_2(\max_t f(\omega(t)))$ and

$$F(\omega, \alpha) = (\alpha - 1)\, \varepsilon(\omega) + \max_{|t - t'| \leq \alpha} \rho(\omega(t), \omega(t')).$$

$F(\omega, \cdot)$ is strictly increasing, negative at $\alpha = 0$ and non-negative at $\alpha = 1$; its zero is called $2A(\omega)$, and A depends continuously on ω. At the zero, two points of ω with parameters at most $2A$ apart are at distance at most $(1 - 2A)\varepsilon(\omega) < \varepsilon_1$, so they are joined by a unique minimal geodesic. Let k be the largest integer $\leq 1/A(\omega)$. The path $h(\omega)$ agrees with ω at $t = 0, A, 2A, \ldots, kA$ and at $t = 1$, and is a minimal geodesic on each of the intervals between. Milnor then writes "Clearly h is continuous" (p. 94), and says that arguments as in §16 give homotopies $h \circ i \simeq \mathrm{id}$ and $i \circ h \simeq \mathrm{id}$.

What fails. As we read it, the claim does not hold for the metric d of Ω. The number of intervals of $h(\omega)$ changes when $1/A(\omega)$ passes through an integer. Suppose $1/A(\omega_0) = k$ exactly. Then $kA(\omega_0) = 1$, the last interval $[kA, 1]$ is empty, and $h(\omega_0)$ has k segments. For ω near ω_0 with $A(\omega)$ slightly smaller, still $k(\omega) = k$, but now $kA(\omega) = 1 - \delta$ with a small $\delta > 0$. So $h(\omega)$ has one more segment, from $\omega(1 - \delta)$ to $\omega(1) = q$ in time δ. Its length $\ell = \rho(\omega(1 - \delta), q)$ is small, but its energy is ℓ^2/δ. For a piecewise smooth ω this is harmless: by Cauchy–Schwarz ℓ^2/δ is at most the energy of ω on the last δ of time, and along a sequence converging in Ω that energy is uniformly small (Lemma 16.B of the exposition, Part VIII). But h must be defined and continuous on all of Ω^*, and for a path that is merely continuous ℓ can be as large as $\delta^{1/3}$, so that $\ell^2/\delta \approx \delta^{-1/3} \to \infty$. Since the energy is continuous on (Ω, d), $h(\omega)$ then cannot converge to $h(\omega_0)$ in Ω.

The formalization turns this into a theorem, for Milnor's own construction with $M = \mathbb{R}$. Everything below is in `Chapter10/MilnorH`, most of it in the namespace `MilnorH`.

Milnor's choices for $M = \mathbb{R}$. Take $p = q = 0$. Then Ω^* is the space of continuous $\omega : [0, 1] \to \mathbb{R}$ with $\omega(0) = \omega(1) = 0$, with the uniform metric, and Ω the space of piecewise smooth such paths, with Milnor's metric (§IX.1, with $N = \mathbb{R}$).

- *The distance* is $\rho(x, y) = |x - y|$ `Chapter10/MilnorH · riemDist_real`. The segment $t \mapsto x + t(y - x)$ has length $|x - y|$, which bounds ρ from above `Chapter7/Distance · riemDist_le_pathLength`, and $\rho(x, y) \geq |x - y|$ always (§IX.1).
- *The exhaustion* is $f(x) = |x|$ `Chapter10/MilnorH · exhaust`, with compact sublevel sets $[-r, r]$.
- $\varepsilon_2 \equiv 1$ `Chapter10/MilnorH · epsTwo`. Any two points of $\mathbb{R}$ are joined by a unique minimal geodesic, the segment, which depends affinely on its end points; so $\varepsilon_1 \equiv \infty$, and the constant 1 is an admissible continuous ε_2. This admissibility is elementary and is recorded in the file's documentation, not as a separate theorem; the formal definitions simply make these choices. Then $\varepsilon(\omega) = \varepsilon_2(\sup_t |\omega(t)|) = 1$ for every ω `Chapter10/MilnorH · eps` `Chapter10/MilnorH · eps_eq`.

The modulus. Milnor's maximum is the modulus of continuity

$$m_\omega(\alpha) = \sup\{|\omega(t) - \omega(t')| : t, t' \in [0,1],\ |t - t'| \le \alpha\}$$

`Chapter10/MilnorH · modulus`. The set is bounded by $2 \max |\omega|$ `Chapter10/MilnorH · bddAbove_modulus`, so every term is at most the supremum `Chapter10/MilnorH · le_modulus`, and a common bound for the terms bounds it `Chapter10/MilnorH · modulus_le`. For $0 \le \alpha \le \beta$: $m_\omega(\alpha) \ge 0$, m_ω is non-decreasing, and $m_\omega(0) = 0$ `Chapter10/MilnorH · modulus_nonneg` `Chapter10/MilnorH · modulus_mono` `Chapter10/MilnorH · modulus_zero`. It is subadditive, $m_\omega(\alpha + \beta) \le m_\omega(\alpha) + m_\omega(\beta)$: a step $|t - t'| \le \alpha + \beta$ longer than α is split at the point at distance α from t towards t' `Chapter10/MilnorH · modulus_add_le`. Hence $|m_\omega(\alpha) - m_\omega(\beta)| \le m_\omega(|\alpha - \beta|)$ `Chapter10/MilnorH · abs_modulus_sub_le`. By uniform continuity of ω on the compact $[0,1]$, $m_\omega(\delta) \to 0$ as $\delta \to 0$ `Chapter10/MilnorH · exists_modulus_le` `Mathlib/Topology/UniformSpace/HeineCantor · CompactSpace.uniformContinuous_of_continuous`. So m_ω is continuous on $[0, \infty)$ `Chapter10/MilnorH · continuousOn_modulus`.

Milnor's F, A and k. With these choices

$$F(\omega, \alpha) = (\alpha - 1)\,\varepsilon(\omega) + m_\omega(\alpha) = \alpha - 1 + m_\omega(\alpha)$$

`Chapter10/MilnorH · milnorF`, which is strictly increasing in $\alpha \ge 0$, since m_ω is non-decreasing `Chapter10/MilnorH · milnorF_lt`. The formal definition of $2A(\omega)$ makes sense for every ω:

$$2A(\omega) = \inf\{\alpha \in [0,1] : F(\omega, \alpha) \ge 0\}$$

`Chapter10/MilnorH · twoA`. Milnor's claims about it are then proved, so this is exactly his $2A$.

- 1 belongs to the set, because $F(\omega, 1) = m_\omega(1) \ge 0$ `Chapter10/MilnorH · one_mem_twoA_set`.
- A zero $\alpha_0 \in [0,1]$ of $F(\omega, \cdot)$ equals $2A(\omega)$: it lies in the set, and a smaller element α would give $0 \le F(\omega, \alpha) < F(\omega, \alpha_0) = 0$ `Chapter10/MilnorH · twoA_eq`.
- $2A(\omega) \in (0,1]$ and $F(\omega, 2A(\omega)) = 0$ `Chapter10/MilnorH · twoA_spec`. $F(\omega, \cdot)$ is continuous on $[0,1]$ with $F(\omega, 0) = -1 < 0 \le F(\omega, 1)$, so the intermediate value theorem gives a zero α_0 `Mathlib/Topology/Order/IntermediateValue · intermediate_value_Icc`. By the previous point $\alpha_0 = 2A(\omega)$, and $\alpha_0 \ne 0$ because $F(\omega, 0) = -1$.

Then $A(\omega) = 2A(\omega)/2$ `Chapter10/MilnorH · milnorA` is positive `Chapter10/MilnorH · milnorA_pos` and at most $\frac{1}{2}$, since $2A(\omega) \le 1$. (Milnor writes $A : \Omega^* \to (0,1]$; the values lie in $(0, \frac{1}{2}]$.) Milnor's k is $\lfloor 1/A(\omega) \rfloor$ `Chapter10/MilnorH · milnorK`, and $kA \le 1$ `Chapter10/MilnorH · milnorK_mul_le`. The continuity of A in ω, which Milnor also claims, is true (Lemma 17.A of the exposition) but is not formalized; it is not involved in the failure of continuity.

Milnor's h. For a function x, a number $A > 0$ and an integer k with $kA \le 1$, let $\mathcal{L}_{A,k}(x)$ be the path through the points $(jA, x(jA))$, $0 \le j \le k$, and $(1, x(1))$, affine in between `Chapter10/MilnorH · plin`. On $[jA, (j+1)A]$, $j < k$, it is the segment `Chapter10/MilnorH · seg` `Chapter10/MilnorH · plin_eq_seg`, and on $[kA, 1]$ the last segment `Chapter10/MilnorH · lastSeg` `Chapter10/MilnorH · plin_eq_last`, which is a single point when $kA = 1$. It is smooth on every interval free of the break points $0, A, \dots, kA$ `Chapter10/MilnorH · breaks` `Chapter10/MilnorH · plin_piece`, hence piecewise smooth `Chapter10/MilnorH · plin_isPiecewiseSmooth`. It starts at $x(0)$, ends at $x(1)$, and passes through the nodes `Chapter10/MilnorH · plin_zero` `Chapter10/MilnorH · plin_one` `Chapter10/MilnorH · plin_node`. Its acceleration vanishes on every open interval between nodes `Chapter10/MilnorH · deriv_deriv_plin_seg` `Chapter10/MilnorH · deriv_deriv_plin_last`. Its energy is

$$E(\mathcal{L}_{A,k}(x)) = \sum_{j<k} \frac{(x((j+1)A) - x(jA))^2}{A} + \frac{(x(1) - x(kA))^2}{1 - kA}$$

`Chapter10/MilnorH · integral_seg` `Chapter10/MilnorH · integral_last` `Chapter10/MilnorH · energy_plin`. When $kA = 1$ the last term is 0: the integral over a degenerate interval vanishes, and in the formal arithmetic the quotient $0/0$ is 0.

Milnor's $h(\omega)$ is $\mathcal{L}_{A(\omega),k(\omega)}(\omega)$, with ω extended by constants off $[0,1]$ `Chapter10/MilnorH · hPath`. It satisfies Milnor's two conditions: it agrees with ω at $t = jA$, $j \le k$, and at $t = 1$ `Chapter10/MilnorH · hPath_node` `Chapter10/MilnorH · hPath_one`, and on each interval between it has zero acceleration, so it is a geodesic of $\mathbb{R}$ there `Chapter10/MilnorH · deriv_deriv_hPath`; the minimal geodesics of $\mathbb{R}$ are exactly these segments. It is a path of Ω `Chapter10/MilnorH · hPath_mem`, which defines Milnor's map $h : \Omega^* \to \Omega$ `Chapter10/MilnorH · hMap`.

The family ω_λ. Let

$$g(t) = (1-t)^{1/3} - (1-t), \qquad \omega_\lambda = \lambda g$$

`Chapter10/MilnorH · g` `Chapter10/MilnorH · omegaFam` `Chapter10/MilnorH · omegaFam_apply`. Then $g(0) = g(1) = 0$ `Chapter10/MilnorH · g_zero` `Chapter10/MilnorH · g_one`, so $\omega_\lambda \in \Omega^*(\mathbb{R}; 0, 0)$, and $g(1-s) = s^{1/3} - s$ `Chapter10/MilnorH · g_one_sub`. Near $t = 1$ the path is only Hölder continuous of exponent $\frac{1}{3}$; that is the whole point. With $u = (1-t)^{1/3} \in [0,1]$ one has $g = u - u^3$, so $0 \le g$ on $[0,1]$ `Chapter10/MilnorH · g_nonneg`, and

$$c_0 - g = \Big(u - \frac{\sqrt{3}}{3}\Big)^2 \Big(u + \frac{2\sqrt{3}}{3}\Big) \ge 0, \qquad c_0 = \frac{2\sqrt{3}}{9}$$

`Chapter10/MilnorH · c₀` `Chapter10/MilnorH · g_le`. The maximum c_0 is attained at $1 - s_0$, $s_0 = \sqrt{3}/9 = (\sqrt{3}/3)^3$ `Chapter10/MilnorH · s₀` `Chapter10/MilnorH · g_one_sub_s₀`, and $s_0 < \frac{1}{2}$ `Chapter10/MilnorH · s₀_lt_half`. The critical parameter is

$$\lambda_* = \frac{\sqrt{3}}{2}, \qquad \lambda_* c_0 = \frac{1}{3}, \qquad \lambda_*^2 = \frac{3}{4}$$

`Chapter10/MilnorH · lamStar` `Chapter10/MilnorH · lamStar_mul_c₀` `Chapter10/MilnorH · lamStar_sq`.

Uniform convergence. $d^*(\omega_\lambda, \omega_\mu) \le c_0 |\lambda - \mu|$, because $|\lambda g - \mu g| = |\lambda - \mu|\, g \le c_0 |\lambda - \mu|$ `Chapter10/MilnorH · dist_omegaFam_le`. So $\lambda \mapsto \omega_\lambda$ is Lipschitz into Ω^* `Chapter10/MilnorH · continuous_omegaFam`, and $\omega_\lambda \to \omega_{\lambda_*}$ uniformly as $\lambda \to \lambda_*$ `Chapter10/MilnorH · tendsto_omegaFam`.

A and k along the family. For $\lambda \ge 0$ and $\alpha \ge s_0$, $m_{\omega_\lambda}(\alpha) = \lambda c_0$ `Chapter10/MilnorH · modulus_omegaFam`. It is at most λc_0 because all values of ω_λ lie in $[0, \lambda c_0]$, and at least λc_0 because of the pair $(1 - s_0, 1)$, whose parameters are $s_0 \le \alpha$ apart and whose values are λc_0 and 0. Hence, as long as $\lambda c_0 \le \frac{1}{2}$, the number $\alpha = 1 - \lambda c_0 \ge \frac{1}{2} > s_0$ is a zero of F: $F(\omega_\lambda, 1 - \lambda c_0) = -\lambda c_0 + \lambda c_0 = 0$. So

$$2A(\omega_\lambda) = 1 - \lambda c_0, \qquad A(\omega_\lambda) = \frac{1 - \lambda c_0}{2}$$

`Chapter10/MilnorH · twoA_omegaFam` `Chapter10/MilnorH · milnorA_omegaFam`, a continuous function of λ. At λ_*: $A = \frac{1}{3}$, $1/A = 3$, $k = 3$, and $kA = 1$ `Chapter10/MilnorH · milnorA_lamStar` `Chapter10/MilnorH · one_div_milnorA_lamStar` `Chapter10/MilnorH · milnorK_lamStar`. This is the critical configuration: $1/A$ is an integer and the last interval is empty. For $\lambda_* \le \lambda \le \frac{5}{4}\lambda_*$ one has $\frac{1}{3} \le \lambda c_0 \le \frac{5}{12}$ `Chapter10/MilnorH · lam_mul_c₀_mem`, so $1/A = 2/(1 - \lambda c_0) \in [3, \frac{24}{7}]$ and $k = 3$ throughout `Chapter10/MilnorH · milnorK_omegaFam`.

The energy at λ_*. $h(\omega_{\lambda_*})$ has nodes $0, \frac{1}{3}, \frac{2}{3}, 1$ and no last segment, so by the energy formula

$$E(h(\omega_{\lambda_*})) = 3\lambda_*^2 \Big(g(\tfrac{1}{3})^2 + \big(g(\tfrac{2}{3}) - g(\tfrac{1}{3})\big)^2 + g(\tfrac{2}{3})^2 \Big)$$

`Chapter10/MilnorH · energy_hPath_lamStar`, which is at most 1, because the values $\lambda_* g$ lie in $[0, \frac{1}{3}]$ `Chapter10/MilnorH · energy_hPath_lamStar_le`. Numerically it is about 0.44.

The energy blow-up. Let $\delta_\lambda = (3\lambda c_0 - 1)/2 = 1 - 3A(\omega_\lambda)$ `Chapter10/MilnorH · delta`, the length of the new last interval. It is positive for $\lambda > \lambda_*$ `Chapter10/MilnorH · delta_pos` and at most $\frac{1}{8}$ for $\lambda \le \frac{5}{4}\lambda_*$.

Lemma. For $\lambda_* < \lambda \le \frac{5}{4}\lambda_*$ and $\delta = \delta_\lambda$,

$$E(h(\omega_\lambda)) \;\ge\; \frac{\lambda^2 (\delta^{1/3} - \delta)^2}{\delta} \;\ge\; \frac{27}{64}\, \delta^{-1/3}$$

`Chapter10/MilnorH · le_energy_hPath`.

The proof as formalized. Here $k = 3$, $A = (1-\lambda c_0)/2$ and $kA = 1-\delta < 1$, so $h(\omega_\lambda)$ has a fourth segment, from $(1-\delta, \omega_\lambda(1-\delta)) = (1-\delta, \lambda(\delta^{1/3}-\delta))$ to $(1,0)$. In the energy formula every term is non-negative, so the energy is at least the last term, $\lambda^2(\delta^{1/3}-\delta)^2/\delta$. Put $v = \delta^{1/3}$, so $v^3 = \delta$ and $0 < v \le \frac{1}{2}$ (as $\delta \le \frac{1}{8}$). Then

$$\frac{\lambda^2(v-v^3)^2}{v^3} = \frac{\lambda^2(1-v^2)^2}{v} \;\ge\; \frac{\frac{3}{4}\cdot\frac{9}{16}}{v} = \frac{27}{64}\,\delta^{-1/3},$$

using $\lambda^2 \ge \lambda_*^2 = \frac{3}{4}$ and $1 - v^2 \ge \frac{3}{4}$.

As $\lambda \downarrow \lambda_*$, $\delta_\lambda^{1/3} \downarrow 0$ `Chapter10/MilnorH · tendsto_delta_cbrt`, so

$$E(h(\omega_\lambda)) \to \infty \qquad (\lambda \downarrow \lambda_*)$$

`Chapter10/MilnorH · tendsto_energy_hPath`. (Numerically, $E(h(\omega_\lambda))$ is already about 9.7 when $\lambda/\lambda_* - 1 = 10^{-3}$.)

$\sqrt{E}$ is 1-Lipschitz for d. This is Lemma 16.A of the exposition, specialized to the paths of $\mathbb{R}$ `Chapter10/MilnorH · MilnorH.lipschitzWith_sqrt_energy` `Chapter10/PathMetric · lipschitzWith_sqrt_energy`. It follows that

$$d(h(\omega_\lambda), h(\omega_{\lambda_*})) \;\ge\; \sqrt{E(h(\omega_\lambda))} - \sqrt{E(h(\omega_{\lambda_*}))} \;\to\; \infty \qquad (\lambda \downarrow \lambda_*)$$

`Chapter10/MilnorH · tendsto_edist_hMap`.

Theorem (Milnor's h is not continuous). For $M = \mathbb{R}$, $p = q = 0$, $f = |\cdot|$ and $\varepsilon_2 \equiv 1$, Milnor's map $h : \Omega^* \to \Omega$ is not continuous at ω_{λ_*} `Chapter10/MilnorH · milnor_h_not_continuous`; in particular it is not continuous `Chapter10/MilnorH · milnor_h_not_continuous'`.

The proof as formalized. If h were continuous at ω_{λ_*}, then, since $\omega_\lambda \to \omega_{\lambda_*}$, $h(\omega_\lambda) \to h(\omega_{\lambda_*})$ in Ω as $\lambda \downarrow \lambda_*$. The energy is continuous on Ω, so $E(h(\omega_\lambda))$ would converge to the finite number $E(h(\omega_{\lambda_*}))$. But it tends to ∞.

What exactly fails, and what does not.

- **What fails is the continuity of h for d**, and only that. A is continuous, as Milnor says; along the family it is $(1-\lambda c_0)/2$, and $k = 3$ for all $\lambda_* \le \lambda \le \frac{5}{4}\lambda_*$. What moves is the last break point $kA = 3A$, which leaves 1, and the segment born there is short in time and in space, so h is continuous for the uniform metric d^* (an observation of the exposition, not formalized). The failure is in the L^2 term of d: the new segment has speed ℓ/δ on an interval of length δ, so it contributes ℓ^2/δ to the energy, and this is unbounded.
- **It is not an artefact of the choices.** The claim is made for every M, every f and every admissible ε_2, so one admissible instance is a counterexample to it. Other constant choices $\varepsilon_2 \equiv c$ give nothing new: $F_c(c\omega, \alpha) = c\,F_1(\omega,\alpha)$, so $A_c(c\omega) = A_1(\omega)$ and $h_c(c\omega) = c\,h_1(\omega)$, and the family $c\,\omega_\lambda$ fails in the same way (a remark of the file's documentation, not formalized).
- **The rest of the proof falls with h.** The homotopies $h \circ i \simeq \mathrm{id}$ and $i \circ h \simeq \mathrm{id}$ by the arguments of §16 are homotopies through maps into Ω, so they need h continuous for d. A weaker statement, that i is a weak homotopy equivalence, would be enough for Theorem 19.2 (the exposition derives it at once from the sets Y_k below, since every compact subset of Ω^* lies in one of them; this is not formalized), but Theorem 19.6 and the remark after Corollary 18.2 use the full theorem.
- **The theorem is true.** It is proved below by a different route, whose key decision is to keep the subdivision fixed: on each of the open sets Y_k the break points are the dyadic points $j/2^k$, so no segment is ever born, and the sets are glued with the Appendix.

The finding was confirmed by an independent review against the scan of the book, by numerical checks, and by this formal proof (Findings, §§16–19).

IX.13 Convergence in Milnor's metric controls the velocities

The modified proof deforms paths by maps given pointwise by smooth formulas: a point $\gamma(s)$ is moved along a minimal geodesic that depends smoothly on $\gamma(s)$ and on a point of a broken geodesic. Continuity of such a map for d needs more than d visibly provides. d compares the speeds $\|\gamma'\|$, but the velocity of $s \mapsto \Phi(\gamma(s), \dots)$ is $D\Phi(\gamma'(s), \dots)$, which involves the velocity vector itself. The lemmas of this section supply the missing control. Milnor's continuity claim passes over them, and they are needed even for maps that are continuous.

Integration by parts on a smooth piece. Let K be smooth on $[s,t] \subseteq [0,1]$ and $\beta \in \Omega$. Then

$$\int_s^t \langle \beta', K\rangle = \langle \beta(t), K(t)\rangle - \langle \beta(s), K(s)\rangle - \int_s^t \langle \beta, K'\rangle$$

`Chapter10/FullPathSpace · PathSp.integral_inner_deriv_eq`. The function $\langle \beta, K\rangle$ is continuous on $[s,t]$ and differentiable off the finite set of break points of β `Chapter10/FullPathSpace · PathSp.hasDerivAt_of_notMem`, with derivative $\langle \beta', K\rangle + \langle \beta, K'\rangle$. Both terms are integrable, the first because β' is continuous on the pieces of β `Chapter8/Piecewise · intervalIntegrable_of_piecewise`. The fundamental theorem of calculus off a countable set applies `Mathlib/MeasureTheory/Integral/DivergenceTheorem · integral_eq_of_hasDerivAt_off_countable_of_le`.

Uniform convergence. If $\gamma_j \to \gamma$ in Ω, then $\gamma_j \to \gamma$ uniformly on $[0,1]$ in the norm of V, since $\|\cdot\| \le \rho$ and the first term of d is the supremum of ρ `Chapter10/FullPathSpace · PathSp.eventually_norm_sub_le_of_tendsto`.

Weak convergence of the velocities. If $\gamma_j \to \gamma$ in Ω, then $\int_0^1 \langle \gamma_j', \gamma'\rangle \to \int_0^1 \|\gamma'\|^2$ `Chapter10/FullPathSpace · PathSp.tendsto_integral_inner_deriv`. The proof is an induction over the pieces of γ `Chapter8/Piecewise · piecewise_induction`: adjacent intervals add, so it suffices to treat one smooth piece $[s,t]$, on which γ' agrees with a function K smooth on $[s,t]$ `Chapter8/Energy · IsPiecewiseSmooth.derivWithin_piece`. Integration by parts rewrites $\int_s^t \langle \gamma_j', K\rangle$ and $\int_s^t \|\gamma'\|^2 = \int_s^t \langle \gamma', K\rangle$. The boundary terms converge because $\gamma_j \to \gamma$ pointwise, and the integral terms differ by at most $(t-s)\,\sup\|K'\|\,\sup_{[s,t]}\|\gamma_j - \gamma\|$, which tends to 0 by uniform convergence.

L^2 convergence of the velocities. If $\gamma_j \to \gamma$ in Ω, then $\int_0^1 \|\gamma_j' - \gamma'\|^2 \to 0$ `Chapter10/FullPathSpace · PathSp.tendsto_integral_norm_deriv_sub_sq`. Expand

$$\int_0^1 \|\gamma_j' - \gamma'\|^2 = E(\gamma_j) - 2\int_0^1 \langle \gamma_j', \gamma'\rangle + E(\gamma) \;\to\; E(\gamma) - 2E(\gamma) + E(\gamma) = 0,$$

using the continuity of E for d (§IX.1) and the weak convergence.

A criterion for convergence in Ω. Conversely, if $\alpha_j(s) \to \alpha(s)$ for every $s \in [0,1]$ and $\int_0^1 \|\alpha_j' - \alpha'\|^2 \to 0$, then $\alpha_j \to \alpha$ in Ω `Chapter10/FullPathSpace · PathSp.tendsto_of_tendsto_apply_of_integral`.

The proof as formalized. Once $\int_0^1 \|\alpha_j' - \alpha'\|^2 \le 1$, the inequality $\|\alpha_j'\|^2 \le 2\|\alpha'\|^2 + 2\|\alpha_j' - \alpha'\|^2$ gives $E(\alpha_j) \le C = 2E(\alpha) + 2$. Paths of energy at most C are uniformly Hölder of exponent $\frac{1}{2}$ for ρ, by Cauchy–Schwarz, so pointwise convergence at the points of a finite net of $[0,1]$ makes the supremum term of d small `Chapter10/RetractionConv · PathSp.eventually_supDist_le_of_tendsto_apply` (Part VIII). The second term is at most $\big(\int_0^1 \|\alpha_j' - \alpha'\|^2\big)^{1/2}$, because $|\|\alpha_j'\| - \|\alpha'\|| \le \|\alpha_j' - \alpha'\|$.

So convergence in Ω is pointwise convergence together with L^2 convergence of the velocities. The last lemma says that smooth formulas preserve the second.

Smooth maps preserve L^2 convergence of velocities. Let Φ be of class C^1 on an open set U containing the curve $F_0([a,b])$. Let $F_i \to F_0$ uniformly on $[a,b]$, all curves differentiable off finite sets, with veloci-

ties F_i' bounded in L^2, $\int_a^b \|F_i'\|^2 \le C$, and converging to F_0' in L^2. Then $\int_a^b \|(\Phi\circ F_i)' - (\Phi\circ F_0)'\|^2 \to 0$ `Chapter10/FullPathOmega · tendsto_integral_deriv_comp_sub_sq`.

The proof as formalized.

1. The image $F_0([a,b])$ is compact, so a closed δ-thickening K' of it lies in U `Mathlib/Topology/MetricSpace/Thickening · IsCompact.exists_cthickening_subset_open`. On the compact K', $D\Phi$ is bounded by some M and uniformly continuous.
2. Given ε, choose η with $2\eta^2(|C|+1) = \varepsilon/2$, and ρ_0 such that points of K' closer than ρ_0 have $\|D\Phi(x) - D\Phi(x')\| < \eta$. Eventually F_i is within $\min(\delta,\rho_0)$ of F_0 on $[a,b]$, so $F_i(s) \in K'$ and $\|D\Phi(F_i(s)) - D\Phi(F_0(s))\| < \eta$.
3. Off the finitely many exceptional points, by the chain rule,

$$(\Phi\circ F_i)' - (\Phi\circ F_0)' = \big(D\Phi(F_i) - D\Phi(F_0)\big)F_i' + D\Phi(F_0)\,(F_i' - F_0'),$$

   so its squared norm is at most $2\eta^2\|F_i'\|^2 + 2M^2\|F_i' - F_0'\|^2$.
4. Integrating, the first part is at most $2\eta^2 C \le \varepsilon/2$, and the second is eventually $< \varepsilon/2$ by the L^2 convergence.

IX.14 Good balls

Milnor's ε_1 asks for unique minimal geodesics depending differentiably on their end points. The modified proof needs three things more. First, a uniform scale: the same uniqueness for all pairs of a region at once. Second, convexity: the minimal geodesics between points of a region stay in it, so that a deformation along them does not leave the region. Third, one smooth map $(y,z,s) \mapsto \sigma_{yz}(s)$ on each whole region, not only near each pair. Lemma 17.B of the exposition gets these from J. H. C. Whitehead's convex balls. The formalization gets convexity from the ambient distance instead, without normal coordinates.

Corollary 10.8, recalled (Part VI). For a scale δ and points $y, z \in N$, the conclusion of Corollary 10.8 says: there is $v \in T_yN$ with $\|v\| < \delta$, $\exp_y v = z$ and $\rho(y,z) = \|v\|$; it is the only tangent vector at y of norm $< \delta$ with $\exp_y(\cdot) = z$; and it is the value at (y,z) of a map G, smooth on an open set of pairs, which gives such vectors for the nearby pairs at distance $< \delta$ `Chapter7/Distance · IsShortJoin`. Lemma 10.3, read with the distance, gives around each $x \in N$ a number $s > 0$ and an open set $\Omega_x \ni x$ such that points of $N \cap \Omega_x$ are at distance $< s$ from x, and every $y \in N \cap \Omega_x$ and z at distance $< \delta \le s$ from y are joined as in Corollary 10.8 at the scale δ `Chapter7/Distance · exists_isShortJoin`.

The minimal join vector. A vector w is *the minimal join vector* from y to z if $y, z \in N$, $w \in T_yN$, $\exp_y w = z$, $\rho(y,z) = \|w\|$, and the conclusion of Corollary 10.8 holds for y, z at some scale $\delta > \|w\|$ `Chapter10/GoodBalls · IsMinJoin`. It is unique: two such vectors have the same norm $\rho(y,z)$, smaller than the scale of the first, and the uniqueness in Corollary 10.8 identifies both with its vector `Chapter10/GoodBalls · IsMinJoin.eq`. The function $v(y,z)$ is defined as the minimal join vector when there is one, and 0 otherwise `Chapter10/GoodBalls · minJoin`. It is one global function, so it does not depend on the region in which it is found `Chapter10/GoodBalls · minJoin_unique` `Chapter10/GoodBalls · isMinJoin_minJoin`. It is always tangent at y `Chapter10/GoodBalls · minJoin_mem_tangentSubspace`, it is the vector of Corollary 10.8 when that applies `Chapter10/GoodBalls · isMinJoin_of_isShortJoin`, and $v(y,y) = 0$ `Chapter10/GoodBalls · minJoin_self`. The *minimal geodesic* is $\sigma_{yz}(s) = \exp_y(s\,v(y,z))$ `Chapter10/GoodBalls · minGeo`. It starts at y `Chapter10/GoodBalls · minGeo_zero`, is constant when $y = z$ `Chapter10/GoodBalls · minGeo_self`, and has initial velocity $v(y,z)$ `Chapter10/GoodBalls · hasDerivAt_minGeo_zero`. When there is a minimal join vector, it lies in the domain of exp `Chapter10/GoodBalls · IsMinJoin.mem_expDom`, and σ_{yz} is a geodesic on an open interval containing $[0,1]$, of constant speed $\|v(y,z)\|$ `Chapter10/GoodBalls · IsMinJoin.isGeodesicOn_minGeo`.

Convexity from the ambient distance. Three lemmas replace Whitehead's theorem.

- *A bound for the second fundamental form.* For $x \in N$ there is $R > 0$ with $R\,\|\mathrm{II}_y(v,v)\| \le \|v\|^2$ for $y \in N$, $\|y - x\| < R$, $v \in T_yN$ `Chapter10/GoodBalls · exists_secondFundForm_bound`. Near x, $\mathrm{II}_y(v,v)$ is the derivative of the smooth ambient extension of $y \mapsto P_y$ in a slice chart at x, applied to v twice `Chapter6/Projection · projRep` `Chapter6/Projection · dN_tanProj_eq_fderiv`. That derivative is continuous at x, so its norm is below $C = \|D(\text{extension})(x)\|+1$ on some ball $B(x, R_1)$; then $R = \min(R_1, 1/C)$ works, as $R\,\|\mathrm{II}_y(v,v)\| \le C^{-1} \cdot C\|v\|^2$.
- *Convexity of the squared distance.* Under that bound, along a geodesic γ that stays within R of x for $s \in [a,b]$, the function $s \mapsto \|\gamma(s) - x\|^2$ is convex on $[a,b]$ `Chapter10/GoodBalls · convexOn_norm_sub_sq`. Its second derivative is $2\langle\gamma - x, \gamma''\rangle + 2\|\gamma'\|^2$; along a geodesic $\gamma'' = \mathrm{II}(\gamma',\gamma')$ `Chapter7/Geodesic · IsGeodesicOn.deriv_deriv_eq`, and $|\langle\gamma - x, \gamma''\rangle| \le R\,\|\mathrm{II}(\gamma',\gamma')\| \le \|\gamma'\|^2$. A non-negative second derivative gives convexity `Mathlib/Analysis/Convex/Deriv · convexOn_of_hasDerivWithinAt2_nonneg`.
- *Short geodesics stay in small balls.* Under the same bound, a geodesic γ on $(-\eta, 1+\eta)$ with $\gamma(0), \gamma(1) \in B(x,r)$ and $r + \|\gamma'(0)\| \le R$ maps $[0,1]$ into $B(x,r)$ `Chapter10/GoodBalls · dist_lt_of_isGeodesicOn`. The speed is constant `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq`, so $\|\gamma(s) - \gamma(0)\| \le \|\gamma'(0)\|\,s$ and γ stays within R of x on $[0,1]$. There $\|\gamma - x\|^2$ is convex, so it is at most the larger of its values at 0 and 1, which are $< r^2$.

Good balls. An open set $O \subseteq V$ is a *good ball* if (shortJoin) for one $\delta > 0$ any two points y, z of $N \cap O$ are at distance $< \delta$ and joined as in Corollary 10.8 at the scale δ, and (convex) $\sigma_{yz}(s) \in N \cap O$ for $y, z \in N \cap O$ and $0 \le s \le 1$ `Chapter10/GoodBalls · IsGoodBall`. Good balls are balls of V intersected with N, not metric balls of ρ. In a good ball O, for $y, z \in N \cap O$:

- $v(y,z)$ is the minimal join vector `Chapter10/GoodBalls · IsGoodBall.isMinJoin` `Chapter10/GoodBalls · IsGoodBall.join`; it is the only tangent vector of norm $< \delta$, for one δ uniform on O, with $\exp_y(\cdot) = z$ `Chapter10/GoodBalls · IsGoodBall.exists_uniq`; and a tangent vector w at y with $\exp_y w = z$ and $\rho(y,z) = \|w\|$ is $v(y,z)$ `Chapter10/GoodBalls · IsGoodBall.eq_minJoin`. So σ_{yz} is the unique minimal geodesic from y to z; it has length $\|v(y,z)\| = \rho(y,z)$ `Chapter10/GoodBalls · IsGoodBall.norm_minJoin`, ends at z `Chapter10/GoodBalls · IsGoodBall.minGeo_one`, and is a geodesic of constant speed `Chapter10/GoodBalls · IsGoodBall.isGeodesicOn_minGeo`.
- Near each pair, v agrees with the smooth map G of Corollary 10.8 `Chapter10/GoodBalls · IsGoodBall.eventually_minJoin_eq`: G gives tangent vectors of norm $< \delta$ reaching the second point, and uniqueness at the scale δ identifies them with v. Hence v is continuous on $(N \cap O)^2$ `Chapter10/GoodBalls · IsGoodBall.continuousOn`, and so is $(y,z,s) \mapsto \sigma_{yz}(s)$ on $(N \cap O)^2 \times [0,1]$, as exp is continuous near the points of its domain `Chapter7/GeodesicFlow · eventually_mem_expDom_and_continuousWithinAt_exp` `Chapter10/GoodBalls · IsGoodBall.continuousOn_minGeo`.
- Sub-arcs of minimal geodesics are minimal geodesics: $v(y, \sigma_{yz}(s)) = s\,v(y,z)$ for $0 \le s \le 1$, by uniqueness at the uniform scale, since $\sigma_{yz}(s) \in N \cap O$ by convexity `Chapter10/GoodBalls · IsGoodBall.minJoin_minGeo`. So the minimal geodesic from y to $\sigma_{yz}(s)$ is $\tau \mapsto \sigma_{yz}(\tau s)$, and $\rho(y, \sigma_{yz}(s)) = s\,\rho(y,z)$ `Chapter10/GoodBalls · IsGoodBall.minGeo_minGeo` `Chapter10/GoodBalls · IsGoodBall.riemDist_minGeo`. A deformation of Milnor's kind, which moves a break point along the path, needs these facts to fix broken geodesics; the deformation actually used (§IX.16) does not.

Smooth dependence on the end points. Near each pair (y_0, z_0) of points of $N \cap O$ there are an open set $U \ni (y_0, z_0)$ of $V \times V$, a number $\varepsilon > 0$, and a map Φ smooth on $U \times (-\varepsilon, 1+\varepsilon)$, with $\Phi((y,z),s) = \sigma_{yz}(s)$ for all s and all $y, z \in N \cap O$ with $(y,z) \in U$ `Chapter10/GoodBalls · IsGoodBall.exists_smooth_minGeo`.

The proof as formalized. Take the smooth G of Corollary 10.8 near (y_0, z_0), and a smooth retraction r onto N near y_0, read off a slice chart `Chapter11/ExpSmooth · exists_retraction`. The initial data $c(y,z) = r(y)$ and $\bar w(y,z) = P_{r(y)}G(y,z)$ are smooth and tangent, and on pairs of the good ball they are y and $v(y,z)$. The radial geodesics $(u,t) \mapsto \exp_{c(u)}(t\,\bar w(u))$ `Chapter11/ExpSmooth · raysE` depend smoothly on (u,t) on a neighbourhood of $\{(y_0,z_0)\} \times [0,1]$, because the base geodesic $\sigma_{y_0 z_0}$ exists on an open interval containing $[0,1]$, and geodesics depend smoothly on their initial data over compact intervals `Chapter11/`

`ExpSmooth · raysGoodE_of_isGeodesicOn` (Part X).

Lemma 17.B: every point is the centre of good balls. For $x \in N$ there is $r_0 > 0$ such that the ambient balls $B(x, r)$ are good for all $r \le r_0$ `Chapter10/GoodBalls · exists_isGoodBall_ball`.

The proof as formalized.

1. Take R from the bound on the second fundamental form at x, and s, Ω_x from Lemma 10.3 read with the distance. Put $t = \min(R/3, s/2)$.
2. Choose r_1 such that the points of $N \cap B(x, r_1)$ are at distance $< t$ from x, which is possible since the ρ-balls form a basis of neighbourhoods (§IX.1), and r_2 with $B(x, r_2) \subseteq \Omega_x$. Put $r_0 = \min(r_1, r_2, R/3)$, and let $r \le r_0$.
3. *Short joins.* For $y, z \in N \cap B(x, r)$, $\rho(y, z) \le \rho(y, x) + \rho(x, z) < 2t \le s$, and $y \in \Omega_x$, so y and z are joined as in Corollary 10.8 at the scale s.
4. *Convexity.* The minimal geodesic σ_{yz} is a geodesic with $\sigma(0) = y$, $\sigma(1) = z$ and $\|\sigma'(0)\| = \rho(y, z) < 2t \le 2R/3$. As $r \le R/3$, $r + \|\sigma'(0)\| < R$, and the third convexity lemma keeps $\sigma([0, 1])$ in $B(x, r)$; it lies in N, being a geodesic of N.

Two variants follow: good balls inside any given ball $B(x, \varepsilon)$ `Chapter10/GoodBalls · exists_isGoodBall`, and a Lebesgue number: for compact $K \subseteq N$ there is $\varepsilon > 0$ such that every ball $B(x, \varepsilon)$, $x \in K$, lies in a good ball `Chapter10/GoodBalls · exists_lebesgue_goodBall` `Mathlib/Topology/MetricSpace/Pseudo/Lemmas · lebesgue_number_lemma_of_metric`. The proof of 17.1 uses a refinement of the latter (§IX.15).

IX.15 Systems of joins

The topological part of the proof only uses a few properties of the minimal geodesics, and the formalization isolates them.

Definition. A *system of joins* for N consists of a map $(y, z, u) \mapsto \sigma_{yz}(u)$ from $V \times V \times \mathbb{R}$ to V, the *joins*, and a family of *good sets*, such that `Chapter10/FullPathDeform · JoinSystem`:

- every good set O is open;
- (convexity) for $y, z \in N \cap O$ and $u \in [0, 1]$, $\sigma_{yz}(u) \in N \cap O$;
- $\sigma_{yz}(0) = y$ for $y \in N$; $\sigma_{yz}(1) = z$ for $y, z \in N \cap O$, O good; and $\sigma_{yy}(u) = y$;
- (smoothness) for each good set O there are an open $U \supseteq (N \cap O)^2$, a number $\varepsilon > 0$, and a map Φ smooth on $U \times (-\varepsilon, 1 + \varepsilon)$ with $\Phi((y, z), u) = \sigma_{yz}(u)$ for $y, z \in N \cap O$ and $u \in (-\varepsilon, 1 + \varepsilon)$;
- (Lebesgue numbers) for every compact $K \subseteq N$ there is $\varepsilon > 0$ such that each ball $B(x, \varepsilon)$, $x \in K$, lies in a good set.

The smoothness asks for **one** smooth map on a whole good set. Milnor's differentiable dependence on the end points, and the local smoothness of §IX.14, give it only near each pair. The good balls are turned into a system of joins by gluing.

Smooth gluing with a parameter. Let $K \subseteq S$ with K compact in a finite-dimensional space E, and $g : E \times \mathbb{R} \to F$. Suppose every $x \in K$ has an open neighbourhood U_x, a number $\varepsilon_x > 0$ and a map Φ_x smooth on $U_x \times (-\varepsilon_x, 1 + \varepsilon_x)$ with $\Phi_x(y, u) = g(y, u)$ for $y \in S \cap U_x$ and all u. Then there are $\varepsilon > 0$ and one map Φ, smooth on $E \times (-\varepsilon, 1 + \varepsilon)$, with $\Phi(y, u) = g(y, u)$ for $y \in K$ and all u `Chapter10/GoodJoin · exists_smooth_glue`.

The proof as formalized. Finitely many U_{x_i} cover K. Take a smooth partition of unity ψ_i on E subordinate to them on K `Mathlib/Geometry/Manifold/PartitionOfUnity · SmoothPartitionOfUnity.exists_isSubordinate`, let ε be the least ε_{x_i}, and put $\Phi(y, u) = \sum_i \psi_i(y)\, \Phi_{x_i}(y, u)$. Each term is smooth on $E \times (-\varepsilon, 1 + \varepsilon)$: near a point with $y \in U_{x_i}$ because both factors are, and near a point with $y \notin U_{x_i}$ because it vanishes on a neighbourhood, y being outside the closed support of ψ_i. For $y \in K$, every i with $\psi_i(y) \neq 0$ has $y \in U_{x_i}$, so $\Phi_{x_i}(y, u) = g(y, u)$, and $\sum_i \psi_i(y) = 1$.

Local compactness. For $x \in N$ there is $r > 0$ with $N \cap \bar{B}(x,r)$ compact `Chapter10/GoodJoin · exists_isCompact_inter_closedBall`: inside the domain of a slice chart, N is the zero set of the normal component, so $N \cap \bar{B}(x,r)$ is closed in the compact ball.

The good sets. For $x \in N$ and $\varepsilon > 0$ there is $r \in (0,\varepsilon)$ such that the balls $B(x,r')$ are good for all $r' \le 2r$ and $N \cap \bar{B}(x,r)$ is compact `Chapter10/GoodJoin · exists_goodSet`: take $r = \min(r_0/2, r_1, \varepsilon/2)$ with r_0 from Lemma 17.B and r_1 from local compactness.

The system of joins of good balls. The joins are the minimal geodesics σ_{yz} of §IX.14, and the good sets are the balls $B(x,r)$, $x \in N$, of the previous statement `Chapter10/GoodJoin · goodJoin` `Chapter10/GoodJoin · goodJoin_geo`.

The verification as formalized. Openness, convexity, and the end-point conditions hold because $B(x,r)$ itself is a good ball. For smoothness, the compact set $K = (N \cap \bar{B}(x,r))^2$ lies in $(N \cap B(x,2r))^2$, and $B(x,2r)$ is a good ball. Near each pair of K its minimal geodesics have a local smooth extension (§IX.14), valid for all parameters u, and gluing gives one map Φ, smooth on $(V \times V) \times (-\varepsilon, 1+\varepsilon)$, which equals σ on K and so on $(N \cap B(x,r))^2$. This is why the definition asks for good balls up to radius $2r$: the closed ball $\bar{B}(x,r)$, needed for compactness, must sit inside an open good ball. For Lebesgue numbers, cover the compact $K \subseteq N$ by good sets $B(y,r_y)$, $y \in K$, and take a Lebesgue number of this cover.

IX.16 Dyadic pieces, and the deformation inside Ω^*

Fix a system of joins J. The modified proof replaces his variable subdivision $0, A(\omega), 2A(\omega), \ldots$ by the dyadic subdivisions, which are fixed, and restricts each to the open set of paths for which it is fine enough.

Dyadic subdivisions. The dyadic points of level k are the $j/2^k$ `Chapter10/FullPathSpace · dy`, and $j_k(s) = \min(\lfloor s\,2^k \rfloor, 2^k - 1)$ is the index of a dyadic interval of level k containing $s \in [0,1]$, the cap putting $s = 1$ in the last one `Chapter10/FullPathSpace · dseg` `Chapter10/FullPathSpace · mem_Icc_dseg` `Chapter10/FullPathDeform · dseg_dy` `Chapter10/FullPathDeform · dseg_eq_or`. Each dyadic interval of level $k+1$ lies in one of level k `Chapter10/FullPathSpace · Icc_dy_succ_subset`.

Definitions. A function c is *good at level k* if each dyadic piece $c([j/2^k, (j+1)/2^k])$, $j < 2^k$, lies in $N \cap O$ for some good set O `Chapter10/FullPathDeform · JoinSystem.IsGood`. Its *dyadic broken geodesic* is

$$b_k(c)(s) = \sigma_{c(j/2^k)\,c((j+1)/2^k)}(s\,2^k - j), \qquad j = j_k(s)$$

`Chapter10/FullPathDeform · JoinSystem.broken`, and the *deformation* moves each point of c along the join to the corresponding point of $b_k(c)$:

$$H_k(c,u)(s) = \sigma_{c(s)\,b_k(c)(s)}(u)$$

`Chapter10/FullPathDeform · JoinSystem.interp`.

This deformation is not Milnor's r_u of §16, which the exposition's Lemma 17.C reuses and which moves a break point along the path. Moving every point along a join keeps each piece in its good set by convexity, and is given by one smooth formula on each piece, which is what continuity for d needs.

For c good at level k:

- On the j-th piece, $b_k(c)(s) = \sigma_{c(j/2^k)\,c((j+1)/2^k)}(s\,2^k - j)$, also at the right end point, where j_k jumps to the next piece; there both formulas give $c((j+1)/2^k)$, by the end-point conditions `Chapter10/FullPathDeform · JoinSystem.IsGood.broken_eq` `Chapter10/FullPathDeform · JoinSystem.param_mem_Icc`.
- *Convexity keeps everything in place.* On the j-th piece, $b_k(c)(s) \in N \cap O$, since it is a join of two points of $N \cap O$ `Chapter10/FullPathDeform · JoinSystem.broken_mem_of`, and then $H_k(c,u)(s) \in N \cap O$ for $u \in [0,1]$, as a join of $c(s)$ and $b_k(c)(s)$ `Chapter10/FullPathDeform · JoinSystem.interp_mem_of`. So $b_k(c)$ and every $H_k(c,u)$ are good at level k `Chapter10/FullPathDeform · JoinSystem.IsGood.broken` `Chapter10/FullPathDeform · JoinSystem.IsGood.interp`.

- $H_k(c,0) = c$ and $H_k(c,1) = b_k(c)$ on $[0,1]$ Chapter10/FullPathDeform · JoinSystem.interp_zero Chapter10/FullPathDeform · JoinSystem.interp_one. The broken geodesic passes through the dyadic points of c Chapter10/FullPathDeform · JoinSystem.broken_dy, in particular through $c(0)$ and $c(1)$ Chapter10/FullPathDeform · JoinSystem.broken_zero Chapter10/FullPathDeform · JoinSystem.broken_one, so the deformation fixes the end points, by $\sigma_{yy}(u) = y$ Chapter10/FullPathDeform · JoinSystem.interp_zero_right Chapter10/FullPathDeform · JoinSystem.interp_one_right.
- All of this only sees the values of c on $[0,1]$ Chapter10/FullPathDeform · JoinSystem.isGood_congr Chapter10/FullPathDeform · JoinSystem.broken_congr Chapter10/FullPathDeform · JoinSystem.interp_congr, and good at level k implies good at level $k+1$ Chapter10/FullPathDeform · JoinSystem.IsGood.mono.

The sets Y_k (Lemma 17.C (i) of the exposition). A continuous path $\gamma \in \Omega^*$, extended by constants off $[0,1]$ Chapter10/FullPathDeform · JoinSystem.ext, lies in Y_k if it is good at level k Chapter10/FullPathDeform · JoinSystem.YSet, that is, if each piece $\gamma([j/2^k,(j+1)/2^k])$ maps into a good set Chapter10/FullPathDeform · JoinSystem.mem_YSet_iff.

- *Y_k is open in Ω^** Chapter10/FullPathDeform · JoinSystem.isOpen_YSet. For a compact piece and an open good set, the paths mapping the piece into the set form a neighbourhood in the compact-open topology Mathlib/Topology/CompactOpen · ContinuousMap.eventually_mapsTo Chapter10/FullPathDeform · JoinSystem.isCompact_piece; intersect over the 2^k pieces.
- $Y_k \subseteq Y_{k+1}$ Chapter10/FullPathDeform · JoinSystem.YSet_mono, as dyadic pieces refine.
- *Every continuous path lies in some Y_k* Chapter10/FullPathDeform · JoinSystem.exists_mem_YSet. The image $\gamma([0,1])$ is a compact subset of N; let ε be a Lebesgue number for it. By uniform continuity there is δ with $|t-t'| < \delta \Rightarrow \|\gamma(t)-\gamma(t')\| < \varepsilon$; take k with $2^{-k} < \delta$. Then each piece maps into $B(\gamma(j/2^k),\varepsilon)$, which lies in a good set.

The deformation is continuous on Y_k. The map $(\gamma,u,s) \mapsto H_k(\gamma,u)(s)$ is continuous on $Y_k \times [0,1] \times [0,1]$ Chapter10/FullPathDeform · JoinSystem.continuous_interp_star.

The proof as formalized. The closed sets $\{s \in [j/2^k,(j+1)/2^k]\}$ form a finite cover, so it suffices to prove continuity on each Mathlib/Topology/LocallyFinite · LocallyFinite.continuous. Near a point (γ_0,u_0,s_0) of the j-th one, let O be the good set containing the j-th piece of γ_0, and Φ its smooth map. Nearby paths still map the j-th piece into O, by the compact-open argument, and for them

$$H_k(\gamma,u)(s) = \Phi\Big((\gamma(s),\ \Phi((\gamma(j/2^k),\gamma((j+1)/2^k)),\ s\,2^k - j)),\ u\Big),$$

a composite of continuous maps, since evaluation $(\gamma,s) \mapsto \gamma(s)$ is continuous on $\Omega^* \times \mathbb{R}$ Chapter10/FullPathDeform · JoinSystem.continuous_ext.

Since each $H_k(\gamma,u)$ is good at level k and runs from p to q, the deformation is a continuous map $H_k^* : Y_k \times [0,1] \to Y_k$ Chapter10/FullPathDeform · JoinSystem.interpStar Chapter10/FullPathDeform · JoinSystem.interpStar_apply, with $H_k^*(\gamma,0) = \gamma$ Chapter10/FullPathDeform · JoinSystem.interpStar_zero.

IX.17 The deformation inside Ω: X_k is homotopy equivalent to Y_k

This is the analytic heart of the modified proof, the step at which Milnor's h falls short: the maps must be continuous for d.

Smooth formulas on pieces. An interval $[a,b] \subseteq [0,1]$ with no dyadic point of level k inside lies in one dyadic piece Chapter10/FullPathOmega · JoinSystem.exists_piece_of_Icc. On the j-th piece of a function c good at level k, with good set O and smooth map Φ,

$$b_k(c)(s) = \Phi((c(j/2^k),c((j+1)/2^k)),\ s\,2^k - j), \qquad H_k(c,u)(s) = \Phi((c(s),\ b_k(c)(s)),\ u)$$

Chapter10/FullPathOmega · JoinSystem.IsGood.exists_formula. So $b_k(c)$ is smooth on every interval free of the finite set of dyadic points Chapter10/FullPathOmega · JoinSystem.dyPts Chapter10/FullPathOmega · JoinSystem.

IsGood.contDiffOn_broken, and $H_k(c,u)$ on every such interval on which c is smooth Chapter10/FullPathOmega · JoinSystem.IsGood.contDiffOn_interp. If $c(0) = p$ and $c(1) = q$, then $b_k(c) \in \Omega$, with break points among the dyadic points Chapter10/FullPathOmega · JoinSystem.IsGood.broken_mem_pathSpace; and if $c \in \Omega$, then $H_k(c,u) \in \Omega$, with break points among the dyadic points and those of c Chapter10/FullPathOmega · JoinSystem.IsGood.interp_mem_pathSpace.

The sets X_k. $X_k \subseteq \Omega$ is the set of piecewise smooth paths good at level k Chapter10/FullPathOmega · JoinSystem.XSet, which is $i^{-1}(Y_k)$ Chapter10/FullPathOmega · JoinSystem.XSet_eq_preimage. The inclusion $i : X_k \to Y_k$ is continuous Chapter10/FullPathOmega · JoinSystem.toY. To produce elements of Ω the maps below are normalized to be constant off $[0,1]$ Chapter10/FullPathOmega · JoinSystem.clampF Chapter10/FullPathOmega · JoinSystem.clampF_eqOn; since d only sees $[0,1]$, this changes nothing but makes the values definite.

The broken geodesic $b : Y_k \to X_k$ Chapter10/FullPathOmega · JoinSystem.brokenΩ Chapter10/FullPathOmega · JoinSystem.brokenΩ_eqOn is continuous from d^* to d Chapter10/FullPathOmega · JoinSystem.continuous_brokenΩ.

The proof as formalized. By the criterion of §IX.13 it suffices to prove pointwise convergence and L^2 convergence of the velocities as $\gamma \to \gamma_0$ in Y_k. Pointwise convergence: $b(\gamma)(s) = H_k^*(\gamma,1)(s)$, which is continuous in γ (§IX.16). For the velocities, cut $[0,1]$ at the dyadic points and add over adjacent intervals Chapter10/FullPathOmega · JoinSystem.integral_deriv_sub_sq_add Chapter10/FullPathOmega · JoinSystem.tendsto_integral_brokenΩ. On an interval inside the j-th piece, $b(\gamma) = \Phi \circ F_\gamma$ with $F_\gamma(s) = ((\gamma(j/2^k), \gamma((j+1)/2^k)), s\,2^k - j)$ for all γ near γ_0, since nearby paths keep the piece in the good set. The curves F_γ converge uniformly to F_{γ_0}, because their vertices converge, and their velocities are all the constant $((0,0),2^k)$. So smooth maps preserving L^2 convergence of velocities (§IX.13) gives the claim on the interval.

The deformation $H : X_k \times [0,1] \to X_k$ Chapter10/FullPathOmega · JoinSystem.interpΩ Chapter10/FullPathOmega · JoinSystem.interpΩ_eqOn is continuous for d Chapter10/FullPathOmega · JoinSystem.continuous_interpΩ.

The proof as formalized. Let $(\gamma,u) \to (\gamma_0,u_0)$ in $X_k \times [0,1]$.

1. *Pointwise convergence* goes through Ω^*: $H(\gamma,u)(s) = H_k^*(i\gamma,u)(s)$ Chapter10/FullPathOmega · JoinSystem.interpΩ_eq_star, which is continuous in (γ,u) because i and H_k^* are.
2. *The ingredients converge in Ω.* $\gamma \to \gamma_0$ in Ω, and $b(i\gamma) \to b(i\gamma_0)$ in Ω by the continuity of b Chapter10/FullPathOmega · JoinSystem.brokenΩ_toY_eqOn. By §IX.13 both converge uniformly, their velocities converge in L^2, and their energies converge.
3. *On a piece.* Cut $[0,1]$ at the dyadic points and at the break points of γ_0. On an interval inside the j-th piece, $H(\gamma,u) = \Phi \circ F_{\gamma,u}$ with $F_{\gamma,u}(s) = ((\gamma(s), b(i\gamma)(s)), u)$, for all (γ,u) near (γ_0,u_0). The velocity of $F_{\gamma,u}$ is $((\gamma', b(i\gamma)'),0)$ off finitely many points Chapter10/FullPathSpace · PathSp.hasDerivAt_of_notMem. Its squared norm is at most $\|\gamma'\|^2 + \|b(i\gamma)'\|^2$ Chapter10/FullPathOmega · JoinSystem.norm_pair_zero_sq_le, so $\int \|F'_{\gamma,u}\|^2 \le E(\gamma) + E(b(i\gamma)) \le C$ eventually. Likewise $\int \|F'_{\gamma,u} - F'_{\gamma_0,u_0}\|^2$ is at most the sum of the L^2 distances of the two velocities, which tends to 0 by step 2 (the integrands are integrable, as continuous functions of four piecewise smooth velocities Chapter10/FullPathOmega · JoinSystem.intervalIntegrable_comp_deriv₄). $F_{\gamma,u} \to F_{\gamma_0,u_0}$ uniformly, by step 2 and $u \to u_0$. Smooth maps preserve L^2 convergence of velocities (§IX.13), which gives the L^2 convergence of the velocities of $H(\gamma,u)$ on the interval.
4. Adding over the intervals and applying the criterion of §IX.13 concludes.

Lemma 17.C (ii) of the exposition. For every k, the inclusion $i : X_k \to Y_k$ is a homotopy equivalence, with homotopy inverse b Chapter10/FullPathOmega · JoinSystem.isHomotopyEquiv_toY.

The proof as formalized. Both homotopies are the deformation, run backwards.

- $b \circ i \simeq \mathrm{id}$ *on* X_k. At time t take $H(\gamma, 1-t)$, except at $t = 1$, where take γ itself. At $t = 0$

this is $H(\gamma, 1) = b(i\gamma)$ Chapter10/FullPathOmega · JoinSystem.interpΩ_one. At $t = 1$, $H(\gamma, 0)$ agrees with γ on $[0, 1]$, so it is at distance 0 from γ Chapter10/FullPathOmega · JoinSystem.edist_interpΩ_zero but need not equal it as a function on $\mathbb{R}$. The modified homotopy is at distance 0 from the continuous $(t, \gamma) \mapsto H(\gamma, 1-t)$, hence continuous (§IX.1). This is the one place where the pseudometric needs care.

- $i \circ b \simeq \mathrm{id}$ *on* Y_k. $(t, \gamma) \mapsto H_k^*(\gamma, 1-t)$, which is $i(b(\gamma))$ at $t = 0$ Chapter10/FullPathOmega · JoinSystem.toY_brokenΩ and γ at $t = 1$.

IX.18 Theorem 17.1

It remains to glue the homotopy equivalences $X_k \to Y_k$. The sets X_k and Y_k are open, increasing, and exhaust Ω and Ω^*. Milnor's Appendix provides the gluing, with two adjustments.

Example 1, pulled back. Let Y be paracompact and Hausdorff, $B_0 \subseteq B_1 \subseteq \cdots$ subsets of Y such that every point lies in the interior of some B_n, and $g : X \to Y$ continuous. Then X is the homotopy direct limit of the sets $g^{-1}(B_n)$, with no assumption on X Chapter10/FullPathSpace · isHomotopyDirectLimit_preimage.

The proof as formalized. Take a partition of unity (ρ_n) on Y subordinate to the open sets $\operatorname{int} B_n$ Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate, and $f = \sum_n n\,\rho_n$, continuous by local finiteness. If $x \notin g^{-1}(B_i)$, then $\rho_j(g(x)) = 0$ for $j \le i$, since the support of ρ_j lies in $B_j \subseteq B_i$. Hence $f(g(x)) = \sum_{j>i} j\,\rho_j(g(x)) \ge (i+1)\sum_j \rho_j(g(x)) = i+1$. Lemma 2.1 of the exposition, the mechanism of Example 1, then applies to $f \circ g$ on X Appendix/Telescope · isHomotopyDirectLimit_of_level. Example 1 itself is the case $g = \mathrm{id}$ Appendix/Examples · example1.

The point is that Ω is not Hausdorff: its pseudometric does not separate paths that agree on $[0, 1]$. So Example 1, which the Appendix states for paracompact Hausdorff spaces, is applied to Ω^*, a metric space, which is paracompact by A. H. Stone's theorem (provided by Mathlib), and the partition of unity is pulled back to Ω along i.

Theorem A on open pieces. If X and Y are the homotopy direct limits of $\{X_n\}$ and $\{Y_n\}$, and $f : X \to Y$ carries each X_n into Y_n by a homotopy equivalence, then f is a homotopy equivalence Appendix/MonotoneUnion · theoremA_general (Part XI). This is Milnor's statement of Theorem A, with no closedness hypothesis. His proof needs closed pieces, and the X_k and Y_k here are open. The Appendix's formal proof compares X and Y with the mapping telescopes of their inclusions Appendix/MonotoneUnion · isHomotopyEquiv_telescopeProj', whose truncations are closed, runs Milnor's proof there Appendix/TheoremA · theoremA, and transfers the result through the projections, which are homotopy equivalences.

Theorem 17.1 from a system of joins. For every system of joins J and all p, q, the inclusion $i : \Omega(N; p, q) \to \Omega^*(N; p, q)$ is a homotopy equivalence Chapter10/FullPathTheorem · JoinSystem.isHomotopyEquiv_toPathC.

The proof as formalized.

1. Every $\gamma \in \Omega^*$ lies in some Y_k, which is open, hence in its interior.
2. Ω^* is the homotopy direct limit of the Y_k by Example 1, and Ω is that of the $X_k = i^{-1}(Y_k)$ by Example 1 pulled back along i.
3. i maps X_k into Y_k, and its restriction there is the inclusion $X_k \to Y_k$ of §IX.17, a homotopy equivalence Appendix/Telescope · restrictTo.
4. Theorem A on open pieces concludes.

Theorem 17.1 (Milnor p. 93). For every slice submanifold N of V, complete or not, and all p, q, the inclusion $i : \Omega(N; p, q) \to \Omega^*(N; p, q)$ of the piecewise smooth paths, with Milnor's metric, into the continuous paths, with the compact-open topology, is a homotopy equivalence Chapter10/FullPathTheorem · theorem_17_1. It is the previous theorem for the system of joins of good balls (§IX.15).

No completeness is used: good balls exist around every point of every slice submanifold, and the Lebesgue numbers are taken on compact images of paths. Completeness is needed for Theorem 17.3, and its absence is the subject of §IX.11.

IX.19 What the formalization taught us

- **The one step of the book that does not hold as we read it.** Milnor's claim that h is clearly continuous (p. 94) fails for the metric d of Ω. When $1/A(\omega)$ crosses an integer from below, $h(\omega)$ acquires a new last segment whose duration δ and length ℓ tend to 0 but whose energy ℓ^2/δ need not, and for $\omega_\lambda = \lambda((1-t)^{1/3} - (1-t))$ it is at least $\frac{27}{64}\delta^{-1/3}$. The formal proof of non-continuity follows Milnor's definitions exactly, for $M = \mathbb{R}$ with the admissible choices $f = |\cdot|$ and $\varepsilon_2 \equiv 1$. Theorem 17.1 is true and is proved by another route.
- **What the modified proof needs that Milnor's construction does not provide.** Regions convex for the minimal geodesics, here from the ambient distance and a bound on the second fundamental form rather than from Whitehead's normal-coordinate argument. One smooth minimal-geodesic map on each whole region, obtained by gluing local ones with a partition of unity. A uniform scale of uniqueness. The control of velocities, not only speeds, by d. Care with Ω not being Hausdorff, in Example 1 and in one homotopy. And Theorem A for open pieces, through telescopes. No completeness is needed.
- **Theorem 17.3.** The levels must tend to ∞. Milnor's requirement of exactly one critical value per interval is impossible when there are finitely many geodesics, and even at most one is not needed for the homotopy type, only for reading homology off a slab. The ladder needs to commute only up to homotopy, so no nested subdivisions, no Whitehead theorem and no Corollary 17.2 are needed, but a relative Theorem 3.5, for pairs of sublevel sets below a compact one, is.
- **The sphere.** The list of geodesics must be shown complete, which uses uniqueness of geodesics after extending past the end points. The multiplicity $n-1$ must be exact and the only conjugate points along a great circle must be the multiples of π, which determining all Jacobi fields vanishing at 0 gives. Non-conjugacy of p and q must be proved, not assumed, and the index comes from Theorem 15.1.
- **Loops and completeness.** Milnor's proof of 17.4 gives the path space $\Omega(S^n; p, q)$, $q \neq \pm p$, not the loop space. Corollary 17.5 needs the completeness it omits, as $\mathbb{R}^{n+1} \setminus \{0\}$ shows.
- **Homology from the filtration.** No cell structure on Ω is needed for its homology: singular chains have compact supports, and compact subsets of Ω have bounded energy. That is true for Milnor's metric, because E is continuous for it, and false for the topology of uniform convergence. The vanishing of the steps is needed in two consecutive degrees.

Part X. Non-conjugate points, topology and curvature (Milnor §§18–19)

In §§18–19 Milnor draws the global consequences of the index theorem and of the fundamental theorem of §17. Section 18 identifies the points conjugate to p with the critical values of $\exp_p$ (Theorem 18.1). Sard's theorem then shows that almost every q is conjugate to p along no geodesic (Corollary 18.2), which is the hypothesis under which Theorem 17.3 describes $\Omega(M; p, q)$. Section 19 relates curvature to topology. Non-positive sectional curvature rules out conjugate points (Lemma 19.1), and this leads to the theorem of Cartan and Hadamard (19.2) and to the vanishing of the higher homotopy groups (19.3). A positive lower bound $(n-1)/r^2$ on the Ricci curvature forces conjugate points on every geodesic longer than πr (Myers, 19.4), bounds the diameter (19.5), and leaves the path space only finitely many cells in each dimension (19.6).

Every result of this Part is proved for a slice submanifold N of a finite-dimensional real inner product space V, modelled on Z, with the induced metric, as in Parts VI–IX; $n = \dim Z$ is the dimension of N. No statement assumes N closed, compact, connected or complete unless it says so. Every Riemannian manifold is of this kind by Nash's embedding theorem, which is quoted, not proved. The formal route leaves Milnor's at five places, each explained where it occurs.

1. Milnor treats $\exp_p$ as a smooth map on all of TM_p, but §10 gives smoothness only near the zero section. The smooth dependence of geodesics on their initial data over a compact time interval is therefore proved first (X.2–X.4). It also removes the completeness assumption from §18.
2. Theorem 18.1 is proved with the explicit Jacobi fields $W_X(t) = t\, d(\exp_q)_{tv}(X)$. It comes with the multiplicity, which is the dimension of the kernel of $d(\exp_q)_v$.
3. Cartan–Hadamard is proved with covering spaces rather than with the path space. In non-positive curvature $\exp_p : T_pN \to N$ is a covering map, and simple connectivity makes it injective. Milnor's statements about the path space are then derived from the covering (X.17).
4. The first half of Corollary 19.3 is proved by lifting cubes through $\exp_p$. No abstract universal covering appears.
5. For Theorem 19.6 the continuity of the Ricci curvature on the unit tangent bundle, which Milnor uses, is proved. The homology of Ω is computed from the energy filtration, not from cells.

Two statements are not proved: that the fundamental group of a complete manifold of non-positive curvature has no element of finite order other than the identity (the second half of 19.3), and Theorem 19.6 when p and q are conjugate along some geodesic (X.20).

Map.

- `Chapter11/ExpSmooth`: geodesics that depend smoothly on a parameter in a normed space; exp smooth on its whole domain; $\exp_q$, its domain and its derivative; Theorem 18.1 with the multiplicity.

- `Chapter11/NonConjugate`: the conjugate locus; Corollary 18.2; the density of non-conjugate points; the complete case.
- `Chapter11/Curvature`: Lemma 19.1; index and nullity 0; E_{**} positive definite; $\exp_p$ nowhere critical.
- `Chapter11/Myers`: the Ricci tensor, its symmetry and Milnor's Assertion; a parallel frame along a geodesic; Theorem 19.4, Corollary 19.5, the unit sphere.
- `Chapter11/LongGeodesics`: test fields on pieces of $[0,1]$; the index bound in the proof of Theorem 19.6.
- `Chapter11/ExpCovering`: the Gauss lemma on the whole domain of $\exp_p$; $\exp_p$ a local diffeomorphism; lifting curves; $\exp_p$ a covering map.
- `Chapter11/CartanHadamard`: Theorem 19.2; the first half of 19.3; one geodesic in each homotopy class; the path space in non-positive curvature.
- `Chapter11/PositiveRicci`: the continuity and the minimum of the Ricci curvature; Theorem 19.6 for non-conjugate end points.
- `Chapter11/RicciHomology`: the homology of $\Omega(N;p,q)$ is finite-dimensional in each degree under the hypotheses of 19.6.

X.1 The setting

The objects of this Part are those of §0.3, §0.8 and §0.9 and of Parts V–IX. This section recalls them in the form in which the proofs below use them.

The submanifold and its curvature. $N \subseteq V$ is a slice submanifold modelled on Z with complement W `Milnor/Chapter2/Submanifold · IsSliceSubmanifold`. Its tangent space $T_qN \subseteq V$ at $q \in N$ is the range of the derivative of the inclusion `Chapter1/ManifoldSubmanifold · tangentSubspace`, of dimension n `Chapter9/JacobiSystem · finrank_tanAt Chapter9/JacobiSystem · tanAt_of_mem`. P_q is the orthogonal projection of V onto T_qN `Chapter6/Projection · tanProj`. Along curves the proofs use a total version P_x, which is the projection for $x \in N$ and 0 for $x \notin N$ `Chapter8/Energy · prj`; it is idempotent and self-adjoint `Chapter8/Energy · prj_prj Chapter8/Energy · inner_prj_left`, and it fixes exactly the tangent vectors `Chapter8/Energy · prj_eq_self_iff`. The second fundamental form is $\mathrm{II}(a,b) = (dP)_q(a)\,b$, the derivative of $q \mapsto P_q$ along N `Chapter6/Submanifold · secondFundForm`. Milnor's curvature tensor, with his sign, is

$$R(x,y)z = \mathrm{II}(y,\mathrm{II}(x,z)) - \mathrm{II}(x,\mathrm{II}(y,z))$$

`Chapter6/Submanifold · curvOp Chapter6/Submanifold · curvOp_apply`. On T_qN it is a trilinear map `Chapter6/Submanifold · curvTensor` with the identities of Lemma 9.3 `Chapter6/Submanifold · isCurvatureTensor_curvTensor`, and along curves it is used in a total form that agrees with it at points of N `Chapter8/SecondVariation · curvR Chapter8/SecondVariation · curvR_of_mem`.

Geodesics and the exponential map. A geodesic on an open set $s \subseteq \mathbb{R}$ is a curve $\gamma : \mathbb{R} \to V$, smooth on s, with values in N there, whose acceleration is normal to N `Chapter7/Geodesic · IsGeodesicOn`. Its velocity is tangent `Chapter7/Geodesic · IsGeodesicOn.deriv_mem`, the tangential part of its acceleration vanishes `Chapter7/Geodesic · IsGeodesicOn.tanProj_deriv_deriv`, and on an interval its speed is constant `Chapter7/Geodesic · IsGeodesicOn.norm_deriv_eq`. A pair (q,v) lies in the domain of exp if some geodesic γ on an interval $(-\delta, 1+\delta)$ has $\gamma(0) = q$ and $\gamma'(0) = v$ `Chapter7/Exp · IsGeodesicFrom Chapter7/Exp · expDom`. Then $\exp_q(v) = \gamma(1)$ `Chapter7/Exp · exp`, whatever γ is chosen `Chapter7/Exp · IsGeodesicFrom.exp_eq`, and $\exp_q(v) \in N$ `Chapter7/HopfRinow · exp_mem_of_mem_expDom`. Further, $\exp_q(0) = q$ `Chapter7/Exp · exp_zero`, and $\exp_q$, as a map from the vector space T_qN to V, has as derivative at 0 the inclusion $T_qN \to V$ `Chapter7/Exp · hasFDerivAt_exp_zero`. The curve $t \mapsto \exp_y(tw)$ has velocity w at $t = 0$ `Chapter7/GeodesicFlow · hasDerivAt_exp_smul_zero`. N is *geodesically complete* if (q,v) lies in the domain for every $q \in N$ and $v \in T_qN$ `Chapter7/GeodesicFlow · IsGeodesicallyComplete`, and then every $t \mapsto \exp_q(tv)$ is a geodesic on all of $\mathbb{R}$ `Chapter7/GeodesicFlow · isGeodesicOn_univ_exp_smul`. Completeness of N for Milnor's

distance ρ Chapter7/Distance · riemDist Chapter7/Metric · IsRiemComplete implies geodesic completeness Chapter7/GeodesicFlow · isGeodesicallyComplete_of_isRiemComplete, and a closed $N \subseteq V$ is complete for ρ Chapter7/Metric · isRiemComplete_of_isClosed.

Curves, Jacobi fields and conjugate points. A *smooth curve on* $[0,1]$ is a curve that is smooth on the closed interval (with one-sided derivatives at the ends) and has values in N there Chapter9/JacobiSystem · IsSmoothCurveIn. Its velocity $\gamma'(t)$ is taken within $[0,1]$ Chapter9/JacobiSystem · velAlong, and so is the covariant derivative $DX/dt = P_{\gamma(t)}(dX/dt)$ of a field along it Chapter9/JacobiSystem · covAlong. In the interior of $[0,1]$ the two-sided version is also used Chapter8/FirstVariation · covD, with the same notation. A *Jacobi field* along γ is a field that is smooth on $[0,1]$, tangent to N, and satisfies $D^2J/dt^2 + R(V,J)V = 0$ on $(0,1)$, where $V = d\gamma/dt$ Chapter8/Jacobi · jacobiExpr Chapter8/Jacobi · IsJacobiField. Being a Jacobi field depends only on the values of J on $[0,1]$ Chapter9/JacobiSpace · IsJacobiField.congr. The Jacobi fields form a vector space $\mathcal{J}(\gamma)$, each normalized to be constant off $[0,1]$ Chapter9/JacobiSpace · jacobiSpace. A Jacobi field is determined by $J(s)$ and $DJ/dt(s)$ at any $s \in [0,1]$, and every pair of tangent vectors occurs Chapter9/JacobiSpace · jacEval_bijective. In particular $J(s) = 0$ and $DJ/dt(s) = 0$ force $J = 0$ on $[0,1]$ Chapter9/JacobiSystem · IsJacobiField.eqOn_zero. Parameters $a \neq b$ of $[0,1]$ are *conjugate* along γ if some Jacobi field vanishes at a and at b without vanishing on all of $[0,1]$ Chapter9/JacobiSpace · IsConjugate. Their *multiplicity* is the dimension of the space of fields of $\mathcal{J}(\gamma)$ that vanish at a and b Chapter9/JacobiSpace · conjSpace Chapter9/JacobiSpace · mem_conjSpace Chapter9/JacobiSpace · conjMultiplicity, and a, b are conjugate exactly when it is positive Chapter9/JacobiSpace · isConjugate_iff. As in Part VII, "$\gamma(t)$ is conjugate to $\gamma(0)$" means that the parameters 0 and t are conjugate.

The path space and E_{**}. $\Omega(N;p,q)$ is the set of piecewise smooth paths in N from p to q Chapter8/Energy · pathSpace, with energy $E(\omega) = \int_0^1 \|d\omega/dt\|^2\,dt$ Chapter8/Energy · energy and length $L(\omega)$ Chapter7/Path · pathLength. An *unbroken geodesic* of Ω is a path smooth on $[0,1]$ that is a geodesic on $(0,1)$ Chapter8/Critical · IsUnbrokenGeodesic, and such a path is a smooth curve in the sense above Chapter9/JacobiSystem · IsUnbrokenGeodesic.isSmoothCurveIn. $T\Omega_\gamma$ is the space of piecewise smooth tangent fields along γ that vanish at 0 and 1 Chapter8/Energy · pathTangent. E_{**} is a symmetric bilinear form on it Chapter8/Hessian · energyHessian Chapter8/Hessian · energyHessian_isSymm, given by the integral form of the second variation formula

$$\tfrac{1}{2}E_{**}(W_1,W_2) = \int_0^1 \Big(\Big\langle \frac{DW_1}{dt}, \frac{DW_2}{dt} \Big\rangle - \langle R(V,W_1)V, W_2\rangle \Big)\,dt$$

Chapter8/Hessian · energyHessian_eq_integral. Its *index* is the supremum, in $\mathbb{N} \cup \{\infty\}$, of the dimensions of the finite-dimensional subspaces on which it is negative definite Chapter9/FormIndex · IsNegDefOn Chapter9/FormIndex · formIndex, so one such subspace of dimension k gives index $\geq k$ Chapter9/FormIndex · le_formIndex. Its *nullity* Chapter9/JacobiSpace · hessNullity is positive exactly when 0 and 1 are conjugate (Theorem 14.1) Chapter9/JacobiSpace · hessNullity_pos_iff, and a field W with $E_{**}(W,\cdot) = 0$ is a Jacobi field Chapter8/Jacobi · theorem_14_1. The index theorem 15.1 says that the index is the number of $t \in (0,1)$ conjugate to 0, counted with multiplicity, a finite sum Chapter9/IndexTheorem · theorem_15_1. At a minimal geodesic Chapter8/Hessian · IsMinimalPath the form E_{**} is positive semi-definite (Lemma 13.6) Chapter8/Hessian · lemma_13_6.

The topology of the path space. $\Omega(N;p,q)$ carries Milnor's metric Chapter10/PathMetric · PathSp. $\Omega^*(N;p,q)$ is the space of continuous paths with the compact-open topology Chapter10/FullPathSpace · pathCSet Chapter10/FullPathSpace · PathC. The inclusion $\Omega \to \Omega^*$ is a homotopy equivalence, with no completeness assumed (Theorem 17.1; Part IX, §IX.18) Chapter10/FullPathTheorem · theorem_17_1. Theorem 17.3 (Part IX, §IX.6) applies when N is closed in V and every geodesic from p to q is non-degenerate. It produces lists $\mathrm{geo}_0, \mathrm{geo}_1, \ldots$ of geodesics from p to q with three properties. Each geodesic from p to q agrees on $[0,1]$ with an entry of exactly one list, and the entries of a list are distinct on $[0,1]$. The energies increase strictly from one list to the next. $\Omega(N;p,q)$ has the homotopy type of the countable cell complex with one cell of dimension $\mathrm{index}(\gamma)$ for each entry γ, attached list by list Chapter10/FundamentalTheorem · theorem_17_3. Here the index of a geodesic is taken as a natural number Chapter10/FundamentalTheorem · geodesicIndex Chapter10/FundamentalTheorem · geodesicIndex_eq. "Has the homotopy type of a countable cell complex" means: homo-

topy equivalent to the mapping telescope of a tower of cell attachments with the given cells `Appendix/MonotoneUnion · IsCountableCellComplexType` (§0.5; Part XI, §XI.15).

X.2 Radial geodesics depending smoothly on a parameter

The analytic core of §18 is a statement Milnor never formulates: geodesics depend smoothly on their initial data over a compact time interval. Part VII proved it for a parameter $u \in \mathbb{R}$, for Lemma 14.4 `Chapter9/GeodesicVariation · raysGood_of_isGeodesicOn`. The smoothness of exp needs parameters in $V \times V$, so the argument is repeated with parameters in an arbitrary real normed space E.

The data. Let $U_0 \subseteq E$ be open, $u_0 \in U_0$, and let $c, w : E \to V$ be smooth on U_0 with $c(u) \in N$ and $P_{c(u)} w(u) = w(u)$ for $u \in U_0$. The *rays* of the family are

$$\mathrm{rays}(u,t) = \exp_{c(u)}(t\, w(u))$$

`Chapter11/ExpSmooth · raysE`. Say that the rays are *good up to* b if there are $\varepsilon, \eta > 0$ with the following three properties. The ball $B(u_0,\varepsilon)$ lies in U_0. rays is smooth on $B(u_0,\varepsilon) \times (-\eta, b+\eta)$. For every u in the ball, $t \mapsto \mathrm{rays}(u,t)$ is a geodesic on $(-\eta, b+\eta)$ `Chapter11/ExpSmooth · RaysGoodE`. Good up to b implies good up to every $b' \le b$ `Chapter11/ExpSmooth · RaysGoodE.mono`. The time derivative $\partial G/\partial t$ of a map G on $E \times \mathbb{R}$ is the derivative applied to $(0,1)$ `Chapter11/ExpSmooth · partialT`. It is the derivative of $t \mapsto G(u,t)$ `Chapter11/ExpSmooth · hasDerivAt_partialT` and is smooth where G is, on an open set `Chapter11/ExpSmooth · contDiffOn_partialT`. On an open set where rays is smooth, the velocity of the u-ray at time t is $\partial\,\mathrm{rays}/\partial t\,(u,t)$ `Chapter11/ExpSmooth · deriv_raysE_eq`.

The local flow. For $x_1 \in N$ and $R > 0$ there are an open $U \ni x_1$ of V, an $h > 0$ and a map $\Phi(y,w,t)$, smooth on $U \times B_R(0) \times (-h,h)$, with the following property. For $y \in N \cap U$ and $w \in T_y N$ with $\|w\| < R$, the curve $t \mapsto \exp_y(tw)$ is a geodesic on $(-h,h)$ and equals $\Phi(y,w,t)$ there `Chapter9/GeodesicVariation · exists_smooth_ray`. This is Lemma 10.2 with smooth dependence on the initial data `Chapter7/GeodesicLocal · lemma_10_2`, rescaled by $\exp_y(tw) = \exp_y((t/\kappa)(\kappa w))$ so that every speed below R is admitted.

The start: the rays are good up to 0 `Chapter11/ExpSmooth · raysGoodE_zero`.

The proof as formalized. Apply the local flow at $c(u_0)$ with $R = \|w(u_0)\| + 1$. The map $u \mapsto (c(u), w(u))$ is continuous at u_0, so there is $\varepsilon > 0$ such that every $u \in B(u_0,\varepsilon)$ satisfies $u \in U_0$, $c(u) \in U$ and $\|w(u)\| < R$. On $B(u_0,\varepsilon) \times (-h/2, h/2)$ one then has $\mathrm{rays}(u,t) = \Phi(c(u), w(u), t)$, a composite of smooth maps, and each ray is a geodesic on $(-h,h) \supseteq (-h/2, h/2)$. Take $\eta = h/2$.

One step `Chapter11/ExpSmooth · RaysGoodE.step`. Suppose the rays are good up to $b \ge 0$. Let Φ on $U_1 \times B_R(0) \times (-h,h)$ be a local flow as above, such that the base ray at time b lies in U_1 and has speed $< R$. Then the rays are good up to $b + h/2$.

The proof as formalized.

1. Let ε, η be the constants for b. Put $F(u) = \mathrm{rays}(u,b)$ and $G(u) = \partial\,\mathrm{rays}/\partial t\,(u,b)$, the position and the velocity of the u-ray at time b. Both are smooth on $B(u_0,\varepsilon)$, as restrictions of smooth maps to the slice $t = b$.
2. By continuity at u_0 there is $\varepsilon' > 0$ with $B(u_0,\varepsilon') \subseteq B(u_0,\varepsilon)$, $F(u) \in U_1$ and $\|G(u)\| < R$ for $u \in B(u_0,\varepsilon')$.
3. For such u, $F(u) \in N$ and $G(u) \in T_{F(u)}N$, since they are the position and velocity of a geodesic. $(F(u), G(u))$ is the geodesic flow of $(c(u), w(u))$ at time b `Chapter7/GeodesicFlow · geoFlow` `Chapter9/GeodesicVariation · geoFlow_eq_exp`. The local flow makes $t \mapsto \exp_{F(u)}(t\,G(u))$ a geodesic on $(-h,h)$. The gluing lemma for geodesics `Chapter7/GeodesicFlow · isGeodesicOn_exp_smul_union` then shows that the u-ray is a geodesic on $(-\eta, b+\eta) \cup (b-h, b+h)$, and that its position at time $t \in (b-h, b+h)$ is the position at time $t - b$ of the flow from $(F(u), G(u))$, that is $\Phi(F(u), G(u), t-b)$.

4. Put $\eta' = \min(\eta, h/4)$. Then $(-\eta', b + h/2 + \eta') \subseteq (-\eta, b + \eta) \cup (b - h, b + h)$, so every ray with $u \in B(u_0, \varepsilon')$ is a geodesic on the new interval.
5. Smoothness is checked near each point (u, t). If $t < b + \eta$, the point lies in the old box, where rays is smooth. If $t \geq b + \eta$, it lies in $B(u_0, \varepsilon') \times (b - h, b + h)$, where $\text{rays}(u, t) = \Phi(F(u), G(u), t - b)$ is smooth. A map smooth near every point of a set is smooth on it `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_of_locally_contDiffOn`.

Smooth dependence over a compact interval `Chapter11/ExpSmooth · raysGoodE_of_isGeodesicOn`. If the base ray $t \mapsto \exp_{c(u_0)}(t\, w(u_0))$ is a geodesic on $(-\delta, B + \delta)$, with $B \geq 0$ and $\delta > 0$, then the rays are good up to B.

The proof as formalized. Let S be the set of $b \in [0, B]$ up to which the rays are good. It contains 0; let $b_0 = \sup S \in [0, B]$. The base ray γ_0 is a geodesic near b_0. Apply the local flow at $\gamma_0(b_0)$ with $R = \|\gamma_0'(b_0)\| + 1$, obtaining U_1, h and Φ. Since $t \mapsto (\gamma_0(t), \gamma_0'(t))$ is continuous, there is $r > 0$ such that $\gamma_0(t) \in U_1$ and $\|\gamma_0'(t)\| < R$ for $|t - b_0| < r$. Choose $s \in S$ with $s > b_0 - \min(r, h/4)$. The step at s shows that the rays are good up to $s + h/2$, hence up to $\min(s + h/2, B)$, which therefore belongs to S and is at most b_0. But $s + h/2 > b_0 + h/4 > b_0$, so the minimum is B, and $B \in S$. A supremum argument takes the place of a finite cover of $[0, B]$.

X.3 The exponential map is smooth on its whole domain

Milnor applies the notion of a critical point to $\exp_p$ on all of TM_p (p. 98), so he treats it as smooth there. Part VI proves smoothness of exp only near the zero section, from Lemma 10.2 (Part VI, §VI.7) `Chapter7/Exp · contMDiffOn_exp`. This section proves it on the whole domain, without completeness.

A smooth retraction onto N `Chapter11/ExpSmooth · exists_retraction`. Every $p \in N$ has an open neighbourhood U in V with a map r, smooth on U, with values in N and fixing the points of $N \cap U$.

Proof. Take the slice chart σ chosen at p `Chapter6/Projection · sliceAt` and put $r(y) = \sigma^{-1}(\sigma(y)_Z, 0)$, the local parametrization of N `Chapter1/ManifoldSubmanifold · sliceParam` applied to the Z-component of $\sigma(y)$. On the open set where σ is defined and $\sigma(y)_Z$ lies in the target of the induced chart of N, r is smooth `Chapter1/ManifoldSubmanifold · contDiffOn_sliceParam` and has values in N `Chapter1/ManifoldSubmanifold · sliceParam_mem`. For $y \in N$ it gives y back `Chapter1/ManifoldSubmanifold · sliceParam_toChart`.

The tangent bundle is $TN = \{(y, w) : y \in N,\ w \in T_y N\}$ `Milnor/Chapter8/TangentBundle · tangentBundleSet`. It is a slice submanifold of $V \times V$ modelled on $Z \times Z$ `Milnor/Chapter8/TangentBundle · isSliceSubmanifold_tangentBundle`. The domain of exp lies in TN, because a geodesic from (q, v) has $q = \gamma(0) \in N$ and $v = \gamma'(0)$ tangent `Chapter11/ExpSmooth · expDom_subset_tangentBundleSet`. Conversely, a point (y, w) of TN lies in the domain as soon as $t \mapsto \exp_y(tw)$ is a geodesic on some $(-\eta, 1 + \eta)$. The curve itself is then the required geodesic, with value y and velocity w at 0 `Chapter11/ExpSmooth · mem_expDom_of_isGeodesicOn`.

Theorem (exp **is smooth on its domain)** `Chapter11/ExpSmooth · exists_contDiffOn_exp`. For every (p, v) in the domain of exp there are an open $O \subseteq V \times V$ containing (p, v) and a map F, smooth on O, such that every point z of $TN \cap O$ lies in the domain and $\exp(z) = F(z)$.

The proof as formalized.

1. $(p, v) \in TN$. Take a retraction r on $U \ni p$.
2. The parameter space is $E = V \times V$ with $U_0 = U \times V$. The data are $c(y, x) = r(y)$ and $w(y, x) = P_{r(y)} x$. The first is smooth with values in N. The second is smooth, because the projection along a smooth map into N is smooth `Chapter8/Energy · ContDiffOn.prj_comp`, and it is tangent at $r(y)$. On $TN \cap U_0$ the data are the identity: $r(y) = y$ for $y \in N$, and $P_y x = x$ for $x \in T_y N$.
3. The base ray $t \mapsto \exp_p(tv)$ is a geodesic on some $(-\delta, 1 + \delta)$ `Chapter7/Exp · IsGeodesicFrom.isGeodesicOn_exp_smul`. By X.2 with $B = 1$ there are $\varepsilon, \eta > 0$ such that rays is smooth on $B((p, v), \varepsilon) \times (-\eta, 1 + \eta)$ and every ray from the ball is a geodesic on $(-\eta, 1 + \eta)$.

4. Take $O = B((p, v), \varepsilon)$ and $F(z) = \text{rays}(z, 1)$, smooth on O. For $z = (y, x) \in O \cap TN$, the ray of the data is $t \mapsto \exp_y(tx)$, a geodesic on $(-\eta, 1 + \eta)$. So z lies in the domain, and $\exp_y(x) = \text{rays}(z, 1) = F(z)$.

Two consequences follow. The domain of exp is $O' \cap TN$ for an open $O' \subseteq V \times V$, namely the union of the sets O chosen at the points of the domain `Chapter11/ExpSmooth · exists_isOpen_expDom`. And for the manifold structure of TN, exp is smooth at every point of its domain `Chapter11/ExpSmooth · contMDiffOn_exp_expDom`: near such a point it agrees, on the open set of points of TN lying in O, with F composed with the smooth inclusion $TN \to V \times V$.

X.4 The exponential map at one point

For $q \in N$ the domain of $\exp_q$ is $\mathcal{D}_q = \{v \in T_qN : (q, v) \in \text{domain of } \exp\}$ `Chapter11/ExpSmooth · expDomAt`. It contains 0 `Chapter11/ExpSmooth · zero_mem_expDomAt`. It is star-shaped about 0 `Chapter11/ExpSmooth · smul_mem_expDomAt`: if γ is a geodesic from q with velocity v, then for $0 \le t \le 1$ the curve $s \mapsto \gamma(ts)$ is a geodesic from q with velocity tv `Chapter7/Exp · IsGeodesicFrom.exp_smul`. Near each $v \in \mathcal{D}_q$, $\exp_q$ is the restriction of a smooth map `Chapter11/ExpSmooth · exists_contDiffAt_expAt`: compose the F of X.3 with the affine map $y \mapsto (q, y)$, which sends T_qN into TN, and take the preimage of O. Hence $\mathcal{D}_q$ is open in T_qN `Chapter11/ExpSmooth · isOpen_expDomAt`, and $\exp_q$ is smooth on it as a map from the vector space T_qN to V `Chapter11/ExpSmooth · contDiffOn_expAt`. For $v \in \mathcal{D}_q$ the *radial geodesic* $\gamma_v(t) = \exp_q(tv)$ is a geodesic on some $(-\delta, 1 + \delta)$ `Chapter11/ExpSmooth · isGeodesicOn_ray` and a smooth curve on $[0, 1]$ `Chapter11/ExpSmooth · isSmoothCurveIn_ray`. If N is geodesically complete, $\mathcal{D}_q = T_qN$ `Chapter11/NonConjugate · expDomAt_eq_univ`, which is Milnor's setting.

The derivative of $\exp_q$ at v is the linear map $d(\exp_q)_v : T_qN \to V$ `Chapter11/ExpSmooth · dExpAt` `Chapter11/ExpSmooth · dExpAt_apply`. This is Milnor's $\exp_*$ on $T(TM_p)_v$, with $T(TM_p)_v$ identified with TM_p, as he does on p. 99 when he differentiates along paths $v(u)$ in TM_p. $\exp_q$ is *critical* at v when $d(\exp_q)_v$ is not one-to-one. Milnor's $\exp_*$ takes values in the tangent space of M at $\exp v$, the formal one in V; whether a linear map is one-to-one does not depend on the space that receives its values.

X.5 Theorem 18.1

Milnor's proof (pp. 98–99) has two halves. If $\exp_* X = 0$ with $X \neq 0$, a path $v(u)$ in TM_p with $v(0) = v$ and $dv/du(0) = X$ gives the variation $\alpha(u, t) = \exp(t\, v(u))$ through geodesics. Its variation field is a Jacobi field that vanishes at both ends and has non-zero initial derivative. Conversely, if $\exp_*$ is non-singular at v, the Jacobi fields built from a basis $X_1, \dots, X_n$ have independent values at $\exp v$, and a count of dimensions shows that no non-trivial Jacobi field vanishes at both ends. The formal proof takes the straight path $v(u) = v + uX$ and writes its variation field down.

Definition `Chapter11/ExpSmooth · expJacobi`. For $q \in N$ and $v, X \in T_qN$ put

$$W_X(t) = t\, d(\exp_q)_{tv}(X) \qquad (0 \le t \le 1),$$

extended to be constant off $[0, 1]$ (the formula is applied to the point of $[0, 1]$ nearest to t) `Chapter11/ExpSmooth · expJacobi_apply_of_mem`. The definition packages $X \mapsto W_X$ as a linear map from T_qN to fields along γ_v. Directly from the formula, $W_X(0) = 0$ `Chapter11/ExpSmooth · expJacobi_zero` and $W_X(1) = d(\exp_q)_v(X)$ `Chapter11/ExpSmooth · expJacobi_one`.

Lemma (W_X is a Jacobi field) `Chapter11/ExpSmooth · isJacobiField_expJacobi`. For $v \in \mathcal{D}_q$ and $X \in T_qN$, W_X is a Jacobi field along γ_v.

The proof as formalized.

1. Let $g(u,t) = t(v+uX)$, a smooth map $\mathbb{R}^2 \to T_qN$, and $\alpha(u,t) = \exp_q(g(u,t))$, Milnor's variation. The compact set $\{0\} \times [0,1]$ is mapped by g into the open set $\mathcal{D}_q$, because $g(0,t) = tv$ and $\mathcal{D}_q$ is star-shaped. The generalized tube lemma `Mathlib/Topology/Compactness/Compact · generalized_tube_lemma` gives open sets $U' \ni 0$ and $T' \supseteq [0,1]$ with $g(U' \times T') \subseteq \mathcal{D}_q$. Shrinking U' to an interval $(-\varepsilon, \varepsilon)$ gives a box $(-\varepsilon,\varepsilon) \times [0,1]$ on which g lands in $\mathcal{D}_q$. Milnor does not mention this step, but without it his variation is not defined on any such box.
2. On the box $(-\varepsilon,\varepsilon) \times [0,1]$, α is smooth, as a composite of g with $\exp_q$, which is smooth on $\mathcal{D}_q$. It has values in N, because exp maps its domain into N.
3. For $|u| < \varepsilon$ the point $v + uX = g(u,1)$ lies in $\mathcal{D}_q$. Its radial geodesic, which is $\alpha(u,\cdot)$, is a geodesic on an interval containing $[0,1]$, hence on $(0,1)$.
4. Lemma 14.3 `Chapter9/JacobiVariation · lemma_14_3` (Part VII) now says that $t \mapsto \partial\alpha/\partial u\,(0,t)$ is a Jacobi field along $\alpha(0,\cdot) = \gamma_v$.
5. For $t \in [0,1]$, $u \mapsto t(v+uX)$ has derivative tX at $u = 0$, and $\exp_q$ is differentiable at $tv \in \mathcal{D}_q$. The chain rule gives $\partial\alpha/\partial u\,(0,t) = d(\exp_q)_{tv}(tX) = W_X(t)$. A Jacobi field depends only on its values on $[0,1]$, so W_X is one.

Lemma ($DW_X/dt\,(0) = X$) `Chapter11/ExpSmooth · covAlong_expJacobi_zero`. For every $v, X \in T_qN$ the covariant derivative of W_X along γ_v at $t = 0$ is X.

The proof as formalized. The domain $\mathcal{D}_q$ is open and contains 0, so $\exp_q$ is smooth near 0, and $t \mapsto d(\exp_q)_{tv}(X)$ is differentiable at $t = 0$. By the product rule, the derivative of $t \mapsto t\,d(\exp_q)_{tv}(X)$ at 0 is $d(\exp_q)_0(X)$; the second term carries the factor $t = 0$. This is also the derivative within $[0,1]$. Since $d(\exp_q)_0$ is the inclusion `Chapter7/Exp · hasFDerivAt_exp_zero`, the derivative is X. The covariant derivative projects it by $P_{\gamma_v(0)} = P_q$, which fixes X.

Milnor obtains $DW/dt(0) = D/du\,v(u)$ from the symmetry lemma 8.7 and the covariant derivative along the constant curve $u \mapsto \alpha(u,0) = p$. The formal proof needs neither, because W_X is known explicitly and $d(\exp_q)_0$ is the identity.

So W_X belongs to $\mathcal{J}(\gamma_v)$ `Chapter11/ExpSmooth · expJacobi_mem_jacobiSpace`; it was built to be constant off $[0,1]$. The linear map $X \mapsto W_X$ is injective `Chapter11/ExpSmooth · expJacobi_injective`, since X is recovered as $DW_X/dt(0)$.

Lemma (every Jacobi field vanishing at 0 is a W_X) `Chapter11/ExpSmooth · eq_expJacobi`. If $J \in \mathcal{J}(\gamma_v)$ and $J(0) = 0$, then $J = W_X$ with $X = DJ/dt(0)$.

Proof. $X = DJ/dt(0)$ is a projection onto $T_{\gamma_v(0)}N = T_qN$ `Chapter8/Energy · prj_mem`, so W_X is defined. J and W_X both vanish at 0 and have covariant derivative X there. The map $J \mapsto (J(0), DJ/dt(0))$ is injective on $\mathcal{J}(\gamma_v)$ `Chapter9/JacobiSpace · jacEval_bijective`, which is Milnor's statement that a Jacobi field is determined by these two vectors (p. 77; Corollary 14.B of the exposition). Hence $J = W_X$.

Theorem (the fields vanishing at both ends) `Chapter11/ExpSmooth · conjSpace_eq_map_ker`. For $v \in \mathcal{D}_q$, the fields of $\mathcal{J}(\gamma_v)$ vanishing at 0 and 1 are exactly the W_X with $X \in \ker d(\exp_q)_v$.

Proof. If J vanishes at 0 and 1, then $J = W_X$ by the last lemma, and $0 = J(1) = W_X(1) = d(\exp_q)_v(X)$. Conversely, for X in the kernel, $W_X \in \mathcal{J}(\gamma_v)$, $W_X(0) = 0$ and $W_X(1) = d(\exp_q)_v(X) = 0$.

Theorem 18.1, with the multiplicity `Chapter11/ExpSmooth · theorem_18_1_multiplicity`. Let $q \in N$ and $v \in \mathcal{D}_q$. The multiplicity of 0 and 1 as conjugate parameters along $\gamma_v(t) = \exp_q(tv)$ equals $\dim \ker d(\exp_q)_v$.

Proof. The space whose dimension is the multiplicity is the image of $\ker d(\exp_q)_v$ under the injective linear map $X \mapsto W_X$, and an injective linear map preserves dimension.

Theorem 18.1 `Chapter11/ExpSmooth · theorem_18_1`. Let $q \in N$ and $v \in \mathcal{D}_q$. Then $\exp_q(v)$ is conjugate to q along γ_v, that is, the parameters 0 and 1 are conjugate, if and only if $\exp_q$ is critical at v.

Proof. The parameters $0 \neq 1$ of $[0,1]$ are conjugate if and only if the multiplicity is positive `Chapter9/JacobiSpace · isConjugate_iff`. By the previous theorem this means $\ker d(\exp_q)_v \neq 0$, that is, $d(\exp_q)_v$ is not one-to-one.

Milnor's dimension count for the converse is replaced by the bijectivity of $J \mapsto (J(0), DJ/dt(0))$, and completeness is not assumed: v ranges over the open, star-shaped domain $\mathcal{D}_q$. The equality of the multiplicity with the nullity of $\exp_*$ is a strengthening that Milnor does not state.

X.6 The conjugate locus and Corollary 18.2

Definition `Chapter11/NonConjugate · conjLocus`. For $p \in V$, the *conjugate locus* $C(p)$ is the set of q for which there is a smooth curve γ on $[0,1]$, a geodesic on $(0,1)$, with $\gamma(0) = p$ and $\gamma(1) = q$, along which 0 and 1 are conjugate. Conjugacy depends only on γ on $[0,1]$ `Chapter11/NonConjugate · IsConjugate.congr_curve` `Chapter11/NonConjugate · IsConjugate.congr_curve_iff`, because the Jacobi equation does `Chapter9/JacobiVariation · IsJacobiField.congr_curve`.

Milnor states that Corollary 18.2 follows at once from 18.1 and Sard's theorem (p. 99). Three facts are hidden in this. Every geodesic from p must be a radial geodesic, so that its conjugate points are critical values of $\exp_p$. $\exp_p$ must be smooth on its whole domain (X.3). And without completeness the domain must be open (X.4).

Theorem (18.1 along any geodesic from p) `Chapter11/NonConjugate · theorem_18_1_of_geodesic`. Let $p \in N$ and let γ be a smooth curve on $[0,1]$, a geodesic on $(0,1)$, with $\gamma(0) = p$. Then its initial velocity $v = \gamma'(0)$ lies in T_pN and in $\mathcal{D}_p$, $\exp_p(v) = \gamma(1)$, and $\gamma(1)$ is conjugate to p along γ if and only if $\exp_p$ is critical at v.

The proof as formalized. A geodesic segment extends past its end points and is radial `Chapter9/GeodesicVariation · exists_initial_ray` (Part VII): there is $\eta > 0$ such that $t \mapsto \exp_p(tv)$ is a geodesic on $(-\eta, 1+\eta)$ that agrees with γ on $[0,1]$. The velocity of a smooth curve in N is tangent `Chapter9/JacobiSystem · IsSmoothCurveIn.velAlong_mem`, so $v \in T_pN$. The ray is a geodesic on an interval containing $[0,1]$, so $v \in \mathcal{D}_p$, and $\exp_p(v)$, the ray at $t = 1$, is $\gamma(1)$. Conjugacy along γ is conjugacy along the ray, which agrees with γ on $[0,1]$, and Theorem 18.1 applies to the ray.

Theorem (the conjugate locus is the set of critical values of $\exp_p$) `Chapter11/NonConjugate · conjLocus_eq`. For $p \in N$,

$$C(p) = \exp_p\left(\{x \in \mathcal{D}_p : d(\exp_p)_x \text{ is not one-to-one}\}\right).$$

Proof. If $q \in C(p)$ through γ, the previous theorem gives $v \in \mathcal{D}_p$, critical, with $\exp_p(v) = \gamma(1) = q$. Conversely, for a critical $x \in \mathcal{D}_p$, the radial geodesic γ_x is a smooth curve on $[0,1]$ and a geodesic on $(0,1)$, running from p to $\exp_p(x)$, along which 0 and 1 are conjugate by Theorem 18.1. If N is geodesically complete, $C(p)$ is the set of critical values of $\exp_p : T_pN \to N$ `Chapter11/NonConjugate · conjLocus_eq_of_complete`.

Corollary 18.2 `Chapter11/NonConjugate · corollary_18_2`. Let $p \in N$, let σ be any slice chart of N, and let μ be any Haar measure on Z for the Borel σ-algebra. Then the image of $C(p) \cap \operatorname{dom}\sigma$ under the chart $y \mapsto \sigma(y)_Z$ of N has μ-measure 0.

The book never defines "measure zero" in a manifold. Its Sard theorem (6.1) states the conclusion in these words, but §6 applies it only to maps into $\mathbb{R}^n$, and §18 is the first place where the target is a manifold. The formal statement reads the phrase in every chart: almost every q means every q outside a set whose image in every slice chart is null for every Haar measure.

The proof as formalized.

1. $T = T_pN$ has dimension $n = \dim Z$, so there is a linear isomorphism $e : Z \to T$ `Mathlib/Topology/Algebra/Module/FiniteDimension · ContinuousLinearEquiv.ofFinrankEq`.
2. Let $U' = e^{-1}(\mathcal{D}_p) \cap (\exp_p \circ e)^{-1}(\operatorname{dom}\sigma)$, an open subset of Z by continuity. On it the map $g(z) = \sigma(\exp_p(ez))_Z$ is smooth, because the slice chart is smooth on its domain `Milnor/Chapter8/TangentBundle · sliceChart_contDiffOn`.
3. Sard's theorem of the companion project `Milnor/Chapter3/Sard · sard` says that g maps the set of points of U' where dg is not onto `Milnor/Chapter3/Basic · Sard.critSet` to a μ-null set. It remains to show that the chart image of $C(p) \cap \operatorname{dom}\sigma$ lies in this image.
4. Let $y \in C(p) \cap \operatorname{dom}\sigma$. By the previous theorem $y = \exp_p(x)$ with $x \in \mathcal{D}_p$ critical. Then $z = e^{-1}x \in U'$ and $g(z) = \sigma(y)_Z$. By the chain rule $dg_z = \mathrm{pr}_Z \circ d\sigma_y \circ d(\exp_p)_x \circ e$. If dg_z were onto, then, being an endomorphism of the finite-dimensional space Z, it would also be one-to-one. But $d(\exp_p)_x(a) = d(\exp_p)_x(b)$ gives $dg_z(e^{-1}(a-b)) = 0$, hence $a = b$, so $d(\exp_p)_x$ would be one-to-one, against the choice of x. So dg_z is not onto and $\sigma(y)_Z \in g(\text{critical set})$.

Milnor's Sard theorem is stated for C^1 maps; the companion project proves it for smooth maps, which is the case here.

Theorem (non-conjugate points are dense) `Chapter11/NonConjugate · dense_compl_conjLocus`. For $p \in N$, the points of N not conjugate to p along any geodesic are dense in N.

This is what §18 is for: Milnor's justification of Theorem 17.3 rests on the existence of such points. From measure zero to existence one more fact is needed: a null set has empty interior.

The proof as formalized. Give Z its Borel σ-algebra and the Haar measure μ of a basis. Let U be an open subset of N containing a point q, and σ the slice chart at q. The chart of N induced by σ is a partial homeomorphism, so the image O of $U \cap \operatorname{dom}\sigma$ is open in Z, and it is non-empty. A non-empty open set has positive Haar measure `Mathlib/MeasureTheory/Measure/OpenPos · IsOpen.measure_pos`, so O is not contained in the null set of Corollary 18.2. A point of O outside that set is the image of some $y \in U \cap \operatorname{dom}\sigma$, and $y \notin C(p)$, since otherwise its image would lie in the null set.

Theorem (non-degeneracy off the conjugate locus) `Chapter11/NonConjugate · hessNullity_eq_zero_of_notMem_conjLocus`. If $q \notin C(p)$, then E_{**} has nullity 0 at every unbroken geodesic $\gamma \in \Omega(N; p, q)$.

Proof. A positive nullity makes 0 and 1 conjugate (Theorem 14.1), and γ, a smooth curve from p to q and a geodesic on $(0,1)$, would put q in $C(p)$.

This is the hypothesis of Theorems 16.3 and 17.3, and Part IX, §IX.6, states Theorem 17.3 in this form `Chapter10/FundamentalTheorem · theorem_17_3_nonconj`. With the density theorem it holds for a dense set of q, and with Corollary 18.2 for almost every q.

X.7 Lemma 19.1 and non-positive curvature

Definition `Chapter11/Curvature · NonposCurvature`. N has *non-positive curvature* if $\langle R(A,B)A, B\rangle \le 0$ for all $A, B \in T_qN$ and all $q \in N$, which is Milnor's hypothesis in Lemma 19.1. This says exactly that every sectional curvature is ≤ 0 `Chapter11/Curvature · nonposCurvature_iff_sectionalCurvature`, for the sectional curvature

$$K(A,B) = \frac{\langle R(A,B)A, B\rangle}{\|A\|^2\|B\|^2 - \langle A, B\rangle^2}$$

of independent A, B `Chapter6/CurvatureAlgebra · sectionalCurvature` (Part V, §V.8). For independent A, B the denominator is positive `Chapter6/CurvatureAlgebra · gram_pos`, so $K(A,B)$ has the sign of $\langle R(A,B)A, B\rangle$. For dependent A, B the numerator vanishes `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_apply_self_eq_zero_of_not_linearIndependent`.

Lemma (the derivative of $\langle DJ/dt, J\rangle$) `Chapter11/Curvature · hasDerivAt_inner_covAlong`. Let γ be a smooth curve on $[0,1]$ and J a Jacobi field along it. For $t \in (0,1)$,

$$\frac{d}{dt}\Big\langle \frac{DJ}{dt}, J\Big\rangle = \Big\|\frac{DJ}{dt}\Big\|^2 - \langle R(V,J)V, J\rangle.$$

The proof as formalized. On the open interval $(0,1)$ the derivative within $[0,1]$ is the ordinary one, so $DJ/dt = PJ'$ there, a smooth function. The product rule gives

$$\frac{d}{dt}\langle PJ', J\rangle = \langle (PJ')', J\rangle + \langle PJ', J'\rangle.$$

For the first term, $J = PJ$ because J is tangent, and P is self-adjoint, so $\langle (PJ')', J\rangle = \langle P(PJ')', J\rangle = \langle D^2J/dt^2, J\rangle$. By the Jacobi equation this equals $-\langle R(V,J)V, J\rangle$. For the second term, P is idempotent and self-adjoint, so $\langle PJ', J'\rangle = \langle PJ', PJ'\rangle = \|DJ/dt\|^2$. Milnor writes the same derivative as $\langle D^2J/dt^2, J\rangle + \|DJ/dt\|^2$.

The covariant derivative within $[0,1]$ of a field smooth on $[0,1]$ is continuous on $[0,1]$, ends included `Chapter11/Curvature · continuousOn_covAlong`, because $t \mapsto P_{\gamma(t)}$ is smooth on $[0,1]$ `Chapter9/JacobiSystem · IsSmoothCurveIn.contDiffOn_prj`.

Lemma (the key step of 19.1) `Chapter11/Curvature · IsJacobiField.eqOn_zero_of_nonpos`. Let N have non-positive curvature, let γ be a smooth curve on $[0,1]$ that is a geodesic on $(0,1)$, and let J be a Jacobi field along γ with $J(a) = J(b) = 0$ for some $0 \le a < b \le 1$. Then $J = 0$ on $[0,1]$.

The proof as formalized.

1. Let $g = \langle DJ/dt, J\rangle$. For $t \in (a,b) \subseteq (0,1)$, $V(t)$ is the velocity of a geodesic, so it is tangent, and $J(t)$ is tangent. The hypothesis gives $\langle R(V,J)V, J\rangle \le 0$, and the last lemma gives $g'(t) \ge \|DJ/dt(t)\|^2 \ge 0$.
2. g is continuous on $[a,b]$, by the continuity of DJ/dt and of J, and its derivative inside is ≥ 0, so it is monotone there `Mathlib/Analysis/Calculus/Deriv/MeanValue · monotoneOn_of_deriv_nonneg`. As $g(a) = g(b) = 0$, g vanishes on $[a,b]$.
3. On (a,b), g is locally constant, so $g' = 0$ and $\|DJ/dt\|^2 \le g' = 0$. Hence $DJ/dt = 0$ on (a,b).
4. At a: the interval (a,b) is non-empty and accumulates at a, and DJ/dt is continuous at a. Its limit from the right is therefore both $DJ/dt(a)$ and 0, so $DJ/dt(a) = 0$.
5. $J(a) = 0$ and $DJ/dt(a) = 0$, so $J = 0$ on $[0,1]$ `Chapter9/JacobiSystem · IsJacobiField.eqOn_zero`.

Milnor observes that g is strictly increasing where $DJ/dt \neq 0$; only the inequality $g' \ge \|DJ/dt\|^2$ is used here. He takes $J(0) = J(t_0) = 0$. Step 4, which passes from $DJ/dt = 0$ on the open interval to the end point, is left implicit in his proof.

Lemma 19.1 `Chapter11/Curvature · lemma_19_1`. If N has non-positive curvature, then along a smooth curve on $[0,1]$ that is a geodesic on $(0,1)$, no two parameters a, b are conjugate.

Proof. Conjugacy would give a Jacobi field vanishing at $a \neq b$ but not on all of $[0,1]$. Ordering a and b, the key step makes it vanish on $[0,1]$.

The lemma is local along each geodesic: it needs neither completeness nor compactness. Its consequences for the index form are the first steps of Milnor's proof of 19.2.

Theorem (index and nullity 0) `Chapter11/Curvature · formIndex_eq_zero_of_nonpos`. In non-positive curvature every unbroken geodesic $\gamma \in \Omega(N; p, q)$ has index 0 and nullity 0.

Proof. By the index theorem the index is $\sum_{t\in(0,1)}$ (multiplicity of 0 and t). Each multiplicity is 0, since a positive one would make 0 and t conjugate, against Lemma 19.1. So the sum is 0. A positive nullity would make 0 and 1 conjugate (Theorem 14.1).

Theorem (E_{**} is positive definite) `Chapter11/Curvature · energyHessian_pos_of_nonpos`. In non-positive curvature, $E_{**}(X,X) > 0$ for every $X \in T\Omega_\gamma$ that does not vanish identically on $[0,1]$, at every unbroken geodesic $\gamma \in \Omega(N;p,q)$.

The proof as formalized. By the integral form, $E_{**}(Y,Y) = 2\int_0^1 (\|DY/dt\|^2 - \langle R(V,Y)V,Y\rangle)\,dt$. The curvature term is ≤ 0 at every $t \in (0,1)$ by hypothesis, since $Y(t)$ and $V(t)$ are tangent, and it vanishes at $t = 0,1$, where $Y = 0$. So E_{**} is positive semi-definite. If $E_{**}(X,X) = 0$, then X lies in the kernel of E_{**}, by the Cauchy–Schwarz inequality for symmetric semi-definite forms `Mathlib/LinearAlgebra/SesquilinearForm/Basic · apply_apply_same_eq_zero_iff`. That is, $E_{**}(X,\cdot) = 0$, so X is a Jacobi field (Theorem 14.1). It vanishes at 0 and 1 and not everywhere on $[0,1]$, so 0 and 1 would be conjugate, against Lemma 19.1. Milnor does not state this positivity.

Theorem ($\exp_p$ is nowhere critical) `Chapter11/Curvature · injective_dExpAt_of_nonpos`. In non-positive curvature, $d(\exp_p)_v$ is one-to-one for every $p \in N$ and every $v \in \mathcal{D}_p$.

Proof. Otherwise Theorem 18.1 makes 0 and 1 conjugate along γ_v, against Lemma 19.1. This is the last step of Milnor's proof of 19.2 (p. 103).

X.8 The Ricci tensor and Milnor's Assertion

Definition `Chapter11/Myers · ricci`. The *Ricci tensor* at $q \in N$ is the bilinear form on T_qN

$$K(U_1,U_2) = \operatorname{trace}\,(W \mapsto R(U_1,W)U_2),$$

the trace of a linear map of T_qN. This is Milnor's definition (p. 104) taken literally. In an orthonormal basis $E_1,\dots,E_n$ of T_qN,

$$K(U_1,U_2) = \sum_i \langle R(U_1,E_i)U_2,E_i\rangle$$

`Chapter11/Myers · ricci_eq_sum`, because the trace of a linear map T is $\sum_i \langle E_i, TE_i\rangle$ `Mathlib/Analysis/InnerProductSpace/Trace · LinearMap.trace_eq_sum_inner`.

Symmetry `Chapter11/Myers · ricci_comm`. $K(U_1,U_2) = K(U_2,U_1)$. Milnor says this follows easily from 9.3. The formal proof computes in an orthonormal basis and uses, term by term, the pair symmetry $\langle R(X,Y)Z,W\rangle = \langle R(Z,W)X,Y\rangle$ of Lemma 9.3 (4) `Chapter6/CurvatureAlgebra · IsCurvatureTensor.inner_comm_pairs`, which holds for the curvature tensor of N `Chapter6/Submanifold · isCurvatureTensor_curvTensor`. With $X = U_1$, $Y = W = E_i$ and $Z = U_2$ it gives $\langle R(U_1,E_i)U_2,E_i\rangle = \langle R(U_2,E_i)U_1,E_i\rangle$.

Milnor's Assertion `Chapter11/Myers · ricci_eq_sum_erase`. For an orthonormal basis (b_i) of T_qN and an index i_0,

$$K(b_{i_0},b_{i_0}) = \sum_{i\neq i_0} \langle R(b_{i_0},b_i)b_{i_0},b_i\rangle,$$

the sum of the sectional curvatures of the planes spanned by b_{i_0} and the other b_i.

Proof. As Milnor says, the term $i = i_0$ of the sum in the basis vanishes, because $R(x,x) = 0$ `Chapter11/Myers · curvOp_self`: $R(x,x)$ is a difference of two equal compositions of second fundamental forms.

The hypothesis of Myers' theorem is the bound $K(U,U) \ge (n-1)/r^2$ for every unit vector $U \in T_qN$ and every $q \in N$, with $n = \dim Z$ `Chapter11/Myers · RicciLowerBound`.

X.9 Myers' theorem (Theorem 19.4)

Milnor's proof (pp. 104–105) runs as follows. Take parallel fields $P_1,\dots,P_n$ along the geodesic $\gamma : [0,1] \to M$ of length L, orthonormal at one point and hence everywhere, with P_n pointing along γ,

so that $V = LP_n$. Put $W_i(t) = \sin(\pi t)P_i(t)$, compute $\frac{1}{2}E_{**}(W_i, W_i)$, and sum over $i < n$. The sum is $\int_0^1 \sin^2(\pi t)((n-1)\pi^2 - L^2K(P_n, P_n))\,dt$, which is negative when $L > \pi r$. Some $E_{**}(W_i, W_i)$ is then negative, so the index is positive and the index theorem produces a conjugate point. The formal proof follows this outline. It first has to make the parallel frame available. Parallel transport (Part V, §V.19) is built for smooth curves defined on all of $\mathbb{R}$ `Chapter6/Parallel · IsParallel` `Chapter6/Parallel · parallelField` `Chapter6/Parallel · parallelField_spec`, so γ is extended beyond $[0, 1]$ first.

A smooth clamp `Chapter11/Myers · exists_smooth_clamp`. For $a > 0$ there is a smooth $\varphi : \mathbb{R} \to [-a, 1 + a]$ with $\varphi(t) = t$ on $[0, 1]$, namely

$$\varphi(t) = t + (1 + a - t)\,\tau\Big(\frac{t-1}{a}\Big) + (-a - t)\,\tau\Big(\frac{-t}{a}\Big),$$

where τ is Mathlib's smooth transition function: 0 on $(-\infty, 0]$, 1 on $[1, \infty)$, with values in $[0, 1]$ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`. On $[0, 1]$ both τ-terms vanish. For $t \le 0$ only the last term can be non-zero; it moves t towards $-a$ by a fraction $\tau \in [0, 1]$ of the distance, and $\varphi(t) = -a$ for $t \le -a$. The case $t \ge 1$ is symmetric. The formal proof checks these inequalities case by case.

Extension of a geodesic segment `Chapter11/Myers · exists_smooth_extension`. Let $\gamma \in \Omega(N; p, q)$ be an unbroken geodesic of length L. There are a smooth curve $c : \mathbb{R} \to N$ with $c = \gamma$ on $[0, 1]$ and a smooth field V_f along c, tangent to N, with $V_f = \gamma'$ on $(0, 1)$ and $\|V_f(t)\| = L$ for all t, such that $L^{-1}V_f$ is parallel along c.

The proof as formalized.

1. The radial geodesic $\mathrm{ray}(t) = \exp_{\gamma(0)}(t\gamma'(0))$ is a geodesic on some $(-\eta, 1 + \eta)$ that agrees with γ on $[0, 1]$ `Chapter9/GeodesicVariation · exists_initial_ray`.
2. $L = \|\mathrm{ray}'(0)\|$. The length of a geodesic on an interval is its constant speed times the length of the interval `Chapter7/Shortest · IsGeodesicOn.pathLength_eq`, and $\gamma' = \mathrm{ray}'$ on $(0, 1)$. By constancy of the speed, $\|\mathrm{ray}'(t)\| = L$ on $(-\eta, 1 + \eta)$.
3. Let φ be the clamp for $a = \eta/2$, with values in $(-\eta, 1+\eta)$. Put $c = \mathrm{ray} \circ \varphi$ and $V_f(t) = \mathrm{ray}'(\varphi(t))$. Both are smooth, $c = \gamma$ on $[0, 1]$, $V_f = \gamma'$ on $(0, 1)$, $\|V_f\| = L$, and $V_f(t)$ is tangent at $c(t)$.
4. $L^{-1}V_f$ is parallel along c. Its derivative is $L^{-1}\varphi'(t)\,\mathrm{ray}''(\varphi(t))$, and the acceleration of a geodesic has no tangential part `Chapter7/Geodesic · IsGeodesicOn.tanProj_deriv_deriv`, so the covariant derivative vanishes. Outside $[0, 1]$ the curve c runs along the geodesic at a varying speed and is in general not a geodesic, but V_f is still parallel along it.

A parallel frame `Chapter11/Myers · exists_parallel_frame`. Let $\gamma \in \Omega(N; p, q)$ be an unbroken geodesic of length $L > 0$. Take c and V_f as above. Then there are a finite index set I, a subset $s \subseteq I$ with $|s| + 1 = n$, and parallel fields P_i along c with $\|P_i(t)\| = 1$, such that for every r with the Ricci bound and every t,

$$\sum_{i \in s} \langle R(V_f, P_i)V_f, P_i\rangle(t) \;\ge\; L^2\,\frac{|s|}{r^2}.$$

The proof as formalized.

1. $e_0 = L^{-1}V_f(0)$ is a unit vector of $T_{c(0)}N$. Extend $\{e_0\}$ to an orthonormal basis b of $T_{c(0)}N$ `Mathlib/Analysis/InnerProductSpace/PiL2 · Orthonormal.exists_orthonormalBasis_extension`, indexed by a finite set $I \ni e_0$; let i_0 be the index of e_0.
2. Let P_i be the parallel field along c with $P_i(0) = b_i$. Two parallel fields with the same value at one time coincide `Chapter6/Parallel · IsParallel.eq`, so $P_{i_0} = L^{-1}V_f$ and $V_f = LP_{i_0}$. This uses that V_f itself is parallel, which Milnor's identity $V = LP_n$ takes for granted.

3. At each t, $(P_i(t))_i$ is the image of b under parallel transport from $c(0)$ to $c(t)$, a linear isometry onto $T_{c(t)}N$ `Chapter6/Parallel · parallelTransport` `Chapter6/Parallel · parallelTransport_apply`. So it is an orthonormal basis of $T_{c(t)}N$, and $\|P_i(t)\| = 1$. This is why Milnor's frame, orthonormal at one point, is orthonormal everywhere. The Assertion needs it, since the trace is computed in the basis $(P_i(t))$.
4. Let $s = I \setminus \{i_0\}$. Then $|s| + 1 = |I| = \dim T_{c(0)}N = n$.
5. At time t, apply the Ricci bound to the unit vector $P_{i_0}(t)$ and the Assertion in the basis $(P_i(t))$: $\sum_{i\in s}\langle R(P_{i_0}, P_i)P_{i_0}, P_i\rangle = K(P_{i_0}, P_{i_0}) \geq (n-1)/r^2 = |s|/r^2$. Since $V_f = LP_{i_0}$, each term $\langle R(V_f, P_i)V_f, P_i\rangle$ equals $L^2\langle R(P_{i_0}, P_i)P_{i_0}, P_i\rangle$ by linearity `Chapter6/Submanifold · curvOp_smul_left`. Multiply by L^2.

For smooth c and smooth tangent fields V_f, P along it, the function $t \mapsto \langle R(V_f, P)V_f, P\rangle$ is continuous `Chapter11/Myers · continuous_inner_curvR`, because R depends smoothly on smooth data `Chapter8/Jacobi · contDiffAt_curvR`. So the integrals below exist. Two integrals are needed: $\int_0^1 \sin^2(\pi t)\,dt = \int_0^1 \cos^2(\pi t)\,dt = \frac{1}{2}$ `Chapter11/Myers · integral_sin_sq_pi_mul` `Chapter11/Myers · integral_cos_sq_pi_mul`, by the substitution $x = \pi t$ and the primitives of $\sin^2$ and $\cos^2$ `Mathlib/Analysis/SpecialFunctions/Integrals/Basic · integral_sin_sq` `Mathlib/Analysis/SpecialFunctions/Integrals/Basic · integral_cos_sq`.

The test fields `Chapter11/Myers · energyHessian_sin_parallel`. Let $\gamma \in \Omega(N; p, q)$ agree on $[0, 1]$ with a smooth curve $c : \mathbb{R} \to N$. Let P be parallel along c with $\|P\| = 1$, and let V_f be a field with $V_f = \gamma'$ on $(0, 1)$. Then $W(t) = \sin(\pi t)P(t)$ lies in $T\Omega_\gamma$ and

$$\tfrac{1}{2}E_{**}(W, W) = \int_0^1 \left(\pi^2\cos^2(\pi t) - \sin^2(\pi t)\,\langle R(V_f, P)V_f, P\rangle\right) dt.$$

The proof as formalized. W is smooth, hence piecewise smooth. It is tangent along γ on $[0, 1]$, because there $P(t) \in T_{c(t)}N = T_{\gamma(t)}N$. It vanishes at 0 and 1, because $\sin 0 = \sin \pi = 0$. In the integral form of E_{**} take $t \in (0, 1)$. Then $W'(t) = \sin(\pi t)P'(t) + \pi\cos(\pi t)P(t)$. The projection kills $P'(t)$, P being parallel, and fixes $P(t)$, so $DW/dt = \pi\cos(\pi t)P(t)$ and $\|DW/dt\|^2 = \pi^2\cos^2(\pi t)$. The curvature term is $\sin^2(\pi t)\langle R(V_f, P)V_f, P\rangle$ by linearity `Chapter6/Submanifold · curvOp_smul_right`.

Milnor uses the form $-\int\langle W, D^2W/dt^2 + R(V, W)V\rangle$ of $\frac{1}{2}E_{**}$ and obtains $\pi^2\sin^2(\pi t)$ where the integral form used here gives $\pi^2\cos^2(\pi t)$. Both integrate to $\pi^2/2$, so nothing else changes.

The key estimate `Chapter11/Myers · exists_energyHessian_neg`. Let $n \geq 2$, $r > 0$, and let the Ricci curvature satisfy $K(U, U) \geq (n-1)/r^2$ for all unit vectors U. If $\gamma \in \Omega(N; p, q)$ is an unbroken geodesic with $L(\gamma) > \pi r$, there is $X \in T\Omega_\gamma$ with $E_{**}(X, X) < 0$.

The proof as formalized.

1. $L > \pi r > 0$, so the parallel frame exists. Let $W_i = \sin(\pi t)P_i$ for $i \in s$ and $k_i(t) = \langle R(V_f, P_i)V_f, P_i\rangle(t)$.
2. By the test-field formula, $\sum_{i\in s} E_{**}(W_i, W_i) = 2\int_0^1 \sum_{i\in s} f_i$ with $f_i = \pi^2\cos^2(\pi t) - \sin^2(\pi t)k_i(t)$. The f_i are continuous, so the sum passes under the integral.
3. Pointwise, $\sum_i f_i = |s|\pi^2\cos^2(\pi t) - \sin^2(\pi t)\sum_i k_i \leq |s|\pi^2\cos^2(\pi t) - |s|(L^2/r^2)\sin^2(\pi t) = g(t)$, by the frame estimate and $\sin^2 \geq 0$.
4. $\int_0^1 g = |s|(\pi^2 - L^2/r^2)/2$. Here $|s| = n - 1 \geq 1$, and $\pi^2 < L^2/r^2$ because $L > \pi r > 0$. So the sum of the $E_{**}(W_i, W_i)$ is negative, and one of its terms is negative.

Theorem 19.4 (Myers) `Chapter11/Myers · theorem_19_4`. Let $n \geq 2$, $r > 0$, and $K(U, U) \geq (n-1)/r^2$ for every unit vector U at every point of N. Then every unbroken geodesic $\gamma \in \Omega(N; p, q)$ of length $> \pi r$ has a parameter $t \in (0, 1)$ conjugate to 0, and it is not minimal: some path of $\Omega(N; p, q)$ is shorter.

The proof as formalized. Take X with $E_{**}(X,X) < 0$; then $X \neq 0$. On the line spanned by X, $E_{**}(aX,aX) = a^2 E_{**}(X,X) < 0$ for $a \neq 0$, so E_{**} is negative definite there and the index is at least 1. By the index theorem, the index is the sum over $t \in (0,1)$ of the multiplicities of 0 and t. If no t were conjugate to 0, every multiplicity would be 0 and so would the index. At a minimal geodesic E_{**} is positive semi-definite (Lemma 13.6), which $E_{**}(X,X) < 0$ excludes. Milnor argues instead with a variation along W_i that decreases the energy; Lemma 13.6 packages the same argument.

A hidden hypothesis: $n \geq 2$. For $n = 1$ the Ricci tensor vanishes, the sum in the Assertion being empty, and the hypothesis $0 \geq 0$ holds for every r. But a segment of a line of length $> \pi r$ has no conjugate points and is minimal. The formal statement assumes $n \geq 2$, and the proof uses it at step 4 of the key estimate.

X.10 Corollary 19.5 and the sphere

A *minimal geodesic* in Milnor's sense (p. 61) is the restriction to $[0,1]$ of a geodesic on an open set containing $[0,1]$, whose length is at most that of every piecewise smooth path in N with the same end points `Chapter7/HopfRinow · IsMinimalGeodesic`. Such a γ is an unbroken geodesic of $\Omega(N;\gamma(0),\gamma(1))$ of minimal length `Chapter11/Myers · IsMinimalGeodesic.isMinimalPath`: being smooth on a neighbourhood of $[0,1]$, it is piecewise smooth, and the comparison paths are the same.

Corollary 19.5 `Chapter11/Myers · corollary_19_5`. Let N be connected and complete for ρ, let $r > 0$ with $(n-1)/r^2 > 0$, and let $K(U,U) \geq (n-1)/r^2$ for all unit vectors U. Then $\rho(p,q) \leq \pi r$ for all $p, q \in N$, and N is compact.

The proof as formalized. $(n-1)/r^2 > 0$ forces $n \geq 2$. The Hopf–Rinow theorem for connected complete N `Chapter7/HopfRinow · hopf_rinow_of_isRiemComplete` (Part VI, §VI.19) gives two things. First, any p, q are joined by a minimal geodesic γ on $[0,1]$ with $\rho(p,q) = L(\gamma)$. Second, closed ρ-bounded subsets of N are compact. If $L(\gamma) > \pi r$, Theorem 19.4 says that γ is not minimal in Ω, against the first point; so $\rho(p,q) \leq \pi r$. Then N, closed in itself, is ρ-bounded `Chapter7/Metric · IsRiemBounded`, hence compact. Connectedness is a hidden hypothesis, needed for minimal geodesics to exist; it is Milnor's standing assumption.

The unit sphere. Let $S^n \subseteq \mathbb{R}^{n+1}$ be the unit sphere, a level set `Milnor/Chapter5/HairyBall · Sph` and a slice submanifold `Milnor/Chapter6/EulerSphere · sphSub`. Its curvature is $R(x,y)z = \langle x,z\rangle y - \langle y,z\rangle x$ for tangent x, y, z `Chapter6/SphereCurvature · curvOp_sph` (Part V, §V.21, from the Gauss equation). So $K(U,U) = (n-1)\|U\|^2$ `Chapter11/Myers · ricci_sph`: in an orthonormal basis (b_i) of T_qS^n the terms are $\langle R(U,b_i)U,b_i\rangle = \|U\|^2 - \langle U,b_i\rangle^2$, and their sum is $n\|U\|^2 - \|U\|^2$ by Parseval's identity, since T_qS^n has dimension n. Hence the Ricci bound holds with $r = 1$ `Chapter11/Myers · ricciLowerBound_sph`. For $n \geq 2$, every unbroken geodesic of $\Omega(S^n;p,q)$ of length $> \pi$ has a parameter in $(0,1)$ conjugate to 0 and is not minimal, and $\rho(p,q) \leq \pi$ for all p, q `Chapter11/Myers · sphere_myers`. This is Theorem 19.4 with $r = 1$, and Corollary 19.5. S^n is connected, a sphere in a space of dimension $n+1 \geq 2$ `Mathlib/Analysis/Normed/Module/Connected · isConnected_sphere`. It is complete because it is compact `Milnor/Chapter6/EulerSphere · isCompact_sph`, hence closed in $\mathbb{R}^{n+1}$. Milnor's sphere of radius r and his remark that the result is best possible are not formalized.

X.11 Long geodesics have large index

In the proof of 19.6 (p. 106) Milnor shows that a geodesic of length $> k\pi r$ has index $\geq k$. Arguing as for 19.4, he finds, for each $i = 1, \dots, k$, a field X_i that vanishes outside $((i-1)/k, i/k)$ and has $E_{**}(X_i,X_i) < 0$, and he calls the orthogonality $E_{**}(X_i,X_j) = 0$, $i \neq j$, clear. The formal proof constructs these fields on an arbitrary piece $[a,b] \subseteq [0,1]$.

Integrals over pieces. Let $0 \leq a \leq b \leq 1$. If $G = 0$ on $(0,a)$ and on $(b,1)$, and $G = F$ on (a,b) with F continuous, then $\int_0^1 G = \int_a^b F$ `Chapter11/LongGeodesics · integral_eq_of_eq0n_pieces`: split the integral at

a and b, and use that integrals over an interval only see the integrand on its interior. If $\kappa(b-a)=\pi$, then $\int_a^b \sin^2(\kappa(t-a))\,dt = \int_a^b \cos^2(\kappa(t-a))\,dt = (b-a)/2$ Chapter11/LongGeodesics · integral_sin_sq_shift Chapter11/LongGeodesics · integral_cos_sq_shift.

Definition (the field on a piece) Chapter11/LongGeodesics · sinBump. For $a<b$, $\kappa=\pi/(b-a)$ and a field P, put

$$X(t)=\sin\big(\kappa(\max(a,\min(t,b))-a)\big)\,P(t).$$

The clamp makes continuity evident. On $[a,b]$, $X(t)=\sin(\kappa(t-a))P(t)$ Chapter11/LongGeodesics · sinBump_of_mem. For $t\le a$, $X(t)=\sin(0)P(t)=0$ Chapter11/LongGeodesics · sinBump_of_le. For $t\ge b$, $X(t)=\sin(\pi)P(t)=0$ Chapter11/LongGeodesics · sinBump_of_ge. So X vanishes on a neighbourhood of every $t\notin[a,b]$ Chapter11/LongGeodesics · sinBump_eventually_zero, and near every $t\in(a,b)$ it agrees with the smooth function $\sin(\kappa(t-a))P(t)$ Chapter11/LongGeodesics · sinBump_eventuallyEq. Off $[a,b]$ both X and DX/dt vanish Chapter11/LongGeodesics · covD_sinBump_of_notMem, and $R(x,0)=0$ Chapter11/LongGeodesics · curvOp_zero_right.

Lemma Chapter11/LongGeodesics · sinBump_mem_pathTangent. Let $\gamma=c$ on $[0,1]$ for a smooth $c:\mathbb{R}\to N$, let P be parallel along c, and let $0\le a<b\le 1$. Then the field X on $[a,b]$ lies in $T\Omega_\gamma$.

Proof. X is continuous. It is piecewise smooth with break points $\{a,b\}$: on an interval whose interior contains neither a nor b, it is either identically 0 (left of a or right of b) or equal to the smooth function $\sin(\kappa(t-a))P(t)$. It is tangent, as P is, and it vanishes at $0\le a$ and at $1\ge b$.

Lemma (the energy of a field on a piece) Chapter11/LongGeodesics · energyHessian_sinBump. With P parallel of norm 1, and V_f smooth, tangent and equal to γ' on $(0,1)$,

$$\tfrac12 E_{**}(X,X)=\int_a^b \big(\kappa^2\cos^2(\kappa(t-a))-\sin^2(\kappa(t-a))\,\langle R(V_f,P)V_f,P\rangle\big)\,dt.$$

Proof. In the integral form, the integrand vanishes on $(0,a)$ and $(b,1)$, where X and DX/dt vanish. On (a,b) it is computed as for the test fields of X.9, with κ in place of π: $DX/dt=\kappa\cos(\kappa(t-a))P$.

Lemma (orthogonality) Chapter11/LongGeodesics · energyHessian_sinBump_disjoint. For fields X on $[a,b]$ and X' on $[a',b']$, built from parallel P, P', with $b\le a'$: $E_{**}(X,X')=0$.

Proof. The integrand $\langle DX/dt,DX'/dt\rangle-\langle R(V,X)V,X'\rangle$ vanishes on $(0,a)$ and on $(b,1)$, where X and DX/dt vanish. It also vanishes on (a,b), where $t<b\le a'$, so that X' and DX'/dt vanish. Only the common end point, a null set, is left. The integral form has no jump terms at the break points, which is why Milnor can treat the orthogonality as evident.

Lemma (a negative field on each long piece) Chapter11/LongGeodesics · exists_sinBump_neg. Let $n\ge 2$, $r>0$, and let $c,V_f,(P_i)_{i\in s}$ be a parallel frame with the estimate of X.9. If $0\le a<b\le 1$ and $L(b-a)>\pi r$, then the field on $[a,b]$ built from some P_i, $i\in s$, has $E_{**}(X_i,X_i)<0$.

Proof. As in X.9, $\sum_{i\in s}\frac12 E_{**}(X_i,X_i)$ is at most

$$|s|\Big(\kappa^2\,\frac{b-a}{2}-\frac{L^2}{r^2}\,\frac{b-a}{2}\Big)=|s|\,\frac{b-a}{2}\,\Big(\kappa^2-\frac{L^2}{r^2}\Big),$$

which is negative because $|s|\ge 1$ and $\kappa=\pi/(b-a)<L/r$.

Theorem (the index bound of 19.6) Chapter11/LongGeodesics · theorem_19_6_index. Let $n\ge 2$, $r>0$, $K(U,U)\ge(n-1)/r^2$ for every unit vector U, and let $\gamma\in\Omega(N;p,q)$ be an unbroken geodesic with $L(\gamma)>k\pi r$, $k\in\mathbb{N}$. Then the index of E_{**} at γ is at least k.

The proof as formalized.

1. For $k=0$ there is nothing to prove. Otherwise $L>0$; take the parallel frame of X.9.

2. The pieces $[i/k, (i+1)/k]$, $i = 0, \dots, k-1$, have length $1/k$, and $L/k > \pi r$. For each i the last lemma gives an index $j(i) \in s$ such that the field Y_i on the i-th piece built from $P_{j(i)}$ has $E_{**}(Y_i, Y_i) < 0$.
3. For $i < i'$ the pieces satisfy $(i+1)/k \le i'/k$, so $E_{**}(Y_i, Y_{i'}) = 0$ by orthogonality, and by symmetry also for $i > i'$.
4. Let $\Phi : \mathbb{R}^k \to T\Omega_{\gamma'}$, $\Phi(x) = \sum_i x_i Y_i$. Expanding bilinearly, the cross terms vanish and $E_{**}(\Phi x, \Phi x) = \sum_i x_i^2 E_{**}(Y_i, Y_i)$. For $x \neq 0$, some $x_{i_0} \neq 0$ contributes a negative term and the others contribute terms ≤ 0, so $E_{**}(\Phi x, \Phi x) < 0$.
5. Hence Φ is injective, since $\Phi x = 0$ gives $E_{**}(\Phi x, \Phi x) = 0$, and E_{**} is negative definite on its k-dimensional range. The index is at least k.

That the X_i span a k-dimensional space, which Milnor takes as evident, is a consequence here, not an extra observation: the fields are non-zero with disjoint supports, and step 4 gives both their independence and the negativity.

Two reformulations follow. A geodesic of index $< k$ has length $\le k\pi r$ `Chapter11/LongGeodesics · pathLength_le_of_formIndex_lt`, the form used in the proof of 19.6. And with the index theorem, a geodesic of length $> k\pi r$ has at least k parameters in $(0,1)$ conjugate to 0, counted with multiplicity `Chapter11/LongGeodesics · le_finsum_conjMultiplicity_of_ricci`.

X.12 The Gauss lemma on the whole domain, and $\exp_p$ as a local diffeomorphism

Milnor proves Theorem 19.2 through the path space (pp. 102–103). There are no conjugate points, so by 17.3 $\Omega(M; p, q)$ has the homotopy type of a discrete set with one point for each geodesic. Ω is connected because M is simply connected, so there is exactly one geodesic from p to q. Then $\exp_p$ is one-to-one and onto, and non-critical by 18.1, hence a diffeomorphism. For a manifold that is not simply connected he passes to the universal covering, and asserts without proof that it inherits a complete metric of curvature ≤ 0 (p. 103). The formal proof takes the standard covering-space route instead, in four steps.

1. $\exp_p$ is a local diffeomorphism, by Lemma 19.1, Theorem 18.1 and the inverse function theorem (this section).
2. Curves in N lift through $\exp_p$, by the completeness of N and the Gauss lemma, which keeps lifts bounded (X.13).
3. Hence $\exp_p$ is a covering map (X.14).
4. If N is simply connected, a covering $\exp_p$ is injective (X.15).

The space T_pN with the map $\exp_p$ is then Milnor's universal covering, and the metric pulled back to it is never introduced.

$\exp_p$ is smooth at every point of its domain `Chapter11/ExpCovering · contDiffAt_expAt`. Radial geodesics have constant speed `Chapter11/ExpCovering · norm_fderiv_expAt_apply`: for $u \in \mathcal{D}_p$ and $0 \le r \le 1$, $\|d(\exp_p)_{ru}(u)\| = \|u\|$. The left side is the velocity at $t = r$ of $t \mapsto \exp_p(tu)$, by the chain rule. The speed of a geodesic is constant, and at $t = 0$ it is $\|u\|$.

The Gauss lemma on the whole domain `Chapter11/ExpCovering · gauss_lemma_expDomAt`. For $w \in \mathcal{D}_p$ and $x \in T_pN$,

$$\langle d(\exp_p)_w(w),\ d(\exp_p)_w(x)\rangle = \langle w, x\rangle.$$

Milnor states the Gauss lemma only inside a normal neighbourhood (Lemma 10.5 `Chapter7/GaussLemma · gauss_lemma`; Part VI, §VI.9). The covering argument needs it on the whole domain, and his completeness claim for the universal cover rests on it.

The proof as formalized. Write D_y for $d(\exp_p)_y$ and D_y^2 for the second derivative; $\exp_p$ is smooth on the open set $\mathcal{D}_p$, which contains the segment $\{rw : 0 \le r \le 1\}$. Let $g(r) = \langle D_{rw}(w), D_{rw}(rx)\rangle$. Then g is continuous on $[0,1]$, $g(0) = 0$, and $g(1)$ is the left side. It suffices to show $g'(r) = \langle w, x\rangle$ on $(0,1)$, because the mean value theorem `Mathlib/Analysis/Calculus/Deriv/MeanValue · exists_hasDerivAt_eq_slope` then gives $g(1) - g(0) = \langle w, x\rangle$. By the product rule,

$$g'(r) = \big\langle D_{rw}^2(w)(w),\ D_{rw}(rx)\big\rangle + \big\langle D_{rw}(w),\ D_{rw}^2(w)(rx) + D_{rw}(x)\big\rangle.$$

1. *The first term vanishes.* $D_{rw}^2(w)(w)$ is the acceleration at $t = r$ of the radial geodesic $t \mapsto \exp_p(tw)$, so it is normal to N at $\exp_p(rw)$. $D_{rw}(rx)$ is the velocity at $s = 0$ of $s \mapsto \exp_p(rw + s\,rx)$. For small s this curve stays in the open domain, hence in N, so its velocity is tangent `Chapter7/Geodesic · hasDerivAt_mem_tangentSubspace`. A normal vector and a tangent vector are orthogonal.
2. *The second term is* $\langle w, x\rangle$. For s near 0, $w + sx \in \mathcal{D}_p$, and the constant speed of its ray gives $\|D_{r(w+sx)}(w + sx)\|^2 = \|w + sx\|^2$. Differentiating at $s = 0$ yields $2\langle D_{rw}(w), D_{rw}^2(rx)(w) + D_{rw}(x)\rangle = 2\langle w, x\rangle$. The second derivative of a C^2 map is symmetric `Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffAt.isSymmSndFDerivAt`, so $D_{rw}^2(rx)(w) = D_{rw}^2(w)(rx)$, which turns this identity into the claim.

This is Milnor's computation for Lemma 10.5, $\langle \partial f/\partial r, \partial f/\partial s\rangle$ with f the radial map, carried out along the whole segment. With $x = w$ it gives $\|d(\exp_p)_w(w)\| = \|w\|$ `Chapter11/ExpCovering · norm_fderiv_expAt_self`.

$\exp_p$ *as a map into* N. $\exp_p$ is regarded as a map $T_pN \to N$ `Chapter7/NormalNbhd · expToN`, with the default value p where $\exp_p(v) \notin N$, which never happens on the domain `Chapter11/ExpCovering · coe_expToN_of_mem`. It is smooth, for the manifold structure of N, at every point of the domain `Chapter11/ExpCovering · contMDiffAt_expToN`: a map into N is smooth at a point if its composite with the inclusion agrees near that point with a smooth map into V `Chapter7/NormalNbhd · contMDiffAt_of_val_eventuallyEq`.

Lemma ($\exp_p$ is a local diffeomorphism at non-critical points) `Chapter11/ExpCovering · isLocalDiffeomorphAt_expToN`. If $v \in \mathcal{D}_p$ and $d(\exp_p)_v$ is one-to-one, then $\exp_p : T_pN \to N$ is a local diffeomorphism at v.

The proof as formalized. Apply the inverse function theorem between manifolds of the companion project `Milnor/Chapter1/InverseFunction · isLocalDiffeomorphAt_of_isInvertible_mfderiv`. Its first hypothesis, smoothness at every point near v, holds because $\mathcal{D}_p$ is open. Its second is the invertibility of the manifold derivative $A : T_pN \to Z$ of $\exp_p$ at v. Near v, the inclusion of N composed with $\exp_p$ is the map $\exp_p : T_pN \to V$. By the chain rule, the derivative of the inclusion composed with A is $d(\exp_p)_v$, which is one-to-one, so A is one-to-one. As $\dim T_pN = \dim Z$, A is bijective `Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · LinearMap.injective_iff_surjective_of_finrank_eq_finrank`, hence invertible. This is Milnor's step from non-critical to locally diffeomorphic.

If N is geodesically complete and $d(\exp_p)_v$ is one-to-one for every $v \in T_pN$, then $\exp_p : T_pN \to N$ is a local diffeomorphism `Chapter11/ExpCovering · isLocalDiffeomorph_expToN`. In non-positive curvature the second hypothesis always holds (X.7) `Chapter11/CartanHadamard · injective_dExpAt_of_nonpos'`.

X.13 Lifting curves through $\exp_p$

In this section and the next, N is geodesically complete and $d(\exp_p)_v$ is one-to-one for every $v \in T_pN$. Then $\exp_p$ is continuous on T_pN `Chapter11/ExpCovering · continuous_expAt`.

Lemma (the derivative of a lift) `Chapter11/ExpCovering · hasDerivAt_expLift`. Let $c : \mathbb{R} \to V$ be smooth with values in N, and let $\Gamma : \mathbb{R} \to T_pN$ be continuous at t_0 with $\exp_p(\Gamma(t)) = c(t)$ for t near t_0. Then Γ is

differentiable at t_0, and $d(\exp_p)_{\Gamma(t_0)}(\Gamma'(t_0)) = c'(t_0)$.

Proof. Let Φ be a local diffeomorphism of $\exp_p$ at $\Gamma(t_0)$. For t near t_0, $\Gamma(t)$ lies in the source of Φ, by continuity, and $\Phi(\Gamma(t)) = c(t)$. So $\Gamma(t) = \Phi^{-1}(c(t))$ near t_0. The curve c is smooth as a map into N `Milnor/Chapter2/Corestrict · contMDiff_codRestrict`, so $\Phi^{-1} \circ c$ is smooth at t_0, and Γ is differentiable there. The chain rule for $\exp_p \circ \Gamma = c$ and the uniqueness of derivatives give the formula.

Lemma (lifts do not escape to infinity) `Chapter11/ExpCovering · norm_expLift_le`. Let c be as above with $\|c'(t)\| \le \Lambda$ on $[0,T]$, and let Γ be continuous on $[0,T]$ with $\exp_p \circ \Gamma = c$ there. Then $\|\Gamma(t)\| \le \|\Gamma(0)\| + 1 + \Lambda t$ on $[0,T]$.

The proof as formalized. Let $\psi = \sqrt{\|\Gamma\|^2 + 1}$, which is differentiable wherever Γ is, unlike $\|\Gamma\|$ at 0. On $(0,T)$, by the last lemma, $\psi' = \langle \Gamma, \Gamma' \rangle / \psi$. The Gauss lemma with $w = \Gamma$ and $x = \Gamma'$ gives

$$\langle \Gamma, \Gamma' \rangle = \langle d(\exp_p)_\Gamma(\Gamma),\ d(\exp_p)_\Gamma(\Gamma') \rangle = \langle d(\exp_p)_\Gamma(\Gamma),\ c' \rangle \le \|\Gamma\|\,\|c'\| \le \|\Gamma\|\,\Lambda \le \psi\Lambda,$$

using $\|d(\exp_p)_\Gamma(\Gamma)\| = \|\Gamma\|$. So $\psi' \le \Lambda$. The mean value inequality `Mathlib/Analysis/Calculus/Deriv/MeanValue · Convex.image_sub_le_mul_sub_of_deriv_le` gives $\psi(t) \le \psi(0) + \Lambda t$. Finally $\psi(0) \le \|\Gamma(0)\| + 1$ and $\|\Gamma(t)\| \le \psi(t)$.

The radial part of the velocity of a lift is at most the speed of the curve below it. This inequality is the whole content of Milnor's claim that the pulled-back metric is complete, and it is all that the proof below uses.

Theorem (curves lift) `Chapter11/ExpCovering · exists_expLift`. Let $c : \mathbb{R} \to V$ be smooth with values in N, and $v_0 \in T_pN$ with $\exp_p(v_0) = c(0)$. Then there is Γ, continuous on $[0,1]$, with $\Gamma(0) = v_0$ and $\exp_p(\Gamma(t)) = c(t)$ for $t \in [0,1]$.

The proof as formalized.

1. Let Λ bound $\|c'\|$ on the compact interval $[0,1]$. Let S be the set of $T \in [0,1]$ such that a lift from v_0 exists on $[0,T]$. The constant v_0 shows $0 \in S$. Let $T_0 = \sup S$, and choose $u_m \in S$ with $u_m \to T_0$ `Mathlib/Topology/Order/IsLUB · exists_seq_tendsto_sSup`, with lifts Γ_m on $[0,u_m]$.
2. By the last lemma all $\Gamma_m(u_m)$ lie in the closed ball of radius $\|v_0\| + 1 + \Lambda$. In the finite-dimensional space T_pN a subsequence $\Gamma_{m_k}(u_{m_k})$ converges to some w `Mathlib/Topology/MetricSpace/Sequences · tendsto_subseq_of_bounded`. By continuity of $\exp_p$ and of c, $\exp_p(w) = c(T_0)$.
3. Let Φ be a local diffeomorphism of $\exp_p$ at w. Then $c(T_0)$ lies in its target, and there is $\eta > 0$ with $c(t)$ in the target for $|t - T_0| < \eta$. Choose k so large that $\Gamma_{m_k}(u_{m_k})$ lies in the source of Φ and $a := u_{m_k} > T_0 - \eta$.
4. Let $T_1 = \min(1, T_0 + \eta/2) \ge a$. Define Γ to be Γ_{m_k} on $[0,a]$ and $\Phi^{-1} \circ c$ on $(a, T_1]$. It lifts c, since $\exp_p \circ \Phi^{-1} = \mathrm{id}$ on the target. It is continuous on $[0,T_1]$: at the junction, $\Gamma_{m_k}(a)$ lies in the source of Φ and lies over $c(a)$, so $\Phi^{-1}(c(a)) = \Gamma_{m_k}(a)$.
5. So $T_1 \in S$ and $T_1 \le T_0$. As $T_0 + \eta/2 > T_0$, the minimum must be 1. Hence $1 \in S$.

Theorem (lifts are unique) `Chapter11/ExpCovering · expLift_unique`. Two lifts of the same curve that are continuous on $[0,1]$ and agree at one time agree on $[0,1]$.

Proof. $\exp_p$ is a local homeomorphism, hence locally injective, and T_pN is Hausdorff, so $\exp_p$ is a separated map `Mathlib/Topology/SeparatedMap · T2Space.isSeparatedMap`. For such a map, two continuous maps on a preconnected set whose composites agree, and which agree at one point, agree everywhere `Mathlib/Topology/SeparatedMap · IsSeparatedMap.eqOn_of_comp_eqOn`.

X.14 $\exp_p$ is a covering map

Theorem `Chapter11/ExpCovering · isCoveringMap_expToN`. If N is geodesically complete and $\exp_p$ is nowhere critical, then $f = \exp_p : T_pN \to N$ is a covering map `Mathlib/Topology/Covering/Basic · IsCoveringMap`: every point of N has an evenly covered open neighbourhood, with a discrete fibre `Mathlib/Topology/Covering/Basic · IsEvenlyCovered`.

The proof as formalized. Fix $q \in N$.

1. *A normal ball and its radial geodesics.* Take normal coordinates at q `Chapter7/NormalNbhd · exists_isNormalRadius` `Chapter7/NormalNbhd · IsNormalRadius` `Chapter7/NormalNbhd · NormalCoord` (Part VI, §VI.8). They give an $\varepsilon > 0$ and an open $U \subseteq V$ such that $\exp_q$ maps the ε-ball of T_qN bijectively onto $B = N \cap U$, with a smooth inverse G. For $y \in B$ let $\ell(y) = G(y) \in T_qN$, which is continuous in y, and let $\rho_y(t) = \exp_q(t\,\ell(y))$. This is a geodesic on $\mathbb{R}$, by completeness, from $\rho_y(0) = q$ to $\rho_y(1) = y$.
2. *The sections.* For e in the fibre $f^{-1}(q)$ and $y \in B$, let $\Gamma_{e,y}$ be the lift of ρ_y from e (X.13). The map $(t, y) \mapsto \Gamma_{e,y}(t)$ on $[0,1] \times B$ lifts the continuous map $(t, y) \mapsto \rho_y(t)$. It is continuous at $t = 0$, where it is the constant e, and continuous in t for each y. By the homotopy lifting lemma for local homeomorphisms that are separated maps `Mathlib/Topology/Homotopy/Lifting · IsLocalHomeomorph.continuous_lift`, it is jointly continuous. So the section $\sec_e(y) = \Gamma_{e,y}(1)$ is continuous, and $f(\sec_e(y)) = y$.
3. *Every point over B lies on a section.* Let $f(v) = z \in B$. Lift the reversed geodesic $t \mapsto \rho_z(1-t)$ from v. Its end point e lies over $\rho_z(0) = q$. Reversed, this lift is a lift of ρ_z from e, so by uniqueness it is $\Gamma_{e,z}$. Hence $\sec_e(z) = v$.
4. *The fibre is discrete*, because f is locally injective and constant on it `Mathlib/Topology/SeparatedMap · discreteTopology_iff_locallyInjective`. If the fibre is empty, step 3 shows that nothing lies over B, and B is evenly covered with empty fibre `Mathlib/Topology/Covering/Basic · IsEvenlyCovered.of_preimage_eq_empty`.
5. *The sheets.* Let $U_e = \sec_e(B)$. The sheets have the following properties.
   - U_e is open. Given $z_0 \in B$, let ϕ be a local diffeomorphism of f at $\sec_e(z_0)$. Consider the points y of the source of ϕ such that $f(y) = z$ for some z with $\sec_e(z)$ in the source of ϕ. They form an open neighbourhood of $\sec_e(z_0)$. Each such y satisfies $f(y) = f(\sec_e(z))$ with both points in the source of ϕ, where f is injective, so $y = \sec_e(z) \in U_e$.
   - f is injective on U_e, because $f \circ \sec_e = \mathrm{id}$, and f maps U_e onto B.
   - The sheets are pairwise disjoint. If $\sec_{e_1}(z_1) = \sec_{e_2}(z_2)$, applying f gives $z_1 = z_2 = z$. The lifts $\Gamma_{e_1,z}$ and $\Gamma_{e_2,z}$ of ρ_z then agree at $t = 1$, hence everywhere by uniqueness, and at $t = 0$ this says $e_1 = e_2$.
   - The sheets cover $f^{-1}(B)$, by step 3.
   - A subset $W \subseteq B$ is open if and only if $f^{-1}(W) \cap U_e$ is open. One direction is continuity. For the other, $W = f(f^{-1}(W) \cap U_e)$, and a local homeomorphism is an open map.
6. These are the hypotheses of Mathlib's construction of a trivialization with discrete fibre over an open set `Mathlib/Topology/Covering/Basic · IsOpen.trivializationDiscrete`, so q is evenly covered `Mathlib/Topology/Covering/Basic · IsEvenlyCovered.of_trivialization`.

Corollary `Chapter11/CartanHadamard · isCoveringMap_expToN_of_nonpos`. If N is geodesically complete and has non-positive curvature, then $\exp_p : T_pN \to N$ is a covering map for every $p \in N$.

T_pN is a vector space, hence contractible, so this covering is the universal covering of Milnor's p. 103. No abstract covering space and no metric on it are needed.

X.15 Theorem 19.2 (Cartan–Hadamard)

Lemma (injectivity) `Chapter11/CartanHadamard · injective_expToN_of_simplyConnected`. If $\exp_p$ is a covering map and N is simply connected, then $\exp_p$ is one-to-one.

Proof. Let $f = \exp_p$ and suppose $f(v_1) = f(v_2)$. The segment $s(t) = v_1 + t(v_2 - v_1)$ projects to a loop $f \circ s$ at $f(v_1)$. In a simply connected space this loop is homotopic, relative to its end points, to the constant loop `Mathlib/AlgebraicTopology/FundamentalGroupoid/SimplyConnected · SimplyConnectedSpace.paths_homotopic`. Lifts from v_1 of paths homotopic relative to the end points end at the same point `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.liftPath_apply_one_eq_of_homotopicRel`. The lift of $f \circ s$ from v_1 is s itself, by the uniqueness of lifts `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.eq_liftPath_iff'`, and the lift of the constant loop is constant `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.liftPath_const`. So $v_2 = s(1) = v_1$.

Lemma (surjectivity) `Chapter11/CartanHadamard · surjective_expToN`. If N is connected and geodesically complete, $\exp_p$ is onto. This is the Hopf–Rinow theorem 10.9 `Chapter7/HopfRinow · theorem_10_9` (Part VI, §VI.18): $\rho(p, q) < \infty$ in a connected N `Chapter7/Metric · riemDist_ne_top`, and 10.9 gives v with $\exp_p(v) = q$.

Two facts about geodesics from p are used. If N is geodesically complete, every radial geodesic $t \mapsto \exp_p(tv)$ is an unbroken geodesic of $\Omega(N; p, \exp_p(v))$ `Chapter11/CartanHadamard · ray_mem_pathSpace`. Conversely, every unbroken geodesic $\gamma \in \Omega(N; p, q)$ is radial `Chapter11/CartanHadamard · exists_eqOn_ray`: it agrees on $[0, 1]$ with $t \mapsto \exp_p(tv)$ for its initial velocity v, and $\exp_p(v) = q$ (X.6). A simply connected space is connected `Chapter11/CartanHadamard · isPreconnected_of_simplyConnected`, and a closed $N \subseteq V$ is geodesically complete `Chapter11/CartanHadamard · isGeodesicallyComplete_of_isClosed`.

The diffeomorphism `Chapter11/CartanHadamard · expDiffeomorph` `Chapter11/CartanHadamard · coe_expDiffeomorph`. If N is geodesically complete, simply connected and of non-positive curvature, then $\exp_p : T_pN \to N$ is a diffeomorphism `Mathlib/Geometry/Manifold/Diffeomorph · Diffeomorph`. It is a local diffeomorphism (X.12) and bijective (the two lemmas), and a bijective local diffeomorphism is a diffeomorphism `Mathlib/Geometry/Manifold/LocalDiffeomorph · IsLocalDiffeomorph.diffeomorphOfBijective`, since its inverse agrees near each point with a smooth local inverse.

Theorem 19.2 (Cartan) `Chapter11/CartanHadamard · theorem_19_2`. Let N be geodesically complete (for instance closed in V), simply connected, with $\langle R(A, B)A, B\rangle \leq 0$ everywhere. Then any two points p, q of N are joined by an unbroken geodesic $\gamma \in \Omega(N; p, q)$, which is unique: every unbroken geodesic of $\Omega(N; p, q)$ agrees with it on $[0, 1]$. And N is diffeomorphic to $\mathbb{R}^n$.

The proof as formalized.

1. Let v be the preimage of q under the diffeomorphism $\exp_p$. The radial geodesic of v is an unbroken geodesic from p to q.
2. Any unbroken geodesic γ' from p to q is the radial geodesic of some v' with $\exp_p(v') = q$. Injectivity gives $v' = v$, so γ' agrees with the radial geodesic of v on $[0, 1]$.
3. N is non-empty, which is part of being simply connected. For $p \in N$, T_pN is linearly isomorphic to $\mathbb{R}^n$, since both have dimension n. Composing this isomorphism with $\exp_p$ gives a diffeomorphism $\mathbb{R}^n \to N$.

The order of the argument is reversed. Milnor derives the uniqueness of geodesics from the path space and then the injectivity of $\exp_p$. This route needs Theorem 17.1 to identify the components of Ω with the homotopy classes of paths, a step he does not state. The formal proof derives injectivity from the covering, uniqueness from injectivity, and Milnor's statements about the path space afterwards (X.17). Both routes rest on Lemma 19.1. In place of 17.1, the covering route needs the bound on lifts of X.13.

X.16 Corollary 19.3: the higher homotopy groups

Milnor's proof of the first half of 19.3 is one line: $\pi_i(M) = \pi_i(\tilde{M}) = 0$ for $i > 1$, as $\tilde{M} \cong \mathbb{R}^n$ is contractible. The formal proof lifts the maps that define π_i (§0.7) directly. In this section i is the dimension of a cube I^i, and k, l index its coordinates.

The cube I^i is contractible, by the straight-line contraction $(s, y) \mapsto (1 - s)y$ to the corner 0 `Chapter11/CartanHadamard · contractibleSpace_cube` `Mathlib/Topology/Homotopy/Contractible · contractible_iff_id_nullhomotopic`. Its faces $\{y : y_k = a\}$ are preconnected `Chapter11/CartanHadamard · isPreconnected_cube_face`, being products of intervals and a point.

Lemma `Chapter11/CartanHadamard · genLoop_homotopic_const`. Let N be geodesically complete with non-positive curvature, $i \geq 2$, $x \in N$, and let $\gamma : I^i \to N$ be continuous and send the boundary of the cube `Mathlib/Topology/Homotopy/HomotopyGroup · Cube.boundary` to x, that is, a generalized loop `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop`. Then γ is homotopic to the constant map relative to the boundary `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.Homotopic`.

The proof as formalized.

1. $f = \exp_x$ is a covering map with $f(0) = x$. The cube is contractible, hence simply connected, and locally path connected. The corner 0 of the cube lies on its boundary, so $\gamma(0) = x = f(0)$. Hence γ lifts to a continuous $F : I^i \to T_xN$ with $F(0) = 0$ `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.existsUnique_continuousMap_lifts`.
2. On each face $\{y_k = a\}$ with $a \in \{0, 1\}$, $f \circ F = \gamma$ is constant, equal to x. The face is preconnected and the fibre over x is discrete, so F is constant on the face `Mathlib/Topology/Covering/Basic · IsCoveringMap.constOn_of_comp`.
3. Let y lie on the face $\{y_k = a\}$, and let z be the point with $z_k = y_k$ and all other coordinates 0. Then $F(y) = F(z)$, as both lie on that face. Because $i \geq 2$ there is $l \neq k$, and z and 0 both lie on the face $\{y_l = 0\}$, so $F(z) = F(0) = 0$. Hence $F = 0$ on the boundary of the cube.
4. $H(s, y) = f((1 - s)F(y))$ is a homotopy from $f \circ F = \gamma$ to the constant $f(0) = x$. For y on the boundary, $H(s, y) = f(0) = x$ for every s.

Step 3 is where $i \geq 2$ enters: the boundary of the cube is connected only then. For $i = 1$ the boundary $\{0, 1\}$ is not connected, and the argument stops, as the statement does.

Corollary 19.3, first half `Chapter11/CartanHadamard · corollary_19_3`. If N is geodesically complete with non-positive curvature, then $\pi_i(N, x) = 0$ `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.Pi` for every $i > 1$ and every $x \in N$.

Proof. Every class is represented by a generalized loop, and by the lemma every generalized loop is homotopic, relative to the boundary, to the constant loop. So any two classes coincide.

X.17 Homotopy classes of paths and the path space

Two further statements of Milnor's concern a complete M of non-positive curvature. On p. 102: $\Omega(M; p, q)$ has the homotopy type of a 0-dimensional complex with one vertex for each geodesic. On p. 103: each homotopy class of paths from p to q contains exactly one geodesic. Both follow from the covering. A path $\gamma \in \Omega(N; p, q)$ is regarded as a continuous path in N from p to q `Chapter11/CartanHadamard · pathOfMem`.

Theorem (one geodesic in each homotopy class) `Chapter11/CartanHadamard · homotopy_class_unique_geodesic`. Let N be geodesically complete with non-positive curvature, $p, q \in N$, and c a continuous path from p to q in N. Then some unbroken geodesic $\gamma \in \Omega(N; p, q)$ is homotopic to c relative to the end points. Two unbroken geodesics of $\Omega(N; p, q)$ that are both homotopic to c relative to the end points agree on $[0, 1]$.

The proof as formalized. Let $f = \exp_p$, a covering map with $f(0) = p$.

1. *Existence.* Let Γ be the lift of c from 0 `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.liftPath` and $v = \Gamma(1)$, so that $f(v) = q$. The radial geodesic of v is an unbroken geodesic of $\Omega(N; p, q)$. The segment $t \mapsto tv$ and Γ are paths in T_pN from 0 to v. T_pN is contractible, hence simply connected, so the two paths are homotopic relative to the end points. Such homotopies in T_pN correspond exactly to homotopies of the projected paths in N `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.homotopicRel_iff_comp`. The projections are the radial geodesic of v and c.
2. *Uniqueness.* Let γ, γ' be two such geodesics. Both are radial, of vectors v and v' (X.15), and their lifts from 0 are the segments to v and to v', by the uniqueness of lifts. γ and γ' are homotopic relative to the end points, through c, so their lifts from 0 end at the same point. Hence $v = v'$ and $\gamma = \gamma'$ on $[0, 1]$.

Milnor obtains this by applying 19.2 to the universal covering with the lifted metric. Here the covering is $\exp_p$ itself.

The path space through Theorem 17.3 `Chapter11/CartanHadamard · omega_cells_of_nonpos`. Let N be closed in V, of non-positive curvature, and $p, q \in N$. Then $\Omega(N; p, q)$ has the homotopy type of a countable cell complex all of whose cells are 0-cells, one for each geodesic from p to q.

Proof. Every geodesic from p to q has nullity 0 and index 0 (X.7). So Theorem 17.3 applies, and its cells, which have the dimensions of the indices, all have dimension 0 `Chapter10/FundamentalTheorem · geodesicIndex_eq`. This is the first step of Milnor's proof of 19.2.

One cell `Chapter11/CartanHadamard · omega_single_cell`. If moreover N is simply connected, the complex has exactly one cell.

Proof. Let γ_0 be the unique geodesic of Theorem 19.2 from p to q, and let n_0 be the list of Theorem 17.3 in which it occurs. Every entry of every list is a geodesic from p to q, so it agrees with γ_0 on $[0, 1]$. An entry of a list $m \neq n_0$ would therefore agree on $[0, 1]$ with an entry of list n_0, which Theorem 17.3 excludes, so the other lists are empty. Two entries of list n_0 would agree on $[0, 1]$, which is also excluded. So list n_0 has exactly one entry, giving one 0-cell at stage n_0 and none elsewhere.

What remains of Milnor's sentence is that a complex with only 0-cells is a discrete set. For the formal cell complexes, which are mapping telescopes, this is not formalized. The homotopy type is computed directly from the covering instead.

Theorem (Ω^* and a fibre of $\exp_p$) `Chapter11/CartanHadamard · nonempty_homotopyEquiv_pathC_fiber`. Let N be geodesically complete with non-positive curvature, and $p, q \in N$. Then $\Omega^*(N; p, q)$ is homotopy equivalent `Mathlib/Topology/Homotopy/Equiv · ContinuousMap.HomotopyEquiv` to the fibre $f^{-1}(q)$ of $f = \exp_p$.

The proof as formalized.

1. The evaluation $H(t, \gamma) = \gamma(t)$ is continuous on $[0, 1] \times \Omega^*$, and $H(0, \gamma) = p = f(0)$. The homotopy lifting property of coverings `Mathlib/Topology/Homotopy/Lifting · IsCoveringMap.liftHomotopy` gives a continuous $\tilde{H} : [0, 1] \times \Omega^* \to T_pN$ with $f \circ \tilde{H} = H$ and $\tilde{H}(0, \gamma) = 0$. Thus $\tilde{H}(\cdot, \gamma)$ is the lift of γ from 0.
2. $\mathrm{end}(\gamma) = \tilde{H}(1, \gamma)$ is a continuous map $\Omega^* \to f^{-1}(q)$.
3. The fibre is discrete `Mathlib/Topology/Covering/Basic · IsEvenlyCovered.discreteTopology_fiber`, so $\mathrm{geo}(v) = (t \mapsto f(tv))$, a path from p to q, depends continuously on v.
4. $\mathrm{end} \circ \mathrm{geo} = \mathrm{id}$. The lift of $t \mapsto f(tv)$ from 0 is $t \mapsto tv$, by the uniqueness of lifts `Mathlib/Topology/Covering/Basic · IsCoveringMap.eq_of_comp_eq`, and it ends at v.
5. $\mathrm{geo} \circ \mathrm{end} \simeq \mathrm{id}$, through

$$G(s, \gamma)(t) = f\big((1 - s)\tilde{H}(t, \gamma) + s\, t\, \tilde{H}(1, \gamma)\big),$$

the straight-line homotopy in T_pN from the lift of γ to the segment with the same end points, projected by f. At $s = 0$ it is γ; at $s = 1$ it is geo(end(γ)). For every s it runs from $f(0) = p$ to $f(\tilde{H}(1, \gamma)) = q$, so it stays in Ω^*.

The points of the fibre are the geodesics from p to q `Chapter11/CartanHadamard · geodesic_iff_mem_fiber`, for complete N. For v in the fibre, the radial geodesic of v is an unbroken geodesic of $\Omega(N; p, q)$. Every unbroken geodesic of $\Omega(N; p, q)$ agrees on $[0, 1]$ with the radial geodesic of a point of the fibre. And two points of the fibre whose radial geodesics agree on $[0, 1]$ are equal, since both are the velocity at 0 within $[0, 1]$, which is unique.

Theorem (the path space has the homotopy type of a discrete set) `Chapter11/CartanHadamard · omega_homotopyEquiv_fiber`. Let N be geodesically complete with non-positive curvature, and $p, q \in N$. The fibre of $\exp_p$ over q is discrete and countable, and $\Omega(N; p, q)$ is homotopy equivalent to it.

Proof. $\Omega \simeq \Omega^*$ by Theorem 17.1, composed with the last theorem. The fibre is a discrete subspace of the separable space T_pN, hence countable `Mathlib/Topology/Bases · TopologicalSpace.separableSpace_iff_countable`.

Together with the description of the fibre, this is Milnor's statement on p. 102, for every geodesically complete N of non-positive curvature, closed in V or not, and for all p, q.

Contractibility. If $D : T \to N$ is a homeomorphism from a real normed space with $D(0) = p$, then $\Omega^*(N; p, q)$ is contractible `Chapter11/CartanHadamard · contractibleSpace_pathC Mathlib/Topology/Homotopy/Contractible · ContractibleSpace`. The homotopy

$$H(s, \gamma)(t) = D\big((1 - s)D^{-1}(\gamma(t)) + s\, t\, D^{-1}(q)\big)$$

contracts it onto $\gamma_0(t) = D(t\, D^{-1}(q))$. At $t = 0$ its value is $D(0) = p$, and at $t = 1$ it is $D(D^{-1}(q)) = q$. With $D = \exp_p$ (X.15) and Theorem 17.1 this gives **Theorem 19.2 for the path space** `Chapter11/CartanHadamard · contractibleSpace_pathSp`: if N is geodesically complete, simply connected and of non-positive curvature, $\Omega(N; p, q)$ is contractible. It has the homotopy type of a point, as Milnor's proof asserts.

X.18 Theorem 19.6 for non-conjugate end points

Milnor's proof (p. 106) has four steps. The continuous function $K(U, U)$ on the compact space of unit vectors takes a minimum $(n - 1)/r^2 > 0$. A geodesic of length $> k\pi r$ has index $\geq k$ (X.11). If p and q are not conjugate along any geodesic, Theorem 16.3 leaves only finitely many geodesics of length $\leq k\pi r$, hence only finitely many of index $< k$. Theorem 17.3 then finishes the proof. The formal proof supplies the continuity of $K(U, U)$, which Milnor does not discuss: the Ricci tensor is a trace over the moving subspace T_qN.

The Ricci curvature as a trace on V. For $q \in N$ and $u \in V$ let $\mathrm{Ric}_{q,u}$ be the linear map $W \mapsto R(P_q u, P_q W)P_q u$ of V `Chapter11/PositiveRicci · ricciMap Chapter11/PositiveRicci · ricciMap_apply`. For $U \in T_qN$, $K(U, U) = \mathrm{tr}_V \mathrm{Ric}_{q,U}$ `Chapter11/PositiveRicci · ricci_eq_trace_ricciMap`. Indeed $\mathrm{Ric}_{q,U} = \iota \circ A \circ \pi$, where A is $W \mapsto R(U, W)U$ on T_qN, ι the inclusion and π the orthogonal projection onto T_qN. A trace does not change when the factors of a composite are cycled `Mathlib/LinearAlgebra/Trace · LinearMap.trace_comp_comm'`, and $\pi \circ \iota = \mathrm{id}$, so the trace is tr A.

Lemma (continuity) `Chapter11/PositiveRicci · continuous_trace_ricciMap`. $(q, u) \mapsto \mathrm{tr}_V \mathrm{Ric}_{q,u}$ is continuous on $N \times V$.

The proof as formalized. Near q_0, take the slice chart σ at q_0 and the smooth map Q, defined on an open subset of V, that agrees with $q \mapsto P_q$ on N `Chapter6/Projection · projRep Chapter6/Projection · projRep_eq Chapter6/Projection · contDiffAt_projRep`. For tangent a, $\mathrm{II}(a, b) = dQ(a)\, b$ `Chapter6/Projection · dN_tanProj_eq_fderiv`.

Fix an orthonormal basis (e_i) of V. Near (q_0, u_0) the trace is

$$\sum_i \Big\langle e_i,\ dQ_q(Q_q e_i)(dQ_q(Q_q u)(Q_q u)) - dQ_q(Q_q u)(dQ_q(Q_q e_i)(Q_q u))\Big\rangle,$$

which is continuous in (q, u) because Q and dQ are.

Lemma (the minimum) `Chapter11/PositiveRicci · exists_ricciLowerBound`. If N is compact, $n \geq 2$, and $K(U,U) > 0$ for every non-zero tangent vector U, there is $r > 0$ with $K(U,U) \geq (n-1)/r^2$ for every unit tangent vector U.

The proof as formalized. The unit tangent bundle $\{(q,u) \in N \times V : P_q u = u,\ \|u\| = 1\}$ is closed, because $q \mapsto P_q$ is continuous `Chapter6/Projection · contMDiff_tanProj`. It lies in the compact set $N \times S$, S the unit sphere of V, so it is compact; this is Milnor's compact space of unit vectors. If it is empty, any r will do. Otherwise the continuous trace attains a minimum m on it `Mathlib/Topology/Order/Compact · IsCompact.exists_isMinOn`, at a unit tangent vector, so $m > 0$. Put $r = \sqrt{(n-1)/m}$; then $(n-1)/r^2 = m \leq K(U,U)$.

Energy and index. The energy of an unbroken geodesic is the square of its length `Chapter11/PositiveRicci · energy_eq_pathLength_sq_of_geodesic`: its speed is constant, which is the case of equality in $L^2 \leq E$ `Chapter8/Energy · pathLength_sq_eq_energy_iff`. Hence, under the Ricci bound, a geodesic of index $\leq d$ has energy $\leq ((d+1)\pi r)^2$ `Chapter11/PositiveRicci · energy_le_of_geodesicIndex_le`: its index is $< d+1$, so its length is $\leq (d+1)\pi r$ (X.11).

Theorem 19.6, under a Ricci bound `Chapter11/PositiveRicci · theorem_19_6_of_ricciLowerBound`. Let N be closed in V, $n \geq 2$, $r > 0$ with $K(U,U) \geq (n-1)/r^2$ for all unit vectors, and $p, q \in N$ with $q \notin C(p)$. Then $\Omega(N; p, q)$ has the homotopy type of a countable cell complex in which, for each d, only finitely many stages contain a cell of dimension d. Each stage is a finite list, so there are only finitely many cells of each dimension.

The proof as formalized.

1. As $q \notin C(p)$, every geodesic from p to q has nullity 0 (X.6). Theorem 17.3 gives the lists geo_m, the cells of stage m having the dimensions of the indices of its entries, and the energies increasing strictly from one list to the next.
2. Fix d and put $e_d = ((d+1)\pi r)^2$. For each stage m containing a cell of dimension d, choose an entry $g(m)$ of geo_m of index d. Then $E(g(m)) \leq e_d$.
3. $m \mapsto E(g(m))$ is injective on these stages, because energies increase strictly across lists. Its values lie in the set of energies $\leq e_d$ of geodesics from p to q, which is finite by Theorem 16.3 `Chapter10/FundamentalTheorem · finite_geodesicEnergies` (Part VIII, §VIII.14). So there are finitely many such stages.

Theorem 19.6 `Chapter11/PositiveRicci · theorem_19_6`. Let N be compact, $n \geq 2$, with positive definite Ricci tensor, and $p, q \in N$ with $q \notin C(p)$. Then $\Omega(N; p, q)$ has the homotopy type of a countable cell complex with only finitely many cells in each dimension.

Proof. A compact N is closed in V, and the last two lemmas give r.

Three things differ from the book. Milnor's CW-complex is here a countable cell complex in the sense of §0.5, a mapping telescope (Part XI, §XI.15). Its identification with the union of its stages with the weak topology is Example 2 of the Appendix (Part XI, §XI.7), which needs Whitehead's theorem and is not formalized. The end points are assumed not conjugate along any geodesic, which is all that Milnor's proof covers, although his statement is for all p, q (X.20). And $n \geq 2$ is assumed. It makes $(n-1)/r^2$ positive and provides a field perpendicular to γ; in dimension 1 the Ricci tensor vanishes, so it cannot be positive definite there anyway.

X.19 The homology of the path space

With cellular homology, Theorem 19.6 would make $H_i(\Omega;\Bbbk)$ finite-dimensional at once, and Milnor computes the homology of ΩS^n from its cells in this way on p. 96. The formal cell complexes have attaching maps that need not be cellular, so cellular homology is not available. The homology is computed instead from the filtration of Ω by the sublevel sets $\Omega^a = \{E \le a\}$ `Chapter2/Crossing · sublevel` of the energy, a continuous function on Ω `Chapter10/FundamentalTheorem · PathSp.energyC`, as in Milnor's §§3 and 5 (Parts II and IV), carried over to the path space in Part IX. Homology is singular homology (§0.6) with coefficients in a field $\Bbbk$, of spaces `Homology/MorseHomologyBasic · spaceHomology` and of pairs of subsets `Homology/TripleFiltration · subsetHomology`; $\Bbbk$ is taken in the same universe as V, a formal condition without mathematical content. Let $\mathcal{E}(a,b)$ be the set of energies in (a,b) of geodesics from p to q `Chapter11/RicciHomology · slabEnergies`.

Theorem (the homology of a slab) `Chapter11/RicciHomology · slab_homology`. Let N be closed in V, $p,q \in N$, every geodesic from p to q non-degenerate, and let $a \le b$ be levels that are not energies of geodesics. Then for every j, $H_j(\Omega^b,\Omega^a;\Bbbk)$ is finite-dimensional, and it is 0 if no geodesic with energy in (a,b) has index j.

The proof as formalized. By strong induction on the number m of elements of $\mathcal{E}(a,b)$, which is finite by Theorem 16.3.

1. *At most one energy.* This is the slab computation of Part IX, §IX.9 `Chapter10/PathHomology · BrokenSetup.finrank_subsetHomology_Omega`. A setup of broken geodesics `Chapter10/Broken · BrokenSetup` at level $\max(b,0)+1$ `Chapter10/Broken · exists_brokenSetup_uniform`, available because closed ρ-balls of a closed N are compact `Chapter10/Broken · isCompact_closedRiemBall_of_isClosed`, transports the slab to the energy on a finite-dimensional manifold of broken geodesics. There $H_j(\Omega^b,\Omega^a;\Bbbk)$ is finite-dimensional, of dimension the number of geodesics with energy in (a,b) and index j. If none has index j, the dimension is 0.
2. *At least two energies.* Let $e_1 < e_2$ be the two smallest elements of $\mathcal{E}(a,b)$ and $c = (e_1+e_2)/2$. The level c is not an energy, since every other energy in (a,b) is $\ge e_2 > c$. The slab (a,c) contains exactly the energy e_1 and the slab (c,b) the other $m-1$, so the induction hypothesis applies to both. The triple $\Omega^a \subseteq \Omega^c \subseteq \Omega^b$ has an exact sequence `Homology/TripleFiltration · subsetTripleLongExactSeq`, containing $H_j(\Omega^c,\Omega^a) \to H_j(\Omega^b,\Omega^a) \to H_j(\Omega^b,\Omega^c)$. The middle term is finite-dimensional when the outer terms are `Chapter3/Inequalities · finiteDimensional_middle`, and 0 when they are `Chapter10/SublevelHomology · LongExactSeq.subsingleton_B`. A hypothesis that no geodesic in (a,b) has index j passes to both halves.

Theorem (finite-dimensional homology) `Chapter11/RicciHomology · finiteDimensional_spaceHomology_omega`. Let N be closed in V, $n \ge 2$, $r > 0$ with $K(U,U) \ge (n-1)/r^2$ for all unit vectors, and $p,q \in N$ with $q \notin C(p)$. Then $H_i(\Omega(N;p,q);\Bbbk)$ is finite-dimensional for every i and every field $\Bbbk$.

The proof as formalized.

1. *Levels.* For $k \ge 1$ choose $a_k \in (k-1,k)$ that is not the energy of a geodesic from p to q; there are only finitely many such energies $\le k$. Put $a_0 = -1$. The levels a_k increase, are never energies of geodesics, and satisfy $a_k \ge k-1$. Let $A_k = \Omega^{a_k}$. Every compact subset of Ω lies in some A_k, because the energy is continuous and bounded on it `Chapter10/SublevelHomology · exists_subset_sublevel_of_isCompact`.
2. *High slabs have no homology in degrees i and $i+1$.* Choose $k_0 > ((i+2)\pi r)^2 + 1$, so that $a_{k_0} > ((i+2)\pi r)^2$. For $k \ge k_0$, every geodesic with energy in (a_k, a_{k+1}) has energy $> ((i+2)\pi r)^2$. Its index is therefore $\ge i+2$, since an index $\le i+1$ would bound the energy by $((i+2)\pi r)^2$ (X.18). By the slab theorem the slab has zero homology in degrees i and $i+1$.
3. *Passage to Ω.* Hence $H_i(A_{k_0},\emptyset) \cong H_i(\Omega)$ `Chapter10/SublevelHomology · spaceHomologyEquivOfSteps`. This is the argument of Part IX, §IX.8: singular chains have compact support, and steps with no homology in degrees i and $i+1$ change nothing in degree i.

4. *The bottom.* Energies are $\geq 0 > -1$ `Chapter10/PathMetric · PathSp.energy_nonneg_of_le`, so $A_0 = \emptyset$. Thus $H_i(A_{k_0}, \emptyset)$ is $H_i(A_{k_0}, A_0)$ `Homology/TripleFiltration · finiteDimensional_subsetHomology_congr`, which is finite-dimensional by the slab theorem.

Under Milnor's hypotheses of 19.6 (N compact, $n \geq 2$, positive definite Ricci tensor, $q \notin C(p)$) the same conclusion holds `Chapter11/RicciHomology · finiteDimensional_spaceHomology_omega_of_compact`, with r from the minimum of X.18.

X.20 What is not done

- **The second half of Corollary 19.3**: in a complete N of non-positive curvature, $\pi_1(N)$ has no element of finite order other than the identity. Milnor's proof uses the cohomology of groups. For M with contractible universal covering, $H^k(M)$ is the cohomology of $\pi_1(M)$ (he cites Hu), and a finite cyclic group has non-zero cohomology in arbitrarily high degrees. As `Chapter11/CartanHadamard` records, neither is available to the formalization.
- **Theorem 19.6 when p and q are conjugate along some geodesic.** Milnor states 19.6 for all p, q, but his proof treats only non-conjugate end points. The general case needs a q' not conjugate to p, which the density theorem of X.6 provides, and a homotopy equivalence $\Omega(N; p, q) \simeq \Omega(N; p, q')$ obtained by changing the end point through Theorem 17.1 (Lemma 17.D of the exposition). That lemma is not formalized. The same restriction applies to the finiteness of the homology in X.19.
- **Cell complexes and spaces.** That a mapping telescope with only 0-cells is homotopy equivalent to a discrete space is not formalized; X.17 obtains the discreteness from the covering instead. Milnor's CW-complexes are the formal telescopes only up to Example 2 of the Appendix (Part XI, §XI.7), which needs Whitehead's theorem.
- **The finiteness of π_1 under positive Ricci curvature** (p. 105, after 19.5). It needs the universal covering of N with the lifted metric, to which 19.5 would apply. The covering of X.14 needs $\exp_p$ to be nowhere critical, which fails in positive curvature.
- **Side remarks of §19**: the sectional curvature as the Gaussian curvature of the image under $\exp_p$ of a plane (p. 100); the optics paragraph and the examples of pp. 101–102; the sphere of radius r and the claim that 19.4 is best possible (p. 105); the remarks of pp. 106–107 on complete non-compact manifolds, on $S^m \times S^k$ and on Synge's theorem.
- **Abstract Riemannian manifolds**: everything is proved for submanifolds of a Euclidean space with the induced metric; Nash's theorem is quoted.

The module docstrings of `Chapter11/Curvature`, `Chapter11/Myers` and `Chapter11/LongGeodesics` still list Theorem 19.2, Corollary 19.3 and the rest of Theorem 19.6 as not done. They predate the four later files, as `DocstringErrata.md` records in its one open row. This Part follows the proofs, not those notes.

What the formalization taught us

- **§18 needs no completeness**, but it needs exp to be smooth on its whole domain, which §10 does not provide. The smooth dependence of geodesics on their initial data over a compact interval supplies it `Chapter11/ExpSmooth · raysGoodE_of_isGeodesicOn` `Chapter11/ExpSmooth · exists_contDiffOn_exp`.
- **Theorem 18.1 is a statement about multiplicities** `Chapter11/ExpSmooth · theorem_18_1_multiplicity`. The explicit fields $t\, d(\exp_q)_{tv}(X)$ replace both the symmetry lemma and the dimension count `Chapter11/ExpSmooth · covAlong_expJacobi_zero`.
- **Corollary 18.2 is not immediate.** Milnor's one-line derivation hides the radiality of geodesics from p `Chapter11/NonConjugate · theorem_18_1_of_geodesic`, an undefined notion of measure zero in a manifold `Chapter11/NonConjugate · corollary_18_2`, and the step from measure zero to existence `Chapter11/NonConjugate · dense_compl_conjLocus`.
- **Hidden hypotheses**: $n \geq 2$ in Myers' theorem `Chapter11/Myers · theorem_19_4`; connectedness in Corollary 19.5 `Chapter11/Myers · corollary_19_5`; the parallelism of the velocity field in the frame of 19.4 `Chap–`

ter11/Myers · exists_parallel_frame; the continuity of the Ricci curvature on the unit tangent bundle in 19.6 Chapter11/PositiveRicci · continuous_trace_ricciMap; and, in $\pi_i(M) = \pi_i(\tilde{M})$, the connectedness of the boundary of the cube, true only for $i \geq 2$ Chapter11/CartanHadamard · genLoop_homotopic_const.

- **The completeness of the universal covering** is one inequality: lifts through $\exp_p$ grow no faster than the length of the curve below them Chapter11/ExpCovering · norm_expLift_le. It rests on the Gauss lemma on the whole domain of $\exp_p$ Chapter11/ExpCovering · gauss_lemma_expDomAt.
- **Cartan–Hadamard needs no path space**: the covering gives the theorem Chapter11/CartanHadamard · theorem_19_2, and Milnor's path-space statements follow from it Chapter11/CartanHadamard · omega_homotopyEquiv_fiber. On the way, E_{**} turns out to be positive definite in non-positive curvature Chapter11/Curvature · energyHessian_pos_of_nonpos.
- **The homology of Ω under positive Ricci curvature** comes from the energy filtration and the index bound, without cellular homology Chapter11/RicciHomology · finiteDimensional_spaceHomology_omega.

Part XI. The Appendix: the homotopy type of a monotone union

Milnor's Appendix (pp. 149–153) gives another way to finish the proof of Theorem 17.3. The path space Ω is the union of its increasing sublevel sets $\Omega^{a_0} \subseteq \Omega^{a_1} \subseteq \cdots$, each of which has the homotopy type of a CW-complex. The question is when this information about the pieces determines the homotopy type of the union. Milnor's tool is the *telescope* $X_\Sigma \subseteq X \times \mathbb{R}$, which stacks the pieces $X_n \times [n, n+1]$ on top of one another. His main result, Theorem A, says that a map which is a homotopy equivalence piece by piece is a homotopy equivalence, provided both spaces are *homotopy direct limits* of their pieces. This part retells the formal proofs of the whole Appendix except Example 2 and a remark that is stated without proof. The setting is arbitrary topological spaces, with no separation axiom anywhere except in Example 1. Theorem A is proved twice: once by Milnor's own proof, with his two cases, his explicit homotopies and a possible misprint corrected, and once exactly as he states it.

The formal route departs from the book in five places, each explained where it occurs.

- **Closedness.** Every map Milnor writes on a telescope is given strip by strip. Pasting such pieces is guaranteed to give a continuous map when the strips $X_n \times [n, n+1]$ are closed in X_Σ, as they are when the X_n are closed in X, and in general they are not (§XI.4). Milnor never says so. The formal Case 1 assumes the X_n closed, and the hypothesis is not an artifact of the formal proof: without it, Milnor's homotopy inverse g can be discontinuous (§XI.10). His proof of Theorem A as a whole needs only the Y_n closed, because Case 2 applies Case 1 only to truncations of telescopes, which are always closed (§XI.11). Theorem A as Milnor states it, with no closedness at all, follows by running the same proof on *mapping telescopes*, where a map given block by block is always continuous (§§XI.12–11.13).
- **A silent lemma.** Case 2 ends by passing from the telescope of the truncations X_Σ^n back to X_Σ. This needs X_Σ to be the homotopy direct limit of its truncations, which Milnor does not mention. It is Lemma 2.2 (§XI.6).
- **A possible misprint.** The third line of Milnor's homotopy h_u has $(4-3t)u$ where $(4t-3)u$ is meant (§XI.10).
- **Sublevel sets need no paracompactness.** Milnor's application rests on Example 1, which needs Ω paracompact. For the sublevel sets of a continuous function the formal proof needs neither paracompactness nor Hausdorffness (§XI.7), and this is the form Theorem 17.3 uses (§XI.15).
- **CW-complexes become mapping telescopes.** Example 2, that a CW-complex is the homotopy direct limit of its subcomplexes, rests on Whitehead's theorem, which Mathlib does not have. So the Corollary and the topological part of Theorem 17.3 conclude with a homotopy equivalence to the mapping telescope of a tower of iterated adjunction spaces, which stands for a countable CW-complex (§§XI.14–11.15).

The Appendix serves Part IX twice. Theorem 17.1 uses Theorem A for open, not closed, pieces of path spaces, so it needs the version without closedness. Theorem 17.3 uses the countable cell complexes of §XI.15.

Map.

- `Appendix/Telescope` (630 lines): homotopy equivalences as a property of a map; the telescope, its strips, (T1); the projection and homotopy direct limits; (T3) and maps given strip by strip; the induced map f_Σ and the reduction to it; truncations and (T2); Lemmas 2.1 and 2.2.
- `Appendix/Examples` (239 lines): Example 1; sublevel sets; Example 3.
- `Appendix/TheoremA` (765 lines): reparametrizations of the strips; Milnor's maps h_1, their inverse g and the homotopies H_u, K_u; Case 1; Case 2; Theorem A with closed Y_i.
- `Appendix/MappingTelescope` (517 lines): mapping telescopes, their truncations, and (T2) for them.
- `Appendix/MonotoneUnion` (569 lines): the comparison with mapping telescopes; Theorem A without closedness; monotone unions and the Corollary; cell towers and the topological part of Theorem 17.3.

The names (T1)–(T3), Lemma 2.1 and Lemma 2.2 are those of the retelling (`Exposition/Appendix.md`, §§1–2). Milnor gives these facts no names, and most of them he uses without stating them.

XI.1 Conventions: homotopy equivalences, restrictions, clamps

Spaces are arbitrary topological spaces and maps are continuous. A subset carries the subspace topology, and a point of a subset $S \subseteq X$ is a point of X together with a proof that it lies in S. I is the interval $[0,1]$. A sequence $X_0, X_1, \ldots$ of subsets is *increasing* when $X_0 \subseteq X_1 \subseteq \cdots$. Formally this is a monotone map from $\mathbb{N}$ to the subsets of X. Milnor's sequences are always increasing, and the statements below say when they assume it.

Homotopies. A homotopy from f_0 to $f_1 : X \to Y$ is Mathlib's: a continuous $F : I \times X \to Y$, time first, with $F(0,\cdot) = f_0$ and $F(1,\cdot) = f_1$ `Mathlib/Topology/Homotopy/Basic · ContinuousMap.Homotopy`. *Homotopic* means that a homotopy exists `Mathlib/Topology/Homotopy/Basic · ContinuousMap.Homotopic`; this is an equivalence relation, compatible with composition on either side. A homotopy of maps into a subset takes its values in that subset. So "$f_n \simeq \mathrm{id}$ in X_n" means a homotopy through maps $X_n \to X_n$.

Homotopy equivalences. A map $f : X \to Y$ is a **homotopy equivalence** if some $g : Y \to X$ has $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$ `Appendix/Telescope · IsHomotopyEquiv` (§0.4). This is a property of the map. Mathlib's homotopy equivalence is instead a structure that carries both maps f, g and the two facts $gf \simeq \mathrm{id}$, $fg \simeq \mathrm{id}$ `Mathlib/Topology/Homotopy/Equiv · ContinuousMap.HomotopyEquiv`. The two notions convert into each other `Appendix/Telescope · IsHomotopyEquiv.of_homotopyEquiv` `Appendix/Telescope · IsHomotopyEquiv.exists_homotopyEquiv`. Two spaces are *homotopy equivalent* when such a structure exists between them, as it does whenever some map between them is a homotopy equivalence `Appendix/Telescope · IsHomotopyEquiv.nonempty`. Milnor uses four closure properties without comment, and a fifth is his Assertion of p. 22. Each has a short proof:

1. identities and homeomorphisms are homotopy equivalences `Appendix/Telescope · IsHomotopyEquiv.id` `Appendix/Telescope · IsHomotopyEquiv.of_homeomorph`;
2. a composite of homotopy equivalences is one, with the composite of the inverses as inverse `Appendix/Telescope · IsHomotopyEquiv.comp`;
3. a map homotopic to a homotopy equivalence is one, with the same inverse: if $gf \simeq \mathrm{id}$, $fg \simeq \mathrm{id}$ and $f \simeq f'$, then $gf' \simeq gf \simeq \mathrm{id}$ and $f'g \simeq fg \simeq \mathrm{id}$ `Appendix/Telescope · IsHomotopyEquiv.of_homotopic`;
4. a homotopy inverse is itself a homotopy equivalence `Appendix/Telescope · IsHomotopyEquiv.of_inverse`;
5. **Milnor's Assertion** (p. 22, Part II): a map F with a left homotopy inverse L ($LF \simeq \mathrm{id}$) and a right homotopy inverse R ($FR \simeq \mathrm{id}$) is a homotopy equivalence. The proof: $L \simeq L(FR) = (LF)R \simeq R$, so $RF \simeq LF \simeq \mathrm{id}$, and R is a two-sided homotopy inverse `Chapter2/Attaching · homotopic_of_leftInv_rightInv` `Appendix/Telescope · IsHomotopyEquiv.of_leftInv_rightInv`. Case 2 ends with it (p. 153).

Restrictions. If $f : X \to Y$ carries $S \subseteq X$ into $T \subseteq Y$, then $f| : S \to T$ is its restriction `Appendix/Telescope · restrictTo`. When f carries each X_n into Y_n, f_n denotes the restriction $X_n \to Y_n$, as in Milnor.

Clamps. Milnor defines maps by formulas on pieces: $(x, n + 2t)$ for $0 \le t \le \frac{1}{2}$, another formula for $\frac{1}{2} \le t \le 1$. The formal text writes each piece as a map on the whole strip by clamping its parameter to $[0, 1]$:

$$\mathrm{clamp}(s) = \min(1, \max(0, s))$$

`Chapter2/Attaching · clampI` `Mathlib/Order/Interval/Set/ProjIcc · Set.projIcc`. It is continuous and is the identity on $[0, 1]$ `Appendix/Telescope · clampI_coe`. It sends $s \le 0$ to 0 and $s \ge 1$ to 1 `Appendix/Telescope · clampI_of_le_zero` `Appendix/Telescope · clampI_of_one_le`. A map defined by cases, "$F(q)$ if $a(q) \le b(q)$, $G(q)$ otherwise", is continuous when a, b, F and G are continuous on the whole domain and $F = G$ wherever $a = b$ `Mathlib/Topology/Order/OrderClosed · Continuous.if_le`. Clamping is what makes each branch continuous on the whole domain. So this lemma reduces the continuity of Milnor's piecewise formulas to their agreement at the break points, which is what Milnor checks when he says that a map is well defined. On the range where a branch is actually used its clamp does nothing, and the formulas below are written without clamps.

Some statements ask that certain spaces lie in the same *universe*, a size bound on types; the condition has no mathematical content and is not mentioned again.

XI.2 The telescope and its strips

Definition (the telescope; Milnor p. 149). For subsets $X_0, X_1, \ldots$ of a space X,

$$X_\Sigma = X_0 \times [0, 1] \cup X_1 \times [1, 2] \cup X_2 \times [2, 3] \cup \cdots \subseteq X \times \mathbb{R},$$

with the subspace topology of $X \times \mathbb{R}$ `Appendix/Telescope · telescope` `Appendix/Telescope · mem_telescope`.

This is exactly Milnor's definition, including his choice of topology. Every point (x, τ) of X_Σ has $\tau \ge 0$ `Appendix/Telescope · nonneg_of_mem_telescope`, and τ is called its *level*.

(T1) The telescope of an increasing sequence. If $X_0 \subseteq X_1 \subseteq \cdots$, then $(x, \tau) \in X_\Sigma$ if and only if $x \in X_n$ for some integer $0 \le n \le \tau$ `Appendix/Telescope · mem_telescope_iff`.

Proof. One direction is the definition. Conversely, let $x \in X_n$ with $n \le \tau$, and let $m = \lfloor \tau \rfloor$. Then $m \ge n$, so $x \in X_m$ by monotonicity, and $m \le \tau < m + 1$ puts (x, τ) on $X_m \times [m, m + 1]$. □

Two consequences are used constantly. *Upward closure:* if $(x, \tau) \in X_\Sigma$ and $\tau' \ge \tau$, then $(x, \tau') \in X_\Sigma$, so a homotopy that only raises levels stays in the telescope `Appendix/Telescope · mem_telescope_of_le`. *The level determines the piece:* a point of X_Σ at a level $\tau < n + 1$ lies over X_n, because (T1) gives $x \in X_m$ with $m \le \tau < n + 1$, so $m \le n$ `Appendix/Telescope · mem_of_mem_telescope`. Together they say that, for $\tau \ge 0$, $(x, \tau) \in X_\Sigma$ exactly when $x \in X_{\lfloor \tau \rfloor}$. Milnor uses all of this without comment.

The strips as maps. Let $i_n : X_n \to X_{n+1}$ be the inclusion `Appendix/Telescope · up`, and let

$$\sigma_n : X_n \times I \to X_\Sigma, \qquad \sigma_n(x, t) = (x, n + t),$$

be the n-th strip `Appendix/Telescope · stripMap`. Milnor writes $(x, n + t)$ for this point, and so does this part. The top of each strip is the bottom of the next: $\sigma_n(x, 1) = \sigma_{n+1}(i_n x, 0)$ `Appendix/Telescope · stripMap_one`. Every point lies on a strip `Appendix/Telescope · exists_stripMap`. For an increasing sequence it lies on a canonical one with $t < 1$, namely $n = \lfloor \tau \rfloor$ and $t = \tau - \lfloor \tau \rfloor$ `Appendix/Telescope · exists_stripMap_lt_one`, and a point at an integer level m is $\sigma_m(x, 0)$ `Appendix/Telescope · eq_stripMap_of_level`; both facts use (T1) to place x in X_n. Two functions on X_Σ that agree on every strip are equal `Appendix/Telescope · telescope_ext` `Appendix/Telescope · telescope_ext'`. Every identity between maps of telescopes in this part is proved this way, by evaluating both sides at $(x, n + t)$, as Milnor does.

XI.3 Homotopy direct limits and the reduction to f_Σ

Definition (Milnor p. 149). The *projection* is $p : X_\Sigma \to X$, $p(x,\tau) = x$ `Appendix/Telescope · telescopeProj`. X is the **homotopy direct limit** of $\{X_i\}$ when p is a homotopy equivalence `Appendix/Telescope · IsHomotopyDirectLimit`.

Neither Milnor's definition nor the formal one asks that the X_i cover X. The sufficient condition of Lemma 2.1 below forces it, and nothing in this part needs it separately.

The induced map. Let $f : X \to Y$ carry each X_n into Y_n. Then $f_\Sigma(x,\tau) = (f(x),\tau)$ maps X_Σ into Y_Σ, because $(f(x),\tau)$ lies on the same strip as (x,τ) `Appendix/Telescope · mem_telescope_map` `Appendix/Telescope · telescopeMap`. On the strips, $f_\Sigma(x,n{+}t) = (f_n(x),n{+}t)$ `Appendix/Telescope · telescopeMap_stripMap`, and $p_Y f_\Sigma = f p_X$ holds by definition `Appendix/Telescope · telescopeProj_comp_telescopeMap`. Since f_Σ is the restriction of $f \times \mathrm{id}_{\mathbb{R}}$, it is continuous without any pasting. This matters in §XI.11.

Theorem (the reduction; Milnor p. 150). Let X and Y be the homotopy direct limits of $\{X_i\}$ and $\{Y_i\}$, and let f carry each X_n into Y_n. If f_Σ is a homotopy equivalence, so is f. `Appendix/Telescope · IsHomotopyEquiv.of_telescopeMap`

Milnor dismisses this step in a single sentence. The formal statement assumes neither monotonicity nor closedness.

Proof. Let s be a homotopy inverse of p_X, so $sp_X \simeq \mathrm{id}$ and $p_X s \simeq \mathrm{id}$. Then s is itself a homotopy equivalence `Appendix/Telescope · IsHomotopyEquiv.of_inverse`, and $p_Y f_\Sigma s$ is a composite of three homotopy equivalences `Appendix/Telescope · IsHomotopyEquiv.comp`. But

$$p_Y f_\Sigma s = f p_X s \simeq f,$$

the equality by $p_Y f_\Sigma = f p_X$ and the homotopy by $p_X s \simeq \mathrm{id}$. A map homotopic to a homotopy equivalence is one `Appendix/Telescope · IsHomotopyEquiv.of_homotopic`. □

The same three lines close the proof of Theorem A without closedness (§XI.13), with mapping telescopes in place of telescopes.

XI.4 Pasting along the strips: (T3) and the hypothesis it needs

Every map and every homotopy that Milnor writes on a telescope is given by formulas in $(x, n+t)$, strip by strip, and he never says why the result is continuous. The reason is the following lemma, and the lemma has a hypothesis.

Theorem (T3). Let every X_n be closed in X, and let P be any space. A function $F : P \times X_\Sigma \to Z$ is continuous as soon as, for every n, the function $(q,x,t) \mapsto F(q,(x,n{+}t))$ is continuous on $P \times X_n \times I$. `Appendix/Telescope · continuous_of_stripMap`

Proof as formalized. Let $S_n = \{(q,(x,\tau)) : x \in X_n,\ n \le \tau \le n+1\}$, the n-th strip times P.

1. The S_n cover $P \times X_\Sigma$, by the definition of X_Σ.
2. Each S_n is closed. It is the preimage of X_n under the continuous coordinate x, intersected with two closed inequalities in the coordinate τ. This is the one place where the closedness of X_n is used.
3. The family is locally finite. The open neighbourhood $\{|\tau - \tau_0| < 1\}$ of a point at level τ_0 meets S_n only if $n \le \tau < \tau_0 + 1$, so only for the finitely many $n < \lceil \tau_0 + 1 \rceil$.
4. F is continuous on each S_n. The map $S_n \to P \times X_n \times I$, $(q,(x,\tau)) \mapsto (q,x,\tau - n)$, is continuous, and on S_n the function F is the given continuous function composed with it, since $(x, n{+}(\tau{-}n)) = (x,\tau)$.

A function continuous on each member of a locally finite cover by closed sets is continuous `Mathlib/Topology/LocallyFinite · LocallyFinite.continuous`. □

Why closedness is needed. Without it a strip need not be closed in X_Σ. In $X = [0,1]$ take $X_0 = (0,1]$ and $X_1 = [0,1]$. The point $(0,1)$ lies in X_Σ, on the strip $X_1 \times [1,2]$. It is the limit of the points $(\frac{1}{k}, 1 - \frac{1}{k})$ of the strip $X_0 \times [0,1]$, which does not contain it. So a function that is continuous on each strip is free to do anything on $X_0 \times [0,1]$ near $(0,1)$, whatever its value at $(0,1)$. §XI.10 shows that Milnor's map g does exactly this for a suitable choice of homotopy.

Maps given strip by strip. The formal text builds all of Milnor's maps through one construction. Let $\Phi_n : P \times X_n \times I \to Z$ be continuous maps that agree at the integer levels:

$$\Phi_{n+1}(q, i_n x, 0) = \Phi_n(q, x, 1).$$

Define $F(q,(x,\tau)) = \Phi_{\lfloor\tau\rfloor}(q, x, \tau - \lfloor\tau\rfloor)$, using the canonical strip. Here $x \in X_{\lfloor\tau\rfloor}$ by (T1), and $\tau - \lfloor\tau\rfloor \in [0,1)$ `Appendix/Telescope · telescopeLiftFun Appendix/Telescope · mem_floor Appendix/Telescope · fract_mem`. Then $F(q,(x,n+t)) = \Phi_n(q,x,t)$ for every n and every $t \in I$ `Appendix/Telescope · telescopeLiftFun_stripMap`. For $t < 1$ the floor of $n+t$ is n. For $t = 1$ it is $n+1$, and the compatibility condition gives the value. So F is continuous on every strip, and (T3) makes it continuous `Appendix/Telescope · telescopeLift Appendix/Telescope · telescopeLift_stripMap`.

With P a point this gives maps $X_\Sigma \to Z$ `Appendix/Telescope · telescopeDesc Appendix/Telescope · telescopeDesc_stripMap`. With $P = I$ it gives homotopies, whose two ends are checked strip by strip `Appendix/Telescope · telescopeHomotopy`. All three constructions call (T3), so all three need the X_n closed. Every map of Case 1, the reparametrizations of §XI.8 and Milnor's map G of Case 2 are built with them. These constructions are the only way that closedness hypotheses enter this part.

XI.5 Truncations: (T2)

Definition (Milnor p. 152). The n-th truncation is $X_\Sigma^n = \{(x,\tau) \in X_\Sigma : \tau \le n\}$ `Appendix/Telescope · truncation`. For an increasing sequence it is $X_0 \times [0,1] \cup \cdots \cup X_{n-1} \times [n-1,n] \cup X_n \times \{n\}$.

Whatever the X_i, X_Σ^n is closed in X_Σ, being cut out by $\tau \le n$ `Appendix/Telescope · isClosed_truncation`, and $X_\Sigma^0 \subseteq X_\Sigma^1 \subseteq \cdots$ `Appendix/Telescope · monotone_truncation`. The first of these facts is what saves Case 2.

For an increasing sequence, let $\iota_n : X_n \to X_\Sigma^n$, $\iota_n(x) = (x,n)$ `Appendix/Telescope · stripMap_zero_mem_truncation Appendix/Telescope · truncIncl`. Let $r_n : X_\Sigma^n \to X_n$, $r_n(x,\tau) = x$, which lands in X_n by (T1) since $\tau \le n < n+1$ `Appendix/Telescope · truncProj`. Both are restrictions of continuous maps of $X \times \mathbb{R}$, and $r_n \iota_n = \mathrm{id}$ by definition `Appendix/Telescope · truncProj_comp_truncIncl`.

Theorem (T2). For an increasing sequence, $\iota_n r_n \simeq \mathrm{id}$ in X_Σ^n. Hence $\iota_n : X_n \to X_\Sigma^n$ and r_n are homotopy equivalences. `Appendix/Telescope · truncIncl_comp_truncProj Appendix/Telescope · isHomotopyEquiv_truncIncl Appendix/Telescope · isHomotopyEquiv_truncProj`

Milnor states without proof that X_Σ^n contains $X_n \times \{n\}$ as a deformation retract (p. 152).

Proof. Put $D(s,(x,\tau)) = (x, (1-s)n + s\tau)$. As $\tau \le n$, the new level lies between τ and n. It is at least τ, so D stays in X_Σ by upward closure, and it is at most n, so D stays in X_Σ^n. $D(0,\cdot) = \iota_n r_n$ and $D(1,\cdot) = \mathrm{id}$, and D is continuous as a formula in the coordinates. □

No closedness is needed: the deformation is a single formula on all of X_Σ^n, not a map pasted from strips. D also fixes $X_n \times \{n\}$, so it is a deformation retraction, as Milnor says. The formal statement records only what is used later, the homotopy.

Lemma (levels). Let $X_0 \subseteq X_1 \subseteq \cdots$, and let $F : X_\Sigma \to Z$ and $\ell : Z \to \mathbb{R}$ be functions. Suppose that F sends the n-th strip to points with $\ell \le n+1$ and its bottom $X_n \times \{n\}$ to points with $\ell \le n$. Then F sends X_Σ^m to points with $\ell \le m$, for every m. `Appendix/Telescope · level_le_of_stripMap`

Proof. Write a point of X_Σ^m as $(x, n+t)$ on its canonical strip, with $t < 1$ and $n + t \le m$. If $t = 0$, then $\ell \le n \le m$. If $t > 0$, then $n < m$, so $n + 1 \le m$ as both are integers, and $\ell \le n + 1 \le m$. □

In Case 2, ℓ is the level and F is Gf_Σ or $f_\Sigma G$. Milnor asserts without argument that these maps carry each truncation into itself, and this lemma is the proof.

XI.6 Lemma 2.1 and Lemma 2.2: when the graph of a function is a deformation retract

Lemma 2.1. Let $X_0 \subseteq X_1 \subseteq \cdots \subseteq X$. Let $f : X \to \mathbb{R}$ be continuous with $f \geq 0$ and $f(x) \geq n+1$ whenever $x \notin X_n$. Then X is the homotopy direct limit of $\{X_i\}$. `Appendix/Telescope · isHomotopyDirectLimit_of_level`

This is the mechanism of Milnor's Example 1, where he asserts without argument that the image of $x \mapsto (x, f(x))$ is a deformation retract of X_Σ (p. 149). Once f is given, it needs neither paracompactness nor closedness.

Proof as formalized.

1. *The graph lies in X_Σ.* Let $m = \lfloor f(x) \rfloor$. If $x \notin X_m$, then $f(x) \geq m+1 > f(x)$, which is absurd. So $x \in X_m$ with $m \leq f(x)$, and $(x, f(x)) \in X_\Sigma$ by (T1).
2. *A section.* $\sigma_f(x) = (x, f(x))$ is continuous, and $p\sigma_f = \mathrm{id}_X$ by definition.
3. *The deformation.* Put $D_f(s, (x, \tau)) = (x, (1-s)f(x) + s\tau)$. It stays in X_Σ. If $\tau \leq f(x)$, the new level is at least τ, and upward closure applies to (x, τ). If $\tau \geq f(x)$, it is at least $f(x)$, and upward closure applies to $(x, f(x))$, which lies in X_Σ by step 1. $D_f(0, \cdot) = \sigma_f p$ and $D_f(1, \cdot) = \mathrm{id}$, so $\sigma_f p \simeq \mathrm{id}$.

So p is a homotopy equivalence with homotopy inverse σ_f. □

Each $D_f(s, \cdot)$ fixes the graph of f, so the graph is a deformation retract, as Milnor says; the formal proof needs only the two homotopies. The hypothesis forces $\bigcup_n X_n = X$, since a point outside every X_n would have $f(x) \geq n+1$ for all n.

Lemma 2.1 does most of the work of this part. Example 1, its form for sublevel sets, Lemma 2.2 and the corresponding fact for mapping telescopes all reduce to it, each by writing down a suitable f (§§XI.7 and 11.12). So does the version of Example 1 pulled back along a map, which Part IX uses for Theorem 17.1 (§XI.13).

Lemma 2.2. For any sequence $X_0, X_1, \ldots$ of subsets of X, increasing or not, closed or not, the telescope X_Σ is the homotopy direct limit of its truncations $\{X_\Sigma^n\}$. `Appendix/Telescope · isHomotopyDirectLimit_truncation`

Proof. Apply Lemma 2.1 to the space X_Σ, the increasing sequence $X_\Sigma^0 \subseteq X_\Sigma^1 \subseteq \cdots$, and $f(x, \tau) = \tau + 1 \geq 1$. A point outside X_Σ^n has $\tau > n$, hence $f > n+1$. □

The telescope of the truncations is a telescope of a telescope: a subspace of $X_\Sigma \times \mathbb{R}$, whose points are pairs $((x, \tau), \tau')$ with $\tau \leq \lfloor \tau' \rfloor$. Milnor needs Lemma 2.2 at the end of Case 2 and does not mention it (§XI.11). Example 1 would give it only if X_Σ were paracompact, and nothing guarantees that.

XI.7 Milnor's three examples

Example 1 (Milnor p. 149). Let X be paracompact and Hausdorff, and let $X_0 \subseteq X_1 \subseteq \cdots$ be subsets such that every point of X lies in the interior of some X_n. Then X is the homotopy direct limit of $\{X_i\}$. `Appendix/Examples · example1`

Hidden hypothesis: Hausdorff. Milnor says only paracompact. Mathlib's partitions of unity subordinate to an open cover need a space that is normal as well as paracompact `Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate` `Mathlib/Topology/Compactness/`

`Paracompact · ParacompactSpace`. A paracompact Hausdorff space is normal by Dieudonné's theorem, which Mathlib provides as an instance. So the formal statement reads paracompactness with the Hausdorff axiom included, as Bourbaki and Dugundji do.

Proof as formalized. Milnor's sentence, that a partition of unity gives the function, becomes the following.

1. The interiors $\operatorname{int} X_n$ form an open cover of X. Take a partition of unity $\{\beta_n\}$ subordinate to it: $\beta_n \geq 0$, the supports are locally finite, the support of β_n lies in $\operatorname{int} X_n$, and $\sum_n \beta_n = 1$.
2. Put $f = \sum_n n\,\beta_n$. It is a locally finite sum, hence continuous `Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.continuous_finsum_smul`, and $f \geq 0$.
3. Let $x \notin X_i$. For $j \leq i$, $\beta_j(x) = 0$: otherwise x lies in the support of β_j, hence in $X_j \subseteq X_i$. The sums are finite at x, so
$$i + 1 = \sum_j \beta_j(x)(i+1) \leq \sum_j \beta_j(x)\, j = f(x),$$
term by term: for $j \leq i$ both terms vanish, and for $j > i$ one has $i + 1 \leq j$.
4. Lemma 2.1 concludes. □

Sublevel sets. Let $E : X \to \mathbb{R}$ be continuous, on any space X, and let $a_0 \leq a_1 \leq \cdots$ with $a_n \to \infty$. Then X is the homotopy direct limit of the sublevel sets $X^{a_n} = \{E \leq a_n\}$ (§0.4). `Appendix/Examples · isHomotopyDirectLimit_sublevel` `Chapter2/Crossing · sublevel`

Deviation (stronger). No paracompactness, no Hausdorffness and no interiors are needed. This is the situation of Theorem 17.3, where Milnor's application (p. 150) appeals to Example 1 and hence to the paracompactness of the path space; here a function is written down directly.

Proof as formalized.

1. For each j let $\theta_j(x) = \operatorname{clamp}(E(x) - a_j + 1)$. It is continuous, with values in $[0, 1]$; it is 0 where $E \leq a_j - 1$ and 1 where $E \geq a_j$.
2. *The supports are locally finite.* Given x_0, choose N with $a_j \geq E(x_0) + 2$ for all $j \geq N$, which is possible as a is increasing and tends to ∞. On the open neighbourhood $\{E < E(x_0) + 1\}$ of x_0, every $j \geq N$ has $E - a_j + 1 < 0$, so $\theta_j = 0$. Only the θ_j with $j < N$ can be nonzero near x_0.
3. So $f = \sum_j \theta_j$ is locally a finite sum of continuous functions, hence continuous `Mathlib/Topology/Algebra/Monoid · continuous_finprod` (in its additive form), and $f \geq 0$.
4. Let $x \notin X^{a_i}$. Then $E(x) > a_i \geq a_j$ for every $j \leq i$, so $E(x) - a_j + 1 > 1$ and $\theta_j(x) = 1$. The remaining terms are ≥ 0, so $f(x) \geq i + 1$.
5. The sublevel sets increase with a_n `Chapter2/Crossing · sublevel_mono`, and Lemma 2.1 concludes. □

Example 2 (Milnor p. 149) is not formalized. It says that a CW-complex is the homotopy direct limit of an increasing sequence of subcomplexes with union the whole complex. Milnor's proof is that p induces isomorphisms of all homotopy groups, so that Whitehead's theorem applies. That needs Whitehead's theorem, the fact that compact subsets of a CW-complex lie in finite subcomplexes, and the CW topology of $X \times \mathbb{R}$. None of them is in Mathlib. This one gap shapes the end of the Appendix. Wherever Milnor takes a union $K = \bigcup K_i$ of complexes with the weak topology, the formal text takes a mapping telescope of the K_i instead (§§XI.12–11.15).

Example 3 (Milnor p. 150). Let $X_n = \{0\} \cup [\frac{1}{n+1}, 1] \subseteq [0, 1]$ `Appendix/Examples · example3Set`. Milnor writes $[0] \cup [1/i, 1]$, indexing from 1. These sets are closed `Appendix/Examples · isClosed_example3Set` and increasing `Appendix/Examples · monotone_example3Set`. Their union is $[0, 1]$, since every $x > 0$ exceeds some $\frac{1}{n+1}$ `Appendix/Examples · iUnion_example3Set`. Then $[0, 1]$ is **not** the homotopy direct limit of $\{X_n\}$. `Appendix/Examples · example3`

Milnor gives no proof.

Proof as formalized. Suppose that $p : X_\Sigma \to [0,1]$ has a homotopy inverse s, and let Λ be a homotopy from sp to the identity of X_Σ. Only this half of the hypothesis will be used. Let $C = \{(x,\tau) \in X_\Sigma : x = 0\}$, the ray $\{0\} \times [0,\infty)$.

1. *C is open and closed in X_Σ.* It is closed as the preimage of 0. For openness, let $(0,\tau_0) \in C$ and consider the open set
$$W = \{(x,\tau) \in X_\Sigma : x < \tfrac{1}{\tau_0+2},\ \tau < \tau_0 + 1\},$$
which contains $(0,\tau_0)$. Suppose a point (x,τ) of W had $x \neq 0$. It would lie on some strip n, with $n \le \tau$ and $x \ge \frac{1}{n+1}$. Then $\frac{1}{n+1} < \frac{1}{\tau_0+2}$ gives $n > \tau_0 + 1 > \tau \ge n$, which is absurd. So $W \subseteq C$. In words: below the level $\tau_0 + 1$, every point of the telescope off the ray has first coordinate at least $\frac{1}{\tau_0+2}$.
2. *Paths cannot leave C.* For a path $\gamma : I \to X_\Sigma$, the set $\gamma^{-1}(C)$ is open and closed in the connected interval, hence empty or everything `Mathlib/Topology/Connected/Clopen · isClopen_iff`. So $\gamma(0) \in C$ if and only if $\gamma(1) \in C$.
3. *Three paths.* The path $u \mapsto \Lambda(u,(0,0))$ runs from $sp(0,0) = s(0)$ to $(0,0)$. The map s itself is a path from $s(0)$ to $s(1)$, its domain being $[0,1]$. The path $u \mapsto \Lambda(u,(1,0))$ runs from $s(1)$ to $(1,0)$. As $(0,0) \in C$, step 2 gives $s(0) \in C$, then $s(1) \in C$, then $(1,0) \in C$, which is false. □

The proof shows more than the statement: p has no left homotopy inverse at all. The point 0 lies in every X_n but in the interior of none, so the hypothesis of Example 1 fails at a single point, and the telescope separates that point from the rest.

XI.8 Theorem A, and reparametrizing the strips

Theorem A (Milnor p. 150). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and Y that of $Y_0 \subseteq Y_1 \subseteq \cdots$. Let $f : X \to Y$ carry each X_i into Y_i by a homotopy equivalence $f_i : X_i \to Y_i$. Then f is a homotopy equivalence.

The formal text proves it twice. The first version assumes the Y_i closed and follows Milnor's proof `Appendix/TheoremA · theoremA` (§XI.11). The second is exactly Milnor's statement `Appendix/MonotoneUnion · theoremA_general` (§XI.13). Milnor's proof has three steps:

1. the reduction to f_Σ (§XI.3);
2. Case 1, where $X = Y$, $X_i = Y_i$ and each f_i is homotopic to the identity (§§XI.9–11.10);
3. Case 2, the general case, which builds a homotopy inverse G of f_Σ and applies Case 1 to Gf_Σ and $f_\Sigma G$ (§XI.11).

In this section and in §§XI.9–11.10, $X_0 \subseteq X_1 \subseteq \cdots$ are **closed** in X, so that (T3) is available.

Reparametrizing the strips. Twice Milnor declares two maps homotopic without argument when they differ by a change of speed inside the strips (pp. 151 and 152). Let $\rho : I \to I$ be continuous with $\rho(0) = 0$ and $\rho(1) = 1$, and put
$$R_\rho(x, n+t) = (x, n + \rho(t)).$$
It is pasted from the strips: at an integer level both neighbouring formulas give $(x, n+1)$, because ρ fixes 0 and 1 `Appendix/TheoremA · reparam` `Appendix/TheoremA · reparam_stripMap`.

Lemma. $R_\rho \simeq \mathrm{id}$. `Appendix/TheoremA · reparam_homotopic`

Proof. Take the straight-line homotopy $(s,(x,n+t)) \mapsto (x, n + (1-s)\rho(t) + st)$. It stays inside each strip, since a convex combination of two numbers in $[0,1]$ lies in $[0,1]$. It agrees at the integer levels for every s, since both ρ and the identity fix 0 and 1. It is R_ρ at $s = 0$ and the identity at $s = 1$, and it is assembled strip by strip as in §XI.4. □

The case that is used is $R_c(x, n+t) = (x, n + \min(ct, 1))$ for $c \geq 1$, from $\rho(t) = \mathrm{clamp}(ct)$ Appendix/TheoremA · scaleI Appendix/TheoremA · scaleMap Appendix/TheoremA · scaleMap_stripMap Appendix/TheoremA · scaleMap_homotopic. Milnor's h_0 is $f_\Sigma R_2$ (§XI.10), and his homotopies H_u end at R_4 (§XI.9).

XI.9 Milnor's maps h_1, their inverse g, and the homotopies H_u and K_u

Milnor's proof of Case 1 deforms f_Σ to a map h_1 with two properties. He then shows that *any* map with these two properties is a homotopy equivalence (p. 151). The formal text isolates the class of such maps.

Definition (Milnor maps). A map $\psi : X_\Sigma \to X_\Sigma$ is a *Milnor map* if

- (low) $\psi(x, n+t) = (x, n+2t)$ for $0 \leq t \leq \frac{1}{2}$, and
- (high) $\psi(x, n+t)$ has level $n+1$ for $\frac{1}{2} \leq t \leq 1$. Appendix/TheoremA · IsMilnorMap

In the second condition Milnor asks that $h_1(x, n+t) \in X_{n+1} \times \{n+1\}$. Asking only for the level is the same, since a point of X_Σ at level $n+1$ lies over X_{n+1} by (T1).

Elementary properties. A Milnor map fixes every point at an integer level: such a point is $(x, m+0)$, and (low) at $t = 0$ returns it Appendix/TheoremA · IsMilnorMap.fix. Milnor uses this without saying so when he computes $h_1 g$. It follows that $\psi(\psi(x, n+t)) = \psi(x, n+t)$ for $t \geq \frac{1}{2}$, the inner value being at level $n+1$ Appendix/TheoremA · IsMilnorMap.fix_high. It also follows that $\psi(x, n+1) = (x, n+1)$ Appendix/TheoremA · IsMilnorMap.apply_one, and (low) gives $\psi(x, n+\frac{1}{2}) = (x, n+1)$. The formal text packages the last two facts as $\psi(x, n+\frac{1}{2}+\frac{\mu}{2}) = (x, n+1)$ for $\mu \in \{0, 1\}$ Appendix/TheoremA · IsMilnorMap.apply_half_or_one.

So for each $x \in X_n$ the path $s \mapsto \psi(x, n+s)$, $\frac{1}{2} \leq s \leq 1$, is a **loop** at $(x, n+1)$. Every homotopy below shrinks this loop. It is convenient to parametrize the loop by $\mu \in [0, 1]$, with $s = \frac{1}{2} + \frac{\mu}{2}$.

Milnor's inverse g (p. 151).

$$g(x, n+t) = \begin{cases} (x, n+2t) & 0 \leq t \leq \frac{1}{2}, \\ \psi(x, n+\frac{3}{2}-t) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

Appendix/TheoremA · milnorInvStrip Appendix/TheoremA · milnorInv

At $t = \frac{1}{2}$ both lines give $(x, n+1)$, because $\psi(x, n+1) = (x, n+1)$. Between strips, the top of strip n gives $\psi(x, n+\frac{1}{2}) = (x, n+1)$ and the bottom of strip $n+1$ gives $(x, n+1)$. This is Milnor's remark that g is well defined because h_1 takes the value $(x, n+1)$ at both $n+\frac{1}{2}$ and $n+1$. The map is pasted from the strips by (T3).

g is a Milnor map itself Appendix/TheoremA · milnorInv_low Appendix/TheoremA · milnorInv_high Appendix/TheoremA · milnorInv_isMilnorMap. Property (low) is its first line. For (high): at $t = \frac{1}{2}$ the first line has level $n+1$, and for $t > \frac{1}{2}$ the value $\psi(x, n+\frac{3}{2}-t)$ has level $n+1$ because $\frac{3}{2} - t \geq \frac{1}{2}$. In particular g also fixes the integer levels.

The two composites. Using only these facts,

$$\psi g(x, n+t) = \begin{cases} (x, n+4t) & t \leq \frac{1}{4}, \\ \psi(x, n+2t) & \frac{1}{4} \leq t \leq \frac{1}{2}, \\ \psi(x, n+\frac{3}{2}-t) & \frac{1}{2} \leq t \leq 1, \end{cases} \qquad g\psi(x, n+t) = \begin{cases} (x, n+4t) & t \leq \frac{1}{4}, \\ \psi(x, n+\frac{3}{2}-2t) & \frac{1}{4} \leq t \leq \frac{1}{2}, \\ \psi(x, n+t) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

For ψg: if $t \leq \frac{1}{2}$, g gives $(x, n+2t)$, to which ψ applies (low) when $2t \leq \frac{1}{2}$; if $t \geq \frac{1}{2}$, g gives a point at level $n+1$, which ψ fixes. For $g\psi$: if $t \leq \frac{1}{4}$ both maps act by (low); if $\frac{1}{4} \leq t \leq \frac{1}{2}$, ψ gives $(x, n+2t)$ with $2t \geq \frac{1}{2}$, and $g(x, n+2t) = \psi(x, n+\frac{3}{2}-2t)$; if $t \geq \frac{1}{2}$, ψ gives a point at level $n+1$, which g fixes. On

$[\frac{1}{4}, 1]$, ψg runs along the loop forwards, μ going from 0 to 1, and then backwards. $g\psi$ runs backwards and then forwards.

The loop homotopies. For a continuous function $\mu(u, t)$, define on each strip

$$L_u^\mu(x, n+t) = \begin{cases} (x, n+4t) & t \le \frac{1}{4}, \\ \psi(x, n+\frac{1}{2}+\frac{1}{2}\mu(u,t)) & t \ge \frac{1}{4}. \end{cases}$$

`Appendix/TheoremA · loopFun`

This is continuous on each strip times I when the two lines agree at $t = \frac{1}{4}$, where the first gives $(x, n+1)$. By the packaged property above, that holds when $\mu(u, \frac{1}{4}) \in \{0, 1\}$ `Appendix/TheoremA · continuous_loopFun`. The strips fit together when $\mu(u, 1) \in \{0, 1\}$, since then the top of strip n is $(x, n+1)$, the bottom of strip $n+1$. (T3) then makes $(u, z) \mapsto L_u^\mu(z)$ a homotopy `Appendix/TheoremA · loopHomotopy`. If $\mu(1, t) \in \{0, 1\}$ for $\frac{1}{4} < t \le 1$, the end L_1^μ is R_4, since every point with $t \ge \frac{1}{4}$ goes to $(x, n+1)$ `Appendix/TheoremA · loopFun_one`. Milnor's two homotopies are two choices of μ.

Milnor's H_u. Take $\mu_H(u, t) = \min(4t{-}1,\ 1{-}u,\ 2{-}2t)$ `Appendix/TheoremA · muH` `Appendix/TheoremA · continuous_muH`.

- At $u = 0$ it is $4t{-}1$ on $[\frac{1}{4}, \frac{1}{2}]$ and $2{-}2t$ on $[\frac{1}{2}, 1]$ `Appendix/TheoremA · muH_zero_low` `Appendix/TheoremA · muH_zero_high`. Then $\frac{1}{2} + \frac{1}{2}\mu_H$ is $2t$, respectively $\frac{3}{2} - t$, and $L_0^{\mu_H} = \psi g$.
- At $u = 1$ it vanishes on $[\frac{1}{4}, 1]$ `Appendix/TheoremA · muH_one`, so $L_1^{\mu_H} = R_4$.
- It vanishes at $t = \frac{1}{4}$ and at $t = 1$ for every u `Appendix/TheoremA · muH_quarter` `Appendix/TheoremA · muH_end`.

Hence there is a homotopy $\psi g \simeq R_4$ `Appendix/TheoremA · milnorH`.

This is Milnor's H_u of p. 151, written as one formula. Milnor's parameter runs over $[0, \frac{1}{2}]$. With $u_M = u/2$, the cap $\mu \le 1 - u$ means $s \le 1 - u_M$: his homotopy cuts the excursion along the loop at the point $\psi(x, n + 1 - u_M)$. The minimum switches from $4t - 1$ to $1 - u$ at $t = \frac{1-u_M}{2}$, and from $1 - u$ to $2 - 2t$ at $t = \frac{1}{2} + u_M$; these are exactly his break points. His end $H_{1/2}$ is R_4, which he declares homotopic to the identity without argument; the lemma of §XI.8 is the proof.

The analogous homotopy K_u. For $g\psi$, Milnor says only that an analogous argument applies (p. 152). The formal text takes $\mu_K(u, t) = \max(2 - 4t,\ 2t - 1,\ u)$ `Appendix/TheoremA · muK` `Appendix/TheoremA · continuous_muK`.

- At $u = 0$ it is $2{-}4t$ on $[\frac{1}{4}, \frac{1}{2}]$ and $2t{-}1$ on $[\frac{1}{2}, 1]$ `Appendix/TheoremA · muK_zero_low` `Appendix/TheoremA · muK_zero_high`. Then $\frac{1}{2} + \frac{1}{2}\mu_K$ is $\frac{3}{2} - 2t$, respectively t, so $L_0^{\mu_K} = g\psi$.
- At $u = 1$ it is 1 on $[\frac{1}{4}, 1]$ `Appendix/TheoremA · muK_one`, so $L_1^{\mu_K} = R_4$, as $\psi(x, n+1) = (x, n+1)$.
- It equals 1 at $t = \frac{1}{4}$ and at $t = 1$ for every u `Appendix/TheoremA · muK_quarter` `Appendix/TheoremA · muK_end`.

Hence there is a homotopy $g\psi \simeq R_4$ `Appendix/TheoremA · milnorK`. Where H_u caps the loop from above, K_u cuts it from below: the part of the loop with $\mu < u$ is never visited, and at $u = 1$ nothing of the loop remains.

Theorem (Milnor's maps are homotopy equivalences; pp. 151–152). For closed X_n, every Milnor map ψ is a homotopy equivalence, with homotopy inverse g. `Appendix/TheoremA · isHomotopyEquiv_of_milnorMap`

Proof. $g\psi \simeq R_4 \simeq \mathrm{id}$ and $\psi g \simeq R_4 \simeq \mathrm{id}$, by the two loop homotopies and §XI.8. □

The formal proofs that the loop homotopies start at ψg and at $g\psi$ are the case analyses displayed above for the two composites, over the three ranges $t \le \frac{1}{4}$, $\frac{1}{4} < t \le \frac{1}{2}$ and $t > \frac{1}{2}$.

XI.10 Case 1

Theorem (Case 1; Milnor pp. 150–152). Let $X_0 \subseteq X_1 \subseteq \cdots$ be closed in X. Let $\varphi : X \to X$ carry each X_n into itself by a map φ_n that is homotopic, in X_n, to the identity of X_n. Then $\varphi_\Sigma : X_\Sigma \to X_\Sigma$ is a homotopy equivalence. Appendix/TheoremA · theoremA_case1

Milnor's Case 1 assumes $X_i = Y_i$; the formal statement takes $Y = X$ and a self-map φ. It does not assume that X is a homotopy direct limit.

Choose homotopies $h_u^n : X_n \to X_n$, $0 \le u \le 1$, with $h_0^n = \varphi_n$ and $h_1^n = \mathrm{id}$. The formal proof picks one for each n from the hypothesis.

Milnor's h_u, corrected. On the n-th strip,

$$h_u(x, n+t) = \begin{cases} (h_u^n(x),\ n+2t) & 0 \le t \le \frac{1}{2}, \\ (h^n_{(3-4t)u}(x),\ n+1) & \frac{1}{2} \le t \le \frac{3}{4}, \\ (h^{n+1}_{(4t-3)u}(x),\ n+1) & \frac{3}{4} \le t \le 1. \end{cases}$$

Appendix/TheoremA · case1Fun

The misprint. Milnor prints the third line as $h^{n+1}_{(4-3t)u}$ (p. 151). At $t = 1$ the printed index happens to give the right value u. At $t = \frac{3}{4}$, however, it gives $h^{n+1}_{7u/4}$, which is not even defined for $u > \frac{4}{7}$, while continuity at $t = \frac{3}{4}$ requires h_0^{n+1} there. The correct index $(4t-3)u$ runs from 0 at $t = \frac{3}{4}$ to u at $t = 1$. The second line, $h^n_{(3-4t)u}$, is printed correctly.

Read along the strip, h_u first rises to the top of the strip while applying h_u^n. It then runs the homotopy h^n back from time u to time 0, arriving at $\varphi(x)$. Finally it runs the next homotopy h^{n+1} forward from time 0 to time u, arriving at the value from which the next strip starts.

Continuity on a strip Appendix/TheoremA · continuous_case1Fun. At $t = \frac{1}{2}$, the first line gives $(h_u^n x, n+1)$ as a point at the top of strip n. The second line gives the same point, since $(3-2)u = u$, as a point at the bottom of strip $n+1$. They are equal by the gluing of strips. At $t = \frac{3}{4}$ both indices are 0. Since $h_0^n = \varphi_n$ and $h_0^{n+1} = \varphi_{n+1}$, both lines give $(\varphi(x), n+1)$; here one uses that φ_{n+1} and φ_n are restrictions of the same φ. The clamps make each line continuous on $I \times X_n \times I$, and the case-splitting lemma of §XI.1 does the rest.

Compatibility between strips Appendix/TheoremA · case1Fun_compat. At the top of strip n the third line gives $(h_u^{n+1} x, n+1)$, as $(4-3)u = u$. At the bottom of strip $n+1$ the first line gives the same point.

By (T3) with $P = I$, $(u, z) \mapsto h_u(z)$ is continuous on $I \times X_\Sigma$ Appendix/TheoremA · case1Family Appendix/TheoremA · case1Map Appendix/TheoremA · case1Map_stripMap. So $h_0 \simeq h_1$.

$h_0 = \varphi_\Sigma R_2$ Appendix/TheoremA · case1Map_zero. At $u = 0$ all three indices are 0, and $h_0^n = \varphi_n$. So

$$h_0(x, n+t) = (\varphi(x), n + \min(2t, 1)) = \varphi_\Sigma(x, n + \min(2t, 1)) = \varphi_\Sigma R_2(x, n+t).$$

As $R_2 \simeq \mathrm{id}$, $h_0 \simeq \varphi_\Sigma$. Milnor takes this homotopy as evident (p. 151).

h_1 is a Milnor map Appendix/TheoremA · case1Map_one. At $u = 1$ the first line is $(h_1^n x, n+2t) = (x, n+2t)$, which is (low). For $t \ge \frac{1}{2}$ every line is at level $n+1$, which is (high).

Proof of Case 1. $\varphi_\Sigma \simeq h_0 \simeq h_1$, and h_1 is a homotopy equivalence by §XI.9. A map homotopic to a homotopy equivalence is one. □

Why Case 1 needs closed sets. All of its maps are pasted from strips of X_Σ: h_u, g, H_u, K_u, R_2 and R_4. Without closedness the construction itself can fail, not only its formal proof. The retelling gives an example, which is not formalized. Let $X = [0, 1]$, $X_0 = (0, 1]$, $X_n = X$ for $n \ge 1$ and $\varphi = \mathrm{id}$. Take $h_u^n = \mathrm{id}$

for $n \geq 1$, and for h_u^0 a homotopy from the identity of $(0,1]$ to itself that, at the time $u = 1 - \frac{x}{2}$, pushes the point x all the way to 1:

$$h_u^0(x) = x + (1-x)\max\Big(0,\ 1 - \frac{|u - 1 + x/2|}{x/2}\Big).$$

For $t = 1 - \frac{x}{8}$, Milnor's g gives

$$g(x,t) = h_1(x, \tfrac{1}{2} + \tfrac{x}{8}) = (h^0_{1-x/2}(x), 1) = (1,1).$$

But $(x, 1 - \frac{x}{8}) \to (0,1)$ as $x \to 0$, and $g(0,1) = (0,1)$. So g is discontinuous at $(0,1)$: the strip $X_0 \times [0,1]$ is not closed, exactly as in §XI.4. (The formal text does not ask whether the conclusion of Case 1 survives without closedness: below, Case 1 is applied only to truncations, which are closed.)

The Remark (p. 150). Milnor observes that under the hypotheses of Case 1 one might expect φ_Σ to be homotopic to the identity, and says without an example that it need not be. **This is not formalized.** The retelling gives a counterexample built from a phantom map. The map goes from the telescope of $S^1 \xrightarrow{z^2} S^1 \xrightarrow{z^2} \cdots$ to $\mathbb{CP}^\infty$; it is null-homotopic on every finite stage but detected by the second cohomology of the telescope. The example needs cellular cohomology and $\mathbb{CP}^\infty$, which are not available. What is formalized is the positive statement, that φ_Σ is a homotopy equivalence.

XI.11 Case 2, and Theorem A with closed Y_i

The heart of Case 2 is a statement about self-maps of a single telescope, which the formal text isolates.

Theorem (self-maps that preserve the truncations). Let $X_0 \subseteq X_1 \subseteq \cdots$ be any increasing sequence in X, closed or not. Let $F : X_\Sigma \to X_\Sigma$ carry each truncation X_Σ^m into itself, and suppose that $F(x,m) = (e_m(x), m)$ for maps $e_m : X_m \to X_m$ homotopic to the identity. Then F is a homotopy equivalence. `Appendix/TheoremA · isHomotopyEquiv_of_truncation`

Proof as formalized. Apply the reduction of §XI.3 with the space X_Σ in place of X, its sequence of truncations $\{X_\Sigma^m\}$ on both sides, and the self-map F. Both sides are homotopy direct limits by Lemma 2.2. So it suffices that the induced map F_Σ of the telescope of the truncations, a subspace of $X_\Sigma \times \mathbb{R}$, is a homotopy equivalence. That is Case 1 for the space X_Σ, the closed sets X_Σ^m (§XI.5) and the self-map F. Case 1 asks that each restriction $F_m : X_\Sigma^m \to X_\Sigma^m$ be homotopic to the identity. With $\iota = \iota_m$ and $r = r_m$ the maps of (T2),

$$F_m \simeq F_m\, \iota\, r = \iota\, e_m\, r \simeq \iota\, r \simeq \mathrm{id}.$$

The first homotopy is $\iota r \simeq \mathrm{id}$ (T2); the equality is the hypothesis $F(x,m) = (e_m x, m)$; the second homotopy is $e_m \simeq \mathrm{id}$; the last is (T2) again. □

This is the step that Milnor compresses into one sentence (p. 152). He says there that X_Σ^n deformation retracts onto $X_n \times \{n\}$, that Gf_Σ restricted there is $g_n f_n$ and hence homotopic to the identity, and that Case 1 therefore applies to the sequence $\{X_\Sigma^n\}$. Two things are left implicit. First, Case 1 concludes that the induced map of the telescope of $\{X_\Sigma^n\}$ is a homotopy equivalence; to come back to Gf_Σ itself one needs X_Σ to be the homotopy direct limit of its truncations, which is Lemma 2.2. Second, Case 1 needs closed sets, and the truncations are closed whatever the X_n are.

Milnor's map G. Now let $f : X \to Y$ carry each X_n into Y_n by a homotopy equivalence f_n, and choose homotopy inverses $g_n : Y_n \to X_n$. Let $i_n : X_n \to X_{n+1}$ and $j_n : Y_n \to Y_{n+1}$ be the inclusions.

The squares commute up to homotopy `Appendix/TheoremA · homotopic_square`:

$$i_n g_n \simeq g_{n+1} f_{n+1}\, i_n g_n = g_{n+1}\, j_n f_n\, g_n \simeq g_{n+1} j_n.$$

The steps use $g_{n+1}f_{n+1} \simeq \mathrm{id}$, then $f_{n+1}i_n = j_n f_n$ (both are restrictions of f), then $f_n g_n \simeq \mathrm{id}$. This is Milnor's computation (p. 152). Choose homotopies $k^n_u : Y_n \to X_{n+1}$ with $k^n_0 = i_n g_n$ and $k^n_1 = g_{n+1}j_n$; these are Milnor's h^n_u, renamed to avoid a clash with Case 1. Define on the strips of Y_Σ

$$G(y, n+t) = \begin{cases} (g_n(y),\ n+2t) & 0 \le t \le \tfrac{1}{2}, \\ (k^n_{2t-1}(y),\ n+1) & \tfrac{1}{2} \le t \le 1. \end{cases}$$

`Appendix/TheoremA · case2InvStrip` `Appendix/TheoremA · case2Inv` `Appendix/TheoremA · case2Inv_stripMap`

At $t = \frac{1}{2}$ both lines give $(g_n y, n+1)$, as $k^n_0 = i_n g_n$. At the top of the strip the second line gives $(g_{n+1}y, n+1)$, as $k^n_1 = g_{n+1}j_n$, and this is the bottom of the next strip. G is pasted from the strips of Y_Σ, so it needs the Y_n closed. It has two properties `Appendix/TheoremA · case2Inv_level` `Appendix/TheoremA · case2Inv_zero`: it carries the n-th strip of Y_Σ to levels $\le n+1$, and $G(y, n) = (g_n(y), n)$. From here on the formal proof forgets the formula for G and uses only these two properties.

Theorem (Case 2: f_Σ is a homotopy equivalence). Let $X_0 \subseteq X_1 \subseteq \cdots$ in X and $Y_0 \subseteq Y_1 \subseteq \cdots$ in Y, with the Y_n closed. Let $f : X \to Y$ carry each X_n into Y_n by a homotopy equivalence. Then $f_\Sigma : X_\Sigma \to Y_\Sigma$ is a homotopy equivalence. `Appendix/TheoremA · theoremA_telescope`

Deviation. This is narrower than Milnor's Case 2 by the closedness of the Y_n. It is stronger in two ways: there is no homotopy direct limit hypothesis, and no closedness of the X_n.

Proof as formalized.

1. *Gf_Σ is a homotopy equivalence.* f_Σ carries the n-th strip of X_Σ to the n-th strip of Y_Σ, and G carries that to levels $\le n+1$. The bottom point (x, n) goes to $G(f(x), n) = (g_n f_n x, n)$, at level n. By the lemma on levels (§XI.5), Gf_Σ carries each X^m_Σ into itself `Appendix/Telescope · level_le_of_stripMap`. Moreover $Gf_\Sigma(x, m) = (g_m f_m(x), m)$ with $g_m f_m \simeq \mathrm{id}$. The theorem above applies, with $e_m = g_m f_m$.
2. *$f_\Sigma G$ is a homotopy equivalence*, in the same way on Y_Σ. G carries strip n to levels $\le n+1$, and f_Σ preserves levels. Moreover $f_\Sigma G(y, m) = (f_m g_m(y), m)$ with $f_m g_m \simeq \mathrm{id}$.
3. Let k_1 be a homotopy inverse of Gf_Σ. Then $(k_1 G)f_\Sigma \simeq \mathrm{id}$, so $k_1 G$ is a left homotopy inverse of f_Σ. Let k_2 be a homotopy inverse of $f_\Sigma G$. Then $f_\Sigma(Gk_2) \simeq \mathrm{id}$, so Gk_2 is a right homotopy inverse. By Milnor's Assertion (§XI.1), f_Σ is a homotopy equivalence `Appendix/Telescope · IsHomotopyEquiv.of_leftInv_rightInv`. □

Theorem A, by Milnor's proof. Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and Y that of $Y_0 \subseteq Y_1 \subseteq \cdots$, with the Y_i closed in Y. Let $f : X \to Y$ carry each X_i into Y_i by a homotopy equivalence. Then f is a homotopy equivalence. `Appendix/TheoremA · theoremA`

Proof. Case 2 and the reduction of §XI.3. □

Why only the Y_i need to be closed. Follow the pasting. In Case 2 the only map pasted from the strips of a telescope of the original sets is G, on Y_Σ. Case 1 is used only for the truncations of X_Σ and of Y_Σ, which are closed in any case. f_Σ is not pasted at all, and (T2), Lemma 2.2 and the lemma on levels need nothing. So the one closedness hypothesis of Milnor's proof concerns the target sets. In his application they are subcomplexes K_i of a CW-complex, hence closed. In Theorem 17.1, however, Part IX applies Theorem A to open subsets of path spaces, and there the hypothesis has to go.

XI.12 The mapping telescope

Two things make the formal text replace Milnor's telescope by a mapping telescope. The first is closedness. In X_Σ a map given strip by strip is known to be continuous only when the strips are closed (§XI.4), while the mapping telescope carries the topology in which every such map is continuous. The second is that the complexes K_i of Theorem 17.3 and of the Corollary are not subspaces of one space in this project. Attaching a cell is a pushout in the category of topological spaces `Chapter2/Attaching · AttachCell`. It comes

with a map $K \to K \cup e^\lambda$ Chapter2/Attaching · cellInl rather than an inclusion of subsets, so a sequence of complexes is a sequence of spaces and maps.

Sequences. A sequence $K_0 \xrightarrow{c_0} K_1 \xrightarrow{c_1} K_2 \to \cdots$ is a functor from $\mathbb{N}$ to topological spaces. For $m \le n$ write $c_{m,n} : K_m \to K_n$ for the composite Appendix/MappingTelescope · tmap, so that $c_{n,n} = \mathrm{id}$ and $c_{m,n}c_{l,m} = c_{l,n}$ Appendix/MappingTelescope · tmap_refl Appendix/MappingTelescope · tmap_tmap. Thus $c_n = c_{n,n+1}$.

Definition (the mapping telescope). $\mathrm{Tel}(K)$ is the disjoint union of the blocks $K_n \times I$, each with the product topology, divided by the relation $(k,1)_n \sim (c_n k, 0)_{n+1}$, with the quotient topology Appendix/MappingTelescope · MapTelRel Appendix/MappingTelescope · MappingTelescope.

Write $[k,t]_n$ for the class of (k,t) in the n-th block Appendix/MappingTelescope · MappingTelescope.mk. So $[k,1]_n = [c_n k, 0]_{n+1}$ Appendix/MappingTelescope · MappingTelescope.mk_one, every point has this form Appendix/MappingTelescope · MappingTelescope.exists_mk, and $(k,t) \mapsto [k,t]_n$ is continuous Appendix/MappingTelescope · MappingTelescope.isQuotientMap_quotMk Appendix/MappingTelescope · MappingTelescope.continuous_mk. The mapping telescope is not in the book; the retelling uses it to prove the Corollary.

Maps out of the mapping telescope, block by block. Let $\Phi_n : K_n \times I \to Z$ be continuous with $\Phi_{n+1}(c_n k, 0) = \Phi_n(k,1)$. Then $[k,t]_n \mapsto \Phi_n(k,t)$ is a well-defined continuous map $\mathrm{Tel}(K) \to Z$ Appendix/MappingTelescope · MappingTelescope.desc Appendix/MappingTelescope · MappingTelescope.desc_mk. The Φ_n together are continuous on the disjoint union Mathlib/Topology/Constructions · continuous_sigma and respect the relation, so they descend to the quotient Mathlib/Topology/Constructions · continuous_quot_lift. There is no hypothesis: this is (T3) without closedness.

With a parameter. Let P be a locally compact space and $\Phi_n : P \times K_n \times I \to Z$ continuous and compatible in the same way. Then $(q,[k,t]_n) \mapsto \Phi_n(q,k,t)$ is continuous on $P \times \mathrm{Tel}(K)$ Appendix/MappingTelescope · MappingTelescope.lift Appendix/MappingTelescope · MappingTelescope.lift_mk. The formal proof goes through the exponential law. Each Φ_n curries to a continuous map $K_n \times I \to C(P,Z)$, the space of maps with the compact-open topology Mathlib/Topology/CompactOpen · ContinuousMap.curry. These are compatible, so they descend to a continuous map $\mathrm{Tel}(K) \to C(P,Z)$. Uncurrying it gives a continuous map on $\mathrm{Tel}(K) \times P$, because P is locally compact Mathlib/Topology/CompactOpen · ContinuousMap.uncurry. This is J. H. C. Whitehead's lemma that the product of a quotient map with a locally compact space is a quotient map, in the form the formal proof uses. With $P = I$ it gives homotopies, block by block Appendix/MappingTelescope · MappingTelescope.homotopy. Two maps that agree on every block are equal Appendix/MappingTelescope · MappingTelescope.ext.

Levels and truncations. The level $\ell[k,t]_n = n + t$ is well defined, as $(n+1) + 0 = n + 1$; it is continuous and ≥ 0 Appendix/MappingTelescope · MappingTelescope.level Appendix/MappingTelescope · MappingTelescope.level_mk Appendix/MappingTelescope · MappingTelescope.level_nonneg. The truncations $\mathrm{Tel}^n(K) = \{\ell \le n\}$ are closed and increasing Appendix/MappingTelescope · MappingTelescope.trunc Appendix/MappingTelescope · MappingTelescope.isClosed_trunc Appendix/MappingTelescope · MappingTelescope.monotone_trunc. $\mathrm{Tel}(K)$ is the homotopy direct limit of $\{\mathrm{Tel}^n(K)\}$, by Lemma 2.1 with $f = \ell + 1$ Appendix/MappingTelescope · MappingTelescope.isHomotopyDirectLimit_trunc.

Theorem ((T2) for mapping telescopes). The map $\kappa_n : K_n \to \mathrm{Tel}^n(K)$, $\kappa_n(k) = [k,0]_n$, is a homotopy equivalence. Appendix/MappingTelescope · MappingTelescope.mk_zero_mem_trunc Appendix/MappingTelescope · MappingTelescope.levelIncl Appendix/MappingTelescope · MappingTelescope.isHomotopyEquiv_levelIncl

In X_Σ this was a one-line deformation, because every point has a coordinate x that does not change as the level rises. A mapping telescope has no such coordinate. A point $[k,t]_m$ of $\mathrm{Tel}^n(K)$ must be pushed up through the blocks $m, m+1, \dots, n$, and its first coordinate changes on the way, from k to $c_m k$, to $c_{m+1}c_m k$, and so on. The formal proof takes four steps.

1. Trajectories. For $k \in K_m$ and a height $s \in [0, d+1]$, the point at height s above $[k,0]_m$ along the maps

c is

$$\mathrm{traj}_d(m,k,s) = \begin{cases} [k,s]_m & s \le 1, \\ \mathrm{traj}_{d-1}(m+1, c_m k, s-1) & s > 1, \end{cases}$$

with $\mathrm{traj}_0(m,k,s) = [k, \mathrm{clamp}(s)]_m$ Appendix/MappingTelescope · MappingTelescope.traj Appendix/MappingTelescope · MappingTelescope.traj_of_le_one Appendix/MappingTelescope · MappingTelescope.traj_zero_right. It is continuous in (k,s), by induction on d; the two cases agree at $s=1$ because $[k,1]_m = [c_m k, 0]_{m+1}$ Appendix/MappingTelescope · MappingTelescope.continuous_traj. Its level is $m+s$ for $0 \le s \le d+1$ Appendix/MappingTelescope · MappingTelescope.level_traj.

2. The push. On the block m put

$$\mathrm{push}_n(s, [k,t]_m) = \mathrm{traj}_d\,(m, k,\ t + s\max(0,\ n-m-t)), \qquad d = \max(n-m, 0)$$

Appendix/MappingTelescope · MappingTelescope.pushFun. For a point at level $m+t \le n$, the height rises linearly from t at $s=0$ to $n-m$ at $s=1$, that is, to level n. A point at level $\ge n$ does not move. Each block formula is continuous Appendix/MappingTelescope · MappingTelescope.continuous_pushFun, and the formulas agree at the gluing Appendix/MappingTelescope · MappingTelescope.pushFun_compat. For $m < n$: the trajectory of k from height 1 in block m is the trajectory of $c_m k$ from height 0 in block $m+1$. If $s\max(0, n-m-1) = 0$, both are the point $[k,1]_m = [c_m k, 0]_{m+1}$; otherwise the recursion steps into the next block. For $m \ge n$, both sides are the point itself. By the parameter version of block-by-block maps, $\mathrm{push}_n : I \times \mathrm{Tel}(K) \to \mathrm{Tel}(K)$ is continuous Appendix/MappingTelescope · MappingTelescope.push. It starts at the identity Appendix/MappingTelescope · MappingTelescope.push_zero and fixes each $[k,0]_n$ Appendix/MappingTelescope · MappingTelescope.push_bottom. On $\mathrm{Tel}^n(K)$ it moves levels by $\ell(\mathrm{push}_n(s,z)) = \ell(z) + s(n-\ell(z))$ Appendix/MappingTelescope · MappingTelescope.level_push. So it keeps $\mathrm{Tel}^n(K)$ in itself Appendix/MappingTelescope · MappingTelescope.push_mem_trunc and ends at level n Appendix/MappingTelescope · MappingTelescope.level_push_one.

3. Reading off the point at level n. Every point at level exactly n is $[k,0]_n$ for some k Appendix/MappingTelescope · MappingTelescope.exists_mk_zero_of_level. Indeed, if it is $[k,t]_m$ with $m+t=n$, then either $t=1$, and it is $[c_m k, 0]_{m+1}$ with $m+1=n$; or $t<1$, and then $m=n$ and $t=0$. The k is unique. Consider the function that sends $[k,t]_m$ to $c_{m,n}(k)$ when $m+t=n$, and to nothing otherwise. It respects the gluing, because $c_{m+1,n}c_m = c_{m,n}$, so it is defined on the mapping telescope Appendix/MappingTelescope · MappingTelescope.atLevel. At $[k,0]_n$ it returns k Appendix/MappingTelescope · MappingTelescope.atLevel_mk_zero, so κ_n is injective Appendix/MappingTelescope · MappingTelescope.mk_zero_injective.

4. κ_n is an embedding Appendix/MappingTelescope · MappingTelescope.isInducing_mk_zero: every open $V \subseteq K_n$ is $\kappa_n^{-1}(W)$ for an open $W \subseteq \mathrm{Tel}(K)$. Let S be the set of points (k',t) of the blocks $m \le n$ with $|m+t-n| < \frac{1}{2}$ and $c_{m,n}(k') \in V$. In each block S is open: it is an open condition on t intersected with the preimage of V under the continuous $c_{m,n}$, and it is empty for $m > n$. It is saturated for the gluing. A top point $(k,1)_m$ within $\frac{1}{2}$ of level n has $m+1=n$, and then it lies in S exactly when $(c_m k, 0)_{m+1}$ does, as $c_{m+1,n}c_m = c_{m,n}$. So S is the preimage of an open set W of the quotient. At level exactly n the condition defining S reads $k \in V$, so $\kappa_n^{-1}(W) = V$.

Proof of the theorem. Using step 3, define $q_n : \mathrm{Tel}^n(K) \to K_n$ by $[q_n(z), 0]_n = \mathrm{push}_n(1, z)$ Appendix/MappingTelescope · MappingTelescope.push_one_eq Appendix/MappingTelescope · MappingTelescope.levelRetract Appendix/MappingTelescope · MappingTelescope.mk_levelRetract. It is continuous because $\kappa_n q_n = \mathrm{push}_n(1,\cdot)$ is continuous and κ_n is an embedding Mathlib/Topology/Maps/Basic · IsInducing.continuous_iff. Next, $q_n\kappa_n = \mathrm{id}$, by injectivity and because push_n fixes $[k,0]_n$ Appendix/MappingTelescope · MappingTelescope.levelRetract_levelIncl Appendix/MappingTelescope · MappingTelescope.levelRetract_comp_levelIncl. Finally, $s \mapsto \mathrm{push}_n(s,\cdot)$ is a homotopy in $\mathrm{Tel}^n(K)$ from the identity to $\kappa_n q_n$ Appendix/MappingTelescope · MappingTelescope.levelIncl_comp_levelRetract. So κ_n and q_n are homotopy equivalences Appendix/MappingTelescope · MappingTelescope.isHomotopyEquiv_levelRetract. □

In X_Σ the retraction r_n was simply $(x,\tau) \mapsto x$. Here the K_n are not subspaces of a common space, so the retraction has to be read off at level n, and the embedding is what makes it continuous. Part XII

uses the same push to compute the homotopy groups of mapping telescopes Homotopy/Telescope · bijective_homotopyGroupMap_telescopeIncl.

XI.13 Theorem A with no closedness

The mapping telescope of the inclusions. For an increasing sequence $X_0 \subseteq X_1 \subseteq \cdots$, the inclusions form a sequence of spaces and maps Appendix/MonotoneUnion · subspaceFunctor. Write T_X for its mapping telescope and T_X^m for its truncations, with the maps κ_m^X and q_m^X of §XI.12. The comparison map

$$c_X : T_X \to X_\Sigma, \qquad [x,t]_n \mapsto (x, n+t),$$

is continuous block by block, the gluing being respected because $\sigma_n(x,1) = \sigma_{n+1}(x,0)$ Appendix/MonotoneUnion · telescopeComparison Appendix/MonotoneUnion · telescopeComparison_mk. It is a bijection, an elementary check that the formal text never needs. Through it, T_X is the set X_Σ with the finest topology in which every strip $X_n \times I \to X_\Sigma$ is continuous.

Theorem. For closed X_n, c_X is a homeomorphism. Appendix/MonotoneUnion · telescopeComparisonHomeomorph

Proof. The inverse $(x, n+t) \mapsto [x,t]_n$ is given strip by strip, compatibly because $[x,1]_n = [x,0]_{n+1}$, and it is continuous by (T3). The two composites are the identity, as one checks on blocks and on strips. □

So for closed sets the mapping telescope of the inclusions *is* Milnor's telescope, and (T3) says exactly that the subspace topology of X_Σ is then the colimit topology of the strips. For arbitrary sets the two topologies may differ, but not up to homotopy.

Theorem. For any increasing sequence, c_X is a homotopy equivalence. Appendix/MonotoneUnion · isHomotopyEquiv_telescopeComparison

Proof. Apply Theorem A of §XI.11, the version with closed targets. The source is the space T_X with its truncations T_X^m. The target is the space X_Σ with its truncations X_Σ^m, which are closed. The map is c_X. Both spaces are homotopy direct limits of their truncations (§XI.12 and Lemma 2.2), and c_X carries T_X^m into X_Σ^m because it preserves levels. It remains to see that each restriction $R_m : T_X^m \to X_\Sigma^m$ is a homotopy equivalence. Now $R_m \kappa_m^X = \iota_m$, as both send x to (x,m), so

$$R_m \simeq R_m\, \kappa_m^X\, q_m^X = \iota_m\, q_m^X,$$

using $\kappa_m^X q_m^X \simeq \mathrm{id}$. And $\iota_m q_m^X$ is a composite of homotopy equivalences, by (T2) for both telescopes. □

Consequently, if X is the homotopy direct limit of $\{X_n\}$, then $\pi_X = p\, c_X : T_X \to X$, $[x,t]_n \mapsto x$, is a homotopy equivalence Appendix/MonotoneUnion · telescopeProj' Appendix/MonotoneUnion · isHomotopyEquiv_telescopeProj'.

The induced map. Let f carry each X_n into Y_n. Then $\Phi_f[x,t]_n = [f(x),t]_n$ is a continuous map $T_X \to T_Y$, built block by block with no hypothesis Appendix/MonotoneUnion · mappingTelescopeMap Appendix/MonotoneUnion · mappingTelescopeMap_mk. It preserves levels, hence the truncations Appendix/MonotoneUnion · mapsTo_mappingTelescopeMap.

Theorem A, as Milnor states it. Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$ and Y that of $Y_0 \subseteq Y_1 \subseteq \cdots$, arbitrary subsets. Let $f : X \to Y$ carry each X_i into Y_i by a homotopy equivalence. Then f is a homotopy equivalence. Appendix/MonotoneUnion · theoremA_general

Proof as formalized.

1. *Φ_f is a homotopy equivalence.* Apply Theorem A with closed targets to T_X with its truncations T_X^m, to T_Y with its truncations T_Y^m, which are closed, and to Φ_f. Both spaces are homotopy direct limits of their truncations. The restriction $R_m : T_X^m \to T_Y^m$ satisfies $R_m \kappa_m^X = \kappa_m^Y f_m$, so

$$R_m \simeq R_m\, \kappa_m^X\, q_m^X = \kappa_m^Y\, f_m\, q_m^X,$$

a composite of three homotopy equivalences.
2. *Commutation.* $f\,\pi_X = \pi_Y\,\Phi_f$, since both send $[x,t]_n$ to $f(x)$; this is checked block by block.
3. *Conclusion*, as in §XI.3. Let s be a homotopy inverse of π_X. Then $\pi_Y \Phi_f s$ is a composite of homotopy equivalences, and it equals $f\pi_X s \simeq f$. □

Where the closedness went. Milnor's proof is run with the truncations $T_X^m \subseteq T_X$ and $T_Y^m \subseteq T_Y$ in the roles of his $X_i \subseteq X$ and $Y_i \subseteq Y$. Inside it, Case 2 pastes G on the telescope of the closed sets T_Y^m, and Case 1 works with truncations of telescopes; nothing else is pasted. The closedness that Milnor's proof needs is thus a property of the subspace topology on X_Σ, not of the theorem.

Where it is used. In Part IX, Theorem 17.1 says that the inclusion of the piecewise smooth paths Ω into all continuous paths Ω^* is a homotopy equivalence. It is proved by Theorem A for open sets: the sets $Y_k \subseteq \Omega^*$ of paths whose dyadic pieces of level k lie in good balls, and their preimages $X_k \subseteq \Omega$. Ω^* is metric, hence paracompact, so it is the homotopy direct limit of the Y_k by Example 1. In the formal model Ω is not Hausdorff (its paths are functions on $\mathbb{R}$, and the metric sees only $[0,1]$), so for Ω Example 1 is pulled back along the inclusion instead `Chapter10/FullPathSpace · isHomotopyDirectLimit_preimage`. The X_k and Y_k are open, not closed, so the version without closedness is the one applied `Chapter10/FullPathTheorem · JoinSystem.isHomotopyEquiv_toPathC Chapter10/FullPathTheorem · theorem_17_1`.

XI.14 Monotone unions and the Corollary

Theorem (subspace form). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$. Let $Y_0 \subseteq Y_1 \subseteq \cdots$ be closed in Y, and let $f : X \to Y$ carry each X_i into Y_i by a homotopy equivalence. Then X is homotopy equivalent to Y_Σ. `Appendix/MonotoneUnion · nonempty_homotopyEquiv_telescope`

Proof. $p : X_\Sigma \to X$ is a homotopy equivalence by hypothesis, and so is $f_\Sigma : X_\Sigma \to Y_\Sigma$ by Case 2 (§XI.11). Compose an inverse of the first with the second. □

Here Y need not be a homotopy direct limit, since the conclusion names Y_Σ, not Y. By (T3) the topology of Y_Σ is the colimit topology of the strips (§XI.13), so it depends only on the topologies of the Y_i. If Y is a CW-complex and the Y_i are subcomplexes, then $Y_\Sigma \simeq Y$ is Example 2, which is not formalized. That is why the telescope, not Y, appears in the conclusion.

Lemma (squares of inverses). Suppose $ja \simeq a'i$, and b, b' satisfy $ab \simeq \mathrm{id}$ and $b'a' \simeq \mathrm{id}$. Then $ib \simeq b'j$, because

$$ib \simeq b'a'\,ib \simeq b'\,j\,ab \simeq b'j.$$

`Appendix/MonotoneUnion · homotopic_inverse_square`

This is the computation of Milnor's square in Case 2, made abstract.

Theorem (mapping telescope form). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$, closed or not. Let $K_0 \xrightarrow{c_0} K_1 \xrightarrow{c_1} \cdots$ be any sequence of spaces and maps, and let $g_n : X_n \to K_n$ be homotopy equivalences with $c_n g_n \simeq g_{n+1} i_n$ on X_n. Then X is homotopy equivalent to $\mathrm{Tel}(K)$. `Appendix/MonotoneUnion · nonempty_homotopyEquiv_mappingTelescope`

Deviation (stronger). In Milnor's application the ladder commutes strictly, and his K_i are subcomplexes of one complex. Here the squares need only commute up to homotopy, and the K_n are arbitrary spaces.

Proof as formalized.

1. Choose homotopy inverses g_n' of g_n. By the lemma on squares, $i_n g_n' \simeq g_{n+1}' c_n : K_n \to X_{n+1}$; choose homotopies k_u^n from $i_n g_n'$ to $g_{n+1}' c_n$.
2. Define $\Psi : \mathrm{Tel}(K) \to X_\Sigma$ block by block, as Milnor defined G:

$$\Psi[k,t]_n = \begin{cases} (g_n'(k),\ n+2t) & 0 \le t \le \tfrac{1}{2}, \\ (k_{2t-1}^n(k),\ n+1) & \tfrac{1}{2} \le t \le 1. \end{cases}$$

Appendix/MonotoneUnion · compareStrip Appendix/MonotoneUnion · compareMap Appendix/MonotoneUnion · compareMap_mk The checks at $t = \frac{1}{2}$ and at $t = 1$ are those for G, with c_n in place of j_n. Since Ψ is defined out of a mapping telescope, it is continuous with no closedness at all.

3. $\Psi[k,0]_n = (g'_n(k), n)$ Appendix/MonotoneUnion · compareMap_mk_zero, and Ψ carries $\mathrm{Tel}^m(K)$ into X^m_Σ Appendix/MonotoneUnion · compareMap_mapsTo. A point $[k,t]_n$ of $\mathrm{Tel}^m(K)$ with $t = 0$ goes to level $n \le m$. If $t > 0$, then $n < m$, so $n+1 \le m$, and Ψ lands at a level $\le n+1$.
4. Ψ is a homotopy equivalence Appendix/MonotoneUnion · isHomotopyEquiv_compareMap. This is Theorem A with closed targets, for $\mathrm{Tel}(K)$ with its truncations and X_Σ with its closed truncations; both spaces are homotopy direct limits of their truncations. The restriction $R_m : \mathrm{Tel}^m(K) \to X^m_\Sigma$ satisfies $R_m \kappa_m = \iota_m g'_m$, so $R_m \simeq \iota_m g'_m q_m$, a composite of homotopy equivalences.
5. So $X \simeq X_\Sigma \simeq \mathrm{Tel}(K)$, the first by hypothesis. □

The direction of Ψ is chosen so that it starts from the mapping telescope, where pasting is free. The closedness that G needed in Case 2 is then never needed. In particular no closedness of the X_n is assumed, here or in the Corollary.

The Corollary (Milnor p. 153). If X is the homotopy direct limit of $\{X_i\}$ and each X_i has the homotopy type of a CW-complex, then so does X. Milnor gives no proof, remarking only that it is not hard. The formal statements replace the CW-complexes in two ways.

Theorem (the Corollary, in the telescope model). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$, and let $e_n : X_n \to K_n$ be homotopy equivalences, with homotopy inverses e'_n. With the maps $c_n = e_{n+1}\, i_n\, e'_n : K_n \to K_{n+1}$, X is homotopy equivalent to the mapping telescope of $K_0 \to K_1 \to \cdots$. Appendix/MonotoneUnion · corollary_appendix Mathlib/CategoryTheory/Functor/OfSequence · Functor.ofSequence

Proof. $c_n e_n = e_{n+1} i_n e'_n e_n \simeq e_{n+1} i_n$, as $e'_n e_n \simeq \mathrm{id}$. The mapping telescope form applies with $g_n = e_n$. □

Theorem (the Corollary for finite cell complexes). If each X_n has the homotopy type of a finite iterated adjunction space with cells cells_n (§0.5) Chapter2/CWType · IsFiniteCellComplexType Chapter2/CWType · IsCellAttachment, then X is homotopy equivalent to the mapping telescope of a sequence of such spaces. Appendix/MonotoneUnion · corollary_cellComplex

Proof. Choose the spaces and the homotopy equivalences, and apply the previous theorem. □

What is not proved is that the mapping telescope is itself a CW-complex. That needs cellular approximation, to replace the c_n by cellular maps, and the CW structure of the mapping telescope of cellular maps. Neither is in Mathlib, and the retelling's proof of the Corollary uses both. The formal Corollary therefore ends with a mapping telescope.

XI.15 Cell towers and the topological part of Theorem 17.3

The countable CW-complexes of Theorem 17.3 are built by attaching cells, finitely many at each stage. The formal model keeps the stages and the maps between them, and takes their mapping telescope for the union.

Definition (relative cell attachment). K is obtained from L by attaching cells of dimensions $(\lambda_1, \ldots, \lambda_k)$, with map $j : L \to K$, if it is obtained by attaching a λ_1-cell to L, then a λ_2-cell to the result, and so on, each along an arbitrary continuous map of its boundary sphere into the space built so far. The map j is the composite of the maps $K' \to K' \cup e^\lambda$ Appendix/MonotoneUnion · IsRelCellAttachment.

If L is a finite iterated adjunction space with cells c, and K is obtained from L by attaching cells c', then K is a finite iterated adjunction space with cells c followed by c' Appendix/MonotoneUnion · IsCellAttachment.of_rel. The proof is by induction on the attachments. With no cells there is nothing to do. With one more cell, attach it to the space that has cells c followed by the earlier part of c'.

Definition (cell towers). A *cell tower* with cells $\text{cells}_0, \text{cells}_1, \ldots$ (finite lists of dimensions) is a sequence $K_0 \to K_1 \to \cdots$ with two properties. K_0 is a finite iterated adjunction space with cells cells_0. Each K_{n+1} is obtained from K_n by attaching the cells cells_{n+1}, and the map $K_n \to K_{n+1}$ is the attachment map `Appendix/MonotoneUnion · CellTower`.

Each stage K_n is then a finite iterated adjunction space with the cells $\text{cells}_0, \ldots, \text{cells}_n$, in this order `Appendix/MonotoneUnion · CellTower.isCellAttachment`, by induction on n with the previous lemma. A tower can be given by its stages and maps, which is how an inductive construction produces it `Appendix/MonotoneUnion · CellTower.ofSequence` `Appendix/MonotoneUnion · CellTower.tmap_ofSequence`.

Definition (§0.5). *X has the homotopy type of a countable cell complex* with cells $\text{cells}_0, \text{cells}_1, \ldots$ if it is homotopy equivalent to the mapping telescope of some cell tower with these cells `Appendix/MonotoneUnion · IsCountableCellComplexType`. The property passes to homotopy equivalent spaces `Appendix/MonotoneUnion · IsCountableCellComplexType.of_homotopyEquiv`.

Deviation (narrower). This is weaker than having the homotopy type of a CW-complex in two ways. The attaching maps need not be cellular. And the mapping telescope of the tower stands for the union of its stages with the weak topology, to which it is homotopy equivalent by Example 2, which is not formalized.

Theorem (towers). Let X be the homotopy direct limit of $X_0 \subseteq X_1 \subseteq \cdots$. Let $g_n : X_n \to K_n$ be homotopy equivalences to the stages of a cell tower with $c_n g_n \simeq g_{n+1} i_n$. Then X has the homotopy type of the countable cell complex. `Appendix/MonotoneUnion · isCountableCellComplexType_of_tower`

Proof. The mapping telescope form of §XI.14. □

Theorem 17.3, the topological part (Milnor p. 150). Let $E : X \to \mathbb{R}$ be continuous and $a_0 \le a_1 \le \cdots$ with $a_n \to \infty$. Let $g_n : X^{a_n} \to K_n$ be homotopy equivalences from the sublevel sets $X^{a_n} = \{E \le a_n\}$ to the stages of a cell tower, each extending the previous one: $c_n g_n = g_{n+1}$ on X^{a_n}. Then X has the homotopy type of the countable cell complex with cells $\text{cells}_0, \text{cells}_1, \ldots$. `Appendix/MonotoneUnion · isCountableCellComplexType_of_sublevel`

Proof. The sublevel sets increase, and X is their homotopy direct limit by §XI.7, with no paracompactness. Equal maps are homotopic, so the squares commute up to homotopy. The previous theorem applies. □

The variant with the tower given by its stages and maps is the form that an inductive construction produces `Appendix/MonotoneUnion · isCountableCellComplexType_of_sublevel'`.

Comparison with Milnor's alternative step (p. 150). Milnor has homotopy equivalences $\Omega^{a_i} \to K_i$ that form a strictly commutative ladder with the inclusions. He takes the union map $\Omega \to K = \bigcup K_i$ and applies Theorem A with Examples 1 and 2. The formal proof differs at three points.

- The hypotheses Milnor leaves implicit are stated: $a_i \to \infty$, so that the sublevel sets exhaust X; E continuous; and each map extending the previous one, without which no union map exists.
- No union map is formed, since $K = \bigcup K_i$ with its weak topology is not available. The homotopy equivalence $X \simeq \text{Tel}(K)$ is built in the other direction, from homotopy inverses, by Milnor's map G of Case 2.
- Example 1 is replaced by its sublevel form of §XI.7.

Where it is used. In Part IX, Theorem 17.3 is proved with towers that commute only up to homotopy, through the theorem on towers above `Chapter10/RelativeCells · isCountableCellComplexType_of_rel` `Chapter10/RelativeCells · isCountableCellComplexType_of_relSublevel` `Chapter10/FundamentalTheorem · theorem_17_3`. So even the strict extension property is not needed there.

XI.16 What the formalization taught us

- **Pasting needs closed strips.** Every map Milnor defines on a telescope is given strip by strip, and it is continuous when the X_n are closed (T3) `Appendix/Telescope · continuous_of_stripMap`. Case 1 as

written needs them `Appendix/TheoremA · theoremA_case1`, and otherwise his g can be discontinuous.

- **Milnor's proof of Theorem A needs only the Y_i closed.** Case 2 applies Case 1 only to truncations of telescopes, which are always closed `Appendix/Telescope · isClosed_truncation`, and it pastes only G on a telescope of the original sets `Appendix/TheoremA · theoremA_telescope`.
- **Theorem A is true as Milnor states it.** Run on mapping telescopes, where pasting is automatic, his proof needs no closedness at all `Appendix/MonotoneUnion · theoremA_general`. Theorem 17.1 needs exactly this, for open pieces `Chapter10/FullPathTheorem · JoinSystem.isHomotopyEquiv_toPathC`.
- **A silent lemma.** Case 2 returns from the telescope of the truncations to X_Σ through Lemma 2.2, which holds with no hypothesis at all `Appendix/Telescope · isHomotopyDirectLimit_truncation`.
- **A possible misprint.** The third line of h_u is $h^{n+1}_{(4t-3)u'}$, not $h^{n+1}_{(4-3t)u}$ `Appendix/TheoremA · case1Fun`.
- **Evident steps become formulas.** $h_0 = f_\Sigma R_2$ `Appendix/TheoremA · case1Map_zero`. Milnor's H_u and the analogous K_u are single formulas with $\mu = \min(4t-1, 1-u, 2-2t)$ and $\mu = \max(2-4t, 2t-1, u)$ `Appendix/TheoremA · milnorH` `Appendix/TheoremA · milnorK`. The computation of $h_1 g$ uses that h_1 fixes the integer levels `Appendix/TheoremA · IsMilnorMap.fix`. That Gf_Σ preserves the truncations is a lemma about levels `Appendix/Telescope · level_le_of_stripMap`.
- **Paracompactness is not needed for sublevel sets** `Appendix/Examples · isHomotopyDirectLimit_sublevel`, and Example 1 needs the Hausdorff axiom as well as paracompactness `Appendix/Examples · example1`.
- **Example 3 has a short proof** once one sees that the ray over 0 is open and closed in the telescope `Appendix/Examples · example3`.
- **CW-complexes become mapping telescopes.** Without Whitehead's theorem, Example 2 is out of reach, so the Corollary and Theorem 17.3 conclude with a mapping telescope of iterated adjunction spaces `Appendix/MonotoneUnion · corollary_cellComplex` `Appendix/MonotoneUnion · isCountableCellComplexType_of_sublevel`. The squares need only commute up to homotopy `Appendix/MonotoneUnion · nonempty_homotopyEquiv_mappingTelescope`.
- **(T2) is harder for mapping telescopes.** Points must be pushed up through the blocks, and the continuity of the retraction rests on $k \mapsto [k,0]_n$ being an embedding `Appendix/MappingTelescope · MappingTelescope.isInducing_mk_zero`.

Part XII. Homotopy groups, cells and fibre bundles

Milnor's Part IV rests on facts of homotopy theory that he quotes rather than proves. Theorem 22.1 turns a lower bound λ_0 on the indices of the non-minimal geodesics into the statement that the space Ω^d of minimal geodesics carries the low homotopy groups of the path space: Ω has the homotopy type of Ω^d with cells of dimension $\geq \lambda_0$ attached, and attaching such cells changes no π_i with $i \leq \lambda_0 - 2$ (p. 118). Bott periodicity (§§23–24) passes between path spaces, loop spaces and the manifold itself, $\pi_i\Omega \cong \pi_{i+1}M$, and compares the homotopy groups of unitary groups, spheres, Stiefel and Grassmann manifolds through the exact homotopy sequences of fibre bundles, with $\pi_i S^n = 0$ for $i < n$ as input. Mathlib defines the homotopy groups $\pi_n(X, x)$ and their group structure, but not the maps that continuous maps induce on them, nor what is built on those: change of basepoint, invariance under homotopy equivalence, cellular approximation, Serre fibrations and their exact homotopy sequence. The vanishing of $\pi_i S^n$ for $i < n$ is on its list of things to do. This part retells the formal proofs of all of these, from the twelve files of `Morse/Homotopy/` that do not concern particular bundles. The bundles themselves (coset maps of unitary groups, Stiefel and Grassmann manifolds) are Part XIII's.

The setting is arbitrary topological spaces, with a Hausdorff hypothesis only where a cell is embedded in an ambient space. The formal route departs from the textbook one (Hatcher, *Algebraic Topology*, Sections 4.1–4.2) in six places, each explained where it occurs.

- **No relative homotopy groups.** Milnor states Theorem 22.1 as $\pi_i(\Omega, \Omega^d) = 0$ for $i < \lambda_0$ and reads off the inclusion from the exact sequence of the pair. Mathlib has no relative homotopy groups. The formal text uses the *compression* form of the vanishing and proves surjectivity and injectivity of the induced maps directly from it (§XII.2). This treats $i = 0$ and $i = 1$, where the terms are not abelian groups or not groups at all, with the same argument as every other degree, at every basepoint.
- **The push-off by smooth approximation.** The key step of cellular approximation, pushing a map of a cube off the centre of a cell of larger dimension (Hatcher, Lemma 4.10), is proved with the Stone–Weierstrass theorem for smooth maps of the companion project and the fact that a C^1 map from a space of smaller dimension misses a dense set (§XII.5). No simplicial approximation is used.
- **No separation axiom on the space to which a cell is attached.** A continuous "probe" on $Y \cup_\varphi e^n$ measures how deep a point lies in the cell and reads off its coordinate (§XII.6), and the radial deformation of the punctured cell onto its boundary is replaced by a radial *expansion* of a small ball, defined on the whole cell (§XII.7).
- **Countable complexes are mapping telescopes**, as in Part XI, and their homotopy groups are computed by compactness (§§XII.9–12.10). The deformation of a truncation of the telescope moves the basepoint, so change of basepoint (§XII.3) is needed even for this.
- **Fibre bundles as a property of a map.** Local triviality is a property of an arbitrary map $p : E \to B$ (§XII.13), so that it applies to maps between spaces built independently. Locally trivial maps are Serre fibrations by Hatcher's subdivision argument, with an explicit order on the boxes and an explicit retraction of a box onto the faces where the lift is already known (§XII.14).

- **Exactness as pointed sets, at every place.** The exact sequence is proved at the fibre, the total space and the base for every finite index set, π_0 included (§XII.16). Exactness of pointed sets gives only trivial kernels, so the injectivity statements used in §23 need two further arguments (§XII.17).

The results are stated for Mathlib's homotopy groups $\pi_N(X, x)$, indexed by an arbitrary finite set N; for $N = \{0, \dots, n-1\}$ they are statements about π_n.

Map.

- `Homotopy/Basic` (169 lines): induced maps; compressions; surjectivity and injectivity from compressions.
- `Homotopy/ChangeOfBase` (498 lines): the frame construction; change of basepoint; free homotopies; homotopy equivalences; homotopy-commutative squares.
- `Homotopy/PushOff` (184 lines): the push-off lemma in the coordinates of a cell.
- `Homotopy/CellAttach` (729 lines): the point-set topology of $Y \cup_\varphi e^n$; the push-off and the radial push there; cells and homotopy groups; other pushout squares and embedded cells.
- `Homotopy/Telescope` (327 lines): exhaustions by compact sets; increasing unions; mapping telescopes; cell towers.
- `Homotopy/RelCells` (177 lines): relative cell complexes, with finitely and countably many cells; the form of Theorem 22.1.
- `Homotopy/LoopSpace` (416 lines): reindexing cubes; $\pi_M(\Omega X) \cong \pi_{M+1}(X)$, as groups; path spaces with different end points.
- `Homotopy/Fibration` (317 lines): locally trivial maps; the open box; Serre fibrations; the absolute lifting property.
- `Homotopy/HomotopyLifting` (694 lines): the retraction onto faces of a cube; the grid; one box; the order of the boxes; locally trivial maps are Serre fibrations.
- `Homotopy/ExactSequence` (587 lines): boundary lifts; the boundary map; exactness at the three places.
- `Homotopy/ExactSequenceIso` (425 lines): homomorphisms; injectivity; bijectivity; two out of three; the forms with π_i.
- `Homotopy/Spheres` (174 lines): contractible spaces; $\pi_i S^n = 0$ for $i < n$.

XII.1 Conventions: cubes, generalized loops and the extra coordinate

Spaces are topological spaces, maps are continuous, and $I = [0, 1]$. Homotopies are Mathlib's, with time first, and a homotopy *relative to* S fixes every point of S at all times (§0.4) `Mathlib/Topology/Homotopy/Basic · ContinuousMap.Homotopy` `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel` `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopicRel`. Such homotopies compose with maps on the left `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel.compContinuousMap` and can be concatenated `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel.trans`, and all their stages agree on S `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel.eq_fst` `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel.fst_eq_snd`.

Cubes. Mathlib indexes cubes by an arbitrary type N. The cube I^N is the space of functions $N \to I$ with the product topology, and its boundary ∂I^N is the set of points with some coordinate equal to 0 or 1 `Mathlib/Topology/Homotopy/HomotopyGroup · Cube.boundary`. For $N = \{0, \dots, n-1\}$ this is the usual cube; for $N = \emptyset$ it is a point with empty boundary. Most statements assume N finite, with $|N|$ elements.

Generalized loops and homotopy groups (§0.7). A generalized loop at $x \in X$ is a continuous $\gamma : I^N \to X$ equal to x on ∂I^N `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop` `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.boundary`. They form a space $\Omega^N(X, x)$, which contains the constant loop `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.const`. Two are equivalent when they are homotopic relative to ∂I^N `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.Homotopic`. The set of classes $[\gamma]$ is $\pi_N(X, x)$ `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup`, and $\pi_n(X, x) = \pi_{\{0,\dots,n-1\}}(X, x)$ `Mathlib/Topology/Homotopy/`

HomotopyGroup · HomotopyGroup.Pi. The class of the constant loop is written 0, or 1 where the group is written multiplicatively. For $N = \emptyset$ a generalized loop is a point of X and a homotopy relative to $\emptyset$ is a path, so $\pi_\emptyset(X, x)$ is the set $\pi_0(X)$ of path components, whatever x is Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.pi0EquivZerothHomotopy. "$\pi_N(X, x)$ is trivial" means that it has at most one element.

The group structure. For nonempty N, $\pi_N(X, x)$ is a group Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.group, abelian when $|N| \geq 2$ Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.commGroup, and the fundamental group when $|N| = 1$ Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.pi1EquivFundamentalGroup. Products are concatenations along a coordinate $i \in N$. The loop $\mathrm{trans}_i(f, g)$ is f on $\{t_i \leq \frac{1}{2}\}$ and g on $\{t_i \geq \frac{1}{2}\}$, each rescaled in the coordinate i Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.transAt, and $\mathrm{symm}_i f$ is f with t_i replaced by $1 - t_i$ Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.symmAt. The class of $\mathrm{trans}_i(f, g)$ does not depend on i Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.transAt_indep. One convention of Mathlib matters below: the product $[p] \cdot [q]$ is the class of $\mathrm{trans}_i(q, p)$, so the *second* factor runs first Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.mul_spec. Further $[p]^{-1} = [\mathrm{symm}_i p]$ Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.inv_spec, and the identity is the class of the constant loop Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.one_def.

The extra coordinate. Write N^+ for N with one new element $\star$ adjoined (formally `Option N`, with $\star$ = `none`). Then $|N^+| = |N| + 1$, and π_{N^+} is the homotopy group one degree above π_N. A point of I^{N^+} is a pair (t, z): its $\star$-coordinate $t \in I$ and the point $z \in I^N$ of its other coordinates Homotopy/LoopSpace · optionPoint Homotopy/ExactSequence · optionTail Homotopy/LoopSpace · optionPoint_eta. The correspondence is continuous both ways Homotopy/LoopSpace · continuous_optionPoint Homotopy/ExactSequence · continuous_optionTail. The point (t, z) lies on ∂I^{N^+} exactly when $t \in \{0, 1\}$ or $z \in \partial I^N$ Homotopy/ExactSequence · mem_boundary_option_iff Homotopy/LoopSpace · optionPoint_mem_boundary_of_left Homotopy/LoopSpace · optionPoint_mem_boundary_of_right. The extra coordinate is the loop parameter in §XII.12 and the time of a lifting problem in §§XII.15–12.17. Reindexing along a bijection $M \cong N$ identifies π_M with π_N (§XII.12), so $\pi_{\{0,\dots,n\}}$ and $\pi_{\{0,\dots,n-1\}^+}$ are the same group.

Paths. A path from x to y is Mathlib's Mathlib/Topology/Path · Path, with reversal γ^{-1} and concatenation $\gamma \cdot \omega$ Mathlib/Topology/Path · Path.symm Mathlib/Topology/Path · Path.trans. Formulas below evaluate paths at real numbers outside $[0, 1]$. This means the extension of γ to $\mathbb{R}$ that is constant on $(-\infty, 0]$ and on $[1, \infty)$ Mathlib/Topology/Path · Path.extend. A homotopy H from f to g moves each point x along the path $t \mapsto H(t, x)$ from $f(x)$ to $g(x)$ Mathlib/Topology/Homotopy/Path · ContinuousMap.Homotopy.evalAt.

Clamps and cases. As in Part XI (§XI.1), $\mathrm{clamp}(s) = \min(1, \max(0, s))$ Mathlib/Order/Interval/Set/ProjIcc · Set.projIcc. A map defined by one formula on $\{a \leq b\}$ and another on $\{a \geq b\}$ is continuous when both formulas are continuous everywhere and agree on $\{a = b\}$ Mathlib/Topology/Order/OrderClosed · Continuous.if_le. Nearly every homotopy of this part is built this way, and the formulas are written without the clamps on the range where they do nothing. $\bar{s} = 1 - s$ is the reversal of I Mathlib/Topology/UnitInterval · unitInterval.symm.

Statements about adjunction spaces ask that the spaces involved lie in one universe, as in Part II. The condition has no mathematical content and is not mentioned again.

XII.2 Induced maps and the compression criteria

Definition (induced maps). A map $f : X \to Y$ carries a generalized loop γ at x to the generalized loop $f \circ \gamma$ at $f(x)$ Homotopy/Basic · genLoopMap Homotopy/Basic · genLoopMap_apply, and a homotopy H relative to ∂I^N to the homotopy $f \circ H$, again relative to ∂I^N Homotopy/Basic · genLoopMap_homotopic. So it induces

$$f_* : \pi_N(X, x) \to \pi_N(Y, f(x)), \qquad f_*[\gamma] = [f \circ \gamma]$$

Homotopy/Basic · homotopyGroupMap Homotopy/Basic · homotopyGroupMap_mk.

It is functorial, $(g \circ f)_* = g_* \circ f_*$ and $\mathrm{id}_* = \mathrm{id}$, as one sees on representatives `Homotopy/Basic · homotopyGroupMap_comp` `Homotopy/Basic · homotopyGroupMap_id`, and it sends the class of the constant loop to the class of the constant loop `Homotopy/ExactSequence · homotopyGroupMap_const`. It is defined as a map of sets, for every N, so that one notion covers π_0, π_1 and the higher groups. For nonempty N it is a homomorphism (§XII.17), and a bijective f_* is then an isomorphism of groups.

Definition (compressions). Let $j : Y \to X$ be a map and P a space. *j compresses maps from P* if every map $f : P \to X$ is homotopic, relative to $f^{-1}(j(Y))$, to a map with values in $j(Y)$ `Homotopy/Basic · Compresses`.

The relative clause says that the points of P already mapped into $j(Y)$ stay where they are. This is Hatcher's compression criterion for $\pi_i(X, A) = 0$, with the cube replaced by an arbitrary space P. It is what the push-off argument produces for $X = Y \cup e^\lambda$ and P of dimension $< \lambda$ (§XII.7).

Two small facts are used throughout. A homotopy relative to S is a homotopy relative to every $T \subseteq S$ `Homotopy/Basic · homotopyRelMono`. And when j is an embedding, a map $g : P \to X$ with values in $j(Y)$ factors as $g = j \circ g'$ with g' continuous `Homotopy/Basic · exists_lift_of_isEmbedding`. The preimage $g'(p)$ of $g(p)$ is unique, and g' is continuous because Y carries the topology induced by j `Mathlib/Topology/Maps/Basic · IsInducing.continuous_iff`.

Theorem (compression gives surjectivity). Let $j : Y \to X$ be an embedding that compresses maps from the cube I^N. Then $j_* : \pi_N(Y, y) \to \pi_N(X, j(y))$ is onto for every $y \in Y$. `Homotopy/Basic · surjective_homotopyGroupMap_of_compresses`

The proof as formalized.

1. Let γ be a generalized loop in X at $j(y)$. Compress it: there are $g : I^N \to X$ with values in $j(Y)$ and a homotopy F from γ to g relative to $\gamma^{-1}(j(Y))$. Write $g = j \circ g'$.
2. $\partial I^N \subseteq \gamma^{-1}(j(Y))$, because $\gamma = j(y)$ there. So F fixes the boundary, and for $z \in \partial I^N$, $j(g'(z)) = g(z) = \gamma(z) = j(y)$, whence $g'(z) = y$ by injectivity. Thus g' is a generalized loop in Y at y.
3. $j \circ g' = g$, and F run backwards is a homotopy from g to γ relative to the boundary. So $j_*[g'] = [\gamma]$. □

Theorem (compression gives injectivity). Let $j : Y \to X$ be an embedding that compresses maps from the cylinder $I \times I^N$. Then $j_* : \pi_N(Y, y) \to \pi_N(X, j(y))$ is one-to-one for every $y \in Y$. `Homotopy/Basic · injective_homotopyGroupMap_of_compresses`

The proof as formalized.

1. Let α, α' be generalized loops in Y at y with $j_*[\alpha] = j_*[\alpha']$, and let $F : I \times I^N \to X$ be a homotopy from $j \circ \alpha$ to $j \circ \alpha'$ relative to ∂I^N.
2. Compress F itself, as a map of the cylinder: there is $g = j \circ g'$ with a homotopy from F to g that fixes every point where F lands in $j(Y)$. At those points $g = F$.
3. F lands in $j(Y)$ on the bottom, the top and the sides of the cylinder: $F(0, z) = j(\alpha(z))$, $F(1, z) = j(\alpha'(z))$, and $F(s, z) = j(\alpha(z))$ for $z \in \partial I^N$. There $g = F$, and injectivity of j gives $g'(0, z) = \alpha(z)$, $g'(1, z) = \alpha'(z)$ and $g'(s, z) = \alpha(z)$ on the sides.
4. So g' is a homotopy in Y from α to α' relative to ∂I^N, and $[\alpha] = [\alpha']$. □

Neither proof uses a group structure, a finiteness hypothesis on N, or any property of X. For $N = \emptyset$ they say that j is onto on path components when it compresses points, and one-to-one when it compresses paths. Milnor passes from $\pi_i(\Omega, \Omega^d) = 0$ to the isomorphism $\pi_i\Omega^d \cong \pi_i\Omega$, $i \leq \lambda_0 - 2$, through the exact sequence of the pair (p. 118), which for $i = 0$ needs a separate argument because the terms are not groups (the retelling, `Exposition/Chapter13.md`, records this). The two criteria replace both the relative groups and the exact sequence, uniformly in i.

XII.3 The frame construction and the change of basepoint

A map that is only *freely* homotopic to another, the basepoint being allowed to move, does not induce the same map on homotopy groups: the two differ by a change of basepoint. Mathlib has no change of basepoint for higher homotopy groups. The formal text builds one, with a single device that also produces all the homotopies of this section and the next. In this section, and in every later statement that goes through a change of basepoint, N is finite, because the device uses the largest of the coordinates.

The sup-distance from the centre. For $y \in I^N$ let

$$d(y) = \max_{j \in N} \left| y_j - \tfrac{1}{2} \right| \in [0, \tfrac{1}{2}],$$

the sup norm of $y - (\frac{1}{2}, \dots, \frac{1}{2})$ `Homotopy/ChangeOfBase · cubeDist`. It is continuous `Homotopy/ChangeOfBase · continuous_cubeDist`, bounds every $|y_j - \frac{1}{2}|$ `Homotopy/ChangeOfBase · abs_le_cubeDist`, is at most $\frac{1}{2}$ `Homotopy/ChangeOfBase · cubeDist_le_half` and equals $\frac{1}{2}$ on ∂I^N `Homotopy/ChangeOfBase · cubeDist_of_mem_boundary`. It vanishes when N is empty `Homotopy/ChangeOfBase · cubeDist_of_isEmpty`, and otherwise the maximum is attained at some coordinate `Homotopy/ChangeOfBase · exists_abs_eq_cubeDist`.

Rescaling. For $r > 0$ the map

$$\rho_r(y)_j = \operatorname{clamp}\left(\frac{y_j - \frac{1}{2}}{2r} + \frac{1}{2} \right)$$

stretches the cube $\{d \le r\}$ of radius r about the centre affinely onto I^N `Homotopy/ChangeOfBase · cubeScale`, and depends continuously on y and on r when r is a positive continuous function of a parameter `Homotopy/ChangeOfBase · continuous_cubeScale`. On $\{d \le r\}$ the clamp does nothing `Homotopy/ChangeOfBase · coe_cubeScale`, and there $d(\rho_r(y)) = d(y)/2r$ `Homotopy/ChangeOfBase · cubeDist_cubeScale`. The level set $\{d = r\}$ goes to ∂I^N: a coordinate with $|y_j - \frac{1}{2}| = r$ goes to 0 or 1, and such a coordinate exists because N is nonempty when $d(y) = r > 0$ `Homotopy/ChangeOfBase · cubeScale_mem_boundary`. Finally $\rho_{1/2}$ is the identity `Homotopy/ChangeOfBase · cubeScale_half`, and $\rho_r \circ \rho_{r'} = \rho_{2rr'}$ on $\{d \le 2rr'\}$ when $r \le \frac{1}{2}$ `Homotopy/ChangeOfBase · cubeScale_cubeScale`.

Definition (frames). Let r be a function of a parameter q with $0 < r(q) \le \frac{1}{2}$, and let $\mathrm{in}(q, y)$, $y \in I^N$, and $\mathrm{out}(q, \delta)$, $\delta \in \mathbb{R}$, take values in Y. The frame is

$$\mathrm{Fr}(q, y) = \begin{cases} \mathrm{in}(q, \rho_{r(q)}(y)) & d(y) \le r(q), \\ \mathrm{out}(q, d(y)) & d(y) \ge r(q), \end{cases}$$

`Homotopy/ChangeOfBase · frame` `Homotopy/ChangeOfBase · frame_apply`. It is the map $\mathrm{in}(q, \cdot)$ shrunk into the cube of radius $r(q)$ about the centre, surrounded on the "annulus" $\{r(q) \le d \le \frac{1}{2}\}$ by a map that is constant on each level set of d.

It is continuous when r, in and out are, provided $\mathrm{in}(q, y) = \mathrm{out}(q, r(q))$ for $y \in \partial I^N$ `Homotopy/ChangeOfBase · continuous_frame`. Indeed on $\{d = r\}$ the point $\rho_r(y)$ lies on ∂I^N, so the two formulas agree there. Under the same compatibility the frame equals $\mathrm{out}(q, \frac{1}{2})$ on ∂I^N `Homotopy/ChangeOfBase · frame_of_mem_boundary`. If $d(y) = \frac{1}{2} \le r(q)$, then $r(q) = \frac{1}{2}$, $\rho_{1/2}$ is the identity and $\mathrm{in}(q, y) = \mathrm{out}(q, \frac{1}{2})$; otherwise the second formula applies directly. Hence:

Lemma (a frame with a parameter in I is a homotopy). If $\mathrm{out}(s, \frac{1}{2}) = y_1$ for all $s \in I$, and the frames at $s = 0$ and $s = 1$ are generalized loops α and β at y_1, then the frame is a homotopy from α to β relative to ∂I^N. `Homotopy/ChangeOfBase · homotopic_of_frame`

Definition (change of basepoint). For a path γ from y_0 to y_1 and a generalized loop α at y_0, let $\gamma_\# \alpha$ be

the frame with $r = \frac{1}{4}$, inner part α and outer part $\delta \mapsto \gamma(4\delta - 1)$:

$$\gamma_{\#}\alpha(y) = \begin{cases} \alpha(\rho_{1/4}(y)) & d(y) \le \frac{1}{4}, \\ \gamma(4d(y) - 1) & d(y) \ge \frac{1}{4}. \end{cases}$$

Homotopy/ChangeOfBase · changeBase Homotopy/ChangeOfBase · changeBase_apply

The two formulas agree at $d = \frac{1}{4}$, because $\alpha = y_0 = \gamma(0)$ on ∂I^N Homotopy/ChangeOfBase · continuous_changeBaseOuter, and on ∂I^N the value is $\gamma(1) = y_1$. So $\gamma_{\#}\alpha$ is a generalized loop at y_1: the loop α shrunk into the middle of the cube, with the path γ running radially from its boundary out to ∂I^N. A homotopy F relative to ∂I^N from α to α' gives the frame with inner part $F(s, \cdot)$, a homotopy from $\gamma_{\#}\alpha$ to $\gamma_{\#}\alpha'$ Homotopy/ChangeOfBase · changeBase_homotopic. So $\gamma_{\#}$ is defined on classes Homotopy/ChangeOfBase · changeBaseMap Homotopy/ChangeOfBase · changeBaseMap_mk.

Lemma (changing back). $(\gamma^{-1})_{\#}(\gamma_{\#}\alpha) \simeq \alpha$ relative to ∂I^N. Homotopy/ChangeOfBase · changeBase_symm_changeBase

The proof as formalized. First the composite, by the rules for ρ. On $\{d \le \frac{1}{8}\}$ it is $\alpha(\rho_{1/4}\rho_{1/4}(y)) = \alpha(\rho_{1/8}(y))$. On $\{\frac{1}{8} \le d \le \frac{1}{4}\}$ it is $\gamma(8d-1)$, since $d(\rho_{1/4}(y)) = 2d(y)$. On $\{d \ge \frac{1}{4}\}$ it is $\gamma^{-1}(4d-1) = \gamma(2-4d)$. So the composite runs out along γ and back again on an annulus. The homotopy is the frame with

$$r(s) = \frac{1+3s}{8}, \qquad \text{in} = \alpha, \qquad \text{out}(s, \delta) = \gamma\big(\min(8(\delta - r(s)),\ 2 - 4\delta)\big).$$

At $s = 0$ it is the composite, case by case: the minimum is $8d - 1$ for $\frac{1}{8} \le d \le \frac{1}{4}$, and $2 - 4d$ for $d \ge \frac{1}{4}$. At $s = 1$, $r = \frac{1}{2} \ge d(y)$ for every y, and the frame is $\alpha \circ \rho_{1/2} = \alpha$. Compatibility holds because $\text{out}(s, r(s)) = \gamma(\min(0, 2 - 4r(s))) = \gamma(0) = y_0$, and the boundary value is $\text{out}(s, \frac{1}{2}) = \gamma(0) = y_0$ because $8(\frac{1}{2} - r(s)) \ge 0$. As s grows the inner cube grows, and the excursion along γ shrinks to nothing. □

Theorem (change of basepoint is a bijection). For a path γ from y_0 to y_1, $\gamma_{\#} : \pi_N(Y, y_0) \to \pi_N(Y, y_1)$ is a bijection, with inverse $(\gamma^{-1})_{\#}$. Homotopy/ChangeOfBase · changeBaseMap_symm_comp Homotopy/ChangeOfBase · changeBaseEquiv

One composite is the identity by the lemma, and the other by the lemma for γ^{-1}, since $(\gamma^{-1})^{-1} = \gamma$.

XII.4 Free homotopies and homotopy equivalences

Theorem (a free homotopy changes the induced map by a change of basepoint). Let H be a homotopy from f to $g : X \to Y$, let $x_0 \in X$, and let $\omega(t) = H(t, x_0)$, a path from $f(x_0)$ to $g(x_0)$. Then $g_* = \omega_{\#} \circ f_*$ on $\pi_N(X, x_0)$. Homotopy/ChangeOfBase · changeBase_genLoopMap_homotopic Homotopy/ChangeOfBase · homotopyGroupMap_eq_changeBaseMap

The proof as formalized. For a generalized loop α at x_0 take the frame with

$$r(s) = \frac{1+s}{4}, \qquad \text{in}(s, y) = H(s, \alpha(y)), \qquad \text{out}(s, \delta) = \omega(4\delta - 1).$$

Compatibility: on ∂I^N, $\text{in}(s, y) = H(s, x_0) = \omega(s)$, and $\text{out}(s, r(s)) = \omega(1 + s - 1) = \omega(s)$. Boundary value: $\text{out}(s, \frac{1}{2}) = \omega(1) = g(x_0)$. At $s = 0$ the frame is $\omega_{\#}(f \circ \alpha)$ by definition. At $s = 1$ the inner cube has radius $\frac{1}{2}$, the annulus has disappeared, and the frame is $H(1, \alpha(\rho_{1/2}(y))) = g(\alpha(y))$. So the frame is a homotopy from $\omega_{\#}(f \circ \alpha)$ to $g \circ \alpha$ relative to ∂I^N. □

Since $\omega_\#$ is a bijection, homotopic maps have bijective, one-to-one or onto induced maps together, at every basepoint `Homotopy/ChangeOfBase · bijective_homotopyGroupMap_iff` `Homotopy/ChangeOfBase · injective_homotopyGroupMap_iff` `Homotopy/ChangeOfBase · surjective_homotopyGroupMap_iff`.

Theorem (homotopy equivalences). A homotopy equivalence $e : X \to Y$ (§0.4) induces a bijection $e_* : \pi_N(X, x_0) \to \pi_N(Y, e(x_0))$ for every $x_0 \in X$. `Homotopy/ChangeOfBase · bijective_homotopyGroupMap_of_homotopyEquiv` `Homotopy/ChangeOfBase · homotopyGroupEquivOfHomotopyEquiv`

The proof as formalized. Let e' be the homotopy inverse. The difficulty is the basepoint: $e' \circ e$ is homotopic to the identity, but $e'(e(x_0))$ need not be x_0, so e'_* at $e(x_0)$ is not literally an inverse of e_* at x_0.

1. $e' \circ e \simeq \mathrm{id}_X$, so $(e' \circ e)_* = e'_* \circ e_*$ is a bijection at x_0, by the previous theorem and $\mathrm{id}_* = \mathrm{id}$.
2. $e \circ e' \simeq \mathrm{id}_Y$, so $(e \circ e')_* = e_* \circ e'_*$ is a bijection at $e(x_0)$. Here e_* is taken at $e'(e(x_0))$.
3. By 2, e'_* at $e(x_0)$ is one-to-one, being the first map of a one-to-one composite. By 1 it is onto, being the last map of an onto composite. So it is a bijection, and $e_* = (e'_*)^{-1} \circ (e'_* \circ e_*)$ is a bijection at x_0. □

A homeomorphism is a homotopy equivalence `Mathlib/Topology/Homotopy/Equiv · Homeomorph.toHomotopyEquiv` and so induces bijections `Homotopy/CellAttach · bijective_homotopyGroupMap_homeomorph`. The theorem also holds for maps that are homotopy equivalences in the sense of the Appendix (Part XI, §XI.1), which are Mathlib homotopy equivalences `Appendix/Telescope · IsHomotopyEquiv.exists_homotopyEquiv` `Homotopy/RelCells · bijective_homotopyGroupMap_of_isHomotopyEquiv`.

Theorem (transfer along homotopy-commutative squares). Let $\iota : A \to X$ and $j : B \to K$ be maps, and $a : A \to B$ and $e : X \to K$ homotopy equivalences with $e \circ \iota \simeq j \circ a$. Then ι_* is bijective (one-to-one, onto) on π_N at $x_0 \in A$ if and only if j_* is at $a(x_0)$. `Homotopy/ChangeOfBase · bijective_homotopyGroupMap_iff_of_square` `Homotopy/ChangeOfBase · injective_homotopyGroupMap_iff_of_square` `Homotopy/ChangeOfBase · surjective_homotopyGroupMap_iff_of_square`

Proof. Compose ι_* on the left with the bijection e_* at $\iota(x_0)$, and j_* on the right with the bijection a_* at x_0; neither changes bijectivity. By functoriality the composites are $(e \circ \iota)_*$ and $(j \circ a)_*$, which are bijective together because the two maps are homotopic. □

The square need only commute up to a free homotopy, and this is exactly how relative cell complexes are defined (Part IX, §IX.2). The theorem is what lets the cell theorems below, proved for adjunction spaces, apply to spaces that only have their homotopy type.

XII.5 The push-off lemma in coordinates

The key step of cellular approximation is Hatcher's Lemma 4.10: a map from a space of dimension i into a space with a cell of dimension $\lambda > i$ can be deformed, without moving the part mapped far from the centre of the cell, so as to miss a whole ball about the centre. Hatcher proves it by simplicial approximation on a subdivision of the domain. The formal proof uses smoothness instead. It is possible because every domain that occurs, a cube or a cube times I, embeds in a vector space of smaller dimension than the cell. This section does the work in the coordinates of the cell; §XII.7 transports it to an attached cell.

The cut-off. Let

$$\beta(\delta) = \max\big(0, \min(1, 3 - 4\delta)\big),$$

so that $\beta = 1$ on $(-\infty, \frac{1}{2}]$, $\beta = 0$ on $[\frac{3}{4}, \infty)$, β is linear in between, continuous, and takes values in $[0, 1]$ `Homotopy/PushOff · pushCutoff` `Homotopy/PushOff · continuous_pushCutoff` `Homotopy/PushOff · pushCutoff_nonneg` `Homotopy/PushOff · pushCutoff_le_one` `Homotopy/PushOff · pushCutoff_of_le` `Homotopy/PushOff · pushCutoff_of_ge`.

Lemma (smooth approximation through an injection). Let P be compact, $\iota : P \to V$ a continuous injection into a finite-dimensional real normed space, $C \subseteq P$ closed, $h : P \to E$ continuous on C with E finite-dimensional, and $\delta > 0$. Then there is a smooth $Q : V \to E$ with $\|Q(\iota(p)) - h(p)\| < \delta$ for all $p \in C$. `Homotopy/PushOff · exists_contDiff_near`

The proof as formalized. If P is empty there is nothing to prove. Otherwise ι is a continuous injection of a compact space into a Hausdorff space, hence a closed embedding `Mathlib/Topology/Separation/Hausdorff · Continuous.isClosedEmbedding`, and $\iota(C)$ is compact. A left inverse ι^{-1} of ι is continuous on $\iota(C)$, because ι is an embedding, and maps $\iota(C)$ into C. So $h \circ \iota^{-1}$ is continuous on the compact set $\iota(C) \subseteq V$, and the Stone–Weierstrass theorem for smooth maps of the companion project gives a smooth Q within δ of it on $\iota(C)$ `Milnor/Chapter8/Approx · Problems.exists_contDiff_approx_of_isCompact`. (That theorem approximates each coordinate of a map in a basis of E by the restriction of a smooth function; these restrictions form an algebra that separates points, because linear functionals do `Mathlib/Topology/ContinuousMap/StoneWeierstrass · ContinuousMap.exists_mem_subalgebra_near_continuous_of_separatesPoints`.) Evaluating at $\iota(p)$ for $p \in C$ gives the claim. □

Theorem (the push-off lemma in coordinates). Let P be compact with a continuous injection $\iota : P \to V$, where V and E are finite-dimensional real normed spaces with $\dim V < \dim E$. Let $W \subseteq P$, and let $h : P \to E$ be continuous on W with $\|h(p)\| < 1$ for $p \in W$, such that $C = \{p \in W : \|h(p)\| \le \frac{3}{4}\}$ is closed in P. Then there are a map $H : I \times P \to E$, continuous on $I \times W$, and $\varepsilon > 0$ such that:

1. $H(0, p) = h(p)$ for all p;
2. $H(t, p) = h(p)$ for all t whenever $\|h(p)\| \ge \frac{3}{4}$;
3. $\|H(t, p)\| < 1$ for $p \in W$;
4. $\|H(1, p)\| \ge \varepsilon$ for $p \in W$.

`Homotopy/PushOff · exists_pushOff`

In the application $E = \mathbb{R}^\lambda$ is the cell, W is the set of points mapped into the open cell, and h is the coordinate in the cell. The homotopy moves nothing that lies outside the ball of radius $\frac{3}{4}$, stays in the open cell, and ends missing the ball of radius ε. The set W need not be open; closedness of C is all that is used.

The proof as formalized.

1. *Approximation.* By the lemma, with $\delta = \frac{1}{8}$, there is a smooth $Q : V \to E$ with $\|Q(\iota(p)) - h(p)\| < \frac{1}{8}$ on C.
2. *A missed point.* Since $\dim V < \dim E$, the complement of the range of the C^1 map Q is dense `Mathlib/Topology/MetricSpace/HausdorffDimension · ContDiff.dense_compl_range_of_finrank_lt_finrank`. Mathlib proves this through Hausdorff dimension: the range of a C^1 map on V has Hausdorff dimension at most $\dim V$ `Mathlib/Topology/MetricSpace/HausdorffDimension · ContDiff.dimH_range_le`, smaller than that of any open subset of E. It is the easiest case of Sard's theorem, every point being critical. So some p_0 with $\|p_0\| < \frac{1}{8}$ is not a value of Q.
3. *A ball missing the image.* $Q(\iota(P))$ is compact, hence closed, and does not contain p_0. So there is $\varepsilon_0 > 0$ with $\|Q(\iota(p)) - p_0\| \ge \varepsilon_0$ for every $p \in P$.
4. *The homotopy.* Put
$$H(t, p) = h(p) + t\,\beta(\|h(p)\|)\,(Q(\iota(p)) - p_0 - h(p)), \qquad \varepsilon = \min(\varepsilon_0, \tfrac{1}{4}).$$
It is continuous on $I \times W$, since h is continuous there and β, Q, ι are continuous. Clearly $H(0, \cdot) = h$, and $\beta(\|h(p)\|) = 0$ when $\|h(p)\| \ge \frac{3}{4}$, which is clause 2.
5. *It stays in the open ball over W.* If $\|h(p)\| \ge \frac{3}{4}$ then $H(t, p) = h(p)$ has norm < 1. Otherwise $p \in C$, and
$$\|Q(\iota(p)) - p_0 - h(p)\| \le \|Q(\iota(p)) - h(p)\| + \|p_0\| < \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$
The coefficient $t\beta$ lies in $[0, 1]$, so $\|H(t, p)\| < \frac{3}{4} + \frac{1}{4} = 1$.
6. *It misses the ball of radius ε at $t = 1$.* If $\|h(p)\| \le \frac{1}{2}$, then $\beta = 1$ and $H(1, p) = Q(\iota(p)) - p_0$, of norm $\ge \varepsilon_0 \ge \varepsilon$. If $\|h(p)\| \ge \frac{3}{4}$, then $H(1, p) = h(p)$, of norm $\ge \frac{3}{4} > \varepsilon$. In between, the correction $\beta(\cdots)(Q(\iota(p)) - p_0 - h(p))$ has norm $< \frac{1}{4}$ by step 5, so $\|H(1, p)\| > \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \ge \varepsilon$. □

The translation by $-p_0$ is the whole trick: Q approximates h near the centre, and subtracting a point not in the range of Q moves the hole of $Q \circ \iota$ to the centre of the cell.

XII.6 The point-set topology of $Y \cup_\varphi e^n$

Recall from §0.5 and Part II: for a space Y and a map $\varphi : S^{n-1} \to Y$, the space $X = Y \cup_\varphi e^n$ is the pushout, in the category of topological spaces, of φ and the inclusion $S^{n-1} \to D^n$ `Chapter2/Attaching · AttachCell`. It comes with the inclusion $Y \to X$ and the characteristic map $D^n \to X$ `Chapter2/Attaching · cellInl Chapter2/Attaching · cellInr`, and a map out of X is a pair of maps on Y and on D^n that agree along φ `Chapter2/Attaching · cellDesc`. Here D^n and S^{n-1} are the closed unit ball and the unit sphere of $\mathbb{R}^n$ `Mathlib/Topology/Category/TopCat/Sphere · TopCat.disk Mathlib/Topology/Category/TopCat/Sphere · TopCat.diskBoundary Mathlib/Topology/Category/TopCat/Sphere · TopCat.diskBoundaryInclusion`. A point of D^n is a vector z with $\|z\| \le 1$ `Chapter2/Attaching · diskPt Chapter2/Attaching · dvec Chapter2/Attaching · norm_dvec_le`, called "the point z of the cell" when viewed in X. For $u \in S^{n-1}$ the points $\varphi(u)$ of Y and u of the cell coincide in X `Chapter2/Attaching · cellInl_apply_eq`. Nothing is assumed about Y: it need not be Hausdorff, or even T_1. Everything the push-off needs is extracted from the pushout itself.

The underlying set. The forgetful functor to sets preserves pushouts, so the set underlying X is the pushout of the underlying sets `Homotopy/CellAttach · isPushout_forget`. Since $S^{n-1} \to D^n$ is injective `Homotopy/CellAttach · injective_diskBoundaryInclusion`, Mathlib's description of pushouts of sets gives three facts. Every point of X comes from Y or from D^n `Homotopy/CellAttach · exists_eq_cellInl_or_cellInr Mathlib/CategoryTheory/Limits/Types/Pushouts · Types.eq_or_eq_of_isPushout`. The inclusion $Y \to X$ is injective `Homotopy/CellAttach · cellInl_injective Mathlib/CategoryTheory/Limits/Types/Pushouts · Types.pushoutCocone_inr_injective_of_isColimit`. And the point z of the cell equals the point y of Y exactly when $z \in S^{n-1}$ and $\varphi(z) = y$ `Homotopy/CellAttach · cellInr_eq_cellInl_iff Mathlib/CategoryTheory/Limits/Types/Pushouts · CategoryTheory.Limits.Types.pushoutCocone_inl_eq_inr_iff_of_isColimit`.

The topology. A set $S \subseteq X$ is closed if and only if its preimages in Y and in D^n are closed `Homotopy/CellAttach · isClosed_attachCell_iff`. A colimit carries the final topology for the maps from the objects of the diagram `Mathlib/Topology/Category/TopCat/Limits/Basic · TopCat.isClosed_iff_of_isColimit`; the third object S^{n-1} adds no condition, since its preimage is the preimage under φ of the preimage in Y.

Lemma. The inclusion $Y \to X$ is a closed embedding. `Homotopy/CellAttach · isClosedEmbedding_cellInl`

Proof. A continuous injective closed map is a closed embedding `Mathlib/Topology/Maps/Basic · IsClosedEmbedding.of_continuous_injective_isClosedMap`. Let $C \subseteq Y$ be closed. The preimage of its image in Y is C, by injectivity. The preimage in D^n is the image of $\varphi^{-1}(C)$ under $S^{n-1} \to D^n$, by the description of the identifications. This is the image of a compact set, a closed subset of the compact sphere, so it is compact, hence closed in the Hausdorff ball. □

The probe. Define $\Pi : X \to \mathbb{R}^n \times \mathbb{R}$ by

$$\Pi = (0,0) \text{ on } Y, \qquad \Pi(z) = \big((1-\|z\|)\,z,\ 1-\|z\|\big) \text{ at the point } z \text{ of the cell}$$

`Homotopy/CellAttach · cellProbe Homotopy/CellAttach · cellProbe_inl Homotopy/CellAttach · cellProbe_inr`. The two formulas agree along φ, as $\|z\| = 1$ on S^{n-1}, so Π is continuous by the universal property. Its second coordinate is the **height** η `Homotopy/CellAttach · cellHeight`, continuous `Homotopy/CellAttach · continuous_cellHeight`, equal to 0 on Y and to $1-\|z\|$ at the point z of the cell `Homotopy/CellAttach · cellHeight_inl Homotopy/CellAttach · cellHeight_inr`, and never negative `Homotopy/CellAttach · cellHeight_nonneg`. The **coordinate** is $\zeta = (\text{first coordinate of } \Pi)/\eta$ `Homotopy/CellAttach · cellCoord`, which by the formal convention $0^{-1} = 0$ is 0 on Y. It is continuous on the open cell $\{\eta > 0\}$ `Homotopy/CellAttach · continuousOn_cellCoord`, and equals z at the point z of the cell when $\|z\| < 1$ `Homotopy/CellAttach · cellCoord_inr`.

Four facts follow.

- Y is exactly the zero set of η Homotopy/CellAttach · mem_range_cellInl_iff: the point z of the cell has height 0 only if $\|z\| = 1$, and then it lies in Y Homotopy/CellAttach · cellInr_mem_range_of_norm_eq_one.
- A point of positive height is the point z of the cell for some $\|z\| < 1$ Homotopy/CellAttach · exists_cellInr_of_cellHeight_pos, and there $\|\zeta\| = 1 - \eta$ Homotopy/CellAttach · norm_cellCoord.
- So the open cell $\{\eta > 0\}$ is open, its coordinate ζ is continuous on it, and for $r < 1$ the closed ball $\{\|\zeta\| \le r\}$ of the cell is the closed set $\{\eta \ge 1 - r\}$.
- The characteristic map extends to all of $\mathbb{R}^n$. Let $c(z) = z/\max(1, \|z\|)$ be the radial retraction of $\mathbb{R}^n$ onto its closed unit ball Homotopy/CellAttach · clampBall, continuous Homotopy/CellAttach · continuous_clampBall, with $\|c(z)\| = \min(1, \|z\|)$ Homotopy/CellAttach · norm_clampBall Homotopy/CellAttach · norm_clampBall_le and $c(z) = z$ on the ball Homotopy/CellAttach · clampBall_of_norm_le. Let $\chi(z)$ be the point $c(z)$ of the cell Homotopy/CellAttach · cellChar. Then χ is continuous Homotopy/CellAttach · continuous_cellChar and is the point z of the cell when $\|z\| \le 1$ Homotopy/CellAttach · cellChar_of_norm_le. It inverts the coordinate: $\chi(\zeta(x)) = x$ when $\eta(x) > 0$ Homotopy/CellAttach · cellChar_cellCoord, and $\zeta(\chi(z)) = z$, $\eta(\chi(z)) = 1 - \|z\|$ for $\|z\| < 1$ Homotopy/CellAttach · cellCoord_cellChar Homotopy/CellAttach · cellHeight_cellChar.

The usual proof that the open cell is open and homeomorphic to the open ball works in the quotient topology directly. The probe replaces it by two continuous functions, and with them the continuity of every map below is a matter of continuous functions on open sets.

XII.7 The push-off and the radial push in $Y \cup_\varphi e^n$

The radial push of the ball. For $0 < \varepsilon \le 1$ and $t \in I$ let

$$R_t(z) = c\big((1 + t(\varepsilon^{-1} - 1))\, z\big)$$

Homotopy/CellAttach · radialPush, continuous in (t, z) Homotopy/CellAttach · continuous_radialPush, with values in the closed ball Homotopy/CellAttach · norm_radialPush_le. R_0 is the identity of the ball Homotopy/CellAttach · radialPush_zero. Every R_t fixes the unit sphere, because the factor $1 + t(\varepsilon^{-1} - 1)$ is at least 1 and the clamp divides it out again Homotopy/CellAttach · radialPush_of_norm_eq_one. And $R_1(z) = c(z/\varepsilon)$ lies on the unit sphere whenever $\|z\| \ge \varepsilon$ Homotopy/CellAttach · norm_radialPush_one.

The radial push of X. On the cell, $z \mapsto (t \mapsto$ the point $R_t(z)$ of the cell$)$ is a continuous map from D^n to the space of paths in X Homotopy/CellAttach · radialDisk Mathlib/Topology/CompactOpen · ContinuousMap.curry. On Y take the constant paths. The two agree along φ, because R_t fixes the sphere Homotopy/CellAttach · radialDisk_compat. The universal property gives a continuous map from X to the space of paths in X, and uncurrying it, which is legitimate because I is locally compact Mathlib/Topology/CompactOpen · ContinuousMap.uncurry, gives a homotopy $\mathcal{R} : I \times X \to X$ Homotopy/CellAttach · cellRadial. This is the same use of the exponential law as for homotopies out of an adjunction space (§0.5) Chapter2/Attaching · cellHomotopy. Its properties:

- $\mathcal{R}_t$ is the identity on Y for every t Homotopy/CellAttach · cellRadial_inl Homotopy/CellAttach · cellRadial_of_mem_range, and is R_t on the cell Homotopy/CellAttach · cellRadial_inr;
- $\mathcal{R}_0$ is the identity of X Homotopy/CellAttach · cellRadial_zero;
- $\mathcal{R}_1$ maps into Y every point x such that $\eta(x) > 0$ implies $\|\zeta(x)\| \ge \varepsilon$ Homotopy/CellAttach · cellRadial_one_mem. A point of Y stays in Y. A point z of the cell with $\|z\| < 1$ has positive height, hence $\|z\| \ge \varepsilon$, and goes to the sphere, which lies in Y. A point with $\|z\| = 1$ is fixed and already lies in Y.

Hatcher's proof finishes by projecting radially from a point of the open cell that the deformed map misses, a deformation of the punctured cell onto its boundary sphere that is undefined at the puncture. The formal proof never removes a point: $\mathcal{R}$ is defined on all of X, and it expands the small ball $\{\|z\| < \varepsilon\}$ over the whole cell. It is enough that the map to be pushed avoids that ball, and the push-off provides a ball, not just a point, because its image is compact.

A homotopy relative to S can be given by its values: a continuous F on $I \times P$ with the right ends that fixes S Homotopy/CellAttach · homotopyRelOfFun.

Theorem (the push-off lemma; Hatcher, Lemma 4.10). Let P be compact with a continuous injection $\iota : P \to V$ into a finite-dimensional real vector space with $\dim V < n$. Every map $f : P \to X = Y \cup_\varphi e^n$ is homotopic, relative to $\{p : \eta(f(p)) < \frac{1}{4}\}$, to a map g that misses a ball of the cell: for some $\varepsilon > 0$, $\|\zeta(g(p))\| \geq \varepsilon$ whenever $\eta(g(p)) > 0$. `Homotopy/CellAttach · exists_homotopicRel_avoid_ball`

The set $\{\eta(f(p)) < \frac{1}{4}\}$ consists of the points that f maps into Y or outside the closed ball of radius $\frac{3}{4}$ of the cell.

The proof as formalized.

1. *Coordinates.* Let $W = \{p : \eta(f(p)) > 0\}$, an open set, and $h = \zeta \circ f$, continuous on W with $\|h\| = 1 - \eta \circ f < 1$ there. The set $\{p \in W : \|h(p)\| \leq \frac{3}{4}\}$ equals $\{p : \eta(f(p)) \geq \frac{1}{4}\}$, which is closed.
2. *The push-off in coordinates* (§XII.5), with $E = \mathbb{R}^n$, gives H and ε.
3. *Back in X.* Put $F(t,p) = \chi(H(t,p))$ if $\eta(f(p)) > 0$, and $F(t,p) = f(p)$ otherwise.
4. *F fixes $\{\eta \circ f < \frac{1}{4}\}$.* For height 0 this is the definition. For $0 < \eta(f(p)) < \frac{1}{4}$, $\|h(p)\| = 1 - \eta(f(p)) > \frac{3}{4}$, so $H(t,p) = h(p)$ and $F(t,p) = \chi(\zeta(f(p))) = f(p)$.
5. *F is continuous.* Near a point (t,p) with $\eta(f(p)) > 0$, F agrees with $\chi \circ H$ on the open set $\{\eta \circ f > 0\}$, and $\chi \circ H$ is continuous at (t,p) because H is continuous on the neighbourhood $I \times W$. Near a point with $\eta(f(p)) = 0$, F agrees with $(t,p) \mapsto f(p)$ on the open set $\{\eta \circ f < \frac{1}{4}\}$, by step 4.
6. *The ends.* $F(0,p) = f(p)$, since $H(0,p) = h(p)$ and $\chi(\zeta(f(p))) = f(p)$ at positive height. Let $g = F(1,\cdot)$. By steps 4 and 5, F is a homotopy from f to g relative to $\{\eta \circ f < \frac{1}{4}\}$.
7. *g misses the ball.* If $\eta(g(p)) > 0$, then $\eta(f(p)) > 0$, for otherwise $g(p) = f(p)$ would have height 0. So $g(p) = \chi(H(1,p))$ with $\|H(1,p)\| < 1$, hence $\zeta(g(p)) = H(1,p)$, of norm $\geq \varepsilon$. □

Theorem (the radial push into Y). Let $g : P \to X$ and $\varepsilon > 0$ be such that $\|\zeta(g(p))\| \geq \varepsilon$ whenever $\eta(g(p)) > 0$. Then g is homotopic, relative to $g^{-1}(Y)$, to a map into Y. `Homotopy/CellAttach · exists_homotopicRel_into_range`

Proof. Let $\varepsilon' = \min(\varepsilon, \frac{1}{4})$, which lies in $(0,1]$ as the radial push requires, and take $(t,p) \mapsto \mathcal{R}_t(g(p))$ for ε'. It starts at g, fixes the points mapped into Y, and at $t = 1$ lands in Y because $\|\zeta(g(p))\| \geq \varepsilon \geq \varepsilon'$ at positive height. □

Theorem (compression into Y). Let P be compact with a continuous injection into a finite-dimensional real vector space V with $\dim V < n$. Then the inclusion $Y \to Y \cup_\varphi e^n$ compresses maps from P: every $f : P \to Y \cup_\varphi e^n$ is homotopic, relative to $f^{-1}(Y)$, to a map into Y. `Homotopy/CellAttach · compresses_cellInl`

The proof as formalized. The push-off gives F from f to g, relative to $\{\eta \circ f < \frac{1}{4}\}$. This set contains $f^{-1}(Y)$, where $\eta = 0$, so F is also relative to $f^{-1}(Y)$. The radial push gives G from g to a map into Y, relative to $g^{-1}(Y)$. This set contains $f^{-1}(Y)$, because F fixes those points and so $g = f$ there. The concatenation of F and G is relative to $f^{-1}(Y)$ `Homotopy/Basic · homotopyRelMono` `Mathlib/Topology/Homotopy/Basic · ContinuousMap.HomotopyRel.trans`. □

XII.8 Attaching cells does not change the low homotopy groups

The cube and the cylinder embed in vector spaces of their own dimension: $I^N \to \mathbb{R}^N$ `Homotopy/CellAttach · cubeEmb` `Homotopy/CellAttach · continuous_cubeEmb` `Homotopy/CellAttach · injective_cubeEmb` and $I \times I^N \to \mathbb{R} \times \mathbb{R}^N$ `Homotopy/CellAttach · cylEmb` `Homotopy/CellAttach · continuous_cylEmb` `Homotopy/CellAttach · injective_cylEmb`, of dimensions $|N|$ and $|N| + 1$. So the inclusion $Y \to Y \cup_\varphi e^n$ compresses maps from I^N when $|N| < n$, and from $I \times I^N$ when $|N| + 1 < n$ `Homotopy/CellAttach · compresses_cellInl_cube` `Homotopy/CellAttach · compresses_cellInl_cyl`.

Theorem (one cell). Let N be finite, Y any space, $\varphi : S^{n-1} \to Y$ and $y \in Y$. The inclusion $Y \to Y \cup_\varphi e^n$ induces a map $\pi_N(Y, y) \to \pi_N(Y \cup_\varphi e^n, y)$ that is

- onto if $|N| < n$ `Homotopy/CellAttach · surjective_homotopyGroupMap_cellInl`,
- one-to-one if $|N| + 1 < n$ `Homotopy/CellAttach · injective_homotopyGroupMap_cellInl`,
- bijective if $|N| + 2 \le n$ `Homotopy/CellAttach · bijective_homotopyGroupMap_cellInl`.

In Milnor's terms: attaching a λ-cell induces isomorphisms on π_i for $i \le \lambda - 2$ and epimorphisms for $i = \lambda - 1$.

Proof. The inclusion is a closed embedding (§XII.6), in particular an embedding, and the compression criteria of §XII.2 apply with the two compressions above. □

Theorem (finitely many cells). Let K be obtained from L by attaching cells of dimensions $c_1, \dots, c_k$ one at a time, each along an arbitrary map, with $j : L \to K$ the composite of the inclusions (§0.5; Part IX, §IX.2) `Appendix/MonotoneUnion · IsRelCellAttachment`. If all $c_i \ge |N| + 2$, then j_* is bijective on π_N at every $x \in L$; if all $c_i > |N|$, it is onto. `Homotopy/CellAttach · bijective_homotopyGroupMap_of_isRelCellAttachment` `Homotopy/CellAttach · surjective_homotopyGroupMap_of_isRelCellAttachment`

Proof. Induction on the attachment. With no cells j is the identity, and $\mathrm{id}_* = \mathrm{id}$. If $K = K_1 \cup_\psi e^c$ and j is $j_1 : L \to K_1$ followed by the inclusion, then j_* is $(j_1)_*$ followed by the map of the one-cell theorem at the basepoint $j_1(x)$, a composite of bijections (surjections). □

Other pushout squares. Suppose $j : A \to Q$ and $\chi : D^n \to Q$ form a pushout square with some $\psi : S^{n-1} \to A$ and the inclusion $S^{n-1} \to D^n$ `Mathlib/CategoryTheory/Limits/Shapes/Pullback/IsPullback/Defs · CategoryTheory.IsPushout`. Then Q is canonically homeomorphic to $A \cup_\psi e^n$, with j corresponding to the inclusion `Mathlib/CategoryTheory/Limits/Shapes/Pullback/IsPullback/Defs · CategoryTheory.IsPushout.isoPushout`. Compressions are transported along homeomorphisms of the target `Homotopy/CellAttach · Compresses.comp_homeomorph`, so j compresses maps from every compact P that embeds in dimension $< n$ `Homotopy/CellAttach · compresses_of_isPushout`. On homotopy groups the inclusion of A into $A \cup_\psi e^n$ is j_* followed by the bijection induced by the homeomorphism, so j_* is bijective for $|N| + 2 \le n$ and onto for $|N| < n$ `Homotopy/CellAttach · bijective_homotopyGroupMap_of_isPushout` `Homotopy/CellAttach · surjective_homotopyGroupMap_of_isPushout`.

Theorem (an embedded cell). Let M be a Hausdorff space, $A \subseteq M$ closed, and $\chi : \mathbb{R}^n \to M$ continuous and injective on the closed unit ball D^n, with $\chi(z) \in A$ exactly when $\|z\| = 1$, for $z \in D^n$. Then:

- every map from a compact P that embeds continuously in dimension $< n$ into $A \cup \chi(D^n)$ is homotopic, relative to the preimage of A, to a map into A `Homotopy/CellAttach · compresses_union_cell`;
- the inclusion $A \to A \cup \chi(D^n)$ is bijective on π_N for $|N| + 2 \le n$ and onto for $|N| < n$, at every basepoint $a \in A$ `Homotopy/CellAttach · bijective_homotopyGroupMap_union_cell` `Homotopy/CellAttach · surjective_homotopyGroupMap_union_cell`.

Proof. Lemma D in a Hausdorff space (Part II, §II.11) makes $A \cup \chi(D^n)$ the pushout of $\chi|S^{n-1} : S^{n-1} \to A$ and $S^{n-1} \to D^n$, with the inclusion of A as one side of the square `Chapter2/CellPatch · exists_isPushout_union_cell_of_t2`. Then the results for pushout squares apply. □

This is the form in which Theorem 3.2 and Remark 3.3 produce cells (Part II), and the form used for spheres in §XII.18. The Hausdorff hypothesis and the closedness of A are those of Lemma D; they make the cell a closed set on which χ is a homeomorphism.

XII.9 Compact sets lie in a finite stage

Milnor's countable complexes are increasing unions, and their homotopy groups are computed by compactness: a map from a cube, or a homotopy on a cylinder, has compact image, and a compact set lies in

a finite stage. The formal statements take this as a hypothesis on the stages, so that it applies equally to sublevel sets and to truncations of telescopes.

Lemma. If $A_0 \subseteq A_1 \subseteq \cdots \subseteq X$ and every point of X has some A_n as a neighbourhood, then every compact $K \subseteq X$ lies in some A_n. `Homotopy/Telescope · exists_subset_of_isCompact`

Proof. The interiors of the A_n form an increasing open cover of K, so finitely many of them, hence one, cover it `Mathlib/Topology/Compactness/Compact · IsCompact.elim_directed_cover`. □

For a map $f : Y \to X$ with values in a subset S, its corestriction is the same map as a map $Y \to S$ `Homotopy/Telescope · corestrict`, and the inclusion of S is a continuous map $S \to X$ `Homotopy/Telescope · valC`.

Theorem (compactness). Let $A_0, A_1, \ldots$ be subsets of X such that every compact subset of X lies in some A_n, and let $\iota : Y \to X$ be a map with values in every A_n. If for every n the corestriction $\iota_n : Y \to A_n$ is onto (one-to-one, bijective) on π_N at y_0, then so is ι. `Homotopy/Telescope · surjective_homotopyGroupMap_of_exhaustion` `Homotopy/Telescope · injective_homotopyGroupMap_of_exhaustion` `Homotopy/Telescope · bijective_homotopyGroupMap_of_exhaustion`

The proof as formalized. ι is ι_n followed by the inclusion of A_n, so $\iota_* = (\text{inclusion})_* \circ (\iota_n)_*$.

- *Onto.* Let γ be a generalized loop in X at $\iota(y_0)$. Its image is compact, so it lies in some A_n, and γ is a generalized loop in A_n at $\iota_n(y_0)$. Choose $[\alpha]$ with $(\iota_n)_*[\alpha] = [\gamma \text{ in } A_n]$. The inclusion carries $[\gamma \text{ in } A_n]$ to $[\gamma]$, so $\iota_*[\alpha] = [\gamma]$.
- *One-to-one.* Let F be a homotopy relative to ∂I^N in X from $\iota \circ \alpha$ to $\iota \circ \alpha'$. Its image $F(I \times I^N)$ is compact, so it lies in some A_n, and F is a homotopy in A_n from $\iota_n \circ \alpha$ to $\iota_n \circ \alpha'$, still relative to ∂I^N. Since $(\iota_n)_*$ is one-to-one, $[\alpha] = [\alpha']$. □

No finiteness of N and no group structure is involved. The hypothesis that ι lands in every A_n is satisfied in practice by maps into A_0, the A_n being increasing.

Theorem (increasing unions). Let $A_0 \subseteq A_1 \subseteq \cdots \subseteq X$ be such that every compact subset of X lies in some A_n. If every inclusion $A_n \to A_{n+1}$ is bijective (onto) on π_N at every basepoint, then so is the inclusion $A_0 \to X$. `Homotopy/Telescope · bijective_homotopyGroupMap_valC_of_step` `Homotopy/Telescope · surjective_homotopyGroupMap_valC_of_step`

Proof. By the compactness theorem it suffices that each corestriction $A_0 \to A_n$ is bijective (onto). It is the identity for $n = 0$, and for $n + 1$ it is the corestriction to A_n followed by the inclusion $A_n \to A_{n+1}$; induction on n. □

XII.10 Mapping telescopes and countable cell complexes

The countable complexes of the project are mapping telescopes of towers of iterated adjunction spaces (Part IX, §IX.4; Part XI, §XI.15). Recall from Part XI (§XI.12) the mapping telescope $\mathrm{Tel}(K)$ of $K_0 \to K_1 \to \cdots$ `Appendix/MappingTelescope · MappingTelescope`, the maps $c_{m,n} : K_m \to K_n$ of the sequence `Appendix/MappingTelescope · tmap` with $c_{n,n} = \mathrm{id}$ and $c_{m,n}c_{l,m} = c_{l,n}$ `Appendix/MappingTelescope · tmap_refl` `Appendix/MappingTelescope · tmap_tmap`, its points $[k,t]_n$ with $[k,1]_n = [c_n k, 0]_{n+1}$ `Appendix/MappingTelescope · MappingTelescope.mk` `Appendix/MappingTelescope · MappingTelescope.mk_one`, the level $\ell[k,t]_n = n + t$ `Appendix/MappingTelescope · MappingTelescope.level`, the truncations $\mathrm{Tel}^n(K) = \{\ell \le n\}$ `Appendix/MappingTelescope · MappingTelescope.trunc`, the push push_n, which deforms $\mathrm{Tel}^n(K)$ within itself up to level n along the trajectories of the maps `Appendix/MappingTelescope · MappingTelescope.traj` `Appendix/MappingTelescope · MappingTelescope.pushFun` `Appendix/MappingTelescope · MappingTelescope.push` `Appendix/MappingTelescope · MappingTelescope.push_mk`, and (T2): $\kappa_n : K_n \to \mathrm{Tel}^n(K)$, $k \mapsto [k,0]_n$, is a homotopy equivalence `Appendix/MappingTelescope · MappingTelescope.levelIncl` `Appendix/MappingTelescope · MappingTelescope.isHomotopyEquiv_levelIncl`.

Let $\kappa : K_0 \to \mathrm{Tel}(K)$, $k \mapsto [k,0]_0$, be the inclusion of the bottom `Homotopy/Telescope · telescopeIncl`. Its values have level 0, so they lie in every truncation `Homotopy/Telescope · telescopeIncl_mem_trunc`.

Lemma (pushing the bottom of a block). For $m \le n$ and $k \in K_m$, $\mathrm{push}_n(1, [k,0]_m) = [c_{m,n}k, 0]_n$. `Homotopy/Telescope · push_one_mk_zero`

Proof. At time 1 the push moves a point $[k,t]_m$ of level $\le n$ to height $t + \max(0, n-m-t)$ along the trajectory of k, here to height $n-m$. The trajectory of k reaches height d at $[c_{m,m+d}k, 0]_{m+d}$ `Homotopy/Telescope · traj_self`, by induction on d. For $d = 0$ this is the starting point `Appendix/MappingTelescope · MappingTelescope.traj_zero_right`. For $d = 1$ it is the top $[k,1]_m = [c_m k, 0]_{m+1}$ of the block. For $d \ge 2$ the trajectory steps into the next block and continues from $c_m k$, and $c_{m+1,m+d}\, c_m = c_{m,m+d}$. □

Lemma. The corestriction $K_0 \to \mathrm{Tel}^n(K)$ of κ is homotopic, as a map into $\mathrm{Tel}^n(K)$, to $\kappa_n \circ c_{0,n}$. `Homotopy/Telescope · homotopic_corestrict_telescopeIncl`

Proof. The homotopy is $(s,k) \mapsto \mathrm{push}_n(s, [k,0]_0)$. It stays in $\mathrm{Tel}^n(K)$ `Appendix/MappingTelescope · MappingTelescope.push_mem_trunc`, starts at $[k,0]_0$ `Appendix/MappingTelescope · MappingTelescope.push_zero`, and ends at $[c_{0,n}k, 0]_n$ by the previous lemma. □

This homotopy moves the basepoint: $[x_0, 0]_0$ travels up to $[c_{0,n}x_0, 0]_n$. It is a free homotopy, and the comparison of the induced maps goes through the change of basepoint of §XII.4.

Theorem (mapping telescopes). Let N be finite. If every composite $c_{0,n} : K_0 \to K_n$ is bijective (onto) on π_N at x_0, so is $\kappa : K_0 \to \mathrm{Tel}(K)$. `Homotopy/Telescope · bijective_homotopyGroupMap_telescopeIncl` `Homotopy/Telescope · surjective_homotopyGroupMap_telescopeIncl`

The proof as formalized.

1. Every compact subset of $\mathrm{Tel}(K)$ lies in a truncation: the continuous level is bounded on it `Mathlib/Topology/Order/Compact · IsCompact.bddAbove_image`, say by n.
2. By the compactness theorem of §XII.9 it suffices that each corestriction $K_0 \to \mathrm{Tel}^n(K)$ is bijective (onto) at x_0.
3. That map is freely homotopic to $\kappa_n \circ c_{0,n}$, so the two are bijective (onto) together (§XII.4).
4. κ_n is a homotopy equivalence, so $(\kappa_n)_*$ is bijective at $c_{0,n}(x_0)$ (§XII.4), and $(\kappa_n \circ c_{0,n})_* = (\kappa_n)_* \circ (c_{0,n})_*$ is bijective (onto) by hypothesis. □

If every step $c_n : K_n \to K_{n+1}$ is bijective (onto) on π_N at every basepoint, so is every composite $c_{0,n}$, by induction, since $c_{0,n+1} = c_n \circ c_{0,n}$ `Homotopy/Telescope · bijective_homotopyGroupMap_tmap_of_step` `Homotopy/Telescope · surjective_homotopyGroupMap_tmap_of_step`.

Theorem (countable cell complexes). Let T be a cell tower `Appendix/MonotoneUnion · CellTower`: K_0 is an iterated adjunction space and each K_{n+1} is obtained from K_n by attaching finitely many cells, the map $K_n \to K_{n+1}$ being the composite inclusion (Part IX, §IX.4). If every cell attached after K_0 has dimension $\ge |N|+2$ (resp. $> |N|$), then the inclusion $\kappa : K_0 \to \mathrm{Tel}(K)$ is bijective (onto) on π_N at every basepoint. `Homotopy/Telescope · bijective_homotopyGroupMap_of_cellTower` `Homotopy/Telescope · surjective_homotopyGroupMap_of_cellTower`

Proof. Each step is a relative cell attachment with cells of the required dimension, hence bijective (onto) at every basepoint by the theorem on finitely many cells (§XII.8). So are the composites, and then κ by the previous theorem. □

Milnor's statement concerns a countable CW-complex $K = \bigcup K_i$ with the weak topology. The mapping telescope stands for it, as everywhere in the project; the comparison of the two is Example 2 of the Appendix, which rests on Whitehead's theorem and is not formalized (Part XI, §§XI.14–11.15).

XII.11 Relative cell complexes: the form of Theorem 22.1

Recall from Part IX (§IX.2) that $\iota : A \to X$ *has the homotopy type of the inclusion of A with cells $(c_1, \dots, c_k)$ attached* if there are a relative cell attachment $j : A \to K$ with these cells and a homotopy equivalence

$e : X \to K$ with $e \circ \iota \simeq j$ `Chapter10/RelativeCells · IsRelCellComplexType`. This is the form in which the relative Theorem 3.5 describes a step $M^{a_n} \subseteq M^{a_{n+1}}$ of a filtration by sublevel sets.

Theorem (relative cell complexes, finitely many cells). Let A and X be spaces and suppose $\iota : A \to X$ has the homotopy type of the inclusion of A with cells $(c_1, \dots, c_k)$ attached. If all $c_i \geq |N| + 2$ (resp. $> |N|$), then $\iota_* : \pi_N(A, a) \to \pi_N(X, \iota(a))$ is bijective (onto) for every $a \in A$. `Homotopy/RelCells · bijective_homotopyGroupMap_of_isRelCellComplexType` `Homotopy/RelCells · surjective_homotopyGroupMap_of_isRelCellComplexType`

Proof. Take the data $j : A \to K$ and $e : X \to K$. The square with the identity of A on the left, $e \circ \iota \simeq j \circ \mathrm{id}_A$, is a homotopy-commutative square of homotopy equivalences, so ι_* is bijective at a if and only if j_* is (§XII.4). j_* is bijective by the theorem on finitely many cells (§XII.8). □

Theorem (countably many cells). Let $B_0 \subseteq B_1 \subseteq \cdots \subseteq X$ be such that every compact subset of X lies in some B_n, and suppose each inclusion $B_n \to B_{n+1}$ `Chapter10/RelativeCells · inclC` has the homotopy type of a relative cell complex with cells of dimension $\geq |N|+2$ (resp. $> |N|$). Then $B_0 \to X$ is bijective (onto) on π_N at every basepoint. `Homotopy/RelCells · bijective_homotopyGroupMap_of_seq` `Homotopy/RelCells · surjective_homotopyGroupMap_of_seq`

Proof. Each step is bijective (onto) at every basepoint by the previous theorem; the theorem on increasing unions (§XII.9) finishes. □

Sublevel sets satisfy the compactness hypothesis: if $E : X \to \mathbb{R}$ is continuous and $a_n \to \infty$, every compact $K \subseteq X$ lies in some $\{E \leq a_n\}$, because E is bounded on K `Homotopy/RelCells · exists_subset_sublevel`.

Theorem (the setting of Theorem 22.1). Let $E : X \to \mathbb{R}$ be continuous and $a_0 \leq a_1 \leq \cdots$ a sequence tending to ∞. Suppose each inclusion $X^{a_n} \to X^{a_{n+1}}$ of sublevel sets (§0.4) `Chapter2/Crossing · sublevel` `Chapter2/Crossing · sublevel_mono` has the homotopy type of a relative cell complex with cells of dimension $\geq |N| + 2$ (resp. $> |N|$). Then $X^{a_0} \to X$ is bijective (onto) on π_N at every basepoint. `Homotopy/RelCells · bijective_homotopyGroupMap_sublevel` `Homotopy/RelCells · surjective_homotopyGroupMap_sublevel`

With $X = \Omega$, E the energy, $a_0 = d$ and all cells of dimension $\geq \lambda_0$, this is the conclusion Milnor draws on p. 118: $\pi_i \Omega^d \to \pi_i \Omega$ is bijective for $i \leq \lambda_0 - 2$. It is also onto for $i = \lambda_0 - 1$, which Milnor does not state and the retelling (`Exposition/Chapter13.md`) records beside his statement. The hypothesis, a relative cell structure on every step of the energy filtration with cells of dimension at least λ_0, is the geometric content of Theorem 22.1. Providing it for Ω is not part of this library, which supplies the step from cells to homotopy groups.

Theorem (countable complexes up to homotopy). Let T be a cell tower whose cells attached after K_0 have dimension $\geq |N|+2$ (resp. $> |N|$). Let $\iota : A \to Z$ be a map, and $a : A \to K_0$ and $e : Z \to \mathrm{Tel}(K)$ homotopy equivalences with $e \circ \iota \simeq \kappa \circ a$. Then ι_* is bijective (onto) on π_N at every basepoint. `Homotopy/RelCells · bijective_homotopyGroupMap_of_telescope` `Homotopy/RelCells · surjective_homotopyGroupMap_of_telescope`

Proof. Transfer along the square (§XII.4), and apply the theorem on countable cell complexes (§XII.10). □

This is the form that matches the conclusion of Theorem 17.3 as formalized, where a space is homotopy equivalent to the mapping telescope of a cell tower `Appendix/MonotoneUnion · IsCountableCellComplexType`.

XII.12 Loop spaces and path spaces

Reindexing. A bijection $e : M \to N$ identifies I^M with I^N, and carries generalized loops on I^M to generalized loops on I^N `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.congr`. It carries homotopies relative to the boundary to such homotopies, because a point of I^N with a boundary coordinate n corresponds to a point of I^M with the boundary coordinate $e^{-1}(n)$ `Homotopy/LoopSpace · genLoop_homotopic_congr`, and reindexing back undoes it `Homotopy/LoopSpace · genLoop_congr_symm_congr`. So $\pi_M(X, x) \cong \pi_N(X, x)$

Homotopy/LoopSpace · homotopyGroupCongr. Concatenation along m becomes concatenation along $e(m)$ Homotopy/LoopSpace · genLoop_congr_transAt, so for nonempty M this is an isomorphism of groups Homotopy/LoopSpace · homotopyGroupCongrMulEquiv.

The loop space. ΩX is the space of loops at x, Mathlib's paths from x to x with the compact-open topology Mathlib/Topology/Homotopy/HomotopyGroup · LoopSpace, based at the constant loop Mathlib/Topology/Path · Path.refl. A generalized loop γ on I^M in ΩX assigns to each $z \in I^M$ a loop $\gamma(z)$ at x, the constant loop when $z \in \partial I^M$.

Uncurrying. Put $\hat{\gamma}(t, z) = \gamma(z)(t)$ on I^{M^+} Homotopy/LoopSpace · loopUncurry. It is continuous because evaluation of paths is, and it is a generalized loop at x: for $t \in \{0, 1\}$ the value is an end point of a loop at x, and for $z \in \partial I^M$ the loop $\gamma(z)$ is constant. Conversely, a generalized loop β on I^{M^+} gives for each z the loop $t \mapsto \beta(t, z)$ Homotopy/LoopSpace · loopCurryPath. These loops depend continuously on z, because a family of paths is continuous exactly when the map of two variables is Homotopy/LoopSpace · continuous_loopCurryPath Mathlib/Topology/Path · Path.continuous_uncurry_iff, and they are constant for $z \in \partial I^M$; this is a generalized loop in ΩX Homotopy/LoopSpace · loopCurry. The two constructions are inverse Homotopy/LoopSpace · loopCurry_loopUncurry Homotopy/LoopSpace · loopUncurry_loopCurry. The same uncurrying, applied to a homotopy $F(s, z)$ relative to ∂I^M, gives the homotopy $(s, (t, z)) \mapsto F(s, z)(t)$ relative to ∂I^{M^+}, and conversely Homotopy/LoopSpace · loopUncurry_homotopic Homotopy/LoopSpace · loopCurry_homotopic.

Theorem. For every index set M, uncurrying is a bijection $\pi_M(\Omega X, \text{const}) \cong \pi_{M^+}(X, x)$. In particular $\pi_i(\Omega X) \cong \pi_{i+1}(X)$ for every $i \geq 0$. Homotopy/LoopSpace · homotopyGroupLoopEquiv Homotopy/LoopSpace · homotopyGroupPiLoopEquiv

The second statement reindexes $\{0, \dots, i\}$ as $\{0, \dots, i-1\}^+$ Mathlib/Logic/Equiv/Fin/Basic · finSuccEquiv.

Theorem (as groups). For nonempty M the bijection is an isomorphism of groups; in particular $\pi_{i+1}(\Omega X) \cong \pi_{i+2}(X)$ as groups. Homotopy/LoopSpace · homotopyGroupLoopMulEquiv Homotopy/LoopSpace · homotopyGroupPiLoopMulEquiv

Proof. Uncurrying carries concatenation along a coordinate $m \in M$ of the family of loops to concatenation along the same coordinate m of I^{M^+}, because changing the coordinate m of (t, z) does not touch t Homotopy/LoopSpace · update_some_comp Homotopy/LoopSpace · loopUncurry_transAt. By Mathlib's description of products, $[p] \cdot [q] = [\text{trans}_m(q, p)]$ on both sides, so products correspond. □

Path spaces. Milnor's $\Omega(M; p, q)$ has two different end points, and §§21–24 pass freely between it and the loop space (`Exposition/Findings.md` records that the proof of Theorem 21.7 treats $\Omega(G; p, q)$ and needs a lemma to change the end points). The formal statement is a homotopy equivalence, and it needs a homotopy that depends continuously on the path.

Lemma (cancelling a path, continuously). For a path ω from b to c, the map $\gamma \mapsto (\gamma \cdot \omega) \cdot \omega^{-1}$ of the space of paths from a to b to itself is homotopic to the identity. Homotopy/LoopSpace · homotopic_trans_trans_symm

The proof as formalized. At time s, with $u(s) = (1 + 3s)/4$, let

$$\theta_s(\gamma)(t) = \begin{cases} \gamma(t/u(s)) & t \leq u(s), \\ \omega(\,\min(4(t - u(s)),\ 2 - 2t)) & t \geq u(s). \end{cases}$$

Homotopy/LoopSpace · cancelFamily It runs through γ on $[0, u(s)]$, then out along ω and back to its start. The two formulas agree at $t = u(s)$, both giving b, and $\theta_s(\gamma)$ ends at b: for $u(s) < 1$ the second formula gives $\omega(\min(4(1 - u(s)), 0)) = \omega(0)$. The map $(s, \gamma) \mapsto \theta_s(\gamma)$ is continuous Homotopy/LoopSpace · continuous_cancelFamily. At $s = 1$, $u = 1$ and $\theta_1(\gamma) = \gamma$ Homotopy/LoopSpace · cancelFamily_one. At $s = 0$, $u = \frac{1}{4}$, and $\theta_0(\gamma)$ is $\gamma(4t)$ on $[0, \frac{1}{4}]$, $\omega(4t - 1)$ on $[\frac{1}{4}, \frac{1}{2}]$ (the minimum is $4t - 1 \leq 1 \leq 2 - 2t$ there), and $\omega(2 - 2t)$ on $[\frac{1}{2}, 1]$. This is $(\gamma \cdot \omega) \cdot \omega^{-1}$, since Mathlib's concatenation runs its first path at double speed Homotopy/LoopSpace · cancelFamily_zero. □

Theorem (changing the end point). For a path ω from b to c, $\gamma \mapsto \gamma \cdot \omega$ is a homotopy equivalence from the space of paths from a to b to the space of paths from a to c, with homotopy inverse $\delta \mapsto \delta \cdot \omega^{-1}$. If X is path-connected, the space of paths from p to q is homotopy equivalent to the loop space at p. `Homotopy/LoopSpace · pathTransHomotopyEquiv` `Homotopy/LoopSpace · pathHomotopyEquivLoop`

Proof. One composite is the lemma for ω, the other the lemma for ω^{-1}, as $(\omega^{-1})^{-1} = \omega$. For path-connected X take for ω a chosen path from q to p `Mathlib/Topology/Connected/PathConnected · PathConnectedSpace.somePath`. □

With §XII.4, the homotopy groups of $\Omega(X; p, q)$ at a path γ are those of the loop space at the loop $\gamma \cdot \omega$; at the constant loop, the loop space has the homotopy groups of X shifted by one degree. These are the pieces from which Milnor's identification $\pi_i \Omega(M; p, q) \cong \pi_{i+1} M$ (p. 118, and throughout §§23–24) is assembled. Note that the shift is stated at the constant loop only; the formal text says nothing about the other path components of ΩX beyond what change of basepoint gives inside a component.

XII.13 Locally trivial maps and Serre fibrations

In §§23–24 Milnor quotes the exact homotopy sequence of a fibre bundle from Steenrod. The formal text sets up the two notions it rests on, local triviality and the homotopy lifting property, and then proves the sequence (§§XII.15–12.17).

Definition (local triviality). Let $p : E \to B$ be any function and F a space. p is *locally trivial with fibre* F if every $b \in B$ has an open neighbourhood U with a homeomorphism $\varphi : p^{-1}(U) \to U \times F$ whose first component is p `Homotopy/Fibration · IsLocallyTrivial`.

Mathlib's fibre bundles are a structure on a total space assembled from the fibres `Mathlib/Topology/FiberBundle/Basic · FiberBundle`. Here local triviality is a property of a given map, so that it applies directly to maps between spaces constructed independently, such as a unitary group mapping onto a sphere (Part XIII). Continuity of p is not part of the definition, because the lifting argument uses only the homeomorphisms over the sets U.

The definition is checked in the expected cases and is invariant in the expected ways. A map covered by Mathlib's local trivializations is locally trivial `Homotopy/Fibration · isLocallyTrivial_of_trivializations` `Mathlib/Topology/FiberBundle/Trivialization · Trivialization`. In particular the projection of a Mathlib fibre bundle is `Homotopy/Fibration · isLocallyTrivial_proj`, and so is the projection $B \times F \to B$, with $U = B$ `Homotopy/Fibration · isLocallyTrivial_fst`. If p is locally trivial and $h : E' \to E$ is a homeomorphism, $p \circ h$ is locally trivial, with the trivializations restricted along h `Homotopy/Fibration · IsLocallyTrivial.comp_homeomorph`. A homeomorphism of the fibre changes nothing `Homotopy/Fibration · IsLocallyTrivial.congr_fiber`.

The fibres. A point of the fibre $p^{-1}(b)$ lies in $p^{-1}(U)$ for any $U \ni b$ `Homotopy/Fibration · mem_preimage_of_mem_fiber`, and a trivialization over U gives a homeomorphism $p^{-1}(b) \to F$: $x \mapsto \mathrm{pr}_F\, \varphi(x)$, with inverse $f \mapsto \varphi^{-1}(b, f)$ `Homotopy/Fibration · fiberHomeomorphOfTriv`. So every fibre of a locally trivial map is homeomorphic to F `Homotopy/Fibration · IsLocallyTrivial.nonempty_fiberHomeomorph` `Homotopy/Fibration · IsLocallyTrivial.fiberHomeomorph`.

Definition (the open box). $J^N = I^N \times \{0\} \cup \partial I^N \times I \subseteq I^N \times I$ `Homotopy/Fibration · openBox` `Homotopy/Fibration · mem_openBox` `Homotopy/Fibration · mk_zero_mem_openBox` `Homotopy/Fibration · mk_mem_openBox_of_mem_boundary`, the boundary of the cylinder without its top face (Hatcher's notation).

Definition (Serre fibrations). $p : E \to B$ is a *Serre fibration* if for every n, every map $H : I^n \times I \to B$ and every $g : I^n \times I \to E$ that is continuous on J^n with $p \circ g = H$ on J^n, there is a map $G : I^n \times I \to E$ with $p \circ G = H$ everywhere and $G = g$ on J^n `Homotopy/Fibration · IsSerreFibration`.

This is the homotopy lifting property for cubes in its *relative* form: the lift is prescribed on the whole open box, not only on the bottom. That is the form the boundary map needs, since it lifts a cube relative to the part of its boundary that goes to the basepoint (§XII.15). In the uses, g is given by a formula on

the whole cylinder that is right only on J^n, which is why only continuity on J^n and agreement with H there are asked.

The definition quantifies over the cubes I^n, $n \in \mathbb{N}$, so that it stays in one universe. It extends to cubes indexed by any finite set N `Homotopy/Fibration · IsSerreFibration.exists_lift`: choose a bijection $N \cong \{0, \dots, n-1\}$, reindex the cube by it `Homotopy/Fibration · cubeCongr` `Homotopy/Fibration · cubeCongr_apply`, which preserves boundaries `Homotopy/Fibration · cubeCongr_mem_boundary_iff` and hence open boxes, solve the reindexed problem, and reindex back.

Theorem (the absolute homotopy lifting property). Let p be a Serre fibration and N finite. Every homotopy $H : I^N \times I \to B$ lifts to E with any prescribed lift g_0 of $H(\cdot, 0)$ at time 0: there is G with $p \circ G = H$ and $G(\cdot, 0) = g_0$. `Homotopy/Fibration · IsSerreFibration.exists_lift_of_bot`

The proof as formalized. The relative property for $(I^N \times I, J^N)$ gives the absolute one for $(I^N \times I, I^N \times \{0\})$ through a map of pairs with a section.

1. Let $\phi(x,t) = (\rho_{1/4}(x),\ t \cdot \mathrm{clamp}(2 - 4d(x)))$ `Homotopy/Fibration · boxFold` `Homotopy/Fibration · continuous_boxFold`, with ρ and d as in §XII.3. The cube of radius $\frac{1}{4}$ about the centre is stretched onto I^N, and time is scaled by a factor that is 1 on that cube and 0 on ∂I^N. So ϕ maps J^N into $I^N \times \{0\}$ `Homotopy/Fibration · boxFold_snd_of_mem_openBox`.
2. Let $\psi(x,t) = (\frac{x}{2} + \frac{1}{4}, t)$, shrinking the cube by half about its centre `Homotopy/Fibration · cubeShrink` `Homotopy/Fibration · continuous_cubeShrink`. Then $d(\frac{x}{2} + \frac{1}{4}) = d(x)/2 \le \frac{1}{4}$ `Homotopy/Fibration · cubeDist_cubeShrink`, so $\rho_{1/4}(\frac{x}{2} + \frac{1}{4}) = x$ `Homotopy/Fibration · cubeScale_cubeShrink` and the time factor is $\mathrm{clamp}(2 - 2d(x)) = 1$. Hence $\phi \circ \psi = \mathrm{id}$ `Homotopy/Fibration · boxFold_shrink`.
3. Lift $H \circ \phi$ relative to J^N with the prescribed values $g(z) = g_0(\phi(z)_1)$, the first component. On J^N, $\phi(z) = (\phi(z)_1, 0)$, so $p(g(z)) = H(\phi(z)_1, 0) = H(\phi(z))$. Let G' be the lift.
4. $G = G' \circ \psi$ lifts H, since $p \circ G' \circ \psi = H \circ \phi \circ \psi = H$. At time 0, $\psi(x,0)$ lies on the bottom of J^N, so $G(x,0) = g(\psi(x,0)) = g_0(x)$. □

XII.14 Homotopy lifting for locally trivial maps

Theorem (Hatcher, Proposition 4.48). A locally trivial map is a Serre fibration. In particular the projection of a Mathlib fibre bundle is a Serre fibration. `Homotopy/HomotopyLifting · IsLocallyTrivial.isSerreFibration` `Homotopy/HomotopyLifting · isSerreFibration_proj`

The proof subdivides the cylinder into small boxes, each mapped into a set over which p is trivial, and lifts box by box. Two things need care, and the textbook passes over both: the order in which the boxes are lifted, and the shape of the part of each box where the lift is already known. The formal proof first puts the time coordinate on the same footing as the others, so that one statement about cubes covers every stage.

The statement proved. Identify $I^n \times I$ with the cube I^{N^+} for $N = \{0, \dots, n-1\}$, time becoming the coordinate $\star$ `Homotopy/HomotopyLifting · optionCubeHomeo`. For a finite N and $D \subseteq N$, let $\mathcal{F}_D \subseteq I^N$ be the union of the faces $\{w_c = 0\}$ for all $c \in N$ and the faces $\{w_c = 1\}$ for $c \in D$ `Homotopy/HomotopyLifting · cubeFaces`, a closed set `Homotopy/HomotopyLifting · isClosed_cubeFaces`. Under the identification, the open box J^n is exactly $\mathcal{F}_D$ for $D = \{c \neq \star\}$ `Homotopy/HomotopyLifting · optionCubeHomeo_mem_cubeFaces_iff`: the bottom is the face $\{w_\star = 0\}$, and the sides are the faces $\{w_c = 0\}$ and $\{w_c = 1\}$ for $c \neq \star$.

Theorem (lifting relative to faces). Let $p : E \to B$ be locally trivial with fibre F, N finite, and $D \subseteq N$ with some $c_0 \notin D$. Let $H : I^N \to B$ be a map and g a lift of H given and continuous on $\mathcal{F}_D$. Then there is a map $G : I^N \to E$ with $p \circ G = H$ and $G = g$ on $\mathcal{F}_D$. `Homotopy/HomotopyLifting · IsLocallyTrivial.exists_lift_cubeFaces`

The coordinate $c_0 \notin D$ is needed. For $D = N$ the faces form the whole boundary of the cube; with B a point and E that boundary, the statement would make the boundary a retract of the cube.

(This remark is not formalized.) Applied with $c_0 = \star$, the theorem is the Serre property `Homotopy/HomotopyLifting · IsLocallyTrivial.isSerreFibration`.

The retraction onto faces. For $u \in I^N$ and $c \in N$ put

$$\tau_c(u) = \begin{cases} 4\,|u_c - \tfrac{1}{2}| & c \in D, \\ 2 - u_c & c \notin D, \end{cases} \qquad M(u) = \max_c \tau_c(u)$$

`Homotopy/HomotopyLifting · faceWeight` `Homotopy/HomotopyLifting · faceWeight_of_mem` `Homotopy/HomotopyLifting · faceWeight_of_notMem` `Homotopy/HomotopyLifting · faceMax`. Each weight lies in $[0,2]$ and depends continuously on u `Homotopy/HomotopyLifting · faceWeight_nonneg` `Homotopy/HomotopyLifting · faceWeight_le_two` `Homotopy/HomotopyLifting · continuous_faceWeight`. M is the sup norm of the vector of weights, so it is continuous, bounds every weight and is attained `Homotopy/HomotopyLifting · continuous_faceMax` `Homotopy/HomotopyLifting · faceWeight_le_faceMax` `Homotopy/HomotopyLifting · exists_faceWeight_eq_faceMax`, and $M \le 2$ `Homotopy/HomotopyLifting · faceMax_le_two`. Because $c_0 \notin D$, $M(u) \ge \tau_{c_0}(u) = 2 - u_{c_0} \ge 1$ `Homotopy/HomotopyLifting · one_le_faceMax`. On $\mathcal{F}_D$, $M = 2$: a coordinate with $u_c = 0$ has weight 2 in both cases, and one with $c \in D$, $u_c = 1$ has weight $4 \cdot \tfrac{1}{2} = 2$ `Homotopy/HomotopyLifting · faceMax_of_mem`. Let P be the point with $P_c = \tfrac{1}{2}$ for $c \in D$ and $P_c = 2$ for $c \notin D$, outside the cube when $D \neq N$, and let

$$r_D(u) = \operatorname{clamp}\Big(P + \frac{2}{M(u)}\,(u - P)\Big),$$

coordinatewise `Homotopy/HomotopyLifting · faceRetract`: radial projection from P. It is continuous because $M \ge 1$ `Homotopy/HomotopyLifting · continuous_faceRetract`.

Lemma. r_D maps I^N into $\mathcal{F}_D$ and is the identity on $\mathcal{F}_D$. `Homotopy/HomotopyLifting · faceRetract_mem` `Homotopy/HomotopyLifting · faceRetract_of_mem`

Proof. Choose c with $\tau_c(u) = M(u)$. If $c \in D$, then $|u_c - \tfrac{1}{2}| = M/4 > 0$ and $\tfrac{2}{M}(u_c - \tfrac{1}{2}) = \pm\tfrac{1}{2}$, so the coordinate c of $r_D(u)$ is $\operatorname{clamp}(1) = 1$ or $\operatorname{clamp}(0) = 0$: the point lies on a face $\{w_c = 1\}$ with $c \in D$ or on a face $\{w_c = 0\}$. If $c \notin D$, then $2 - u_c = M$, so $\tfrac{2}{M}(2 - u_c) = 2$ and the coordinate c is $\operatorname{clamp}(2-2) = 0$. On $\mathcal{F}_D$, $M = 2$ and the formula returns u. □

The grid. For $m \ge 1$ and $\kappa : N \to \{0, \dots, m-1\}$ let $B_\kappa = \prod_c [\kappa_c/m, (\kappa_c + 1)/m]$ `Homotopy/HomotopyLifting · gridBox`, a closed box `Homotopy/HomotopyLifting · isClosed_gridBox`, parametrized by $e_\kappa(u) = (\kappa + u)/m$ `Homotopy/HomotopyLifting · gridEmb` `Homotopy/HomotopyLifting · coe_gridEmb` `Homotopy/HomotopyLifting · natCast_add_one_le` `Homotopy/HomotopyLifting · continuous_gridEmb` `Homotopy/HomotopyLifting · gridEmb_mem`. The inverse on the box is $e_\kappa^{-1}(w) = \operatorname{clamp}(mw - \kappa)$ `Homotopy/HomotopyLifting · gridCoord` `Homotopy/HomotopyLifting · continuous_gridCoord` `Homotopy/HomotopyLifting · coe_gridCoord_of_mem` `Homotopy/HomotopyLifting · gridEmb_gridCoord` `Homotopy/HomotopyLifting · gridCoord_gridEmb`. Every point w lies in a box, with $\kappa_c = \min(m-1, \lfloor m w_c \rfloor)$ `Homotopy/HomotopyLifting · exists_mem_gridBox`.

Lemma (subdivision). For every open cover of I^N there is $m \ge 1$ such that every box of the grid of mesh $\tfrac{1}{m}$ lies in a member of the cover. `Homotopy/HomotopyLifting · exists_gridBox_subset`

Proof. The cube is compact; let δ be a Lebesgue number of the cover for the sup metric `Mathlib/Topology/MetricSpace/Pseudo/Lemmas · lebesgue_number_lemma_of_metric`, and m with $\tfrac{1}{m} < \delta$. Every point of B_κ is within $\tfrac{1}{m}$ of the corner $e_\kappa(0)$ in every coordinate, so B_κ lies in the ball of radius δ about that corner, hence in a member of the cover. □

Lifting over one box. Let $\varphi : p^{-1}(U) \to U \times F$ be a trivialization over U. Moving a point $x \in p^{-1}(U)$ into the fibre over $b \in U$ while keeping its F-coordinate gives

$$\mu_b(x) = \varphi^{-1}(b, \operatorname{pr}_F \varphi(x))$$

`Homotopy/HomotopyLifting · fiberMove`. It lies over b `Homotopy/HomotopyLifting · fiberMove_proj`, is x itself when $b = p(x)$ `Homotopy/HomotopyLifting · fiberMove_self`, and depends continuously on (b, x) `Homotopy/HomotopyLifting · continuous_fiberMove`.

Lemma (one box). Let $\varphi : p^{-1}(U) \to U \times F$ be a trivialization, $H : I^N \to B$ with $H(B_\kappa) \subseteq U$, and $A \subseteq I^N$ closed. Suppose that A meets the box exactly in the image of a set of faces: $e_\kappa^{-1}(w) \in \mathcal{F}_{D'}$ for $w \in B_\kappa \cap A$, and $e_\kappa(\mathcal{F}_{D'}) \subseteq A$, for some D' missing a coordinate c_0. Let G' be continuous on A with $p \circ G' = H$ on A. Then G' extends to G, continuous on $A \cup B_\kappa$, with $p \circ G = H$ there. `Homotopy/HomotopyLifting · exists_lift_union_gridBox`

The proof as formalized.

1. *The retraction of the box.* $r(w) = e_\kappa(r_{D'}(e_\kappa^{-1}(w)))$ is continuous, takes values in A (they lie in $e_\kappa(\mathcal{F}_{D'})$) and in B_κ, and is the identity on $B_\kappa \cap A$, by the lemma on $r_{D'}$ and $e_\kappa e_\kappa^{-1} = \mathrm{id}$ on the box.
2. *A continuous candidate.* $G' \circ r$ is continuous on the whole cube, G' being continuous on A, and its values lie over $H(r(w)) \in U$, so in $p^{-1}(U)$.
3. *The new lift.* On the box put $L(w) = \mu_{H(w)}(G'(r(w)))$: the point of $G'(r(w))$'s F-coordinate in the fibre over $H(w)$. It is continuous and lies over $H(w)$. On $B_\kappa \cap A$, $r(w) = w$ and $p(G'(w)) = H(w)$, so $L(w) = \mu_{p(G'(w))}(G'(w)) = G'(w)$.
4. *Pasting.* Let $G = L$ on B_κ and $G = G'$ elsewhere. By step 3, $G = G'$ on A. So G is continuous on the closed set A and on the closed set B_κ, hence on their union `Mathlib/Topology/ContinuousOn · ContinuousOn.union_of_isClosed`, and it lifts H on both. □

The order of the boxes. After the boxes with indices in a finite set S have been lifted, the lift is known on

$$\Lambda(S) = \mathcal{F}_D \cup \bigcup_{\kappa \in S} B_\kappa$$

`Homotopy/HomotopyLifting · liftedPart`, a closed set `Homotopy/HomotopyLifting · isClosed_liftedPart`, and $\Lambda(S) = \Lambda(S \setminus \{\kappa\}) \cup B_\kappa$ for $\kappa \in S$ `Homotopy/HomotopyLifting · liftedPart_insert`. The boxes are added along the pointwise order of the indices, so that S is always a *lower set*: with κ, it contains every $\kappa' \le \kappa$. When the box B_κ is added, κ maximal in S, the prescribed part of the box consists of its lower faces and of those upper faces that lie on the top faces $\{w_c = 1\}$, $c \in D$, of the cube. In the coordinates of the box these are the faces $\mathcal{F}_{D'}$ with

$$D' = \{c \in D : \kappa_c = m - 1\}$$

`Homotopy/HomotopyLifting · upperFaces`, which misses c_0 because D does. Two lemmas make this precise.

Lemma (the known part of the new box). Let κ be maximal in S and $w \in B_\kappa \cap \Lambda(S \setminus \{\kappa\})$. Then $e_\kappa^{-1}(w) \in \mathcal{F}_{D'}$. `Homotopy/HomotopyLifting · gridCoord_mem_cubeFaces`

Proof. If w lies on a face of $\mathcal{F}_D$ with $w_c = 0$, then $\kappa_c = 0$, since $w \in B_\kappa$ forces $\kappa_c/m \le 0$; so the coordinate c of $e_\kappa^{-1}(w)$ is 0. If $w_c = 1$ with $c \in D$, then $(\kappa_c + 1)/m \ge 1$ forces $\kappa_c = m - 1$, so $c \in D'$ and the coordinate is $m - \kappa_c = 1$. If $w \in B_{\kappa'}$ with $\kappa' \in S$, $\kappa' \ne \kappa$, then $\kappa \not\le \kappa'$, for otherwise maximality would give $\kappa' = \kappa$. So $\kappa'_c < \kappa_c$ for some c, and $w_c \le (\kappa'_c + 1)/m \le \kappa_c/m \le w_c$: the point lies on the lower face c of $B_{\kappa'}$, where the box coordinate is 0. □

Lemma (the prescribed faces are known). Let S be a lower set and $\kappa \in S$. Then $e_\kappa(\mathcal{F}_{D'}) \subseteq \Lambda(S \setminus \{\kappa\})$. `Homotopy/HomotopyLifting · gridEmb_mem_of_mem_cubeFaces`

Proof. Let $u \in \mathcal{F}_{D'}$ and $w = e_\kappa(u)$. If $u_c = 0$ and $\kappa_c = 0$, then $w_c = 0$ and $w \in \mathcal{F}_D$. If $u_c = 0$ and $\kappa_c > 0$, let κ' be κ with κ_c lowered by one. Then $\kappa' \le \kappa$, so $\kappa' \in S$ as S is a lower set, and $\kappa' \ne \kappa$; and w lies in $B_{\kappa'}$, its coordinate c being $\kappa_c/m = (\kappa'_c + 1)/m$. If $u_c = 1$ with $c \in D'$, then $w_c = (\kappa_c + 1)/m = 1$ with $c \in D$, so $w \in \mathcal{F}_D$. □

Proof of the theorem on lifting relative to faces.

1. For each $b \in B$ choose an open $U_b \ni b$ with a trivialization. By subdivision there is m such that H maps each box B_κ into some $U_{b(\kappa)}$.

2. *Claim:* for every lower set S of indices there is G, continuous on $\Lambda(S)$, lifting H there and equal to g on $\mathcal{F}_D$. The proof is a strong induction on S, removing one element at a time `Mathlib/Data/Finset/Card · Finset.eraseInduction`.
3. For $S = \emptyset$, $\Lambda(\emptyset) = \mathcal{F}_D$ and $G = g$.
4. For nonempty S choose a maximal $\kappa \in S$ `Mathlib/Order/Preorder/Finite · Finset.exists_maximal`. Then $S \setminus \{\kappa\}$ is again a lower set: if $a \le a'$ with $a' \in S \setminus \{\kappa\}$, then $a \in S$, and $a = \kappa$ would give $\kappa \le a'$, hence $a' = \kappa$ by maximality. The induction hypothesis gives G' on $A = \Lambda(S \setminus \{\kappa\})$. The two lemmas are the hypotheses of the one-box lemma with $D' = \{c \in D : \kappa_c = m - 1\}$, which gives G on $A \cup B_\kappa = \Lambda(S)$. It equals $G' = g$ on $\mathcal{F}_D \subseteq A$.
5. The set of all indices is a lower set, and Λ of it is the whole cube, since every point lies in a box. The resulting G is continuous on I^N. □

Proof that locally trivial maps are Serre fibrations. Transport the lifting problem on $I^n \times I$ to I^{N^+}, $N = \{0, \dots, n-1\}$, by the homeomorphism, which carries J^n onto $\mathcal{F}_D$ with $D = \{c \neq \star\}$, apply the theorem with $c_0 = \star$, and transport back `Homotopy/HomotopyLifting · IsLocallyTrivial.isSerreFibration`. □

The induction along lower sets is what makes the textbook's "lift box by box" correct. In a careless order, a box could be reached after boxes above it in some coordinate, and the part of it where the lift is already known would not be a set of faces onto which the box retracts.

XII.15 The boundary map

The setting of this section and the next two: $p : E \to B$ is a continuous Serre fibration, $e_0 \in E$, and $F = p^{-1}(p(e_0))$ is the fibre through e_0, a subspace of E `Homotopy/ExactSequence · fiber`, based at e_0 `Homotopy/ExactSequence · fiberPt`, with inclusion $i : F \to E$ `Homotopy/ExactSequence · fiberIncl` `Homotopy/ExactSequence · fiberIncl_apply` and $p = p(e_0)$ on F `Homotopy/ExactSequence · proj_fiber`. N is finite, possibly empty. The group $\pi_{n+1}(B)$ is $\pi_{N^+}(B, p(e_0))$, and its extra coordinate $\star$ is the *time* of the lifting problems. A generalized loop α on I^{N^+} gives the map $(z,t) \mapsto \alpha(t,z)$ of $I^N \times I$ `Homotopy/ExactSequence · cubeOfLoop`, which sends the open box J^N to $p(e_0)$, because $(0,z)$ and (t,z) with $z \in \partial I^N$ lie on ∂I^{N^+}. Simplification rules for points (t,z) are recorded once `Homotopy/ExactSequence · optionPoint_none` `Homotopy/ExactSequence · optionPoint_some` `Homotopy/ExactSequence · optionTail_optionPoint` `Homotopy/ExactSequence · optionPoint_optionTail` `Homotopy/ExactSequence · update_none_eq_optionPoint` `Homotopy/ExactSequence · update_some_optionPoint`, as are the clamps of numbers ≤ 0 and ≥ 1 `Homotopy/ExactSequence · projIcc_eq_zero` `Homotopy/ExactSequence · projIcc_eq_one`.

Definition (boundary lifts). A map $G : I^N \times I \to E$ is a *boundary lift* of α if

$$p(G(z,t)) = \alpha(t,z), \qquad G(z,0) = e_0, \qquad G(z,t) = e_0 \text{ for } z \in \partial I^N$$

`Homotopy/ExactSequence · IsBoundaryLift`. Its *top face* $z \mapsto G(z,1)$ lies in F, because $\alpha(1,z) = p(e_0)$ `Homotopy/ExactSequence · IsBoundaryLift.mem_fiber`, and is e_0 on ∂I^N. So it is a generalized loop in F at e_0 `Homotopy/ExactSequence · topFace` `Homotopy/ExactSequence · coe_topFace`.

A generalized loop β on I^{N^+} in E at e_0, read as the map $(z,t) \mapsto \beta(t,z)$, is a boundary lift of $p \circ \beta$ `Homotopy/ExactSequence · isBoundaryLift_genLoopMap`, with constant top face. Boundary lifts always exist `Homotopy/ExactSequence · exists_isBoundaryLift`: lift $(z,t) \mapsto \alpha(t,z)$ relative to J^N with the constant e_0 prescribed there, which lies over $p(e_0) = \alpha(t,z)$ on J^N `Homotopy/Fibration · IsSerreFibration.exists_lift`.

Theorem (homotopic loops have homotopic boundaries). Let $\alpha \simeq \alpha'$ relative to ∂I^{N^+}, and let G, G' be boundary lifts of α, α'. Then their top faces are homotopic in F relative to ∂I^N. `Homotopy/ExactSequence · homotopic_topFace`

The proof as formalized. Let $K(s,(t,z))$ be the homotopy from α to α'; it equals $p(e_0)$ on ∂I^{N^+} for every s. Lift it, as one lifting problem on the cube I^{N^+} of the variables (s,z), the homotopy parameter s taking the place of $\star$, times the time t:

1. The map to lift is $(s, z, t) \mapsto K(s, (t, z))$.
2. The prescribed lift on the open box is

$$g(s,z,t) = \begin{cases} G(z, \operatorname{clamp}(t-2s)) & s \le \frac{1}{2}, \\ G'(z, \operatorname{clamp}(t-2+2s)) & s \ge \frac{1}{2}, \end{cases}$$

   continuous because both formulas give e_0 at $s = \frac{1}{2}$, where both clamps vanish.
3. It lies over K on the open box. At $t = 0$ both clamps vanish, so $g = e_0$, over $K(s, (0, z)) = p(e_0)$. At $s = 0$, $g = G(z, t)$, over $\alpha(t, z) = K(0, (t, z))$. At $s = 1$, $g = G'(z, t)$, over $\alpha'(t, z)$. For $z \in \partial I^N$, $g = e_0$ by the side conditions, over $K(s, (t, z)) = p(e_0)$.
4. Let L be the lift. Its top $(s, z) \mapsto L(s, z, 1)$ lies in F, since $K(s, (1, z)) = p(e_0)$. On the open box $L = g$, so at $s = 0$ it is $G(z, 1)$, at $s = 1$ it is $G'(z, \operatorname{clamp}(1)) = G'(z, 1)$, and for $z \in \partial I^N$ it is e_0. This is the required homotopy of top faces. □

Definition (the boundary map). $\partial : \pi_{N^+}(B, p(e_0)) \to \pi_N(F, e_0)$ sends $[\alpha]$ to the class of the top face of a chosen boundary lift of α `Homotopy/ExactSequence · boundaryLoop Homotopy/ExactSequence · boundaryMap`. By the theorem it is well defined, and $\partial[\alpha]$ is the class of the top face of *any* boundary lift of α `Homotopy/ExactSequence · boundaryMap_mk`, the case $\alpha' = \alpha$ `Mathlib/Topology/Homotopy/HomotopyGroup · GenLoop.Homotopic.refl`.

Textbook proofs of exactness glue a homotopy of the top face onto a lift and reparametrize the cube. The formal proof does such gluing once, inside a lifting problem, in the following lemma, and never reparametrizes cubes afterwards.

Lemma (changing the top face). Let G be a boundary lift of α whose top face is homotopic in F, relative to ∂I^N, to γ. Then some boundary lift G' of the same α has top face exactly γ. `Homotopy/ExactSequence · exists_isBoundaryLift_topFace_eq`

The proof as formalized. Let Γ be the homotopy in F from the top face of G to γ.

1. *A path for each z.* For $r \in \mathbb{R}$ let $P(z, r) = G(z, \operatorname{clamp} r)$ for $r \le 1$ and $P(z, r) = \Gamma(\operatorname{clamp}(r-1), z)$ for $r \ge 1$: the path $G(z, \cdot)$ followed by the path $\Gamma(\cdot, z)$ in F. It is continuous, since both give $G(z, 1)$ at $r = 1$. It satisfies $P(z, 0) = e_0$; $P(z, r) = e_0$ for $z \in \partial I^N$, because Γ is relative to ∂I^N; $p(P(z, r)) = \alpha(\operatorname{clamp} r, z)$ for $r \le 1$; and $P(z, r) \in F$ for $r > 1$.
2. *The lifting problem.* On the cube I^{N^+} of the variables (t, z), with a new time s, lift the constant homotopy $(t, z, s) \mapsto \alpha(t, z)$, prescribing on the open box $g(t, z, s) = P(z, t(1 + s))$.
3. *It lies over α on the open box.* At $s = 0$, $g = P(z, t) = G(z, t)$, over $\alpha(t, z)$. On ∂I^{N^+}, $\alpha = p(e_0)$, and g lies over it: if $t(1 + s) > 1$, P is in F; otherwise $p(g) = \alpha(\operatorname{clamp}(t(1 + s)), z)$, and this is $p(e_0)$ because $t = 0$, or $t = 1$ and then $s = 0$, or $z \in \partial I^N$.
4. *The new lift.* Let L be the lift and $G'(z, t) = L(t, z, 1)$. It lies over $\alpha(t, z)$. On the open box $L = g$, so $G'(z, 0) = P(z, 0) = e_0$, $G'(z, t) = P(z, 2t) = e_0$ for $z \in \partial I^N$, and the top face is $G'(z, 1) = P(z, 2) = \Gamma(1, z) = \gamma(z)$. □

At $s = 1$ the prescribed values run through $G(z, \cdot)$ at double speed and then through the homotopy Γ: the lift of the constant homotopy of α moves G to a lift that has absorbed Γ.

XII.16 Exactness at every place

Theorem (the exact homotopy sequence). In the setting of §XII.15, consider

$$\pi_{N^+}(B, p(e_0)) \xrightarrow{\partial} \pi_N(F, e_0) \xrightarrow{i_*} \pi_N(E, e_0) \xrightarrow{p_*} \pi_N(B, p(e_0)).$$

At each of the three middle places the kernel, the preimage of the class of the constant loop, is the image:

- at $\pi_N(F)$: $i_* c = 0$ if and only if c is in the image of ∂ `Homotopy/ExactSequence · exact_fiber`;
- at $\pi_N(E)$: $p_* c = 0$ if and only if c is in the image of i_* `Homotopy/ExactSequence · exact_total`;

- at $\pi_{N^+}(B)$: $\partial c = 0$ if and only if c is in the image of p_* Homotopy/ExactSequence · exact_base.

As N runs over $\emptyset, \{0\}, \{0,1\}, \ldots$ this is exactness of the whole sequence

$$\cdots \to \pi_{n+1}B \to \pi_n F \to \pi_n E \to \pi_n B \to \pi_{n-1}F \to \cdots \to \pi_0 F \to \pi_0 E \to \pi_0 B,$$

as pointed sets, including its end, $N = \emptyset$, where the terms are sets of path components. This is the sequence Milnor quotes from Steenrod for §§23–24.

Exactness at the fibre.

- *An element of the image of ∂ dies in E.* Let G be a boundary lift of α, so $\partial[\alpha]$ is the class of its top face. Run G backwards in time: $(s,z) \mapsto G(z, 1-s)$ is a homotopy in E from the top face to $G(z,0) = e_0$, relative to ∂I^N, where $G = e_0$ throughout.
- *An element that dies in E is in the image.* Let γ be a generalized loop in F and Λ a homotopy in E, relative to ∂I^N, from $i \circ \gamma$ to the constant loop. Put $\alpha(t,z) = p(\Lambda(1-t,z))$. It is a generalized loop in B at $p(e_0)$: at $t=0$ it is $p(e_0)$; at $t=1$ it is $p(\gamma(z)) = p(e_0)$ because γ lies in F; on $z \in \partial I^N$ it is $p(e_0)$. Then $G(z,t) = \Lambda(1-t,z)$ is a boundary lift of α, by construction, and its top face is $\Lambda(0,z) = \gamma(z)$. So $\partial[\alpha] = [\gamma]$. □

Exactness at the total space.

- *The composite is trivial*: $p \circ i$ is constant, so $p_* i_*$ sends everything to the constant class Homotopy/ExactSequence · homotopyGroupMap_proj_fiberIncl.
- *An element that dies in B comes from F.* Let β be a generalized loop in E and Θ a homotopy in B, relative to ∂I^N, from $p \circ \beta$ to the constant loop. Lift $(z,t) \mapsto \Theta(t,z)$ relative to J^N, prescribing $\beta(z)$ there; this lies over $\Theta(0,z) = p(\beta(z))$ at $t=0$, and over $\Theta(t,z) = p(\beta(z))$ for $z \in \partial I^N$. Let L be the lift. Its top $\gamma(z) = L(z,1)$ lies in F, since $\Theta(1,z) = p(e_0)$, and equals $\beta(z) = e_0$ for $z \in \partial I^N$; so γ is a generalized loop in F. Running L backwards in time, $(s,z) \mapsto L(z,1-s)$ is a homotopy in E from $i \circ \gamma$ to $L(z,0) = \beta(z)$, relative to ∂I^N, where $L = \beta = e_0$. So $[\beta] = i_*[\gamma]$. □

Exactness at the base.

- *The composite is trivial.* For a generalized loop β in E, β itself is a boundary lift of $p \circ \beta$, and its top face $\beta(1,z) = e_0$ is constant. So $\partial p_*[\beta] = 0$.
- *An element with trivial boundary comes from E.* Let $\partial[\alpha] = 0$, and let G be a boundary lift of α. Its top face is homotopic in F to the constant loop, so by the lemma on changing the top face some boundary lift G' of α has constant top face e_0. Then $\beta(t,z) = G'(z,t)$ is a generalized loop in E at e_0: it is e_0 at $t=0$ (bottom), at $t=1$ (constant top face) and for $z \in \partial I^N$ (sides). And $p \circ \beta = \alpha$. So $[\alpha] = p_*[\beta]$. □

XII.17 Homomorphisms, injectivity, and the two forms used in §23

Induced maps are homomorphisms. A map f commutes with concatenation along any coordinate, $f \circ \mathrm{trans}_i(\gamma,\gamma') = \mathrm{trans}_i(f\circ\gamma, f\circ\gamma')$, both being given by the same case distinction Homotopy/ExactSequenceIso · genLoopMap_transAt. So for nonempty N, $f_*([\gamma][\gamma']) = f_*[\gamma]\, f_*[\gamma']$ Homotopy/ExactSequenceIso · homotopyGroupMap_mul, and f_* is a group homomorphism Homotopy/ExactSequenceIso · homotopyGroupHom.

Theorem (the boundary map is a homomorphism). For nonempty N, $\partial : \pi_{N^+}(B) \to \pi_N(F)$ is a group homomorphism. Homotopy/ExactSequenceIso · boundaryMap_mul Homotopy/ExactSequenceIso · boundaryHom

The proof as formalized. Let $a = [\alpha]$, $b = [\alpha']$, with boundary lifts G, G', and choose a coordinate $j \in N$. By Mathlib's description of products, taken along the coordinate j of N^+, $ab = [\mathrm{trans}_j(\alpha', \alpha)]$. Put the two lifts side by side in the coordinate j:

$$GG(z,t) = \begin{cases} G'(z \text{ with } z_j \mapsto 2z_j,\ t) & z_j \le \tfrac{1}{2}, \\ G(z \text{ with } z_j \mapsto 2z_j - 1,\ t) & z_j \ge \tfrac{1}{2}. \end{cases}$$

At $z_j = \frac{1}{2}$ both formulas give e_0, by the side conditions, so GG is continuous. It is a boundary lift of $\mathrm{trans}_j(\alpha', \alpha)$. It lies over it, since concatenation along j does not touch the time coordinate. It is e_0 at $t = 0$. And it is e_0 for $z \in \partial I^N$: if the boundary coordinate is j, the rescaled point is on the boundary of the corresponding lift; otherwise the rescaled point keeps that boundary coordinate. Its top face is trans_j of the top faces of G' and G, whose class is $\partial a\, \partial b$ by the same description of products. The identity goes to the identity because ∂ kills the image of p_*, which contains the constant class (§XII.16). □

Injectivity. Exactness of pointed sets says only that the kernel is trivial, and in the low degrees, $\partial : \pi_1(B) \to \pi_0(F)$ and $i_* : \pi_0(F) \to \pi_0(E)$, the targets are not groups, so a trivial kernel does not give injectivity. Two arguments valid in every degree supply it.

Lemma. If $\partial a = \partial b$, then $b^{-1}a$ lies in the image of $p_* : \pi_{N^+}(E) \to \pi_{N^+}(B)$. `Homotopy/ExactSequenceIso · exists_homotopyGroupMap_eq_inv_mul`

This holds for every finite N, including $N = \emptyset$, since π_{N^+} is always a group.

The proof as formalized. Let $a = [\alpha]$ and $b = [\alpha']$ with boundary lifts G and G_1. Since $\partial a = \partial b$, the top faces of G_1 and G are homotopic in F, and the lemma on changing the top face gives a boundary lift G' of α' with the same top face as G. Follow G by G' run backwards in time:

$$\beta(t, z) = \begin{cases} G(z, 2t) & t \le \frac{1}{2}, \\ G'(z, 2 - 2t) & t \ge \frac{1}{2}. \end{cases}$$

The formulas agree at $t = \frac{1}{2}$ because the top faces agree, and β is e_0 at $t = 0$, at $t = 1$ and on $z \in \partial I^N$, so it is a generalized loop in E at e_0. Its projection $p \circ \beta$ is, on the nose, $\mathrm{trans}_\star(\alpha, \mathrm{symm}_\star\, \alpha')$. By Mathlib's conventions this represents $[\mathrm{symm}_\star\, \alpha'] \cdot [\alpha] = b^{-1}a$ `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.inv_spec` `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.mul_spec`. □

Hence, if $\pi_{N^+}(E, e_0)$ is trivial, ∂ is one-to-one: $b^{-1}a = p_*(c) = p_*(1) = 1$ `Homotopy/ExactSequenceIso · injective_boundaryMap`.

Theorem. If $\pi_{N^+}(B, p(e_0))$ is trivial, $i_* : \pi_N(F) \to \pi_N(E)$ is one-to-one. `Homotopy/ExactSequenceIso · injective_homotopyGroupMap_fiberIncl`

The proof as formalized. Let γ, γ' be generalized loops in F and Λ a homotopy in E from $i \circ \gamma$ to $i \circ \gamma'$ relative to ∂I^N.

1. Its projection $\delta(t, z) = p(\Lambda(t, z))$ is a generalized loop on I^{N^+} in B at $p(e_0)$: at $t = 0$ and $t = 1$ it is p of a point of F, and on $z \in \partial I^N$ it is $p(e_0)$.
2. $\pi_{N^+}(B)$ is trivial, so there is a homotopy Q from δ to the constant loop relative to ∂I^{N^+}.
3. Lift $(t, z, s) \mapsto Q(s, (t, z))$ on $I^{N^+} \times I$, prescribing $\Lambda(t, z)$ on the open box. It lies over Q there: at $s = 0$, $Q = \delta = p \circ \Lambda$, and on ∂I^{N^+}, $Q(s, \cdot) = \delta$.
4. The lift R ends, at $s = 1$, over the constant loop, so $(t, z) \mapsto R(t, z, 1)$ lies in F. On ∂I^{N^+} it equals Λ: at $t = 0$ it is $\gamma(z)$, at $t = 1$ it is $\gamma'(z)$, and for $z \in \partial I^N$ it is e_0. So it is a homotopy in F from γ to γ' relative to ∂I^N. □

Surjectivity and bijectivity. If $\pi_N(E)$ is trivial, every $c \in \pi_N(F)$ has $i_* c = 0$, so ∂ is onto by exactness at the fibre `Homotopy/ExactSequenceIso · surjective_boundaryMap`. If $\pi_N(B)$ is trivial, i_* is onto by exactness at the total space `Homotopy/ExactSequenceIso · surjective_homotopyGroupMap_fiberIncl`. Together with injectivity:

- $\partial : \pi_{N^+}(B) \to \pi_N(F)$ is bijective when $\pi_{N^+}(E)$ and $\pi_N(E)$ are trivial `Homotopy/ExactSequenceIso · bijective_boundaryMap`;
- $i_* : \pi_N(F) \to \pi_N(E)$ is bijective when $\pi_{N^+}(B)$ and $\pi_N(B)$ are trivial `Homotopy/ExactSequenceIso · bijective_homotopyGroupMap_fiberIncl`.

Two out of three. If two of the groups vanish in the right degrees, so does the third. If $\pi_N(F)$ and $\pi_N(B)$ are trivial, so is $\pi_N(E)$, because i_* is onto `Homotopy/ExactSequenceIso · subsingleton_homotopyGroup_total`. If

$\pi_{N^+}(E)$ and $\pi_N(F)$ are trivial, so is $\pi_{N^+}(B)$: every c has $\partial c = 0$, so $c = p_*(c')$ with c' the constant class, and p_* preserves it `Homotopy/ExactSequenceIso · subsingleton_homotopyGroup_base`. If $\pi_{N^+}(B)$ and $\pi_N(E)$ are trivial, so is $\pi_N(F)$, because i_* is one-to-one `Homotopy/ExactSequenceIso · subsingleton_homotopyGroup_fiber`.

The forms with π_i. Reindexing turns $\pi_{\{0,\dots,j-1\}^+}$ into π_{j+1}, so a vanishing π_{j+1} gives a vanishing $\pi_{\{0,\dots,j-1\}^+}$ `Homotopy/ExactSequenceIso · subsingleton_homotopyGroup_option`, and the boundary map becomes $\partial : \pi_{j+1}(B, p(e_0)) \to \pi_j(F, e_0)$ `Homotopy/ExactSequenceIso · boundaryMapPi`. The two forms that Milnor's §23 uses follow.

Theorem (a). If $\pi_i(E, e_0)$ is trivial for all $i \le k$, then $\partial : \pi_{j+1}(B) \to \pi_j(F)$ is bijective for $j+1 \le k$ and onto for $j \le k$. `Homotopy/ExactSequenceIso · bijective_boundaryMapPi_of_le` `Homotopy/ExactSequenceIso · surjective_boundaryMapPi_of_le` `Homotopy/ExactSequenceIso · boundaryEquivPi`

So $\pi_i(B) \cong \pi_{i-1}(F)$ for $1 \le i \le k$, and $\pi_{k+1}(B) \to \pi_k(F)$ is onto. Bijectivity for $j+1 \le k$ uses the vanishing of $\pi_{j+1}(E)$ and $\pi_j(E)$; surjectivity uses $\pi_j(E)$ only.

Theorem (b). If $\pi_i(B, p(e_0))$ is trivial for all $i \le k$, then $i_* : \pi_i(F) \to \pi_i(E)$ is bijective for $i+1 \le k$ and onto for $i \le k$. `Homotopy/ExactSequenceIso · bijective_homotopyGroupMap_fiberIncl_of_le` `Homotopy/ExactSequenceIso · surjective_homotopyGroupMap_fiberIncl_of_le`

The two-out-of-three statements have π_i forms as well `Homotopy/ExactSequenceIso · subsingleton_pi_base` `Homotopy/ExactSequenceIso · subsingleton_pi_fiber`.

In §23 (pp. 128–129) the stability of $\pi_i U(m)$ is form (b) for the bundle $U(m) \to U(m+1) \to S^{2m+1}$: the base has $\pi_i S^{2m+1} = 0$ for $i \le 2m$ (§XII.18), so $\pi_i U(m) \to \pi_i U(m+1)$ is bijective for $i \le 2m-1$ and onto for $i = 2m$. And $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i-1} U(m)$ for $1 \le i \le 2m$ is form (a) for the bundle $U(m) \to V \to G_m(\mathbb{C}^{2m})$ of the Stiefel manifold over the Grassmannian, once $\pi_i V = 0$ for $i \le 2m$. The bundles, and their local triviality, are Part XIII's. Both forms are proved in every degree, so they also hold where they concern π_0 or π_1, at the low end of the sequence, where exactness of pointed sets alone would not give injectivity.

XII.18 The homotopy groups of spheres below the dimension

$\pi_i S^n = 0$ for $i < n$ is on Mathlib's list of things to do. It feeds the exact sequences of the bundles $U(m) \to U(m+1) \to S^{2m+1}$ in §§23–24. The formal proof is the cell theorem of §XII.8 applied to the two hemispheres, the same decomposition that computes the homology of spheres in Part IV.

Lemma (contractible spaces). If X is contractible `Mathlib/Topology/Homotopy/Contractible · ContractibleSpace` and N finite, $\pi_N(X, x)$ is trivial for every x. `Homotopy/Spheres · subsingleton_homotopyGroup_of_contractible`

Proof. X is homotopy equivalent to a point `Mathlib/Topology/Homotopy/Contractible · ContractibleSpace.hequiv_unit`. A point has exactly one generalized loop, so its homotopy groups are trivial. The homotopy equivalence induces a bijection (§XII.4), in particular an injection into a trivial set. □

Lemma (basepoints). If x and y are joined by a path `Mathlib/Topology/Connected/PathConnected · Joined` and $\pi_N(X, y)$ is trivial, so is $\pi_N(X, x)$. `Homotopy/Spheres · subsingleton_homotopyGroup_of_joined`

Proof. Change of basepoint along a path from x to y `Mathlib/Topology/Connected/PathConnected · Joined.somePath` is a bijection (§XII.3). □

The hemispheres. For $s = \pm 1$ and z in the closed unit ball of $\mathbb{R}^n$ let

$$h_s(z) = (z,\ s\sqrt{1 - \|z\|^2}) \in \mathbb{R}^{n+1}$$

`Homology/Spheres · hemisphere`, continuous and injective, since it keeps the first n coordinates `Homology/Spheres · continuous_hemisphere` `Homology/Spheres · hemisphere_injective`, with values in the unit sphere `Homology/Spheres · hemisphere_mem_sphere`. The sphere S^n is the union of the lower hemisphere $D_- =$

$h_{-1}(D^n)$ and the upper hemisphere $D_+ = h_1(D^n)$ Homology/Spheres · sphere_eq_union_hemisphere Homotopy/Spheres · lowerHemisphere_union: a point x of the sphere is $h_s(z)$ with z its first n coordinates and s the sign of its last one. A point $h_1(z)$ of the upper hemisphere lies in the lower one exactly when $\|z\| = 1$, on the equator Homology/Spheres · hemisphere_one_mem_iff. So D_+ is an n-cell embedded in the Hausdorff space $\mathbb{R}^{n+1}$, meeting D_- exactly along its boundary.

D_- is closed, being a compact image Homotopy/Spheres · lowerHemisphere Homotopy/Spheres · isClosed_lowerHemisphere. It is contractible Homotopy/Spheres · contractibleSpace_lowerHemisphere: h_{-1} restricted to the closed ball is a continuous injection of a compact space into a Hausdorff space, hence a homeomorphism onto D_-, and the closed ball is convex and nonempty, hence contractible Mathlib/Analysis/Convex/Contractible · Convex.contractibleSpace Mathlib/Topology/Homotopy/Contractible · Homeomorph.contractibleSpace. The south pole $h_{-1}(0) = -e_{n+1}$ lies in D_- Homotopy/Spheres · southPole.

Theorem. $\pi_N(S^n, x)$ is trivial for $|N| < n$ and every $x \in S^n$, where S^n is the unit sphere of $\mathbb{R}^{n+1}$. In particular $\pi_i S^n = 0$ for $i < n$. Homotopy/Spheres · subsingleton_homotopyGroup_sphere Homotopy/Spheres · subsingleton_pi_sphere

The proof as formalized.

1. *At the south pole* Homotopy/Spheres · subsingleton_homotopyGroup_sphere_southPole. By the theorem on embedded cells (§XII.8), with $M = \mathbb{R}^{n+1}$, $A = D_-$ and $\chi = h_1$, the inclusion $D_- \to D_- \cup D_+$ is onto on π_N at the south pole, as $|N| < n$. The source is trivial, D_- being contractible, so the target is trivial. The equality $D_- \cup D_+ = S^n$ is a homeomorphism of subspaces, which induces bijections (§XII.4), so $\pi_N(S^n, -e_{n+1})$ is trivial.
2. *At every point.* $|N| < n$ forces $n \geq 1$, so $\mathbb{R}^{n+1}$ has dimension at least 2 and its unit sphere is path-connected Mathlib/Analysis/Normed/Module/Connected · isPathConnected_sphere. Every $x \in S^n$ is joined to the south pole, and the lemma on basepoints finishes. □

The theorem transfers to other models of the sphere. A linear isometry carries unit sphere to unit sphere homeomorphically Homotopy/Spheres · sphereHomeomorph, and every real inner product space of dimension $n + 1$ is isometric to $\mathbb{R}^{n+1}$ through an orthonormal basis Mathlib/Analysis/InnerProductSpace/PiL2 · stdOrthonormalBasis. So π_i of its unit sphere vanishes for $i < n$ Homotopy/Spheres · subsingleton_pi_sphere_of_finrank. The unit sphere of $\mathbb{C}^m$, of real dimension $2m = (2m - 1) + 1$, is S^{2m-1}, and $\pi_i S^{2m-1} = 0$ for $i < 2m - 1$ Homotopy/Spheres · subsingleton_pi_sphere_complex. This is the form the unitary groups use: $U(m + 1)$ acts on the unit sphere of $\mathbb{C}^{m+1}$, and $\pi_i S^{2m+1} = 0$ for $i \leq 2m$.

The retelling of §23 (`Exposition/Chapter14.md`) quotes this fact from the companion volume (Problem 5 of its Chapter 8: a map of a manifold of smaller dimension into a sphere is homotopic to a constant), and adds that a free null-homotopy suffices because π_1 acts on π_i fixing the zero element. The formal proof needs neither the passage from free to based homotopies nor smooth manifolds as domains: the compression of §XII.7 is already relative to the boundary of the cube. The decomposition into hemispheres is the one Part IV uses for the homology of spheres Homology/Spheres · sphereRelIso.

XII.19 What the formalization taught us

- **Compression replaces relative homotopy groups.** Mathlib has no relative homotopy groups and no exact sequence of a pair. Compressing cubes gives surjectivity and compressing cylinders gives injectivity Homotopy/Basic · surjective_homotopyGroupMap_of_compresses Homotopy/Basic · injective_homotopyGroupMap_of_compresses, for every index set and at every basepoint. Milnor's step from $\pi_i(\Omega, \Omega^d) = 0$ to $\pi_i\Omega^d \cong \pi_i\Omega$ becomes a direct proof, with π_0 and π_1 treated like the rest and the surjectivity at $\lambda_0 - 1$ stated Homotopy/RelCells · bijective_homotopyGroupMap_sublevel Homotopy/RelCells · surjective_homotopyGroupMap_sublevel.
- **Homotopy invariance needs change of basepoint.** Homotopic maps induce the same map on homotopy groups only up to a change of basepoint Homotopy/ChangeOfBase · homotopyGroupMap_eq_changeBaseMap, and even the invariance under homotopy equivalence has to manage basepoints

that do not return `Homotopy/ChangeOfBase · bijective_homotopyGroupMap_of_homotopyEquiv`. One construction, the frame, gives all the homotopies `Homotopy/ChangeOfBase · homotopic_of_frame`. The mapping telescope needs it too, because the push to the top of a truncation moves the basepoint `Homotopy/Telescope · homotopic_corestrict_telescopeIncl`.

- **Cellular approximation without simplices.** The push-off lemma follows from smooth approximation and the null image of a smooth map from a smaller dimension `Homotopy/PushOff · exists_pushOff`, because the domains that matter embed in vector spaces of their own dimension `Homotopy/CellAttach · compresses_cellInl_cube` `Homotopy/CellAttach · compresses_cellInl_cyl`.
- **The attached-to space needs no separation axiom.** The probe, with its height and coordinate, makes the open cell open and its coordinate continuous for any Y `Homotopy/CellAttach · cellProbe` `Homotopy/CellAttach · isClosedEmbedding_cellInl`. The radial push needs no puncture `Homotopy/CellAttach · cellRadial`. Hausdorffness enters only for cells embedded in an ambient space, through Lemma D `Homotopy/CellAttach · bijective_homotopyGroupMap_union_cell`.
- **Countable complexes by compactness.** A compact set lies in a finite stage, of a filtration `Homotopy/Telescope · bijective_homotopyGroupMap_of_exhaustion` or of a mapping telescope `Homotopy/Telescope · bijective_homotopyGroupMap_telescopeIncl`, and the homotopy-commutative squares of Part IX are exactly what the transfer theorem needs `Homotopy/ChangeOfBase · bijective_homotopyGroupMap_iff_of_square`.
- **Fibre bundles as a property.** Local triviality of a map, not a bundle structure on a sigma type `Homotopy/Fibration · IsLocallyTrivial`; Serre fibrations lift relative to the open box `Homotopy/Fibration · IsSerreFibration`, and the absolute property follows by a map of pairs `Homotopy/Fibration · IsSerreFibration.exists_lift_of_bot`.
- **"Lift box by box" needs an order and a retraction.** The boxes are added along lower sets of indices, so that the known part of each new box is a union of faces missing one coordinate, onto which it retracts radially `Homotopy/HomotopyLifting · gridCoord_mem_cubeFaces` `Homotopy/HomotopyLifting · gridEmb_mem_of_mem_cubeFaces` `Homotopy/HomotopyLifting · faceRetract_mem`. Without the missing coordinate the statement is false.
- **Exactness of pointed sets is not injectivity.** The sequence is exact at every place, π_0 included `Homotopy/ExactSequence · exact_fiber` `Homotopy/ExactSequence · exact_total` `Homotopy/ExactSequence · exact_base`, but the forms Milnor uses need injectivity, which takes two extra arguments `Homotopy/ExactSequenceIso · exists_homotopyGroupMap_eq_inv_mul` `Homotopy/ExactSequenceIso · injective_homotopyGroupMap_fiberIncl`. One lemma, changing the top face of a boundary lift inside a lifting problem, replaces every reparametrization of cubes `Homotopy/ExactSequence · exists_isBoundaryLift_topFace_eq`.
- **Mathlib's order of multiplication shows.** $[p] \cdot [q]$ is q followed by p, so the lift built from α and α' represents $b^{-1}a$ rather than ab^{-1} `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.mul_spec`, and the side-by-side lifts of the homomorphism property are put down in the order G', G `Homotopy/ExactSequenceIso · boundaryMap_mul`.
- $\pi_i S^n = 0$ **is a cell theorem.** The sphere is a contractible hemisphere with a cell attached `Homotopy/Spheres · subsingleton_homotopyGroup_sphere_southPole`; no manifold structure on the domain is needed, and no passage from free to based homotopies.

Part XIII. Lie groups, symmetric spaces and periodicity (Milnor §§20–24)

Milnor's Part IV applies the Morse theory of the path space to compact Lie groups. A Lie group with a metric invariant under left and right translations is a symmetric space; its geodesics through the identity are its one-parameter subgroups, its curvature is a bracket, and the conjugate points along a geodesic can be read off from the eigenvalues of one linear map (§§20–21). Theorem 22.1 turns a lower bound λ_0 on the indices of the non-minimal geodesics from p to q into the statement that the space Ω^d of minimal geodesics carries the homotopy groups of the path space below $\lambda_0 - 1$ (§22). For $p = I$, $q = -I$ in $SU(2m)$ the minimal geodesics form the Grassmannian $G_m(\mathbb{C}^{2m})$, every other geodesic has index $\geq 2m+2$, and Bott's periodicity $\pi_{i-1}U \cong \pi_{i+1}U$ follows with the help of three fibre bundles (§23). This part retells the formal proofs of §§20–23, together with the bundles, which Milnor quotes from Steenrod. Milnor's §24 (the orthogonal group) is retold in the sections that follow it.

The setting is that of the rest of the project: a manifold is a slice submanifold N of a finite-dimensional real inner product space V with the induced metric (§§0.1, 0.3), and the matrix groups are slice submanifolds of the space $M_n(\mathbb{K})$ of matrices, $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Everything Milnor proves with left-invariant vector fields is computed in this embedding. The formal route departs from Milnor's in eight places, each explained where it occurs.

- **Matrix groups instead of abstract Lie groups.** The groups are given as compact submanifolds of $M_n(\mathbb{K})$ with an explicit Lie algebra of skew-Hermitian matrices (§XIII.2). The curvature is computed by the Gauss equation of Part V, not from left-invariant fields (§XIII.4); the algebraic part of Milnor's §21 is proved separately for an abstract Lie algebra with an invariant form (§XIII.3), and the two are identified.
- **No eigenbasis.** Milnor computes conjugate points in a parallel orthonormal eigenframe of K_V and quotes the block normal form of a skew-symmetric map. The formal text writes the parallel transport along $\exp(tA)$ explicitly as $U \mapsto e^{tA/2}Ue^{tA/2}$, solves the Jacobi equation entrywise, and replaces the normal form by a complex structure on each eigenspace (§§XIII.3, 13.5, 13.16).
- **Lemma 22.4 by compactness.** The "uniform continuity argument which will be left to the reader" (p. 121) is a compactness argument with negative definite subspaces; no continuity of eigenvalues is used (§XIII.6).
- **Hanner's theorem is replaced by an explicit neighbourhood retraction.** Theorem 22.1 assumes that Ω^d is a topological manifold, and Milnor uses this only through Hanner's theorem (a manifold is a neighbourhood retract). The formal Theorem 22.1 assumes the retraction directly (§XIII.10). It is then written down for the sphere (§XIII.12) and for $SU(2m)$, by a spectral projection (§XIII.18).
- **The separation quotient.** Paths are functions on $\mathbb{R}$, and Milnor's metric does not see their values off $[0,1]$. So "Ω^d is homeomorphic to the equator, or to the Grassmannian" holds only after paths agreeing on $[0,1]$ are identified; the homotopy equivalence, which is what the homotopy groups need, holds without the quotient (§§XIII.12, 13.18).
- **Compressions instead of relative homotopy groups.** As in Part XII, $\pi_i(\Omega, \Omega^d) = 0$ is replaced by the statement that maps of compact spaces of small dimension can be pushed into Ω^d relative to

the part already there (§XIII.7). Theorem 22.1 then gives bijections $\pi_i\Omega^d \to \pi_i\Omega$ uniformly in i, π_0 included.

- **Bijections first, group isomorphisms for $i \geq 1$.** Every isomorphism of §§22–23 is first proved as a bijection of pointed sets. The group isomorphisms, in the degrees where both sides are groups, need one more fact, that change of basepoint is a homomorphism (§XIII.11).
- **Concrete bundles.** Stiefel manifolds, Grassmannians and spheres are concrete spaces of matrices, not coset spaces. The bundles are orbit maps with explicit local sections (§§XIII.19–13.20); all homotopy groups come from the exact sequence of Part XII and one input, $\pi_i S^n = 0$ for $i < n$.

Map.

- `Chapter12/MatrixInner` (359 lines): the Frobenius inner product; isometries; the exponential; the determinant and Jacobi's formula.
- `Chapter12/MatrixGroups` (793 lines): $\mathfrak{u}(n)$, $\mathfrak{su}(n)$; matrix groups and their tangent spaces; $U(n)$, $O(n)$, $SO(n)$, $SU(n)$ as slice submanifolds.
- `Chapter12/BiInvariant` (557 lines): Theorem 21.3 and Corollaries 21.4–21.6 for a Lie algebra with an invariant form; the operator K_V and its eigenspaces.
- `Chapter12/GroupGeodesics` (593 lines): the tangent projection; geodesics, Lemmas 21.1 and 21.2; the second fundamental form; Theorem 21.3 by the Gauss equation.
- `Chapter12/GroupJacobi` (276 lines): the parallel frame along $\exp(tA)$; Theorem 20.5 for matrix groups.
- `Chapter13/Perturbation` (364 lines): Lemma 22.4, in a vector space and on a submanifold.
- `Chapter13/Compression` (358 lines): pushing maps into a subset; pushing off cells; Lemma D.
- `Chapter13/ManifoldOfMinima` (338 lines): Lemma 22.5.
- `Chapter13/MinimalDeform` (210 lines): minimal paths are geodesics; deforming a neighbourhood of the minima.
- `Chapter13/Theorem22` (306 lines): Theorem 22.1 and its corollary on π_i.
- `Homotopy/ChangeOfBaseGroup` (137 lines): change of basepoint is a homomorphism.
- `Chapter13/Corollaries` (141 lines), `Chapter13/CorollaryGroups` (142 lines): Corollary 22.2, as bijections and as group isomorphisms; Corollary 22.3 as groups.
- `Chapter13/SphereMinimal` (389 lines), `Chapter13/Freudenthal` (257 lines): Lemmas A and B for the sphere; Freudenthal's theorem.
- `Chapter13/Componentwise` (385 lines): Theorem 22.1 for an open and closed set of paths.
- `Chapter14/SUMatrices` (318 lines), `Chapter12/SUMinimal` (444 lines): the matrix computations of §23; the minimal geodesics and Lemma 23.1.
- `Chapter14/IndexCount` (710 lines): the index from a spectrum; Lemma 23.2 combinatorially; the counts of §24.
- `Chapter12/SUJacobi` (370 lines), `Chapter12/SUIndex` (530 lines): the Jacobi fields of $SU(n)$; the index by Theorems 15.1 and 20.5.
- `Chapter13/SUInstance` (158 lines), `Chapter14/MinimalPaths` (311 lines), `Chapter14/MinimalRetract` (337 lines): Theorem 22.1 for $SU(2m)$.
- `Homotopy/Homogeneous` (137 lines), `Homotopy/Stiefel` (530 lines), `Homotopy/Grassmannian` (595 lines): the bundles.
- `Homotopy/BundleHomotopy` (408 lines), `Homotopy/BundlesUnitary` (595 lines), `Homotopy/BundlesOrthogonal` (229 lines): their homotopy groups; Lemma 23.4.
- `Chapter14/Periodicity` (198 lines), `Chapter14/PeriodicityGroups` (139 lines), `Chapter14/CircleGroup` (93 lines): Theorem 23.3, the periodicity theorem, Theorem 23.5.

XIII.1 The space of matrices

Milnor treats $U(n)$ as a subset of the space of complex $n \times n$ matrices, "a real vector space of dimension $2n^2$" (p. 125), with the inner product $\langle A, B\rangle = \operatorname{Re}\operatorname{tr}(AB^*)$. Mathlib puts no norm on matrices by default, because several are natural. The formal text chooses one.

Definition (the Frobenius structure). For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and a finite index set ι, the space $M_\iota(\mathbb{K})$ carries the norm $\|A\| = (\sum_{i,j} |a_{ij}|^2)^{1/2}$ Chapter12/MatrixInner · MatrixInner.normedAddCommGroup Chapter12/MatrixInner · MatrixInner.norm_def, a real normed-space structure Chapter12/MatrixInner · MatrixInner.normedSpace, and the real inner product

$$\langle A, B\rangle = \operatorname{Re}\operatorname{tr}(A^* B)$$

Chapter12/MatrixInner · MatrixInner.innerProductSpace Chapter12/MatrixInner · inner_def, which is Milnor's up to the order of the factors. That its norm is the Frobenius norm is the identity $\operatorname{Re}\operatorname{tr}(A^*A) = \sum_{i,j} |a_{ij}|^2$ Chapter12/MatrixInner · MatrixInner.re_trace_conjTranspose_mul_self Chapter12/MatrixInner · MatrixInner.norm_sq_eq. The norm is submultiplicative, so $M_\iota(\mathbb{K})$ is a normed ring and a normed real algebra Chapter12/MatrixInner · MatrixInner.normedRing Chapter12/MatrixInner · MatrixInner.normedAlgebra, and the commutator $[A, B] = AB - BA$ makes it a Lie ring. These structures are switched on only where they are opened.

One technical point is not visible in the mathematics. Mathlib's Frobenius norm induces the usual topology of matrices only up to a proof, and the charted-space structure needed for slice submanifolds of $M_\iota(\mathbb{K})$ cannot be found through such a proof. The formal norm therefore keeps the uniform structure of the matrices and changes only the distance (the "replace uniformity" construction).

Isometries. Hermitian and skew-Hermitian matrices are orthogonal: for $A^* = A$ and $B^* = -B$, $\langle A^*, B^*\rangle = \langle A, B\rangle$ Chapter12/MatrixInner · inner_conjTranspose gives $\langle A, -B\rangle = \langle A, B\rangle$, so $\langle A, B\rangle = 0$ Chapter12/MatrixInner · inner_eq_zero_of_hermitian_of_skew. If $U^*U = I$, then $\langle UA, UB\rangle = \operatorname{Re}\operatorname{tr}(A^*U^*UB) = \langle A, B\rangle$ and, by cycling the trace, $\langle AU, BU\rangle = \langle A, B\rangle$ Chapter12/MatrixInner · inner_mul_left_of_unitary Chapter12/MatrixInner · inner_mul_right_of_unitary. So left and right multiplication by a unitary matrix are linear isometries of $M_\iota(\mathbb{K})$, with inverses multiplication by U^* Chapter12/MatrixInner · mulLeftIso Chapter12/MatrixInner · mulRightIso. ($U^*U = I$ implies $UU^* = I$ for square matrices Chapter12/MatrixInner · mul_conjTranspose_eq_one.)

The exponential. Milnor quotes $\exp(A^*) = (\exp A)^*$, $\exp(A + B) = \exp A \exp B$ for commuting A, B, and the smoothness of exp (p. 125, (1)–(4)). All three are Mathlib's. The exponential is analytic, hence smooth Chapter12/MatrixInner · contDiff_exp_matrix, and so is $t \mapsto \exp(tX)$ Chapter12/MatrixInner · contDiff_exp_smul, with derivative $\exp(tX)X$ Mathlib/Analysis/SpecialFunctions/Exponential · hasDerivAt_exp_smul_const. Since sX and tX commute, $\exp(sX)\exp(tX) = \exp((s + t)X)$ Chapter12/MatrixInner · exp_smul_mul_exp_smul Mathlib/Analysis/Normed/Algebra/MatrixExponential · Matrix.exp_add_of_commute, in particular $\exp(tX) = \exp(tX/2)^2$ Chapter12/MatrixInner · exp_half_mul_exp_half and $\exp X \exp(-X) = I$ Chapter12/MatrixInner · exp_mul_exp_neg, and $\exp(tX)$ commutes with X Chapter12/MatrixInner · exp_smul_mul_self. For skew-Hermitian X, $(\exp X)^* \exp X = \exp(X^*)\exp X = \exp(-X)\exp X = I$ Mathlib/Analysis/Normed/Algebra/MatrixExponential · Matrix.exp_conjTranspose, so $\exp X$ is unitary Chapter12/MatrixInner · exp_conjTranspose_mul_exp_of_skew.

The determinant. Milnor uses $\det(\exp A) = e^{\operatorname{tr} A}$ for $SU(n)$ (p. 126). Mathlib does not have it, and only its consequence for traceless X is needed, which follows from Jacobi's formula.

Lemma (Jacobi's formula). The determinant is smooth Chapter12/MatrixInner · contDiff_det, and $d(\det)_A(AX) = \det A \cdot \operatorname{tr} X$ for all square A, X Chapter12/MatrixInner · fderiv_det_mul.

The proof as formalized. The determinant is the polynomial $\sum_\sigma \operatorname{sgn}\sigma \prod_i a_{\sigma(i)i}$ in the entries, each of which is a real-linear function of the matrix Chapter12/MatrixInner · contDiff_entry, so it is smooth. For the formula, Mathlib expands $\det(1 + sX) = 1 + s\operatorname{tr} X + s^2 p(s)$ with p a polynomial Mathlib/LinearAlgebra/Matrix/Charpoly/Coeff · Matrix.det_one_add_smul, so $s \mapsto \det(1 + sX)$ has derivative $\operatorname{tr} X$ at 0 Chapter12/MatrixInner · hasDerivAt_det_one_add_smul. The line $s \mapsto A + sAX$ has velocity AX, and $\det(A + sAX) = \det A \det(1 + sX)$. Comparing the chain rule with this product gives the two derivatives at $s = 0$, which are equal by uniqueness. □

Corollary. $\frac{d}{dt}\det(\exp tX) = \det(\exp tX)\operatorname{tr} X$ Chapter12/MatrixInner · hasDerivAt_det_exp_smul, by the chain rule and Jacobi's formula at $A = \exp tX$. So if $\operatorname{tr} X = 0$, then $t \mapsto \det(\exp tX)$ has derivative 0, is constant, and equals $\det I = 1$ Chapter12/MatrixInner · det_exp_smul_of_trace_eq_zero.

XIII.2 Compact matrix groups as submanifolds

Milnor's §21 concerns an arbitrary Lie group with a bi-invariant metric; his examples in §§23–24 are $U(n)$, $SU(n)$, $O(n)$ and $SO(n)$, with the metric $\operatorname{Re}\operatorname{tr}(AB^*)$ on the Lie algebra (pp. 125–126, 133). The formal text works only with groups of matrices, as submanifolds of $M_\iota(\mathbb{K})$ with the induced metric. Then the theory of geodesics, Jacobi fields and the index of Parts VI and VII applies to them as it stands.

Definition (the Lie algebras). $\mathfrak{u}(\iota)$ is the set of skew-Hermitian matrices, $X^* = -X$; it is a real Lie subalgebra of $M_\iota(\mathbb{K})$, closed under the commutator since $[X,Y]^* = [Y^*,X^*] = [-Y,-X] = -[X,Y]$ `Chapter12/MatrixGroups · skewHermitian` `Chapter12/MatrixGroups · mem_skewHermitian`. For $\mathbb{K} = \mathbb{R}$ it is $\mathfrak{o}(\iota)$, and a real skew-symmetric matrix has trace 0 `Chapter12/MatrixGroups · trace_eq_zero_of_skew_real`. $\mathfrak{su}(\iota)$ is the set of skew-Hermitian complex matrices of trace 0, again a Lie subalgebra since $\operatorname{tr}[X,Y] = 0$ `Chapter12/MatrixGroups · specialSkewHermitian` `Chapter12/MatrixGroups · mem_specialSkewHermitian`. The Hermitian matrices form a real subspace `Chapter12/MatrixGroups · hermitian`.

The Hermitian part $A \mapsto \frac{1}{2}(A + A^*)$ is a real-linear map onto the Hermitian matrices `Chapter12/MatrixGroups · hermPart` `Chapter12/MatrixGroups · range_hermPart` whose kernel is $\mathfrak{u}(\iota)$ `Chapter12/MatrixGroups · ker_hermPart`. So $\dim\mathfrak{u}(\iota) + \dim\mathrm{Herm}(\iota) = \dim M_\iota(\mathbb{K})$ `Chapter12/MatrixGroups · finrank_skewHermitian_add_finrank_hermitian`.

Definition (matrix group). Let $G \subseteq M_\iota(\mathbb{K})$ be a slice submanifold modelled on Z, and $\mathfrak{g}$ a real Lie subalgebra of $M_\iota(\mathbb{K})$. $(G,\mathfrak{g})$ is a *matrix group* if every $g \in G$ is unitary, G is closed under products and under $g \mapsto g^*$, every $X \in \mathfrak{g}$ is skew-Hermitian, $g\mathfrak{g}g^* \subseteq \mathfrak{g}$ for $g \in G$, $\exp\mathfrak{g} \subseteq G$, and $\dim Z = \dim\mathfrak{g}$ `Chapter12/MatrixGroups · IsMatrixGroup`.

These are exactly the properties used in §§20–23. Every compact Lie group is a closed subgroup of a unitary group (Peter–Weyl), but this is not needed, since the groups of Part IV are given as matrices. Simple consequences: $I = \exp 0 \in G$ `Chapter12/MatrixGroups · IsMatrixGroup.one_mem`, the curves $t \mapsto g\exp(tX)$, $X \in \mathfrak{g}$, lie in G `Chapter12/MatrixGroups · IsMatrixGroup.mul_exp_smul_mem`, and $g^*\mathfrak{g}g \subseteq \mathfrak{g}$ `Chapter12/MatrixGroups · IsMatrixGroup.conj_mem'`.

Theorem (the tangent space). For a matrix group and $g \in G$, the tangent space of G at g (§0.3) is $T_gG = g\mathfrak{g}$ `Chapter12/MatrixGroups · IsMatrixGroup.tangentSubspace_eq`. Equivalently $v \in T_gG$ iff $g^*v \in \mathfrak{g}$ `Chapter12/MatrixGroups · IsMatrixGroup.mem_tangentSubspace_iff` iff $vg^* \in \mathfrak{g}$ `Chapter12/MatrixGroups · IsMatrixGroup.mem_tangentSubspace_iff'`.

The proof as formalized. $g\mathfrak{g}$ is the image of $\mathfrak{g}$ under the isometry $X \mapsto gX$ `Chapter12/MatrixGroups · IsMatrixGroup.leftAlg`, and $v \in g\mathfrak{g}$ iff $g^*v \in \mathfrak{g}$ `Chapter12/MatrixGroups · IsMatrixGroup.mem_leftAlg_iff`.

1. $g\mathfrak{g} \subseteq T_gG$: for $v = gX$, the curve $t \mapsto g\exp(tX)$ lies in G and has velocity gX at $t = 0$. The velocity of a curve in a submanifold is tangent `Chapter7/Geodesic · hasDerivAt_mem_tangentSubspace`.
2. Both spaces have dimension $\dim\mathfrak{g}$: $g\mathfrak{g}$ because $X \mapsto gX$ is injective `Chapter12/MatrixGroups · IsMatrixGroup.finrank_leftAlg`, and T_gG because its dimension is that of the model Z `Chapter9/JacobiSystem · finrank_tanAt`, which is $\dim\mathfrak{g}$ by hypothesis. A subspace of the same finite dimension is the whole space.
3. For the second form, $g^*v \in \mathfrak{g}$ iff $g(g^*v)g^* = vg^* \in \mathfrak{g}$, because conjugation by g and by g^* preserves $\mathfrak{g}$. □

Milnor identifies the tangent space at I with $\mathfrak{u}(n)$ (p. 125, (6)); the formal statement holds at every point, through the curves $g\exp(tX)$ and a dimension count.

Theorem (bi-invariance). For $g \in G$, left and right multiplication by g are linear isometries of $M_\iota(\mathbb{K})$ mapping G onto G `Chapter12/MatrixGroups · IsMatrixGroup.mulLeftIso_image` `Chapter12/MatrixGroups · IsMatrixGroup.mulRightIso_image`. ($gG \subseteq G$ by closure under products, and $x = g(g^*x)$ with $g^*x \in G$; the same on the right.) So the induced metric is invariant under left and right translations.

Theorem ($U(n)$ and $O(n)$). The unitary group $U(\iota) = \{A : A^*A = I\}$ ($O(\iota)$ for $\mathbb{K} = \mathbb{R}$) is a slice submanifold of $M_\iota(\mathbb{K})$ modelled on $\mathfrak{u}(\iota)$ `Chapter12/MatrixGroups · isSliceSubmanifold_unitaryGroup`, and with $\mathfrak{u}(\iota)$ it is a matrix group `Chapter12/MatrixGroups · isMatrixGroup_unitaryGroup`. It is compact `Chapter12/MatrixGroups · isCompact_unitaryGroup`.

The proof as formalized. Milnor's chart is the exponential near 0 (p. 125, (5)), which needs that a small A with $\exp A$ unitary is skew-Hermitian. The formal proof uses a regular level set instead.

1. Let $\Phi(A) = \frac{1}{2}((A^*A - I) + (A^*A - I)^*)$, the Hermitian part of $A^*A - I$, with values in the Hermitian matrices `Chapter12/MatrixGroups · unitaryMap`. Since $A^*A - I$ is already Hermitian, $\Phi(A) = 0$ iff $A^*A = I$ `Chapter12/MatrixGroups · unitaryMap_eq_zero_iff`, so $U(\iota) = \Phi^{-1}(0)$ `Chapter12/MatrixGroups · preimage_unitaryMap`. Φ is smooth `Chapter12/MatrixGroups · contDiff_unitaryMap` with derivative $H \mapsto$ the Hermitian part of $H^*A + A^*H$ `Chapter12/MatrixGroups · hasFDerivAt_unitaryMap`.
2. 0 is a regular value: at a unitary A and a Hermitian S, take $H = \frac{1}{2}AS$. Then $H^*A + A^*H = \frac{1}{2}SA^*A + \frac{1}{2}A^*AS = S$, whose Hermitian part is S `Chapter12/MatrixGroups · isRegularValue_unitaryMap`.
3. The companion project's preimage theorem `Milnor/Chapter2/LevelSet · isSliceSubmanifold_preimage` makes $\Phi^{-1}(0)$ a slice submanifold modelled on any space of dimension $\dim M - \dim \mathrm{Herm} = \dim \mathfrak{u}(\iota)$.
4. The matrix-group axioms are the closure of $U(\iota)$ under products and adjoints, $(gXg^*)^* = gX^*g^* = -gXg^*$, and the unitarity of $\exp X$ for skew-Hermitian X (§XIII.1).
5. Compactness: $U(\iota)$ is closed, as the zero set of a continuous map `Chapter12/MatrixGroups · isClosed_unitaryGroup`, and bounded, since $\|A\|^2 = \mathrm{Re}\,\mathrm{tr}(A^*A) = |\iota|$ `Chapter12/MatrixGroups · norm_sq_of_conjTranspose_mul_self`. □

Theorem ($SO(n)$). $SO(\iota) = \{A \in O(\iota) : \det A = 1\}$ is a slice submanifold of $M_\iota(\mathbb{R})$ modelled on $\mathfrak{o}(\iota)$ `Chapter12/MatrixGroups · isSliceSubmanifold_specialOrthogonalGroup`, a matrix group with $\mathfrak{o}(\iota)$ `Chapter12/MatrixGroups · isMatrixGroup_specialOrthogonalGroup`, and compact `Chapter12/MatrixGroups · isCompact_specialOrthogonalGroup`.

Proof. On $O(\iota)$, $\det A^2 = 1$ `Chapter12/MatrixGroups · det_sq_of_orthogonal`, so $SO(\iota)$ is the intersection of $O(\iota)$ with the open set $\{\det > 0\}$ `Chapter12/MatrixGroups · specialOrthogonalGroup_inter`. An open piece of a slice submanifold is one `Milnor/Chapter8/FramedOps · isSliceSubmanifold_of_locally_eq`. For the axioms, $\exp X$ has determinant 1 because $\mathrm{tr}\, X = 0$ (§XIII.1). Compactness: a closed subset of $O(\iota)$ `Chapter12/MatrixGroups · isClosed_det_eq_one`. □

Theorem ($SU(n)$). For ι nonempty, $SU(\iota) = \{A : A^*A = I, \det A = 1\}$ is a slice submanifold of $M_\iota(\mathbb{C})$ modelled on $\mathfrak{su}(\iota)$ `Chapter12/MatrixGroups · isSliceSubmanifold_specialUnitaryGroup`, a matrix group with $\mathfrak{su}(\iota)$ `Chapter12/MatrixGroups · isMatrixGroup_specialUnitaryGroup`, and compact `Chapter12/MatrixGroups · isCompact_specialUnitaryGroup`.

The proof as formalized. $SU(\iota)$ is **not** a regular level set of $(A^*A - I, \det A - 1)$: on $U(\iota)$ the values of det have modulus 1, so the derivative of det along $U(\iota)$ is never onto $\mathbb{C}$. The formal proof uses the imaginary part of the determinant.

1. Let $\Psi(A) = (\Phi(A), \mathrm{Im}\det A)$ `Chapter12/MatrixGroups · suMap`. On $U(\iota)$, $(\mathrm{Re}\det A)^2 + (\mathrm{Im}\det A)^2 = 1$ `Chapter12/MatrixGroups · normSq_det_of_unitary`. So $\Psi^{-1}(0) = \{A^*A = I, \det A = \pm 1\}$, and $SU(\iota)$ is its open piece $\{\mathrm{Re}\det A > 0\}$ `Chapter12/MatrixGroups · mem_specialUnitaryGroup_iff_level`.
2. The linear map $A \mapsto (\text{Hermitian part of } A, \mathrm{Im}\,\mathrm{tr}\, A)$ `Chapter12/MatrixGroups · suPart` has kernel $\mathfrak{su}(\iota)$ `Chapter12/MatrixGroups · ker_suPart`, because the trace of a skew-Hermitian matrix is imaginary `Chapter12/MatrixGroups · re_trace_of_skew`, and is onto: (S, c) is the image of $S + \frac{c}{n}iI$, since the trace of a Hermitian matrix is real `Chapter12/MatrixGroups · im_trace_of_hermitian` `Chapter12/MatrixGroups · range_suPart`. So $\dim \mathfrak{su} + \dim \mathrm{Herm} + 1 = \dim M_\iota(\mathbb{C})$ `Chapter12/MatrixGroups · finrank_specialSkewHermitian_add`. This needs $n \geq 1$.
3. 0 is a regular value of Ψ `Chapter12/MatrixGroups · isRegularValue_suMap`. Its derivative at A is $H \mapsto (\text{Herm part of } H^*A + A^*H,\ \mathrm{Im}\, d(\det)_A H)$ `Chapter12/MatrixGroups · hasFDerivAt_suMap`. At a point of the level set, $\det A = \delta = \pm 1$. Given (S, c), let $r = c\delta/n$ and $H = AY$ with $Y = \frac{1}{2}S + riI$. Then

$H^*A + A^*H = Y^* + Y = S$, and by Jacobi's formula $d(\det)_A(AY) = \delta \operatorname{tr} Y = \delta(\frac{1}{2}\operatorname{tr} S + rni)$, whose imaginary part is $\delta^2 c = c$.

4. The preimage theorem and the open piece, as for $SO(n)$. The axioms: $\operatorname{tr}(gXg^*) = \operatorname{tr} X$, and $\det\exp X = 1$ for $\operatorname{tr} X = 0$. □

XIII.3 The curvature of a bi-invariant metric, algebraically

Everything Milnor proves about the curvature of a Lie group with a bi-invariant metric (Theorem 21.3, Corollaries 21.4–21.6, the proof of 21.7) is a computation in the Lie algebra, once two geometric facts are granted: left-invariant fields have constant inner products, and their integral curves are geodesics. The formal text does this computation for an abstract real Lie algebra L with a bilinear form B; each hypothesis on B (symmetric, positive definite, ad-invariant) is an explicit assumption of each theorem. §XIII.4 identifies the result with the curvature of the matrix groups.

Definition. B is *ad-invariant* if $B([X,Y],Z) = B(X,[Y,Z])$ for all X, Y, Z (Milnor's 21.3(a)) `Chapter12/BiInvariant · IsAdInvariant`; this is Mathlib's Lie invariance `Chapter12/BiInvariant · isAdInvariant_iff_lieInvariant`. A *connection on left-invariant fields* is a bilinear map $D(X,Y) = X \vdash Y$ on L; it is *Levi-Civita for* B if $X \vdash Y - Y \vdash X = [X,Y]$ (Milnor 8.5) and $B(X \vdash Y, Z) + B(Y, X \vdash Z) = 0$ (8.4 for fields with constant inner products) `Chapter12/BiInvariant · IsLeviCivita`.

That D is a map $L \to L \to L$, i.e. that $\nabla_X Y$ is again left-invariant, is a fact Milnor uses without comment when he applies (d) twice in the curvature; here it is built into the definition.

Theorem 21.3(d) and (a), Milnor's proof. If D is Levi-Civita for B and $X \vdash X = 0$ for all X, then $X \vdash Y = \frac{1}{2}[X,Y]$ `Chapter12/BiInvariant · IsLeviCivita.conn_eq` and B is ad-invariant `Chapter12/BiInvariant · IsLeviCivita.isAdInvariant`.

Proof. $(X+Y) \vdash (X+Y) = 0$ expands to $X \vdash Y + Y \vdash X = 0$; adding the torsion identity $X \vdash Y - Y \vdash X = [X,Y]$ gives $2X \vdash Y = [X,Y]$. For (a), the metric identity for Y, X, Z reads $B(\frac{1}{2}[Y,X],Z) + B(X, \frac{1}{2}[Y,Z]) = 0$, and $[Y,X] = -[X,Y]$. □

The converse route is also proved. The Koszul formula for left-invariant fields, $2B(X \vdash Y, Z) = B([X,Y],Z) - B([Y,Z],X) + B([Z,X],Y)$, follows from three metric identities and three torsion identities for symmetric B `Chapter12/BiInvariant · IsLeviCivita.koszul`. For an ad-invariant, symmetric, non-degenerate B the last two terms cancel, so the Levi-Civita connection is forced to be $\frac{1}{2}[\,,\,]$ `Chapter12/BiInvariant · IsLeviCivita.conn_eq_of_isAdInvariant`. Conversely $\frac{1}{2}[\,,\,]$ `Chapter12/BiInvariant · halfLie` `Chapter12/BiInvariant · halfLie_apply` is Levi-Civita for every ad-invariant B `Chapter12/BiInvariant · isLeviCivita_halfLie`.

Theorem 21.3(b), (c) and Corollary 21.4. With Milnor's sign, $R(X,Y)Z = -X \vdash (Y \vdash Z) + Y \vdash (X \vdash Z) + [X,Y] \vdash Z$ `Chapter12/BiInvariant · curv`. For $D = \frac{1}{2}[\,,\,]$:

$$R(X,Y)Z = \tfrac{1}{4}[[X,Y],Z],$$

`Chapter12/BiInvariant · curv_halfLie` by substitution and the Jacobi identity. For ad-invariant B, $B(R(X,Y)Z,W) = \frac{1}{4}B([X,Y],[Z,W])$ `Chapter12/BiInvariant · curv_inner`, so the sectional curvature is $B(R(X,Y)X,Y) = \frac{1}{4}B([X,Y],[X,Y])$ `Chapter12/BiInvariant · sectional`, which is ≥ 0 for positive definite B `Chapter12/BiInvariant · sectional_nonneg` and vanishes iff $[X,Y] = 0$ `Chapter12/BiInvariant · sectional_eq_zero_iff`. In the now usual sign convention, $R_{\text{std}} = -R$, the formula reads $B(R_{\text{std}}(X,Y)Y,X) = \frac{1}{4}|[X,Y]|^2$ `Chapter12/BiInvariant · sectional_std`. Milnor's footnote, that $(X,Y,Z) \mapsto B([X,Y],Z)$ is alternating (Cartan's 3-form), is the two sign changes under $X \leftrightarrow Y$ and, using invariance, $Y \leftrightarrow Z$ `Chapter12/BiInvariant · cartan_form_swap₁₂` `Chapter12/BiInvariant · cartan_form_swap₂₃`.

The operator K_V. For the proof of 21.7, Milnor considers $K_V(W) = R(V,W)V$. Here

$$K_V = -\tfrac{1}{4}\operatorname{ad}_V \circ \operatorname{ad}_V$$

Chapter12/BiInvariant · jacobiOp Chapter12/BiInvariant · jacobiOp_apply, and $R(V,W)V = K_V(W)$ for $D = \frac{1}{2}[\,,\,]$ Chapter12/BiInvariant · curv_halfLie_eq_jacobiOp. Let B be ad-invariant. Then ad_V is skew-symmetric, $B([V,X],Y) = -B(X,[V,Y])$ Chapter12/BiInvariant · ad_skew, hence K_V is self-adjoint Chapter12/BiInvariant · jacobiOp_symm and $B(K_V W, W) = \frac{1}{4}B([V,W],[V,W])$ Chapter12/BiInvariant · jacobiOp_self. For positive definite B: K_V is positive semi-definite Chapter12/BiInvariant · jacobiOp_self_nonneg, its kernel is the centralizer of V Chapter12/BiInvariant · jacobiOp_eq_zero_iff, its eigenvalues are ≥ 0 Chapter12/BiInvariant · eigenvalue_nonneg, and the only real eigenvalue of ad_V is 0 Chapter12/BiInvariant · ad_eigenvalue_eq_zero (if $[V,X] = aX$ then $aB(X,X) = B([V,X],X) = -B(X,[V,X]) = -aB(X,X)$).

Theorem (the non-zero eigenvalues of K_V occur in pairs). Let L be finite-dimensional and B ad-invariant and positive definite. For every $V \in L$ and $\mu \neq 0$, the eigenspace E_μ of K_V has even dimension Chapter12/BiInvariant · even_finrank_eigenspace.

The proof as formalized. Milnor quotes a normal form: an orthonormal basis in which $\mathrm{Ad}\, V$ is block diagonal with blocks $\begin{pmatrix} 0 & a_i \\ -a_i & 0 \end{pmatrix}$ (p. 117). The formal proof uses only what "occur in pairs" needs.

1. $K_V W = \mu W$ iff $\mathrm{ad}_V^2 W = -4\mu W$ Chapter12/BiInvariant · lie_lie_of_jacobiOp Chapter12/BiInvariant · jacobiOp_of_lie_lie, and ad_V maps E_μ into itself, since it commutes with ad_V^2 Chapter12/BiInvariant · lie_mem_eigenspace.
2. If $\mu < 0$, $E_\mu = 0$ because the eigenvalues are ≥ 0.
3. If $\mu > 0$, let $J = (2\sqrt{\mu})^{-1}\,\mathrm{ad}_V$ on E_μ Chapter12/BiInvariant · eigenJ Chapter12/BiInvariant · eigenJ_apply. Then $J^2 = (4\mu)^{-1}\,\mathrm{ad}_V^2 = -1$ on E_μ Chapter12/BiInvariant · eigenJ_comp_self. A real vector space with $J^2 = -1$ has even dimension, since $\det(J)^2 = \det(-1) = (-1)^{\dim}$ and a square is not negative Chapter12/BiInvariant · even_finrank_of_comp_self_eq_neg_id. □

J is moreover orthogonal for B Chapter12/BiInvariant · eigenJ_isometry, so an orthonormal basis of E_μ can be chosen in pairs w, Jw, on which ad_V has Milnor's blocks with $a = 2\sqrt{\mu}$. Combined with the evenness of the index computed from a spectrum with even multiplicities (§XIII.15), this is 21.7's "the index of every geodesic is even".

Corollary 21.5, the algebraic step. Let L be finite-dimensional, B symmetric, positive definite and ad-invariant, and let the centre of L be trivial. Then the Ricci curvature $\mathrm{tr}\, K_V$ is > 0 for $V \neq 0$ Chapter12/BiInvariant · trace_jacobiOp_pos.

Proof. $\mathrm{tr}\, K_V = -\frac{1}{4}\kappa(V,V)$ with κ the Killing form Chapter12/BiInvariant · trace_jacobiOp. Take a B-orthogonal basis b_i (Mathlib) Mathlib/LinearAlgebra/QuadraticForm/Basic · LinearMap.BilinForm.exists_orthogonal_basis. The i-th diagonal entry of K_V is $B(K_V b_i, b_i)/B(b_i,b_i)$ Chapter12/BiInvariant · trace_eq_sum_of_isOrtho, and each is ≥ 0. If $[V,b_i] = 0$ for all i, then $\mathrm{ad}_V = 0$ and V is central, so $V = 0$. Hence some $[V,b_i] \neq 0$, and that diagonal entry is $\frac{1}{4}B([V,b_i],[V,b_i])/B(b_i,b_i) > 0$. □

Equivalently, the Killing form is negative definite Chapter12/BiInvariant · killingForm_neg. Milnor's "bounded away from zero" and the application of Myers' theorem are not formalized.

Corollary 21.6, the algebraic step. For B ad-invariant, the orthogonal complement $\mathfrak{g}' = \mathfrak{c}^\perp$ of the centre is a Lie *ideal* Chapter12/BiInvariant · centerCompl Chapter12/BiInvariant · mem_centerCompl (Mathlib's orthogonal of an ideal for an invariant form) Mathlib/Algebra/Lie/InvariantForm · LieAlgebra.InvariantForm.orthogonal. Milnor shows it is a subalgebra; the same line shows it is an ideal, which is what the splitting needs. For B also symmetric and positive definite, $L = \mathfrak{c} \oplus \mathfrak{g}'$ Chapter12/BiInvariant · isCompl_center_centerCompl (a vector in both is orthogonal to itself), and $\mathfrak{g}'$ has trivial centre Chapter12/BiInvariant · eq_zero_of_mem_centerCompl: an $X \in \mathfrak{g}'$ commuting with $\mathfrak{g}'$ commutes with $\mathfrak{c} + \mathfrak{g}' = L$, so lies in $\mathfrak{c} \cap \mathfrak{g}' = 0$. The splitting of the simply connected group is not formalized.

XIII.4 Geodesics and curvature of a matrix group

For a matrix group $(G, \mathfrak{g})$ every object of §21 is computed directly in the embedding $G \subseteq M_\iota(\mathbb{K})$, without left-invariant vector fields.

Two matrix identities. For all square matrices,

$$\operatorname{tr}\big((XZ+ZX)(YW+WY)\big) - \operatorname{tr}\big((XW+WX)(YZ+ZY)\big) = -\operatorname{tr}\big([[X,Y],Z]W\big)$$

`Chapter12/GroupGeodesics · trace_gauss_identity`, by expanding and cycling traces. For skew-Hermitian X, Z the anticommutator $XZ + ZX$ is Hermitian `Chapter12/GroupGeodesics · conjTranspose_anticomm` and $[[X,Y],Z]$ is skew-Hermitian `Chapter12/GroupGeodesics · conjTranspose_lie_lie`. So for skew-Hermitian X, Y, Z, W the identity becomes, in inner products,

$$\langle XZ+ZX, YW+WY\rangle - \langle XW+WX, YZ+ZY\rangle = \langle [[X,Y],Z],W\rangle$$

`Chapter12/GroupGeodesics · inner_gauss_identity`. And for skew-Hermitian X, Y, $\langle [X,Y],Z\rangle = \langle X,[Y,Z]\rangle$ `Chapter12/GroupGeodesics · inner_lie_left`.

Definition (the projection onto $\mathfrak{g}$). $P_{\mathfrak{g}}$ is the orthogonal projection of $M_\iota(\mathbb{K})$ onto $\mathfrak{g}$ `Chapter12/GroupGeodesics · algProj` `Chapter12/GroupGeodesics · algProj_mem` `Chapter12/GroupGeodesics · algProj_of_mem`. If $\mathfrak{g}$ consists of skew-Hermitian matrices, $P_{\mathfrak{g}}$ kills the Hermitian matrices `Chapter12/GroupGeodesics · algProj_hermitian`, so $P_{\mathfrak{g}}(X+H) = X$ for $X \in \mathfrak{g}$, H Hermitian `Chapter12/GroupGeodesics · algProj_add_hermitian`. Writing $XY = \frac{1}{2}[X,Y] + \frac{1}{2}(XY+YX)$ gives $P_{\mathfrak{g}}(XY) = \frac{1}{2}[X,Y]$ for $X, Y \in \mathfrak{g}$ `Chapter12/GroupGeodesics · algProj_mul`.

Definition. Milnor's $\langle\,,\rangle$ on $\mathfrak{g}$ is the Frobenius inner product as a bilinear form on $\mathfrak{g}$ `Chapter12/GroupGeodesics · frobForm` `Chapter12/GroupGeodesics · frobForm_apply`. It is ad-invariant `Chapter12/GroupGeodesics · isAdInvariant_frobForm` (this is Milnor's $\langle SAS^{-1}, SBS^{-1}\rangle = \langle A, B\rangle$, differentiated) and positive definite `Chapter12/GroupGeodesics · frobForm_pos`. So all of §XIII.3 applies to $(\mathfrak{g}, \langle\,,\rangle)$.

Theorem (the tangent projection). For $g \in G$ the orthogonal projection of $M_\iota(\mathbb{K})$ onto T_gG (§0.3) is $M \mapsto g\,P_{\mathfrak{g}}(g^*M)$ `Chapter12/GroupGeodesics · IsMatrixGroup.tanProj_eq` `Chapter12/GroupGeodesics · IsMatrixGroup.prj_eq`.

Proof. $gP_{\mathfrak{g}}(g^*M) \in g\mathfrak{g} = T_gG$. For $w \in T_gG$, write $w = g(g^*w)$ and $M = g(g^*M)$; left multiplication by g is an isometry, so $\langle M - gP_{\mathfrak{g}}(g^*M), w\rangle = \langle g^*M - P_{\mathfrak{g}}(g^*M), g^*w\rangle = 0$, since $g^*w \in \mathfrak{g}$. The characterization of the orthogonal projection does the rest. □

Theorem (one-parameter subgroups and their translates are geodesics). For $g \in G$ and $X \in \mathfrak{g}$, $t \mapsto g\exp(tX)$ is a geodesic of G on all of $\mathbb{R}$ (Part VI) `Chapter12/GroupGeodesics · IsMatrixGroup.isGeodesicOn_mul_exp_smul`; in particular so is $t \mapsto \exp(tX)$ `Chapter12/GroupGeodesics · IsMatrixGroup.isGeodesicOn_exp_smul`.

Proof. A geodesic of a submanifold is a smooth curve in it whose acceleration is normal `Chapter7/Geodesic · IsGeodesicOn`. The curve lies in G and is smooth. Its acceleration is $\gamma(t)X^2$ `Chapter12/GroupGeodesics · deriv_deriv_mul_exp_smul`. For $u \in T_{\gamma(t)}G$, $u = \gamma(t)(\gamma(t)^*u)$ with $\gamma(t)^*u \in \mathfrak{g}$, and $\langle\gamma(t)X^2, u\rangle = \langle X^2, \gamma(t)^*u\rangle = 0$, because X^2 is Hermitian, $(X^2)^* = (X^*)^2 = (-X)^2 = X^2$, and Hermitian and skew-Hermitian matrices are orthogonal. □

Milnor obtains $X \vdash X = 0$ from "the integral curves of X are left translates of one-parameter subgroups, and therefore are geodesics" (p. 113), which uses Lemma 21.2 and that left translations are isometries. In the embedding the acceleration is simply computed.

Theorem (every geodesic is a translate of a one-parameter subgroup). Let γ be a geodesic of G on a preconnected open set $s \ni t_0$. Then $X = \gamma(t_0)^*\gamma'(t_0) \in \mathfrak{g}$, and $\gamma(t) = \gamma(t_0)\exp((t-t_0)X)$ on s `Chapter12/GroupGeodesics · IsMatrixGroup.eqOn_mul_exp_smul`.

Proof. $X \in \mathfrak{g}$ because $\gamma'(t_0) \in T_{\gamma(t_0)}G$ `Chapter7/Geodesic · IsGeodesicOn.deriv_mem`. The curve $t \mapsto g'\exp(tX)$ with $g' = \gamma(t_0)\exp(-t_0X) \in G$ is a geodesic on $\mathbb{R}$ and equals $\gamma(t_0)\exp((t-t_0)X)$; at

t_0 it has the value $\gamma(t_0)$ and the velocity $\gamma(t_0)X = \gamma'(t_0)$. Two geodesics on a preconnected open set with the same value and velocity at one point agree `Chapter7/Geodesic · IsGeodesicOn.eqOn`. □

Lemma 21.2. (i) A geodesic $\gamma : \mathbb{R} \to G$ with $\gamma(0) = I$ satisfies $\gamma'(0) \in \mathfrak{g}$, $\gamma(t) = \exp(t\gamma'(0))$, and $\gamma(s+t) = \gamma(s)\gamma(t)$ `Chapter12/GroupGeodesics · IsMatrixGroup.lemma_21_2`. (ii) A differentiable one-parameter subgroup θ of G (with $\theta(0) = I$ and $\theta(s+t) = \theta(s)\theta(t)$) is a geodesic `Chapter12/GroupGeodesics · IsMatrixGroup.isGeodesicOn_of_oneParam`.

The proof as formalized. (i) is the previous theorem at $t_0 = 0$, and the subgroup property is $\exp((s+t)X) = \exp(sX)\exp(tX)$. (ii): first, a differentiable one-parameter subgroup of matrices is $t \mapsto \exp(t\theta'(0))$ `Chapter12/GroupGeodesics · IsMatrixGroup.eq_exp_of_oneParam`. Let $X = \theta'(0)$. Differentiating $\theta(t)\theta(s) = \theta(s)\theta(t)$ in s at 0 gives $\theta(t)X = X\theta(t)$; differentiating $\theta(t+u) = \theta(t)\theta(u)$ in u gives $\theta'(t) = \theta(t)X$. Then $F(s) = \exp(-sX)\theta(s)$ has derivative $-\exp(-sX)X\theta(s) + \exp(-sX)\theta(s)X = 0$, so $F \equiv F(0) = I$ and $\theta(t) = \exp(tX)F(t) = \exp(tX)$. Second, $X \in T_IG = \mathfrak{g}$, because θ is a curve in G through I. So θ is the geodesic $t \mapsto \exp(tX)$. □

Milnor's proof of (i) passes from $\gamma(nt) = \gamma(t)^n$ through commensurable parameters to all of $\mathbb{R}$ "by continuity", which needs the continuity of multiplication as well. The formal proof reads the subgroup property off the exponential.

Consequences. The Riemannian exponential at g (Part VI) is $\exp_g(gX) = g\exp X$ for $X \in \mathfrak{g}$ `Chapter12/GroupGeodesics · IsMatrixGroup.exp_eq`: the geodesic $t \mapsto g\exp(tX)$ starts at g with velocity gX `Chapter7/Exp · IsGeodesicFrom.exp_eq`. And a matrix group is geodesically complete `Chapter12/GroupGeodesics · IsMatrixGroup.isGeodesicallyComplete`: the geodesic with initial vector $v \in T_qG$ is $t \mapsto q\exp(tq^*v)$, defined on $\mathbb{R}$.

Lemma 21.1 (G is a symmetric space). For unitary τ, $I_\tau(A) = \tau A^*\tau$ is an involutive linear isometry of $M_\iota(\mathbb{K})$ `Chapter12/GroupGeodesics · IsMatrixGroup.symmIso` `Chapter12/GroupGeodesics · IsMatrixGroup.symmIso_apply`. On G it is Milnor's $\sigma \mapsto \tau\sigma^{-1}\tau$, since $\sigma^{-1} = \sigma^*$. For $\tau \in G$ it maps G into G `Chapter12/GroupGeodesics · IsMatrixGroup.symmIso_mem`, fixes τ `Chapter12/GroupGeodesics · IsMatrixGroup.symmIso_self`, and reverses the geodesics through τ: $I_\tau(\tau\exp(tX)) = \tau\exp(tX)^*\tau = \tau\exp(-tX)$ for $X \in \mathfrak{g}$ `Chapter12/GroupGeodesics · IsMatrixGroup.symmIso_mul_exp_smul`. That I_τ is an isometry of M_ι is the composite of the isometries $A \mapsto A^*$, $A \mapsto A\tau$, $A \mapsto \tau A$.

Theorem (the second fundamental form). For $g \in G$ and $X, Y \in \mathfrak{g}$, the second fundamental form of G (Part V) is

$$\mathrm{II}_g(gX, gY) = \tfrac{1}{2}\, g(XY + YX)$$

`Chapter12/GroupGeodesics · IsMatrixGroup.secondFundForm_mul`.

Proof. Along a geodesic $\gamma'' = \mathrm{II}(\gamma', \gamma')$ `Chapter7/Geodesic · IsGeodesicOn.deriv_deriv_eq`. For $\gamma(t) = g\exp(tX)$ at $t = 0$ this gives $\mathrm{II}_g(gX, gX) = gX^2$ `Chapter12/GroupGeodesics · IsMatrixGroup.secondFundForm_mul_self`. Apply it to $X + Y$, expand by bilinearity, and use the symmetry of II `Chapter6/Submanifold · secondFundForm_comm`: $gX^2 + 2\mathrm{II}(gX, gY) + gY^2 = g(X+Y)^2$. □

Theorem 21.3, for matrix groups. (d) The covariant derivative (§0.9) of the left-invariant field $q \mapsto qY$ in the direction gX is $g \cdot \frac{1}{2}[X, Y]$ `Chapter12/GroupGeodesics · IsMatrixGroup.covAt_mul_right`. (b) $R(gX, gY)gZ = \frac{1}{4}g[[X, Y], Z]$ for $X, Y, Z \in \mathfrak{g}$, with Milnor's sign `Chapter12/GroupGeodesics · IsMatrixGroup.curvOp_mul`. (c) $\langle R(gX, gY)gZ, gW\rangle = \frac{1}{4}\langle[X, Y], [Z, W]\rangle$ `Chapter12/GroupGeodesics · IsMatrixGroup.inner_curvOp_mul`. Corollary 21.4: $\langle R(gX, gY)gX, gY\rangle = \frac{1}{4}\|[X, Y]\|^2$ `Chapter12/GroupGeodesics · IsMatrixGroup.sectional_mul`, which is ≥ 0 and vanishes iff $XY = YX$ `Chapter12/GroupGeodesics · IsMatrixGroup.sectional_mul_nonneg`.

The proof as formalized. (d) The covariant derivative of a field on a submanifold is the tangential part of its ordinary derivative `Chapter6/Submanifold · covAt` `Chapter6/Projection · dN_comp_val`. The derivative of $q \mapsto qY$ in the direction gX is gXY, and its projection is $gP_{\mathfrak{g}}(g^*gXY) = gP_{\mathfrak{g}}(XY) = g \cdot \frac{1}{2}[X, Y]$.

(b) Milnor substitutes (d) into the formula for R, which needs (d) for all left-invariant fields. The formal proof uses instead the Gauss equation of Part V, $\langle R(x,y)z,w\rangle = \langle \mathrm{II}(x,z),\mathrm{II}(y,w)\rangle - \langle \mathrm{II}(x,w),\mathrm{II}(y,z)\rangle$ `Chapter6/Submanifold · inner_curvOp`.

1. Both $R(gX,gY)gZ$ `Chapter6/Submanifold · curvOp_mem` and $\frac{1}{4}g[[X,Y],Z]$ are tangent at g. A tangent vector orthogonal to T_gG is 0 `Chapter6/Projection · eq_zero_of_mem_of_mem_orthogonal`, so it suffices to compare inner products with every $w = gW$, $W \in \mathfrak{g}$.
2. By the second fundamental form and the isometry $A \mapsto gA$, the Gauss equation gives $\frac{1}{4}(\langle XZ + ZX, YW + WY\rangle - \langle XW + WX, YZ + ZY\rangle)$, which is $\frac{1}{4}\langle [[X,Y],Z],W\rangle$ by the Gauss-equation identity.

(c) and 21.4 follow from (b), the isometry, and $\langle [X,Y],Z\rangle = \langle X,[Y,Z]\rangle$. □

So the curvature of the submanifold G *is* the algebraic curvature of §XIII.3: $R(gX,gY)gZ = g\cdot R_{\frac{1}{2}[,]}(X,Y)Z$ for $X,Y,Z \in \mathfrak{g}$ `Chapter12/GroupGeodesics · IsMatrixGroup.curvOp_eq_curv_halfLie`, also for the trilinear curvature tensor of Lemma 9.1 `Chapter12/GroupGeodesics · IsMatrixGroup.coe_curvTensor_eq_curv_halfLie`, and $R(gA,gW)gA = g\,K_A(W)$ `Chapter12/GroupGeodesics · IsMatrixGroup.curvOp_eq_jacobiOp`.

XIII.5 Jacobi fields along one-parameter subgroups: Theorem 20.5 for matrix groups

Milnor's Theorem 20.5 (p. 110) says that in a locally symmetric space the conjugate points along a geodesic γ with velocity V are governed by the eigenvalues e_i of $K_V(W) = R(V,W)V$: they are the $\gamma(\pi k/\sqrt{e_i})$, and the multiplicity of $\gamma(t)$ is the number of e_i for which t is a multiple of $\pi/\sqrt{e_i}$. His proof extends an eigenbasis U_i of K_V by parallel transport, uses local symmetry to keep $R(V,U_i)V = e_iU_i$, and reduces the Jacobi equation to $w_i'' + e_iw_i = 0$. Milnor's §20 precedes §21, but for a matrix group the formal proof of 20.5 needs the curvature of §XIII.4, so it comes here. For $\gamma(t) = \exp(tA)$, $A \in \mathfrak{g}$, the parallel transport is explicit.

Definition. $h(t) = \exp(tA/2)$ `Chapter12/GroupJacobi · halfExp`, so that $h(t)^2 = \exp(tA)$ `Chapter12/GroupJacobi · halfExp_mul_halfExp`, $h(0) = I$ `Chapter12/GroupJacobi · halfExp_zero`, $h'(t) = h(t)\cdot\frac{1}{2}A$ `Chapter12/GroupJacobi · hasDerivAt_halfExp`, $h(t)$ commutes with A `Chapter12/GroupJacobi · halfExp_mul_comm`, and so does $h(t)^*$ `Chapter12/GroupJacobi · IsMatrixGroup.conjTranspose_halfExp_mul_comm`. $h(t) \in G$ `Chapter12/GroupJacobi · IsMatrixGroup.halfExp_mem`. A field along γ is written $J(t) = h(t)U(t)h(t)$ with $U(t) \in \mathfrak{g}$. Since $hUh = \gamma(t)\cdot h^*Uh$ and $h^*Uh \in \mathfrak{g}$, such a field is tangent. The operator of 20.5 is, as a matrix expression,

$$K_A(U) = \tfrac{1}{4}[[A,U],A]$$

`Chapter12/GroupJacobi · jacK` `Chapter12/GroupJacobi · smul_mul_eq_mul_jacK`.

Lemma (the covariant derivative in this frame). For a differentiable $\mathfrak{g}$-valued U, the covariant derivative (§0.9) along $\exp(tA)$ of $t \mapsto h(t)U(t)h(t)$ is $h(t)U'(t)h(t)$ `Chapter12/GroupJacobi · IsMatrixGroup.covD_conj`.

Proof. The ordinary derivative is $h(\frac{1}{2}(AU + UA) + U')h$ `Chapter12/GroupJacobi · hasDerivAt_conj`, by the product rule and $hA = Ah$. $U'(t) \in \mathfrak{g}$, as the derivative of a curve in a subspace `Chapter12/GroupJacobi · deriv_mem_of_forall_mem`. The covariant derivative is the tangential projection `Chapter8/FirstVariation · covD`, $\gamma P_{\mathfrak{g}}(\gamma^*\cdot)$. Here $\gamma^*\cdot h(\ldots)h = h^*U'h + h^*\frac{1}{2}(AU + UA)h$. The second term is Hermitian ($AU + UA$ is), and $P_{\mathfrak{g}}$ kills it; the first lies in $\mathfrak{g}$. So the projection is $\gamma\, h^*U'h = hU'h$. □

Lemma (the curvature term). For $M \in \mathfrak{g}$, $R(\gamma', hMh)\gamma' = h\,K_A(M)\,h$ along $\gamma(t) = \exp(tA)$ `Chapter12/GroupJacobi · IsMatrixGroup.curvR_conj`.

Proof. $\gamma'(t) = \gamma(t)A$ `Chapter12/GroupJacobi · deriv_exp_smul` and $hMh = \gamma\cdot(h^*Mh)$. By Theorem 21.3(b) for G, $R(\gamma A,\gamma M')\gamma A = \gamma K_A(M')$ with $M' = h^*Mh$ `Chapter8/SecondVariation · curvR`. Since h is unitary and

commutes with A, $K_A(h^*Mh) = h^*K_A(M)h$ `Chapter12/GroupJacobi · IsMatrixGroup.jacK_conj`, and $\gamma h^* = h$. □

Theorem 20.5 for matrix groups, the Jacobi equation. Let $A \in \mathfrak{g}$ and U a smooth $\mathfrak{g}$-valued curve with $U'' + K_A(U) = 0$. Then $J(t) = h(t)U(t)h(t)$ is a Jacobi field along $t \mapsto \exp(tA)$ `Chapter12/GroupJacobi · IsMatrixGroup.isJacobiField_conj`, and its covariant derivative at 0 is $U'(0)$ `Chapter12/GroupJacobi · IsMatrixGroup.covAlong_conj_zero`.

Proof. A Jacobi field along a curve in a submanifold is a smooth tangent field with $D^2J/dt^2 + R(\gamma', J)\gamma' = 0$ `Chapter8/Jacobi · IsJacobiField` `Chapter8/Jacobi · jacobiExpr`. J is smooth and tangent (its projection is itself, as computed above). By the lemma applied twice, $D^2J/dt^2 = hU''h$, and the curvature term is $hK_A(U)h$; their sum is $h(U'' + K_A(U))h = 0$. At $t = 0$, $h(0) = I$, so $DJ/dt(0) = U'(0)$. □

The Jacobi equation has become a linear equation with *constant* coefficients, $U'' + K_A(U) = 0$ on $\mathfrak{g}$: this is Milnor's $w_i'' + e_i w_i = 0$, without choosing the eigenbasis. The formal statement is the implication from the equation to the Jacobi field, which is all that the count of §XIII.17 uses: there every Jacobi field vanishing at 0 is shown to be one of these by uniqueness. The parameter of a conjugate point is a time, and K is computed for the velocity A of the geodesic actually used on $[0, 1]$.

XIII.6 Jiggling a function does not lower the index: Lemma 22.4

Milnor's Lemma 22.4 (p. 119): let K be a compact subset of an open $U \subseteq \mathbb{R}^n$ and $f : U \to \mathbb{R}$ smooth with every critical point in K of index $\geq \lambda_0$; if g has first and second partial derivatives within ε of those of f on K, ε small, then every critical point of g in K has index $\geq \lambda_0$. His proof uses the continuity of the λ-th eigenvalue of the Hessian and ends with "a uniform continuity argument which will be left to the reader" (p. 121). The formal proof uses no eigenvalues. The index of a symmetric bilinear form is the largest dimension of a subspace on which it is negative definite (§0.2) `Chapter1/Nondegenerate · index_isGreatest`, and negative definiteness on a fixed subspace survives small perturbations.

Lemma (a negative definite form has a margin). Let E be finite-dimensional, H a bilinear form on E and $W \subseteq E$ a subspace with $H(v, v) < 0$ for $0 \neq v \in W$. Then there is $\mu > 0$ with $H(v, v) \leq -\mu\|v\|^2$ on W `Chapter13/Perturbation · exists_negDef_margin`.

Proof. The unit sphere S of W is compact. If it is empty, $W = 0$ and any μ works. Otherwise $v \mapsto H(v, v)$ attains its maximum on S at some $v_0 \neq 0$; take $\mu = -H(v_0, v_0) > 0$. For $v \neq 0$ in W, $H(v, v) = \|v\|^2 H(v/\|v\|, v/\|v\|) \leq \|v\|^2 H(v_0, v_0)$. □

Lemma (forms near a form of index $\geq \lambda$ have index $\geq \lambda$). If $H(v, v) \leq -\mu\|v\|^2$ on W and $\|H' - H\| < \mu$ (operator norm), then $\dim W \leq \operatorname{index} H'$ `Chapter13/Perturbation · le_index_of_norm_sub_lt`.

Proof. For $0 \neq v \in W$, $|(H' - H)(v, v)| \leq \|H' - H\|\|v\|^2 < \mu\|v\|^2$, so $H'(v, v) < H(v, v) + \mu\|v\|^2 \leq 0$. A form negative definite on W has index $\geq \dim W$ `Chapter1/Nondegenerate · finrank_le_index_of_negDef`. □

Lemma 22.4. Let E be a finite-dimensional real normed space, $U \subseteq E$ open, $K \subseteq U$ compact, $f : E \to \mathbb{R}$ of class C^2 on U, and suppose every critical point of f in K has index $\geq \lambda_0$. Then there is $\varepsilon > 0$ such that every $g : E \to \mathbb{R}$ with $\|dg_x - df_x\| < \varepsilon$ and $\|d^2g_x - d^2f_x\| < \varepsilon$ at each $x \in K$ has all its critical points in K of index $\geq \lambda_0$ `Chapter13/Perturbation · lemma_22_4`.

The closeness is measured by operator norms of the differences of derivatives, equivalent to Milnor's partial derivatives up to constants, and f is only C^2. No regularity is assumed of g beyond what the hypotheses on its derivatives say.

The proof as formalized.

1. Let $P(\varepsilon, y)$ be the statement: every g with $\|dg_y - df_y\| < \varepsilon$ and $\|d^2g_y - d^2f_y\| < \varepsilon$ that is critical at y has index $\geq \lambda_0$ at y. It is monotone in ε. df and d^2f are continuous on U.
2. *Locally.* For each $x \in K$, $P(\varepsilon, y)$ holds for all (ε, y) near $(0, x)$.

- If $df_x = 0$, x is critical, with index $\geq \lambda_0$. Take a subspace W of dimension index d^2f_x on which d^2f_x is negative definite `Chapter1/Nondegenerate · exists_negDef_finrank_eq_index`, and a margin μ for it. For y near x, $\|d^2f_y - d^2f_x\| < \mu/2$; for $\varepsilon < \mu/2$ and g as in $P(\varepsilon, y)$, $\|d^2g_y - d^2f_x\| < \mu$, so index $d^2g_y \geq \dim W = \text{index}\, d^2f_x \geq \lambda_0$.
- If $df_x \neq 0$, let $\eta = \|df_x\|/2$. For y near x, $\|df_y - df_x\| < \eta$. For $\varepsilon < \eta$, a g critical at y would have $\|df_y\| = \|dg_y - df_y\| < \eta$, so $\|df_x\| < 2\eta$, which is false; $P(\varepsilon, y)$ holds vacuously.

3. *Uniformly.* A property holding near $(0, x)$ for every x of a compact K holds for all $y \in K$ and all ε in a neighbourhood of 0 `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually`. Choose such an $\varepsilon > 0$. □

This compactness argument is the "uniform continuity argument" Milnor leaves to the reader.

Lemma 22.4 for a perturbation $g = f + h$. With the same hypotheses there is $\varepsilon > 0$ such that for every h of class C^2 on U with $\|dh\|$ and $\|d^2h\|$ (as iterated derivatives) $< \varepsilon$ on K, every critical point of $f + h$ in K has index $\geq \lambda_0$ `Chapter13/Perturbation · lemma_22_4_add`. Indeed $d(f + h) - df = dh$ and $d^2(f + h) - d^2f = d^2h$ where both are C^2 `Chapter13/Perturbation · fderiv_hessian_add`. This is the form in which the approximation theorems of Part III deliver g.

Lemma 22.4 on a submanifold, for linear perturbations. Let $N \subseteq V$ be a slice submanifold, $F : V \to \mathbb{R}$ smooth and $K \subseteq N$ compact, and suppose every critical point of $F|N$ in K has index $\geq \lambda_0$ (§0.2). Then there is $\delta > 0$ such that for every linear form a on V with $\|a\| < \delta$, every critical point of $(F + a)|N$ in K has index $\geq \lambda_0$ `Chapter13/Perturbation · lemma_22_4_submanifold`.

The proof as formalized. Milnor applies 22.4 "in a coordinate neighbourhood"; derivatives on a manifold exist only in charts.

1. *In one slice chart* `Chapter13/Perturbation · exists_perturb_slice`. Let σ be a slice chart with parametrization ϕ on its open target T, and $z_0 \in T$. Choose $r > 0$ with $\bar{B}(z_0, r) \subseteq T$, and let $K_z = \bar{B}(z_0, r) \cap \phi^{-1}(K)$, compact. Critical points of $F|N$ correspond to critical points of $F \circ \phi$ `Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_sliceParam` with the same index `Chapter1/ManifoldSubmanifold · index_hessian_comp_sliceParam`. So Lemma 22.4 for a perturbation applies to $F \circ \phi$ on K_z and gives ε.
2. The first two derivatives of ϕ are bounded on the closed ball, by C_1 and C_2; let $C = |C_1| + |C_2| + 1$. For linear a, $d^i(a \circ \phi) = a \circ d^i\phi$, of norm $\leq \|a\|C$. So $\|a\| < \varepsilon/C$ makes both derivatives of $a \circ \phi$ smaller than ε on K_z, and every critical point of $(F + a)|N$ in K whose chart value lies in $B(z_0, r)$ has index $\geq \lambda_0$. Take $\delta = \varepsilon/C$.
3. *Globally.* Every $q \in K$ lies in a slice chart; step 1 at $z_0 = \sigma(q)$ gives a neighbourhood of q and a δ. The compactness of K, through the same uniformity lemma as above, gives one δ. □

XIII.7 Pushing maps past critical levels of high index

In the proof of Lemma 22.5 (p. 122) Milnor says: $g^{-1}(-\infty, c + \delta]$ has the homotopy type of $g^{-1}(-\infty, 2\delta]$ with cells of dimension $\geq \lambda_0$ attached, and since $r < \lambda_0$ the map $h : (I^r, \partial I^r) \to (M, M^0)$ is homotopic to a map into $g^{-1}(-\infty, 2\delta]$. A homotopy type is not enough for this: the argument needs deformations that fix the lower level, and the push-off lemma of cellular approximation. The formal text proves the step in the following form.

Definition (pushing maps). Let $A, X \subseteq T$ and P a space. *A pushes maps from P out of X* if every map $h : P \to T$ with values in X is homotopic, relative to $h^{-1}(A)$, to a map with values in A `Chapter13/Compression · PushesInto`. The homotopy may move in all of T.

For $P = I^r$ and maps sending ∂I^r into A this is Milnor's "$\pi_r(X, A) = 0$" in the strong form of the exposition (relative to the boundary and at all base points at once), and it is the compression of Part XII (§XII.2) for the inclusion, when $X = T$. Every set pushes maps out of itself `Chapter13/Compression · pushesInto_self`;

pushing out of X implies pushing out of any $X' \subseteq X$ `Chapter13/Compression · PushesInto.mono`; and pushing is transitive.

Lemma (pushing in two steps). If $A \subseteq B$, A pushes maps from P out of B, and B pushes maps from P out of X, then A pushes maps from P out of X `Chapter13/Compression · PushesInto.trans`.

Proof. Push h into B by F, relative to $h^{-1}(B)$, then the result g into A by G, relative to $g^{-1}(A)$. A homotopy relative to a set is relative to every subset `Homotopy/Basic · homotopyRelMono`. $h^{-1}(A) \subseteq h^{-1}(B)$, and on $h^{-1}(A)$, $g = h$ (since F fixes $h^{-1}(B)$), so $h^{-1}(A) \subseteq g^{-1}(A)$. The concatenation of F and G is relative to $h^{-1}(A)$. □

Lemma (a strong deformation retract pushes maps). If $A \subseteq X$ and A is a strong deformation retract of X (§0.4), A pushes maps from any P out of X `Chapter13/Compression · pushesInto_of_isStrongDeformationRetract`. *Proof*: compose h with the deformation G; at time 1 the map lies in A, and G fixes A at all times. The homotopy is assembled from its two ends and its fixed set `Homotopy/CellAttach · homotopyRelOfFun`. □

Lemma (pushing off cells). Let T be Hausdorff, $A \subseteq T$ closed, P compact and injected continuously into a real vector space E. Let $\chi_1, \dots, \chi_k$ be embedded closed cells of dimensions $n_i > \dim E$, pairwise disjoint, each meeting A exactly in its boundary sphere. Then A pushes maps from P out of $A \cup e_1 \cup \cdots \cup e_k$ `Chapter13/Compression · pushesInto_union_cells`.

Proof. For one cell `Chapter13/Compression · pushesInto_union_cell` this is Part XII's compression for $A \subseteq A \cup e$ `Homotopy/CellAttach · compresses_union_cell` (§XII.8), translated from maps into the subspace $A \cup e$ to maps into T. For several cells, induct over a list of them without repetition: $A \cup e_2 \cup \cdots \cup e_k$ is closed (a finite union of compact images in a Hausdorff space), the cell e_1 meets it exactly in its boundary because the cells are disjoint, so one cell is pushed off, and the induction hypothesis together with transitivity finishes. □

Lemma D (passing critical levels of high index). Let $N \subseteq V$ be a slice submanifold of a finite-dimensional space, $F : V \to \mathbb{R}$ smooth, $f = F|N$, with $N^{a_0} = f^{-1}(-\infty, a_0]$ compact. Let a be a regular value of f, and suppose every critical point q with $a < f(q) \le a_0$ is non-degenerate of index $\ge \lambda_0$. Let P be compact and injected continuously into a real vector space of dimension $< \lambda_0$. Then for $a \le b < a_0$, N^a pushes maps from P out of N^b `Chapter13/Compression · lemma_D`.

The proof as formalized.

1. *Finitely many critical points.* The slabs $f^{-1}[b, b']$, $b' \le a_0$, are compact. The critical points in $f^{-1}[a, a_0]$ form a compact set `Chapter1/ManifoldMorse · isClosed_setOf_isCriticalPointM` which is discrete, since non-degenerate critical points are isolated and none lies at the regular level a `Chapter1/ManifoldMorse · isDiscrete_inter_setOf_isCriticalPointM`. So there are finitely many, a finite set C, and their values form a finite set.
2. *Induction on the number m of points of C at level $\le b$.* If there are none, f has no critical point in $f^{-1}[a, b]$, so N^a is a strong deformation retract of N^b by Theorem 3.1 on a submanifold `Chapter2/Submanifold · theorem_3_1_submanifold` (Part II), and it pushes maps.
3. Otherwise let c be the largest critical value $\le b$, attained on the set $S \subseteq C$; $a < c < a_0$. A finite set of reals has a gap η around c `Chapter13/Compression · exists_gap_of_finite`. Choose $\varepsilon_0 < \min(\eta, (c - a)/2, (a_0 - c)/2)$; then the only critical points with values in $[c - \varepsilon_0, c + \varepsilon_0]$ are those of S. Remark 3.3 on a submanifold `Chapter2/SeveralCritical · remark_3_3_submanifold` gives, for small $\varepsilon < \varepsilon_0$, disjoint embedded cells χ_q, $q \in S$, of dimension $\text{index}_q \ge \lambda_0$, meeting $N^{c-\varepsilon}$ in their boundaries, such that $X_1 = N^{c-\varepsilon} \cup \bigcup_q \chi_q$ is a strong deformation retract of $N^{c+\varepsilon}$.
4. The chain of pushes: (i) $N^{c+\varepsilon}$ pushes out of N^b, by Theorem 3.1 if $b > c + \varepsilon$ (no critical value in $[c + \varepsilon, b]$, by the choice of c); (ii) X_1 pushes out of $N^{c+\varepsilon}$, a strong deformation retract; (iii) $N^{c-\varepsilon}$ pushes out of X_1, by pushing off the cells, whose dimensions exceed $\dim E$; (iv) N^a pushes out of $N^{c-\varepsilon}$, by the induction hypothesis, since fewer critical points lie below $c - \varepsilon$. Transitivity, three times, gives the claim. □

The critical points above a are finitely many because they are isolated in a compact set and cannot accumulate at the regular level a; Milnor does not say so.

XIII.8 A Morse theorem for a manifold of minima: Lemma 22.5

Milnor's Lemma 22.5 (p. 121): let $f : M \to \mathbb{R}$ be smooth with minimum 0 and compact sublevel sets; if the set M^0 of minimal points is a manifold and every critical point in $M - M^0$ has index $\geq \lambda_0$, then $\pi_r(M, M^0) = 0$ for $0 \leq r < \lambda_0$. His proof: by Hanner, M^0 is a retract of a neighbourhood U which deforms into M^0; a map h into M^c is pushed down, by a smooth approximation g of f with non-degenerate critical points and derivatives close enough for Lemma 22.4, into $g^{-1}(-\infty, 2\delta] \subseteq U$, and then into M^0.

The formal version differs in three ways. Hanner's theorem is not formalized: the conclusion of that step, a neighbourhood U from which M^0 pushes maps, is a hypothesis. The approximation g is a linear perturbation cut off by a bump function, defined on all of N. And the lower level is chosen to be regular.

Lemma (a neighbourhood that deforms into A pushes maps). If $D : I \times U \to T$ is continuous with $D(0, x) = x$, $D(1, x) \in A$, and $D(t, x) = x$ for $x \in A$, then A pushes maps from any P out of U `Chapter13/ManifoldOfMinima · pushesInto_of_deformation`. (Compose h with D.)

Lemma (germs). Critical points and Hessians of $F|N$ at q depend only on the germ of F at q `Chapter13/ManifoldOfMinima · criticalPoint_hessianM_congr`, since the manifold derivative and the local representative do.

Lemma 22.5 below a level. Let $N \subseteq V$ be a slice submanifold of a finite-dimensional space, $F : V \to \mathbb{R}$ smooth, $f = F|N \geq d$, with N^{a_0} compact. Suppose every critical point q with $d < f(q) < a_0$ has index $\geq \lambda_0$, and $M^0 = f^{-1}(d)$ has an open neighbourhood U such that M^0 pushes maps from P out of U. Let P be compact and injected continuously into a real vector space of dimension $< \lambda_0$. Then for every $b < a_0$, M^0 pushes maps from P out of N^b `Chapter13/ManifoldOfMinima · lemma_22_5_sublevel`.

This is the form used for Theorem 22.1, where the function is only defined below a level. For $P = I^r$, $r < \lambda_0$, or $P = I \times I^r$, $r + 1 < \lambda_0$, it gives the surjectivity and injectivity statements on π_r.

The proof as formalized.

1. If $b < d$, N^b is empty and there is nothing to do. Let $b \geq d$, so $d < a_0$.
2. *f is bounded below off U.* $N^{a_0} \cap U^c$ is compact. If it is empty, put $m_1 = a_0$; otherwise f attains a minimum m_1 on it, and $m_1 > d$ because the points of level d lie in $M^0 \subseteq U$. In either case $f(q) < m_1$ implies $q \in U$.
3. Let $\delta = \min((m_1 - d)/4, (a_0 - b)/5)$ and $K_{st} = f^{-1}[d + \delta, b + 4\delta]$, compact. Lemma 22.4 on the submanifold, applied to K_{st} (where all critical points have index $\geq \lambda_0$), gives $\delta_4 > 0$.
4. *The approximation.* N^{a_0} is bounded, by $R_0 \geq 0$. A smooth bump function ρ equals 1 on the ball of radius $R_0 + 1$ and vanishes outside radius $R_0 + 2$. Through a linear isometry L of V with a Euclidean space, the companion project's measure-zero theorem (the proof of Corollary 6.8, Part III) says that $(F + \langle a, L\cdot\rangle)|N$ is a Morse function for almost every a `Chapter4/Whitney · ae_isMorseFunction_add_inner_comp`. A ball of positive radius has positive measure, so such an a exists with $\|a\|$ as small as we like. Let $\ell = \langle a, L\cdot\rangle$, chosen with $\|\ell\| < \min(\delta_4, \delta/(R_0 + 3))$, and put $G = F + \rho\ell$, $g = G|N$.
5. *Estimates.* $|g - f| = \rho|\ell| < \delta$ everywhere on N: where $\|q\| < R_0 + 2$, $|\ell(q)| \leq \|\ell\|(R_0 + 2) < \delta$, and elsewhere $\rho = 0$. Let $a_1 = b + 3\delta$. Then $g(q) \leq a_1$ implies $f(q) \leq a_0$, so $g^{-1}(-\infty, a_1]$ is compact, and its points have $\|q\| \leq R_0$, where $G = F + \ell$ near q. By the germ lemma the critical points of g below a_1 are those of the Morse function $(F + \ell)|N$, with the same Hessians, so they are non-degenerate.
6. *A regular level.* The critical points of g below a_1 are finitely many (compact and discrete, as in Lemma D), so the interval $(d + 2\delta, d + 3\delta)$ contains a regular value α of g. Milnor's level 2δ might be critical.

7. *Lemma D for g.* Every critical point q of g with $\alpha < g(q) \le a_1$ satisfies $d+\delta \le f(q) \le b+4\delta$, so lies in K_{st}, and is a critical point of $(F+\ell)|N$ with $\|\ell\| < \delta_4$; so its index is $\ge \lambda_0$. Lemma D gives: g^α pushes maps from P out of g^β, $\beta = \max(b+\delta, \alpha) < a_1$.
8. *The chain.* $M^0 \subseteq g^\alpha$ (there $g < f+\delta = d+\delta < \alpha$), $g^\alpha \subseteq U$ (there $f < g+\delta < d+4\delta \le m_1$), and $N^b \subseteq g^\beta$. So M^0 pushes out of $U \supseteq g^\alpha$, g^α pushes out of $g^\beta \supseteq N^b$, and transitivity concludes. □

Milnor's g lives only on the compact $M^{c+2\delta}$; a function on a compact set is not enough to apply Theorems 3.1 and 3.2, which need it on a manifold. The bump function makes g a smooth function on all of N, uniformly close to f.

Lemma 22.5. If moreover all sublevel sets are compact, and every critical point of f outside M^0 has index $\ge \lambda_0$, then M^0 pushes maps from P out of all of N `Chapter13/ManifoldOfMinima · lemma_22_5` (Milnor's $\pi_r(N, M^0) = 0$, $r < \lambda_0$): the image of a compact P lies in some N^b.

XIII.9 Minimal paths, and deforming a neighbourhood of them

Theorem 22.1 is proved in the finite-dimensional model of Part IX: the manifold B of broken geodesics with break points $t_1, \dots, t_{k-1}$, of energy $< c$, identified with an open subset of N^m by its break points (Parts VIII and IX). Two ingredients are proved first.

Theorem (a path of least energy is an unbroken geodesic). If $\gamma \in \Omega(N; p, q)$ has energy $\le$ that of every path from p to q, then γ is an unbroken geodesic `Chapter13/MinimalDeform · isUnbrokenGeodesic_of_minimal`.

Proof. By Corollary 12.3 `Chapter8/Critical · corollary_12_3` (Part VII), it suffices that γ is a critical point of the energy: for every variation α of γ, $\frac{d}{du}E(\alpha_u)|_{u=0} = 0$. Each α_u is a path from p to q, so $u \mapsto E(\alpha_u)$ has a local minimum at 0, and its derivative vanishes there. □

So the minimal paths are the paths of energy $\le d$, d the least energy, and they are broken geodesics for every subdivision; in particular Milnor's deformation r_u of Part IX fixes them.

Theorem (deforming a neighbourhood of the minima). Let B be a broken-geodesic setup (Part VIII) with energy bound c and m break points `Chapter10/Broken · BrokenSetup`, $O \subseteq V^m$ open with $N^m \cap O$ contained in the set of admissible break points, and $d < c$. Suppose $\{\gamma \in \Omega : E(\gamma) \le d\}$ is a retract of an open set $O_r \subseteq \Omega$ containing it, by a retraction ρ. Then $M^0 = \{y \in N^m \cap O : E'(y) \le d\}$ has an open neighbourhood U in $N^m \cap O$ from which it pushes maps from any P `Chapter13/MinimalDeform · BrokenSetup.exists_pushesInto_nodes`.

This is Milnor's "taking U small, we may assume that each point of U is joined to its image under the retraction by a unique minimal geodesic; hence U can be deformed into M^0 within M" (p. 121), in the manifold of break points.

The proof as formalized.

1. *The retraction on break points.* A point y of $S = N^m \cap O$ determines the broken geodesic ω_y through its break points; $y \mapsto \omega_y$ is continuous and $E(\omega_y) = E'(y)$. On $U_1 = \{y : \omega_y \in O_r\}$, open, let $s(y)$ be the break points of $\rho(\omega_y)$. It is continuous on U_1, and lies in the admissible set with $E'(s(y)) \le d$, because the break points of a path of energy $< c$ are admissible with no larger energy `Chapter10/Finite · nodeVal_mem_nodes_of_energy_lt`. On M^0, $s(y) = y$.
2. *Where short joins exist.* The break points of an admissible family lie in the closed Riemannian ball of radius $\sqrt{c}$ about p `Chapter13/MinimalDeform · BrokenSetup.mem_ball_of_mem_nodes`. Let $U_2 \subseteq U_1$ be the open set where each y_i is within the distance δ of the setup from $s(y)_i$ (the distance is continuous `Chapter7/Metric · continuousOn_riemDist`). It contains M^0, where the distance is 0.
3. *The deformation.* $D(y,\theta)_i = \exp_{y_i}(\theta\, v(y_i, s(y)_i))$, where $v(x,x')$ is the initial vector of the unique short geodesic from x to x' (Corollary 10.8, Part VI) `Chapter10/Broken · BrokenSetup.joinVec_spec`. It is continuous on $U_2 \times [0,1]$ `Chapter10/RetractionCont · BrokenSetup.continuousWithinAt_shortGeo`, starts at y,

ends at $s(y)$, stays in N^m, and fixes M^0, because the short join of a point with itself is 0 `Chapter13/MinimalDeform · BrokenSetup.joinVec_self`.

4. *Staying in O.* For $y \in M^0$ the whole track is $y \in O$. By the tube lemma (compactness of $[0,1]$), the set U_3 of $y \in U_2$ whose track stays in O is a neighbourhood of each point of M^0. Take $U =$ the interior of U_3; D restricted to $I \times U$ is a deformation into M^0 fixing M^0, and the neighbourhood lemma of §XIII.8 applies. □

The deformation must stay in the open manifold $N^m \cap O$, which holds near M^0 by the tube lemma; and the retraction, assumed on a neighbourhood in Ω, is transported to B by taking break points. Milnor says neither.

XIII.10 Theorem 22.1

Definition. Ω^d is the set of paths of least energy from p to q in Milnor's path space $\Omega = \Omega(N; p, q)$ with Milnor's metric (Part IX) `Chapter13/Theorem22 · minPaths`. By Lemma 12.1 these are the minimal geodesics, parametrized proportionally to arc length, and $d = \rho(p,q)^2$; formally d is just the least energy, which exists by Hopf–Rinow.

Theorem 22.1. Let N be a slice submanifold of a finite-dimensional inner product space, $p, q \in N$, with all closed Riemannian balls about p compact. Suppose that every geodesic from p to q which is not of least energy has index $\geq \lambda_0$ (the index of the Hessian of E, Part VII), and that Ω^d is a retract of an open neighbourhood in Ω. Let P be compact and injected continuously into a real vector space of dimension $< \lambda_0$. Then every map $h : P \to \Omega$ is homotopic, relative to $h^{-1}(\Omega^d)$, to a map into Ω^d `Chapter13/Theorem22 · theorem_22_1`.

With $P = I^i$, $i < \lambda_0$, this is Milnor's $\pi_i(\Omega, \Omega^d) = 0$ in the strong form, relative to ∂I^i and at all base points.

The hypothesis "Ω^d is a topological manifold" is replaced by its consequence, through Hanner's theorem, that Ω^d is a neighbourhood retract. In Milnor's proof the manifold structure is used only for this. Hanner's theorem in the metrizable B, composed with Milnor's r_1, would give a retraction of a neighbourhood in Ω, and conversely.

The proof as formalized. If P is empty there is nothing to do. Fix $x_0 \in P$.

1. *An energy bound.* E is continuous on Ω `Chapter10/PathMetric · PathSp.continuous_energy` and P is compact, so $E(h(x)) \leq c_0$ for all x. Let $c = c_0 + 3$ and take a broken-geodesic setup B for energy $< c$ `Chapter10/Broken · exists_brokenSetup_uniform`, which exists because the closed ball of radius $\sqrt{c}$ is compact.
2. *The least energy d.* B contains a path of least energy, an energy minimizer among all paths from p to q `Chapter10/Finite · BrokenSetup.exists_energy_minimizing`; let d be its energy, $d \leq c_0$. Then $\gamma \in \Omega^d$ iff $E(\gamma) \leq d$, and every $\gamma \in \Omega^d$ is an unbroken geodesic (§XIII.9), hence a broken geodesic of B `Chapter10/Finite · BrokenSetup.isBroken_of_isUnbrokenGeodesic`.
3. *Milnor's deformation r_u.* Let $K_1(t,x) = r_t(h(x))$, Milnor's deformation of $\operatorname{Int}\Omega^c$ onto the broken geodesics (Part IX) `Chapter10/Deformation · BrokenSetup.deform`. It does not raise energy `Chapter10/Deformation · BrokenSetup.energy_deform_le`, ends in the broken geodesics `Chapter10/Deformation · BrokenSetup.deform_one_mem_brokenPaths`, and fixes the broken geodesics, in particular Ω^d `Chapter10/Deformation · BrokenSetup.deform_of_mem_brokenPaths`.
4. *The manifold.* Let $c_1 = c_0 + 2$. The energy E' of break points extends to a smooth function F on V^m agreeing with E' below $c_1 + \frac{1}{2}$ `Chapter10/Finite · BrokenSetup.exists_contDiff_eq_energy'`, and $\{y : E'(y) < c_1\} = N^m \cap O$ for an open O `Chapter10/Finite · BrokenSetup.exists_isOpen_sublevel_lt`. $S = N^m \cap O$ is a slice submanifold of V^m `Chapter10/Product · isSliceSubmanifold_pi_inter`; let $f = F|S$. The break points $h_2(x)$ of $K_1(1,x)$ lie in S, with $E'(h_2(x)) \leq c_0$.

5. *Lemma 22.5 on S.* $f \geq d$. $f^{-1}(-\infty, a_0]$, $a_0 = c_0 + 1$, is the compact sublevel set of E' on B Chapter10/BrokenEnergy · BrokenSetup.isCompact_sublevel. A critical point y of f is a broken-geodesic setup point whose path is an unbroken geodesic Chapter10/BrokenVariation · BrokenSetup.isCriticalPointM_iff, and its index on S equals the index of that geodesic in Ω (Theorem 16.2) Chapter10/BrokenIndex · BrokenSetup.theorem_16_2_index. If $f(y) > d$ the geodesic is not of least energy, so its index is $\geq \lambda_0$. The neighbourhood hypothesis is §XIII.9 applied to the given retraction. So $\{f \leq d\}$ pushes h_2 (with values in $f^{-1}(-\infty, c_0]$) into $\{f \leq d\}$ by a homotopy G_2 relative to $h_2^{-1}\{f \leq d\}$.
6. *Back to Ω.* Let $K_2(t, x)$ be the broken geodesic of $G_2(t, x)$. $K_1(1, x)$ is a broken geodesic, so it agrees on $[0, 1]$ with the broken geodesic of its break points $h_2(x)$ (Lemma 16.1) Chapter10/BrokenPaths · BrokenSetup.lemma_16_1. They need not be *equal*: paths are functions on $\mathbb{R}$, and Ω is not Hausdorff. Define $K_2'(t, x) = K_1(1, x)$ if $t = 0$ or $h(x) \in \Omega^d$, and $K_2(t, x)$ otherwise. It is at distance 0 from K_2 everywhere Chapter10/Omega · PathSp.edist_eq_zero_of_eq0n, hence continuous Chapter10/Deformation · PathSp.continuous_of_edist_eq_zero, and it starts at $K_1(1, \cdot)$ and fixes the points with $h(x) \in \Omega^d$ (where $h_2(x) \in \{f \leq d\}$ and G_2 is fixed).
7. K_1 followed by K_2' is a homotopy from h, relative to $h^{-1}(\Omega^d)$, to a map into Ω^d: for $h(x) \in \Omega^d$ the end is $h(x)$; otherwise it is the broken geodesic of a point with $f \leq d$, of energy $\leq d$. □

No reparametrization $F \circ E$ of Milnor's is needed: Lemma 22.5 below a level only uses a compact sublevel set. And "it is sufficient to prove $\pi_i(\operatorname{Int}\Omega^c, \Omega^d) = 0$ for large c" is the energy bound of step 1.

Corollary (Milnor, p. 118). Under the same hypotheses, the inclusion $\Omega^d \to \Omega$ Chapter13/Theorem22 · minPathsIncl compresses maps from every such P (§XII.2) Chapter13/Theorem22 · compresses_minPaths, and so induces bijections $\pi_i(\Omega^d, \gamma) \to \pi_i(\Omega, \gamma)$ for $i \leq \lambda_0 - 2$ Chapter13/Theorem22 · theorem_22_1_bijective and surjections for $i \leq \lambda_0 - 1$ Chapter13/Theorem22 · theorem_22_1_surjective, at every $\gamma \in \Omega^d$ and for every finite index set of size i.

Proof. The cube I^K embeds in $\mathbb{R}^K$ Homotopy/CellAttach · continuous_cubeEmb and the cylinder $I \times I^K$ in $\mathbb{R} \times \mathbb{R}^K$ Homotopy/CellAttach · continuous_cylEmb. Compressing cubes gives surjectivity and compressing cylinders gives injectivity (§XII.2) Homotopy/Basic · surjective_homotopyGroupMap_of_compresses Homotopy/Basic · injective_homotopyGroupMap_of_compresses, for the embedding $\Omega^d \to \Omega$. The cylinder needs $i + 1 < \lambda_0$. □

Milnor passes through the exact sequence of the pair; here π_0 is treated like the rest.

XIII.11 Corollary 22.2, and change of basepoint as a homomorphism

Milnor's Corollary 22.2 (p. 118): with the hypotheses of Theorem 22.1, $\pi_i(\Omega^d) \cong \pi_{i+1}(M)$ for $0 \leq i \leq \lambda_0 - 2$. Proof: $\pi_i\Omega^d \cong \pi_i\Omega$ by 22.1, and $\pi_i\Omega \cong \pi_{i+1}M$. The second step hides a basepoint: $\pi_i(\Omega, \gamma)$ is taken at a minimal geodesic γ from p to q, while $\pi_i(\Omega M) \cong \pi_{i+1}M$ concerns the loops at p, at the constant loop.

Definition. The space Ω^* of continuous paths of Part IX, as functions into V with values in N, is homeomorphic to Mathlib's space of paths from p to q in the subspace N Chapter13/Corollaries · pathCHomeomorph. For a path γ from x_0 to x_1, the loops $\gamma|_{[0,s]} \cdot \gamma|_{[0,s]}^{-1}$, $t \mapsto \gamma(s \min(2t, 2 - 2t))$ Chapter13/Corollaries · contractLoop, depend continuously on s Chapter13/Corollaries · continuous_contractLoop, are the constant loop at $s = 0$ Chapter13/Corollaries · contractLoop_zero and $\gamma \cdot \gamma^{-1}$ at $s = 1$ Chapter13/Corollaries · contractLoop_one. Run backwards, they form a path in the loop space from $\gamma \cdot \gamma^{-1}$ to the constant loop Chapter13/Corollaries · contractPath.

Corollary 22.2. Under the hypotheses of Theorem 22.1, for every minimal path γ and $i + 2 \leq \lambda_0$ there is a bijection $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(N, p)$ Chapter13/Corollaries · corollary_22_2.

The proof as formalized. The bijection is a composite of six: 1. $\pi_i(\Omega^d, \gamma) \to \pi_i(\Omega, \gamma)$, Theorem 22.1; 2. $\pi_i(\Omega, \gamma) \to \pi_i(\Omega^*, \gamma)$, induced by the inclusion, a homotopy equivalence by Theorem 17.1 Chapter10/FullPathTheorem · theorem_17_1 (Part IX), and bijective by §XII.4 Homotopy/RelCells · bijective_homotopyGroupMap_

of_isHomotopyEquiv; 3. the homeomorphism of Ω^* with Mathlib's path space Homotopy/CellAttach · bijective_homotopyGroupMap_homeomorph; 4. $\delta \mapsto \delta \cdot x^{-1}$, x the image of γ, a homotopy equivalence from paths $p \to q$ to loops at p Homotopy/LoopSpace · pathTransHomotopyEquiv Homotopy/ChangeOfBase · bijective_homotopyGroupMap_of_homotopyEquiv, sending x to $x \cdot x^{-1}$; 5. change of basepoint from $x \cdot x^{-1}$ to the constant loop along the contraction above (§XII.3) Homotopy/ChangeOfBase · changeBaseEquiv; 6. $\pi_i(\Omega N, \text{const}) \cong \pi_{i+1}(N, p)$ (§XII.12) Homotopy/LoopSpace · homotopyGroupPiLoopEquiv. □

No connectedness of N is needed: using γ itself to pass from $\Omega(N; p, q)$ to $\Omega(N; p, p)$ lands in the component of the constant loop.

Groups. For $i \geq 1$ Milnor's isomorphisms are of groups. Steps 1–4 are induced by continuous maps, hence homomorphisms (§XII.17) Homotopy/ExactSequenceIso · homotopyGroupHom, and step 6 is a group isomorphism Homotopy/LoopSpace · homotopyGroupPiLoopMulEquiv. What remains is step 5.

Theorem (change of basepoint is a homomorphism). For a path γ from y_0 to y_1 and nonempty finite N, the change of basepoint $c_\gamma : \pi_N(Y, y_0) \to \pi_N(Y, y_1)$ of §XII.3 Homotopy/ChangeOfBase · changeBaseMap satisfies $c_\gamma(ab) = c_\gamma(a)c_\gamma(b)$ Homotopy/ChangeOfBaseGroup · changeBaseMap_mul, so it is a group isomorphism Homotopy/ChangeOfBaseGroup · changeBaseMulEquiv.

The proof as formalized. It avoids manipulating the frames of §XII.3.

1. On the free path space $C(I, Y)$, evaluation at s Homotopy/ChangeOfBaseGroup · evalPath gives two maps ev_0, ev_1 to Y, and $(s, \omega) \mapsto \omega(s)$ is a free homotopy from ev_0 to ev_1 Homotopy/ChangeOfBaseGroup · evalHomotopy. ev_0 is a homotopy equivalence, with inverse the constant paths Homotopy/ChangeOfBaseGroup · constPath and the contraction $(t, \omega) \mapsto (s \mapsto \omega(ts))$ Homotopy/ChangeOfBaseGroup · pathContraction Homotopy/ChangeOfBaseGroup · evalZeroHomotopyEquiv.
2. At the basepoint ω, homotopic maps differ by the change of basepoint along the track of ω: $(ev_1)_* = c_\omega \circ (ev_0)_*$ (§XII.4) Homotopy/ChangeOfBase · homotopyGroupMap_eq_changeBaseMap. $(ev_0)_*$ is onto and both $(ev_i)_*$ are homomorphisms Homotopy/ExactSequenceIso · homotopyGroupMap_mul. Writing $a = (ev_0)_* a'$, $b = (ev_0)_* b'$: $c_\omega(ab) = c_\omega(ev_0)_*(a'b') = (ev_1)_*(a'b') = (ev_1)_* a' \, (ev_1)_* b' = c_\omega(a)c_\omega(b)$ Homotopy/ChangeOfBaseGroup · changeBaseMap_evalAt_mul.
3. A path γ in Y is such an ω; its track is γ itself, up to the identifications of $\gamma(0)$ with y_0 and $\gamma(1)$ with y_1, which commute with change of basepoint and with products Homotopy/ChangeOfBaseGroup · changeBaseMap_cast Homotopy/ChangeOfBaseGroup · cast_mul. □

Corollary 22.2 as groups. For every path γ in $\Omega(N; p, q)$, $\pi_{j+1}(\Omega, \gamma) \cong \pi_{j+2}(N, p)$ as groups Chapter13/CorollaryGroups · nonempty_mulEquiv_pathSp (steps 2–6). Under the hypotheses of Theorem 22.1, $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(N, p)$ as groups for $1 \leq i \leq \lambda_0 - 2$ Chapter13/CorollaryGroups · corollary_22_2_mulEquiv, and there is a surjective homomorphism $\pi_i(\Omega^d, \gamma) \to \pi_{i+1}(N, p)$ for $1 \leq i \leq \lambda_0 - 1$ Chapter13/CorollaryGroups · corollary_22_2_surjective_hom. As sets, the surjection exists for $0 \leq i \leq \lambda_0 - 1$ Chapter13/Freudenthal · corollary_22_2_surjective, by the surjectivity in Theorem 22.1 followed by the bijections of steps 2–6.

XIII.12 The sphere: Lemmas A and B and Freudenthal's theorem

Milnor's Corollary 22.3 (p. 119): $\pi_i(S^n) \cong \pi_{i+1}(S^{n+1})$ for $i \leq 2n - 2$. Proof: for antipodal $p, -p$ on S^{n+1}, the space of minimal geodesics "can be identified with the equator S^n", every non-minimal geodesic "must wind one and a half times around S^{n+1}, and therefore has index $\geq 2n$", and the hypotheses of Theorem 22.1 hold with $\lambda_0 = 2n$. Each clause needs a proof. The exposition (`Exposition/Chapter13.md`) states them as Lemmas A and B.

S^{n+1} is the unit sphere of $\mathbb{R}^{n+2}$ with the companion project's slice structure, and $\gamma_{u,L}(t) = \cos(Lt)p + \sin(Lt)u$, for a unit vector $u \perp p$, is a great circle (Part IX) Chapter10/SphereGeodesics · greatCircle.

Lemma A, first half (the geodesics from p to $-p$). Every geodesic from p to $-p$ agrees on $[0, 1]$ with $\gamma_{u,L}$

for a unit $u \perp p$ and $L = (2m+1)\pi$, $m \geq 0$ Chapter13/SphereMinimal · exists_eqOn_greatCircle_antipode. Conversely each of these is such a geodesic Chapter13/SphereMinimal · greatCircle_mem_pathSpace_antipode Chapter13/SphereMinimal · isUnbrokenGeodesic_greatCircle.

Proof. A geodesic from p is a great circle $\gamma_{u,L}$ on $[0,1]$ Chapter10/SphereGeodesics · exists_eqOn_greatCircle, $L \geq 0$. $\gamma(1) = -p$ and $\langle p, \gamma(1)\rangle = \cos L$ give $\cos L = -1$, so $L = \pi + 2\pi k$, and $k \geq 0$ since $L \geq 0$. □

Theorem ($d = \pi^2$). Every path from p to $-p$ has energy $\geq \pi^2$ Chapter13/SphereMinimal · pi_sq_le_energy_sph.

Proof. The closed balls are compact because the sphere is closed Chapter13/SphereMinimal · isClosed_sph Chapter10/Broken · isCompact_closedRiemBall_of_isClosed. A broken-geodesic setup contains a path of least energy, which is a geodesic, so it is some $\gamma_{u,(2m+1)\pi}$, of energy $((2m+1)\pi)^2 \geq \pi^2$ Chapter10/SphereGeodesics · energy_greatCircle. □

Lemma B. The geodesic $\gamma_{u,(2m+1)\pi}$ from p to $-p$ has index $2mn$ Chapter13/SphereMinimal · formIndex_greatCircle_antipode. Every geodesic from p to $-p$ that does not have least energy has index $\geq 2n$ Chapter13/SphereMinimal · le_formIndex_of_not_minimal_sph.

Proof. By the Index Theorem 15.1 (Part VII) Chapter9/IndexTheorem · theorem_15_1, the index is the sum over $0 < t < 1$ of the multiplicities of the conjugate points. Along a great circle of speed L the parameter t is conjugate to 0, with multiplicity n, exactly when $Lt \in \pi\mathbb{N}$ Chapter10/SphereGeodesics · conjMultiplicity_greatCircle. For $k\pi < L \leq (k+1)\pi$ the number of such $t \in (0,1)$ is k, so the sum is kn Chapter13/SphereMinimal · finsum_mem_Ioo_ite_eq'. With $L = (2m+1)\pi$, $k = 2m$: the passages through $\pm p$ at $t = j/(2m+1)$, $1 \leq j \leq 2m$. The end point $t = 1$ is also conjugate but does not count. For the second statement: the index is a property of the restriction to $[0,1]$ Chapter10/SphereGeodesics · formIndex_energyHessian_congr; if $m = 0$ the energy would be π^2, the least, so $m \geq 1$ and $2mn \geq 2n$. □

Lemma A, second half (the minimal geodesics). The *equator* is $\{u : \|u\| = 1, u \perp p\}$ Chapter13/SphereMinimal · equator, and the *semicircle* through u is $\gamma_{u,\pi}$, a point of Ω Chapter13/SphereMinimal · semicircle of energy π^2 Chapter13/SphereMinimal · energy_semicircle, hence in Ω^d Chapter13/SphereMinimal · semicircle_mem_minPaths. Every path of least energy agrees on $[0,1]$ with a semicircle Chapter13/SphereMinimal · exists_eqOn_semicircle: it is a geodesic (§XIII.9), so $\gamma_{u,(2m+1)\pi}$, and its energy $((2m+1)\pi)^2 \leq \pi^2$ forces $m = 0$.

Theorem ("identified with the equator"). $u \mapsto$ the semicircle through u is continuous for Milnor's metric Chapter13/SphereMinimal · continuous_semicircle. It is a homotopy equivalence from the equator to Ω^d, with homotopy inverse $\gamma \mapsto \gamma(\frac{1}{2})$ Chapter13/SphereMinimal · sphMinPathsHomotopyEquiv, and a homeomorphism of the equator onto the separation quotient of Ω^d Chapter13/SphereMinimal · sphMinPathsHomeomorph.

The proof as formalized.

1. *Continuity.* Milnor's metric has a uniform part and an L^2 part on the speeds (Part IX). Every great circle of speed L has speed $|L|$ at every time Chapter13/SphereMinimal · speed_greatCircle, so on the semicircles the L^2 part vanishes. The uniform part is controlled by pointwise convergence of paths of bounded energy Chapter10/RetractionConv · PathSp.eventually_supDist_le_of_tendsto_apply, and $u \mapsto \gamma_{u,\pi}(s)$ is continuous for each s.
2. *The inverse.* $\gamma \mapsto \gamma(\frac{1}{2})$ is continuous Chapter13/SphereMinimal · continuous_apply_half_sph and maps Ω^d into the equator, since $\gamma_{u,\pi}(\frac{1}{2}) = u$ Chapter13/SphereMinimal · greatCircle_pi_half Chapter13/SphereMinimal · apply_half_mem_equator Chapter13/SphereMinimal · minPathsToEquator. Starting from the equator one returns exactly Chapter13/SphereMinimal · minPathsToEquator_equatorToMinPaths. Starting from $\gamma \in \Omega^d$ one returns to a path agreeing with γ on $[0,1]$, at distance 0 Chapter13/SphereMinimal · edist_equatorToMinPaths_minPathsToEquator.
3. *The homotopy.* The map that is the round trip for $t < 1$ and the identity at $t = 1$ is at distance 0 from the projection $(t, \gamma) \mapsto \gamma$, hence continuous; it is a homotopy from the round trip to the identity. Points at distance 0 are inseparable, which gives the homeomorphism with the separation quotient. □

As paths are functions on $\mathbb{R}$, Ω^d is homeomorphic to the equator only after identifying paths agreeing on $[0,1]$; for homotopy groups the homotopy equivalence is what is used.

Theorem (the neighbourhood retraction for the sphere). $\Omega^d(S^{n+1}; p, -p)$ is a retract of an open neighbourhood in Ω `Chapter13/Freudenthal · exists_retraction_minPaths_sph`.

The proof as formalized. This replaces Hanner's theorem by a formula. 1. The equatorial component $x - \langle p, x\rangle p$ `Chapter13/Freudenthal · eqComp` is continuous, and its normalization lies on the equator where it is nonzero `Chapter13/Freudenthal · eqComp_mem_equator`; on the equator it is the identity `Chapter13/Freudenthal · eqComp_of_mem_equator`. 2. O = the paths ω with $\omega(\frac{1}{2}) \neq \pm p$, i.e. with nonzero equatorial component, is open `Chapter13/Freudenthal · sphNbhd` `Chapter13/Freudenthal · isOpen_sphNbhd`. On O, $\omega \mapsto$ the semicircle through the normalized equatorial component of $\omega(\frac{1}{2})$ `Chapter13/Freudenthal · sphRetrPath` is continuous `Chapter13/Freudenthal · continuousOn_sphRetrPath`. 3. A minimal γ lies in O, and its image agrees with γ on $[0,1]$ `Chapter13/Freudenthal · mem_sphNbhd_and_eqOn`, but need not equal it. So the retraction is $\gamma \mapsto \gamma$ on Ω^d and the semicircle elsewhere on O; it is continuous on O because it is at distance 0 from the continuous map of step 2. □

Corollary 22.3 (Freudenthal). For $i + 2 \leq 2n$ and all $x \in S^n$, $y \in S^{n+1}$, $\pi_i(S^n, x) \cong \pi_{i+1}(S^{n+1}, y)$ as sets `Chapter13/Freudenthal · corollary_22_3`, and as groups for $1 \leq i \leq 2n-2$ `Chapter13/CorollaryGroups · corollary_22_3_mulEquiv`. For $i + 1 \leq 2n$, in particular $i = 2n-1$, there is a surjection $\pi_i(S^n, x) \to \pi_{i+1}(S^{n+1}, y)$ `Chapter13/Freudenthal · corollary_22_3_surjective`, a surjective homomorphism for $i \geq 1$ `Chapter13/CorollaryGroups · corollary_22_3_surjective_hom`.

The proof as formalized. Take the pole $p = y$; the base point of S^{n+1} needs no change, since p is arbitrary. The equator is homeomorphic to S^n, by an isometry of $p^\perp$ with $\mathbb{R}^{n+1}$ through an orthonormal basis `Chapter13/Freudenthal · equatorHomeomorph`. The hypotheses of Theorem 22.1 hold with $\lambda_0 = 2n$: the balls are compact `Chapter13/Freudenthal · hball_sph`, the index bound is Lemma B `Chapter13/Freudenthal · hidx_sph`, and the retraction is the one above. So Corollary 22.2 applies at the semicircle through the image of x, and the bijection is: the homeomorphism $S^n \cong$ equator, the homotopy equivalence equator $\simeq \Omega^d$ `Homotopy/ChangeOfBase · homotopyGroupEquivOfHomotopyEquiv`, and Corollary 22.2. Every base point of S^n is reached, since every point of the equator gives a semicircle. The surjection uses the surjective form of Corollary 22.2. □

The second half of Freudenthal's theorem, $i = 2n-1$, is not stated by Milnor in 22.3; it follows from the surjectivity in Theorem 22.1 (his remark on $\pi_{\lambda_0 - 1}$, p. 118).

XIII.13 Theorem 22.1 for an open and closed set of paths

In §24 (p. 147, the case $k \equiv 2 \pmod 4$) Milnor needs "a variant of Theorem 22.1 for the component $\Omega^{(c)}$ of the path space": the proof applies verbatim to an open and closed subset Ω' of Ω, provided the index hypothesis holds for the non-minimal geodesics in Ω'. The exposition (`Exposition/Chapter15.md`) states it as Lemma 6.2. It is proved here, with Theorem 22.1.

Open sets of Ω are saturated for Milnor's pseudometric: a path at distance 0 from a point of an open set lies in it `Chapter13/Componentwise · PathSp.mem_of_edist_eq_zero`. And a path in any space that starts in an open and closed set stays in it `Chapter13/Componentwise · mem_of_isClopen_of_continuous` `Homotopy/BundlesUnitary · range_subset_of_isClopen`.

Lemma 6.2 (Theorem 22.1 componentwise). In the setting of Theorem 22.1, let $\Omega' \subseteq \Omega$ be open and closed, and suppose every geodesic *in* Ω' that is not of least energy has index $\geq \lambda_0$. Then every map $h : P \to \Omega'$ is homotopic, relative to $h^{-1}(\Omega^d)$, to a map into $\Omega^d \cap \Omega'$ `Chapter13/Componentwise · theorem_22_1_clopen`.

The proof as formalized. The proof of Theorem 22.1, with Ω' seen at three places. 1. Milnor's deformation r_u moves each path continuously, so keeps it in Ω'. 2. The manifold of break points is cut down to $S = N^m \cap O \cap O''$, where O'' is open with $\{y : \omega_y \in \Omega'\}$ = its trace on the admissible break points

(the broken geodesic depends continuously on its break points). The break points of $K_1(1,x)$ lie in S, because their broken geodesic is at distance 0 from $K_1(1,x) \in \Omega'$ and Ω' is saturated. The sublevel set of f on S is compact, because Ω' is closed. The index hypothesis is needed only at critical points of S, whose geodesics lie in Ω'. 3. The final homotopy is inside S, so its broken geodesics lie in Ω'. □

d is the least energy on all of Ω, and no hypothesis that Ω' contain a minimal geodesic is needed: if it contains none, its energy minimum is a non-minimal geodesic of index 0, so the hypothesis forces $\lambda_0 = 0$ and the statement is empty.

Corollary (componentwise, on π_i). Pushing into $A \cap \Omega'$ within X is compressing along the embedding $A \cap \Omega' \to \Omega'$ `Chapter13/Componentwise · clopenIncl` `Chapter13/Componentwise · isEmbedding_clopenIncl` `Chapter13/Componentwise · compresses_clopenIncl`, because the homotopies start in Ω' and stay there. So the inclusion $\Omega^d \cap \Omega' \to \Omega'$ is bijective on π_i for $i \le \lambda_0 - 2$, π_0 included `Chapter13/Componentwise · bijective_clopenIncl` `Chapter13/Componentwise · theorem_22_1_clopen_bijective`. And the inclusion $\Omega^d \to \Omega$ is bijective on π_i, $1 \le i \le \lambda_0 - 2$, at every minimal geodesic in Ω' `Chapter13/Componentwise · theorem_22_1_clopen_bijective_minPathsIncl` `Chapter13/Componentwise · bijective_val_of_pushesInto`, because the inclusion of an open and closed set containing the base point is bijective on π_N for $N \neq \emptyset$ (§XIII.22).

XIII.14 Geodesics from I to $-I$ in $SU(n)$, and Lemma 23.1

Milnor's §23 begins with matrix computations (pp. 125–127). After conjugation by a unitary matrix a skew-Hermitian A is $\mathrm{diag}(ia_1, \dots, ia_n)$; $\exp A = -I$ exactly when $A = \mathrm{diag}(i\pi k_1, \dots, i\pi k_n)$ with all k_j odd; the geodesic $t \mapsto \exp tA$ has length $|A| = \pi(\sum k_j^2)^{1/2}$, so it is minimal iff all $k_j = \pm 1$; and a minimal A is "completely determined by" Eigen$(i\pi)$, an arbitrary subspace, in $SU(2m)$ an arbitrary m-dimensional one. That is Lemma 23.1: the minimal geodesics form $G_m(\mathbb{C}^{2m})$.

Definition. For $k : \iota \to \mathbb{Z}$, $D_k = \mathrm{diag}(i\pi k_1, \dots, i\pi k_n)$ `Chapter14/SUMatrices · suDiag`, and more generally $D_a = \mathrm{diag}(ia_1, \dots, ia_n)$ for real a `Chapter12/SUJacobi · diagI`; $D_k = D_{\pi k}$ `Chapter12/SUJacobi · suDiag_eq_diagI`.

Lemma (the diagonal computations). (i) $\exp D_a = -I$ iff every a_j is an odd multiple of π `Chapter14/SUMatrices · exp_diagonal_eq_neg_one_iff`. (ii) $\|D_k\|^2 = \mathrm{tr}(D_k D_k^*) = \pi^2 \sum k_j^2$ `Chapter14/SUMatrices · trace_suDiag_mul_conjTranspose` `Chapter12/SUMinimal · norm_suDiag_sq`. (iii) $\mathrm{tr}\, D_k = i\pi \sum k_j$ `Chapter14/SUMatrices · trace_suDiag`.

Proof. (i) The exponential of a diagonal matrix is the diagonal of exponentials `Chapter14/SUMatrices · exp_diagonal_mul_I`, and $e^{ia} = -1$ iff $e^{i(a-\pi)} = 1$ iff $a - \pi \in 2\pi\mathbb{Z}$ `Chapter14/SUMatrices · exp_ofReal_mul_I_eq_neg_one_iff`. (ii), (iii) are entrywise. □

Lemma (every geodesic from I to $-I$ is of Milnor's form). If $A \in \mathfrak{su}(\iota)$ and $\exp A = -I$, there are a unitary T and odd integers k_j with $\sum k_j = 0$ and $A = T D_k T^*$ `Chapter12/SUIndex · exists_unitary_suDiag`. Conversely, for k odd, $\exp(T D_k T^*) = -I$ `Chapter12/SUIndex · exp_conj_suDiag`, and $T D_k T^* \in \mathfrak{su}$ when $\sum k_j = 0$ `Chapter12/SUIndex · conj_suDiag_mem`.

Proof. iA is Hermitian; Mathlib's spectral theorem `Mathlib/Analysis/Matrix/Spectrum · Matrix.IsHermitian.spectral_theorem` gives a unitary U and real eigenvalues λ_j with $iA = U\,\mathrm{diag}(\lambda)U^*$, so $A = UDU^*$ with $D = \mathrm{diag}(-i\lambda_j)$. Conjugation commutes with exp `Mathlib/Analysis/Normed/Algebra/MatrixExponential · Matrix.exp_conj`, so $\exp D = U^*(-I)U = -I$, and by (i) $-\lambda_j = \pi k_j$ with k_j odd. $\mathrm{tr}\, A = \mathrm{tr}\, D = i\pi \sum k_j = 0$. □

Lemma (lengths). Conjugation by a unitary matrix preserves the norm `Chapter12/SUMinimal · norm_conj_unitary`, so $\|T D_k T^*\| = \pi(\sum k_j^2)^{1/2}$ `Chapter12/SUMinimal · norm_conj_suDiag_sq` `Chapter12/SUMinimal · norm_eq_pi_sqrt`. The length of $t \mapsto \exp(tA)$ on $[0,1]$ is $\|A\|$ for skew-Hermitian A `Chapter12/SUMinimal · pathLength_exp_smul`, since its speed $\|\exp(tA)A\| = \|A\|$ is constant (left multiplication by a unitary is an isometry) `Chapter7/Path · pathLength_eq_of_norm_deriv`. For odd k_j, $k_j^2 \ge 1$, so $\sum k_j^2 \ge n$ `Chapter14/IndexCount · card_`

le_sum_sq, with equality iff all $k_j = \pm 1$ Chapter14/IndexCount · sum_sq_eq_card_iff. Hence every geodesic $t \mapsto \exp tA$ from I to $-I$ in $SU(n)$ has length $\geq \pi\sqrt{n}$ Chapter12/SUMinimal · pi_sqrt_card_le Chapter12/SUMinimal · pi_sqrt_card_le_norm.

In $SU(2m)$, $k^0 = (1, \dots, 1, -1, \dots, -1)$, m of each sign Chapter12/SUMinimal · kHalf, consists of ± 1's Chapter12/SUMinimal · kHalf_pm, odd Chapter12/SUMinimal · odd_kHalf, with sum 0 Chapter12/SUMinimal · sum_kHalf. So $D_{k^0} \in \mathfrak{su}(2m)$ and $\exp D_{k^0} = -I$: a geodesic of length $\pi\sqrt{2m}$. Each $t \mapsto \exp tA$ with $\exp A = -I$ lies in the path space $\Omega(SU(n); I, -I)$ Chapter12/SUIndex · mem_pathSpace_exp and is an unbroken geodesic Chapter12/SUIndex · isUnbrokenGeodesic_exp.

Theorem (the distance from I to $-I$). In $SU(2m)$, $m \geq 1$, the Riemannian distance from I to $-I$ is $\pi\sqrt{2m}$ Chapter12/SUMinimal · riemDist_one_neg_one.

The proof as formalized. "$\leq$": the distance is at most the length of any path (Part VI) Chapter7/Distance · riemDist_le_pathLength, here the geodesic $\exp(tD_{k^0})$. "$\geq$": the distance is finite. $SU(2m)$ is geodesically complete (§XIII.4), so Hopf–Rinow at the point I (Theorem 10.9, Part VI) Chapter7/HopfRinow · theorem_10_9 gives a tangent vector $v \in T_I SU(2m) = \mathfrak{su}$ with $\exp_I(v) = -I$ and distance $= \|v\|$. The Riemannian exponential at I is the matrix exponential (§XIII.4), so $\exp v = -I$, and $\|v\| \geq \pi\sqrt{2m}$ by the length bound. □

Milnor's "it is minimal iff all $k_j = \pm 1$" needs this: every geodesic has length $\geq \pi\sqrt{n}$, but that the ones of length $\pi\sqrt{n}$ are *minimal* needs a shortest geodesic to exist (Hopf–Rinow) and every geodesic from I to be $\exp(tA)$ (Lemma 21.2 and its converse).

Theorem (minimal geodesics). Let $|\iota| = 2m$, $A \in \mathfrak{su}(\iota)$, $\exp A = -I$. The geodesic $t \mapsto \exp(tA)$ is a minimal geodesic from I to $-I$ (no piecewise smooth path in $SU(2m)$ between them is shorter) Chapter7/HopfRinow · IsMinimalGeodesic iff $A^2 = -\pi^2 I$ Chapter12/SUMinimal · isMinimalGeodesic_exp_iff. For $A = TD_kT^*$, $A^2 = -\pi^2 I$ iff all $k_j = \pm 1$ Chapter12/SUMinimal · sq_conj_suDiag_iff.

Proof. $A^2 = TD_k^2T^* = T\,\mathrm{diag}(-\pi^2 k_j^2)T^*$, which is $-\pi^2 I$ iff all $k_j^2 = 1$. If the geodesic is minimal, its length $\|A\| = \pi(\sum k_j^2)^{1/2}$ is at most that of the geodesic of k^0, $\pi\sqrt{2m}$, and at least $\pi\sqrt{2m}$, so $\sum k_j^2 = 2m$ and all $k_j = \pm 1$. Conversely if all $k_j = \pm 1$, its length $\pi\sqrt{2m}$ is the distance, which is at most the length of every path. □

Lemma 23.1, as sets. A skew-Hermitian A satisfies $A^2 = -\pi^2 I$ iff $A = i\pi(2P - I)$ for a Hermitian projection P Chapter14/SUMatrices · sq_eq_neg_pi_sq_iff, and then $\mathrm{tr}\, A = i\pi(2\,\mathrm{tr}\, P - n)$, which vanishes iff $\mathrm{tr}\, P = n/2$ Chapter14/SUMatrices · trace_pi_smul_projection.

Proof. Given $A^2 = -\pi^2 I$, $P = \frac{1}{2}(I - \frac{i}{\pi}A)$ is Hermitian (as $A^* = -A$) and $P^2 = \frac{1}{4}(I - \frac{2i}{\pi}A - \frac{1}{\pi^2}A^2) = \frac{1}{4}(2I - \frac{2i}{\pi}A) = P$, and $i\pi(2P - I) = A$. Conversely $(i\pi(2P - I))^2 = -\pi^2(4P^2 - 4P + I) = -\pi^2 I$. □

P is the projection onto $\mathrm{Eigen}(i\pi)$. This is Milnor's "A is completely determined by $\mathrm{Eigen}(i\pi)$", which uses that a skew-Hermitian matrix is normal; the identity above gives it without the spectral theorem.

Lemma 23.1. Let $|\iota| = 2m$. Let $\mathrm{minVel} = \{A \in \mathfrak{su}(\iota) : A^2 = -\pi^2 I\}$ Chapter12/SUMinimal · minVel and let $G_m(\mathbb{C}^\iota)$ be the set of Hermitian projections of trace m, with the subspace topology Chapter12/SUMinimal · grassmannian. Then $A \mapsto \frac{1}{2}(I - \frac{i}{\pi}A)$ is a homeomorphism $\mathrm{minVel} \cong G_m(\mathbb{C}^\iota)$, with inverse $P \mapsto i\pi(2P - I)$ Chapter12/SUMinimal · lemma_23_1. And minVel is exactly the set of initial velocities of the minimal geodesics from I to $-I$: $A \in \mathrm{minVel}$ iff $A \in \mathfrak{su}$, $\exp A = -I$ and $t \mapsto \exp tA$ is a minimal geodesic Chapter12/SUMinimal · mem_minVel_iff.

Proof. Both maps land where they should Chapter12/SUMinimal · projOf_mem Chapter12/SUMinimal · velOf_mem, by the computation above and the trace formula with $n = 2m$. They are mutually inverse by scalar algebra and are continuous (affine). For the characterization: $A^2 = -\pi^2 I$ gives $\exp A = -I$ Chapter12/SUMinimal · exp_eq_neg_one_of_sq (diagonalize iA; the eigenvalues satisfy $\lambda^2 = \pi^2$, so are $\pm\pi$, odd multiples of π), and then minimality is the previous theorem. □

Mathlib has no Grassmannian; the projections of trace m stand for the m-dimensional subspaces, and the identification as topological spaces is not formalized. The identification of the set of velocities with the space Ω^d of minimal geodesics in the path space is §XIII.18.

Milnor's eigenvectors. For the index (§§XIII.15–13.17) Milnor computes $K_A(W) = \frac{1}{4}[[A,W],A]$ for diagonal A `Chapter14/SUMatrices · jacobiMat`: for $A = D_k$, $[A,W]$ has entries $(d_j - d_l)w_{jl}$ `Chapter14/SUMatrices · commutator_diagonal_apply`, so

$$K_{D_k}(W)_{jl} = \tfrac{\pi^2}{4}(k_j - k_l)^2 w_{jl}$$

`Chapter14/SUMatrices · jacobiMat_suDiag_apply`. Milnor's E_{jl} ($+1$ at (j,l), -1 at (l,j)) `Chapter14/SUMatrices · eReal` and E'_{jl} ($+i$ at both) `Chapter14/SUMatrices · eImag` are eigenvectors with eigenvalue $\frac{\pi^2}{4}(k_j - k_l)^2$ `Chapter14/SUMatrices · jacobiMat_eReal` `Chapter14/SUMatrices · jacobiMat_eImag`, and the diagonal matrices with eigenvalue 0 `Chapter14/SUMatrices · jacobiMat_diagonal`. For $j \neq l$ they are skew-Hermitian `Chapter14/SUMatrices · eReal_conjTranspose` `Chapter14/SUMatrices · eImag_conjTranspose` and traceless `Chapter14/SUMatrices · trace_eReal` `Chapter14/SUMatrices · trace_eImag`. Milnor does not say why they span; every skew-Hermitian W is its diagonal plus $\sum_{j<l}(\operatorname{Re} w_{jl}E_{jl} + \operatorname{Im} w_{jl}E'_{jl})$ `Chapter14/SUMatrices · eq_diagonal_add_sum_eReal_eImag`. Their linear independence is not formalized; the formal index computation (§XIII.17) needs no eigenbasis.

XIII.15 The index from a spectrum: the combinatorial core of Lemma 23.2

Milnor computes the index of $t \mapsto \exp tA$, $A = D_k$, in three steps (pp. 130–132): Theorem 20.5 turns the eigenvalues e of K_A into conjugate points $t = \pi/\sqrt{e}, 2\pi/\sqrt{e}, \ldots$; the Index Theorem adds their multiplicities over $0 < t < 1$; and the result is summed over the eigenvalues. This section does the counting for an arbitrary list of eigenvalues; §XIII.17 proves that the conjugate points of $SU(n)$ are those the list predicts.

Definition. For a real number e, the *conjugate parameters* of e are the $t \in (0,1)$ of the form $\pi s/\sqrt{e}$, $s \in \mathbb{Z}$, and there are none if $e \leq 0$ `Chapter14/IndexCount · conjParams`. They form a finite set `Chapter14/IndexCount · conjParams_finite` (the image of the integers in $(0, \lceil\sqrt{e}/\pi\rceil)$ `Chapter14/IndexCount · conjParams_subset`). For a spectrum $e : \kappa \to \mathbb{R}$ (eigenvalues with multiplicity, κ finite), the multiplicity of t is $\#\{i : t \in \text{conjParams}(e_i)\}$ `Chapter14/IndexCount · conjMult`, and the *spectral index* is $\sum_t \text{conjMult}(e,t)$ `Chapter14/IndexCount · spectralIndex`: Theorems 20.5 and 15.1 applied to the spectrum.

Lemma ("evidently"). For an integer $d \geq 1$, the conjugate parameters of $(\pi d)^2$ are $1/d, \ldots, (d-1)/d$ `Chapter14/IndexCount · conjParams_sq`, so there are $d-1$ of them `Chapter14/IndexCount · ncard_conjParams_sq`.

Lemma (adding over the eigenvalues). $\text{spectralIndex}(e) = \sum_i \#\text{conjParams}(e_i)$ `Chapter14/IndexCount · spectralIndex_eq`.

Proof. The multiplicity function is supported in the finite union of the sets of conjugate parameters; on it the double sum $\sum_t \sum_i [t \in \text{conjParams}(e_i)]$ may be summed in the other order. □

This interchange of two finite sums is Milnor's "adding over all j,l". Consequences: every eigenvalue $> \pi^2$ produces the conjugate parameter $\pi/\sqrt{e} \in (0,1)$, so the index is at least the number of such eigenvalues `Chapter14/IndexCount · card_lt_le_spectralIndex`; a spectrum in which every eigenvalue is doubled has twice the index `Chapter14/IndexCount · spectralIndex_double` (Bott's "conjugate points occur in pairs"); and if every positive eigenvalue has even multiplicity, the index is even `Chapter14/IndexCount · even_spectralIndex`. With the even eigenspaces of §XIII.3, this is 21.7's "the index of every geodesic is even".

Definition (the index formula). For $k : \iota \to \mathbb{Z}$, $\lambda(k) = \sum_{k_i > k_j}(k_i - k_j - 2)$, summed over ordered pairs `Chapter14/IndexCount · bottIndex`. For odd k_j every term is ≥ 0, since two distinct odd integers dif-

fer by at least 2 Chapter14/IndexCount · two_le_sub_of_odd Chapter14/IndexCount · bottIndex_term_nonneg Chapter14/IndexCount · bottIndex_nonneg. The formula is invariant under $k \mapsto -k$ Chapter14/IndexCount · bottIndex_neg and under reordering Chapter14/IndexCount · bottIndex_comp_equiv, and can be written as a sum over the pairs $k_j < k_i$ Chapter14/IndexCount · bottIndex_eq_sum_subtype.

Theorem (the index formula from Milnor's spectrum). For odd k_j, let suSpectrum(k) consist of the numbers $\frac{\pi^2}{4}(k_i - k_j)^2$, $k_i > k_j$, each twice Chapter14/IndexCount · pairSpectrum Chapter14/IndexCount · suSpectrum. Then spectralIndex(suSpectrum(k)) $= \lambda(k)$ Chapter14/IndexCount · spectralIndex_suSpectrum.

Proof. For odd $b < a$, $\frac{\pi^2}{4}(a-b)^2 = (\pi d)^2$ with $d = (a-b)/2 \geq 1$ an integer, which has $d-1$ conjugate parameters, and $2(d-1) = a-b-2$ Chapter14/IndexCount · two_mul_ncard_conjParams. Summing over pairs Chapter14/IndexCount · two_mul_spectralIndex_pairSpectrum and doubling gives $\lambda(k)$. □

Milnor's "evidently $(k_j - k_l)/2 - 1$" uses that $k_j - k_l$ is even.

Minimal geodesics. If all $k_j = \pm 1$, every term vanishes and $\lambda = 0$ Chapter14/IndexCount · bottIndex_eq_zero_of_minimal. In $SU(2m)$, with $\sum k_j = 0$, a minimal k has exactly m entries of each sign Chapter14/IndexCount · card_eq_of_minimal (Eigen$(i\pi)$ is m-dimensional).

Lemma 23.2 (combinatorial form). Let $k_1, \dots, k_{2m}$ be odd with $\sum k_j = 0$, not all ± 1. Then $\lambda(k) \geq 2m + 2$ Chapter14/IndexCount · bottIndex_ge.

The proof as formalized. Milnor treats two cases (p. 132): at least $m+1$ of the k_j "(say) negative", and m of each sign. The formal proof handles both through one inequality.

1. Let P, N be the numbers of positive and negative k_j, S_p the sum of the positive ones and S_n the sum of $|k_j|$ over the negative ones. Then $P + N = 2m$, $S_p - S_n = \sum k_j = 0$, $S_p \geq P$, $S_n \geq N$, and, restricting λ to the pairs (positive k_i, negative k_j), $NS_p + PS_n - 2PN \leq \lambda(k)$ Chapter14/IndexCount · sum_pos_neg_le_bottIndex Chapter14/IndexCount · sum_pos_neg_eq. If some $k_i \geq 3$ then $S_p \geq P + 2$ and $P \geq 1$; if some $k_i \leq -3$ then $S_n \geq N + 2$ and $N \geq 1$ Chapter14/IndexCount · exists_pos_neg_data.
2. Some k_i is ≥ 3 or ≤ -3, since it is odd and not ± 1.
3. If $P < N$: $N \geq m+1$, and $S_p = S_n \geq N$, so $NS_p + PS_n - 2PN \geq N^2 + PN - 2PN = N(N-P) \geq 2N \geq 2m + 2$, using $N - P \geq 2$. If $P > N$: symmetric. If $P = N = m$: $S_p = S_n \geq m + 2$ (one of them is $\geq$ its count $+2$, and they are equal), so the bound is $2m(m+2) - 2m^2 = 4m \geq 2m + 2$ when $m \geq 1$. □

This reproduces Milnor's bounds $2N$ and $4m$ exactly. His "(say) negative" is the invariance under $k \mapsto -k$; his "one of the positive k_i must be ≥ 3" and "one is ≥ 3 and one is ≤ -3" use $\sum k_j = 0$, which here is $S_p = S_n$.

The unitary group. Milnor calls $U(n)$ "rather inconvenient" (p. 127). Worse: $k = (3, 1, \dots, 1)$ Chapter14/IndexCount · remarkK is odd Chapter14/IndexCount · odd_remarkK, not minimal Chapter14/IndexCount · not_minimal_remarkK, of index 0 Chapter14/IndexCount · bottIndex_remarkK, with $\sum k_j = n + 2$ Chapter14/IndexCount · sum_remarkK, while every minimal k has $|\sum k_j| \leq n$ Chapter14/IndexCount · abs_sum_le_card_of_minimal. Since $\sum k_j$ is the degree of det along the geodesic, a homotopy invariant (not formalized), this geodesic lies in a component of the path space without minimal geodesics, and Theorem 22.1 cannot be applied to $U(n)$ at all.

Counts for §24. The same file proves the combinatorial parts of §24, retold with it: for $O(2m)$ the index formula $\lambda_O(a) = \sum_{i<j}(a_i + a_j - 2) + \sum_{a_i > a_j}(a_i - a_j - 2)$ Chapter14/IndexCount · bottIndex0 is the spectral index of the spectrum $\frac{\pi^2}{4}(a_i \pm a_j)^2$, each twice Chapter14/IndexCount · oSpectrum Chapter14/IndexCount · spectralIndex_oSpectrum, is 0 for the minimal geodesics Chapter14/IndexCount · bottIndex0_eq_zero, and is $\geq 2m-2$ otherwise (Lemma 24.2) Chapter14/IndexCount · bottIndex0_ge; the bounds for $k \not\equiv 2$ Chapter14/IndexCount · card_sub_one_le_spectralIndex and $k \equiv 2 \pmod 4$ Chapter14/IndexCount · card_sub_abs_sum_le_two_mul_bottIndex Chapter14/

IndexCount · card_sub_abs_sum_le_four_mul_spectralIndex are the inputs of Milnor's pp. 145–148, including the correction $4\lambda \geq r - |c|$ of his displayed chain on p. 148.

XIII.16 The Jacobi fields of $SU(n)$, explicitly

By §XIII.5 the Jacobi fields along $\exp(tA)$ are hUh with $U'' + K_A(U) = 0$. For A unitarily conjugate to a diagonal matrix this equation is solved entrywise. Milnor reads the spectrum of K_A off the matrix units, which are not skew-Hermitian; no eigenbasis is used here.

Definition. $s_b(t) = \sin(bt)/b$, and $s_0(t) = t$ Chapter12/SUJacobi · jacSin: the solution of $w'' + b^2 w = 0$, $w(0) = 0$, $w'(0) = 1$ (Milnor's s_i, p. 111). It is even in b Chapter12/SUJacobi · jacSin_neg, smooth Chapter12/SUJacobi · contDiff_jacSin, vanishes at 0 Chapter12/SUJacobi · jacSin_zero, has derivative $\cos(bt)$ Chapter12/SUJacobi · hasDerivAt_jacSin, and $\cos(bt)$ has derivative $-b^2 s_b(t)$ Chapter12/SUJacobi · hasDerivAt_cos_mul_jacSin. For $b \neq 0$, $s_b(t) = 0$ iff $bt \in \pi\mathbb{Z}$ Chapter12/SUJacobi · jacSin_eq_zero_iff.

For $a : \iota \to \mathbb{R}$ let $b_{jl} = (a_j - a_l)/2$, and let $S(t) \odot W$ be the matrix with entries $s_{b_{jl}}(t) w_{jl}$ Chapter12/SUJacobi · hadSin, with derivative the matrix of $\cos(b_{jl}t) w_{jl}$ Chapter12/SUJacobi · hadCos Chapter12/SUJacobi · hasDerivAt_hadSin and second derivative minus the matrix of $b_{jl}^2 s_{b_{jl}}(t) w_{jl}$ Chapter12/SUJacobi · hadSin2 Chapter12/SUJacobi · hasDerivAt_hadCos. (Each is a real combination of constant matrices Chapter12/SUJacobi · of_mul_eq_sum.) $S(0) \odot W = 0$ Chapter12/SUJacobi · hadSin_zero and the derivative at 0 is W Chapter12/SUJacobi · hadCos_zero.

Lemma (K for diagonal A, and conjugation). $K_{D_a}(M)_{jl} = b_{jl}^2 M_{jl}$ Chapter12/SUJacobi · jacK_diagI_apply, so $K_{D_a}(S(t) \odot W)$ is the matrix of $b_{jl}^2 s_{b_{jl}}(t) w_{jl}$ Chapter12/SUJacobi · jacK_diagI_hadSin. For unitary T, $K_{TDT^*}(TMT^*) = TK_D(M)T^*$ Chapter12/SUJacobi · jacK_conj_unitary.

Definition (the solutions). For $A = TD_aT^*$, T unitary, and C a matrix, $U_C(t) = T(S(t) \odot T^*CT)T^*$ Chapter12/SUJacobi · suSol.

Theorem. $U_C'' + K_A(U_C) = 0$ Chapter12/SUJacobi · suSol_ode, $U_C(0) = 0$ Chapter12/SUJacobi · suSol_zero, $U_C'(0) = C$ Chapter12/SUJacobi · deriv_suSol_zero, and $U_C(t) \in \mathfrak{su}(\iota)$ for $C \in \mathfrak{su}(\iota)$ Chapter12/SUJacobi · suSol_mem.

Proof. $U_C'' = -T(\text{matrix of } b^2 s_b w)T^*$ Chapter12/SUJacobi · deriv_deriv_suSol $= -TK_{D_a}(S \odot T^*CT)T^* = -K_A(U_C)$ by the lemma. The initial values: $S(0) \odot W = 0$, and the derivative at 0 is $T(T^*CT)T^* = C$. For membership: conjugation by a unitary preserves $\mathfrak{su}$ Chapter12/SUJacobi · conj_mem_specialSkewHermitian Chapter12/SUJacobi · conj_mem_specialSkewHermitian', and $W \mapsto S(t) \odot W$ preserves $\mathfrak{su}$ Chapter12/SUJacobi · hadSin_mem: $S(t)$ is real and symmetric (since $s_{-b} = s_b$), and its diagonal is t, so $\operatorname{tr}(S \odot W) = t \operatorname{tr} W = 0$. □

Theorem (the Jacobi fields J_C). For $A = TD_aT^* \in \mathfrak{su}(\iota)$ and $C \in \mathfrak{su}(\iota)$, $J_C(t) = e^{tA/2} U_C(t) e^{tA/2}$ Chapter12/SUJacobi · suJac is a Jacobi field along $t \mapsto \exp(tA)$ in $SU(\iota)$ Chapter12/SUJacobi · isJacobiField_suJac, with $J_C(0) = 0$ Chapter12/SUJacobi · suJac_zero and $DJ_C/dt(0) = C$ Chapter12/SUJacobi · covAlong_suJac_zero. Moreover $J_C(t) = 0$ iff $S(t) \odot T^*CT = 0$ Chapter12/SUJacobi · suJac_eq_zero_iff.

Proof. §XIII.5 applied to U_C. For the zeros, $J_C(t) = hTS(t) \odot (T^*CT)\,T^*h$ with h, T unitary, so it vanishes iff the middle factor does. □

XIII.17 The index of a geodesic from I to $-I$, by the Index Theorem: Lemma 23.2

§XIII.15 computed the index from a spectrum. Here the spectrum is shown to be the right one: Theorem 20.5 for $SU(n)$ is proved, as a theorem about the submanifold $SU(n) \subseteq M_n(\mathbb{C})$, and combined with

the project's Index Theorem 15.1.

Definition. $\mathrm{hadKer}(a,t) = \{W \in \mathfrak{su}(\iota) : S(t) \odot W = 0\}$, a real subspace `Chapter12/SUIndex · hadKer` `Chapter12/SUIndex · mem_hadKer` ($W \mapsto S(t) \odot W$ is linear `Chapter12/SUIndex · hadSin_add` `Chapter12/SUIndex · hadSin_smul`). For unitary T, $C \mapsto T^*CT$ is a linear isomorphism `Chapter12/SUIndex · conjEquiv`, and $\{C : T^*CT \in \mathrm{hadKer}(a,t)\}$ `Chapter12/SUIndex · conjRange` `Chapter12/SUIndex · mem_conjRange` has the same dimension `Chapter12/SUIndex · finrank_conjRange`.

Theorem 20.5 for $SU(n)$. Let $A = TD_aT^* \in \mathfrak{su}(\iota)$, T unitary, and $t_0 \in [0,1]$. The multiplicity of t_0 as a parameter conjugate to 0 along $t \mapsto \exp(tA)$, i.e. the dimension of the space of Jacobi fields on $[0,1]$ vanishing at 0 and t_0 (Milnor's definition, Part VII) `Chapter9/JacobiSpace · conjMultiplicity` `Chapter9/JacobiSpace · conjSpace`, equals $\dim \mathrm{hadKer}(a,t_0)$ `Chapter12/SUIndex · conjMultiplicity_su`.

The proof as formalized. Milnor's proof of 20.5 describes all Jacobi fields vanishing at 0 in an eigenbasis. Here they are the J_C, by uniqueness.

1. Each J_C, restricted to $[0,1]$ (composed with the clamp), lies in the space of Jacobi fields along γ `Chapter9/JacobiSpace · IsJacobiField.comp_projIcc_mem` `Chapter9/JacobiSpace · jacobiSpace`, with covariant derivative C at 0.
2. Let D map a Jacobi field J vanishing at 0 and t_0 to $DJ/dt(0)$. D is injective: a Jacobi field with $J(0) = 0$ and $DJ/dt(0) = 0$ vanishes `Chapter9/JacobiSpace · jacobiSpace_eq_zero`.
3. The range of D is $\{C : T^*CT \in \mathrm{hadKer}(a,t_0)\}$. If J is such a field, $C = DJ/dt(0)$ is tangent at I, so $C \in \mathfrak{su}$. A Jacobi field is determined by its value and derivative at 0 `Chapter9/JacobiSpace · jacEval_bijective`, so $J = J_C$ on $[0,1]$, and $J(t_0) = 0$ gives $S(t_0) \odot T^*CT = 0$. Conversely, for such C, J_C vanishes at 0 and t_0 and maps to C.
4. So the multiplicity is the dimension of the range, which is $\dim \mathrm{hadKer}(a,t_0)$. □

Theorem (the count). For $a = \pi k$ and $0 < t < 1$, $\dim \mathrm{hadKer}(\pi k, t) = \mathrm{conjMult}(\mathrm{suSpectrum}(k), t)$ `Chapter12/SUIndex · finrank_hadKer`.

The proof as formalized. Coordinates: for each pair $k_l < k_j$ whose eigenvalue $\frac{\pi^2}{4}(k_j - k_l)^2$ has t as a conjugate parameter, the real and imaginary parts of w_{jl} `Chapter12/SUIndex · hadCoord`. These are Milnor's E_{jl}, E'_{jl} components. 1. For $k_l < k_j$ and $0 < t < 1$: $s_{b_{jl}}(t) = 0$ iff t is a conjugate parameter of $\frac{\pi^2}{4}(k_j - k_l)^2$ `Chapter12/SUIndex · jacSin_eq_zero_iff_mem_conjParams`, since $b_{jl} = \pi(k_j - k_l)/2 > 0$ and $\sqrt{b^2} = b$. 2. *Injective* `Chapter12/SUIndex · hadCoord_injective`. Let $W \in \mathrm{hadKer}$ have zero coordinates. Every entry satisfies $s_{b_{jl}}(t) w_{jl} = 0$ `Chapter12/SUIndex · hadKer_entry`. For $k_l < k_j$: if t is a conjugate parameter the coordinates give $w_{jl} = 0$; otherwise $s_{b_{jl}}(t) \neq 0$ and $w_{jl} = 0$. For $k_l = k_j$: $s_0(t) = t \neq 0$, so $w_{jl} = 0$. For $k_l > k_j$: $w_{jl} = -\bar{w}_{lj} = 0$ by skewness `Chapter12/SUIndex · hadKer_skew`. 3. *Surjective* `Chapter12/SUIndex · hadCoord_surjective`. Given coordinates c, put $f_{jl} = c_{jl,0} + ic_{jl,1}$ for the admissible pairs and 0 elsewhere, and $W = f - f^*$. W is skew-Hermitian, has zero diagonal, and $S(t) \odot W = 0$ entry by entry, by step 1. 4. So hadKer is isomorphic to $\mathbb{R}^Q$, Q the set of (pair, $\{0,1\}$) with t a conjugate parameter, whose size is $\mathrm{conjMult}(\mathrm{suSpectrum}(k), t)$ by definition. □

Theorem (Theorems 15.1 + 20.5 for $SU(n)$). Let k be odd with $\sum k_j = 0$, T unitary, $A = TD_kT^*$, so that $\gamma(t) = \exp(tA)$ is a geodesic in $\Omega(SU(n); I, -I)$. Its index, the index of the Hessian E_{**} of the energy (Part VII) `Chapter9/FormIndex · formIndex` `Chapter8/Hessian · energyHessian`, is $\mathrm{spectralIndex}(\mathrm{suSpectrum}(k)) = \lambda(k)$ `Chapter12/SUIndex · formIndex_su_geodesic`.

Proof. The Index Theorem 15.1 `Chapter9/IndexTheorem · theorem_15_1` gives the index as $\sum_{0<t<1}$ of the conjugate multiplicities. The multiplicity function of the spectrum is supported in $(0,1)$ `Chapter12/SUIndex · support_conjMult_subset`, and on $(0,1)$ the two multiplicities agree by the two previous theorems. □

Theorem (the index formula in $SU(n)$). Every geodesic $t \mapsto \exp(tA)$ from I to $-I$ in $SU(n)$ ($A \in \mathfrak{su}$, $\exp A = -I$) has, for odd k_j with $\sum k_j = 0$ and A unitarily conjugate to D_k, index $\lambda(k) = \sum_{k_i > k_j}(k_i -$

$k_j - 2$) Chapter12/SUIndex · index_su_eq_bottIndex. (The diagonalization of §XIII.14 provides k and T.)

Lemma 23.2. In $SU(2m)$, a geodesic $t \mapsto \exp(tA)$ from I to $-I$ has index 0 if all $k_j = \pm 1$ (a minimal geodesic), and index $\geq 2m+2$ otherwise Chapter12/SUIndex · lemma_23_2_index. This is §XIII.15's combinatorial lemma applied to the index formula.

Milnor's reduction to diagonal A by an isometry $X \mapsto TXT^{-1}$ fixing $\pm I$ (p. 130) is replaced by carrying T through the computation.

XIII.18 Theorem 22.1 for $SU(2m)$

Milnor (p. 128): by Lemma 23.1, Ω^d is homeomorphic to $G_m(\mathbb{C}^{2m})$, by Lemma 23.2 every non-minimal geodesic has index $\geq 2m+2$, and "together with Theorem 22.1, this proves" $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m)$ for $i \leq 2m$. Three things must be supplied: the index hypothesis in the form Theorem 22.1 asks (for all geodesics from I to $-I$, not only one-parameter subgroups), the neighbourhood retraction, and the identification of $\Omega^d \subseteq \Omega$ with the set of velocities.

Lemma (geodesics from I). Every geodesic $\gamma : [0,1] \to SU(n)$ with $\gamma(0) = I$ agrees on $[0,1]$ with $t \mapsto \exp(tA)$ for some $A \in \mathfrak{su}$ Chapter13/SUInstance · eqOn_exp_smul_of_isUnbrokenGeodesic_su.

Proof. The geodesic is defined on the open interval $(0,1)$. There it equals $\gamma(\frac{1}{2})\exp((t-\frac{1}{2})X)$ (§XIII.4). Both sides are continuous on $[0,1]$, so they agree on the closure. At $t=0$ this gives $\gamma(\frac{1}{2})\exp(-\frac{1}{2}X) = I$, hence $\gamma(t) = \exp(tX)$. □

Theorem (the index hypothesis for $SU(2m)$). The energy of $t \mapsto \exp(tA)$ on $[0,1]$ is $\|A\|^2$ for skew-Hermitian A Chapter13/SUInstance · energy_exp_smul Chapter8/Energy · energy_eq_of_norm_deriv. Every geodesic from I to $-I$ in $SU(2m)$ that does not have least energy has index $\geq 2m+2$ Chapter13/SUInstance · le_formIndex_of_not_minimal_su.

Proof. The geodesic is $\exp(tA)$ on $[0,1]$ with $\exp A = -I$, and the index only depends on $[0,1]$. By Lemma 23.2 the index is $\geq 2m+2$ unless all $k_j = \pm 1$. In that case $A^2 = -\pi^2 I$ and the geodesic is minimal, so for every path w from I to $-I$, $\|A\| \leq L(w)$, and $E(\gamma) = \|A\|^2 \leq L(w)^2 \leq E(w)$ Chapter8/Energy · pathLength_sq_le_energy: γ has least energy, a contradiction. □

$-I \in SU(2m)$ Chapter13/SUInstance · neg_one_mem_su, since $\det(-I) = (-1)^{2m}$. With the compactness of $SU(2m)$, this gives Corollary 22.2 for $SU(2m)$ conditionally on the retraction Chapter13/SUInstance · corollary_22_2_su.

Ω^d as a subspace of the path space. Lemma 23.1 identifies the set minVel of velocities with the Grassmannian. Theorem 22.1 is about $\Omega^d \subseteq \Omega$ with Milnor's metric.

1. For $A \in$ minVel, $\exp(A/2) = A/\pi$ Chapter14/MinimalPaths · exp_half_smul_of_mem_minVel: in the diagonal form, $e^{\pm i\pi/2} = \pm i$. So a minimal geodesic remembers its velocity at its midpoint. The matrix exponential is continuous Chapter14/MinimalPaths · continuous_matrix_exp.
2. On minVel, $\|A\| = \pi\sqrt{2m}$ Chapter14/MinimalPaths · norm_of_mem_minVel, and the speed of $\exp(tA)$ is $\|A\|$ at every t Chapter14/MinimalPaths · speed_exp_smul. Every path from I to $-I$ has energy $\geq (\pi\sqrt{2m})^2 = \rho(I,-I)^2$ Chapter14/MinimalPaths · pi_sq_le_energy. So $t \mapsto \exp(tA)$, $A \in$ minVel Chapter14/MinimalPaths · expPath Chapter14/MinimalPaths · expPath_apply, of energy $(\pi\sqrt{2m})^2$ Chapter14/MinimalPaths · energy_expPath, lies in Ω^d Chapter14/MinimalPaths · expPath_mem_minPaths.
3. Conversely every $\gamma \in \Omega^d$ agrees on $[0,1]$ with $\exp(tA)$ for some $A \in$ minVel Chapter14/MinimalPaths · exists_minVel_of_mem_minPaths. It is a geodesic (§XIII.9), so $\exp(tA)$ with $\exp A = -I$; its energy $\|A\|^2$ is at most that of the geodesic of k^0 ($D_{k^0} \in$ minVel Chapter14/MinimalPaths · suDiag_kHalf_mem_minVel), so $\|A\| \leq \pi\sqrt{2m}$, which is the distance; so it is a minimal geodesic, and $A^2 = -\pi^2 I$. Then $A = \pi\gamma(\frac{1}{2})$ Chapter14/MinimalPaths · pi_smul_apply_half.

4. $A \mapsto \exp(\cdot A)$ is continuous for Milnor's metric Chapter14/MinimalPaths · continuous_expPath: the speeds are all $\pi\sqrt{2m}$, so the L^2 part vanishes, and the uniform part (measured with the Riemannian distance) is controlled by pointwise convergence at bounded energy, as for the sphere. The inverse $\gamma \mapsto \pi\gamma(\frac{1}{2})$ Chapter14/MinimalPaths · continuous_apply_half only needs the value at $\frac{1}{2}$.

Theorem (Lemma 23.1 for the path space). $A \mapsto (t \mapsto \exp tA)$ Chapter14/MinimalPaths · minVelToPaths is a homotopy equivalence minVel $\simeq$ Ω^d with homotopy inverse $\gamma \mapsto \pi\gamma(\frac{1}{2})$ Chapter14/MinimalPaths · pathsToMinVel Chapter14/MinimalPaths · minPathsHomotopyEquiv, and a homeomorphism onto the separation quotient of Ω^d Chapter14/MinimalPaths · minVelHomeomorph. The proof is the one of §XIII.12: the round trip from minVel is the identity Chapter14/MinimalPaths · pathsToMinVel_minVelToPaths, and from Ω^d it is at distance 0 Chapter14/MinimalPaths · edist_minVelToPaths_pathsToMinVel. With Lemma 23.1 this is Milnor's $\Omega^d \cong G_m(\mathbb{C}^{2m})$.

Theorem (the neighbourhood retraction for $SU(2m)$). $\Omega^d(SU(2m); I, -I)$ is a retract of an open neighbourhood in Ω Chapter14/MinimalRetract · exists_retraction_minPaths_su.

The proof as formalized. Read off a matrix near the projection P from $\omega(\frac{1}{2})$, and project it to the nearest Hermitian projection by the functional calculus.

1. *The spectral projection.* For Hermitian H, specProj(H) is the spectral projection onto the eigenvalues $> \frac{1}{2}$, Mathlib's continuous functional calculus applied to the indicator of $(\frac{1}{2}, \infty)$ Chapter14/MinimalRetract · halfInd Chapter14/MinimalRetract · specProj. The indicator is continuous off $\frac{1}{2}$ Chapter14/MinimalRetract · continuousOn_halfInd, so $H \mapsto$ specProj(H) is continuous along any continuous family of Hermitian matrices that never have $\frac{1}{2}$ as an eigenvalue Chapter14/MinimalRetract · continuousOn_specProj Mathlib/Analysis/CStarAlgebra/ContinuousFunctionalCalculus/Continuity · ContinuousOn.cfc_of_mem_nhdsSet. It is a Hermitian projection Chapter14/MinimalRetract · specProj_isSelfAdjoint Chapter14/MinimalRetract · specProj_mul_self, and the identity on Hermitian projections, whose spectrum is in $\{0, 1\}$ Chapter14/MinimalRetract · specProj_eq_self Chapter14/MinimalRetract · half_notMem_spectrum_of_idem. "$\frac{1}{2}$ is not an eigenvalue" is an open condition, $\det(\frac{1}{2} - H) \neq 0$ Chapter14/MinimalRetract · isOpen_half_notMem_spectrum. The trace of a Hermitian projection is a natural number, the number of eigenvalues 1 Chapter14/MinimalRetract · exists_trace_eq_natCast.
2. *The matrix read off.* hermOf(X) is the Hermitian part of $\frac{1}{2}(I - iX)$ Chapter14/MinimalRetract · hermOf, Hermitian Chapter14/MinimalRetract · hermOf_isSelfAdjoint and continuous Chapter14/MinimalRetract · continuous_hermOf. At $X = \omega(\frac{1}{2}) = A/\pi$ for a minimal geodesic it is $P = \frac{1}{2}(I - \frac{i}{\pi}A)$ Chapter14/MinimalRetract · hermOf_of_mem_minVel. Let $Q(\omega) =$ specProj(hermOf $\omega(\frac{1}{2})$) Chapter14/MinimalRetract · minProj.
3. *The neighbourhood.* O = the paths with $\frac{1}{2}$ not an eigenvalue of hermOf $\omega(\frac{1}{2})$ and $|\operatorname{tr} Q(\omega) - m| < \frac{1}{2}$ Chapter14/MinimalRetract · minNbhd. Q is continuous on the first set Chapter14/MinimalRetract · continuousOn_minProj, so O is open Chapter14/MinimalRetract · isOpen_minNbhd. On O, $Q(\omega)$ is a Hermitian projection whose trace is a natural number within $\frac{1}{2}$ of m, so equal to m: $Q(\omega) \in G_m$ Chapter14/MinimalRetract · minProj_mem.
4. *The retraction.* $\omega \mapsto \exp(t \cdot i\pi(2Q(\omega) - I))$ on O Chapter14/MinimalRetract · retrPath is continuous (Lemma 23.1 and the continuity of $A \mapsto \exp(\cdot A)$) Chapter14/MinimalRetract · continuousOn_retrPath with values in Ω^d Chapter14/MinimalRetract · retrPath_mem_minPaths. For $\omega \in \Omega^d$: $\omega(\frac{1}{2}) = A/\pi$, hermOf $= P$ has no eigenvalue $\frac{1}{2}$ and is its own spectral projection, of trace m; so $\omega \in O$ and the retraction agrees with ω on $[0, 1]$ Chapter14/MinimalRetract · mem_minNbhd_and_eqOn.
5. As for the sphere, the retraction is $\omega \mapsto \omega$ on Ω^d and retrPath elsewhere on O, continuous because the two are at distance 0 on Ω^d. □

The trace is locally constant on projections, so the rank m is preserved near Ω^d. Milnor needs the retraction (through Hanner's theorem) but never writes it down.

Corollary 22.2 for $SU(2m)$, unconditionally. For $i \leq 2m$ and every minimal geodesic γ from I to $-I$, $\pi_i(\Omega^d, \gamma) \cong \pi_{i+1}(SU(2m), I)$ Chapter14/MinimalRetract · corollary_22_2_su', and as groups for $1 \leq i \leq 2m$

`Chapter14/PeriodicityGroups · corollary_22_2_su_mulEquiv`. ($\lambda_0 = 2m+2$, so $i \le \lambda_0 - 2 = 2m$.)

XIII.19 Local trivializations from orbit sections; Stiefel manifolds

Milnor uses three fibrations in §23 (pp. 128–129), $U(m) \to U(m+1) \to S^{2m+1}$, $U(m) \to U(2m) \to U(2m)/U(m)$ and $U(m) \to U(2m)/U(m) \to G_m(\mathbb{C}^{2m})$, and their orthogonal analogues in §24, all quoted from Steenrod (*The Topology of Fibre Bundles*, §7). Steenrod's argument: a coset map $G \to G/H$ with local cross-sections is a bundle. The formal text isolates that argument for an equivariant map between spaces on which a group acts, and builds the bundles on concrete spaces of matrices, so that no coset space appears.

Definition (a trivialization from inverse maps). If $p : E \to B$ is continuous, $U \subseteq B$, $\varphi : p^{-1}(U) \to F$ and $\psi : U \times F \to E$ are continuous, $p(\psi(b,y)) = b$, and φ, ψ are inverse in the evident sense, then $x \mapsto (p(x), \varphi(x))$ is a homeomorphism $p^{-1}(U) \cong U \times F$ over U `Homotopy/Homogeneous · trivOfInverse` `Homotopy/Homogeneous · trivOfInverse_apply_fst`.

Theorem (local triviality from an orbit section). Let a topological group G act continuously on E and B, $p : E \to B$ continuous and equivariant, $b_0 \in B$. If $s : U \to G$ is continuous with $s(b)b_0 = b$ on U, then $p^{-1}(U) \cong U \times p^{-1}(b_0)$ over U, by $x \mapsto (p(x), s(p(x))^{-1}x)$ with inverse $(b,y) \mapsto s(b)y$ `Homotopy/Homogeneous · orbitTriv` `Homotopy/Homogeneous · orbitTriv_apply_fst`. If moreover G acts transitively on B and such a section exists on an open neighbourhood U_0 of b_0, then every $b \in B$ has an open neighbourhood U with $p^{-1}(U) \cong U \times p^{-1}(b_0)$ over U `Homotopy/Homogeneous · exists_triv_of_orbitSection`, and the same with $p^{-1}(b_0)$ replaced by any space homeomorphic to it `Homotopy/Homogeneous · exists_triv_of_orbitSection'`.

Proof. $s(p(x))^{-1}x$ lies over b_0, by equivariance `Homotopy/Homogeneous · smul_mem_fiber`; the two maps are continuous (inversion and the action are) and mutually inverse. For transitivity, write $b = gb_0$ and move the section: $U = g\,U_0$, $s'(b') = g\,s_0(g^{-1}b')$; then $s'(b')b_0 = g\,g^{-1}b' = b'$. □

The conclusion is the local triviality of Part XII (§XII.13) `Homotopy/Fibration · IsLocallyTrivial`, so such maps are Serre fibrations with the exact homotopy sequence of §§XII.15–17.

Definition (Stiefel manifolds). $V_\kappa(\mathbb{K}^\iota) = \{F \in M_{\iota\times\kappa}(\mathbb{K}) : F^*F = I\}$, the orthonormal κ-frames (columns of F) in $\mathbb{K}^\iota$, with the subspace topology `Homotopy/Stiefel · Stiefel` `Homotopy/Stiefel · mem_stiefel`. For $\kappa = \iota$ it is the unitary group `Homotopy/Stiefel · coe_unitaryGroup_eq_stiefel`; for κ a point it is the unit sphere; for $\iota = \{1..2m\}$, $\kappa = \{1..m\}$ it is Milnor's complex Stiefel manifold $U(2m)/U(m)$ (a frame F is the coset $gU(m)$ of any unitary g whose last m columns are F). $U(\iota)$ acts continuously by left multiplication `Homotopy/Stiefel · mul_mem_stiefel` `Homotopy/Stiefel · coe_smul_stiefel`.

Split the coordinates as $\iota = \iota_1 \sqcup \kappa$. Write a frame as $F = \binom{W}{C}$, W the first $|\iota_1|$ rows, C the last $|\kappa|$, so that $W^*W + C^*C = I$ `Homotopy/Stiefel · conjTranspose_mul_self_fromRows` `Homotopy/Stiefel · stiefel_blocks`. The *base frame* is $F_0 = \binom{0}{I}$, the last $|\kappa|$ standard basis vectors `Homotopy/Stiefel · baseFrame` `Homotopy/Stiefel · baseFrame_mem` `Homotopy/Stiefel · stiefelBase`.

Theorem (the local section). Where $I + C$ is invertible,

$$s(F) = \begin{pmatrix} I - W(I+C^*)^{-1}W^* & W \\ -(I+C)(I+C^*)^{-1}W^* & C \end{pmatrix}$$

`Homotopy/Stiefel · secMat` is unitary `Homotopy/Stiefel · secMat_mem_unitaryGroup` and its last $|\kappa|$ columns are F, i.e. $s(F)F_0 = F$ `Homotopy/Stiefel · secMat_mul_baseFrame`. The set $\{F : I + C \text{ invertible}\}$ `Homotopy/Stiefel · secDomain` is open `Homotopy/Stiefel · isOpen_secDomain` and contains F_0 (where $C = I$, $\det 2I \neq 0$) `Homotopy/Stiefel · stiefelBase_mem_secDomain`, and s is continuous there `Homotopy/Stiefel · continuousAt_secMat` `Homotopy/Stiefel · localSection` `Homotopy/Stiefel · continuous_localSection` `Homotopy/Stiefel · localSection_smul`.

For one column ($C = c$ a scalar) $s(F)$ is the unitary map of the plane spanned by e and F sending e to F, with determinant $(1+c)/(1+\bar{c})$: a matrix version of the rotation taking one unit vector to another.

The proof as formalized. $s(F)F_0$ is the last block column, $\binom{W}{C} = F$. For unitarity `Homotopy/Stiefel · secMat_unitary_aux`, let $M = (I + C^*)^{-1}$, $N = (I + C)^{-1}$, so $M^* = N$, $N(I + C) = I$, $(I + C^*)M = I$, and $W^*W = I - C^*C$. Multiplying out $s(F)^*s(F)$ blockwise: - the (ι_1, ι_1) block is $I - WMW^* - WNW^* + W\,N(W^*W + (I+C^*)(I+C))M\,W^*$, and $N(W^*W + (I+C^*)(I+C))M = N(I+C)M + N(I+C^*)M$ (using $W^*W = I - C^*C$) $= M + N$, so the block is I; - the (ι_1, κ) block is $W - W\,N(W^*W + (I + C^*)C) = W - WN(I + C) = 0$; - the (κ, ι_1) block is $W^* - (W^*W + C^*(I + C))MW^* = W^* - (I + C^*)MW^* = 0$; - the (κ, κ) block is $W^*W + C^*C = I$. □

Theorem (transitivity). $U(\iota)$ acts transitively on orthonormal frames: every $F \in V_\kappa(\mathbb{K}^{\iota_1 \sqcup \kappa})$ is gF_0 for a unitary g `Homotopy/Stiefel · exists_unitary_smul_baseFrame`. The columns of F are orthonormal in the Euclidean space `Homotopy/Stiefel · orthonormal_cols`; Mathlib extends an orthonormal family to an orthonormal basis `Mathlib/Analysis/InnerProductSpace/PiL2 · Orthonormal.exists_orthonormalBasis_extension_of_card_eq`, placed so that the given vectors are the last $|\kappa|$; the matrix of that basis is unitary and its last columns are F.

The bundles. (i) *Restriction to the last vectors.* $V_{\kappa_1 \sqcup \kappa_2}(\mathbb{K}^{\iota_1 \sqcup \kappa_2}) \to V_{\kappa_2}(\mathbb{K}^{\iota_1 \sqcup \kappa_2})$, $F \mapsto$ its last $|\kappa_2|$ columns `Homotopy/Stiefel · stiefelRestrict`, is continuous `Homotopy/Stiefel · continuous_stiefelRestrict` and equivariant `Homotopy/Stiefel · stiefelRestrict_smul`. A frame whose last columns are F_0 is block diagonal, $\mathrm{diag}(A, I)$ with $A^*A = I$ `Homotopy/Stiefel · eq_fromBlocks_of_toCols₂` (the off-diagonal blocks vanish by orthonormality), and conversely `Homotopy/Stiefel · fromBlocks_mem_stiefel Homotopy/Stiefel · toCols₂_fromBlocks`; so the fibre over F_0 is $V_{\kappa_1}(\mathbb{K}^{\iota_1})$ `Homotopy/Stiefel · stiefelFibreHomeomorph`. The map is locally trivial with this fibre `Homotopy/Stiefel · stiefelRestrict_locallyTrivial`. (ii) *The orbit map.* $U(\iota_1 \sqcup \kappa) \to V_\kappa$, $g \mapsto gF_0$, the last $|\kappa|$ columns of g `Homotopy/Stiefel · unitaryOrbit Homotopy/Stiefel · coe_unitaryOrbit`, continuous `Homotopy/Stiefel · continuous_unitaryOrbit` and equivariant `Homotopy/Stiefel · unitaryOrbit_mul`. Milnor's embedding $S \mapsto \mathrm{diag}(S, I)$ is a continuous homomorphism $U(\iota_1) \to U(\iota_1 \sqcup \kappa)$ `Homotopy/Stiefel · fromBlocks_mem_unitaryGroup Homotopy/Stiefel · unitaryIncl Homotopy/Stiefel · coe_unitaryIncl Homotopy/Stiefel · continuous_unitaryIncl`, and it is a homeomorphism onto the fibre over F_0 `Homotopy/Stiefel · fromBlocks_toBlocks_of_mem_fiber Homotopy/Stiefel · unitaryFibreHomeomorph Homotopy/Stiefel · unitaryFibreHomeomorph_symm_apply`. The orbit map is locally trivial with fibre $U(\iota_1)$ `Homotopy/Stiefel · unitaryOrbit_locallyTrivial`.

Both are instances of the orbit-section theorem, with the section s above. For κ a point (ii) is $U(m) \to U(m+1) \to S^{2m+1}$ (or $O(m) \to O(m+1) \to S^m$), and for $\iota_1 = \kappa$ of size m it is $U(m) \to U(2m) \to U(2m)/U(m)$.

XIII.20 The Grassmannian and the bundle $U(m) \to V_m \to G_m$

Milnor (p. 129): "$G_m(\mathbb{C}^{2m})$ can be identified with the coset space $U(2m)/U(m) \times U(m)$", and the exact sequence of $U(m) \to U(2m)/U(m) \to G_m(\mathbb{C}^{2m})$ gives $\pi_i G_m \cong \pi_{i-1} U(m)$.

Definition. $G_k(\mathbb{K}^\iota)$ is the set of Hermitian projections P ($P^* = P$, $P^2 = P$) of trace k, with the subspace topology `Homotopy/Grassmannian · Grassmannian`. This is the model of Lemma 23.1: Milnor's Grassmannian of §XIII.14 is the same set `Homotopy/BundlesUnitary · grassmannian_eq_Grassmannian`. $U(\iota)$ acts continuously by $g \cdot P = gPg^*$ `Homotopy/Grassmannian · conj_mem_grassmannian Homotopy/Grassmannian · coe_smul_grassmannian`. The *frame-to-span map* $V_m(\mathbb{K}^\iota) \to G_m(\mathbb{K}^\iota)$, $F \mapsto FF^*$ `Homotopy/Grassmannian · mul_conjTranspose_mem_grassmannian Homotopy/Grassmannian · frameSpan`, sends an orthonormal frame to the projection onto its span; it is continuous `Homotopy/Grassmannian · continuous_frameSpan` and equivariant `Homotopy/Grassmannian · frameSpan_smul`.

Lemma (comparison of projections by traces). If P, Q are Hermitian projections with $PQ = Q$ and $\mathrm{tr}\,P = \mathrm{tr}\,Q$, then $P = Q$ `Homotopy/Grassmannian · eq_of_mul_eq_of_trace_eq`. *Proof*: $QP = Q$ by taking adjoints, so $R = P - Q$ satisfies $R^*R = P - Q - Q + Q = R$, and $\mathrm{tr}(R^*R) = \mathrm{tr}\,R = 0$ forces $R = 0$. □

Theorem (every projection is spanned by a frame). Every $P \in G_m(\mathbb{K}^\iota)$ is FF^* for an orthonormal m-frame F `Homotopy/Grassmannian · exists_frame_of_mem_grassmannian`.

Proof. Take an orthonormal basis $b_1, \dots, b_r$ of the space of vectors fixed by P Homotopy/Grassmannian · fixSubmodule, the columns of a frame F Homotopy/Grassmannian · colMatrix Homotopy/Grassmannian · colMatrix_mem_stiefel. $Q = FF^*$ fixes that space, and $PQ = Q$ since the b_k are fixed by P; $QP = P$ since the columns of P are fixed by P. Taking adjoints, $PQ = P$, so $P = Q$, and $m = \operatorname{tr} P = \operatorname{tr}(F^*F) = r$. □

The Gram–Schmidt section. Split $\iota = \iota_1 \sqcup \{1..m\}$, with base frame F_0 and base point $P_0 = F_0F_0^*$ Homotopy/Grassmannian · grassBase. For P near P_0 the columns of PF_0 Homotopy/Grassmannian · gsVecs Homotopy/Grassmannian · colMatrix_gsVecs are linearly independent, since their Gram matrix $F_0^*PF_0$ is invertible Homotopy/Grassmannian · linearIndependent_gsVecs. The Gram–Schmidt process turns them into an orthonormal frame $\sigma(P)$ Homotopy/Grassmannian · gsFrame Homotopy/Grassmannian · gsFrame_mem.

1. *Continuity.* The Gram–Schmidt process is continuous at every linearly independent family Homotopy/Grassmannian · continuousAt_gramSchmidt Homotopy/Grassmannian · continuousAt_gramSchmidtNormed: by well-founded induction, the n-th vector is $f_n - \sum_{i<n}(\langle u_i, f_n\rangle/\|u_i\|^2)u_i$ Mathlib/Analysis/InnerProductSpace/GramSchmidtOrtho · gramSchmidt_def'', built from earlier vectors by operations continuous where $u_i \neq 0$, which holds at independent families. So σ is continuous where $F_0^*PF_0$ is invertible Homotopy/Grassmannian · continuousAt_gsFrame.
2. *$\sigma(P)$ spans the range of P.* The columns of PF_0 are fixed by P (as $P^2 = P$), hence so is their Gram–Schmidt span Homotopy/Grassmannian · mul_gsFrame. So $Q = \sigma\sigma^*$ satisfies $PQ = Q$, and $\operatorname{tr} Q = m = \operatorname{tr} P$; by the comparison lemma $\sigma(P)\sigma(P)^* = P$ Homotopy/Grassmannian · gsFrame_mul_conjTranspose.
3. At P_0 the columns F_0 are already orthonormal, so $\sigma(P_0) = F_0$ Homotopy/Grassmannian · gsFrame_base.
4. On the open set $\{P : F_0^*PF_0$ invertible and $I + C(\sigma(P))$ invertible$\}$ Homotopy/Grassmannian · grassDomain Homotopy/Grassmannian · isOpen_grassDomain, which contains P_0 Homotopy/Grassmannian · grassBase_mem_grassDomain, $g(P) = s(\sigma(P))$ Homotopy/Grassmannian · grassSection is a continuous Homotopy/Grassmannian · continuous_grassSection unitary with $g(P)P_0g(P)^* = s(\sigma)F_0F_0^*s(\sigma)^* = \sigma\sigma^* = P$ Homotopy/Grassmannian · grassSection_smul.

Theorem (the bundle). $U(\iota)$ acts transitively on G_m Homotopy/Grassmannian · exists_unitary_smul_grassBase (span by a frame, move the frame). A frame spanning the range of P_0 is $\binom{0}{S}$ with S unitary Homotopy/Grassmannian · eq_fromRows_of_mul_conjTranspose Homotopy/Grassmannian · fromRows_zero_mem_stiefel Homotopy/Grassmannian · fromRows_zero_mul_conjTranspose, so the fibre over P_0 is $U(m)$ Homotopy/Grassmannian · frameSpanFibreHomeomorph, included by $S \mapsto \binom{0}{S}$ Homotopy/Grassmannian · frameSpanFibreHomeomorph_symm_apply. The frame-to-span map is locally trivial with fibre $U(m)$ Homotopy/Grassmannian · frameSpan_locallyTrivial, by the orbit-section theorem.

XIII.21 Homotopy groups of Stiefel manifolds, unitary groups and Grassmannians

The bundles are locally trivial in the sense of Part XII Homotopy/BundleHomotopy · isLocallyTrivial_stiefelRestrict Homotopy/BundleHomotopy · isLocallyTrivial_unitaryOrbit Homotopy/BundleHomotopy · isLocallyTrivial_frameSpan, hence Serre fibrations (§XII.14) Homotopy/HomotopyLifting · IsLocallyTrivial.isSerreFibration, and the exact sequence (§§XII.15–17) gives everything else from one input. Let $d = \dim_{\mathbb{R}} \mathbb{K}$ (2 for $\mathbb{C}$, 1 for $\mathbb{R}$).

Definition. *The spheres are connected up to their dimension*: $\pi_i(V_1(\mathbb{K}^n)) = 0$ whenever $i + 2 \leq dn$, i.e. $\pi_i S^{dn-1} = 0$ for $i < dn - 1$ Homotopy/BundleHomotopy · SphereConnected. It holds for $\mathbb{C}$ with $d = 2$ and for $\mathbb{R}$ with $d = 1$ (§§XIII.22, 13.23), by Part XII's $\pi_i S^n = 0$ for $i < n$.

General facts. A one-point space has trivial homotopy groups Homotopy/BundleHomotopy · subsingleton_homotopyGroup_of_subsingleton; homeomorphisms transport them Homotopy/BundleHomotopy · subsingleton_homotopyGroup_of_homeomorph; every fibre of a locally trivial map is homeomorphic to the model fibre Homotopy/BundleHomotopy · IsLocallyTrivial.nonempty_homeomorph_fiber (restrict a trivialization over a neighbourhood of b to $\{b\} \times F$); a map identified by a homeomorphism with the inclusion of a fibre is

bijective, or surjective, on π_N when the inclusion is Homotopy/BundleHomotopy · bijective_homotopyGroupMap_of_fiber Homotopy/BundleHomotopy · surjective_homotopyGroupMap_of_fiber; and if every $\pi_{i'}$, $i' \le i$, of the base vanishes and π_i of the fibre does, then π_i of the total space vanishes Homotopy/BundleHomotopy · subsingleton_pi_total, since the fibre inclusion is then onto on π_i (§XII.17) Homotopy/ExactSequenceIso · surjective_homotopyGroupMap_fiberIncl_of_le. Reindexing coordinates is a homeomorphism of Stiefel manifolds and of unitary groups Homotopy/BundleHomotopy · stiefelCongr Homotopy/BundleHomotopy · unitaryCongr; a Stiefel manifold of empty frames is a point.

Theorem (connectivity of Stiefel manifolds). If the spheres are connected with constant d, then $\pi_i(V_k(\mathbb{K}^n)) = 0$ for $k \le n$ and $i + 1 \le d(n-k) + d - 1$ Homotopy/BundleHomotopy · subsingleton_pi_stiefel. So $V_k(\mathbb{C}^n)$ is $2(n-k)$-connected and $V_k(\mathbb{R}^n)$ is $(n-k-1)$-connected.

Proof. Induction on k. For $k = 0$ the Stiefel manifold is a point. For $k + 1 \le n + 1$, reindex $V_{k+1}(\mathbb{K}^{n+1})$ as frames in $\mathbb{K}^{n \sqcup 1}$ indexed by $k \sqcup 1$, and restrict to the last vector: a locally trivial map onto the sphere $V_1(\mathbb{K}^{n+1})$ with fibre $V_k(\mathbb{K}^n)$. The base has $\pi_{i'} = 0$ for $i' \le i$ (since $i + 2 \le d(n+1)$), and the fibre has $\pi_i = 0$ by induction ($n - k$ is unchanged). □

Milnor (p. 129) gets the case $V_m(\mathbb{C}^{2m})$, $i \le 2m$, from the stability of $\pi_i U(m)$ through the exact sequence of $U(m) \to U(2m) \to U(2m)/U(m)$. The induction avoids that sequence and gives all n, k.

Theorem (stability). If the spheres are connected with constant d, the inclusion $U(m) \to U(m+k)$, $S \mapsto \mathrm{diag}(S, I_k)$ Homotopy/BundleHomotopy · unitaryInclC, induces bijections on π_i for $i + 2 \le dm + d - 1$ Homotopy/BundleHomotopy · bijective_homotopyGroupMap_unitaryIncl and surjections for $i + 1 \le dm + d - 1$ Homotopy/BundleHomotopy · surjective_homotopyGroupMap_unitaryIncl, at every base point S.

Proof. The orbit map $U(m+k) \to V_k(\mathbb{K}^{m+k})$ Homotopy/BundleHomotopy · unitaryOrbitC is locally trivial with fibre $U(m)$; the fibre through $\mathrm{diag}(S, I)$ is $U(m)$, included by Milnor's embedding Homotopy/BundleHomotopy · unitaryOrbit_unitaryIncl Homotopy/BundleHomotopy · unitaryFibreAt. The base is $(dm + d - 2)$-connected Homotopy/BundleHomotopy · subsingleton_pi_unitaryOrbit_base. In the exact sequence, the fibre inclusion is bijective on π_i when the base has $\pi_{i'} = 0$ for $i' \le i + 1$ Homotopy/ExactSequenceIso · bijective_homotopyGroupMap_fiberIncl_of_le, and surjective when $i' \le i$. □

For $\mathbb{C}$ this is bijective for $i < 2m$ and onto for $i \le 2m$ (Lemma 23.4); for $\mathbb{R}$, bijective for $i < m - 1$ and onto for $i \le m - 1$.

Theorem ($\pi_{j+1}G_m \cong \pi_j U(m)$). If the spheres are connected with constant d, then for $j+2 \le dm'+d-1$ there is a bijection $\pi_{j+1}(G_m(\mathbb{K}^{m'+m}), P_0) \cong \pi_j(U(m), I)$ Homotopy/BundleHomotopy · grassmannianEquiv, and for $j + 3 \le dm' + d - 1$ a group isomorphism $\pi_{j+2}G_m \cong \pi_{j+1}U(m)$ Homotopy/BundleHomotopy · nonempty_mulEquiv_grassmannian.

Proof. The frame-to-span map $V_m(\mathbb{K}^{m'+m}) \to G_m$ Homotopy/BundleHomotopy · frameSpanC is a Serre fibration Homotopy/BundleHomotopy · isSerreFibration_frameSpanC whose total space is $(dm' + d - 2)$-connected Homotopy/BundleHomotopy · subsingleton_pi_stiefel_sum. When the total space has $\pi_i = 0$ for $i \le j + 1$, the boundary map $\pi_{j+1}(\text{base}) \to \pi_j(\text{fibre})$ is a bijection (§XII.17) Homotopy/ExactSequenceIso · boundaryEquivPi. Compose with the identification of the fibre with $U(m)$ Homotopy/BundleHomotopy · grassFibre, which sends the base point of the fibre to I Homotopy/BundleHomotopy · grassFibre_fiberPt. For the group version, the boundary map is a homomorphism Homotopy/ExactSequenceIso · boundaryHom and bijective when π_{j+1} and π_{j+2} of the total space vanish Homotopy/ExactSequenceIso · bijective_boundaryMap Homotopy/ExactSequenceIso · subsingleton_homotopyGroup_option; the reindexing of cube coordinates is a group isomorphism Homotopy/LoopSpace · homotopyGroupCongrMulEquiv. □

XIII.22 Lemma 23.4 for the unitary group

Theorem (spheres). $V_1(\mathbb{K}^\iota)$ is the unit sphere of $\mathbb{K}^\iota$, $F \mapsto$ its column Homotopy/BundlesUnitary · stiefelSphere (a one-frame is a unit vector Homotopy/BundlesUnitary · mem_stiefel_fin_one Homotopy/BundlesUnitary · norm_

eq_one_iff_sum_star_mul_self). So the complex spheres are connected with $d = 2$ Homotopy/BundlesUnitary · sphereConnected_complex, transported from Part XII's $\pi_i S^{2n-1} = 0$ for $i < 2n-1$ (§XII.18) Homotopy/Spheres · subsingleton_pi_sphere_complex. And $U(m) \to U(m+1) \to S^{d(m+1)-1}$, $g \mapsto$ its last column, is locally trivial with the honest unit sphere as base Homotopy/BundlesUnitary · sphereOrbit Homotopy/BundlesUnitary · isLocallyTrivial_sphereOrbit, because local triviality is invariant under homeomorphisms of the base Homotopy/BundlesUnitary · IsLocallyTrivial.homeomorph_comp (Milnor's $U(m+1)/U(m) \cong S^{2m+1}$, p. 128).

Lemma 23.4 (Milnor, pp. 128–129), for $\mathbb{C}$. - The inclusion $U(m) \to U(m+k)$ is bijective on π_i for $i < 2m$ Homotopy/BundlesUnitary · bijective_homotopyGroupMap_unitaryIncl_complex and onto for $i \le 2m$ Homotopy/BundlesUnitary · surjective_homotopyGroupMap_unitaryIncl_complex. - $V_k(\mathbb{C}^n)$ is $2(n-k)$-connected Homotopy/BundlesUnitary · subsingleton_pi_stiefel_complex; for $V_m(\mathbb{C}^{2m}) = U(2m)/U(m)$ this is Milnor's $\pi_i(U(2m)/U(m)) = 0$, $i \le 2m$. - $\pi_{j+1}(G_m(\mathbb{C}^{2m}), P_0) \cong \pi_j(U(m), I)$ for $j+1 \le 2m$ Homotopy/BundlesUnitary · grassmannianEquiv_complex, as groups for $j+2 \le 2m$ Homotopy/BundlesUnitary · nonempty_mulEquiv_grassmannian_complex.

These are the general theorems of §XIII.21 at $d = 2$ and $m' = m$. Milnor's isomorphisms are basepoint-free and of groups. Here $\pi_1 G_m \cong \pi_0 U(m)$ is a bijection of pointed sets (both are trivial); the group isomorphism is for $i \ge 2$.

The bridge to Lemma 23.1 and Theorem 22.1. Milnor's $G_m(\mathbb{C}^{2m})$ of Lemma 23.1 is the Grassmannian of the bundle, for $|\iota| = 2m$. Reindexing $\iota \cong \{1..m\} \sqcup \{1..m\}$ is a homeomorphism of Grassmannians Homotopy/BundlesUnitary · reindex_mem_grassmannian Homotopy/BundlesUnitary · grassCongr Homotopy/BundlesUnitary · card_eq_of_card_eq_two_mul, and the Grassmannian is homogeneous: the action of a unitary matrix moves any P to P_0 Homotopy/BundlesUnitary · grassMove Homotopy/BundlesUnitary · grassMove_self. Their composite Homotopy/BundlesUnitary · grassmannianToBase Homotopy/BundlesUnitary · grassmannianToBase_self and the transport of homotopy groups by a homeomorphism, as sets and as groups Homotopy/BundlesUnitary · piEquivOfHomeomorph Homotopy/BundlesUnitary · nonempty_mulEquiv_of_homeomorph, give

$$\pi_{j+1}(G_m(\mathbb{C}^{2m}), P) \cong \pi_j(U(m), I) \quad (j+1 \le 2m)$$

at every point P Homotopy/BundlesUnitary · grassmannianPiEquiv, as groups for $j+2 \le 2m$ Homotopy/BundlesUnitary · nonempty_mulEquiv_grassmannianPi. Through Lemma 23.1 the same holds for the space minVel of initial velocities of minimal geodesics Homotopy/BundlesUnitary · minVelPiEquiv Homotopy/BundlesUnitary · nonempty_mulEquiv_minVelPi.

Theorem ($\pi_j SU(n) \cong \pi_j U(n)$ for $j \ge 2$). The inclusion $SU(n) \to U(n)$ Homotopy/BundlesUnitary · suInclC induces bijections on π_{j+2} at every point Homotopy/BundlesUnitary · bijective_homotopyGroupMap_suInclC (the last clause of Lemma 23.4).

The proof as formalized. 1. *The trivial bundle $SU(n) \to U(n) \to S^1$.* det maps $U(n)$ to the circle Homotopy/BundlesUnitary · det_mem_circle Homotopy/BundlesUnitary · detCircle. $D(z) = \mathrm{diag}(z, 1, \dots, 1)$ is unitary Homotopy/BundlesUnitary · diagonal_update_mem Homotopy/BundlesUnitary · diagCircle, continuous in z Homotopy/BundlesUnitary · continuous_diagCircle, with $\det D(z) = z$ Homotopy/BundlesUnitary · det_diagCircle; so $gD(\det g)^{-1}$ has determinant 1 Homotopy/BundlesUnitary · det_mul_inv_diagCircle, and $g \mapsto (\det g, gD(\det g)^{-1})$ is a homeomorphism $U(n) \cong S^1 \times SU(n)$ Homotopy/BundlesUnitary · unitaryProd. A globally trivial map is locally trivial Homotopy/BundlesUnitary · isLocallyTrivial_of_homeomorph Homotopy/BundlesUnitary · isLocallyTrivial_detCircle, and $SU(n)$ is the fibre through each of its points Homotopy/BundlesUnitary · suFibre. 2. *$\pi_i S^1 = 0$ for $i \ge 2$* Homotopy/BundlesUnitary · subsingleton_pi_circle. A covering map with homeomorphic fibres is locally trivial Homotopy/BundlesUnitary · IsCoveringMap.isLocallyTrivial; $t \mapsto e^{it}$ is a covering (Mathlib) Mathlib/Analysis/SpecialFunctions/Complex/Circle · Circle.isCoveringMap_exp whose fibre over $e^{i\theta}$ is $\theta + 2\pi\mathbb{Z} \cong \mathbb{Z}$ Homotopy/BundlesUnitary · circleFibreEquiv, discrete, so $\mathbb{R} \to S^1$ is locally trivial with fibre $\mathbb{Z}$ Homotopy/BundlesUnitary · isLocallyTrivial_circleExp. A discrete space has trivial higher homotopy groups Homotopy/BundlesUnitary · subsingleton_homotopyGroup_discrete, and $\mathbb{R}$ is contractible Homotopy/Spheres · subsingleton_homotopyGroup_of_contractible; the exact sequence gives $\pi_{j+2} S^1 = 0$ Homotopy/ExactSequenceIso · subsingleton_pi_base. 3. In the exact sequence of det, the fibre inclusion is bijective

on π_{j+2} because π_{j+2} and π_{j+3} of the base vanish Homotopy/ExactSequenceIso · bijective_homotopyGroupMap_fiberIncl. □

Milnor's $j = 0$ (both groups connected) is not formalized.

Clopen sets. The path component of a base point carries all higher homotopy groups: the inclusion of an open and closed set $s \ni x$ induces bijections $\pi_N(s, x) \to \pi_N(X, x)$ for N nonempty Homotopy/BundlesUnitary · bijective_homotopyGroupMap_clopen. A connected image that meets a clopen set lies in it Homotopy/BundlesUnitary · range_subset_of_isClopen; a generalized loop at x meets s at the corner of the cube Homotopy/BundlesUnitary · zero_mem_cube_boundary Homotopy/BundlesUnitary · genLoop_mem_of_isClopen, so restricts to s Homotopy/BundlesUnitary · genLoopRestrict (surjectivity); a homotopy, being connected, also stays in s (injectivity). This is what §XIII.13 used for Ω'.

XIII.23 The orthogonal bundles

The real case of §XIII.21 serves Milnor's §24, retold after this part; it is collected here with the other bundles. The real spheres are connected with $d = 1$ Homotopy/BundlesOrthogonal · sphereConnected_real ($\pi_i S^{n-1} = 0$ for $i < n-1$ Homotopy/Spheres · subsingleton_pi_sphere_of_finrank), and $O(m) \to O(m+1) \to S^m$ is locally trivial Homotopy/BundlesOrthogonal · isLocallyTrivial_sphereOrbit_real. Hence:

- stability for $O(m)$: $T \mapsto \mathrm{diag}(T, I_k)$ is bijective on π_i for $i < m-1$ Homotopy/BundlesOrthogonal · bijective_homotopyGroupMap_orthogonalIncl and onto for $i \le m-1$ Homotopy/BundlesOrthogonal · surjective_homotopyGroupMap_orthogonalIncl. With $k = 16$ this is Milnor's stabilization $\Omega_0(16r) \to \Omega_0(16r+16)$, $T \mapsto T \oplus I_{16}$ (p. 138), bijective for $i < 16r - 1$ Homotopy/BundlesOrthogonal · bijective_homotopyGroupMap_omegaZeroStab;
- $V_k(\mathbb{R}^n)$ is $(n-k-1)$-connected Homotopy/BundlesOrthogonal · subsingleton_pi_stiefel_real;
- $\pi_{j+1} G_m(\mathbb{R}^{m'+m}) \cong \pi_j O(m)$ for $j + 2 \le m'$ Homotopy/BundlesOrthogonal · grassmannianEquiv_real, as groups for $j + 3 \le m'$ Homotopy/BundlesOrthogonal · nonempty_mulEquiv_grassmannian_real, at every point of a Grassmannian of $\mathbb{R}^\iota$, $\iota \cong \{1..m'\} \sqcup \{1..m\}$ Homotopy/BundlesOrthogonal · grassmannianToBase_real Homotopy/BundlesOrthogonal · grassmannianPiEquiv_real Homotopy/BundlesOrthogonal · nonempty_mulEquiv_grassmannianPi_real. For the middle Grassmannian $G_r(\mathbb{R}^{2r})$ of $\Omega_7(16r)$ this gives $\pi_i G_r(\mathbb{R}^{2r}) \cong \pi_{i-1} O(r)$ for $1 \le i \le r-1$.

Theorem ($\pi_i SO(n) \cong \pi_i O(n)$ for $i \ge 1$). The inclusion $SO(n) \to O(n)$ Homotopy/BundlesOrthogonal · soInclC induces bijections on π_N for every nonempty finite N Homotopy/BundlesOrthogonal · bijective_homotopyGroupMap_soInclC (Milnor's "$\pi_{i+1} SO(n) = \pi_{i+1} O(n)$" in Theorem 24.3). *Proof*: on $O(n)$, $\det^2 = 1$ Homotopy/BundlesOrthogonal · det_mul_self_of_mem_orthogonalGroup, so $SO(n)$ is the set $\{\det = 1\} = \{\det > 0\}$ Homotopy/BundlesOrthogonal · soSet, open and closed in $O(n)$ Homotopy/BundlesOrthogonal · isClopen_soSet, and homeomorphic to its copy there Homotopy/BundlesOrthogonal · soHomeomorph; the clopen lemma of §XIII.22 applies. □

What §24 needs beyond this (the colimits O and Ω_k, the stabilizations of $\Omega_k(16r)$ for $k \ge 1$, the coset descriptions $\Omega_1(n) \cong O(n)/U(n/2)$ and the quaternionic bundles, and $\pi_0 O(n) = \mathbb{Z}/2$) is not in these files; the module header lists it.

XIII.24 Theorem 23.3, the periodicity theorem, and Theorem 23.5

Theorem 23.3 (Bott). Let $|\iota| = 2m$. For every $P \in G_m(\mathbb{C}^\iota)$ and $i \le 2m$, $\pi_i(G_m(\mathbb{C}^\iota), P) \cong \pi_{i+1}(SU(2m), I)$ Chapter14/Periodicity · theorem_23_3, as groups for $1 \le i \le 2m$ Chapter14/PeriodicityGroups · theorem_23_3_mulEquiv.

Proof. Three steps: the homeomorphism $G_m \cong$ minVel of Lemma 23.1, sending P to A; the homotopy equivalence minVel $\simeq \Omega^d$ of §XIII.18, sending A to the geodesic $\exp(tA)$; and the unconditional Corol-

lary 22.2 for $SU(2m)$ at that geodesic. For groups, the first two are induced by continuous maps, hence homomorphisms, and the third is the group form. □

Milnor states 23.3 as: the inclusion $G_m(\mathbb{C}^{2m}) \to \Omega(SU(2m); I, -I)$ induces isomorphisms in dimensions $\leq 2m$. Theorem 22.1 is proved here as a compression statement, which gives bijections $\pi_i\Omega^d \to \pi_i\Omega$, and the loop-space step is a bijection too; so 23.3 and the periodicity theorem are first stated as bijections, and then as group isomorphisms where both sides are groups.

Milnor's chain. For $1 \leq i \leq 2m$ (written $i = j + 1$), $\pi_{i-1}U(m) \cong \pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1}SU(2m) \cong \pi_{i+1}U(2m)$ `Chapter14/Periodicity · nonempty_equiv_pi_unitary_two_mul`, as groups for $2 \leq i \leq 2m$ `Chapter14/PeriodicityGroups · nonempty_mulEquiv_pi_unitary_two_mul`. *Proof*: Lemma 23.4 at the point \$P = \$ the image of D_{k^0} ($|\{1..m\} \sqcup \{1..m\}| = 2m$ `Chapter14/Periodicity · card_sum_fin`), Theorem 23.3 at P, and $\pi_{i+1}SU \cong \pi_{i+1}U$ ($i + 1 \geq 2$). □

The periodicity theorem. For $1 \leq i$ and $i + 2 \leq 2m$, $\pi_{i-1}U(m) \cong \pi_{i+1}U(m)$ `Chapter14/Periodicity · periodicity_theorem`, as groups for $i \geq 2$ `Chapter14/PeriodicityGroups · periodicity_theorem_mulEquiv`.

Proof. The chain lands in $U(2m)$. Stability $U(m) \to U(m + m)$ is bijective on π_{i+1} because $i + 1 < 2m$, and sends I to I (a homomorphism of groups). □

Milnor's "for $i \leq 2m$" in the chain gives $\pi_{i-1}U(m) \cong \pi_{i+1}U(2m)$; to land back in $U(m)$ stability needs $i + 1 < 2m$. Both are recorded.

Theorem 23.5 (Bott). For $j + 1 \leq 2m$: $\pi_j U(m)$ is trivial for even j, and $\pi_j U(m) \cong \mathbb{Z}$ (as sets) for odd j `Chapter14/Periodicity · theorem_23_5`. For $2k + 2 \leq 2m$, $\pi_{2k+1}U(m) \cong \mathbb{Z}$ as groups `Chapter14/CircleGroup · theorem_23_5_mulEquiv`.

The proof as formalized. 1. *π_0 and π_1 of the circle.* π_0 of a discrete space is the space `Chapter14/Periodicity · Homotopy.zerothHomotopyEquivOfDiscrete`, and π_0 of a path-connected space is trivial `Chapter14/Periodicity · Homotopy.subsingleton_pi_zero`. $\pi_1(S^1) \cong \mathbb{Z}$ as sets `Chapter14/Periodicity · Homotopy.nonempty_pi_one_circle_equiv_int`: in the exact sequence of the covering $\mathbb{Z} \to \mathbb{R} \to S^1$, $\mathbb{R}$ is contractible, so the boundary map $\pi_1(S^1) \to \pi_0(\text{fibre}) = \mathbb{Z}$ is bijective. 2. *Stability from $U(1) = S^1$.* $\det : U(1) \to S^1$ is a homeomorphism `Chapter14/Periodicity · Homotopy.unitaryOneHomeomorph` sending I to 1 `Chapter14/Periodicity · Homotopy.unitaryOneHomeomorph_one`. For $m \geq 1$ and $i < 2$, stability $U(1) \to U(1 + k)$ and reindexing give $\pi_i U(m) \cong \pi_i S^1$ `Chapter14/Periodicity · Homotopy.nonempty_pi_unitary_equiv_circle`. (For $m = 0$ the range $j + 1 \leq 0$ is empty.) 3. *Induction.* $j = 0$: trivial; $j = 1$: $\mathbb{Z}$; $j + 2$: the periodicity theorem $\pi_j U(m) \cong \pi_{j+2}U(m)$, valid since $j + 3 \leq 2m$. 4. *Groups.* In the stable range, $\pi_j U(m) \cong \pi_1 U(m)$ for odd j `Chapter14/PeriodicityGroups · nonempty_mulEquiv_pi_unitary_odd` and $\pi_j U(m) \cong \pi_2 U(m)$ for even $j \geq 2$ `Chapter14/PeriodicityGroups · nonempty_mulEquiv_pi_unitary_even`, by the group form of periodicity. $\pi_1 U(m) \cong \pi_1 S^1$ as groups for $m \geq 1$ `Chapter14/CircleGroup · Homotopy.nonempty_mulEquiv_pi_one_unitary_circle`, all steps being induced by continuous maps `Chapter14/CircleGroup · Homotopy.piMulEquivOfHomeomorph`. Finally $\pi_1(S^1, 1) \cong \mathbb{Z}$ as groups `Chapter14/CircleGroup · Homotopy.piOneCircleMulEquiv`: Mathlib's covering-space theory makes $t \mapsto e^{it}$ the quotient covering by the deck group $2\pi\mathbb{Z}$ `Mathlib/Analysis/SpecialFunctions/Complex/Circle · Circle.isAddQuotientCoveringMap_exp`, and for the simply connected $\mathbb{R}$ the monodromy at 0 is an anti-isomorphism $\pi_1(S^1, 1) \cong (2\pi\mathbb{Z})^{op}$ `Mathlib/Topology/Homotopy/Lifting · IsAddQuotientCoveringMap.fundamentalGroupEquiv`. The deck group is abelian, so the "op" is harmless, and $2\pi\mathbb{Z} \cong \mathbb{Z}$ `Chapter14/CircleGroup · Homotopy.intAddEquivZMultiplesTwoPi`. Mathlib's fundamental group is identified with the cube model of π_1 `Mathlib/Topology/Homotopy/HomotopyGroup · HomotopyGroup.pi1MulEquivFundamentalGroup`, and $\mathbb{Z}$ is written multiplicatively to match. □

The even groups are trivial, and π_0 is not a group; Milnor's $\pi_1 S^1 \cong \mathbb{Z}$ is quoted, here it is the boundary map of the covering (as sets) and Mathlib's monodromy (as groups).

XIII.25 The Grassmannian of projections is the Grassmannian of subspaces

Lemma 23.1 (§XIII.14) and the bundles (§XIII.20) model $G_m(\mathbb{C}^\iota)$ as the set of Hermitian projections of trace m, with the topology of the matrices. Milnor's $G_m(\mathbb{C}^{2m})$ is "the space of m-dimensional subspaces" (p. 127), with no topology named; the standard one (Steenrod) is the quotient topology from the Stiefel manifold of orthonormal m-frames, $F \mapsto \operatorname{span} F$. A late file identifies the two.

Lemma (the Stiefel manifold is compact). Every entry of an orthonormal frame has modulus ≤ 1 `Homotopy/GrassmannianQuotient · norm_entry_le_one_of_mem_stiefel`, and the frames form a closed set, so $V_\kappa(\mathbb{K}^\iota)$ is compact `Homotopy/GrassmannianQuotient · isCompact_stiefel`.

Theorem ($G_m = V_m/U(m)$). For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, the map $F \mapsto FF^*$ from m-frames to projections of trace m (§XIII.20) has as fibres exactly the orbits of the right action of $U(m)$: $FF^* = F'F'^*$ iff $F' = FS$ for some $S \in U(m)$ `Homotopy/GrassmannianQuotient · frameSpan_eq_iff`. It is onto `Homotopy/GrassmannianQuotient · surjective_frameSpan` and a quotient map `Homotopy/GrassmannianQuotient · isQuotientMap_frameSpan`, so the quotient by the orbit relation `Homotopy/GrassmannianQuotient · frameSetoid` `Homotopy/GrassmannianQuotient · frameSetoid_eq_ker` is homeomorphic to the Grassmannian of projections, $[F] \mapsto FF^*$ `Homotopy/GrassmannianQuotient · stiefelQuotientHomeomorph` `Homotopy/GrassmannianQuotient · stiefelQuotientHomeomorph_mk`.

Proof. If $FF^* = F'F'^*$, put $S = F^*F'$. Then $FS = FF^*F' = F'F'^*F' = F'$ (as $F'^*F' = I$), and $S^*S = F'^*FF^*F' = F'^*F'F'^*F' = I$. Conversely $(FS)(FS)^* = FSS^*F^* = FF^*$. A continuous surjection from a compact space onto a Hausdorff space is a quotient map, and the fibres being the orbits, the induced map on the orbit space is a continuous bijection which is a homeomorphism. □

Theorem (projections are subspaces). The fixed vectors of FF^* are the span of the columns of F `Homotopy/GrassmannianQuotient · fixSubmodule_colMatrix`; the range of a projection of trace m has dimension m `Homotopy/GrassmannianQuotient · finrank_fixSubmodule` and determines the projection `Homotopy/GrassmannianQuotient · eq_of_fixSubmodule_eq`, and every m-dimensional subspace is such a range, the projection FF^* onto the span of an orthonormal basis `Homotopy/GrassmannianQuotient · exists_fixSubmodule_eq`. So $P \mapsto \operatorname{range} P$ is a bijection from $G_m(\mathbb{K}^\iota)$ onto the m-dimensional subspaces `Homotopy/GrassmannianQuotient · grassmannianEquivSubmodule` `Homotopy/GrassmannianQuotient · coe_grassmannianEquivSubmodule`, also for subspaces of $\iota \to \mathbb{K}$, where $\operatorname{range} P = \{x : Px = x\}$ `Homotopy/GrassmannianQuotient · grassmannianEquivSubmodule'` `Homotopy/GrassmannianQuotient · mem_grassmannianEquivSubmodule'`. The composite $V_m \to G_m \to \{\text{subspaces}\}$ is $F \mapsto \operatorname{span} F$ `Homotopy/GrassmannianQuotient · grassmannianEquivSubmodule_frameSpan`. For $\mathbb{K} = \mathbb{C}$ these are stated for the Grassmannian of Lemma 23.1 `Homotopy/GrassmannianQuotient · grassmannian_eq` `Homotopy/GrassmannianQuotient · stiefelQuotientHomeomorph'` `Homotopy/GrassmannianQuotient · grassmannianSubmoduleEquiv`.

So the Grassmannian of §§XIII.14–13.24, with the topology of the matrices, is the space of m-dimensional subspaces with Steenrod's quotient topology: transport the topology along the bijection and both theorems say the same thing. Milnor's Lemma 23.1 and Theorem 23.3 are therefore about his space, not a substitute.

XIII.26 Milnor's §24: the plan, and the formal route

Milnor's §24 proves Bott periodicity for the orthogonal group, $\pi_i O \cong \pi_{i+8} O$. Fix anticommuting complex structures $J_1, \dots, J_{k-1}$ on $\mathbb{R}^n$ and let $\Omega_k(n)$ be the complex structures anticommuting with them ($\Omega_0(n) = O(n)$). The minimal geodesics from J_k to $-J_k$ in $\Omega_k(n)$ are the semicircles through the points of $\Omega_{k+1}(n)$ (Lemma 24.4), every other geodesic has index growing with n (pp. 144–148), and Theorem 22.1 gives $\pi_i\Omega_{k+1}(n) \cong \pi_{i+1}\Omega_k(n)$ in a range growing with n (Theorem 24.5). Eight steps lead from $O(16r)$ to $\Omega_8(16r) \cong O(r)$ (Lemma 24.6), and the colimit gives Theorem 24.7. The algebra behind the index bounds and the eight spaces is that of Clifford modules (Lemma 24.8, Bott–Shapiro).

The formal proof follows this plan, with $n = 16r$ and explicit structures, and stops short of the colimit: the periodicity theorem is proved in the *stable range*, $\pi_i O(n) \cong \pi_{i+8} O(n')$ for $n \geq i+2$ and $n' \geq i+10$ (§XIII.44). The places where it departs from Milnor are these.

- **Algebra in an arbitrary ring.** The sign bookkeeping with words in the J_α is done once, in any ring, and applied to matrices and to endomorphisms of abstract spaces (§XIII.27). Lemma 24.8 is proved from Mathlib's Clifford algebra by a cyclic-vector argument, for $k \leq 8$, with a linear (not isometric) isomorphism (§§XIII.30–13.31).
- **Strata instead of $\Omega_k(n)$.** The components of $\Omega_k(n)$ have different dimensions, so $\Omega_k(n)$ is not a manifold modelled on one space. Theorem 22.1 is applied to the open and closed stratum of J_k, where the tangent space has constant dimension (§XIII.32).
- **No curvature of $\Omega_k(n)$.** Milnor computes the index through the curvature of the totally geodesic $\Omega_k(n)$ and the eigenvalues of K_A. The formal proof needs only a lower bound, and obtains it from Killing fields: isometries of $\Omega_k(n)$ fixing J_k produce Jacobi fields vanishing at the conjugate points Milnor predicts (§XIII.35).
- **An explicit retraction.** As for $SU(2m)$, Theorem 22.1's manifold hypothesis is replaced by a neighbourhood retraction, here by orthogonal projection onto $\mathcal{T}_{J_k}$ followed by the polar decomposition (§XIII.34).
- **The winding number.** For $k = 2, 6$ the path space is split by the winding number of $\det_{\mathbb{C}}$, which must be shown to be an integer in Milnor's normalization; for $k = 6$ this needs $\det_{\mathbb{C}}$ to be a fourth power on $\Omega_6(n)$ (§XIII.39).
- **Bijections at a finite stage.** Every isomorphism is a bijection of pointed sets at a finite stage $n = 16r$. The colimits Ω_k and the group structure along the chain are not formalized (§XIII.44).

Map for §24.

- `Chapter15/Relations` (189 lines): words in anticommuting square roots of -1; the θ's.
- `Chapter15/ComplexStructure` (455 lines): complex structures; $\Omega_k(n)$; the algebra of Lemmas 24.4 and 24.6.
- `Chapter15/Clifford` (372 lines): explicit structures on $\mathbb{R}^{16}$ and $\mathbb{R}^{16r}$; Milnor's imbedding; vector fields on spheres.
- `Chapter15/CliffordModules` (595 lines), `Chapter15/MinimalModules` (643 lines): dimensions of Clifford modules; Lemma 24.8.
- `Chapter15/TotallyGeodesic` (344 lines), `Chapter15/OmegaManifold` (554 lines): Lemma 24.4 (i).
- `Chapter15/OmegaMinimal` (611 lines), `Chapter15/OmegaRetract` (298 lines): Lemmas 24.1 and 24.4 (iii); the retraction.
- `Chapter15/OmegaKilling` (272 lines), `Chapter15/OmegaIntertwiners` (298 lines), `Chapter15/InvariantSplitting` (181 lines), `Chapter15/SplittingClifford` (527 lines), `Chapter15/SplittingData` (275 lines), `Chapter15/IndexBound` (270 lines): the index for $k \not\equiv 2 \pmod 4$ and Lemma 24.2.
- `Chapter15/FiniteStage` (257 lines): Theorems 24.3 and 24.5 at a finite stage, $k \not\equiv 2$.
- `Chapter15/Winding` (396 lines), `Chapter15/OmegaWinding` (188 lines), `Chapter15/OmegaComponents` (488 lines): the components $\Omega^{(c)}$.
- `Chapter15/SplittingTwo` (283 lines), `Chapter15/SplittingTwoData` (295 lines), `Chapter15/KillingTwo` (291 lines), `Chapter15/IndexBoundTwo` (360 lines), `Chapter15/FiniteStageTwo` (178 lines): the case $k = 2, 6$.
- `Chapter15/Stabilization` (193 lines), `Chapter15/EightCommutant` (211 lines), `Chapter15/OmegaEight` (324 lines), `Chapter15/BottChain` (146 lines), `Chapter15/StablePeriodicity` (182 lines): stabilization; $\Omega_8(16r) \cong O(r)$; Theorem 24.7.

XIII.27 Words in anticommuting square roots of -1

Throughout §24 Milnor multiplies anticommuting complex structures and says what commutes with what ("clearly", "it is easy to see"). The formal text does this bookkeeping once, in an arbitrary ring R.

Definition. $J : \mathbb{N} \to R$ satisfies the relations of the Clifford algebra C_k if $J_\alpha^2 = -1$ for $1 \le \alpha \le k$ and $J_\alpha J_\beta = -J_\beta J_\alpha$ for $1 \le \alpha \ne \beta \le k$ `Chapter15/Relations · IsClifford`. From the definition: the swap of two factors, also inside a longer word `Chapter15/Relations · IsClifford.swap` `Chapter15/Relations · IsClifford.swap_assoc`, the cancellation of a square `Chapter15/Relations · IsClifford.sq'` `Chapter15/Relations · IsClifford.sq_assoc`, and restriction to fewer generators `Chapter15/Relations · IsClifford.mono`. Two tactics normalize words with these rules (sorted indices, squares cancelled, signs in front), so that each identity below is a computation. x and y *anticommute* if $xy = -yx$ `Chapter15/Relations · AntiCommute`.

Definition (Milnor's involutions, pp. 139–141). $\theta_3 = J_1J_2J_3$, $\theta_5 = J_1J_4J_5$, $\theta_6 = J_2J_4J_6$, $\theta_7 = J_1J_6J_7$ `Chapter15/Relations · θ₃` `Chapter15/Relations · θ₅` `Chapter15/Relations · θ₆` `Chapter15/Relations · θ₇`, and $\omega_7 = J_1J_2\cdots J_7$ `Chapter15/Relations · ω₇`.

Lemma (the table of the exposition, §4). With the relations of C_k for k large enough:

- $\theta_3^2 = 1$, and θ_3 commutes with J_1, J_2, J_3 `Chapter15/Relations · θ₃_spec` and anticommutes with every J_α, $\alpha \ge 4$ `Chapter15/Relations · θ₃_anticommute`;
- $\theta_5^2 = 1$; θ_5 commutes with J_1, J_4, J_5 and with θ_3, and anticommutes with J_2, J_3 `Chapter15/Relations · θ₅_spec` and with J_α, $\alpha \ge 6$ `Chapter15/Relations · θ₅_anticommute`;
- $\theta_6^2 = 1$; θ_6 commutes with J_2, J_4, J_6, θ_3, θ_5, and anticommutes with J_1, J_3, J_5 `Chapter15/Relations · θ₆_spec` and with J_α, $\alpha \ge 7$ `Chapter15/Relations · θ₆_anticommute`;
- $\theta_7^2 = 1$; θ_7 commutes with J_1, J_6, J_7, θ_3, θ_5, θ_6, and anticommutes with $J_2, \dots, J_5$ `Chapter15/Relations · θ₇_spec` and with J_α, $\alpha \ge 8$ `Chapter15/Relations · θ₇_anticommute`;
- J_7J_8 commutes with $\theta_3, \theta_5, \theta_6$ and anticommutes with θ_7 `Chapter15/Relations · J₇J₈_spec`; J_3J_4 anticommutes with θ_3 and commutes with J_1, J_2 `Chapter15/Relations · J₃J₄_spec`;
- $\omega_7 = -\theta_3\theta_5\theta_7$, and ω_7 commutes with every J_α, $\alpha \le 7$ `Chapter15/Relations · ω₇_spec`.

Proof. Each item is a normalization of words. For instance $\theta_3 J_4 = J_1J_2J_3J_4 = -J_4J_1J_2J_3$, three swaps. That J_α commutes with θ_{abc} iff $\alpha \in \{a, b, c\}$ is the count of swaps: J_α passes three factors, one of which is itself if $\alpha \in \{a, b, c\}$. □

This is the "commute/anticommute" table of the exposition (Chapter 15, §4), complete for $J_1, \dots, J_8$. Milnor states these facts without proof.

XIII.28 Complex structures and the spaces $\Omega_k(n)$

Matrices are real, indexed by a finite type ι with $|\iota| = n$; $J : \mathbb{N} \to M_n(\mathbb{R})$ is the sequence of fixed structures (J_0 unused).

Definition (Milnor p. 133). A *complex structure* is an orthogonal J with $J^2 = -I$ `Chapter15/ComplexStructure · IsComplexStructure`.

Lemma. (a) J is a complex structure iff $J^{\mathsf{T}} = -J$ and $J^2 = -I$ `Chapter15/ComplexStructure · isComplexStructure_iff`. (b) Then $J^{\mathsf{T}}J = I$ `Chapter15/ComplexStructure · IsComplexStructure.transpose_mul_self`, $-J$ is one `Chapter15/ComplexStructure · IsComplexStructure.neg`, n is even `Chapter15/ComplexStructure · IsComplexStructure.even_card` and $|J|^2 = \operatorname{tr}(J^{\mathsf{T}}J) = n$ `Chapter15/ComplexStructure · IsComplexStructure.trace_transpose_mul_self`.

Proof. (a) If J is orthogonal, $J^{\mathsf{T}} = J^{-1} = -J$; if J is skew, $JJ^{\mathsf{T}} = -J^2 = I$. (b) $(\det J)^2 = \det(J^2) = \det(-I) = (-1)^n$ is positive, so n is even: Milnor's "now suppose n is even". □

Definition (Milnor p. 136). $\Omega_0(n) = O(n)$, and for $k \ge 1$, $\Omega_k(n)$ is the set of complex structures anticommuting with $J_1, \dots, J_{k-1}$ `Chapter15/ComplexStructure · Omega`. Milnor's standing assumption, that $J_1, \dots, J_k$

are anticommuting complex structures, is a separate predicate `Chapter15/ComplexStructure · IsAnticommFamily`; it puts the base point J_k in $\Omega_k(n)$ `Chapter15/ComplexStructure · IsAnticommFamily.mem_omega`.

Lemma (p. 136). $\Omega_{k+1}(n) \subseteq \Omega_k(n)$ `Chapter15/ComplexStructure · omega_succ_subset`, and every $\Omega_k(n)$ is compact `Chapter15/ComplexStructure · isCompact_omega`.

Proof. $O(n)$ is closed `Chapter15/ComplexStructure · isClosed_orthogonalGroup` and its entries are bounded by 1 `Chapter15/ComplexStructure · abs_apply_le_one_of_mem_orthogonalGroup`, so it is compact `Chapter15/ComplexStructure · isCompact_orthogonalGroup`; the conditions $J^2 = -I$ and $JJ_\alpha = -J_\alpha J$ are closed `Chapter15/ComplexStructure · isClosed_omega`. □

Lemma (pp. 134–135). If $J^2 = -I$, then $\exp(\theta J) = \cos\theta\, I + \sin\theta\, J$ `Chapter15/ComplexStructure · exp_smul_of_mul_self_eq_neg_one`; in particular $\exp(\pi J) = -I$ `Chapter15/ComplexStructure · exp_pi_smul`.

Proof. $z \mapsto \operatorname{Re} z\, I + \operatorname{Im} z\, J$ is an algebra homomorphism $\mathbb{C} \to M_n(\mathbb{R})$, continuous `Chapter15/ComplexStructure · complexLift_continuous`, and it commutes with the exponential series; $e^{i\theta} = \cos\theta + i\sin\theta$. □

Lemma 24.4 (iii), algebraic part. Let $J_1, \dots, J_k$ be anticommuting complex structures, $k \geq 1$. For $J' \in \Omega_{k+1}(n)$, $A = -J_k J'$ is a complex structure which anticommutes with J_k, commutes with $J_1, \dots, J_{k-1}$, and satisfies $J_k A = J'$ `Chapter15/ComplexStructure · omega_succ_decomp`. Conversely, for such an A, $J_k A \in \Omega_{k+1}(n)$ `Chapter15/ComplexStructure · mul_mem_omega_succ`. The semicircle $\cos t\, J_k + \sin t\, J'$ lies in $\Omega_k(n)$ for every t `Chapter15/ComplexStructure · semicircle_mem_omega` and equals $J_k \exp(tA)$ `Chapter15/ComplexStructure · mul_exp_eq_semicircle`: it starts at J_k, passes through J' at $t = \pi/2$ and ends at $-J_k$.

Proof. $A^2 = J_k J' J_k J' = -J_k^2 J'^2 = -I$; A is a product of two orthogonal matrices, so orthogonal `Chapter15/ComplexStructure · IsComplexStructure.mul_mem_orthogonalGroup`; the commutation rules are word computations. The semicircle is $J_k \exp(tA)$ by the formula for $\exp(tA)$, and lies in $\Omega_k(n)$ by Assertions 1 and 2 below. □

Assertions 1 and 2 (p. 137), the "if" halves. If J is a complex structure and A is skew and anticommutes with J, then $J \exp A$ is a complex structure `Chapter15/ComplexStructure · isComplexStructure_mul_exp`. If A commutes with J_α and J_α anticommutes with J, then J_α anticommutes with $J \exp A$ `Chapter15/ComplexStructure · anticomm_mul_exp`.

Proof. $J \exp A = \exp(-A) J$ when $AJ = -JA$ `Chapter15/ComplexStructure · mul_exp_eq_exp_neg_mul`, and $\exp(A)J = J\exp(-A)$, so $(J\exp A)(J\exp A) = J(\exp(A)J)\exp A = J\,J\exp(-A)\exp A = -I$ `Chapter15/ComplexStructure · exp_neg_mul_exp`. $J\exp A$ is orthogonal because $\exp A$ is, A being skew. For Assertion 2, J_α commutes with $\exp A$ and anticommutes with J. □

The "only if" halves need A small and are §XIII.32.

Lemma 24.4 (ii) (p. 136). For $P, J' \in \Omega_{k+1}(n)$, the reflection $I_P(J') = PJ'^{-1}P = -PJ'P$ lies in $\Omega_{k+1}(n)$ `Chapter15/ComplexStructure · reflection_mem_omega`. *Proof*: it is orthogonal, $(PJ'P)^2 = PJ'P^2J'P = P^2 = -I$, and it anticommutes with each J_α since each of its three factors does. □ The symmetric-space structure of the components (a geodesic-reversing isometry at each point) is not formalized beyond this.

Remark (p. 138). Anticommuting complex structures are orthogonal for the trace form, $\operatorname{tr}(J'^{\mathsf{T}}J) = 0$ `Chapter15/ComplexStructure · trace_transpose_mul_eq_zero`: all of $\Omega_k(n)$ lies on the sphere $|J|^2 = n$, and the minimal geodesics are great semicircles from the pole J_k through the equator $\Omega_{k+1}(n)$.

Lemma 24.6, odd steps (pp. 139–141). Let $1 \leq a < b \leq k$. For $J' \in \Omega_{k+1}(n)$, $\theta = J_a J_b J'$ is a symmetric orthogonal involution commuting with J_a, J_b and anticommuting with every other J_c, $c \leq k$ `Chapter15/ComplexStructure · theta_of_mem_omega`. Conversely every such θ comes from $J' = J_b J_a \theta \in \Omega_{k+1}(n)$ `Chapter15/ComplexStructure · mem_omega_of_theta`, and the two maps are inverse `Chapter15/ComplexStructure · mul_theta_mul`.

Proof. Word computations: $\theta^{\mathsf{T}} = J'^{\mathsf{T}} J_b^{\mathsf{T}} J_a^{\mathsf{T}} = -J' J_b J_{a'}$ and moving J' past J_b, J_a costs two signs, giving θ; $\theta^2 = 1$ similarly. Conversely $(J_b J_a \theta)^2 = (J_b J_a)^2 \theta^2 = -I$ uses that θ commutes with J_a, J_b. □

An involution is the same as its $+1$-eigenspace, which is how Milnor phrases (3), (5), (6), (7): Ω_3 is the set of quaternionic subspaces $\{\theta_3 = 1\}$, and so on. The even steps (4) and (8), $J \mapsto J_k J$, are Lemma 24.4 (iii) with $A = -J_k J$. Milnor's "it is not hard to see that J_5 is uniquely determined" is the inverse map $\theta \mapsto J_b J_a \theta$. Of the identifications of Lemma 24.6 with Grassmannians and classical groups, only (8) is formalized as a homeomorphism (§XIII.43).

XIII.29 Explicit structures on $\mathbb{R}^{16}$ and $\mathbb{R}^{16r}$

Milnor assumes n "divisible by a high power of 2" and never writes the structures down. The formal text builds them from three 2×2 matrices.

Definition. $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\tau = \mathrm{diag}(1, -1)$, $\kappa = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ `Chapter15/Clifford · σ₂` `Chapter15/Clifford · τ₂` `Chapter15/Clifford · κ₂`, with $\sigma^2 = -I$, $\tau^2 = \kappa^2 = I$ `Chapter15/Clifford · σ₂_mul_σ₂` `Chapter15/Clifford · τ₂_mul_τ₂` `Chapter15/Clifford · κ₂_mul_κ₂`, any two anticommuting `Chapter15/Clifford · τ₂_mul_σ₂` `Chapter15/Clifford · κ₂_mul_σ₂` `Chapter15/Clifford · κ₂_mul_τ₂`, σ skew and τ, κ symmetric `Chapter15/Clifford · σ₂_transpose` `Chapter15/Clifford · τ₂_transpose` `Chapter15/Clifford · κ₂_transpose`. Kronecker products ("words") of them multiply slot by slot; signs and transposes pass through `Chapter15/Clifford · neg_kronecker'` `Chapter15/Clifford · kronecker_neg'` `Chapter15/Clifford · transpose_kronecker'`. So two words anticommute iff they carry different non-identity letters in an odd number of slots, and a word with exactly one σ is a complex structure.

Definition. One structure σ on $\mathbb{R}^2$ `Chapter15/Clifford · bott2`; three on $\mathbb{R}^4 = \mathbb{R}^2 \otimes \mathbb{R}^2$, $1 \otimes \sigma$, $\sigma \otimes \tau$, $\sigma \otimes \kappa$ (the quaternions) `Chapter15/Clifford · bott4`; seven on $\mathbb{R}^8$ (the octonions) `Chapter15/Clifford · bott8`; and eight on $\mathbb{R}^{16} = (\mathbb{R}^2)^{\otimes 4}$,

$$J'_1 = 1 \otimes 1 \otimes \sigma \otimes \tau,\ J'_2 = 1 \otimes \sigma \otimes \tau \otimes \tau,\ J'_3 = \sigma \otimes 1 \otimes \kappa \otimes \tau,\ J'_4 = \sigma \otimes \tau \otimes \tau \otimes \tau,$$

$$J'_5 = \sigma \otimes \kappa \otimes \tau \otimes \tau,\ J'_6 = \tau \otimes \sigma \otimes \kappa \otimes \tau,\ J'_7 = \kappa \otimes \sigma \otimes \kappa \otimes \tau,\ J'_8 = 1 \otimes 1 \otimes 1 \otimes \sigma$$

`Chapter15/Clifford · bott16`, the first seven being the seven of $\mathbb{R}^8$ tensored with τ `Chapter15/Clifford · bott16_eq_kronecker`.

Theorem. These are anticommuting families of complex structures: one on $\mathbb{R}^2$, three on $\mathbb{R}^4$, seven on $\mathbb{R}^8$, eight on $\mathbb{R}^{16}$ `Chapter15/Clifford · isAnticommFamily_bott2` `Chapter15/Clifford · isAnticommFamily_bott4` `Chapter15/Clifford · isAnticommFamily_bott8` `Chapter15/Clifford · isAnticommFamily_bott16`. *Proof*: the 28 pairs on $\mathbb{R}^{16}$ are checked slot by slot. □

Transport. An anticommuting family can be restricted to fewer members `Chapter15/Clifford · IsAnticommFamily.mono` and satisfies the relations of C_k `Chapter15/Clifford · IsAnticommFamily.isClifford`. It is carried along any ring homomorphism of matrix algebras compatible with transposition `Chapter15/Clifford · IsAnticommFamily.of_ringHom`, in particular along a renaming of coordinates `Chapter15/Clifford · IsAnticommFamily.reindex` and to the block sums $J_\alpha \oplus \cdots \oplus J_\alpha$ `Chapter15/Clifford · IsAnticommFamily.blockDiagonal`. So $\mathbb{R}^{16r}$ carries Milnor's eight structures $J_\alpha = J'_\alpha \oplus \cdots \oplus J'_\alpha$ `Chapter15/Clifford · isAnticommFamily_bott16r`. Two families on $\mathbb{R}^n$ and $\mathbb{R}^{n'}$ give one on $\mathbb{R}^{n+n'}$, $J_\alpha \oplus J'_\alpha$ `Chapter15/Clifford · IsAnticommFamily.fromBlocks`, and **Milnor's imbedding** (p. 138) $J \mapsto J \oplus J'$ carries $\Omega_k(n) \times \Omega_k(n')$ into $\Omega_k(n + n')$ for these structures `Chapter15/Clifford · fromBlocks_mem_omega`. With $J' = J'_k$ this is $\Omega_k(n) \to \Omega_k(n + n')$; Milnor's imbedding needs $\mathbb{R}^{n'}$ to carry k structures itself.

Definition (Milnor's m_k, p. 143). $m_0, \ldots, m_8 = 1, 2, 4, 4, 8, 8, 8, 8, 16$ and $m_{k+8} = 16 m_k$ `Chapter15/Clifford · milnorM`. For $k \le 8$, $\mathbb{R}^{m_k}$ carries k anticommuting complex structures `Chapter15/Clifford · exists_isAnticommFamily_fin`: the families above, cut down and reindexed.

Vector fields on spheres (Remark, pp. 143–144). $\langle Au, Bv\rangle = \langle u, A^{\mathsf T}Bv\rangle$ `Chapter15/Clifford · dotProduct_mulVec_eq`, and $\langle u, Su\rangle = 0$ for skew S `Chapter15/Clifford · dotProduct_mulVec_self_eq_zero`. Hence, for a complex structure, $Ju \perp u$ `Chapter15/Clifford · IsComplexStructure.dotProduct_mulVec_self` and $|Ju| = |u|$ `Chapter15/Clifford · IsComplexStructure.dotProduct_mulVec_mulVec`; for two anticommuting ones $Ju \perp J'u$, as $J^{\mathsf T}J' = -JJ'$ is skew `Chapter15/Clifford · dotProduct_mulVec_of_anticomm`. So for an anticommuting family, $\langle J_\alpha u, J_\beta u\rangle = \delta_{\alpha\beta}|u|^2$ `Chapter15/Clifford · IsAnticommFamily.dotProduct_mulVec`: $J_1u, \dots, J_ku$ are orthonormal tangent vectors to the unit sphere at u. On S^{16r-1} the eight $u \mapsto J_\alpha u$ are orthonormal tangent fields `Chapter15/Clifford · bott16r_vectorFields`. Adams' upper bound is not formalized.

XIII.30 The dimensions of Clifford modules

Milnor needs two facts about spaces with anticommuting complex structures: the number of minimal summands of $\mathbb{R}^n$ tends to infinity with n, and minimal summands are isomorphic (Lemma 24.8, used on p. 145). This section proves the dimension facts, for an abstract finite-dimensional real vector space V with endomorphisms J_α satisfying the relations of C_k; no inner product is needed.

Lemma (divisibility for $k \le 2$). A complex structure makes V a complex vector space, so $2 \mid \dim V$ `Chapter15/CliffordModules · two_dvd_finrank_of_sq`. Two anticommuting ones J_1, J_2 are the units i, j of a quaternionic structure ($k = J_1J_2$) `Chapter15/CliffordModules · quatBasis`, so V is a module over the quaternions (Mathlib's `QuaternionAlgebra.lift`) and $4 \mid \dim V$ `Chapter15/CliffordModules · four_dvd_finrank_of_anticomm`.

Lemma (halving). If $\theta^2 = 1$ and an injective P anticommutes with θ, then $\dim V = 2\dim\{\theta = 1\}$ `Chapter15/CliffordModules · finrank_eq_two_mul_of_anticomm`; the same inside a subspace U preserved by θ and P `Chapter15/CliffordModules · finrank_submodule_eq_two_mul`.

Proof. $V = \{\theta = 1\} \oplus \{\theta = -1\}$, since $v = \frac12(v + \theta v) + \frac12(v - \theta v)$ `Chapter15/CliffordModules · mem_ker_sub_one_iff`. $P\theta = -\theta P$ makes P map each eigenspace into the other, injectively, so they have the same dimension. □

A complex structure is injective `Chapter15/CliffordModules · injective_of_mul_self_eq_neg_one`; an operator commuting with θ preserves $\{\theta = 1\}$ `Chapter15/CliffordModules · mapsTo_ker_of_commute`, and intersections of invariant subspaces are invariant `Chapter15/CliffordModules · mapsTo_inf`; restrictions keep the relations `Chapter15/CliffordModules · restrict_mul_self` `Chapter15/CliffordModules · restrict_anticomm`.

Definition (Milnor's chain of subspaces, pp. 139–141). $V_1 = \{\theta_3 = 1\}$, $W = V_1 \cap \{\theta_5 = 1\}$, $X = W \cap \{\theta_6 = 1\}$, $X_1 = X \cap \{\theta_7 = 1\}$ `Chapter15/CliffordModules · eigV₁` `Chapter15/CliffordModules · eigW` `Chapter15/CliffordModules · eigX` `Chapter15/CliffordModules · jointEigen`.

Theorem (the chain of Lemma 24.6). With the relations of C_8, $\dim V = 2\dim V_1$, $\dim V_1 = 2\dim W$, $\dim W = 2\dim X$, $\dim X = 2\dim X_1$ `Chapter15/CliffordModules · finrank_chain`, so $\dim V = 16\dim X_1$ `Chapter15/CliffordModules · finrank_eq_sixteen_mul`. Each halving needs fewer structures: the first $k \ge 4$ `Chapter15/CliffordModules · finrank_eq_two_mul_eigV₁`, the second $k \ge 5$ `Chapter15/CliffordModules · finrank_eigV₁_eq`, the third $k \ge 6$ `Chapter15/CliffordModules · finrank_eigW_eq`, the fourth $k = 8$ `Chapter15/CliffordModules · finrank_eigX_eq`.

Proof. Halving four times, with the operator that anticommutes with the new involution and commutes with the old ones (the table of §XIII.27): J_4 for θ_3; J_2 for θ_5 inside V_1; J_1 for θ_6 inside W; J_7J_8 for θ_7 inside X. Each is injective, the last as a product of two complex structures. □

Milnor uses these operators in Lemma 24.6 without comment; the halving is where "the proof of 24.8 follows that of 24.6".

Theorem (m_k divides the dimension). For $k \ge 4$, $8 \mid \dim V$ `Chapter15/CliffordModules · eight_dvd_finrank`: V_1 is preserved by J_1, J_2 (which commute with θ_3), so $4 \mid \dim V_1$, and $\dim V = 2\dim V_1$. Hence for $k \le 8$ every $(J_1, \dots, J_k)$-space has dimension divisible by m_k `Chapter15/CliffordModules · milnorM_dvd_finrank`

(case by case: $k = 1, 2, 3, 4$–$7, 8$), in particular $m_k \mid n$ if $\mathbb{R}^n$ carries k anticommuting complex structures `Chapter15/CliffordModules · IsAnticommFamily.isClifford_toLin'` `Chapter15/CliffordModules · milnorM_dvd_card`. As $m_k > 0$ `Chapter15/CliffordModules · milnorM_pos` and $\mathbb{R}^{m_k}$ carries k structures (§XIII.29), m_k is the least positive dimension of such a space `Chapter15/CliffordModules · isLeast_milnorM`.

Corollary (the dimensions in $\mathbb{R}^{16r}$). With Milnor's structures on $\mathbb{R}^{16r}$, X_1, X, W, V_1 have dimensions $r, 2r, 4r, 8r$ `Chapter15/CliffordModules · finrank_jointEigen_bott16r`. These are Milnor's $V_1 \cong \mathbb{H}^{2r}$, $X \cong \mathbb{R}^{2r}$ and $\dim X_1 = r$ for the base point J_7.

The Clifford algebra. Let $Q_k(x) = -\sum x_i^2$ on $\mathbb{R}^k$ `Chapter15/CliffordModules · cliffQ` `Chapter15/CliffordModules · cliffQ_apply` and C_k Mathlib's Clifford algebra of Q_k. The linear map $x \mapsto \sum x_i J_{i+1}$ `Chapter15/CliffordModules · cliffGen` squares to $Q_k(x)$ (symmetrize the double sum) `Chapter15/CliffordModules · cliffGen_mul_self`, so by the universal property it extends to an algebra homomorphism $C_k \to \operatorname{End} V$ sending e_i to J_{i+1} `Chapter15/CliffordModules · cliffordLift` `Chapter15/CliffordModules · cliffordLift_ι`: Milnor's footnote on p. 136, "these structures make $\mathbb{R}^n$ a module over a Clifford algebra". Mathlib identifies C_k linearly with the exterior algebra, whence $\dim C_k = 2^k$ `Chapter15/CliffordModules · finrank_cliffordAlgebra`.

Definition (Milnor p. 143). V is a *minimal* $(J_1, \dots, J_k)$-space if it is non-zero and has no invariant subspace other than 0 and V `Chapter15/CliffordModules · IsMinimalCliffordModule`. An invariant subspace is again a $(J_1, \dots, J_k)$-space `Chapter15/CliffordModules · IsClifford.exists_restrict`.

Theorem. (a) A minimal space has dimension $\le 2^k$ `Chapter15/CliffordModules · finrank_le_two_pow`. (b) For $k \le 8$, a space of dimension m_k is minimal `Chapter15/CliffordModules · isMinimal_of_finrank_eq_milnorM`, and a minimal space has $m_k \le \dim \le 2^k$ `Chapter15/CliffordModules · milnorM_le_finrank_le`. (c) For $k \le 2$ a minimal space has dimension exactly m_k `Chapter15/CliffordModules · finrank_eq_milnorM_of_le_two`: the lines over $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$.

Proof. (a) For $u \neq 0$ the image of $c \mapsto c \cdot u$, $C_k \to V$, is invariant and non-zero, so it is V; its dimension is at most 2^k. (b) An invariant subspace of V is a $(J_1, \dots, J_k)$-space, so its dimension is a multiple of m_k, hence 0 or m_k. (c) For $k \le 2$, $m_k = 2^k$. □

(a) already gives Milnor's first need: the number of minimal summands of $\mathbb{R}^n$ is at least $n/2^{k+1}$, which tends to infinity with n. Mathlib has the universal property, $C_1 \cong \mathbb{C}$, $C_2 \cong \mathbb{H}$, $\dim C_k = 2^k$ and the graded tensor product, but nothing on the matrix-algebra structure of C_k ($\mathbb{H} \oplus \mathbb{H}$, $\mathbb{H}(2)$, ..., $\mathbb{R}(16)$) or on $C_{k+8} \cong C_k \otimes \mathbb{R}(16)$; the algebra column of the exposition's table is not derived, and $m_{k+8} = 16 m_k$ is the definition of m_k.

XIII.31 Lemma 24.8

Lemma 24.8 (Bott and Shapiro), dimensions. For $k \le 8$, every minimal $(J_1, \dots, J_k)$-space has dimension exactly m_k `Chapter15/MinimalModules · finrank_eq_milnorM_of_isMinimal`.

Lemma 24.8, uniqueness. For $k \le 8$ and $k \not\equiv 3 \pmod 4$, any two minimal $(J_1, \dots, J_k)$-spaces V, V' are isomorphic by a linear isomorphism T with $TJ_\alpha = J'_\alpha T$ for all α `Chapter15/MinimalModules · exists_iso_of_isMinimal`.

Milnor's proof "follows that of 24.6". What carries over is the splitting by the θ's; uniqueness needs a cyclic vector in addition. The formal proof makes C_k itself the model.

The proof as formalized.

1. *The algebra as a module over itself.* Let $e_\alpha \in C_k$ be the generators ($e_\alpha = 0$ outside $[1, k]$) `Chapter15/MinimalModules · cliffE`. They satisfy the relations of C_k `Chapter15/MinimalModules · isClifford_cliffE`, from $Q_k(e_i) = -1$ and $Q_k(e_i + e_j) = -2$ `Chapter15/MinimalModules · cliffQ_single` `Chapter15/MinimalModules · cliffQ_single_add`. Right multiplications by $-e_\alpha$ `Chapter15/MinimalModules · cliffR`

compose as $R_\alpha R_\beta = (\cdot e_\beta e_\alpha)$ Chapter15/MinimalModules · cliffR_mul, so they also satisfy the relations Chapter15/MinimalModules · isClifford_cliffR, and their θ's are the right multiplications by the θ's of C_k (three sign changes and one reversal of three anticommuting factors cancel) Chapter15/MinimalModules · θ₃_cliffR Chapter15/MinimalModules · θ₅_cliffR Chapter15/MinimalModules · θ₆_cliffR Chapter15/MinimalModules · θ₇_cliffR. So the halving chain of §XIII.30 applies to C_k with these operators.

2. *The fixed subspace.* Let $F \subseteq C_k$ be the elements c with $c\theta = c$ for the θ's available for k Chapter15/MinimalModules · mem_ker_mulRight_sub_one. It is a left ideal Chapter15/MinimalModules · mem_of_fixed_mul Chapter15/MinimalModules · ker_mulRight_mul. By the halving chain, $\dim F = 2^k/2^t$ where t is the number of θ's: $16/2 = 8$ for $k = 4$ (one θ), $32/4 = 8$ for $k = 5$, $64/8 = 8$ for $k = 6$, $256/16 = 16$ for $k = 8$. In each case $\dim F = m_k$.
3. *The orbit map.* For $u \in V$ the orbit map $c \mapsto c \cdot u$ Chapter15/MinimalModules · cyclicMap Chapter15/MinimalModules · cyclicMap_apply sends e_α to J_α Chapter15/MinimalModules · cliffordLift_cliffE and is C_k-linear, $(e_\alpha c) \cdot u = J_\alpha(c \cdot u)$ Chapter15/MinimalModules · cyclicMap_mul. For V minimal and $u \neq 0$ it is onto Chapter15/MinimalModules · range_cyclicMap_eq_top.
4. *A fixed cyclic vector.* The θ's of C_k act on V as the θ's of the J_α Chapter15/MinimalModules · cliffordLift_θ₃ Chapter15/MinimalModules · cliffordLift_θ₅ Chapter15/MinimalModules · cliffordLift_θ₆ Chapter15/MinimalModules · cliffordLift_θ₇. The joint $+1$-eigenspace of V is $1/2^t$ of V (§XIII.30), so non-zero; take $u \neq 0$ in it Chapter15/MinimalModules · exists_ne_zero_mem. If $\theta u = u$, the orbit map does not see right multiplication by θ, and every value is taken on F: multiply by the idempotents $\frac{1}{2}(1 + \theta)$ Chapter15/MinimalModules · exists_fixed. So the orbit map maps F onto V Chapter15/MinimalModules · map_cyclicMap_eq_top and $\dim V \leq \dim F$ Chapter15/MinimalModules · finrank_le_of_fixed.
5. *Dimension and isomorphism.* $m_k \mid \dim V$ and $0 < \dim V \leq \dim F = m_k$ force $\dim V = m_k$, and then the orbit map is an isomorphism $F \cong V$ Chapter15/MinimalModules · exists_linearEquiv_of_fixed Chapter15/MinimalModules · minimal_of_fixed. This is carried out for $k \leq 2$ with $F = C_k$ Chapter15/MinimalModules · minimal_le_two and for $k = 4, 5, 6, 8$ Chapter15/MinimalModules · minimal_four Chapter15/MinimalModules · minimal_five Chapter15/MinimalModules · minimal_six Chapter15/MinimalModules · minimal_eight.
6. $k = 3, 7$. An involution commuting with all J_α is ± 1 on a minimal space, its eigenspaces being invariant Chapter15/MinimalModules · central_involution: this is Milnor's "$J_3 = \pm J_1 J_2$", "$J_7 = \pm J_1 \cdots J_6$". If $J_{k+1} = w\theta$ with w a word in $J_1, \ldots, J_k$ and $\theta = \pm 1$, a subspace invariant under $J_1, \ldots, J_k$ is invariant under J_{k+1}, so a minimal $(J_1, \ldots, J_{k+1})$-space is minimal for $J_1, \ldots, J_k$ Chapter15/MinimalModules · isMinimal_pred. With $\theta = \theta_3$, $w = J_2 J_1$ for $k = 3$, and $\theta = \omega_7$, $w = J_6 \cdots J_1$ for $k = 7$: minimal for 3 structures is minimal for 2, of dimension $4 = m_3$ Chapter15/MinimalModules · minimal_three; minimal for 7 is minimal for 6, of dimension $8 = m_7$ Chapter15/MinimalModules · minimal_seven.
7. *Uniqueness.* Two minimal spaces both isomorphic to the same left ideal F by orbit maps are isomorphic to each other, $T = e' \circ e^{-1}$, and T commutes with the J_α because both orbit maps are C_k-linear and F is a left ideal Chapter15/MinimalModules · exists_iso_of_equiv. For $k \not\equiv 3 \pmod 4$, $k \leq 8$, steps 5 give the two isomorphisms. □

What is not proved. The isomorphism is linear, not an isometry as in Milnor's definition (for orthogonal J_α the polar decomposition would give one); the use on p. 145 needs only the linear one (§XIII.36). That there are exactly two classes for $k = 3, 7$, and anything for $k > 8$, are not formalized. Theorem 24.5 needs Lemma 24.8 for $k + 1 \leq 8$ structures, so it is proved for $k \leq 7$, which is all periodicity needs; Milnor states it for all k.

XIII.32 Lemma 24.4 (i): the strata of $\Omega_k(n)$ are totally geodesic submanifolds

Milnor (p. 137): near $J \in \Omega_k(n)$ the points of $O(n)$ are $J \exp A$ with A small and skew; those in $\Omega_k(n)$ are the $J \exp A$ with A in

$$\mathcal{T}_J = \{A \in \mathfrak{o}(n) : AJ = -JA,\ AJ_\alpha = J_\alpha A\ (\alpha < k)\};$$

so $\Omega_k(n)$ is a smooth submanifold, and "this clearly implies" that it is totally geodesic. Two things are missing: the converse halves of Assertions 1 and 2 hold only for small A, and $\Omega_k(n)$ is not a manifold of one dimension.

Lemma (slice submanifolds from parametrizations). Let $S \subseteq V$. Suppose that near each $x \in S$ there is a smooth $\Phi : Z \times W \to V$ with $\Phi(0) = x$ and invertible derivative at 0, such that near 0 the points $\Phi(z, 0)$ lie in S, and every point $\Phi(q) \in S$ is also $\Phi(g(q), 0)$ for some $g(q) \to 0$. Then S is a slice submanifold modelled on Z `Chapter15/TotallyGeodesic · isSliceSubmanifold_of_param` (§0.1). *Proof*: the slice chart is a local inverse of Φ, from the companion project's inverse function theorem; the hypotheses say that S corresponds to $Z \times 0$ near x. □

Definition (totally geodesic sets of orthogonal matrices). A slice submanifold N of orthogonal matrices with, at each $x \in N$, a subspace T_x of skew matrices of dimension $\dim N$ such that $x \exp A \in N$ and $A \in T_{x \exp A}$ for $A \in T_x$ `Chapter15/TotallyGeodesic · IsTotGeo`. For $O(n)$, $T_x = \mathfrak{o}(n)$; for $\Omega_k(n)$, $T_x = \mathcal{T}_x$.

Theorem (geodesics of a totally geodesic set). Let N be such a set. (a) $xx^* = I$ and $x \exp(tA) \in N$ for $A \in T_x$ `Chapter15/TotallyGeodesic · IsTotGeo.mul_conjTranspose` `Chapter15/TotallyGeodesic · IsTotGeo.mul_exp_smul_mem`. (b) The tangent space of N at x is xT_x `Chapter15/TotallyGeodesic · IsTotGeo.leftT` `Chapter15/TotallyGeodesic · IsTotGeo.mem_leftT_iff` `Chapter15/TotallyGeodesic · IsTotGeo.tangentSubspace_eq` `Chapter15/TotallyGeodesic · IsTotGeo.mem_tangentSubspace_iff`. (c) For $A \in T_x$, $t \mapsto x \exp(tA)$ is a geodesic of N on all of $\mathbb{R}$ `Chapter15/TotallyGeodesic · IsTotGeo.isGeodesicOn_mul_exp_smul`. (d) Every geodesic of N on a preconnected set is of this form, $t \mapsto \gamma(t_0) \exp((t - t_0)X)$ with $X = \gamma(t_0)^* \gamma'(t_0) \in T_{\gamma(t_0)}$ `Chapter15/TotallyGeodesic · IsTotGeo.eqOn_mul_exp_smul`; every unbroken geodesic from p is $s \mapsto p \exp(sV)$ on $[0, 1]$, $V \in T_p$ `Chapter15/TotallyGeodesic · IsTotGeo.eqOn_of_isUnbrokenGeodesic`. (e) The Riemannian exponential is $\exp_x(xA) = x \exp A$ `Chapter15/TotallyGeodesic · IsTotGeo.exp_eq`, and N is geodesically complete `Chapter15/TotallyGeodesic · IsTotGeo.isGeodesicallyComplete`. (f) For orthogonal x and skew V, $t \mapsto x \exp(tV)$ has speed $\|V\|$, energy $\|V\|^2$ and length $\|V\|$ on $[0, 1]$ `Chapter15/TotallyGeodesic · IsTotGeo.norm_deriv_mul_exp_smul` `Chapter15/TotallyGeodesic · IsTotGeo.energy_mul_exp_smul` `Chapter15/TotallyGeodesic · IsTotGeo.pathLength_mul_exp_smul`, lies in $\Omega(N; p, p \exp V)$ `Chapter15/TotallyGeodesic · IsTotGeo.mem_pathSpace_mul_exp` and is an unbroken geodesic `Chapter15/TotallyGeodesic · IsTotGeo.isUnbrokenGeodesic_mul_exp`.

Proof. (b) The curves $t \mapsto x \exp(tA)$ lie in N with velocity xA at 0, so xT_x is contained in the tangent space; both have dimension $\dim N$. (c) The acceleration of $\gamma(t) = x \exp(tA)$ is $\gamma(t)A^2$. At $y = \gamma(t)$ the tangent space is yT_y, consisting of $y \cdot$ (skew), and A^2 is symmetric, so $\langle yA^2, yB \rangle = \operatorname{tr}(A^2 B^{\mathsf{T}}) = 0$ for B skew: the acceleration is normal, which is the geodesic equation of a submanifold (Part VI). (d) By (c) and the uniqueness of geodesics with given initial point and velocity. □

"Totally geodesic" is never used as a definition: Theorem 22.1 needs the list of the geodesics of N itself, and it follows from the normality of the acceleration, without comparing the connections of N and $O(n)$ (the exposition's Lemma 2.2).

The exponential near 0**.** In the Frobenius-normed algebra, exp has derivative the identity at 0 `Chapter15/OmegaManifold · exists_hasStrictFDerivAt_exp_zero`. So it is injective on a ball `Chapter15/OmegaManifold · exists_exp_injective` and has a local inverse near I `Chapter15/OmegaManifold · exists_log` (Mathlib's inverse function theorem). The Frobenius norm is invariant under transposition `Chapter15/OmegaManifold · norm_conjTranspose_real` and orthogonal conjugation `Chapter15/OmegaManifold · norm_conj_orth`, which commutes with exp `Chapter15/OmegaManifold · exp_conj_orth`.

Definition. Milnor's $\mathcal{T}_x$: the skew A anticommuting with x (for $k \geq 1$) and commuting with $J_1, \dots, J_{k-1}$ `Chapter15/OmegaManifold · tanAlg` `Chapter15/OmegaManifold · mem_tanAlg`; for $k = 0$ it is $\mathfrak{o}(n)$, the same at every point `Chapter15/OmegaManifold · tanAlg_zero`. (Membership in $\Omega_k(n)$ unpacked: `Chapter15/OmegaManifold · omega_orth` `Chapter15/OmegaManifold · omega_sq` `Chapter15/OmegaManifold · omega_anti` `Chapter15/OmegaManifold · mem_omega_of`.)

Assertions 1 and 2, both halves. (if) For $x \in \Omega_k(n)$ and $A \in \mathcal{T}_x$, $x \exp A \in \Omega_k(n)$ `Chapter15/OmegaManifold · mul_exp_mem_omega`, and $A \in \mathcal{T}_{x \exp A}$ `Chapter15/OmegaManifold · mem_tanAlg_mul_exp`. (only if) Let ε be a radius on which exp is injective. If x and $x \exp C$ lie in $\Omega_k(n)$ and $\|C\| < \varepsilon$, then $C \in \mathcal{T}_x$ `Chapter15/OmegaManifold · mem_tanAlg_of_mul_exp_mem`.

Proof of "only if". Three applications of injectivity, each to two matrices of norm $< \varepsilon$. (Skew) $\exp C$ is orthogonal, as x and $x \exp C$ are, so $\exp(C^\mathsf{T}) = (\exp C)^\mathsf{T} = \exp(-C)$, and $C^\mathsf{T} = -C$. (Anticommutes with x) $(x \exp C)^2 = -I$ and $x^{-1} = -x$ give $\exp(x^\mathsf{T} C x) \exp C = I$, so $\exp(x^\mathsf{T} C x) = \exp(-C)$ and $x^\mathsf{T} C x = -C$; the norms agree by orthogonal invariance. (Commutes with J_α) J_α anticommutes with both x and $x \exp C$, so J_α commutes with $\exp C$, $\exp(J_\alpha C J_\alpha^\mathsf{T}) = \exp C$ and $J_\alpha C J_\alpha^\mathsf{T} = C$. □

This is where Milnor's ball B_ε must be invariant under $A \mapsto -A$, A^T and orthogonal conjugation: the Frobenius norm is.

Lemma (the local logarithm, p. 137). Near $x \in \Omega_k(n)$, every $y \in \Omega_k(n)$ is $x \exp(L(y))$ with $L(y) \in \mathcal{T}_x$, L continuous at x with $L(x) \to 0$ `Chapter15/OmegaManifold · exists_local_log`. *Proof*: $L(y) = \log(x^\mathsf{T} y)$, small near x, and the "only if" halves. □

Lemma ($\dim \mathcal{T}$ **is locally constant).** Conjugation $B \mapsto g^\mathsf{T} B g$ by an orthogonal g commuting with the J_α is an isometry `Chapter15/OmegaManifold · conjIso` `Chapter15/OmegaManifold · conjIso_apply` carrying $\mathcal{T}_x$ into $\mathcal{T}_{g^\mathsf{T} x g}$ `Chapter15/OmegaManifold · conj_mem_tanAlg`, so $\dim \mathcal{T}_{g^\mathsf{T} x g} = \dim \mathcal{T}_x$ `Chapter15/OmegaManifold · finrank_tanAlg_conj_le` `Chapter15/OmegaManifold · finrank_tanAlg_conj`. For $A \in \mathcal{T}_x$, $x \exp A = g^\mathsf{T} x g$ with $g = \exp(A/2)$ (as A anticommutes with x), so $\dim \mathcal{T}$ is constant along geodesics `Chapter15/OmegaManifold · finrank_tanAlg_mul_exp`, and by the local logarithm it is locally constant on $\Omega_k(n)$ `Chapter15/OmegaManifold · eventually_finrank_tanAlg`.

Definition (the stratum). The stratum of $\Omega_k(n)$ through p is the set of $x \in \Omega_k(n)$ with $\dim \mathcal{T}_x = \dim \mathcal{T}_p$ `Chapter15/OmegaManifold · omegaStratum` `Chapter15/OmegaManifold · omegaStratum_subset` `Chapter15/OmegaManifold · self_mem_omegaStratum`. It is open in $\Omega_k(n)$ `Chapter15/OmegaManifold · eventually_mem_omegaStratum`, closed `Chapter15/OmegaManifold · isClosed_omegaStratum`, hence compact `Chapter15/OmegaManifold · isCompact_omegaStratum`.

Lemma 24.4 (i). The stratum through p is a slice submanifold of $M_n(\mathbb{R})$ modelled on $\mathcal{T}_p$ `Chapter15/OmegaManifold · isSliceSubmanifold_omegaStratum`, and a totally geodesic set of orthogonal matrices with $T = \mathcal{T}$ `Chapter15/OmegaManifold · isTotGeo_omegaStratum`: its geodesics are the $t \mapsto x \exp(tA)$, $A \in \mathcal{T}_x$.

The proof as formalized. Near x in the stratum, parametrize by $\Psi(a, w) = x(\exp a + w)$, $a \in \mathcal{T}_x$, $w \in \mathcal{T}_x^\perp$, where $\mathcal{T}_p \cong \mathcal{T}_x$ (same dimension) is identified by a linear isomorphism. Ψ is smooth with $\Psi(0) = x$; its derivative at 0 is $(a, w) \mapsto x(a + w)$, an isomorphism (left multiplication by an orthogonal matrix, after $\mathcal{T}_x \oplus \mathcal{T}_x^\perp = M_n(\mathbb{R})$). The points $\Psi(a, 0) = x \exp a$ lie in the stratum (Assertions 1, 2 and the constancy of $\dim \mathcal{T}$ along geodesics). A point $\Psi(q)$ of the stratum near x is $x \exp(L(\Psi q))$ with $L(\Psi q) \in \mathcal{T}_x$ small, so it is $\Psi(g(q), 0)$ with $g(q)$ the projection of $L(\Psi q)$, and $g(q) \to 0$. The slice lemma applies. Total geodesy is the definition checked with these facts. □

Why strata. The components of $\Omega_k(n)$ have different dimensions: $\Omega_3(16r)$ is a union of quaternionic Grassmannians $G_j(\mathbb{H}^{4r})$ of all dimensions $4j(4r - j)$. Milnor says so (p. 139) and restricts attention to "the component of largest dimension". The formal Theorem 22.1 sees only the stratum of J_k; the paths from J_k stay in it, and, being open and closed in $\Omega_k(n)$, it has the homotopy groups of $\Omega_k(n)$ at J_k (§XIII.38).

XIII.33 Lemmas 24.1 and 24.4 (iii): the minimal geodesics

Milnor's Lemma 24.1 (pp. 133–135) puts a skew A in the block normal form $\operatorname{diag}(a_j J_0)$ (quoted): $\exp(\pi A) = -I$ forces odd a_j, the length $\pi(2\sum a_j^2)^{1/2} \geq \pi\sqrt{n}$, with equality iff $A^2 = -I$. The formal proof needs no normal form.

Lemma (the eigenvalues of $-V^2$). A solution of $c' = Vc$ is $c(t) = \exp(tV)c(0)$ `Chapter15/OmegaMinimal · eq_exp_smul_mul_of_hasDerivAt`. Let V be skew with $\exp V = -I$, $E \neq 0$ a matrix and $V^2E = -\lambda E$. Then $\lambda \geq \pi^2$ `Chapter15/OmegaMinimal · pi_sq_le_of_sq_mul`.

Proof. $\|VE\|^2 = \langle E, V^{\mathsf{T}}VE\rangle = \lambda\|E\|^2$ `Chapter15/OmegaMinimal · inner_mul_left_adj`, so $\lambda \geq 0$. If $\lambda = 0$ then $VE = 0$, $c \equiv E$ solves $c' = Vc$, and $E = \exp(V)E = -E$, a contradiction. If $\lambda > 0$, put $\mu = \sqrt{\lambda}$: $c(t) = \cos(\mu t)E + \mu^{-1}\sin(\mu t)VE$ solves $c' = Vc$, so $c(1) = \exp(V)E = -E$. Pairing with E and using $\langle E, VE\rangle = 0$ `Chapter15/OmegaMinimal · inner_self_mul_skew` gives $\cos\mu = -1$, and so $\mu \geq \pi$ (as $\cos > -1$ on $(0,\pi)$). □

Lemma 24.1, the length. For V skew with $\exp V = -I$: $V^{\mathsf{T}}V = U\operatorname{diag}(\lambda)U^{\mathsf{T}}$ with U orthogonal and every $\lambda_i \geq \pi^2$ `Chapter15/OmegaMinimal · exists_eigen_ge` (Mathlib's spectral theorem for the symmetric $V^{\mathsf{T}}V = -V^2$, each column of U an eigenvector). As $\|V\|^2 = \operatorname{tr} V^{\mathsf{T}}V = \sum\lambda_i$ `Chapter15/OmegaMinimal · norm_sq_eq_trace`, $\|V\|^2 \geq \pi^2 n$ `Chapter15/OmegaMinimal · pi_sq_mul_card_le_norm_sq`, and if $\|V\|^2 \leq \pi^2 n$ then every $\lambda_i = \pi^2$ and $V^2 = -\pi^2 I$ `Chapter15/OmegaMinimal · mul_self_eq_of_norm_sq_le`.

Definition. For $p \in \Omega_k(n)$, $\operatorname{minVel}_O(p) = \{A \in \mathcal{T}_p : A^2 = -I\}$ `Chapter15/OmegaMinimal · minVelO`, and $\gamma_A(t) = p\exp(\pi tA)$ `Chapter15/OmegaMinimal · omegaGeo`. For $A \in \operatorname{minVel}_O$, $\exp(\pi A) = -I$, $\|A\|^2 = n$ and $\|\pi A\| = \pi\sqrt{n}$ `Chapter15/OmegaMinimal · exp_pi_smul_of_minVelO` `Chapter15/OmegaMinimal · norm_sq_of_minVelO` `Chapter15/OmegaMinimal · norm_pi_smul_of_minVelO`. Assume from now on that $\operatorname{minVel}_O(p)$ is non-empty (for $p = J_k$ this is $\Omega_{k+1}(n) \neq \emptyset$, below). Then $-p$ lies in the stratum of p `Chapter15/OmegaMinimal · neg_mem_omegaStratum`, $\gamma_A \in \Omega(\text{stratum}; p, -p)$ `Chapter15/OmegaMinimal · omegaGeo_mem_pathSpace`, with energy $\pi^2 n$ `Chapter15/OmegaMinimal · energy_omegaGeo`.

Theorem (the distance from p to $-p$ is $\pi\sqrt{n}$, p. 138). In the stratum of p, $\rho(p,-p) = \pi\sqrt{n}$ `Chapter15/OmegaMinimal · riemDist_omegaStratum`. Every path from p to $-p$ has energy $\geq \pi^2 n$ `Chapter15/OmegaMinimal · pi_sq_card_le_energy`.

Proof. "$\leq$": the distance is at most the length of γ_A `Chapter7/Distance · riemDist_le_pathLength`, which is $\|\pi A\|$. "$\geq$": the stratum is geodesically complete (§XIII.32), so Hopf–Rinow at p (Theorem 10.9, Part VI) `Chapter7/HopfRinow · theorem_10_9` gives a tangent vector $v = pV$, $V \in \mathcal{T}_p$, with $\exp_p(v) = -p$ and $\rho(p,-p) = \|v\|$. The Riemannian exponential is $p\exp V$, so $\exp V = -I$, and $\|v\| = \|V\| \geq \pi\sqrt{n}$ by Lemma 24.1. The energy bound follows from length$^2 \leq$ energy. □

Milnor argues "minimal in $O(n)$, hence certainly minimal in $\Omega_k(n)$" (p. 138). What is used instead is that a shortest geodesic exists in the stratum and is an $O(n)$-geodesic $p\exp(tV)$, whose length is $\geq \pi\sqrt{n}$ by the spectral argument; distances in $\Omega_k(n)$ and $O(n)$ are never compared.

Theorem (the minimal geodesics). As a point of $\Omega(\text{stratum}; p, -p)$ `Chapter15/OmegaMinimal · omegaPath` `Chapter15/OmegaMinimal · omegaPath_apply`, γ_A has least energy `Chapter15/OmegaMinimal · omegaPath_mem_minPaths` (the set Ω^d of paths of least energy is §XIII.10's `Chapter13/Theorem22 · minPaths`). Conversely every path of least energy agrees on $[0,1]$ with some γ_A, $A \in \operatorname{minVel}_O(p)$ `Chapter15/OmegaMinimal · exists_minVelO_of_mem_minPaths`.

Proof. A path of least energy is an unbroken geodesic `Chapter13/MinimalDeform · isUnbrokenGeodesic_of_minimal`, hence $t \mapsto p\exp(tV)$ on $[0,1]$ with $V \in \mathcal{T}_p$ (§XIII.32); $\exp V = -I$ from the end point. Its energy $\|V\|^2$ is at most that of γ_{A_0}, $\pi^2 n$, so $V^2 = -\pi^2 I$ by Lemma 24.1, and $A = V/\pi \in \operatorname{minVel}_O$. □

Remark (p. 138). The midpoint of γ_A is pA `Chapter15/OmegaMinimal · omegaGeo_half` ($\exp(\pi A/2) = A$ when $A^2 = -I$).

Theorem (Lemma 24.4 (iii) in the path space). All the γ_A have constant speed $\pi\sqrt{n}$ `Chapter15/OmegaMinimal · speed_omegaGeo`, so $A \mapsto \gamma_A$ is continuous for Milnor's metric `Chapter15/OmegaMinimal · continuous_omegaPath` (the L^2 term of the metric vanishes, and pointwise convergence at bounded energy controls the uniform term) `Chapter15/OmegaMinimal · minVelToOmegaPaths`. The midpoint map $\gamma \mapsto p^{\mathsf{T}}\gamma(\frac{1}{2})$ is continuous `Chapter15/OmegaMinimal · continuous_apply_half_omega`, lands in minVel_O on Ω^d `Chapter15/OmegaMinimal · conjTranspose_mul_half` `Chapter15/OmegaMinimal · half_mem_minVelO` `Chapter15/OmegaMinimal · omegaPathsToMinVel`, and is a left inverse of $A \mapsto \gamma_A$ `Chapter15/OmegaMinimal · omegaPathsToMinVel_minVelToOmegaPaths`; the other composite is at distance 0 from the identity `Chapter15/OmegaMinimal · edist_minVelToOmegaPaths`. Hence $A \mapsto \gamma_A$ is a homotopy equivalence $\mathrm{minVel}_O(p) \simeq \Omega^d$ `Chapter15/OmegaMinimal · omegaMinPathsHomotopyEquiv`, and a homeomorphism onto the separation quotient of Ω^d `Chapter15/OmegaMinimal · omegaMinPathsHomeomorph`.

The proof is that of §XIII.12 for the sphere and §XIII.18 for $SU(2m)$: points of Ω at distance 0 agree on $[0,1]$ and are inseparable, which gives the homotopy and the homeomorphism with the quotient. Paths are functions on $\mathbb{R}$, so Milnor's "homeomorphic" holds only after paths agreeing on $[0,1]$ are identified.

Definition. Milnor's base point: $J_0 = I$ and J_k for $k \geq 1$ `Chapter15/OmegaMinimal · basePt` `Chapter15/OmegaMinimal · basePt_mem_omega`.

Lemma 24.4 (iii). Let $J_1, \dots, J_k$ be anticommuting complex structures. $A \in \mathrm{minVel}_O(J_k)$ iff $J_k A \in \Omega_{k+1}(n)$ `Chapter15/OmegaMinimal · mem_minVelO_iff`, and $A \mapsto J_k A$ is a homeomorphism $\mathrm{minVel}_O(J_k) \cong \Omega_{k+1}(n)$, with inverse $J' \mapsto J_k^{\mathsf{T}} J'$ `Chapter15/OmegaMinimal · minVelOHomeomorph`. The geodesic of J' is the semicircle $t \mapsto \cos(\pi t) J_k + \sin(\pi t) J'$ `Chapter15/OmegaMinimal · omegaGeo_eq_semicircle`.

Proof. For $k = 0$: $J_0 = I$, $\mathcal{T}_I = \mathfrak{o}(n)$, and a skew A with $A^2 = -I$ is a complex structure (§XIII.28): Lemma 24.1. For $k \geq 1$: the algebraic part of §XIII.28 in both directions. Both maps are multiplications by fixed orthogonal matrices. □

Lemma 24.4 (iii) needs $\Omega_{k+1}(n) \neq \emptyset$, which here is the hypothesis that $J_1, \dots, J_{k+1}$ be anticommuting complex structures; Milnor's version "for $0 \leq l < k$" uses his standing assumption tacitly.

XIII.34 The neighbourhood retraction for $\Omega_k(n)$

The formal Theorem 22.1 asks that Ω^d be a retract of an open neighbourhood in Ω (§XIII.10), the conclusion of Hanner's theorem for Milnor's "Ω^d is a topological manifold". For $\Omega_k(n)$ the retraction is the nearest complex structure in $\mathcal{T}_p$, found by orthogonal projection onto $\mathcal{T}_p$ and the polar decomposition.

Definition. $S^{-1/2}$ for a symmetric S, by Mathlib's continuous functional calculus applied to $x \mapsto x^{-1/2}$ `Chapter15/OmegaRetract · invSqrtR` `Chapter15/OmegaRetract · invSqrtM`. It is continuous where the spectrum stays in $(0,\infty)$ `Chapter15/OmegaRetract · continuousOn_invSqrtR` `Chapter15/OmegaRetract · continuousOn_invSqrtM`, symmetric `Chapter15/OmegaRetract · invSqrtM_isSelfAdjoint`, commutes with everything S commutes with `Chapter15/OmegaRetract · commute_invSqrtM`, satisfies $S^{-1/2}S^{-1/2}S = I$ `Chapter15/OmegaRetract · invSqrtM_mul_self_mul` and $I^{-1/2} = I$ `Chapter15/OmegaRetract · invSqrtM_one`. If $\det(Y^{\mathsf{T}}Y) \neq 0$, the spectrum of $Y^{\mathsf{T}}Y$ is in $(0,\infty)$ `Chapter15/OmegaRetract · spectrum_subset_Ioi`.

Definition. For a matrix Y (the midpoint of a path), $Y_1 = \mathrm{midDir}(Y)$ is the orthogonal projection of $p^{\mathsf{T}}Y$ onto $\mathcal{T}_p$ `Chapter15/OmegaRetract · midDir`, and $\mathrm{polarDir}(Y) = Y_1(Y_1^{\mathsf{T}}Y_1)^{-1/2}$ its polar part `Chapter15/OmegaRetract · polarDir`.

Lemma. If $\det(Y_1^{\mathsf{T}}Y_1) \neq 0$, $\mathrm{polarDir}(Y) \in \mathrm{minVel}_O(p)$ `Chapter15/OmegaRetract · polarDir_mem`. On a midpoint pA of a minimal geodesic, $\mathrm{midDir}(pA) = A$ and $\mathrm{polarDir}(pA) = A$ `Chapter15/OmegaRetract · midDir_mul` `Chapter15/OmegaRetract · conjTranspose_mul_self_of_minVelO` `Chapter15/OmegaRetract · polarDir_mul`. midDir is continuous `Chapter15/OmegaRetract · continuous_midDir`.

Proof. $Y_1 \in \mathcal{T}_p$ is skew, so $S = Y_1^{\mathsf{T}}Y_1 = -Y_1^2$ `Chapter15/OmegaRetract · isSelfAdjoint_conjTranspose_mul_self'`

commutes with Y_1, and with p and the J_α: Y_1 anticommutes with p, so Y_1^2 commutes with it, and Y_1 commutes with J_α. Hence $f = S^{-1/2}$ commutes with Y_1, p and the J_α, and $Y_1 f$ is skew, anticommutes with p and commutes with the J_α: it lies in $\mathcal{T}_p$. Finally $(Y_1 f)^2 = Y_1^2 f^2 = -f^2 S = -I$. On pA with $A \in \mathrm{minVel}_O$: $p^\mathsf{T} pA = A$ is already in $\mathcal{T}_p$ and $A^\mathsf{T} A = I$. □

The constraints of $\mathcal{T}_p$ must be imposed first, by projection, and the square root must commute with them; this is why the polar part lands in $\mathcal{T}_p$.

Theorem (the retraction). Let $O = \{\omega \in \Omega(\text{stratum}; p, -p) : \det(Y_1^\mathsf{T} Y_1) \neq 0,\ Y = \omega(\tfrac{1}{2})\}$ `Chapter15/OmegaRetract · omegaNbhd`. Then O is open `Chapter15/OmegaRetract · continuous_midDir_half` `Chapter15/OmegaRetract · isOpen_omegaNbhd`, contains Ω^d, and Ω^d is a retract of O `Chapter15/OmegaRetract · exists_retraction_minPaths_omega`.

Proof. On O, $\omega \mapsto \mathrm{polarDir}(\omega(\tfrac{1}{2}))$ is continuous `Chapter15/OmegaRetract · continuousOn_polarDir_half`, so $\omega \mapsto \gamma_{\mathrm{polarDir}(\omega(1/2))}$ `Chapter15/OmegaRetract · omegaRetrPath` is continuous on O `Chapter15/OmegaRetract · continuousOn_omegaRetrPath` with values in Ω^d `Chapter15/OmegaRetract · omegaRetrPath_mem_minPaths`. A minimal γ lies in O and its image agrees with γ on $[0,1]$ `Chapter15/OmegaRetract · mem_omegaNbhd_and_eqOn` (its midpoint is pA), but need not equal it. So the retraction is $\gamma \mapsto \gamma$ on Ω^d and the formula elsewhere on O; it is at distance 0 from the continuous formula `Chapter10/Omega · edist_eq_zero_of_eqOn`, hence continuous `Chapter10/Deformation · continuous_of_edist_eq_zero`. □

This is the third explicit retraction of the project (after the equator of the sphere, §XIII.12, and the spectral projection for $SU(2m)$, §XIII.18); Hanner's theorem is never needed.

XIII.35 Killing fields and lower bounds for the index

Milnor bounds the index from below (pp. 145–146) through the curvature of $\Omega_k(n)$ (Lemma 2.2 (b) of the exposition: a totally geodesic submanifold has the curvature of $O(n)$) and the eigenvectors of $K_A(B) = -\frac{1}{4}[A,[A,B]]$ on $\mathcal{T}$: an eigenvalue $e^2 \geq 4$ gives a conjugate point at $t = 1/e \leq \frac{1}{2}$. For a lower bound only eigenvectors are needed, and each of Milnor's eigenvectors B is $[X, A]$ for an X in the isotropy algebra of J_k. So the formal proof produces the Jacobi fields from isometries, as Milnor does for the sphere on p. 80.

Definition. The isotropy algebra of p: the skew X commuting with p and with $J_1, \dots, J_{k-1}$ `Chapter15/OmegaKilling · isoAlg`. For such X and every u, $g = \exp(uX)$ is orthogonal and commutes with p and the J_α `Chapter15/OmegaKilling · isoAlg_exp`, with $g^\mathsf{T} = \exp(-uX)$ `Chapter15/OmegaKilling · exp_smul_conjTranspose`; so $y \mapsto gyg^{-1}$ is an isometry of $\Omega_k(n)$ fixing p. The *Killing field* of X along $\gamma(t) = p\exp(tV)$ is $K(t) = X\gamma(t) - \gamma(t)X$ `Chapter15/OmegaKilling · killing`, the derivative at $u = 0$ of $\exp(uX)\gamma(t)\exp(-uX)$ `Chapter15/OmegaKilling · hasDerivAt_conj_exp`.

Theorem (Killing fields are Jacobi fields). For $V \in \mathcal{T}_p$ and X in the isotropy algebra, K is a Jacobi field along $t \mapsto p\exp(tV)$ in the stratum of p `Chapter15/OmegaKilling · isJacobiField_killing`.

Proof. $\alpha(u,t) = \exp(uX)\,p\exp(tV)\,\exp(-uX) = p\exp(t\,gVg^{-1})$, $g = \exp(uX)$, because g commutes with p and conjugation commutes with exp. And $gVg^{-1} \in \mathcal{T}_p$ (conjugation by g preserves $\mathcal{T}$, §XIII.32, and fixes p). So each $\alpha(u,\cdot)$ is a geodesic of the stratum (§XIII.32), α is smooth, and Lemma 14.3 (Part VII) `Chapter9/JacobiVariation · lemma_14_3` makes its variation field $\partial_u\alpha(0,t) = K(t)$ a Jacobi field. □

Lemma. $K'(t) = X\gamma(t)V - \gamma(t)VX$ `Chapter15/OmegaKilling · hasDerivAt_killing`; $K(0) = 0$ `Chapter15/OmegaKilling · killing_zero`; $K(\tau) = 0$ if X commutes with $\exp(\tau V)$ `Chapter15/OmegaKilling · killing_eq_zero` (then $K(\tau) = p(X\exp(\tau V) - \exp(\tau V)X)$); and if K vanishes on $[0,1]$ then X commutes with V `Chapter15/OmegaKilling · commute_of_killing_eqOn` (from $K'(0) = p(XV - VX)$).

Theorem (the multiplicity of a conjugate point). Let γ be an unbroken geodesic from p equal on $[0,1]$ to $p\exp(tV)$, $V \in \mathcal{T}_p$, $\tau \in [0,1]$, and $\mathcal{X}$ a space of X in the isotropy algebra commuting with $\exp(\tau V)$,

on which $X \mapsto [X,V]$ is injective. Then $\gamma(\tau)$ is conjugate to $\gamma(0)$ with multiplicity $\geq \dim \mathcal{X}$ Chapter15/OmegaKilling · finrank_le_conjMultiplicity (the multiplicity of Part VII Chapter9/JacobiSpace · conjMultiplicity). *Proof*: $X \mapsto K$ maps $\mathcal{X}$ linearly into the Jacobi fields vanishing at 0 and τ, injectively by the last lemma. □

Theorem (the index bound from Killing fields). For a finite set $T \subseteq (0,1)$ and spaces $\mathcal{X}_\tau$ as above, the index of γ is at least $\sum_{\tau\in T} \dim \mathcal{X}_\tau$ Chapter15/OmegaKilling · sum_finrank_le_formIndex. *Proof*: the Index Theorem 15.1 (Part VII) Chapter9/IndexTheorem · theorem_15_1 writes the index as the sum of the multiplicities of the conjugate points in $(0,1)$, a finite sum, which dominates the sum over T Chapter15/OmegaKilling · sum_le_finsum_mem. □

Intertwiners. If $A^2D = -b^2D$, then $\exp(tA)D = \cos(bt)D + b^{-1}\sin(bt)AD$ Chapter15/OmegaIntertwiners · exp_smul_mul_of_sq; on the right, $D\exp(tA) = \cos(at)D + a^{-1}\sin(at)DA$ when $DA^2 = -a^2D$ Chapter15/OmegaIntertwiners · eq_mul_exp_smul_of_hasDerivAt Chapter15/OmegaIntertwiners · mul_exp_smul_of_sq (uniqueness for $c' = cA$). D is an *intertwiner* from the $-a^2$- to the $-b^2$-eigenspace of A^2 if $bDA = -aAD$, $A^2D = -b^2D$, $DA^2 = -a^2D$ Chapter15/OmegaIntertwiners · IsInterRel: D maps M_h to M_j, anticommuting with the complex structures A/a and A/b. Then D^T intertwines back Chapter15/OmegaIntertwiners · IsInterRel.conjTranspose, and D commutes with $\exp(sA)$ when $s(a+b) = 2\pi$ Chapter15/OmegaIntertwiners · IsInterRel.commute_exp: $D\exp(sA) = \cos(as)D + a^{-1}\sin(as)DA = \cos(as)D - b^{-1}\sin(as)AD$, and $as = 2\pi - bs$ turns this into $\cos(bs)D + b^{-1}\sin(bs)AD = \exp(sA)D$.

Theorem (the count, pp. 145–146). Let A be skew, $a_0 > 1$, $b_j \geq 1$ ($j \in \kappa$), and intertwiners $D_{j,l}$ ($l < r_0$) from the $-a_0^2$-eigenspace M_0 to pieces M_j, with $D_{j,l} - D_{j,l}^\mathsf{T}$ in the isotropy algebra, detected by projections ($DR = D$, $D^\mathsf{T}R = 0$ for R onto M_0; Q_j onto M_j isolating $AD_{j,l}$), and the $AD_{j,l}$ independent for each j. Then with $\tau_j = 2/(a_0+b_j) \in (0,1)$ there are spaces $\mathcal{X}_\tau$ as in the index bound for $V = \pi A$, of total dimension $r_0|\kappa|$ Chapter15/OmegaIntertwiners · exists_killing_spaces.

Proof. Put $X_{j,l} = D_{j,l} - D_{j,l}^\mathsf{T}$ and $Y_{j,l} = [A, X_{j,l}]$ Chapter15/OmegaIntertwiners · adR. Multiplying on the right by R kills the D^T terms and gives $Y_{j,l}R = -\frac{a_0+b_j}{b_j}AD_{j,l}$; applying Q_j isolates the pieces. So a linear relation among the Y's gives one among the $AD_{j,l}$ for each j, which is trivial: the Y's, hence the X's, are independent. $X_{j,l}$ commutes with $\exp(\tau_j\pi A)$, since D and D^T do ($\tau_j\pi(a_0+b_j) = 2\pi$). Group the X's by τ; a combination commuting with A has $\sum cY = 0$, so all coefficients vanish. □

The independence of the eigenvectors of different pairs, which Milnor does not discuss, is the projection argument. Only $a_0 > 1$ and $b_j \geq 1$ are used: $\tau_j < 1$ follows, and the oddness of the a_h is not needed.

XIII.36 The splitting into minimal pieces

Lemma (invariant subspaces, p. 144). Let $\mathcal{G}$ be a set of operators on a finite-dimensional inner product space, each self-adjoint or skew-adjoint Chapter15/InvariantSplitting · IsStarSigned. A subspace is invariant Chapter15/InvariantSplitting · IsInvar or minimal invariant (non-zero, no invariant subspace in between) Chapter15/InvariantSplitting · IsMinInvar. Then: the orthogonal complement of an invariant subspace is invariant Chapter15/InvariantSplitting · IsInvar.orthogonal; intersections and the whole space are invariant Chapter15/InvariantSplitting · IsInvar.inf Chapter15/InvariantSplitting · isInvar_top; every non-zero invariant subspace contains a minimal one (of least dimension) Chapter15/InvariantSplitting · exists_minInvar_le; and every invariant subspace is the orthogonal sum of a list of minimal ones, pairwise orthogonal, dimensions adding up Chapter15/InvariantSplitting · exists_minInvar_decomp. The orthogonal projection onto an invariant subspace commutes with $\mathcal{G}$ Chapter15/InvariantSplitting · IsInvar.starProjection_comm. A self-adjoint operator commuting with $\mathcal{G}$ and preserving a minimal M is a scalar on M Chapter15/InvariantSplitting · IsMinInvar.exists_scalar (its eigenspaces in M are invariant).

Proof of the splitting. Induction on dimension: take a minimal $M \leq U$, then split $U \cap M^\perp$, invariant by

the complement lemma. □

The operators of §6. On E (later $\mathbb{R}^n$), $a = A$ skew, $q = J$ ($q = 1$ for $k = 0$), $j_\alpha = J_\alpha$ ($\alpha < k$), with Milnor's relations: q a skew square root of -1 anticommuting with a, the j_α skew anticommuting square roots of -1 anticommuting with q and commuting with a `Chapter15/SplittingClifford · IsOmegaOps`. The generators $\{a, q, j_1, \dots, j_{k-1}\}$ `Chapter15/SplittingClifford · omegaGens` `Chapter15/SplittingClifford · IsOmegaOps.a_mem` `Chapter15/SplittingClifford · IsOmegaOps.q_mem` `Chapter15/SplittingClifford · IsOmegaOps.j_mem` are skew or self-adjoint `Chapter15/SplittingClifford · IsOmegaOps.isStarSigned`, $q^* q = 1$ `Chapter15/SplittingClifford · IsOmegaOps.adjoint_q_mul_q`, and $-a^2$ commutes with all of them `Chapter15/SplittingClifford · IsOmegaOps.a_sq_comm`.

Lemma 6.1 (the exposition). On a minimal invariant piece M, $A^2 = -c$ for a scalar c `Chapter15/SplittingClifford · IsOmegaOps.exists_sq_eq`: Milnor's "the eigenvalues are all equal, except for sign". And if $\exp(\pi A) = -I$, then $c \geq 1$ on every non-zero piece `Chapter15/IndexBound · one_le_of_sq_mulVec` `Chapter15/IndexBound · one_le_of_minInvar`: for a vector $w \neq 0$ with $A^2 w = -cw$, the matrix E with every column w `Chapter15/IndexBound · colRep` `Chapter15/IndexBound · mul_colRep` satisfies $(\pi A)^2 E = -\pi^2 c E$, and the eigenvalue lemma of §XIII.33 gives $\pi^2 c \geq \pi^2$.

The $k+1$ structures on a piece (p. 145). $(qa)^2 = a^2$ `Chapter15/SplittingClifford · IsOmegaOps.qa_qa`, qa anticommutes with every j_α `Chapter15/SplittingClifford · IsOmegaOps.j_qa` and with q for $k \geq 1$ `Chapter15/SplittingClifford · IsOmegaOps.q_qa`. So on a piece M with $A^2 = -c$, the restrictions of $J_1, \dots, J_{k-1}, J, s\,JA$ `Chapter15/SplittingClifford · cliffTop` `Chapter15/SplittingClifford · cliffTop_lt` `Chapter15/SplittingClifford · cliffTop_k` `Chapter15/SplittingClifford · cliffTop_succ` `Chapter15/SplittingClifford · IsOmegaOps.cliffTop_anti` `Chapter15/SplittingClifford · cliffFam` `Chapter15/SplittingClifford · cliffFam_apply` `Chapter15/SplittingClifford · cliffFam_lt` `Chapter15/SplittingClifford · cliffFam_k` `Chapter15/SplittingClifford · cliffFam_succ` satisfy the relations of C_{k+1} when $s^2 c = 1$ `Chapter15/SplittingClifford · IsOmegaOps.isClifford_cliffFam`, and M is a minimal space for them `Chapter15/SplittingClifford · IsOmegaOps.isMinimal_cliffFam`: a subspace invariant under them is invariant under $A = s^{-1} J^*(sJA)$. This is Milnor's $J'_h = a_h^{-1} JA$, with $s = \pm a_h^{-1}$.

Intertwiners between two pieces. $T : M_0 \to M_1$ *intertwines* the structures with parameters s_0 on M_0 and s_1 on M_1 if it carries each of $J_1, \dots, J_{k-1}, J, s_0 JA$ to the corresponding $J_1, \dots, J_{k-1}, J, s_1 JA$ `Chapter15/SplittingClifford · IsInter` `Chapter15/SplittingClifford · IsInter.apply_j` `Chapter15/SplittingClifford · IsInter.apply_q`. Then $T(Ax) = (s_1/s_0) A(Tx)$ `Chapter15/SplittingClifford · IsInter.apply_a`. The operator $D = \iota_{M_1} \circ T \circ \pi_{M_0}$ on E `Chapter15/SplittingClifford · interOp` `Chapter15/SplittingClifford · interOp_apply` `Chapter15/SplittingClifford · IsInvar.orthogonalProjectionOnto_comm` commutes with J and the J_α `Chapter15/SplittingClifford · interOp_comm_q` `Chapter15/SplittingClifford · interOp_comm_j`, satisfies $DA = (s_1/s_0) AD$ `Chapter15/SplittingClifford · interOp_a`, $A^2 D = -c_1 D$, $DA^2 = -c_0 D$ `Chapter15/SplittingClifford · a_a_interOp` `Chapter15/SplittingClifford · interOp_a_a`, the projection identities `Chapter15/SplittingClifford · interOp_starProjection` `Chapter15/SplittingClifford · adjoint_interOp_starProjection` `Chapter15/SplittingClifford · starProjection_a_interOp` `Chapter15/SplittingClifford · starProjection_a_interOp_of_orth` `Chapter15/SplittingClifford · interOp_apply_mem`, and $AD \neq 0$ `Chapter15/SplittingClifford · eq_zero_of_a_eq_zero` `Chapter15/SplittingClifford · a_interOp_ne_zero`; moreover AD and ADA are independent `Chapter15/SplittingClifford · linearIndependent_a_interOp` (the doubling for $k = 0$).

Lemma 24.8 between two pieces (p. 145). If $A^2 = -a_0^2$ on M_0 and $-b^2$ on M_1, both minimal, and $k+1 \leq 8$, $k+1 \not\equiv 3 \pmod 4$, there is an isomorphism $T : M_0 \cong M_1$ intertwining the parameters a_0^{-1} and $-b^{-1}$ `Chapter15/SplittingData · IsOmegaOps.exists_inter`: T commutes with J_α and J, and $T(JA/a_0) = -(JA/b)T$. *Proof*: both pieces are minimal spaces for $k+1$ structures (the last one $a_0^{-1} JA$, resp. $-b^{-1} JA$), and Lemma 24.8 (§XIII.31) applies. □ Milnor flips the sign of J ("an isomorphism from M_h to $\bar{M}_j$"); flipping the last structure instead is equivalent, and keeps T commuting with J, so that $D - D^{\mathsf{T}}$ is in the isotropy algebra. That the isomorphism is only linear does not matter: $D - D^{\mathsf{T}}$ is skew for any D, and D^{T} intertwines backwards because all the structures are skew.

Theorem (the data, pp. 144–146). Suppose every piece has $c \geq 1$ and $A^2 \neq -I$. Then some minimal

piece M_0 has $c_0 > 1$ Chapter15/SplittingData · IsOmegaOps.exists_piece_gt (otherwise $A^2 = -I$ on the span of all pieces, which is everything), and there are: $a_0 = \sqrt{c_0} > 1$; for each of the other pieces M_j, $b_j = \sqrt{c_j} \geq 1$ and operators $D_{j,l} = \iota_{M_j} T_j \pi_{M_0} A^l$ ($l < r_0$, where $r_0 = 1$, or $r_0 = 2$ for $k = 0$) with all the relations of the count of §XIII.35 Chapter15/SplittingData · OmegaPieceData; and $(s+1)m_{k+1} = n$, s the number of other pieces Chapter15/SplittingData · IsOmegaOps.exists_omegaPieceData, since every piece is a minimal space for $k+1$ structures, of dimension m_{k+1} (Lemma 24.8).

For $k = 0$ (Lemma 24.2) each pair contributes two eigenvectors, D and DA: the two blocks $C_\pm$ of the exposition's computation, which give Milnor's $2m-2$ rather than $m-1$.

XIII.37 The index of a non-minimal geodesic, $k \not\equiv 2 \pmod 4$, and Lemma 24.2

Matrices and operators are exchanged by Mathlib's star-algebra isomorphism $M_n(\mathbb{R}) \cong \mathrm{End}(\mathbb{R}^n)$ Chapter15/IndexBound · matE, which respects products, scalars and transposes Chapter15/IndexBound · matE_mul Chapter15/IndexBound · matE_conjTranspose Chapter15/IndexBound · adjoint_matE_of_skew; under it $A \in \mathcal{T}_{J_k}$, J_k, J_α satisfy the relations of §XIII.36 Chapter15/IndexBound · isOmegaOps_matE, and an operator commuting with a skew or symmetric one has adjoint commuting with it Chapter15/IndexBound · commute_adjoint.

Lemma. For $A \in \mathcal{T}_{J_k}$ with $\exp(\pi A) = -I$ and $A^2 \neq -I$, $k+1 \leq 8$, $k+1 \not\equiv 3 \pmod 4$, there are a finite $T \subseteq (0,1)$ and spaces $\mathcal{X}_\tau$ in the isotropy algebra of J_k as in §XIII.35, of total dimension $r_0 s$, where $(s+1)m_{k+1} = n$ Chapter15/IndexBound · exists_killing_spaces_omega. *Proof*: the data of §XIII.36 transported to matrices, and the count of §XIII.35. □

Theorem (the index bound, pp. 145–146; Lemma 24.2). Let $J_1, \dots, J_k$ be anticommuting complex structures with $\Omega_{k+1}(n) \neq \emptyset$, $k+1 \leq 8$, $k+1 \not\equiv 3 \pmod 4$. Every geodesic from J_k to $-J_k$ in the stratum of J_k which does not have least energy has index $\geq r_0 s$, where $n = (s+1)m_{k+1}$ and $r_0 = 1$, or $r_0 = 2$ for $k = 0$ Chapter15/IndexBound · le_formIndex_of_not_minimal_omega.

Proof. The geodesic is $t \mapsto J_k \exp(tV)$ on $[0,1]$, $V \in \mathcal{T}_{J_k}$, $\exp V = -I$ (§XIII.32). Put $A = V/\pi$. If $A^2 = -I$, then $V^2 = -\pi^2 I$, its energy $\|V\|^2 = \pi^2 n$ Chapter15/IndexBound · norm_sq_of_sq_eq is the least (§XIII.33), contradicting the hypothesis. So $A^2 \neq -I$, and the lemma and §XIII.35 give the bound. □

For $k = 0$, $m_1 = 2$, $s = n/2 - 1$ and $r_0 = 2$: index $\geq n - 2 = 2m - 2$ in $O(2m)$, Lemma 24.2. For $k \geq 1$, $r_0 = 1$: index $\geq n/m_{k+1} - 1$, Milnor's p. 146. "Non-minimal" means "not of least energy", which is the hypothesis of Theorem 22.1. The exact index formula of Lemma 24.2, $\lambda = \sum_{i<j}(a_i + a_j - 2) + \sum_{a_i > a_j}(a_i - a_j - 2)$, and the counts for §24 are §XIII.15's combinatorics; the geodesic bound here is proved without normal forms or the full spectrum of K_A.

XIII.38 Theorems 24.3 and 24.5 at a finite stage, $k \not\equiv 2 \pmod 4$

Let $J_1, \dots, J_{k+1}$ be anticommuting complex structures on $\mathbb{R}^n$, J_k the base point ($J_0 = I$) Chapter15/FiniteStage · basePt_mem. Then $J_k^{\mathsf{T}} J_{k+1} \in \mathrm{minVel}_O(J_k)$ Chapter15/FiniteStage · minVel_base, and $-J_k$ is in the stratum of J_k Chapter15/FiniteStage · neg_basePt_mem.

Definition (Milnor's φ_r, p. 138). $\varphi : \Omega_{k+1}(n) \to \Omega(\Omega_k(n); J_k, -J_k)$, $\varphi(J')(t) = \cos(\pi t) J_k + \sin(\pi t) J'$ Chapter15/FiniteStage · omegaPhi Chapter15/FiniteStage · omegaPhi_apply, the path space being that of the stratum of J_k. It is a homotopy equivalence onto Ω^d: the composite of the homeomorphism of Lemma 24.4 (iii) and the homotopy equivalence of §XIII.33 Chapter15/FiniteStage · isHomotopyEquiv_omegaPhi_minPaths.

Theorem (the index hypothesis). For $k+1 \leq 8$, $k+1 \not\equiv 3 \pmod 4$: every non-minimal geodesic from J_k to $-J_k$ has index $\geq r_0(n/m_{k+1} - 1)$ Chapter15/FiniteStage · omega_hidx (§XIII.37, with $n/m_{k+1} = s+1$).

Theorem (Theorem 22.1 for $\Omega_k(n)$). In the same range, the inclusion $\Omega^d \to \Omega(\Omega_k(n); J_k, -J_k)$ is bijective on π_i for $i+2 \le r_0(n/m_{k+1}-1)$, at every minimal geodesic `Chapter15/FiniteStage · omega_theorem_22_1_bijective`. *Proof*: Theorem 22.1 in its bijective form (§XIII.10) `Chapter13/Theorem22 · theorem_22_1_bijective`, with the stratum (a slice submanifold, §XIII.32), its closed Riemannian balls compact because it is closed `Chapter10/Broken · isCompact_closedRiemBall_of_isClosed`, the index hypothesis, and the retraction of §XIII.34. □

Theorem 24.5 at a finite stage, $k \not\equiv 2 \pmod 4$. For $k \le 7$, $k \not\equiv 2 \pmod 4$, φ induces bijections $\pi_i(\Omega_{k+1}(n), J') \to \pi_i(\Omega(\Omega_k(n); J_k, -J_k), \varphi(J'))$ at every J', for $i+2 \le r_0(n/m_{k+1}-1)$ `Chapter15/FiniteStage · omegaPhi_bijective`. With $r_0 = 1$ this is Milnor's $i \le n/m_{k+1} - 3$.

Proof. φ is the inclusion of Ω^d composed with a homotopy equivalence; maps on homotopy groups compose `Homotopy/Basic · homotopyGroupMap_comp`, a homotopy equivalence is bijective on them `Homotopy/ChangeOfBase · bijective_homotopyGroupMap_of_homotopyEquiv`, and Theorem 22.1 does the rest. □

From the stratum to $\Omega_k(n)$. The stratum, as a subset of $\Omega_k(n)$ `Chapter15/FiniteStage · stratumIn`, is open and closed `Chapter15/FiniteStage · isClopen_stratumIn` and homeomorphic to the stratum in $M_n(\mathbb{R})$ `Chapter15/FiniteStage · stratumInHomeomorph`; so the stratum carries the higher homotopy groups of $\Omega_k(n)$ at each of its points `Chapter15/FiniteStage · nonempty_pi_stratum_equiv`, by the clopen lemma of §XIII.22 `Homotopy/BundlesUnitary · bijective_homotopyGroupMap_clopen`.

Theorem 24.5 as Milnor's isomorphism. For $k \le 7$, $k \not\equiv 2 \pmod 4$ and $i+2 \le r_0(n/m_{k+1}-1)$: $\pi_i(\Omega_{k+1}(n), J') \cong \pi_{i+1}(\Omega_k(n), J_k)$, as sets, at every J' `Chapter15/FiniteStage · omega_pi_equiv`. *Proof*: the homotopy equivalence $\Omega_{k+1}(n) \simeq \Omega^d$ `Homotopy/ChangeOfBase · homotopyGroupEquivOfHomotopyEquiv`, Corollary 22.2 (§XIII.11) `Chapter13/Corollaries · corollary_22_2` for the stratum, and the stratum lemma. □

Theorem 24.3 (Bott, p. 136). For any complex structure J_1 on $\mathbb{R}^n$ (n is then even) and $i+4 \le n$: $\varphi : \Omega_1(n) \to \Omega(O(n); I, -I)$, $J \mapsto \cos(\pi t)I + \sin(\pi t)J$, is bijective on π_i at every point, and $\pi_i(\Omega_1(n), J') \cong \pi_{i+1}(O(n), I)$ `Chapter15/FiniteStage · theorem_24_3`. *Proof*: $k=0$, $r_0=2$, $m_1=2$, $r_0(n/2-1) = n-2$; the stratum of I in $\Omega_0(n) = O(n)$ is all of $O(n)$ `Chapter15/FiniteStage · omegaStratum_zero`. □

All isomorphisms are bijections of pointed sets; $\pi_{i+1}SO(n) = \pi_{i+1}O(n)$, which Milnor uses to pass from $SO(n)$ to $O(n)$, is not needed, since the path space of $O(n)$ is used directly (its stratum of I is $O(n)$).

XIII.39 The winding number of $\det_{\mathbb{C}}$ and the sets $\Omega^{(c)}$

For $k \equiv 2 \pmod 4$ ($k = 2, 6$) the index bound of §XIII.37 fails: Lemma 24.8 is not available for $k+1 \equiv 3$ structures, and indeed $\Omega(\Omega_k(n); J, -J)$ has infinitely many components while $\Omega_{k+1}(n)$ has finitely many. Milnor (p. 147) makes $\mathbb{R}^n$ complex by $i = J_1 \cdots J_{k-1}$, puts $f(J') = \det_{\mathbb{C}}(J^{-1}J') \in S^1$, and sorts the paths from J to $-J$ by the degree c of the loop $f \circ \omega$, suitably normalized; Theorem 22.1 is then applied to each class $\Omega^{(c)}$ separately (Lemma 6.2, §XIII.13).

Angles. $\theta : [0,1] \to \mathbb{R}$ is the *angle* of $t \mapsto F(\omega(t))$ if $\theta(0) = 0$ and $F(\omega(t)) = e^{i\theta(t)}$ `Chapter15/Winding · IsAngleLift`. It is unique, $\exp : \mathbb{R} \to S^1$ being a covering `Chapter15/Winding · IsAngleLift.unique`. It exists when $|F \circ \omega| = 1$ and $F(\omega(0)) = 1$ `Chapter15/OmegaComponents · exists_isAngleLift`, by path lifting for the covering $t \mapsto e^{it}$ `Mathlib/Analysis/SpecialFunctions/Complex/Circle · Circle.isCoveringMap_exp`, and $n\theta$ is the angle of F^n `Chapter15/OmegaComponents · IsAngleLift.pow`. If ω' is uniformly close to ω, $|F(\omega'(t)) - F(\omega(t))| < 1$, and the two loops start together, then $\theta + \arg(F \circ \omega' / F \circ \omega)$ is the angle of $F \circ \omega'$, with the same end value when the loops end together `Chapter15/Winding · re_pos_of_norm_sub_one_lt` `Chapter15/Winding · IsAngleLift.perturb` (the quotient has positive real part, so its argument is continuous): Milnor's "$|\vartheta_\omega - \vartheta_{\omega'}| < \pi$".

Definition. For a path space $\Omega(N; p, q)$, windSet(F, w) is the set of paths whose angle of $F \circ \omega$ ends at w `Chapter15/Winding · windSet`.

Theorem (p. 147). If F is continuous with $|F| = 1$ on a compact set containing N, every windSet(F, w) is open and closed in $\Omega(N; p, q)$ `Chapter15/Winding · isClopen_windSet`, and two of them with different w are disjoint `Chapter15/OmegaComponents · disjoint_windSet`.

Proof. $t \mapsto F(\omega(t))$ is continuous `Chapter15/Winding · continuous_comp_path`. Paths close for Milnor's metric are uniformly close on $[0, 1]$ `Chapter15/Winding · norm_sub_lt_of_edist_lt`, and F is uniformly continuous on the compact set; so there is δ such that paths at distance $< \delta$ have $|F \circ \omega' - F \circ \omega| < 1$, start at p and end at q together, and by the perturbation lemma have the same end angle `Chapter15/Winding · exists_mem_windSet_of_close`. The statement is symmetric in ω, ω', which gives open and closed. Disjointness is uniqueness of the angle. □

Neither the closedness of the loop $f \circ \omega$ nor the existence of the angle is needed for this; they are needed for every path to lie in some $\Omega^{(c)}$ with c an integer (below).

$\det_{\mathbb{C}}$ of a real matrix. For a complex structure $\mathbf{i}$ on $\mathbb{R}^n$, let $M \mapsto M_{\mathbb{C}}$ be complexification `Chapter15/Winding · cplx` `Chapter15/Winding · cplx_apply` `Chapter15/Winding · continuous_cplx` `Chapter15/Winding · cplx_transpose` `Chapter15/Winding · cplx_smul` `Chapter15/Winding · trace_cplx` and $P_{\pm} = \frac{1}{2}(1 \mp \sqrt{-1}\,\mathbf{i}_{\mathbb{C}})$ the projections onto the $\pm\sqrt{-1}$-eigenspaces of $\mathbf{i}$ in $\mathbb{C}^n$ `Chapter15/Winding · projP` `Chapter15/Winding · projM`: complementary idempotents `Chapter15/Winding · cplx_sq` `Chapter15/Winding · projP_add_projM` `Chapter15/Winding · projP_mul_projM` `Chapter15/Winding · projM_mul_projP` `Chapter15/Winding · projP_mul_projP` `Chapter15/Winding · projM_mul_projM` commuting with every $M_{\mathbb{C}}$ for M commuting with $\mathbf{i}$ `Chapter15/Winding · cplx_comm_projP` `Chapter15/Winding · cplx_comm_projM`. Define

$$\det_{\mathbb{C}} M = \det(M_{\mathbb{C}} P_+ + P_-)$$

`Chapter15/Winding · cdet` `Chapter15/Winding · continuous_cdet`: M acts on the $+\sqrt{-1}$-eigenspace, which is the complex vector space $(\mathbb{R}^n, \mathbf{i})$, and $M_{\mathbb{C}} P_+ + P_-$ is M there and 1 on the other eigenspace. No complex basis is chosen.

Lemma. On matrices commuting with $\mathbf{i}$: $\det_{\mathbb{C}}$ is multiplicative `Chapter15/Winding · cdet_mul`, $\det_{\mathbb{C}} 1 = 1$ `Chapter15/Winding · cdet_one`, $\overline{\det_{\mathbb{C}} M} = \det_{\mathbb{C}} M^{\mathsf{T}}$ `Chapter15/Winding · conj_cdet`, so $|\det_{\mathbb{C}} M| = 1$ for orthogonal M `Chapter15/Winding · norm_cdet`. For a skew V commuting with $\mathbf{i}$ (trace 0), $\det_{\mathbb{C}} \exp(tV) = e^{-\sqrt{-1}\,t\,\mathrm{tr}(V\mathbf{i})/2}$ `Chapter15/Winding · cdet_exp`.

Proof of the last. A solution of $c' = cX$ is $c(0)\exp(tX)$ `Chapter15/Winding · eq_mul_exp_smul_of_hasDerivAt_cplx`, which gives Jacobi's formula $\det\exp(tX) = e^{t\,\mathrm{tr}\,X}$ `Chapter15/Winding · det_exp_smul_cplx`. Here $\exp(tV)_{\mathbb{C}} P_+ + P_- = \exp(tV_{\mathbb{C}} P_+)$ and $\mathrm{tr}(V_{\mathbb{C}} P_+) = \frac{1}{2}(\mathrm{tr}\,V - \sqrt{-1}\,\mathrm{tr}(V\mathbf{i}))$. □

Milnor's i and f. $\mathbf{i}$ is *central* for k if it is a complex structure commuting with every matrix that commutes with $J_1, \ldots, J_{k-1}$ and anticommuting with every matrix that anticommutes with them `Chapter15/OmegaWinding · IsCentralI`. With $\mathbf{i} = J_1 \cdots J_{k-1}$ `Chapter15/OmegaWinding · iWordM` this holds for $k = 2$ ($\mathbf{i} = J_1$) `Chapter15/OmegaWinding · iWordM_two` `Chapter15/OmegaWinding · isCentralI_two` and $k = 6$ ($\mathbf{i} = J_1 J_2 J_3 J_4 J_5$, five factors give a complex structure by the sign rules) `Chapter15/OmegaWinding · iWordM_six` `Chapter15/OmegaWinding · isCentralI_six` `Chapter15/FiniteStageTwo · isCentralI_iWordM`. Milnor's $f(X) = \det_{\mathbb{C}}(p^{\mathsf{T}} X)$, $p = J_k$ `Chapter15/OmegaWinding · omegaF` `Chapter15/OmegaWinding · continuous_omegaF`. For $X \in \Omega_k(n)$, X and p anticommute with the J_α `Chapter15/OmegaWinding · anti_of_mem_omega`, so $p^{\mathsf{T}} X$ commutes with $\mathbf{i}$ `Chapter15/OmegaWinding · IsCentralI.comm_transpose_mul` and is orthogonal: $|f| = 1$ on $\Omega_k(n)$ `Chapter15/OmegaWinding · norm_omegaF`. $f(p) = 1$ `Chapter15/OmegaWinding · omegaF_self`, and along a geodesic $f(p\exp(tV)) = e^{-\sqrt{-1}\,t\,\mathrm{tr}(V\mathbf{i})/2}$ for $V \in \mathcal{T}_p$ `Chapter15/OmegaWinding · omegaF_geo`.

Definition (Milnor's $\Omega^{(c)}$). omegaComp(w): the paths from J_k to $-J_k$ in the stratum of J_k whose f-angle ends at w `Chapter15/OmegaWinding · omegaComp`. The component $\Omega^{(c)}$, $c \in \mathbb{Z}$, is omegaComp$(\pi m_{k+1} c/2)$.

Theorem. (a) Each omegaComp(w) is open and closed Chapter15/OmegaWinding · isClopen_omegaComp, and they are pairwise disjoint Chapter15/OmegaComponents · disjoint_omegaComp. (b) A path equal on $[0,1]$ to $t \mapsto J_k \exp(tV)$, $V \in \mathcal{T}$, lies in omegaComp(w) iff $w = -\operatorname{tr}(V\mathbf{i})/2$ Chapter15/OmegaWinding · mem_omegaComp_geo_iff. (c) For $k = 2, 6$, every path from J_k to $-J_k$ lies in exactly one $\Omega^{(c)}$, $c \in \mathbb{Z}$ Chapter15/OmegaComponents · existsUnique_mem_omegaComp.

Proof. (a) The theorem on windSet above, with $F = f$, N the compact stratum. (b) By the geodesic formula the angle is $t \mapsto -t \operatorname{tr}(V\mathbf{i})/2$. (c) Existence of c for $k = 2$ Chapter15/OmegaComponents · exists_mem_omegaComp_two: the angle θ exists, and $f(-J_2) = \det_{\mathbb{C}}(-1) = \det_{\mathbb{C}} \exp(\pi J_1) = e^{\sqrt{-1}\pi n/2} = 1$, because $\operatorname{tr}(J_1 J_1) = -n$ and $4 = m_2$ divides n (§XIII.30). So $\theta(1) \in 2\pi\mathbb{Z} = \pi m_3 \mathbb{Z}/2$ ($m_3 = 4$). For $k = 6$ see below Chapter15/OmegaComponents · exists_mem_omegaComp_six. Uniqueness is (a). □

The integer for $k = 6$. Milnor's normalization puts the end angle in $\pi m_7 \mathbb{Z}/2 = 4\pi\mathbb{Z}$, while $f(-J_6) = 1$ only gives $2\pi\mathbb{Z}$. The extra factor 2 comes from $\det_{\mathbb{C}}$ being a fourth power on $\Omega_6(n)$.

- Let $\operatorname{pdet}(Q, A) = \det(AQ + 1 - Q)$, the determinant of A on the range of a projection Q commuting with it Chapter15/OmegaComponents · pdet Chapter15/OmegaComponents · continuous_pdet Chapter15/OmegaComponents · pdet_one; $\operatorname{pdet}(Q, -1)^2 = 1$ for an idempotent Q Chapter15/OmegaComponents · pdet_neg_one_sq; and $\det_{\mathbb{C}} M = \operatorname{pdet}(P_+, M_{\mathbb{C}})$ Chapter15/OmegaComponents · cdet_eq_pdet. Commuting real matrices have commuting projections Chapter15/OmegaComponents · commute_projP Chapter15/OmegaComponents · commute_projP_projP.
- **Splitting off a swapped eigenspace.** If complex structures g, s commute with Q and A and anticommute with each other, then $\operatorname{pdet}(Q, A) = \operatorname{pdet}(QP_g, A)^2$ Chapter15/OmegaComponents · pdet_split. *Proof*: with $B = A - 1$, $Q_1 = QP_g$, $Q_2 = QP_{-g}$: $Q_1 Q_2 = 0$, $Q_1 + Q_2 = Q$, so $(1 + BQ_1)(1 + BQ_2) = 1 + BQ$; and s conjugates $1 + BQ_1$ to $1 + BQ_2$ (as $sP_g = P_{-g}s$), so the two factors have the same determinant. □
- On $\Omega_6(n)$, $J_6^{\mathsf{T}} X$ commutes with $J_1, \dots, J_5$ Chapter15/OmegaComponents · commute_transpose_mul_of_mem_omega. Among $\mathbf{i} = J_1 \cdots J_5$, $J_1 J_2$, $J_3 J_4$, J_1, J_3 the needed relations hold Chapter15/OmegaComponents · isCentralI_six' Chapter15/OmegaComponents · six_relations: split first by $g = J_1 J_2$ with $s = J_1$, then by $g = J_3 J_4$ with $s = J_3$. With $Q = P_{\mathbf{i}} P_{J_1 J_2} P_{J_3 J_4}$ Chapter15/OmegaComponents · q6, idempotent Chapter15/OmegaComponents · isIdempotentElem_q6, $f(X) = \operatorname{pdet}(Q, (J_6^{\mathsf{T}} X)_{\mathbb{C}})^4$ Chapter15/OmegaComponents · omegaF_eq_pow_four.
- *Proof of existence for $k = 6$.* $G = \operatorname{pdet}(Q, (J_6^{\mathsf{T}} \cdot)_{\mathbb{C}})$ has $|G|^4 = |f| = 1$ along the path, and $G = 1$ at J_6; let φ be its angle. The angle of $f = G^4$ is 4φ. At the end $e^{\sqrt{-1}\varphi(1)} = \operatorname{pdet}(Q, -1) = \pm 1$, so $\varphi(1) = c\pi$ and $4\varphi(1) = 4\pi c = \pi m_7 c/2$. □

The commutant of $J_1, \dots, J_5$ is $M_{n/8}(\mathbb{C})$ acting on $\mathbb{C}^4 \otimes \mathbb{C}^{n/8}$, which is why $\det_{\mathbb{C}}$ is a fourth power; Milnor does not normalize c and does not need it to be an integer for a single component, but the index bound in terms of $|c|$ and the stabilization of c (§XIII.42) do.

XIII.40 The case $k \equiv 2 \pmod 4$: the pieces, Killing fields and the index

The pieces (p. 147). With $\mathbf{i} = J_1 \cdots J_{k-1}$ as an operator Chapter15/SplittingTwo · iWord Chapter15/SplittingTwo · mem_iWord_list: a subspace invariant under the J_α is invariant under $\mathbf{i}$ Chapter15/SplittingTwo · iWord_mem Chapter15/SplittingTwo · iWord_mem_of_isInvar, an operator commuting with the J_α commutes with $\mathbf{i}$ Chapter15/SplittingTwo · commute_iWord, and the relations of $\mathbf{i}$ with A and J are recorded Chapter15/SplittingTwo · IsIOps, for matrices through §XIII.37's isomorphism Chapter15/IndexBoundTwo · iWord_matE Chapter15/IndexBoundTwo · isIOps_matE. The relations survive $A \mapsto -A$ and $J \mapsto -J$ Chapter15/SplittingTwo · IsIOps.neg_a Chapter15/SplittingTwo · IsOmegaOps.neg_a Chapter15/SplittingTwo · IsOmegaOps.neg_q, with the same invariant and minimal subspaces Chapter15/SplittingTwo · isInvar_omegaGens_neg_a Chapter15/SplittingTwo · isInvar_omegaGens_of_neg_a Chapter15/SplittingTwo · isMinInvar_omegaGens_neg_a Chapter15/

SplittingTwo · isInvar_omegaGens_neg_q.

Lemma (p. 147). On a minimal piece M, $A = c\,\mathbf{i}$ for a scalar c Chapter15/SplittingTwo · IsOmegaOps.exists_eq_smul_iWord, so $A^2 = -c^2$ there Chapter15/SplittingTwo · a_a_of_eq_smul. If $\exp(\pi A) = -I$ and $Av = c\,\mathbf{i}v$ for some $v \neq 0$, c is an odd integer Chapter15/IndexBoundTwo · exists_odd_of_mulVec ($\exp(tA)v = \cos(ct)v + \sin(ct)\mathbf{i}v$, so $\cos(\pi c) = -1$) Chapter15/IndexBoundTwo · colRep_smul.

Proof. $A\mathbf{i}$ is symmetric (A and $\mathbf{i}$ are commuting skew matrices) and commutes with A, J and the J_α, so it is a scalar on a minimal piece (§XIII.36); multiply by $\mathbf{i}^{-1} = -\mathbf{i}$. □

Milnor finds the a_h from the complex eigenspaces of A; the scalar lemma avoids them.

Lemma 24.8 between two pieces, for k structures (p. 148). A minimal piece with $A = c\,\mathbf{i}$ is a minimal space for $J_1, \dots, J_{k-1}, J'$ whenever J'-invariance implies J-invariance ($J' = \pm J$) Chapter15/SplittingTwo · isMinimal_cliffFam_k, since A is then a word in the J_α. For $k \le 8$, $k \not\equiv 3 \pmod 4$, two pieces M_0, M_1 with $A = c_0\mathbf{i}$, $c_1\mathbf{i}$ ($c_0, c_1 \neq 0$) are isomorphic by a T commuting with the J_α and anticommuting with J Chapter15/SplittingTwo · exists_inter_two: Lemma 24.8 for the structures $(J_1, \dots, J_{k-1}, J)$ on M_0 and $(J_1, \dots, J_{k-1}, -J)$ on M_1.

Theorem (the splitting, p. 147). For $k \neq 0$, $k+1 \le 8$, and A with odd scalars on minimal pieces: $\mathbb{R}^n = M_1 \oplus \cdots \oplus M_\nu$, pairwise orthogonal minimal pieces, $A = a_h\mathbf{i}$ on M_h with a_h odd, $\nu m_{k+1} = n$, $\operatorname{tr}(A\mathbf{i}) = -m_{k+1}\sum a_h$, and $A^2 = -I$ if all $a_h = \pm 1$ Chapter15/SplittingTwoData · exists_decomp_two. (The trace splits along the internal direct sum Chapter15/SplittingTwoData · foldr_sup_le_iSup; $A\mathbf{i} = -a_h$ on M_h; transported to matrices by Chapter15/IndexBoundTwo · trace_matE.)

So the geodesic $t \mapsto J\exp(\pi t A)$ lies in $\mathrm{omegaComp}(-\pi\operatorname{tr}(A\mathbf{i})/2) = \Omega^{(c)}$ with $c = \sum a_h$: Milnor's $c = a_1 + \cdots + a_\nu$. For $k = 2, 6$, $m_{k+1} = m_k$, Milnor's dimension of M_h.

Intertwiners for multiples of i. $D : M_a \to M_b$ intertwines $A = a\mathbf{i}$ with $A = b\mathbf{i}$ if $DA = aD\mathbf{i}$, $AD = b\mathbf{i}D$, $D\mathbf{i} = \mathbf{i}D$ Chapter15/KillingTwo · IsInterRelI. Then $D\exp(sA) = \cos(as)D + \sin(as)D\mathbf{i}$ and $\exp(sA)D = \cos(bs)D + \sin(bs)\mathbf{i}D$ Chapter15/KillingTwo · IsInterRelI.mul_exp Chapter15/KillingTwo · IsInterRelI.exp_mul, so $D\mathbf{i}$ commutes with $\exp(sA)$ whenever $s(a-b) \in 2\pi\mathbb{Z}$ Chapter15/KillingTwo · IsInterRelI.commute_exp: the conjugate points $t = 2m/(a-b)$. $D^{\mathsf T}$ intertwines back Chapter15/KillingTwo · IsInterRelI.conjTranspose.

The data (p. 148). From a piece M_0 with $a_0 \ge 3$, Lemma 24.8 gives intertwiners $D_j = \iota_{M_j} T_j \pi_{M_0}$ to every piece with $a_j < 0$ Chapter15/SplittingTwoData · TwoPieceData, with $DA = a_0 D\mathbf{i}$, $AD = a_j\mathbf{i}D$, $DJ = -JD$, $DJ_\alpha = J_\alpha D$, the projections, and the count

$$\nu - \sum_h a_h \le 4\sum_j \Big(\frac{a_0 - a_j}{2} - 1\Big)$$

Chapter15/SplittingTwoData · exists_twoPieceData_of_decomp. *Proof of the count*: each term is $\ge (|a_j|+1)/2$; summing gives at least $q/2$, q the sum of $|a_j|$ over $a_j < 0$, and $2q = \sum|a_h| - \sum a_h \ge \nu - \sum a_h$. □

Killing fields with several parameters. For $X_j \in$ isotropy algebra with $[X_j, V]$ independent, and distinct parameters $\tau_{j,s} \in (0,1)$, $s < n_j$, at which X_j commutes with $\exp(\tau V)$, there are spaces $\mathcal{X}_\tau$ as in §XIII.35 of total dimension $\sum n_j$ Chapter15/KillingTwo · exists_killing_spaces_gen. Here $X_j = D_j\mathbf{i} - (D_j\mathbf{i})^{\mathsf T}$, which commutes with J (D_j and $\mathbf{i}$ both anticommute with it) and the J_α Chapter15/IndexBoundTwo · sub_conjTranspose_mem_isoAlg, with $\tau_{j,s} = 2(s+1)/(a_0 - a_j)$, $s < (a_0-a_j)/2 - 1$ Chapter15/KillingTwo · exists_killing_spaces_two Chapter15/IndexBoundTwo · killing_of_twoPieceData. Milnor's eigenvector B, with eigenvalue $(a_h - a_j)^2/4$, is again $[X, A]$ for a Killing field.

Theorem (the index bound, p. 148). Let $k \neq 0$, $k+1 \le 8$, $k \not\equiv 3 \pmod 4$, with Milnor's $\mathbf{i}$ central. A geodesic from J_k to $-J_k$ in $\Omega^{(c)}$ which is not of least energy has index $\ge \lfloor(\nu - |c|)/4\rfloor$, where $\nu m_{k+1} = n$ Chapter15/IndexBoundTwo · le_formIndex_of_not_minimal_omega_two.

The proof as formalized. 1. The geodesic is $J_k \exp(tV)$, $V = \pi A$, $\exp V = -I$, and $A^2 \neq -I$ as in §XIII.37. 2. From $\Omega^{(c)}$ and (b) of §XIII.39: $\operatorname{tr}(A\mathbf{i}) = -m_{k+1}c$. The splitting gives odd a_h with $\operatorname{tr}(A\mathbf{i}) = -m_{k+1}\sum a_h$, so $\sum a_h = c$. 3. Not all $a_h = \pm 1$ (else $A^2 = -I$), so some $|a_{h_0}| \geq 3$, the a_h being odd. 4. If $a_{h_0} \geq 3$: the data and Killing fields for the pairs (h_0, j), $a_j < 0$, give index $\geq \sum_j ((a_0 - a_j)/2 - 1) \geq (\nu - c)/4 \geq (\nu - |c|)/4$. 5. If $a_{h_0} \leq -3$: the same for $-A$, whose scalars are $-a_h$, with $-c$. The Killing fields for $-A$ serve for A: X commutes with $\exp(-\tau V)$ iff it commutes with $\exp(\tau V)$. 6. The index bound of §XIII.35. □

Milnor writes "suppose for example that some a_h is equal to -3" and derives $\lambda \geq (r - c)/4$; the formal statement is $4\lambda \geq \nu - |c|$, both signs covered by $A \mapsto -A$. Only the pairs (h_0, j) with a_j of the opposite sign are used, but each contributes all of its $(a_0 - a_j)/2 - 1$ conjugate points.

XIII.41 Theorem 24.5 at a finite stage, $k = 2, 6$

Let $J_1, \ldots, J_{k+1}$ be anticommuting complex structures, $k \neq 0$, $k + 1 \leq 8$, $k \not\equiv 3 \pmod 4$, and $\mathbf{i} = J_1 \cdots J_{k-1}$ central (true for $k = 2, 6$).

Lemma (the component of $\varphi(J')$). The f-angle along the semicircle through J' is $-\pi \operatorname{tr}(J_k^{\mathsf{T}} J' \mathbf{i})/2$ `Chapter15/FiniteStageTwo · omegaPhi_mem_omegaComp`. So $\Omega_{k+1}(n)^{(c)} = \{J' : -\operatorname{tr}(J_k^{\mathsf{T}} J' \mathbf{i}) = m_{k+1}c\}$, open and closed in $\Omega_{k+1}(n)$.

Theorem (Lemma 6.2's hypothesis on $\Omega^{(c)}$). Every geodesic in $\Omega^{(c)}$ not of least energy has index $\geq (n/m_{k+1} - |c|)/4$ `Chapter15/FiniteStageTwo · omega_hidx_two` (§XIII.40).

Theorem (Lemma 6.2 on $\Omega^{(c)}$, with π_0). The inclusion of the minimal geodesics in $\Omega^{(c)}$ into $\Omega^{(c)}$ is bijective on π_j for $j + 2 \leq (\nu - |c|)/4$, $\nu = n/m_{k+1}$ `Chapter15/FiniteStageTwo · omega_minPaths_bijective_two`. *Proof*: Theorem 22.1 for an open and closed set of paths (§XIII.13) `Chapter13/Componentwise · theorem_22_1_clopen_bijective` `Chapter13/Componentwise · clopenIncl`, with the stratum, the clopen $\Omega^{(c)}$, the index hypothesis and the retraction of §XIII.34. □

Theorem 24.5 at a finite stage, $k \equiv 2 \pmod 4$. For J' with $\varphi(J') \in \Omega^{(c)}$ and $j + 3 \leq (\nu - |c|)/4$, φ is bijective on π_{j+1} at J' `Chapter15/FiniteStageTwo · omegaPhi_bijective_two` `Chapter15/FiniteStageTwo · theorem_24_5_two_six`, and $\pi_{j+1}(\Omega_{k+1}(n), J') \cong \pi_{j+2}(\Omega_k(n), J_k)$ as sets `Chapter15/FiniteStageTwo · omega_pi_equiv_two`.

Proof. $\varphi = (\Omega^d \to \Omega) \circ e$ with e the homotopy equivalence of §XIII.38. The image of J' is a minimal geodesic in $\Omega^{(c)}$, and the inclusion $\Omega^d \to \Omega$ is bijective on π_{j+1} there (§XIII.13) `Chapter13/Componentwise · theorem_22_1_clopen_bijective_minPathsIncl`: $\Omega^{(c)}$ is open and closed, so it carries the higher homotopy groups of the whole path space. For the isomorphism, the loop-space step is Corollary 22.2's group form (§XIII.11) `Chapter13/CorollaryGroups · nonempty_mulEquiv_pathSp`, $\pi_{j+1}\Omega(N; p, q) \cong \pi_{j+2}(N, p)$ along any path, and then the stratum lemma of §XIII.38. □

The π_0 part (pp. 147–148). For $(n/m_{k+1} - |c|)/4 \geq 2$: (i) every path in $\Omega^{(c)}$ is joined, in $\Omega(\Omega_k(n); J_k, -J_k)$, to $\varphi(J')$ for some $J' \in \Omega_{k+1}(n)$; (ii) if $\varphi(J') \in \Omega^{(c)}$, then J', J'' are joined in $\Omega_{k+1}(n)$ iff $\varphi(J'), \varphi(J'')$ are joined `Chapter15/OmegaComponents · theorem_24_5_pi0_two_six`.

Proof. (i) Push the constant map at ω (from the 0-cube) into the minimal geodesics of $\Omega^{(c)}$ (§XIII.13) `Chapter13/Componentwise · pushesInto_cube_clopen`; the push is a path from ω to a minimal γ, and γ is joined to $\varphi(e^{-1}\gamma)$ because $e \circ e^{-1}$ is homotopic to the identity `Chapter15/OmegaComponents · joined_homotopyEquiv`. (ii) "Only if" is continuity. "If": a path from $\varphi(J')$ to $\varphi(J'')$ stays in the clopen $\Omega^{(c)}$; push it (a map from the cylinder $I \times I^0$) into the minimal geodesics relative to its ends `Chapter13/Componentwise · pushesInto_cyl_clopen`, carry the result back by e^{-1}, and join the ends to J', J'' by the homotopies of e. □

With §XIII.39 (c) (every path lies in exactly one $\Omega^{(c)}$), φ identifies $\pi_0\Omega_{k+1}(n)^{(c)}$ with $\pi_0\Omega^{(c)}$ for every c

with $|c| \le \nu - 8$. The components with $|c| > \nu - 8$ are not controlled at a finite stage: Milnor's bijection of path components is a statement about the limit $n \to \infty$, where every c is eventually in range, and the limit is not formalized. Milnor's "$\Omega_k(n)$ has an infinite cyclic fundamental group" is not needed; the integer c replaces it.

XIII.42 Stabilization

From here on $n = 16r$ and the structures are Milnor's $J_\alpha = J'_\alpha \oplus \cdots \oplus J'_\alpha$ `Chapter15/BottChain · bottJ`, eight anticommuting complex structures on $\mathbb{R}^{16r}$ `Chapter15/BottChain · isAnticommFamily_bottJ` `Chapter15/BottChain · card_bottJ`.

Definition. $\mathbb{R}^{16r} \oplus \mathbb{R}^{16} \cong \mathbb{R}^{16(r+1)}$, the added block last `Chapter15/Stabilization · stabEquiv` `Chapter15/Stabilization · stabEquiv_symm_castSucc` `Chapter15/Stabilization · stabEquiv_symm_last`. Under it $M^{\oplus r} \oplus M = M^{\oplus(r+1)}$ `Chapter15/Stabilization · reindex_fromBlocks_blockDiagonal`, so $J_\alpha \oplus J'_\alpha$ is J_α at stage $r+1$ `Chapter15/Stabilization · reindex_fromBlocks_bottJ`. Milnor's stabilization is $J \mapsto J \oplus J'_k$, and $T \mapsto T \oplus I_{16}$ for $k = 0$ `Chapter15/Stabilization · stab`.

Lemma (p. 138). $J \mapsto J \oplus J'_k$ maps $\Omega_k(16r)$ into $\Omega_k(16r+16)$ for $k \le 8$ `Chapter15/Stabilization · mem_omega_reindex` `Chapter15/Stabilization · bott16_basePt_mem` `Chapter15/Stabilization · stab_mem_omega` and carries J_k to J_k `Chapter15/Stabilization · stab_basePt`.

Theorem (compatibility with φ; "not difficult to see", p. 138). $\varphi_{r+1}(J' \oplus J'_{k+1})(t) = \varphi_r(J')(t) \oplus \delta_k(t)$, where $\delta_k(t) = \cos(\pi t)J'_k + \sin(\pi t)J'_{k+1}$ `Chapter15/Stabilization · smul_reindex_add` `Chapter15/Stabilization · omegaPhi_stab`. *Proof*: block sums are linear, $\cos(\pi t)(J_k \oplus J'_k) + \sin(\pi t)(J' \oplus J'_{k+1}) = (\cos(\pi t)J_k + \sin(\pi t)J') \oplus \delta_k(t)$. □

Theorem (the label does not change). Traces are invariant under reindexing `Chapter15/Stabilization · trace_reindex_equiv`, and the label trace is additive under stabilization `Chapter15/Stabilization · trace_stab`. For $k = 2, 6$ the contribution of the added block, $\mathrm{tr}(J'^{\mathsf{T}}_k J'_{k+1}\mathbf{i}')$, vanishes `Chapter15/BottChain · trace_bott16_two` `Chapter15/BottChain · trace_bott16_six` (in each slot of the Kronecker word some letter is traceless), so $-\mathrm{tr}(J_k^{\mathsf{T}} J'\mathbf{i})$, hence c, does not change `Chapter15/Stabilization · omegaLabel_stab`: the exposition's "δ_k has $c = 0$". In particular the base point J_{k+1} lies in $\Omega_{k+1}(16r)^{(0)}$ `Chapter15/BottChain · omegaPhi_bottJ_mem_omegaComp`.

Not formalized: the colimits $\Omega_k = \bigcup_r \Omega_k(16r)$ with the colimit topology, and the naturality of the isomorphisms of Theorem 24.5 under stabilization (which would need the loop-space identification $\pi_j\Omega(N;p,q) \cong \pi_{j+1}N$ to be natural). The formal periodicity theorem avoids the colimit by working in the stable range of $O(n)$ (§XIII.44).

XIII.43 $\Omega_8(16r) \cong O(r)$: Lemma 24.6 (8)

Milnor (p. 142): $J_8 \mapsto J_7 J_8 | X_1$ identifies $\Omega_8(16r)$ with the isometries $X_1 \to X_2$. The formal proof replaces the joint eigenspace X_1 by the commutant of $J'_1, \dots, J'_7$ on $\mathbb{R}^{16}$.

The Pauli basis. The letters $1, \sigma, \tau, \kappa$ `Chapter15/EightCommutant · kletter` and their Kronecker words $L(w_0) \otimes L(w_1) \otimes L(w_2) \otimes L(w_3)$, $w : \{0..3\} \to \{0..3\}$ `Chapter15/EightCommutant · kword`, are orthogonal for the trace form ($\mathrm{tr}(L_a^{\mathsf{T}} L_b) = 2\delta_{ab}$, $\mathrm{tr}(W_u^{\mathsf{T}} W_v) = 16\delta_{uv}$) `Chapter15/EightCommutant · trace_letter_pair` `Chapter15/EightCommutant · trace_word_pair`, hence independent `Chapter15/EightCommutant · linearIndependent_word`; there are $256 = \dim M_{16}(\mathbb{R})$ of them `Chapter15/EightCommutant · card_word`, so they form a basis `Chapter15/EightCommutant · wordBasis` `Chapter15/EightCommutant · wordBasis_apply`. Two letters commute or anticommute `Chapter15/EightCommutant · lanti` `Chapter15/EightCommutant · bsgn` `Chapter15/EightCommutant · bsgn_xor` `Chapter15/EightCommutant · letter_comm`, so two words commute or anticommute according to the parity of the slots with different non-identity letters `Chapter15/EightCommutant · wanti` `Chapter15/EightCommutant · word_`

comm. J'_α is the word with code code(α) Chapter15/EightCommutant · bottCode Chapter15/EightCommutant · bott16_eq_word.

Theorem (the commutant). A matrix X on $\mathbb{R}^{16}$ commuting with $J'_1, \dots, J'_7$ is $b \cdot 1 + c \cdot W$, $W = 1 \otimes 1 \otimes 1 \otimes \tau$ Chapter15/EightCommutant · wCode Chapter15/EightCommutant · W16 Chapter15/EightCommutant · eq_of_commute_bott16. $W^2 = 1$, $W^{\mathsf{T}} = W$, W commutes with $J'_1, \dots, J'_7$ and anticommutes with J'_8 Chapter15/EightCommutant · W16_mul_self Chapter15/EightCommutant · W16_transpose Chapter15/EightCommutant · W16_comm Chapter15/EightCommutant · W16_anti.

Proof. Write $X = \sum_u c_u W_u$ in the word basis. $X = -J'_\alpha X J'_\alpha$ (from $J'^2_\alpha = -1$ and commutation), and conjugation by J'_α multiplies W_u by ∓ 1 according as they commute or anticommute; so $c_u = 0$ unless W_u commutes with all seven. A finite check (by `decide`) shows that only the codes of 1 and W do Chapter15/EightCommutant · eq_of_not_wanti_aux Chapter15/EightCommutant · eq_of_not_wanti Chapter15/EightCommutant · word_zero. □

So the commutant of the Clifford action of $J'_1, \dots, J'_7$ on $\mathbb{R}^{16}$ is $\mathbb{R} \oplus \mathbb{R}$, the two eigenspaces of $W = \pm J'_1 \cdots J'_7$.

On $\mathbb{R}^{16r} = \mathbb{R}^{16} \otimes \mathbb{R}^r$. Let $P_\pm = \frac{1}{2}(1 \pm W)$ Chapter15/OmegaEight · Pp Chapter15/OmegaEight · Pm: complementary symmetric idempotents Chapter15/OmegaEight · Pp_add_Pm Chapter15/OmegaEight · Pp_mul_Pp Chapter15/OmegaEight · Pm_mul_Pm Chapter15/OmegaEight · Pp_mul_Pm Chapter15/OmegaEight · Pm_mul_Pp Chapter15/OmegaEight · Pp_transpose Chapter15/OmegaEight · Pm_transpose, commuting with $J'_1, \dots, J'_7$ Chapter15/OmegaEight · comm_Pp Chapter15/OmegaEight · comm_Pm and exchanged by J'_8 Chapter15/OmegaEight · eight_Pp Chapter15/OmegaEight · eight_Pm. Two basis points x_0, x_1 of $\mathbb{R}^{16}$ with $W = 1, -1$ there Chapter15/OmegaEight · x0 Chapter15/OmegaEight · x1 Chapter15/OmegaEight · W16_x0 Chapter15/OmegaEight · W16_x1 Chapter15/OmegaEight · Pp_x0 Chapter15/OmegaEight · Pm_x0 Chapter15/OmegaEight · Pp_x1 Chapter15/OmegaEight · Pm_x1 read off coordinates. Put $q(U,V) = P_+ \otimes U + P_- \otimes V$ for $U, V \in M_r(\mathbb{R})$ Chapter15/OmegaEight · qOf, with coordinates \$U = \$ the (x_0, x_0) block, \$V = \$ the (x_1, x_1) block Chapter15/OmegaEight · ext0 Chapter15/OmegaEight · ext1 Chapter15/OmegaEight · ext0_qOf Chapter15/OmegaEight · ext1_qOf. Then $q(U,V)q(U',V') = q(UU', VV')$, $q(U,V)^{\mathsf{T}} = q(U^{\mathsf{T}}, V^{\mathsf{T}})$, $q(1,1) = 1$ Chapter15/OmegaEight · qOf_mul Chapter15/OmegaEight · transpose_qOf Chapter15/OmegaEight · qOf_one; $J_\alpha = J'_\alpha \otimes 1$ Chapter15/OmegaEight · bottJ_eq_kronecker commutes with $q(U,V)$ for $\alpha \le 7$ Chapter15/OmegaEight · comm_qOf, and $J_8 q(U,V) = q(V,U) J_8$ Chapter15/OmegaEight · eight_qOf. Every matrix commuting with $J_1, \dots, J_7$ is $q(U,V)$ Chapter15/OmegaEight · eq_qOf_of_commute (the commutant theorem block by block).

Lemma 24.6 (8). $\Omega_8(16r)$ is homeomorphic to $O(r)$ Chapter15/OmegaEight · omegaEightHomeomorph, by $J \mapsto U$ where $-J_8 J = q(U, U^{\mathsf{T}})$, with inverse $U \mapsto J_8 q(U, U^{\mathsf{T}})$; the base point J_8 goes to I Chapter15/OmegaEight · omegaEightHomeomorph_basePt.

Proof. $J_8^2 = -1$, $J_8^{\mathsf{T}} = -J_8$ Chapter15/OmegaEight · bottJ_eight_sq Chapter15/OmegaEight · bottJ_eight_transpose. For $J \in \Omega_8(16r)$, $Q = -J_8 J$ commutes with $J_1, \dots, J_7$ (both factors anticommute with them), so $Q = q(U,V)$ Chapter15/OmegaEight · omegaEightU Chapter15/OmegaEight · qOf_of_mem_omega_eight. Q is orthogonal, so U is Chapter15/OmegaEight · omegaEightU_mem. $J^2 = -1$ gives $q(V,U)q(U,V) = 1$ (using $J_8 q(U,V) = q(V,U) J_8$), so $UV = 1$ and $V = U^{\mathsf{T}}$: $J = J_8 q(U, U^{\mathsf{T}})$. Conversely, for U orthogonal, $J_8 q(U, U^{\mathsf{T}})$ is orthogonal, squares to -1 and anticommutes with $J_1, \dots, J_7$ Chapter15/OmegaEight · omegaEight_inv_mem. Both maps are continuous Chapter15/OmegaEight · continuous_ext0 Chapter15/OmegaEight · continuous_qOf_transpose. □

The anticommutation of J with $J_1, \dots, J_7$ is automatic once Q is in the commutant; the conditions on $Q = q(U,V)$ are only that U, V are orthogonal and $UV = 1$. Milnor's X_1, X_2 are $P_\pm$ applied to a line of $\mathbb{R}^{16}$, tensored with $\mathbb{R}^r$. The other identifications of Lemma 24.6 ($\Omega_4 \cong Sp(2r)$, the Grassmannians for $k = 3, 7$, the coset spaces) are not formalized; the chain does not need them.

XIII.44 Bott's chain and Theorem 24.7

Theorem (Bott's chain at the finite stage $n = 16r$). For $2j + 12 \le r$, $\pi_{j+1}(\Omega_8(16r), J_8) \cong \pi_{j+9}(O(16r), I)$ Chapter15/BottChain · bott_chain, and in all degrees, π_0 included: for $2i + 7 \le r$,

$\pi_i(\Omega_8(16r), J_8) \cong \pi_{i+8}(O(16r), I)$ Chapter15/StablePeriodicity · bott_chain_of_le, as sets.

Proof (of the second; the first is the same with $i = j + 1$). Here $n/m_8 = r$, $n/m_k = 2r$ for $k = 4, \dots, 7$, $n/m_k = 4r$ for $k = 2, 3$. The eight steps of Theorem 24.5, each at the base points: 1. $\Omega_8 \to \Omega_7$ ($k = 7$, §XIII.38): $\pi_i\Omega_8 \cong \pi_{i+1}\Omega_7$, needing $i + 2 \le r - 1$. 2. $\Omega_7 \to \Omega_6$ ($k = 6$, §XIII.41, $c = 0$, the base point J_7 being in $\Omega_7^{(0)}$, §XIII.42): $\pi_{i+1}\Omega_7 \cong \pi_{i+2}\Omega_6$, needing $i + 3 \le \lfloor 2r/4 \rfloor$. 3. $\Omega_6 \to \Omega_5 \to \Omega_4 \to \Omega_3$ ($k = 5, 4, 3$, §XIII.38), needing $i + 6 \le 2r - 1$. 4. $\Omega_3 \to \Omega_2$ ($k = 2$, §XIII.41, $c = 0$): $\pi_{i+5}\Omega_3 \cong \pi_{i+6}\Omega_2$, needing $i + 7 \le \lfloor 4r/4 \rfloor = r$. 5. $\Omega_2 \to \Omega_1$ ($k = 1$): needing $i + 8 \le 4r - 1$. 6. $\Omega_1 \to O(16r)$, Theorem 24.3: $\pi_{i+7}\Omega_1 \cong \pi_{i+8}O(16r)$, needing $i + 11 \le 16r$.

The steps through Ω_6 and Ω_2 are the binding ones, $r \ge 2i + 6$ and $r \ge i + 7$; $2i + 7 \le r$ covers all. The $k = 2, 6$ steps are used only in degrees ≥ 1 ($\pi_{i+1}\Omega_7$ and $\pi_{i+5}\Omega_3$), so $\pi_0\Omega_8$ comes from the $k = 7$ step, which includes π_0. □

Theorem 24.7 at a finite stage. For $2i + 7 \le r$: $\pi_i(O(r), I) \cong \pi_{i+8}(O(16r), I)$ Chapter15/StablePeriodicity · bott_periodicity_finite_of_le; for $2j + 12 \le r$ also $\pi_{j+1}(O(r), I) \cong \pi_{j+9}(O(16r), I)$ Chapter15/OmegaEight · bott_periodicity_finite. *Proof*: the chain and $\Omega_8(16r) \cong O(r)$ (§XIII.43), base point to base point, through the transport of homotopy groups by a homeomorphism Homotopy/BundlesUnitary · piEquivOfHomeomorph. □

Stability. For $\mathbb{K} = \mathbb{R}$ ($d = 1$) and $\mathbb{C}$ ($d = 2$) the unit sphere of $\mathbb{K}^m$ has $\pi_i = 0$ for $i + 2 \le dm$ (§XIII.21) Homotopy/BundlesOrthogonal · sphereConnected_real Homotopy/BundlesUnitary · sphereConnected_complex, and the inclusion $U_{\mathbb{K}}(m) \to U_{\mathbb{K}}(m')$ (O for $\mathbb{R}$, U for $\mathbb{C}$) gives $\pi_i U_{\mathbb{K}}(m) \cong \pi_i U_{\mathbb{K}}(m')$ for $m \le m'$ when $i+2 \le dm+d-1$ Chapter15/StablePeriodicity · nonempty_pi_unitary_equiv_of_le, hence whenever both are in the stable range Chapter15/StablePeriodicity · nonempty_pi_unitary_equiv_stable. Reindexing the coordinates transports homotopy groups Chapter15/StablePeriodicity · unitaryCongr_one Chapter15/StablePeriodicity · piUnitaryCongr. For $O(n)$: $\pi_i O(n) \cong \pi_i O(n')$ when $i + 2 \le n, n'$.

Theorem 24.7 (Bott), in the stable range. For every $i \ge 0$, $n \ge i + 2$ and $n' \ge i + 10$: $\pi_i(O(n), I) \cong \pi_{i+8}(O(n'), I)$, as sets Chapter15/StablePeriodicity · theorem_24_7.

Proof. With $r = 2i + 7$: $\pi_i O(n) \cong \pi_i O(r)$ (stability, $i + 2 \le n$ and $i + 2 \le r$), $\cong \pi_{i+8}O(16r)$ (the finite stage), and $\pi_{i+8}O(16r) \cong \pi_{i+8}O(n')$ (reindexing $\mathbb{R}^{16} \otimes \mathbb{R}^r \cong \mathbb{R}^{16r}$, then stability with $i + 10 \le 16r$ and $i + 10 \le n'$). □

$\pi_i O(n)$ for $i < n - 1$ is the stable value Milnor calls $\pi_i O$, so this is Theorem 24.7 for the stable groups, without the colimit O.

The unitary periodicity theorem in the stable range. For $i < 2n$ and $i + 2 < 2n'$: $\pi_i(U(n), I) \cong \pi_{i+2}(U(n'), I)$ Chapter15/StablePeriodicity · periodicity_theorem_stable. *Proof*: stability to $U(i+1)$; Milnor's chain $\pi_i U(i + 1) \cong \pi_{i+2}U(2i + 2)$ (§XIII.24) Chapter14/Periodicity · nonempty_equiv_pi_unitary_two_mul; reindexing $\{1..i + 1\} \sqcup \{1..i + 1\}$; stability to $U(n')$. □ This removes the range question of p. 129 (§XIII.24): stated between stable groups, no bound tied to one m is needed.

What is not formalized of §24. The colimit spaces O, Ω_k and the limit maps $\Omega_{k+1} \to \Omega\Omega_k$; the group structure along the chain (each step of Theorem 24.5 as a homomorphism), so Theorem 24.7 is a bijection of sets; the homotopy types ($O \simeq \Omega^8 O$, Whitehead's theorem, CW types); the table of $\pi_i O$ and $\pi_i Sp$; the identifications of $\Omega_1, \dots, \Omega_7$ with coset spaces and Grassmannians beyond the bijections of §XIII.28; the symmetric-space structure; the two classes for $k = 3, 7$; and Adams' theorem.

XIII.45 Late additions to earlier parts: complex projective space (Part III, Milnor §4)

Sections §§XIII.45–XIII.47 retell files added to the formalization after Parts III, V, IX and X were written. Those parts are not revised; where they say that something is not formalized, these sections supersede them.

Milnor's §4 (pp. 26–27) puts the Morse function $f = \sum c_j |z_j|^2$ on $\mathbb{C}P^n$, finds one critical point of each index $0, 2, \dots, 2n$, and concludes that $\mathbb{C}P^n$ is a CW complex $e^0 \cup e^2 \cup \cdots \cup e^{2n}$. Part III did not treat $\mathbb{C}P^n$.

The model. $\mathbb{C}P^n$ is the set of Hermitian projections of trace one in $M_{n+1}(\mathbb{C})$, the Grassmannian of §XIII.14 with $m = 1$: the class of a unit vector z is $P = zz^*$, $P_{ab} = z_a \bar{z}_b$, and $|z_a|^2 = P_{aa}$. Every point is of this form `Chapter3/ProjectiveSpace · CP.exists_vec_of_mem` `Homotopy/Grassmannian · exists_frame_of_mem_grassmannian`. Milnor's coordinates then become entries of P, visibly independent of the representative.

Definition (Milnor's coordinates, p. 26). The model space is $\mathbb{R}^{2n}$, pairs (x_j, y_j) `Chapter3/ProjectiveSpace · CP.Model`, with $z_j = x_j + iy_j$ `Chapter3/ProjectiveSpace · CP.cz` and $|z|^2$ `Chapter3/ProjectiveSpace · CP.nsq` `Chapter3/ProjectiveSpace · CP.nsq_eq` `Chapter3/ProjectiveSpace · CP.nsq_nonneg`. For $|z| < 1$ and an index k, let v be the unit vector with $v_k = \sqrt{1 - |z|^2}$ and $v_j = z_j$ for $j \neq k$ `Chapter3/ProjectiveSpace · CP.vec` `Chapter3/ProjectiveSpace · CP.sum_vec_mul_star`; the parametrization of U_k is $z \mapsto vv^*$ `Chapter3/ProjectiveSpace · CP.param` `Chapter3/ProjectiveSpace · CP.param_mem` `Chapter3/ProjectiveSpace · CP.param_kk` `Chapter3/ProjectiveSpace · CP.param_succAbove_k`. Milnor's coordinates $(|z_k|/z_k) z_j$ are $P_{jk}/\sqrt{P_{kk}}$ on $U_k = \{\operatorname{Re} P_{kk} > 0\}$ `Chapter3/ProjectiveSpace · CP.coordC` `Chapter3/ProjectiveSpace · CP.coord` `Chapter3/ProjectiveSpace · CP.cz_coord`.

Lemma. $\operatorname{coord}_k \circ \operatorname{param}_k = \mathrm{id}$ on $|z| < 1$ `Chapter3/ProjectiveSpace · CP.coord_param`, and on U_k, $|\operatorname{coord}_k P| < 1$ and $\operatorname{param}_k(\operatorname{coord}_k P) = P$ `Chapter3/ProjectiveSpace · CP.nsq_coord_lt_and_param_coord`. Both are smooth `Chapter3/ProjectiveSpace · CP.contDiff_nsq` `Chapter3/ProjectiveSpace · CP.contDiff_cz` `Chapter3/ProjectiveSpace · CP.contDiffOn_vec` `Chapter3/ProjectiveSpace · CP.ofCLM` `Chapter3/ProjectiveSpace · CP.contDiffOn_param` `Chapter3/ProjectiveSpace · CP.contDiffOn_coordC` `Chapter3/ProjectiveSpace · CP.contDiffOn_coord`. The U_k cover $\mathbb{C}P^n$, since $\sum_k P_{kk} = \operatorname{tr} P = 1$ `Chapter3/ProjectiveSpace · CP.exists_re_diag_pos`.

Lemma (slice charts from parametrizations). A local parametrization $\Phi : Z \times W \to V$, smooth near 0 with invertible derivative there, $\Phi(z, 0) \in S$ for small z, and every $\Phi(q) \in S$ equal to $\Phi(g(q), 0)$ with $g(q) \to 0$, gives a slice chart of S at $\Phi(0)$ whose parametrization is $z \mapsto \Phi(z, 0)$ near 0 `Chapter3/ProjectiveSpace · exists_sliceChart_of_localParam`: the variant of §XIII.32's lemma that keeps track of the chart.

Theorem. $\mathbb{C}P^n$ is a slice submanifold of $M_{n+1}(\mathbb{C}) = \mathbb{R}^{2(n+1)^2}$ of dimension $2n$ `Chapter3/ProjectiveSpace · CP.Normal` `Chapter3/ProjectiveSpace · CP.finrank_add` `Chapter3/ProjectiveSpace · CP.isSliceSubmanifold`, with a slice chart at every $x \in U_k$ whose parametrization is Milnor's, $z \mapsto \operatorname{param}_k(\operatorname{coord}_k x + z)$ `Chapter3/ProjectiveSpace · CP.exists_sliceChart`. It is compact: closed, and on the unit sphere of the Frobenius norm ($\|P\|^2 = \operatorname{tr} P^2 = 1$) `Chapter3/ProjectiveSpace · CP.norm_eq_one` `Chapter3/ProjectiveSpace · CP.isCompact`.

Proof. At $x \in U_k$ take $\Phi(z, w) = \operatorname{param}_k(\operatorname{coord}_k x + z) + w$, w in a complement of its image. The derivative of param_k is injective (it has the left inverse coord_k), $\Phi(z, 0) \in \mathbb{C}P^n$, and points of $\mathbb{C}P^n$ near x are in U_k, where coord_k inverts param_k. □

That Milnor's numbers "are coordinates" (p. 26) needs exactly this: the parametrization is an immersion and a homeomorphism onto the open U_k.

Definition (Milnor's function). $f(P) = \sum_j c_j \operatorname{Re} P_{jj}$ $(= \sum c_j |z_j|^2)$ `Chapter3/ProjectiveMorse · CP.height` `Chapter3/ProjectiveMorse · CP.contDiff_height`. The scan of p. 26 prints $f = c_j |z_j|$; the coordinate formula shows that the sum of squares is meant.

Lemma (Milnor's formula). On U_k, $f(\operatorname{param}_k z) = c_k + \sum_{j \neq k} (c_j - c_k)(x_j^2 + y_j^2)$ `Chapter3/ProjectiveMorse · CP.weight` `Chapter3/ProjectiveMorse · CP.wform` `Chapter3/ProjectiveMorse · CP.wform_apply` `Chapter3/ProjectiveMorse · CP.model` `Chapter3/ProjectiveMorse · CP.height_param`. This quadratic polynomial has derivative and Hessian `Chapter3/ProjectiveMorse · CP.hasFDerivAt_model` `Chapter3/ProjectiveMorse · CP.hessian_model` $2 \sum_p (c_{j(p)} - c_k) v_p u_p$. For distinct c_j the weights are non-zero `Chapter3/ProjectiveMorse · CP.weight_ne_zero` `Chapter3/ProjectiveMorse · CP.eq_zero_of_forall`, so its only critical point is 0 `Chapter3/ProjectiveMorse · CP.`

isCriticalPoint_model_iff, non-degenerate Chapter3/ProjectiveMorse · CP.isNondegenerate_hessian_model, of index $2\#\{j : c_j < c_k\}$ Chapter3/ProjectiveMorse · CP.rank Chapter3/ProjectiveMorse · CP.index_hessian_model.

Proposition B (p. 26). For distinct c_j: the critical points of f are exactly the $n+1$ coordinate points $p_k = \mathrm{param}_k(0)$, the projections onto the axes Chapter3/ProjectiveMorse · CP.nsq_zero Chapter3/ProjectiveMorse · CP.vec_zero Chapter3/ProjectiveMorse · CP.param_zero Chapter3/ProjectiveMorse · CP.param_zero_mem Chapter3/ProjectiveMorse · CP.re_param_zero_kk Chapter3/ProjectiveMorse · CP.isCriticalPointM_height_iff; p_k is non-degenerate of index $2\#\{j : c_j < c_k\}$ Chapter3/ProjectiveMorse · CP.isNondegenerate_indexM_height; and f is a Morse function Chapter3/ProjectiveMorse · CP.isNondegenerate_of_isCriticalPointM.

Proof. Critical points, non-degeneracy and the index may be computed in any slice parametrization (Part I) Chapter1/ManifoldSubmanifold · isCriticalPointM_comp_val_iff_sliceParam Chapter1/ManifoldSubmanifold · index_hessian_comp_sliceParam. At $q \in U_k$ the parametrization is $z \mapsto \mathrm{param}_k(\mathrm{coord}_k\, q + z)$, and f in it is the model polynomial at $\mathrm{coord}_k\, q$ Chapter3/ProjectiveMorse · CP.crit_index_of_mem: q is critical iff $\mathrm{coord}_k\, q = 0$, i.e. $q = p_k$. □

Theorem (the Morse numbers and the cells, p. 27). $k \mapsto \#\{j : c_j < c_k\}$ is a bijection of $\{0, \dots, n\}$ Chapter3/ProjectiveMorse · CP.rank_lt Chapter3/ProjectiveMorse · CP.rank_lt_rank Chapter3/ProjectiveMorse · CP.bijective_rank (Milnor's tacit "renumber so that $c_0 < \cdots < c_n$"), so f has one critical point of each index $0, 2, \dots, 2n$ and none other Chapter3/ProjectiveMorse · CP.critCount_height. By Theorem 3.5 (Part II) Chapter2/CWType · theorem_3_5_count', $\mathbb{C}P^n$ has the homotopy type of an iterated adjunction space Chapter2/CWType · IsFiniteCellComplexType with one cell in each even dimension $\le 2n$ Chapter3/ProjectiveMorse · CP.exists_cells. Over any field, $\dim H_\lambda(\mathbb{C}P^n) = 1$ for $\lambda = 0, 2, \dots, 2n$ and 0 otherwise Chapter3/ProjectiveMorse · CP.lacunary Chapter3/ProjectiveMorse · CP.bettiNumber_eq Chapter3/ProjectiveMorse · bettiNumber_complexProjectiveSpace, by the lacunary principle (Corollary 5.4, Part IV) Homology/MorseHomology · bettiNumber_eq_critCount_of_lacunary.

Integral homology is not formalized (the project's homology has field coefficients).

XIII.46 Late additions to earlier parts: the Lefschetz theorems (Part V, Milnor §7)

Part V (§V.6) recorded that none of the four topological statements of §7 was formalized. All four now are, over a field and with the hypotheses made explicit.

Complex submanifolds. The ambient space is a real inner product space with an orthogonal J, $J^2 = -1$ Chapter5/ComplexIndex · IsOrthComplexStructure Chapter5/ComplexIndex · IsOrthComplexStructure.inner_left ($\mathbb{C}^m = \mathbb{R}^{2m}$, multiplication by i), and a slice submanifold N is *complex* if every tangent space is J-invariant Chapter5/ComplexIndex · IsComplexSubmanifold; its normal spaces are then J-invariant too Chapter5/ComplexIndex · IsComplexSubmanifold.inner_J_eq_zero. A complex submanifold in Milnor's sense (bianalytic) has this property; the converse is not needed.

The index bound (pp. 40–41). At every critical point q of $L_p = \|\cdot - p\|^2$ on a complex submanifold, $2 \cdot \mathrm{index} \le \dim_{\mathbb{R}} N$, degenerate or not Chapter5/ComplexIndex · two_mul_indexM_normSq_sub_le.

The proof as formalized (the exposition's "shorter route", without focal points). 1. Read J in the coordinates x of a slice chart: \$J'(u) = \$ the first component of $d\sigma \circ J \circ dx_u$, so $dx_u \circ J'(u) = J \circ dx_u$ Chapter5/ComplexIndex · Jc Chapter5/ComplexIndex · fderiv_sliceParam_Jc, and J' is differentiable Chapter5/ComplexIndex · differentiableAt_Jc_apply. 2. Differentiating, for ν normal at $x(u)$: $\langle \nu, d^2x(\eta, J'\xi)\rangle = \langle \nu, J\, d^2x(\eta, \xi)\rangle$ Chapter5/ComplexIndex · inner_fderiv_fderiv_sliceParam_Jc; the derivative of J' contributes a tangent vector, orthogonal to ν. 3. With $\nu = q - p$, the Hessian of L_p is $H(\xi,\xi) = 2(\|dx\,\xi\|^2 + \langle d^2x(\xi,\xi), \nu\rangle)$, and by step 2 $H(J'\xi, J'\xi) = 4\|dx\,\xi\|^2 - H(\xi,\xi)$. So on a negative definite subspace W, $J'W$ is positive definite and $W \cap J'W = 0$, whence $2\dim W \le \dim N$ Chapter5/IndexBound · two_mul_index_le_finrank. □

This replaces Milnor's Assertions 1 and 2 and Lemma 6.9 (focal points in pairs symmetric about q), and needs no non-degeneracy.

Homology of slabs. For a slice submanifold N, F smooth with N^{a_0} compact and non-degenerate critical points there, and regular values $a \le b \le a_0$: if no critical point with value in (a,b) `Chapter5/Lefschetz · critValuesIn` has index j, then $H_j(N^b, N^a) = 0$ `Chapter5/Lefschetz · subsingleton_subsetHomology_sublevel` (Theorems 3.1, 3.2 and Remark 3.3 with the exact sequences of triples, by induction on the number of critical values). A proper Morse function ≥ 0 has regular levels $a_0 < 0 \le a_1 \le \cdots \to \infty$ `Chapter5/Lefschetz · exists_regular_levels`. The same statements hold when only a slab $f^{-1}[a', a]$ is compact `Chapter5/LefschetzSlab · sublevel_step_critical_of_slab` `Chapter5/LefschetzSlab · sublevel_step_of_slab` `Chapter5/LefschetzSlab · subsingleton_subsetHomology_slab`: Theorems 3.1 and 3.2 only use the compactness of the slab, which is what §7 needs for $-L_p$ and $1/L_p$, whose sublevel sets are not compact.

Theorem 7.2 (p. 39). Let $M \subseteq \mathbb{R}^n$ be a closed slice submanifold of real dimension $2k$, complex for an orthogonal complex structure J. Then every critical point of L_p on M has index $\le k$ `Chapter5/Lefschetz · indexM_normSq_sub_le`, and M has the homotopy type of a countable cell complex (Part XI) `Appendix/MonotoneUnion · IsCountableCellComplexType` all of whose cells have dimension $\le k$ `Chapter5/Lefschetz · theorem_7_2`.

Proof. Choose p with L_p a Morse function on M (Theorem 6.6, Part III) `Chapter4/Focal · exists_isMorseFunction_normSq_sub`; its sublevel sets are compact because M is closed `Chapter4/Existence · isCompact_sublevel_normSq_sub`. Take regular levels $a_m \to \infty$. Between consecutive levels the sublevel set is obtained by attaching cells, one for each critical point, of dimension its index `Chapter10/RelativeCells · isRelCellComplexType_sublevel_of_isCompact`; the tower of these attachments is a countable cell complex of homotopy type M `Chapter10/RelativeCells · isCountableCellComplexType_of_relSublevel`, and every index is $\le k$. □

Theorem 7.1 (p. 39), over a field. Such an M has $H_i(M;\Bbbk) = 0$ for $i > k$ `Chapter5/Lefschetz · theorem_7_1`. *Proof*: each slab has $H_i = 0$ for $i > k$ (no critical point of index i), the first sublevel set is empty, and the homology of the exhaustion is assembled from the steps `Chapter10/SublevelHomology · spaceHomologyEquivOfSteps` `Chapter10/SublevelHomology · exists_subset_sublevel_of_isCompact`. □ It is computed from the sublevel filtration, not from cellular chains.

"Non-singular affine algebraic variety" and "bianalytically embedded closed complex manifold" are replaced by "closed slice submanifold with complex tangent spaces"; the real dimension $2k$ is fixed by the model space, so M has pure dimension; "k-dimensional CW complex" is "of dimension at most k".

Hyperplane sections: the abstract pair. X compact, $A \subseteq X$ closed, $e : M \cong X \setminus A$ with M a closed complex submanifold of $\mathbb{C}^n$ of complex dimension k, and $\|e^{-1}y\| \to \infty$ as $y \to A$. **Hypothesis (N)** (p. 42): a neighbourhood U of A which deforms into A within X, keeping A pointwise fixed `Chapter5/HyperplaneSection · HypothesisN`. Milnor obtains it from triangulability; here it is a hypothesis, and the deformation must fix A (the exposition's "hidden hypothesis").

Lemma. For a non-degenerate symmetric form, $\operatorname{index}(-H) + \operatorname{index} H = \dim$ `Chapter5/HyperplaneSection · index_neg_add_index`; Hessians, critical points and non-degeneracy pass to $-f$ `Chapter5/HyperplaneSection · hessianM_neg` `Chapter5/HyperplaneSection · isCriticalPointM_neg_iff` `Chapter5/HyperplaneSection · isNondegenerate_neg_iff` `Chapter5/HyperplaneSection · hessian_neg`. So at a non-degenerate critical point of L_p on M, $-L_p$ has index $2k -$ index $\ge k$ `Chapter5/HyperplaneSection · le_indexM_neg_normSq_sub`. A homeomorphism of pairs transports vanishing relative homology `Chapter5/HyperplaneSection · subsingleton_subsetHomology_of_homeomorph`, and a deformation of B into A fixing A makes $H_*(B,A) \to H_*(C,A)$ zero `Chapter5/HyperplaneSection · relHomology_map_eq_zero_of_deform`.

Theorem (the vanishing behind 7.3 and 7.4). Under these hypotheses and (N), $H_r(X, A;\Bbbk) = 0$ for $r < k$ `Chapter5/HyperplaneSection · subsingleton_subsetHomology_univ_of_section`.

Proof (the exposition's, without duality). Choose p with L_p Morse, and a regular value R so large that $V_R = A \cup e\{L_p \ge R\}$ lies in U. (1) V_R deforms into A within X fixing A, so $H_r(V_R, A) \to H_r(X, A)$ is

zero. (2) A lies in the interior of V_R (as $L_p \to \infty$ at A); excising it and transporting by e, $H_r(X, V_R) \cong H_r(M, \{L_p \geq R\})$, which vanishes for $r < k$ by Morse theory for $-L_p$ on the compact slab $\{L_p \leq R\}$, all of whose critical points have index $\geq k$. (3) The exact sequence of the triple (X, V_R, A). □

Corollary 7.3 (Lefschetz), over a field. $H_r(A) \to H_r(X)$ is bijective for $r < k-1$ and onto for $r = k-1$ Chapter5/HyperplaneSection · corollary_7_3_of_section (the exact sequence of the triple $\emptyset \subseteq A \subseteq X$). Milnor's proof quotes Lefschetz duality, with unstated hypotheses; this one uses the ingredients of his proof of 7.4.

The projective case. $\mathbb{C}P^n$ as in §XIII.45, $P = \{Z_0 = 0\} = \{Q_{00} = 0\}$. The affine chart $Q \mapsto (Q_{j0}/Q_{00})_j$ Chapter5/ProjectiveSection · CP.Cn Chapter5/ProjectiveSection · CP.aff has inverse $u \mapsto$ the projection onto the line of $(1, u)$ Chapter5/ProjectiveSection · CP.lift Chapter5/ProjectiveSection · CP.nc Chapter5/ProjectiveSection · CP.nc_ne_zero Chapter5/ProjectiveSection · CP.proj Chapter5/ProjectiveSection · CP.sum_lift Chapter5/ProjectiveSection · CP.proj_mem Chapter5/ProjectiveSection · CP.proj_zero_zero Chapter5/ProjectiveSection · CP.aff_proj Chapter5/ProjectiveSection · CP.zero_zero_eq, and $\|\operatorname{aff} Q\|^2 = (1 - Q_{00})/Q_{00}$ Chapter5/ProjectiveSection · CP.proj_aff_and_norm. $\mathbb{C}^n$ is $\mathbb{R}^{2n}$ with multiplication by i Chapter5/ProjectiveSection · CP.finrank_Cn Chapter5/ProjectiveSection · CP.theta Chapter5/ProjectiveSection · CP.mulI Chapter5/ProjectiveSection · CP.cpxJ Chapter5/ProjectiveSection · CP.cpxJ_apply Chapter5/ProjectiveSection · CP.isOrthComplexStructure_cpxJ; the chart and its inverse are continuous Chapter5/ProjectiveSection · CP.continuous_lift Chapter5/ProjectiveSection · CP.continuous_proj Chapter5/ProjectiveSection · CP.continuousOn_aff. For $V \subseteq \mathbb{C}P^n$: its affine part Chapter5/ProjectiveSection · CP.affinePart and hyperplane section Chapter5/ProjectiveSection · CP.hyperplaneSection, $V - P \cong$ affine part Chapter5/ProjectiveSection · CP.proj_zero_zero_ne Chapter5/ProjectiveSection · CP.affineHomeomorph, and the affine coordinates tend to ∞ at $V \cap P$ Chapter5/ProjectiveSection · CP.tendsto_norm_affineHomeomorph_symm.

Corollary 7.3 (p. 41). Let $V \subseteq \mathbb{C}P^n$ be closed, with affine part a complex submanifold of complex dimension k (V of pure dimension k, singular points in P), and assume (N). Then $H_r(V \cap P; \Bbbk) \to H_r(V; \Bbbk)$ is an isomorphism for $r < k-1$ and onto for $r = k-1$ Chapter5/ProjectiveSection · corollary_7_3. Pure dimension is a hypothesis Milnor omits (the exposition's counterexample $\Pi \cup L$ in $\mathbb{C}P_4$).

Theorem 7.4 (Lefschetz, p. 42). Under the same hypotheses, $\pi_r(V, V \cap P) = 0$ for $r < k$: every map $I^r \to V$ is homotopic, relative to the preimage of $V \cap P$, to a map into $V \cap P$ Chapter5/LefschetzHomotopy · theorem_7_4, and the same for any compact P' embedded in a vector space of dimension $< k$ Chapter5/LefschetzHomotopy · theorem_7_4_of_embedding. Consequently $\pi_i(V \cap P, x) \to \pi_i(V, x)$ is bijective for $i < k-1$ and onto for $i = k-1$, at every $x \in V \cap P$ Chapter5/LefschetzHomotopy · theorem_7_4_bijective Chapter5/LefschetzHomotopy · theorem_7_4_surjective.

$\pi_r(V, V \cap P)$ is not in Mathlib; the statement is its compression criterion (Hatcher §IV.1), in the strong form "relative to the preimage", the language of pushing maps (§XIII.7) Chapter13/Compression · PushesInto, with $r = 0$ included.

The proof as formalized (abstract pair, then projective). 1. *Lemma D on a compact slab.* Lemma D (§XIII.7) Chapter13/Compression · lemma_D pushes maps from compact P' of dimension $< \lambda_0$ down past critical levels of index $\geq \lambda_0$; it holds when only the slab $f^{-1}[a, a_0]$ is compact Chapter5/LefschetzHomotopy · lemma_D_slab. 2. *Milnor's function.* $\psi = 1/(1 + L_p)$ on $X \setminus A \cong M$, 0 on A Chapter5/LefschetzHomotopy · sectionFun Chapter5/LefschetzHomotopy · sectionFun_of_mem Chapter5/LefschetzHomotopy · sectionFun_apply Chapter5/LefschetzHomotopy · sectionFun_nonneg, continuous because $L_p \to \infty$ at A Chapter5/LefschetzHomotopy · continuous_sectionFun. Milnor's $1/L_p$ is infinite at p. 3. *Levels.* $L_p \circ e^{-1}$ is bounded by some B on the compact $X \setminus U$; take a regular value $R > B$. Then $V_R = \{\psi \leq 1/(1 + R)\}$ contains A and lies in U, and $C = \{\psi \geq 1/(2 + R)\}$ is closed with $\overline{X \setminus C} \subseteq V_R$. 4. *Pushing maps into C down.* A map into C lands in $X \setminus A$; read in M, Lemma D for $-L_p$ on the compact slab $\{-R \leq -L_p \leq 1\}$, where every critical point has index $\geq k$ (the lemma above), pushes it into $\{L_p \geq R\} \subseteq V_R$, relative to the part already there. 5. *Localizing.* If maps into a closed C can be pushed into $Y \supseteq \overline{X \setminus C}$, then every map can Chapter5/LefschetzHomotopy · pushesInto_univ_of_

isClosed (glue the push with the identity off C: the exposition's "extended by the identity on $V - K$"). So every map $P' \to X$ is pushed into V_R. 6. *Hypothesis (N).* The deformation of $U \supseteq V_R$ into A fixing A pushes from V_R into A Chapter13/ManifoldOfMinima · pushesInto_of_deformation, and pushes compose Chapter13/Compression · PushesInto.trans. This is the abstract Theorem 7.4 Chapter5/LefschetzHomotopy · theorem_7_4_of_section. 7. *Homotopy groups.* A push of all maps from P' into A is a compression by the inclusion $A \to X$ Chapter5/LefschetzHomotopy · compresses_val_of_pushesInto; compressing cubes and cylinders makes $\pi_i(A, a) \to \pi_i(X, a)$ bijective (Part XII) Chapter5/LefschetzHomotopy · bijective_homotopyGroupMap_val_of_pushesInto. The projective case is the abstract one through the affine chart. □

XIII.47 Late additions to earlier parts: Corollaries 17.2 and 17.4, Theorem 19.6 for all end points (Parts IX and X)

Part IX left Corollary 17.2 and the loop-space form of 17.4 unformalized; Part X proved Theorem 19.6 only for p, q not conjugate. One device, moving the end point, completes all three.

Lemma (moving the end point). Points at finite Riemannian distance are joined by a path in N Chapter10/CWType · joined_of_riemDist_lt_top. Every $q \in N$ is joined by a path in N to a point q' not conjugate to p along any geodesic Chapter10/CWType · exists_joined_notMem_conjLocus: the non-conjugate points are dense (Corollary 18.2, Part X) Chapter11/NonConjugate · dense_compl_conjLocus, so there is one in a small Riemannian ball around q, which is path connected.

Lemma 17.D with Theorem 17.1. A path σ from q to q' gives a homotopy equivalence $\Omega(N; p, q) \simeq \Omega(N; p, q')$ Chapter10/CWType · pathSpHomotopyEquiv: $\Omega \simeq \Omega^*$ by Theorem 17.1 (Part IX) Chapter10/FullPathTheorem · theorem_17_1, Ω^* is homeomorphic to Mathlib's space of paths Chapter13/Corollaries · pathCHomeomorph, and $\omega \mapsto \omega \cdot \sigma$ is a homotopy equivalence of path spaces (Part XII) Homotopy/LoopSpace · pathTransHomotopyEquiv.

Corollary 17.2 (p. 95). For N closed in V, $\Omega(N; p, q)$ with Milnor's metric has the homotopy type of a countable cell complex Chapter10/CWType · corollary_17_2, and so does $\Omega^*(N; p, q)$ Chapter10/CWType · corollary_17_2_pathC.

Proof. Move q to a non-conjugate q'; Theorem 17.3 (Part IX) gives an explicit countable cell complex for $\Omega(N; p, q')$ Chapter10/FundamentalTheorem · theorem_17_3_nonconj, and the cell-complex type is invariant under homotopy equivalence Appendix/MonotoneUnion · IsCountableCellComplexType.of_homotopyEquiv. □

Milnor quotes his theorem that loop spaces of CW spaces have CW type (Trans. AMS 90, 1959). The formal route needs N complete (closed in V), which Milnor's does not, but gives the countable complex of 17.3 rather than an abstract CW type.

Homotopy invariance of homology. Homotopy equivalent spaces have isomorphic homology over a field Chapter10/CWType · spaceHomologyEquivOfHomotopyEquiv and the same Betti numbers Chapter10/CWType · bettiNumber_eq_of_homotopyEquiv.

Corollary 17.4 for the loop space (p. 96). For $n \geq 2$ and $p \in S^n$, the loop space $\Omega(S^n; p, p)$ is homotopy equivalent to $\Omega(S^n; p, q)$ for some $q \neq \pm p$ ($q \perp p$; S^n is path connected) Chapter10/CWType · exists_loop_homotopyEquiv_sph, so it has the homotopy type of the countable cell complex with one cell in each dimension $0, n-1, 2(n-1), \ldots$ Chapter10/CWType · corollary_17_4_loop (from §IX.10's path-space form Chapter10/SpherePaths · corollary_17_4), and for $n \geq 3$ its Betti numbers over any field are 1 in the degrees divisible by $n-1$ and 0 otherwise Chapter10/CWType · bettiNumber_sphere_loops Chapter10/SpherePaths · bettiNumber_sphere_paths. This was the hidden hypothesis of 17.4: p is conjugate to itself along every great circle, so 17.3 does not apply to loops directly.

Theorem 19.6 for all end points (p. 106). Let N be compact, of dimension ≥ 2, with positive definite Ricci tensor. For all $p, q \in N$, $\Omega(N; p, q)$ has the homotopy type of a countable cell complex with finitely

many cells in each dimension `Chapter11/PositiveRicciAll · theorem_19_6_all`, and its homology over any field is finite-dimensional in each degree `Chapter11/PositiveRicciAll · finiteDimensional_spaceHomology_omega_all`.

Proof. Move q to a non-conjugate q' in a small ball (no connectedness of N is needed); Theorem 19.6 for non-conjugate end points (Part X) `Chapter11/PositiveRicci · theorem_19_6` `Chapter11/RicciHomology · finiteDimensional_spaceHomology_omega_of_compact` and the invariance of both conclusions under homotopy equivalence. □

Milnor states 19.6 for all p, q but proves it for non-conjugate ones; the exposition's completion is the one formalized.

XIII.48 What the formalization taught us

- **Hanner's theorem is never needed; the retractions are explicit.** Theorem 22.1 uses "Ω^d is a manifold" only through Hanner's theorem, so the formal version assumes a neighbourhood retraction (§XIII.10), and each application writes one down: the equator of the sphere (§XIII.12), a spectral projection for $SU(2m)$ (§XIII.18), and for $\Omega_k(n)$ the nearest complex structure, by projection onto $\mathcal{T}_{J_k}$ followed by the polar decomposition (§XIII.34) `Chapter15/OmegaRetract · exists_retraction_minPaths_omega`. The constraints of $\mathcal{T}$ must be imposed before the square root, and the square root must commute with them.
- **"Homeomorphic" means "homeomorphic after identifying paths that agree on $[0,1]$".** Paths are functions on $\mathbb{R}$ and Milnor's metric only sees $[0,1]$. Lemmas 23.1 and 24.4 (iii) hold as homeomorphisms onto the separation quotient of Ω^d `Chapter14/MinimalPaths · minPathsHomotopyEquiv` `Chapter15/OmegaMinimal · omegaMinPathsHomeomorph`; what the homotopy groups need is the homotopy equivalence without the quotient `Chapter15/OmegaMinimal · omegaMinPathsHomotopyEquiv`. The Grassmannian of projections is Milnor's Grassmannian of subspaces with Steenrod's topology (§XIII.25) `Homotopy/GrassmannianQuotient · stiefelQuotientHomeomorph`.
- **The ranges are part of the theorem.** Milnor's chain on p. 129 lands in $U(2m)$, and returning to $U(m)$ needs $i+1 < 2m$ (§XIII.24); stated between stable groups the question disappears `Chapter15/StablePeriodicity · periodicity_theorem_stable`. For O, the eight steps of Theorem 24.5 at $n = 16r$ need $2i + 7 \le r$, the factor 4 lost in the case $k \equiv 2 \pmod 4$ being the binding one, and stability needs $i + 2 \le n$ `Chapter15/StablePeriodicity · theorem_24_7`.
- $\Omega_k(n)$ **is not a manifold of one dimension.** Its components have different dimensions ($\Omega_3(16r)$ contains every $G_j(\mathbb{H}^{4r})$). The formal objects are the strata where $\dim \mathcal{T}$ is constant, open and closed in $\Omega_k(n)$ `Chapter15/OmegaManifold · isClosed_omegaStratum` `Chapter15/OmegaManifold · isSliceSubmanifold_omegaStratum`; the paths from J_k stay in its stratum, which carries the homotopy groups of $\Omega_k(n)$ at J_k `Chapter15/FiniteStage · nonempty_pi_stratum_equiv`. Milnor's Assertions 1 and 2 are "iff" only for small A, and the "only if" halves need a ball on which exp is injective and which is invariant under transposition and orthogonal conjugation `Chapter15/OmegaManifold · mem_tanAlg_of_mul_exp_mem`.
- **No normal forms.** Lemma 24.1's bound $|A| \ge \pi\sqrt{n}$ comes from one eigenvalue at a time: on an eigenvector of $-V^2$ the exponential is a rotation, so $\exp V = -I$ forces the eigenvalue to be $\ge \pi^2$ `Chapter15/OmegaMinimal · pi_sq_le_of_sq_mul`. As for $SU(2m)$ (§XIII.14), minimality needs a shortest geodesic to exist (Hopf–Rinow) and every geodesic of the stratum to be $J_k \exp(tV)$; distances in $\Omega_k(n)$ and $O(n)$ are never compared.
- **Killing fields instead of curvature.** Milnor's index bounds go through the curvature of the totally geodesic $\Omega_k(n)$ and the eigenvectors of K_A. Each of his eigenvectors is $[X, A]$ for an infinitesimal isometry X fixing J_k, so the Jacobi fields are Killing fields and Lemma 14.3 suffices `Chapter15/OmegaKilling · isJacobiField_killing` `Chapter15/OmegaKilling · sum_finrank_le_formIndex`; "totally geodesic" is only used through the list of geodesics. The independence of the eigenvectors of different pairs, which Milnor does not discuss, is a projection argument `Chapter15/OmegaIntertwiners · exists_killing_spaces`.
- **Lemma 24.8 is proved only for $k \le 8$, so Theorem 24.5 only for $k \le 7$.** Milnor states 24.5 for

all k but treats Lemma 24.8 only for ≤ 8 structures; $k \leq 7$ is all periodicity needs `Chapter15/MinimalModules · exists_iso_of_isMinimal`. The isomorphism is linear, not isometric, and that suffices: $D - D^{\mathsf{T}}$ is skew for any D. "The proof of 24.8 follows that of 24.6" is, formally, the halving by the θ's applied to the Clifford algebra itself, plus a cyclic vector fixed by the θ's (§XIII.31).

- **The winding number must be shown to be an integer.** Milnor's $f(-J) = \det_{\mathbb{C}}(-1) = (-1)^{n/2}$ closes the loop only for $4 \mid n$. With his normalization of c, the end angle must lie in $\pi m_{k+1}\mathbb{Z}/2$: for $k = 2$ this is $2\pi\mathbb{Z}$ and follows from $4 \mid n$ `Chapter15/OmegaComponents · exists_mem_omegaComp_two`; for $k = 6$ it is $4\pi\mathbb{Z}$, and needs $\det_{\mathbb{C}}$ to be a fourth power on $\Omega_6(n)$ `Chapter15/OmegaComponents · omegaF_eq_pow_four Chapter15/OmegaComponents · exists_mem_omegaComp_six`. That c survives stabilization is a trace computation on $\mathbb{R}^{16}$ `Chapter15/Stabilization · omegaLabel_stab`. Milnor's "$\Omega_k(n)$ has an infinite cyclic fundamental group" is not needed.
- **Both signs in the index bound for $k = 2, 6$.** Milnor's "suppose for example that some a_h is equal to -3" yields $\lambda \geq (\nu - c)/4$; the formal bound is $4\lambda \geq \nu - |c|$, the other sign by $A \mapsto -A$, whose Killing fields serve for A `Chapter15/IndexBoundTwo · le_formIndex_of_not_minimal_omega_two`.
- **π_0 at a finite stage is only partly controlled.** For $k = 2, 6$, φ_r matches the path components of $\Omega_{k+1}(n)^{(c)}$ and $\Omega^{(c)}$ only for $|c| \leq n/m_{k+1} - 8$ `Chapter15/OmegaComponents · theorem_24_5_pi0_two_six`; Milnor's bijection of path components is a statement about the limit, where every c is eventually in range. The chain uses the $k = 2, 6$ steps only in degrees ≥ 1, so $\pi_0 O(r) \cong \pi_8 O(16r)$ is still proved `Chapter15/StablePeriodicity · bott_periodicity_finite_of_le`.
- **Explicit structures and a commutant.** "n divisible by a high power of 2" is replaced by eight Kronecker words on $\mathbb{R}^{16}$ `Chapter15/Clifford · isAnticommFamily_bott16`, and $\Omega_8(16r) \cong O(r)$ by the computation that the commutant of $J'_1, \ldots, J'_7$ is spanned by 1 and $W = 1 \otimes 1 \otimes 1 \otimes \tau$ `Chapter15/EightCommutant · eq_of_commute_bott16 Chapter15/OmegaEight · omegaEightHomeomorph`, instead of joint eigenspaces.
- **Bijections first, groups where they come cheap.** Every isomorphism of §§22–24 is first a bijection of pointed sets; the group forms need change of basepoint to be a homomorphism (§XIII.11), and are proved for §§22–23. For §24 the eight-step chain, and so Theorem 24.7, remains a bijection of sets: upgrading it would need each step of Theorem 24.5 as a group map and the loop-space identification to be natural under stabilization. The colimits O, Ω_k and the homotopy types ($O \simeq \Omega^8 O$) are not formalized; Theorem 24.7 is proved for the stable groups $\pi_i O(n)$, $i < n - 1$.
- **Concrete bundles, one input.** Stiefel manifolds, Grassmannians and spheres are spaces of matrices, the bundles are orbit maps with explicit local sections, and all the homotopy groups of §23 come from the exact sequence of Part XII and $\pi_i S^n = 0$ for $i < n$ (§§XIII.19–13.23).
- **The late additions had hidden hypotheses of their own.** The Lefschetz theorems need pure dimension and a Hypothesis (N) whose deformation fixes $V \cap P$ (§XIII.46) `Chapter5/HyperplaneSection · HypothesisN`, and Theorems 3.1–3.2 on compact slabs rather than sublevel sets; Corollary 7.3 is proved without Lefschetz duality and Theorem 7.4 by compression. Corollaries 17.2, 17.4 for loops and Theorem 19.6 for all end points all follow by moving the end point off the conjugate locus (§XIII.47) `Chapter10/CWType · pathSpHomotopyEquiv`.

Concordance

Every declaration cited in this book, by file, with its line in the source. 5665 declarations in 427 files. The web edition also shows each exact formal statement.

Morse/Appendix/Examples.lean. `example1` (68); `isHomotopyDirectLimit_sublevel` (101); `example3Set` (149); `monotone_example3Set` (152); `isClosed_example3Set` (160); `iUnion_example3Set` (165); `example3` (176).

Morse/Appendix/MappingTelescope.lean. `tmap` (53); `tmap_refl` (56); `tmap_tmap` (62); `MapTelRel` (71); `MappingTelescope` (77); `MappingTelescope.mk` (86); `MappingTelescope.mk_one` (89); `MappingTelescope.isQuotientMap_quotMk` (93); `MappingTelescope.continuous_mk` (97); `MappingTelescope.exists_mk` (101); `MappingTelescope.desc` (110); `MappingTelescope.desc_mk` (119); `MappingTelescope.lift` (125); `MappingTelescope.lift_mk` (132); `MappingTelescope.homotopy` (139); `MappingTelescope.ext` (151); `MappingTelescope.level` (162); `MappingTelescope.level_mk` (170); `MappingTelescope.level_nonneg` (173); `MappingTelescope.trunc` (179); `MappingTelescope.isClosed_trunc` (182); `MappingTelescope.monotone_trunc` (185); `MappingTelescope.isHomotopyDirectLimit_trunc` (191); `MappingTelescope.traj` (207); `MappingTelescope.traj_of_le_one` (212); `MappingTelescope.traj_zero_right` (218); `MappingTelescope.continuous_traj` (221); `MappingTelescope.level_traj` (237); `MappingTelescope.pushFun` (257); `MappingTelescope.continuous_pushFun` (260); `MappingTelescope.pushFun_compat` (267); `MappingTelescope.push` (296); `MappingTelescope.push_mk` (299); `MappingTelescope.push_zero` (303); `MappingTelescope.push_bottom` (309); `MappingTelescope.level_push` (315); `MappingTelescope.push_mem_trunc` (330); `MappingTelescope.level_push_one` (337); `MappingTelescope.atLevel` (346); `MappingTelescope.atLevel_mk_zero` (359); `MappingTelescope.mk_zero_injective` (363); `MappingTelescope.exists_mk_zero_of_level` (369); `MappingTelescope.isInducing_mk_zero` (392); `MappingTelescope.mk_zero_mem_trunc` (456); `MappingTelescope.levelIncl` (461); `MappingTelescope.push_one_eq` (466); `MappingTelescope.levelRetract` (470); `MappingTelescope.mk_levelRetract` (481); `MappingTelescope.levelRetract_levelIncl` (485); `MappingTelescope.levelRetract_comp_levelIncl` (492); `MappingTelescope.levelIncl_comp_levelRetract` (497); `MappingTelescope.isHomotopyEquiv_levelIncl` (509); `MappingTelescope.isHomotopyEquiv_levelRetract` (512).

Morse/Appendix/MonotoneUnion.lean. `homotopic_inverse_square` (108); `nonempty_homotopyEquiv_telescope` (135); `compareStrip` (154); `compareMap` (175); `compareMap_mk` (190); `compareMap_mk_zero` (196); `compareMap_mapsTo` (205); `isHomotopyEquiv_compareMap` (232); `nonempty_homotopyEquiv_mappingTelescope` (256); `corollary_appendix` (274); `corollary_cellComplex` (298); `IsRelCellAttachment` (313); `IsCellAttachment.of_rel` (318); `CellTower` (331); `CellTower.isCellAttachment` (339); `CellTower.ofSequence` (349); `CellTower.tmap_ofSequence` (359); `IsCountableCellComplexType` (370); `IsCountableCellComplexType.of_homotopyEquiv` (373); `isCountableCellComplexType_of_tower` (387); `isCountableCellComplexType_of_sublevel` (400); `isCountableCellComplexType_of_sublevel'` (417); `subspaceFunctor` (439); `telescopeComparison` (447); `telescopeComparison_mk` (451); `telescopeComparisonHomeomorph` (457); `isHomotopyEquiv_telescopeComparison` (474); `telescopeProj'` (494); `isHomotopyEquiv_telescopeProj'` (499); `mappingTelescopeMap` (506); `mappingTelescopeMap_mk`

(514); mapsTo_mappingTelescopeMap (520); theoremA_general (533).

Morse/Appendix/Telescope.lean. IsHomotopyEquiv (81); IsHomotopyEquiv.of_homotopyEquiv (87); IsHomotopyEquiv.exists_homotopyEquiv (90); IsHomotopyEquiv.nonempty (95); IsHomotopyEquiv.id (100); IsHomotopyEquiv.of_homeomorph (103); IsHomotopyEquiv.comp (106); IsHomotopyEquiv.of_homotopic (113); IsHomotopyEquiv.of_leftInv_rightInv (121); IsHomotopyEquiv.of_inverse (127); restrictTo (134); clampI_coe (146); clampI_of_le_zero (158); clampI_of_one_le (161); telescope (172); mem_telescope (175); nonneg_of_mem_telescope (179); mem_telescope_iff (184); mem_telescope_of_le (196); mem_of_mem_telescope (202); up (212); stripMap (221); stripMap_one (242); exists_stripMap (247); eq_stripMap_of_level (254); exists_stripMap_lt_one (260); telescope_ext (273); telescope_ext' (279); continuous_of_stripMap (288); mem_floor (329); fract_mem (333); telescopeLiftFun (338); telescopeLiftFun_stripMap (353); telescopeLift (373); telescopeLift_stripMap (385); telescopeDesc (392); telescopeDesc_stripMap (398); telescopeHomotopy (406); telescopeProj (424); IsHomotopyDirectLimit (433); mem_telescope_map (439); telescopeMap (446); telescopeMap_stripMap (453); telescopeProj_comp_telescopeMap (457); IsHomotopyEquiv.of_telescopeMap (464); truncation (482); isClosed_truncation (485); monotone_truncation (489); stripMap_zero_mem_truncation (494); truncIncl (499); truncProj (509); truncProj_comp_truncIncl (519); truncIncl_comp_truncProj (525); isHomotopyEquiv_truncIncl (550); isHomotopyEquiv_truncProj (554); level_le_of_stripMap (561); isHomotopyDirectLimit_of_level (584); isHomotopyDirectLimit_truncation (616).

Morse/Appendix/TheoremA.lean. reparam (101); reparam_stripMap (110); reparam_homotopic (118); scaleI (142); scaleMap (154); scaleMap_stripMap (158); scaleMap_homotopic (163); IsMilnorMap (175); IsMilnorMap.fix (183); IsMilnorMap.fix_high (188); IsMilnorMap.apply_one (193); IsMilnorMap.apply_half_or_one (198); milnorInvStrip (211); milnorInv (230); milnorInv_low (244); milnorInv_high (251); milnorInv_isMilnorMap (258); loopFun (271); continuous_loopFun (276); loopHomotopy (292); muH (309); muK (313); continuous_muH (316); continuous_muK (321); muH_zero_low (325); muH_zero_high (330); muH_one (335); muH_quarter (341); muH_end (348); muK_zero_low (356); muK_zero_high (361); muK_one (366); muK_quarter (369); muK_end (376); loopFun_one (384); milnorH (397); milnorK (420); isHomotopyEquiv_of_milnorMap (447); case1Fun (466); continuous_case1Fun (474); case1Fun_compat (505); case1Family (514); case1Map (521); case1Map_stripMap (526); case1Map_zero (534); case1Map_one (550); theoremA_case1 (572); isHomotopyEquiv_of_truncation (597); case2InvStrip (625); case2Inv (646); case2Inv_stripMap (660); case2Inv_zero (668); case2Inv_level (675); homotopic_square (687); theoremA_telescope (712); theoremA (757).

Morse/Chapter1/Examples.lean. not_isNondegenerate_of_frequently (46); frequently_of_line (53); Examples.hasFDerivAt_cube (74); Examples.isCriticalPoint_cube_iff (80); Examples.hessian_cube (88); Examples.cube_degenerate (100); Examples.monkey (110); Examples.hasFDerivAt_monkey (112); Examples.isCriticalPoint_monkey_iff (120); Examples.hessian_monkey (139); Examples.monkey_degenerate (155); Examples.sqFst (164); Examples.hasFDerivAt_sqFst (166); Examples.isCriticalPoint_sqFst_iff (172); Examples.not_isNondegenerate_sqFst (181); Examples.sqMul (193); Examples.hasFDerivAt_sqMul (195); Examples.isCriticalPoint_sqMul_iff (202); Examples.not_isNondegenerate_sqMul (220).

Morse/Chapter1/Hadamard.lean. exists_tube_bound (81); contDiffOn_intervalIntegral_aux (108); contDiffOn_intervalIntegral (178); contDiffOn_intervalIntegral_infty (201); contDiffOn_integral_smul_comp (209); exists_hadamard_on (224); exists_hadamard_coord (253); exists_hadamard_two_on (280).

Morse/Chapter1/Manifold.lean. exists_fderiv_eq_of_localInverse (126); hessian_comp_apply (154); nullSpace_bilinearComp (186); nullity_bilinearComp (199); index_le_finrank (206); localRep (227); IsCriticalPointM (232); hessianM (239); IsNondegenerateCriticalPointM (243); indexM (247); nullityM (251); IsMorseFunction (256); IsMorseFunction.isNondegenerateCriticalPointM (262); localRep_eq (283); localRep_chartAt (291); localRep_self (295); writtenInExtChartAt_eq_localRep (299); eventuallyEq_localRep_comp (305); mdifferentiableAt_iff_differentiableAt_localRep

(312); contMDiffAt_iff_contDiffAt_localRep (323); mfderiv_eq_fderiv_localRep (338); isCriticalPointM_iff_fderiv (353); isCriticalPointM_iff (360); isCriticalPointM_iff_hasMFDerivAt (365); surjective_iff_ne_zero (371); isCriticalPointM_iff_milnor (388); hessianM_symm (406); indexM_le_finrank (411); isNondegenerate_hessianM_iff_nullityM_eq_zero (414); localRep_model (420); hessianM_model (428); isCriticalPointM_model_iff (431); indexM_model (436); IsSmoothCoord (454); IsSmoothCoord.toPartialDiffeomorph (460); isSmoothCoord_toOpenPartialHomeomorph (473); isSmoothCoord_iff_mem_maximalAtlas (484); isSmoothCoord_chartAt (494); IsSmoothCoord.trans (505); IsSmoothCoord.isOpen_trans_source (518); IsSmoothCoord.contDiffOn_trans (528); IsSmoothCoord.contDiffAt_trans (533); IsSmoothCoord.exists_fderiv_trans_eq (544); IsSmoothCoord.comp_symm_eventuallyEq (567); IsSmoothCoord.isCriticalPoint_trans_iff (575); IsSmoothCoord.hessian_trans (602); IsSmoothCoord.exists_hessian_trans_eq_bilinearComp (617); IsSmoothCoord.isNondegenerate_trans_iff (629); IsSmoothCoord.index_trans_eq (638); IsSmoothCoord.nullity_trans_eq (646); hessianM_eq_chartAt (665); contDiffAt_comp_chartAt_symm (670); fderiv_comp_chartAt_symm_eq_zero (676); IsSmoothCoord.mfderiv_symm_eq (686); IsSmoothCoord.mfderiv_eq (707); IsSmoothCoord.exists_mfderiv_eq (716); IsSmoothCoord.exists_mfderiv_symm_eq (722); IsSmoothCoord.isCriticalPointM_iff (730); IsSmoothCoord.hessian_comp_symm (744); IsSmoothCoord.isNondegenerate_iff (756); IsSmoothCoord.index_eq (764); IsSmoothCoord.nullity_eq (772); IsSmoothCoord.isNondegenerateCriticalPointM_iff (780); IsSmoothCoord.contDiffAt_comp_symm (793); IsSmoothCoord.differentiableAt_comp_symm (803); IsSmoothCoord.isCriticalPointM_iff_forall_partial (814); IsSmoothCoord.isNondegenerate_iff_det_ne_zero (830); isCriticalPointM_iff_of_mem_maximalAtlas (846); hessian_comp_symm_of_mem_maximalAtlas (852); isNondegenerate_iff_of_mem_maximalAtlas (860); indexM_eq_of_mem_maximalAtlas (866); nullityM_eq_of_mem_maximalAtlas (872).

Morse/Chapter1/ManifoldMorse.lean. eventually_isCriticalPointM_iff (80); eventually_not_isCriticalPointM (95); IsNondegenerateCriticalPointM.exists_nhds_isCriticalPointM_eq (120); isClosed_setOf_isCriticalPointM (135); isDiscrete_inter_setOf_isCriticalPointM (158); finite_inter_setOf_isCriticalPointM (177); IsMorseFunction.finite_setOf_isCriticalPointM (188); IsMorseFunction.finite_inter_setOf_isCriticalPointM (195); morse_lemmaM_aux (214); exists_morse_chartM (236); morse_lemmaM (268); isNondegenerate_diagForm_morseSign (308); morseNormalForm (322); morseNormalForm_eq (326); contDiff_morseNormalForm (332); isCriticalPoint_morseNormalForm (338); hessian_morseNormalForm (351); isNondegenerateCriticalPointM_of_morse_chart (369); indexM_eq_of_morse_chart (407).

Morse/Chapter1/ManifoldSubmanifold.lean. sliceParam (98); isOpen_toChart_target (103); sliceParam_mem (106); val_toChart_symm (109); toChart_apply_mem_target (113); sliceParam_toChart (118); contDiffOn_sliceParam (122); contDiffAt_sliceParam (129); comp_val_comp_toChart_symm_eventuallyEq (135); toChart_mem_maximalAtlas (154); isSmoothCoord_toChart (162); contMDiffAt_comp_val (170); contMDiff_comp_val (179); mdifferentiableAt_comp_val (186); isCriticalPointM_comp_val_iff_sliceParam (199); hessian_comp_sliceParam (211); isNondegenerate_hessian_comp_sliceParam_iff (226); index_hessian_comp_sliceParam (241); nullity_hessian_comp_sliceParam (254); isNondegenerateCriticalPointM_comp_val_iff_sliceParam (267); tangentSubspace (294); mem_tangentSubspace_iff (299); tangentSubspace_eq_range_fderiv_sliceParam (306); isCriticalPointM_comp_val_iff_forall_tangent (332); isCriticalPointM_comp_val_iff_gradient_mem_orthogonal (364); isCriticalPointM_comp_val_iff_starProjection_gradient_e (375); hasFDerivAt_normSq_sub (384); fderiv_normSq_sub_apply (392); hessian_normSq_sub (398); hessian_normSq_sub_apply (413); contDiff_normSq_sub (418); isCriticalPointM_normSq_sub_iff (423); hessian_normSq_sub_comp (441); indexM_normSq_sub_eq (455); nullityM_normSq_sub_eq (467); isNondegenerate_hessianM_normSq_sub_iff (477); sliceParam_tub (501); tangentSubspace_eq_tanN (503); indexM_comp_val_eq_tub (507); isNondegenerateCriticalPointM_comp_val_iff_tub (518).

Morse/Chapter1/MorseLemma.lean. morseSign (76); diagForm (80); diagForm_apply (84); diagForm_symm (88); diagForm_morseSign_apply_self (94); quadForm_diagForm (107); in-

dex_diagForm_morseSign (114); exists_perm_neg_first (131); exists_linearEquiv_diagForm (168); fderiv_translate (211); hessian_translate (215); exists_morse_chart (227); morse_lemma (306); index_eq_of_morse_chart (358).

Morse/Chapter1/Nondegenerate.lean. IsCriticalPoint (91); IsCriticalPoint.fderiv_eq (94); isCriticalPoint_iff (98); IsCriticalPoint.comp (105); hessian (113); hessian_symm (117); hessian_congr (122); hessian_bilinear (127); quadForm (145); nullSpace (152); mem_nullSpace (155); nullity (159); index (165); IsNondegenerate (169); isNondegenerate_iff (172); isNondegenerate_iff_nullity_eq_zero (176); IsNondegenerateCriticalPoint (181); negDef_iff_posDef_restrict (184); index_isGreatest (196); exists_negDef_finrank_eq_index (207); finrank_le_index_of_negDef (211); quadForm_bilinearComp_equivalent (219); index_bilinearComp (225); isNondegenerate_bilinearComp_iff (230); index_smul (243); isNondegenerate_smul_iff (254); hessian_comp (264); isNondegenerate_hessian_comp_iff (286); index_hessian_comp (297); isNondegenerate_iff_det_ne_zero (311); isNondegenerate_hessian_iff_det_ne_zero (335); eventually_not_isCriticalPoint (346); IsNondegenerateCriticalPoint.exists_nhds_isCriticalPoint_eq (359).

Morse/Chapter1/SquareRoot.lean. exists_openPartialHomeomorph_of_contDiffOn (76); precomp (124); sqrtAux (130); sqrtAux_apply (135); sqrtAux_apply_self (140); sqrtAux_id (145); contDiff_sqrtAux (149); sqrtAuxDeriv (156); hasFDerivAt_sqrtAux (163); finrank_dual_eq (181); dualEquiv (189); sqrtAuxDerivEquiv (200); exists_sqrt_family (227).

Morse/Chapter10/Broken.lean. Subdivision (88); Subdivision.len (100); Subdivision.len_pos (102); Subdivision.t_nonneg (105); Subdivision.t_le_one (108); Subdivision.t_lt_iff (113); Subdivision.sum_len (116); Subdivision.breaks (123); Subdivision.mem_breaks (125); Subdivision.breaks_subset (128); Subdivision.segOf (136); Subdivision.segOf_eq (140); Subdivision.segOf_of_one_le (154); Subdivision.mem_Icc_segOf (161); Subdivision.exists_seg_of_piece (185); Subdivision.uniform (213); Subdivision.uniform_len (221); vert (236); vert_zero (238); vert_last (240); vert_inner (243); exists_isShortJoin' (258); corollary_10_8' (273); closedRiemBall (299); closedRiemBall_subset (303); isCompact_closedRiemBall_of_isClosed (309); isCompact_closedRiemBall_of_isRiemComplete (335); mem_closedRiemBall_of_energy_le (376); riemDist_le_sum_filter (404); riemDist_le_sum (424); sum_integral_fin (429); BrokenSetup (450); exists_brokenSetup (466); exists_brokenSetup_uniform (480); BrokenSetup.p_mem_ball (494); BrokenSetup.joinVec (502); BrokenSetup.joinVec_spec (508); BrokenSetup.mem_expDom_joinVec (528); BrokenSetup.energy' (549); BrokenSetup.nodes (554); BrokenSetup.vert_mem (561); BrokenSetup.energy'_nonneg_terms (570); BrokenSetup.riemDist_vert_lt (576); BrokenSetup.vert_mem_ball (596); BrokenSetup.segPath (619); BrokenSetup.path (625); BrokenSetup.nodeVal (628); BrokenSetup.IsBroken (632); BrokenSetup.joinVec_vert (641); BrokenSetup.segPath_start (653); BrokenSetup.segPath_end (658); BrokenSetup.segPath_isGeodesicOn (665); BrokenSetup.path_eq_segPath (704); BrokenSetup.contDiffOn_path (719); BrokenSetup.path_mem (726); BrokenSetup.path_mem_pathSpace (733); BrokenSetup.isBroken_path (752); BrokenSetup.nodeVal_path (758); BrokenSetup.norm_deriv_path (764); BrokenSetup.energy_path (780).

Morse/Chapter10/BrokenEnergy.lean. contDiff_vert (65); BrokenSetup.gaps (87); BrokenSetup.continuousOn_gaps (91); BrokenSetup.exists_isOpen_nodes (100); BrokenSetup.exists_smooth_energy' (131); BrokenSetup.continuousOn_energy' (160); BrokenSetup.contMDiff_energy' (172); BrokenSetup.isCompact_sublevel (184).

Morse/Chapter10/BrokenIndex.lean. hasDerivAt_deriv_line (66); eq_of_quadratic_eq (89); injective_fderiv_sliceParam' (110); BrokenSetup.eqOn_of_isJacobiField_seg (135); chartTangentEquiv (167); chartTangentEquiv_apply (191); BrokenSetup.chartBrokenEquiv (205); BrokenSetup.chartBrokenEquiv_node (216); BrokenSetup.hessian_apply_self (228); BrokenSetup.hessian_eq_energyHessian (350); BrokenSetup.theorem_16_2_index (378).

Morse/Chapter10/BrokenPaths.lean. contDiffOn_Icc_of_isGeodesicOn (52); Bro-

kenSetup.brokenPaths (73); BrokenSetup.vert_nodeVal (78); BrokenSetup.exists_segVec (91); BrokenSetup.nodeVal_mem_nodes (186); BrokenSetup.path_nodeVal (211); BrokenSetup.lemma_16_1 (238).

Morse/Chapter10/BrokenSplitting.lean. deriv_comp_affine (71); curvR_smul_smul (78); IsJacobiField.comp_affine (89); nonneg_of_isLocalMin_of_hasDerivAt (121); BrokenSetup.energy'_nodeVal_le (146); BrokenSetup.path_affine_eq (177); BrokenSetup.isSmoothCurveIn_path (199); BrokenSetup.not_isConjugate_seg (206); BrokenSetup.brokenSpace_eq_zero (245); BrokenSetup.exists_brokenSpace_node (287); BrokenSetup.brokenSpace_node_mem (376); BrokenSetup.brokenNodeMap (383); BrokenSetup.brokenNodeMap_bijective (391); BrokenSetup.brokenEquiv (407); BrokenSetup.brokenEquiv_apply (411); BrokenSetup.energyHessian_brokenSpace_eq (420); BrokenSetup.energyHessian_brokenSpace_nodeZero (437); BrokenSetup.energyHessian_nodeZero_nonneg (448); BrokenSetup.energyHessian_nodeZero_pos (483); BrokenSetup.isCompl_brokenSpace_nodeZero (553); BrokenSetup.formIndex_brokenSpace (580); BrokenSetup.finrank_ker_brokenSpace (608).

Morse/Chapter10/BrokenVariation.lean. exists_curve_of_mem_tangentSubspace (61); BrokenSetup.exists_variation_path (98); BrokenSetup.varField_path_node (199); BrokenSetup.firstVariation_path (213); BrokenSetup.prj_jump_path (272); BrokenSetup.fderiv_eq_jumps (285); BrokenSetup.isCriticalPointM_iff (317).

Morse/Chapter10/CWType.lean. joined_of_riemDist_lt_top (64); exists_joined_notMem_conjLocus (88); pathSpHomotopyEquiv (110); corollary_17_2 (122); corollary_17_2_pathC (130); spaceHomologyEquivOfHomotopyEquiv (141); bettiNumber_eq_of_homotopyEquiv (151); exists_loop_homotopyEquiv_sph (159); corollary_17_4_loop (184); bettiNumber_sphere_loops (192).

Morse/Chapter10/Deformation.lean. PathSp.pathSpace_congr (62); pathSpace_congr (62); PathSp.supDist_congr (73); PathSp.l2Dist_congr (80); PathSp.continuous_of_edist_eq_zero (89); continuous_of_edist_eq_zero (89); BrokenSetup.deformFun (110); BrokenSetup.deformFun_mem (115); BrokenSetup.deform (123); BrokenSetup.coe_deform (126); BrokenSetup.deform_of_not (130); BrokenSetup.eqOn_deform (140); BrokenSetup.edist_deform (145); BrokenSetup.deform_zero (155); BrokenSetup.deform_of_mem_brokenPaths (165); BrokenSetup.energy_deform_le (174); BrokenSetup.deform_one_mem_brokenPaths (183); BrokenSetup.continuousOn_deform (196); BrokenSetup.theorem_16_2_deformationRetract (237); BrokenSetup.continuous_deform_sublevel (250); BrokenSetup.retrMap (262); BrokenSetup.inclMap (274); BrokenSetup.homotopyEquivBroken (282).

Morse/Chapter10/Finite.lean. BrokenSetup.exists_isOpen_sublevel_lt (75); BrokenSetup.exists_contDiff_eq_energy' (94); BrokenSetup.geodIndex (104); BrokenSetup.isBroken_of_isUnbrokenGeodesic (109); BrokenSetup.eqOn_path_nodeVal_of_isUnbrokenGeodesic (115); BrokenSetup.theorem_16_3 (132); BrokenSetup.theorem_16_3_count (240); BrokenSetup.theorem_16_3_nonconj (266); BrokenSetup.nodeVal_mem_nodes_of_energy_lt (287); nodeVal_mem_nodes_of_energy_lt (287); BrokenSetup.exists_energy_minimizing (304).

Morse/Chapter10/FullPathDeform.lean. dseg_dy (53); dseg_eq_or (59); JoinSystem (91); JoinSystem.IsGood (110); JoinSystem.broken (115); JoinSystem.interp (120); JoinSystem.IsGood.broken_eq (132); JoinSystem.param_mem_Icc (156); JoinSystem.broken_mem_of (168); JoinSystem.interp_mem_of (176); JoinSystem.IsGood.broken (182); JoinSystem.IsGood.interp (186); JoinSystem.interp_zero (191); JoinSystem.interp_one (195); JoinSystem.broken_dy (202); JoinSystem.broken_zero (225); JoinSystem.broken_one (229); JoinSystem.interp_zero_right (233); JoinSystem.interp_one_right (236); JoinSystem.isGood_congr (240); JoinSystem.broken_congr (252); JoinSystem.interp_congr (256); JoinSystem.IsGood.mono (260); JoinSystem.ext (273); JoinSystem.YSet (285); JoinSystem.isCompact_piece (290); JoinSystem.mem_YSet_iff (293); JoinSystem.isOpen_YSet (310); JoinSystem.YSet_mono (325); JoinSystem.exists_mem_YSet (330); JoinSystem.continuous_ext

(352); JoinSystem.continuous_interp_star (363); JoinSystem.interpStar (466); JoinSystem.interpStar_apply (486); JoinSystem.interpStar_zero (491).

Morse/Chapter10/FullPathOmega.lean. tendsto_integral_deriv_comp_sub_sq (62); JoinSystem.exists_piece_of_Icc (206); JoinSystem.IsGood.exists_formula (231); JoinSystem.dyPts (252); JoinSystem.IsGood.contDiffOn_broken (259); JoinSystem.IsGood.contDiffOn_interp (273); JoinSystem.IsGood.broken_mem_pathSpace (306); JoinSystem.IsGood.interp_mem_pathSpace (318); JoinSystem.clampF (344); JoinSystem.clampF_eqOn (347); JoinSystem.XSet (352); JoinSystem.XSet_eq_preimage (359); JoinSystem.brokenΩ (367); JoinSystem.brokenΩ_eqOn (374); JoinSystem.integral_deriv_sub_sq_add (380); JoinSystem.tendsto_integral_brokenΩ (392); JoinSystem.continuous_brokenΩ (485); JoinSystem.toY (503); JoinSystem.interpΩ (509); JoinSystem.interpΩ_eqOn (515); JoinSystem.interpΩ_eq_star (519); JoinSystem.brokenΩ_toY_eqOn (526); JoinSystem.norm_pair_zero_sq_le (533); JoinSystem.intervalIntegrable_comp_deriv₄ (544); JoinSystem.continuous_interpΩ (574); JoinSystem.interpΩ_one (814); JoinSystem.edist_interpΩ_zero (823); JoinSystem.toY_brokenΩ (830); JoinSystem.isHomotopyEquiv_toY (840).

Morse/Chapter10/FullPathSpace.lean. dy (52); dseg (55); mem_Icc_dseg (94); Icc_dy_succ_subset (121); isHomotopyDirectLimit_preimage (139); pathCSet (180); PathC (184); PathSp.toC (191); PathSp.toPathC (195); toPathC (195); PathSp.edist_toPathC_le (202); edist_toPathC_le (202); PathSp.continuous_toPathC (218); continuous_toPathC (218); PathSp.hasDerivAt_of_notMem (232); PathSp.eventually_norm_sub_le_of_tendsto (241); PathSp.integral_inner_deriv_eq (256); PathSp.tendsto_integral_inner_deriv (290); PathSp.tendsto_integral_norm_deriv_sub_sq (379); PathSp.tendsto_of_tendsto_apply_of_integral (409).

Morse/Chapter10/FullPathTheorem.lean. pathIncl (71); JoinSystem.isHomotopyEquiv_toPathC (80); theorem_17_1 (110).

Morse/Chapter10/FundamentalTheorem.lean. PathSp.energyC (92); energyC (92); BrokenSetup.isRelCellComplexType_nodes (116); BrokenSetup.omegaEquivNodes (229); BrokenSetup.omegaEquivNodes_apply (234); BrokenSetup.isRelCellComplexType_Omega (240); BrokenSetup.finite_geodesics (261); BrokenSetup.eq_of_eqOn_path (319); geodesicIndex (331); geodesicIndex_eq (335); BrokenSetup.geodIndex_eq_geodesicIndex (340); theorem_17_3_of_levels (351); exists_mem_Ioo_succ_of_lt (431); finite_geodesicEnergies (446); theorem_17_3 (467); theorem_17_3_nonconj (549).

Morse/Chapter10/GoodBalls.lean. IsMinJoin (102); minJoin (110); minGeo (114); IsMinJoin.eq (120); minJoin_unique (135); isMinJoin_minJoin (141); minJoin_mem_tangentSubspace (148); isMinJoin_of_isShortJoin (161); minJoin_self (169); minGeo_zero (178); minGeo_self (183); hasDerivAt_minGeo_zero (188); IsMinJoin.mem_expDom (194); IsMinJoin.isGeodesicOn_minGeo (210); exists_secondFundForm_bound (229); convexOn_norm_sub_sq (261); dist_lt_of_isGeodesicOn (301); IsGoodBall (344); IsGoodBall.isMinJoin (357); IsGoodBall.join (367); IsGoodBall.exists_uniq (378); IsGoodBall.eq_minJoin (390); IsGoodBall.norm_minJoin (402); IsGoodBall.minGeo_one (407); IsGoodBall.isGeodesicOn_minGeo (413); IsGoodBall.eventually_minJoin_eq (421); IsGoodBall.continuousOn (435); IsGoodBall.continuousOn_minGeo (447); IsGoodBall.minJoin_minGeo (469); IsGoodBall.minGeo_minGeo (483); IsGoodBall.riemDist_minGeo (490); IsGoodBall.exists_smooth_minGeo (502); exists_isGoodBall_ball (554); exists_isGoodBall (611); exists_lebesgue_goodBall (621).

Morse/Chapter10/GoodJoin.lean. exists_smooth_glue (42); exists_isCompact_inter_closedBall (107); exists_goodSet (131); goodJoin (152); goodJoin_geo (192).

Morse/Chapter10/Incomplete.lean. zeroGraph (97); openSliceChart (117); isSliceSubmanifold_of_isOpen (128); tangentSubspace_of_isOpen (138); eq_add_smul_of_hasDerivAt (155); eqOn_affine_of_hasDerivAt (178); tanProj_eq_id_of_top (208); prj_eq_id_of_top (214); secondFundForm_eq_zero_of_top (221); curvR_eq_zero_of_top (231); IsGeodesicOn.deriv_deriv_eq_zero_of_top (241); IsGeodesicOn.eq_add_smul_of_top (250); IsUnbroken-

Geodesic.eqOn_segment_of_top (261); IsJacobiField.deriv_deriv_eq_zero_of_top (273); IsJacobiField.eqOn_affine_of_top (286); not_isConjugate_of_top (301); notMem_conjLocus_of_top (321); segment_mem_pathSpace (354); isUnbrokenGeodesic_segment (361); unique_geodesic_of_top (373); formIndex_energyHessian_eq_zero_of_top (384); hessNullity_eq_zero_of_top (396); puncturedSub (416); tangentSubspace_puncturedSub (421); curvR_puncturedSub (426); notMem_conjLocus_puncturedSub (430); formIndex_energyHessian_puncturedSub (434); eqOn_segment_puncturedSub (441); unique_geodesic_puncturedSub (449); neg_notMem_expDom_puncturedSub (459); not_isGeodesicallyComplete_puncturedSub (474); not_isClosed_punctured (481); isPathConnected_punctured (485); radialProj (492); sphIncl (507); radialCoeff_pos (512); puncturedHomotopyEquiv (525); remark_17_5_completeness (567).

Morse/Chapter10/MilnorH.lean. riemDist_real (96); exhaust (113); epsTwo (116); eps (119); modulus (122); milnorF (126); twoA (130); milnorA (133); milnorK (136); eps_eq (138); bddAbove_modulus (140); le_modulus (149); modulus_le (153); modulus_nonneg (157); modulus_mono (160); modulus_zero (164); modulus_add_le (170); abs_modulus_sub_le (192); exists_modulus_le (204); continuousOn_modulus (213); milnorF_lt (225); one_mem_twoA_set (231); twoA_eq (237); twoA_spec (247); milnorA_pos (261); milnorK_mul_le (264); seg (277); lastSeg (281); plin (286); plin_eq_seg (294); plin_eq_last (320); breaks (328); plin_piece (331); plin_isPiecewiseSmooth (358); plin_zero (362); plin_one (369); plin_node (402); deriv_deriv_plin_seg (410); deriv_deriv_plin_last (417); integral_seg (423); integral_last (433); energy_plin (446); hPath (472); hPath_node (475); hPath_one (480); deriv_deriv_hPath (486); hPath_mem (493); hMap (502); g (508); c_0 (511); s_0 (514); lamStar (517); s_0_lt_half (538); lamStar_mul_c_0 (542); lamStar_sq (546); g_one_sub (551); g_zero (553); g_one (555); g_nonneg (558); g_le (567); g_one_sub_s_0 (578); omegaFam (585); omegaFam_apply (592); dist_omegaFam_le (600); continuous_omegaFam (607); tendsto_omegaFam (614); modulus_omegaFam (618); twoA_omegaFam (633); milnorA_omegaFam (640); lam_mul_c_0_mem (644); milnorK_omegaFam (651); milnorA_lamStar (662); one_div_milnorA_lamStar (667); milnorK_lamStar (671); energy_hPath_lamStar (676); energy_hPath_lamStar_le (687); delta (697); delta_pos (699); le_energy_hPath (707); tendsto_delta_cbrt (745); tendsto_energy_hPath (760); MilnorH.lipschitzWith_sqrt_energy (770); tendsto_edist_hMap (775); milnor_h_not_continuous (796); milnor_h_not_continuous' (807).

Morse/Chapter10/Omega.lean. PathSp.edist_eq_zero_of_eqOn (65); edist_eq_zero_of_eqOn (65); BrokenSetup.tendsto_path_apply (78); BrokenSetup.pathP (99); BrokenSetup.continuous_pathP (102); BrokenSetup.continuous_nodeVal (138); BrokenSetup.nodeMap (145); BrokenSetup.pathMap (153); BrokenSetup.homotopyEquivNodes (162); BrokenSetup.theorem_16_3_Omega (192); BrokenSetup.theorem_16_3_Omega_count (212).

Morse/Chapter10/PathHomology.lean. BrokenSetup.finrank_subsetHomology_Omega (79); windowLevels (224); lt_windowLevels_succ (239); windowLevels_succ_lt (246); mem_window_iff (254); windowLevels_monotone (277); tendsto_windowLevels (286); List.eq_singleton_of_pairwise_not (296); oneCellEach (308); omega_cells_of_unique (316); finrank_window (368); bettiNumber_omega_of_unique (456).

Morse/Chapter10/PathMetric.lean. integral_mul_sq_le (58); sqrt_integral_add_sq_le (80); IsPiecewiseSmooth.exists_deriv_piece (110); intervalIntegrable_comp_deriv (119); intervalIntegrable_comp_deriv$_3$ (134); IsPiecewiseSmooth.exists_bound_deriv (153); PathSp (185); PathSp.mem (193); PathSp.speed (196); PathSp.supDist (200); supDist (200); PathSp.l2Dist (203); l2Dist (203); PathSp.l2Dist_nonneg (205); PathSp.l2Dist_comm (207); PathSp.l2Dist_self (210); PathSp.intervalIntegrable_speed (212); PathSp.l2Dist_triangle (221); PathSp.supDist_self (237); PathSp.supDist_comm (242); PathSp.supDist_triangle (245); PathSp.riemDist_le_supDist (251); PathSp.instPseudoEMetricSpace (257); instPseudoEMetricSpace (257); PathSp.edist_def (270); PathSp.ofReal_l2Dist_le_edist (273); PathSp.supDist_le_edist (276); PathSp.energy_eq_integral_speed (280); PathSp.abs_sqrt_energy_sub_le (285); PathSp.lipschitzWith_sqrt_energy (315); lipschitzWith_sqrt_energy (315); PathSp.energy_nonneg_of_le (322); PathSp.continuous_energy (326); continuous_energy (326); PathSp.isClosed_sublevel (333); PathSp.isOpen_sublevel_lt (337); PathSp.energy_le_of_bound

(342); PathSp.exists_energy_lt_of_tendsto (355).

Morse/Chapter10/Product.lean. SliceChart.pi (43); SliceChart.pi_source (88); sliceParam_pi (95); isSliceSubmanifold_pi (99); SliceChart.inter (108); sliceParam_inter (124); IsSliceSubmanifold.inter_isOpen (129); isSliceSubmanifold_pi_inter (135); piInterChart (141); mem_piInterChart_source (147); tangentSubspace_pi_inter (157); contMDiffAt_of_locally_eq (207).

Morse/Chapter10/RelativeCells.lean. inclC (72); IsRelCellAttachment.transport (83); IsRelCellComplexType (100); IsRelCellComplexType.of_homotopyEquiv (111); IsRelCellComplexType.congr (116); IsRelCellComplexType.of_eq (122); IsRelCellComplexType.comp_homotopyEquiv (127); IsRelCellComplexType.of_square (140); IsRelCellComplexType.attach (167); IsRelCellComplexType.mono (182); isRelCellComplexType_union_cells (194); IsRelCellComplexType.sublevel_step_of_isCompact (266); isRelCellComplexType_sublevel_of_isCompact (344); TowerStage (505); TowerStep (511); nonempty_towerStep (520); towerStages (535); towerStepAt (543); isCountableCellComplexType_of_rel (553); isCountableCellComplexType_of_relSublevel (573).

Morse/Chapter10/Retraction.lean. energy_congr_Icc (71); riemDist_sq_le_energy (81); BrokenSetup.sgeoOver (97); BrokenSetup.sgeoOver_left (99); BrokenSetup.sgeoOver_right (109); BrokenSetup.sgeoOver_isGeodesicOn (115); BrokenSetup.energy_sgeoOver (154); BrokenSetup.retrSeg (173); BrokenSetup.retr (178); BrokenSetup.retrSeg_eqOn_left (182); BrokenSetup.path_t (189); BrokenSetup.energy_path_seg (197); BrokenSetup.riemDist_lt_of_seg (219); BrokenSetup.nodeVal_mem (244); BrokenSetup.path_nodeVal_t (247); BrokenSetup.u_mem (254); BrokenSetup.retrSeg_mem_pathSpace (257); BrokenSetup.retrSeg_eqOn_mid (311); BrokenSetup.retrSeg_eqOn_right (322); BrokenSetup.energy_retrSeg_mid_le (338); BrokenSetup.energy_retrSeg_le (353); BrokenSetup.path_nodeVal_eq_sgeoOver (402); BrokenSetup.retrSeg_eq_retr (412); BrokenSetup.retr_mem_pathSpace (443); BrokenSetup.energy_retr_le (447); BrokenSetup.retr_zero (458); BrokenSetup.retr_one (471); BrokenSetup.retr_of_mem_brokenPaths (488).

Morse/Chapter10/RetractionCont.lean. mem_expDom_smul (45); tendsto_of_tendsto_riemDist (56); BrokenSetup.joinable (76); BrokenSetup.continuousWithinAt_joinVec (83); BrokenSetup.continuousWithinAt_shortGeo (102); PathSp.riemDist_le_sqrt (133); PathSp.tendsto_riemDist_apply (158); PathSp.eventually_energy_lt (183); PathSp.speed_congr (219); PathSp.integral_speed_sub_sq_le (228); PathSp.integral_speed_sub_sq_le_energy (240); BrokenSetup.integral_speed_path_sub_sq (267); BrokenSetup.tendsto_integral_speed_path (298).

Morse/Chapter10/RetractionConv.lean. PathSp.eventually_supDist_le_of_tendsto_apply (56); BrokenSetup.tendsto_sgeoOver (123); BrokenSetup.mem_joinable (149); BrokenSetup.speed_retrSeg_mid (159); BrokenSetup.energy_retrSeg_mid (169); BrokenSetup.tendsto_l2Dist (186); BrokenSetup.tendsto_apply (195); BrokenSetup.tendsto_nodeVal (202); BrokenSetup.tendsto_retrSeg_apply (221); BrokenSetup.eventually_supDist_retrSeg_le (294); BrokenSetup.eventually_integral_retrSeg_le (305); BrokenSetup.tendsto_retrSeg (482).

Morse/Chapter10/SphereGeodesics.lean. formIndex_energyHessian_congr (109); greatCircle (134); hasDerivAt_greatCircle (144); deriv_deriv_greatCircle (160); isGeodesicOn_greatCircle (217); energy_greatCircle (239); exists_unit_orthogonal (250); exists_eqOn_greatCircle (266); finrank_orthogonal_span_pair (334); isJacobiField_sin_smul (360); isJacobiField_smul_deriv_greatCircle (392); conjMultiplicity_greatCircle (408); sphAngle (554); sphDir (557); sphLen (561); sphGeod (567); sphAngle_pos (586); sphAngle_lt_pi (589); inner_sphDir (617); norm_sphDir (622); eq_sphDir (633); sphGeod_one (644); sphGeod_mem_pathSpace (657); isUnbrokenGeodesic_sphGeod (664); mul_pi_lt_sphLen (670); sphLen_lt (678); sphLen_pos (685); sphLen_strictMono (689); energy_sphGeod (696); exists_eqOn_sphGeod (704); finsum_mem_Ioo_ite_eq (764); conjMultiplicity_sphGeod (826); formIndex_sphGeod (836); formIndex_eq_of_eqOn_sphGeod (847); notMem_conjLocus_sph (858).

Morse/Chapter10/SpherePaths.lean. strictMono_energy_sphGeod (63); tendsto_energy_sphGeod (72); corollary_17_4 (83); bettiNumber_sphere_paths (96).

Morse/Chapter10/SublevelCW.lean. exists_contDiff_eq_of_locally (54); IsFiniteCellComplex-Type.sublevel_step_of_isCompact (95); isFiniteCellComplexType_sublevel_of_isCompact (170).

Morse/Chapter10/SublevelHomology.lean. LongExactSeq.bijective_f_of_subsingleton (110); LongExactSeq.subsingleton_B (118); exactAt_of_injective_of_exists (141); subsingle-ton_subsetHomology_iff_exactAt (172); subsingleton_subsetHomology_of_exhaustion (183); subsingleton_subsetHomology_univ_of_exhaustion (229); subsingleton_subsetHomology_of_steps (239); subsetHomologyEquivOfExhaustion (258); spaceHomologyEquivOfExhaustion (271); nonempty_linearEquiv_spaceHomology_of_exhaustion (282); spaceHomologyEquivOfExhaustion' (291); bettiNumber_eq_of_exhaustion (300); finiteDimensional_spaceHomology_of_exhaustion (309); spaceHomologyEquivOfSteps (320); bettiNumber_eq_of_steps (330); ex-ists_subset_sublevel_of_isCompact (339); Homology.sublevel_step_critical_of_isCompact (366); Homology.sublevel_step_of_isCompact (465).

Morse/Chapter11/CartanHadamard.lean. injective_dExpAt_of_nonpos' (121); isCov-eringMap_expToN_of_nonpos (126); injective_expToN_of_simplyConnected (133); sur-jective_expToN (154); ray_mem_pathSpace (162); exists_eqOn_ray (172); isPrecon-nected_of_simplyConnected (185); isGeodesicallyComplete_of_isClosed (190); expDiffeo-morph (196); coe_expDiffeomorph (205); theorem_19_2 (214); contractibleSpace_cube (254); isPreconnected_cube_face (264); genLoop_homotopic_const (286); corollary_19_3 (325); omega_cells_of_nonpos (340); nonempty_homotopyEquiv_pathC_fiber (364); geodesic_iff_mem_fiber (453); omega_homotopyEquiv_fiber (480); contractibleSpace_pathC (493); contractibleSpace_pathSp (547); omega_single_cell (561); pathOfMem (603); homotopy_class_unique_geodesic (615).

Morse/Chapter11/Curvature.lean. hasDerivAt_inner_covAlong (95); continuousOn_covAlong (133); NonposCurvature (144); nonposCurvature_iff_sectionalCurvature (150); IsJacob-iField.eqOn_zero_of_nonpos (179); lemma_19_1 (234); formIndex_eq_zero_of_nonpos (246); energyHessian_pos_of_nonpos (267); injective_dExpAt_of_nonpos (300).

Morse/Chapter11/ExpCovering.lean. contDiffAt_expAt (79); norm_fderiv_expAt_apply (85); gauss_lemma_expDomAt (103); coe_expToN_of_mem (224); contMDiffAt_expToN (230); isLo-calDiffeomorphAt_expToN (240); isLocalDiffeomorph_expToN (287); hasDerivAt_expLift (297); norm_fderiv_expAt_self (332); norm_expLift_le (343); continuous_expAt (403); exists_expLift (412); expLift_unique (525); isCoveringMap_expToN (547).

Morse/Chapter11/ExpSmooth.lean. partialT (89); hasDerivAt_partialT (91); contDif-fOn_partialT (97); raysE (112); RaysGoodE (117); RaysGoodE.mono (127); raysGoodE_zero (136); deriv_raysE_eq (167); RaysGoodE.step (174); raysGoodE_of_isGeodesicOn (280); ex-ists_retraction (328); expDom_subset_tangentBundleSet (344); mem_expDom_of_isGeodesicOn (354); exists_contDiffOn_exp (365); exists_isOpen_expDom (406); contMDiffOn_exp_expDom (418); expDomAt (441); zero_mem_expDomAt (444); smul_mem_expDomAt (448); exists_contDiffAt_expAt (455); isOpen_expDomAt (467); contDiffOn_expAt (473); isGeodesicOn_ray (481); isSmoothCurveIn_ray (488); expJacobi (500); expJacobi_apply_of_mem (513); dExpAt (521); dExpAt_apply (525); expJacobi_zero (529); expJacobi_one (533); isJacobiField_expJacobi (539); covAlong_expJacobi_zero (595); expJa-cobi_mem_jacobiSpace (619); expJacobi_injective (625); eq_expJacobi (635); conjSpace_eq_map_ker (658); theorem_18_1_multiplicity (677); theorem_18_1 (686).

Morse/Chapter11/LongGeodesics.lean. integral_eq_of_eqOn_pieces (78); integral_sin_sq_shift (98); integral_cos_sq_shift (110); sinBump (126); sinBump_of_mem (134); sinBump_of_le (139); sin-Bump_of_ge (144); sinBump_eventually_zero (149); sinBump_eventuallyEq (156); curvOp_zero_right (163); sinBump_mem_pathTangent (172); covD_sinBump_of_notMem (201); energyHessian_sinBump (211); energyHessian_sinBump_disjoint (256); exists_sinBump_neg (295); theorem_19_6_index (378); path-Length_le_of_formIndex_lt (456); le_finsum_conjMultiplicity_of_ricci (465).

Morse/Chapter11/Myers.lean. ricci (98); ricci_eq_sum (103); ricci_comm (113); curvOp_self (123); ricci_eq_sum_erase (130); exists_smooth_clamp (141); integral_sin_sq_pi_mul (188); integral_cos_sq_pi_mul (197); energyHessian_sin_parallel (212); RicciLowerBound (258); exists_smooth_extension (266); exists_parallel_frame (328); continuous_inner_curvR (394); exists_energyHessian_neg (410); theorem_19_4 (486); IsMinimalGeodesic.isMinimalPath (520); corollary_19_5 (533); ricci_sph (564); ricciLowerBound_sph (586); sphere_myers (595).

Morse/Chapter11/NonConjugate.lean. conjLocus (76); IsConjugate.congr_curve (82); IsConjugate.congr_curve_iff (89); theorem_18_1_of_geodesic (97); conjLocus_eq (118); corollary_18_2 (140); dense_compl_conjLocus (200); hessNullity_eq_zero_of_notMem_conjLocus (221); expDomAt_eq_univ (231); conjLocus_eq_of_complete (236).

Morse/Chapter11/PositiveRicci.lean. ricciMap (74); ricciMap_apply (80); ricci_eq_trace_ricciMap (86); continuous_trace_ricciMap (109); exists_ricciLowerBound (159); energy_eq_pathLength_sq_of_geodesic (200); energy_le_of_geodesicIndex_le (211); theorem_19_6_of_ricciLowerBound (226); theorem_19_6 (258).

Morse/Chapter11/PositiveRicciAll.lean. theorem_19_6_all (52); finiteDimensional_spaceHomology_omega_all (63).

Morse/Chapter11/RicciHomology.lean. slabEnergies (61); slab_homology (68); finiteDimensional_spaceHomology_omega (170); finiteDimensional_spaceHomology_omega_of_compact (244).

Morse/Chapter12/BiInvariant.lean. even_finrank_of_comp_self_eq_neg_id (122); IsAdInvariant (136); isAdInvariant_iff_lieInvariant (138); IsLeviCivita (152); IsLeviCivita.conn_eq (160); IsLeviCivita.isAdInvariant (174); IsLeviCivita.koszul (183); IsLeviCivita.conn_eq_of_isAdInvariant (199); halfLie (214); halfLie_apply (216); isLeviCivita_halfLie (220); curv (233); curv_halfLie (236); curv_inner (243); sectional (248); ad_skew (252); jacobiOp (259); jacobiOp_apply (262); curv_halfLie_eq_jacobiOp (266); jacobiOp_symm (270); jacobiOp_self (277); jacobiOp_self_nonneg (289); sectional_nonneg (296); sectional_std (303); sectional_eq_zero_iff (309); jacobiOp_eq_zero_iff (320); eigenvalue_nonneg (332); ad_eigenvalue_eq_zero (342); jacobiOp_of_lie_lie (355); lie_lie_of_jacobiOp (361); lie_mem_eigenspace (370); eigenJ (379); eigenJ_apply (385); eigenJ_comp_self (389); even_finrank_eigenspace (407); eigenJ_isometry (425); trace_jacobiOp (442); trace_eq_sum_of_isOrtho (450); trace_jacobiOp_pos (462); killingForm_neg (498); centerCompl (508); mem_centerCompl (512); isCompl_center_centerCompl (517); eq_zero_of_mem_centerCompl (532); cartan_form_swap$_{12}$ (550); cartan_form_swap$_{23}$ (554).

Morse/Chapter12/GroupGeodesics.lean. trace_gauss_identity (84); conjTranspose_anticomm (98); conjTranspose_lie_lie (105); inner_gauss_identity (114); inner_lie_left (123); algProj (135); algProj_mem (138); algProj_of_mem (142); algProj_hermitian (147); algProj_add_hermitian (155); algProj_mul (160); frobForm (175); frobForm_apply (182); isAdInvariant_frobForm (187); frobForm_pos (194); deriv_deriv_mul_exp_smul (212); IsMatrixGroup.tanProj_eq (231); IsMatrixGroup.prj_eq (244); IsMatrixGroup.isGeodesicOn_mul_exp_smul (251); IsMatrixGroup.isGeodesicOn_exp_smul (273); IsMatrixGroup.eqOn_mul_exp_smul (280); IsMatrixGroup.lemma_21_2 (310); IsMatrixGroup.eq_exp_of_oneParam (324); IsMatrixGroup.isGeodesicOn_of_oneParam (361); IsMatrixGroup.exp_eq (378); IsMatrixGroup.isGeodesicallyComplete (390); IsMatrixGroup.symmIso (406); IsMatrixGroup.symmIso_apply (430); IsMatrixGroup.symmIso_mem (435); IsMatrixGroup.symmIso_self (440); IsMatrixGroup.symmIso_mul_exp_smul (446); IsMatrixGroup.secondFundForm_mul_self (457); IsMatrixGroup.secondFundForm_mul (469); IsMatrixGroup.curvOp_mul (496); IsMatrixGroup.sectional_mul (523); IsMatrixGroup.sectional_mul_nonneg (534); IsMatrixGroup.covAt_mul_right (544); IsMatrixGroup.curvOp_eq_curv_halfLie (560); IsMatrixGroup.coe_curvTensor_eq_curv_halfLie (569); IsMatrixGroup.inner_curvOp_mul (577); IsMatrixGroup.curvOp_eq_jacobiOp (585).

Morse/Chapter12/GroupJacobi.lean. halfExp (62); halfExp_mul_halfExp (64); halfExp_zero (70); hasDerivAt_halfExp (73); halfExp_mul_comm (77); deriv_exp_smul (87); hasDerivAt_conj (92); de-

riv_mem_of_forall_mem (107); jacK (119); smul_mul_eq_mul_jacK (123); IsMatrixGroup.halfExp_mem (140); IsMatrixGroup.conjTranspose_halfExp_mul_comm (144); IsMatrixGroup.covD_conj (158); IsMatrixGroup.jacK_conj (187); IsMatrixGroup.curvR_conj (199); IsMatrixGroup.isJacobiField_conj (223); IsMatrixGroup.covAlong_conj_zero (261).

Morse/Chapter12/MatrixGroups.lean. skewHermitian (87); hermitian (102); specialSkewHermitian (113); mem_skewHermitian (132); mem_specialSkewHermitian (140); trace_eq_zero_of_skew_real (148); hermPart (154); ker_hermPart (188); range_hermPart (194); finrank_skewHermitian_add_finrank_hermitian (200); IsMatrixGroup (224); IsMatrixGroup.one_mem (243); IsMatrixGroup.mul_exp_smul_mem (250); IsMatrixGroup.conj_mem' (255); IsMatrixGroup.leftAlg (260); IsMatrixGroup.mem_leftAlg_iff (264); IsMatrixGroup.finrank_leftAlg (277); IsMatrixGroup.tangentSubspace_eq (286); IsMatrixGroup.mem_tangentSubspace_iff (312); IsMatrixGroup.mem_tangentSubspace_iff' (322); IsMatrixGroup.mulLeftIso_image (336); IsMatrixGroup.mulRightIso_image (349); unitaryMap (376); unitaryMap_eq_zero_iff (381); preimage_unitaryMap (385); hasFDerivAt_unitaryMap (409); contDiff_unitaryMap (421); isRegularValue_unitaryMap (428); isSliceSubmanifold_unitaryGroup (451); isMatrixGroup_unitaryGroup (461); norm_sq_of_conjTranspose_mul_self (477); isClosed_unitaryGroup (482); isCompact_unitaryGroup (489); isClosed_det_eq_one (497); det_sq_of_orthogonal (514); specialOrthogonalGroup_inter (521); isSliceSubmanifold_specialOrthogonalGroup (536); isMatrixGroup_specialOrthogonalGroup (546); isCompact_specialOrthogonalGroup (567); suMap (590); normSq_det_of_unitary (593); mem_specialUnitaryGroup_iff_level (605); suPart (623); im_trace_of_hermitian (631); re_trace_of_skew (638); ker_suPart (644); range_suPart (655); finrank_specialSkewHermitian_add (675); hasFDerivAt_suMap (683); isRegularValue_suMap (696); isSliceSubmanifold_specialUnitaryGroup (741); isMatrixGroup_specialUnitaryGroup (758); isCompact_specialUnitaryGroup (782).

Morse/Chapter12/MatrixInner.lean. MatrixInner.normedAddCommGroup (78); MatrixInner.norm_def (87); MatrixInner.normedSpace (92); MatrixInner.norm_sq_eq (98); MatrixInner.re_trace_conjTranspose_mul_self (102); MatrixInner.innerProductSpace (113); MatrixInner.normedRing (148); MatrixInner.normedAlgebra (157); inner_def (175); inner_conjTranspose (177); inner_eq_zero_of_hermitian_of_skew (183); inner_mul_left_of_unitary (192); inner_mul_right_of_unitary (198); mul_conjTranspose_eq_one (203); mulLeftIso (208); mulRightIso (227); contDiff_exp_matrix (246); contDiff_exp_smul (252); exp_smul_mul_exp_smul (257); exp_half_mul_exp_half (263); exp_smul_mul_self (269); exp_conjTranspose_mul_exp_of_skew (274); exp_mul_exp_neg (280); contDiff_entry (288); contDiff_det (294); hasDerivAt_det_one_add_smul (305); fderiv_det_mul (325); hasDerivAt_det_exp_smul (342); det_exp_smul_of_trace_eq_zero (351).

Morse/Chapter12/SUIndex.lean. hadSin_add (72); hadSin_smul (77); hadKer (84); mem_hadKer (92); conjEquiv (96); conjRange (110); mem_conjRange (114); finrank_conjRange (117); jacSin_eq_zero_iff_mem_conjParams (130); hadCoord (157); hadKer_entry (172); hadKer_skew (177); hadCoord_injective (183); hadCoord_surjective (211); finrank_hadKer (262); conjMultiplicity_su (277); conj_suDiag_mem (361); exp_conj_suDiag (375); exists_unitary_suDiag (389); mem_pathSpace_exp (440); isUnbrokenGeodesic_exp (448); support_conjMult_subset (458); formIndex_su_geodesic (472); index_su_eq_bottIndex (493); lemma_23_2_index (508).

Morse/Chapter12/SUJacobi.lean. jacSin (50); jacSin_zero (57); jacSin_neg (60); hasDerivAt_jacSin (68); hasDerivAt_cos_mul_jacSin (82); contDiff_jacSin (93); jacSin_eq_zero_iff (104); hadSin (116); hadCos (120); hadSin2 (124); of_mul_eq_sum (140); hasDerivAt_hadSin (147); hasDerivAt_hadCos (158); hadSin_zero (183); hadCos_zero (187); diagI (193); jacK_diagI_apply (197); jacK_diagI_hadSin (204); jacK_conj_unitary (212); suDiag_eq_diagI (221); suSol (230); deriv_deriv_suSol (245); suSol_ode (258); suSol_zero (263); deriv_suSol_zero (266); conj_mem_specialSkewHermitian (277); conj_mem_specialSkewHermitian' (285); hadSin_mem (292); suSol_mem (308); suJac (316); suJac_zero (319); isJacobiField_suJac (326); covAlong_suJac_zero (335); suJac_eq_zero_iff (346).

Morse/Chapter12/SUMinimal.lean. norm_suDiag_sq (65); norm_conj_unitary (71);

norm_conj_suDiag_sq (80); pathLength_exp_smul (85); kHalf (99); kHalf_pm (103); sum_kHalf (108); odd_kHalf (121); pi_sqrt_card_le (130); norm_eq_pi_sqrt (137); pi_sqrt_card_le_norm (144); riemDist_one_neg_one (151); sq_conj_suDiag_iff (201); isMinimalGeodesic_exp_iff (240); minVel (302); grassmannian (307); exp_eq_neg_one_of_sq (312); mem_minVel_iff (367); projOf_mem (379); velOf_mem (397); lemma_23_1 (419).

Morse/Chapter13/Componentwise.lean. PathSp.mem_of_edist_eq_zero (59); mem_of_isClopen_of_continuous (65); theorem_22_1_clopen (71); clopenIncl (280); isEmbedding_clopenIncl (283); compresses_clopenIncl (294); bijective_clopenIncl (311); bijective_val_of_pushesInto (321); pushesInto_cube_clopen (357); pushesInto_cyl_clopen (362); theorem_22_1_clopen_bijective (369); theorem_22_1_clopen_bijective_minPathsIncl (377).

Morse/Chapter13/Compression.lean. PushesInto (68); pushesInto_self (73); PushesInto.mono (76); PushesInto.trans (80); pushesInto_of_isStrongDeformationRetract (94); pushesInto_union_cell (131); pushesInto_union_cells (161); exists_gap_of_finite (216); lemma_D (232).

Morse/Chapter13/Corollaries.lean. pathCHomeomorph (55); contractLoop (76); continuous_contractLoop (82); contractLoop_zero (87); contractLoop_one (91); contractPath (105); corollary_22_2 (116).

Morse/Chapter13/CorollaryGroups.lean. nonempty_mulEquiv_pathSp (54); corollary_22_2_mulEquiv (74); corollary_22_2_surjective_hom (91); corollary_22_3_mulEquiv (113); corollary_22_3_surjective_hom (127).

Morse/Chapter13/Freudenthal.lean. corollary_22_2_surjective (64); eqComp (100); eqComp_mem_equator (106); eqComp_of_mem_equator (114); sphNbhd (120); isOpen_sphNbhd (123); sphRetrPath (129); continuousOn_sphRetrPath (133); mem_sphNbhd_and_eqOn (147); exists_retraction_minPaths_sph (172); equatorHomeomorph (198); hidx_sph (222); hball_sph (228); corollary_22_3 (234); corollary_22_3_surjective (246).

Morse/Chapter13/ManifoldOfMinima.lean. pushesInto_of_deformation (67); criticalPoint_hessianM_congr (88); lemma_22_5_sublevel (120); lemma_22_5 (319).

Morse/Chapter13/MinimalDeform.lean. isUnbrokenGeodesic_of_minimal (53); BrokenSetup.mem_ball_of_mem_nodes (74); BrokenSetup.joinVec_self (80); BrokenSetup.exists_pushesInto_nodes (90).

Morse/Chapter13/Perturbation.lean. exists_negDef_margin (68); le_index_of_norm_sub_lt (109); lemma_22_4 (125); fderiv_hessian_add (194); lemma_22_4_add (215); exists_perturb_slice (245); lemma_22_4_submanifold (333).

Morse/Chapter13/SUInstance.lean. eqOn_exp_smul_of_isUnbrokenGeodesic_su (50); energy_exp_smul (83); le_formIndex_of_not_minimal_su (96); neg_one_mem_su (130); corollary_22_2_su (139).

Morse/Chapter13/SphereMinimal.lean. finsum_mem_Ioo_ite_eq' (69); isClosed_sph (122); greatCircle_mem_pathSpace_antipode (133); isUnbrokenGeodesic_greatCircle (143); exists_eqOn_greatCircle_antipode (151); pi_sq_le_energy_sph (177); formIndex_greatCircle_antipode (194); le_formIndex_of_not_minimal_sph (211); equator (233); semicircle (241); energy_semicircle (244); semicircle_mem_minPaths (249); speed_greatCircle (253); continuous_semicircle (262); exists_eqOn_semicircle (287); greatCircle_pi_half (307); continuous_apply_half_sph (312); apply_half_mem_equator (323); minPathsToEquator (330); minPathsToEquator_equatorToMinPaths (334); edist_equatorToMinPaths_minPathsToEquator (338); sphMinPathsHomotopyEquiv (350); sphMinPathsHomeomorph (374).

Morse/Chapter13/Theorem22.lean. minPaths (72); theorem_22_1 (76); minPathsIncl (244); compresses_minPaths (250); theorem_22_1_bijective (270); theorem_22_1_surjective (291).

Morse/Chapter14/CircleGroup.lean. Homotopy.intAddEquivZMultiplesTwoPi (49); Homotopy.piOneCircleMulEquiv (56); Homotopy.piMulEquivOfHomeomorph (64); Homotopy.nonempty_mulEquiv_pi_one_unitary_circle (71); theorem_23_5_mulEquiv (87).

Morse/Chapter14/IndexCount.lean. conjParams (143); conjParams_subset (146); conjParams_finite (161); conjParams_sq (165); ncard_conjParams_sq (196); conjMult (212); spectralIndex (217); spectralIndex_eq (221); card_lt_le_spectralIndex (247); spectralIndex_double (265); even_spectralIndex (272); bottIndex (294); two_le_sub_of_odd (298); bottIndex_term_nonneg (304); bottIndex_nonneg (311); bottIndex_neg (315); bottIndex_comp_equiv (323); bottIndex_eq_sum_subtype (330); two_mul_ncard_conjParams (340); pairSpectrum (356); two_mul_spectralIndex_pairSpectrum (359); suSpectrum (366); spectralIndex_suSpectrum (371); card_le_sum_sq (389); sum_sq_eq_card_iff (395); card_eq_of_minimal (404); bottIndex_eq_zero_of_minimal (422); sum_pos_neg_le_bottIndex (432); sum_pos_neg_eq (449); exists_pos_neg_data (459); bottIndex_ge (507); remarkK (541); odd_remarkK (543); bottIndex_remarkK (547); not_minimal_remarkK (552); sum_remarkK (557); abs_sum_le_card_of_minimal (564); card_sub_abs_sum_le_two_mul_bottIndex (574); card_sub_abs_sum_le_four_mul_spectralIndex (596); bottIndex0 (613); oSpectrum (618); spectralIndex_oSpectrum (624); bottIndex0_eq_zero (645); bottIndex0_ge (650); card_sub_one_le_spectralIndex (693).

Morse/Chapter14/MinimalPaths.lean. exp_half_smul_of_mem_minVel (63); continuous_matrix_exp (100); norm_of_mem_minVel (107); speed_exp_smul (121); pi_sq_le_energy (129); expPath (142); expPath_apply (146); energy_expPath (149); expPath_mem_minPaths (155); continuous_expPath (162); suDiag_kHalf_mem_minVel (187); exists_minVel_of_mem_minPaths (195); pi_smul_apply_half (223); continuous_apply_half (231); minVelToPaths (238); pathsToMinVel (243); pathsToMinVel_minVelToPaths (253); edist_minVelToPaths_pathsToMinVel (258); minPathsHomotopyEquiv (270); minVelHomeomorph (295).

Morse/Chapter14/MinimalRetract.lean. halfInd (67); specProj (70); continuousOn_halfInd (72); continuousOn_specProj (89); specProj_isSelfAdjoint (102); specProj_mul_self (105); specProj_eq_self (117); half_notMem_spectrum_of_idem (123); isOpen_half_notMem_spectrum (127); exists_trace_eq_natCast (145); hermOf (165); hermOf_isSelfAdjoint (169); continuous_hermOf (176); hermOf_of_mem_minVel (182); minProj (199); minNbhd (206); continuousOn_minProj (211); isOpen_minNbhd (218); minProj_mem (224); retrPath (244); continuousOn_retrPath (250); retrPath_mem_minPaths (262); mem_minNbhd_and_eqOn (269); exists_retraction_minPaths_su (302); corollary_22_2_su' (328).

Morse/Chapter14/Periodicity.lean. theorem_23_3 (62); card_sum_fin (73); nonempty_equiv_pi_unitary_two_mul (79); periodicity_theorem (92); Homotopy.zerothHomotopyEquivOfDiscrete (110); Homotopy.subsingleton_pi_zero (121); Homotopy.nonempty_pi_one_circle_equiv_int (129); Homotopy.unitaryOneHomeomorph (140); Homotopy.unitaryOneHomeomorph_one (151); Homotopy.nonempty_pi_unitary_equiv_circle (155); theorem_23_5 (170).

Morse/Chapter14/PeriodicityGroups.lean. corollary_22_2_su_mulEquiv (55); theorem_23_3_mulEquiv (69); nonempty_mulEquiv_pi_unitary_two_mul (86); periodicity_theorem_mulEquiv (105); nonempty_mulEquiv_pi_unitary_odd (119); nonempty_mulEquiv_pi_unitary_even (130).

Morse/Chapter14/SUMatrices.lean. suDiag (73); jacobiMat (76); commutator_diagonal_apply (81); jacobiMat_suDiag_apply (87); eReal (95); eImag (98); jacobiMat_eReal (101); jacobiMat_eImag (118); jacobiMat_diagonal (135); eReal_conjTranspose (143); eImag_conjTranspose (147); trace_eReal (151); trace_eImag (154); eq_diagonal_add_sum_eReal_eImag (160); trace_suDiag_mul_conjTranspose (232); trace_suDiag (242); exp_ofReal_mul_I_eq_neg_one_iff (246); exp_diagonal_mul_I (267); exp_diagonal_eq_neg_one_iff (277); sq_eq_neg_pi_sq_iff (285); trace_pi_smul_projection (311).

Morse/Chapter15/BottChain.lean. bottJ (46); isAnticommFamily_bottJ (49); card_bottJ (52); trace_bott16_two (55); trace_bott16_six (60); omegaPhi_bottJ_mem_omegaComp (67); bott_chain (90).

Morse/Chapter15/Clifford.lean. σ₂ (73); τ₂ (75); κ₂ (77); σ₂_mul_σ₂ (79); τ₂_mul_τ₂ (81); κ₂_mul_κ₂ (83); τ₂_mul_σ₂ (85); κ₂_mul_σ₂ (87); κ₂_mul_τ₂ (89); σ₂_transpose (91); τ₂_transpose (93); κ₂_transpose (95); neg_kronecker' (98); kronecker_neg' (100); transpose_kronecker' (102); bott2 (114); bott4 (119); bott8 (129); bott16 (144); bott16_eq_kronecker (155); isAnticommFamily_bott2 (158); isAnticommFamily_bott4 (164); isAnticommFamily_bott8 (172); isAnticommFamily_bott16 (181); IsAnticommFamily.mono (195); IsAnticommFamily.isClifford (200); IsAnticommFamily.of_ringHom (205); IsAnticommFamily.reindex (216); IsAnticommFamily.blockDiagonal (222); isAnticommFamily_bott16r (234); IsAnticommFamily.fromBlocks (241); fromBlocks_mem_omega (256); milnorM (285); exists_isAnticommFamily_fin (297); dotProduct_mulVec_eq (320); dotProduct_mulVec_self_eq_zero (325); IsComplexStructure.dotProduct_mulVec_self (333); IsComplexStructure.dotProduct_mulVec_mulVec (338); dotProduct_mulVec_of_anticomm (343); IsAnticommFamily.dotProduct_mulVec (353); bott16r_vectorFields (362).

Morse/Chapter15/CliffordModules.lean. two_dvd_finrank_of_sq (95); quatBasis (109); four_dvd_finrank_of_anticomm (120); mem_ker_sub_one_iff (137); finrank_eq_two_mul_of_anticomm (143); injective_of_mul_self_eq_neg_one (190); mapsTo_ker_of_commute (198); finrank_submodule_eq_two_mul (206); mapsTo_inf (231); restrict_mul_self (236); restrict_anticomm (243); eight_dvd_finrank (251); eigV₁ (279); eigW (282); eigX (285); jointEigen (288); finrank_eq_two_mul_eigV₁ (291); finrank_eigV₁_eq (298); finrank_eigW_eq (308); finrank_eigX_eq (321); finrank_chain (337); finrank_eq_sixteen_mul (345); milnorM_dvd_finrank (354); IsAnticommFamily.isClifford_toLin' (380); milnorM_dvd_card (387); milnorM_pos (392); isLeast_milnorM (408); finrank_jointEigen_bott16r (420); cliffQ (439); cliffQ_apply (442); cliffGen (446); cliffGen_mul_self (452); finrank_cliffordAlgebra (486); cliffordLift (501); cliffordLift_ι (507); IsMinimalCliffordModule (524); finrank_le_two_pow (530); IsClifford.exists_restrict (554); isMinimal_of_finrank_eq_milnorM (568); milnorM_le_finrank_le (580); finrank_eq_milnorM_of_le_two (588).

Morse/Chapter15/ComplexStructure.lean. IsComplexStructure (97); isComplexStructure_iff (100); IsComplexStructure.transpose_mul_self (112); IsComplexStructure.neg (121); IsComplexStructure.even_card (128); IsComplexStructure.trace_transpose_mul_self (138); Omega (144); omega_succ_subset (149); isClosed_orthogonalGroup (154); abs_apply_le_one_of_mem_orthogonalGroup (160); isCompact_orthogonalGroup (172); isClosed_omega (180); isCompact_omega (194); IsAnticommFamily (202); IsAnticommFamily.mem_omega (207); IsComplexStructure.mul_mem_orthogonalGroup (212); complexLift_continuous (216); exp_smul_of_mul_self_eq_neg_one (222); exp_pi_smul (236); omega_succ_decomp (244); mul_mem_omega_succ (267); semicircle_mem_omega (288); mul_exp_eq_semicircle (316); mul_exp_eq_exp_neg_mul (323); exp_neg_mul_exp (330); isComplexStructure_mul_exp (337); anticomm_mul_exp (353); reflection_mem_omega (361); trace_transpose_mul_eq_zero (380); theta_of_mem_omega (392); mem_omega_of_theta (422); mul_theta_mul (451).

Morse/Chapter15/EightCommutant.lean. kletter (44); kword (47); trace_letter_pair (50); trace_word_pair (55); linearIndependent_word (64); card_word (73); wordBasis (77); wordBasis_apply (80); lanti (84); wanti (87); bsgn (91); bsgn_xor (93); letter_comm (96); word_comm (102); bottCode (110); bott16_eq_word (121); wCode (125); W16 (128); eq_of_not_wanti_aux (132); eq_of_not_wanti (137); word_zero (146); eq_of_commute_bott16 (150); W16_mul_self (195); W16_transpose (199); W16_comm (203); W16_anti (207).

Morse/Chapter15/FiniteStage.lean. basePt_mem (73); minVel_base (77); neg_basePt_mem (83); omegaPhi (88); omegaPhi_apply (94); isHomotopyEquiv_omegaPhi_minPaths (101); omega_hidx (110); omega_theorem_22_1_bijective (129); omegaPhi_bijective (148); stratumIn (166); isClopen_stratumIn (169); stratumInHomeomorph (180); nonempty_pi_stratum_equiv (191); omega_pi_equiv (211); omegaStratum_zero (233); theorem_24_3 (242).

Morse/Chapter15/FiniteStageTwo.lean. isCentralI_iWordM (51); omegaPhi_mem_omegaComp (64); omega_hidx_two (81); omega_minPaths_bijective_two (102); omegaPhi_bijective_two (123); theo-

rem_24_5_two_six (151); omega_pi_equiv_two (161).

Morse/Chapter15/IndexBound.lean. matE (45); matE_mul (49); matE_conjTranspose (56); adjoint_matE_of_skew (60); isOmegaOps_matE (66); colRep (92); mul_colRep (95); one_le_of_sq_mulVec (101); one_le_of_minInvar (120); commute_adjoint (130); exists_killing_spaces_omega (144); norm_sq_of_sq_eq (216); le_formIndex_of_not_minimal_omega (226).

Morse/Chapter15/IndexBoundTwo.lean. colRep_smul (44); exists_odd_of_mulVec (48); iWord_matE (90); isIOps_matE (95); sub_conjTranspose_mem_isoAlg (109); trace_matE (143); killing_of_twoPieceData (153); le_formIndex_of_not_minimal_omega_two (203).

Morse/Chapter15/InvariantSplitting.lean. IsInvar (50); IsMinInvar (53); IsStarSigned (57); IsInvar.orthogonal (64); IsInvar.inf (76); isInvar_top (80); exists_minInvar_le (83); exists_minInvar_decomp (99); IsMinInvar.exists_scalar (134); IsInvar.starProjection_comm (169).

Morse/Chapter15/KillingTwo.lean. IsInterRelI (49); IsInterRelI.mul_exp (57); IsInterRelI.exp_mul (76); IsInterRelI.commute_exp (96); IsInterRelI.conjTranspose (105); exists_killing_spaces_gen (134); exists_killing_spaces_two (198).

Morse/Chapter15/MinimalModules.lean. cliffE (68); cliffQ_single (72); cliffQ_single_add (79); isClifford_cliffE (89); cliffR (106); cliffR_mul (109); isClifford_cliffR (113); θ₃_cliffR (123); θ₅_cliffR (132); θ₆_cliffR (140); θ₇_cliffR (148); mem_ker_mulRight_sub_one (156); exists_fixed (162); mem_of_fixed_mul (188); ker_mulRight_mul (193); cyclicMap (203); cyclicMap_apply (207); cliffordLift_cliffE (210); cyclicMap_mul (217); range_cyclicMap_eq_top (223); map_cyclicMap_eq_top (238); finrank_le_of_fixed (251); exists_linearEquiv_of_fixed (261); cliffordLift_θ₃ (278); cliffordLift_θ₅ (284); cliffordLift_θ₆ (290); cliffordLift_θ₇ (296); exists_ne_zero_mem (304); minimal_of_fixed (311); minimal_le_two (326); minimal_four (340); minimal_five (370); minimal_six (405); minimal_eight (449); central_involution (496); isMinimal_pred (518); minimal_three (533); minimal_seven (548); finrank_eq_milnorM_of_isMinimal (563); exists_iso_of_equiv (580); exists_iso_of_isMinimal (597).

Morse/Chapter15/OmegaComponents.lean. exists_isAngleLift (64); IsAngleLift.pow (79); disjoint_windSet (92); pdet (108); continuous_pdet (110); pdet_one (113); pdet_neg_one_sq (116); cdet_eq_pdet (124); commute_projP (127); commute_projP_projP (130); pdet_split (139); commute_transpose_mul_of_mem_omega (207); q6 (224); isCentralI_six' (231); six_relations (235); isIdempotentElem_q6 (248); omegaF_eq_pow_four (260); exists_mem_omegaComp_six (285); exists_mem_omegaComp_two (331); disjoint_omegaComp (369); existsUnique_mem_omegaComp (374); joined_homotopyEquiv (397); theorem_24_5_pi0_two_six (419).

Morse/Chapter15/OmegaEight.lean. Pp (48); Pm (50); Pp_add_Pm (52); Pp_mul_Pp (56); Pm_mul_Pm (61); Pp_mul_Pm (66); Pm_mul_Pp (71); Pp_transpose (76); Pm_transpose (79); comm_Pp (82); comm_Pm (86); eight_Pp (90); eight_Pm (94); x0 (99); x1 (100); W16_x0 (102); W16_x1 (104); Pp_x0 (106); Pm_x0 (107); Pp_x1 (108); Pm_x1 (109); qOf (116); ext0 (120); ext1 (124); ext0_qOf (127); ext1_qOf (130); qOf_mul (133); transpose_qOf (138); qOf_one (141); bottJ_eq_kronecker (144); comm_qOf (149); eight_qOf (154); eq_qOf_of_commute (161); bottJ_eight_sq (196); bottJ_eight_transpose (199); omegaEightU (203); qOf_of_mem_omega_eight (206); omegaEightU_mem (215); omegaEight_inv_mem (228); continuous_ext0 (245); continuous_qOf_transpose (249); omegaEightHomeomorph (257); omegaEightHomeomorph_basePt (304); bott_periodicity_finite (314).

Morse/Chapter15/OmegaIntertwiners.lean. exp_smul_mul_of_sq (49); eq_mul_exp_smul_of_hasDerivAt (74); mul_exp_smul_of_sq (97); IsInterRel (124); IsInterRel.conjTranspose (130); IsInterRel.commute_exp (151); adR (174); exists_killing_spaces (185).

Morse/Chapter15/OmegaKilling.lean. sum_le_finsum_mem (47); isoAlg (68); killing (83); hasDerivAt_conj_exp (87); isoAlg_exp (96); exp_smul_conjTranspose (105); isJacobiField_killing (112);

hasDerivAt_killing (174); killing_zero (180); killing_eq_zero (184); commute_of_killing_eqOn (190); finrank_le_conjMultiplicity (206); sum_finrank_le_formIndex (247).

Morse/Chapter15/OmegaManifold.lean. exists_hasStrictFDerivAt_exp_zero (59); exists_exp_injective (68); exists_log (80); norm_conjTranspose_real (93); norm_conj_orth (98); omega_orth (120); omega_sq (124); omega_anti (128); mem_omega_of (133); tanAlg (156); mem_tanAlg (170); mul_exp_mem_omega (180); mem_tanAlg_mul_exp (194); exp_conj_orth (202); mem_tanAlg_of_mul_exp_mem (211); conjIso (283); conjIso_apply (286); conj_mem_tanAlg (292); finrank_tanAlg_conj_le (314); finrank_tanAlg_conj (325); tanAlg_zero (342); finrank_tanAlg_mul_exp (349); exists_local_log (380); eventually_finrank_tanAlg (400); omegaStratum (413); omegaStratum_subset (416); self_mem_omegaStratum (419); eventually_mem_omegaStratum (423); isClosed_omegaStratum (429); isCompact_omegaStratum (438); isSliceSubmanifold_omegaStratum (447); isTotGeo_omegaStratum (543).

Morse/Chapter15/OmegaMinimal.lean. eq_exp_smul_mul_of_hasDerivAt (63); inner_mul_left_adj (86); inner_self_mul_skew (90); pi_sq_le_of_sq_mul (100); exists_eigen_ge (164); norm_sq_eq_trace (205); pi_sq_mul_card_le_norm_sq (211); mul_self_eq_of_norm_sq_le (220); minVel0 (249); omegaGeo (252); exp_pi_smul_of_minVel0 (255); norm_sq_of_minVel0 (259); norm_pi_smul_of_minVel0 (263); neg_mem_omegaStratum (271); omegaGeo_mem_pathSpace (277); energy_omegaGeo (284); riemDist_omegaStratum (294); pi_sq_card_le_energy (338); omegaPath (352); omegaPath_apply (355); omegaPath_mem_minPaths (359); exists_minVel0_of_mem_minPaths (369); omegaGeo_half (398); speed_omegaGeo (404); continuous_omegaPath (413); continuous_apply_half_omega (439); minVelToOmegaPaths (445); conjTranspose_mul_half (450); half_mem_minVel0 (454); omegaPathsToMinVel (463); omegaPathsToMinVel_minVelToOmegaPaths (470); edist_minVelToOmegaPaths (475); omegaMinPathsHomotopyEquiv (487); omegaMinPathsHomeomorph (512); basePt (537); basePt_mem_omega (539); mem_minVel0_iff (548); minVel0Homeomorph (579); omegaGeo_eq_semicircle (597).

Morse/Chapter15/OmegaRetract.lean. invSqrtR (52); invSqrtM (55); continuousOn_invSqrtR (57); continuousOn_invSqrtM (60); invSqrtM_isSelfAdjoint (71); commute_invSqrtM (74); invSqrtM_mul_self_mul (77); invSqrtM_one (91); spectrum_subset_Ioi (98); midDir (122); polarDir (127); isSelfAdjoint_conjTranspose_mul_self' (131); polarDir_mem (136); midDir_mul (175); conjTranspose_mul_self_of_minVel0 (180); polarDir_mul (184); continuous_midDir (189); omegaNbhd (197); continuous_midDir_half (201); isOpen_omegaNbhd (206); continuousOn_polarDir_half (210); omegaRetrPath (221); continuousOn_omegaRetrPath (228); omegaRetrPath_mem_minPaths (240); mem_omegaNbhd_and_eqOn (248); exists_retraction_minPaths_omega (272).

Morse/Chapter15/OmegaWinding.lean. IsCentralI (45); iWordM (55); iWordM_two (57); iWordM_six (59); isCentralI_two (63); isCentralI_six (73); omegaF (103); continuous_omegaF (105); anti_of_mem_omega (110); IsCentralI.comm_transpose_mul (115); norm_omegaF (129); omegaF_self (139); omegaF_geo (145); omegaComp (157); isClopen_omegaComp (162); mem_omegaComp_geo_iff (169).

Morse/Chapter15/Relations.lean. IsClifford (62); IsClifford.swap (71); IsClifford.swap_assoc (74); IsClifford.sq' (78); IsClifford.sq_assoc (80); IsClifford.mono (83); AntiCommute (99); θ_3 (106); θ_5 (108); θ_6 (110); θ_7 (112); ω_7 (114); θ_3_spec (119); θ_3_anticommute (123); θ_5_spec (129); θ_5_anticommute (135); θ_6_spec (141); θ_6_anticommute (148); θ_7_spec (154); θ_7_anticommute (162); J_7J_8_spec (168); J_3J_4_spec (175); ω_7_spec (181).

Morse/Chapter15/SplittingClifford.lean. IsOmegaOps (55); omegaGens (71); IsOmegaOps.a_mem (79); IsOmegaOps.q_mem (81); IsOmegaOps.j_mem (83); IsOmegaOps.isStarSigned (86); IsOmegaOps.adjoint_q_mul_q (95); IsOmegaOps.a_sq_comm (100); IsOmegaOps.exists_sq_eq (117); IsOmegaOps.qa_qa (137); IsOmegaOps.j_qa (142); IsOmegaOps.q_qa (145); cliffFam (153); cliffTop (163); cliffTop_lt (167); cliffTop_k (171); cliffTop_succ (175); IsOmegaOps.cliffTop_anti (178); cliffFam_apply (208); cliffFam_lt (214); cliffFam_k (219); cliffFam_succ (223); IsOmegaOps.isClifford_cliffFam (230); IsOmegaOps.isMinimal_cliffFam (263); IsInter (321); IsInter.apply_j (329); IsInter.apply_q (335); IsInter.apply_a (347); IsInvar.orthogonalProjectionOnto_comm (371); interOp (380); interOp_apply (383); interOp_comm_q

(386); interOp_comm_j (395); interOp_a (405); a_a_interOp (415); interOp_a_a (421); interOp_starProjection (433); adjoint_interOp_starProjection (440); starProjection_a_interOp (448); starProjection_a_interOp_of_orth (454); interOp_apply_mem (463); eq_zero_of_a_eq_zero (467); a_interOp_ne_zero (475); linearIndependent_a_interOp (488).

Morse/Chapter15/SplittingData.lean. OmegaPieceData (48); IsOmegaOps.exists_piece_gt (67); IsOmegaOps.exists_inter (104); IsOmegaOps.exists_omegaPieceData (118).

Morse/Chapter15/SplittingTwo.lean. iWord (59); mem_iWord_list (61); iWord_mem (67); commute_iWord (82); IsIOps (92); IsIOps.neg_a (99); IsOmegaOps.neg_a (102); IsOmegaOps.neg_q (114); isInvar_omegaGens_neg_a (128); isInvar_omegaGens_of_neg_a (138); isMinInvar_omegaGens_neg_a (144); isInvar_omegaGens_neg_q (150); iWord_mem_of_isInvar (160); IsOmegaOps.exists_eq_smul_iWord (165); a_a_of_eq_smul (201); isMinimal_cliffFam_k (212); exists_inter_two (252).

Morse/Chapter15/SplittingTwoData.lean. foldr_sup_le_iSup (45); exists_decomp_two (62); TwoPieceData (148); exists_twoPieceData_of_decomp (170).

Morse/Chapter15/Stabilization.lean. stabEquiv (46); stabEquiv_symm_castSucc (61); stabEquiv_symm_last (65); reindex_fromBlocks_blockDiagonal (70); reindex_fromBlocks_bottJ (92); stab (98); mem_omega_reindex (102); bott16_basePt_mem (119); stab_mem_omega (123); stab_basePt (132); smul_reindex_add (138); omegaPhi_stab (144); trace_reindex_equiv (156); trace_stab (162); omegaLabel_stab (184).

Morse/Chapter15/StablePeriodicity.lean. unitaryCongr_one (52); piUnitaryCongr (57); nonempty_pi_unitary_equiv_of_le (63); nonempty_pi_unitary_equiv_stable (73); bott_chain_of_le (85); bott_periodicity_finite_of_le (146); theorem_24_7 (158); periodicity_theorem_stable (172).

Morse/Chapter15/TotallyGeodesic.lean. isSliceSubmanifold_of_param (59); IsTotGeo (109); IsTotGeo.mul_conjTranspose (123); IsTotGeo.mul_exp_smul_mem (126); IsTotGeo.leftT (132); IsTotGeo.mem_leftT_iff (137); IsTotGeo.tangentSubspace_eq (153); IsTotGeo.mem_tangentSubspace_iff (178); IsTotGeo.isGeodesicOn_mul_exp_smul (185); IsTotGeo.eqOn_mul_exp_smul (208); IsTotGeo.exp_eq (244); IsTotGeo.isGeodesicallyComplete (255); IsTotGeo.eqOn_of_isUnbrokenGeodesic (268); IsTotGeo.norm_deriv_mul_exp_smul (300); IsTotGeo.energy_mul_exp_smul (310); IsTotGeo.pathLength_mul_exp_smul (319); IsTotGeo.mem_pathSpace_mul_exp (329); IsTotGeo.isUnbrokenGeodesic_mul_exp (335).

Morse/Chapter15/Winding.lean. IsAngleLift (54); IsAngleLift.unique (58); re_pos_of_norm_sub_one_lt (67); IsAngleLift.perturb (74); windSet (116); continuous_comp_path (120); norm_sub_lt_of_edist_lt (125); exists_mem_windSet_of_close (134); isClopen_windSet (155); cplx (177); cplx_apply (179); continuous_cplx (181); cplx_transpose (184); cplx_smul (186); trace_cplx (189); projP (193); projM (196); cdet (199); continuous_cdet (201); cplx_sq (206); projP_add_projM (209); projP_mul_projM (213); projM_mul_projP (219); projP_mul_projP (225); projM_mul_projM (230); cplx_comm_projP (235); cplx_comm_projM (240); cdet_mul (246); cdet_one (265); conj_cdet (269); norm_cdet (297); eq_mul_exp_smul_of_hasDerivAt_cplx (313); det_exp_smul_cplx (336); cdet_exp (358).

Morse/Chapter2/Attaching.lean. AttachCell (81); cellInl (85); cellInr (89); cellInl_apply_eq (94); attachCell_hom_ext (98); cellDesc (108); cellHomotopy (128); diskPt (166); dvec (174); norm_dvec_le (181); continuous_dite_le (233); radial (260); radial_of_le (268); radial_of_ge (273); radial_inclusion (288); continuous_radial (298); radial_congr (329); radial_of_eq_one (339); clampI (353); kOuter (368); kOuter_half (377); kOuter_one (381); kCell (386); lemma36Map (401); lemma36Map_inl (408); aH (418); hOuter (431); hOuter_aH (434); hOuter_one (443); continuous_hOuter (451); lemma36_zero (471); lemma36Homotopy (513); lemma_3_6 (536); attachMap (555); attachMap_inl (559); attachMap_inr (563); homotopyAttach (568); aQ (573); qOuter (585); qOuter_aQ (589); qOuter_one (596); lemma37Homotopy (612); homotopic_of_leftInv_rightInv (650); lemma_3_7 (671).

Morse/Chapter2/CWType.lean. IsCellAttachment (110); IsFiniteCellComplexType (119); IsFiniteCellComplexType.of_isEmpty (125); IsFiniteCellComplexType.of_homotopyEquiv (130); IsFiniteCellComplexType.attach (139); isCriticalPointM_of_forall_le (151); IsFiniteCellComplexType.sublevel_step (173); isFiniteCellComplexType_sublevel (226); theorem_3_5 (351); theorem_3_5_count (381); isFiniteCellComplexType_union_cells (414); IsFiniteCellComplexType.sublevel_step_finset (479); isFiniteCellComplexType_sublevel' (551); theorem_3_5' (684); theorem_3_5_count' (712).

Morse/Chapter2/CellAdjunction.lean. isPushout_of_cell (65); exists_isPushout_union_cell (148); theorem_3_2_attach (212).

Morse/Chapter2/CellAttach.lean. homotopyRelMono (102); IsStrongDeformationRetract.trans (109); isCriticalPoint_congr (137); splitCoord (147); norm_sq_splitCoord_fst (166); norm_sq_splitCoord_snd (173); negIdxEquiv (180); IsSplitMorseChart (197); exists_isSplitMorseChart (207); IsSplitMorseChart.isCriticalPoint_comp_iff (240); milnorBump (262); milnorF (267); ellipsoidE (271); cellE (275); handleE (279); isCompact_ellipsoid (284); ellipsoid_subset_target (300); ellipsoidE_subset_source (304); mem_ellipsoidE_iff (309); isCompact_ellipsoidE (317); isClosed_ellipsoidE (323); milnorBump_eq_zero (328); contDiff_milnorBump (337); contDiff_milnorF (356); milnorF_eq (361); milnorF_eq_of_notMem (367); milnorF_le (372); milnorF_apply_self (377); sublevel_milnorF_add (383); isCriticalPoint_milnorF_iff (393); sublevel_milnorF_eq_union (412); cellE_subset_handleE (419); cellE_subset_sublevel_milnorF (434); cellE_inter_sublevel (443); handleRetract (467); handleRetract_of_mem (471); handleRetract_of_notMem (476); retractModel_mem_target (482); milnorF_handleRetract_le (491); handleRetract_zero (503); handleRetract_of_mem_sublevel (510); handleRetract_of_mem_cellE (519); handleRetract_of_notMem_ellipsoidE (527); handleRetract_one_mem (538); continuous_handleRetract (558); handleRetract_mem (594); handleRetractOne (602); handleHomotopy (611); isStrongDeformationRetract_handle (624); isStrongDeformationRetract_milnorF (651); isStrongDeformationRetract_union_cellE (677); exists_forall_ball_subset_target (705); cellE_parametrization (718); exists_morseChart_forall_small (781); theorem_3_2 (812); theorem_3_2_homotopyEquiv (853).

Morse/Chapter2/CellAttachSubmanifold.lean. exists_contDiff_comp_val_eq_of_chart (95); exists_isMorsePatch_submanifold (139); contMDiff_patchBump (172); isCriticalPointM_patchF_iff (198); theorem_3_2_submanifold (236); theorem_3_2_attach_submanifold (318).

Morse/Chapter2/CellModel.lean. eq_union_closure_diff (115); IsMilnorMu (127); IsMilnorMu.antitone (145); IsMilnorMu.nonneg (149); IsMilnorMu.monotone_add (157); muBump (168); contDiff_muBump (171); muBump_nonneg (178); muBump_le_one (181); muBump_of_le (184); muBump_of_ge (189); milnorMu (195); hasDerivAt_milnorMu (199); contDiff_milnorMu (211); milnorMu_of_ge (216); lt_milnorMu_zero (224); isMilnorMu_milnorMu (239); CellModel.xi (259); CellModel.eta (263); CellModel.morseQuad (285); CellModel.morseF (289); CellModel.morseQuad_eq_morseF (292); CellModel.ellipsoid (297); CellModel.cellModel (301); CellModel.handleModel (305); CellModel.isClosed_ellipsoid (316); CellModel.ellipsoid_subset_ball (319); CellModel.cellModel_subset_ellipsoid (324); CellModel.morseF_le_morseQuad (331); CellModel.morseF_eq_of_notMem_ellipsoid (337); CellModel.morseF_le_iff (344); CellModel.morseF_lt_of_mem_cellModel (355); CellModel.cellModel_inter_sublevel (365); CellModel.sublevel_morseF_eq_union (378); CellModel.hasFDerivAt_xi (393); CellModel.hasFDerivAt_eta (397); CellModel.hasFDerivAt_morseF (403); CellModel.eq_zero_of_hasFDerivAt_morseF (418); CellModel.hasFDerivAt_morseF_zero (444); CellModel.hasFDerivAt_morseF_zero_iff (450); CellModel.hasFDerivAt_morseQuad_zero_iff (457); CellModel.retractScale (474); CellModel.retractModel (479); CellModel.retractScale_zero (485); CellModel.retractScale_nonneg (489); CellModel.retractScale_le_one (495); CellModel.retractModel_zero (507); CellModel.xi_retractModel (510); CellModel.eta_retractModel (513); CellModel.eta_retractModel_le (518); CellModel.retractModel_of_snd_eq_zero (525); CellModel.retractModel_of_xi_le

(529); CellModel.retractModel_of_case2 (543); CellModel.retractModel_of_le_xi (550); CellModel.retractModel_of_morseQuad_le (558); CellModel.retractModel_of_mem_cellModel (563); CellModel.retractModel_one_mem (568); CellModel.retractModel_mem_ellipsoid (593); CellModel.continuous_retractModel (601); CellModel.morseF_retractModel_le (635); CellModel.retractModel_of_notMem_ellipsoid (647); CellModel.mem_diff_of_mem_cellModel (663); CellModel.tendsto_smul_cellModel (673); CellModel.cellModel_subset_handleModel (693); CellModel.retractModel_mem_sublevel (708); CellModel.retractOne (716); CellModel.retractHomotopy (725); CellModel.isStrongDeformationRetract_model (741); CellModel.morseQuad_retractModel_zero_one_le (759); CellModel.isStrongDeformationRetract_model_level (791).

Morse/Chapter2/CellPatch.lean. IsMorsePatch (68); patchBump (75); patchF (79); patchEllipsoid (83); patchCell (87); isCompact_patchEllipsoid (106); isClosed_patchEllipsoid (112); sublevel_patchF_add (147); patchCell_subset_sublevel_patchF (157); patchRetract (170); retractModel_mem_target_of_patchF_le (186); continuous_patchRetract (270); isStrongDeformationRetract_patch_handle (316); exists_forall_ball_subset_patch_target (343); patchCell_parametrization (356); patch_theorem_3_2 (412); exists_isPushout_union_cell_of_t2 (465); exists_homotopyEquiv_pushout_of_cell (521).

Morse/Chapter2/CriticalLevel.lean. IsStrongDeformationRetract.of_retract (59); isStrongDeformationRetract_level (97); isStrongDeformationRetract_milnorF_level (121); isStrongDeformationRetract_sublevel_crit (145); isStrongDeformationRetract_union_cellE_crit (159); remark_3_4 (187).

Morse/Chapter2/Crossing.lean. sublevel (73); sublevel_mono (80); IsStrongDeformationRetract (90); IsStrongDeformationRetract.exists_homotopyEquiv (97); eventually_lt_of_hasDerivAt_pos (121); UnitSpeedInSlab (134); UnitSpeedInSlab.reverse (143); UnitSpeedInSlab.le_of_le (155); UnitSpeedInSlab.le_of_le' (172); UnitSpeedInSlab.eq_add_of_le (180); UnitSpeedInSlab.eq_add (208); deformation (228); deformation_of_ge (231); IsSlabFlow (241); IsSlabFlow.deformation_of_le (252); IsSlabFlow.deformation_zero (256); IsSlabFlow.continuous_deformation (265); IsSlabFlow.unitSpeed (271); IsSlabFlow.apply_eq_add (275); IsSlabFlow.le_of_apply_le (283); IsSlabFlow.apply_le_of_le (290); IsSlabFlow.le_of_apply_le' (306); IsSlabFlow.image_sublevel (326); IsSlabFlow.apply_deformation (338); IsSlabFlow.deformation_mem (346); IsSlabFlow.deformation_one_mem (354); IsSlabFlow.retraction (362); IsSlabFlow.retraction_mem (367); IsSlabFlow.homotopyRel (373); IsSlabFlow.isStrongDeformationRetract (384); IsSlabFlow.exists_homotopyEquiv (390).

Morse/Chapter2/Regular.lean. contDiff_smul_of_tsupport (84); exists_cutoff_mul (101); contDiff_gradient (135); fderiv_apply_gradient (141); isCriticalPoint_iff_gradient_eq_zero (147); exists_cutoff (158); gradField (170); contDiff_gradField (176); hasCompactSupport_gradField (180); isBddLipschitz_gradField (185); fderiv_apply_gradField (191); isSlabFlow_gradField (198); apply_flow_gradField (222); flow_gradField_image_sublevel (228); flow_gradField_of_notMem (236); theorem_3_1 (253); exists_diffeomorph_image_sublevel_eq (273).

Morse/Chapter2/Remark.lean. RemarkP13.eq_of_forall_eq_intCast (49); RemarkP13.not_nullhomotopic (71); RemarkP13.M (128); RemarkP13.F (131); RemarkP13.sublevelHomeomorph (153); RemarkP13.j (169); RemarkP13.γ (175); RemarkP13.γ_zero_eq_one (187); RemarkP13.not_contractibleSpace (198); RemarkP13.remark_p13_not_isCriticalPoint (208); RemarkP13.remark_p13_not_isCompact (220); RemarkP13.remark_p13 (254); RemarkP13.remark_p13_not_isStrongDeformationRetract (260); RemarkP13.remark_p13_not_homeomorph (265); RemarkP13.remark_p13_plane (271).

Morse/Chapter2/SeveralCritical.lean. isStrongDeformationRetract_of_eq_univ (64); IsStrongDeformationRetract.union_of_isClosed (72); multiF (152); contDiff_multiF (155); multi_induction (171); exists_isMorsePatch_submanifold_subset (297); patchCell_props (329); remark_3_3_submanifold (366).

Morse/Chapter2/Submanifold.lean. contDiffOn_sliceChart (87); contDiffOn_sliceChart_invFun (90); fderiv_sliceChart_fderiv_invFun (104); fderiv_sliceParam_apply (118); chartField (137); contDiffOn_chartField (140); chartField_neg (146); chartField_eq_fderiv_sliceParam (150); chartField_mem_tangentSubspace (159); exists_chartField_pos (166); chartDomain (190); isOpen_chartDomain (193); subset_chartDomain (196); isOpen_chartDomain_diff (202); exists_tangent_field (220); exists_integralCurve_mem (275); flow_mem_of_tangent (336); theorem_3_1_submanifold (372); theorem_3_1_submanifold_of_isCompact (442).

Morse/Chapter3/EulerChar.lean. critCount (106); finsum_neg_one_pow_indexM_eq_sum_critCount (111); IsEulerNumber (146); exists_isEulerNumber (155); IsEulerNumber.finsum_eq_of_ambient (162); exists_contDiff_comp_val_eq (203); IsEulerNumber.finsum_eq (236); IsEulerNumber.sum_critCount_eq (247); exists_eulerNumber_eq_sum_critCount (263); sum_critCount_eq_of_isMorseFunction (275); IsEulerNumber.eq_sum_critCount (286).

Morse/Chapter3/EulerCharAlgebra.lean. sgn_prod (72); prod_sgn_eq_neg_one_pow (81); inner_apply_self_eq_sum_eigenvalues (101); quadForm_equivalent_weightedSumSquares (114); index_eq_ncard_eigenvalues_neg (131); eigenvalues_ne_zero (136); isNondegenerate_iff_bijective (151); sgn_det_eq_neg_one_pow_index (173); isSymmetric_of_symm (182); sgn_det_eq_neg_one_pow_index_of_symm (189); index_eq_finrank_of_negDef (197); index_eq_zero_of_nonneg (205).

Morse/Chapter3/EulerCharGradient.lean. tangentialGradient (94); tangentialGradient_mem (98); inner_tangentialGradient (104); tangentialGradient_eq_zero_iff (110); inner_fderiv_gradient (124); contDiffAt_gradient (136); tubRet (156); tubRet_of_mem (158); contDiffAt_tubRet (163); dret_eq_starProjection (171); comp_tubRet_comp_val (180); fderiv_comp_tubRet (190); tangentialGradient_eq_gradient (201); fderiv_comp_tubRet_eq_zero (210); contMDiff_tangentialGradient (223); isTangent_tangentialGradient (236); tangentialGradient_mem_range (242); dvr_tangentialGradient (252); hessT (273); hessT_symm (282); hessT_apply_eq_inner_dvr (290); inner_dvT_tangentialGradient (302); tanN_eq_range (313); tanEquiv (319); hessian_comp_φ_eq (342); index_hessT (368); isNondegenerate_hessT_iff (377); bijective_dvT_tangentialGradient (392); indexN_tangentialGradient (415).

Morse/Chapter3/EulerCharSphere.lean. isEulerNumber_sph (67); sum_critCount_sph (73); heightFun (81); gradient_heightFun (87); tangentialGradient_heightFun (95); isCriticalPointM_heightFun_iff (110); hessT_heightFun (118); isNondegenerate_hessT_heightFun (127); isMorseFunction_heightFun (139); indexM_heightFun_pos (153); indexM_heightFun_neg (170); critCount_heightFun (187); sum_critCount_heightFun (248); IsEulerNumber.eq_sph (265).

Morse/Chapter3/Inequalities.lean. altSum (210); altSum_zero (214); altSum_succ (218); altSum_succ_add_altSum (229); altSum_congr (234); altSum_add (237); altSum_sub (241); altSum_sum (247); altSum_eq_of_telescope (254); sum_neg_one_pow_mul_eq (264); finrank_range_eq_altSum (287); altSum_finrank_eq_zero_of_exact (304); LongExactSeq (320); LongExactSeq.finiteDimensional_middle (352); finiteDimensional_middle (352); LongExactSeq.finrank_le (365); LongExactSeq.finrank_range_δ (383); LongExactSeq.altSum_le (419); LongExactSeq.altSum_eq_of_finrank_eq_zero (429); IsSubadditive (452); IsAdditive (456); IsAdditive.isSubadditive (459); le_sum_of_chain (465); eq_sum_of_chain (479); chain_le (491); IsSubadditive.le_sum (499); IsAdditive.eq_sum (506); FiltrationHomology (528); morseNumber (545); FiltrationHomology.isSubadditive_finrank (558); FiltrationHomology.isSubadditive_altSum (568); FiltrationHomology.isAdditive_eulerChar (581); FiltrationHomology.finrank_le_morseNumber (596); FiltrationHomology.altSum_le_morseNumber (606); FiltrationHomology.sum_neg_one_pow_finrank_eq (616); le_of_altSum_le (651); altSum_eq_of_altSum_le (664); sum_neg_one_pow_eq_of_altSum_le (676); eq_of_altSum_le_succ (683); eq_of_altSum_le_zero (699); eq_of_altSum_le_of_lacunary (707); lacunary_of_odd (719); FiltrationHomology.finrank_eq_morseNumber_of_succ (736); FiltrationHomology.finrank_eq_morseNumber_of_zero (747); FiltrationHomol-

ogy.finrank_eq_morseNumber_of_lacunary (758); finiteDimensional_of_steps (779); FiltrationHomology.ofSteps (796); morseNumber_eq_card (818); morseNumber_eq_card_of_single (836); FiltrationHomology.weak_morse_inequality (846); FiltrationHomology.morse_inequality (856); FiltrationHomology.finrank_eq_zero_of_forall_ne (866); FiltrationHomology.euler_characteristic_eq (879).

Morse/Chapter3/ProjectiveMorse.lean. CP.height (74); CP.contDiff_height (76); CP.weight (82); CP.wform (85); CP.wform_apply (89); CP.model (94); CP.height_param (96); CP.hasFDerivAt_model (114); CP.hessian_model (124); CP.weight_ne_zero (135); CP.eq_zero_of_forall (140); CP.isCriticalPoint_model_iff (148); CP.isNondegenerate_hessian_model (165); CP.rank (175); CP.index_hessian_model (177); CP.nsq_zero (199); CP.vec_zero (201); CP.param_zero (208); CP.param_zero_mem (213); CP.crit_index_of_mem (218); CP.re_param_zero_kk (277); CP.isCriticalPointM_height_iff (282); CP.isNondegenerate_indexM_height (299); CP.isNondegenerate_of_isCriticalPointM (309); CP.rank_lt (319); CP.rank_lt_rank (327); CP.bijective_rank (336); CP.critCount_height (346); CP.lacunary (389); CP.bettiNumber_eq (402); CP.exists_cells (413); bettiNumber_complexProjectiveSpace (426).

Morse/Chapter3/ProjectiveSpace.lean. exists_sliceChart_of_localParam (72); CP.Model (127); CP.cz (130); CP.nsq (133); CP.vec (136); CP.param (140); CP.coordC (144); CP.coord (148); CP.nsq_eq (151); CP.nsq_nonneg (155); CP.cz_coord (157); CP.sum_vec_mul_star (162); CP.param_mem (171); CP.param_kk (182); CP.param_succAbove_k (187); CP.coord_param (192); CP.exists_vec_of_mem (203); CP.nsq_coord_lt_and_param_coord (216); CP.contDiff_nsq (253); CP.contDiff_cz (256); CP.contDiffOn_vec (263); CP.ofCLM (276); CP.contDiffOn_param (280); CP.contDiffOn_coordC (288); CP.contDiffOn_coord (302); CP.exists_re_diag_pos (312); CP.Normal (323); CP.finrank_add (325); CP.exists_sliceChart (335); CP.isSliceSubmanifold (418); CP.norm_eq_one (426); CP.isCompact (435).

Morse/Chapter3/Reeb.lean. IsSlabFlow.image_level (123); exists_diffeomorph_image_sublevel_level (154); exists_min_max_of_ncard_eq_two (230); reeb_of_finrank_pos (302); reeb_of_finrank_eq_zero (388); reeb (426); reeb_of_contMDiff (442).

Morse/Chapter3/ReebGluing.lean. alexanderCone (80); norm_alexanderCone (93); alexanderCone_of_mem_sphere (100); alexanderCone_alexanderCone (109); alexanderCone_id (126); continuous_alexanderCone (134); alexanderConeHomeomorph (159); hemiProj (182); hemiHeight (186); hemiLift (190); norm_sq_eq_hemiProj_hemiHeight (203); norm_hemiProj_le (225); hemiHeight_sq (232); hemiLift_mem_sphere (238); eq_of_hemiProj_eq_of_sameSign (252); nonempty_homeomorph_sphere_of_union_closedBall (282).

Morse/Chapter3/ReebMorse.lean. isLocalMin_localRep (67); isLocalMax_localRep (81); isCriticalPointM_of_isLocalMin (95); isCriticalPointM_of_isLocalMax (100); morseNormalForm_single (112); tendsto_and_eventually_morse_axis (130); norm_sq_toLp (150); IsSmoothCoord.transHomeomorph_euclidean (156); indexM_eq_zero_of_isLocalMin (170); indexM_eq_finrank_of_isLocalMax (197); exists_morse_chart_of_isLocalMin (226); exists_morse_chart_of_isLocalMax (242); exists_image_closedBall_eq_sublevel (274).

Morse/Chapter3/Torus.lean. isCriticalPointM_comp_val_of_fderiv_eq_zero (111); hessian_comp_sliceParam_of_fderiv_eq_zero (121); index_eq_one_of_finrank_eq_two (143); isNondegenerate_and_index_of_diag (165); height (212); pointP (222); pointQ (224); pointR (226); pointS (228); height_pointP (243); height_pointQ (245); height_pointR (247); height_pointS (249); coef_ne_zero (257); height_mem_Icc (268); sublevel_height_eq_empty (278); sublevel_height_eq_univ (285); isCriticalPointM_height_iff_forall (301); isCriticalPointM_height_iff (311); isCriticalPointM_height_iff' (360); onAxis_spec (372); pointP_spec (468); pointQ_spec (480); pointR_spec (492); pointS_spec (505); isMorseFunction_height (517); criticalValues_height (528); Torus.critCount_height (550); critCount_height (550); isEulerNumber_torus (618); IsEulerNumber.eq_torus (629); sum_critCount_torus (637); Torus.rotField (645); IsEulerNumber.eq_torus_of_rotation (652); theorem_3_1_torus (686).

Morse/Chapter3/TorusSurface.lean. torus (75); quad (89); poly (93); dpoly (100); d2poly (105); hasFDerivAt_quad (115); hasFDerivAt_poly (123); hasFDerivAt_coef (132); hasFDerivAt_dpoly (143); hessian_poly (160); contDiff_poly (164); dpoly_apply (168); poly_eq_mul (176); mem_torus_iff (186); quad_pos (194); dpoly_ne_zero (202); poly1 (245); preimage_poly1 (251); isRegularValue_poly1 (258); isSliceSubmanifold_torus (285); isClosed_torus (291); norm_le_of_mem_torus (296); isCompact_torus (310); onAxis (320); onAxis_mem_iff (327); onAxis_mem (331); dpoly_onAxis (336); d2poly_onAxis (344); nonempty_torus (352); finrank_tangentSubspace_torus (360); mem_tangentSubspace_torus_iff (369).

Morse/Chapter4/Approximation.lean. rieszEquiv (88); negGrad (114); inner_negGrad (116); contDiffOn_negGrad (119); hasFDerivAt_negGrad (124); inner_fderiv_negGrad (132); finrank_ker_pos_iff_not_surjective (148); hessian_comp_xParam (156); hessian_add_inner (165); fderiv_add_inner (173); mem_image_critPoints_of_not_isNondegenerate (186); exists_local_extension_of_contMDiff (275); ae_isMorseFunction_add_inner (302); exists_mem_isMorseFunction_add_inner (342); exists_bound_iteratedFDeriv_val_comp_symm (363); exists_isMorseFunction_approx (383).

Morse/Chapter4/Compact.lean. ae_isMorseFunction_normSq_sub_finite_critPoints (38); exists_isMorseFunction_finite_critPoints (51).

Morse/Chapter4/Curvature.lean. tanSpace (89); dParamEquiv (92); finrank_tanSpace (98); d2Param_symm (101); secondFFTan (107); shapeOp (119); inner_shapeOp_dParamEquiv (126); isSymmetric_shapeOp (130); inner_dParamEquiv (146); mem_nullSpace_firstFF_sub_iff (154); map_nullSpace_firstFF_sub_eq_eigenspace (174); nullity_firstFF_sub_smul_secondFF (191); nullity_firstFF_sub_zero_smul_secondFF (197); focalMult (212); isFocalPoint_Mult_iff_focalMult (215); isFocalPoint_iff_focalMult_pos (229); focalMult_add_smul (242); isFocalPoint_add_smul_iff (263); focalMult_add_inv_smul (282); principalCurvatures (297); finrank_eigenspace_shapeOp (301); focalMult_add_smul_eq_card (308); sum_focalMult_add_smul_le (321); index_firstFF_sub_secondFF (343); indexM_normSq_sub_eq_card (386); indexM_normSq_sub_eq_finsum_focalMult (404).

Morse/Chapter4/Existence.lean. isCompact_setOf_normSq_sub_le (62); isCompact_sublevel_normSq_sub (74); ae_isMorseFunction_normSq_sub_and_isCompact_sublevel (80); exists_isMorseFunction_isCompact_sublevel (92); exists_isMorseFunction_finite_critPoints_sublevel (102).

Morse/Chapter4/Focal.lean. nullity_smul_of_ne_zero (95); slice_dim_add_eq_of_nonempty (107); endpointNullity (126); IsFocalPointMult (133); IsFocalPoint (137); focalSet (142); endpointNullity_pos_iff (148); isFocalPoint_iff (163); focalSet_eq_image_critPoints (174); measure_focalSet_eq_zero (201); endpointNullity_eq_nullity (220); isFocalPointMult_add_smul_iff (231); not_isFocalPoint_self (250); isCriticalPointM_normSq_sub_iff_mem_normalBundleSet (273); hessian_normSq_sub_xParam (290); nullityM_normSq_sub_eq_endpointNullity (305); isFocalPointMult_iff (319); isDegenerate_normSq_sub_iff_isFocalPoint (335); mem_focalSet_of_not_isMorseFunction (358); ae_isMorseFunction_normSq_sub (372); ae_volume_isMorseFunction_normSq_sub (387); exists_mem_isMorseFunction_normSq_sub (395); exists_isMorseFunction_normSq_sub (405).

Morse/Chapter4/NormalBundle.lean. mem_normalBundleSet_iff (93); xParam (116); uParam (119); dParam (122); d2Param (125); xParam_uParam (128); contDiffAt_xParam (130); injective_dParam (139); fderiv_σφ_comp_dParam (144); injective_fderiv_σφ (164); hasFDerivAt_B (173); B_self_inner (186); finrank_normalBundle_add (200); mem_range_mfd_val_normalBundle_iff (210); firstFF (290); secondFF (296); normalForm (302); normalForm_id (317); secondFF_smul (320); normalEnd (334); normalEnd_id (336); hasMFDerivAt_normalEnd (342); contMDiffOn_normalEnd (363); finrank_ker_mfd_normalEnd (379).

Morse/Chapter4/Sard.lean. mem_critSet_of_mem_critPoints_of_mdifferentiableAt (65); mea-

sure_image_inter_critPoints_eq_zero (91); measure_image_critPoints_eq_zero (123).

Morse/Chapter4/Uniform.lean. ballDiffeo (72); norm_ballDiffeo_lt_one (98); ballSliceChart (117); isSliceSubmanifold_ballImage (151); toBall (167); fromBall (180); contMDiff_fromBall (187); ballSliceChart_toChart_toBall (202); fromBall_ballSliceChart_toChart_symm (214); isNondegenerateCriticalPoint_iff_of_eventuallyEq (222); isCriticalPoint_iff_of_eventuallyEq (230); isMorseFunction_of_ballImage (236); contDiff_inner_ballDiffeo (290); ae_isMorseFunction_add_inner_ballDiffeo (296); exists_isMorseFunction_approx_uniform (319).

Morse/Chapter4/Whitney.lean. finrank_le_of_injective_clm (127); exists_isSliceSubmanifold_whitney (135); exists_embedding_euclidean (156); exists_diffeomorph_sliceSubmanifold (180); exists_isMorseFunction_of_compactSpace (201); ae_isMorseFunction_add_inner_comp (216); exists_isMorseFunction_approx_of_compactSpace (257); isCriticalPointM_of_finrank_eq_zero (341); sum_critCount_of_finrank_eq_zero (351); eulerNumberM (365); sum_critCount_eq_of_isMorseFunction_of_compactSpace (376); sum_critCount_eq_eulerNumberM (402); exists_eulerNumber_of_compactSpace (412); IsEulerNumber.eq_eulerNumberM (422); eulerNumberM_eq_of_diffeomorph (438); exists_isEulerNumber_eulerNumberM (458); reeb_of_compactSpace (481); IsStrongDeformationRetract.preimage_homeomorph (510); theorem_3_1_of_compactSpace (536).

Morse/Chapter4/WhitneyInvariance.lean. isCriticalPointM_comp_symm_iff (74); setOf_isCriticalPointM_comp_symm (104); diffeoCoord (118); isSmoothCoord_diffeoCoord (135); hessianM_comp_symm_eq (149); mfderiv_chartAt_symm_self_apply (161); hessianM_comp_symm_apply (172); isNondegenerate_hessianM_comp_symm_iff (198); isNondegenerateCriticalPointM_comp_symm_iff (206); indexM_comp_symm (219); nullityM_comp_symm (227); IsMorseFunction.comp_diffeomorph_symm (237); isMorseFunction_comp_symm_iff (246); critCount_comp_symm (255); ncard_setOf_isCriticalPointM_comp_symm (275).

Morse/Chapter4/WhitneySlice.lean. exists_continuousLinearEquiv_coprod (85); exists_sliceChart_range (141); isSliceSubmanifold_range (259); contMDiff_ofInjective_symm (271); diffeomorphRange (314); exists_diffeomorph_range (336).

Morse/Chapter5/ComplexIndex.lean. IsOrthComplexStructure (60); IsOrthComplexStructure.inner_left (64); IsComplexSubmanifold (74); IsComplexSubmanifold.inner_J_eq_zero (80); Jc (92); fderiv_sliceParam_Jc (96); differentiableAt_Jc_apply (115); inner_fderiv_fderiv_sliceParam_Jc (134); two_mul_indexM_normSq_sub_le (169).

Morse/Chapter5/HyperplaneSection.lean. index_neg_add_index (72); hessianM_neg (93); isCriticalPointM_neg_iff (102); subsingleton_subsetHomology_of_homeomorph (124); relHomology_map_eq_zero_of_deform (156); HypothesisN (195); isNondegenerate_neg_iff (199); hessian_neg (206); le_indexM_neg_normSq_sub (215); subsingleton_subsetHomology_univ_of_section (252); corollary_7_3_of_section (402).

Morse/Chapter5/IndexBound.lean. nullity_bilinearComp_equiv (139); nullity_neg (156); complexStructureEquiv (163); nullity_sub_smul_eq_nullity_add_smul (174); nullity_sub_smul_eq_nullity_add_smul' (187); apply_J_apply_J_eq (200); two_mul_index_le_finrank (210); realMulI (242); realMulI_realMulI (248); realMulI_injective (251); index_le_finrank_complex (257); reBilin (265); reBilin_apply (269); reBilin_I_smul (275); index_le_finrank_of_eq_sub_re (282); apply_I_smul_apply_I_smul_eq (293); exists_pos_apply_of_eq_sub_re (302); reInner (325); reInner_apply (329); reInner_self (332); re_inner_real_smul_left (337); reInner_I_smul (343); inner_eq_zero_of_re_inner_eq_zero (351); nullity_reBilin_sub_smul_reInner_eq (363); eigenspace_eq_nullSpace_sub_smul_reInner (371); finrank_eigenspace_neg_eq (391); apply_I_smul_apply_I_smul_eq_two_mul_norm_sq_sub (406); index_le_finrank_of_hermitian (416); index_hessian_le_finrank_complex (437); reInner_comm (452); hessian_normSq_sub_holomorphic (462); index_hessian_normSq_sub_holomorphic_le (508);

nullity_hessian_normSq_sub_add_eq_sub (527).

Morse/Chapter5/Lefschetz.lean. critValuesIn (82); subsingleton_subsetHomology_sublevel (92); exists_regular_levels (201); indexM_normSq_sub_le (262); theorem_7_2 (276); theorem_7_1 (314).

Morse/Chapter5/LefschetzHomotopy.lean. lemma_D_slab (84); pushesInto_univ_of_isClosed (216); sectionFun (281); sectionFun_of_mem (284); sectionFun_apply (288); sectionFun_nonneg (293); continuous_sectionFun (300); theorem_7_4_of_section (349); compresses_val_of_pushesInto (491); bijective_homotopyGroupMap_val_of_pushesInto (507); theorem_7_4_of_embedding (531); theorem_7_4 (551); theorem_7_4_bijective (559); theorem_7_4_surjective (573).

Morse/Chapter5/LefschetzSlab.lean. sublevel_step_critical_of_slab (63); sublevel_step_of_slab (162); subsingleton_subsetHomology_slab (250).

Morse/Chapter5/ProjectiveSection.lean. CP.Cn (61); CP.aff (65); CP.lift (69); CP.nc (72); CP.nc_ne_zero (74); CP.proj (78); CP.sum_lift (81); CP.proj_mem (90); CP.proj_zero_zero (108); CP.aff_proj (111); CP.zero_zero_eq (119); CP.proj_aff_and_norm (127); CP.finrank_Cn (166); CP.theta (170); CP.mulI (174); CP.cpxJ (178); CP.cpxJ_apply (182); CP.isOrthComplexStructure_cpxJ (184); CP.continuous_lift (196); CP.continuous_proj (201); CP.continuousOn_aff (209); CP.affinePart (218); CP.hyperplaneSection (221); CP.proj_zero_zero_ne (225); CP.affineHomeomorph (229); CP.tendsto_norm_affineHomeomorph_symm (253); corollary_7_3 (291).

Morse/Chapter6/Connection.lean. cov_smul_left (91); cov_add_right (96); cov_smul_right (104); IsSymmetric (120); isSymmetric_iff_torsion_eq_zero (125); milnorCurvature (140); nomizuCurvature (146); milnorCurvature_eq_neg (151); milnorCurvature_swap (157); IsCompatible (178); isCompatible_iff_isMetricCompatible (185); koszul (194); two_mul_inner_cov_eq_koszul (204); cov_eq_of_isSymmetric_of_isCompatible (216); cov_eq_of_torsion_eq_zero_of_isMetricCompatible (234).

Morse/Chapter6/CurvatureAlgebra.lean. octahedron (108); inner_add_inner_eq_zero_iff (139); IsCurvatureTensor (151); IsCurvatureTensor.apply_swap (163); IsCurvatureTensor.apply_self (166); IsCurvatureTensor.inner_swap (172); IsCurvatureTensor.inner_apply_self (176); IsCurvatureTensor.inner_bianchi (180); IsCurvatureTensor.inner_comm_pairs (185); IsCurvatureTensor.neg (192); IsCurvatureTensor.smul (203); IsCurvatureTensor.inner_neg_apply_swap (212); sectionalCurvature (225); gram_smul_add (229); gram_pos (237); IsCurvatureTensor.apply_smul_add (262); IsCurvatureTensor.inner_apply_smul_add (273); IsCurvatureTensor.inner_apply_self_eq (282); IsCurvatureTensor.sectionalCurvature_smul_add (290); IsCurvatureTensor.inner_apply_self_eq_zero_of_not_linearIndependent (297); mixedDiff (322); inner_apply_self_eq_zero_or_exists_orthonormal (330); IsCurvatureTensor.mixedDiff_eq (372); IsCurvatureTensor.polarization (379); IsCurvatureTensor.ext (392); IsCurvatureTensor.ext_of_sectionalCurvature (404); IsCurvatureTensor.ext_of_orthonormal (418); curvatureOne (431); curvatureOne_apply (438); isCurvatureTensor_curvatureOne (443); inner_curvatureOne_self (454); sectionalCurvature_smul_curvatureOne (460); neg_smul_curvatureOne_apply (468); IsCurvatureTensor.eq_smul_curvatureOne_of_orthonormal (479); IsCurvatureTensor.apply_eq_of_sectionalCurvature_eq (489).

Morse/Chapter6/LinearODE.lean. exists_nnnorm_le_Icc (55); linearODE_eqOn (63); linearODE_eq (72); radTrunc (92); radTrunc_of_norm_le (94); norm_radTrunc_le (97); lipschitzWith_radTrunc (104); exists_skewLinearODE_Ioo (151); exists_skewLinearODE (205).

Morse/Chapter6/Parallel.lean. velocity (94); IsFieldAlong (97); covDeriv (101); covDeriv_mem (106); covDeriv_add (112); covDeriv_smul (119); covDeriv_comp (128); hasDerivAt_inner_covDeriv (136); IsCovDerivAlong (148); isCovDerivAlong_covDeriv (158); IsCovDerivAlong.eq_sum (170); IsCovDerivAlong.eq_covDeriv (215); IsParallel (224); dProjAlong (230); hasDerivAt_tanProj_along (235); deriv_tanProj_along (239); contDiff_dProjAlong (242); continuous_dProjAlong (246); IsParallel.hasDerivAt (250); IsParallel.eq (260); skewAlong (265); inner_skewAlong_self (269);

exists_isParallel (275); exists_unique_isParallel (317); IsParallel.contDiff (323); IsParallel.inner_eq (340); IsParallel.add (351); IsParallel.const_smul (360); parallelField (376); parallelField_spec (379); IsParallel.eq_parallelField (386); parallelTransportL (392); parallelTransportL_apply (417); surjective_parallelTransportL (421); parallelTransport (430); parallelTransport_apply (435); parallelTransport_self (440); parallelTransport_trans (446); inner_parallelTransport (456).

Morse/Chapter6/Projection.lean. transposeL (87); transposeL_apply (90); gramL (94); gramL_apply (96); gramL_injective (98); finrank_clmDual_eq (108); gramL_bijective (112); gramEquiv (119); coe_gramEquiv (123); gramL_inverse_apply (127); gramProj (134); gramProj_eq_starProjection (137); contDiffAt_gramProj (151); gramProj_eq_starProjection' (163); sliceAt (184); mem_sliceAt_source (186); chartAt_eq_sliceAt (188); tanCoord (194); tanCoord_mfderiv_val (197); mfderiv_val_tanCoord (220); dN (230); dN_apply (234); dN_congr (238); dN_const (243); dN_comp_val (250); dN_eq_fderiv (265); fderiv_eq_zero_of_eventually (271); exists_differentiableAt_rep (280); exists_contDiffOn_rep (299); dN_add (326); dN_neg (336); dN_sub (343); dN_clm_comp (353); dN_bilinear (366); dN_smul (389); dN_inner (397); dN_clm_apply (405); dN_clm_comp' (413); dN_val (422); tanProj (433); tanProj_apply_mem (435); tanProj_eq_self (438); tanProj_eq_self_iff (441); tanProj_tanProj (444); tanProj_comp_tanProj (447); inner_tanProj_left (452); inner_tanProj_of_mem (455); tanProj_eq_zero_iff (459); sub_tanProj_mem_orthogonal (463); tanProj_sub_tanProj (467); eq_zero_of_mem_of_mem_orthogonal (471); injective_fderiv_sliceParam (478); tanProj_eq_gramProj (503); projRep (512); projRep_eq (515); contDiffAt_projRep (520); contMDiff_tanProj (534); mdifferentiableAt_tanProj (547); dN_tanProj_eq_fderiv (554); inner_dN_tanProj_comm (568); dN_tanProj_comp_add (587); dN_tanProj_apply_add (596); tanProj_dN_tanProj (603); tanProj_dN_tanProj_of_mem (612); dN_tanProj_mem_of_mem_orthogonal (618); dN_tanProj_mem_orthogonal_of_mem (626); contMDiff_curve (640); deriv_mem_tangentSubspace (646); hasDerivAt_comp_curve (661); contDiff_comp_curve (676); contDiff_tanProj_comp (684); hasDerivAt_tanProj_comp (689); contMDiff_map (706); fderiv_mem_tangentSubspace (712); hasFDerivAt_comp_map (726); contDiff_comp_map (746).

Morse/Chapter6/SphereCurvature.lean. mem_tangentSubspace_sph_iff (41); inner_self_sph (45); tanProj_sph (49); tanProj_sph_self (58); secondFundForm_sph (63); curvOp_sph (92); curvTensor_sph (111); sectionalCurvature_sph (118); inner_curvTensor_sph_orthonormal (125).

Morse/Chapter6/Submanifold.lean. IsTangentField (117); projField (120); isTangentField_projField (122); contMDiff_projField (125); projField_apply_of_mem (133); covAt (141); leviCivita (145); bracket (150); covAt_mem (152); leviCivita_mem (155); isTangentField_leviCivita (158); leviCivita_add_left (162); leviCivita_smul_left (167); leviCivita_add_right (172); leviCivita_smul_right (180); dN_inner_eq_covAt (191); contMDiff_dN_apply (203); contMDiff_leviCivita (224); secondFundForm (241); secondFundForm_eq_sub_tanProj (246); sub_tanProj_dN_eq_of_eq (258); tanProj_secondFundForm (267); secondFundForm_mem_orthogonal (272); inner_secondFundForm_comm (277); secondFundForm_add_left (282); secondFundForm_smul_left (287); secondFundForm_add_right (292); secondFundForm_smul_right (297); secondFundForm_sub_right (302); tanProj_secondFundForm_dN (308); secondFundForm_comm (330); bracket_mem (382); leviCivita_sub_leviCivita (394); dN_bracket (404); IsSymmCompat (449); isSymmCompat_leviCivita (464); IsSymmCompat.two_mul_inner (479); IsSymmCompat.eq_leviCivita (497); curvature (518); curvOp (524); curvOp_apply (529); curvature_eq (540); inner_curvOp (600); gauss_equation (609); curvOp_mem (623); curvOp_add_left (632); curvOp_smul_left (639); curvOp_add_right (646); curvOp_smul_right (653); curvTensor (661); coe_curvTensor_apply (682); curvature_eq_curvTensor (687); curvature_congr (697); isCurvatureTensor_curvTensor (715); curvature_add_swap (733); curvature_bianchi (740); inner_curvature_add_inner (752); inner_curvature_comm_pairs (767).

Morse/Chapter6/Surface.lean. partialX (61); partialY (64); hasDerivAt_partialX (67); hasDerivAt_partialY (73); contDiff_partialX (78); contDiff_partialY (81); partialY_partialX_eq (85); partialX_partialY_eq (93); partialY_partialX (102); partialX_clm_apply (109); par-

tialY_clm_apply (116); velX (135); velY (138); covDerivX (142); covDerivY (146); covDerivX_eq (150); covDerivY_eq (155); velX_mem (163); velY_mem (167); covDerivX_velY (172); covDerivY_covDerivX_sub (181).

Morse/Chapter7/Distance.lean. pathLengths (71); riemDist (77); riemDist_le_pathLength (80); riemDist_self (86); riemDist_eq_top_of_notMem (94); pathLengths_comm (100); riemDist_comm (108); riemDist_triangle (113); NormalCoord.ofReal_le_riemDist (157); NormalCoord.riemDist_exp_le (167); NormalCoord.riemDist_exp (186); NormalCoord.riemDist_lt_iff (197); eq_of_riemDist_eq_zero (229); hasBasis_nhdsWithin_riemDist (250); IsShortJoin (302); exists_isShortJoin (314); corollary_10_8 (369).

Morse/Chapter7/Exp.lean. IsGeodesicFrom (72); expDom (77); exp (82); IsGeodesicFrom.mem_expDom (87); IsGeodesicFrom.mem (92); isGeodesicFrom_of_isGeodesicOn (100); IsGeodesicFrom.eqOn (115); IsGeodesicFrom.exp_eq (129); exp_eq_of_mem_expDom (136); isGeodesicFrom_const (141); exp_zero (146); zero_mem_expDom (150); IsGeodesicFrom.exp_smul (156); IsGeodesicFrom.isGeodesicOn_exp_smul (189); exists_exp_eq (202); exists_isOpen_exp (221); contMDiffOn_exp (245); hasFDerivAt_exp_zero (270).

Morse/Chapter7/GaussLemma.lean. hasDerivAt_norm_of_ne (67); hasDerivAt_mem_of_eventually_mem (76); NormalCoord.contDiffAt_F (91); NormalCoord.hasFDerivAt_F (96); NormalCoord.eventually_F_eq (102); NormalCoord.norm_fderiv_F_apply (116); NormalCoord.fderiv_G_fderiv_F (153); NormalCoord.injective_fderiv_F (175); NormalCoord.inner_fderiv_F (181); NormalCoord.hasDerivAt_radius (306); gauss_lemma (390); gauss_lemma_orthogonal (413).

Morse/Chapter7/Geodesic.lean. tangentSubspace_congr (75); IsGeodesicOn (86); IsGeodesicOn.mono (114); IsGeodesicOn.congr (122); IsGeodesicOn.comp_mul_add (133); isGeodesicOn_const (157); hasDerivAt_mem_tangentSubspace (170); hasDerivAt_projRep_comp (191); IsGeodesicOn.deriv_mem (206); IsGeodesicOn.tanProj_deriv_deriv (212); IsGeodesicOn.deriv_deriv_eq (221); IsGeodesicOn.hasDerivAt_norm_sq (239); IsGeodesicOn.norm_deriv_eq (245); IsGeodesicOn.eventuallyEq (256); IsGeodesicOn.eqOn (291); isGeodesicOn_of_deriv_deriv_eq (310); isGeodesicOn_univ_iff_covDeriv (326); geodesicSpray (340); IsGeodesicOn.mem_tangentBundleSet (344); isGeodesicOn_iff_hasDerivAt_geodesicSpray (352).

Morse/Chapter7/GeodesicFlow.lean. isPreconnected_preimage_mul_add (75); IsGeodesicOn.exp_smul_eq (88); IsGeodesicOn.eqOn_exp_smul (111); geoFlow (122); exists_localFlow_exp (131); exists_isGeodesicOn_exp_smul (241); hasDerivAt_exp_smul_zero (249); geoFlow_zero (254); isGeodesicOn_exp_smul_union (267); eventually_mem_expDom_and_continuousWithinAt_exp (365); continuousOn_exp (550); IsGeodesicallyComplete (562); isGeodesicOn_univ_exp_smul (567); isGeodesicallyComplete_iff (607); isGeodesicallyComplete_of_isRiemComplete (643).

Morse/Chapter7/GeodesicLocal.lean. pinvL (70); gramProj_eq_comp_pinvL (73); pinvL_apply_self (75); contDiffAt_pinvL (79); sliceChart_sliceParam (109); fderiv_sliceChart_comp_fderiv_sliceParam (115); fst_fderiv_sliceChart_fderiv_sliceParam (131); tangentSubspace_sliceParam (140); fderiv_sliceParam_fst_fderiv (149); chartAcc (162); chartSpray (166); chartAcc_smul (169); chartAcc_zero (173); contDiffOn_chartSpray (178); hasDerivAt_sliceParam_chart (196); isGeodesicOn_chart (208); LocalSpray (247); nonempty_localSpray (260); LocalSpray.c (273); LocalSpray.good (279); LocalSpray.isOpen_good (282); LocalSpray.mem_good (285); LocalSpray.mem_ball_of_good (289); LocalSpray.mul_mem_Ioo (300); LocalSpray.chartCurve (306); LocalSpray.geo (311); LocalSpray.hasDerivAt_chartCurve (320); LocalSpray.chartCurve_zero (338); LocalSpray.isGeodesicOn_geo (344); LocalSpray.geo_zero (351); LocalSpray.hasDerivAt_geo_zero (357); LocalSpray.contDiffOn_geo (367); lemma_10_2 (383).

Morse/Chapter7/HopfRinow.lean. IsMinimalGeodesic (101); exp_mem_of_mem_expDom (109); NormalCoord.exists_riemDist_eq_add (129); broken_geodesic_eq (208); isMinimalGeodesic_exp_smul (281); theorem_10_9 (308); theorem_10_9' (449); isCompact_closure_of_isRiemBounded (464); isRiemComplete_of_forall_isCompact_closure (513); corollary_10_10 (548); isCom-

pact_of_isClosed_of_isRiemBounded (571); hopf_rinow (590); hopf_rinow_of_isRiemComplete (607); hopf_rinow_of_isClosed (625); hopf_rinow_of_isCompact (638); sphere_hopf_rinow (651).

Morse/Chapter7/Metric.lean. eventually_riemDist_lt (69); ofReal_norm_sub_le_riemDist (78); continuousOn_riemDist (90); continuousOn_riemDist_left (126); continuous_riemDist (136); riemDist_ne_top (147); IsRiemComplete (177); IsRiemBounded (185); tendsto_riemDist_of_tendsto (190); isRiemComplete_of_isClosed (203); riemEMetricSpace₀ (222); topology_eq_riemEMetricSpace₀ (232); riemEMetricSpace (262); riemMetricSpace (269); riemMetricSpace_dist (274); riemMetricSpace_edist (278); riemMetricSpace_topology (282); isRiemComplete_iff_completeSpace (288).

Morse/Chapter7/Minimal.lean. isGeodesicOn_of_locally (61); minimal_segment (76); corollary_10_7 (207); corollary_10_7_extend (238).

Morse/Chapter7/NormalNbhd.lean. NormalCoord (83); IsNormalRadius (108); LocalSpray.fst_mem_target (117); LocalSpray.mem_ball_smul (122); LocalSpray.φ_smul (134); LocalSpray.chartExp (167); LocalSpray.contDiff_chartExp (170); LocalSpray.chartExp_mem (175); LocalSpray.chartExp_zero (180); LocalSpray.isGeodesicFrom_geo (201); LocalSpray.exp_eq_chartExp (207); LocalSpray.hasDerivAt_chartExp_line (213); LocalSpray.chartF (233); LocalSpray.jacobian (236); LocalSpray.hasFDerivAt_chartF (246); lemma_10_3 (320); lemma_10_3_geodesic (611); exists_isNormalRadius (626); contMDiffAt_of_val_eventuallyEq (635); expToN (661); coe_expToN (665); NormalCoord.exp_mem (674); NormalCoord.coe_orthogonalProjectionOnto_G (681); NormalCoord.expDiffeo (690); exists_expDiffeo (748).

Morse/Chapter7/ODE.lean. eqOn_of_eventuallyEq (56); hasDerivAt_fst_comp (83); hasDerivAt_snd_comp (87); ode_eventuallyEq (94); ode_eqOn (115); exists_localFlow (134); theorem_10_1 (193); theorem_10_1_unique (221).

Morse/Chapter7/Path.lean. ae_uIoc_of_forall_Ioo_diff (57); pathLength (83); IsAdmissible (87); IsAdmissible.mono (102); IsAdmissible.integral_deriv (117); IsAdmissible.pathLength_nonneg (122); IsAdmissible.norm_sub_le_pathLength (127); IsAdmissible.pathLength_add (132); IsAdmissible.pathLength_le (137); IsAdmissible.eq_of_pathLength_eq_zero (147); IsAdmissible.continuousOn_pathLength (156); IsAdmissible.monotoneOn_pathLength (163); pathLength_eq_of_norm_deriv (175); IsPiecewiseSmooth (191); IsPiecewiseSmooth.intervalIntegrable_of_contDiffOn (200); IsPiecewiseSmooth.isAdmissible (214); IsPiecewiseSmooth.of_contDiffOn (270); IsPiecewiseSmooth.mono (276); IsPiecewiseSmooth.concat (283); IsPiecewiseSmooth.reverse (309); IsPiecewiseSmooth.translate (323); pathLength_concat (338); pathLength_reverse (376); pathLength_translate (386).

Morse/Chapter7/Shortest.lean. IsGeodesicOn.pathLength_eq (81); NormalCoord.mem_U_self (92); NormalCoord.G_self (96); NormalCoord.G_ne_zero (100); NormalCoord.ballSet (109); NormalCoord.isOpen_ballSet (112); NormalCoord.mem_ballSet (116); NormalCoord.closedBallSet (120); NormalCoord.isCompact_closedBallSet (124); NormalCoord.mem_closedBallSet (130); NormalCoord.radial (145); NormalCoord.pathLength_exp_smul (159); NormalCoord.radius_sub_le_pathLength (170); NormalCoord.direction_eq_of_pathLength_eq (199); NormalCoord.exists_crossing (283); NormalCoord.norm_le_pathLength (375); NormalCoord.eq_exp_of_pathLength_eq (393); NormalCoord.image_eq_of_pathLength_eq (476); theorem_10_4 (513).

Morse/Chapter8/Critical.lean. IsCriticalPath (72); IsUnbrokenGeodesic (78); isCriticalPath_iff (83); IsUnbrokenGeodesic.firstVariation_eq_zero (102); contDiff_smul_of_tsupport_subset (120); bumpAt (134); isGeodesicOn_of_accel (138); accel_eq_zero_of_critical (153); prj_jump (230); jump_eq_zero_of_critical (253); liftField (339); IsGeodesicOn.hasDerivAt_lift (342); contDiffAt_liftField (353); exists_isGeodesicOn_through (365); isGeodesicOn_of_glue (410); corollary_12_3 (517).

Morse/Chapter8/Energy.lean. IsPiecewiseSmooth.map₂ (102); IsPiecewiseSmooth.add (114); IsPiecewiseSmooth.smul (118); IsPiecewiseSmooth.clm_apply (122); IsPiecewiseSmooth.derivWithin_piece

(135); jump (152); jump_eq_zero (154); jump_congr (158); derivWithin_Ici_eq_of_piece (164); derivWithin_Iic_eq_of_piece (170); derivWithin_eq_deriv_of_piece (178); prj (195); prj_mem (206); prj_eq_self_iff (210); prj_prj (215); inner_prj_left (221); inner_prj_right_of_eq (228); hasFDerivWithinAt_mem_tangentSubspace (239); hasDerivWithinAt_mem_tangentSubspace (265); contDiffWithinAt_prj_comp (275); ContDiffOn.prj_comp (291); IsPiecewiseSmooth.prj_comp (298); pathSpace (313); energy (318); IsPiecewiseSmooth.intervalIntegrable_norm_deriv_sq (327); integral_norm_deriv_sub_sq (337); pathLength_sq_le_energy (351); energy_eq_of_norm_deriv (367); pathLength_sq_eq_energy_iff (382); pathTangent (441); mem_pathTangent (457); IsVariation (473); IsVariation.isPiecewiseSmooth (488); IsVariation.mem_pathSpace (499); IsVariation.comp (507); varField (523); IsVariation.hasDerivAt_param (527); IsVariation.prj_varDeriv (537); IsVariation.varField_mem (549).

Morse/Chapter8/FirstVariation.lean. covD (75); accel (79); covD_eq_covDeriv (83); firstVariation (90); firstVariation_congr_field (95); mem_pathTangent_congr (107); piece_deriv (121); piece_varDeriv (126); piece_deriv_deriv (132); piece_deriv_varDeriv (142); piece_derivWithin_Ici (154); piece_derivWithin_Iic (159); IsVariation.differentiableAt (184); finsum_mem_Ioo_eq_sum (190); finsum_jump_eq_sum (204); IsVariation.differentiableAt_varDeriv (211); IsVariation.hasDerivAt_energy (225); first_variation (377); exp_mem_of_mem_expDom (410); exists_variation (429); IsPiecewiseSmooth.smulRight (508); exists_variation_one (515); exists_variation_two (537).

Morse/Chapter8/Hessian.lean. continuousWithinAt_curvR (78); hessForm (110); PieceData (120); PieceData.D (134); PieceData.contDiffOn_D (143); PieceData.D_eq (148); PieceData.covD_covD_eq (153); PieceData.continuousOn_curv (168); PieceData.integral_eq (184); exists_pieceData (226); IsPiecewiseSmooth.ae_differentiableAt (242); secondVariation_eq_hessForm (259); inner_curvR_comm (321); intervalIntegrable_hessIntegrand (329); hessForm_comm (342); covD_add (361); covD_smul (368); hessForm_add_right (375); hessForm_smul_right (387); energyHessian (405); energyHessian_apply (432); energyHessian_isSymm (438); energyHessian_eq_integral (447); IsVariation.partial₁_mem (460); IsVariation.partial₂_mem (468); hasDerivAt_energyHessian (477); secondVariation_congr (486); hasDerivAt_deriv_energy (510); IsMinimalPath (523); energy_congr (528); energy_le_of_isMinimalPath (538); lemma_13_6 (557).

Morse/Chapter8/Jacobi.lean. contDiffOn_of_ode (80); hasDerivAt_of_hasDerivAt_punctured (106); contDiffAt_curvR (135); jacobiExpr (165); IsJacobiField (171); IsJacobiField.mem_pathTangent (180); IsJacobiField.secondVariation_eq_zero (187); IsUnbrokenGeodesic.prj_deriv (208); prj_jacobiExpr (213); contDiffOn_jacobiExpr (224); jacobiExpr_eq_zero_of_null (246); jump_eq_prj_jump (317); jump_eq_zero_of_null (342); deriv_deriv_eq_of_jacobiExpr (439); smooth_near_of_null (502); theorem_14_1 (681).

Morse/Chapter8/Piecewise.lean. piecewise_induction (47); sum_filter_Ioo_split (87); exists_piece_left (113); exists_piece_right (129); exists_piece_around (145); exists_piece_mem (156); continuousOn_of_piecewise (172); intervalIntegrable_of_piecewise (189); integral_eq_of_piecewise (209); pdU (235); pdT (239); uniqueDiffOn_strip (243); contDiffOn_pdU (253); contDiffOn_pdT (257); hasDerivAt_pdU (263); hasDerivWithinAt_pdT (276); hasDerivAt_pdT (285); derivWithin_Ici_eq_pdT (291); derivWithin_Iic_eq_pdT (298); deriv_eq_pdU (310); pdU_pdT (317); hasDerivAt_integral_strip (348); hasDerivAt_integral_of_piecewise (382).

Morse/Chapter8/SecondVariation.lean. curvR (80); curvR_of_mem (83); partialX_partialY_of_contDiffOn (102); curvature_identity (143); secondVariation (247); IsVariation.hasDerivAt_jump (256); IsVariation.exists_param₂ (297); second_variation (337); second_variation_eq (520); remark_13_5 (560).

Morse/Chapter9/BrokenJacobi.lean. fieldsUpTo (72); fieldsUpTo_mono (79); fieldsUpTo_apply_self (84); IsFine (106); exists_node_piece (112); exists_isFine (178); IsBrokenJacobi (228); IsBrokenJacobi.eqOn_zero (235); exists_isBrokenJacobi (256); brokenSpace (319); nodeZero (335); brokenSpace_le_fieldsUpTo (345); nodeZero_le_fieldsUpTo (349); IsBrokenJacobi.mem_pathTangent

(354); IsBrokenJacobi.eventuallyEq (386); energyHessian_broken_eq (408); energyHessian_broken_nodeZero (428); energyHessian_nodeZero_pos (436).

Morse/Chapter9/FormIndex.lean. IsNegDefOn (50); IsPosDefOn (54); formIndex (59); le_formIndex (62); formIndex_le (67); finrank_map_of_injOn (73); formIndex_le_of_map (85); formIndex_restrict_le (101); formIndex_smul (106); formIndex_eq_restrict_of_isCompl (116); ker_eq_map_ker_restrict (156); exists_isNegDefOn_of_formIndex_eq (195); formIndex_eq_of_equiv (215); formIndex_eq_sigNeg (246); radical_toQuadraticMap (256); exists_isPosDefOn_add (280); finrank_add_finrank_le (292); eventually_isNegDefOn (312); eventually_isPosDefOn (368).

Morse/Chapter9/GeodesicVariation.lean. exists_smooth_ray (86); geoFlow_eq_exp (131); exists_midpoint_ray (137); rays (269); RaysGood (274); RaysGood.mono (284); raysGood_zero (293); deriv_rays_eq (326); RaysGood.step (336); raysGood_of_isGeodesicOn (446); exists_initial_ray (494); lemma_14_4 (529); jacEvalEnds_bijective_of_not_isConjugate (659).

Morse/Chapter9/IndexAssertions.lean. le_formIndex_of_compound (62); max_lt_of_isFine (155); assertion_3 (162); assertion_4_le (187); energyHessian_cut (218); exists_dual (267); assertion_4_ge (298); assertion_4 (414).

Morse/Chapter9/IndexContinuity.lean. continuousWithinAt_of_injective (65); validFor (111); brokenOf (117); brokenOf_mem (121); brokenOf_apply_node (126); Hform (137); Hform_apply (145); Hform_isSymm (152); formIndex_Hform (161); finrank_ker_Hform (169); isFine_of_isFine (196); Hform_eq_jumps (218); brokenOf_eqOn (230); continuousWithinAt_jump (272); continuousWithinAt_Hform (379); eventually_isNegDefOn_within (401); eventually_isPosDefOn_within (411); mem_validFor_of_near (422).

Morse/Chapter9/IndexForm.lean. jacobiExpr_congr (65); isJacobiField_zero (74); energyHessian_eq_jumps (87); IsSmoothCurveIn.mem_pathSpace (127); exists_jacBound (130); jacBound (142); jacBound_nonneg (144); le_jacBound (146); fineMesh (152); fineMesh_pos (154); integrand_ge (160); integral_norm_sq_le (208); hessForm_ge (241); eqOn_zero_of_hessForm (276); isPiecewiseSmooth_indicator (314); indicator_mem_pathTangent (349); eq_zero_of_short (365); jacEvalEnds (425); jacEvalEnds_bijective (435); exists_jacobi_ends (449).

Morse/Chapter9/IndexReduction.lean. exists_brokenSpace_eq (54); isCompl_broken_nodeZero (85); lemma_15_4 (112); lemma_15_4_ker (129); nodeSpace (143); nodeEval (150); nodeEval_bijective (159); nodeEquiv (180); finiteDimensional_brokenSpace (184); eq_jacobi_of_jumps_zero (194); jacobiSpace_eq_zero_of_eqOn (255); finrank_ker_broken (269); indexUpTo (360); indexUpTo_mono (365); indexUpTo_le_finrank (368); indexUpTo_lt_top (376); indexUpTo_eq_zero (381); indexUpTo_one (419).

Morse/Chapter9/IndexTheorem.lean. eq_finsum_of_jumps (58); indexUpTo_toNat (207); indexUpTo_one_eq (213); theorem_15_1 (243); corollary_15_2 (267); remark_14_6 (277).

Morse/Chapter9/JacobiODE.lean. norm_radTrunc_le_norm (45); norm_le_of_norm_deriv_le (51); exists_linearODE_Ioo (63); exists_linearODE (137); linSol (172); linSol_self (181); hasDerivAt_linSol (184); eq_linSol (189); linSol_add (193); linSol_smul (201); continuous_linSol (213); eqOn_Icc_linSol (219); linFlow (242); continuous_linFlow (251).

Morse/Chapter9/JacobiSpace.lean. IsJacobiField.congr (76); isJacobiField_iff (98); jacobiSpace (112); mem_jacobiSpace (134); IsJacobiField.comp_projIcc_mem (138); jacobiSpace_eq_zero (143); covAlong_add (150); covAlong_smul (157); jacEval (162); jacEval_bijective (178); jacEquiv (207); finrank_jacobiSpace (216); exists_jacobi (222); IsConjugate (235); conjSpace (241); mem_conjSpace (245); conjMultiplicity (259); isConjugate_iff (263); restrictIcc (287); hessNullSpace (298); hessNullity (304); hessNullSpace_eq (309); theorem_14_1_nullity (335); hessNullity_pos_iff (343); finrank_jacobiSpace_zero (351); isJacobiField_smul_vel (379); isJacobiField_const_curve (416); eq_const_of_velAlong_one_eq_zero (435); remark_14_2 (474).

Morse/Chapter9/JacobiSystem.lean. IsSmoothCurveIn (71); IsUnbrokenGeodesic.isSmoothCurveIn (77); tanAt (85); tanAt_of_mem (88); mem_tanAt (92); finrank_tanAt (102); velAlong (117); dpA–

long (120); covAlong (124); curvMid (128); curvAlong (134); jacS (138); jacSys (142); jacA (150); jacSys_apply (160); curvAlong_apply (169); IsSmoothCurveIn.contDiffOn_prj (179); IsSmoothCurveIn.velAlong_mem (211); IsSmoothCurveIn.dp_eq (218); IsSmoothCurveIn.prj_dp_prj (233); IsSmoothCurveIn.jacS_of_tangent (241); IsSmoothCurveIn.prj_jacS_of_tangent (249); IsSmoothCurveIn.jacS_sub_prj (257); IsSmoothCurveIn.prj_curvAlong (267); IsSmoothCurveIn.contDiffOn_curvAlong (276); IsSmoothCurveIn.contDiffOn_jacSys (312); IsSmoothCurveIn.continuous_jacA (327); contDiffOn_Icc_of_hasDerivAt (335); jacSol (359); jacSol_add (373); jacSol_smul (377); jacSol_zero (381); jacSol_mem_tanAt (387); contDiffOn_jacSol (449); isJacobiField_jacSol (456); IsJacobiField.eqOn_jacSol (494); IsJacobiField.eqOn_zero (561).

Morse/Chapter9/JacobiVariation.lean. IsJacobiField.congr_curve (58); lemma_14_3 (83); IsNormalRadius.isGeodesicOn_ray (166); IsNormalRadius.isSmoothCurveIn_ray (174); fderiv_extension_exp_zero (184); remark_14_5_of_isNormalRadius (207); remark_14_5 (355); remark_14_5_bijective (368).

Morse/Flow/Global.lean. IsBddLipschitz (77); IsBddLipschitz.of_contDiff (84); IsBddLipschitz.smul (94); eqOn_Ioo_of_hasDerivAt (105); IsBddLipschitz.isIntegralCurve_eq (114); IsBddLipschitz.exists_local (125); IsBddLipschitz.exists_isIntegralCurve (143); flow (171); IsBddLipschitz.flow_spec (176); IsBddLipschitz.flow_zero (183); IsBddLipschitz.isIntegralCurve_flow (190); IsBddLipschitz.hasDerivAt_flow (194); IsBddLipschitz.eq_flow (199); IsBddLipschitz.flow_add (206); IsBddLipschitz.flow_comp (211); IsBddLipschitz.flow_neg_flow (216); IsBddLipschitz.flow_flow_neg (221); IsBddLipschitz.flow_nat_mul (226); IsBddLipschitz.flow_smul (238); IsBddLipschitz.flow_of_eq_zero (247); IsBddLipschitz.flow_of_notMem_tsupport (253); dist_le_of_hasDerivAt (261); IsBddLipschitz.dist_flow_le (271); IsBddLipschitz.lipschitzWith_flow (284); IsBddLipschitz.dist_flow_flow_le (292); IsBddLipschitz.continuous_flow (301); IsBddLipschitz.flowHomeomorph (336); IsBddLipschitz.flow_mem_of_eventually (359); IsBddLipschitz.flow_mem (378).

Morse/Flow/Nemytskii.lean. nem (48); nem_apply (52); continuous_nem (56); lipschitzWith_nem (60); norm_nem_le (67); pmul□ (77); norm_pmul□ _le (86); pmul (94); pmul_apply (100); norm_pmul_le (104); contDiff_pmul (107); dmap (111); dmap_apply (115); hasFDerivAt_nem (120); norm_pmul_nem_dmap_le (148); contDiff_nem_nat (158); contDiff_nem (177).

Morse/Flow/Smooth.lean. IsBddLipschitz.flowCurve (87); IsBddLipschitz.flowCurve_apply (90); IsBddLipschitz.volt_flowCurve (96); constCLM (114); constCLM_apply (124); contDiff_flow_one_of_lipschitz (131); contDiff_iterate (151); contDiff_flow (159); contDiff_flow_uncurry (185); flowDiffeomorph (237); coe_flowDiffeomorph (245); coe_flowDiffeomorph_symm (250); IsOneParameterGroup (259); Generates (267); IsOneParameterGroup.zero (272); IsOneParameterGroup.isIntegralCurve (286); lemma_2_4 (300).

Morse/Flow/Volterra.lean. primFun (47); primFun_apply (53); norm_primFun_le (57); prim□ (70); prim (84); prim_apply (88); norm_prim_le (92); norm_prim_apply_le (96); contDiff_prim (99); volt (111); volt_apply (114); dist_prim_nem_le (118); dist_le_dist_volt (126); volt_injective (138); contractingWith_volt (148); voltInv (154); volt_voltInv (158); voltInv_volt (166); voltHomeomorph (171); coe_voltHomeomorph (187); contDiff_volt (197); hasFDerivAt_volt (202); norm_prim_comp_pmul_lt (209); voltDeriv (222); coe_voltDeriv (228); contDiff_voltHomeomorph_symm (235).

Morse/Homology/AffineSimplex.lean. sum_smul_mem (54); aff (59); aff_vertex (71); convex_simplex (77); verts (80); aff_comp_aff (88); aff_verts (99); simplexMap_eq_aff (105); aff_comp_simplexMap (117); bary (125); coe_bary (127); aff_bary (133); realize (140); lreal (147); lreal_single (153); sbd_lreal (158); push_aff_lreal (163); bsd (171); bsdH (175); lmap_aff_bsd (178); lmap_aff_bsdH (183); map_mem_supported (197); mapDomain_mem_supported (214); vset (229); range_aff_subset (240); diam_range_aff_le (247); bary_mem_convexHull (252); dist_bary_le (258); Shrunk (284); Shrunk.mono (287); convexHull_vset_comp_subset (297); diam_vset_comp_le (302); vset_cons (307); bsd_mem_supported (317); bsd_pow_mem_supported (367); dist_aff_le (386); diam_range_aff (393);

diam_range_aff_le_of_mem_support (403); diam_vset_verts_le (411).

Morse/Homology/CellAttach.lean. TopTriple.isoBiprodOfRetraction (79); cellPairHom (106); cellAttachMap (114); relHomology_cellAttach_iso (126); relHomologyCellAttachIso (139); relHomologyCellAttachTopIso (149); isZero_relHomology_cellAttach (157); finite_and_finrank_relHomology_cellAttach (166); finrank_relHomology_cellAttach (179); cellsUnion (197); cellsUnion_insert (202); isClosed_cellsUnion (210); inter_cellsUnion_subset (217); cellsUnion_subset_insert (227); relHomologyCellsUnionInsertIso (238); mem_cellsUnion_iff (310); isZero_relHomology_cellsUnion_empty (321); finite_and_finrank_relHomology_cellsUnion (329); finrank_relHomology_cellsAttach (373); finiteDimensional_relHomology_cellsAttach (385); finrank_relHomology_cellsAttach_univ (399); isZero_relHomology_cellsUnion (411); isZero_relHomology_cellsAttach (441); nonempty_relHomology_cellsUnion_iso (454); relHomology_cellsAttach_iso (506).

Morse/Homology/CellExcision.lean. continuousOn_invFunOn_image (77); radialPush (104); radialPush_zero (106); radialPush_of_norm_eq_one (108); norm_radialPush (112); radialPush_ne_zero (120); radialPush_mem_closedBall (129); norm_radialPush_one (135); continuousOn_radialPush (140); isStrongDeformationRetract_sdiff_center (162); isStrongDeformationRetract_singleton_image (261); isIso_relHomology_map_of_isStrongDeformationRetract_snd (310); subsetPairHom_comp (323); excision_of_eq (338); cellPair (350); image_sphere_subset (355); isIso_relHomology_map_cellPair (366).

Morse/Homology/Excision.lean. exists_comp_eq_of_range_subset (62); push_injective (71); range_push (78); push_mem_supported_iff (92); coverOf (115); iUnion_interior_coverOf (117); mem_smallSet_coverOf (124); push_mem_inSet_preimage (137); excisionMap (147); push_mem_smallSet (154); pushSmall (161); pushSmall_ι (171); excisionMapSmall (174); excisionMapSmall_comp (179); bijective_excisionMapSmall_f (186); isIso_excisionMapSmall (233); quasiIso_excisionMap (241); quotπ (257); pushHom_comp_quotπ (261); concreteRelShortComplex_shortExact (274); mathlibRelShortComplexIso (300); mathlibRelShortComplex_shortExact (305); isColimitMathlibRelShortComplex (311); relChainsIsoQuot (320); relπ_relChainsIsoQuot_hom (325); relChainsIsoQuot_naturality (333); excision_of_eq_inter (370); excision_of_union (405); excision (415); excision_of_closure_subset_interior (436).

Morse/Homology/LinearChains.lean. tuples (65); tuples_δ (72); LinChain (89); lbd (94); lbd_single (96); lbd_lbd (102); lmap (106); lmap_lbd (114); cone (125); cone_single (129); aug (134); aug_lbd (140); cons_comp_succAbove_succ (148); lbd_cone_zero (156); lbd_cone_succ (170); lmap_cone (185); sd (206); sdH (213); sd_succ_single (227); sdH_succ_single (232); lbd_sd (239); lbd_sdH_zero (257); lbd_sdH_of_key (261); lbd_sdH (279); lmap_sd (291); lmap_sdH (306).

Morse/Homology/MayerVietoris.lean. chainsIn (59); chainsInIncl (63); chainsInToSmall (73); mvShortComplex (93); biprod_X_eq_add (103); inSet_inter (135); mvShortComplex_shortExact (140); subspaceChainsHom (212); bijective_subspaceChainsHom_f (227); subspaceChainsIso (244); exact_mk_of_iso (267); mathlibChainsIso (281); mathlibChainsIso_incl (291); mathlibChainsIso_val (299); isIso_homologyMap_smallChainsι (322); mvIso₁ (335); mvIso₂ (340); mvIso₃ (348); mvMap₁ (355); mvMap₂ (363); mvIso₁_hom_comp_mvMap₁ (391); mvIso₂_hom_comp_mvMap₂ (402); mvδ (412); mv_exact₂ (436); mv_exact₃ (444); mv_exact₁ (452).

Morse/Homology/ModuleComplex.lean. natHomMk (42); natHomotopyMk (51); subcomplex (87); subcomplexι (108); quotient (120); quotientπ (136); quotientπ_comp_eq_zero (149); shortExact_quotient (156); isColimitQuotient (168); subcomplexMap (178); quotientMap (197); quotientMap_id (215); quotientMap_comp (219); quotientHomotopy (229).

Morse/Homology/MorseHomology.lean. sublevel_step_critical (119); sublevel_step (211); exists_morse_filtration (274); exists_cells_filtration (342); critCount_eq_card (390); finiteDimensional_spaceHomology (404); bettiNumber_le_critCount (416); altSum_bettiNumber_le (430); sum_bettiNumber_eq_sum_critCount (447); bettiNumber_eq_zero_of_finrank_lt (473);

subsingleton_spaceHomology_of_finrank_lt (490); bettiNumber_eq_critCount_of_succ (500); bettiNumber_eq_critCount_of_zero (517); bettiNumber_eq_critCount_of_lacunary (532); exists_morse_filtration' (556); bettiNumber_le_critCount' (578); altSum_bettiNumber_le' (587); sum_bettiNumber_eq_sum_critCount' (598).

Morse/Homology/MorseHomologyBasic.lean. exists_separating_seq (95); exists_mem_Ioo_of_lt (153); eq_of_mem_Ioo_of_monotone (169); isCriticalPointM_of_forall_ge (180); LongExactSeq.bijective_g_of_subsingleton (197); LongExactSeq.equivOfSubsingleton (207); IsStrongDeformationRetract.preimage_subtype (216); card_filter_equiv (224); spaceHomology (243); bettiNumber (250); spaceHomologyEquivOfHomeomorph (256); bettiNumber_eq_of_homeomorph (262); subsetHomologyUnivEquiv (267); finrank_subsetHomology_eq_bettiNumber (276); finiteDimensional_spaceHomology_of_subsetHomology (282); CellAttachHomology (301); subsetHomologyEquivOfIsSDR (318); subsingleton_subsetHomology_of_isSDR (324); subsetHomologyEquivOfIsSDRBottom (333); finrank_subsetHomology_of_cells (346).

Morse/Homology/MorseHomologyCells.lean. cellAttachHomology (47).

Morse/Homology/MorseHomologyManifold.lean. contDiff_normSq_sub (105); finiteDimensional_spaceHomology_of_isCompact (111); bettiNumber_eq_zero_of_isCompact (120); IsEulerNumber.eq_sum_bettiNumber (143); exists_whitney_normSq (168); finiteDimensional_spaceHomology_of_compactSpace (190); bettiNumber_eq_zero_of_compactSpace (199); eulerNumberM_eq_sum_bettiNumber (208); exists_ambient_of_isMorseFunction (224); sum_bettiNumber_eq_sum_critCount_of_compactSpace (252); altSum_bettiNumber_le_of_compactSpace (260); bettiNumber_le_critCount_of_compactSpace (283); bettiNumber_eq_critCount_of_lacunary_of_compactSpace (291); bettiNumber_of_ncard_eq_two (305); torus_bettiNumber (371); sphere_sum_bettiNumber (404); sphere_bettiNumber (414); circle_bettiNumber (435).

Morse/Homology/Pretheory.lean. singularHomologyPretheory (42); singularHomologyPretheory_H□ (50); singularHomologyPretheory_δ_app (63); isHomotopyInvariant_singularHomologyPretheory (69).

Morse/Homology/Relative.lean. cokernelSequence (93); cokernelSequence_shortExact (105); singularChains (118); relShortComplex (135); relShortComplex_shortExact (159); relChains (165); relChains_obj (167); relπ (171); map_relπ (189); relπ_relChains_map (195); relChains_hom_ext (202); relHomology (207); relHomology_obj (210); relHomology_map (213); relδ (224); relδ_comp (229); comp_relδ (233); homologyMap_map_comp_relπ (237); relHomology_exact₁ (242); relHomology_exact₂ (247); relHomology_exact₃ (252); relHomologySequence_exact (259); epi_homologyMap_relπ_zero (265); relδ_naturality (271); relδNatTrans (280); relδNatTrans_app_succ (286); isZero_relHomology_of_epi_of_mono (291); isZero_relHomology_of_isIso_homologyMap (299); isIso_relδ (305); homologyToRelHomology (313); isZero_singularChains_X_of_isEmpty (326); isZero_singularChains_of_isEmpty (334); isIso_relπ_of_isEmpty (345); isIso_homologyMap_relπ_of_isEmpty (350); relHomologyInclIso (355); isZero_relChains_of_isIso (372); isZero_relHomology_of_isIso (376); isZero_relHomology_diag (381); quasiIso_relChains_map (390); isIso_relHomology_map_of_quasiIso (399); isIso_relHomology_map (406); isIso_relHomology_map_of_degree (423); subsetInclusion (439); subsetPair (454); subsetPairHom (463); subsetMap (476); subsetPairMap (486); subsetPairIsoOfEq (502).

Morse/Homology/RelativeHomotopy.lean. singularChainsHomotopyEquivOfHomotopy (62); singularChainsHomotopyEquiv (74); quasiIso_map_of_homotopyEquiv (84); isIso_homologyMap_of_homotopyEquiv (92); isIso_relHomology_map_of_homotopyEquiv (100); isZero_homology_of_contractibleSpace (108); isIso_relδ_of_contractibleSpace (118); cylinderContraction (131); isIso_homologyMap_fst (152); cylinder (162); cylinderι₀ (172); cylinderι₁ (175); cylinderπ (178); cylinderι₀_π (186); cylinderι₁_π (190); isIso_relHomology_map_cylinderπ (195); relHomology_map_cylinderι₀_eq_ι₁ (200); cylinderHom (210); cylinderι₀_cylinderHom

(214); cylinderι₁_cylinderHom (219); relHomology_map_eq_of_homotopy (227); relHomology_map_eq_of_homotopic (235).

Morse/Homology/RelativeRetract.lean. isIso_homologyMap_of_isStrongDeformationRetract (38); isIso_relHomology_map_of_isStrongDeformationRetract (49); relHomologyIsoOfIsStrongDeformationRetract (61); isZero_relHomology_of_isStrongDeformationRetract (68).

Morse/Homology/Simplex.lean. freeFunctor (72); freeChainsFunctor (81); freeChains (86); bd (93); freeChains_d (96); bd_single (109); bd_bd (116); lmapDomain_bd (134); Simplex (146); simplexMap (150); simplexMap_id (158); simplexMap_comp (162); faceMap (168); sing (180); sing_δ (199); singFunctor (204); Chains (218); sbd (224); sbd_single (227); sbd_sbd (234); singChains (241); push (256); push_single (260); push_id (265); push_comp (271); push_sbd (279); pushHom (289); singChainsFunctor (301); freeChainsIso (320); sigmaIsoFinsupp (337); sigmaConstIsoFree (350); ι_sigmaConstIsoFree_hom_app (363); chainComplexFunctorIso (370); toSSetIsoSing (378); singChainsIso (385); ι_singChainsIso_hom_app_f (393).

Morse/Homology/SmallChains.lean. inSet (60); isPrecompClosed_inSet (62); smallSet (68); isPrecompClosed_smallSet (70); inSet_subset_smallSet (73); sbd_mem_supported (79); sbd_ssd_pow (87); subdivRadius (101); subdivRadius_mono_left (106); IsFine (115); isFine_zero_of_mem_smallSet (119); exists_isFine (126); dist_simplexMap_le (139); IsFine.comp_faceMap (160); IsFine.ssd_pow_mem (171); fineOrder (195); isFine_fineOrder (197); fineOrder_le (201); fineOrder_comp_faceMap_le (205); fineOrder_eq_zero (209); ssd_pow_mem_of_le (214); partialH (231); partialH_apply (234); partialH_telescope (239); partialH_zero_degree (255); ssdH_ssd_pow_mem (259); smallH (266); smallH_single (269); smallRetr (276); smallRetr_zero_apply (280); smallRetr_succ_apply (283); smallH_zero_degree (288); sbd_smallRetr (295); smallRetr_mem_supported (308); smallH_eq_zero (344); smallRetr_eq_self (354); smallH_mem_supported (364); smallRetr_mem_supported_of (373); supp (396); suppChains (407); quotChains (412); smallChains (419); smallChainsι (423); smallRetraction (428); smallChainsι_smallRetraction (444); smallRetractionHomotopy (450); smallChainsHomotopyEquiv (470); quasiIso_smallChainsι (478); suppSmall (485); smallQuotChains (501); smallQuotMap (506); smallQuotRetraction (510); smallQuotHomotopyEquiv (516); quasiIso_smallQuotMap (538).

Morse/Homology/Spheres.lean. hemisphere (73); continuous_hemisphere (87); hemisphere_injective (99); norm_sq_hemisphere (105); hemisphere_mem_sphere (113); sphere_eq_union_hemisphere (127); hemisphere_one_mem_iff (158); homologyIsoBiprodOfRetraction (192); isZero_homology_singleton (223); homologyZeroSingletonIso (228); retractionSingleton (244); map_retractionSingleton (249); homologyIsoBiprodSingleton (256); finrank_eq_one_of_iso (272); finrank_eq_zero_of_isZero (276); finrank_biprod (290); finite_and_finrank_eq_ite (301); isZero_relHomology_cell_singleton (319); image_closedBall_fin_zero (332); image_sphere_fin_zero (342); relHomologyCellPairZeroIso (346); continuousOn_comp_hemisphere (362); injOn_comp_hemisphere (367); hemisphere_neg_one_zero_mem_sphere (372); singleton_subset_image_sphere (378); spherePointPair (383); lowerHemisphere_subset (387); singleton_subset_lowerHemisphere (393); image_sphere_eq_union (399); upperHemisphere_mem_iff (405); isClosed_lowerHemisphere (414); sphereRelIso (425); cellTriple (444); isIso_cellTriple_δ (450); cellPairIso (466); isZero_relHomology_cellPair (479); finrank_relHomology_cellPair (500); spherePointPairIso (519); isZero_relHomology_spherePointPair (525); homologyImageSphereZeroIso (532); homologyImageSphereTopIso (540); isZero_homology_imageSphere (547); finrank_homology_imageSphere (555); homologyImageSphereZeroIsoBiprod (574); diskPair (591); diskPairIsoCellPair (595); relHomologyDiskPairIso (599); isZero_relHomology_diskPair (605); finrank_relHomology_diskPair (611); sphereHomology (619); sphereHomologyZeroIso (624); sphereHomologyTopIso (629); sphereHomologyZeroIsoBiprod (634); isZero_sphereHomology (639); finrank_sphereHomology (645); isEmbedding_diskBoundaryInclusion (657); topDiskPair (670); homotopyEquivOfBijective (675); diskToCell (702); isIso_relHomology_map_diskToCell (710); relHomologyTopDiskPairIso (737); isZero_relHomology_topDiskPair (746); finrank_relHomology_topDiskPair (756).

Morse/Homology/Subdivision.lean. modelOp (63); modelOp_single (67); push_modelOp (72); idSimplex (80); push_idSimplex (83); modelOp_idSimplex (88); modelOp_add_model (94); modelOp_sub_model (101); modelOp_zero_model (109); modelOp_modelOp (117); sbd_modelOp (125); modelOp_sbd (133); IsPrecompClosed (146); modelOp_mem_supported (151); sdModel (170); sdHModel (176); aff_comp_verts (182); push_aff_sdModel (188); push_aff_sdHModel (194); lreal_verts (199); sbd_idSimplex (204); sbd_sdModel (211); sbd_sdHModel (218); sdModel_zero (228); sdHModel_zero (231); ssd (244); ssdH (248); sbd_ssd (251); sbd_ssdH (256); ssd_zero (263); ssdH_zero (268); push_ssd (272); push_ssdH (277); push_ssd_pow (281); ssd_mem_supported (289); ssdH_mem_supported (296); ssd_pow_mem_supported (301); ssd_lreal (312); ssd_pow_lreal (321); ssd_pow_single (330); subdivisionHom (345); subdivision (354); subdivisionHomotopy (362).

Morse/Homology/Triple.lean. cokernel_comp_shortExact' (59); cokernel_comp_shortExact (69); TopTriple (83); TopTriple.isEmbedding_k (107); TopTriple.pairYZ (112); TopTriple.pairXZ (115); TopTriple.pairXY (118); TopTriple.map₁ (121); TopTriple.map₂ (127); TopTriple.ofSubsets (143); TopTriple.ofSubsets_pairYZ (159); TopTriple.ofSubsets_pairXZ (160); TopTriple.ofSubsets_pairXY (161); TopTriple.relChains_map_map₁_comp_map₂ (171); TopTriple.shortComplex (180); TopTriple.singularChains_map_fac (184); TopTriple.relChains_map_map₁_eq (189); TopTriple.relChains_map_map₂_eq (197); TopTriple.shortComplex_shortExact (207); TopTriple.δ (215); TopTriple.δ_comp (219); TopTriple.comp_δ (223); TopTriple.map₁_comp_map₂ (227); TopTriple.exact₁ (233); TopTriple.exact₂ (237); TopTriple.exact₃ (241); TopTriple.epi_map₂_zero (245); TopTriple.fromPair (252); TopTriple.δ_eq (266).

Morse/Homology/TripleFiltration.lean. TopTriple.longExactSeq (73); subsetHomology (103); finrank_subsetHomology_congr (109); finiteDimensional_subsetHomology_congr (115); subsingleton_subsetHomology_self (122); subsetTripleLongExactSeq (130); subsetTripleLongExactSeq' (137); filtrationGroup (154); filtrationHomology (161); finrank_filtrationGroup (180); singularMorseNumber (186); monotone_truncate (192); finiteDimensional_truncate_step (197); finrank_truncate_total (211); morseNumber_truncate (218); truncatedFiltrationHomology (231); finiteDimensional_subsetHomology_of_steps (239); singular_weak_morse_inequality (247); singular_morse_inequality (257); singular_euler_characteristic (269); singularMorseNumber_eq_card (288); singular_weak_morse_inequality_of_cells (299); singular_morse_inequality_of_cells (309); singular_euler_characteristic_of_cells (321).

Morse/Homotopy/Basic.lean. genLoopMap (51); genLoopMap_apply (55); genLoopMap_homotopic (59); homotopyGroupMap (64); homotopyGroupMap_mk (68); homotopyGroupMap_comp (72); homotopyGroupMap_id (78); homotopyRelMono (87); Compresses (94); exists_lift_of_isEmbedding (98); surjective_homotopyGroupMap_of_compresses (108); injective_homotopyGroupMap_of_compresses (135).

Morse/Homotopy/BundleHomotopy.lean. subsingleton_homotopyGroup_of_subsingleton (62); subsingleton_homotopyGroup_of_homeomorph (75); IsLocallyTrivial.nonempty_homeomorph_fiber (80); bijective_homotopyGroupMap_of_fiber (107); surjective_homotopyGroupMap_of_fiber (120); subsingleton_pi_total (134); stiefelCongr (155); unitaryCongr (181); isLocallyTrivial_stiefelRestrict (207); isLocallyTrivial_unitaryOrbit (213); isLocallyTrivial_frameSpan (220); SphereConnected (234); subsingleton_pi_stiefel (241); unitaryOrbit_unitaryIncl (280); unitaryOrbitC (286); unitaryInclC (291); unitaryFibreAt (295); subsingleton_pi_unitaryOrbit_base (302); bijective_homotopyGroupMap_unitaryIncl (314); surjective_homotopyGroupMap_unitaryIncl (325); frameSpanC (343); isSerreFibration_frameSpanC (347); subsingleton_pi_stiefel_sum (351); grassFibre (361); grassFibre_fiberPt (365); grassmannianEquiv (372); nonempty_mulEquiv_grassmannian (383).

Morse/Homotopy/BundlesOrthogonal.lean. sphereConnected_real (74); isLocallyTrivial_sphereOrbit_real (82); bijective_homotopyGroupMap_orthogonalIncl (92); surjective_homotopyGroupMap_orthogonalIncl (98); bijective_homotopyGroupMap_omegaZeroStab

(105); subsingleton_pi_stiefel_real (111); grassmannianEquiv_real (123); nonempty_mulEquiv_grassmannian_real (129); grassmannianToBase_real (138); grassmannianPiEquiv_real (148); nonempty_mulEquiv_grassmannianPi_real (155); det_mul_self_of_mem_orthogonalGroup (172); soSet (177); isClopen_soSet (181); soHomeomorph (199); soInclC (211); bijective_homotopyGroupMap_soInclC (218).

Morse/Homotopy/BundlesUnitary.lean. piEquivOfHomeomorph (98); nonempty_mulEquiv_of_homeomorph (104); IsLocallyTrivial.homeomorph_comp (113); mem_stiefel_fin_one (134); norm_eq_one_iff_sum_star_mul_self (153); stiefelSphere (176); sphereConnected_complex (190); sphereOrbit (203); isLocallyTrivial_sphereOrbit (209); bijective_homotopyGroupMap_unitaryIncl_complex (224); surjective_homotopyGroupMap_unitaryIncl_complex (231); subsingleton_pi_stiefel_complex (238); grassmannianEquiv_complex (244); nonempty_mulEquiv_grassmannian_complex (251); reindex_mem_grassmannian (265); grassCongr (274); grassMove (292); grassMove_self (296); grassmannian_eq_Grassmannian (309); card_eq_of_card_eq_two_mul (312); grassmannianToBase (319); grassmannianToBase_self (325); grassmannianPiEquiv (332); nonempty_mulEquiv_grassmannianPi (340); minVelPiEquiv (351); nonempty_mulEquiv_minVelPi (357); range_subset_of_isClopen (372); zero_mem_cube_boundary (378); genLoop_mem_of_isClopen (381); genLoopRestrict (389); bijective_homotopyGroupMap_clopen (395); IsCoveringMap.isLocallyTrivial (424); circleFibreEquiv (432); isLocallyTrivial_circleExp (447); subsingleton_homotopyGroup_discrete (455); subsingleton_pi_circle (462); isLocallyTrivial_of_homeomorph (472); det_mem_circle (483); detCircle (489); diagonal_update_mem (500); diagCircle (512); det_diagCircle (515); continuous_diagCircle (519); det_mul_inv_diagCircle (523); unitaryProd (530); isLocallyTrivial_detCircle (557); suInclC (563); suFibre (567); bijective_homotopyGroupMap_suInclC (585).

Morse/Homotopy/CellAttach.lean. clampBall (87); continuous_clampBall (90); norm_clampBall (94); norm_clampBall_le (101); clampBall_of_norm_le (105); radialPush (110); continuous_radialPush (113); radialPush_zero (118); radialPush_of_norm_eq_one (121); norm_radialPush_one (129); norm_radialPush_le (135); injective_diskBoundaryInclusion (150); isPushout_forget (155); exists_eq_cellInl_or_cellInr (160); cellInl_injective (165); cellInr_eq_cellInl_iff (170); isClosed_attachCell_iff (176); isClosedEmbedding_cellInl (187); cellProbe (209); cellHeight (221); cellCoord (225); continuous_cellHeight (228); continuousOn_cellCoord (231); cellProbe_inl (235); cellProbe_inr (239); cellHeight_inl (244); cellHeight_inr (247); cellCoord_inr (250); cellHeight_nonneg (254); cellInr_mem_range_of_norm_eq_one (260); mem_range_cellInl_iff (267); exists_cellInr_of_cellHeight_pos (279); norm_cellCoord (287); cellChar (294); continuous_cellChar (297); cellChar_of_norm_le (301); cellChar_cellCoord (305); cellCoord_cellChar (311); cellHeight_cellChar (315); radialDisk (326); radialDisk_compat (334); cellRadial (346); cellRadial_inl (351); cellRadial_inr (358); cellRadial_zero (366); cellRadial_of_mem_range (372); cellRadial_one_mem (377); homotopyRelOfFun (397); exists_homotopicRel_avoid_ball (410); exists_homotopicRel_into_range (495); compresses_cellInl (512); Compresses.comp_homeomorph (530); compresses_of_isPushout (548); compresses_union_cell (562); cubeEmb (581); continuous_cubeEmb (585); injective_cubeEmb (589); cylEmb (593); continuous_cylEmb (597); injective_cylEmb (601); compresses_cellInl_cube (605); compresses_cellInl_cyl (609); surjective_homotopyGroupMap_cellInl (617); injective_homotopyGroupMap_cellInl (623); bijective_homotopyGroupMap_cellInl (631); bijective_homotopyGroupMap_of_isRelCellAttachment (639); surjective_homotopyGroupMap_of_isRelCellAttachment (653); bijective_homotopyGroupMap_homeomorph (670); bijective_homotopyGroupMap_of_isPushout (679); surjective_homotopyGroupMap_of_isPushout (689); bijective_homotopyGroupMap_union_cell (706); surjective_homotopyGroupMap_union_cell (717).

Morse/Homotopy/ChangeOfBase.lean. cubeDist (61); continuous_cubeDist (64); abs_le_cubeDist (68); cubeDist_le_half (72); cubeDist_of_mem_boundary (77); cubeDist_of_isEmpty (83); exists_abs_eq_cubeDist (87); cubeScale (97); continuous_cubeScale (101); coe_cubeScale (108); cubeScale_half (123); cubeDist_cubeScale (130); cubeScale_mem_boundary (139);

cubeScale_cubeScale (162); frame (179); frame_apply (183); continuous_frame (188); frame_of_mem_boundary (198); homotopic_of_frame (210); continuous_changeBaseOuter (230); changeBase (237); changeBase_apply (257); changeBase_homotopic (262); changeBaseMap (283); changeBaseMap_mk (286); changeBase_symm_changeBase (291); changeBaseMap_symm_comp (341); changeBaseEquiv (349); changeBase_genLoopMap_homotopic (362); homotopyGroupMap_eq_changeBaseMap (392); bijective_homotopyGroupMap_iff (400); injective_homotopyGroupMap_iff (409); surjective_homotopyGroupMap_iff (418); bijective_homotopyGroupMap_of_homotopyEquiv (429); homotopyGroupEquivOfHomotopyEquiv (445); bijective_homotopyGroupMap_iff_of_square (459); injective_homotopyGroupMap_iff_of_square (472); surjective_homotopyGroupMap_iff_of_square (485).

Morse/Homotopy/ChangeOfBaseGroup.lean. evalPath (38); constPath (41); evalHomotopy (44); pathContraction (51); evalZeroHomotopyEquiv (59); changeBaseMap_evalAt_mul (83); changeBaseMap_cast (98); cast_mul (107); changeBaseMap_mul (114); changeBaseMulEquiv (130).

Morse/Homotopy/ExactSequence.lean. optionTail (66); continuous_optionTail (69); optionPoint_none (73); optionPoint_some (77); optionTail_optionPoint (81); optionPoint_optionTail (84); mem_boundary_option_iff (87); update_none_eq_optionPoint (98); update_some_optionPoint (105); projIcc_eq_zero (120); projIcc_eq_one (124); fiber (130); fiberPt (134); fiberIncl (138); fiberIncl_apply (142); proj_fiber (145); IsBoundaryLift (157); IsBoundaryLift.mem_fiber (163); topFace (170); coe_topFace (177); cubeOfLoop (182); isBoundaryLift_genLoopMap (188); exists_isBoundaryLift (197); homotopic_topFace (213); exists_isBoundaryLift_topFace_eq (304); boundaryLoop (400); boundaryMap (406); boundaryMap_mk (412); homotopyGroupMap_const (422); homotopyGroupMap_proj_fiberIncl (434); exact_fiber (441); exact_total (505); exact_base (554).

Morse/Homotopy/ExactSequenceIso.lean. genLoopMap_transAt (65); homotopyGroupMap_mul (73); homotopyGroupHom (86); boundaryMap_mul (99); boundaryHom (169); exists_homotopyGroupMap_eq_inv_mul (180); injective_boundaryMap (238); injective_homotopyGroupMap_fiberIncl (250); surjective_boundaryMap (316); bijective_boundaryMap (321); surjective_homotopyGroupMap_fiberIncl (326); bijective_homotopyGroupMap_fiberIncl (331); subsingleton_homotopyGroup_total (340); subsingleton_homotopyGroup_base (346); subsingleton_homotopyGroup_fiber (355); subsingleton_homotopyGroup_option (364); boundaryMapPi (371); bijective_boundaryMapPi_of_le (377); surjective_boundaryMapPi_of_le (385); boundaryEquivPi (391); bijective_homotopyGroupMap_fiberIncl_of_le (398); surjective_homotopyGroupMap_fiberIncl_of_le (407); subsingleton_pi_base (414); subsingleton_pi_fiber (421).

Morse/Homotopy/Fibration.lean. IsLocallyTrivial (60); isLocallyTrivial_of_trivializations (65); isLocallyTrivial_proj (73); isLocallyTrivial_fst (80); IsLocallyTrivial.comp_homeomorph (90); IsLocallyTrivial.congr_fiber (111); mem_preimage_of_mem_fiber (118); fiberHomeomorphOfTriv (126); IsLocallyTrivial.nonempty_fiberHomeomorph (148); IsLocallyTrivial.fiberHomeomorph (154); openBox (162); mem_openBox (165); mk_zero_mem_openBox (169); mk_mem_openBox_of_mem_boundary (172); IsSerreFibration (179); cubeCongr (185); cubeCongr_apply (193); cubeCongr_mem_boundary_iff (197); IsSerreFibration.exists_lift (208); cubeShrink (233); continuous_cubeShrink (237); cubeDist_cubeShrink (241); cubeScale_cubeShrink (250); boxFold (263); continuous_boxFold (266); boxFold_snd_of_mem_openBox (273); boxFold_shrink (279); IsSerreFibration.exists_lift_of_bot (291).

Morse/Homotopy/Grassmannian.lean. continuousAt_gramSchmidt (61); continuousAt_gramSchmidtNormed (82); eq_of_mul_eq_of_trace_eq (102); Grassmannian (125); conj_mem_grassmannian (129); coe_smul_grassmannian (156); mul_conjTranspose_mem_grassmannian (174); frameSpan (183); continuous_frameSpan (187); frameSpan_smul (191); colMatrix (205); colMatrix_mem_stiefel (208); fixSubmodule (230); exists_frame_of_mem_grassmannian (252); grassBase (330); gsVecs (333); gsFrame (337); colMatrix_gsVecs (341); linearIndependent_gsVecs (349); gsFrame_mem (361); mul_gsFrame (367); gsFrame_mul_conjTranspose (387); gsFrame_base

(400); continuousAt_gsFrame (416); grassDomain (449); isOpen_grassDomain (453); grassBase_mem_grassDomain (477); grassSection (486); continuous_grassSection (489); grassSection_smul (496); exists_unitary_smul_grassBase (503); eq_fromRows_of_mul_conjTranspose (513); fromRows_zero_mem_stiefel (527); fromRows_zero_mul_conjTranspose (534); frameSpanFibreHomeomorph (563); frameSpanFibreHomeomorph_symm_apply (577); frameSpan_locallyTrivial (586).

Morse/Homotopy/GrassmannianQuotient.lean. norm_entry_le_one_of_mem_stiefel (60); isCompact_stiefel (68); frameSpan_eq_iff (92); surjective_frameSpan (118); isQuotientMap_frameSpan (124); frameSetoid (129); frameSetoid_eq_ker (138); stiefelQuotientHomeomorph (144); stiefelQuotientHomeomorph_mk (152); fixSubmodule_colMatrix (166); finrank_fixSubmodule (184); eq_of_fixSubmodule_eq (194); exists_fixSubmodule_eq (212); grassmannianEquivSubmodule (223); coe_grassmannianEquivSubmodule (232); grassmannianEquivSubmodule_frameSpan (240); grassmannianEquivSubmodule' (256); mem_grassmannianEquivSubmodule' (268); grassmannian_eq (286); stiefelQuotientHomeomorph' (290); grassmannianSubmoduleEquiv (295).

Morse/Homotopy/Homogeneous.lean. trivOfInverse (35); trivOfInverse_apply_fst (49); smul_mem_fiber (63); orbitTriv (73); orbitTriv_apply_fst (95); exists_triv_of_orbitSection (104); exists_triv_of_orbitSection' (126).

Morse/Homotopy/HomotopyLifting.lean. cubeFaces (65); isClosed_cubeFaces (68); faceWeight (82); faceWeight_of_mem (85); faceWeight_of_notMem (89); faceWeight_nonneg (93); faceWeight_le_two (100); continuous_faceWeight (110); faceMax (122); faceWeight_le_faceMax (125); faceMax_le_two (130); one_le_faceMax (135); exists_faceWeight_eq_faceMax (140); continuous_faceMax (148); faceMax_of_mem (151); faceRetract (167); continuous_faceRetract (172); faceRetract_mem (186); faceRetract_of_mem (229); gridBox (248); isClosed_gridBox (251); natCast_add_one_le (261); gridEmb (265); coe_gridEmb (275); gridCoord (280); continuous_gridEmb (283); continuous_gridCoord (287); gridEmb_mem (291); coe_gridCoord_of_mem (301); gridEmb_gridCoord (310); gridCoord_gridEmb (319); exists_mem_gridBox (326); exists_gridBox_subset (350); fiberMove (381); fiberMove_proj (386); fiberMove_self (390); continuous_fiberMove (396); exists_lift_union_gridBox (405); upperFaces (470); liftedPart (474); isClosed_liftedPart (477); liftedPart_insert (481); gridCoord_mem_cubeFaces (500); gridEmb_mem_of_mem_cubeFaces (540); IsLocallyTrivial.exists_lift_cubeFaces (592); optionCubeHomeo (643); optionCubeHomeo_mem_cubeFaces_iff (652); IsLocallyTrivial.isSerreFibration (673); isSerreFibration_proj (689).

Morse/Homotopy/LoopSpace.lean. genLoop_homotopic_congr (47); genLoop_congr_symm_congr (62); homotopyGroupCongr (68); genLoop_congr_transAt (76); homotopyGroupCongrMulEquiv (98); optionPoint (118); continuous_optionPoint (121); optionPoint_mem_boundary_of_left (127); optionPoint_mem_boundary_of_right (131); optionPoint_eta (136); loopUncurry (141); loopCurryPath (160); continuous_loopCurryPath (167); loopCurry (172); loopCurry_loopUncurry (177); loopUncurry_loopCurry (182); loopUncurry_homotopic (187); loopCurry_homotopic (210); homotopyGroupLoopEquiv (239); homotopyGroupPiLoopEquiv (251); update_some_comp (255); loopUncurry_transAt (264); homotopyGroupLoopMulEquiv (281); homotopyGroupPiLoopMulEquiv (293); cancelFamily (308); continuous_cancelFamily (333); cancelFamily_zero (352); cancelFamily_one (382); homotopic_trans_trans_symm (390); pathTransHomotopyEquiv (401); pathHomotopyEquivLoop (410).

Morse/Homotopy/PushOff.lean. pushCutoff (51); continuous_pushCutoff (54); pushCutoff_nonneg (58); pushCutoff_le_one (61); pushCutoff_of_le (64); pushCutoff_of_ge (68); exists_contDiff_near (80); exists_pushOff (108).

Morse/Homotopy/RelCells.lean. bijective_homotopyGroupMap_of_isHomotopyEquiv (55); bijective_homotopyGroupMap_of_isRelCellComplexType (70); surjective_homotopyGroupMap_of_isRelCellComplexType (81); bijective_homotopyGroupMap_of_seq (100); surjective_homotopyGroupMap_of_seq (109); exists_subset_sublevel (119); bijective_homotopyGroupMap_sublevel (130); surjective_homotopyGroupMap_sublevel (141); bijec-

tive_homotopyGroupMap_of_telescope (154); surjective_homotopyGroupMap_of_telescope (165).

Morse/Homotopy/Spheres.lean. subsingleton_homotopyGroup_of_contractible (47); subsingleton_homotopyGroup_of_joined (59); lowerHemisphere (70); isClosed_lowerHemisphere (73); contractibleSpace_lowerHemisphere (77); lowerHemisphere_union (94); southPole (99); subsingleton_homotopyGroup_sphere_southPole (104); subsingleton_homotopyGroup_sphere (121); subsingleton_pi_sphere (135); sphereHomeomorph (143); subsingleton_pi_sphere_of_finrank (155); subsingleton_pi_sphere_complex (165).

Morse/Homotopy/Stiefel.lean. Stiefel (67); mem_stiefel (71); coe_unitaryGroup_eq_stiefel (82); mul_mem_stiefel (87); coe_smul_stiefel (101); baseFrame (121); conjTranspose_mul_self_fromRows (124); baseFrame_mem (129); stiefelBase (134); stiefel_blocks (137); secMat (144); secMat_mul_baseFrame (149); secMat_unitary_aux (157); secMat_mem_unitaryGroup (207); secDomain (222); isOpen_secDomain (225); stiefelBase_mem_secDomain (236); continuousAt_secMat (244); localSection (272); continuous_localSection (275); localSection_smul (281); orthonormal_cols (295); exists_unitary_smul_baseFrame (306); stiefelRestrict (355); continuous_stiefelRestrict (359); stiefelRestrict_smul (364); eq_fromBlocks_of_toCols₂ (373); fromBlocks_mem_stiefel (395); toCols₂_fromBlocks (403); stiefelFibreHomeomorph (409); stiefelRestrict_locallyTrivial (428); unitaryOrbit (442); continuous_unitaryOrbit (444); unitaryOrbit_mul (447); coe_unitaryOrbit (451); fromBlocks_mem_unitaryGroup (458); unitaryIncl (466); coe_unitaryIncl (474); continuous_unitaryIncl (478); fromBlocks_toBlocks_of_mem_fiber (482); unitaryFibreHomeomorph (498); unitaryFibreHomeomorph_symm_apply (511); unitaryOrbit_locallyTrivial (520).

Morse/Homotopy/Telescope.lean. exists_subset_of_isCompact (55); corestrict (63); valC (67); surjective_homotopyGroupMap_of_exhaustion (73); injective_homotopyGroupMap_of_exhaustion (94); bijective_homotopyGroupMap_of_exhaustion (115); bijective_homotopyGroupMap_valC_of_step (125); surjective_homotopyGroupMap_valC_of_step (148); telescopeIncl (179); traj_self (184); push_one_mk_zero (205); telescopeIncl_mem_trunc (215); homotopic_corestrict_telescopeIncl (221); bijective_homotopyGroupMap_telescopeIncl (235); surjective_homotopyGroupMap_telescopeIncl (252); bijective_homotopyGroupMap_tmap_of_step (269); surjective_homotopyGroupMap_tmap_of_step (285); bijective_homotopyGroupMap_of_cellTower (309); surjective_homotopyGroupMap_of_cellTower (319).

Mathlib/Algebra/BigOperators/Finprod.lean. finsum (103).

Mathlib/Algebra/Category/ModuleCat/Colimits.lean. finsuppCoconeIsColimit (176).

Mathlib/Algebra/Homology/HomologySequence.lean. ShortExact.δ (292); ShortExact.homology_exact₁ (303); ShortExact.homology_exact₂ (309); ShortExact.homology_exact₃ (324).

Mathlib/Algebra/Homology/HomologySequenceLemmas.lean. HomologySequence.δ_naturality (58); HomologySequence.quasiIso_τ₃ (173).

Mathlib/Algebra/Homology/ShortComplex/Exact.lean. Splitting.ofExactOfRetraction (1014).

Mathlib/Algebra/Homology/ShortComplex/HomologicalComplex.lean. epi_homologyMap_of_epi_of_not_rel (631).

Mathlib/Algebra/Lie/InvariantForm.lean. LieAlgebra.InvariantForm.orthogonal (73).

Mathlib/Algebra/Order/Archimedean/Basic.lean. exists_pow_lt_of_lt_one (251).

Mathlib/Algebra/Order/BigOperators/Ring/Finset.lean. sq_sum_div_le_sum_sq_div (169).

Mathlib/Algebra/QuadraticDiscriminant.lean. discrim_le_zero (124).

Mathlib/AlgebraicTopology/EilenbergSteenrod.lean. HomologyPretheory (36); IsHomotopyInvariant (144).

Mathlib/AlgebraicTopology/FundamentalGroupoid/SimplyConnected.lean. SimplyConnectedSpace.paths_homotopic (79).

Mathlib/AlgebraicTopology/SingularHomology/Basic.lean. AlgebraicTopology.singularChainComplexFunctor (36); singularChainComplexFunctor (36); AlgebraicTopology.singularHomologyFunctor (55); singularHomologyFunctor (55); isZero_singularHomologyFunctor_of_totallyDisconnectedSpace (126).

Mathlib/AlgebraicTopology/SingularHomology/HomologyZero.lean. singularHomology$_0\varepsilon$ (42).

Mathlib/AlgebraicTopology/SingularHomology/HomotopyInvariance.lean. singularChainComplexFunctorObjMap (49).

Mathlib/Analysis/CStarAlgebra/ContinuousFunctionalCalculus/Continuity.lean. ContinuousOn.cfc_of_mem_nhdsSet (345).

Mathlib/Analysis/Calculus/BumpFunction/Basic.lean. ContDiffBump (70); ContDiffBump.one_of_mem_closedBall (137).

Mathlib/Analysis/Calculus/BumpFunction/FiniteDimension.lean. exists_contDiff_tsupport_subset (45).

Mathlib/Analysis/Calculus/ContDiff/Defs.lean. ContDiffOn (472); contDiffOn_of_locally_contDiffOn (587); contDiffOn_succ_iff_fderiv_of_isOpen (892); ContDiffAt (937); ContDiffAt.eventually (1036); ContDiff (1068); contDiff_succ_iff_hasFDerivAt (1172).

Mathlib/Analysis/Calculus/ContDiff/Operations.lean. contDiffAt_map_inverse (850); OpenPartialHomeomorph.contDiffAt_symm (887); Homeomorph.contDiff_symm (946).

Mathlib/Analysis/Calculus/ContDiff/RCLike.lean. ContDiffAt.exists_lipschitzOnWith (131).

Mathlib/Analysis/Calculus/Deriv/Basic.lean. derivWithin (145); deriv (153).

Mathlib/Analysis/Calculus/Deriv/MeanValue.lean. exists_hasDerivAt_eq_slope (122); Convex.image_sub_le_mul_sub_of_deriv_le (354); monotoneOn_of_deriv_nonneg (410).

Mathlib/Analysis/Calculus/DerivativeTest.lean. isLocalMax_of_deriv_deriv_neg (409).

Mathlib/Analysis/Calculus/FDeriv/Basic.lean. hasFDerivAt_iff_isLittleO_nhds_zero (815); norm_fderiv_le_of_lipschitz (947).

Mathlib/Analysis/Calculus/FDeriv/Bilinear.lean. ContinuousLinearMap.hasFDerivAt_of_bilinear (120).

Mathlib/Analysis/Calculus/FDeriv/Comp.lean. fderiv_comp (177).

Mathlib/Analysis/Calculus/FDeriv/CompCLM.lean. HasFDerivAt.clm_comp (61).

Mathlib/Analysis/Calculus/FDeriv/Defs.lean. HasFDerivAt (121); fderivWithin (149); fderiv (158).

Mathlib/Analysis/Calculus/FDeriv/Equiv.lean. HasFDerivAt.eventually_ne (355).

Mathlib/Analysis/Calculus/FDeriv/Extend.lean. hasDerivWithinAt_Ici_of_tendsto_deriv (107).

Mathlib/Analysis/Calculus/FDeriv/Symmetric.lean. ContDiffAt.isSymmSndFDerivAt (533); ContDiffWithinAt.isSymmSndFDerivWithinAt (559).

Mathlib/Analysis/Calculus/Gradient/Basic.lean. gradient (82); inner_gradient_left (291).

Mathlib/Analysis/Calculus/InverseFunctionTheorem/ContDiff.lean. ContDiffAt.toOpenPartialHomeomorph (31).

Mathlib/Analysis/Calculus/LocalExtr/Basic.lean. IsLocalMin.fderiv_eq_zero (181).

Mathlib/Analysis/Calculus/MeanValue.lean. constant_of_has_deriv_right_zero (360); lipschitzWith_of_nnnorm_fderiv_le (516); Convex.norm_image_sub_le_of_norm_hasFDerivWithin_le' (528); IsOpen.is_const_of_deriv_eq_zero (768).

Mathlib/Analysis/Calculus/ParametricIntervalIntegral.lean. intervalIntegral.hasFDerivAt_integral_of_dominated_of_fderiv_le (56); hasDerivAt_integral_of_dominated_loc_of_deriv_le (97).

Mathlib/Analysis/Calculus/TangentCone/Real.lean. uniqueDiffWithinAt_Iic (135).

Mathlib/Analysis/Calculus/TaylorIntegral.lean. map_add_eq_sum_add_integral_iteratedFDeriv (69).

Mathlib/Analysis/Complex/CoveringMap.lean. Complex.isCoveringMap_exp (40).

Mathlib/Analysis/Convex/Contractible.lean. Convex.contractibleSpace (34).

Mathlib/Analysis/Convex/Deriv.lean. convexOn_of_hasDerivWithinAt2_nonneg (232).

Mathlib/Analysis/Convex/StdSimplex.lean. stdSimplex (35); stdSimplex.barycenter (397).

Mathlib/Analysis/InnerProductSpace/Basic.lean. ext_inner_right (328); abs_real_inner_le_norm (465); inner_eq_norm_mul_iff_real (787).

Mathlib/Analysis/InnerProductSpace/Calculus.lean. hasStrictFDerivAt_norm_sq (197); HasFDerivAt.norm_sq (211).

Mathlib/Analysis/InnerProductSpace/Dual.lean. InnerProductSpace.continuousLinearMapOfBilin (192).

Mathlib/Analysis/InnerProductSpace/GramSchmidtOrtho.lean. gramSchmidt_def'' (68).

Mathlib/Analysis/InnerProductSpace/Orthogonal.lean. Submodule.orthogonal (43).

Mathlib/Analysis/InnerProductSpace/PiL2.lean. EuclideanSpace (113); Orthonormal.exists_orthonormalBasis_extension (1034); Orthonormal.exists_orthonormalBasis_extension_of_card_eq (1047); stdOrthonormalBasis (1076).

Mathlib/Analysis/InnerProductSpace/Projection/Basic.lean. Submodule.starProjection (124); Submodule.eq_starProjection_of_mem_of_inner_eq_zero (173); Submodule.starProjection_apply_eq_zero_iff (639).

Mathlib/Analysis/InnerProductSpace/Spectrum.lean. LinearMap.IsSymmetric.card_filter_eigenvalues_eq (289); LinearMap.IsSymmetric.eigenvectorBasis (300); LinearMap.IsSymmetric.eigenvectorBasis_apply_self_apply (332); det_eq_prod_eigenvalues (391).

Mathlib/Analysis/InnerProductSpace/Trace.lean. LinearMap.trace_eq_sum_inner (27).

Mathlib/Analysis/Matrix/Spectrum.lean. Matrix.IsHermitian.spectral_theorem (141).

Mathlib/Analysis/Normed/Algebra/MatrixExponential.lean. Matrix.exp_conjTranspose (95); Matrix.exp_add_of_commute (130); Matrix.exp_conj (187).

Mathlib/Analysis/Normed/Module/Ball/Homeomorph.lean. OpenPartialHomeomorph.univUnitBall (46).

Mathlib/Analysis/Normed/Module/Connected.lean. isPathConnected_compl_singleton_of_one_lt_rank (120); isPathConnected_sphere (209); isConnected_sphere (239).

Mathlib/Analysis/Normed/Module/Convex.lean. convexHull_exists_dist_ge (123); convexHull_diam (159).

Mathlib/Analysis/Normed/Module/FiniteDimension.lean. LinearMap.exists_antilipschitzWith (238).

Mathlib/Analysis/Normed/Operator/LinearIsometry.lean. `LinearIsometryEquiv.ofSurjective` (951).

Mathlib/Analysis/ODE/Basic.lean. `IsIntegralCurve` (54).

Mathlib/Analysis/ODE/ExistUnique.lean. `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt₀` (71); `ContDiffAt.exists_forall_mem_closedBall_exists_eq_forall_mem_Ioo_hasDerivAt₀` (157); `ODE_solution_unique_of_mem_Ioo` (279); `ODE_solution_unique_univ` (341).

Mathlib/Analysis/ODE/Gronwall.lean. `norm_le_gronwallBound_of_norm_deriv_right_le` (134); `dist_le_of_trajectories_ODE` (228).

Mathlib/Analysis/ODE/PicardLindelof.lean. `IsPicardLindelof` (79); `IsPicardLindelof.of_time_independent` (662).

Mathlib/Analysis/SpecialFunctions/Complex/Circle.lean. `Circle.isAddQuotientCoveringMap_exp` (450); `Circle.isCoveringMap_exp` (456).

Mathlib/Analysis/SpecialFunctions/Complex/Log.lean. `Complex.exp_eq_exp_iff_exists_int` (171).

Mathlib/Analysis/SpecialFunctions/Exponential.lean. `hasDerivAt_exp_smul_const` (382).

Mathlib/Analysis/SpecialFunctions/Integrals/Basic.lean. `integral_sin_sq` (482); `integral_cos_sq` (555).

Mathlib/Analysis/SpecialFunctions/SmoothTransition.lean. `Real.smoothTransition` (146).

Mathlib/Analysis/SpecialFunctions/Trigonometric/Complex.lean. `Real.cos_eq_cos_iff` (309).

Mathlib/Analysis/SpecificLimits/Normed.lean. `Units.oneSub` (333).

Mathlib/CategoryTheory/Abelian/DiagramLemmas/KernelCokernelComp.lean. `kernelCokernelCompSequence_exact` (219).

Mathlib/CategoryTheory/Functor/OfSequence.lean. `Functor.ofSequence` (105).

Mathlib/CategoryTheory/Limits/Shapes/Pullback/IsPullback/Defs.lean. `CategoryTheory.IsPushout` (78); `CategoryTheory.IsPushout.isoPushout` (333).

Mathlib/CategoryTheory/Limits/Types/Pushouts.lean. `CategoryTheory.Limits.Types.pushoutCocone_inl_eq_inr_if` (225); `Types.pushoutCocone_inr_injective_of_isColimit` (243); `Types.eq_or_eq_of_isPushout` (264).

Mathlib/Data/Finset/Card.lean. `Finset.eraseInduction` (936).

Mathlib/Data/Set/Card.lean. `Set.ncard` (614).

Mathlib/Geometry/Manifold/ChartedSpace.lean. `ChartedSpace` (139); `chartAt` (153).

Mathlib/Geometry/Manifold/ContMDiff/Atlas.lean. `IsManifold.mem_maximalAtlas_iff_contMDiffOn` (181).

Mathlib/Geometry/Manifold/ContMDiff/Defs.lean. `ContMDiffAt` (181); `contMDiffAt_iff` (184); `ContMDiffOn` (200); `ContMDiff` (211); `contMDiffAt_iff_contMDiffAt_nhds` (836).

Mathlib/Geometry/Manifold/Diffeomorph.lean. `Diffeomorph` (82).

Mathlib/Geometry/Manifold/IsManifold/Basic.lean. `ModelWithCorners` (166); `modelWithCornersSelf` (209); `ModelWithCorners.Boundaryless` (587); `IsManifold` (780); `IsManifold.maximalAtlas` (871); `IsManifold.subset_maximalAtlas` (878); `TangentSpace` (1034).

Mathlib/Geometry/Manifold/IsManifold/ExtChartAt.lean. `extChartAt` (454); `writtenInExtChartAt` (801).

Mathlib/Geometry/Manifold/LocalDiffeomorph.lean. `PartialDiffeomorph` (80); `IsLocalDiffeomorph.diffeomorphOfBijective` (355).

Mathlib/Geometry/Manifold/MFDeriv/Basic.lean. `mfderiv_zero_of_not_mdifferentiableAt` (525); `mfderiv_comp` (1195).

Mathlib/Geometry/Manifold/MFDeriv/Defs.lean. `mdifferentiableAt_iff` (251); `mfderiv` (333).

Mathlib/Geometry/Manifold/MFDeriv/NormedSpace.lean. `mvfderiv` (451).

Mathlib/Geometry/Manifold/PartitionOfUnity.lean. `SmoothPartitionOfUnity.exists_isSubordinate` (554); `exists_contMDiff_support_eq_eq_one_iff` (758).

Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Basic.lean. `IsCovariantDerivativeOn` (91); `CovariantDerivative` (365); `ContMDiffCovariantDerivative` (411).

Mathlib/Geometry/Manifold/VectorBundle/CovariantDerivative/Torsion.lean. `CovariantDerivative.torsion` (120); `CovariantDerivative.torsion_eq_zero_iff` (143).

Mathlib/Geometry/Manifold/VectorBundle/MDifferentiable.lean. `mdifferentiableAt_extend` (736).

Mathlib/Geometry/Manifold/VectorBundle/Riemannian.lean. `IsContMDiffRiemannianBundle` (66).

Mathlib/Geometry/Manifold/VectorField/LieBracket.lean. `VectorField.mlieBracket` (77); `VectorField.mlieBracket_swap` (135).

Mathlib/LinearAlgebra/Complex/FiniteDimensional.lean. `finrank_real_of_complex` (57).

Mathlib/LinearAlgebra/Dimension/Finrank.lean. `Module.finrank` (62).

Mathlib/LinearAlgebra/FiniteDimensional/Lemmas.lean. `Submodule.finrank_sup_add_finrank_inf_eq` (53); `finrank_range_add_finrank_ker` (173); `LinearMap.injective_iff_surjective_of_finrank_eq_finrank` (197).

Mathlib/LinearAlgebra/Matrix/BilinearForm.lean. `LinearMap.BilinForm.nondegenerate_iff_det_ne_zero` (550).

Mathlib/LinearAlgebra/Matrix/Charpoly/Coeff.lean. `Matrix.det_one_add_smul` (208).

Mathlib/LinearAlgebra/QuadraticForm/Basic.lean. `QuadraticMap.associated_left_inverse` (907); `LinearMap.BilinForm.exists_orthogonal_basis` (1342); `QuadraticMap.weightedSumSquares` (1419).

Mathlib/LinearAlgebra/QuadraticForm/Real.lean. `QuadraticForm.equivalent_one_neg_one_weighted_sum_squared` (66).

Mathlib/LinearAlgebra/QuadraticForm/Signature.lean. `sigPos` (65); `sigPos_le_finrank` (73); `exists_finrank_eq_sigPos_and_posDef` (87); `sigNeg` (98); `sigNeg_isGreatest` (101); `QuadraticMap.Equivalent.sigNeg_eq` (133); `QuadraticForm.sigPos_add_finrank_le_of_nonpos` (146); `QuadraticForm.sigNeg_weightedSumSquares` (205); `QuadraticForm.sigPos_add_sigNeg_add_radical` (232); `QuadraticForm.sigNeg_of_equiv_weightedSumSquares` (251).

Mathlib/LinearAlgebra/SesquilinearForm/Basic.lean. `apply_apply_same_eq_zero_iff` (864).

Mathlib/LinearAlgebra/Trace.lean. `LinearMap.trace_comp_comm'` (265).

Mathlib/Logic/Equiv/Fin/Basic.lean. `finSuccEquiv` (120).

Mathlib/MeasureTheory/Integral/DivergenceTheorem.lean. `integral_eq_of_hasDerivAt_off_countable_of_le` (373).

Mathlib/MeasureTheory/Integral/DominatedConvergence.lean. `intervalIntegral.continuousAt_of_dominated_interval` (265); `intervalIntegral.continuous_primitive` (479).

Mathlib/MeasureTheory/Integral/IntervalIntegral/Basic.lean. ContinuousLinearMap.intervalIntegral_comp_comm (870); intervalIntegral.integral_add_adjacent_intervals (1095); integral_eq_zero_iff_of_le_of_nonneg_ae (1287).

Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus.lean. intervalIntegral.integral_hasDerivAt_left (753); intervalIntegral.integral_eq_sub_of_hasDerivAt (1148).

Mathlib/MeasureTheory/Measure/OpenPos.lean. IsOpen.measure_pos (47).

Mathlib/MeasureTheory/Measure/Restrict.lean. Measure.exists_mem_of_measure_ne_zero_of_ae (416).

Mathlib/Order/Interval/Set/ProjIcc.lean. Set.projIcc (48).

Mathlib/Order/Preorder/Finite.lean. Finset.exists_maximal (44).

Mathlib/Topology/Algebra/MetricSpace/Lipschitz.lean. continuousAt_of_locally_lipschitz (49).

Mathlib/Topology/Algebra/Module/Equiv.lean. ContinuousLinearEquiv.ofUnit (813).

Mathlib/Topology/Algebra/Module/FiniteDimension.lean. ContinuousLinearEquiv.ofFinrankEq (425).

Mathlib/Topology/Algebra/Monoid.lean. continuous_finprod (995).

Mathlib/Topology/Bases.lean. TopologicalSpace.separableSpace_iff_countable (454); TopologicalSpace.countable_cover_nhds (937); TopologicalSpace.countable_cover_nhdsWithin (946).

Mathlib/Topology/Category/TopCat/Limits/Basic.lean. TopCat.isClosed_iff_of_isColimit (269).

Mathlib/Topology/Category/TopCat/Sphere.lean. TopCat.disk (28); disk (28); TopCat.diskBoundary (32); diskBoundary (32); TopCat.diskBoundaryInclusion (58).

Mathlib/Topology/Category/TopPair.lean. TopPair (31); TopPair.Homotopy (126).

Mathlib/Topology/CompactOpen.lean. ContinuousMap.compactOpen (51); ContinuousMap.eventually_mapsTo (65); ContinuousMap.continuous_postcomp (139); ContinuousMap.curry (430); ContinuousMap.uncurry (466).

Mathlib/Topology/Compactness/Compact.lean. IsCompact.elim_directed_cover (189); IsCompact.elim_finite_subcover (202); IsCompact.eventually_forall_of_forall_eventually (464); generalized_tube_lemma (752); IsCompact.finite (1060); isCompact_univ_pi (1178).

Mathlib/Topology/Compactness/LocallyCompact.lean. exists_compact_between (172).

Mathlib/Topology/Compactness/Paracompact.lean. ParacompactSpace (63).

Mathlib/Topology/Connected/Basic.lean. IsPreconnected.union (120); IsPreconnected.subset_of_closure_inter_subset (434).

Mathlib/Topology/Connected/Clopen.lean. isClopen_iff (114); IsClopen.eq_univ (122).

Mathlib/Topology/Connected/PathConnected.lean. Joined (62); Joined.somePath (70); PathConnectedSpace.somePath (570).

Mathlib/Topology/Constructions.lean. continuous_quot_lift (726); continuous_sigma (1299).

Mathlib/Topology/ContinuousMap/Algebra.lean. ContinuousMap.evalCLM (623).

Mathlib/Topology/ContinuousMap/StoneWeierstrass.lean. ContinuousMap.exists_mem_subalgebra_near_continuous
(313).

Mathlib/Topology/ContinuousOn.lean. ContinuousOn.union_of_isClosed (879).

Mathlib/Topology/Covering/Basic.lean. IsEvenlyCovered (40); IsEvenlyCovered.discreteTopology_fiber (60); IsEvenlyCovered.of_trivialization (120); IsEvenlyCovered.of_preimage_eq_empty (136); IsCoveringMap (289); IsCoveringMap.eq_of_comp_eq (369); IsCoveringMap.constOn_of_comp (381); IsOpen.trivializationDiscrete (440).

Mathlib/Topology/DiscreteSubset.lean. isDiscrete_iff_forall_mem_exists_isOpen (70).

Mathlib/Topology/EMetricSpace/Defs.lean. PseudoEMetricSpace (110); EMetricSpace.replaceTopology (804).

Mathlib/Topology/FiberBundle/Basic.lean. FiberBundle (186); FiberBundle.extend (906); FiberBundle.extend_apply_self (912).

Mathlib/Topology/FiberBundle/Trivialization.lean. Trivialization (377).

Mathlib/Topology/Homeomorph/Lemmas.lean. Continuous.homeoOfEquivCompactToT2 (476).

Mathlib/Topology/Homotopy/Basic.lean. ContinuousMap.Homotopy (77); ContinuousMap.Homotopic (333); ContinuousMap.HomotopyWith (385); ContinuousMap.HomotopyRel (540); ContinuousMap.HomotopyRel.eq_fst (549); ContinuousMap.HomotopyRel.fst_eq_snd (555); ContinuousMap.HomotopyRel.trans (589); ContinuousMap.HomotopyRel.compContinuousMap (618); ContinuousMap.HomotopicRel (628).

Mathlib/Topology/Homotopy/Contractible.lean. ContinuousMap.Nullhomotopic (26); ContractibleSpace (49); ContractibleSpace.hequiv_unit (52); contractible_iff_id_nullhomotopic (62); Homeomorph.contractibleSpace (87); ContractibleSpace.hequiv (114).

Mathlib/Topology/Homotopy/Equiv.lean. ContinuousMap.HomotopyEquiv (43); HomotopyEquiv (43); Homeomorph.toHomotopyEquiv (75).

Mathlib/Topology/Homotopy/HomotopyGroup.lean. Cube.boundary (58); LoopSpace (85); GenLoop (96); GenLoop.boundary (139); GenLoop.const (143); GenLoop.congr (158); GenLoop.Homotopic (232); GenLoop.Homotopic.refl (240); GenLoop.transAt (401); GenLoop.symmAt (416); HomotopyGroup (440); HomotopyGroup.Pi (457); HomotopyGroup.pi0EquivZerothHomotopy (488); HomotopyGroup.pi1EquivFundamentalGroup (529); HomotopyGroup.group (546); HomotopyGroup.transAt_indep (569); HomotopyGroup.one_def (582); HomotopyGroup.mul_spec (586); HomotopyGroup.inv_spec (594); HomotopyGroup.commGroup (602); HomotopyGroup.pi1MulEquivFundamentalGroup (623).

Mathlib/Topology/Homotopy/Lifting.lean. IsLocalHomeomorph.continuous_lift (127); IsCoveringMap.liftPath (259); IsCoveringMap.eq_liftPath_iff' (272); IsCoveringMap.liftPath_const (276); IsCoveringMap.liftHomotopy (299); IsCoveringMap.homotopicRel_iff_comp (356); IsCoveringMap.liftPath_apply_one_eq_of_homotopicRel (370); IsCoveringMap.existsUnique_continuousMap_lifts (470); IsAddQuotientCoveringMap.fundamentalGroupEquiv (762).

Mathlib/Topology/Homotopy/Path.lean. ContinuousMap.Homotopy.evalAt (452).

Mathlib/Topology/LocallyFinite.lean. LocallyFinite.continuous (111).

Mathlib/Topology/Maps/Basic.lean. IsInducing.continuous_iff (124); IsClosedEmbedding.of_continuous_injective_isClosedMap (842).

Mathlib/Topology/MetricSpace/Contracting.lean. ContractingWith.fixedPoint (274); ContractingWith.fixedPoint_isFixedPt (278).

Mathlib/Topology/MetricSpace/HausdorffDimension.lean. ContDiff.dimH_range_le (554); ContDiff.dense_compl_range_of_finrank_lt_finrank (571).

Mathlib/Topology/MetricSpace/Pseudo/Basic.lean. finite_cover_balls_of_compact (265).

Mathlib/Topology/MetricSpace/Pseudo/Lemmas.lean. lebesgue_number_lemma_of_metric (120).

Mathlib/Topology/MetricSpace/Sequences.lean. `tendsto_subseq_of_bounded` (38).

Mathlib/Topology/MetricSpace/Thickening.lean. `IsCompact.exists_cthickening_subset_open` (443).

Mathlib/Topology/Neighborhoods.lean. `isOpen_setOfPred_eventually_nhds` (242).

Mathlib/Topology/OpenPartialHomeomorph/Defs.lean. `OpenPartialHomeomorph` (53).

Mathlib/Topology/OpenPartialHomeomorph/IsImage.lean. `OpenPartialHomeomorph.restrOpen` (216).

Mathlib/Topology/Order/Compact.lean. `IsCompact.exists_isMinOn` (230); `IsCompact.bddAbove_image` (334).

Mathlib/Topology/Order/IntermediateValue.lean. `IsClosed.Icc_subset_of_forall_exists_gt` (332); `IsClosed.Icc_subset_of_forall_mem_nhdsWithin` (408); `intermediate_value_Icc` (559).

Mathlib/Topology/Order/IsLUB.lean. `exists_seq_tendsto_sSup` (329).

Mathlib/Topology/Order/OrderClosed.lean. `Continuous.if_le` (678).

Mathlib/Topology/PartitionOfUnity.lean. `PartitionOfUnity.continuous_finsum_smul` (271); `PartitionOfUnity.exists_isSubordinate` (622).

Mathlib/Topology/Path.lean. `Path` (58); `Path.refl` (140); `Path.symm` (150); `Path.continuous_uncurry_iff` (182); `Path.extend` (189); `Path.trans` (274).

Mathlib/Topology/SeparatedMap.lean. `T2Space.isSeparatedMap` (69); `discreteTopology_iff_locallyInjective` (160); `IsSeparatedMap.eqOn_of_comp_eqOn` (209).

Mathlib/Topology/Separation/Hausdorff.lean. `Set.Finite.t2_separation` (110); `Set.EqOn.of_subset_closure` (529); `Continuous.isClosedEmbedding` (677).

Mathlib/Topology/Sequences.lean. `IsCompact.tendsto_subseq` (298).

Mathlib/Topology/UniformSpace/Cauchy.lean. `cauchySeq_tendsto_of_complete` (433).

Mathlib/Topology/UniformSpace/HeineCantor.lean. `CompactSpace.uniformContinuous_of_continuous` (38); `IsCompact.uniformContinuousAt_of_continuousAt` (58).

Mathlib/Topology/UnitInterval.lean. `unitInterval` (32); `unitInterval.symm` (75).

Mathlib/Topology/VectorBundle/Riemannian.lean. `RiemannianBundle` (421).

../Milnor/Chapter1/InverseFunction.lean. `hasFDerivAt_writtenInExtChartAt` (82); `eventually_isInvertible_fderiv` (95); `isLocalDiffeomorphAt_of_isInvertible_mfderiv` (119); `ContMDiff.isLocalDiffeomorphAt_of_isInvertible_mfderiv` (254).

../Milnor/Chapter1/RegularValue.lean. `IsCriticalPoint` (69); `Milnor.IsCriticalPoint` (69); `IsRegularValue` (74).

../Milnor/Chapter2/Brown.lean. `critPoints` (48).

../Milnor/Chapter2/Corestrict.lean. `IsSliceSubmanifold.contMDiff_codRestrict` (41); `contMDiff_codRestrict` (41).

../Milnor/Chapter2/LevelSet.lean. `isSliceSubmanifold_preimage` (251); `injective_mfderiv_val` (282).

../Milnor/Chapter2/Submanifold.lean. `SliceChart` (107); `IsSliceSubmanifold` (116); `SliceChart.symm_mem` (125); `SliceChart.mk_zero_mem_target` (138); `SliceChart.toChart` (146);

SliceChart.toChart_symm_apply (187); IsSliceSubmanifold.chartedSpace (197); IsSliceSubmanifold.mem_atlas_iff (206); IsSliceSubmanifold.isManifold (236); IsSliceSubmanifold.contMDiff_val (247).

../Milnor/Chapter2/WithBoundary.lean. isHalfSliceSubmanifold_of_regularValue (96).

../Milnor/Chapter3/Basic.lean. Sard.critSet (51).

../Milnor/Chapter3/Sard.lean. sard (112).

../Milnor/Chapter5/HairyBall.lean. sphG (30); Sphere.inner_self_eq_one_of_mem (49); preimage_eq_sphere (60); Sph (81); Sphere.Sph (81).

../Milnor/Chapter5/Orient.lean. mfd (35).

../Milnor/Chapter6/ChartIndex.lean. Tub.tanN (118); dret_mem_tanN (154); dret_dret (196); ker_dret (220); dvr (265); dvr_mem_tanN (298); dvT (421); det_fderiv_w₀ (439); Tub.indexN_eq_sgn_det (790); indexN_eq_sgn_det (790).

../Milnor/Chapter6/EulerSphere.lean. Sphere.sphSub (147); sphSub (147); isCompact_sph (154); Sphere.mem_range_mfd_val_iff (186); dvr_gradSph_apply (430); indexSum_sph (579).

../Milnor/Chapter6/Extend.lean. exists_extension (23).

../Milnor/Chapter6/Hadamard.lean. contDiff_intervalIntegral (115); contDiff_intervalIntegral_infty (139); exists_hadamard (148).

../Milnor/Chapter6/Nondegenerate.lean. Sphere.exists_radius_of_nondegenerate (69).

../Milnor/Chapter6/PoincareHopf.lean. Tub.indexN (88); indexN (88); Tub.poincare_hopf (114); poincare_hopf (114).

../Milnor/Chapter6/Tubular1.lean. Tub.σ (26); Tub.mem_source (29); Tub.φ (39); φ (39).

../Milnor/Chapter6/Tubular3.lean. Tub.z₀ (38); z₀ (38); Tub.tubular (146); tubular (146).

../Milnor/Chapter6/Tubular4.lean. Tub.tubε (28); Tub.tubr (33); tubr (33); Nε (63); Tub.Nε (63).

../Milnor/Chapter8/Approx.lean. Problems.exists_contDiff_approx_of_isCompact (59).

../Milnor/Chapter8/FramedOps.lean. isSliceSubmanifold_of_locally_eq (89).

../Milnor/Chapter8/LocalSlice.lean. IsSliceSubmanifold.of_locally_preimage (104); IsSliceSubmanifold.range_mfderiv_val_eq_ker (138).

../Milnor/Chapter8/NormalBundle.lean. normalBundleSet (75); NormalAux.B (105); NormalAux.g (113); NormalAux.mem_normalBundleSet_iff (124); NormalAux.fderiv_g_apply (199); NormalAux.surjective_fderiv_g (206); isSliceSubmanifold_normalBundle (218); normalSum (265); contMDiff_normalSum (269).

../Milnor/Chapter8/TangentBundle.lean. Problems.sliceChart_contDiffOn (70); sliceChart_contDiffOn (70); Problems.tangentBundleSet (133); tangentBundleSet (133); Problems.sliceChart_mem_tangentSpace_iff (138); sliceChart_mem_tangentSpace_iff (138); Problems.isSliceSubmanifold_tangentBundle (161); isSliceSubmanifold_tangentBundle (161).